

Tempus Text Command Reference

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About This Manual

The Cadence® Tempus™ Timing Solution provides a sign-off timing and signal integrity solution for a design flow. This manual describes the syntax for the Tempus Timing Signoff Solution text commands.

Audience

This manual is written for experienced designers of digital integrated circuits. Such designers must be familiar with design planning, placement and routing, block implementation, chip assembly, and design verification. Designers must also have a solid understanding of UNIX and Tcl/Tk programming.

Conventions Used in This Manual

This section describes the typographic and syntax conventions used in this manual.

<i>text</i>	Indicates text that you must type exactly as shown. For example: <code>report_annotated_check -missing_setup</code>
<i>text</i>	Indicates information for which you must substitute a name or value. In the following example, you must substitute the name of a specific file for <i>worst_entries</i> : <code>report_clock_timing -type skew -nworst worst_entries</code>
<i>text</i>	Indicates the following: • Text found in the graphical user interface (GUI), including form names, button labels, and field names • Terms that are new to the manual, are the subject of discussion, or need special emphasis • Titles of manuals
[]	Indicates optional arguments. In the following example, you can specify none, one, or both of the bracketed arguments: <code>command [-arg1] [-arg2 value]</code>

[]	Indicates an optional choice from a mutually exclusive list. In the following example, you can specify any of the arguments or none of the arguments, but you cannot specify more than one: <pre>command [arg1 arg2 arg3 arg4]</pre>
{ }	Indicates a required choice from a mutually exclusive list. In the following example, you must specify one, and only one, of the arguments: <pre>command {arg1 arg2 arg3}</pre>
[[] []]	Indicates a required choice of one or more items in a list. In the following example, you must choose one argument from the list, but you can choose more than one: <pre>command {[arg1] [arg2] [arg3]}</pre>
{ }	Indicates curly braces that must be entered with the command syntax. In the following example, you must type the curly braces: <pre>command arg1 {x y}</pre>
...	Indicates that you can repeat the previous argument.
.	Indicates an omission in an example of computer output or input.

Related Documents

For more information about Tempus, see the following documents. You can access these and other Cadence documents with the Cadence Help online documentation system.

- [New Features in Tempus](#)
Provides information about new features in this release of Tempus.
- [Tempus Known Problems and Solutions](#)
Describes important Cadence Change Requests (CCRs) for Tempus, including solutions for working around known problems.
- [Tempus User Guide](#)
Provides information about using various Tempus features.
- [Tempus Menu Reference](#)
Describes Tempus graphical user interface.

- [*Tempus Foundation Flows User Guide*](#)
Describes the Cadence-recommended procedures for performing basic STA and SI signoff, MMC signoff, and CPF-based MMC signoff using the Tempus Timing Solution.
- README file
Contains installation, compatibility, and other prerequisite information, including a list of Cadence change requests (CCRs) that were resolved in this release. You can read this file online at downloads.cadence.com.

System Commands - Basic Tool-Tcl Commands

- [find_global](#)
- [getenv](#)
- [help](#)
- [man](#)
- [Puts](#)
- [redirect](#)
- [set_limited_access_feature](#)
- [setenv](#)
- [start_gui](#)
- [stop_gui](#)
- [tempus](#)

find_global

```
find_global  
[-help]  
[pattern]  
[[-tcl] | [[-name] [-value] [-type] [-default] [-info ] [-non-default] [-range_or_enum ] [-  
transient]]]
```

Returns all the public variables. If you want to retrieve specific global variables, you can use patterns with this command.

Parameters

-help	Outputs a brief description that includes type and default information for each <code>find_global</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man find_global</code> .
<i>pattern</i>	Specifies a string with wildcards that can be used to retrieve specific global variables.
-default	Displays the default value of variables. If a pattern is specified, it displays the default value for all variables that match the specified pattern.
-info	Displays the description of variables. If a pattern is specified, it displays the descriptions of all variables that match the specified pattern.
-name	Displays the name of all or specified global variables.
-non_default	Returns the list of variables whose values are not the same as their default values.
- range_or_enum	Displays the range or legal enum value list, if they exist, for all or specified global variables.
-tcl	Returns a TCL list that has every field value of the specified variables. If a pattern is not specified, it displays this information for all global variables.
-transient	Displays whether the specified variable is transient. If a pattern is not specified, it displays this information for all global variables.
-type	Displays the type (<code>Boolean</code> , <code>Integer</code> , <code>String</code> , <code>Enum</code>) of all or specified variables.
-value	Displays the current value of all or specified variables.

Examples

- The following command returns all public variables and private variables with a non-default value:

```
find_global
```

- The following command returns all public variables (and private variables with a non-default value) beginning with `timing*pess*`:

```
find_global timing*pess*
```

Output:

```
timing_remove_clock_reconvergence_pessimism 1    # (Boolean, default=0, persistent)
# Enables Clock Path Pessimism Removal (CPPR) analysis during timing analysis
```

- The following command returns information on a specific variable, `lefDefOutVersion`.

```
find_global lefDefOutVersion
```

The output is as follows:

```
lefDefOutVersion ""          # (String, default="", persistent)
# Specify LEF/DEF output version. Possible string values are 5.3, 5.4, 5.5,
# 5.6 and 5.7
```

- The following command returns names and default information for all global variables starting with the pattern `def`.

```
find_global def* -name -default
```

The output is as follows:

```
defHierChar          # (default="")
defInShapeBasedDefFile      # (default=0)
defOutLefVia          # (default=0)
defOutLefNDR          # (default=0)
defOverlapWireReportFileName    # (default="")
defOutPolygonDieArea      # (default=1)
defOutBumpAsInstance      # (default=1)
```

Related Commands

- [set_global](#)
- [get_global](#)

getenv

getenv
variable_name

Searches the system environment for the specified *variable_name* and returns its value. If the variable is not defined in the environment, the software returns an empty string "".

Parameters

<i>variable_name</i>	Indicates the name of the environment variable to be retrieved.
----------------------	---

Examples

- The following command returns the value of the variable DISPLAY:

```
getenv DISPLAY
```

help

help

`[commandName | msg_id | -k keyword | string ]`

Usage: `help [-help] [-k | -cmd | -var ] [<pattern>]`

Displays one of the following types of information based on the specified argument:

- Syntax of a command or global variable
- Summary of the message text based on the specified message ID
- List of commands based on keywords
- List of commands based on a search string

Parameters

`-help`: Prints out the command usage

`-k`: perform keyword search (bool, optional)

`-cmd`: perform command search only (bool, optional)

`-var`: perform variable search only (bool, optional)

`<pattern>`: command pattern, message ID, or keyword

Command Syntax

To display the syntax of a command or a variable, use the following command:

`help commandName`

The command syntax output is written in the log file and displayed in the Tempus console.

Message Summary

To display a summary of the message text for a particular message ID, use the following command:

`help msg_id`

Shows the summary of the message text for the message ID specified.

Keyword Searches

To perform a keyword search, use the following command:

```
help -k keyword
```

Shows a list of commands with the specified keyword.

Note: Keyword-based searches are case-insensitive.

Partial String Searches

To perform a partial string search, use the following command:

```
help *string*
```

Shows all commands containing the specified string. The search results vary depending on whether you use wildcards (*) in the beginning, end, or middle of the string.

Note: Partial string searches are case-insensitive.

Parameters

<code>-cmd</code> <i>commandName</i>	Displays help for the specific command name. If you do not specify a command name, all supported Tempus commands are displayed.
<code>-k</code> <i>keyword</i>	Searches for the specified keyword and displays commands associated with that keyword.
<i>pattern</i>	Specifies command pattern, message ID, or keyword.
<code>-var</code>	Performs variable search only.

Command Order

You can use this command after starting the Tempus software.

Examples

- The following command prints the command syntax for the `save_design` command:

```
help save_design
Usage: save_design
save_design <session_name>
[-rc] [-overwrite]
```

- The following command shows the message summary for the `IMPSYC-2` message ID:

```
help IMPSYC-2
```

Timing information is not defined for cell %s; Check the timing library (.lib) file and make sure the timing information exists for the cell, and you can run the check_library command to verify if the timing library has complete information after the design is loaded.

- The following command shows all commands that include "timing" as a keyword:

```
help -k timing
```

- The following command shows how you can use a wildcard at the beginning and end of the string to get a list of all the commands that contain the string:

```
tempus 5> help *timing*
```

Single match found:

```
Usage: timing_enable_timing_window_pessimism_removal
```

man

man
command_name | *msg_id*

Displays information for a command or variable from the *Tempus Text Command Reference*, or prints detailed information for a specified message ID. If no manual page is located, man prints an error message.

The man pages for Tempus commands and variables are located in the following directory:

<install_dir>/<platform>/share/tempus/man

Parameter

<i>command_name</i>	Indicates the name of the text command (or variable) for which the manual information is required.
<i>msg_id</i>	Specifies the unique message ID for which you want to display detailed information. The output is categorized into three sections: • NAME: Displays the message ID and its severity level. • SUMMARY: Displays the summary of the message text. • DESCRIPTION: Displays detailed information about the message. The detailed information can include the message background, likely causes, and alternative steps to troubleshoot the problem.

Example

- The following command returns the manual description of the command `report_timing`:
`man report_timing`
- The following command returns the manual description of the `ENCSYC-2` message ID:
`man IMPSYC-2`

Puts

Puts
[-nonewline]
string

Equivalent to the TCL procedure puts. Additionally, the software outputs the command to both standard output and the log file.

redirect

```
redirect
[-help]
[file_or_var_name]
[command]
[-tee]
[-stdin | -append | -variable ]
[-variable ]
[-stderr | -stdin]
```

Redirects the output of a command to a file.

Parameters

-help	Prints a brief description that includes type and default information for each redirect parameter.
<i>file_or_var_name</i>	Specifies the name of the file in which all the results are directed.
<i>command</i>	Specifies the command to start. Intermediate output from this command, as well as the result of the command, are redirected to the file specified using <i>file_or_var_name</i> .
-append	Appends stdout/stderr to file or variable.
-stderr	Redirects stderr.
-stdin	Redirects stdin.
-tee	Redirects output to both the screen and file or # variable.
-variable	Redirects to a TCL variable.

Examples

- The following command redirects the output of the command, `report_timing`, to the file named `output.log`:
`tempus > redirect output.log {report_timing}`
- The following command redirects the output of the command, `get_multi_cpu_usage`, to the variable `x`:

```
tempus > redirect x {get_multi_cpu_usage} -variable
```

Here the value of variable `x` would be:

```
tempus > puts $x
Total CPU(s) Enabled: 16
Current License(s): 1 Tempus_Timing_Signoff_XL
keepLicense: true
licenseList: tpsmp tpsl tpsxl
```

set_limited_access_feature

```
set_limited_access_feature  
[-help]  
featureName  
{1 | 0}
```

Enables or disables a limited-access feature.

The Tempus software provides certain limited-access features that showcase advanced technology integrated into the software. These features have been internally qualified at Cadence but have had only limited customer testing.

Note: Limited-access features are enabled by a variable specified through the `set_limited_access_feature` command. To use the limited-access features, you can contact your Cadence representative to qualify your usage and ensure your requirements are met before deploying them widely.

Parameters

<code>-help</code>	Prints a brief description that includes type and default information for each <code>set_limited_access_feature</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <pre>man set_limited_access_feature</pre>
<code>featureName</code>	Specifies the name of the variable that enables the limited access feature.
<code>{1 0}</code>	Enables/disables a feature - 1 enables a feature, 0 disables a feature.

setenv

```
setenv  
variable_name  
value
```

Sets the value of a system environment variable.

Parameters

<i>value</i>	Specifies the value of the variable.
<i>variable_name</i>	Specifies the name of the variable.

start_gui

start_gui

Displays the graphical user interface (GUI) of Tempus.

Parameters

- help	Prints out the command usage. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man start_gui</code>
-----------	---

stop_gui

stop_gui

Closes the Tempus graphical user interface (GUI)

Parameters

- help	Prints out the command usage. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man stop_gui</code>
-----------	--

tempus

```
tempus
[-help]
[-abort_on_error]
[-cds_lib_file in_file]
[-stylus]
[-cpus integer]
[-disable_user_startup]
[-distributed]
[-execute Tcl_commands]
[-files in_file_list]
[-lic_multi_cpu lic_list]
[-lic_options lic_list]
[-lic_startup lic_list]
[-lic_startup_options lic_list]
[-lic_startup_extra lic_list]
[-log prefix]
[-no_gui]
[-no_logv]
[-overwrite]
[-eco]
[-version]
[-wait minutes]
```

Starts the Tempus software. If you do not specify any parameters, the following actions occur:

- A log file and a command file are created in the form `tempus.log#` and `tempus.cmd#`.
- The first available license is checked out.
- The Tempus main window opens.

Parameters

-help	Prints out the command usage. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man tempus</code>
-abort_on_error	Exits if an error is detected at any point. If any Tcl command returns <code>TCL_ERROR</code> to the top level, or any <code>Error (XXX) : ...</code> message occurs, the process will exit.

<code>-cds_lib_file in_file</code>	Loads the specified file instead of the default cds.lib file in the current directory to obtain the names and locations of OpenAccess libraries. The cds.lib file lists the OpenAccess library names and their physical locations.
<code>-cpus integer</code>	Specifies the number of CPUs to use for multi-threading on the current host. This calls the appropriate TCL command to set the local CPU count. This happens before any <code>-execute</code> or <code>-files</code> options are processed. This may require extra multi-CPU licenses to be checked out (using the <code>-lic_multi_cpu</code> option).
<code>- disable_user_startup</code>	Reads only the init file in the installation hierarchy. Does not read init files in the home or current directory. This option suppresses the default loading from the home and current directories, which can be used to guarantee there are no local overrides to various project settings. For more information, see " Initialization Files ".
<code>-distributed</code>	Starts Tempus in the Distributed Static Timing Analysis (DSTA) mode. DSTA supports all the command-line parameters that are supported by non-distributed Tempus, except for the following parameters: • <code>-cds_lib_file</code> • <code>-no_logv</code> • <code>-no_gui</code> This is because DSTA does not support OA, GUI, and verbose log. When you specify these parameters with <code>tempus -distributed</code>, an error message will be displayed.
<code>-execute Tcl_commands</code>	Specifies the Tcl commands to be executed as a quoted string at startup.
<code>-files in_file_list</code>	Specifies one or more Tcl files to be sourced at startup, after the commands specified using the <code>-execute</code> parameter are processed.
<code>-lic_multi_cpu lic_list</code>	Controls which licenses are to be used for higher multi-CPU usage in priority order from left to right. The allowed license names are: • <code>tpsm</code>: Tempus Timing Signoff Solution MP • <code>tps1</code>: Tempus Timing Signoff Solution L • <code>tpsxl</code>: Tempus Timing Signoff Solution XL Note: For information on licenses, see "Product and Licensing Information".

<code>-lic_options</code> <code>lic_list</code>	Specifies the order in which extra option licenses should be checked out if needed. In some cases, more than one option can be used to enable a given feature. You can control which options are allowed to be checked out and, when required during the run, their priority (from left to right). The allowed license names are: • <code>tpstso</code>: Tempus Timing Signoff Solution TSO • <code>tpspi</code>: Tempus Timing Signoff Solution - PI (Power Integrity) Option • <code>tpsng</code>: Tempus Timing Signoff Solution Next Generation • <code>vtssl</code>: Voltus IC Power Integrity Solution • <code>vtsl</code>: Voltus IC Power Integrity Solution • <code>vtcaa</code>: Voltus IC Power Integrity Solution Advanced Analysis GXL Option • <code>invs_ms</code>: Innovus Mixed Signal Option The option licenses can be dynamically checked out when a specific TCL command requires them. This may cause the TCL script to stall and wait for a license, or to exit if the wait period expires. For example, <code>-lic_options "tpstso tpsxl"</code> will only allow these two options to be checked out dynamically, and in the given priority order. Note: For more information, see "Product and Licensing Information".
<code>-lic_startup</code> <code>lic_list</code>	Requests one of these base licenses at startup in priority order from left to right. The supported licenses are: • <code>tpsl</code>: Tempus Timing Signoff Solution L • <code>tpsxl</code>: Tempus Timing Signoff Solution XL • <code>vdsl</code>: Virtuoso Digital Signoff Timing Solution For example, <code>-lic_startup "tpsxl tpsl"</code> will try to check-out <code>tpsxl</code> first and then <code>tpsl</code>. The check-outs will be attempted repeatedly until it succeeds or the specified wait (<code>-wait</code>) period expires.

<pre>-lic_startup_extra lic_list</pre>	Checks out the specified Innovus license to enable saving physical changes. By default, Tempus cannot save physical changes to the database, but if you check out the correct extra Innovus license at startup, you can save physical changes. The supported licenses are: • <code>invs</code> - Innovus Implementation System • <code>invsb</code> - Innovus Implementation System Basic • <code>vdixl</code> - Virtuoso® Digital Implementation XL Note: The <code>vdixl</code> license can only be used with the Tempus base license <code>vdst</code>. For more information, see "Product and Licensing Information".
<pre>-lic_startup_options lic_list</pre>	Specifies additional option licenses needed at startup. If not provided, they will be checked out when needed during the run. The allowed license names are: • <code>tpstso</code>: Tempus Timing Signoff Solution TSO • <code>tpspi</code>: Tempus Timing Signoff Solution - PI (Power Integrity) Option • <code>tpsng</code>: Tempus Timing Signoff Solution Next Generation • <code>vtssl</code>: Voltus IC Power Integrity Solution • <code>vtssl</code>: Voltus IC Power Integrity Solution • <code>vtcaa</code>: Voltus IC Power Integrity Solution Advanced Analysis GXL Option • <code>invs_ms</code>: Innovus Mixed Signal Option For example, <code>-lic_startup_options "tpsl, tpsxl"</code> will try to check-out both <code>tpsl</code> and <code>tpsxl</code> when Tempus starts up. The checkout attempts will be repeated until it succeeds or the specified <code>-wait</code> period expires.

<code>-log prefix</code>	Specifies the prefix for the <code>.log</code>, <code>.cmd</code>, and <code>.logv</code> files. If three arguments are given, such as <code>-log "a b c"</code>, then the three names are used exactly without any extension added. So, <code>-log "mylog mycmd mylogv"</code> will result in <code>mylog mycmd mylogv</code> files. The default for <i>prefix</i> is <code>tempus</code>. So if <code>-log</code> is not specified, <code>tempus.log</code>, <code>tempus.cmd</code>, and <code>tempus.logv</code> files will be created. If you specify <code>-log test</code>, the tool creates <code>test.log</code>, <code>test.cmd</code>, and <code>test.logv</code> files. The <code>.log</code> file contains the normal logging output, the <code>.cmd</code> file contains the Tcl commands that were executed, and the <code>.logv</code> file contains a super-set of the <code>.log</code> output plus "verbose" debugging output that is needed only for detailed debugging. If the <code>.log</code> file already exists, an integer is added to the end of all the files to make them unique. This means if <code>-log test</code> is specified, the tool will create <code>test.log test.cmd test.logv</code> for the first run, <code>test.log1 test.cmd1 test.logv1</code> for the next run, and so on. Only the existence of <code>.log</code> is checked; the existence of the <code>.cmd</code> or <code>.logv</code> files is not checked. The <code>-overwrite</code> option can be used to disable this behavior and overwrite an existing <code>.log</code> file. If the prefix has a period in it, the last extension is stripped off for the <code>.cmd</code> and <code>.logv</code> usage. For example, <code>-log out.log</code> will result in <code>out.log out.cmd out.logv</code> files, and <code>-log out.a.log</code> will result in <code>out.a.log out.a.cmd out.a.logv</code> files. You can disable any given file by using <code>/dev/null</code>. For example, <code>-log "my.log /dev/null /dev/null"</code> will only create <code>my.log</code>. Default: <code>tempus.log</code> Note: If you provide the log file name under a directory path and the directory path does not exist, Tempus will create the directory path and save the log file under the directory.
<code>-no_gui</code>	Disables GUI and Tk support. When you use this parameter, the GUI will not be available. You can launch the GUI any time during the session by issuing the <code>start_gui</code> command.
<code>-no_logv</code>	Specifies that a verbose log file will not be created.
<code>-overwrite</code>	Overwrites an existing <code>.log</code> file (and associated <code>.cmd</code> and <code>.logv</code> files) instead of adding an integer to make the files unique.

<code>-eco</code>	Needed for running the Tempus ECO flow using the <code>eco_opt_design</code> command
<code>-version</code>	Returns the program version information.
<code>-wait <i>minutes</i></code>	Specifies the number of minutes the system waits for a license to become available before exiting. <i>Default:</i> 0 (no wait time) <i>Value Range:</i> 0 to 10,000

Initialization Files

By default, various initialization files are loaded at startup if they exist. They can be used to configure the GUI, load utility Tcl files, or configure Tempus settings. The initialization files are read in the following order:

1. `~/ {.tempus.pref.tcl, .tempus.color.tcl}` in the home directory if the GUI is enabled
2. `./ {tempus.pref.tcl, tempus.color.tcl}` in the current directory if the GUI is enabled
3. `~/ .tempusrc` in the home directory
4. `<install_dir>/etc/ssv/tempus.tcl` in the installation directory
5. `~/ tempus.tcl` in the home directory
6. `./ tempus.tcl` in the current directory

If the `-stylus` parameter is specified, then the initialization files are loaded in the following order instead:

1. `~/ .cadence/tempus/ {gui.pref.tcl, gui.color.tcl, workspaces}` in the home directory if the GUI is enabled
2. `.cadence/tempus/ {gui.pref.tcl, gui.color.tcl, workspaces/}` in the current directory if the GUI is enabled
3. `<install_dir>/etc/ssv/tempus.tcl` in the installation directory
4. `~/ .cadence/tempus/tempus.tcl` in the home directory
5. `./ .cadence/tempus/tempus.tcl` in the current directory
6. `./ tempus.tcl` in the current directory

Setting Up the Run Time Environment

If `install_dir` is the location of the Tempus installation, then you should set up the run time environment as follows:

- Add the `install_dir/bin` directory to the path. The `bin` directory has links to all the public executable files in the install hierarchy.
- If you want the Tempus man pages to be available with the Unix `man` command, then you can add the `install_dir/share/tempus/man` to the `MANPATH` envvar.
- If you want the Tcl man pages to be available with the Unix `man` command, then you can add the `install_dir/share/tcltools/man` to the `MANPATH` envvar.

For example, you can add the following to the startup shell script:

```
set install_dir = /tools/tempus16.2/lxx86
set path = ($install_dir/bin $path)
setenv MANPATH $install_dir/share/tempus/man:$install_dir/share/tcltools/man:$MANPATH
```

Note: When Tempus launches, it automatically adds the legacy or the CUI man pages (if the `-stylus` option is specified), and the TCL man pages at the beginning of the current `MANPATH` inside Tempus. Therefore, from within Tempus, the `man` command will show both the sets of man pages before any other man pages.

Temporary File Locations

Each Tempus session creates its own temporary directory to store temporary files at the beginning of the run.

By default, the `tmp_dir` is created in the `/tmp` directory. If the Unix environment variable (`TMPDIR`) is set, then `tmp_dir` is created inside `$TMPDIR`.

The name of the `tmp_dir` will appear as given below:

```
tempus_temp_[pid]_[hostname]_[user]_xxxxxx
```

Where `_xxxxxx` is a string that is added to make the directory unique. For example:

```
tempus_temp_10233_farm254_bob_nfp9ez
```

The temporary directory is automatically removed on exit or if the run is terminated with a catchable signal (e.g., `SIGSEGV`).

Examples

- The following command starts the Tempus software with a log file named `myLog.log` and a command log file named `myCmdLog.log`:

```
tempus -log myLog.log -cmd myCmdLog.log
```
- The following command starts Tempus and enables the Tempus ECO feature of the software:

```
tempus -eco
```
- The following command launches Tempus and executes the script, `myScript.tcl`, upon starting:

```
tempus -int myScript.tcl
```

Global Configuration Commands

- [get_global](#)
- [get_var](#)
- [report_globals](#)
- [set_global](#)
- [set_var](#)

get_global

```
get_global  
global_variable_name
```

Retrieves the current value of the specified timing global variable. To set the value of a timing global variable, use the [set_global](#) command.

For a list of timing global variables, see [Analysis Global Variables](#).

Example

The following command retrieves the setting for the [timing_cpnr_threshold_ps](#) global.

```
get_global timing_cpnr_threshold_ps
```

get_var

get_var
[-cte]
pattern

Prints the values of one or more variables. The command and output are echoed to the log file.

Parameters

-cte	Prints the value of the timing global variable. This option is equivalent to using the <code>get_global</code> command.
<i>pattern</i>	Prints the variable names that match the specified pattern.

Example

- The following command prints the value of the variable `timing_crpr_threshold_ps`:
`get_var -cte timing_crpr_threshold_ps`

report_globals

```
report_globals  
[-help]  
[pattern]  
[-add global_name_list]
```

Generates a list of the officially supported global variables, with their current and default values.

Note: The `report_globals` command is now obsolete. This command will continue to be supported in the current release, but will be removed in the next major release of the software. To search for global variables, you can use the `find_global` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-add</code> <code><i>global_name_list</i></code>	Adds the specified global variables to the generated list of supported global variables. When you use this parameter, the global variables are included in the generated list, along with their registered and current values.
<code><i>pattern</i></code>	Reports the global variables that match the specified pattern. Pattern matching can be used for applications as well as user-specified global variables (registered using the <code>report_globals -add</code> parameter).

Related Commands

- `find_global`

set_global

```
set_global  
global_variable_name  
value
```

Sets the specified timing global variable to a specified value. To retrieve the current value of a timing global variable, use the [get_global](#) command.

For a list of timing global variables, see [Global Configuration Variables](#).

Example

- The following command sets the value of the `timing_cppr_threshold_ps` global variable to 25:

```
set_global timing_cppr_threshold_ps 25
```

Related Commands

- [get_global](#)
- [find_global](#)

set_var

```
set_var  
variable_name  
value
```

Sets the values of one or more variables.

Parameters

<i>value</i>	Specifies the value of the variable.
<i>variable_name</i>	Specifies the name of the variable.

The following global variables can be set using the `set_var` command:

- `load_netlist_ignore_undefined_cell`
- `distributed_child_license_checkout_list`
- `enable_wire_load_model_support`
- `search_path`

Global Configuration Variables

- [General Global Variables](#)
- [Analysis Global Variables](#)
- [AOCV Global Variables](#)
- [SOCV Global Variables](#)
- [Timer Global Variables](#)
- [Library Global Variables](#)
- [Reporting Global Variables](#)
- [Constraint Global Variables](#)
- [Timing Model Global Variables](#)
- [Paradime ECO Global Variables](#)
- [Delay Calculation Global Variables](#)
- [Import and Export Global Variables](#)

General Global Variables

Using General Globals

This chapter describes the general global variables that can be set to change the default behavior in Tempus. You can set general global variables as follows:

```
set_var variable_namevalue
```

For example:

```
tempus > set_var load_netlist_ignore_undefined_cell 1
```

Global Descriptions

- [auto_file_dir](#)

- `auto_file_prefix`
- `distributed_child_license_checkout_list`
- `enable_legacy_metric`
- `enable_wire_load_model_support`
- `enc_source_continue_on_error`
- `enc_source_echo_filename`
- `enc_source_verbose_cmd_print_length_limit`
- `enc_tcl_error_info`
- `enc_tcl_report_header_with_value`
- `gui_verbose`
- `jedai_catalog_token`
- `jedai_catalog_url`
- `load_netlist_ignore_undefined_cell`
- `metric_capture_reg2reg_metrics`
- `metric_capture_vth_per_power_domain`
- `metric_enable`
- `restore_db_directory`
- `restore_db_file_check`
- `restore_db_stop_at_design_in_memory`
- `restore_db_tool`
- `script_search_path`
- `soft_stack_size_limit`
- `enc_resolve_show_tcl_command_with_value`

auto_file_dir

`auto_file_dir` *directoryName*

Default: `./`

Data_type: String

Represents the top-level directory to store files/sub-directories that are auto-generated in a session by the software.

When specified, the software redirects or stores files/sub-directories auto-generated by the software to the specified location. This allows multiple simultaneous runs from the same run directory to direct auto-generated files to user-specified directories to avoid over-writing of such files.

If a user-specified directory does not exist, it will be created automatically. In case you do not have the permission to access the specified directory, or create a new directory at the specified location, then the software will use the default value of the variable and display an error message.

To set this global variable, use the `set` command.

Related Global Variables

- `auto_file_prefix`

Related Topics

- "[Design Import](#)" chapter in the *Tempus User Guide*:

Other Reference Information

- Click [here](#) for more information on this global variable.

auto_file_prefix

`auto_file_prefix` *string*

Default: ""

Data_type: String

Represents the prefix to be applied to all files/sub-directories that are auto-generated in a session by the software.

When specified, the software adds a prefix to the files. For example, a default name like "CTE_loops.rpt" will change to "<prefix>CTE_loops.rpt". These files are created by Tempus automatically and are retained after the run completes. This includes report files or other output files with default names that are auto-generated in the current run directory.

To set this global variable, use the `set` command.

Related Global Variables

- `auto_file_dir`

Related Topics

- "Design Import" chapter in the *Tempus User Guide*:
 - [Auto-Generated Files Management](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

distributed_child_license_checkout_list

`distributed_child_license_checkout_list {tpsl tpsxl vdst}`

Controls license that is being checked out by the Distributed MMMC client. This enables Distributed MMMC design analysis to utilize advanced features.

To set this variable, use the `set_var` command.

Default: `tpsl tpsxl vdst`

Data_type: String

Related Topics

- "[Product and Licensing Information](#)" chapter in the *Tempus User Guide*:
 - [Tempus Timing Solution Product Options](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

enable_legacy_metric

`enable_legacy_metric {true | false}`

Default: `false`

Data_type: Boolean

Enables the legacy `get_metric` command. If set to `true`, the `get_metric` command used is old legacy.

Related Commands

- `get_metric`

Other Reference Information

- Click [here](#) for more information on this global variable.

enable_wire_load_model_support

`enable_wire_load_model_support {1 | 0}`

Enables the wire load model capability, when set to 1. If set to 0, the wire load model capability will be disabled.

To set this variable, use the `set_var` command.

Default: 1

Data_type: Boolean

Other Reference Information

- Click [here](#) for more information on this global variable.

enc_source_continue_on_error

`enc_source_continue_on_error value`

Default: 0

Data_type: Boolean

Enables/disables the 'source' to continue the script on TCL_ERROR. If `true`, the `source` command will ignore Tcl errors, and continue processing the script files. If `false`, then the Tcl error will halt the script.

Related Commands

- [source](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

enc_source_echo_filename

`enc_source_echo_filename`
{true | false}

Default: false

Data_type: Boolean

When set to `true`, the log file will display the file name being sourced before the actual sourcing of the file. By default, the file name of the file being sourced is not displayed in the log file.

Related Commands

- [source](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

enc_source_verbose_cmd_print_length_limit

`enc_source_verbose_cmd_print_length_limit value`

Default: -1

Data_type: Integer

Limits the command printed to stdout and the .log file to the specified number of characters for verbose script source.

enc_tcl_error_info

```
enc_tcl_error_info {true | false}
```

Default: false

Data_type: Boolean

When set to true, `enc_tcl_error_info` prints the error information if there is an issue in the Tcl script.

enc_tcl_report_header_with_value

`enc_tcl_report_header_with_value {true | false}`

Default: `false`

Data_type: Boolean

When set to `true`, the software will print the TCL variable value in the report header.

Other Reference Information

- Click [here](#) for more information on this global variable.

gui_verbose

`gui_verbose {all db none}`

Default: `db`

Data_type: Enum

Controls which GUI actions are logged. You can select:

- `all`: Logs all GUI actions.
- `db`: Logs the GUI actions that affect the DB, such as `gui_select`, but does not capture display-only actions such as `gui_dim_foreground`.
- `none`: Does not log any GUI actions. So, the `.log` and `.cmd` files are incomplete.

jedai_catalog_token

`jedai_catalog_token` *token*

Type: string

Default: ""

Authenticates and provides access to the OpenAI services.

jedai_catalog_url

```
jedai_catalog_url <jedai_server_url>
```

Default: ""

Data_type: string

Enables the AI chat box in the Tempus GUI. You need to set the following setting in `enc.tcl`.

```
#set server url  
set jedai_catalog_url <server url>
```


load_netlist_ignore_undefined_cell

```
load_netlist_ignore_undefined_cell { true | false }
```

Default: false

Data_type: Boolean

Creates a dummy .lib file and a user-defined noise model (UDN) for cells that are not present in the loaded .lib or cdB. The dummy .lib file is used for timing analysis, and the dummy UDN file is used for signal integrity analysis. To generate the dummy .lib and UDN, set this variable to 1.

This global variable can be used for non CDB flows also.

To set this variable, use the `set_var` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

metric_capture_reg2reg_metrics

metric_capture_reg2reg_metrics {true | false}

Default: false

Data_type: Boolean

Forces the capture of reg2reg metrics regardless of path groups.

metric_capture_vth_per_power_domain

`metric_capture_vth_per_power_domain {true | false}`

Default: false

Data_type: Boolean

Captures vth metrics per power domain.

metric_enable

`metric_enable {true | false}`

Default: `false`

Data_type: Boolean

Enables the metric snapshot capture.

Other Reference Information

- Click [here](#) for more information on this global variable.

restore_db_directory

`restore_db_directory` *directory_name*

Data_type: `string`

Default: ""

Specifies the absolute path of the database, that is used to restore the session. In case of a non-restored session, this variable returns a blank value.

The value of this global variable is set internally by the software.

Other Reference Information

- Click [here](#) for more information on this global variable.

restore_db_file_check

`restore_db_file_check {true | false}`

Default: `true`

Data_type: Boolean

By default, `read_design` will check all the files inside the saved DB directory. If any file is edited, renamed, or deleted, the tool will give an error. However, `read_design` will not perform this check if `restore_db_file_check` is set to `false`.

Other Reference Information

- Click [here](#) for more information on this global variable.

restore_db_stop_at_design_in_memory

```
restore_db_stop_at_design_in_memory {true | false}
```

Data_type: Boolean

Default: true

By default, tool gives error and stops if you call [read_design](#) more than once in the same session unless `restore_db_stop_at_design_in_memory` is set to 0. Using `read_design` twice in one session is not recommended as settings from the first database may not be cleaned up properly and can cause problems in the second database.

restore_db_tool

restore_db_tool

Data_type: String

Default: ""

Indicates the name of the tool that created the restored db on the disk database.

Other Reference Information

- Click [here](#) for more information on this global variable.

script_search_path

script_search_path

Default: "."

Data_type: String

Provides a set of directories for the software to search for files that are sourced using the Tcl [source](#) command. The software searches each search path directory for the specified file.

The default value represents the current application run directory.

Other Reference Information

- Click [here](#) for more information on this global variable.

soft_stack_size_limit

soft_stack_size_limit

Default: 8

Data_type: String

Specifies the soft stack size limit in mbytes. When the tool is launched, the soft stack size limit is auto-enlarged to 0.2% RAM. If 0.2%RAM is larger than the hard limit, the tool sets it to the hard limit.

Other Reference Information

- Click [here](#) for more information on this global variable.

enc_resolve_show_tcl_command_with_value

```
enc_resolve_show_tcl_command_with_value {true | false}
```

Type: boolean

Default: false

When set to `true`, it prints the values of Tcl variables in the resolve log.

Analysis Global Variables

- `distributed_mmmc_disable_reports_auto_redirection`
- `distributed_sync_master_client`
- `lib_build_async_de_assert_arc`
- `timing_all_registers_filter_clock_pins_by_clock`
- `timing_all_registers_include_icg_cells`
- `timing_allow_input_delay_on_clock_source`
- `timing_analysis_precision_ps`
- `timing_apply_check_derate_to_external_output_delay`
- `timing_apply_default_primary_input_assertion`
- `timing_apply_exceptions_to_data_check_related_pin`
- `timing_apply_setup_hold_exceptions_to_data_check_related_pin`
- `timing_cap_unit`
- `timing_case_analysis_for_icg_propagation`
- `timing_case_analysis_for_sequential_propagation`
- `timing_case_analysis_propagation`
- `timing_check_timing_report_all_checks`
- `timing_clock_phase_propagation`
- `timing_clock_source_use_driving_cell`
- `timing_clock_uncertainty_from_to_precedence`
- `timing_collection_all_fanin_fanout_traversal_mode`
- `timing_collection_result_display_limit`
- `timing_collection_variable_assignment_compatibility`
- `timing_consider_late_arrival_for_ssi_derate`
- `timing_constraint_disable_min_max_input_delay_worst_casing`
- `timing_context_apply_default_primary_input_assertion`
- `timing_context_apply_port_sdc_exceptions`
- `timing_context_clock_mapping_check_waveform`
- `timing_context_clock_mapping_percentage_tolerance`

- `timing_context_clock_phase_based_clock_mapping`
- `timing_context_continue_on_invalid_module_list`
- `timing_context_data_phase_based_clock_mapping`
- `timing_context_port_based_clock_mapping`
- `timing_continue_on_error`
- `timing_cppr_opposite_edge_sigma_scale_factor_cell`
- `timing_cppr_opposite_edge_sigma_scale_factor_net`
- `timing_cppr_propagate_thru_latches`
- `timing_cppr_remove_clock_to_data_crp`
- `timing_cppr_self_loop_mode`
- `timing_cppr_skip_clock_reconvergence`
- `timing_cppr_skip_clock_reconvergence_for_unmatched_clocks`
- `timing_cppr_threshold_ps`
- `timing_cppr_transition_sense`
- `timing_create_clock_default_propagated`
- `timing_create_clock_use_ideal_slew`
- `timing_default_opcond_per_lib`
- `timing_defer_mmmc_object_updates`
- `timing_derate_aocv_dynamic_delays`
- `timing_derate_aocv_reference_point`
- `timing_derate_dynamic_compatibility`
- `timing_derate_incremental_adjust_additive_mode`
- `timing_derate_negative_delay_backward_compatibility`
- `timing_derate_ocv_reference_point`
- `timing_derate_spatial_distance_unit`
- `timing_derate_voltage_scaling_mode`
- `timing_disable_black_box_endpoints`
- `timing_disable_constant_propagation_for_sequential_cells`
- `timing_disable_get_clocks_for_current_instance`
- `timing_disable_model_parasitics`

- `timing_disable_pulsewidth_same_edge_si_cpnr_mode`
- `timing_driving_cell_override_library`
- `timing_drv_analysis_ignore_latency_phases`
- `timing_dynamic_loop_breaking`
- `timing_enable_genclock_source_path_register_limit`
- `timing_enable_get_objects_regexp_compatibility`
- `timing_enable_hier_context_unmapped_clock_analysis`
- `timing_enable_hierarchical_get_nets_support`
- `timing_enable_individual_distance_inter_die_analysis`
- `timing_enable_latch_thru_mode`
- `timing_enable_latency_through_clock_gating`
- `timing_enable_model_constraint_warnings`
- `timing_enable_path_delay_to_unconstrained_endpoints_compatibility`
- `timing_enable_spatial_derate_mode`
- `timing_enable_vtskew_derate_mode`
- `timing_enable_wire_variation`
- `timing_enable_zero_delay_analysis_mode`
- `timing_full_context_flow`
- `timing_generate_normalized_driver_waveform`
- `timing_generated_clocks_inherit_ideal_latency`
- `timing_get_of_objects_hier_compatibility`
- `timing_hier_object_name_compatibility`
- `timing_hierarchical_context_continue_on_top_block_clock_mismatch`
- `timing_ignore_clock_based_path_delay_on_data_checks`
- `timing_ignore_lumped_rc_assertions`
- `timing_io_use_clock_network_latency`
- `timing_ipd_allow_paths_for_asynchronous_clocks`
- `timing_ipd_xdomain_path_filtering_threshold`
- `timing_max_transition_use_si_transition`
- `timing_model_netlist_exclude_dummy`

- `timing_multifrequency_clock_rounding_factor`
- `timing_normalized_driver_waveform_clip_linear_part`
- `timing_normalized_driver_waveform_weight_factor`
- `timing_null_collection_return_compatibility`
- `timing_path_based_enable_bounding_box_for_io_paths`
- `timing_path_based_enable_closest_common_pin_driver_for_bounding_box`
- `timing_path_based_report_analysis_summary_max_paths_limit`
- `timing_pba_exhaustive_path_nworst_limit`
- `timing_pll_clock_use_driving_cell`
- `timing_prefix_module_name_with_library_genclk`
- `timing_propagate_latch_data_uncertainty`
- `timing_property_arrival_clocks_consider_clock_source_data_phase`
- `timing_property_arrival_clocks_consider_data_phase`
- `timing_property_arrival_clocks_consider_latency_phase`
- `timing_property_clock_used_as_data_unconstrained_clock_source_paths`
- `timing_rcdb_allow_mismatch_option`
- `timing_recompute_sdf_in_setuphold_mode`
- `timing_reduce_multi_drive_net_arcs`
- `timing_reduce_multi_drive_net_arcs_threshold`
- `timing_remove_clock_reconvergence_pessimism`
- `timing_report_clock_pin_as_begin_point`
- `timing_report_constraint_extended_cell_em_flow`
- `timing_resolve_driver_conflicts`
- `timing_sdf_adjust_negative_setuphold`
- `timing_sdf_enable_setuphold_scond_ccond`
- `timing_self_loop_paths_no_skew`
- `timing_self_loop_paths_no_skew_max_depth`
- `timing_self_loop_paths_no_skew_max_slack`
- `timing_set_clock_source_to_output_as_data`
- `timing_set_scaling_for_negative_checks`

- `timing_set_scaling_for_negative_delays`
- `timing_socv_sigma_spatial_multiplier`
- `timing_socv_view_based_nsigma_multiplier_mode`
- `timing_spatial_derate_chip_size`
- `timing_spatial_derate_distance_mode`
- `timing_suppress_escape_characters`
- `timing_suppress_ilm_constraint_mismatches`
- `timing_time_unit`
- `timing_timing_window_pessimism_removal_include_si_delay`
- `timing_use_clock_pin_attribute_for_clock_net_marking`
- `timing_use_latch_early_launch_edge`
- `timing_use_latch_time_borrow`
- `timing_use_verilog_for_model_netlist`
- `timing_write_sdf_no_escape_backslash_control`
- `write_global_slack_worst_trigger_path_on_clocks`

distributed_mmmc_disable_reports_auto_redirection

distributed_mmmc_disable_reports_auto_redirection
{true | false}

Default: false

Data_type: String

Allows reports to be generated in the specified location.

When set to `true`, the software does not allow the reports for different views to get overwritten. When set to `false`, the reports are generated automatically in the DMMMC view directory structure.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

distributed_sync_master_client

`distributed_sync_master_client {true | false}`

Default: `true`

Data_type: Boolean

Enables synchronous execution of commands on the Tempus master and clients in the Paradime flow (when `set_distribute_mmmc_mode -eco` parameter is set to `true`).

When set to `false`, the commands will run only on the Tempus master.

Some of the commands that are handled in the Paradime flow, such as, `set_delay_cal_mode`, `change_cell`, `add_repeater`, can have an impact on the design settings and netlist connectivity. This global variable affects only those commands that are executed both on the master and clients, when the commands are issued on the master.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

lib_build_asynch_de_assert_arc

```
lib_build_asynch_de_assert_arc  
{true | false}
```

Default: true

Data_type: Boolean

Controls whether input to output arcs from the preset or clear pins transitioning to inactive state are included when the timing system is initialized.

When the control signal to the preset or clear pins transitions to inactive state (deassertion), in some cases, the output pin can subsequently transition. In the case of edge-triggered flip-flops, the `removal` timing check (a type of hold check) prevents the preset or clear from deasserting too close to the active edge of the clock; therefore, there is usually no chance of the output changing, and no need to have an additional arc to model this transition.

However, in the case of latches, it is possible that the `removal` constraint will still allow the signal to deassert when the latch is in transparent mode. If the input state of the latch is not equivalent to the state previously forced by the preset or clear, an output transition results. In these situations, an additional arc is required to model this transition. A similar corner-case can result when both preset and clear signals are asserted, and one of them is subsequently released.

Note: The `timing_enable_preset_clear_arcs` global variable has higher priority than `lib_build_asynch_de_assert_arc`. If both of these global variables are set to `true`, then all the asynchronous arcs are recognized. If `timing_enable_preset_clear_arcs` is set to `true` and `lib_build_asynch_de_assert_arc` is set to `false`, then only the assert asynchronous arcs are recognized. If `timing_enable_preset_clear_arcs` is set to `false`, then both the assert and de-assert asynchronous arcs will not be recognized.

 Because the `timing_enable_preset_clear_arcs` and `lib_build_asynch_de_assert_arc` global variables affect how the timing graph is built, it is important to set the desired state before performing any timing analysis.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_all_registers_filter_clock_pins_by_clock

```
timing_all_registers_filter_clock_pins_by_clock  
{true | false}
```

Default: false

Data_type: Boolean

Reports clock pins filtered based on the specified clocks.

When set to `true`, the software will filter and return only those clock pins where at least one clock is from the specified list of clocks.

This global variable applies only when any of the `all_registers` command parameters, `-clock`, `-rise_clock`, `-fall_clock`, is specified.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_all_registers_include_icg_cells

```
timing_all_registers_include_icg_cells  
{true | false}
```

Default: true

Data_type: Boolean

Includes integrated clock gating (ICG) cell information from the `all_registers` report output.

When set to `false`, the global does not allow the `all_registers` report output to display ICG cell details.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_allow_input_delay_on_clock_source

```
timing_allow_input_delay_on_clock_source  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, this global variable allows the software to apply input delay constraints on a pin where a clock was previously asserted.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_analysis_precision_ps

```
timing_analysis_precision_ps  
{0.1 | 0.01}
```

Default: 0.01

Data_type: Double

Allows you to set the delay precision value (in ps) for the `write_sdf` command.

The `write_sdf` command shows a double digit value after the decimal point by default. You can use this global variable for more accuracy. 0.1ps or 0.01ps

To set this global variable, you can use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_apply_check_derate_to_external_output_delay

```
timing_apply_check_derate_to_external_output_delay  
{true | false}
```

Default: false

Data_type: Boolean

Applies check derate to output external delay.

The `set_timing_derate -cell_check` parameter by default does not apply check derate to output external delay (e.g., `set_output_delay`). To enable this feature, you can set the `timing_apply_check_derate_to_external_output_delay` timing global variable to `true`. The global applies cell (check) derate to the top boundary port.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_apply_default_primary_input_assertion

```
timing_apply_default_primary_input_assertion  
{true | false}
```

Default: true

Data_type: Boolean

When set to `true`, primary input and bidirectional ports that do not have an explicit arrival time specified (through the `set_input_delay` constraint), are provided a default arrival time. The arrival is referenced to an internal default clock that is indicated by the @ sign in various timing reports.

If a `set_input_delay` constraint is assigned to the input port, but the `-clock` parameter was not specified, the software assumes the constraint to be with reference to the default clock.

When set to `false`, inputs ports with no explicit `set_input_delay` are unconstrained for sequential timing analysis.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_apply_exceptions_to_data_check_related_pin

```
timing_apply_exceptions_to_data_check_related_pin  
{true | false}
```

Default: true

Data_type: Boolean

Controls if false path assertions which block the data path to the related (-from) reference pin of the data-to-data check disable the data-to-data check.

When set to `false`, the data-to-data checks remain valid when a false path assertion blocks the data path to the related (-from) pin.

When set to `true`, any false path assertion which blocks the data path to the related (-from) pin of the data-to-data check also causes the check to be disabled.

This global variable has no effect on the data-to-data checks with a reference clock, where an exception blocks the clock reference signal at the reference pin. The data-to-data check will always be disabled in this situation.

To set this global variable, use the `set` command.

Examples

- In the following example, the data-to-data check is disabled:

```
set timing_apply_exceptions_to_data_check_related_pin true  
set_data_check -from portA -to portB  
set_false_path -to portA
```

- In the following example, the data-to-data check is valid. This is the default situation:

```
set timing_apply_exceptions_to_data_check_related_pin false  
set_data_check -from portA -to portB  
set_false_path -to portA
```

- In this case, the data check is always disabled. and is not affected by this global variables' settings:

```
set_data_check -from portA -to portB  
set_false_path -to [get_clocks ClkA]
```

where the path to `portA` is clocked by `ClkA`.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_apply_setup_hold_exceptions_to_data_check_related_pin

`timing_apply_setup_hold_exceptions_to_data_check_related_pin {true | false}`

Default: `true`

Data_type: Boolean

Disables the early signal data-to-data paths, when false path is applied on the late reference path.

When set to `true`, the `set_false_path -setup` parameter settings on the reference path disables early signal d2d path, and vice versa. If both the `-setup` and `-hold` parameters are set, the signal early/late d2d path is disabled, irrespective of this global variable's settings.

When set to `false`, the `set_false_path` exceptions set on d2d reference path do not affect data path slack. When the `set_false_path` command is set on the reference d2d path with only `-setup` (or only `-hold`) parameter, the signal path of d2d check is not affected by these exceptions.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_cap_unit

timing_cap_unit *string*

Default: " "

Data_type: String

Specifies the capacitance units to be used for timing constraints and timing reports. The default unit for capacitance is inferred from the timing libraries. You can override this inference by explicitly setting the value to 1pf or 1ff. Setting this variable explicitly is the same as executing the `set_library_unit -cap` command to set the capacitance units.

To set this variable, use the `set` command.

For example,

```
set timing_cap_unit 1pf
```

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_case_analysis_for_icg_propagation

```
timing_case_analysis_for_icg_propagation
{false | require_seq_prop | always | case_only | constant_only}
```

Default: false

Data_type: Enum

Controls whether constants, which propagate to the inputs of integrated clock gating (ICG) cells, are further propagated to the output pins of these cells. The ICG cells typically contain an internal register or latch structure which synchronizes the devices' enable signal. If a constant propagates to this enable, it can be selectively propagated through the internal sequential element to the related output pins.

This global setting can uniquely control the handling of sequential propagation through ICG cells, or it can work in conjunction with the more general `timing_case_analysis_for_sequential_propagation` global variable or `set_analysis_mode -sequentialConstProp` settings.

The constants can be present in the analysis due to several different reasons, broadly breaking down to netlist constants and case analysis. The netlist constants come from connections to tie-hi or tie-lo cells, a 1'b0/1'b1 assignment in the netlist, or explicit power rail connections. The constants derived from these sources are considered to be immutable. The constants due to case analysis come strictly from the `set_case_analysis` constraint and are considered to be temporary - configuring the analysis to align to a certain mode. You may choose to handle constants from any source in a similar manner or customize the handling based on the constant type.

If you choose to allow propagation of only a specific type of constant, the output pin's state function must resolve to a constant by only considering inputs pins with the required constant type. Otherwise, the constant will not be propagated.

The propagation of constants to the output of a sequential device may affect how the timing checks on related input pins are analyzed. If all the possible transitions on a given input pin are unable to change the constant state of the output pin, the timing checks on the input pin will be disabled by default.

Changing the state of this global variable will change where constants are present in the analysis, typically resulting in sets of timing arcs and timing checks being enabled or disabled. As a result, a full timing update, including a fresh delay calculation, will be initiated when this setting is toggled.

You can use one of the following options:

Option	Description
false	Disables sequential propagation from any constant source (default setting).
require_seq_prop	When specified, the software will defer ICG sequential propagation handling to what is more generally set for all sequential devices, via either the <code>timing_case_analysis_for_sequential_propagation</code> global variable or the related <code>set_analysis_mode -sequentialConstProp</code> mode setting. For all the other options, the state of these external settings is not consequential.

always	Allows sequential propagation from any constant source - netlist or constraint-based.
constant_only	Allows sequential propagation only for constants that are due to the propagation of netlist-based constants.
case_only	Allows sequential propagation only for constants that are due to the propagation of constraint-based constants from <code>set_case_analysis</code> settings.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_case_analysis_for_sequential_propagation

timing_case_analysis_for_sequential_propagation
{false | true | case_only | constant_only}

Default: false

Data_type: Enum

Controls whether constants, which propagate to the inputs of sequential devices like registers or flip-flops, are propagated to the output pins of those devices. This global variable will also affect the propagation of constants through integrated clock gating (ICG) cells - depending on the setting of the related global variable `timing_case_analysis_for_icg_propagation`. Refer to the documentation for this global variable for more information on ICG handling.

The constants can be present in the analysis due to several different reasons - broadly breaking down to netlist constants and case analysis. The netlist constants come from connections to tie-hi or tie-lo cells, a 1'b0/1'b1 assignment in the netlist, or explicit power rail connections. The constants derived from these sources are considered to be immutable. The constants due to case analysis come strictly from the `set_case_analysis` constraint and are considered to be temporary - configuring the analysis to align to a certain mode. You may choose to handle constants from any source in a similar manner or customize the handling based on the constant type.

If you choose to allow propagation of only a specific type of constant, the output pin's state function must resolve to a constant by only considering inputs pins with the required constant type. Otherwise, the constant will not be propagated. The propagation of constants to the output of a sequential device may affect how the timing checks on related input pins are analyzed. If all the possible transitions on a given input pin are unable to change the constant state of the output pin, the timing checks on the input pin will be disabled by default.

Changing the state of this global variable will change where constants are present in the analysis, typically resulting in sets of timing arcs and timing checks being enabled or disabled. As a result, a full timing update, including a fresh delay calculation, will be initiated when this setting is toggled.

Note: The `set_analysis_mode -sequentialConstProp` parameter provides overlapping functionality with this global variable. Changing the state of `-sequentialConstProp` will also change the state of this global variable. You should refrain from using both controls at the same time.

You can use one of the following options:

Option	Description
false	Disables sequential propagation from any constant source (default setting).
true	Allows sequential propagation from any constant source - netlist or constraint-based.
constant_only	Allows sequential propagation only for constants that are due to the propagation of netlist-based constants.

<code>case_only</code>	Allows sequential propagation only for constants that are due to the propagation of constraint-based constants from the <code>set_case_analysis</code> command.
------------------------	---

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_case_analysis_propagation

timing_case_analysis_propagation
{true | false}

Default: true

Data_type: Boolean

Enables case analysis propagation.

The `timing_case_analysis_propagation` global variable is the top-level control for enabling or disabling all case propagation in the current session. When set to `true`, case propagation will be enabled and other global variables can then be used to selectively control case propagation in specific circumstances. The `timing_case_analysis_for_sequential_propagation` global variable can be used to control whether case is propagated across sequential devices. Propagation through integrated clock-gating (ICG) cells can be controlled with the `timing_case_analysis_for_icg_propagation` global variable.

When set to `false`, the software disables case analysis propagation.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_check_timing_report_all_checks

```
timing_check_timing_report_all_checks  
{true | false}
```

Default: false

Data_type: Boolean

Enables reporting of all the checks performed by the `check_timing` command.

When set to `true`, the `check_timing` command reports all the checks performed, even if there is no violation(s) for the corresponding check type.

When set to `false`, only the check types that have any violations will be reported.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_clock_phase_propagation

timing_clock_phase_propagation
{positive | negative | both}

Default: both

Data_type: Enum

Allows you to the select appropriate clock phases at register clock pins, when both the positive and negative phases of the same clock signal are seen in the clock network. For example, a register whose clock pin is fed by a two-input XOR gate where one input is a clock port and the other input is a data port. When the value of the data signal of the XOR gate is unknown, the XOR gate can see both positive and negative phases of the clock signal.

both	Selects both positive and negative clock phases at the register clock pins.
positive	Selects positive clock phases.
negative	Selects negative clock phases.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_clock_source_use_driving_cell

```
timing_clock_source_use_driving_cell  
{true | false}
```

Default: true

Data_type: Boolean

When set to `false`, delay calculation in propagated clock mode will use an idealized driver model at the `create_clock` root pin or port when it calculates delays for the first stage of the clock tree. The idealized model is a ramp input with the transition time set by the system default transition, which is normally 0ns.

When this variable is set to `true`, the drive characteristics of the cell output pin that `create_clock` is asserted on will be used to calculate the first stage clock tree delays and slews. The input slew used for this driver model is whatever slew has been propagated to the cell's input pins from upstream logic. The arrival time of the clock at the root pin is not modified in any way.

If you wish to specify the input slews to be used for the clock driver cell, that is, not use the upstream propagated slews - you can use the `set_annotated_transition` constraint on the cell's input pins to control this. Setting an input transition of 0ns on the input pins can be useful when trying to correlate analysis results with other applications which do not use the full propagated transition times.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_clock_uncertainty_from_to_precedence

`timing_clock_uncertainty_from_to_precedence {true | false}`

Default: `false`

Data_type: Boolean

Sets the clock uncertainty precedence.

When set to `true`, the software gives higher precedence to clock-to-clock uncertainty.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_collection_all_fanin_fanout_traversal_mode

`timing_collection_all_fanin_fanout_traversal_mode {hierarchical | flat}`

Default: `flat`

Data_type: Enum

Controls how the software processes netlists during `all_fanin/all_fanout` command processing.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_collection_result_display_limit

`timing_collection_result_display_limit` *integer*

Default: 100

Data_type: Integer

Limits the number of objects of a collection to be displayed in the output report.

When set to an integer value, all the commands, such as, `get_cells`, `add_to_collection`, `query_objects` etc, that display collection objects in the output report will be impacted.

When set to `-1`, all the elements in the collection will be displayed in the output.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_collection_variable_assignment_compatibility

timing_collection_variable_assignment_compatibility
{true | false}

Default: true

Data_type: Boolean

Disables printing the names of the output of `get_*` commands.

When set to `false`, the software prints the `get_*` command output on the shell. This global variable should be used for debugging purposes only.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_consider_late_arrival_for_ssi_derate

timing_consider_late_arrival_for_ssi_derate
{true | false}

Default: false

Data_type: Boolean

Controls constant propagation across sequential cells.

When set to `true`, the software will use the late arrival time for window overlapping evaluation during simultaneous switching input (SSI) analysis.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_constraint_disable_min_max_input_delay_worst_casing

timing_constraint_disable_min_max_input_delay_worst_casing
{true | false}

Default: false

Data_type: Boolean

Disables the override of the maximum value (set using the `set_input_delay -max` parameter) from the specified `-min` value.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_context_apply_default_primary_input_assertion

```
timing_context_apply_default_primary_input_assertion  
{true | false}
```

Default: false

Data_type: Boolean

Propagates the phase at block level for unmapped clocks.

When set to `true`, the nets in the fanout of data ports with unmapped clocks are considered to have infinite timing windows during SI context analysis.

When set to `false`, the unmapped top clock arrivals on data ports are dropped during context analysis.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_context_apply_port_sdc_exceptions

`timing_context_apply_port_sdc_exceptions {true | false}`

Default: `false`

Data_type: Boolean

When set to `true`, the software will consider path exceptions on block constraints in the hierarchical context flow.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_context_clock_mapping_check_waveform

timing_context_clock_mapping_check_waveform
{true | false}

Default: true

Data_type: Boolean

Controls the waveform checking.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_context_clock_mapping_percentage_tolerance

timing_context_clock_mapping_percentage_tolerance *double*

Default: 0

Range: {}

Controls the tolerance value of period and waveform in clock mapping.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_context_clock_phase_based_clock_mapping

```
timing_context_clock_phase_based_clock_mapping  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, the port-based clock mapping search is enhanced by additionally attempting to map originally unmapped top clocks.

Such unmapped top clock on data port is to be mapped to block clock if top clock was mapped to block clock as a clock on clock port.

Note: This global variable can be used when `timing_context_port_based_clock_mapping` is set to `true`.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_context_continue_on_invalid_module_list

`timing_context_continue_on_invalid_module_list {true | false}`

Default: `false`

Data_type: Boolean

Enables to continue timing analysis without generating timing context.

When set to `true`, the software will continue with timing analysis without generating timing context, when all the specified modules/instances are invalid.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_context_data_phase_based_clock_mapping

```
timing_context_data_phase_based_clock_mapping  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, the port-based clock mapping search is enhanced by additionally trying to map originally unmapped top clocks.

Such unmapped top clock on data port is to be mapped to block clock if top clock was mapped to block clock on other data ports.

Note: This variable can be applied when `timing_context_port_based_clock_mapping` is set to `true`.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_context_port_based_clock_mapping

timing_context_port_based_clock_mapping {true | false}

Default: false

Data_type: Boolean

Enables clock mapping based on block ports.

When set to `true`, clock mapping is done depending solely on clock property of the top and block clock for each clock and data port separately. For enabling N-to-1 or 1-to-N clock mapping, this global variable should be set `true`.

When set to `false`, only 1-to-1 clock mapping is done. In such cases, the following rules are used to determine clock mapping:

- Rule1: Clocks with matching property (period/duty cycle)
- Rule2: Clocks with matching name
- Rule3: Clocks present on maximum number common ports
- Rule4: Any one of the matching clocks

For example,

- Consider the following illustration:



When set to `true`, the following mapping will be seen:

Block Port	Block Clock	Mapped top clock
P1	CLK	CLK, TCLK

When set to `false`, CLK will be mapped to CLK according to Rule 2.

Notes:

- The `context_data` property of block port can be accessed only when this global variable is set `true`.
- When set to `true`, the software may increase the pessimism as the worst latency/arrival will be picked for block context analysis.
- If there are multiple block clocks that can be mapped to same top clock on the same block port,

then Tempus does not perform auto mapping (when set to `true`). Whereas for the `false` setting, clock mapping is done depending on the above mentioned rules.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_continue_on_error

```
timing_continue_on_error  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, the software continues to process if an error occurs when any constraints are used, or during timing analysis, or when a library is read. This global variable does not have any effect on other commands, except constraints or commands related to timing analysis.

When set to `false`, if an error occurs, during timing analysis, library reading, or defined constraints, the software stops processing and exits.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_cppr_opposite_edge_sigma_scale_factor_cell

timing_cppr_opposite_edge_sigma_scale_factor_cell *double*

Default: 1

Data_type: Double

Value Range: 0 to 1

Controls the cell sigma delay component for CPPR credit for opposite transition edges at the common pin.

The scale factors are applicable only when the common pin has mismatched transitions for the launch and capture paths.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_cppr_opposite_edge_sigma_scale_factor_net

timing_cppr_opposite_edge_sigma_scale_factor_net *double*

Default: 1

Data_type: Double

Value Range: 0 to 1

Controls the net sigma delay component for CPPR credit for opposite transition edges at the common pin.

The scale factors are applicable only when the common pin has mismatched transitions for the launch and capture paths.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_cppr_propagate_thru_latches

```
timing_cppr_propagate_thru_latches  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, the CPPR of a timing path crossing a latch (or latches), for example, ff1 -> lat1 -> ff2, is calculated using the common point of the origin device (ff1) and the capture device at the end of the path (ff2).

When set to `false`, the segment-based CPPR calculation is performed, considering the common point of latch (lat1) and the capture device (ff2). This CPPR setting is considered only when the latch thru analysis mode is enabled and a latch thru path exists.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_cppr_remove_clock_to_data_crp

```
timing_cppr_remove_clock_to_data_crp  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, removes clock reconvergence pessimism (CRP) for clock source paths.

To set this global variable, use the `set_global` command.

Related Topics

- "[Analysis and Reporting](#)" chapter in the *Tempus User Guide*:
 - [Supported CPPR Global Variables](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_cppr_self_loop_mode

timing_cppr_self_loop_mode
{true | false}

Default: true

Data_type: Boolean

When set to `true` in case of self-loop paths, computes CPPR adjustment by taking the difference between early and late clock arrival time of the common point; the clock pin of the flop. A self-loop path is a path that goes from a clock pin of a flop and ends at the data pin of the same flop. However detecting self-loop paths requires runtime, and is design dependent.

When set to `false`, does not detect self-loop paths. In this case, the software computes CPPR adjustment by using the common point as the output pin driving the clock pin of the flop. This might result in better runtime, but less accurate results. The loss of accuracy is the difference between early and late delay of the interconnect portion driving the clock pin.

To set this global variable, use the `set_global` command.

Related Topics

- "[Analysis and Reporting](#)" chapter in the *Tempus User Guide*:
 - [Supported CPPR Global Variables](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_cppr_skip_clock_reconvergence

timing_cppr_skip_clock_reconvergence
{true | false}

Default: false

Data_type: Boolean

Specifies the branch point to use for computing clock path pessimism removal (CPPR) adjustment when there is reconvergence in the clock tree.

When set to `false`, the software uses the branch point closest to the register clock pins where the clocks reconverge. When set to `true`, the software uses the farthest branch point from the register clock pins (that is, the branch point closest to the clock root pin where the clocks diverge).

Note: The `timing_cppr_skip_clock_reconvergence` and `timing_enable_pessimistic_cppr_for_reconvergent_clock_paths` global variables are mutually exclusive.

Also, the `timing_cppr_skip_clock_reconvergence` global variable has precedence over `timing_enable_pessimistic_cppr_for_reconvergent_clock_paths`.

To set this global variable, use the `set_global` command.

Related Topics

- "[Analysis and Reporting](#)" chapter in the *Tempus User Guide*:
 - [Supported CPPR Global Variables](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_cppr_skip_clock_reconvergence_for_unmatched_clocks

`timing_cppr_skip_clock_reconvergence_for_unmatched_clocks {true | false}`

Default: `false`

Data_type: Boolean

Allows the software to skip re-convergence of unmatched clocks.

When set to `true`, the software uses the farthest branch point from the register clock pins when the launching and capturing clocks are different - the re-converge points are ignored and not searched.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_cppr_threshold_ps

timing_cppr_threshold_ps *value*

Default: 20

Data_type: Double

Specifies the maximum amount of pessimism that clock path pessimism removal (CPPR) analysis is allowed to leave in the path. The unit is pico seconds (ps).

When this global variable is set, all the paths might be reported without having their pessimism removed. This global variable can be used to speed up CPPR analysis runtime - the larger the value, the faster the runtime.

To set this global variable, use the [set_global](#) command.

Related Topics

- "[Analysis and Reporting](#)" chapter in the *Tempus User Guide*:
 - [Supported CPPR Global Variables](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_cppr_transition_sense

```
timing_cppr_transition_sense  
{normal | same_transition | same_transition_expanded}
```

Default: normal

Data_type: Enum

Specifies the transition sense of the launching and capturing clocks at the common node, to calculate clock path pessimism removal (CPPR).

When set to `normal`, the CPPR analysis computes the CPPR value even if the launching and capturing clock transitions at the common node are in opposite directions. If the transition sense at the common node are in opposite directions, the pessimism is the minimum of the difference between late and early arrival times for rise and fall clock transitions.

When set to `same_transition`, the CPPR analysis computes the CPPR value only if the launching clock transition is same as the capturing clock transition at the common node. If the transitions are different, the CPPR value is zero.

When set to `same_transition_expanded`, the CPPR analysis computes the CPPR value only if the launching clock transition is the same as the capturing clock transition at the common node. If the transitions are different, the software searches until it finds a common point where the transitions are the same, and calculates the CPPR value from that point.

To set this global variable, use the `set_global` command.

Related Topics

- "[Analysis and Reporting](#)" chapter in the *Tempus User Guide*:
 - [Supported CPPR Global Variables](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_create_clock_default_propagated

```
timing_create_clock_default_propagated  
{true | false}
```

Default: false

Data_type: Boolean

Enables clocks to be created in propagated mode.

By default, all clocks, created using `create_clock` and `create_generated_clock`, are in ideal mode.

When the global is set to `true`, all subsequently created regular clocks, derived clocks, and library generated clocks are in propagated mode.

There is no effect on the functionality of `set_propagated_clock` and `set_clock_latency` commands.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_create_clock_use_ideal_slew

```
timing_create_clock_use_ideal_slew  
{true | false}
```

Default: false

Data_type: Boolean

Enables the use of ideal slew at the inputs of the clock driver cells.

When set to `true`, the software detects created clocks, and uses zero input slew. Other slew values, including those set using `set_annotated_transition` command, will be ignored.

When set to `false`, the software honors `set_annotated_transition` command settings.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_default_opcond_per_lib

```
timing_default_opcond_per_lib  
{true | false}
```

Default: true

Data_type: Boolean

Determines the default operating condition behavior when there is no explicit user operating condition specified through the `set_op_cond` command.

When set to `true`, each timing library uses its own internally-defined default operating condition specification.

When set to `false`, the default operating condition of the master library (that is, the first library in the library search list) is used to set the PVT operation point for the entire design. All timing libraries are accessed and derated to that operating point.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_defer_mmmc_object_updates

```
timing_defer_mmmc_object_updates  
{true | false}
```

Default: false

Data_type: Boolean

Allows a sequence of MMMC updates to be accumulated before new data is loaded and analyzed.

When set to `false`, the multi-mode multi-corner (MMMC) update commands, `update_constraint_mode` and `update_analysis_view`, actively load new data upon execution. Since all the timing information is reloaded with each update command, executing a sequence of updates can result in multiple reloads and a subsequent degradation of performance and turn-around-time.

If the `set_analysis_view -update_timing` parameter is specified, the software indicates that the MMMC object updates are complete and new configuration can be loaded and analyzed.

For example:

```
tempus > update_constraint_mode -name ioTiming -sdc_files {postCtsIO.sdc}  
tempus > update_constraint_mode -name coreTiming -sdc_files {postCtsCore.sdc}  
tempus > set_analysis_view -update_timing
```

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_derate_aocv_dynamic_delays

```
timing_derate_aocv_dynamic_delays  
{true | false}
```

Default: true

Data_type: Boolean

Controls AOCV derating of SI-induced delays.

When set to `false`, AOCV derates will not be applied on dynamic, SI-induced delays, that is, a value of `false` ensures AOCV derating is only applied to the static component of the delay.

When set to `true`, AOCV derating is applied to both static and dynamic arc delay components.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_derate_aocv_reference_point

`timing_derate_aocv_reference_point {1 | 0}`

Default: 1

Data_type: Integer

Allows specification of reference point for AOCV factors. As per the current AOCV library definitions, the AOCV reference point is always expected to be set at 1.0.

To set this global variable, use the `set_global` command.

Related Topics

- "[Analysis and Reporting](#)" chapter in the *Tempus User Guide*:
 - [Handling OCV Derating With AOCV](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_derate_dynamic_compatibility

timing_derate_dynamic_compatibility
{true | false}

Default: true

Data_type: Boolean

Controls how the [set_timing_derate](#) command factors that are not specified, using either `-static` or `-dynamic` options, are applied to the delay.

When this global is set to 1, the behavior will be compatible with previous releases. In this mode, the static and dynamic components will be summed up before applying derating.

When set to 0, the static and dynamic components will be derated separately and then combined. A setting of 0 is equivalent to using separate derate assertions with the `-dynamic` and `-static` options explicitly specified.

To set the global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_derate_incremental_adjust_additive_mode

timing_derate_incremental_adjust_additive_mode
{true | false}

Default: false

Data_type: Boolean

Allows the software to accumulate incremental derate values instead of overwriting them.

When set to `false`, the successive application of incremental derating will overwrite any existing derate value with the new value.

When set to `true`, the successive applications of the incremental derates will be accumulated into a final derate value.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_derate_negative_delay_backward_compatibility

`timing_derate_negative_delay_backward_compatibility {true | false}`

Default: `true`

Data_type: Boolean

Controls the effective derate calculation for negative delays. When effective derate is requested for negative delays, the value will be converted according to the negative delay scaling rule, e.g., 2 - positiveVal.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_derate_ocv_reference_point

```
timing_derate_ocv_reference_point {1 | 0}
```

Default: 1

Data_type: Integer

Allows specification of reference point for OCV factors. As per the current OCV library definitions, the OCV reference point is always expected to be set at 1.0.

To set this global variable, use the `set_global` command.

Related Topics

- "[Analysis and Reporting](#)" chapter in the *Tempus User Guide*:
 - [Handling OCV Derating With AOCV](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_derate_spatial_distance_unit

timing_derate_spatial_distance_unit
{default | 1um | 1nm}

Default: default

Data_type: Enum

Specifies distance units in the SOCV and AOCV library formats.

This global variable supports 3 distance units - default, 1um, and 1nm. The default value is the default units that are derived from the current system units for distance (μm /nm overrides the default).

To set this global variable, use the `set` command.

Note: The `timing_derate_spatial_distance_unit` global variable should be set before a design is loaded.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_derate_voltage_scaling_mode

timing_derate_voltage_scaling_mode {first_library | snap_to_nearest | interpolate}

Default: first_library

Data_type: Enum

Controls how the timing derates are computed when analysis voltage is not at the characterization voltage of any of the currently loaded timing libraries. When timing analysis environment is configured for interpolation, the MMMC library set will contain a list of timing libraries at different voltage characterizations points.

You can specify the following options:

first_library	Derates associated with the first library in the timing library list will be used.
snap_to_nearest	Allows the software to choose one set of derates from the library closest to the required analysis voltage.
interpolate	Allows the timer to perform interpolation of the derates associated with the two bounding reference libraries.

Consider the following example:

```
create_library_set -name ls_interp_1.50V_1.80V  
-timing [list stdcell_1.50V.lib stdcell_1.75V.lib stdcell_1.80V.lib]
```

In this example, by default (*first_library*) the derates associated with the library `stdcell_1.50V.lib` will be used.

When set to *snap_to_nearest*, if the analysis voltage is 1.70V, the derates from the library `stdcell_1p75V.lib` will be used.

When set to *interpolate*, if the analysis voltage is 1.60V, then interpolation of derates from the libraries `stdcell_1.50V.lib` and `stdcell_1.75V.lib` will be used.

If interpolation mode is selected and the analysis voltage is beyond either the smallest or largest voltage, then the library derates will not be interpolated. The derates associated with the nearest bounding library will be used. If the analysis voltage is 1.90V, then only the derates from `stdcell_1.80V.lib` will be used.

If interpolation mode is enabled, but not all the bounding libraries have derate values, a default derate value of 1.0 will be assumed for the missing derates and used for interpolation. A TA-960 warning message will also be issued.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_black_box_endpoints

`timing_disable_black_box_endpoints {true | false}`

Default: `false`

Data_type: Boolean

Ignores any boundary model endpoints on black box instances.

When set to `false`, the fanout logic from the block's input ports is traced for storing as timed logic within the boundary model. If a pin is reached with no outgoing timing arcs, this is considered an endpoint of the model and the fanin to that endpoint will be included in the model.

When set to `true`, such pins are ignored when they are on cells for which no timing or LEF cell definition was loaded. This is intended to make the model generation more tolerant of missing libraries in cases where the power/ground connectivity is included in the netlist.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_constant_propagation_for_sequential_cells

timing_disable_constant_propagation_for_sequential_cells
{true | false}

Default: false

Data_type: Boolean

Controls constant propagation across sequential cells.

When set to `true`, the software disables constant propagation across sequential cells.

When set to `false`, the software enables constant propagation across sequential cells.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_get_clocks_for_current_instance

```
timing_disable_get_clocks_for_current_instance  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, the software disables the `current_instance` command support on `get_clocks`.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_model_parasitics

timing_disable_model_parasitics *value*

Default: 0

Data_type: Integer

Disables saving parasitics data into a model.

When set to `true`, the software will not save parasitics data into the boundary model, so the generated boundary models will occupy a smaller disk space. However, while reading this boundary model at the top level, the original block parasitics (which was used while generating the model) should be read along with the top level parasitics.

When set to `false`, the parasitic data is not stored in the boundary models. Hence, the user does not need to read the block parasitics, since it is read from the boundary models. By default, this global variable is set to `false`.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_pulsewidth_same_edge_si_cppr_mode

```
timing_disable_pulsewidth_same_edge_si_cppr_mode  
{true | false}
```

Default: true

Data_type: Boolean

Enables SI CPPR credit up to the last divergent pin for pulse width checks for zero width pulse clocks. By default, the software credits minimum of rise and fall transition CPPR up to the common pin as the transitions mismatch at the common pin.

The software computes base delay CPPR up to the common pin as CPPR credit for the path. The software then finds an alternate common timing pin in the clock network where the transition for launch and capture match. If a match is found, the software includes the SI delay up to this pin in the CPPR credit also.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_driving_cell_override_library

```
timing_driving_cell_override_library  
{true | false}
```

Default: false

Data_type: Boolean

Controls the selection of library cells specified using the [set_driving_cell](#) command.

When set to `true`, a driving cell will be picked up from the Max Library for max drive cell adjustment and Min Library for min drive cell adjustment, irrespective of the library specified with the `set_driving_cell - library` parameter.

When set to `false`, the libraries are selected as specified with the `set_driving_cell` command.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_drv_analysis_ignore_latency_phases

timing_drv_analysis_ignore_latency_phases
{true | false}

Default: false

Data_type: Boolean

Enables ignoring latency phases in DRV analysis.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_dynamic_loop_breaking

timing_dynamic_loop_breaking
{true | false}

Default: false

Data_type: Boolean

Enables dynamic loop breaking.

By default (*false*), when the timing engine encounters combinational loops in the design, it breaks the loop by disabling a timing arc along the path. The heuristics used in breaking the loop cannot guarantee that a real path through the disabled arc will not be blocked as a consequence. Therefore, users must carefully investigate the arcs that have been disabled to ascertain that the loop break was done safely. If loop breaks are not then added to the master constraint file, they could move to other locations as the design matures.

When set to *true*, the timing system performs a more sophisticated analysis of the design, and ensures that no valid paths are broken inadvertently due to loops.

Note: Setting this global variable to *true* will add additional run time to the analysis.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_genclk_source_path_register_limit

```
timing_enable_genclk_source_path_register_limit  
{true | false}
```

Default: false

Data_type: Boolean

Limits the generated clock source latency path to traverse across one register only.

When set to `false`, the software allows master clock latency path to traverse across any number of sequential devices to reach the generated clock target pin.

When set to `true`, the software restricts the master clock latency path for a generated clock to traverse exactly one sequential device.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_get_objects_regexp_compatibility

timing_enable_get_objects_regexp_compatibility {true | false}

Default: false

Data_type: Boolean

Defines the pattern search scope to boundaries deduced from hierarchy separation characters specified in the search pattern.

When set to `false`, the commands with `-regexp` parameter will search at the top level only (unless used with the `-hierarchical` parameter).

When set to `true`, the commands will search the specified pattern within the hierarchies based upon the given pattern hierarchy (similar to pattern search without the `-regexp` parameter). The `-hierarchical` along with the `-regexp` parameter shows the same behavior similar to when the command is used without the `-regexp` option - only the leaf level object name is compared to the pattern match.

To set this global variable, use the `set` command.

Examples

Command behavior when this global variable is set to `false`:

```
get_pins -regexp seg2/.*/QN
**WARN: (TCLCMD-513): The software could not find a matching object of the specified type for
the pattern '.*/*.*'
**ERROR: (TCLCMD-917): Cannot find 'pins' that match 'seg2/.*/QN'

get_pins -regexp seg2/.*/QN -hierarchical
seg2/u3/QN seg2/u9/QN seg2/u14/QN

get_pins -regexp .*/QN -hierarchical
seg1/u3/QN seg1/u9/QN seg1/u14/QN seg2/u3/QN seg2/u9/QN seg2/u14/QN seg3/u3/QN seg3/u9/QN
seg3/u14/QN u2/QN u5/QN u8/QN
```

Command behavior when when this global variable is set to `true`:

```
get_pins -regexp seg2/.*/QN
seg2/u3/QN seg2/u9/QN seg2/u14/QN

get_pins -regexp seg2/.*/QN -hierarchical
**WARN: (TCLCMD-513): The software could not find a matching object of the specified type for
the pattern '.*/*.*'
**ERROR: (TCLCMD-917): Cannot find 'pins' that match 'seg2/.*/QN'

get_pins -regexp .*/QN -hierarchical
seg1/u3/QN seg1/u9/QN seg1/u14/QN seg2/u3/QN seg2/u9/QN seg2/u14/QN seg3/u3/QN seg3/u9/QN
seg3/u14/QN u2/QN u5/QN u8/QN
```

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_hier_context_unmapped_clock_analysis

```
timing_enable_hier_context_unmapped_clock_analysis  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, for unmapped block clocks, the software enables timing analysis based on the block SDC constraints set on block ports.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_hierarchical_get_nets_support

```
timing_enable_hierarchical_get_nets_support  
{true | false}
```

Default: false

Data_type: Boolean

Enables hierarchical segment support for the `get_nets` command.

When set to `false`, the `get_nets` command will report only flat nets.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_individual_distance_inter_die_analysis

```
timing_enable_individual_distance_inter_die_analysis  
{true | false}
```

Default: false

Data_type: Boolean

Enables individual inter-die calculation in 3DIC analysis.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_latch_thru_mode

`timing_enable_latch_thru_mode {true | false}`

Default: `false`

Data_type: Boolean

Enables the latch through analysis mode in which timing paths can propagate across latches, depending on their arrival time with respect to the latch transparency window.

When set to `true`, the arrival and required times will be propagated across a latch if it is in transparent mode. The reporting will show the entire flow through the path, rather than individual segments. This allows the overall path slack to be seen through upstream borrowing latch stages. The timing analysis, timing optimization, and reporting will consider the sequence of borrowing latch stages as a single path. As a result, the timing optimization reaches the target slack by viewing the entire path.

This feature provides better visibility into the overall context of latch path timing issues - for both the stage and path-level slacks. Additionally, this improves usability for debugging and manual ECOs, and shows better quality of result for Tempus ECO timing closure.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_latency_through_clock_gating

```
timing_enable_latency_through_clock_gating  
{true | false}
```

Default: true

Data_type: Boolean

Allows propagation of latency phases across clock gating elements.

When set to `true`, the software allows propagation of clock latency phase across clock gating elements.

When set to `false`, the software suppresses propagation of clock latency across clock gating elements from the enable pin. The enable pin of the clock gating element should not be in the clock network for any other clocks. This is not recommended because the enable path (through the 'inferred' clock gates, such as, AND, NAND, OR) is either for clock latency or data paths. If the gated clock check is met, then the enable should not pass through to the output.

In case of ICGs, there is no arc from the EN to the CLKOUT on ICGs that have an incoming enable signal latched. In case of ICG/AND, using the enable path may not be correct for determining latency paths. If no latency paths are found, then the generated clocks definition will need to be re-checked.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_model_constraint_warnings

timing_enable_model_constraint_warnings *integer*

Default: 0

Data_type: Integer

Controls the warning/error messages associated with constraints applied on missing objects in the top level run with boundary models.

A top level flat SDC may contain constraints for block internal pins/nets, which are excluded from the boundary model netlist. To avoid false messages associated with missing objects in a model netlist, Tempus by default suppresses messages associated with all the missing objects - if boundary models are loaded.

However, this may suppress the real warning/error messages as well (which may be reported in a flat run without boundary models). To enable messages associated with missing objects, you can set `timing_enable_model_constraint_warning` global variable to 0. This global variable does not impact sessions without boundary models.

To set this variable, you can use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_path_delay_to_unconstrained_endpoints_compatibility

timing_enable_path_delay_to_unconstrained_endpoints_compatibility

{true | false}

Default: false

Data_type: Boolean

When set to `true`, the software does not constrain paths with path delay - if the path ends at an unconnected input pin of a combination cell.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_spatial_derate_mode

timing_enable_spatial_derate_mode
{true | false}

Default: true

Data_type: Boolean

Enables the spatial derate mode in SOCV analysis.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_vtskew_derate_mode

timing_enable_vtskew_derate_mode
{true | false}

Default: false

Data_type: Boolean

Enables voltage threshold (VT) skew analysis.

When set to `true`, the software analyzes the effects of VT skew in a multi-VT environment.

Due to variations in the manufacturing process, the voltage threshold (Vth) of a given Vth class (e.g., SVT, LVT, ULVT) can show some variations that might result in cells of that class operating either faster or slower than the nominal value. The Tempus Vth skew analysis automates the process of selecting which Vth permutations need to be run, which ensures a robust analysis and optimizes the software to run faster and efficiently.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_wire_variation

```
timing_enable_wire_variation  
{true | false}
```

Default: false

Data_type: Boolean

Enables wire variation analysis in path-based analysis mode.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_zero_delay_analysis_mode

timing_enable_zero_delay_analysis_mode
{true | false}

Default: false

Data_type: Boolean

When set to `true`, the software will use zero delays during timing analysis.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_full_context_flow

`timing_full_context_flow integer`

Default: 0

Data_type: Integer

Enables the full chip timing context mode. This variable captures the estimated slews and timing windows as part of the timing context in the boundary model.

When set to 1, the full chip timing analysis (STA or DSTA) run saves the timing context that is used by the block scope timing later in the flow.

To set this variable, you can use the set command.

Related Topics

- "[Distributed Static Timing Analysis](#)" chapter in the *Tempus User Guide*:
 - [Running SmartScope Hierarchical Analysis](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_generate_normalized_driver_waveform

timing_generate_normalized_driver_waveform
{true | false}

Default: true

Data_type: Boolean

Generates the analytical normalized driver waveform for timing library if the waveform is not specified in the library. The waveform is generated by weighed mean of linear and exponential waveform.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_generated_clocks_inherit_ideal_latency

```
timing_generated_clocks_inherit_ideal_latency  
{false | true | actual_partial_latency_path | ideal_partial_latency_path}
```

Default: false

Data_type: Enum

When the `set_clock_latency` command is used to define network latency for ideal clock mode at intermediate pins of a clock network, Tempus considers the latency. The `timing_generated_clocks_inherit_ideal_latency` global variable allows including the closest applied network latency for ideal inheritance also. You can use the following options:

- `false`: Uses only the source latency of generated clocks, if it exists.
- `true`: Allows the generated clocks to inherit the ideal network latency of the parent clock.
- `actual_partial_latency_path`: Uses the network latency of the master clock asserted closest to the target pin and propagated delays for the path from this point to the assertion point of the generated clock.
- `ideal_partial_latency_path`: Uses the network latency of the master clock asserted closest to the target pin and ideal delays for the path from this point to the assertion point of the generated clock.

Note: If the `set_clock_latency -source` parameter is used to specify latency of a generated clock, the software will completely override any inheritance.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_get_of_objects_hier_compatibility

```
timing_get_of_objects_hier_compatibility  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, the `get_pins -of_objects` parameter will return leaf pins connected to the specified objects.

When set to `false`, the `get_pins -of_objects` parameter will return hierarchical pins and leaf pins connected to the specified objects. In order to return leaf pins only when the global is set to `false`, the `-leaf` option must be specified along with the `get_pins -of_objects` parameter.

When set to `true`, the `get_cells -objects` parameter returns cells connected to the net at the same level of hierarchy as the net itself. Hierarchical ports will be returned if the net is connected to a hierarchical module.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_hier_object_name_compatibility

```
timing_hier_object_name_compatibility  
{true | false}
```

Default: true

Data_type: Boolean

Controls how hierarchical delimiters are interpreted in the search pattern, when the `-hier` parameter is used with the `get_*` collection commands.

When set to `true`, the software interprets the hierarchical delimiter as part of the search pattern itself, and tries to match names that include the delimiter. For example, if you specify:

```
get_cells -hier L1/R*
```

The software searches at every level of the hierarchy for instances with the name `L1/R*`, completely ignoring the hierarchy separator:

When set to `false`, hierarchical delimiters are interpreted as hierarchy separators. For example, if you specify:

```
get_cells -hier L1/R*
```

The software searched for instances with names matching `R*` in the hierarchy level below `L1`:

```
TL2/L1/RX1 TL2/L1/RX2
```

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_hierarchical_context_continue_on_top_block_clock_mismatch

```
timing_hierarchical_context_continue_on_top_block_clock_mismatch  
{true | false}
```

Default: false

Data_type: Boolean

Allows loading of the timing context model with unmapped clocks.

When set to `true`, the software allows the script to process even if there are any mismatches between the top and block-level clocks. The mismatches are reported in the *timing_context_clock_mapping_data.rpt* file, which shows data that allows you to determine if corrections or additions are required to the clock mapping file.

When set to `false`, the software will show an error if any of the top-level clocks cannot be matched with a block-level clock.

Note: This global variable should not be used for signoff.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_ignore_clock_based_path_delay_on_data_checks

```
timing_ignore_clock_based_path_delay_on_data_checks  
{true | false}
```

Default: false

Data_type: Boolean

Disables clock-based path delay application on data checks.

When set to `true`, the Tempus software enables the data checks to be reported by disabling the applied path delay. The "Data Check Setup" details are displayed in the timing report output.

When set to `false`, the timing reports display the "Path Delay" details. By default, this global variable is set to `false`.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_ignore_lumped_rc_assertions

```
timing_ignore_lumped_rc_assertions  
{true | false}
```

Default: false

Data_type: Boolean

When set to `false`, the `set_load` and `set_resistance` values override the actual extracted representation. A warning message is issued stating that the specified constraints on internal nets will override extracted parasitics on those nets.

When set to `true`, instructs the software to ignore `set_load` and `set_resistance` assertions on internal nets in the design. Parasitic data for internal nets normally should be provided by Tempus RC extraction, or external SPEF annotation.

Using `set_load` to specify I/O boundary loading on ports is supported by either setting of this global.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_io_use_clock_network_latency

timing_io_use_clock_network_latency
[always | ideal]

Default: ideal

Data_type: Enum

Controls whether network latency of a reference clock is added or not to the data arrival time on the port.

When set to `ideal`, the `set_input_delay` and the `set_output_delay` constraints will not add the network latency of the reference clock to the data arrival time on the port if the clock is in `propagated` mode.

When set to `always`, network latency is added to the data arrival time on the port regardless of the clock propagation mode.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_ipd_allow_paths_for_asynchronous_clocks

`timing_ipd_allow_paths_for_asynchronous_clocks {true | false}`

Default: `true`

Data_type: Boolean

When set to `true`, the software will ensure that the paths between the asynchronous clocks are allowed to be treated as IPD (inter-power domain) paths.

For more information, refer to the "Inter Power Domain (IPD) Analysis" chapter in the *Tempus User Guide*.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_ipd_xdomain_path_filtering_threshold

`timing_ipd_xdomain_path_filtering_threshold` *double*

Default: 0.003

Data_type: Double

When this global variable is used, the specified threshold (in ns) will be used in the path filtering criteria during native inter-power domain (xdomain) solution.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_max_transition_use_si_transition

`timing_max_transition_use_si_transition {true | false}`

Default: `false`

Data_type: Boolean

Enables using noisy slew for max_transition checks.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_model_netlist_exclude_dummy

timing_model_netlist_exclude_dummy {true | false}

Default: false

Data_type: Boolean

Excludes the dummy cell definitions that are written to the model netlist.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_multifrequency_clock_rounding_factor

timing_multifrequency_clock_rounding_factor
{1e-04 | 1e-05}

Default: 1e-05

Data_type: Double

Rounds off irrational numbers so that they can be read appropriately. The "round off factor" helps improve the level of accuracy of floating point numbers.

To round off clock period numbers accurate to the fourth decimal place, you can use the following command:

```
set_global timing_multifrequency_clock_rounding_factor 1.0e-4
```

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_normalized_driver_waveform_clip_linear_part

```
timing_normalized_driver_waveform_clip_linear_part  
{true | false}
```

Default: false

Data_type: Boolean

When set to 1, the linear ramp component in the normalized driver waveform is saturated to the maximum value, while generating the normalized driver waveform.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_normalized_driver_waveform_weight_factor

timing_normalized_driver_waveform_weight_factor
<float value>

Default: 0.5

Data_type: Double

Defines the value of the weight of linear components in the algorithm for generation of normalized driver waveform.

When set to `1`, the waveform will be purely linear. When set to `0`, the waveform will be purely exponential. You can specify values within the range of 0 to 1.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_null_collection_return_compatibility

```
timing_null_collection_return_compatibility  
{true | false}
```

Default: false

Data_type: Boolean

Allows you to migrate previous release scripts to the new use model.

When the global is set to `true`, the software returns "NA" values that are consistent with the previous release. The nil return values are returned as an empty collection. This allows the return value to be used directly by collection iterators, such as `foreach_in_collection`.

The default value is `false`, which is consistent with the current release.

Note: This compatibility option may be removed from future releases. You should update your scripts to the new use model to avoid future problems.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_path_based_enable_bounding_box_for_io_paths

```
timing_path_based_enable_bounding_box_for_io_paths  
{true | false}
```

Default: false

Data_type: Boolean

Enables the bounding box mode for input-output paths.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_path_based_enable_closest_common_pin_driver_for_bounding_box

```
timing_path_based_enable_closest_common_pin_driver_for_bounding_box  
{true | false}
```

Default: false

Data_type: Boolean

Enables bounding box calculation using closest common pin driver from the clock mesh.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_path_based_report_analysis_summary_max_paths_limit

timing_path_based_report_analysis_summary_max_paths_limit *integer*

Default: 2000000

Data_type: Integer

Specifies the limit on the maximum paths that can be reported in PBA mode with the [report_analysis_summary](#) command.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_pba_exhaustive_path_nworst_limit

timing_pba_exhaustive_path_nworst_limit *value*

Default: 10000

Data_type: Integer

Specifies the maximum number of paths per endpoint (nworst) that will be re-timed in exhaustive path-based analysis (PBA).

In exhaustive PBA mode, the value specified with the [report_timing -nworst](#) parameter is used only during the final reporting. The value specified with the `timing_pba_exhaustive_path_nworst_limit` global variable (and not the `report_timing -nworst` value) is used to specify the maximum number of paths to be re-timed per endpoint.

If the limit specified by `timing_pba_exhaustive_path_nworst_limit` is reached and the `timing_path_based_enable_exhaustive_depth_bounded_by_gba` global variable is set to:

- `true`: The worst of the current PBA slack is used and N+1th GBA path is reported.
- `false`: The current PBA slack is reported - which may be optimistic.

To set this global variable, use the `set` command.

Related Topics

- "[Analysis and Reporting](#)" chapter in the *Tempus User Guide*:
 - [Path-Based Analysis](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_pll_clock_use_driving_cell

```
timing_pll_clock_use_driving_cell  
{true | false}
```

Default: false

Data_type: Boolean

Uses delay calculation of all the enabled arcs to the PLL clock output to compute the transition.

When set to `true`, the delay calculation of all enabled arcs to the PLL clock output will be used to compute the transition. If no driving arcs are available in the model or if they are disabled, the same idealized transition will be used.

When set to `false`, the software considers the PLL clock output to be electrically isolated from the input-side reference clock, and creates the output clock with an idealized transition. You can use the `set_annotated_transition` command to customize the output clock transition.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_prefix_module_name_with_library_genclk

```
timing_prefix_module_name_with_library_genclk  
{true | false}
```

Default: true

Data_type: Boolean

When set to `true`, the software appends the instance name to the clock pin name when creating a generated clock. When set to `false`, the software uses only the clock pin name when creating a generated clock.

For example, assume the cell for instance `a/b` in the timing library contained the following generated clock group:

```
generated_clock(genclk2) {  
  clock_pin : clk1;  
  master_pin : "int/clk";  
  multiplied_by : 2;  
  duty_cycle : 50.0;  
}
```

By default (`true`), the software creates a generated clock with the following name:

```
a/b/int/clk
```

If you set the global to `false`, the software creates a generated clock with the following name:

```
int/clk
```

If you set this global to `false`, and the `timing_library_genclk_use_group_name` global variable is set to `true`, the software creates a generated clock with the following name:

```
genclk2
```

To set this global variable, use the `set_global` command.

Related Topics

- "[Timing Model Generation](#)" in the *Tempus User Guide*:
 - [Basic Elements of Timing Model Extraction](#)
 - [Validation of Generated ETM](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_propagate_latch_data_uncertainty

```
timing_propagate_latch_data_uncertainty  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, uses the clock phase associated with a flush latch's data pin as the "from" clock phase for downstream uncertainty timing calculations.

When set to `false`, the downstream uncertainty calculations are based on the clock phase of the flush latch's clock signal.

To set this global variable, use the `set` command.

timing_property_arrival_clocks_consider_clock_source_data_phase

```
timing_property_arrival_clocks_consider_clock_source_data_phase  
{true | false}
```

Default: true

Data_type: Boolean

Considers the clock source data phases.

The `get_property` and `report_property` commands support the `arrival_clocks` collection property on pin objects. The `arrival_clocks` property considers data phases, clock source data phases, and latency phases.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_property_arrival_clocks_consider_data_phase

timing_property_arrival_clocks_consider_data_phase
{true | false}

Default: true

Data_type: Boolean

Returns the pure data phase.

The `get_property` and `report_property` commands support the `arrival_clocks` collection property on pin objects. The `arrival_clocks` property considers data phases, clock source data phases, and latency phases.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_property_arrival_clocks_consider_latency_phase

timing_property_arrival_clocks_consider_latency_phase
{true | false}

Default: true

Data_type: Boolean

Returns the latency phase.

The `get_property` and `report_property` commands support the `arrival_clocks` collection property on pin objects. The `arrival_clocks` property considers data phases, clock source data phases, and latency phases.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_property_clock_used_as_data_unconstrained_clock_source_path S

```
timing_property_clock_used_as_data_unconstrained_clock_source_paths  
{true | false}
```

Default: true

Data_type: Boolean

Returns a value of `true` for `is_clock_used_as_data` timing property (set using the [get_property](#) command).

When set to `true`, the `is_clock_used_as_data` timing property returns a value of `true` (irrespective of whether the clock source path is constrained or unconstrained), if the pin lies in the clock source path and at least one of the clocks arriving at the pin is used as data in the downstream network of the pin.

When set to `false`, the pins lying in the constrained clock source data paths are reported as `true`. For example, if there is a path from clock source to output port and the output port is unconstrained, then the `is_clock_used_as_data` property will not return `true`.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_rcdb_allow_mismatch_option

`timing_rcdb_allow_mismatch_option {true | false}`

Default: `true`

Data_type: Boolean

Disables errors while reading boundary models.

When set to `true`, the software disables errors when reading the boundary model RCBD for cases where the contents do not match the boundary model netlist.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_recompute_sdf_in_setuphold_mode

```
timing_recompute_sdf_in_setuphold_mode  
{true | false}
```

Default: false

Data_type: Boolean

Controls the recomputing of SDF delays when the software is in simultaneous setup and hold analysis mode.

When the software is not in simultaneous setup and hold analysis mode, only one corner of data is stored at a time; therefore additional delay calculation is required for `write_sdf`. When the software is in simultaneous setup and hold analysis mode, both sets of data are in memory at the same time, and additional delay calculation is not necessary.

If set to false, `write_sdf` will write out the delays from memory, instead of calling delay calculation.

Note: This global variable must be used with `timing_enable_simultaneous_setup_hold_mode` true.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_reduce_multi_drive_net_arcs

```
timing_reduce_multi_drive_net_arcs  
{true | false}
```

Default: true

Data_type: Boolean

Controls whether multi-drive or mesh networks are reduced to simpler, single driver models.

When set to `true`, the nets that meet the drivers' receivers arc threshold (defined using `timing_reduce_multi_drive_net_arcs_threshold` global variable), are reduced. The net arcs are created only from one driver or from the drivers for which you have set the constraints.

If you specify constraints on the net driven by parallel buffers, it might result in increased run time and memory degradation. By specifying timing constraints through nets, the constraints are moved to the drivers. This leads to the creation of net arcs from all the drivers, thus disabling net arc reduction. In such cases, you can select one driver of the net and use that driver to specify the constraints.

When set to `false`, the number of net arcs created for timing analysis for nets driven by parallel buffers are not reduced. If a net is driven by n parallel buffers and is connected to m sink pins, then the total number of net arcs created is $n*m$. For example, if n is 100 and m is 20000, the number of net arcs created for timing analysis will be two million. This may result in increased run time and memory usage.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_reduce_multi_drive_net_arcs_threshold

timing_reduce_multi_drive_net_arcs_threshold *int*

Default: 10000

Data_type: Integer

Sets a threshold number that is used to trigger the reduction of timing arcs of nets driven by parallel buffers. The reduction takes place if the number of arcs is greater than the threshold value and when the `timing_reduce_multi_drive_net_arcs` global variable is set to `true`.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_remove_clock_reconvergence_pessimism

timing_remove_clock_reconvergence_pessimism
{true | false}

Default: false

Data_type: Boolean

Enables Clock Path Pessimism Removal (CPPR) analysis during timing analysis. When set to `true`, the timing analysis commands remove pessimism from slack calculation. Clock path pessimism is the difference of launching and capturing clock arrival times at the common node. This pessimism can be due to difference in minimum and maximum cell delays along the clock paths or due to the reconvergent clock paths. If the launching and capturing registers are triggered by the same clock and there is no common node between the launching and capturing clock paths, the pessimism is the difference between late and early clock source latency.

To set this global variable, use the `set_global` command.

Note: The `timing_remove_clock_reconvergence_pessimism` global should not be used with the `set_analysis_mode` command. Use this global only when the `set_analysis_mode` command is not available. When the `set_analysis_mode` command is available, use the `-cppr` parameter to set or reset the CPPR analysis.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_clock_pin_as_begin_point

```
timing_report_clock_pin_as_begin_point  
{true | false}
```

Default: false

Data_type: Boolean

Allows the software to report clock pins as begin points.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_constraint_extended_cell_em_flow

```
timing_report_constraint_extended_cell_em_flow  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, the software enables extended Cell EM flow.

To set this global variable, use the `set` command.

For more information, refer to the [Cell EM Analysis and ECO](#) section in the *Tempus User Guide*.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_resolve_driver_conflicts

timing_resolve_driver_conflicts
{conservative | aggressive}

Default: aggressive

Data_type: Enum

Resolves constant propagation results on nets with multiple drivers.

When set to `conservative` mode, the net values are as follows:

- If all the drivers are 1, the net will be driven to value 1
- If all the drivers are 0, the net will be driven to value 0
- In other cases, the net value will be `Undefined`

When set to `aggressive` mode, the net values are as follows:

- If some of the drivers are 0, the net value will be 0
- If some of the drivers are 1, but none are 0, then the net value will be 1
- In other cases, the net value will be `Undefined`

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_sdf_adjust_negative_setuphold

timing_sdf_adjust_negative_setuphold
{true | false}

Default: false

Data_type: Boolean

Controls how the setup and hold check values are adjusted when both are initially negative in the SDF file.

When set to `true`, the software sets the smaller check delay value to a negated larger check delay value. For example, if the initial setup/hold value is equal to -1/-2. then the value -2 from the setup is smaller and it is reset to a negated value of the larger one (-1 from hold). So the adjusted setup/hold value is equal to -1/1.

When set to `false`, both the negative setup and hold check values are set to 0.

The command order for using this global variable is as follows:

- `read_sdf`
- `timing_sdf_adjust_negative_setuphold {true | false}`
- `write_sdf`

The timing report (`report_timing`) results are not impacted.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_sdf_enable_setuphold_scond_ccond

timing_sdf_enable_setuphold_scond_ccond
{true | false}

Default: false

Data_type: Boolean

When set to `true`, enables proper generation of SDF (Standard Delay Format) with `scond` and `ccond` qualifiers on `SETUPHOLD` and `RECREM` timing checks.

Since some SDF readers do not accept `scond` and `ccond` qualifiers this global is set to `false` by default.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_self_loop_paths_no_skew

```
timing_self_loop_paths_no_skew  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, factors the skew on self loops. The software ignores the skew on the clock pin for hold analysis for data paths originating on sequential elements (for example, Q-pin of a flip flop) and ending on the data pins of the same element (D-pin of the same flip flop). Use this variable to eliminate pessimism due to skew when the launch and the capture edge of the clock are the same.

To set this global variable, use the `set_global` command.

Note: The `timing_self_loop_paths_no_skew` global should not be used with the `set_analysis_mode` command. Use this global only when the `set_analysis_mode` command is not available. When the `set_analysis_mode` command is available, use the `-timingSelfLoopsNoSkew` parameter to enable skew factoring on self loops.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_self_loop_paths_no_skew_max_depth

timing_self_loop_paths_no_skew_max_depth *value*

Default: 10

Data_type: Integer

Identifies self-loop paths at the specified depth.

To set this global variable, use the [set_global](#) command.

For example:

```
set_global timing_self_loop_paths_no_skew_max_depth 20
```

Allows the timer traverse 20 timing arcs (10 stages of logic) before abandoning the search for a self loop.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_self_loop_paths_no_skew_max_slack

timing_self_loop_paths_no_skew_max_slack *value*

Default: 0

Data_type: Double

Identifies self-loop paths at the specified threshold.

To set this global variable, use the [set_global](#) command.

For example:

```
set_global timing_self_loop_paths_no_skew_max_slack 1.0
```

In this case, all timing endpoints with a worst slack less than 1.0 are considered for self loop skew removal.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_set_clock_source_to_output_as_data

```
timing_set_clock_source_to_output_as_data  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, causes a clock source path leading to an output or bidi port to be treated as a data path if there is a `set_output_delay` or `set_data_check` assertion on the port.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_set_scaling_for_negative_checks

timing_set_scaling_for_negative_checks
{default | divider | multiplier}

Default: default

Data_type: Enum

When set to default, the derated check is the negative check * (2.0 - derating factor), where the derating factor is the value that you set using the `set_timing_derate` command.

Modifies the scaling of the negative timing analysis check value. This variable does not affect the positive checks. When set to `divider`, divides the negative check value by the derating factor. When set to `multiplier`, multiplies the negative check value by the derating factor.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_set_scaling_for_negative_delays

```
timing_set_scaling_for_negative_delays  
{default | divider | multiplier}
```

Default: default

Data_type: Enum

Modifies the scaling of the negative delay value to be used during timing analysis. This variable does not affect the positive delay.

When set to `default`, the derated check/delay = negative check/delay * (2.0 - derating factor), where the derating factor is the value that you set using the `set_timing_derate` command.

When set to `divider`, divides the negative delay value by the derating factor. When set to `multiplier`, multiplies the negative delay value by the derating factor.

Currently, constraints are used during timing in the flattened mode, so the internal ILM instances are seen instead of the LEF pins of the ILMs. Therefore, reading the bounding box constraints causes errors without this variable set. Notice that if you want to see the LEF pins of the ILM in GUI, the design must be in the unflattened mode.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_socv_sigma_spatial_multiplier

timing_socv_sigma_spatial_multiplier
{0 | 1}

Default: 1

Data_type: Double

Provides a fractional percentage multiplier, which allows you to select some (or all, none) of the spatial derate to be applied.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_socv_view_based_nsigma_multiplier_mode

`timing_socv_view_based_nsigma_multiplier_mode {true | false}`

Default: `true`

Data_type: Boolean

Enables setting of view-based separate setup and hold sigma multipliers.

By default or with an explicit `true` setting, the timing system is configured to support different N-sigma multiplier values for Setup vs. Hold – and for different analysis views. You can use the `set_socv_reporting_nsigma_multiplier` command to set specific N-sigma multipliers.

In earlier releases, the `timing_set_nsigma_multiplier` global variable was used to specify a single N-sigma value that was used for both Setup and Hold analysis across all analysis views. To continue using older scripts with this control, you can set the `timing_socv_view_based_nsigma_multiplier_mode` global variable to `false`.

Note: This backward compatibility option will not be supported in future releases. It is recommended to migrate your scripts to use the new view-based methodology.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_spatial_derate_chip_size

`timing_spatial_derate_chip_size value_in_microns`

Default: 1e+30

Data_type: Double

Allows you to define the chip size.

The chip size is the diagonal length of a rectangle that is formed by the chip's length and width. You can obtain the diagonal distance by calculating the square-root of the $\text{Length}^2 + \text{Width}^2$. The diagonal distance should always be specified in units of microns.

The chip size can be picked in the following ways (in order of highest to lowest precedence):

- When set to a value other than 1e+30, the specified value will have the highest precedence.
- If the physical floorplan information is available, the chip size will be derived automatically from the floorplan.
- If the SPEF parasitic files containing the coordinate information are loaded, then the chip size will be automatically derived.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_spatial_derate_distance_mode

timing_spatial_derate_distance_mode
{chip_size | arc_bounding_box}

Default: chip_size

Data_type: Enum

Allows you to select the spatial mode. You can specify one of the following options:

- `chip_size`: Uses graph-based analysis (GBA) spatial mode.
- `arc_bounding_box`: When specified, the software reduces GBA spatial derate pessimism by limiting bounding box calculations only to all the timing paths through an instance. The reduced GBA-PBA gap and GBA pessimism result in improved PBA runtime. You can use the `arc_bounding_box` option when the `timing_enable_pessimistic_cpvr_for_reconvergent_clock_paths` global variable is set to `true`.

Note: The spatial derate mode can be controlled using the `timing_enable_spatial_derate_mode` global variable. By default, this global variable is set to `true`.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_suppress_escape_characters

```
timing_suppress_escape_characters  
{true | false}
```

Default: true

Data_type: Boolean

Suppresses the reporting of escape characters in timing object names.

When the global is set to `true`, all the reporting and collection commands report the timing objects names (e.g., pins, ports, or instances) without the escape characters.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_suppress_ilm_constraint_mismatches

```
timing_suppress_ilm_constraint_mismatches  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, the software suppresses all error and warning messages related to objects not found when loading SDC constraint files for the ILM flow.

In the current ILM flow, SDC constraints originally specified against the complete flat netlist for the design do not get filtered when being applied against the ILM view of the design. Constraints that reference parts of the design that were pruned cause warnings and errors during constraint loading.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_time_unit

timing_time_unit *string*

Default: none

Data_type: String

Specifies the timing units to be used for timing constraints and timing reports. The default units for timing are inferred from the timing libraries. You can override this inference by explicitly setting the value to none, 1ns, 1ps, 10ps, or 100ps.

Setting this variable explicitly is the same as executing the `set_library_unit -time` command to set the time units.

To set this global variable, you can use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_timing_window_pessimism_removal_include_si_delay

`timing_timing_window_pessimism_removal_include_si_delay {true | false}`

Default: `false`

Data_type: Boolean

Controls inclusion of SI delay in timing window pessimism removal between the attacker and victim pins in the design.

The cross-talk analysis of a net depends on the relative positioning of attacker and victim arrival times. To compute the worst case switching times, the arrival windows of attacker and victim are considered. If the attacker and victim share a common clock path, then considering the original arrival windows will be pessimistic, and the common path contribution in these arrival windows needs to be adjusted.

When set to `false`, the software adjusts the elongated timing windows with the difference in the minimum-maximum base delay of the common clock path at the top level - while calculating the closest switching time, and the SI incremental delay differences are not adjusted.

When set to `true`, the software adjusts the timing windows with the SI incremental delay differences as well, in addition to the base delay. When this global variable is enabled, the timing window calculation can be optimistic, especially for cases where the attacker and victim are triggered by different clock edges. But for cases where all the attacker and victim nets are triggered by the same clock edge, then real pessimism will be removed from the timing window calculation.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_use_clock_pin_attribute_for_clock_net_marking

```
timing_use_clock_pin_attribute_for_clock_net_marking  
{true | false}
```

Default: false

Data_type: Boolean

Allows the software to use clock pin attribute setting for clock net marking.

The timing system determines which nets are recognized as clock nets by considering different heuristics - including having an upstream path to clock sources and a downstream path to clock targets.

When set to `true`, the pins that have the Liberty timing library pin attribute setting 'clock:true' will be interpreted as additional clock targets for the clock marking algorithm.

When set to `false`, the set of clock targets includes pins that are the launching point for sequential paths or are the reference pin for sequential timing checks - typically register clock pins.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_use_latch_early_launch_edge

```
timing_use_latch_early_launch_edge  
{true | false}
```

Default: true

Data_type: Boolean

Enables the latch use early launch edge feature.

During timing analysis when time borrowing and clock path pessimism removal (CPPR) is enabled, the given time is calculated using both borrow time and CPPR time. This may result in unnecessary timing violations. Since CPPR is an accurate means of pessimism removal, you are required to set the `timing_use_latch_early_launch_edge` global variable to `false` so that CPPR is applied to all the paths in the design.

When set to `true`, the launch clock path of timing paths starting at a latch is timed in early mode for setup analysis, and for late mode for hold analysis. No CPPR credit is given to the timing path.

When set to `false`, the behavior is like for any other path. The launch clock path of timing paths starting at a latch is timed in late mode for setup analysis, and early mode for hold analysis. The CPPR credit is given to the path according to the global CPPR settings.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_use_latch_time_borrow

```
timing_use_latch_time_borrow  
{true | false}
```

Default: true

Data_type: Boolean

When set to `false`, does not consider time borrowing during timing analysis. Time borrowing is the amount of time borrowed by a previous logic.

To set this global variable, use the `set_global` command.

Note: The `timing_use_latch_time_borrow` global should not be used with the `set_analysis_mode` command. Use this global only when the `set_analysis_mode` command is not available. When the `set_analysis_mode` command is available, use the `-timeBorrowing` parameter to control time borrowing consideration.

Related Topics

- "[Appendix - Base Delay Correlation with SPICE](#)" chapter in the *Tempus User Guide*:
 - [Debugging Correlation Issues](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_use_verilog_for_model_netlist

`timing_use_verilog_for_model_netlist` *integer*

Default: 0

Data_type: Integer

Allows you to choose whether to write the scope and boundary model netlists in Verilog format instead of binary.

When set to 1, the software will use Verilog format for saving model nets.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_write_sdf_no_escape_backslash_control

```
timing_write_ sdf_no_escape_backslash_control  
{true | false}
```

Default: false

Data_type: Boolean

Controls how the `write_sdf` command handles escape characters when the `-no_escape` parameter is specified.

For example, consider the following input Verilog expression:

```
\a\b\c\d
```

When the global is set to `true`, the `write_sdf` command removes the first escape character and outputs `a/b\c\d` in the SDF file.

When set to `false`, the command removes all the escape characters and outputs `a/b/c/d`.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

write_global_slack_worst_trigger_path_on_clocks

```
write_global_slack_worst_trigger_path_on_clocks  
{true | false}
```

Default: false

Data_type: Boolean

Specifies whether the `write_global_slack_report` command reports worst slack on the register clock pins or not. When set to `true`, the `write_global_slack_report` command reports worst slack due to triggering paths.

A CLK pin might drive a Q pin of the same register, and have a setup or hold check arc with respect to the D pin. In the default setting, when the global is set to `false`, the `write_global_slack_report` command reports no slacks along the clock paths. When the global is set to `true`, the command will report the worst slack on register clock pins due to the related trigger paths and not due to the related timing checks.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

AOCV Global Variables

- [aocv_chip_size](#)
- [aocv_core_size](#)
- [timing_aocv_analysis_mode](#)
- [timing_aocv_derate_mode](#)
- [timing_aocv_slack_threshold](#)
- [timing_aocv_stage_count_recalculate_on_timing_reset](#)

aocv_chip_size

```
aocv_chip_size real_number
```

Default: 1e+30

(Will be automatically calculated from the SPEF file, if it is loaded.)

Data_type: Double

Specifies the diagonal length of the chip, in microns.

The chip and core sizes specified with the `aocv_chip_size` and `aocv_core_size` global variables take priority over chip and core size values derived from the SPEF file, or through placement. In the case that there is no SPEF file loaded, or placement information available:

- If you specify `aocv_chip_size` but not `aocv_core_size`, the software considers the core size to be equal to the `aocv_chip_size` value.
- If you specify `aocv_core_size` but not `aocv_chip_size`, the software ends the AOCV analysis and generates an error message.
- If both `aocv_chip_size` and `aocv_core_size` are not specified, the software ends the AOCV analysis and generates an error message.

To set this global variable, use the `set_global` command.

Related Topics

- "[Analysis and Reporting](#)" chapter in the *Tempus User Guide*:
 - [Using AOCV Derating Library Formats](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

aocv_core_size

`aocv_core_size` *real_number*

Default: 1e+30

(Will be automatically calculated from the SPEF file, if it is loaded.)

Data_type: Double

Specifies the diagonal length of the core area, in microns.

The chip and core sizes specified with the `aocv_chip_size` and `aocv_core_size` global variables take priority over chip and core size values derived from the SPEF file, or through placement. In the case that there is no SPEF file loaded, or placement information available:

- If you specify `aocv_chip_size` but not `aocv_core_size`, the software considers the core size to be equal to the `aocv_chip_size` value.
- If you specify `aocv_core_size` but not `aocv_chip_size`, the software ends the AOCV analysis and generates an error message.
- If both `aocv_chip_size` and `aocv_core_size` are not specified, the software ends the AOCV analysis and generates an error message.

To set this global variable, use the `set_global` command.

Related Topics

- "[Analysis and Reporting](#)" chapter in the *Tempus User Guide*:
 - [Using AOCV Derating Library Formats](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_aocv_analysis_mode

timing_aocv_analysis_mode
{launch_capture | clock_only | separate_data_clock | combine_launch_capture}

Default: launch_capture

Data_type: Enum

Controls handling of AOCV analysis modes. You can specify one of the following modes:

- **launch_capture:** AOCV analysis will be performed for both data and clock paths, and the AOCV derating factor will be calculated based on both clock and data path stages.
- **clock_only:** AOCV analysis will be performed for clock paths only, and the AOCV derating factor will be calculated based on the clock path stage depth only.
- **separate_data_clock:** The software calculates the AOCV stage count for clock paths and data paths separately. Both clock and data paths have related AOCV derating factors.
- **combine_launch_capture:** The software calculates AOCV stage count using combined depth for launch and capture paths. This is achieved by summation of path depth on the launch and capture paths.

To set this global variable, use the `set` command.

Related Topics

- "[Analysis and Reporting](#)" chapter in the *Tempus User Guide*:
 - [Stage Counts and their Properties for Launch and Capture Paths](#)
 - [Configuring AOCV Modes](#)
 - [Reporting in AOCV Modes](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_aocv_derate_mode

```
timing_aocv_derate_mode  
{aocv_multiplicative | aocv_additive}
```

Default: aocv_multiplicative

Data_type: Enum

Controls the AOCV derating mode.

When set to `aocv_multiplicative`, the derating factor will be calculated as AOCV derating * OCV derating, which is set using the `set_timing_derate` command.

When set to `aocv_additive`, the derating factor will be calculated as AOCV derating + OCV derating values.

When you use this global variable, the `report_timing` command shows the `total_derate` column in the timing report output, which allows you to view and cross-check the calculated total derate factor.

To set this global variable, use the `set_global` command.

Related Topics

- "[Analysis and Reporting](#)" chapter in the *Tempus User Guide*:
 - [Handling OCV Derating With AOCV](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_aocv_slack_threshold

timing_aocv_slack_threshold *integer*

Default: 0

Data_type: double

Analyzes AOCV slack that is based at the specified threshold.

To use this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_aocv_stage_count_recalculate_on_timing_reset

```
timing_aocv_stage_count_recalculate_on_timing_reset  
{true | false}
```

Default: false

Data_type: Boolean

Re-calculates the AOCV stage counts when timing is reset.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

SOCV Global Variables

- [timing_cppr_opposite_edge_mean_scale_factor](#)
- [timing_cppr_opposite_edge_sigma_scale_factor](#)
- [timing_set_nsigma_multiplier](#)
- [timing_socv_preserve_variation_with_annotations](#)
- [timing_socv_rc_variation_mode](#)
- [timing_socv_statistical_min_max_mode](#)

timing_cppr_opposite_edge_mean_scale_factor

`timing_cppr_opposite_edge_mean_scale_factor` *value*

Default: 1

Value Range: 0 to 1

Data_type: Double

Controls the mean delay component for CPPR credit for opposite transition edges at a common pin.

When this global variable is specified, the software provides a finer level configuration of the CPPR credit for reducing Common Path Pessimism Removal (CPPR) in timing analysis. The software allows a certain amount of CPPR credit in the case of opposite edge transitions to be scaled to a user-specified value, that is between zero credit or full credit provided by the existing controls.

You can use [timing_cppr_opposite_edge_sigma_scale_factor](#) global variable for adjustment of the sigma component of the CPPR credit. In SOCV analysis mode, you can use both of these global variables to separately control the adjustment of the mean and sigma components of the CPPR credit. In regular static-timing analysis (STA) mode (without SOCV), only the mean scale factor is considered for CPPR adjustment scaling.

You can use the `set` command to set this global variable.

Related Topics

- ["Analysis and Reporting"](#) chapter in the *Tempus User Guide*
 - [Supported CPPR Global Variables](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_cppr_opposite_edge_sigma_scale_factor

`timing_cppr_opposite_edge_sigma_scale_factor` *value*

Default: 1

Data_type: Double

Controls the sigma delay component for CPPR credit for opposite transition edges at a common pin.

This global variable provides a finer level configuration of the CPPR credit for reducing Common Path Pessimism Removal (CPPR) in timing analysis.

You can use `timing_cppr_opposite_edge_mean_scale_factor` global variable for adjustment of the mean component of the CPPR credit. In SOCV analysis mode, you can use both of these global variables to separately control the adjustment of the mean and sigma components of the CPPR credit. In static-timing analysis (STA) mode, only the mean scale factor is considered.

You can use the `set` command to set this global variable.

Related Topics

- ["Analysis and Reporting"](#) chapter in the *Tempus User Guide*:
 - [Supported CPPR Global Variables](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_set_nsigma_multiplier

`timing_set_nsigma_multiplier value`

Default: 3

Data_type: Double

Controls the sigma multiplier to be used in SOCV mode.

This global variable allows a single N value for both setup and hold reporting. For example,

```
set timing_set_nsigma_multiplier 4.5
```

To set this global variable, use the `set` command.

Note: The `timing_set_nsigma_multiplier` global variable is obsolete and has been replaced with the `set_socv_reporting_nsigma_multiplier` command. You can use this global variable by setting `timing_socv_view_based_nsigma_multiplier_mode` to `false`. For future compatibility, you should migrate your scripts to use the new view-based methodology.

Related Topics

- "[Analysis and Reporting](#)" chapter in the *Tempus User Guide*:
 - [Path Reporting With Discrete Values](#)
 - [Sample Initialization and Configuration Scripts](#)
- `set_socv_reporting_nsigma_multiplier`

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_socv_preserve_variation_with_annotations

```
timing_socv_preserve_variation_with_annotations  
{true | false}
```

Default: false

Data_type: Boolean

Preserves the sigma value computed by delay calculation along with the annotation from SDC.

In the clock mesh flow, it is possible to preserve sigma values even after applying annotation on slew/delay through SDC.

When this global variable is set to `true`, the software accepts the sigma value computed by delay calculation (for nets that have delay/slew annotations from SDC, e.g., clock-mesh), along with the annotation from SDC.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_socv_rc_variation_mode

```
timing_socv_rc_variation_mode  
{true | false}
```

Default: true

Data_type: Boolean

Enables modeling of interconnect variation in SOCV mode.

The SOCV flow provides support for RC variation on the interconnect.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_socv_statistical_min_max_mode

```
timing_socv_statistical_min_max_mode  
{statistical | mean_and_three_sigma_bounded}
```

Default: mean_and_three_sigma_bounded

Data_type: Enum

Determines how worst cases will be computed in SOCV mode.

You can select one of the following options:

- **statistical:** The mean and standard deviation for the result of worst-cases is computed to get the greatest or smallest of normal distributions.
- **mean_and_three_sigma_bounded:** The worst mean value is chosen as the mean of the resulting distribution. The standard deviation is computed as a bound from the worst (mean+/-3*sigma) value. This ensures that the 3-sigma value of the result always bounds the worst 3-sigma value.

Note: The `mean_and_three_sigma_bounded` option is required for SOCV-based optimization in Innovus. It is also highly recommended for analysis in Tempus due to improved bounding properties.

To set this global variable, use the `set` command.

Note: The `timing_socv_statistical_min_max_mode` global variable is obsolete and will be removed in a future release. The variable still works in this release, but to avoid any warnings and to ensure compatibility with future releases, you should remove it from your scripts.

Related Topics

- "[Analysis and Reporting](#)" chapter in the *Tempus User Guide*:
 - [SOCV Analysis](#)
 - [Sample Initialization and Configuration Scripts](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

Timer Global Variables

- `timing_disable_bidi_output_timing_checks`
- `timing_disable_bus_contention_check`
- `timing_disable_clock_gating_checks`
- `timing_disable_clockperiod_checks`
- `timing_disable_drv_reports_on_constant_nets`
- `timing_disable_floating_bus_check`
- `timing_disable_genclk_combinational_blocking`
- `timing_disable_inferred_clock_gating_checks`
- `timing_disable_internal_inout_cell_paths`
- `timing_disable_internal_inout_net_arcs`
- `timing_disable_lib_pulsewidth_checks`
- `timing_disable_library_data_to_data_checks`
- `timing_disable_library_tiehi_tielo`
- `timing_disable_netlist_constants`
- `timing_disable_nochange_checks`
- `timing_disable_non_sequential_checks`
- `timing_disable_output_as_clock_port`
- `timing_disable_parallel_arcs`
- `timing_disable_recovery_removal_checks`
- `timing_disable_report_header_info`
- `timing_disable_retime_clock_path_slew_propagation`
- `timing_disable_sdf_retain_arc_merging`
- `timing_disable_skew_checks`
- `timing_disable_test_signal_arc`
- `timing_disable_timing_model_latch_inferencing`
- `timing_disable_tristate_disable_arcs`
- `timing_disable_user_data_to_data_checks`
- `timing_enable_all_fanin_fanout_levels_compatibility`

- `timing_enable_aocv_slack_based`
- `timing_enable_case_analysis_conflict_warning`
- `timing_enable_clock2clock_clockgating_check`
- `timing_enable_clock_phase_based_rise_fall_derating`
- `timing_enable_data_through_clock_gating`
- `timing_enable_derating_for_pulsewidth_checks`
- `timing_enable_early_late_data_slews_for_setuphold_mode_checks`
- `timing_enable_genclk_divide_by_inherit_parent_duty_cycle`
- `timing_enable_genclk_edge_based_source_latency`
- `timing_enable_get_object_escaped_name_backward_compatible`
- `timing_enable_mmmc_loop_handling`
- `timing_enable_multi_drive_net_reduction_with_assertions`
- `timing_enable_multifrequency_latch_analysis`
- `timing_enable_pessimistic_cpvr_for_reconvergent_clock_paths`
- `timing_enable_power_ground_constants`
- `timing_enable_preset_clear_arcs`
- `timing_enable_pulsed_latch`
- `timing_enable_sdc_compatible_data_check_mcp`
- `timing_enable_si_cpvr`
- `timing_enable_simultaneous_setup_hold_mode`
- `timing_enable_tristate_clock_gating`
- `timing_enable_uncertainty_for_clock_checks`
- `timing_enable_uncertainty_for_pulsewidth_checks`
- `timing_path_based_enable_verbose_mode`

`timing_disable_bidi_output_timing_checks`

`timing_disable_bidi_output_timing_checks`
{true | false}

Default: false

Data_type: Boolean

When set to `false`, timing checks on both the input and output sides of the bidirectional pin are analyzed.

When set to `true`, the software disables timing checks on the output side of bidirectional pins.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_bus_contention_check

```
timing_disable_bus_contention_check  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, disables propagation of maximum delay along three state disable timing arcs and minimum delay along three state enable arcs. If you know that bus contention conditions do not occur in your design, disable these checks by setting these globals to `true`.

Checks for setup and hold violations in three-state bus designs. Checks the worst delay path to ensure that the latched signals are stable and not unknown values - X. Ensures proper timing checks for bus contention.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_clock_gating_checks

```
timing_disable_clock_gating_checks  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, disables all clock gating checks, including those declared in the library on integrated clock gating cells, and those that are inferred on combinational gates in the clock path.

To set this global variable, use the `set_global` command.

Note: The `timing_disable_clock_gating_checks` global variable should not be used with the `set_analysis_mode` command.

If the `set_analysis_mode -clockGatingCheck` parameter is specified, then you need to set this global variable to `true`.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_clockperiod_checks

```
timing_disable_clockperiod_checks  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, disables timing model clock period checks during timing analysis.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_drv_reports_on_constant_nets

```
timing_disable_drv_reports_on_constant_nets  
{true | false}
```

Default: false

Data_type: Boolean

Disables DRV checks for nets that are disabled by constant propagation.

When set to `true`, the `report_constraint` command does not report DRV violations for constant nets.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_floating_bus_check

```
timing_disable_floating_bus_check  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, disables propagation of minimum delay through three state disable timing arcs and maximum delay through three state enable arcs. These checks are only valid during floating bus conditions.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_genclk_combinational_blocking

```
timing_disable_genclk_combinational_blocking  
{true | false}
```

Default: false

Data_type: Boolean

Disables blocking of latency path for a generated clock.

When set to `true`, allows the latency path to traverse through the upstream generated clock.

When set to `false`, if two back-to-back generated clocks are on the same path, and if the downstream generated clock is set using the `create_generated_clock -combinational` parameter, then the latency path for this generated clock is blocked by the upstream clock. The downstream generated clock is created with zero latency.

The software shows a warning if the source latency path to the upstream generated clock cannot be found. For example,

```
**WARN: (TA-1018): A source latency path to the generated clock GC2 through source pin clk1  
to target pin GC2_1/Y in view default_emulate_view cannot be found. Timing analysis will use  
0 ns source latency for the generated clock and will interpret the master clock based on the  
polarity at the master clock source pin.
```

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_inferred_clock_gating_checks

```
timing_disable_inferred_clock_gating_checks  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, disables clock gating checks that are inferred on combinational elements in the clock path. Explicit clock gating checks that are described in the timing library are not affected by this global variable.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_internal_inout_cell_paths

```
timing_disable_internal_inout_cell_paths  
{true | false}
```

Default: true

Data_type: Boolean

When set to `false`, enables internal bidirectional feedback paths that are completely contained in one instance.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_internal_inout_net_arcs

```
timing_disable_internal_inout_net_arcs  
{true | false}
```

Default: true

Data_type: Boolean

When set to `true`, this global disables internal bidirectional feedback paths that span multiple instances.

When set to `false`, this global enables internal bidirectional feedback paths, which are feedback paths that you might not want to use during analysis and optimization.

This global applies only to paths that span multiple instances. The global has no impact on internal feedback paths that are completely contained in one instance.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_lib_pulsewidth_checks

```
timing_disable_lib_pulsewidth_checks  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, disables timing model pulse width checks during timing analysis.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_library_data_to_data_checks

```
timing_disable_library_data_to_data_checks  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, the software disables the data-to-data checks that are coded in a library as non-sequential timing checks. The data-to-data checks are distinguished from other non-sequential timing checks because both the reference pin and constrained pin side of the timing check are data pins rather than clock pins.

You can use the [timing_disable_non_sequential_checks](#) global variable to control other types of non-sequential checks where the reference pin is a clock pin.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_library_tiehi_tielo

```
timing_disable_library_tiehi_tielo  
{true | false}
```

Default: false

Data_type: Boolean

Disables constant functions in a library.

When set to `false`, the software considers constant propagation from library tie high/low cells prior to those from logical netlist and constraints. There may be cases where programmable tie cells are used whose value can be changed later in the flow, assuming the programmable tie cells are modelled with constant output functions.

When set to `true`, the library constants are not propagated during timing analysis or timing optimization.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_netlist_constants

```
timing_disable_netlist_constants  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, ignores constants defined in the Verilog netlist. This includes all uses of the in-place constant notation, such as `1'b0`, and the use of power and ground nets. Constants from `set_case_analysis` assertions and from tie-hi and tie-lo cells are still recognized.

When set to `false`, the software recognizes all constants.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_nochange_checks

timing_disable_nochange_checks
{true | false}

Default: false

Data_type: Boolean

When set to `true`, disables no change timing model checks during timing analysis.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_non_sequential_checks

timing_disable_non_sequential_checks
{true | false}

Default: false

Data_type: Boolean

When set to `true`, the software disables the timing arcs between any data-to-clock or clock-to-clock checks - when `timing_type` attribute is set to one of the following:

- `non_seq_setup_rising`
- `non_seq_setup_falling`
- `non_seq_hold_rising`
- `non_seq_hold_falling`

This global variable affects non-sequential checks only, where a reference pin of the timing check is a clock pin. The data-to-data checks (where the reference pin of the timing check is a data pin) are not affected by this setting. To disable data-to-data checks you can use the `timing_disable_library_data_to_data_checks` global variable.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_output_as_clock_port

timing_disable_output_as_clock_port
{true | false}

Default: false

Data_type: Boolean

Allows a clock path to be considered as a data path.

When set to `true`, the software allows a clock source path, leading to an output or bidi port, to be treated as a data path, irrespective of constraints, such as, `set_output_delay`, `set_data_check`, or `set_max_delay`, asserted on that port.

When set to `false`, all the clock source paths, leading to output ports, are considered as valid clock end points. Any unconstrained output port may not be considered as data end point.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_parallel_arcs

timing_disable_parallel_arcs
{true | false}

Default: true

Data_type: Boolean

Enables the software to automatically perform reduction of parallel delay arcs and check arcs to improve efficiency.

When set to `false`, parallel delay and check arc reduction will be disabled.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_recovery_removal_checks

```
timing_disable_recovery_removal_checks  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, disables recovery and removal timing model checks during timing analysis.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_report_header_info

`timing_disable_report_header_info {true | false}`

Default: `false`

Data_type: Boolean

Controls whether timing reports are generated using a common report header.

When set to `false`, the software generates timing reports with a report header that includes the following information:

- Tempus software version number
- Platform on which the software is being run
- Time when the report was generated
- Command used to generate the report

Any report-specific banner information is included in the same header box, but listed after the common header information.

When set to `true`, the software generates backward-compatible timing reports without the additional header information.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_retime_clock_path_slew_propagation

timing_disable_retime_clock_path_slew_propagation
{true | false}

Default: true

Data_type: Boolean

When set to `true`, path based analysis is performed on data paths only and the clock segments are not considered.

When set to `false`, analysis is performed on both data and clock paths.

To set this global variable, use the `set` command.

Related Topics

- "[Analysis and Reporting](#)" chapter in the *Tempus User Guide*:
 - [Reporting in Path-Based Analysis](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_sdf_retain_arc_merging

```
timing_disable_sdf_retain_arc_merging  
{true | false}
```

Default: false

Data_type: Boolean

Controls handling of retain arcs in SDF2.1.

When set to `false`, the Retain information is merged into SDF 2.1, as Retain represents a valid minimum path timing for a cell.

When set to `true`, the Retain arc merging will be disabled.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_skew_checks

```
timing_disable_skew_checks  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, disables library skew checks for timing analysis.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_test_signal_arc

```
timing_disable_test_signal_arc  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, timing analysis will not analyze the signal arcs coming from or going to the test pin. Such pins are marked in the timing library with either the `test_scan_in`, the `test_scan_enable` attribute for the input pin, the `test_scan_clock` attribute for the input clock pin, or the `test_output_only` attribute for the output pin.

The following are the equivalent Liberty and TLF pin attributes:

Liberty and TLF Pin Attributes

Type	Liberty	TLF
enable	test_scan_enable	SCAN_PINTYPE (ENABLE)
input	test_scan_in	SCAN_PINTYPE (INPUT)
input clock	test_scan_clock	SCAN_PINTYPE (CLOCK)
output	test_output_only	SCAN_PINTYPE (OUTPUT)

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_timing_model_latch_inferencing

timing_disable_timing_model_latch_inferencing
{true | false}

Default: true

Data_type: Boolean

Controls whether latch behavior is inferred for cell descriptions tagged with the Liberty `timing_model_type` attribute, including all values: `abstracted`, `extracted`, or `qtm`.

When set to `false`, the software infers latch behavior from the timing check and timing delay arc relationships in the model's description.

When set to `true`, latch inferencing does not happen for cells with the `timing_model_type` attribute. Explicit Liberty latch modelling constructs must be coded into the description in order to model latch behavior.

This global does not affect Liberty timing model cells that do not contain the `timing_model_type` attribute. Standard cell core libraries generally do not have this attribute specified; therefore, they typically are not be affected by this global setting.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_tristate_disable_arcs

```
timing_disable_tristate_disable_arcs  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, disables all 0/1->Z transitions of tristate arcs during timing analysis. By default, all tristate arcs are enabled.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_disable_user_data_to_data_checks

```
timing_disable_user_data_to_data_checks  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, disables data to data checks that are created by the `set_data_check` command. This global variable does not affect non-sequential timing checks described in the timing library.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_all_fanin_fanout_levels_compatibility

```
timing_enable_all_fanin_fanout_levels_compatibility  
{true | false}
```

Default: true

Data_type: Boolean

Enables computation of gate level count (for `all_fanout/all_fanin` commands) when an input pin is specified using the `all_fanout -from` parameter and output pin is specified using the `all_fanin -to` parameter. When the initial gate is traversed, the count is computed as 1.

When set to `false`, the global computes the gate level count as 0.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_aocv_slack_based

timing_enable_aocv_slack_based {true | false}

Default: false

Data_type: Boolean

Enables AOCV analysis using slack-based method.

When set to `true`, the software will try to remove excess pessimism from the graph-based analysis (GBA) AOCV stage counts. The paths to endpoints, which are under the specified slack criteria (default 0.0 ns), are considered when computing stage counts. This allows the stage counts and subsequent AOCV derates to be aligned with the failing paths – and not unnecessarily punished by derates coming from related short paths that meet timing requirements.

Note: This setting is not recommended for signoff timing analysis in Tempus. For signoff, path-based analysis (PBA) is the recommended flow for removing excess GBA pessimism.

See [timing_aocv_slack_threshold](#).

To use this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_case_analysis_conflict_warning

timing_enable_case_analysis_conflict_warning
{true | false}

Default: true

Data_type: Boolean

Displays a warning when propagated and asserted constant values are not the same when a full timing update is performed.

When set to `true`, a report file, named `CTE_constant_mismatch.rpt`, is generated that contains details of pins and the reason for constant mismatch. This report is over-written each time a full timing update takes place.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_clock2clock_clockgating_check

```
timing_enable_clock2clock_clockgating_check  
{true | false}
```

Default: false

Data_type: Boolean

Enables clock-to-clock clock gating checks.

When set to `true`, the software performs clock to clock gating checks at gating elements. For clock to clock gating checks, the frequency of a clock arriving at the reference side of a check should be equal to or greater than the frequency of the clock arriving at the signal side of the check.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_clock_phase_based_rise_fall_derating

```
timing_enable_clock_phase_based_rise_fall_derating  
{true | false}
```

Default: false

Data_type: Boolean

Determines the interpretation of edge-specific derating factors for clock paths, specified using the `set_timing_derate -rise/-fall` parameters.

When set to `false`, the `set_timing_derate -rise/-fall` parameter refer to the direction of the transition from the perspective of each individual output pin along the clock path.

When set to `true`, the `-rise/-fall` parameter refers to which edge (or phase) of the clock is triggering the transition as viewed from the source of the clock where the clock waveform is defined.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_data_through_clock_gating

```
timing_enable_data_through_clock_gating  
{true | false}
```

Default: true

Data_type: Boolean

When set to `false`, blocks signals arriving on the enable of the clock-gating check.

When set to `true`, allows both data and clock signals at the enable pin of a clock-gating check to propagate forward past the check.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_derating_for_pulsewidth_checks

```
timing_enable_derating_for_pulsewidth_checks  
{true | false}
```

Default: false

Data_type: Boolean

Controls whether the cell check derates, set by `set_timing_derate -cell_check` command, can be applied for pulse width checks.

When set to `true`, the pulse width checks honor cell check derate factors.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_early_late_data_slews_for_setuphold_mode_checks

timing_enable_early_late_data_slews_for_setuphold_mode_checks
{true | false}

Default: false

Data_type: Boolean

Controls calculated delays.

When set to `true`, the software allows use of both the minimum and maximum data slews for setup and hold checks. By default, the minimum or maximum slew values for clock pins are considered, and for data pins only the maximum slew value is considered while calculating the setup/hold timing check values.

This global variable enables an eight-way timing check for:

- Early/late libraries
- Clock/data pins
- Min/max slews

You can use this global variable in simultaneous setup or hold mode only. To enable this mode, you need to set `timing_enable_simultaneous_setup_hold_mode` to `true`. In this mode, you can use the `report_delay_calculation` command (`-from` and `-to` parameters) to report all the setup/hold check values corresponding to minimum/maximum combination of clock and data slews.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_genclock_divide_by_inherit_parent_duty_cycle

timing_enable_genclock_divide_by_inherit_parent_duty_cycle
{true | false}

Default: false

Data_type: Boolean

Controls inheritance of duty cycle from master clock for generated clocks (generated using the `create_generated_clock -divide_by` parameter).

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_genclock_edge_based_source_latency

```
timing_enable_genclock_edge_based_source_latency  
{true | false}
```

Default: true

Data_type: Boolean

Controls how the software chooses generated clock source latency paths. Certain specifications of the [create_generated_clock](#) constraint imply a cause-and-effect relationship between the edges of the master clock and those of the generated clock.

When set to `true`, the software checks whether a path being analyzed has the correct cause-effect relationship consistent with the specified constraint. Paths that are found to be inconsistent are ignored. Only the paths that are in agreement with the required edge relationship are considered for source latency. If no such path exists, the software uses a source latency of 0 ns. This selection is done independently for each edge of the generated clock; therefore the result might be different for each edge.

The edge-to-edge sufficiency checks are performed assuming a Rise-to-Rise, Rise-to-Fall relationship between the master clock and the generated clock with an even number `-divide_by` factor. For the odd `-divide_by` factors, the sufficiency check is performed assuming a Rise-to-Rise, Fall-to-Fall relationship. The sufficiency checking is not performed for `-multiply_by` options.

When set to `false`, the software does not check paths for the correct cause-effect relationship.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_get_object_escaped_name_backward_compatible

```
timing_enable_get_object_escaped_name_backward_compatible  
{true | false}
```

Default: false

Data_type: Boolean

Enables printing of escape characters in the `get_object_name` command output.

When set to `false`, the `get_object_name` command does not print escape characters (starting with a backslash) in the object name.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_mmmc_loop_handling

timing_enable_mmmc_loop_handling
{true | false}

Default: true

Data_type: Boolean

When set to `true`, loop breaking is handled independently per analysis view, according to the following rules in order of precedence:

- If an analysis view does not contain a combinational loop, it does not inherit any loop breaking done for other views.
- If a loop exists, and the user has supplied specific case analysis and `set_disable_timing` constraints to break the loop at a certain place, the specified loop break is honored.
- If a loop is still detected, the software automatically selects a point to break the loop, and applies the loop break consistently to any views where the loop exists.

When set to `false`, if a combinational loop is detected in any analysis view, the timing system breaks the loop by disabling an arc along the looping path. These combinational loop breaks are reflected in all analysis views, even if those views by themselves do not contain these looping paths.

 Cadence strongly recommends that all combinational loops detected by the software be evaluated, to ensure that the loop is broken at the most ideal location. Automatic loop breaking decisions are made based on how the design is traversed. Small changes in the design can influence where loop breaking decisions are made.

To reduce risk and the need to re-evaluate loop-breaking decisions, it is highly recommended that once combinational loop breaks have been evaluated, the user add constraints to ensure that the loop is always broken in that location.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_multi_drive_net_reduction_with_assertions

timing_enable_multi_drive_net_reduction_with_assertions
{none | delay | all}

Default: none

Data_type: Enum

Allows reduction of multi-drive networks when constraint assertions are present on the network elements:

The following options are available:

none: Reduction is not performed on portions of the network where constraints have been asserted.

delay: Reduction of the multi-drive network will still be applied if only `set_annotated_delay` and `set_annotated_transition` constraints have been applied. Other constraints present on the multi-drive network can still prevent full reduction.

all: The software ignores the presence of any assertions on the multi-drive network and forces reduction to occur.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_multifrequency_latch_analysis

```
timing_enable_multifrequency_latch_analysis  
{true | false}
```

Default: false

Data_type: Boolean

Enables multi-frequency latch timing analysis of the latch time borrowing for when a data signal coming to a latch is controlled by a clock with a frequency different to a clock of the latch enabling signal.

When set to `true`, the edges of all sending and receiving clocks are analyzed at every latch, which results in less pessimistic, and more detailed timing analysis.

When set to `false`, latch time borrowing analysis uses the worst case relationship between sending and receiving clocks at the latch, which can produce paths with pessimistic slack.

Note: Enabling advanced latch analysis can result in some performance penalty.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_pessimistic_cppr_for_reconvergent_clock_paths

timing_enable_pessimistic_cppr_for_reconvergent_clock_paths
{true | false}

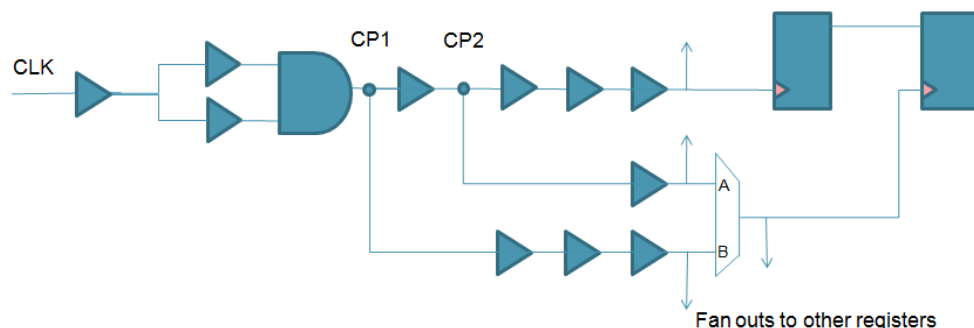
Default: false

Data_type: Boolean

When set to `true`, the software will consider all combinations of launch and capture clock path pairs to identify all possible common pins, and then selects the pin with minimum CPPR adjustment over all such common pins as the common pin.

When set to `false`, the default, the software will choose the CPPR common pin based on the commonality of the worst launch and worst capture path only.

Consider the following example,



In the above diagram, both capture paths are considered along with the single launch path, and the common pin is chosen as the pin with the least CPPR adjustment. The pin CP1 is chosen as a common pin and CPPR is reported up to this pin. The same common pin, CP1, will be used for CPPR adjustment whether the capture clock path goes through the mux A pin or the mux B pin. When this global variable is set to false, CPPR common pin is chosen by the commonality in the worst-case launch and worst-case capture paths. so CP2 is chosen as the common pin and CPPR is reported up to this pin.

Note: The `timing_cppr_skip_clock_reconvergence` and `timing_enable_pessimistic_cppr_for_reconvergent_clock_paths` global variables are mutually exclusive.

Also, the `timing_cppr_skip_clock_reconvergence` global variable has precedence over `timing_enable_pessimistic_cppr_for_reconvergent_clock_paths`.

To set this global variable, use the `set_global` command.

Related Topics

- "Analysis and Reporting" chapter in the *Tempus User Guide*:
 - Supported CPPR Global Variables

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_power_ground_constants

```
timing_enable_power_ground_constants  
{true | false}
```

Default: false

Data_type: Boolean

Determines if case analysis is inferred from power and ground rail connections/nets, e.g., VDD and GND.

When set to `true`, the software recognizes the constants that come from power and ground rail connections.

When set to `false`, the software ignores these constants. This provides an intuitive behavior with logical style netlist (and other tools that do not read physical data).

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_preset_clear_arcs

```
timing_enable_preset_clear_arcs  
{true | false}
```

Default: false

Data_type: Boolean

Determines whether the timing arcs are created to model the transition to active state (assertion) of the preset or clear pin, and the subsequent transition of the output to a controlled state.

When set to `true`, the software builds the arcs and the timer automatically analyzes paths through them (enable clock propagation through preset clear pins).

When set to `false`, the software does not build these arcs, because they are associated with analyzing the reset mode timing paths. Filtering them prevents unwanted interaction with functional mode timing analysis.

You can use the `lib_build_async_de_assert_arc` global variable to control whether input to output arcs from the preset or clear pins transitioning to inactive state are included when the timing system is initialized. If both the `timing_enable_preset_clear_arcs` and the `lib_build_async_de_assert_arc` global variables are set to `true`, all asynchronous arcs are recognized. However, if `timing_enable_preset_clear_arcs` is set to `true` and `lib_build_async_de_assert_arc` is set to `false`, only assert asynchronous arcs are recognized.

Note: The `timing_enable_preset_clear_arcs` global variable can also impact delay calculation results of a design. When an arc is disabled, the software will calculate delay using the linear table lookup. When an arc is not disabled, the delay calculations are made based on the quadratic table lookup.

Note: This global variable should be set to `true` when reset checks are performed on a dedicated reset SDC.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_pulsed_latch

```
timing_enable_pulsed_latch  
{true | false}
```

Default: true

Data_type: Boolean

Enables pulse latch analysis.

When set to `true`, the pulse latch analysis mode provides several additional pulse-clock specific constraint specifications and reporting capabilities.

When in pulse-latch mode, you can use the following pulse clock constraints:

- `set_pulse_clock_max_width`
- `set_pulse_clock_min_width`
- `set_pulse_clock_max_transition`
- `set_pulse_clock_min_transition`

Additionally, in this mode, the `report_constraint` command supports the following pulse clock check types:

- `pulse_clock_max_width`
- `pulse_clock_min_width`
- `pulse_clock_max_transition`
- `pulse_clock_min_transition`

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_sdc_compatible_data_check_mcp

```
timing_enable_sdc_compatible_data_check_mcp  
{true | false}
```

Default: false

Data_type: Boolean

Enables checking for SDC compatible data checks when `set_multicycle_path -start` parameter is specified.

When set to `true`, the `-start` parameter shifts (or reverses) the reference clock back by one clock cycle.

When set to `false`, the `set_multicycle_path -start` and `-end` parameters use the same number of cycles, even though the actual adjustment is computed using the signal clock for `-start` and reference clock for `-end` parameters. Additionally, the software computes adjustments for `set_data_check -hold` checks between multiple frequency clocks with the `set_multicycle_path -start -setup` setting.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_si_cppr

timing_enable_si_cppr
{true | false}

Default: true

Data_type: Boolean

Enables more accurate CPPR analysis when incremental delays are present and the [timing_remove_clock_reconvergence_pessimism](#) global variable is set to true.

When set to `true`, the incremental delay is not included in the CPPR calculation during setup analysis, but is included in the CPPR calculation during hold analysis.

When set to `false`, the incremental delay is included in both setup and hold CPPR calculations.

To set this global variable, use the [set_global](#) command.

Related Topics

- ["Analysis and Reporting"](#) chapter in the *Tempus User Guide*:
 - [Supported CPPR Global Variables](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_simultaneous_setup_hold_mode

timing_enable_simultaneous_setup_hold_mode
{true | false}

Default: true

Data_type: Boolean

Controls whether setup and hold checks are analyzed separately, or together on the same timing graph.

When set to `false`, the `set_analysis_mode -checkType setup` and `hold` parameters are used to switch the system from analyzing setup slacks to analyzing hold slacks.

When set to `true`, setup and hold slacks are analyzed concurrently. As a consequence, both `-early` and `-late` queries for reporting commands such as `report_timing` can be satisfied immediately, without additional calculation. Simultaneous mode is useful for immediately understanding the consequences to both setup and hold slacks from what-if operations and manual ECOs to the design. The `set_analysis_mode -checkType setup/hold` do not function in simultaneous mode.

Note: Simultaneous setup and hold mode can be used only for timing analysis; it cannot be used for any physical implementation steps.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_tristate_clock_gating

```
timing_enable_tristate_clock_gating  
{true | false}
```

Default: false

Data_type: Boolean

The timing system infers gated-clock checks in the design where a data and clock path intersect at a combinational device. When this global is set to `true`, inferred gated-clock checks are added when clock and data signals converge through the tristate enable and data input of tristate buffers. By default (a setting of `false`), gated-clock checks are not added for these situations.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_uncertainty_for_clock_checks

```
timing_enable_uncertainty_for_clock_checks  
{true | all | false | none | pulse_width | period | skew}
```

Default: true

Data_type: String

Considers the clock uncertainty when performing clock checks.

You can specify the following options:

- `true`: The `set_clock_uncertainty` command supports clock uncertainty for period, pulse width, skew, and separation checks.
- `all`
- `false`
- `none`
- `pulse_width`
- `period`
- `skew`

Note: The `timing_enable_uncertainty_for_clock_checks` global variable overrides `timing_enable_uncertainty_for_pulsewidth_checks` setting.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_uncertainty_for_pulsewidth_checks

timing_enable_uncertainty_for_pulsewidth_checks
{true | false}

Default: false

Data_type: Boolean

When set to `true`, considers clock uncertainty when performing minimum pulse width checks. Clock uncertainty is considered during checks of both user-defined minimum pulse widths (using the `.lib`).

Note: Specifying a value of `true` affects the following commands that take minimum pulse width checks into account:

- `report_timing -check_type pulse_width`
- `report_timing -check_clocks`
- `report_min_pulse_width`
- `report_analysis_coverage`

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_path_based_enable_verbose_mode

timing_path_based_enable_verbose_mode *value*

Default: 1

Data_type: Enum

Controls printing of verbose messages during and at the end of path-based analysis (PBA).

You can specify one of the three values - 0, 1, and 2. These are described below:

- 0: Verbose messages are not reported during exhaustive PBA analysis. For default PBA analysis, the path count will be reported.
- 1: Reports the final message for PBA path count and max depth reached during exhaustive PBA. Progress messages are not reported.
- 2: Reports multiple progress messages after regular intervals (in addition to the final message for PBA path count and max depth).

This global variable impacts messages when the `report_timing` and `report_constraint` commands are used in PBA mode.

To set this global variable, use the `set` command.

Examples

- The following commands print messages during and after PBA analysis:

```
set timing_path_based_enable_verbose_mode 2
report_timing -retime path_slew_propagation -retime_mode exhaustive -max_paths 500000 -
nworst 1
```

The following progress messages are displayed during PBA analysis:

```
INFO: Path Based Analysis (PBA) currently performed on '108455' paths
INFO: Path Based Analysis (PBA) currently performed on '217574' paths
INFO: Path Based Analysis (PBA) currently performed on '327511' paths
INFO: Path Based Analysis (PBA) currently performed on '437472' paths
....
```

The following messages are displayed at the end of PBA analysis mentioning the PBA path count and max depth reached during exhaustive PBA:

```
INFO: Path Based Analysis (PBA) performed on total '2660526' paths
INFO: Max depth 9460 reached for end point 'top/hier1/hier2/flop/d' during Path Based
Analysis
```

Other Reference Information

- Click [here](#) for more information on this global variable.

Library Global Variables

- [timing_library_convert_async_setuphold_to_recrem](#)
- [timing_library_genclk_use_group_name](#)
- [timing_library_hold_constraint_corner_sigma_multiplier](#)
- [timing_library_hold_sigma_multiplier](#)
- [timing_library_ignore_combinational_arc_mismatch_tables](#)
- [timing_library_infer_async_pins_from_timing_arcs](#)
- [timing_library_infer_cap_range_from_ccs_receiver_model](#)
- [timing_library_infer_cap_range_from_ecsm_receiver_model](#)
- [timing_library_infer_socv_from_aocv](#)
- [timing_library_read_ccs_noise_data](#)
- [timing_library_read_without_power](#)
- [timing_library_scale_aocv_to_socv_to_n_sigma](#)
- [timing_library_setup_constraint_corner_sigma_multiplier](#)
- [timing_library_setup_sigma_multiplier](#)
- [timing_library_use_trilib_drv_values](#)
- [timing_library_use_two_piece_receiver_cap](#)

timing_library_convert_async_setuphold_to_recrem

```
timing_library_convert_async_setuphold_to_recrem  
[1 | 0]
```

Default: 1

Data_type: Integer

Controls whether single-edged setup and hold checks on asynchronous pins in the Liberty library are inferred as recovery and removal checks.

This global variable provides support for older modeling styles in legacy libraries.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_library_genclk_use_group_name

timing_library_genclk_use_group_name
{true | false}

Default: false

Data_type: Boolean

Uses generated_clock group name when creating a generated clock from a library-generated clock group.

When set to `false`, the software uses the pin name when creating the generated clock. For example, assume the cell for instance `a/b` in the timing library contained the following generated clock group:

```
generated_clock(genclk2) {  
  clock_pin : clk1;  
  master_pin : "int/clk";  
  multiplied_by : 2;  
  duty_cycle : 50.0;  
}
```

By default (`false`), the software creates a generated clock with the following name:

`a/b/clk1`

If the `timing_prefix_module_name_with_library_genclk` global variable is set to `true`, and you set this global variable to `true`, the software creates a generated clock with the following name:

`a/b/genclk2`

If the `timing_prefix_module_name_with_library_genclk` global variable is set to `false`, and you set this global variable to `true`, the software creates a generated clock with the following name:

`genclk2`

To set this global variable, use the `set_global` command.

Related Topics

- "Timing Model Generation" chapter in the *Tempus User Guide*:
 - [Basic Elements of Timing Model Extraction](#)
 - [Validation of Generated ETM](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_library_hold_constraint_corner_sigma_multiplier

timing_library_hold_constraint_corner_sigma_multiplier *double*

Default: 0

Data_type: Double

Applies user control multiplier to generate a constraint table using sigma values applied for hold arc.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_library_hold_sigma_multiplier

`timing_library_hold_sigma_multiplier` *double*

Default: 0

Data_type: Double

Applies user control multiplier to generate a constraint table applied for hold arc.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_library_ignore_combinational_arc_mismatch_tables

timing_library_ignore_combinational_arc_mismatch_tables
{true | false}

Default: false

Data_type: Boolean

When set to `true`, the software will give preference to `timing_type` over arc tables. This will change infer arcs based on the `timing_type` (liberty format) or timing tables settings. A warning message will be displayed when the preference (`timing_type` or arc tables) is changed.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_library_infer_async_pins_from_timing_arcs

```
timing_library_infer_async_pins_from_timing_arcs  
{true | false}
```

Default: *false*

Data_type: Boolean

When set to `true`, a pin is marked clear if it has timing arcs with type “clear” to output pins. The sequential block is not required to be present in the cell for marking a pin as clear.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_library_infer_cap_range_from_ccs_receiver_model

```
timing_library_infer_cap_range_from_ccs_receiver_model  
{true | false}
```

Default: false

Data_type: Boolean

Infers the rise or fall of capacitance range values from CCS receiver capacitance model.

When set to `false`, the rise/fall CCS capacitance range values are not inferred. The NLDM pin capacitance range values defined in the library are honored.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_library_infer_cap_range_from_ecsm_receiver_model

```
timing_library_infer_cap_range_from_ecsm_receiver_model {true | false}
```

Default: `false`

Data_type: Boolean

Infers the rise or fall of NLDM pin capacitance range from the ECSM capacitance range while calculating the total loading capacitance during switching power computation. If the global is set to `true` and the capacitance range is not defined in the library, the rise/fall capacitance range is inferred from the ECSM receiver capacitance models. The capacitance range values are derived either from the pin-level ECSM cap models or the timing-level ECSM cap models.

When this global is set to `false`, the rise/fall capacitance range values are not inferred, and the library values are honored.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_library_infer_socv_from_aocv

`timing_library_infer_socv_from_aocv {true | false}`

Default: `false`

Data_type: Boolean

Controls the transformation of AOCV data.

This global variable infers sensitivity data from AOCV libraries, and must be set before loading any timing library data.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_library_read_ccs_noise_data

```
timing_library_read_ccs_noise_data  
{true | false}
```

Default: true

Data_type: Boolean

Allows CCS noise constructs to be read from a library.

This global variable can be used when the CCS library contains noise models, and SI glitch analysis is being performed.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_library_read_without_power

```
timing_library_read_without_power  
{true | false}
```

Default: true

Data_type: Boolean

When set to `true`, this global variable allows the software to read timing libraries without power information.

When set to `false`, the software reads power data also.

Note: The `find_global` command shows the default value as `false`. However, this global variable is set to `true` by default in Tempus.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_library_scale_aocv_to_socv_to_n_sigma

timing_library_scale_aocv_to_socv_to_n_sigma
<n_sigma value>

Default: 3

Data_type: Double

Derives AOCV derates based on 3-sigma variation.

To set this global variable, use the `set` command.

Related Topics

- "[Analysis and Reporting](#)" chapter in the *Tempus User Guide*:
 - [Using AOCV Derating Libraries with SOCV Analysis](#)
 - [Sample Initialization and Configuration Scripts](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_library_setup_constraint_corner_sigma_multiplier

timing_library_setup_constraint_corner_sigma_multiplier *double*

Default: 0

Data_type: Double

Applies user control multiplier to generate a constraint table using sigma values applied for setup arc.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_library_setup_sigma_multiplier

`timing_library_setup_sigma_multiplier` *double*

Default: 0

Applies a user control multiplier to generate a constraint table using sigma values applied for setup arcs.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_library_use_trilib_drv_values

```
timing_library_use_trilib_drv_values  
{true | false}
```

Default: false

Data_type: Boolean

Allows the software to use a range of trilib DRV values for performing delay calculations.

When set to `false`, the DRV values are picked up from the first library of a trilib set and are used for DRV reporting. In some cases, the DRV constraints may not be the same across multi-voltage libraries. For this you can use a range of trilib DRV values, by setting the `timing_library_use_trilib_drv_values` global variable to `true`.

When set to `true`, the trilib (or voltage scaling) mechanism allows more accurate delay calculation in case timing libraries are not characterized at operating voltage. Through this mechanism, the bounding libraries of the operating voltage are obtained from the trilib set and the worst value is used in delay calculations.

This global variable is supported for `report_constraint` based DRVs. The global will impact the `report_constraint` command when there is a voltage scaling setup and the command is issued with the `-drv_violation_type` parameter.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_library_use_two_piece_receiver_cap

timing_library_use_two_piece_receiver_cap {true | false}

Default: false

Data_type: Boolean

Enables the use of 2-piece receiver capacitance constructs, if a library contains both the N-piece and 2-piece receiver capacitance construct.

When set to `true`, if a library has 2-piece and multi-piece pin capacitance, then the software gives priority to 2-piece models.

When set to `false`, the software reads the N-piece receiver capacitance constructs.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

Reporting Global Variables

- `report_precision`
- `report_precision_capacitance`
- `report_precision_derate`
- `report_precision_power`
- `report_precision_sensitivity`
- `report_timing_format`
- `timing_check_timing_signal_level_high_to_low_threshold`
- `timing_check_timing_signal_level_low_to_high_threshold`
- `timing_enable_db_attribute_query`
- `timing_generated_clocks_allow_nested_assertions`
- `timing_path_based_enable_exhaustive_depth_bounded_by_gba`
- `timing_path_based_enable_report_launch_clock_path`
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- `timing_property_is_clock_pin_consider_clock_source`
- `timing_property_return_null_collection_with_quiet`
- `timing_rail_swing_checks_high_voltage_threshold`
- `timing_rail_swing_checks_low_voltage_threshold`
- `timing_report_analysis_summary_csv_extended_new_format`
- `timing_report_arrival_property_worstcase_mode`
- `timing_report_begin_end_pair_max_path_limit`
- `timing_report_check_timing_enable_clock_crossing_verbose_clock_groups`
- `timing_report_check_timing_unconstrained_endpoints_due_to_constants`
- `timing_report_constraint_enable_drv_on_ideal_network`
- `timing_report_constraint_enable_extended_drv_format`
- `timing_report_constraint_format`
- `timing_report_constraint_report_all_drv_violation`

- `timing_report_constraint_rise_fall_clock_period_check`
- `timing_report_constraint_use_infinity_slack_for_unconstrained`
- `timing_report_default_frequency_for_unconstrained_nets`
- `timing_report_disable_max_paths_per_group`
- `timing_report_drv_enable_clock_source_as_clock`
- `timing_report_drv_enable_frequency_per_view`
- `timing_report_drv_enable_ghz_notation_for_drv_fields`
- `timing_report_drv_enable_slew_threshold_scaling`
- `timing_report_drv_per_frequency_per_input_slew`
- `timing_report_drv_use_worst_timing_slack`
- `timing_report_enable_clock_to_for_unconstrained_paths`
- `timing_report_enable_clock_unrolling`
- `timing_report_enable_cppr_point`
- `timing_report_enable_em_index_clipping_report`
- `timing_report_enable_lead_trail_to_rise_fall_map`
- `timing_report_enable_markers`
- `timing_report_enable_max_capacitance_drv_for_constants`
- `timing_report_enable_max_path_limit_crossed`
- `timing_report_enable_qpin_as_start_for_begin_end_pair`
- `timing_report_enable_report_clock_timing_across_clock_pin`
- `timing_report_enable_si_debug`
- `timing_report_enable_slack_summary_at_bottom`
- `timing_report_enable_unique_pins_multiple_capture_clock_paths`
- `timing_report_enable_use_valid_start_end_points`
- `timing_report_enable_write_twf_third_party_for_input_pins`
- `timing_report_generated_clock_info`
- `timing_report_group_based_mode`
- `timing_report_limit_max_path_count`
- `timing_report_max_transition_check_using_nsigma_slew`
- `timing_report_print_normalized_slack`

- [timing_report_property_fastest_clock_consider_data_phase](#)
- [timing_report_pulse_width_matching_launch_capture_paths](#)
- [timing_report_redirect_message_types](#)
- [timing_report_retime_formatting_mode](#)
- [timing_report_skip_constraint_loop_check](#)
- [timing_report_skip_icg_from_all_fanin_fanout_collection](#)
- [timing_report_socv_summary_mean_sigma](#)
- [timing_report_timing_header_detail_info](#)
- [timing_report_unconstrained_path_early_late_header](#)
- [timing_report_unconstrained_paths](#)
- [timing_report_use_receiver_model_capacitance](#)
- [timing_report_use_worst_parallel_cell_arc](#)
- [timing_use_incremental_si_transition](#)
- [timing_waveform_aware_pulse_width_checks_high_voltage_level](#)
- [timing_waveform_aware_pulse_width_checks_low_voltage_level](#)

report_precision

`report_precision` *integer*

Default: 3

Data_type: Integer

Controls the number of digits appearing after the decimal point in the [report_timing](#), [report_net](#), [report_cell_instance_timing](#), [report_clocks](#), [get_property](#), and [report_property](#) command output reports. The maximum allowed value of *integer* is 8.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

report_precision_capacitance

`report_precision_capacitance` *integer*

Default: 3

Data_type: Integer

Controls the number of digits appearing after the decimal point in load values in the timing report output.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

report_precision_derate

`report_precision_derate` *integer*

Default: 3

Data_type: Integer

Controls the number of digits appearing after the decimal point in derate values in the timing report output.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

report_precision_power

`report_precision_power` *integer*

Default: 3

Data_type: Integer

Controls the number of digits appearing after the decimal point in power values in the timing report output.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

report_precision_sensitivity

`report_precision_sensitivity` *integer*

Default: 3

Data_type: Integer

Controls the number of digits appearing after the decimal point in sensitivity values in the timing report output.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

report_timing_format

`report_timing_format {column_list}`

Default: instance arc cell delay arrival required

Data_type: String

Specifies a `report_timing` format. See `report_timing -format` for more information.

Valid format columns are:

adjustment, abs_delay_err, abs_slew_err, annotation, aocv_adj_stages, aocv_derate, aocv_weight, arc, arrival, arrival_mean, arrival_sigma, cell, delay, delay_mean, delay_sigma, direction, edge, fanin, fanout, flags, hpin, incr_delay, instance, instance_location, latch_window, load, module, net, pct_delay_err, pct_slew_err, phase, phys_info, pin, pin_load, pin_location, power_domain, required, retime_delay, retime_delay_mean, retime_delay_sigma, retime_mis_delta, retime_mis_derate, retime_slew, retime_slew_mean, retime_slew_sigma, stage_count, slew, slew_mean, slew_sigma, socv_derate, spice_arrival, spice_delay, spice_slew, stolen, timing_point, total_derate, user_derate, voltage, vt_skew_derate, when_cond, wire_load, wlmodel

For details on these format columns, refer to section [Report Timing Column List](#).

Depending on the options specified, this global variable also enables the display of slew mean and retime slew mean details in the timing report.

For example, the following command tells `report_timing` to only list the instance, arc, delay, slew, arrival times in the report:

```
set_global report_timing_format {instance arc delay slew arrival}
```

To set this global variable, use the `set_global` command.

Related Topics

- ["Analysis and Reporting"](#) chapter in the *Tempus User Guide*:
- Reporting with OCV/AOCV Derating
- SOCV Analysis Reporting
- Reporting in Path-Based Analysis
- MIS Delta Delay Reporting
- VT Skew Reporting
- Base Delay Reporting
 - [Reporting in Path-Based Analysis](#)
 - [Sample Initialization and Configuration Scripts](#)
 - [SI Delay Reporting](#)

- [Base Delay Reporting](#)
- [AOCV Reporting](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_check_timing_signal_level_high_to_low_threshold

timing_check_timing_signal_level_high_to_low_threshold *double*

Default: 0

Data_type: Double

Sets the threshold value for warning the signal level difference between driver and receiver voltages, when the driver is at a higher voltage than the receiver voltage.

The software shows the signal level warnings ([check_timing](#) `signal_level`) that report the net driver/receiver pin pairs for which the pin voltage does not match. The signal level warning messages can be filtered based on these threshold values.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_check_timing_signal_level_low_to_high_threshold

timing_check_timing_signal_level_low_to_high_threshold *double*

Default: 0

Data_type: Double

Sets the threshold value for warning the signal level difference between driver and receiver voltages, when the driver is at a lower voltage than the receiver voltage.

The software shows the signal level warnings ([check_timing](#) `signal_level`) that report the net driver/receiver pin pairs for which the pin voltage does not match. The signal level warning messages can be filtered based on these threshold values.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_enable_db_attribute_query

```
timing_enable_db_attribute_query  
{true | false}
```

Default: true

Data_type: Boolean

Allows to query DB attributes with the [get_property](#) command.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_generated_clocks_allow_nested_assertions

`timing_generated_clocks_allow_nested_assertions {true | false}`

Default: `false`

Data_type: Boolean

Controls clock latency handling for nested generated clocks that are created on the same timing pin as the master clock.

When set to `true`, the software handles generation of multiple nested generated clocks that are asserted on the same pin, such that the child clock inherits the clock latency value of its parent clock.

When set to `false`, the nested generated clocks asserted on the same pin do not inherit the latency values and are created with 0ns master clock latency. Any full clock timing path reported using these clocks will show the clock path up to the target pin of the generated clock.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_path_based_enable_exhaustive_depth_bounded_by_gba

```
timing_path_based_enable_exhaustive_depth_bounded_by_gba  
{true | false}
```

Default: true

Data_type: Boolean

Enables the timing report to bound PBA results by GBA slacks of the respective leftover nworst paths, whenever exhaustive PBA depth for an endpoint is exhausted.

When set to `true`, during exhaustive PBA, if you specify exhaustive depth as N, and an endpoint has nworst paths greater than N, then the timing report will show the worst path corresponding to worst slack out of worst PBA slack (out of first N paths) and GBA slack (for the (N+1)th path). Here, the variable N refers to exhaustive PBA depth and can be set using the `timing_pba_exhaustive_path_nworst_limit` global variable.

When set to `false`, during exhaustive PBA, if you specify exhaustive depth as N, and an endpoint has nworst paths greater than N, then the software will ignore the paths with depth greater than N, which may result in reporting an optimistic PBA slack for the endpoint reaching the depth.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_path_based_enable_report_launch_clock_path

`timing_path_based_enable_report_launch_clock_path {true | false}`

Default: `true`

Data_type: Boolean

Allows the reporting launch clock path in path-based analysis (PBA) path type full report.

When set to `false`, the software does not report these paths in the timing reports.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_path_based_exhaustive_enable_design_coverage

```
timing_path_based_exhaustive_enable_design_coverage  
{true | false}
```

Default: false

Data_type: Boolean

Enables exhaustive path-based analysis (PBA) to automatically pick up and analyze all the violating endpoints in the design, irrespective of the specified maximum paths.

When set to `true`, exhaustive PBA analysis automatically picks up and analyzes the violating design endpoints instead of N endpoints (specified with the `report_timing -max_paths` parameter). After analyzing the violating endpoints, the software reports the top N worst endpoints - depending on the specified value of N.

When set to `false`, exhaustive PBA analysis restricts analysis (using the `-max_paths` parameter) to the specified number of N endpoints.

This global variable is valid for both EPBA as well as [infinite depth path-based analysis \(IPBA\)](#).

Note: When this global variable is enabled, there may be an increase in run time because the software will analyze all the violating endpoints and analysis will not be restricted to the specified N endpoints.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_path_based_exhaustive_max_paths_limit

`timing_path_based_exhaustive_max_paths_limit` *integer*

Default: 2000000

Data_type: Integer

Specifies the maximum number of paths that can be internally analyzed during exhaustive path-based analysis (PBA).

If the number of candidate paths during exhaustive PBA is greater than the specified max paths limit, then the software will skip path-based analysis of these paths beyond this limit. The paths exceeding the specified max paths limit will then be included as GBA in the final report output so that real violations are not missed.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_property_arrival_window_enable_tcl_dict_format

timing_property_arrival_window_enable_tcl_dict_format
{true | false}

Default: false

Data_type: Boolean

Allows the software to print new format of the `arrival_window` property (see [get_property](#) command).

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_property_is_clock_pin_consider_clock_source

timing_property_is_clock_pin_consider_clock_source
{true | false}

Default: true

Data_type: Boolean

When set to `true`, the software considers the clock target pins as clock pins.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_property_return_null_collection_with_quiet

```
timing_property_return_null_collection_with_quiet  
{true | false}
```

Default: false

Data_type: Boolean

Allows you to query values of all the properties.

When set to `true`, the software enables returning values of properties of all the objects, when the `get_property -quiet` parameter is specified.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_rail_swing_checks_high_voltage_threshold

timing_rail_swing_checks_high_voltage_threshold *double*

Default: 0.95

Data_type: Double

Specifies the voltage threshold values for high rail swing checks.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_rail_swing_checks_low_voltage_threshold

timing_rail_swing_checks_low_voltage_threshold *double*

Default: 0.05

Data_type: Double

Specifies the voltage threshold values for low rail swing checks.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_analysis_summary_csv_extended_new_format

```
timing_report_analysis_summary_csv_extended_new_format  
{true | false}
```

Default: false

Data_type: Boolean

Keeps the analysis summary CSV extended report in sync with the summary timing report. These reports are available when the respective [report_timing](#) parameters are specified:

- `-analysis_summary_csv_extended_file`
- `-analysis_summary_file`

When set to `true` (and when the [report_timing](#) `-analysis_summary_csv_extended_file` parameter is specified), the column name sequence will be from fromClk to toClk, that is, fromClk will be printed first, and then toClk. Also, the timing report will show "-", if toClk or fromClk is not present (and when [report_analysis_summary](#) `-expanded_summary_clocks` is set to `false`).

When set to `false`, the missing toClk or fromClk column will be represented as "*".

To set this global variable, use the `set` command.

Related Topics

- "[Analysis and Reporting](#)" chapter in the *Tempus User Guide*:
 - [Supported CPPR Global Variables](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_arrival_property_worstcase_mode

`timing_report_arrival_property_worstcase_mode` *integer*

Default: 0

Data_type: Integer

Differentiates the clock and data phases based on the `arrival_*` properties (see [get_property](#) command) . The accepted values are 0, 1, 2, or 3.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_begin_end_pair_max_path_limit

`timing_report_begin_end_pair_max_path_limit` *integer*

Default: 2000000

Data_type: Integer

Sets a limit on the maximum number of paths that can be reported when the `report_timing - begin_end_pair` parameter is specified.

Note: Higher maximum paths limit may lead to high memory usage and runtime.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_check_timing_enable_clock_crossing_verbose_clock_groups

timing_report_check_timing_enable_clock_crossing_verbose_clock_groups
{true | false}

Default: false

Data_type: Boolean

When set to `true`, the software displays clock groups (if present) in the `check_timing -check_only clock_crossing -verbose` command output.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_check_timing_unconstrained_endpoints_due_to_constants

```
timing_report_check_timing_unconstrained_endpoints_due_to_constants {true | false}
```

Default: false

Data_type: Boolean

Enables the software to flag warnings for unconstrained signals arriving at the end points due to case analysis.

When set to `false`, the `check_timing` command output will show warnings for the unconstrained endpoints.

When set to `true`, the command output will also show warnings for unconstrained endpoints due to case analysis under the subcategory `uncons_endpoints_due_to_case_analysis`. For more details, see [check_timing](#).

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_constraint_enable_drv_on_ideal_network

`timing_report_constraint_enable_drv_on_ideal_network {true | false}`

Default: `true`

Data_type: Boolean

Enables DRV reporting of ideal nets.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_constraint_enable_extended_drv_format

timing_report_constraint_enable_extended_drv_format
{true | false}

Default: false

Data_type: Boolean

Enables the software to report data as per clock/data and rise/fall phases.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_constraint_format

`timing_report_constraint_format` *string*

Default: ""

Data_type: String

Specifies the DRV fields that will be reported with the `report_constraint` command. For valid format column options, refer to `report_constraint -drv_fields`.

If DRV fields are not specified with this global variable (or with `report_constraint -drv_fields`), then the default set of DRV fields will be displayed. By default, the constraint report will display the pin names, the required and actual DRC values, and the view names.

Note: The reporting fields specified with the `report_constraint -drv_fields` parameter will override the global variable settings.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_constraint_report_all_drv_violation

`timing_report_constraint_report_all_drv_violation {true | false}`

Default: `true`

Data_type: Boolean

Enables reporting of all the [design rule violations](#) (DRV).

When set to `false`, the software reports the worst 1000 DRV violations per DRV check (specified with the [report_constraint](#) `-all_violators -drv_violation_type` command).

When set to `true`, the software will report all the DRV violations. You can use the `report_constraint -max_pins_per_drv` parameter to specify the maximum number of DRV violations to be reported.

Note: The `-max_pins_per_drv` parameter has higher precedence over the `timing_report_constraint_report_all_drv_violation` global variable setting.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_constraint_rise_fall_clock_period_check

```
timing_report_constraint_rise_fall_clock_period_check  
{true | false}
```

Default: false

Data_type: Boolean

Enables reporting of rise-to-rise and fall-to-fall clock period checks separately for the [report_constraint](#) command.

When set to `false`, the software reports the rise-to-rise and fall-to-fall clock period checks under the same table in the report. With the `false` setting, when the `report_constraint -check_type clock_period` option is specified, the software reports violations that are the worst amongst all the rise-fall transitions for each point.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_constraint_use_infinity_slack_for_unconstrained

`timing_report_constraint_use_infinity_slack_for_unconstrained {true | false}`

Default: false

Data_type: Boolean

Reports the timing slacks of unconstrained paths through the DRV violating pin as “INFINITY”. By default, the timing slacks of unconstrained paths are reported as * (asterisk).

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_default_frequency_for_unconstrained_nets

timing_report_default_frequency_for_unconstrained_nets *value*

Default: 1e+09

Data_type: Double

Defines the frequency for unconstrained nets to be used in maximum capacitance DRV reporting.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_disable_max_paths_per_group

timing_report_disable_max_paths_per_group {true | false}

Default: false

Data_type: Boolean

Controls the interpretation of the `report_timing -max_paths` parameter when you request reporting across a set of path groups. By default (`false` setting), the specified `-max_paths` will be reported per path group. With the `true` setting, a total of the specified `-max_paths` will be reported across all of the specified path groups.

For Stylus flow cross-tool compatibility – you should specify a setting of `true`. For backward compatibility with the Tempus legacy UI flow – you should use a value of `false`.

For example,

When set to `false`:

Legacy UI: `report_timing -path_group {reg2reg in2reg} -nworst 2 ...`

Stylus UI: `report_timing -group {reg2reg in2reg} -nworst 2 ...`

The `report_timing` command will return up to 2 paths for the `reg2reg` group and also up to 2 paths for the `in2reg` group.

When set to `true`:

Legacy UI: `set timing_report_disable_max_paths_per_group true`

Stylus UI: `set_db timing_report_disable_max_paths_per_group true`

A total of 2 paths will be reported across both path groups. The paths may come entirely from one or the other group – or both, based on the slack value.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_drv_enable_clock_source_as_clock

`timing_report_drv_enable_clock_source_as_clock {true | false}`

Default: `false`

Data_type: Boolean

Enables the clock DRV for pure clock source path instead of data DRV.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_drv_enable_frequency_per_view

timing_report_drv_enable_frequency_per_view
{true | false}

Default: true

Data_type: Boolean

Enables view-based frequency DRV flow.

When set to `true`, in the MMMC DRV flow, the software performs frequency computation for design pins as per the clock period of specific analysis views reaching a pin. In case of non-activity based flow, the frequency is computed as per the fastest clock of all the views.

When set to `false`, in the propagate activity flow, the software performs frequency computation for design pins from propagated activity on that pin.

For other flows, the software performs frequency computation for design pins as per the smallest clock period (largest clock frequency) from all the active analysis views.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_drv_enable_ghz_notation_for_drv_fields

`timing_report_drv_enable_ghz_notation_for_drv_fields {true | false}`

Default: `false`

Data_type: Boolean

Enables reporting of clock frequency in GigaHertz (Ghz). The default frequency unit is MegaHertz (MHz).

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_drv_enable_slew_threshold_scaling

`timing_report_drv_enable_slew_threshold_scaling {true | false}`

Default: `false`

Data_type: Boolean

Enables the transition constraints scaling as per the slew threshold value.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_drv_per_frequency_per_input_slew

```
timing_report_drv_per_frequency_per_input_slew  
{true | false}
```

Default: false

Data_type: Boolean

Enables frequency dependent DRV lookup from a library.

When set to `true`, the software will enable the `report_constraint` command to honor `max_transition` or `max_capacitance` limit after worst-casing due to the following cases:

- One-dimensional frequency DRV commands, such as, `set_max_tran_per_freq` or `set_max_cap_per_freq`.
- Two-dimensional maximum transition or capacitance frequency/electro-migration tables.
- Other frequency independent DRV limit sources, e.g., `set_max_transition` from the SDC or library limit.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_drv_use_worst_timing_slack

`timing_report_drv_use_worst_timing_slack {true | false}`

Default: `false`

Data_type: Boolean

Enables reporting of the worst DRV slack across all the views for configurable DRV report.

When set to `true`, the software shows a non-default special case report, where timing slacks for MMMC flow is the worst among all active views.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_enable_clock_to_for_unconstrained_paths

timing_report_enable_clock_to_for_unconstrained_paths
{true | false}

Default: false

Data_type: Boolean

Enables unconstrained path reporting when the `report_timing -clock_to` parameter is specified.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_enable_clock_unrolling

```
timing_report_enable_clock_unrolling  
{true | false}
```

Default: false

Data_type: Boolean

Enables the clock unrolled reporting format that shows more intuitive timing reports with respect to constraints.

When set to `true`, the `report_timing` command provides an optional format where the clock edge arrival times in the detailed path section show the worst-case clock edge pair used for the analysis. This format can provide a more intuitive understanding of the clock edges used - especially in cases where the launch and capture clocks have different frequencies.

When set to `false`, the detailed path section shows clock edge arrival times with respect to the base cycle of the clocks. Any required cycle-accounting adjustments are provided by the Phase Shift and Cycle Adjustment entries in the timing report summary section.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_enable_cppr_point

`timing_report_enable_cppr_point { true | false }`

Default: `false`

Data_type: Boolean

Enables the timing report to output CPPR point details.

When set to `true`, the `report_timing` command output shows the CPPR point field.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_enable_em_index_clipping_report

`timing_report_enable_em_index_clipping_report {true | false}`

Default: `false`

Data_type: Boolean

Enables index clipping reporting for EM (electro-migration) DRV (design rule violation) reports.

For more information, refer to the [Cell EM Analysis and ECO](#) chapter in the *Tempus User Guide*.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_enable_lead_trail_to_rise_fall_map

```
timing_report_enable_lead_trail_to_rise_fall_map  
{true | false}
```

Default: false

Data_type: Boolean

Enables reporting of rise/fall edge times, instead of lead/trail edge times of a clock (that are reported using the [report_clocks](#) command).

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_enable_markers

timing_report_enable_markers
{true | false}

Default: true

Data_type: Boolean

Enables display of markers in reports.

When set to `true`, the `report_timing` output format displays markers that identify pins specified using the `-from`, `-to`, or `-through` parameters. The markers, displayed as ‘->’, show up in the ‘timing point’ or ‘hpin’ column (if the ‘timing point’ column is not available) of the timing report. This feature provides better readability. When set to `false`, the timing report does not display the markers.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_enable_max_capacitance_drv_for_constants

timing_report_enable_max_capacitance_drv_for_constants
{true | false}

Default: false

Data_type: Boolean

When set to `true`, performs max_capacitance design rule violation check on constant elements.

Note: This global variable will be honored only when `timing_disable_drv_reports_on_constant_nets` is set to `true`.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_enable_max_path_limit_crossed

```
timing_report_enable_max_path_limit_crossed  
{true | false}
```

Default: false

Data_type: Boolean

Determines if the number of reported paths for a particular clock group equals the specified maximum paths limit.

When set to `true`, the software will issue a warning message in case the number of reported paths for a clock group is equal to the given maximum paths limit. This applies to both graph- and path-based reporting modes.

This variable is supported in clock group-based non-statistical reporting flows only. The clock group-based reporting mode can be enabled by setting `timing_report_group_based_mode` global variable to `true`.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_enable_qpin_as_start_for_begin_end_pair

```
timing_report_enable_qpin_as_start_for_begin_end_pair  
{true | false}
```

Default: false

Data_type: Boolean

Enables reporting of multiple paths with different q-pins having the same clock pin.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_enable_report_clock_timing_across_clock_pin

`timing_report_enable_report_clock_timing_across_clock_pin {true | false}`

Default: `false`

Data_type: Boolean

Enables fanout tracing across clock pins if any combinational arc is present.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_enable_si_debug

```
timing_report_enable_si_debug  
{true | false}
```

Default: false

Data_type: Boolean

Allows the software to capture signal integrity (SI) related details for each stage of a path being analyzed during graph-based analysis or path-based analysis in the form of attributes.

To set this global variable, use the `set_global` command.

Related Topics

- "[Analysis and Reporting](#)" chapter in the *Tempus User Guide*:
 - [Using Signal Integrity Attributes with get_property Command](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_enable_slack_summary_at_bottom

```
timing_report_enable_slack_summary_at_bottom {true | false}
```

Default: false

Data_type: Boolean

Prints additional time details, such as, Required time, Arrival, and Slack, at the bottom of the timing report. For example,

```
Req Time = 0.262  (-3.00S) | 0.544   | 0.094 | 0.141  
Arr Time = 2.238  (+3.00S) | 1.774   | 0.155 | 0.131  
Slk Time = -1.773 (-3.00S) | -1.230  | 0.181 | 0.181
```

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_enable_unique_pins_multiple_capture_clock_paths

timing_report_enable_unique_pins_multiple_capture_clock_paths {true | false}

Default: false

Data_type: Boolean

Enables reporting of different paths with respect to the reference clocks in the unique pins.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_enable_use_valid_start_end_points

`timing_report_enable_use_valid_start_end_points {true | false}`

Default: `false`

Data_type: Boolean

Shows valid start/end points in the timing report.

When set to `true`, the valid start/end points are included in the timing report. The paths are reported with the `report_timing` command - when the pins passed (-from and -to) are the actual begin and end pins of the data path. For example, for a register the begin pin is register's CK pin and the end pin is register's D pin. Similarly, an input pin in a path is the path's begin pin and output pin is the path's end pin.

When set to `false`, any point in a path can be used as a -from point and is considered as a -through point internally. All the pins using -from/-to are treated as -through pins. Therefore, a path is reported with register's Q pin as a from pin, even though the correct start pin is a CK pin. Or any intermediate pin belonging to a path can be passed as -from/-to and the path will still be reported to correct start/end pins with the given pin lying somewhere in between of path.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_enable_write_twf_third_party_for_input_pins

```
timing_report_enable_write_twf_third_party_for_input_pins  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, the software writes timing information for input pins in the third-party TWF file.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_generated_clock_info

timing_report_generated_clock_info {true | false}

Default: true

Data_type: Boolean

Generated clock information is automatically added to the report, if generated clocks are encountered in either the launching or capturing clock paths.

The generated clock information column is displayed if the `Delay` field exists. If the `Delay` field does not exist, then the generated clock adjustment column is added to the report only when explicitly specified using the `report_timing -format` command or by setting the `report_timing_format` global variable.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_group_based_mode

timing_report_group_based_mode
{true | false}

Default: false

Data_type: Boolean

Groups paths by clock domain.

When set to `true`, the behavior of the following reporting commands will change as follows:

report_timing

The behavior of the following `report_timing` parameters changes as follows:

- `-max_points`: In this mode you are not required to specify this parameter. If specified, the parameter is mapped to `-max_paths` option.
- `-max_paths M`: Reports `M` paths per clock group.
- `-nworst N`: Limits the number of paths per endpoint per clock group. By default, the value of `-nworst` is 1.

The `report_timing` command will have the following four types of groups (if design does not have user-defined path groups):

- Asynch group: Recovery/removal checks.
- Clock gating group: Inferred and library clock gating checks.
- Default group: Combinational max delay or min delay paths.
- Clock Groups: Groups for all design clocks. For a design with three clocks, three groups will be reported, one group for each clock. Endpoints sharing the same reference clock will be part of the same group. If an endpoint belongs to multiple groups then it will be reported separately in each group.

Note: If user-defined path groups are specified, the software will override the path groups described above.

report_min_pulse_width

The `report_min_pulse_width` command reports pulse width slacks of all the clocks for a pin. In summary mode, the "clock" column will be displayed. In verbose mode (with `path_type` option), the behavior will be similar to the `report_timing` output in group-based mode.

report_constraint

The `report_constraint` command will report the following checks:

- For max_delay/setup and min_delay/hold type checks paths will be reported for each clock group separately.
- The behavior of pulse width check types will be the same as `report_min_pulse_width` command.

report_analysis_coverage

The `report_analysis_coverage` command will report a check at end points for all the clocks. In verbose mode, the "clock" column will be displayed.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_limit_max_path_count

`timing_report_limit_max_path_count` *value*

Default: 2000000

Data_type: integer

Sets the maximum number of violations (of data/clock checks) to be reported.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_max_transition_check_using_nsigma_slew

```
timing_report_max_transition_check_using_nsigma_slew  
{true | false}
```

Default: false

Data_type: Boolean

Enables the `report_constraint` command to use mean and sigma slew values for reporting transition values in DRV checks.

When set to `true`, in the SOCV flow the `report_constraint` command uses the mean plus n-sigma slew values for DRV checks.

When set to `false`, the `report_constraint` command will report only the mean slews.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_print_normalized_slack

```
timing_report_print_normalized_slack  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, the software displays normalized slack based on the worst slack of different views.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_property_fastest_clock_consider_data_phase

timing_report_property_fastest_clock_consider_data_phase

{true | false}

Default: false

Data_type: Boolean

Considers the data signals while determining the fastest clock arriving on any given pin.

When set to `false`, the clocks arriving in clock phase form will be considered for determining the fastest clock.

When set to `true`, the clocks arriving in clock phase as well as data phase form will be considered.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_pulse_width_matching_launch_capture_paths

timing_report_pulse_width_matching_launch_capture_paths {true | false}

Default: false

Data_type: Boolean

Reports pulse width paths that have matching launch and capture paths - the count of pins and the pins must match exactly. The paths that do not meet this condition will not be reported.

When set to `true`, the software will impact the minimum pulse width report that is generated with the `report_min_pulse_width` command along with the `-path_type short` and `-path_type full_clock` options.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_redirect_message_types

```
timing_report_redirect_message_types  
{none | info}
```

Default: none

Data_type: Enum

Redirects reporting messages to a report file.

When set to `info`, the path count and max depth (displayed when the `report_timing` command is run in PBA mode) will be appended to the final report output.

When set to `none`, the reporting messages are not redirected to a report file.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_retime_formatting_mode

timing_report_retime_formatting_mode
{manual | retime_compare | retime_replace}

Default: manual

Data_type: Enum

Controls retiming fields to be displayed in the timing report.

You can set one of the following options:

- `manual`: Does not change the `report_timing` format.
- `retime_compare`: If GBA columns are present in the report format, then the corresponding retime options are added. If PBA columns are already present, then there is no change in the format.
- `retime_replace`: If GBA columns are present in the report format, then the GBA columns are replaced by the corresponding PBA columns. If PBA columns are already in the report, then there is no change in the format.

Note: The report format is also controlled through the `report_timing -retime_format` parameter. The `report_timing` command setting has precedence over this global variable.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_skip_constraint_loop_check

timing_report_skip_constraint_loop_check
{true | false}

Default: true

Data_type: Boolean

Allows you to skip loop checking when the [check_timing](#) command is used.

When set to `true`, the software skips reporting of loops in case the timing graph inside a loop is broken. This may happen if some of the commands, such as, `create_clock`, `set_input_delay`, or `set_output_delay` are defined inside a loop.

Note: When this global variable is set to `true`, the `check_timing` command will report lesser loops than the ones reported in the CTE_loops.rpt report - which reports all the loops.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_skip_icg_from_all_fanin_fanout_collection

```
timing_report_skip_icg_from_all_fanin_fanout_collection  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, the software skips ICG clock pins from the `all_fanin` (or `all_fanout`) command collection, when queried with startpoints/endpoints.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_socv_summary_mean_sigma

```
timing_report_socv_summary_mean_sigma {true | false}
```

Default: false

Data_type: Boolean

Prints the report header in new format when the [report_timing](#) command is run in SOCV mode. The new header contains nsigma, mean, and sigma details.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_timing_header_detail_info

timing_report_timing_header_detail_info
{default | extended}

Default: default

Data_type: Enum

Controls the timing report (generated using the [report_timing](#) command) header format. You can specify one of the following options:

- When set to `default`, the software generates timing reports with a report header that includes the following information:
 - Tempus software version number
 - Platform on which the software is being run
 - Time when the report was generated
 - Command used to generate the report
- When set to `extended`, the software generates timing reports with additional header information given below:
 - The analysis mode - best case worst case or on chip variation
 - The active views (displayed only in MMMC mode)
 - The clock reconvergence pessimism (CPPR)
 - The latch slack mode
 - The specified clock propagation value
 - The specified delay calculation engine

Note: You can use this global variable when [timing_disable_report_header_info](#) is set to `false`. To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_unconstrained_path_early_late_header

```
timing_report_unconstrained_path_early_late_header  
{true | false}
```

Default: false

Data_type: Boolean

Provides information in the timing report header that indicates whether an unconstrained path is a late or early path.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_unconstrained_paths

timing_report_unconstrained_paths
{true | false}

Default: false

Data_type: Boolean

When set to `true`, the `report_timing` command reports unconstrained paths if the software cannot find any constrained paths to report.

When set to `false`, the `report_timing` command reports only the constrained paths.

Note: The behavior of `timing_report_unconstrained_paths` global variable is not the same as the behavior of `report_timing -unconstrained` command. If there are several paths to the same endpoint from different begin points, including constrained and unconstrained paths, the `report_timing -unconstrained` parameter always reports unconstrained paths, even if a constrained path can be reported. Whereas the `timing_report_unconstrained_paths` global reports constrained paths but not unconstrained paths.

To set this global variable, use the `set_global` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_use_receiver_model_capacitance

`timing_report_use_receiver_model_capacitance {true | false}`

Default: `false`

Data_type: Boolean

Enables the max_capacitance violations based on dynamic pin capacitance.

When set to `true`, the software checks for both the input level and timing arc level dynamic pin capacitance tables for worst-case pin capacitance table for given input slew. If conditional pin capacitance tables (or `timing_arcs`) exist, the worst-case is used among all the conditions.

This global variable performs static pin capacitance inference after delay calculation, so the delay computation is not impacted. The pin capacitance is used only for reporting and max_cap violations.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_report_use_worst_parallel_cell_arc

```
timing_report_use_worst_parallel_cell_arc  
{true | false}
```

Default: false

Data_type: Boolean

When set to `false`, setting the `-nworst` parameter reports several paths by selecting parallel arcs in the library cell between two pins.

When set to `true`, only a single worst path is reported by selecting the worst delay arc among all parallel arcs of a cell. For example, for reporting timing through A to Y pins (where the cell is a 2-input cell), the parallel arc is applicable for the same input and output pin. So the worst arc for A->Y is chosen from all arcs of A->Y.

To set this global variable, use the [set_global](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_use_incremental_si_transition

`timing_use_incremental_si_transition {true | false}`

Default: `false`

Data_type: Boolean

Enables usage of incremental slew during DRV violation reporting.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_waveform_aware_pulse_width_checks_high_voltage_level

timing_waveform_aware_pulse_width_checks_high_voltage_level *double*

Default: 0.95

Data_type: Double

Specifies the high voltage level to be used for waveform-aware pulse width checks.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_waveform_aware_pulse_width_checks_low_voltage_level

timing_waveform_aware_pulse_width_checks_low_voltage_level *double*

Default: 0.05

Data_type: Double

Specifies the low voltage level to be used for waveform-aware pulse width checks.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

Constraint Global Variables

- [timing_all_registers_identify_macros_include_is_macro_cell](#)
- [timing_collection_all_fanout_fanin_enable_trace_through_traversal_latch_data_pin](#)
- [timing_constraint_enable_detailed_report_invalid_begin_end_points](#)
- [timing_constraint_enable_logging](#)
- [timing_constraint_enable_report_invalid_begin_end_points](#)
- [timing_constraint_enable_search_path](#)
- [timing_constraint_enable_separate_multicycle_data_checks](#)
- [timing_constraint_load_minimal_set_for_automated_eco](#)
- [timing_constraint_path_delay_exclude_check_delay_from_ignore_clock_latency](#)
- [timing_constraint_path_delay_exclude_io_delay_from_ignore_clock_latency](#)
- [timing_constraint_path_delay_exclude_unconstrained_endpoints](#)
- [timing_constraint_path_delay_include_clock_pin_endpoints](#)
- [timing_constraint_update_io_latency_averaging_mode](#)
- [timing_constraint_warn_for_timing_derate_exceeding_max_limit](#)
- [timing_derate_enable_incremental_multiplication_mode](#)
- [timing_derate_incremental_multiply_accumulative_mode](#)

timing_all_registers_identify_macros_include_is_macro_cell

```
timing_all_registers_identify_macros_include_is_macro_cell {true | false}
```

Default: true

Data_type: Boolean

Enables to include cells as macro cells.

When set to `true`, the software includes cells as macro cells if the `is_macro_cell` property (see [get_property](#)) is set to `true` in the library.

When set to `false`, the `all_registers` command will not return cells for which the `is_macro_cell` property is `true`.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_collection_all_fanout_fanin_enable_trace_through_traversal_latch_data_pin

```
timing_collection_all_fanout_fanin_enable_trace_through_traversal_latch_data_pin  
{true | false}
```

Default: false

Data_type: Boolean

Enables traversal across latch data pins, when a collection is specified using the [all_fanout](#) or [all_fanin](#) command with the `-trace_through all` setting.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_constraint_enable_detailed_report_invalid_begin_end_points

timing_constraint_enable_detailed_report_invalid_begin_end_points
{true | false}

Default: false

Data_type: Boolean

Enables detailed reporting of invalid begin/end points specified with the [set_false_path](#)/[set_multicycle_path](#) constraints.

When set to `false`, the software checks only the structural end points.

To set this global variable, use the [set](#) command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_constraint_enable_logging

`timing_constraint_enable_logging {true | false}`

Default: `false`

Data_type: Boolean

Enables saving of the constraint commands in the log file.

Note: The impact of this global variable is visible only when the constraints are loaded using the `read_sdc` command or the view definition file.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_constraint_enable_report_invalid_begin_end_points

timing_constraint_enable_report_invalid_begin_end_points
{true | false}

Default: false

Data_type: Boolean

Enables reporting of invalid begin or end points that are specified with the [set_false_path](#) or [set_multicycle_path](#) constraint.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_constraint_enable_search_path

`timing_constraint_enable_search_path {true | false}`

Default: `false`

Data_type: Boolean

Enables search of constraint files from the search path.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_constraint_enable_separate_multicycle_data_checks

`timing_constraint_enable_separate_multicycle_data_checks {true | false}`

Default: `false`

Data_type: Boolean

Controls multicycle path constraint application on data checks.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_constraint_load_minimal_set_for_automated_eco

`timing_constraint_load_minimal_set_for_automated_eco {true | false}`

Default: `false`

Data_type: Boolean

Skips loading the SDC constraints that are not required in the Tempus session, when the `eco_opt_design` command is used. The timing data is obtained from the ECO timing database - this global variable reduces the TAT for designs with complex constraint mode files.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_constraint_path_delay_exclude_check_delay_from_ignore_clock_latency

```
timing_constraint_path_delay_exclude_check_delay_from_ignore_clock_latency {true | false}
```

Default: false

Data_type: Boolean

Excludes check arc delays from slack computation.

When set to `true`, the software excludes check arc delays from slack computation for paths constrained by the `set_max_delay/set_min_delay -ignore_clock_latency` command.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_constraint_path_delay_exclude_io_delay_from_ignore_clock_latency

timing_constraint_path_delay_exclude_io_delay_from_ignore_clock_latency {true | false}

Default: false

Data_type: Boolean

Excludes the input/output delays from slack computation.

When set to `true`, the software excludes the input/output delays from slack computation for paths constrained by the `set_max_delay/set_min_delay -ignore_clock_latency` command.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_constraint_path_delay_exclude_unconstrained_endpoints

```
timing_constraint_path_delay_exclude_unconstrained_endpoints  
{true | false}
```

Default: false

Data_type: Boolean

Allows you to configure the valid endpoints.

When path delay exceptions are asserted (using the `set_min_delay` or `set_max_delay` commands) with a complete `-from` and `-to` specification, the exception can be applied between any two data path pins. If the constraint assertion includes only a `-from` and/or `-through` argument, then a set of valid endpoints must be established to define the constrained paths. This global variable allows you to configure the set of valid endpoints (inferred `-to` pins) that are selected.

When set to `false`, the set of allowed endpoints is simply all of the endpoints in the combinational fanout of the `-from` and/or `-through` arguments - essentially, wherever the path tracing stops. The path delay constraint will be applied as if there were a `-to` argument specified for each endpoint.

When set to `true`, the software uses different rules to determine the valid set of inferred `-to` endpoints. Only constrained pins in the fanout of `-from` and/or `-through` arguments are considered as valid `-to` endpoints for the constraint. A constrained endpoint will have some type of timing check - such as Setup or Hold already in force on that pin. Additionally, each transition at the endpoints will be examined to determine if each is individually constrained. The path exception will only be applied to the edges that are constrained. As an example, if a register preset or a clear pin was a potential endpoint, then only the edge constrained by the recovery or removal check will have the path delay applied. To apply the path delay constraint to unconstrained endpoints while in this mode, you must specify them explicitly with the `-to`, `-rise_to`, or `-fall_to` parameters of the `set_min_delay` and `set_max_delay` constraints.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_constraint_path_delay_include_clock_pin_endpoints

timing_constraint_path_delay_include_clock_pin_endpoints
{true | false}

Default: false

Data_type: Boolean

Considers register clock pins as end-points for path delay constraints applied upstream.

When set to `true`, the `set_max_delay` command considers unconstrained paths ending at the register clock pins.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_constraint_update_io_latency_averaging_mode

```
timing_constraint_update_io_latency_averaging_mode  
{rise_fall | rise_only}
```

Default: rise_fall

Data_type: Enum

Controls the analysis to update IO latency for both rise and fall, or for rise only. You can set one of the options - rise_fall, rise_only.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_constraint_warn_for_timing_derate_exceeding_max_limit

`timing_constraint_warn_for_timing_derate_exceeding_max_limit` *double*

Default: 100

Data_type: Double

Value Range: {0 to 100}

Sets a limit on derate values. If a larger derate value is applied, the derates are still processed. However, the software issues a warning.

Note: It is recommended to specify derate values within the acceptable value range, because large derate values may cause errors during processing.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_derate_enable_incremental_multiplication_mode

```
timing_derate_enable_incremental_multiplication_mode  
{true | false}
```

Default: false

Controls the overwrite or accumulative mode settings when the `set_timing_derate -multiply` parameter is specified.

When set to `false`, the latest `set_timing_derate -multiply` command will overwrite the previous constraints at the same level.

When set to `true`, all the values in history will be accumulatively multiplied, when multiple `set_timing_derate -multiply` settings are made on the same object.

To set this global variable, use the set command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_derate_incremental_multiply_accumulative_mode

`timing_derate_incremental_multiply_accumulative_mode {true | false}`

Default: false

Data_type: Boolean

Controls the overwrite or accumulative mode settings when the `set_timing_derate -multiply` parameter is specified. You can use this global variable when the `timing_derate_enable_incremental_multiplication_mode` is set to `true` (that is, the `-multiply` setting must be enabled).

When set to `false`, the last specified `set_timing_derate -multiply` command will overwrite the previous constraints at the same level.

When set to `true`, all the values saved in history will be accumulatively multiplied, when multiple `-multiply` settings are made on the same object.

To set this global variable, use the `set` command.

Examples

- The following example shows that in overwrite mode, the multiplicative adjustment will be updated to 0.5:

```
set_global timing_derate_enable_incremental_multiplication_mode true
set_global timing_derate_incremental_multiply_accumulative_mode false
set_timing_derate -multiply 1.2
set_timing_derate -multiply 0.5
```

- The following example shows that in accumulative mode, the multiplicative adjustment will be updated to $1.2 \times 0.5 = 0.6$:

```
set_global timing_derate_enable_incremental_multiplication_mode true
set_global timing_derate_incremental_multiply_accumulative_mode true
set_timing_derate -multiply 1.2
set_timing_derate -multiply 0.5
```

Other Reference Information

- Click [here](#) for more information on this global variable.

Timing Model Global Variables

- [timing_extract_model_aocv_mode](#)
- [timing_extract_model_case_analysis_in_library](#)
- [timing_extract_model_check_arcs_as_lvf](#)
- [timing_extract_model_consider_design_level_drv](#)
- [timing_extract_model_disable_cycle_adjustment](#)
- [timing_extract_model_enable_combinational_arc_to_clock_source_on_output_ports](#)
- [timing_extract_model_enable_output_pin_attributes](#)
- [timing_extract_model_enable_unconstrained_port_modelling](#)
- [timing_extract_model_exhaustive_validation_dir](#)
- [timing_extract_model_exhaustive_validation_mode](#)
- [timing_extract_model_gating_as_nochange_arc](#)
- [timing_extract_model_half_cycle_non_interface_paths](#)
- [timing_extract_model_ideal_clock_latency_arc](#)
- [timing_extract_model_include_applied_load_in_characterization_range](#)
- [timing_extract_model_include_applied_slew_in_characterization_range](#)
- [timing_extract_model_max_feedthrough_characterization_load](#)
- [timing_extract_model_non_borrowing_latch_path_as_setup](#)
- [timing_extract_model_slew_propagation_mode](#)
- [timing_extract_model_synchronous_clocks_as_skew_checks](#)
- [timing_extract_model_write_clock_checks_as_arc](#)
- [timing_extract_model_write_clock_checks_as_scalar_tables](#)
- [timing_extract_model_write_function_for_combinational_arcs](#)
- [timing_extract_model_write_lvf](#)
- [timing_extract_model_write_min_max_clock_tree_path](#)

timing_extract_model_aocv_mode

`timing_extract_model_aocv_mode {graph_based | path_based | none}`

Default: none

Data_type: Enum

Range: {none graph_based path_based}

Allows you to set the AOCV derating mode.

The valid values are described below:

- `graph_based`: Specifies that the delays in the timing model are derated using the graph-based stage counts.
- `path_based`: Specifies that the delays in the timing model are derated using the total path stage count of worst path between those pin pairs.
- `none`: Specifies that the delays in the timing model are non-AOCV derated base delays.

Note: Before setting this global to `graph_based` or `path_based` mode, it is mandatory that the AOCV libraries are read in and AOCV analysis is enabled (by setting `set_analysis_mode -aocv true`).

To set this global variable, use the `set` command.

Related Topics

- "[Extracted Timing Models](#)" chapter in the *Tempus User Guide*:
 - [AOCV Derating Mode](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_extract_model_case_analysis_in_library

```
timing_extract_model_case_analysis_in_library  
{true | false}
```

Default: `true`

Data_type: Boolean

Enables the software to model the constants as a function in the extracted timing model.

When set to `true`, the software models constants as functions in the library if these propagate to the output port.

When set to `false`, instead of modeling constants as functions inside the library, these are written as `set_case_analysis` in the constraints file (*.asrt).

To set this global variable, use the `set` command.

Related Topics

- "Timing Model Generation" chapter in the *Tempus User Guide*:
 - [Basic Elements of Timing Model Extraction](#)
 - [Constraint Generation during Model Extraction](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_extract_model_check_arcs_as_lvf

`timing_extract_model_check_arcs_as_lvf {true | false}`

Default: `true`

Data_type: Boolean

Enables check arcs to be modeled as LVF arcs. By default, the check arcs will be written as nSigma numbers.

Tempus allows you to write mean+/-nSigma values for constraint arcs, while writing separate mean and sigma values for delay and transition data. This can be enabled by setting

`timing_extract_model_check_arcs_as_lvf` global variable to `false` and
`timing_extract_model_write_lvf` to `true`.

This global variable will not have any impact if `timing_extract_model_write_lvf` is set to `false`.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_extract_model_consider_design_level_drv

```
timing_extract_model_consider_design_level_drv  
{true | false}
```

Default: true

Data_type: Boolean

Considers `set_max_capacitance` and `set_max_transition` DRV constraints, that have been applied on the current design during model extraction.

When set to `true`, these constraints applied at the design level will be considered during model extraction. However, the maximum capacitance and maximum transition from various DRV constraints applied/inferred on the ports is considered.

When set to `false`, the software extracts the worse-case DRV values and the constraints applied on the current design will be ignored.

To set this global variable, use the `set_global` command.

Related Topics

- "[Timing Model Generation](#)" chapter in the *Tempus User Guide*:
 - [Basic Elements of Timing Model Extraction](#)
 - [Constraint Generation during Model Extraction](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_extract_model_disable_cycle_adjustment

```
timing_extract_model_disable_cycle_adjustment  
{true | false}
```

Default: false

Data_type: Boolean

Controls cycle adjustment for multi-cycle paths.

When set to `false`, if the multi-cycle paths specified in the SDC at the block-level cannot be pushed out to the ports (since there are other paths going through these paths), then the Tempus software adjusts the delay/check numbers in the ETM to take into account the cycle adjustment.

When set to `true`, the cycle adjustment for the multi-cycle paths in ETM arc delay are disabled.

To set this global variable, use the `set` command.

Related Topics

- "[Timing Model Generation](#)" chapter in the *Tempus User Guide*:
 - [Constraint Generation during Model Extraction](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_extract_model_enable_combinational_arc_to_clock_source_on_output_ports

```
timing_extract_model_enable_combinational_arc_to_clock_source_on_output_ports  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, the Tempus software models the combinational arc ending on an end point with the clock asserted on it.

To set this global variable, use the `set` command.

Related Topics

- "[Timing Model Generation](#)" chapter in the *Tempus User Guide*.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_extract_model_enable_output_pin_attributes

timing_extract_model_enable_output_pin_attributes *String*

Default: none

Data_type: String

Specifies the attributes (e.g., always_on, is_isolated) to enable on output pins during ETM generation. You can use this global variable when backward compatibility with older ETM behavior is required.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_extract_model_enable_unconstrained_port_modelling

timing_extract_model_enable_unconstrained_port_modelling
{true | false}

Default: false

Data_type: Boolean

Enables extraction of unconstrained paths to the output ports.

When set to `false`, for unconstrained output ports, which have no inbound timing, there will be no arcs modelled for the paths.

When set to `true`, the software will extract dummy arcs inbound to unconstrained output ports that will show the following characteristics:

- Early delay/slew of last stage connected to the output ports will be modelled.
- Related pins will be specified as unique unconstrained internal pins.
- Extracted slew values will be modelled as unidimensional *_transition tables that are dependent on the output capacitance and real propagated slews inside a block.
- Extracted unidimensional delays will be displayed in the cell_* tables.

To set this global variable, use the `set` command.

Related Topics

- "[Timing Model Generation](#)" chapter in the *Tempus User Guide*.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_extract_model_exhaustive_validation_dir

timing_extract_model_exhaustive_validation_dir *string*

Default: " "

Data_type: String

Specifies the output directory where the additional validation reports generated during ETM Extremity Validation (but not generated during normal validation) will be written.

To set this global variable, use the `set` command.

Related Topics

- "[Timing Model Generation](#)" chapter in the *Tempus User Guide*:
 - [ETM Extremity Validation](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_extract_model_exhaustive_validation_mode

timing_extract_model_exhaustive_validation_mode
{true | false}

Default: false

Data_type: Boolean

Enables extracted timing model (ETM) validation at the minimum and maximum indices of the slew load indices.

When set to `false`, the ETM validation is done only for the input slews/output-loads that have been asserted on the design during timing model extraction.

When set to `true`, the software uses the “Extremity-Validation” (EV-ETM) methodology that allows ETM to model the timing arcs in the design for a range of discrete input slews/output-loads, thus providing a more comprehensive validation.

To set this global variable, use the `set` command.

Related Topics

- "[Timing Model Generation](#)" chapter in the *Tempus User Guide*:
 - [ETM Extremity Validation](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_extract_model_gating_as_nochange_arc

```
timing_extract_model_gating_as_nochange_arc  
{true | false}
```

Default: true

Data_type: Boolean

Controls the conversion of clock gating checks to nochange arcs.

When set to `false`, the software disables conversion of clock gating checks to nochange arcs in the extracted timing model.

To set this global variable, use the set command.

Related Topics

- "[Timing Model Generation](#)" chapter in the *Tempus User Guide*:
 - [Basic Elements of Timing Model Extraction](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_extract_model_half_cycle_non_interface_paths

`timing_extract_model_half_cycle_non_interface_paths {true | false}`

Default: `false`

Data_type: Boolean

When set to `true`, the software models half cycle non-interface paths as MPW (minimum pulse width) checks in ETM.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_extract_model_ideal_clock_latency_arc

```
timing_extract_model_ideal_clock_latency_arc  
{true | false}
```

Default: true

Data_type: Boolean

When set to `true`, the latency arcs for the generated clocks that are defined ideal are extracted with reference as master clock during ETM extraction.

When set to `false`, the arcs are not extracted for generated clocks that are defined ideal during ETM extractions.

To set this global variable, use the `set` command.

Related Topics

- "[Timing Model Generation](#)" chapter in the *Tempus User Guide*:
 - [Latency Arcs Modeling](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_extract_model_include_applied_load_in_characterization_range

timing_extract_model_include_applied_load_in_characterization_range
{true | false}

Default: true

Data_type: Boolean

Specifies that the actual load visible to the timer that will be included in the load characterization range for path end points.

To set this global variable, use the `set` command.

Related Topics

- "[Timing Model Generation](#)" chapter in the *Tempus User Guide*:
 - [Basic Elements of Timing Model Extraction](#)
 - [Constraint Generation during Model Extraction](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_extract_model_include_applied_slew_in_characterization_range

```
timing_extract_model_include_applied_slew_in_characterization_range  
{true | false}
```

Default: true

Data_type: Boolean

Specifies that the actual timer slew must be included in the characterization range timing model.

To set this global variable, use the `set` command.

Related Topics

- "[Timing Model Generation](#)" chapter in the *Tempus User Guide*:
 - [Basic Elements of Timing Model Extraction](#)
 - [Constraint Generation during Model Extraction](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_extract_model_max_feedthrough_characterization_load

timing_extract_model_max_feedthrough_characterization_load *value*

Default: 0

Data_type: Double

Specifies that the load is used by the model extractor for feed-through path characterization.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_extract_model_non_borrowing_latch_path_as_setup

```
timing_extract_model_non_borrowing_latch_path_as_setup  
{true | false}
```

Default: false

Data_type: Boolean

Models the non-borrowing interface latch paths as setup checks.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_extract_model_slew_propagation_mode

```
timing_extract_model_slew_propagation_mode  
{worst_slew | path_based_slew}
```

Default: worst_slew

Data_type: Enum

Specifies the type of slew propagation to use for generating extracted timing model (ETM).

In worst slew propagation mode, the software propagates the worst slew of all the incoming arcs at a converging point for extracting the arcs.

In path based slew propagation mode, the software propagates the actual slew for the path elements for extracting the arcs.

To set this global variable, use the [set_global](#) command.

Related Topics

- "[Timing Model Generation](#)" chapter in the *Tempus User Guide*:
 - [ETM Generation for MMMC Designs](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_extract_model_synchronous_clocks_as_skew_checks

`timing_extract_model_synchronous_clocks_as_skew_checks {true | false}`

Default: `false`

Data_type: Boolean

Enables modelling of skew checks between synchronous clocks on the design boundary.

When set to `true`, the software additionally captures constraints in ETM for reg-to-reg paths, which are launched and captured by different synchronous clocks within the block. The software converts the slack of such paths into a skew-rising/falling constraint requirement between the two clock pins at the IP boundary.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_extract_model_write_clock_checks_as_arc

```
timing_extract_model_write_clock_checks_as_arc  
{true | false}
```

Default: false

Data_type: Boolean

When set to `true`, the clock checks (`min_period` and `min_pulse_width`) are modelled as arcs in the extracted timing models (ETM). Clock check arcs will be modelled as a single dimensional table with constrained and reference pin as clock pin itself.

When set to `false`, clock checks are modelled as scalar numbers in the extracted ETM.

To set this global variable, use `set` command.

Related Topics

- "[Timing Model Generation](#)" chapter in the *Tempus User Guide*:
 - [Basic Elements of Timing Model Extraction](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_extract_model_write_clock_checks_as_scalar_tables

```
timing_extract_model_write_clock_checks_as_scalar_tables  
{true | false}
```

Default: false

Data_type: Boolean

Allows the extracted timing model to write the minimum pulse width or minimum period checks as scalar arcs.

To set this global variable, use the `set` command.

Related Topics

- "[Timing Model Generation](#)" chapter in the *Tempus User Guide*:
 - [Basic Elements of Timing Model Extraction](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_extract_model_write_function_for_combinational_arcs

`timing_extract_model_write_function_for_combinational_arcs {true | false}`

Default: `false`

Data_type: Boolean

When set to `true`, the software writes the function attribute on ports with combinational arcs.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_extract_model_write_lvf

timing_extract_model_write_lvf {true | false}

Default: false

Data_type: Boolean

Generates extracted timing model (ETM) containing Liberty Variation Format (LVF) model.

When set to `true`, this global variable enables writing separate mean and variation tables for all the delay/constraint and transition tables.

When set to `false`, the variation data will be merged with the mean values. The mean +/- nsigma values of delay/constraint/transition data will be written in the ETM.

To set this global variable, use the `set` command.

Other Reference Information

- Click [here](#) for more information on this global variable.

timing_extract_model_write_min_max_clock_tree_path

```
timing_extract_model_write_min_max_clock_tree_path  
{true | false}
```

Default: true

Data_type: Boolean

Writes the minimum/maximum worst latency for each clock source.

To set this global variable, use the `set` command.

Related Topics

- "[Timing Model Generation](#)" chapter in the *Tempus User Guide*:
 - [Latency Arcs Modeling](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

Paradime ECO Global Variables

- [eco_client_tempus_executable](#)
- [eco_def_file_list](#)
- [eco_disable_parasitic_loading_from_clients](#)
- [eco_drv_view_list](#)
- [eco_dynamic_view](#)
- [eco_hold_view_list](#)
- [eco_leakage_view](#)
- [eco_lef_file_list](#)
- [eco_select_partition_list](#)
- [eco_setup_view_list](#)

eco_client_tempus_executable

eco_client_tempus_executable

Default: " "

Data_type: String

Specifies different Tempus version for clients than the Tempus version for master. This is useful in scenarios where timing analysis and ECO need to be run with different Tempus versions.

To set this global variable, use the `set` command.

Note: This global is used only in the Paradime ECO flow.

Related Topics

- "[Signoff ECO](#)" chapter in the *Tempus User Guide*:
 - [Parallel Distributed Interactive MMC Analysis and ECO \(PARADIME\) Flow](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

eco_def_file_list

eco_def_file_list

Default: " "

Data_type: String

Specifies the list of DEF files required for physical Tempus ECO run.

To set this global variable, use the `set` command.

Note: This global is used only in the Paradime ECO flow.

Related Topics

- "[Signoff ECO](#)" chapter in the *Tempus User Guide*:
 - [Parallel Distributed Interactive MMC Analysis and ECO \(PARADIME\) Flow](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

eco_disable_parasitic_loading_from_clients

```
eco_disable_parasitic_loading_from_clients  
{true | false}
```

Default: false

Data_type: Boolean

Disables loading SPEF from clients on the Tempus master, when `set_distribute_mmmc_mode -eco` is set to `true`.

Note: You need to provide parasitics information on the Tempus master after design loading, when this global is used.

To set this global variable, use the `set` command.

Note: This global is used only in the Paradime ECO flow.

Related Topics

- "[Signoff ECO](#)" chapter in the *Tempus User Guide*:
 - [Parallel Distributed Interactive MMMC Analysis and ECO \(PARADIME\) Flow](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

eco_drv_view_list

eco_drv_view_list

Default: " "

Data_type: String

Specifies the list of DRV views for Tempus ECO run.

To set this global variable, use the `set` command.

Note: This global is used only in the Paradime ECO flow. The views specified in `eco_drv_view_list` along with `eco_setup_view_list` will be considered as DRV views.

Related Topics

- "[Signoff ECO](#)" chapter in the *Tempus User Guide*:
 - [Parallel Distributed Interactive MMMC Analysis and ECO \(PARADIME\) Flow](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

eco_dynamic_view

eco_dynamic_view

Default: " "

Data_type: String

Sets the dynamic power view name for the ECO master session.

To set this global variable, use the `set` command.

Note: This global is used only in the Paradime ECO flow.

Related Topics

- "[Signoff ECO](#)" chapter in the *Tempus User Guide*:
 - [Parallel Distributed Interactive MMC Analysis and ECO \(PARADIME\) Flow](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

eco_hold_view_list

eco_hold_view_list

Default: " "

Data_type: String

Specifies the list of hold views for Tempus ECO run.

To set this global variable, use the `set` command.

Note: This global is used only in the Paradime ECO flow.

Related Topics

- "[Signoff ECO](#)" chapter in the *Tempus User Guide*:
 - [Parallel Distributed Interactive MMC Analysis and ECO \(PARADIME\) Flow](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

eco_leakage_view

eco_leakage_view

Default: " "

Data_type: String

Sets the leakage power view name for the ECO master session.

To set this global variable, use the `set` command.

Note: This global is used only in the Paradime ECO flow.

Related Topics

- "[Signoff ECO](#)" chapter in the *Tempus User Guide*:
 - [Parallel Distributed Interactive MMC Analysis and ECO \(PARADIME\) Flow](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

eco_lef_file_list

eco_lef_file_list

Default: " "

Data_type: String

Specifies the list of LEF files required for physical Tempus ECO run.

To set this global variable, use the `set` command.

Note: This global is used only in the Paradime ECO flow.

Related Topics

- "[Signoff ECO](#)" chapter in the *Tempus User Guide*:
 - [Parallel Distributed Interactive MMMC Analysis and ECO \(PARADIME\) Flow](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

eco_select_partition_list

eco_select_partition_list

Default: " "

Data_type: String

Lists the subset of partitions for which the physical data has to be loaded while merging the DEF files although user might point to all available DEF files using the `eco_def_file_list` global variable.

To set this global variable, use the `set` command.

Note: This global is used only in the Paradime ECO flow.

Related Topics

- "[Signoff ECO](#)" chapter in the *Tempus User Guide*:
 - [Parallel Distributed Interactive MMC Analysis and ECO \(PARADIME\) Flow](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

eco_setup_view_list

eco_setup_view_list

Default: " "

Data_type: String

Specifies the list of setup views for Tempus ECO run.

To set this global variable, use the `set` command.

Note: This global is used only in the Paradime ECO flow.

Related Topics

- "[Signoff ECO](#)" chapter in the *Tempus User Guide*:
 - [Parallel Distributed Interactive MMC Analysis and ECO \(PARADIME\) Flow](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

Delay Calculation Global Variables

- [delaycal_default_net_delay](#)
- [delaycal_default_net_load](#)
- [delaycal_input_transition_delay](#)
- [delaycal_use_default_delay_limit](#)

delaycal_default_net_delay

`delaycal_default_net_delay delay_value`

Default: 1000ps

Data_type: String

Sets the default net delay for delay calculation. The software uses this default value for nets that exceed 1,000 terminals. The delay calculation is performed for nets with fewer than 1,000 terminals.

Example

- The following global variable sets the net delay to 3ns for nets that exceed 1,000 terminals:

```
set delaycal_default_net_delay 3ns
```

The software displays the following information:

```
Set Default Net Delay as 3000 ps.
```

Related Global Variables

- [delaycal_default_net_load](#)
- [delaycal_input_transition_delay](#)
- [delaycal_use_default_delay_limit](#)

Other Reference Information

- Click [here](#) for more information on this global variable.

delaycal_default_net_load

`delaycal_default_net_load load_value`

Default: 0.5pF

Data_type: String

Sets the default net load for delay calculation. You can use this global variable to specify the default net load value that is used for nets that exceed 1,000 terminals.

Example

- The following global variable sets the net load to 2 pF for nets that exceed 1,000 terminals:

```
set delaycal_default_net_load 2pF
```

The software displays the following information:

```
Set Default Net Load as 2 pF.
```

Other Reference Information

- Click [here](#) for more information on this global variable.

delaycal_input_transition_delay

delaycal_input_transition_delay

Default: 4ps

Data_type: String

Sets the default input transition delays and the ideal clock transition time for delay calculation. The software uses this default transition time for nets that exceed 1,000 terminals.

Note: It is mandatory to specify time units for this command. Valid time units are ps, ns, etc.

In addition, the default input transition value is applied to those pins to which the slew was not propagated. For example:

- Tiehi or tielo pins
- Macro output pins, which do not have a related input pin

Note: A macro output pin, which has a related input pin will most likely have slew propagation from the related input pin to the output pin.

- Macro output pins for which all the timing arcs are disabled
- Input ports without a user-specified transition value

Example

- The following global variable sets the input transition delay to 300ps for nets that exceed 1,000 terminals:

```
set delaycal_input_transition_delay 300ps
```

The software displays the following information:

```
Set Input Pin Transition Delay as 300 ps.
```

Other Reference Information

- Click [here](#) for more information on this global variable.

delaycal_use_default_delay_limit

delaycal_use_default_delay_limit

Default: 1000

Data_type: Integer

Sets the pin number threshold for nets for delay calculation. Nets that have more than the specified number of pins are excluded from delay calculation, and use the default delay values instead.

Net default delay values are:

- Net delay: 1 ns
- Transition time: 0 ps
- Net load: 0.5 pF

Note: In post route stage, default delay calculation does not honor this variable. Slews and delays are computed based on actual parasitics.

To change the default delay values, set the following global variables:

- delaycal_default_net_delay
- delaycal_default_net_load
- delaycal_input_transition_delay

Example

- The following global variable sets the pin threshold number for nets to 2,000:

```
set delaycal_use_default_delay_limit 2000
```

The software displays the following information:

```
Set Using Default Delay Limit as 2000.
```

Nets that exceed pin count of 2,000 are excluded from delay calculation and use the default delay values.

Other Reference Information

- Click [here](#) for more information on this global variable.

Import and Export Global Variables

- [init_verilog_tolerate_port_mismatch](#)
- [restore_db_version](#)
- [merge_multi_port_to_single_port](#)
- [init_lef_check_antenna](#)
- [check_design_across_hierarchy](#)
- [check_design_report_async_pins](#)
- [check_design_report_scan_pins](#)
- [check_design_threshold_fanout](#)
- [save_db_restrict_under_batch_mode](#)
- [fp_vertical_row](#)
- [init_enable_drc_region_layer](#)
- [restore_db_keep_updated_macro_lefs](#)
- [save_db_restrict_under_eco_batch_mode](#)
- [init_tiers](#)

init_verilog_tolerate_port_mismatch

`init_verilog_tolerate_port_mismatch` *value*

Default: 0

Data_type: Integer

Specifies whether new instance ports can be created during Verilog netlist parsing.

By default, this variable is set to 0. If the ports of the cell and its instantiated instance do not match, the Verilog parser stops processing and the tool errors out. If instead, you would like the tool to continue, set the `init_verilog_tolerate_port_mismatch` variable to 1. When the variable value is set to 1, the Verilog parser ignores the mismatch and adds the new port to the cell. This is intended to be used during prototyping when it may be alright to lose some connections in 'dirty netlists'. Warnings about mismatches will still be generated.

Set this variable before the Verilog file is read in.

Other Reference Information

- Click [here](#) for more information on this global variable.

restore_db_version

`restore_db_version` *string*

Default: ""

Data_type: String

Restores the software database version that generates the current design.

Example

The following command restores database version saved by Tempus 19.10-p003_1:

```
read_design test.enc.dat TOP    # the design is saved within Tempus 19.10-p003_1
puts $restore_db_version
19.10-p003_1
saveDesign test2.enc           # sets restore_db_version to the current version, restores
it to the saved-out database, then sets it back to original 19.10-p003_1 value
puts $restore_db_version
19.10-p003_1
```

Other Reference Information

- Click [here](#) for more information on this global variable.

merge_multi_port_to_single_port

```
merge_multi_port_to_single_port {true | false}
```

Default: false

Data_type: Boolean

When set to `true`, multiple ports in a LEF PIN are merged into a single port. However, the pin should not have the `MUSTJOINALLPORTS` attribute as it specifies that all of the ports must be connected individually.

init_lef_check_antenna

```
init_lef_check_antenna {0 | 1}
```

Default: 1

Data_type: Boolean

If you set `init_lef_check_antenna` to `true`, LEF or OA cells will have process antenna consistency checks done for LEF ANTENNA* data (or OA equivalent data) attached to their pins. Input pins with 0 gate-area, output pins with 0 diff-area, and inout pins with 0 gate-area or 0 diff-area values will generate warnings. If `false`, the tool will not perform consistency checking.

check_design_across_hierarchy

`check_design_across_hierarchy {true | false}`

Default: `true`

Data_type: Boolean

Checks for undriven/multi-driven/unloaded/constant pins/ports by checking across hierarchical boundaries. By default checks are across hierarchy.

Related Information

- `check_design_report_async_pins`
- `check_design_report_scan_pins`
- `check_design_threshold_fanout`

check_design_report_async_pins

`check_design_report_async_pins {true | false}`

Default: `true`

Data_type: `Boolean`

Checks async pins in a netlist design.

Related Information

- [check_design_across_hierarchy](#)
- [check_design_report_scan_pins](#)
- [check_design_threshold_fanout](#)

check_design_report_scan_pins

`check_design_report_scan_pins {true | false}`

Default: `true`

Data_type: `Boolean`

Checks scan pins in a netlist design.

Related Information

- [check_design_across_hierarchy](#)
- [check_design_report_async_pins](#)
- [check_design_threshold_fanout](#)

check_design_threshold_fanout

`check_design_threshold_fanout {true | false}`

Default: 10000

Data_type: Integer

Checks for upper threshold in a netlist design.

Related Information

- [check_design_across_hierarchy](#)
- [check_design_report_async_pins](#)
- [check_design_report_scan_pins](#)

save_db_restrict_under_batch_mode

```
save_db_restrict_under_batch_mode {true | false}
```

Data_type: Boolean

Default: false

Controls whether to save a database under the batch mode. By default, a new database is saved under the batch mode. However, if `save_db_restrict_under_batch_mode` is set to true, the tool gives an error.

fp_vertical_row

```
fp_vertical_row {0 | 1}
```

Default: 0

Data_type: Integer

Specifies whether the rows in the floorplan are vertical or horizontal. By default, the rows in the floorplan are horizontal. Set this variable to 1 to enable vertical rows.

Example

- The following command sets the rows in the floorplan to vertical.

```
set fp_vertical_row 1
```

init_enable_drc_region_layer

`init_enable_drc_region_layer {list_of_drc_region_layer}`

Default: ""

Data_type: String

Specifies a list of `drc_region_layer` to restore the active DRC region layer. The `init_enable_drc_region_layer` global is saved through `saveDesign` in the *.global file.

```
init_enable_drc_region_layer {M1_FB1, M2_FB1}

LAYER M1_FB1

    TYPE ROUTING ;

    PROPERTY LEF58_REGION "REGION FB1 BASEDLAYER Metall ;" ;

    WIDTH 0.460 ;

    MINWIDTH 0.230 ;

    MAXWIDTH 0.60 ;

    SPACING 0.230 ;

    SPACING 0.6 RANGE 10.0 100000.0 ;

    PITCH 0.560 ;

    DIRECTION HORIZONTAL ;

END M1_FB1
```

Tool loads only the DRC region layer which is defined in `init_enable_drc_region_layer`.

If the base layer (Metal3) only has one DRC region layer(M3_FB1) and is not defined in `init_enable_drc_region_layer`, the tool loads the DRC region layer without showing any message. However, if the base layer(Metal3) has multiple DRC region layers(M3_FB1, M3_FB6) and are not defined in `init_enable_drc_region_layer`, the tool will give a warning message and load the first DRC region layer.

restore_db_keep_updated_macro_lefs

```
restore_db_keep_updated_macro_lefs {true | false}
```

Type: Boolean

Default: false

If set to true, the lef files of the updated macros that have been saved in DB would be honored, even if the new lef files are specified in `read_design -lef_files {xxx}` for these macros. If set to false, the lef files of updated macros in db would be ignored. Only the lef files that are specified in `read_design -lef_files {xxx}` would be updated.

Related Information

- [read_design](#)

save_db_restrict_under_eco_batch_mode

`save_db_restrict_under_eco_batch_mode {true | false}`

Type: Boolean

Default: `false`

Controls whether to restrict saving the database in the ECO batch mode.

Related Information

- [save_design](#)

init_tiers

`init_tiers string`

Type: String

Default: ""

Specifies the orientation of each tier, either 'up' or 'down', starting with the bottommost, for a multi-tier design in `init_design`. You can specify the orientation of each tier using the keywords `up` or `down`, starting from the bottommost tier. At least two tiers must be defined; specifying a single tier is not allowed and results in an error.

Examples (two-tier designs):

- { up down } – Face-to-Face (F2F)
- { down up } – Back-to-Back (B2B)
- { up up } – Face-to-Back (F2B)
- { down down } – Face-to-Back (F2B)

For designs with more than two tiers, you need to add additional entries for Tier2, Tier3, and so on. Only the up and down keywords are accepted.

General Commands

- [Message Commands](#)
- [Utility Commands](#)

Message Commands

- [get_message](#)
- [report_message](#)
- [set_message](#)
- [set_message_limit](#)
- [suppress_message](#)
- [unset_message_limit](#)
- [unsuppress_message](#)

get_message

```
get_message
-help
-id msgID
{-severity | -limit | -suppress | -count | -short | -long | -on_error | -
stop_script_limit | -detail}
```

Returns the limit, severity, or suppression information about a message.

Parameters

-help	Outputs a brief description that includes type and default information for each <code>get_message</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man get_message</code>
-count	Returns the number of times this message has been issued this session.
-detail	Returns the message of specified id.
-id <i>msgID</i>	Specifies a single message ID, in the form "ENCFOO-123".
-limit	Returns the limit for the specified message as an integer. -1 is unlimited. 0 is suppressed.
-long	Returns the long message description, as used for the Description section in " <code>man msgID</code> ".
-on_error	Returns the error action for the specified message.
-severity	Returns the severity, <code>error</code> , <code>warn</code> , <code>info</code> , or <code>default</code> , of the specified message as a string.
-short	Returns the short message description. Equivalent to " <code>help msgID</code> ".
-stop_script_limit	Returns the message limit allowed for every public command. Message limit set as -1 means unlimited.
-suppress	Returns 1 if the message is suppressed, 0 if not suppressed.

Examples

The following commands return the suppression, severity, and limit levels for the message ID `IMPDB-1241`:

- `tempus> get_message -id IMPDB-1241 -suppress`
0
- `tempus> get_message -id IMPDB-1241 -severity`
default
- `tempus> get_message -id IMPDB-1241 -limit`
20

Related Commands

- `set_message`

report_message

```
report_message
[-help]
[[{-errors | -warnings | -all } [-count ]] | -suppressed ]
[[-prefix <string> [-detail ] [-summary ]] ]
[-detail | -count | -suppressed ]
[-summary | -count | -suppressed]
```

Reports the messages that have been issued since the software was launched.

When you run the `report_message` command without any parameters, it prints a summary table and does not return any Tcl value. However, when you specify a parameter(s), it returns a list a of message IDs.

Parameters

-help	Prints a brief description that includes type and default information for each <code>report_message</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man report_message</code>
-all	Returns a list of all issued message IDs as a Tcl result. The result does not include suppressed message IDs.
-count	Returns count of every message when the <code>-errors</code> , <code>-warnings</code> , or <code>-all</code> parameters are specified.
-detail	Returns a list of detailed messages of the specified prefix.
-errors	Returns a list of issued error message IDs as a Tcl result.
-prefix <string>	Returns a list of issued error message IDs that match with the specified prefix.
-summary	Returns the message summary for the specified prefix.
-suppressed	Returns a list of suppressed message IDs as a Tcl result.
-warnings	Returns a list of issued warning message IDs as a Tcl result.

Examples

- The following command prints a summary table and returns no Tcl value.

```
tempus > report_message
```

Output:

```
*** Summary of all messages that are not suppressed in this session:
```

Severity	ID	Count	Summary
WARNING	IMPFP-2101	6	Shifter for %s to %s is not defined in s...
WARNING	IMPFP-3961	1	techSite '%s' has no related Cells in LE...
WARNING	IMPMSMV-1610	4	View %s domain %s operating voltage %g d...
WARNING	IMPTS-3	692	Can't create min and max cell relation f...
WARNING	IMPTS-419	119	Timing arc(s) mismatch found in cell '%s'...
ERROR	IMPTS-423	5	No library found for instance '%s', cell...
ERROR	IMPTS-424	1	Missing library for some instance found...
WARNING	IMPEXT-6202	1	In addition to the technology file, capa...
WARNING	IMPSYT-3046	2	Unknown preference name "%s".
WARNING	IMPSYC-1592	2	No default tech site found for power dom...
WARNING	IMPPP-4022	2	Option "-%s" is obsolete and has been re...
WARNING	IMPSC-1001	2	Unable to trace scan chain "%s".
WARNING	IMPSC-1020	1	Instance's output pin "%s/%s" (Cell "%s"...
WARNING	IMPSC-1114	2	The scan chain "%s" is not traced.
WARNING	IMPSC-1143	1	Unable to apply DEF ordered sections for...
WARNING	IMPSC-1144	2	Scan chain "%s" was not traced through. ...
WARNING	IMPOPT-3058	1003	Cell %s/%s has already a dont_use attrib...
WARNING	IMPOAX-571	1	Property '%s' from OA is a hierarchical ...
WARNING	IMPOAX-1257	1	Allowing saveDesign to create OpenAccess...
WARNING	IMPTCM-77	1	Option "%s" for command %s is obsolete a...


```
*** Message Summary: 1843 warning(s), 6 error(s)
```

- The following command reports all the message IDs:

```
report_message -all
```

- The following command reports a list of suppressed message IDs:

```
report_message -suppressed
```

Related Commands

- [set_message](#)
- [suppress_message](#)

set_message

```
set_message  
[-help]  
[-id list_of_msgIDs]  
[-severity {warn | error | info | reset} | -limit number | -no_limit | -suppress | -  
unsuppress | -on_error {msg_only | exit | stop_script | -stop_script_limit <number>}]
```

Control messages that are displayed for a particular design. The command specifies the message severity levels - `INFO`, `WARNING`, or `ERROR` - at the beginning of the message. For example, if there are some warnings that are critical for the design, you can change the message severity to `ERROR`. Similarly, if there are messages that you want to ignore, you can change the message severity level to `INFO`.

Parameters

<code>-help</code>	Outputs a brief description that includes type and default information for each <code>set_message</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man set_message</code>.
<code>-id</code> <code>list_of_msgIDs</code>	Specifies the list of message IDs that require a change in the message severity level. If you do not specify a message ID, the change in severity level is applied to all message IDs.
<code>-severity {warn error info reset}</code>	Specifies the new severity level for the message ID being modified.
<code>-limit number</code>	Sets the message limit for the selected messages. The limit must be greater than zero.
<code>-no_limit</code>	Removes any existing limit on the selected messages. The suppressed messages are not unsuppressed.
<code>-on_error {msg_only exit stop_script}</code>	Takes extra action for error messages specified using the <code>-id list_of_msgIDs</code> parameter. Without the <code>-id</code> parameter, the extra action applies to all the error messages. You can specify any one of the following options: • <code>exit</code>: Exits immediately. • <code>stop_script</code>: Waits for a command to finish and return to the Tcl prompt; • <code>msg_only</code>: Prints an error message. If the message severity is changed from <code>error</code> to <code>warn</code> or <code>info</code>, the software will no longer trigger extra action.
<code>-stop_script_limit <number></code>	Sets the message limit for every public command. If you set the <code>-stop_script_limit</code> parameter to 10, then at the 10th time the message will be triggered for one command. The software will wait for the current Tcl command to finish, the Tcl script will stop executing - and return to the interactive Tcl prompt. <i>Default:</i> -1 (unlimited)
<code>-suppress</code>	Suppresses the message.
<code>-unsuppress</code>	Unsuppresses the message.

Examples

- The following command changes the message `IMPLF-81` from `ERROR` to `WARN`:

```
tempus > set_message -id IMPLF-81 -severity warn
```

Related Commands

- `report_message`
- `suppress_message`

set_message_limit

```
set_message_limit
[-help]
numberToLimit
[<prefix [ID1, ID2, ...]>]
```

Limits error and warning messages to a specified number per unique ID. Any occurrences beyond the limit are suppressed. The default limit is 20 messages per message ID. The default limit is applicable only for `ERROR*` and `WARNING*` messages; there is no default limit for `verify*`, `check*`, and `INFO*` messages.

Parameters

<code>-help</code>	Outputs a brief description that includes type and default information for each <code>set_message_limit</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man set_message_limit</code>.
<code>numberToLimit</code>	Limits error and warning messages to a specified number per unique ID. Any occurrences beyond the limit are suppressed. The default limit is 20 per message ID. The default limit is applicable only for <code>ERROR*</code> and <code>WARNING*</code> messages; there is no default limit for <code>verify*</code>, <code>check*</code> and <code>INFO*</code> messages. <i>Default: 20</i>
<code>[<prefix [ID1, ID2, ...]>]</code>	Specifies the number per unique ID of specified prefix.

Examples

- The following command sets a limit of 10 messages per unique ID for all error and warning messages:

```
tempus > set_message_limit 10
```

- The following command sets a limit of 2 for the `TCLCMD-917` error message.

```
tempus > set_message_limit 2 TCLCMD 917
```

After 2 occurrences of `TCLCMD-917`, which is the same as the limit set, the system announces that the limit has been reached.

```
[DEV]tempus 4> set_message_limit 2 TCLCMD 917
[DEV]tempus 5> set_propagated_clock CLK1
```

```
**ERROR: (TCLCMD-917):    Cannot find 'ports, pins, or clocks' that match 'CLK1'  
[DEV]tempus 6> set_propagated_clock CLK1  
**ERROR: (TCLCMD-917):    Cannot find 'ports, pins, or clocks' that match 'CLK1'  
[DEV]tempus 7> set_propagated_clock CLK1  
Message <TCLCMD-917> has exceeded the message display limit of '2'.  
setMessageLimit/set_message_limit sets the limit. unsetMessageLimit/unset_message_limit  
can be used to reset this.
```

Related Commands

- [unset_message_limit](#)

suppress_message

```
suppress_message alpha_prefix numId1  
[numId2...numIdn]
```

Suppresses error or warning messages identified by a unique ID.

Note: This command suppresses the specified message regardless of the setting specified by the `set_message_limit` command. For example, consider the case where you have used the `set_message_limit` command to specify that a message should be suppressed after a fixed number of occurrences. If you now run the `suppress_message` command on this message, the message will be suppressed, irrespective of the number of times it occurs.

Parameters

alpha_prefix	Specifies the alpha prefix for the error or warning message.
numID	Specifies the unique numerical ID of the message to suppress.
numId2...numIdn	Specifies additional numerical IDs for the same alpha prefix.

Examples

- The following command suppresses the message with the message ID `SI-2190`:

```
tempus > suppress_message SI 2190
```
- This example suppresses the `SI-501` and `SI-818` messages:

```
tempus > suppress_message SI 501 818
```
- Alternately, you can explicitly name each message. This example also suppresses the `SI-501` and `SI-818` messages:

```
tempus > suppress_message SI-501 SI-818
```
- This example suppresses the `SI-501`, `SI-818`, `Tempus-36`, and `Tempus-102` messages:

```
tempus > suppress_message SI 501 818 Tempus 36 102
```
- Alternately, you can explicitly name each message. This example also suppresses the `SI-501`, `SI-818`, `Tempus-36`, and `Tempus-102` messages.

```
tempus > suppress Message SI-501 SI-818 Tempus-36 Tempus-102
```
- This example suppresses all messages from the SI catalog:

```
tempus > suppress_message SI
```
- Multiple parameters can be used in combination on a single line. This example suppresses all SI messages and `Tempus-102` messages.

```
tempus > suppress_message SI Tempus-102
```

Related Commands

- [unsuppress_message](#)

unset_message_limit

```
unset_message_limit  
[-help]  
[prefix [ID1, ID2...]]
```

Removes the default or user-specified limit on the number of error and warning messages to display. After you use this command, all error and warning messages are output to the log file.

Parameters

<code>-help</code>	Outputs a brief description that includes type and default information for each <code>unset_message_limit</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man unset_message_limit</code> .
<code>prefix [ID1, ID2...]</code>	Removes the default or user-specified limit on error and warning messages with <code>prefix</code> .

Examples

- The following command unsets the limit for the TCLCMD-917 error message:

```
tempus > unset_message_limit TCLCMD 917
```

Related Commands

- `set_message_limit`

unsuppress_message

unsuppress_message

[-help]

<prefix numID [*numId2...numIdn*]>

Reverses the effect of the `suppress_message` command.

Parameters

-help	Prints out the command usage.
prefix	Specifies the alpha prefix for the error or warning message.
numID	Specifies the unique numerical ID of the message to unsuppress.
numId2...numIdn	Specifies additional numerical IDs for the same alpha prefix.

Example

- This following command unsuppresses the warning messages with the message ID `SI-2190`:

```
tempus > unsuppress_message SI 2190
```

Related Commands

- `suppress_message`

Utility Commands

- [get_metric](#)
- [get_preference](#)
- [get_version](#)
- [monitor_hosts](#)
- [report_disk](#)
- [report_net_stat](#)
- [report_resource](#)
- [set_preference](#)
- [source](#)
- [write_flow_template](#)
- [alias](#)

get_metric

```
get_metric  
-help  
<pattern>  
[-branch list_of_patterns]  
[-exclude_inherited]  
[-id id_name]  
[-names label_name]  
[-pending]  
[-pending_top]  
[-raw]
```

Gets the current state of metric(s).

You can use this command to query the current state of any metric or metrics from the current specified list of metrics as well as a previous state.

Parameters

<code>-help</code>	Outputs a brief description that includes type and default information for each <code>get_metric</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <pre>man get_metric</pre>
<code>-branch list_of_patterns</code>	Specifies list of patterns to match snapshot branches. This will limit all output to metrics stored in snapshots whose branch matches the specified branch. This requires any inheritance to be re-run to exclude inherited values from non-matching snapshots.
<code>-exclude_inherited</code>	Specifies not to return any inherited value.
<code>-id id_name</code>	Specifies the database ID to output. Default is current database.
<code>-names label_name</code>	List of patterns to match snapshot names Name of label to get metrics from (wildcards supported); no label reports current metrics
<code><pattern></code>	Specifies the list of patterns to match metrics. Supports wildcard *. Use of * gets all metrics
<code>-pending</code>	Returns metrics from the pending (not yet created) snapshot.
<code>-pending_top</code>	Returns metrics from the entire pending (not yet created) snapshot stack.
<code>-raw</code>	Returns metrics as raw value (without normalization to defined units).

get_preference

```
get_preference  
[-help]  
preference_name
```

Returns the current preference settings.

Parameters

<code>-help</code>	Prints out the command usage
<code>preference_name</code>	Specifies the name of a preference.

Related Commands

- [set_preference](#)

get_version

`get_version [-help]`

Returns the Tempus software version number.

Parameters

<code>-help</code>	Prints a brief description of the command. For a detailed description of the command, use the <code>man</code> command: <code>man get_version</code>
--------------------	--

monitor_hosts

```
monitor_hosts
[-help]
[-log string]
[-overwrite]
[-period integer]
```

Monitors hosts and periodically prints the memory or CPU usage, and temporary disk speed of the hosts. This command automatically determines the hosts to be monitored. By default, after every 60 seconds hosts are monitored and the disk information is written to a log file.

This command is helpful to analyze and debug the performance of a session and helps check the machine load, the available memory, and track possible reasons for a slow down or unexpected exit during a session.

The `monitor_hosts` command is supported in STA, DSTA, and DMMMC flows.

The `monitor_hosts` command output is described below:

Status Keys/Tags	Memory (%age of Available Memory on the corresponding host)	CPU Usage	Disk Speed (Mb/sec)
E – Excellent	Memory used less than 75%	CPU Usage less than 85%	Disk speed more than 250
G – Good	Memory used between 75% and 85%	CPU Usage between 85% to 95%	Disk speed between 125 and 250
B – Bad (*)	Memory used between 85% and 99%	CPU Usage between 95% to 105%	Disk speed between 50 and 125
T – Terrible (**)	Memory used above 99%	CPU Usage more than 105%	Disk speed less than 50

The `monitor_hosts` command output will show host/machine specific information and not process specific information. So if multiple processes are running on the same machine, then the command

output will show the combined impact of all the processes, on the machine's memory, machine's CPU usage etc.

The "T" or "B" state denotes that a session may be thrashing or may not give the best results with respect to the performance. For better performance, you must re-run this command with better machines.

Parameters

<code>-help</code>	Prints out the command usage. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man monitor_hosts</code> .
<code>-log string</code>	Specifies the log file name. <i>Default:</i> host-monitor.log
<code>-overwrite</code>	Overwrites the log file, if it already exists.
<code>-period integer</code>	Specifies the sampling period after which hosts will be monitored. <i>Default:</i> 60

Examples

- A sample monitor log file is shown below:

```
Status keys: E excellent, G good, B bad, T terrible.
Memory values:
total = Real memory of host
progs = Real memory used by processes, including swappable cache
used = Real memory used by processes, excluding swappable cache
Started at: 17/04/25 04:24:22
lgi15(0) TMPDIR is : /tmp/tempus_temp_20836_lgi15_ark_lEAcvw

=====
Host          CPU      Memory (GB)          TMPDIR (GB)      (Mb/s)
name(id)      util %   total progs used %used  total  avail   rate
=====
04:24:52
lgi15(0)       6 E     32   24   15   46 E   100.74  94.67 E  262 E
```

- The following example output shows that host "sjfqos011" has very high CPU utilization (123%) because of multiple clients (client 1 and client 3) running on same host. Also, %used more than 99T shows that the memory is ~100% consumed on the "sjfqos011" and "sjfqos010" host machines.

```
Status keys: E excellent, G good, B bad, T terrible.
Memory values:
total = Real memory of host
progs = Real memory used by processes, including swappable cache
used = Real memory used by processes, excluding swappable cache
```

```

Started at: 20/07/08 07:10:10
sjfqos009(0) TMPDIR is : /tmp/ssv_tmpdir_25921_eYAlmJ
sjfqos011(1) TMPDIR is : /tmp/ssv_tmpdir_26114_ZuTGDp
sjfqos010(2) TMPDIR is : /tmp/ssv_tmpdir_26124_pjq6Cp
sjfqos011(3) TMPDIR is : /tmp/ssv_tmpdir_26160_vxwbEp
sjfqos012(4) TMPDIR is : /tmp/ssv_tmpdir_26173_FC8xvo

=====
Host CPU Memory (GB) TMPDIR (GB) (Mb/s)
name(id) util % total progs used %used total avail rate
=====
07:10:10
sjfqos009(0) 12 E 378 262 128 34 E 274.50 254.26 E N/A
sjfqos011(1) 16 E 378 262 138 36 E 274.50 254.25 E N/A
sjfqos010(2) 19 E 378 262 125 33 E 274.50 254.25 E N/A
sjfqos011(3) 16 E 378 262 138 36 E 274.50 254.25 E N/A
sjfqos012(4) 62 E 378 262 118 31 E 274.50 254.25 E N/A
07:11:10
sjfqos009(0) 12 E 378 262 133 36 E 274.50 254.26 E 127
sjfqos011(1) 123 T 378 262 168 44 E 274.50 245 E 145
sjfqos010(2) 91 G 378 262 138 36 E 274.50 254.25 E 103
sjfqos011(3) 123 T 378 262 143 38 E 274.50 245 E 14
sjfqos012(4) 82 E 378 262 131 35 E 274.50 254.25 E 135
07:12:10
sjfqos009(0) 12 E 378 262 138 36 E 274.50 254.26 E 125
sjfqos011(1) 99 B 378 378 378 100 T 274.50 254.25 E 145
sjfqos010(2) 97 B 378 262 373 99 T 274.50 254.25 E 103
sjfqos011(3) 99 B 378 378 378 100 T 274.50 254.25 E 12
sjfqos012(4) 86 G 378 262 138 36 E 274.50 254.25 E 133

```

Related Commands

- [set_multi_cpu_usage](#)
- [set_distribute_mmmc_mode](#)

report_disk

```
report_disk
[-help]
[-dir directory_name]
```

Reports the disk metrics of the current working directory and the /tmp directory. The output of this command allows you to investigate Mb/sec throughput of the disk and disk I/O requirements. This command is supported in both distributed (DSTA) and non-distributed Tempus.

This command also reports the used and free disk space for the directory specified by the `-dir` parameter (default is /tmp). When clients are started, it reports disk throughput and space on each client.

Parameters

<code>-help</code>	Outputs a brief description that includes type and default information for each <code>report_disk</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man report_disk</code>.
<code>-dir</code> <i>directory_name</i>	Specifies the name of the directory in which disk metric information is written. The default directory is /tmp.

Examples

- The following is a sample output for `report_disk` before and after starting clients:

```
dsta> report_disk
DISK_INFO : 206.66 Mb/S    381G total    1.1G used    359G free    /tmp
{206.66 381G 1.1G 359G}

dsta> distribute_start_clients -count 2
INFO (DSTA-1017): Starting 2 client processes.
Connected to sjfsb439 39262 1 ( PID=31660 )
Connected to sjfsb439 35186 0 ( PID=31658 )
INFO (DSTA-1019): Received application variables on master.
0

dsta> report_disk
DISK_INFO (Master)   : 182.53 Mb/S    381G total    1.1G used    359G free    /tmp
DISK_INFO (Client 0) : 94.90 Mb/S    381G total    1.1G used    359G free    /tmp
DISK_INFO (Client 1) : 104.28 Mb/S    381G total    1.1G used    359G free    /tmp
{182.53 381G 1.1G 359G} {94.90 381G 1.1G 359G} {104.28 381G 1.1G 359G}
```

- The following is an example that shows the usage of the -dir parameter:

```
dsta> report_disk

DISK_INFO : 173.85 Mb/S   118G total   4.9G used   107G free   /tmp
{173.85 118G 4.9G 107G}

dsta> report_disk -dir .

DISK_INFO : 98.51 Mb/S   9.6T total   9.3T used   395G free   .
{98.51 9.6T 9.3T 395G}

dsta> set_distribute_host -lsf -queue ssv

INFO (PRL-36): The timeout for a remote job to respond is 3600 seconds.
Detected a queue named ssv in the local LSF cluster.
Submit command for task runs will be: /grid/sfi/farm/bin/bsub -q ssv

dsta> set_multi_cpu_usage -remoteHost 2 -cpuPerRemoteHost 1 -localCpu 1
dsta> distribute_start_clients

Starting 2 client processes.
Job <5810482> is submitted to queue <ssv>. The associated lsf task ID is '0'
Job <5810559> is submitted to queue <ssv>. The associated lsf task ID is '1'
Connected to sjfsb080 43051 0 ( PID=18912 )
Connected to sjfsb080 36921 1 ( PID=19194 )
Received application variables on master.

dsta> report_disk

DISK_INFO (Master)   : 163.19 Mb/S   118G total   4.9G used   107G free   /tmp
DISK_INFO (Client 0) : 93.94 Mb/S   190G total   96G used   84G free   /tmp
DISK_INFO (Client 1) : 82.12 Mb/S   190G total   96G used   84G free   /tmp
{163.19 118G 4.9G 107G} {93.94 190G 96G 84G} {82.12 190G 96G 84G}

dsta> report_disk -dir ./

DISK_INFO (Master)   : 85.83 Mb/S   9.6T total   9.2T used   404G free   ./
DISK_INFO (Client 0) : 33.77 Mb/S   9.6T total   9.2T used   404G free   ./
DISK_INFO (Client 1) : 32.47 Mb/S   9.6T total   9.2T used   404G free   ./
```

{85.83 9.6T 9.2T 404G} {33.77 9.6T 9.2T 404G} {32.47 9.6T 9.2T 404G}

report_net_stat

report_net_stat

Reports the design statistics, such as number of cells, nets, pins, and I/Os.
You can use this command after importing the design.

Parameters

None

report_resource

```
report_resource
[-help]
[-peak]
[-verbose]
[-start <string> | [-end <string> [-session_cpu] [-diff_memory]]]
[-check_MemCpu {true | false} [-period <integer>]
[-memory_threshold <float>]]
```

Displays the peak memory used during a software session. This command is used to measure the peak memory at any given point in the flow and and make the script better for automation flows.

Parameters

-help	Prints out the command usage.
-check_MemCpu {true false}	Checks the machine's physical memory and CPU resources. When specified, the software enables memory/CPU checking and reports a warning if the free memory is below 5 percent of the total memory or if the average CPU load is large.
-diff_memory	Outputs different peak memory and current memory between the step start and step end.
- memory_threshold <float>	Specifies the threshold as ratio of total machine/LSF memory. A warning is issued if the available memory goes below this threshold. <i>Min: 0</i> <i>Max: 1</i>
-peak	Reports current peak memory till a given point in a flow. This parameter also works with the -start and -end parameters of the report_resource command so that peak memory observed by an individual command can be reported.
-period integer	Specifies the period for checking the physical memory and CPU usage. <i>Default: 1800s</i> <i>Min: 1</i> <i>Max: INT_MAX</i>
-start startpoint	Specifies the start point to measure the peak memory attained during a specific command. Note: The -start and -end parameters support any user-specified string, such as report_timing, or read_design. You must specify the same string for both the -start and -end parameters to measure the peak memory attained during a specific command.

<code>-session_cpu</code>	Reports the current CPU and real time after the reports last step status.
<code>-end <i>endpoint</i></code>	Specifies the end point to measure the peak memory attained during a specific command.
<code>-verbose</code>	Gives detailed memory usage information. It displays how much memory is being used at any time and of what form (physical vs. virtual). When specified, the following information is displayed: • <code>Cpu(s)</code> is the number of available processors in the machine. • <code>Load average</code> is the system load averages for the last 1 minute, 5 minutes, and 15 minutes. • <code>Mem</code> and <code>Swap</code> are the current memory information of the machine. • <code>Data Resident Set (DRS)</code> is the amount of physical memory devoted to other than executable code. "<code>current mem</code>" shows this value (Total current DRS) . • <code>Private Dirty (DRT)</code> is the memory which must be written to disk before the corresponding physical memory location can be used for some other virtual page. "<code>peak res</code>" shows this value (Total peak DRT). This is the minimum number that you must reserve to run the program. • <code>Virtual Size (VIRT)</code> is the total amount of virtual memory used by the task. • <code>Resident Size (RES)</code> is the non-swapped physical memory a task has used. The number of "Total Peak RES" is the recommended physical memory to reserve.

Examples

- The following command script specifies to display the peak memory attained during the `report_timing` command:

```
tempus > report_resource -start report_timing
tempus > report_timing -machine_readable -max_paths 100
tempus > report_resource -end report_timing
```

The following message is displayed:

```
Ending "report_timing" (total cpu=0:00:55.9, real=0:00:56.0, peak res=1032.5M, current
mem=1213.2M)
```

- The following command script specifies to display the peak memory attained till the `report_timing` command:


```
tempus > report_timing -machine_readable -max_paths 100
tempus > report_resource -peak
```

The following message is displayed:

```
Current peak res=1032.4M
```

- The following command script specifies to display CPU time/memory information during Test:

```
tempus > report_resource -start Test
tempus > report_resource -start read_design
tempus > read_design -physical_data design.setup
tempus > report_resource -end read_design
tempus > report_resource -start report_timing
tempus > report_timing -machine_readable -max_paths 100
tempus > report_resource -end report_timing
tempus > report_resource -end Test
```

The following message is displayed:

```
Ending "read_design" (total cpu=0:00:41.1, real=0:00:42.0, peak res=924.2M, current
mem=1152.7M)
Ending "report_timing" (total cpu=0:00:55.8, real=0:00:56.0, peak res=1032.4M, current
mem=1213.1M)
Ending "Test" (total cpu=0:01:37, real=0:01:38, peak res=1032.4M, current mem=1213.1M)
```

- The following command specifies to display detailed memory information:

```
report_resource -verbose
```

The following output is displayed:

```
Current (total cpu=0:00:23.5, real=0:10:24, peak res=567.8M, current mem=567.8M)
Cpu(s) 40, load average (past 1 minute): 7.41, load average (past 5 minutes): 7.46, load
average (past 15 minutes): 7.52
Mem: 527321900k total, 317217944k used, 210103956k free, 3752k buffers
Swap: 134217724k total, 0k used, 134217724k free, 183952200k cached

Memory Detailed Usage :
      Data Resident Set (DRS)   Private Dirty (DRT)   Virtual Size (VIRT)
Resident Size (RES)
Total current :    642.3M                155.6M                1743.3M                567.8M
peak :           642.3M                155.6M                1743.3M                567.8M
```

Related Commands

- `read_design`
- `report_timing`

set_preference

```
set_preference  
[-help]  
preference_name  
value
```

Defines preferences, such as, commands issued during a session can be logged to a log or the screen.

Parameters

-help	Outputs a brief description that includes type and default information for each set_preference parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man set_preference</code> .
preference_name	The name of the preference variable you want to set. To see a list of preferences, select Tools - Preferences from the menu, then click <i>Save</i> . In the Save Preferences form, select the required save option and click <i>OK</i> . If you select the default <i>Save to Home Directory</i> option, the list of preferences and their values are saved in .enc file in your home directory.
value	The value for the preference. For example, the <code>CmdLogMode</code> preference has the following values. 0--write to command only 1--write to command and log (default) 2--write to command, log, and screen When writing to a log and screen, the commands are preceded by <CMD>.

Preference Variables

For a complete list of preference variables that you can set using the `set_preference` command and their GUI equivalent in the Preferences form.

Examples

The following command writes log to command, log, and screen:

```
tempus > set_preference CmdLogMode 2
```

Related Topics

For more information on setting preferences from the GUI, see:

- The Preferences form in the "Tools Menu" chapter in the *Tempus Menu Reference*.

Related Commands

- `get_preference`

source

```
source  
[-help]  
<file_name>  
[-echo]  
[-quiet | -verbose]  
[-echo]
```

Reads a file and executes the commands in the file (scripts).

Parameters

-help	Describes the command usage.
-echo	Enables source to be in verbose mode.
<file_name>	Specifies the name of the file to source.
-quiet	Forces source to be normal Tcl source command.
-verbose	Forces source to be in verbose mode.

write_flow_template

```
write_flow_template
[-help]
[-directory directory_name]
[-tempus]
```

Copies the Tempus Foundation flows' templates into the directory you specify, or to the current directory, if you do not specify one with the `-directory` parameter.

Parameters

<code>-help</code>	Outputs a brief description that includes type and default information for each <code>write_flow_template</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man write_flow_template</code> .
<code>-directory</code> <code>directory_name</code>	Specifies the directory in which to save the templates.
<code>-tempus</code>	Writes the Tempus flow template.

Examples

- The following command copies the Tempus foundation flows templates into the `tempusFF` directory:

```
tempus > write_flow_template -directory tempusFF
```

alias

```
alias
[-help]
new_cmd
old_cmd
[-args {{new old} ...}]
```

Creates command name aliases for Tcl commands, except built-in Tcl commands. This command supports argument name aliases for application commands and procs that use `define_proc_arguments` and `parse_proc_arguments` commands for parsing the arguments.

The argument aliases are private, so these do not display with `-help`, and also do not support partial option completion or matching. However, the command aliases are public, so `<tab>` complete works.

You can use this command to migrate an existing Tcl script when some of the commands or argument

names have changed.

Parameters

<code>-help</code>	Prints a brief description that includes type and default information for each <code>bindkey</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man bindkey</code>
<code>new_cmd</code>	Specifies the new command name that is an alias for the existing <code>old_cmd</code> name. It can be the same name as the <code>old_cmd</code> if you only want to create new <code>-arg</code> aliases, and do not need a new command alias.
<code>old_cmd</code>	Specifies the existing old command name.
<code>-args</code> <code>{{new</code> <code>old} ...}</code>	Specifies a list of new and old argument names to create aliases. You can give multiple <code>new</code> names for the same <code>old</code> name. The new aliases are private, and do not show up with <code>-help</code> . You can create more than one alias for the same <code>old</code> name.

Example

The following command adds a new command name alias `new_read_sdf` and some arg name aliases for `read_sdf`:

```
>alias new_read_sdf read_sdf -args {
    {-ignore_error -continue_on_error}
    {-summary -report_summary}
    {-sum -report_summary}
}
```

Then, the following commands

```
>new_read_sdf in.sdf -ignore_error -summary
>new_read_sdf in.sdf -ignore_error -sum
>new_read_s in.sdf -ignore_error -sum
```

would behave the same as

```
>read_sdf in.sdf -continue_on_error -report_summary
```

Note: Tcl partial command matching works if the partial command is unique as shown in the above example, but the alias argument names are private, and will not support partial option matching, or appear in the `-help` output.

Related Topics

- [define_proc_arguments](#)
- [parse_proc_arguments](#)

Design Import Commands

- `check_design`
- `get_design_mode`
- `read_def`
- `read_design`
- `read_instance_temp`
- `read_instance_voltage`
- `read_lib`
- `read_power_domain`
- `read_rcdb`
- `read_sdc`
- `read_verilog`
- `read_view_definition`
- `restore_oa_design`
- `save_design`
- `set_cell_power_domain`
- `set_design_mode`
- `set_import_mode`
- `set_license_check`
- `set_top_module`
- `write_def`
- `write_rcdb`

check_design

```
check_design
[-id violation_id]
[-out_file fileName]
{[-type {library netlist power_intent timing hierarchical pin_assign budget assign_statements
place opt cts route signoff all} [-format {table detail}] |-type_3d {tech_lib netlist
power_intent timing place floorplan route report_only all}}][-no_check ]}
```

Checks that all prerequisites to run all or part of the hierarchical or block implementation flow are met. The `check_design` command checks the preconditions for the major flow steps before they are run. You can run these checks standalone using the `check_design` command. If there are errors in the `check_design` run, the current script will stop.

Parameters

<code>-help</code>	Prints a brief description that includes the type and default information for each <code>check_design</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <pre>man check_design</pre>
<code>-format {table detail}</code>	Specifies the format for reporting violations. <code>table</code>: Prints the results in the table format. <code>detail</code>: Prints details of violations.
<code>-id violation_id</code>	Specifies the violation IDs.

-type

Specifies the category of the flow to be checked:

- `power_intent`: Performs checks related to MSV setup including power domain/fence checks.
- `timing`: Performs checks related to timing setup integrity.
- `hierarchical`: Performs check if the partition has a fence and if the fence is inside the core boundary.
- `pin_assign`: Performs checks related to pin assignment for partitioning flows.
- `budget`: Performs checks related to time budgeting for partitioning flows.
- `assign_statements`: Performs checks for the netlist for assign statements.
- `place`: Performs checks related to placement such as PG rail alignment and pin access issues.
- `opt`: Performs checks related to optimization such as checks for available cells.
- `cts`: Performs checks related to clock tree definition such as constraint targets and route types.
- `route`: Performs checks related to routing such as congestion, placement, and pin access issues.
- `signoff`: Checks related to the signoff activities including extraction and ECOs.
- `all`: Runs all categories.

<pre>-type_3d {tech_lib netlist power_intent timing place floorplan route report_only all}}</pre>	Specifies the 3DIC design categories to check. You can specify single or multiple categories • <code>tech_lib</code> ◦ Checks the library integrity and corrects binding ◦ Checks all timing lib wh • <code>netlist</code> ◦ Checks the library integrity and corrects binding • <code>power_intent</code> ◦ Checks related to MSV setup including power domain/fence checks. ◦ Checks related power domain bind to correct design • <code>timing</code> ◦ Performs checks related to timing setup integrity. • <code>place</code> ◦ Checks related to placement such as pin alignment • <code>floorplan</code> ◦ Checks related to floorplanning • <code>route</code> ◦ Checks related to routing • <code>all</code> ◦ Checks all categories
<pre>-no_check</pre>	Just echo out all the messages that would be checked, don't do any checks Do not run the checks but echo the checks that would run per the categories provided by the user

`-out_file fileName`

Specifies the output report file. Contents dumped to this file in the Tcl format.

get_design_mode

```
get_design_mode
[-help]
[-backsideBottomRoutingLayer layer]
[-backsideTopRoutingLayer layer]
[-bottomRoutingLayer]
[-compressedPGDB]
[-congEffort {low|medium|high|auto}]
[-earlyClockFlow]
[-idealHoldFixing {true|false}]
[-merge_trim_shapes
{{layers layer_name_list gap gap_value max_merge_num number_of_max_merge masks
{mask_num ...}} ...}]
[-node]
[-optimizationDensityScreenMargin margin]
[-quiet]
[-nonDefault]
[-topRoutingLayer]
[-trim_grid_group]
```

Displays the following information in the log file and at the console:

- Parameter name
- Current value
- Type (Boolean, string, and so on)
- Whether the current value was set by user

If you do not specify a parameter, the software displays information for all the parameters.

Parameters

-help	Outputs a brief description that includes type and default information for each <code>get_design_mode</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <pre>man get_design_mode.</pre>
-backsidebottomroutinglayer	Returns the highest backside LEF layer name for global and detail routing. <i>Default:</i> ""

<code>-backsideTopRoutingLayer</code>	Returns the lowest backside LEF layer name for global and detail routing. <i>Default:</i> ""
<code>-bottomRoutingLayer</code>	Returns the lowest lef layer name for global and detail routing. <i>Default:</i> ""
<code>-compressedPGDB</code>	Returns whether the compressed PG feature is enabled. When it's true, <code>read_design</code> will automatically compress the PG data when load a design.
<code>-congEffort</code> {low medium high auto}	Returns the congestion effort level.
<code>-earlyClockFlow</code>	Returns the setting for early clock flow. <i>Default:</i> false
<code>-idealHoldFixing</code>	Returns the value ideal hold fixing flow. <i>Default:</i> false
<code>-merge_trim_shapes</code>	Returns the setting for trim shapes merge.
<code>-node</code>	Returns the design technology node.
<code>-optimizationDensityScreenMargin</code>	Returns the margin on the top of original density screen for optimization.
<code>-quiet</code>	Returns the current settings for the specified parameter in Tcl list format only. If you specify <code>-quiet</code> without a parameter, the software displays the current settings of all <code>set_design_mode</code> parameters in Tcl list format.
<code>-nonDefault</code>	Displays the current settings which are not default settings.
<code>-topRoutingLayer layer_name</code>	Returns the highest lef layer name for global and detail routing. <i>Default:</i> ""

`-trim_grid_group group_name`

Returns the GROUP name of the set from which the trim metal grid will be used for placement and routing.

Default: ""

If the specified group name does not exist in LEF, the tool will return an ERROR.

Examples

- The following command displays the current setting of the `-process` parameter:

```
tempus > get_design_mode -process
```

The software displays the following information:

```
-process 65 # int, default=65, min=10, max=250
```

- The following command displays the current setting for the `-process` parameter in Tcl list format only:

```
tempus > get_design_mode -process -quiet
```

The software displays the following information:

```
65
```

Related Commands

- `set_design_mode`

read_def

```
read_def
[-help]
[<defFile>]
[-force_create_phy_inst]
[-keepPinGeometry]
[-preserve_shape]
[-rdl_def <string>]
[-rdl_orientation <orientation>]
[-rdl_placement <(x y)>]
[-read_flat_data <dir>]
[-skip_pg]
[-skip_signal]
[-top]
[-top_scope]
[-topcell_orientation <orientation>]
[-topcell_placement <(x y)>]
[-write_flat_data <dir>]
```

Reads a single DEF file. This command can be used to supply single DEF to generate a physical database.

The `read_def` command cannot read multiple or hierarchical DEF files. To read multiple or hierarchical DEF files and create a flattened database, use the `merge_hierarchical_def` command.

The `read_lib -lef` command is required in order to load DEFs. This command should never include LEF definition of the block level partitions. It should only include LEF definition for primitive cells like standard cells, IOs and memories.

Note: All hierarchical DEFs are passed to power and rail analysis without flattening and processed by Power Analysis and Rail Analysis engines separately. The flattened DEF is only used for display of layout in the GUI. The flattened DEF should not be used for analysis.

Parameters

<code>-help</code>	Prints out the command usage
<code><defFile></code>	Specifies the def file to be read.
<code>-force_create_phy_inst</code>	Specifies to force create physical cell instances
<code>-keepPinGeometry</code>	Keeps pins shapes equivalent to the block pin shapes, in the form of special route
<code>-preserve_shape</code>	Preserves the shape of pins.
<code>-rdl_def <string></code>	Specifies RDL def file.
<code>-rdl_placement (x y)</code>	Specifies offset of RDL def.
<code>-rdl_orientation <orientation></code>	Specifies orientation for RDL def.
<code>-read_flat_data <dir></code>	Reads flat data from the specified directory.
<code>-skip_pg</code>	Skips merging routing on pg nets.
<code>-skip_signal</code>	Skips reading signal nets.
<code>-top</code>	Reads DEF files with the same top cell name.
<code>-top_scope</code>	Merges routing on top scope nets only.
<code>-topcell_orientation <orientation></code>	Specifies orientation for topcell.
<code>-topcell_placement (x y)</code>	Specifies placement for topcell.
<code>-write_flat_data <dir></code>	Writes flat data to the specified directory.

Examples

- The following command reads the compressed DEF file `dma_mac_filled.def.gz`:
tempus > read_def dma_mac_filled.def.gz

Related Commands

- `merge_hierarchical_def`
- `read_lib` -lef

read_design

```
read_design
[-help]
[-black_box_modules blackBoxModules]
<dir <dir topcell>>
[-cellview {lib | cell | view} ]
[-common]
[-dynamic_view userDynamicView]
[-exclude_path_collection]
[-hold_views userHoldViews]
[-leakage_view userLeakageView]
[-mmmcFile userMmmcFile]
[-lef_files lefFiles]
[-module_dirs moduleDirs]
[-no_timing_graph]
[-oaRef <oaRefLib>]
[-physical_data]
[-setup_views userSetupViews]
```

Loads a design database saved in Tempus and Innovus. You can also read the physical data information such as floorplan, placement, and routing using the `-physical_data` parameter.

Note: You can also use the `read_def` command to read the DEF file for physical information.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-black_box_modules</code> <i>blackBoxModules</i>	Specifies modules for which black box models should be loaded.
<code>-cellview {<i>lib</i> <i>cell</i> <i>view</i>}</code>	Restores an OpenAccess design. It specifies the name of the library, top cell, and cell view for the OA design. This parameter works both for the physical and non-physical flow.
<code>-common</code>	Specifies to read the common-db design directory.
<code><dir <dir topcell>></code>	Specifies the design data directory name and topcell.
<code>-dynamic_view</code> <i>userDynamicView</i>	Specifies the analysis view for the dynamic power analysis.
<code>-exclude_path_collection</code>	Restores design data without archived path collection.

<code>-hold_views userHoldViews</code>	Specifies the hold views for loading, rather than the hold views active during <code>save_design</code> .
<code>-leakage_view userLeakageView</code>	Specifies the analysis view for leakage power analysis.
<code>-lef_files lefFiles</code>	Specifies the LEF files for loading. You can use this parameter, if you need to override the existing LEF files in the design being restored and load different LEF files. • This option is intended to allow upgrading to new versions of LEF files that are compatible with the old versions. The list of all LEF files used by this design must be given. • If any pin locations or cell boundaries or OBS shapes are changed for used cells, new DRC violations may occur. • If pins are added or removed from used cells, or used cells are removed, fatal errors may occur when loading the netlist. • If the removed pins and cells are unused in the design, warnings or errors may still occur if any library-specific data was stored for those cells or pins during the design flow (for example, <code>dont_touch</code> data)
<code>-mmmcFile userMmmcFile</code>	Specifies the name of the path to a <code>viewDefinition.tcl</code> (MMMC) file.
<code>-module_dirs moduleDirs</code>	Specifies module-level sessions that should override the corresponding modules in the top-level hierarchical session.
<code>-no_timing_graph</code>	Restores the design data without a timing graph.
<code>-oaRef oaRefLib</code>	Reads the OA reference library.
<code>-physical_data</code>	Reads the physical data such as floorplan, placement, and routing that you generate in Encounter.
<code>-setup_views userSetupViews</code>	Specifies the setup views for loading.

Examples

- The following command loads the saved database `dtmf_chip`.
`tempus > read_design dtmf_chip.dat dtmf_chip`
- The following command reads a design called `dtmf_chip` with physical data.
`tempus > read_design -physical_data dtmf_chip.dat dtmf_chip`
- The following command loads the 'layout' view of the 'design1' cell from the 'designLib' libraries of an OA database.
`tempus > read_design -cellview {designLib design1 layout}`
- The following command loads the 'layout' view of the 'dtmf_chip' design from the 'designLib' libraries of an OA database in physical mode.
`tempus > read_design -cellview "designLib dtmf_chip layout" -physical_data`

Related Commands

- `save_design`

read_instance_temp

```
read_instance_temp
[-help]
[-defaultTemperature value]
[infile filename]
[-reset]
```

Loads the instance-based temperature information to account for the effects of temperature when calculating the delays and slews.

Parameters

-help	Prints out the command usage.
-defaultTemperature <i>value</i>	Specifies the default temperature value (in celsius) to use. Note: The temperature settings in the temperature file take precedence over this parameter.
-infile <i>filename</i>	Specifies the name of the temperature file that contains the instance-based temperature values (in celsius). When used with the -defaultTemperature option, the instances that do not appear in these files will use the specified default temperature. The format of the temperature file is as follows: <pre>VERSION "1.0" ; CREATION "Date" ; PROGRAM "Vendor" ` DIVIDECHAR "/" ; TEMPERATURE_UNIT "C" ; - instance1 temperature ; - instance2 temperature ; - instance3 temperature ; - instance4 temperature ; - instance5 temperature ; - END;</pre>
-reset	Resets all the temperature settings.

Examples

- The following command loads the temperature file named `sampletemp`:

```
tempus > read_instance_temp -infile sampletemp
```


Related Commands

- `set_inst_temperature_file`

read_instance_voltage

```
read_instance_voltage  
[-help]  
-ir_drop {irDropFile}  
[-early | -late]  
[-min | -max]  
[-list]  
[-reset]
```

Loads the instance-based IR-drop information that is output by power-rail analysis tools, such as Voltus, to account for the effects of IR drop when calculating the delays and slews. You can read in four different files for early-max, early-min, late-max, and late-min IR-drop analysis.

Parameters

-help	Prints out the command usage.
-early -late	Applies the effects of IR-drop to the early path (data hold/clock setup) or late path (data setup/clock hold) in your design. Use the <code>-early</code> or <code>-late</code> parameter in conjunction with the <code>-min</code> or <code>-max</code> parameter to account for the effects of early-max, early-min, late-max, or late-min IR-drop analysis.

<code>-ir_drop</code> <code>{irDropFile}</code>	Specifies the names of the IR-drop files to load. The format of an IR drop file is shown below: <pre> VERSION "2.0" CREATION "Tue Jul 21 02:10:15 2015" CREATOR "user1" PROGRAM "Vendor" DIVIDERCHAR "/" DESIGN "design1" UNITS VOLTAGE VOLT 1 INSTANCE_COUNT 1379138 NOMINAL_VOLTAGE 1.1 POWER_NET vdd GROUND_NET gnd WINDOW SWITCHING_AND_TIMING RP_VALUE IV/EIV RP_FORMAT BRIEF RP_INST_LIMIT 2000000000 RP_THRESHOLD -1e+06 RP_PIN_NAME FALSE INSTANCESUPPLY 1379138 vdd 1.1 EFFECTIVE BEGIN - ICC_PRECTS_33 0.95948 C12T28SOI_LLTHP_XOR2X33_P16 - ICC_PRECTS_5 0.96383 C12T28SOI_LL_CNNAND2X15_P16 - u_stc2/ICC_RTO_TIMING_6 0.96524 C12T28SOI_LLBR1P2NHP_BFX34_P16 - u_ca7_scu_lld_tagrams/U207 0.96705 C12T28SOI_LLBR1P8NHP_BFX48_P16 - u_ca7_scu_lld_tagrams/U244 0.96759 C12T28SOI_LLBR1P2NHP_BFX43_P16 </pre>
<code>-list</code>	Provides a list of all specified IR-drop files.
<code>-min</code> <code>-max</code>	Applies the effects of IR-drop to the minimum operating condition or maximum operating condition. Use the <code>-min</code> or <code>-max</code> parameter in conjunction with the <code>-early</code> or <code>-late</code> parameter to account for the effects of early-max, early-min, late-max, or late-min IR-drop analysis. Note: If the <code>-min</code> or <code>-max</code> parameter is used without the <code>-early</code> or <code>-late</code> parameter, the IR drop is applied to both early and late analysis. To apply the IR drop for late analysis only, use the <code>-min</code> or <code>-max</code> parameter with the <code>-late</code> parameter. To apply the IR drop for early analysis only, use the <code>-min</code> or <code>-max</code> parameter with the <code>-early</code> parameter.
<code>-reset</code>	Resets all IR-drop files.

Examples

- The following commands apply the IR drop for early-min, late-min, early-max, and late-max IR-drop analysis:

```
tempus > read_instance_voltage -early -min -ir_drop { earlymin.ir }  
tempus > read_instance_voltage -late -min -ir_drop { latemin.ir }  
tempus > read_instance_voltage -early -max -ir_drop { earlymax.ir }  
tempus > read_instance_voltage -late -max -ir_drop { latemax.ir }
```

- The following commands apply the IR drop for late-min and late-max IR-drop analysis:

```
tempus > read_instance_voltage -late -min -ir_drop { latemin.ir }  
tempus > read_instance_voltage -late -max -ir_drop { latemax.ir }
```

- The following commands apply the IR drop to early and late IR-drop analysis:

```
tempus > read_instance_voltage -early -ir_drop { early.ir }  
tempus > read_instance_voltage -late -ir_drop { late.ir }
```

read_lib

```
read_lib
[-help]
[<libFiles <library_files>>]
[-library_side_file <<side files>>]
[-aocv <<aocv files>>]
[-cdb <<typ.cdb files>>]
[-lef <<.lef files>>]
[-max <<max.lib files>>]
[-max_aocv <<max_aocv files>>]
[-max_cdb <<max.cdb files>>]
[-min <<min.lib files>>]
[-min_aocv <<min_aocv files>>]
[-min_cdb <<min.cdb files>>]
[-oaRef <<oaRefLib>>]
[-pgv <<power grid lib files>>]
[-socv <<socv files>>]
```

Schedules reading of one or more technology library files. These files either contain timing-related or noise-related information for the primitive cells used in the design. When you use the `read_lib` command, the library information is scheduled for reading but is actually read in after you use the `set_top_module` command.

Timing libraries can be in text format or Cadence binary library format (LDB). The LDB (library database) files generated from other tools, like Innovus, Voltus, or Genus are read in to Tempus with an error message ([TECHLIB-1249](#)). The LDB files are not forward compatible, that is, a LDB generated from 15.2 version cannot be used with 15.1 or prior versions of the software.

Note: The `read_lib` command determines the priority of ECSM vs CCS data (if both are present). The software first checks for ECSM construct, if it is missing then CCS construct is loaded.

Note: You can use wildcards with the `read_lib` command as follows:

```
read_lib [glob *.lib]
```

Parameters

-help	Prints out the command usage.
-------	-------------------------------

<code>-aocv aocv files</code>	Reads the AOCV libraries with timing libraries. In this case, as compared to worst OCV, the tool annotates less pessimistic derate factors from the given AOCV library on to design elements (cells/nets) resulting in less pessimistic timing results. To enable the AOCV flow, you also need to set the following while reading the AOCV libraries: <pre>set_analysis_mode -aocv true</pre>
<code>-cdb typ.cdb files</code>	Specifies the name of the noise library file(s).
<code><libFiles</code> <code><library_files>></code>	Specifies the name of the .lib library or libraries. You can also specify tri-lib bundles using this parameter. ◦ <code>read_lib test.lib</code> This command loads the library file that contains ECSM-noise data. ◦ <code>read_lib -cdb test.sp</code> This command loads the SPICE model and subckt information.
<code>-library_side_file</code> <code><<side files>></code>	Reads the side files.
<code>-lef .lef files</code>	Specifies the name of the LEF files to be loaded.
<code>-max_aocv max_aocv files</code>	Reads the maximum AOCV libraries.
<code>-max max.lib files</code>	Specifies the libraries to be used for maximum timing analysis.
<code>-max_cdb max.cdb files</code>	Reads the maximum noise libraries.
<code>-min_aocv min_aocv files</code>	Reads the minimum AOCV libraries.
<code>-min min.lib files</code>	Specifies the libraries to be used for minimum timing analysis.
<code>-min_cdb min.cdb files</code>	Reads the minimum noise libraries.
<code>-oaRef oaRefLib</code>	Loads OpenAccess reference libraries in the physical flow.
<code>-pgv power grid</code> <code>lib files</code>	Reads the power-grid library files. Note: This parameter is required when you run Voltus.

`-socv socv files`

Reads the SOCV libraries with timing libraries.

To enable the SOCV flow, you also need to set the following while reading the SOCV libraries:

`set_analysis_mode -socv true`

Examples

- The below command reads the LEF files. Note that the first file is a technology LEF.

```
tempus > read_lib -lef tech.lef allcells.lef
```

- The next command reads the minimum libraries.

```
tempus > read_lib min stdcells_min.lib memories_min.lib
```

- The next command reads the maximum libraries.

```
tempus > read_lib max stdcells_min.lib memories_min.lib
```

- The following command reads a list of timing libraries to be used for minimum and maximum analysis.

```
tempus > read_lib -min { fast.lib mem1_min.lib macro2_min.lib }  
      \ -max { slow.lib mem1_max.lib macro2_max.lib }
```

- The following command reads a User Defined Noise model (UDN) and the noise library (cdB):

```
tempus > read_lib -cdb {std_cells.cdB cells.udn}
```

Related Commands

- `set_top_module`

read_power_domain

```
read_power_domain
[-help]
[<powerDomainFile>]
[-1801 <IEEE1801File>]
[-cpf file_name ]
[-msmv file_name]
```

Reads the power domain file created in Innovus System, the MSMV file, or the Common Power Format (CPF) file. The power domain, MSMV, or CPF file is required to perform timing and signal integrity analysis for MSMV designs.

Note: If you read the CPF file with view and commit options, the software switches to Multi-Mode Multi-Corner analysis and resets the parasitics. Use the `read_spef` command to reload the parasitics.

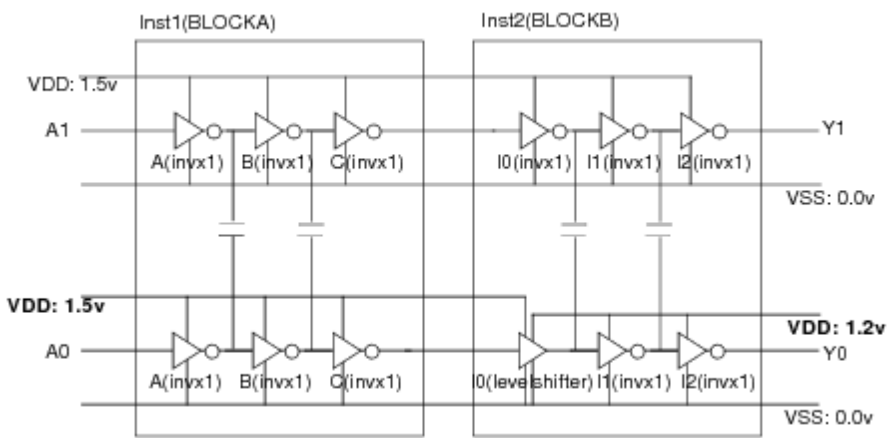
Parameters

<code>-help</code>	Prints out the command usage.
<code><powerDomainFile></code>	Specifies the name of the power domain file created in Innovus System. This file contains the definition of power domain and membership information and uses a <code>.pd</code> extension.
<code>-1801</code> <code><IEEE1801File></code>	Specifies the input file in the IEEE1801 format. While using IEEE 1801 support, note the following: - For Low Power (LP) cell definition: All the LP cells and their related power pin information must be defined in the Liberty file with the Liberty LP attributes. - For Timing Information: Timing information must be defined in Tempus MMMC <code>viewDefinition.tcl</code>.
<code>-cpf file_name</code>	Specifies the name of the Common Power Format (CPF) file, which contains the power domain definitions. This file uses a <code>.cpf</code> extension. The CPF file can be created in Innovus System.
<code>-msmv file_name</code>	Specifies the name of the MSMV file. This file uses a format that is similar to the IR-drop file and can be created in Innovus System. Note: Ensure that the MSMV file contains the complete list of cell instances.

Examples

Here are some examples.

- The following figure displays a circuit for which the power domain file is created in Innovus System.



The following power domain file is created in Innovus System:

```
tempus > createPowerDomain pd_1.2 -timingLibs {ss_1.2v lvlshft}
tempus > createPowerDomain pd_1.5 -default -timingLibs {ss_1.5v}
tempus > modifyPowerDomainMember pd_1.5 -instances *
tempus > modifyPowerDomainMember pd_1.2 -instances Inst2/I0 Inst2/I1 Inst2/I2
```

- The following command reads the power domain file `dma_mac.cpf` :

```
tempus > read_power_domain -cpf dma_mac.cpf
```

The following is the sample script for Tempus IEEE1801 LP support:

```
# read MMC include the library
read_view_definition ../INPUT/viewDefinition.tcl
# read in verilog
read_verilog ../INPUT/dma_mac_postroute.v
set_top_module dma_mac
read_power_domain -1801 <IEEE1801File>
```

The rest of Tempus LP Flow (same as CPF low power flow)

Related Commands

- [report_power_domain](#)

read_rcdb

```
read_rcdb  
[-help]  
dirName
```

Restores the parasitic data that was previously saved with the `write_rcdb` command.

When you use the `read_rcdb` command, the design data must be the same as when you use the `write_rcdb` command. You can restore the parasitic data on systems that are same as those used for saving the data. For example, if the data was saved on Linux in 32bit mode, you can restore it on Linux 32 bit mode only.

 Restoring RCDB that you generated using previous major version(s) of the software is not supported and might result in an error. Use `write_rcdb` to generate a new RCDB.

Parameters

<code>-help</code>	Prints out the command usage.
<code>dirName</code>	Specifies the name of the parasitic file that you saved using the <code>write_rcdb</code> command. Note: The parasitic file must have the <code>.rcdb.d</code> extension.

Examples

- The following command reads an Encounter RC extracted database:

```
tempus > read_rcdb test.rcdb.d
```

Related Commands

- `write_rcdb`

read_sdc

```
read_sdc  
[-help]  
[-reset]  
fileName
```

Loads a timing constraints file. By default, the constraints files are loaded incrementally.

The constraints can be in the design constraints (SDC) format, or they can be more complicated Tcl constraints embedded in a Tcl script using commands such as [get_property](#) and [filter_collection](#).

A list of constraints files also can be specified in the configuration file.

Parameter

-help	Prints out the command usage.
<i>fileName</i>	Specifies the name of the constraint file.
-reset	Replaces all previously loaded constraints information with the information in the specified <i>fileName</i> . Note: Alternatively, you can use the reset_sdc command to reset the constraints files.

Examples

- The following command reads a SDC file called dma_mac_m.sdc:

```
tempus > read_sdc dma_mac_m.sdc
```

Command Order

Use this command after importing the design and loading the timing library.

Related Commands

- [write_sdc](#)
- [reset_sdc](#)

read_verilog

```
read_verilog
[-help]
-disable_name_match
netlist
```

Schedules reading of a structural, gate-level verilog netlist. When you use the `read_lib` command, the verilog information is scheduled for reading but is actually read in after you use the `set_top_module` command. The verilog netlist must be fully mapped to the gate-level, and may not contain any verilog high-level constructs.

To schedule more than one verilog file for reading, you can either list all the files in one single command or use multiple `read_verilog` commands.

Parameter

<code>-help</code>	Prints out the command usage.
<code>-disable_name_match</code>	Disables the file name matching feature. When the file name matching feature is enabled (default) and the specified Verilog file is not present in the working directory, the software searches for the <code>.gz</code> version of the file. For example, if the <code>sample.v</code> file is specified and is not found, the software searches for the <code>sample.v.gz</code> file and uses it if it is available. Similarly, if the <code>sample.v.gz</code> file is specified but is not available, the software searches for <code>sample.v</code> file and uses it if it is available.
<code>netlist</code>	Name of the verilog netlist.

Example

- The following command reads the verilog netlist, `dma_mac.v`:
tempus > read_verilog dma_mac.v
- The following command reads in the verilog netlists defined in `file1.v`, `file2.v` and `file3.v`:
tempus > read_verilog file1.v file2.v file3.v

Related Commands

- `set_top_module`
- `read_lib`

read_view_definition

```
read_view_definition
-help
-cpf
<viewDefinitionFile>
```

Specifies a view definition file for loading an MMMC design.

When using the MMMC flow, it is recommended to use the following flow whereby the view definition file is read before the `set_top_module` command:

```
read_view_definition viewDefinitionFile
read_verilog
set_top_module
report_timing
```

The recommended flow for a design where MMMC configuration is specified in CPF file is as follows:

```
read_view_definition -cpf <cpf_file>
read_verilog
set_top_module
read_power_domain -cpf <cpf_file>
```

Parameters

-help	Outputs a brief description of the <code>read_view_definition</code> command. For a detailed description of the command, use the man command: <code>man read_view_definition</code> .
-cpf	Specifies CPF file with MMMC configuration.
<i>viewDefinitionFile</i>	Specifies MMMC view definition file.

Related Commands

- `read_verilog`
- `set_top_module`
- `report_timing`
- `read_power_domain`

restore_oa_design

```
restore_oa_design
[-help]
lib
cell
view
[-physical_data]
```

Restores all logical design information from the previous design session saved in Innovus System, using the OpenAccess database format. The `restore_oa_design` command cannot restore physical data.

The parameters of this command are order dependent and must be specified in the order shown in the syntax.

Note: In order to restore a design in OpenAccess format, you must have saved the design using the `saveOaDesign` command in the Innovus System session.

Parameters

<code>-help</code>	Prints out the command usage.
<code>cell</code>	Specifies the name of the top cell to restore.
<code>lib</code>	Specifies the name of the OpenAccess library where the design has been saved. When you save a design in OpenAccess format, the library name you specify (along with the path to the library) is saved in a file called <code>libs.def</code> . When you restore a design that was saved in OpenAccess format, the software checks the <code>libs.def</code> file for the library, to find the library's location.
<code>-physical_data</code>	Restores the physical data of the design along with other information.
<code>view</code>	Specifies the name of the cell view to restore.

Examples

- The following command restores the Open Access design called `dma_mac`. The library is called `OAlibrary` and the view is `layout`:

```
tempus > restore_oa_design OAlibrary dma_mac layout -physical_data
```

save_design

```
save_design
[-help]
[-common]
[-keep_partial]
[-libs]
[{{session [-def] [-overwrite ]} | -cellview string}}]
[{-noRc}]
[{-no_timing_graph}]
[-user_path]
```

Saves a snapshot of the design data, including the binary timing data, timing constraints file, power constraints data, netlist, and the timing analysis state. In addition, the `save_design` command saves a Tempus design in the portable mode by default. In this mode, the design and library data do not have any path dependency. As a result, the saved database can be restored from any directory within the same network. The saved database can also be moved to any directory within the same network without any change.

The software saves all library data (`.lib`, `.cdB`, and LEF files) as absolute paths in the configuration file, so that the saved design can be read from any directory.

The command also writes out a power constraints file called `topcellname_power_constraints.tcl` which can be sourced into other Voltus/Tempus sessions.

The `save_design` command does not save physical data (placement, routing, and floorplan information). If the software is in physical mode, the `save_design` command copies the physical data (floorplan, placement, routing and DEF) from a previously read database in the session being saved, to make the saved session complete.

Note: By default, the `save_design` command cannot write out any modified physical data (e.g., from reading DEF files or from any commands used to modify the physical data), unless you include the Innovus license at startup. You can add one of the possible Innovus licenses by using the `tempus -lic_startup_extra` option when Tempus is started.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-cellview</code> <i>string</i>	Saves an OpenAccess design for the Tempus physical flow.
<code>-common</code>	Saves the common-db design data to the specified directory.

<code>-def</code>	Saves the design in the DEF file. To use this option, you need to specify the following parameter: <pre>tempus -lic_startup_extra invs</pre>
<code>-keep_partial</code>	Keeps the partially saved database for debugging purposes only, because restoring this database is not recommended. If the <code>save_design</code> command does not complete successfully, the session data will be removed by default.
<code>-libs</code>	Normally external file references, such as references to <code>.lib</code>, are saved using symbolic links to the original files inside a <code>libs</code> sub-directory inside the DB directory. This option forces copies of the files, rather than symbolic links, in the <code>libs</code> sub-directory so there are no external file references. This makes the DB directory much larger, but it is suitable for sending to a different network where the library and external data files are not available.
<code>-no_timing_graph</code>	Skips the saving of the timing graph by default. This saves time by preventing the timing update to occur in the case of designs where no timing graph is generated for the session.
<code>-overwrite</code>	Overwrites any previously saved session that has the same <code>session_name.dat</code> with the new session. By default, the software checks if the <code>session_name.dat</code> exists in the design. If it does, the software generates an error and does not execute the <code>save_design</code> command. If you specify the <code>-overwrite</code> parameter, the software removes the previously saved session, and then saves the new one.
<code>-noRc</code>	Does not allow RC parasitic information to be saved. You can use this option when input is already <code>rcdb</code> and you do not want to replicate it in the saved session.
<code>session</code>	Specifies the user-defined name of the current Tempus session. The associated data directory is <code>session_name.dat</code>.
<code>-user_path</code>	By default all external file references (such as. library files) are stored as absolute paths without any symbolic links. So, the DB directory is portable anywhere on the same network. This option stores external file references exactly as you entered it For example, full paths like <code>/x/y</code> or relative paths like <code>../x/y</code>, <code>./x/y</code>, and keeps any symbolic links used inside the path names. Example: Consider that there are two library files <code>a.lib</code> and <code>b.lib</code> under:


```
/global/project1/libs/mmmc
```

And two lef files c.lef and d.lef under:

```
/global/project1/libs/lef
```

There are two symlinks to the libs:

```
/usa/db1/libs1.0/mmmc/a.lib → /global/project1/libs/mmmc/a.lib
```

```
/usa/libs/mmmc/b.lib → /global/project1/libs/mmmc/b.lib
```

And two symlinks to the lefs:

```
/usa/db1/libs1.0/lef/c.lef → /global/project1/libs/lef/c.lef
```

```
/usa/libs/lef/d.lef → /global/project1/libs/lef/d.lef
```

Your current work dir is /usa/db1.1/. You have an init script like:

```
set init_mmmc_files ./mmmc.tcl
set init_lef_file {/usa/db1/libs1.0/lef/c.lef ../libs/lef/d.lef}
set init_verilog design.v
init_design
```

Where mmmc.tcl has:

```
create_library_set -name test -timing {/usa/db1/libs1.0/mmmc/a.lib
../libs/mmmc/b.lib }
```

Then after saveDesign, **without -user_path**, the libs and lefs will save link to the real files:

```
> saveDesign db1.2
```

The ./mmmc.tcl file is always stored inside the DB dir, so no external reference is used.

By default, external file references are resolved to their fullpath name without any symlinks.

The .lib files inside the mmmc.tcl file will be referenced

as: {/global/project1/libs/mmmc/a.lib
/global/project1/libs/mmmc/b.lib}

The .lef files will be referenced as: {/global/project1/libs/lef/c.lef
/global/project1/libs/lef/d.lef}

This is portable and can be loaded by restoreDesign anywhere on the same

network.

```
/usa/db1.2/libs1.2/mmmc/a.lib    →    /global/project1/libs/mmmc/a.lib
/usa/db1.2/libs1.2/mmmc/b.lib    →    /global/project1/libs/mmmc/b.lib
/usa/db1.2/libs1.2/lef/c.lef     →    /global/project1/libs/lef/c.lef
/usa/db1.2/libs1.2/lef/d.lef     →    /global/project1/libs/lef/d.lef
```

With -user_path, the libs and lefs keep the symlinks of the current db:

```
> saveDesign db1.2 -user_path
```

The .lib files will be referenced as entered by the

user: {/usa/db1/libs1.0/mmmc/a.lib ../libs/mmmc/b.lib}

The .lef files will be referenced as entered by the user: {/usa/db1/libs1.0/lef/c.lef
../libs/lef/d.lef}

So `restoreDesign` must be run in the same directory as `saveDesign` was run, or in a directory with the same ./y and /x directory structure.

```
/usa/db1.2/libs1.2/mmmc/a.lib    →    /usa/db1/libs1.0/mmmc/a.lib
/usa/db1.2/libs1.2/mmmc/b.lib    →    ../../../libs/mmmc/b.lib
/usa/db1.2/libs1.2/lef/c.lef     →    /usa/db1/libs1.0/lef/c.lef
/usa/db1.2/libs1.2/lef/d.lef     →    ../../../libs/lef/d.lef
```

Examples

Here are some examples.

- The following command saves the design - the database is saved as “test.dat”:

```
tempus 3> save_design test
```

```
#- Begin Save netlist data ... (date=08/21 09:34:28, mem=542.3M)
```

```
Writing Binary DB to test.dat/top.v.bin ...
```

```
#- End Save netlist data ... (date=08/21 09:34:28, total cpu=0:00:00.0, real=0:00:00.0,
peak res=542.8M, current mem=542.8M)
```

```
Parasitic Data of the design is not saved. Design has not been extracted yet.
```

```
#- Begin Save AAE data ... (date=08/21 09:34:28, mem=542.8M)
```

```
Saving AAE Data ...
```

```
AAE_INFO: Saving delay data on disk.
```

```
#- End Save AAE data ... (date=08/21 09:34:29, total cpu=0:00:00.1, real=0:00:01.0, peak
res=543.7M, current mem=543.7M)
```

```
Saving preference file test.dat/gui.pref.tcl ...
```

```
Saving mode setting ...
```

```
Saving global file ...
```

- The following command overwrites an already existing filename:

```
tempus 4> save_design -overwrite test
```

- The following command saves the design without saving the timing graph information:

```
tempus 23> save_design -no_timing_graph test2
```

```
#- Begin Save netlist data ... (date=08/21 09:44:04, mem=557.6M)
```

```
Writing Binary DB to test2.dat/top.v.bin ...
```

```
#- End Save netlist data ... (date=08/21 09:44:04, total cpu=0:00:00.0, real=0:00:00.0,  
peak res=557.7M, current mem=557.7M)
```

```
Parasitic Data of the design is not saved. Design has not been extracted yet.
```

```
#- Begin Save AAE data ... (date=08/21 09:44:04, mem=557.7M)
```

```
Saving AAE Data ...
```

```
#- End Save AAE data ... (date=08/21 09:44:04, total cpu=0:00:00.0, real=0:00:00.0, peak  
res=557.7M, current mem=557.7M)
```

```
Saving preference file test2.dat/gui.pref.tcl ...
```

```
Saving mode setting ...
```

```
Saving global file ...
```

The information related to timing graph is not saved as a part of the saved database. That is, the “AAE/” data will not be saved in the test2.dat/ directory.

Related Commands

- [read_design](#)
- [check_design](#)
- [free_design](#)

set_cell_power_domain

```
set_cell_power_domain
[-help]
[-reset]
[-file file_name]
```

Defines domains for multi-VDD and multi-VSS cells. When reading an instance ASCII power file to perform power-rail analysis, if there are missing power values associated with ground pins, the software uses the information specified with the `set_cell_power_domain` command to derive domain association and distributes power to each ground net accordingly.

This domain information is stored in a side file that is used by power analysis to calculate power using the correct voltage for each domain. If you are specifying an external instance ASCII power file with pin information, this domain information is used to correctly distribute power to each pin.

Use this command before performing power and rail analysis.

Parameters

-help	Prints out the command usage.
-file <i>file_name</i>	Specifies the name of the file containing the cell names and corresponding power-ground pairs. The format of the file is: CELL: <cell name1> <power1> <gnd1> <power2> <gnd2> CELL: <cell name2> <power1> <gnd1> ...
-help	Outputs a brief description for the <code>set_cell_power_domain</code> parameter. For a detailed description of the command, use the <code>man</code> command: <code>man set_cell_power_domain</code> .
-reset	Resets all specified options back to default values.

Examples

- The following command defines two domains, VDD:VSS at 1v and VDDm:VSSm at 0.84v, of the cell mycell in the file domain_file:
tempus > set_cell_power_domain -file domain_file

Related Commands

- `report_power_domain`

set_design_mode

```
set_design_mode
[-help]
[-backsideBottomRoutingLayer layer]
[-backsideTopRoutingLayer layer]
[-bottomRoutingLayer layer]
[-congEffort {low|medium|high|auto}]
[-reset]
[-compressedPGDB {true|false}]
[-earlyClockFlow {true | false}]
[-idealHoldFixing {true|false}]
[-merge_trim_shapes
{{layers layer_name_list gap gap_value max_merge_num number_of_max_merge masks
{mask_num ...}} ...}]
[-node {N12|N10|N7|N7Plus|N6|N5|N3|S11|S10|S8|S7|S5|S4|S3|G7|G5|ICF|I7|C12|C7|unspecified}]
[-optimizationDensityScreenMargin margin]
[-topRoutingLayer layer]
[-trim_grid_group group_name]
```

Specifies the process technology value. You can use this command to change the process technology dependant default settings globally for each application instead of setting several mode options.

With advanced technologies and new design, manufacturing techniques accurate delay calculation becomes more challenging. The main challenges are:

- Delay and slew interpretation of distorted waveforms
- Waveforms are affected by:
 - Distributed RC
 - Back-miller current
 - Crosstalk and input noise
 - Different non-linear characteristics of gates
- Ultra low voltage applications

Tempus STA for advanced nodes has several capabilities for signoff timing accuracy at the respective nodes. The `set_design_mode` command automatically enables the node specific signoff STA setting required to meet the signoff STA accuracy challenges at the specific nodes.

When you specify a process technology value using the `set_design_mode` command, the Tempus software assigns coupling capacitance threshold values to the RC extraction filters automatically. These values determine whether the coupling capacitances of the nets in a design will be lumped to the ground.

The following table lists the coupling capacitance threshold values that are applied based on the

specified process node:

Process Node	Extraction Filters		
	coupling_c_th	relative_c_th	total_c_th
Above 130nm	3.0000	0.03	5.0000
130nm and below	0.4000	1	0.0000
90nm and below	0.2000	1	0.0000
65nm and below	0.1000	1	0.0000
45nm and below	0.1000	1	0.0000

The filtering values set by the Tempus software, which are based on the process node setting, attempt to capture the most significant effect of coupling capacitances on SI analysis, and are strongly recommended. However, you can choose to specify different filtering thresholds using the `-coupling_c_th`, `-relative_c_th`, and `-total_c_th` parameters of the `set_extract_rc_mode` command.

You can use the `get_design_mode` command to return the current settings for the `set_design_mode` command.

Parameters

<code>-help</code>	Outputs a brief description that includes type and default information for each <code>set_design_mode</code> parameter. For a detailed description of the command and its parameters, use the <code>man</code> command: <code>man set_design_mode</code>.
<code>-backsideBottomRoutingLayer layer</code>	Specifies the highest backside LEF layer name for global and detail routing. <i>Default:</i> ""
<code>-backsideTopRoutingLayer layer</code>	Specifies the lowest backside LEF layer name for global and detail routing. <i>Default:</i> ""

<code>-bottomRoutingLayer layer</code>	Specifies the lowest lef layer name for global and detail routing. For a special partial routing on the first routing layer in certain nodes, the routing enablement of that layer must be a global decision in the entire flow of the design. For example, if M0 or M1 routing is enabled, you need to set <code>set_design_mode -bottomRoutingLayer M0/1</code> from the init step. This bottom routing layer setting must not be changed in later stages of the flow. <i>Default:</i> ""
<code>-compressedPGDB {true false}</code>	Specifies whether the compressed PG feature is enabled. When it's true, <code>read_design</code> will automatically compress the PG data when load a design. You can use <code>"dbGet top.isPGCompressed"</code> to check if PG in the design is compressed. <i>Default:</i> false

<pre>-congEffort {low medium high auto}</pre>	Specifies the congestion effort level. It controls placement as well the place_opt_design behavior. It is also possible to have other tuning for other steps in the flow. <i>Default:</i> <code>auto</code> Specify one of the following values: • <code>auto</code>: Automatically determines whether the design is congested and performs extra congestion driven effort for highly congested designs. • <code>high</code>: Runs more iterations of placement in an effort to achieve better congestion results. This parameter increases the run time. This also does congestion-aware buffering. • <code>low</code>: Runs fewer iterations of placement to arrive quickly at a legal placement. This parameter might decrease placement quality. • <code>medium</code>: Runs placement on designs at a normal effort level.
<pre>-earlyClockFlow {true false}</pre>	Enables early clock flow. <i>Default:</i> <code>false</code>
<pre>-idealHoldFixing {true false}</pre>	Enables the ideal hold fixing flow. <i>Default:</i> <code>false</code>
<pre>-merge_trim_shapes {{layers layer_name_list gap gap_value max_merge_num number_of_max_merge masks {mask_num ...}} ...}</pre>	

	Specifies the setting for trim shapes merge. <code>layers</code>: You use this to specify the trim layer list which needs to merge trim shapes. <code>Gap</code>: You use this setting when the end to end gap is less the value. <code>max_merge_num</code>: You use this to specify the max trim shapes number that are merged. <code>masks</code>: You use this to specify the trim shapes color that needs to be merged. <i>Default:</i> ""
<code>-node</code> <code>{N12 N10 N7 N7Plus N6 N5 N3 S11 S10 S8 S7 S5 S4 S3 G7 G5 ICF I7 C12 C7 unspecified}</code>	Sets the design technology node. <i>Default:</i> unspecified
<code>-optimizationDensityScreenMargin</code> <i>margin</i>	Increases margin on the top of original density screen for optimization. <i>Default:</i> 0 <i>Min:</i> 0 <i>Max:</i> 1
<code>-reset</code>	Resets parameters to their default values. The <code>-reset</code> parameter <i>must</i> be the first parameter specified. If you specify <code>-reset</code> by itself, the software resets all <code>set_design_mode</code> parameters to their default values.
<code>-topRoutingLayer</code> <i>layer</i>	Specifies the highest lef layer name for global and detail routing. <i>Default:</i> ""
<code>-trim_grid_group</code> <i>group_name</i>	Specifies the GROUP name of the set from which the trim metal grid will be used for placement and routing. <i>Default:</i> "" If the specified group name does not exist in LEF, the tool will return an ERROR.

Examples

Here are some examples.

- The following command sets the specified process technology value to 80:

```
tempus > set_design_mode -process 80
```

- The following command resets the `-process` parameter to its default value:

```
tempus > set_design_mode -reset -process
```

- The following command also resets the `-process` parameter to its default value:

```
tempus > set_design_mode -reset
```

Related Commands

- `get_design_mode`

set_import_mode

```
set_import_mode
[-help]
[-reset]
[-discardFloatingVNets {true | false}]
[-keepEmptyModule {true | false}]
[-syncRelativePath {true | false}]
```

Controls aspects of how the software reads in the Verilog netlist and other design files.

Parameters

-help	Outputs a brief description for the <code>set_import_mode</code> parameter. For a detailed description of the command, use the <code>man</code> command: <code>man set_import_mode</code>.
-discardFloatingVNets {true false}	Discards floating vnets. <i>Default:</i> <code>true</code>
-keepEmptyModule {true false}	Keeps empty modules. If set to <code>false</code>, the software converts the empty modules into special leaf cells that do not have a physical library associated with them. If the module is a hard macro, using a value of <code>true</code> or <code>false</code> produces identical behavior, that is, the software keeps the module. <i>Default:</i> <code>true</code>
-reset	Resets all or specified options to default value.
-syncRelativePath {true false}	Synchronizes all relative paths in the configuration file to the current working directory. You can load this configuration file from any directory without first changing your current working directory to the previous working directory where the configuration file was saved. <i>Default:</i> <code>false</code> When set to <code>true</code>, you do not need to change your working directory to the directory saved in the previous design session before you load the configuration file.

Examples

- The following command does not synchronize the relative paths in the configuration file to the current working directory:

```
tempus > set_import_mode -syncRelativePath false
```

set_license_check

```
set_license_check  
[-help]  
[-optionList {license1 license2 ...}]  
[-wait time_in_minutes]  
[-default]  
[-checkout license]  
[-status]  
[-existOnServer]
```

Manages licenses for product options. Specifies licenses to check out, license wait time, and the status of all product option licenses.

Note: This command does not manage multi-CPU licenses. For information on the commands that manage multi-CPU licenses, see ["Multiple-CPU Processing Commands"](#).

Parameters

<code>-checkout license</code>	Checks out the licenses for the specified product options when this command runs.
<code>-default</code>	Restores the default. values for this command.
<code>-existOnServer</code>	Outputs a table that provides the following information for each of the licenses that exist on the license server. • License name • Product number • Product name • License string • Version
<code>-help</code>	Outputs a brief description that includes type and default information for each <code>set_license_check</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man set_license_check</code>.
<code>-optionList {license1 license2 ...}</code>	Specifies licenses to check out dynamically. Dynamic licenses are not checked out until they are needed. If you specify a license that is not allowed with your base license, or a license that is not available, the software does not check out that license and instead issues a warning message. If you specify more than one license, begin and end the option list with double quotation marks or braces.
<code>-status</code>	Outputs the following information: • The name of the base license that is checked out • The current wait time • A list of the allowed license options • A list of the available license options • A list of the currently checked-out license options.
<code>-wait time_in_minutes</code>	Specifies the amount of time the system waits for a license to become available. If the license is available in less than the specified wait time, the system checks out the next needed license without waiting. <i>Default:</i> 0 (no wait time) <i>Value range:</i> 0 to 10,000

Examples

Here are some examples.

- The following command outputs a table that provides information about the licenses that exist on the license server:

```
tempus > set_license_check -existOnServer
```

The system outputs the following information:

Name	Prod #	Product Name	License string	Version
tpstso	TPS300	TPS-TSO	Tempus_Timing_Signoff_TSO	13.2
tpsng	TPSENG100	TPS-NG	TPS_NG100	1.0000
vtsexl	VTS200	VTS-XL	Voltus_Power_Integrity_XL	13.2
vtsl	VTS100	VTS-L	Voltus_Power_Integrity_L	13.2
vtjaa	VTS201	VTS-AA	Voltus_Power_Integrity_AA	13.2

- The following command specifies that the license for Tempus Timing Signoff_L is checked out before Tempus Timing Signoff_XL license.

```
tempus > set_license_check -optionList {tpsl tpsxl}
```

- The following command outputs license status and wait time and lists the optional licenses that can be checked out and the licenses that are checked out:

```
tempus > set_license_check -status
```

The system outputs the following information:

```
Current checked out base license: tpsxl(Tempus_Timing_Signoff_XL 13.2).
```

```
When checking-out an optional license, wait for up to: 0 minutes.
```

```
The base license allows the following optional licenses: tpstso tpsng vtsexl vtsl vtjaa
```

```
The automatic optional license check-out is limited by -optionList to: tpstso tpsng  
vtsexl vtsl vtjaa
```

```
Current checked out optional licenses : none
```

```
Summary of all checked out licenses:
```

```
Tempus_Timing_Signoff_XL
```

- The following command sets the wait time to 5 minutes:

```
tempus > set_license_check -wait 5
```


Related Topics

- "[Product and Licensing Information](#)" chapter in the *Tempus User Guide*

set_top_module

```
set_top_module
[-help]
<topCell>
[-ignore_undefined_cell]
[-ignore_timing_library_check]
```

Sets the top module name, and checks for consistency between Verilog netlist and timing libraries. Tempus uses the top module for timing analysis. You must read the libraries (`read_lib`) and Verilog file (`read_verilog`) before using this command. Use this command before reading constraints, parasitic and delay data.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-ignore_timing_library_check</code>	Ignores the check that each cell have a timing library when reading the verilog file.
<code>-ignore_undefined_cell</code>	Creates a user-defined noise model (UDN) for a cell even if the corresponding cdB and .lib files are not available. When you set this option, the software creates a dummy UDN for cells that are not in the loaded cdB or .lib, and uses them during timing and signal integrity analysis.
<code><topCell></code>	Specifies the top cell.

Examples

- The following commands sets the top module to `dma_mac`.

```
tempus > set_top_module dma_mac
```

Related Commands

- `read_lib`
- `read_verilog`

write_def

```
write_def
[-help]
def_file
[-all_layers]
[-bump_as_pin]
[-output_mask_layers string]
[-routing]
[-scan_chain]
[-unit integer]
[-wrong_way_routing_to_specialnet | -no_special_net]
```

Writes the specified information to a DEF file. By default, the `write_def` command writes floorplan, and all placed and unplaced standard cell information to the DEF file.

The `write_def` command does not support multiple groups per region. If you run Trial Route before you run the `write_def` command, the software does not write information about `FIXED` status wires to the DEF file. The `write_def` command supports rectilinear die area.

You can use the `write_def` command after importing a design.

Note: The `write_def` command cannot be used unless you include the Innovus license at startup. You can add one of the possible Innovus licenses by using the `tempus -lic_startup_extra` option when Tempus is started.

Parameters

-help	Prints a brief description that includes the type and default information for each <code>write_def</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man write_def.</code>
-all_layers	Writes out wires, vias, and shapes on non-routing layers to the SPECIALNETS section. This includes the LEF MASTERSLICE layers, IMPLANT layers, and ROUTING or CUT layers with sub-types MIMCAP or PASSIVATION.

<code>-bump_as_pin</code>	Writes bump cells as top-level pins in the <code>PINS</code> section. If a PORT in a block has a PASSIVATION layer, the tool treats it as an embedded bump. The <code>-bump_as_pin</code> parameter outputs all the layer information for embedded bumps in the PIN section as for normal bumps.
<code>def_file</code>	Specifies the DEF output file.
<code>-no_special_net</code>	Excludes special net information from the DEF file.
<code>-output_mask_layers string</code>	Specifies the output MASK data on the specified layers only. The valid values for this option are "metal", "cut", or a list of layers separated by a space character in a form of either a single layer or a range of layers. For example, <code>-output_mask_layers {Metal1:Metal3 Via23}</code>
<code>-routing</code>	Writes the routing information to the <code>NETS</code> section of the DEF file. If the <code>-routing</code> option is not specified, both power and signal vias are not written to the DEF file.
<code>-scan_chain</code>	Writes scan chain information to the DEF file.
<code>-unit integer</code>	Defines the units for the DEF file.
<code>-wrongway_routing_as_specialroute</code>	Writes wrong-way routing in the SPECIALNETS section. This option duplicates any wrong-way routing segments that require extra-width into the SPECIALNETS routing section as a RECT with +SHAPE DRCFILL.

Examples

- The following command writes bump cells as top-level pins in the PINS section of the `test.def` file:

```
write_def -bump_as_pin test.def
```

The PINS section in the `test.def` file is as follows:

```
+++++
PINS 102 ;

- iot_schm_2m16m_3v + NET iot_schm_2m16m_3v + DIRECTION OUTPUT + USE SIGNAL
+ SPECIAL + POLYGON ap ( 93300 65970 ) ( 65970 65970 )
( 65970 93300 ) ( 27330 93300 ) ( 27330 65970 ) ( 0 65970 )
( 0 27330 ) ( 27330 27330 ) ( 27330 0 ) ( 65970 0 )
( 65970 27330 ) ( 93300 27330 ) + FIXED ( 386690 2658905 ) N ;

- tck + NET tck + DIRECTION INPUT + USE SIGNAL
+ SPECIAL + POLYGON ap ( 93300 65970 ) ( 65970 65970 )
( 65970 93300 ) ( 27330 93300 ) ( 27330 65970 ) ( 0 65970 )
( 0 27330 ) ( 27330 27330 ) ( 27330 0 ) ( 65970 0 )
( 65970 27330 ) ( 93300 27330 ) + FIXED ( 53350 2658905 ) N ;
+++++
```

write_rcdb

```
write_rcdb
[-help]
<dirname>
```

Saves the parasitic data that you specify in a Tempus session. Use the `read_rcdb` command to restore the parasitic data.

Parameters

<code>-help</code>	Prints out the command usage.
<code>dirname</code>	Specifies the name of the parasitic file. The parasitic file must have the <code>.rcdb.d</code> extension.

Examples

- The following example writes the parasitic data to a directory called `myparasitics.rcdb`:
`tempus > write_rcdb myparasitics.rcdb.d`

Related Commands

- `read_rcdb`

Multiple-CPU Processing Commands

- [check_multi_cpu_usage](#)
- [get_multi_cpu_usage](#)
- [run_threaded_commands](#)
- [set_multi_cpu_usage](#)

check_multi_cpu_usage

check_multi_cpu_usage

Checks if distributed processing can work in the software environment. This command also ensures that the working directory is accessible from the master machine as well as the Tempus clients. The `check_multi_cpu_usage` command is specified after configuring the multi-CPU settings (`set_distribute_host` and `set_multi_cpu_usage`) to check if all the specified CPUs can be accessed. This command should be invoked before running any commands that use multiple CPUs.

Parameters

None

Examples

Here are some examples.

- The following example command script checks the distributed computing environment when `set_distributed_hosts -local` is specified:

```
tempus > check_multi_cpu_usage
```

```
/it/sjclapa06p6_scr02/clre/exp_2017_05_30/0223/dy/2014/2017/0413
/it/sjclapa06p6_scr02/clre/exp_2017_05_30/0223/dy/2014/2017/0413
/it/sjclapa06p6_scr02/clre/exp_2017_05_30/0223/dy/2014/2017/0413
/it/sjclapa06p6_scr02/clre/exp_2017_05_30/0223/dy/2014/2017/0413
Connected to sjfnl624 35779 0 ( PID=13350 )
Connected to sjfnl624 54715 1 ( PID=13356 )
Connected to sjfnl624 40469 3 ( PID=13397 )
Connected to sjfnl624 50442 2 ( PID=13378 )
Task 0 successfully accessed the current working directory.
Task 3 successfully accessed the current working directory.
Task 1 successfully accessed the current working directory.
Task 2 successfully accessed the current working directory.
Task 3: CPU(workload/CPU): 7.84/12 mem(free/total): 94118680k/98813572k.
Task 1: CPU(workload/CPU): 7.84/12 mem(free/total): 94118432k/98813572k.
Task 2: CPU(workload/CPU): 7.84/12 mem(free/total): 94117068k/98813572k.
Task 0: CPU(workload/CPU): 7.84/12 mem(free/total): 94119084k/98813572k.
Test heartbeat for task 0...
The distributed processing environment has been set up correctly.
```


- The following example command script checks the distributed computing environment when when `set_distributed_hosts -lsf -queue lnx64` is specified:

```
tempus > check_multi_cpu_usage
```

```
Job <5685326> is submitted to queue <lnx64>. The associated lsf task ID is '0'
Executing
/it/sjclapa06p6_scr02/clre/exp_2017_05_30/0223/dy/2014/2017/0413/.clre_sub_22662_0
Job <5685328> is submitted to queue <lnx64>. The associated lsf task ID is '1'
Executing
/it/sjclapa06p6_scr02/clre/exp_2017_05_30/0223/dy/2014/2017/0413/.clre_sub_22662_1
Job <5685329> is submitted to queue <lnx64>. The associated lsf task ID is '2'
Executing
/it/sjclapa06p6_scr02/clre/exp_2017_05_30/0223/dy/2014/2017/0413/.clre_sub_22662_2
Job <5685330> is submitted to queue <lnx64>. The associated lsf task ID is '3'
Executing
/it/sjclapa06p6_scr02/clre/exp_2017_05_30/0223/dy/2014/2017/0413/.clre_sub_22662_3
Connected to sjfnl625 43039 1 ( PID=30833 )
Connected to sjfnl625 34830 2 ( PID=30832 )
Connected to sjfnl625 33887 3 ( PID=30831 )
Connected to sjfib046 60125 0 ( PID=32727 )
Task 1 successfully accessed the current working directory.
Task 3 successfully accessed the current working directory.
Task 2 successfully accessed the current working directory.
Task 0 successfully accessed the current working directory.
Task 3: CPU(workload/CPU): 1.43/12 mem(free/total): 74177208k/82277684k.
Task 2: CPU(workload/CPU): 1.43/12 mem(free/total): 74177208k/82277684k.
Task 1: CPU(workload/CPU): 1.43/12 mem(free/total): 74177208k/82277684k.
Task 0: CPU(workload/CPU): 19.79/20 mem(free/total): 352917068k/396858936k.
Test heartbeat for task 0...
The distributed processing environment has been set up correctly.
```

Related Topics

- ["Appendix - Multi-CPU Usage"](#) in the *Tempus User Guide*
 - ["Checking the Distributed Computing Environment"](#)

Related Commands

- `set_multi_cpu_usage`
- `get_multi_cpu_usage`

get_multi_cpu_usage

```
get_multi_cpu_usage  
[-help]  
[-autoPageFaultMonitor]  
[-cpuPerRemoteHost]  
[-keepLicense]  
[-localCpu]  
[-remoteHost]  
[-threadInfo]  
[-verbose]
```

Displays the requested number of threads or hosts.

To request threads or hosts, use the [set_multi_cpu_usage](#) command.

Parameters

-help	Prints out the command usage.
-autoPageFaultMonitor	Displays the warning rank set for performance issues caused by major page faults
-cpuPerRemoteHost	Displays the number of CPUs available on every remote host for Superthreading.
-keepLicense	Displays the current setting for the set_multi_cpu_usage -keepLicense parameter.
-localCpu	Displays the number of available local CPUs for multi-threading.
-remoteHost	Displays the number of available remote hosts for distributed processing or Superthreading.
-threadInfo	Displays the level of informational messages for each thread.
-verbose	Displays a message with the current multiple-CPU setting.

Examples

Here are some examples.

- The following command displays the requested number of threads for multi-threading:

```
tempus > get_multi_cpu_usage -localCpu
```

- The following command displays the requested number of remote hosts for Superthreading or distributed processing:

```
tempus > get_multi_cpu_usage -remoteHost
```

- The following command displays the requested number of CPUs per remote hosts for Superthreading:

```
tempus > get_multi_cpu_usage -cpuPerRemoteHost
```

Related Topics

- ["Appendix - Multi-CPU Usage"](#) in the *Tempus User Guide*
 - ["Running Distributed Processing"](#)

Related Commands

- [set_multi_cpu_usage](#)
- [check_multi_cpu_usage](#)

run_threaded_commands

```
run_threaded_commands
{
command_1 > file_1
command_2 > file_2
command_3 > file_3
...
}
```

Specifies commands to run in multi-thread mode and executes them in parallel. You can use multi-thread mode for simultaneous setup and hold analysis. Multi-thread mode does use more memory than single-thread mode, but it improves the overall performance.

You can use the `run_threaded_commands` command in distributed MMMC mode to perform super threading.

The `run_threaded_commands` command should only be used when a command is not internally multi-threaded. This is because when this command is run threading takes place at the command level and each command runs with a single thread. So even if a command is multi-threaded internally, it does not utilize more than one thread.

Note: Multiple threads writing to the same log file may cause run time degradation. It is recommended to use the `redirect` command to redirect the command output directly to the respective files in order to reduce the I/O cost. For example,

```
run_threaded_commands {
redirect {source generate_reports.tcl} > logfile_1
redirect {command2 } > logfile_2
.....
}
```

You can redirect the commands inside `run_threaded_commands` to a distinct file, with some exceptions.

Note: You do not need to redirect the following commands to a file because you can specify a file for the output:

- `write_global_slack_report`
- `write_sdc`
- `source`
- `redirect`

Each command inside the `run_threaded_commands` consumes one distributed license. To avoid consuming extra licenses, the software does not pass commented commands to the thread queue. If you specify ten commands in the list, and only four CPUs are available, then the system uses the first four commands for parallel processing. The fifth command is run as soon as any one of the four commands are executed. The software does not wait for all the four commands/jobs to be finish before

scheduling the next four commands/jobs.

Note: The `run_threaded_commands` expects timing to be updated before the commands are launched in parallel. If timing is not updated, then `run_threaded_commands` will internally update the timer.

You can specify one or more of the following timing analysis reporting commands to run in multi-thread mode:

- `check_timing`
- `report_analysis_coverage`
- `report_annotated_check`
- `report_annotated_delay`
- `report_case_analysis`
- `report_cell_instance_timing`
- `report_clock_timing`
- `report_constraint`
- `report_cppr`
- `report_critical_instance`
- `report_inactive_arcs`
- `report_min_pulse_width`
- `report_mode`
- `report_path`
- `report_path_exceptions`
- `report_path_groups`
- `report_ports`
- `report_slack_histogram`
- `report_timing`
- `report_timing_lib`
- `write_global_slack_report`
- `write_sdc`
- `write_sdf`
- `write_timing_windows`

Include this command with caution, as it incurs a penalty on memory.

Include this command with caution, as it incurs a penalty on memory.

In addition, you can also include the following commands:

- `source`
- `redirect`

Note: The `run_threaded_commands` command is recommended for multi-threaded reporting commands. However, this command should be used only on commands that are time consuming.

Command Order

- Specify the following global variable before running `run_threaded_commands`:
`set_global timing_enable_simultaneous_setup_hold_mode true`
- Invoke the `set_multi_cpu_usage` command before `run_threaded_commands`.
- Invoke `run_threaded_commands` before any of the commands to run in multi-thread mode.

Parameters

None

Examples

- The following command will run the first 3 commands in parallel since the number of CPUs is set to 3. The fourth command will launch as soon as any one of the first 3 commands/jobs finish.

```
tempus > set_multi_cpu_usage -localCpu 3
tempus > run_threaded_commands {
check_timing > check_timing.rpt
redirect "report_analysis_coverage" > report_analysis_coverage.rpt
write_sdf base.sdf
write_sdc base.sdc
report_constraint -all_violators -nosplit > report_constraint_all_viol.rpt.gz
report_constraint -max_delay -nosplit >
report_constraint_max_delay.rpt.gz
report_constraint -min_delay -nosplit > report_constraint_min_delay.rpt.gz
source generate_reports.tcl
}
```

Related Commands

- `report_timing`
- `set_multi_cpu_usage`
- `redirect`

set_multi_cpu_usage

```
set_multi_cpu_usage
[-help]
[-acquireLicense integer]
[-keepLicense {true | false}]
[-licenseList {string}]
[-localCpu string]
[-remoteHost integer]
[-cpuPerRemoteHost integer]
[-releaseLicense]
[-threadInfo {0 | 1 | 2}]
[-verbose]
[-autoPageFaultMonitor {0 | 1 | 2 | 3}]
[-reset]
```

Specifies the number of threads to use for multi-threading, or the maximum number of computers to use for distributed processing, or the maximum number of computers and the number of threads to use for Superthreading. Optionally, reports usage information.

 This command is required for multi-threading, distributed processing, and Superthreading.

The `set_multi_cpu_usage` command enables multi-threaded timing reporting under the following conditions:

- `timing_report_group_based_mode` global variable is set to `true`. If there are multiple paths with the same slack then the selection can change on multiple runs. This may result in some changes in the `report_timing` output.
- `report_analysis_coverage` command is in non-verbose mode.
- `check_timing -include clock_crossing` option is specified.

Parameters

<code>-help</code>	Prints out the command usage.
--------------------	-------------------------------

<code>-acquireLicense integer</code>	Acquires licenses to enable the specified number of CPUs. For example, if you specify <code>-acquireLicense 5</code>, the software checks out enough licenses to enable 5 CPUs. After the licenses are checked out, they can be used for multi-threading, distributed processing, or superthreading.
<code>- autoPageFaultMonitor {0 1 2 3}</code>	Specifies to set a warning rank for performance issues caused by major page faults. The parameter supports the following values: • 0 - specifies to disable the reporting of major page faults. • 1 - specifies to set the warning rank as 1. This means $PF/total\ PF > 0.8$, $b > 10$, and $wa > 85\%$ • 2 - specifies to set the warning rank as 2. This means $PF/total\ PF > 0.5$, $b > 5$, and $wa > 65\%$ • 3 - specifies to set the warning rank as 3. This means $PF/total\ PF > 0.2$, $b > 1$, and $wa > 40\%$ where PF is the number of major page faults from the software, total PF is the total number of major page faults in the machine, b is the number of blocked processors, and wa is CPU waiting time.
<code>-cpuPerRemoteHost integer</code>	Specifies the number of CPUs used by clients for multi-threading. Note: For super threading, you must use this parameter in conjunction with the <code>-remoteHost</code> parameter. <i>Default:</i> 1
<code>-keepLicense {true false}</code>	Specifies whether to keep the acquired multiple CPU-licenses until the current session ends. Specify this parameter before running any commands that require multiple-CPU applications. To release all multiple-CPU licenses immediately, use the <code>-releaseLicense</code> option. <i>Default:</i> true
<code>-licenseList {tpsm}</code>	Specifies the list or order of licenses the software should use for checking out licenses during multiple-cpu processing. The parameter does not support an empty license list <code>{}</code>. This parameter allows the <code>tpsm</code> option.

<code>-localCpu</code> <i>string</i>	Specifies the number of CPUs on a local machine. When the <code>-localcpu</code> parameter is set to 1 (default), the multi-threaded timing propagation mode is enabled. Also, both CPU multi-threading and distributed process are enabled. The <code>set_multi_cpu_usage -localCpu</code> command automatically controls the total number of threads used up to the specified number. The number of specified threads cannot be more than the number of CPUs available in the local computer. Consider a computer having 8 CPUs, then the following command will issue a warning that the number of specified local CPUs is greater than the actual number of CPUs available: <pre>set_multi_cpu_usage -localCpu 10</pre> The local CPU count can be the maximum value that is shown as “processors” on the system under “/proc/cpuinfo”. This value can be obtained by running the “checkSysConf” utility in the “tools/bin” folder. <pre>\$> tools/bin/checkSysConf ===== checkSysConf: Version 3.53 ===== OS Version: Red Hat Enterprise Linux Workstation release 7.6 (Maipo) Kernel Version: 3.10.0-957.5.1.el7.x86_64 Hardware Type: x86_64 Memory: 2063415 Megabyte CPU model: Intel(R) Xeon(R) Gold 6254 CPU @ 3.10GHz No. of CPUs: 72 Total Swap Space.....: 131072 Megabyte ===== \$> cat /proc/cpuinfo grep "processor" tail -1 awk -F ":" '{print \$2 + 1}' 72 Default: 1</pre>
<code>-releaseLicense</code>	Releases all multiple-CPU license(s) immediately. By default, the software holds multiple-CPU licenses until the end of the current session. To specify that the software release multiple-CPU licenses after every multiple-CPU command runs, use <code>-keepLicense</code>.
<code>-remoteHost</code> <i>integer</i>	Specifies the number of clients. This parameter is required for distributed processing and super threading. Default: 0
<code>-reset</code>	Resets the specified parameters back to default values.

<pre>-threadInfo {0 1 2}</pre>	Reports usage information. <i>Default: 0</i> Specify one of the following values: 0: Does not write messages to the log file. 1: Writes the final message to the log file. For example, All threaded jobs finished (10 elapsed sec: 0 processor sec (parent), 0 system sec (parent), 0 processor sec (threads), 0 system sec (threads)). 2: Writes additional starting/ending information for each thread. For example, <pre>Starting threaded job 1... Starting threaded job 2... Starting threaded job 3... Ending threaded job 2 (1 elapsed sec, 0 processor sec, 0 system sec, 1.160M). Ending threaded job 3 (2 elapsed sec, 0 processor sec, 0 system sec, 0.895M). Ending threaded job 1 (10 elapsed sec, 0 processor sec, 0 system sec, 1.172M). All threaded jobs finished (10 elapsed sec: 0 processor sec (parent), 0 system sec (parent), 0 processor sec (threads), 0 system sec (threads)).</pre>
<pre>-verbose</pre>	Displays messages when changing the multiple-CPU settings.

Examples

Here are some examples.

- The following command specifies that 8 CPUs can be used on a local machine for multi-threading. This is based on the assumption that 8 CPUs are available on the local machine; otherwise the software will utilize the available CPUs:

```
tempus > set_multi-cpu_usage -localCpu 8
```

- The following command specifies that the number of remote machines to be used is 4 for distributed processing, and also for each of these hosts, 4 CPUs will be used for super threading:

```
tempus > set_multi_cpu_usage -remoteHost 4 -cpuPerRemoteHost 4
```

Here, during DMMMC analysis, in total 16 analysis sessions can be run in parallel.

In addition, multi-CPU configuration settings for distributed processing must be done using the `set_distribute_host` command.

- The following command releases all the multi-CPU licenses immediately:

```
tempus > set_multi_cpu_usage -releaseLicense
```

Related Topics

- ["Appendix - Multi-CPU Usage"](#) in the *Tempus User Guide*
 - ["Running Distributed Processing"](#)

Related Commands

- `get_multi_cpu_usage`
- `check_multi_cpu_usage`

Distributed MMMC Commands and Variables

- [distribute_command](#)
- [distribute_design_views](#)
- [distribute_read_design](#)
- [distribute_views](#)
- [exit_servers](#)
- [get_distribute_host](#)
- [get_distribute_variables](#)
- [merge_timing_reports](#)
- [set_distribute_host](#)
- [set_distribute_mmmc_mode](#)
- [set_distribute_variables](#)

distribute_command

```
distribute_command  
[-help]  
[list_of_commands]  
[-views list_of_views]
```

Distributes interactive analysis commands. When you use this command, multi-mode multi-corner (MMMC) analysis is not run again and the *[list_of_commands]* is run on existing active views.

You can use `distribute_command` when either of the following commands is run:

- `distribute_read_design -restore_design`
- `distribute_views -views {viewnames}`

Parameters

<code>-help</code>	Prints out the command usage.
<code>list_of_commands</code>	Specifies commands to be distributed. The commands that are required to be run only at the Tempus client should be enclosed in {}. You can specify multiple commands in a single <code>distribute_command</code> statement. <pre>tempus > distribute_command { report_timing -through [get_nets net1] set endPin [get_pins u1/a] ; report_timing -to \$endPin }</pre> The commands that are required to take variable value from the master Tempus session should be enclosed in " ". For example, the following commands create two variables <code>testVar1</code> and <code>testVar2</code> on Tempus client with variable values being the same as on master Tempus session. <pre>set testVar1 val1 ; set testVar2 val2 distribute_command "foreach var {testVar1 testVar2} {set \$var \$[subst \$var]} "</pre>
<code>-views list_of_views</code>	Specifies the views to distribute.

Related Commands

- `distribute_read_design`
- `distribute_views`
- `set_multi_cpu_usage`

distribute_design_views

```
distribute_design_views
[-help]
[-client_logfile prefix]
[-create_bm]
[-create_bm_args <string>]
[-load_bm <string>]
[-load_bm_args <string>]
[-save_design {list_of_views | all}]
[-views list_of_views | -restore_dir path]
[-views list_of_views | -restore_design {views | all}]
[-views list_of_views | -restore_cellview {lib | cell}]
[-design_script inputTclCmdFile | -restore_dir path]
[-design_script inputTclCmdFile | -restore_design {views | all}]
[-design_script inputTclCmdFile | -restore_cellview {lib | cell}]
[-restore_design {views | all} -restore_dir path]
[-views list_of_views [-analysis_script list_of_commands]]
[-analysis_script list_of_commands [-views list_of_views]]
[-views list_of_views [-set_analysis_view_log | -combine_num_views integer]]
[-out_dir outputDir [-design_script inputTclCmdFile | -restore_dir path]]
```

Allows the Tempus software to process a view as soon as a client is available.

When this command is used, the software does not have to wait till time out or for all the clients to be available for processing a view. This reduces the turnaround time in situations where machines are not easily available.

Parameters

-help	Prints out the command usage.
-analysis_script <i>list_of_commands</i>	Specifies the file that contains commands for reading in the necessary parasitic files and the reporting commands.
-client_logfile <i>prefix</i>	Specifies the client log filename prefix with the directory.
-combine_num_views <i>integer</i>	Specifies the number of views to combine in a job.
-create_bm	Allows the software to generate boundary models.
-create_bm_args < <i>string</i> >	Specifies the arguments provided to the create_boundary_model command during boundary model generation.

<code>-design_script inputTclCmdFile</code>	Specifies the name of the design and MMMC configuration loading script. The <code>-design_script</code> is mutually exclusive with <code>restore_*</code> options.
<code>-load_bm</code>	Specifies the directory from where boundary model will be loaded.
<code>-load_bm_args <string></code>	Specifies the arguments provided to the <code>read_boundary_model</code> command. For example: <code>distribute_design_views -load_bm {-skip_version_check}</code> When specified, the software skips version checks while loading the boundary model in DMMMC flow.
<code>-out_dir outputDir</code>	Specifies the directory where the run is to be executed.
<code>-restore_cellview {lib cell}</code>	Specifies the name of OA library or cell to be restored.
<code>-restore_design {views all}</code>	Specifies a list of views.
<code>-restore_dir path</code>	Specifies the directory path where the design is to be restored.
<code>-save_design {list_of_views all}</code>	Saves the design session information for the specified (or all) the analysis views.
<code>-set_analysis_view_log</code>	Redirects the <code>set_analysis_view</code> command run log to <code>set_analysis_view_<viewname>.log</code> file.
<code>-views {list_of_views}</code>	Specifies the views to distribute for both setup and hold analysis • The <code>-views</code> parameter is mutually exclusive with <code>restore_*</code> options. • The <code>-views</code> parameter is mandatory when <code>-set_analysis_view_log</code> and <code>-combine_num_views</code> parameters are specified.

Related Commands

- `set_distribute_mmmc_mode`
- `set_multi_cpu_usage`
- `set_distribute_host`

distribute_read_design

```
distribute_read_design  
[-help]  
[-client_logfile prefix]  
-outdir outputDir  
[-restore_cellview {directory cell_name}]  
{-design_script inputTclCmdFile | [-restore_design {views | all} [-restore_dir path]]}
```

Opens up N number of Tempus clients (as set with [set_multi_cpu_usage](#)) for a distributed processing session, and loads the design script on Tempus clients.

By default, the software monitors the health of clients at regular intervals and writes the information about client resources (cpu/memory/tmp) in the host-monitor.log file.

The `distribute_read_design` command is used for distributed multi-mode multi-corner (DMMMC) timing analysis only.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-client_logfile</code> <i>prefix</i>	Specifies the client log filename prefix with directory. The software redirects client logs to a new location. By default, Tempus client logs are created in the directory specified with the <code>-outdir</code> option.
<code>-design_script</code> <i>inputTclCmdFile</i>	Specifies the design loading script. The script contains the common information that needs to be shared by all Tempus client runs. The script must include the design loading commands, as well as all MMMC configuration commands. The script must not include parasitic file loading commands (<code>read_spef</code>), analysis view setting commands (<code>set_analysis_view</code>) or the view specific settings like constraints through <code>set_interactive_constraint_mode</code>. Timing derates settings (<code>set_timing_derate</code>) are corner specific and can be specified in this script.
<code>-outdir</code> <i>outputDir</i>	Specifies the directory where the run is to be executed. All data from the distributed analysis runs will be available in this directory. The software generates view-specific reports in directories with the corresponding view name within this directory.
<code>-</code> <code>restore_cellview</code> { <i>directory</i> <i>cell_name</i> }	Specifies the “ <i>directory</i> ” as a directory name having OpenAccess library name/OA cell and “ <i>cell_name</i> ” refers to the OA cell name.

<code>-restore_design {views all}</code>	Specifies the design views (previously saved using the <code>distribute_views -save_design</code> command) to be restored for both setup and hold analysis. • <code>views</code>: Restores specified design views. • <code>all</code>: Restores all the design views.  The number of sessions specified for restore can not be more than the machines specified with <code>-remoteHost</code> setting.
<code>-restore_dir path</code>	Specifies the directory location which has the saved design views (that were previously saved using the <code>distribute_views -save_design</code> command) to be restored.

Examples

Here are some examples.

- The following example command script lists the commands used to perform distributed multi-mode multi-corner timing analysis:

```
tempus > set_multi_cpu_usage -remoteHost 4
tempus > distribute_read_design \
-design_script /stml/timing/demo_dist_proc/preamble.tcl \
-outdir $output_directory
#preamble.tcl script includes design loading commands and MMMC configuration commands

set Views [list lmin-rmax lmax-rmin lmax-rmax]
tempus > distribute_views \
-views $Views \
-script /stml/timing/demo_dist_proc/commands.tcl
#commands.tcl script contains read_spef and report commands

tempus > exit_servers
```

- The following example command script lists the commands used to save a design in distributed multi-mode multi-corner mode:

```
tempus > set_distribute_host -local | -rsh ...
tempus > set_multicpu_usage -remoteHosts N
tempus > distribute_read_design -design_script dmmmc.tcl -outdir output
set Views [list lmin-rmax lmax-rmin lmax-rmax]
```

```
tempus > distribute_views -views $Views -script dmmmc_report.tcl -save_design all
```

- The following example command script lists the commands used to restore a saved session (with N views) in distributed multi-mode multi-corner mode:

```
tempus > set_distribute_host -local
tempus > set_multicpu_usage -remoteHosts N
set Views [list lmin-rmax lmax-rmin lmax-rmax]
```

```
tempus > distribute_read_design -restore_design $Views -restore_dir output -outdir
new_output
```

output is the same directory used in above example where views were saved.

```
tempus > distribute_command {source dmmmc_reports2.tcl}
# dmmmc_reports2.tcl can have analysis commands but do not need read_spef etc.
```

- The following example command creates the client log files in the aa/bb directory instead of the output directory (specified using the `-outdir` parameter). The log files are prefixed with 'mylog_'. For example, if two clients are launched, the log files will be aa/bb/mylog_0.log and aa/bb/mylog_1.log.

```
tempus > distribute_read_design -client_logfile aa/bb/mylog -outdir output -
design_script script.tcl
```

- The following example restores the dtmf_chip OpenAccess design, which is saved at the path, /home/cdns/oa_dir:
tempus > set outDir "/home/cdns/" distribute_read_design -restore_design {view1 view2} -
restore_dir \$outDir -restore_cellview {oa_dir dtmf_chip} -out_dir output

Related Topics

- ["Distributed Multi-Mode Multi-Corner Timing Analysis"](#) chapter in the *Tempus User Guide*

Related Commands

- `set_multi_cpu_usage`
- `read_spef`
- `set_analysis_view`
- `set_timing_derate`
- `distribute_views`

distribute_views

```
distribute_views  
[-help]  
[-combine_num_views integer]  
[-force_distribute]  
[-save_design list_of_views | all]  
[-script list_of_commands]  
[-set_analysis_view_log]  
[-views list_of_views]
```

Distributes the specified analysis views to the Tempus client(s), then sources the specified command script to perform timing and signal integrity analysis.

The `distribute_views` command is used for distributed multi-mode multi-corner (DMMMC) timing analysis only. Distributed multi-mode multi-corner timing analysis is executed in batch mode. You can use the `distribute_command` for running a single command interactively.

By default, this command distributes a single view per Tempus client. The command can also perform concurrent multi-mode multi-corner timing analysis (CMMMC), within the DMMMC framework. You can combine the analysis views for CMMMC analysis on Tempus client(s). This can be done in the following two ways:

1. Delay corner based auto grouping :

You can specify a simple list of view names in the `-view` option and use the `-combine_num_views` option to specify the maximum number of views to be combined per Tempus client. For example

```
tempus > distribute_views -combine_num_views 4 -views {setup1 setup2 setup3 setup4  
setup5 setup6 setup7 setup8} -script ...
```

Assuming setup1 to setup4 have delay corner dc1 and setup5 to setup8 have delay corner dc2 then above example will group setup1 to setup4 to Tempus client1 and setup5 to setup8 will be grouped to Tempus client2

2. User controlled view grouping:

You can specify a list of list format for combining multiple concurrent views on Tempus client(s). For example,

```
distribute_views -views { {setup1 setup2 setup3 setup4 } { setup5 setup6 setup7 setup8 }  
} -script ...
```

The above command will group setup1 to setup4 views to Tempus client1 will group setup5 to setup8 views to Tempus client2 irrespective of the delay corner. To get optimum runtime while using view grouping, it is recommended to use delay corner based auto grouping as mentioned in example 1 above.

Tempus supports one step method for generating merged MMMC view report from DMMMC master shell. After running the `distribute_views` command, you can issue a single `report_timing/report_constraint` command, which generates reports collectively from all active distributed views (that is, views available for interactive DMMMC analysis). The merged output report is arranged as per slack/criticality order across active views.
All the options of `report_constraint` command, except `-connection_class`, and all the options of `report_timing` command are supported in this mode.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-combine_num_views <i>integer</i></code>	Combines upto N views (having same delay corner) to run at the same time on the same Tempus client.
<code>-force_distribute</code>	Forces the full timing update for given set of views (specified in subsequent <code>distribute_views</code> command) even if those views are already active on Tempus client(s). By default The existing active view(s) will not be re-analyzed and the remaining views from the distribution list will be dispatched for full timing update.
<code>-save_design <i>list_of_views</i> <i>all</i></code>	Saves the design session information for the specified analysis views. You can use the <code>-save_design all</code> option to save all the distributed multi-mode multi-corner (DMMMC) sessions with views that are specified using the <code>-views</code> parameter. The design information is saved in the following location: <pre>\${outdir}/<viewname>/<topcell.enc.dat></pre> where <code>\${outdir}</code> is the location specified using the <code>distribute_read_design -outdir</code> command. Note: When you specify this parameter, any previously saved sessions in <code>outdir</code> will be overwritten. When the <code>-save_design</code> parameter is specified, a distributed OpenAccess (OA) database will be created in the following format during a distributed MMMC run: <pre>LibName = \$outDir CellName = same as OA design Cell Name Views = List of MMMC Views</pre>

<pre>-script list_of_commands</pre>	Specifies the file that contains list of commands for reading in the necessary parasitic files and the reporting commands required for timing and signal integrity analysis. Commands for reading parasitic information must be specified in the script before the reporting commands. You must specify parasitic files specific for all RC corners. For example: <pre>read_spef -rc_corner corner1 rc1.spef read_spef -rc_corner corner2 rc2.spef read_spef -rc_corner corner3 rc3.spef read_spef -rc_corner corner4 rc4.spef</pre> If the file includes report commands that redirect their output to a specified file the software generates the file in the active view directory, in the output directory specified with the <code>distribute_read_design</code> command. (for example: for below command specified in the script <pre>report_timing >rt.rpt</pre> software will generate <code>rt.rpt</code> file in <code>\${outdir}/<viewname></code> directory for each active view where <code>\${outdir}</code> is the location specified using the <code>distribute_read_design -outdir</code> command. For report commands with no redirection specified, the software generates the report in the corresponding log file.
<pre>-set_analysis_view_log</pre>	Redirects <code>set_analysis_view</code> run log to <code>set_analysis_view_<viewname>.log</code> file.
<pre>-views list_of_views</pre>	Specifies views to distribute for analysis. For selectively reporting setup or hold checks for the specified views, you can use <code>all_setup_analysis_views</code> or <code>all_hold_analysis_views</code> commands (specified in the command script using the <code>-script</code> parameter). See example below. Note: The <code>-views</code> parameter is mandatory when the <code>-set_analysis_view_log</code> and <code>-combine_num_views</code> parameters are specified.

Examples

Here are some examples.

- The following example command script `run_dmmmc.tcl` lists the commands used to perform distributed multi-mode multi-corner timing analysis. The script performs distributed analysis for four views on four Tempus clients and then saves the design:

```
tempus > set_multi_cpu_usage -remoteHost 4
tempus > distribute_read_design \
-design_script /stml/timing/demo_dist_proc/preamble.tcl \
-outdir $output_directory

set views [list lmin-rmax lmin-rmin lmax-rmin lmax-rmax]
tempus > distribute_views \
-views $views \
-script /stml/timing/demo_dist_proc/commands.tcl \
-save_design {all}

tempus > exit_servers
```

- The following command script `commands.tcl` generates `setup.rpt` and `hold.rpt` reports for all the four views:

```
tempus > read_spef -rc_corner rmin /stml/timing/demo_dist_proc/min.spef.gz
tempus > read_spef -rc_corner rmax /stml/timing/demo_dist_procmx.spef.gz
tempus > report_timing -late > setup.rpt
tempus > report_timing -early > hold.rpt
```

- The following command script `report_common.tcl` generates `setup.rpt` and `hold.rpt` reports for two views each:

```
set mySetupViews "lmax-rmin lmax-rmax"
set myHoldViews "lmin-rmax lmin-rmin"
```

REPORTS FOR SETUP VIEWS

```
set currentView [all_setup_analysis_views]
set reportSetup 0
foreach vName $currentView { if { [lsearch $mySetupViews $vName ] != -1 } { set
reportSetup 1 ; break; } }
if { $reportSetup } {
## ALL Setup View reporting Commands
report_timing -late > setup.rpt
}
```

REPORTS FOR HOLD VIEWS

```
set currentView [all_hold_analysis_views] ;
```

```
set reportHold 0
foreach vName $currentView { if { [lsearch $myHoldViews $vName ] != -1 } { set
reportHold 1 ; break; } }
if { $reportHold } {
```

ALL Hold View reporting Commands

```
report_timing -early > hold.rpt
}
```

Related Topics

- ["Distributed Multi-Mode Multi-Corner Timing Analysis"](#) chapter in the *Tempus User Guide*

Related Commands

- [set_multi_cpu_usage](#)
- [read_spef](#)
- [report_timing](#)

exit_servers

exit_servers
[-help]
[-view viewName]

Releases the Tempus client(s) and ends the distributed processing session. Specify the `exit_servers` command after analysis is finished.

The `exit_servers` command is used for distributed multi-mode multi-corner timing analysis only.

Parameters

-help	Prints out the command usage.
-view viewName	Removes user-specified view(s) and releases the corresponding Tempus client. This parameter can be used after the <code>distribute_views</code> command is issued. After removing some view(s), you can resume performing interactive analysis for remaining views, however, you cannot run the <code>distribute_views</code> command again in the same session.

Examples

Here are some examples.

- The following example command script will release all Tempus clients after performing distributed multi-mode multi-corner timing analysis for all views:

```
tempus > set_multi_cpu_usage -remoteHost 4

tempus > distribute_read_design \
-design_script /stm1/timing/demo_dist_proc/preamble.tcl \
-outidir $output_directory
set views [list lmin-rmax lmax-rmin lmax-rmax lmin-rmin]
#preamble.tcl script includes design loading commands and MMMC configuration commands

tempus > distribute_views \
-views $views \
-script /stm1/timing/demo_dist_proc/commands.tcl
#commands.tcl script contains read_spef and report commands
```

`exit_servers`

Related Topics

- ["Distributed Multi-Mode Multi-Corner Timing Analysis"](#) chapter in the *Tempus User Guide*

get_distribute_host

```
get_distribute_host  
[-help]  
[-args]  
[-custom_script]  
[-env_script]  
[-hosts]  
[-mode]  
[-queue]  
[-reportLsfInfo file]  
[-resource]  
[-shellTimeout]  
[-single_cpu_lsf_args]  
[-timeOut]  
[-waitForLsfInfo integer]
```

Displays settings for the `set_distribute_host` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-args</code>	Displays the additional arguments when the software is running in the Sun Grid Engine (SGE) or Load Sharing Facility (LSF) mode.
<code>-custom_script</code>	Displays the name of the custom script when the software is running in <code>custom</code> mode.
<code>-env_script</code>	Displays the user-defined shell environment script that is run for each of the slave hosts.
<code>-hosts</code>	Displays the list of host names when the software is running in <code>rsh</code> mode.
<code>-mode</code>	Displays the distributed processing mode. The possible values are <code>local</code> , <code>sge</code> , <code>rsh</code> , <code>ssh</code> , <code>lsf</code> , <code>custom</code> , or <code>0</code> . The command displays <code>0</code> when the software is not in multiple-CPU processing mode.
<code>-queue</code>	Displays the queue name when the software is running in <code>sge</code> or <code>lsf</code> mode.
<code>-reportLsfInfo file</code>	Specifies to restore the LSF information to the specified file. This parameter can only be used when <code>set_distribute_host -reportLsfInfo</code> is specified in the current session. You must specify the <code>get_distribute_host -reportLsfInfo</code> only after the Tempus client exits to obtain the client CPU time. If you specify this command before the Tempus client exits, the command gives a warning message.
<code>-resource</code>	Displays the resource string when the software is running in <code>sge</code> or <code>lsf</code> mode.
<code>-shellTimeout</code>	Displays the number of seconds the host machine waits for <code>rsh/ssh</code> machines to become available for multiple-CPU processing.
<code>-single_cpu_lsf_args</code>	Displays the <code>lsf</code> arguments when the software is running in <code>lsf</code> mode for a single CPU application.
<code>-timeOut</code>	Displays the default timeout for a submitted job that will never start.
<code>-waitForLsfInfo integer</code>	Specifies the timeout period required to wait for task exit and generate the LSF information.

Examples

- The following command displays the current setting for the `-mode` parameter:

```
tempus > get_distribute_host -mode
```

- The following command saves the LSF information to the `lsf.log` file:

```
tempus > get_distribute_host -reportLsfInfo lsf.log
```

Related Topics

- ["Appendix - Multi-CPU Usage"](#) in the *Tempus User Guide*
 - ["Running Distributed Processing"](#)

Related Commands

- `set_distribute_host`

get_distribute_variables

```
get_distribute_variables  
[-help]  
list_of_slave_variables  
[-array_as_multi_dim_array]
```

Retrieves variables from Tempus clients. The results of each variable are stored in the master session as a TCL array with the view name as the key or search index.

When the result is created in the form of TCL variables at the Tempus client, the resulting variables can be retrieved back to the master session anytime after one of the following commands is run:

- `distribute_views -views`
- `distribute_read_design -restore_design`

Parameters

<code>-help</code>	Prints out the command usage.
<code>-array_as_multi_dim_array</code>	Generates tcl array at master for each client array variable with view name as additional index. <i>Default:</i> Array variables retrieved as tcl list indexed only by view name.
<code>list_of_slave_variables</code>	Specifies a list of slave variables to be returned to the master session.

Examples

Here are some examples.

- Assuming `setupView` and `holdView` views are active on two Tempus clients. The following example command script retrieves the value of variables `slackTest` and `arrivalTest` from each Tempus client to master session:

```
# define/set some value for variables at the Tempus client:  
  
1) tempus > distribute_command {  
  array set arrivalTest [list a 1 b 2 c 5 d 7] ;  
}  
  
get_distribute_variables arrivalTest  
  
Puts "ARRIVAL = [array get arrivalTest]"
```


Output of this example will be:

```
ARRIVAL = holdView {d 7 a 1 b 2 c 5} setupView {d 7 a 1 b 2 c 5}
```

```
2) tempus > distribute_command {  
  array set slackTest [list a -1 b -2 c -5 d -7]  
}  
get_distribute_variables -array_as_multi_dim_array slackTest  
Puts "SLACK = [array get slackTest]"
```

Output of this example will be:

```
SLACK = setupView,a -1 holdView,d -7 setupView,b -2 holdView,a -1 setupView,c -5  
holdView,b -2 setupView,d -7 holdView,c -5
```

Related Commands

- [distribute_read_design](#)
- [distribute_views](#)

merge_timing_reports

```
merge_timing_reports
[-help]
[-text]
[-max_paths integer ]
[-nworst <integer>]
[-command {report_constraint | report_timing | report_clock_timing}]
-base_report report_name
[-base_dir output_directory]
-view_dirs view_list
> file_name
```

Merges machine-readable/text timing reports from the specified analysis views into a single report file. The software merges the reports based on the slack value. The path with the most critical slack is listed first in the report.

The `merge_timing_reports` command can be used for both single CPU and distributed multi-mode multi-corner timing analysis.

Note: This command should only be used in DMMMC master script after reports are generated from the `distribute_views` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-command {report_constraint report_timing report_clock_timing}</code>	Merges the text timing reports from the specified file names. The generated timing reports can be from the commands <code>report_constraint</code> , <code>report_timing</code> , or <code>report_clock_timing</code> .
<code>-base_dir <i>output_directory</i></code>	Specifies the path to the output directory where the view-specific directories exist. The software looks in this directory for the base file name in the view specific directories, and then places the merged output in the specified redirection file. If it is not specified then software uses the base directory as the one specified with "distribute_read_design -outdir" option.
<code>-base_report <i>report_name</i></code>	Specifies the base file name of the timing reports to merge.
<code>-max_paths <i>integer</i></code>	Reports the top N worst paths that are picked from multiple views.

<code>-nworst integer</code>	Specifies to have at the most n paths per end point in a merged report.
<code>> redirection_file</code>	Specifies the name of the file in which to merge the timing reports.
<code>-text</code>	Generates merged report in the text format. Note: You can use this option only when the input reports are machine-readable.
<code>-view_dirs view_list</code>	Merges timing reports from the specified directories.

Examples

Here are some examples.

- The following command merges the machine-readable timing reports with the base file name `rt_setup1.rpt` in the analysis views `v1`, `v2`, and `v3` into a single machine readable timing report called `merged_report.rpt`:

```
tempus > merge_timing_reports -view_dirs {v1 v2 v3} -base_report rt_setup1.rpt > merged_report.rpt
```
- The following command merges the machine-readable timing reports with the base file name `rt_setup1.rpt` in the analysis views `v1`, `v2`, and `v3` into a single text timing report called `merged_report.rpt`:

```
tempus > merge_timing_reports -view_dirs {v1 v2 v3} -base_report rt_setup1.rpt -text > merged_report.rpt
```
- The following commands merge text timing reports (from `report_timing` command) with the base file name `topPaths.rpt` in the analysis views `v1` and `v2` into a single timing report called `merged_topPaths.rpt`:

```
tempus > distribute_command {report_timing -max_paths 10 > topPaths.rpt}
tempus > merge_timing_reports -command report_timing -view_dirs {v1 v2} -base_report topPaths.rpt -base_dir dmmmc_run_dir > merged_topPaths.rpt
```
- The following commands merge text timing reports (from `report_constraint` command) with the base file name `allViol.rpt` in the analysis views `v1` and `v2` into a single timing report called `merged_allViol.rpt`:

```
tempus > distribute_command {report_constraint -all_violators > allViol.rpt}
tempus > merge_timing_reports -command report_constraint -view_dirs {v1 v2} -base_report allViol.rpt -base_dir dmmmc_run_dir > merged_allViol.rpt
```

Related Topics

- ["Distributed Multi-Mode Multi-Corner Timing Analysis"](#) chapter in the *Tempus User Guide*
 - ["Aggregating Timing Reports"](#)

Related Commands

- `report_timing`
- `report_constraint`
- `report_clock_timing`
- `distribute_command`
- `distribute_views`

set_distribute_host

```
set_distribute_host
[-help]
[-add string]
[-args string]
[-custom]
[-custom_script string]
[-custom_script_list string]
[-local]
[-lsf]
[-nc]
[-noWaitTaskComeUp]
[-queue string]
[-remove string]
[-reportLsfInfo]
[-resource string]
[-shellTimeout integer]
[-single_cpu_lsf_args string]
[-ssh]
[-timeOut integer]
[-env_script string]
```

Specifies the multiple-CPU processing configuration for distributed processing. The software finds hosts from the last `set_distribute_host` command specified before a distributed processing command starts.

Use `get_distribute_host` to display the current settings for the `set_distribute_host` command.

 This command is required for distributed processing.

Parameters

<code>-help</code>	Prints out the command usage.
--------------------	-------------------------------

<code>-add string</code>	Adds the specified hosts to the <code>rsh/ssh</code> configuration. If you specify more than one host, separate each host name with a space. Each time you specify a host, the software uses it to run one client process. To run more than one client process on a host, specify it more than once. For example, if you have two machines that each has two CPUs and you want to use both CPUs in both machines, type a command like the following: <pre>setDistributeHost -add {machine1 machine2 \ machine1 machine2}</pre>
<code>-args string</code>	Specifies any additional arguments you need to provide to the Sun Grid Engine (SGE) or Load Sharing Facility (LSF) job submission command.
<code>-custom</code>	Specifies to use the custom job submission script for distributed processing.
<code>-custom_script string</code>	Specifies a custom script to run distributed processing. You can use this parameter with <code>-custom</code> only.
<code>-custom_script_list string</code>	Specifies to customize the launch of multiple jobs, based on resources required for these jobs. This parameter allows you to specify multiple launch strings in order of priority with a maximum allowed job count for each string. If you request for N number of jobs, the command starts submitting jobs starting from <code>string1</code>, until the corresponding count is reached, at which point it starts submitting the next string in the list. Therefore, the command submits the first "<code>count1</code>" jobs with <code>string1</code>, the next "<code>count2</code>" jobs with <code>string2</code> and so on. It stops when the total submitted jobs has reached the requested number (N). Note: This parameter can only be specified with the <code>-custom</code> parameter.
<code>-env_script string</code>	Specifies a user-defined shell environment script that is run for each of the client hosts. The parameter allows you to pass these environment settings to the client hosts by including them in the shell environment script. The license variables used by the software are also automatically passed to the client hosts to avoid license errors.
<code>-local</code>	Runs all distributed processing jobs on the master machine.
<code>-lsf</code>	Specifies a Load Sharing Facility configuration. If you do not specify any parameters for <code>-lsf</code>, the software uses default settings. Note: To use this parameter, LSF must already be set up (typically by specifying the <code>LSF_ENVDIR</code> and <code>LSF_SERVERDIR</code> environment variables) and <code>bsub</code> must be in your search path. Contact your LSF administrator for details.

<code>-nc</code>	Uses NC commands (Network Computer from RTDA) for distributed processing.
<code>-noWaitTaskComeUp</code>	Does not allow the software to wait for tasks that are delayed or are not likely to occur.
<code>-queue <i>string</i></code>	Specifies the queue for the LSF or SGE configuration.
<code>-remove <i>string</i></code>	Removes the specified hosts or all hosts from the <code>rsh/ssh</code> configuration. If you specify more than one host, separate each host name with a space.
<code>-reportLsfInfo</code>	Specifies to report LSF information in a single report. This parameter allows you to collect the system statistics on the status of the distributed jobs. It has the following information: • Host name for master and LSF tasks • Number of CPUs on each machine that are being used by a process • Job ID/PID for each task process • Total memory used on each machine • Total virtual memory used on each machine • Total CPU time used on each machine You must specify this parameter with the <code>-lsf</code> parameter. To get the LSF information, specify the <code>getDistributeHost -reportLsfInfo</code> command parameter. You must specify the <code>getDistributeHost -reportLsfInfo</code> only after the LSF tasks exit to obtain the task CPU time. If you specify this command before the task exits, the command gives a warning message.
<code>-resource <i>string</i></code>	Specifies a resource string for the SGE or LSF queue. Note: The correct resource string is specific to your installation. Contact your LSF administrator for the appropriate parameters and values.
<code>-shellTimeout <i>integer</i></code>	Specifies the number of seconds the host machine waits for <code>rsh/ssh</code> machines to become available for multiple-CPU processing. <i>Default:</i> 5
<code>- single_cpu_lsf_args <i>string</i></code>	Specifies the <code>bsub</code> options. For information, see the LSF documentation. Note: You must use this parameter to adjust the LSF argument for a single CPU application. Alternatively, use the <code>-args</code> parameter to adjust the LSF argument for an application that runs in the Superthreading mode.
<code>-ssh</code>	Specifies a secure shell configuration.

`-timeout integer`

Specifies the default timeout for a submitted job that will never start. This functionality prevents the master machine from hanging when the clients do not come up for some reason.

After timeout, a warning message appears informing you that the distributed CPUs did not come online within the expected time. The master will continue waiting for the clients and it is up to you to continue waiting or kill the run if there are problems with the Distributed Resource Management (DRM) system.

A sample warning message is given below:

Task id 2 did not come up within the specified timeout period (300 seconds). Use `set_distribute_host -timeout` to change the value. This may be a real error due to incorrect or incomplete platform/machine specification provided via `set_distributed_hosts` to your distributed resource management system (LSF/Sun Grid Engine etc.). If running locally (`set_distribute_host -local`), the current machine may be overloaded with too many jobs and swapping (check free memory and CPUs with `top`). For local runs, consider reducing the number of CPUs you are using via the `set_multi_cpu_usage` command or free up some memory by killing other memory intensive processes.

Default: 3600 for all the modes (LSF, NC, SGE, RSH, SSH, local and custom

Note: When the `set_distribute_host` command is not specified, the software sets the default mode as local and the EDP timeout is 3600 seconds.

Examples

LSF

The following command uses an existing LSF environment, and relies on the general LSF set up by the system administrator. In this example, the name of the queue implies that all the machines have 4 Gb of memory.

```
tempus > set_distribute_host -lsf -queue mem4G
```

The following command schedules a LSF job named `bigQueue` to run on a Linux machine for a maximum of 30 minutes. No other jobs can run on the machine while `bigQueue` is running.

```
tempus > set_distribute_host -lsf -queue bigQueue -args {-x -W 30} \  
-resource {OSNAME==Linux}
```

The following command schedules a LSF job named `smallQueue` to run on Linux machines with

minimum speed of 2000 MHz.

```
tempus > set_distribute_host -lsf -queue smallQueue -args {-W 20} \  
-resource {OSNAME==Linux && SPEED>2000}
```

Note: Check with your LSF administrator for the correct values for the resource string, as the parameters and their values differ in each installation. For example, the parameter name for the speed of the machine might be `MHZ` instead of `SPEED` (as in the preceding example), and the units might be specified in gigahertz instead of megahertz.

The following command specifies to report LSF information:

```
tempus > set_distribute_host -lsf -args "-q lnx64" -reportLsfInfo
```

You can use the following command to save the LSF information in the specified file:

```
tempus > get_distribute_host -reportLsfInfo lsf.log
```

Local

The following command runs all distributed processing jobs on the machine on which you are running the master version of the Tempus software:

```
tempus > set_distribute_host -local
```

Custom

The following command runs distributed processing with a custom script:

```
tempus > set_distribute_host -custom -custom_script {bsub -q bigQueue -x -R "OS==Linux &&  
memory>20000"}
```

The following command runs distributed processing with a custom script list:

```
tempus > set_distribute_host -custom -custom_script_list {{{bsub -P ICD -W 2 -q admin -R  
"OSNAME==Linux && OSREL==EE50"}1} {{rsh hpc1 -e} 1}}
```

Examples

- The following command allows the Tempus client to source the `my_env_file` before start up on the same machine:

```
set_distribute_host -local -env_script my_env_file
```

This command allows the master to pass on the user-defined environment settings.

- The following command allows the Tempus client to run using the LSF:

```
set_distribute_host -lsf -noWaitTaskComeUp
```

When the `-noWaitTaskComeUp` parameter is specified, the Tempus software will not wait for all the clients to come up when parts of the clients are launching timeout.

Related Topics

- ["Appendix - Multi-CPU Usage"](#) in the *Tempus User Guide*
 - ["Running Distributed Processing"](#)

Related Commands

- `get_distribute_host`

set_distribute_mmmc_mode

```
set_distribute_mmmc_mode
[-help]
[-client_init_execute_script filename]
[-eco {true | false}]
[-enable_high_capacity_eco]
[-enable_smart_mmmc]
[-lic_startup_extra {list_of_options}]
[-load_def_only]
[-merge_view_definitions]
[-min_host_needed integer]
[-set_filler_mode {list_of_options}]
[-use_one_host_per_batch_job {true | false}]
[-interactive {true | false} [-views {list_of_views | all}]]
```

Controls distributed MMMC analysis settings.

Parameters

-help	Prints out the command usage.
- client_init_execute_script <i>filename</i>	Specifies a file containing the TCL code that will be executed on a Tempus client(s) immediately after it is invoked. This allows you to load settings on a client prior to loading a database or design initialization.
-eco {true false}	Enables/disables the timing signoff flow integrated with distributed MMMC.
-enable_high_capacity_eco	Enables database size reduction in the Tempus Paradime flow or Certus for high capacity ECO timing/power closure based on timing violations, list of partition names, or user-defined path group. Note: It is mandatory to apply set_distribute_mmmc_mode -eco true command before set_distribute_mmmc_mode -enable_high_capacity_eco in Tempus Paradime flow. In case of Certus, it should be set before distribute_read_design.

<code>-enable_smart_mmmc</code>	Enables reduced memory or runtime solution when many timing views are enabled in Tempus Paradime flow or Certus. Note: It is mandatory to apply <code>set_distribute_mmmc_mode -eco true</code> command before <code>set_distribute_mmmc_mode -enable_smart_mmmc</code> in Tempus Paradime flow. In case of Certus, it should be set before <code>distribute_read_design</code>.
<code>-lic_startup_extra</code> <code>{list_of_options}</code>	Specifies a list of licenses to be checked out by Tempus clients. When specified, the software shows the extra Innovus startup licenses that are checked out by Tempus clients.
<code>-load_def_only</code>	Specifies if a client database being restored in a flow contains LEF information and needs to load DEF (specified using the <code>eco_def_file_list</code> variable), to make it a fully physical flow.
<code>-interactive {true false}</code>	Enables or disables the interactive mode.
<code>-merge_view_definitions</code>	Creates ECO analysis view(s) from view definition files in restore sessions. The <code>-merge_view_definitions</code> parameter must be used only after <code>set_distribute_mmmc_mode -eco true</code> command is set.
<code>-min_host_needed <i>integer</i></code>	Specifies minimum number of machines required to start a session in case all the machines cannot be acquired after time out.
<code>-set_filler_mode</code> <code>{list_of_options}</code>	Specifies a list of <code>set_filler_mode</code> command options. This parameter allows to pass a list of options that the <code>set_filler_mode</code> command applies during physical data loading in Tempus Paradime.

<pre>- use_one_host_per_batch_job {true false}</pre>	Enables/disables one batch mode job per host. A new client is started for each view or each combined view job. This is used when running flows, such as manual ECO flows, where netlist updates during analysis makes scheduling of subsequent views difficult.
<pre>-views {list_of_views all}</pre>	Specifies a list of views for interactive mode. The <code>all</code> option selects all the view.

Examples

- The following commands enable the distributed MMMC analysis in interactive mode for views `view1` and `view2`:

```
tempus > set viewnames {view1 view2}
tempus > set_distribute_mmmc_mode -views $viewnames -interactive true
```
- The following command disables the interactive mode once all the reporting/ECO on interactive views is done:

```
tempus > set_distribute_mmmc_mode -interactive false
```
- The following command lists all the filler cell names to be applied on `set_filler_mode` command:

```
set_distribute_mmmc_mode -set_filler_mode {-core "FILLX1 FILLX2 FILLX4"}
```
- The following commands enable design loading in Paradime with the minimized set of data to reduce the memory or runtime during ECO fixing stage when many timing views are enabled, and performs Setup timing closure.

```
tempus > set_multi_cpu_usage -localCpu 16 -remoteHost 4 -cpuPerRemoteHost 4
tempus > set_distribute_mmmc_mode -eco true
tempus > set_distribute_mmmc_mode -enable_smart_mmmc
tempus > set_eco_opt_mode -retime path_slew_propagation
tempus > set_distribute_mmmc_mode -merge_view_definitions
tempus > set eco_setup_view_list "<list_of_Setup active_views>"
tempus > set eco_hold_view_list "<list_of_Hold active_views>"
tempus > set eco_lef_file_list "<list_of_LEF_files>"
tempus > set eco_def_file_list "<list_of_DEF_files>"
tempus > distribute_read_design -restore_design all -restore_dir PARADIME_SESSIONS/ -
outdir reports_Paradime
tempus > eco_opt_design -setup
```

Related Commands

- `set_multi_cpu_usage`
- `distribute_read_design`
- `distribute_views`
- `set_filler_mode`

set_distribute_variables

```
set_distribute_variables  
[-help]  
list_of_variables  
[value]
```

Allows you to set variables on the Tempus master and clients.

The variables can be set after issuing the `distribute_read_design` command on the master.

Parameters

<code>-help</code>	Prints out the command usage.
<code>list_of_variables</code>	Specifies a list of variables to be set on master and client(s).
<code>value</code>	Specifies the value which should be set on the master and client.

Examples

- The following command sets a variable `var` having value `1` on the master and client(s):

```
tempus > set_distribute_variables var 1
```
- The following command sets multiple variables with value `1` on master and client(s):

```
tempus > set_distribute_variables "var1 var2 var3" 1
```
- The following command sets multiple variables each with different values on master and client(s):

```
tempus > set var1 1  
1  
tempus > set var2 2  
2  
tempus > set var3 3  
3  
tempus > set_distribute_variables "var1 var2 var3"
```

Related Commands

- `get_distribute_variables`
- `distribute_read_design`
- `distribute_command`

Timing Analysis Commands

- [all_analysis_views](#)
- [all_constraint_modes](#)
- [all_delay_corners](#)
- [all_hold_analysis_views](#)
- [all_library_sets](#)
- [all_op_conds](#)
- [all_rc_corners](#)
- [all_setup_analysis_views](#)
- [apply_boundary_model_constraints](#)
- [check_boundary_model_consistency](#)
- [check_instance_library_in_views](#)
- [check_ipd_logic](#)
- [check_ldb_version](#)
- [check_library](#)
- [check_timing](#)
- [compare_timing_context](#)
- [create_analysis_view](#)
- [create_boundary_model](#)
- [create_constraint_mode](#)
- [create_delay_corner](#)
- [create_glitch_boundary_model](#)
- [create_ipd_configuration](#)
- [create_library_set](#)
- [create_op_cond](#)
- [create_parasitic_coordinate_transform](#)

- `create_pg_net_group`
- `create_property_alias`
- `create_rc_corner`
- `display_timing_map`
- `get_analysis_mode`
- `get_analysis_view`
- `get_capacitance_unit`
- `get_clock_network_objects`
- `get_constant_for_timing`
- `get_constraint_mode`
- `get_delay_corner`
- `get_equivalent_cells`
- `get_interactive_constraint_modes`
- `get_lib_cell_leakage_power`
- `get_library_set`
- `get_op_cond`
- `get_pba_mode`
- `get_propagated_clock`
- `get_resistance_unit`
- `get_socv_rc_variation_factor`
- `get_socv_reporting_nsigma_multiplier`
- `get_time_unit`
- `list_libraries`
- `read_boundary_model`
- `read_ipd_scope`
- `read_path_descriptions`
- `read_sdf`
- `read_spdf`
- `read_spef`
- `read_timing_context`

- [read_twf](#)
- [report_analysis_coverage](#)
- [report_analysis_mode](#)
- [report_analysis_summary](#)
- [report_analysis_views](#)
- [report_annotated_assertions](#)
- [report_annotated_check](#)
- [report_annotated_delay](#)
- [report_annotated_parasitics](#)
- [report_aocv_derate](#)
- [report_boundary_model_constraints](#)
- [report_case_analysis](#)
- [report_cell_instance_timing](#)
- [report_clock_gating_check](#)
- [report_clock_network_transitions](#)
- [report_clock_propagation](#)
- [report_clock_sense](#)
- [report_clock_timing](#)
- [report_clocks](#)
- [report_constraint](#)
- [report_cpvr](#)
- [report_critical_instance](#)
- [report_delay_calculation](#)
- [report_design](#)
- [report_design_rule](#)
- [report_fanin](#)
- [report_fanout](#)
- [report_freq_violation](#)
- [report_inactive_arcs](#)
- [report_instance_library](#)

- `report_ipd_configuration`
- `report_ipd_crossings`
- `report_lib_arcs`
- `report_min_pulse_width`
- `report_mode`
- `report_net`
- `report_parasitic_coordinate_transforms`
- `report_path_exceptions`
- `report_path_groups`
- `report_pba_aocv_derate`
- `report_pg_net_group`
- `report_ports`
- `report_reference`
- `report_settings`
- `report_slack_histogram`
- `report_socv_interconnect_variation`
- `report_socv_library`
- `report_statistical_timing_derate_factors`
- `report_timing`
- `report_timing_context_clock_mapping`
- `report_timing_derate`
- `report_timing_lib`
- `report_timing_robustness`
- `report_wire_load`
- `reset_instance_library`
- `reset_path_adjust`
- `reset_path_adjust_group`
- `reset_sdf_assertions`
- `reset_timing_derate`
- `reset_wire_load_mode`

- `reset_wire_load_model`
- `reset_wire_load_selection_group`
- `select_boundary_model_constraints`
- `set_analysis_mode`
- `set_analysis_view`
- `set_aocv_thresholds`
- `set_boundary_model_clock_mapping`
- `set_boundary_model_view_mapping`
- `set_default_view`
- `set_exclude_net`
- `set_instance_library`
- `set_interactive_constraint_modes`
- `set_io_thresholds`
- `set_ipd_rail_voltage`
- `set_library_unit`
- `set_max_cap_per_freq`
- `set_max_tran_per_freq`
- `set_multi_input_switching_mode`
- `set_op_cond`
- `set_path_adjust`
- `set_path_adjust_group`
- `set_pba_mode`
- `set_socv_constraint_config`
- `set_socv_rc_variation_factor`
- `set_socv_reporting_nsigma_multiplier`
- `set_table_style`
- `set_threshold_voltage_group`
- `set_timing_context_clock_mapping`
- `set_timing_context_view_mapping`
- `set_timing_derate`

- [set_timing_library](#)
- [set_vt_skew_config](#)
- [set_vt_skew_derate](#)
- [set_wire_load_mode](#)
- [set_wire_load_model](#)
- [set_wire_load_selection_group](#)
- [timing_enable_boundary_model_eco](#)
- [update_analysis_view](#)
- [update_constraint_mode](#)
- [update_delay_corner](#)
- [update_io_latency](#)
- [update_library_set](#)
- [update_model_context](#)
- [update_rc_corner](#)
- [update_timing](#)
- [write_global_slack_report](#)
- [write_ipd_scope](#)
- [write_ldb](#)
- [write_loop_break_sdc](#)
- [write_model_context](#)
- [write_path_descriptions](#)
- [write_sdc](#)
- [write_sdf](#)
- [write_timing_context](#)
- [write_timing_windows](#)
- [write_twf](#)

all_analysis_views

```
all_analysis_views  
[-help]  
[-type string]
```

Returns a TCL list of all the defined analysis views in a design.

You can use this command after creating multiple analysis views (using the `create_analysis_view` command).

Parameters

-help	Prints out the command usage.
-type <i>string</i>	Specifies the type of analysis views. The following options are available: • setup: Active setup views in the design. • hold: Active hold views in the design. • active: Active setup and hold views in the design. • dynamic: Dynamic analysis views in the design. • leakage: Leakage analysis views. • em: Electromigration (EM) analysis views. • inactive: Setup/hold views that are not in use. • all: All defined views in the design, including those that are currently inactive.

Examples

- The following commands create analysis views `missionSlow` and `missionFast`:

```
tempus > create_analysis_view -name missionSlow -constraint_mode missionSetup -  
delay_corner dcWCCOM  
tempus > create_analysis_view -name missionFast -constraint_mode missionHold -  
delay_corner dcBCCOM
```

The following command returns a list of the defined analysis views in the design:

```
tempus > all_analysis_views
```

The software returns the following results:

```
{missionSlow missionFast}
```

Related Commands

- `create_analysis_view`
- `set_analysis_view`

all_constraint_modes

```
all_constraint_modes
[-help]
[-active | -active_setup | -active_hold]
```

Returns a TCL list of all defined constraint modes in the design.

Use this command after creating multiple constraint modes ([create_constraint_mode](#)).

Parameters

-help	Prints out the command usage.
-active	Returns a list of the defined constraint modes associated with all active analysis views.
-active_setup	Returns a list of the defined constraint modes associated with active setup analysis views only.
-active_hold	Returns a list of the defined constraint modes associated with active hold analysis views only.

Examples

Here are some examples.

- The following commands create constraint modes `missionSetup` and `missionHold`:

```
tempus > create_constraint_mode -name missionSetup -sdc_files [list io.sdc mission1-
clks.sdc mission1-except.sdc]
tempus > create_constraint_mode -name missionHold -sdc_files [list io.sdc mission1-
clks.sdc mission1-except.sdc dont_use.sdc]
```

- The following command returns a list of all the defined constraint modes in the design:

```
tempus > all_constraint_modes
```

The software returns the following results:

```
{missionSetup missionHold}
```

- The following command returns a list of all the defined constraint modes associated with the active setup analysis views in the design:

```
tempus > all_constraint_modes -active_setup
```

Related Commands

- `create_constraint_mode`

all_delay_corners

all_delay_corners

[-help]

[-active]

Returns a TCL list of all defined delay calculation corner objects in the design.

You can use this command after creating multiple delay calculation corner objects ([create_delay_corner](#)).

Parameters

-help	Prints out the command usage.
-active	Returns a list of the defined delay corners associated with all the active analysis views.

Examples

- The following commands create the delay calculation corners `dcWCCOM` and `dcBCCOM`:

```
tempus > create_delay_corner -name dcWCCOM \
    -library_set IsCOM-1V \
    -opcond_library stdcell_1V \
    -opcond WCCOM \
    -rc_corner rc-cworst
```

```
create_delay_corner -name dcBCCOM \
    -library_set IsCOM-1V \
    -opcond_library stdcell_1V \
    -opcond BCCOM \
    -rc_corner rc-cbest
```

- The following command returns a list of all the delay calculation corners in the design:

```
tempus > all_delay_corners
```

The software returns the following information:

```
{dcWCCOM dcBCCOM}
```

Related Commands

- `create_delay_corner`

all_hold_analysis_views

all_hold_analysis_views
[-help]

Returns a TCL list of all active hold analysis views in the design.

You can use this command after setting active hold analysis views for the design ([set_analysis_view](#)).

Parameters

-help	Prints out the command usage.
-------	-------------------------------

Examples

Here are some examples.

- The following command defines the active setup and hold analysis views for the design:

```
tempus > set_analysis_view -setup {missionWCCOM mission2WCCOM} -hold {missionBCCOM  
testBCCOM}
```

The following command returns a list of the active hold analysis views in the design:

```
tempus > all_hold_analysis_views
```

The software returns the following results:

```
{missionBCCOM testBCCOM}
```

Related Commands

- [set_analysis_view](#)
- [create_analysis_view](#)
- [all_analysis_views](#)

all_library_sets

all_library_sets

Returns a TCL list of all currently defined library sets in the design.

Use this command after creating multiple library sets ([create_library_set](#)).

This command returns a TCL list, which is dumped to a log file. You can use the [redirect](#) command to save this data to a file or variable. For example,

```
redirect -quiet "set somevar [all_library_sets]; echo $somevar" >& somefile
```

Parameters

None

Examples

- The following commands create the library sets IsCOM-1V and IsCOM-2V:

```
tempus > create_library_set -name IsCOM-1V -timing [list stdcell_F_1V.lib ram_F.lib  
pad.lib]  
tempus > create_library_set -name IsCOM-2V -timing [list stdcell_F_2V.lib ram_F.lib  
pad.lib]
```

The following command returns a list of all defined library sets in the design:

```
tempus > all_library_sets
```

The software returns the following results:

```
{IsCOM-1V IsCOM-2V}
```

Related Commands

- [create_library_set](#)
- [set_top_module](#)
- [read_lib](#)

all_op_conds

```
all_op_conds  
[-help]
```

Returns a TCL list of all operating conditions defined for the design.

Use this command after creating multiple operating conditions ([create_op_cond](#)).

This command returns a TCL list, which is dumped to a log file. You can use the [redirect](#) command to save this data to a file or variable. For example,

```
redirect -quiet "set somevar [all_setup_analysis_views]; echo $somevar" >& somefile
```

Parameters

None

Examples

Here are some examples.

- The following commands create operating conditions called `myPVT1` and `myPVT2` for libraries `IsCOM-1V.lib` and `IsCOM-2V.lib`:

```
tempus> create_op_cond -name PVT1  
      -library_file IsCOM-1V.lib  
      -P 1.0  
      -V 1.2  
      -T 120
```

```
tempus> create_op_cond -name PVT2  
      -library_file IsCOM-2V.lib  
      -P 1.0  
      -V 1.2  
      -T 20
```

The following command creates a collection called `myOpConds` that contains the operating conditions `myPVT1` and `myPVT2`:

```
set myOpConds [all_op_conds]
```

Related Commands

- `create_op_cond`
- `set_op_cond`
- `get_op_cond`

all_rc_corners

```
all_rc_corners  
[-help]  
[-active]
```

Returns a TCL list of all currently defined RC corner objects in the design.

Use this command after creating multiple RC corner objects ([create_rc_corner](#)).

Parameters

-help	Prints out the command usage.
-active	Returns a list of the defined RC corners associated with all active analysis views.

Examples

- The following commands create the RC corners `rc-cbest`, `rc-cworst`, and `rc-typical`:

```
tempus> create_rc_corner -name rc-cbest  
tempus> create_rc_corner -name rc-cworst  
tempus> create_rc_corner -name rc-typical
```

The following command returns a list of all defined RC corners in the design:

```
tempus> all_rc_corners
```

The software returns the following results:

```
rc-cbest rc-cworst rc-typical
```

- The following command will read a SPEF file for each of the defined RC corners in the design:

```
tempus > set corners [all_rc_corners]  
  
tempus > foreach corner $corners { read_spef -rc_corner $corner $corner.spef.gz }
```

Related Commands

- [create_rc_corner](#)

all_setup_analysis_views

all_setup_analysis_views

Returns a Tcl list of all active setup analysis views in the design.

Use this command after setting active setup analysis views for the design ([set_analysis_view](#)).

Parameters

-help	Prints out the command usage.
-------	-------------------------------

Example

- The following command defines the active setup and hold analysis views for the design:

```
tempus> set_analysis_view -setup {missionWCCOM mission2WCCOM} -hold {missionBCCOM  
testBCCOM}
```

The following command returns a list of the active setup analysis views in the design:

```
tempus> all_setup_analysis_views
```

The software returns the following results:

```
missionWCCOM mission2WCCOM
```

Related Commands

- [set_analysis_view](#)
- [create_analysis_view](#)
- [all_analysis_views](#)

apply_boundary_model_constraints

apply_boundary_model_constraints

[-help]

Allows the software to pull up a subset of constraints from a block run into the top boundary model session.

When this command is specified, the constraints are applied to instances of a boundary modeled block in a top level boundary model session. For glitch boundary models, glitch constraints, such as [define_bundled_bus](#), [set_glitch_threshold](#), [set_cdb_binding](#), and [set_quiet_attacker](#), are pulled up.

Parameters

-help	Prints out the command usage.
-------	-------------------------------

check_boundary_model_consistency

check_boundary_model_consistency
[-help]

Performs several checks on loaded models.

When this command is run, the software performs the following checks:

- Global variables and modes set during boundary model (BM) generation have the same settings in the loading run.
- If a boundary model sub-module is also instantiated outside of the BM, and the BM definition is used for the module.
- The view definition used to generate the BM is consistent with the loading run's view definition. (BM generation views can be a superset.)
- The loading run's active views are all active during the BM generation.
- Constant logic in the BM generation run has the same constants as in the loading run.
- Power nets/voltages are consistent between BM generation and BM loading.
- Chip size is consistent between BM generation and BM loading.

A report file is written for each loaded boundary model. The default name of the report is `${auto_file_prefix}<cell_name>_bmodel_check.txt`, written in to the `$auto_file_dir`. You can specify the name of the report file using the `read_boundary_model -report_file` command parameter.

Parameters

<code>-help</code>	Prints out the command usage.
--------------------	-------------------------------

Related Commands

- `read_boundary_model`

check_instance_library_in_views

```
check_instance_library_in_views  
[-help]
```

Checks for missing libraries for instances in the active view.

This command checks whether the library for the cell associated with the instance is present in the instance's associated power domain library set in the active views.

Parameters

None

Examples

Here are some examples.

- The following command displays a report showing missing libraries:

```
tempus> check_instance_library_in_views
```

The software displays the following information:

```
Checking the Library binding for instance in active view 'vA'  
Pass. All instances have library definition in view 'vA'  
Checking the Library binding for instance in active view 'vB'  
No library found for instance 'U1', cell 'AND2X1', in powerdomain 'PD_BAD', of view 'vB'  
Fail. Missing Library for some instance found in view 'vB'  
Checking the Library binding for instance in active view 'vAB'  
Pass. All instances have library definition in view 'vAB'
```

check_ipd_logic

```
check_ipd_logic
[-help]
[-outfile <filename>]
{ [-multi_pd_type <enum> [-verbose ]] }
```

Performs IPD logic checks and reports power-domain crossing information based on data and clock path checks.

Parameters

-help	Prints out the command usage.
-multi_pd_type <enum>	Specifies the type of IPD logic check that needs to be performed. You can specify the following options: - clock_divergence: Enables similar crossings on the divergent clock check. - clock_crossing: Enables multiple power domain crossings on clock path checks. - data_crossing: Enables multiple power domain crossings on data path checks. - cppr_aware: Enables checks for multiple power domains post the common point. - all: Enables all the crossing checks, except cppr_aware.
-outfile <filename>	Specifies the name of the output report file (suffix .rpt) to be generated.
-verbose	Allows the report_timing command to generate a detailed output for checks with .tcl suffix.

Examples

- The following command reports all the IPD logic checks in the design:

```
tempus> check_ipd_logic -multi_pd_type all -outfile ipd_logic_check_report -verbose
```

The output of `ipd_logic_check_report.rpt` is as shown below:

```
#####
# Multiple Power Domain Crossings On Data Paths #
#####
1.
Beginpoint: ff1/D (clk1)
Endpoint: ff2/Q (clk1)
PD1->PD2 lvl_cell11/Y
PD2->PD3 lvl_cell12/Y
2.
Beginpoint: ff3/D (clk1)
Endpoint: ff4/Q (clk2)
PD1->PD2 lvl_cell13/Y
PD2->PD3 lvl_cell12/Y

#####
# Clock Divergence Data #
#####
1.
Beginpoint: ff4/D (clk2)
Endpoint: ff5/Q (clk2)
PD1->PD2 lvl_cell15/Y (L) level_cell6/Y (C)

#####
# Multiple Clock Power Domain Crossings #
#####

#####
#Early Check Path Count Summary Output#
#####

Multiple IPD Data Crossing : 2
-----
Count          Power Domain
-----
2              PD1->PD2->PD3

IPD Clock Divergence : 1
-----
Count          Power Domain
-----
1              PD1->PD2

Multiple IPD Clock Crossing : 0
```

```
-----
Count          Power Domain
-----
```

- The output of `ipd_logic_check.tcl` is as shown below:

```
#####
# Multiple Power Domain Crossings On Data Paths  #
#####
report_timing -through lvl_cell1/Y -through lvl_cell2/Y
report_timing -through lvl_cell3/Y -through lvl_cell2/Y

#####
# Clock Divergence Data  #
#####
report_timing -from ff4/D -clock_from clk1 -to ff5/Q -clock_to clk2

#####
# Multiple Clock Power Domain Crossings  #
#####
```

- The following command generates a report showing `cppr_aware` checks:

```
tempus> check_ipd_logic -multi_pd_type cppr_aware -outfile cppr_aware_report
```

cppr_aware_report.rpt

```
#####
# Multiple Power Domain Crossings Paths  #
#####
1.
Beginpoint : inst1/Q(clk1)
Endpoint   : inst2/D(clk2)
PD1 -> PD2 cinst1/Y (L)
PD2 -> PD3 inst3/Y (D)
PD3 -> PD4 inst4/Y (D)
PD1 -> PD4 cins2/Y (C)
2.
Beginpoint : inst11/Q(clk21)
Endpoint    : inst12/D(clk12)
PD1 -> PD2 cinst13/Y (L)
PD2 -> PD3 cinst123/Y (L)
PD3 -> PD4 inst314/Y (D)
PD4 -> PD5 inst414/Y (D)
PD5 -> PD3 inst42/Y (D)
PD1 -> PD2 cins246/Y (C)
PD2 -> PD3 cins26/Y (C)
#####
#Early Check Path Count Summary Output#
#####
Multiple Power Domain Paths : 2
```


Count	Power Domains
1	PD1->PD2 (L) PD2->PD3->PD4 (D) PD1->PD4 (C)
1	PD1->PD2->PD3 (L) PD3->PD4->PD5->PD3 (D) PD1->PD2->PD3 (C)

The types of crossing in the path is indicated by the following abbreviations:

- C : Capture Path
- L : Launch Path
- D: Data Path

Related Commands

- `set_analysis_mode -enable_ipd`
- `set_ipd_rail_voltage`
- `create_pg_net_group`
- `report_pg_net_group`
- `create_ipd_configuration`
- `report_ipd_configuration`
- `write_ipd_scope`
- `read_ipd_scope`
- `report_timing`

Related Information

- For more information, refer to the [Inter Power Domain \(IPD\) Analysis](#) chapter in the *Tempus User Guide*.

check_ldb_version

```
check_ldb_version  
[-help]  
-ldb string
```

Reports the LDB version, size, and the source of the software that is used to generate the library database (LDB) file. The LDB is the Cadence binary library format.

Parameters

-help	Prints out the command usage.
-ldb <i>string</i>	Specifies the LDB (library database) file whose version is to be printed.

Examples

- The following example displays the missing features/recommendations for the library database

```
cell_w.db:  
tempus > check_ldb_version -ldb cell_w.ldb
```

The output is as shown below:

```
LDB file compiled with tool version: v15.20 ((64bit) mm/dd/yyyy 08:40 (Linux 2.6.18-  
371.el5))  
File Size is xxxxx  
Source Tool: Innovus
```

check_library

```
check_library  
[-help]  
[-outfile fileName]  
[-cellName cellName]  
[-library timingLibraryName]  
[-checkPower]  
[-reportMissingPowerOnly]
```

Checks the contents of the timing library and reports any inconsistencies in a log file.

You can use the `check_library` command after importing the design and timing libraries.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-cellName <i>cellName</i></code>	Checks whether timing is defined for all cells with the specified name in all timing libraries. Note: In general, the <code>-cellName</code> parameter is optional. However, if you specify the <code>-library</code> parameter, then you <i>must</i> specify <code>-cellName</code> .
<code>-checkPower</code>	Checks if power constructs are defined for the cells, such as <code>internal_power</code> , <code>leakage_power</code> , or <code>ecsm_power</code> . Note: In general, the <code>-checkPower</code> parameter is optional. However, if you specify <code>-reportMissingPowerOnly</code> , then you <i>must</i> specify <code>-checkPower</code> . If you use the <code>-library</code> parameter, you do not have to specify the <code>-checkPower</code> parameter, because when a library-specific cell is given as input, the software automatically checks for power constructs. <i>Default:</i> Only checks if timing is defined for the cells.
<code>-library <i>timingLibraryName</i></code>	Specifies the timing library to use when checking a particular cell. You must specify the <code>-cellName</code> parameter with <code>-library</code> to specify the particular library cell to check. Note: If you use the <code>-library</code> parameter, you do not have to specify the <code>-checkPower</code> parameter, because when a library-specific cell is given as input, the software automatically checks for power constructs.
<code>-outfile <i>fileName</i></code>	Writes the report to the specified file.
<code>-reportMissingPowerOnly</code>	Reports only the cells that do not have <code>internal_power</code> (at pin level), or <code>leakage_power</code> defined. You must specify the <code>-checkPower</code> parameter with the <code>-reportMissingPowerOnly</code> parameter.

Examples

- The following command checks if timing is defined for all cells that have been read into the design, and writes a report to the log file:

```
tempus> check_library
```
- The following command checks if timing is defined for all cells that have been read into the design, and writes a report to a file named CheckTiming1:

```
tempus> check_library -outfile CheckTiming1
```

- The following command checks if timing and power constructs are defined for all cells that have been read into the design, and writes the report to the log file:

```
tempus> check_library -checkPower
```

- The following command checks if timing and power constructs are defined for all cells that have been read into the design, and writes the report to a file named CheckTiming2:

```
tempus> check_library -outfile CheckTiming2 -checkPower
```

- The following command checks if timing is defined for all cells with the name A01 in all timing libraries, and writes the report to a file named CheckTiming3:

```
tempus> check_library -outfile CheckTiming3 -cellName A01
```

- The following command checks if timing and power constructs are defined for all cells with the name A01, in all timing libraries, and writes the report to a file named AllA01Check:

```
tempus> check_library -outfile AllA01Check -cellName A01 -checkPower
```

- The following command checks if timing and power constructs are defined for the cell with the name A01 in the timing library minFile1.lib, and writes the report to a file named A01Check:

```
tempus> check_library -outfile A01Check -cellName A01 -library minFile1.lib
```

Note: When you use the `-library` parameter, you do not have to specify the `-checkPower` parameter, because when a library-specific cell is given as input, the software automatically checks for power constructs.

- The following command checks if the `internal_power` and/or `leakage_power` constructs are defined for all cells that have been read into the design, and reports only those cells that do not have the constructs defined. The software writes the report to a file named `MissingPower`:

```
tempus> check_library -outfile MissingPower -checkPower -reportMissingPowerOnly
```

Related Commands

- [set_top_module](#)
- [read_lib](#)

check_timing

```
check_timing
[-help]
[-check_only warning_list]
[-clock_crossing_output_format {summary | csv}]
[-exclude_warning warning_list]
[-include_warning warning_list]
[-tcl_list]
[-type type_list]
[-verbose]
[-view view_name]
[> <file_name[.gz]>]
[>> <file_name[.gz]>]
[port_or_pin_list]
```

Performs a variety of consistency and completeness checks on the timing constraints specified for a design.

The checks include arrival time and external delay (or required time) for each clock in a multiple-clock system. In addition, clock and data connectivity are checked to ensure that clock/data is propagated as expected. The clock gating points are also reported.

The `check_timing` command considers constants applied to a pin while reporting warnings.

 Use the `check_timing` command before using any timing analysis commands. Warnings (listed in the [Table](#)) will be displayed when problems occur. For best results, re-run the `check_timing` command after resolving each warning. Redirecting the output to a file is the quickest way to re-run the command.

Parameters

-help	Prints out the command usage.
-------	-------------------------------

<pre>-check_only warning_list</pre>	Allows you to specify warning types. For example, the following command lists the specified warnings only: <pre>check_timing -check_only {uncons_endpoint clock_crossing}</pre> This list can also include warnings that are not listed by default. Valid entries in the <i>warning_list</i> are: <pre>clock_clipping_freq, clock_clipping_gate, clock_crossing, clock_expected, clock_gating_controlling_edge_unknown, clock_gating_no_data, clock_missing_at_sequential_output, clock_not_propagated, clocks_masked_by_another_clock, const_collision, data_check_multiple_reference_signal, data_check_no_reference_signal, ideal_clock_waveform, invalid_pll_configuration, loop, master_clk_edge_not_reaching, missing_clock_groups, missing_drive, multiple_clock, no_drive, no_gen_clock_source, no_input_delay, partial_input_delay, pulse_non-pulse_clock_merge, signal_level, uncons_endpoint, uncons_endpoints_due_to_case_analysis</pre> <pre>clock_expected, clock_not_propagated, no_gen_clock_source, master_clk_edge_not_reaching, data_check_multiple_reference_signal, multiple_clock, data_check_no_reference_signal, clock_gating_no_data, clock_gating_controlling_edge_unknown, clock_missing_at_sequential_output, clocks_masked_by_another_clock, uncons_endpoint, max_delay_constrained_internal_endpoint, no_input_delay, partial_input_delay, no_drive, loop, ideal_clock_waveform, clock_crossing, clock_clipping_gate, clock_clipping_freq, const_collision, signal_level, constant_clock_pin ,missing_drive, reconvergent_clock, unexpandable_clock and missing_clock_groups</pre>
<pre>- clock_crossing_output_format {summary csv}</pre>	Specifies the format of the clock crossing report to be generated - as a summary or in CSV format.

<pre>{> >>} file_name</pre>	When specified with <code>></code>, writes the report to the specified file name. If the file already exists, the software overwrites it. When specified with <code>>></code>, writes the report to the specified file name. If the file already exists, the software concatenates the report to the end of the file. The <code>filename</code> parameter must be the last argument in the list. The <code>filename</code> and <code>-tcl_list</code> parameters are mutually exclusive; you cannot specify them together. Note: To write a compressed report, add the <code>.gz</code> extension to the file name. Default: Report is displayed on standard output without being saved.
<pre>-exclude_warning warning_list</pre>	Excludes the specified warning types when generating the report. Use this parameter to exclude any of the warnings included in the report by default. By default, the software includes the following warnings in the report: <code>clock_expected</code>, <code>loop</code>, <code>ideal_clock_waveform</code>, <code>master_clk_edge_not_reaching</code>, <code>partial_input_delay</code>, <code>no_input_delay</code>, <code>no_drive</code>, <code>no_gen_clock_source</code>, <code>uncons_endpoint</code>, <code>uncons_endpoints_due_to_case_analysis</code>. The complete list of warnings and their descriptions is given in Table.
<pre>-include_warning warning_list</pre>	Includes the specified warning types when generating the report. Use this parameter to add any of the warnings not included in the report by default. By default, the software reports all warnings except the following: <code>clock_clipping_gate</code>, <code>const_collision</code>, <code>clock_crossing</code>, <code>clock_not_propagated</code>, <code>missing_drive</code>, <code>multiple_clock</code>, <code>data_check_multiple_reference_signal</code>, <code>data_check_no_reference_signal</code>, <code>clock_gating_no_data</code>, <code>clock_gating_controlling_edge_unknown</code>, <code>clock_missing_at_sequential_output</code>, <code>signal_level</code>, <code>clocks_masked_by_another_clock</code>. The complete list of warnings and their descriptions is given in Table.
<pre>port_or_pin_list</pre>	Lists pins or ports. For example: <pre>check_timing -verbose [get_ports in*]</pre>

<code>-tcl_list</code>	Produces the report in TCL list format instead of a tabular format. This is useful for integrating timing with custom TCL functions, and also for customizing report generation. The <code>-tcl_list</code> and <code>filename</code> parameters are mutually exclusive; you cannot specify them together.
<code>-type type_list</code>	Limits the checks to the specified type. <i>Default:</i> Check types that have any violations will be reported. To report all the checks, you can use the <code>timing_check_timing_report_all_checks</code> global variable. When set to <code>true</code>, the software reports all checks performed by the <code>check_timing</code> command, even if no violations are found. By default, this global variable is set to <code>false</code>.
	clock_clipping: Warns when there is a problem with clock gating that can result in clock clipping. Reports the following warnings: • <code>clock_clipping_freq</code> • <code>clock_clipping_gate</code>
	clocks: Reports the following clock warnings: • <code>clock_crossing</code> • <code>clock_expected</code> • <code>clock_not_propagated</code> • <code>ideal_clock_waveform</code> • <code>master_clk_edge_not_reaching</code> • <code>no_gen_clock_source</code>
	clock_gating_inferred: Warns when inferred clock-gating checks are not analyzed by the software, even if a clock signal reaches and is used downstream in the clock tree. Reports the following warnings. • <code>clock_gating_controlling_edge_unknown</code> • <code>clock_gating_no_data</code>

	<code>constant_collision</code>: Warns when there is a constant collision on a net connected to the pin or if there is a contradiction on a pin. Constant collision occurs when a net is driven simultaneously by conflicting values, and a constant contradiction occurs when a constant constraint on a pin conflicts with the driven value.
	<code>endpoints</code>: For output ports, warns if no output delay constraint is applied to the port or if a specific (min/max rise/fall) output delay is missing. It also warns when an output delay constraint is applied without clock information. For all endpoints (output ports and register data pins), it warns if any incoming signal is unconstrained. Also warns if a clock arrives where a clock is not expected, or if multiple signals arrive at an endpoint.
	<code>inputs</code>: Reports the following input delay, arrival time assertion, and drive assertion warnings: • <code>partial_input_delay</code> • <code>missing_drive</code> • <code>no_input_delay</code> • <code>no_drive</code> <code>loops</code>: Warns when a combinational loop is detected. Reports the loop and the arc used to break the loop. The <code>-pin</code> option is ignored in this case. Also reports loops caused by generated clocks.
<code>-verbose</code>	Gives detailed information about the warnings. When specified, warnings are grouped and sorted by warning type. <i>Default:</i> Gives a summary of how many warnings of each type exist.
<code>-view view_name</code>	Reports warning(s) that belong to the specified view.

Command Order

Use this command after setting all the constraints but before generating any timing reports.

Warning Messages

The following table shows the `check_timing` warning messages.

Table: check_timing Warning Messages

Warning	Description
<code>clock_clipping_gate</code>	Warning: Clock clipping possible due to wrong gate type or data input.
<code>clock_clipping_freq</code>	Warning: Clock clipping possible due to incompatible clock signal frequencies.
<code>clock_crossing</code>	Warning: Clock domains interact. Comments: Checks clock interactions between multiple clock domains in the design. When you specify this parameter, the Timing Check Crossing shows the interaction between clocks in the design. It indicates whether the paths between the clocks are <code>False</code> or <code>Partial False</code>. If a clock launches one or more paths, which are captured by the report lists all pairs of crossing clocks. - This includes inferred clock gating with a different reference clock. - If a path has been disabled using the <code>set_disable_timing</code> command, crossing clocks on that path are not reported. If all the paths between two clocks are false paths or if they are exclusive/asynchronous clocks, the path is marked as <code>False</code>. This can be defined on clocks, for example, <code>set_false_path -clocks buf1/Y</code>. If only part of the paths are set as false paths or exclusive/asynchronous clocks, the path is marked as <code>Partial False</code>.
	- If a false path is defined only for one check type (say, setup), the clocks are marked as <code>Partial False</code> if the simultaneous setup hold mode is set to <code>on</code>. The crossing clocks are marked <code>False</code> if the simultaneous setup hold mode is set to <code>off</code>. - If only one edge (rise/fall) of a particular clock pair has false paths (using <code>-clock_fall</code>), then the path is marked as a <code>Partial False</code>.

clock_expected	Warning: Clock not found where clock is expected. Comments: Indicates that no clock signal is defined at the See create_clock.
clock_gating_controlling_edge_unknown	Warning: Unknown controlling edge to perform clock gating Comments: Reports library cells with complex functions (I controlling the edge of the signal to perform a clock gating determined automatically. You need <code>detero</code> specify the correct edge by the <code>-high/-low</code> <code>set_clock_gating_check</code> command to resolve this warning Note: The software does not report this warning when either • <code>set_disable_clock_gating_check</code> command is applied • global settings to disable clock gating checks are on.
clock_gating_no_data	Warning: No data signal reaching clock-gating enable pin Comments: Reports clock gating enable pins where no data arrives. You need to do either of the following: • specify a valid data signal to reach the enable pin of clock • remove any path exceptions or disabled checks that block signal from reaching the enable pin Note: The software does not report this warning when either • <code>set_disable_clock_gating_check</code> is applied on such pin • global settings to disable clock gating checks are on.
clocks_masked_by_another_clock	Warning: Clock(s) in fanin cone are dropped due to another Comments: Reported timing pins on which creation of a new generated clock causes clock(s) in fanins of such pins have masked from propagating further. However, for pins having only a master clock waveform on warning is not reported. You can use the detailed mode (<code>-verbose</code> option) to report incoming clock waveforms that have been blocked.

clock_missing_at_sequential_output	Warning: Found latency phase at sequential output on clock phase Comments: Reports output pins of sequential elements in the clock tree for the generated clocks created downstream and a valid clock phase. This indicates possible missing clock (e.g. clock gating) on downstream clock tree. You can add clocks on this pin or downstream clock tree or a clock exists until the downstream generated clock creation point.
clock_not_propagated	Warning: Clock not propagated Comments: Reports ideal clocks reaching the reference pin check.
const_collision	Warning: Constant collision Comments: See set_case_analysis.
data_check_multiple_reference_signal	Warning: Multiple clocked signals are reaching data check Comments: Reports on reference pins for non sequential check) where more than one reference data arrives.
data_check_no_reference_signal	Warning: No clocked signal is reaching data check reference Comments: Reports reference pins for non- sequential timing check) where no valid reference signal (data phase) arrives. You can do one of the following to resolve such warnings: • add valid reference signals to reach such reference pin • remove path exceptions or disabled arcs that are blocking signals
ideal_clock_waveform	Warning: Clock waveform is ideal. Comments: Reports clocks that are ideal and are not propagating. warning type is reported by default (unless <code>-exclude_warnings</code> parameter is specified). You can use the set_propagated_clock command to avoid
invalid_pll_configuration	Warning: No valid path exists from output pin to feedback pin Comments: Shows PLL (phase locked loop) check warning allow you to identify and fix issues in the PLL flow.

loop	Warning: Timing loop found in the design Comments: Indicates that there are timing loops in the design. This includes loops caused by generated clocks.
master_clk_edge_not_reaching	Warning: Master clock edge does not reach the generated clock Comments: This warning means that rise/fall latency edge is not reached by the master clock. You can create generated clock root from the master clock. You can also check that is seeing this warning is actually broken, or edge is so early on this path. There is a global <code>timing_enable_genclock_edge_based_source_latency</code> that can be found at the Tempus text command reference.
partial_input_delay	Warning: Indicates that a rise or fall input delay assertion is not met
pulse_non-pulse_clock_merge	Warning: Merge of pulse and non-pulse clock paths. Comments: Reports points where a pulse clock and a non-pulse clock merge.
missing_clock_groups	Warning: Set clock group related checking. Comments: Reports any potential issues with the defined clock groups.
missing_drive	Warning: Missing specific (rise/fall) drive constraint Comments: Either a rise or fall drive constraint is applied or missing.
multiple_clock	Warning: Multiple clocks are reaching clock reference pin. Comments: Reports clock pins of sequential or clock gate where more than one clock waveform arrives. This warning is for information only, and the software performs timing across all such clocks.
no_input_delay	Warning: No input delay assertion with or without <code>-clock</code> option Comments: See <code>set_input_delay</code>.
no_drive	Warning: No drive constraint Comments: See <code>set_driving_cell</code> and <code>set_drive</code>.

no_gen_clock_source	Warning: No clock source found for generated clock Comments: Indicates that the source pin of the clock generated by the <code>create_clock</code> command is not driven by a clock source.
signal_level	Warning: Driver pin voltage value does not match the sink Comments: Reports net driver/receiver pin pairs for which the driver pin voltage value does not match the sink pin voltage value.
uncons_endpoint	Warning: No constrained signal reaching pin. Comments: For output ports: Primary output port has no input delay constraint. Use <code>set_output_delay</code>, <code>set_max_delay</code>, or <code>set_min_delay</code> to constrain the output port. For register data pins: There is no valid clock reaching the register data pin. Check for missing <code>create_clock</code> for the clock source. The endpoints are reported as unconstrained in both early and late analysis. This does not cover all cases and scenarios that can cause a signal to become unconstrained. To cover all possible cases, you can use the <code>report_analysis_coverage</code> command.
uncons_endpoints_due_to_case_analysis	Warning: Unconstrained signal arriving at endpoint due to case analysis Comments: Reports unconstrained endpoints due to case analysis. To enable this warning, you must set the <code>timing_report_check_timing_unconstrained_endpoint</code> global variable to <code>true</code>.

Examples

Here are some examples.

- Consider the following example:

```
in-----|>-----|>-----out
```

```
buf1 buf2
```

The Timing Check Crossing Clocks reports False or Partial False in some of the scenarios given below:

- `set_input_delay` with respect to CLK1 on in and `set_output_delay` with respect to CLK2 on out:

From Clock Crossing Clocks

CLK1 CLK2

b) `set_input_delay` with respect to CLK1 on in, `set_output_delay` with respect to CLK2 on out, and `set_false_path -to CLK2` (clock-based false path):

From Clock Crossing Clocks

CLK1 CLK2 (False)

c) If the `set_false_path` becomes `set_false_path -rise_to CLK2`, the following is reported:

From Clock Crossing Clocks

CLK1 CLK2 (Partial False)

The following command displays the default summary report:

`tempus> check_timing`

TIMING CHECK SUMMARY		
Warning	Warning Description	Number of Warnings
<code>ideal_clock_waveform</code>	Clock waveform is ideal	1
<code>clock_expected</code>	Clock not found where clock is expected	3
<code>master_clk_edge_not_reaching</code>	Master clock edge does not reach the generated clock target	2
<code>no_drive</code>	No drive assertion	8
<code>no_input_delay</code>	No input delay assertion with respect to clock	4
<code>uncons_endpoint</code>	Unconstrained signal arriving at end point	5

- The following command displays a specified type of warning using the `-type` option if a timing loop is found in the design:

`tempus> check_timing -type loops`

TIMING CHECK SUMMARY		
Warning	Warning Description	Number of

Warnings

```
-----
loop          Timing loop found in the design  0
-----
```

- The following command provides more information about problems and prints a summary using the `-verbose` option:

```
tempus> check_timing -verbose
```

TIMING CHECK SUMMARY

Warning	Warning Description	Number of Warnings
ideal_clock_waveform	Clock waveform is ideal	1
clock_expected	Clock not found where clock is expected	3
master_clk_edge_not_reaching	Master clock edge does not reach the generated clock target	2
no_drive	No drive assertion	8
no_input_delay	No input delay assertion with respect to clock	4
uncons_endpoint	Unconstrained signal arriving at end point	5

TIMING CHECK DETAIL

Pin	Warning
u99/CK	Clock not found where clock is expected
u5/CK	Clock not found where clock is expected
u8/CK	Clock not found where clock is expected
u8/Q	Master clock edge does not reach the target for generated clock GCLK4
CLK1	No drive assertion
in1	No drive assertion
in1	No input delay assertion with respect to clock
out1	Unconstrained signal arriving at end point

TIMING CHECK IDEAL CLOCKS

```
-----
Clock  Waveform
-----
CLK1
-----
```

- When `timing_report_check_timing_unconstrained_endpoints_due_to_constants` global variable is enabled, the case analysis warnings are reported as shown below:

```
tempus 11> check_timing
```

TIMING CHECK SUMMARY		

Warning	Warning Description	Number of Warnings

clock_expected	Clock not found where clock is expected	4202
no_drive	No drive assertion	241
no_input_delay	No input delay assertion with respect to clock	160
uncons_endpoint	Unconstrained signal arriving at end point	145259
uncons_endpoints_due_to_case_analysis		
	>>>>Unconstrained signal arriving at end point due to case analysis	
		5969437

Related Commands

- `report_timing`
- `set_input_delay`
- `create_clock`
- `set_case_analysis`
- `set_output_delay`
- `set_max_delay`
- `set_min_delay`
- `set_drive`
- `set_driving_cell`
- `set_false_path`
- `set_clock_gating_check`
- `set_disable_clock_gating_check`

compare_timing_context

```
compare_timing_context
[-help]
[-abs_thresh float]
[-instances string]
[-module string]
-out_file string
[-pct_thresh float]
-reference string
[-sort_by {abs | pct}]
-target string
[-type {clocks | latency | arrival | required | slew | load | constant | glitch}]
[-verbose]
[-view string]
```

Compares two timing contexts generated by Tempus.

The contents of context, such as clocks, clock latency, arrival time, required, slew, load, constant, and glitch, are compared. To perform context comparison, it is required to generate context using the `-include_text_format` parameter of the `write_timing_context` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-abs_thresh float</code>	Specifies the threshold value to compare the context data. The absolute threshold value is specified in picoseconds (ps). <i>Default:</i> 0
<code>-instances string</code>	Specifies the list of instances for which context comparison will be done.
<code>-module string</code>	Specifies the list of modules for which context comparison will be done.
<code>-out_file string</code>	Specifies the output file that will show the differences between the reference and target context data.
<code>-pct_thresh float</code>	Specifies the percentage threshold as a percentage (%). If the difference is greater than both the specified <code>-abs_thresh</code> and <code>-pct_thresh</code> thresholds, the software considers the comparison as “Fail”. <i>Default:</i> 0
<code>-reference string</code>	Specifies the reference path of the context directory.
<code>-sort_by {abs pct}</code>	Sorts the report by absolute or percent tolerance. By default, the report is sorted by percentage difference.
<code>-target string</code>	Specifies the name of the target context directory used for comparison. Note: The missing entries in the target or reference will be represented by “-”.
<code>-type {clocks latency arrival required slew load constant glitch}</code>	Specifies the type of text data to be compared. The <code>latency</code> option includes comparison of the latency, uncertainty, or incremental delay values. By default, all the text data values are compared.
<code>-verbose</code>	Reports both failed and passed comparison data.
<code>-view string</code>	Specifies the MMMC analysis view for which context comparison is required.

Context Comparison Report Sections and Examples

The following are the sections of the context comparison report, along with examples:

Section 1: Comparison Summary

Shows the summary of total and failed entries/ports in the reference and target context comparison. The header contains the view, module, and instance for which the comparison is reported.

```
-----
Top View: view1
Module: Top
Top Instance: inst1
-----
```

Comparison Summary

Section	Failed Entries	Total Entries	Failed Ports	Total Ports
Clocks	0	1	0	0
Constant Value	0	3	0	3
Latency	48	48	2	2
Arrival Time	0	0	0	0
Required	0	0	0	0
Input Transition	0	12	0	3
Output Load	0	4	0	1
Glitch	2	8	1	3

Section 2: Detailed Clock Comparison

This section provides the details of differences in clock definitions between the reference and target contexts.

Clocks Comparison

Clock Status	Ref Waveform	Ref Period	Target Waveform	Target Period	
CK1	[0 0.000 0.500]	1.000	[0 0.000 0.500]	1.000	Pass

Section 3: Detailed Constant Value Comparison

This section provides the details of differences in constants between the reference and target contexts.

Constants Comparison

Port	Ref Constant	Target Constant	Status
------	--------------	-----------------	--------

prt1 0 1 Fail

Section 4: Detailed Clock Latency Comparison

Provides details of differences in clock latency, pin-based uncertainty, and incremental delays between the reference and target contexts.

Clock Latency Comparison

Port	Clock	Type	Data	Ref	Target	Abs Diff	% Diff	Status
P1	TCK1(P)	latency	early_rise	10.12	10.23	0.11	1.08	Fail
P1	TCK1(P)	incr_delay	early_rise	0.2	0.2	0	0	Pass
P1	TCK1(P)	uncertainty	early_rise	0.1	0.2	0.1	100	Fail
P1	TCK1(P)	latency	late_rise	11.12	11.12	0	0	Pass

Section 5: Detailed Input Arrival Comparison

Provides details of differences in input arrivals between the reference and target contexts.

Input Arrival Comparison

Port	Clock	Data	Ref	Target	Abs Diff	% Diff	Status
test_scanpt	portclk1(P)	early_rise	1206.32	1214.33	8.0017	0.6633	Fail
test_scanpt	portclk0(P)	early_rise	1206.32	1214.33	8.0017	0.6633	Fail
test_scanpt	portclk1(P)	early_fall	1202.58	1210.53	7.9497	0.6610	Fail
test_scanpt	portclk0(P)	early_fall	1202.58	1210.53	7.9497	0.6610	Fail

Section 6: Detailed Required Time Comparison

This section provides the details of differences in required time between the reference and target contexts.

Required Time Comparison

Port	Clock	Data	Ref	Target	Abs Diff	% Diff	Status
tst_scopt	ekph1(P)	early_fall	4.2486	3.2270	-1.0216	-24.0456	Fail
tst_scopt	ekph1(P)	early_rise	-5.9780	-7.0736	-1.0956	18.3272	Fail
tst_scopt	emph1(N)	early_fall	-13.8087	-15.9599	-2.1512	15.5786	Fail
tst_scopt	emph1(N)	early_rise	-23.0335	-25.2794	-2.2459	9.7506	Fail

Section 7: Detailed Input Port Transition Comparison

Provides the details of differences in transition times on input ports between the reference and target contexts.

Input Transition Comparison

Port	Data	Ref	Target	Abs Diff	% Diff	Status
clk2	early_rise	1.510	1.510	0.000	0.000	Pass
clk2	early_fall	0.772	0.772	0.000	0.000	Pass

Section 8: Detailed Output Port Load Comparison

Provides the details of differences in output port load between the reference and target contexts.

Output Load Comparison

Port	Data	Ref	Target	Abs Diff	% Diff	Status
Out	early_rise	171.374	171.374	0.000	0.00	Pass
Out	early_fall	171.374	171.374	0.000	0.000	Pass

Section 9: Detailed Glitch Comparison

Provides the details of differences in glitch value on the pins connected to the input ports between the reference and target contexts.

Glitch Comparison

Port	Pin Name	Data	Ref	Target	Abs Diff	% Diff	Status
clk2	ff3/CK	VH	96.552	96.611	0.059	0.061	Fail
clk2	ff3/CK	VL	95.742	95.776	0.034	0.036	Fail

Related Commands

- [read_timing_context](#)
- [write_timing_context](#)
- [report_timing_context_clock_mapping](#)

create_analysis_view

```
create_analysis_view
[-help]
-name viewName
-constraint_mode modeName
-delay_corner dcCornerObj
```

Creates an analysis view object that associates a delay calculation corner with a constraint mode. An analysis view object provides all of the information necessary to control a single multi-mode multi-corner analysis.

After creating analysis views, use the [set_analysis_view](#) command to specify which views to use for setup and hold optimization or timing analysis.

Use this command after creating constraint modes ([create_constraint_mode](#)) and delay calculation corners ([create_delay_corner](#)).

Parameters

-help	Prints out the command usage.
-name <i>viewName</i>	Specifies the name of the analysis view being defined.
- constraint_mode <i>modeName</i>	Specifies the name of the constraint mode to associate with this analysis view. A constraint mode groups a set of constraints for use with a given analysis. To create the constraint mode, use the create_constraint_mode command.
-delay_corner <i>dcCornerObj</i>	Specifies the name of the delay calculation corner object to associate with this analysis view. A delay calculation corner object contains all of the information needed to control delay calculation for a specific analysis view. To create a delay calculation corner, use the create_delay_corner command.

Examples

- The following command creates an analysis view called `missionSlow` using the constraint mode `missionSetup` and the delay calculation corner `dcWCCOM`:

```
tempus> create_analysis_view
-name missionSlow
-constraint_mode missionSetup
-delay_corner dcWCCOM
```
- The following command creates an analysis view called `missionFast` using the constraint mode

missionHold and the delay calculation corner dcBCCOM:

```
tempus> create_analysis_view  
-name missionFast  
-constraint_mode missionHold  
-delay_corner dcBCCOM
```

Related Commands

- `set_analysis_view`
- `create_constraint_mode`
- `create_delay_corner`

create_boundary_model

```
create_boundary_model  
[-help]  
-dir string  
[-exclude_no_delay_ports]  
[-include_full_parasitics]  
[-include_userpins string]  
[-latch_effort_low]  
[-overwrite]  
[-skip_constant_ports]  
[-skip_constants]  
[-skip_disabled]  
[-skip_ports string]
```

Initiates the creation of a boundary model from a full design.

Parameters

-help	Prints out the command usage.
-dir <i>string</i>	Specifies the directory for boundary model files.
-exclude_no_delay_ports	Skips tracing logic through unconstrained input ports in the boundary model.
-include_full_parasitics	Allows you to write out RCDB parasitics for all lower-level blocks as part of the current boundary model. When specified, the software writes out full parasitics by running the write_rcdb command under the hood.
-include_userpins <i>string</i>	Includes the specified pins in the boundary model. No fanin/fanout tracing is performed for the specified pins.

<code>-latch_effort_low</code>	Skips tracing of logic through latch aggressors that are not part of the interface logic. The aggressors in the fanout of latch aggressors will use an infinite timing window. By default, the entire fanin cone of latch aggressors is preserved to ensure accurate latency adjustment, which may increase the size of the boundary model.
<code>-overwrite</code>	Allows overwrite of the existing directory.
<code>-skip_constant_ports</code>	Skips tracing logic through constant ports in the boundary model.
<code>-skip_constants</code>	Skips tracing through constant logic in the boundary model.
<code>-skip_disabled</code>	Skips tracing of logic through disabled arcs (user or constant disabled) in the boundary model.
<code>-skip_ports <i>string</i></code>	Skips tracing fanin/fanout logic through specified ports in the boundary model.

Examples

- The following command creates a boundary model for the top cell being timed and writes the model files into the `./model_dir` directory. Any existing directory contents are removed.

```
create_boundary_model -dir ./model_dir -overwrite
```

Related Commands and Variables

- `read_boundary_model`

create_constraint_mode

```
create_constraint_mode
[-help]
-name modeName
-sdc_files {file1.sdc file2.sdc ...}
[-ilm_sdc_files {file1.sdc file2.sdc ...}]
[-tcl_vars {{var_name1 value1} {var_name2 value2} {var_name3 value3} ...}]
```

Associates a list of SDC constraint files with a specified constraint mode name, for multi-mode multi-corner analysis. This constraint mode name can be referred to later when creating analysis views. A constraint mode defines one of the possible functional, test, or Dynamic Voltage and Frequency Scaling (DVFS) modes of a design. Many different constraint modes can share SDC files, and the same constraint mode can be associated with multiple analysis views.

SDC files typically contain timing analysis information, such as the clock specifications, case analysis constraints, I/O timings, and path exceptions that make each mode unique. If the ILM is being used, the timing system uses the files specified with `-ilm_sdc_files` during the flatten ILM mode; otherwise, it uses the files specified with `-sdc_files`.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-ilm_sdc_files {file1.sdc file2.sdc ...}</code>	Specifies a TCL list of ILM SDC files to be included in the constraint mode.
<code>-name modeName</code>	Specifies the name of the constraint mode to be created.
<code>-sdc_files {file1.sdc file2.sdc ...}</code>	Specifies a TCL list of SDC files to be included in the constraint mode.
<code>-tcl_vars {{var_name1 value1} {var_name2 value2} {var_name3 value3} ...}</code>	Specifies a list of TCL variables and values to be used in the constraint mode.

Examples

- The following command groups the SDC files `io.sdc`, `mission1-clks.sdc`, and `mission1-except.sdc` to create a mode object named `missionSetup`:

```
tempus> create_constraint_mode -name missionSetup
      -sdc_files [list io.sdc mission1-clks.sdc mission1-except.sdc]
```

- The following command groups the SDC files `io.sdc`, `mission1-clks.sdc`, `mission1-except.sdc`, and `dont_use.sdc` to create a mode object named `missionHold`:

```
tempus> create_constraint_mode -name missionHold  
      -sdc_files [list io.sdc mission1-clks.sdc mission1-except.sdc dont_use.sdc]
```

- The following command groups the SDC files `test-io.sdc`, `test-clks.sdc`, and `test-except.sdc` to create a mode object named `testHold`:

```
tempus> create_constraint_mode -name testHold  
      -sdc_files [list test-io.sdc test-clks.sdc test-except.sdc]
```

Related Commands

- `update_constraint_mode`
- `create_analysis_view`

create_delay_corner

```
create_delay_corner
[-help]
[-early_estimated_worst_irdrop_factor <percent>]
-name delayCornerName
[-si_enabled true | false]
{[-library_set libSetObj] | [-late_library_set libSetObj -early_library_set libSetObj]}
[-opcond_library libName]
[-opcond opcondName]
[-pg_net_voltages voltage_pairs]
[-rc_corner rcCornerObj | [-late_rc_corner rcCornerObj -early_rc_corner rcCornerObj]]
[-irdrop_file list_of_files | -irdrop_data files_or_directories]
[-late_opcond_library libName]
[-early_estimated_worst_irdrop_factor percent]
[-early_opcond_library libName]
[-late_estimated_worst_irdrop_factor percent]
[-late_opcond opCondName]
[-early_opcond opCondName]
{[-late_irdrop_file list_of_files -early_irdrop_file libSetObj] | [-late_irdrop_data files_or_directories -early_irdrop_data files_or_directories]}
[-temp_file tempName | [-late_temp_file tempName -early_temp_file tempName]] }
}
```

Creates a named delay calculation corner object that can be referenced later when creating an analysis view. A delay calculation corner provides all of the information necessary to control delay calculation for a specific view. Each corner contains information on the libraries to use, the operating conditions for accessing them, and the RC extraction parameters to use for calculating parasitic data. Multiple top-level analysis views can share delay corner objects.

Use separate delay-calculation corners to define the Best-Case and Worst-Case differences. Use the `-early_*` and `-late_*` parameters within a single delay calculation corner to control on-chip variation.

Note: A single delay-calculation corner object specifies the delay-calculation rules for the entire design. If a design includes power domains, the delay calculation corner can contain domain-specific subsections that specify the required operating condition information, and any necessary timing library rebinding for the power domain. Use the `update_delay_corner` command to add a power domain definition to a delay calculation corner.

Parameters

<code>-help</code>	Prints out the command usage.
--------------------	-------------------------------

- early_estimated_worst_irDrop_factor <i>percent</i>	Specifies the early voltage derating factor. When this parameter is specified, the software performs delay calculations using the voltage = $\text{nomVolt} * (1 - \text{value})$. <i>Default:</i> 0.0
-early_estimated_worst_irdrop_factor <i>percent</i>	Specifies the early voltage derating factor.
-early_irdrop_data <i>files_or_directories</i>	Specifies the IR drop files or directories to apply when calculating the early arrival times at a single delay corner.
-early_irdrop_file <i>list_of_files</i>	Specifies the list of IR drop files to apply when calculating early arrival times at a single delay corner. This parameter is primarily used when configuring the multi-mode, multi-corner environment for on-chip-variation (OCV) analysis. Note: This parameter is obsolete and will be removed in a future release. To ensure compatibility with future releases, you should update your scripts.
-early_library_set <i>libSetObj</i>	Specifies the library set to associate with this delay corner object for calculating early arrival times at a single delay corner. This parameter is primarily used when configuring the multi-mode multi-corner environment for on-chip-variation (OCV) analysis.
-early_opcond <i>opcondName</i>	Specifies the operating condition to use for calculating early arrival times at a single delay corner. This parameter is primarily used when configuring the multi-mode multi-corner environment for on-chip-variation (OCV) analysis. <i>Default:</i> Each library in the early library set uses its own default operating condition.

<code>-early_opcond_library libName</code>	Specifies the internal library name for the library in which the early operating condition is defined. This is <i>not</i> the library file name. This parameter is primarily used when configuring the multi-mode multi-corner environment for on-chip-variation (OCV) analysis. <i>Default:</i> The software searches the early library set for the specified operating condition (<code>-early_opcond</code>), starting with the master library.
<code>-early_rc_corner rcCornerObj</code>	Specifies the RC corner object to associate with the early corner object.
<code>-early_temp_file tempName</code>	Specifies the name of the temperature file for temperature-aware delay calculation for the early corner.
<code>-irdrop_data files_or_directories</code>	Specifies the IR drop files or directories to apply to both the early and late delay calculation for the specified delay corner object.
<code>-irdrop_file list_of_files</code>	Specifies the list of IR drop files to apply to both early and late delay calculation for this delay corner object. This parameter is primarily used to configure the single-corner or Best-Case Worst-Case (BC-WC) analysis modes. Note: This parameter is obsolete and will be removed in a future release. To ensure compatibility with future releases, you should update your scripts.
<code>-early_estimated_worst_irdrop_factor percent</code>	Specifies the late voltage derating factor. When this parameter is specified, the software performs delay calculations using the voltage = $\text{nomVolt} * (1 - \text{value})$. <i>Default:</i> 0.0
<code>-late_estimated_worst_irdrop_factor <percent></code>	Specifies the late voltage derating factor. When this parameter is specified, the software performs delay calculations using the voltage = $\text{nomVolt} * (1 - \text{value})$. <i>Default:</i> 0.0

<code>-late_irdrop_data files_or_directories</code>	Specifies the IR drop files or directories to apply when calculating the late arrival times at a single delay corner.
<code>-late_irdrop_file list_of_files</code>	Specifies the list of IR drop files to apply when calculating late arrival times at a single delay corner. This parameter is primarily used when configuring the multi-mode multi-corner environment for on-chip-variation (OCV) analysis. Note: This parameter is obsolete and will be removed in a future release. To ensure compatibility with future releases, you should update your scripts.
<code>-late_library_set libSetObj</code>	Specifies the library set to associate with this delay corner object for calculating late arrival times at a single delay corner. This parameter is primarily used when configuring the multi-mode multi-corner environment for on-chip-variation (OCV) analysis.
<code>-late_opcond opcondName</code>	Specifies the operating condition to use for calculating late arrival times at a single delay corner. This parameter is primarily used when configuring the multi-mode multi-corner environment for on-chip-variation (OCV) analysis. <i>Default:</i> Each library in the late library set uses its own default operating condition.
<code>-late_opcond_library libName</code>	Specifies the internal library name for the library in which the late operating condition is defined. This is <i>not</i> the library file name. This parameter is primarily used when configuring the multi-mode multi-corner environment for on-chip-variation (OCV) analysis. <i>Default:</i> The software searches the late library set for the specified operating condition (<code>-late_opcond</code>), starting with the master library.
<code>-late_rc_corner rcCornerObj</code>	Specifies the RC corner object to associate with the late corner object.

<code>-late_temp_file tempName</code>	Specifies the name of the temperature file for temperature aware delay calculation for the late corner.
<code>-library_set libSetObj(s)</code>	Specifies the library set(s) to associate with this delay corner object. To configure an ECSM-based multi-library interpolation, you should specify a Tcl list of library_set, such as: <pre>create_delay_corner -name dc_P1_T1 -library_set {p1_v1_t1 p1_v2_t1 p1_v3_t1}</pre>
<code>-name delayCornerName</code>	Specifies the name for the delay corner object being created.
<code>-opcond opcondName</code>	Specifies the operating condition to use for setup and hold analysis. This parameter is primarily used to configure the single-corner or Best-Case Worst-Case (BC-WC) analysis modes. <i>Default:</i> Each library in the library set uses its own default operating condition.
<code>-opcond_library libName</code>	Specifies the internal library name for the library in which the operating condition is defined. This is <i>not</i> the library file name. This parameter is primarily used to configure the single-corner or Best-Case Worst-Case (BC-WC) analysis modes. <i>Default:</i> The software searches the library set for the specified operating condition (<code>-opcond</code>), starting with the master library.
<code>-pg_net_voltages voltage_pairs</code>	Specifies the power/ground net voltages to be used for both early/late timing analysis.
<code>-rc_corner rcCornerObj</code>	Specifies the RC corner object to associate with this delay corner object. This parameter is primarily used to configure the OCV and Best-Case Worst-Case (BC-WC) analysis modes.

<pre>-si_enabled {true false}</pre>	Controls signal integrity (SI) analysis. When set to <code>true</code>, the analysis of all MMMC views will include SI analysis by default. You can use this parameter to determine further which analysis views receive SI analysis and which do not. When this parameter is set to <code>false</code> on a delay corner, the software disables SI analysis for all the views that reference that corner. By default (or when this parameter is set to <code>true</code>), SI analysis is computed for all the views related to that corner - whenever SI is globally enabled. (The global control of signal integrity (SI) analysis is determined by the <code>set_delay_cal_mode -SIAware</code> setting.)
<pre>-temp_file tempName</pre>	Specifies the name of the temperature file for temperature aware delay calculation for a single corner.

Examples

- The following command creates a delay calculation corner called `dcWCCOM`. This corner uses the libraries from `IsCOM-1V`, sets the operating condition to `WCCOM`, as defined in the `stdcell_1V` timing library, and uses the `rc-cworst` RCcorner:


```
tempus> create_delay_corner -name dcWCCOM
-library_set IsCOM-1V
-opcond_library stdcell_1V
-opcond WCCOM
-rc_corner rc-cworst
```
- The following command creates a delay calculation corner called `dcBCCOM`. This corner uses the libraries from `IsCOM-1V`, sets the operating condition to `BCCOM`, as defined in the `stdcell_1V` timing library, and uses the `rc-cbest` RC corner:


```
tempus> create_delay_corner -name dcBCCOM
-library_set IsCOM-1V
-opcond_library stdcell_1V
-opcond BCCOM
-rc_corner rc-cbest
```
- The following commands create best-case (`dcBCCOM-OCV`) and worst-case (`dcWCCOM-OCV`) delay calculation corners. The OCV variance in each corner is due to different early and late libraries and operating conditions within each delay corner.

```
tempus> create_delay_corner -name dcBCCOM-OCV
-late_library_set BCCOM-SLOW
-early_library_set BCCOM-FAST
-late_opcond_library BCCOM-SLOW
-early_opcond_library BCCOM-FAST
-late_opcond SLOW
-early_opcond FAST
create_delay_corner -name dcWCCOM-OCV
-late_library_set WCCOM-SLOW
-early_library_set WCCOM-FAST
-late_opcond_library WCCOM-SLOW
-early_opcond_library WCCOM-FAST
-late_opcond SLOW
-early_opcond FAST
```

Related Commands

- [get_delay_corner](#)
- [update_delay_corner](#)

create_glitch_boundary_model

```
create_glitch_boundary_model  
[-help]  
[-add_bound_lib_report]  
[-bound_lib_report <string>]  
-dir string  
[-max_stages integer]  
[-overwrite]  
[-skip_constant_ports]  
[-skip_constants]  
[-skip_ports string]
```

Creates a boundary model for the current design for performing glitch analysis.

Parameters

<code>-help</code>	Prints out the command usage.
<code>- add_bound_lib_report</code>	Allows writing a report into the model directory as <code>bound_libs.rpt.gz</code> .
<code>-bound_lib_report <string></code>	Specifies the name of a file to which a report will be written if the bound library/libraries for every instance are included in the model. When this parameter is not specified, a report will not be written.
<code>-dir string</code>	Specifies the directory where the model will be saved. If a directory does not exist, it will be created. If it already exists, and the <code>-overwrite</code> parameter is specified, then the existing directory will be deleted first; otherwise, an error will be displayed.
<code>-max_stages integer</code>	Specifies the maximum number of levels of interface logic to keep from the boundary ports. All attackers on any net along the interface logic path will be kept with the attacker's driver and receiver. When this parameter is specified, the glitch at the interface input paths to the output pins will be retained and fed into the block's internal nets. <i>Default: 2</i>
<code>-overwrite</code>	Allows the software to replace the contents of any existing directory with the generated boundary model.
<code>-skip_constant_ports</code>	Ignores the timing endpoints driven exclusively by constant ports.
<code>- skip_constants</code>	Allows the software to skip any logic that is constant across all views during forward tracing from input ports to find timing endpoints. In addition, the fanin logic is not traced for output ports that are constant in all the views.
<code>-skip_ports string</code>	Specifies primary ports of a block that should be skipped when tracing the interface paths to be included in the model. The ports can be specified as a port name or a list of port names. The forward tracing to find endpoints will not be performed from skipped input ports, and fanin tracing will not be performed from skipped output ports. Note that timing endpoints reachable from both a skipped and a non-skipped input port will be included in the model. In such a case, the skipped input port will be included in the model since it is a part of the fanin logic of the endpoint. Note: This parameter does not support collections.

Examples

- The following command creates a boundary model in the specified directory and overwrites any existing directory contents:

```
create_glitch_boundary_model -dir model_dir -overwrite
```

- The following command creates a boundary model that keeps up to 6 stages of logic from the input ports and to the output ports of the design:

```
create_glitch_boundary_model -dir model_dir -max_stages 6
```

- The following command creates a boundary model that does not include any logic constrained as constant:

```
create_glitch_boundary_model -dir model_dir -skip_constants
```

- The following command creates a boundary model that does not include the fanout logic from input port “scan_in” or fanin logic to the output port “scan_out”:

```
create_glitch_boundary_model -dir model_dir -skip_ports {scan_in scan_out}
```

Related Commands

- [read_boundary_model](#)
- [create_boundary_model](#)

create_ipd_configuration

```
create_ipd_configuration
[-help]
[-no_commit | -load_constraints ]
[-no_commit | -load_derates ]
[-setup | -hold ]
[-setup | -bounded_corners ]
[-hold | -bounded_corners ]
[-min_bounded_corner | -setup | -bounded_corners | -max_bounded_corner ]
[-max_bounded_corner | -hold | -bounded_corners | -min_bounded_corner ]
[-min_bounded_corner | -pd_list <string>]
[-max_bounded_corner | -pd_list <string>]
[-hold | -pd_list <string>]
[-setup | -pd_list <string>]
[-bounded_corners | -pd_list <string>]
```

Creates IPD combinations or the analysis views in both the bounded scope generation and the scope analysis session.

This command automatically generates the views based on the `set_ipd_rail_voltage` settings and loads the view definition for IPD analysis.

Parameters

-help	Prints out the command usage.
-bounded_corners	Creates the minimum/maximum bounding voltage combinations for analysis. This parameter is used only in the scope creation run.
-load_constraints	Loads the constraints stored in the IPD scope during the scope analysis run. (Applicable only in IPD scope analysis.)
-load_derates	Loads the derates stored in the IPD scope during the scope analysis run. (Applicable only in the IPD scope analysis.)
-no_commit	Creates the IPD view definition file named <code>ipdViewDefinition.tcl</code> - but does not load it. By default, the IPD view definition is automatically read in. This parameter is mutually exclusive to the <code>-load_derates</code> and <code>-load_constraints</code> parameters.

Examples

- The following example shows the usage of the `create_ipd_configuration` command in a scope analysis run.

```
create_ipd_configuration -load_derates -load_constraints
```

The `-load_constraints` and `-load_derates` parameters are optionally used in the scope analysis session to load the constraints and derates stored in the IPD scope. This IPD rail voltage specification creates 16 (2^4) views corresponding to 4 PG nets.

- The following command creates bounding IPD views:

```
tempus > set_ipd_rail_voltage -sweep { PGN1 {0.7 0.8} PGN2 {0.9 1.0} PGN3 {1.1 1.2} PGN4  
{0.6 0.7} }  
create_ipd_configuration -bounded_corners
```

This will create bounding IPD views with the following voltage combinations:

- minimum bounding view: PD1=0.8v PD2=1.0v PD3=1.2v PD4=0.7v
- maximum bounding view: PD1=0.7v PD2=0.9v PD3=1.1v PD4=0.6v

Related Commands

- `set_ipd_rail_voltage`
- `create_pg_net_group`
- `report_pg_net_group`
- `report_ipd_configuration`
- `write_ipd_scope`
- `read_ipd_scope`
- `report_ipd_crossings`
- `check_ipd_logic`

Related Information

- For more information, refer to the [Inter Power Domain \(IPD\) Analysis](#) chapter in the *Tempus User Guide*.

create_library_set

```
create_library_set
[-help]
[-aocv string]
[-library_side_file string]
-name libSetName
[-si string]
[-socv string]
[-timing string]
```

Associates a TCL list of timing, AOCV, SOCV, and cdB/UDN libraries with a specified library set name. Library sets allow a group of library files to be treated as a single entity, so that higher-level descriptions (delay calculation corners) can simply refer to the library configuration by name. The same library set can be referenced multiple times by different delay calculation corners. You can create library sets to cover different process corners, different operating voltages for MSMV designs, or for ECSM-based multi-library interpolation.

Note: If you are using the `read_lib`, `read_verilog`, and `set_top_module` commands to specify the design information, you must use the `read_lib` command to read in the basic set of libraries, then specify the `set_top_module` command, before you can use the `create_library_set` command.

Timing libraries can be in text format or Cadence binary library format (LDB). The LDB (library database) files generated from other tools, like Innovus, Voltus, or Genus are read in to Tempus with an error message (TECHLIB-1249). The LDB files are not forward compatible, that is, a LDB generated from 15.2 version cannot be used with 15.1 or prior versions of the software.

Note: The `create_library_set` command determines the priority of ECSM vs CCS data (if both are present). The software first checks for ECSM construct, if it is missing then CCS construct is loaded.

 The order in which you define timing libraries is important. The software considers the first library you specify in the list to be the master library, with each subsequent library having lower priority.

Parameters

-help	Prints out the command usage.
-aocv <i>string</i>	Specifies the AOCV libraries to include in the library set.
-library_side_file <i>string</i>	Specifies the library side file.

<code>-name libSetName</code>	Specifies the name for the library set being created. This name is used to associate the library list with a specific delay calculation corner.
<code>-si string</code>	Specifies cdB libraries and/or user-defined noise (UDN) models to include in the library set. The cdB and/or UDN libraries are required for performing signal integrity analysis. Note: You cannot use this parameter for CCS-N or ECSM-N models, that is, for cdBs only.
<code>-socv string</code>	Specifies the SOCV libraries to include in the library set.
<code>-timing string</code>	Specifies the timing libraries to include in the library set. Note: Libraries intended for use in an ECSM-based multi-library interpolation must be grouped within an additional level of Tcl list structure to identify the intended usage to the library subsystem. Specify three timing libraries that are characterized at different voltages (typically the lower and upper voltage bounds, and a third intermediate point). All voltages required by the software will be interpolated from this set of libraries. The libraries in the set can differ only in the characterization voltage; they must be identical in all other aspects. However, if you want to perform multi-temperature-based library interpolation, you can use two library files that are characterized at different temperatures for each voltage value. See the examples for creating an ECSM-based multi-library set and multi-temperature-based library interpolation in the Examples section.  The ECSM and CCS libraries can be very large, and loading them may take significant time. You can convert the ASCII NLDM (Liberty .lib), ECSM, and CCS libraries to a binary version using the <code>write_ldb</code> command. For converting libraries to a binary format, refer to <code>write_ldb</code>. Instead of the original ASCII libraries, you can specify these binary libraries in the configuration file: • Original MMMC setup file: <code>create_library_set -name libName -timing "AAA.lib BBB.ecsm CCC.ccs"</code> • New MMMC setup file: <code>create_library_set -name libName -timing "AAA.ldb BBB.ldb CCC.ldb"</code>

Examples

- The following command creates a library set that associates timing libraries and cdB libraries with a nominal voltage of 1 volt, with the library name IsCOM-1V:

```
tempus > create_library_set
-name IsCOM-1V -timing [list stdcell_F_1V.lib ram_F.lib pad.lib]
-si [list stdcell_F_2.cdb ram_F.cdb pad.cdb]
```

- The following command creates a library set that can be used to specify the 2-volt version of a core cell library, which could be associated later with a 2-volt MSMV power domain specified in the design:

```
tempus > create_library_set
-name IsCOM-2V
-timing [list stdcell_F_2V.lib]
```

- The following command creates an ECSM-based multi-library set called triLibFast. The libraries are grouped within an additional level of Tcl list structure to identify the intended usage to the library subsystem.

```
tempus > create_library_set
-name triLibFast
-timing [list
    [list stdcell_F_1V.lib stdcell_F_2V.lib stdcell_F_3V.lib]
    ram_F.lib pads.lib
]
```

- The following command performs multi-temperature-based library interpolation for libraries characterized at 0.7, 1.0, and 1.1 volts, each of which is also characterized using -40 and -35 temperature values:

```
tempus > create_library_set
-name Trilib
-timing {{ stdcell_0.7_temp-40.lib stdcell_0.7_temp-35.lib stdcell_1.0_temp-40.lib
stdcell_1.0_temp-35.lib stdcell_1.1_temp-40.lib stdcell_1.1_temp-35.lib } mem.lib io.lib
}
```

Related Commands

- `get_library_set`
- `update_library_set`
- `read_lib`
- `read_verilog`
- `set_top_module`

create_op_cond

```
create_op_cond
[-help]
-name virtualOpcondName
-library_file libraryFileName
-P processValue
-V voltageValue
-T temperatureValue
```

Creates a set of virtual operating conditions in the specified library without actually modifying the library. These virtual operating conditions can then be referenced by a delay calculation corner ([create_delay_corner](#)) as if they actually exist in the library.

Parameters

-help	Prints out the command usage.
-library_file libraryFileName	Specifies the library to use when creating virtual operating conditions.
-name virtualOpcondName	Specifies the name for the operating condition being created.
-P processValue	Specifies the process value for the operating condition.
-T temperatureValue	Specifies the temperature value for the operating condition.
-V voltageValue	Specifies the voltage value for the operating condition.

Examples

- The following command creates a virtual operating condition called `PVT1` for the library `IsCOM-1V.lib`:

```
tempus> create_op_cond -name PVT1
      -library_file IsCOM-1V.lib
      -P 1.0
      -V 1.2
      -T 120
```
- The following command creates a virtual operating condition called `PVT2` for the library `IsCOM-2V.lib`:

```
tempus> create_op_cond -name PVT2
```

```
-library_file IsCOM-2V.lib  
-P 1.0  
-V 1.2  
-T 20
```

Related Commands

- [get_op_cond](#)
- [create_delay_corner](#)

create_parasitic_coordinate_transform

```
create_parasitic_coordinate_transform
[-help]
{-inst inst_name
[-x_block_origin value]
[-x_offset value]
[-y_block_origin value]
[-y_offset value]
[-orient {R0 R90 R180 R270 MY MX90 MX MY90}]}
[-relative_to_parent]} | {-reset}
```

Determines accurate location information of nodes during SPEF reading.

In hierarchical designs, multiple blocks can be implemented separately. During Top level STA analysis, SPEFs of these blocks are stitched together with the top for flat STA analysis. This works well if the pin location of blocks is not required. In the case of AOCV analysis, when spatial analysis is enabled, the accurate location of all nodes (pin/net) in the design is required to calculate accurate derates in the design. This necessitates determining the accurate node location during SPEF reading. This can be achieved by transforming SPEF coordinates as per their physical locations.

Note:

- This command works with all possible SPEF and RCDB input combinations with the `read_parasitics` command, which means for only RCDB(s) as input, a mix of SPEF(s) and RCDB(s) as input, and only SPEF(s) as input.
- If this command is not specified, the `read_spef` command tries to automatically find the orientation and origin of the hierarchical SPEF blocks and applies those identified transforms to the parasitic co-ordinates.
- The `create_parasitic_coordinate_transform` command settings remain in effect for all the subsequent `read_spef` and `read_parasitics` until they are reset using the `-reset` option.
- Whenever the block's origin (in block SPEF/DEF) is non-zero, the x and y coordinates need to be specified for applying the transform. The last two parameters, `-x_block_origin` and `-y_block_origin`, in the following command use-model are only required to be specified for the case of non-zero block origin.

```
create_parasitic_coordinate_transform -inst inst_name
-x_offset x-offset of block origin in top-level DEF/SPEF
-y_offset y-offset of block origin in top-level DEF/SPEF
-orient orientation
-x_block_origin x-coord of non-zero block origin
-y_block_origin y-coord of non-zero block origin
```


For example:

- **For Non-Zero Origin Block:** Specify the block origin as shown below:

```
create_parasitic_coordinate_transform -inst eco_inst1
-x_offset 421.6
-y_offset 485
-orient R180
-x_block_origin 3.4
-y_block_origin 9.7
```

- **For zero origin block:** No need to specify the block origin as shown below:

```
create_parasitic_coordinate_transform
-inst controller
-x_offset 16.8
-y_offset 205
-orient R0
```

or

```
create_parasitic_coordinate_transform
-inst controller
-x_offset 16.8
-y_offset 205
-orient R0
-x_block_origin 0
-y_block_origin 0
```

Both the above commands are equivalent.

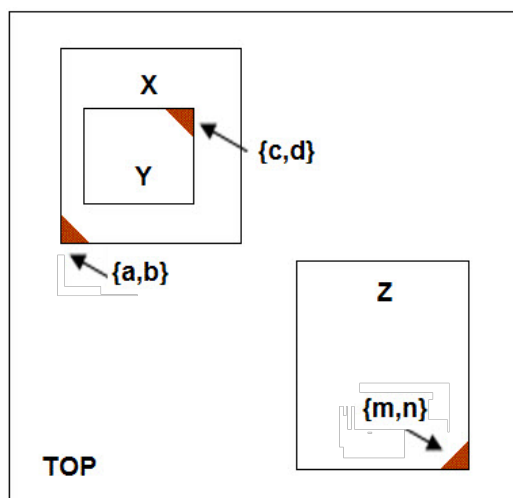
- The impact of this command is valid until the first `read_spef` command is read (in case of MMMC, until all corners SPEFs have been read). If you want to read SPEF again, this command must be applied again. These transformations will be applied during `read_spef` command.

Parameters

<code>-help</code>	Prints out the command usage. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man create_parasitic_coordinate_transform</code>
<code>-inst inst_name</code>	Indicates the hierarchical name of the instance for which SPEF is being read.
<code>-orient {R0 R90 R180 R270 MY MX90 MX MY90}</code>	Indicates the orientation change of instances. The valid values are <code>R0</code> , <code>R90</code> , <code>R180</code> , <code>R270</code> , <code>MX</code> , <code>MX90</code> , <code>MY</code> , and <code>MY90</code> .
<code>-relative_to_parent</code>	When this parameter is specified, the software considers block offsets for the specified instance relative to the immediate parent. By default, the software considers the offset value relative to the top hinst.
<code>-reset</code>	Resets SPEF transform values.
<code>-x_block_origin value</code>	Specifies that in the case of a non-origin block, the offset x-coordinate in microns
<code>-x_offset value</code>	Indicates the actual x-coordinate as exists in the physical floor plan, corresponding to the specified instance.
<code>-y_block_origin value</code>	Specifies that in case of a non-origin block, the offset y-coordinate in microns
<code>-y_offset value</code>	Indicates the actual y-coordinate as exists in the physical floor plan, corresponding to the specified instance.

Examples

Consider the following hierarchical design where X and Z are two blocks inside TOP, and Y is another hierarchical block in X:



Here, X,Y, and Z have the following physical locations in the floor plan:

X - (a,b)

Y- (c,d)

Z- (m,n)

You can apply transformations to all blocks with respect to TOP by using the following commands:

```
tempus> create_parasitic_coordinate_transform -inst X -x_offset a -y_offset b
```

```
tempus> create_parasitic_coordinate_transform -inst Z -x_offset m -y_offset n -orient MY
```

```
tempus> create_parasitic_coordinate_transform -inst X/Y -x_offset c -y_offset d -orient R180
```

These transformations will be applied when the `read_spef` command is issued.

Related Commands

- `set_analysis_view`
- `redirect`
- `read_parasitics`
- `read_spef`
- `report_parasitic_coordinate_transforms`

create_pg_net_group

```
create_pg_net_group
[-help]
-name <name>
-pg_nets <net>
```

Allows you to group a set of PG nets. This helps simplify the overall UI of the IPD flow by specifying sweeps for a PG net group rather than a PG net. Also, the IPD crossings between the PG nets within a PG net group are automatically ignored.

Parameters

-help	Prints out the command usage.
-name <name>	Specifies the PG net group name.
-pg_nets <net>	Specifies a list of power nets.

Examples

- In the following example, the PG nets pgn1, pgn2, and pgn3 are a part of the PG net group PG1.

```
create_pg_net_group -name PG1 -pg_nets {pgn1 pgn2 pgn3}
set_ipd_rail_voltage -sweep { PG1 {0.7 0.8} }
```

The voltage values specified for PG1 apply to the PG nets within the PG net group. The crossings between pgn1 and pgn2, pgn2 and pgn3, and pgn1 and pgn3 will be ignored automatically.

- In the following example, the PG nets pgn1, pgn2, and pgn3 are part of PG net group PG1:

```
tempus > create_pg_net_group -name PG1 -pg_nets {pgn1 pgn2 pgn3}
tempus > create_delay_corners -name dc1 -library_set libset1 -rc_corner rc1 -
pg_net_voltages {PG1@0.7}
```

The voltage values specified for PG1 are applicable for the PG nets that are part of this PG net group (defined using [create_delay_corner](#) -pg_net_voltages).

Related Commands

- `report_pg_net_group`
- `set_ipd_rail_voltage`
- `create_ipd_configuration`
- `report_ipd_configuration`
- `write_ipd_scope`
- `read_ipd_scope`
- `report_ipd_crossings`

Related Information

- For more information, refer to the [Inter Power Domain \(IPD\) Analysis](#) chapter in the *Tempus User Guide*.

create_property_alias

```
create_property_alias
[-help]
-map <alias_property_name_pairs>
-obj_type {pin port cell net clock lib_cell lib_pin design lib timing_path timing_point
timing_arc path_group lib_timing_arc si_victim si_attacker lib_pg_pin pg_pin pg_net}
```

Creates a property alias for the selected objects.

Parameters

-help	Prints out the command usage.
-map <alias_property_name_pairs>	Specifies a pair list of aliases and properties to be associated with that alias, such as {alias1 prop1} or {{ialias1 prop1}{alias2 prop2}...}.
-obj_type {pin port cell net clock lib_cell lib_pin design lib timing_path timing_point timing_arc path_group lib_timing_arc si_victim si_attacker lib_pg_pin pg_pin pg_net}	Specifies the object type for which a property alias needs to be created.

create_rc_corner

```
create_rc_corner
[-help]
[-T float]
-name string
[-postroute_cap string]
[-postroute_res string]
[-postroute_xcap string]
[-qx_tech_file string]
[-via_variation_file string]
```

Creates a named RC corner object that can be referenced later when creating a delay calculation corner object. An RC corner object provides the software with all the information necessary to properly annotate and use the RCs for delay calculation.

Note: If the `read_spef` command is specified without the `-rc_scale` parameter, for example, "`read_spef -rc_corner early_spef_${rc_corner}`", then the software does not apply the capacitance scale factor specified in `-postroute_cap` and `-postroute_xcap` parameters. If the `read_spef` command is specified with the `-rc_scale` parameter, for example, "`read_spef -rc_corner early_spef_${rc_corner} -rc_scale` ", then the software applies the capacitance scale factor. Here, no factor is specified in the `-rc_scale` option.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-name string</code>	Specifies the name for the RC corner object being created. You can choose any name for an RC corner object, but it is recommended that you use a name that can be easily recognized as referring to a specific RC corner (for example, rc-min, rc-max, c-min, c-max, and so on).
<code>-postroute_cap string</code>	Specifies the capacitance scale factor for RC extraction in post-route mode.
<code>-postroute_res string</code>	Specifies the resistance scale factor for RC extraction in post-route mode.
<code>-postroute_xcap string</code>	Specifies the coupling capacitance scale factor for RC extraction in post-route mode.

<code>-qx_tech_file string</code>	Specifies the name of the Quantus technology (.tch) file used by Quantus sign-off RC extraction. TechGen, a built-in functionality of Quantus, generates the technology file. For most technologies, you can obtain this file from the technology vendor. For technologies not available from the technology vendors, you can enter fabrication process information into an ASCII-format interconnect technology (ICT) input file. The ICT file is then used as input to IceCaps, which in turn generates the technology file.
<code>-T float</code>	Specifies the temperature, in degrees Celsius, to use for derating resistance values. Use this parameter to override the default temperature specified in the Quantus technology file. Note: The software does not require that the temperature specified for the PVT operating condition at the delay calculation corner level match the RC nominal or user-specified temperature value.
<code>-via_variation_file string</code>	Specifies a file for modeling via variation. The VIA variation file lists all VIA layer names and their corresponding multipliers. Each RC corner can have its own VIA variation file. For example: <pre>create_rc_corner -name rcCornerName1 \ -via_variation_file viaVariationFileName1 create_rc_corner -name rcCornerName2 \ -via_variation_file viaVariationFileName2</pre>

Examples

- The following command creates an RC corner called `rc-cbest`:

```
tempus > create_rc_corner -name rc-cbest
```

Related Commands

- `update_rc_corner`

display_timing_map

```
display_timing_map  
[-help]  
{[-step float -threshold float] -check_type {setup | hold}} | -clear}
```

Sets color-coded slack threshold values for worst timing paths.

Parameters

-help	Prints a brief description that includes type and default information for each <code>display_timing_map</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man display_timing_map</code>
- check_type {setup hold}	Specifies the check type for timing paths. The possible arguments are <code>setup</code> and <code>hold</code> .
-clear	Clears the timing slack threshold values.
-step	Sets the step value.
-threshold	Sets color-coded slack threshold values for worst timing paths. You can specify a range of threshold values. When specified, the instances shown on the Timing Debug form are color-coded according to the worst timing slack threshold.

Examples

- The following example sets a threshold value of -1:

```
tempus> display_timing_map -step 0.2 -threshold -1
```

get_analysis_mode

```
get_analysis_mode
[-help]
[-aging]
[-analysisType]
[-aocv]
[-asyncChecks]
[-checkType]
[-clkSrcPath]
[-clockGatingCheck]
[-clockPropagation]
[-cppr]
[-enableMultipleDriveNet]
[-graph_pba]
[-honorClockDomains]
[-log]
[-multi_input_switching_mode]
[-nonDefault]
[-propSlew]
[-sequentialConstProp]
[-skew]
[-socv]
[-timeBorrowing]
[-timingEngine]
[-timingSelfLoopsNoSkew]
[-useOutputPinCap]
[-warn]
[-quiet]
```

Displays information for the specified `set_analysis_mode` parameters for the current Tempus session. If you do not specify a parameter, the software displays information for all `set_analysis_mode` parameters.

Parameters

-help	Prints out the command usage.
<i>parameter_names</i>	See <code>set_analysis_mode</code> for descriptions of the timing analysis mode parameters that you can specify.
-quiet	Returns the current settings for the specified parameters in Tcl list format only. If you specify <code>-quiet</code> without any parameters, the software returns the current settings of all <code>set_analysis_mode</code> parameters in Tcl list format.

Examples

- The following command returns the current setting of the `-checkType` parameter:

```
tempus> get_analysis_mode -checkType
```

The software returns the following information:

```
-checkType setup # enums={setup hold}, default=setup  
setup
```

- The following command returns the current setting of the `-checkType` parameter in Tcl list format:

```
tempus> get_analysis_mode -checkType -quiet
```

The software returns the following information:

```
setup
```

- The following command returns the current settings for the `-checkType`, `-skew`, and `-analysisType` parameters:

```
tempus> get_analysis_mode -checkType -skew -analysisType
```

The software returns the following information:

```
-checkType setup # enums={setup hold}, default=setup  
-skew true # bool, default=true  
-analysisType bcwc # enums={single bcwc onChipVariation}, default=bcwc  
{checkType setup} {skew true} {analysisType bcwc}
```

- The following command returns the current settings for the `-checkType`, `-skew`, and `-analysisType` parameters in Tcl list format:

```
tempus> get_analysis_mode -checkType -skew -analysisType -quiet
```

The software returns the following information:

```
{checkType setup} {skew true} {analysisType bcwc}
```

Related Commands

- [set_analysis_mode](#)

get_analysis_view

```
get_analysis_view  
[-help]  
view_Name  
{-constraint_mode | -delay_corner}
```

Returns the constraint mode or delay calculation corner associated with the specified analysis view.

Use this command after creating at least one analysis view (by using the `create_analysis_view` command).

Parameters

-help	Prints out the command usage.
-constraint_mode	Returns the constraint mode associated with the specified analysis view.
-delay_corner	Returns the delay calculation corner associated with the specified analysis view.
view_Name	Specifies the name of the analysis view to query.

Examples

- The following command creates the analysis view `missionSlow`:

```
tempus> create_analysis_view  
-name missionSlow  
-constraint_mode missionSetup  
-delay_corner dcWCCOM
```

The following command returns the delay calculation corner associated with the `missionSlow` analysis view:

```
get_analysis_view missionSlow -delay_corner
```

The software returns the following result:

```
dcBCCOM
```

Related Commands

- `create_analysis_view`
- `set_analysis_view`

get_capacitance_unit

```
get_capacitance_unit
```

```
[-help]
```

Returns the capacitance unit (in picofarads) for the current session, which may be derived from the library file or set with the `set_library_unit -cap` parameter. This unit will be used for generating reports and for setting constraints.

Parameters

<code>-help</code>	Prints out the command usage.
--------------------	-------------------------------

Example

The following command returns the session capacitance unit:

```
get_capacitance_unit
```

```
1.000000pf
```

Related Commands

- `set_library_unit`

get_clock_network_objects

```
get_clock_network_objects  
[-help]  
[-clocks clock_list]  
[-type {cell | pin}]
```

Returns the cells or pins that exist in a clock network of the specified clocks.

Parameters

-help	Prints out the command usage.
-clocks <i>clock_list</i>	Returns the cells or pins present in a clock network of the specified clock(s).
-type {cell pin}	Returns the specified cells or pins in the clock network.

Examples

- The following command returns the pins present in the clock network of all the clocks:
tempus > get_clock_network_objects
- The following command returns the pins that exist in the clock network of the CLK and CLK2 clocks:
tempus > get_clock_network_objects -clocks {CLK CLK2}
- The following command returns the cells that exist in the clock network of the CLK clock.
tempus > get_clock_network_objects -clocks {CLK} -type cell

get_constant_for_timing

```
get_constant_for_timing
[-help]
[-bidi_input | -bidi_output]
pin_name
```

Queries the design database for the state of the specified pin or the state propagated through the combinational logic cone to that pin. The returned value is 0 for logic 0, 1 for logic 1, x for undefined, and z for tristate.

Note: Constants on hierarchical ports are not computed.

✓ The `get_constant_for_timing` command returns the state of only one pin. If you want to know the state on all ports that have a constant set, use the `-type constant` option with the `report_ports` command. For example: `report_ports -type constant`.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-bidi_input -bidi_output</code>	Returns the constant that is set on the input or output part of bidirectional pins. <i>Default:</i> Returns the constant on the output part.
<code>pin_name</code>	Specifies the name of the single pin to query for the pin state. The pin must be an instance pin or primary I/O port.

Examples

- The following command sets the value of the specified pin(s) to either a 1 or 0 for use by the timing engine:
`tempus > set_case_analysis 0 J_block/zbuf0/A`
- The following command queries the design database for the state of the specified pin or the state propagated through the `tempus>` combinational logic cone to that pin:
`tempus > get_constant_for_timing J_block/zbuf0/A`

Related Commands

- `report_ports`
- `set_case_analysis`
- `report_case_analysis`

get_constraint_mode

```
get_constraint_mode
[-help]
modeName
{-sdc_files | -ilm_sdc_files | -tcl_vars}
```

Returns a Tcl list of the regular SDC constraint files, or the ILM SDC constraint files associated with the specified constraint mode.

Use this command after creating constraint modes ([create_constraint_mode](#)).

Parameters

-help	Prints out the command usage.
-ilm_sdc_files	Returns the ILM SDC files associated with the specified constraint mode.
modeName	Specifies the name of the constraint mode to query.
-sdc_files	Returns SDC files associated with the specified constraint mode.
-tcl_vars	Returns a list of TCL variables associated with the specified constraint mode.

Examples

- The following command creates the constraint mode missionSetup:

```
tempus> create_constraint_mode -name missionSetup
-sdc_files [list io.sdc mission1-clks.sdc mission1-except.sdc]
```

The following command returns the SDC constraint files associated with missionSetup:

```
get_constraint_mode missionSetup -sdc_files
```

The software returns the following Tcl list:

```
{io.sdc mission1-clks.sdc mission1-except.sdc}
```

Related Commands

- [create_constraint_mode](#)
- [update_constraint_mode](#)

get_delay_corner

```
get_delay_corner
[-help]
delayCornerName
[-early_irdrop_data]
[-early_irdrop_file]
[-early_library_set]
[-early_opcond]
[-early_opcond_library]
[-early_rc_corner]
[-early_temp_file]
[-irdrop_data]
[-irdrop_file]
[-late_irdrop_data]
[-late_irdrop_file]
[-late_library_set]
[-late_opcond]
[-late_opcond_library]
[-late_rc_corner]
[-late_temp_file]
[-library_set]
[-opcond]
[-opcond_library]
[-power_domain powerDomainName]
[-power_domain_list]
[-rc_corner]
[-si_enabled]
[-supply_set supplySetName]
[-supply_set_list]
[-temp_file]
```

Returns attribute information for the specified delay calculation corner object, or for one of its power domains.

Use this command after creating at least one delay calculation corner object ([create_delay_corner](#)). The delay calculation corner must include a power-domain definition to query power-domain information ([update_delay_corner](#)).

Parameters

<code>-help</code>	Prints a brief description that includes type and default information for each <code>get_delay_corner</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man get_delay_corner</code> .
<code>parameter_names</code>	See create_delay_corner for descriptions of the parameters that you can specify.

Examples

- The following command creates the delay calculation corner `dc1`:

```
tempus> create_delay_corner -name dc1
-library_set libs-2volt
-opcond_library delayvolt_2V
-opcond slow_2V
-rc_corner rc-cworst
```

The following command returns the operating condition for the delay calculation corner `dc1`:

```
tempus> get_delay_corner dc1 -opcond
```

The software returns the following results:

```
slow_2V
```

- The following commands add definitions for power domains `domain-3V` and `domain-5V` to the delay calculation corner `dc1`:

```
tempus> update_delay_corner -name dc1
-power_domain domain-3V
-library_set libs-3volt
-opcond_library delayvolt_3V
-opcond slow_3V
update_delay_corner -name dc1
-power_domain domain-5V
-library_set libs-5volt
-opcond_library delayvolt_5V -opcond slow_5V
```

- The following command returns a list of all the power domains associated with the delay calculation corner `dc1`:

```
tempus> get_delay_corner dc1 -power_domain_list
```

The software returns the following results:

```
{domain-3V domain-5V}
```

- The following command returns the name of the library set associated with the power domain domain-5V:

```
tempus> get_delay_corner dcl -power_domain domain-5V -library_set
```

The software returns the following results:

```
libs-5volt
```

Related Commands

- `create_delay_corner`
- `update_delay_corner`

get_equivalent_cells

```
get_equivalent_cells  
[-help]  
-cell string  
[-delay_corner string]  
[-early_only]  
[-late_only]  
[-library string]  
[-max]  
[-min]  
[-power_domain string]
```

Returns equivalent cells from minimum/maximum libraries or from libraries of specific power domains.

Parameters

-help	Prints out the command usage.
-cell <i>string</i>	Specifies the cell name for which equivalent cells will be fetched.
-delay_corner <i>string</i>	Specifies the delay corner for which equivalent cells from libraries will be fetched.
-early_only	Fetches cells only for early libraries of the specified delay corner.
-late_only	Fetches cells only for late libraries of the specified delay corner.
-library <i>string</i>	Specifies the library for which cells will be fetched.
-max	Fetches equivalent cells for max libraries only.
-min	Fetches equivalent cells for min libraries only.
-power_domain <i>string</i>	Fetches cells for libraries of a specific power domain.

Examples

- The following command will return all the cells equivalent to BUF_X4 (buffers) present in the power domain PD1:

```
tempus> get_equivalent_cells -cell BUF_X4 -power_domain PD1
```

Output:

```
CORELIB/BUF_X1 CORELIB/BUF_X16 CORELIB/BUF_X2 CORELIB/BUF_X32 CORELIB/BUF_X4
```

- The following command will return all the AND cells of different drives present in the library USERLIB:

```
tempus> get_equivalent_cells -cell AND_X8 -library USERLIB
```

get_interactive_constraint_modes

```
get_interactive_constraint_modes  
[-help]
```

Returns a list of the multi-mode multi-corner constraint modes that are in interactive constraint entry mode.

When the software is in interactive constraint entry mode ([set_interactive_constraint_modes](#)), any timing constraints you specify take effect immediately across all active analysis views that are associated with these constraint modes.

Note: Interactive constraint mode only works when the software is in regular multi-mode multi-corner timing analysis mode. You cannot use it for distributed multi-mode multi-corner analysis.

Related Commands

- [set_interactive_constraint_modes](#)

get_lib_cell_leakage_power

```
get_lib_cell_leakage_power
[-help]
-cell string
[-file string]
[-stateleakagegrp]
[ {-view string | -delay_corner string}
[-power_domain string]]
[ -early | -late ]
```

Returns the default leakage power of a cell.

Parameters

-help	Prints out the command usage
-cell <i>string</i>	Specifies the cell name for which the leakage power needs to be fetched (string, required)
-delay_corner <i>string</i>	Selects a cell from the specified delay corner
-early	Selects a cell from the early library of the specified delay corner. If this option is not specified, late will be used by default. (bool, optional)
-file <i>string</i>	Specifies a.Name of the output file. (string, optional)
-late	Selects a cell from the late library of the specified delay corner. If this option is not specified, late will be used by default. (bool, optional)
-power_domain <i>string</i>	Selects a cell from the library of the specified power domain (string, optional)
-stateleakagegrp	Reports state dependent leakage power information.
-view <i>string</i>	Selects the cell from the specified view.

Examples

- The following example reports state-independent cell leakage power along with the return value:
tempus> get_lib_cell_leakage_power -cell xyz
- The following command shows state-dependent groups:
tempus> get_lib_cell_leakage_power -cell xyz -stateleakagegrp

Related Commands

- `report_power`

get_library_set

```
get_library_set
[-help]
libSetName
{-timing | -si | -aocv | -socv}
```

Returns a Tcl list of the timing or cdb libraries for the specified library set.

Use this command after creating at least one library set (by using the [create_library_set](#) command).

Parameters

-help	Prints out the command usage.
-aocv	Returns AOCV library names for the specified library set.
libSetName	Specifies the name of the library set.
-si	Returns the cdb library names for the specified library set.
-socv	Returns the SOCV library names for the specified library set.
-timing	Returns a list of the timing libraries for the specified library set. This list contains the names of files from which the libraries were sourced.

Examples

- The following commands create the library sets IsCOM-1V:

```
tempus> create_library_set -name IsCOM-1V -timing [list stdcell_F_1V.lib ram_F.lib pad.lib]
\ -si [list stdcell_F_2.cdb ram_F.cdb pad.cdb]
```
- The following command returns a TCL list of the timing libraries for the library set IsCOM-1V:

```
tempus> get_library_set -timing IsCOM-1V
```

The software returns the following results:

```
{stdcell_F_1V.lib ram_F.lib pad.lib}
```

Related Commands

- [create_library_set](#)

get_op_cond

```
get_op_cond  
[-help]  
virtualOpcondName  
{-library_file | -P | -V | -T}
```

Returns the process, voltage, or temperature value for the specified operating condition.

Use this command after creating at least one operating condition ([create_op_cond](#)).

Parameters

-help	Prints out the command usage.
-library_file	Specifies the library file in which the operating condition is defined.
-P	Returns the process value for the operating condition.
-T	Returns the temperature value for the operating condition.
-V	Returns the voltage value for the operating condition.
virtualOpcondName	Specifies the name of the operating condition to query

Examples

Here are some examples.

- The following command creates an operating condition called `myPVT2` for the library `IsCOM-2V.lib`:

```
tempus> create_op_cond -name myPVT2  
-library_file IsCOM-2V.lib  
-P 1.0  
-V 1.2  
-T 20
```

The following command creates a collection called `myPVT2-temp` that contains the temperature value of the `myPVT2` operating condition:

```
tempus> set myPVT2-temp [get_op_cond myPVT2 -T]
```

Related Commands

- `create_op_cond`

get_pba_mode

```
get_pba_mode
[-help]
[-epba_bounded_mode]
[-ipba_endpoints_max_limit]
[-quiet]
[-nonDefault]
```

Displays information for the IPBA mode set using the `set_pba_mode` command.

Parameters

-help	Prints out the command usage.
-epba_bounded_mode	Enables or disables IPBA mode with exhaustive PBA analysis (enabled with the <code>report_timing -retime_mode exhaustive</code> command).
-ipba_endpoints_max_limit	Returns the end points' maximum limit for infinite PBA analysis of the top maximum GBA endpoints.
-quiet	Displays the current settings in Tcl list format only.
-nonDefault	Displays the current settings, which are not the default settings.

Related Commands

- `set_pba_mode`
- `report_timing`

get_propagated_clock

```
get_propagated_clock  
[-help]  
[-clock clock_list] [-pin pin_list]  
[> filename]
```

Returns the clock propagation mode, either `ideal` or `propagated`, for the specified clocks or pins.

Parameters

<code>-help</code>	Prints out the command usage.
<code>> <i>filename</i></code>	Specifies the name of a file in which to save the report. You can add the <code>.gz</code> extension to the file name to generate a compressed report. Note: The file name must be the last argument in the list.
<code>-clock</code> <code><i>clock_list</i></code>	Specifies one or more clock waveforms for which you want to retrieve the clock propagation mode information.
<code>-pin</code> <code><i>pin_list</i></code>	Specifies one or more pins or ports for which you want to retrieve the clock propagation mode information. You cannot specify hierarchical pins.

Example

- The following command gets the clock propagation mode information for all clock waveforms:

```
tempus > get_propagated_clock -clock {*}
```

Related Commands

- [reset_propagated_clock](#)
- [set_propagated_clock](#)

get_resistance_unit

`get_resistance_unit`

Returns the resistance unit for the current session.

Example

The following command returns the resistance unit for the current session:

```
tempus> get_resistance_unit
```

```
1.000000 kohm
```


get_socv_rc_variation_factor

```
get_socv_rc_variation_factor  
[-help]  
-view viewName  
{-early | -late}
```

Displays the variation factor for interconnect delays.

Parameterse

-help	Prints out the command usage.
-early	Displays the variation factor for early interconnect delays.
-late	Displays the variation factor for late interconnect delays.
-view <i>viewName</i>	Displays the SOCV interconnect variation factor for the specified view.

Related Commands

- [set_socv_rc_variation_factor](#)

get_socv_reporting_nsigma_multiplier

```
get_socv_reporting_nsigma_multiplier  
[-help]  
-view viewName  
{[-setup | -hold ]}  
[-transition]
```

Allows you to query the sigma multiplier setting for the specified view in setup or hold mode.

Parameters

-help	Prints out the command usage.
-hold	Returns the sigma multiplier for hold mode.
-setup	Returns the sigma multiplier for setup mode.
-transition	Returns the sigma multiplier used for transition-related DRV checks.
-view <i>viewName</i>	Returns the sigma multiplier value for the specified view.

Examples

- The following command displays the sigma multiplier setting for the specified view:

```
tempus > set_socv_reporting_nsigma_multiplier -hold 2.5 -view default_emulate_view_hold  
tempus > get_socv_reporting_nsigma_multiplier -view default_emulate_view_hold -hold  
2.500000
```

Related Commands

- `set_socv_reporting_nsigma_multiplier`

get_time_unit

```
get_time_unit  
[-help]
```

Returns the current time unit (in nanoseconds) for the session derived from the library file or set with the `set_library_unit -time` parameter. This unit will be used during report generation and constraint settings.

Parameters

<code>-help</code>	Prints out the command usage.
--------------------	-------------------------------

Examples

The following command returns the time unit for the session:

```
tempus> get_time_unit
```

```
1.000000ns
```

Related Commands

- `set_library_unit`

list_libraries

```
list_libraries
[-help]
[-power_domain <string>]
[-used]
[-used_analysis]
[-view <string>]
```

Lists all the libraries loaded in the session and used in the design.

Parameters

-help	Prints out the command usage.
-power_domain <string>	Specifies the power domain from which instances are to be selected.
-used	Prints out a list of libraries used in the design and all the other libraries, whose voltages match those of the used libraries (like vt variant libraries). This list is useful for ECO optimization.
- used_analysis	Prints out a list of used libraries in the design.
-view <string>	Specifies the view name whose libraries will be listed.

Examples

Here are some examples.

- The following command reports all the libraries loaded in a session:

```
tempus> list_libraries
```

```

Library      Library      Library
Group        Name         Path
-----
max          fast         /cte//data/libs/t18/fast.lib
min          slow         /cte//data/libs/t18/slow.lib
-----
```

Related Commands

- `current_design`

read_boundary_model

```
read_boundary_model  
[-help]  
[-check_contents]  
[-clock_mapping_file string]  
[-instances string]  
[-report_file string]  
[-skip_version_check]  
[-unconstrained_net_use_inf_tw]  
[-view_mapping_file string]  
{-dir string}
```

Saves the directory information.

This command can be used for each cell boundary model generated by the block-level runs.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-check_contents</code>	Checks for required files in the boundary model directory.
<code>-clock_mapping_file string</code>	Uses the mapping file created by the <code>set_boundary_model_clock_mapping</code> command. The basic syntax of <code>-clock_map_file</code> is: <pre>Block_clock_name(view_name) Top_clock_name(view_name) instance_name</pre> However, both <code>view_name</code> and <code>instance_name</code> are optional. If you want to have the same mapping for all instances and views, then it is specified as follows: <pre>Block_clock_name Top_clock_name</pre> • If it is instance-specific but not view-specific, then it is specified as follows: <pre>Block_clock_name Top_clock_name instance_name</pre> • If it is view-specific but not instance-specific then it is specified as follows: <pre>Block_clock_name(view_name) Top_clock_name(view_name)</pre> Note that no spaces are currently allowed between the clock name and the view name.
<code>-dir string</code>	Specifies the directory for the boundary model.
<code>-instances string</code>	Specifies the instances for which the boundary model must be read.
<code>-report_file string</code>	Specifies the name of the boundary model check report file. After the boundary model is checked and the results are logged into this file. If this parameter is not specified, the report is written to the <code><cell_name>_bmodel_check.txt</code> file by default.
<code>-skip_version_check</code>	Does not check for Tempus version consistency.
<code>-unconstrained_net_use_inf_tw</code>	Forces unconstrained nets to infinite timing windows.
<code>-view_mapping_file string</code>	Uses the mapping file created by the <code>set_boundary_model_view_mapping</code> command.

Examples

- The following command loads the boundary model files from the ./model_dir directory:

```
read_boundary_model -dir ./model_dir
```

- The following command loads the boundary model files from the ./model_dir directory:

```
read_boundary_model -dir ./model_dir -clock_mapping_file clock.data
```

The clock.data file will be read for user-defined updates for clock mappings from the boundary model clock names to the clocks defined in the constraints for the current session.

Related Commands and Variables

- `create_boundary_model`
- `timing_full_context_flow`
- `set_boundary_model_view_mapping`
- `set_boundary_model_clock_mapping`

read_ipd_scope

```
read_ipd_scope  
[-help]  
[<dir_name>]
```

Loads the scope data in the scope analysis run.

Parameters

-help	Prints out the command usage.
<dir_name>	Specifies the scope directory containing the minimum/maximum IPD scopes.

Examples

- The following command shows the usage of `read_ipd_scope` in a scope analysis session.

```
tempus > read_ipd_scope bounded_corner_scope
```

Related Commands

- `set_ipd_rail_voltage`
- `create_pg_net_group`
- `report_pg_net_group`
- `create_ipd_configuration`
- `report_ipd_configuration`
- `write_ipd_scope`
- `report_ipd_crossings`
- `check_ipd_logic`

Related Information

- For more information, refer to the [Inter Power Domain \(IPD\) Analysis](#) chapter in the *Tempus User Guide*.

read_path_descriptions

```
read_path_descriptions
[-help]
filename
[-hpin]
[-path_type {end summary full full_clock end_slack_only summary_slack_only}]
[-retime {aocv ssta path_slew_propagation aocv_path_slew_propagation waveform_propagation}]
```

Reads the file created by the [write_path_descriptions](#) command as an input argument.

The command reads the complete description of the paths specified in the file from another Tempus session and returns the data as a timing path collection. The path's description when read back can be used as a regular path collection for any reporting command or queried using the [get_property](#) command.

Parameters

-help	Prints out the command usage.
<i>filename</i>	Specifies the name of the input file.
-hpin	Prints all the hierarchical crossings of an arc under the "hpin" column in the report_timing table.
-path_type {end summary full full_clock end_slack_only summary_slack_only}	Allows you to choose the report format based on the path type.
-retime {aocv ssta path_slew_propagation aocv_path_slew_propagation waveform_propagation}	Reanalyzes the specified set of paths using the specified analysis method.

Examples

- The following example command reads out the pathfile1.txt. The file contains details of the collection of timing paths with hierarchical pins (`-hpin`) format based on timing and exact path availability in the session in which the path description is read:

```
read_path_descriptions pathfile1.txt -hpin
```

- The following command reads in a path descriptions file and queries the slack:

```
tempus > set read_paths_coll [read_path_descriptions path_desc.txt]
```

```
tempus > get_property $read_paths_coll slack  
-0.123
```

The following command generates a timing report from the path descriptions file:

```
tempus > report_timing [read_path_descriptions path_desc.txt] > paths.rpt
```

Related Commands

- `write_path_descriptions`
- `get_property`
- `report_timing`

read_sdf

```
read_sdf
[-help]
sdf_file_name
[-continue_on_error]
[-ilm_filter]
[-increment]
[-overwrite_incremental_delay]
[-path instanceName]
[-persistent]
[-print_summary]
[-scale float]
[-strict_cond_matching]
[-view viewName]
[-sdf_field {min|typ|max} | [-early_sdf_field {min|typ|max}] [-late_sdf_field {min|typ|max}]
```

Reads an OVI Standard Delay Format (SDF) file and annotates user-specified delay information from the SDF file into the timing system. You can use the `read_sdf` command to supply pre-calculated delays and timing check values from an external delay calculator, or from a previous Tempus session.

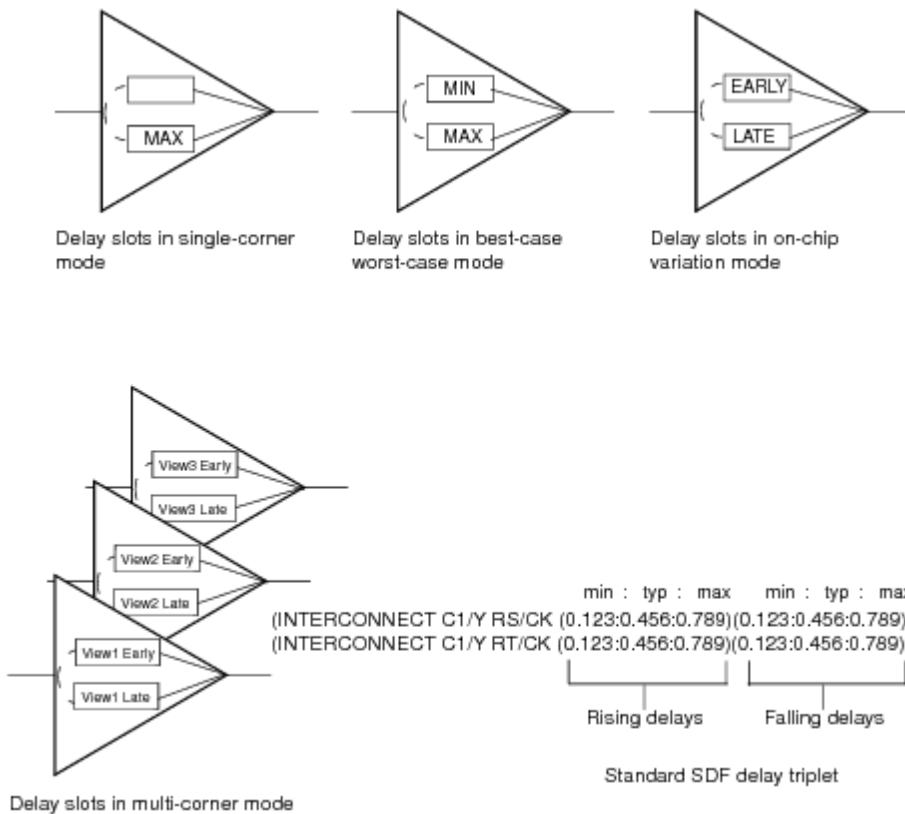
The `read_sdf` command can read SDF files in versions 2.1 and 3.0, and both plain-text and GNU zipped (.gz) formats. The command and the Tempus software do not support the Timing Environment section of the SDF specification.

The SDF file can provide one, two, or three different values for each timing arc delay or timing check value. These values are passed in the min, typ, or max SDF delay fields, represented as min:typ:max.

The Tempus software provides multiple delay placeholders to store the calculated or annotated delay information:

- In single-corner (`set_analysis_mode -analysisType single`) analysis mode, only max delays are available.
- In best-case worst-case (`set_analysis_mode -analysisType bcwc`) mode, a min delay slot is provided for storing min delays used for hold analysis. A max delay slot stores delays used for setup analysis.
- In on-chip variation (`set_analysis_mode -analysisType onChipVariation`) mode, the early delay slots store delays used for calculated early arrival times. The late delay slot stores delays used for calculated late arrival times.
- In multi-corner mode, an early and a late delay slot are provided for each multi-mode multi-corner analysis view.

The `read_sdf` command provides parameters for controlling how the SDF min:typ:max triplet values are mapped into the Tempus software's delay slots. The correct mapping depends on how the delay information in the SDF file was generated and what type of analysis you are performing.



The timing system provides special handling for annotating incremental delays. These delays typically represent plus or minus offsets due to crosstalk effects. Incremental delay annotation to the Tempus software can be done by:

- Importing an incremental SDF file (SDF has an internal `INCREMENT` keyword) using the `read_sdf` command.
- Tcl sourcing a file of `set_annotated_delay -net -delta_only` constraints.

The incremental delay annotations can be viewed in the `report_timing` report by adding the `incr_delay` option to the `report_timing -format` list, or to the `report_timing_format` global variable.

Parameters

<code>-help</code>	Prints out the command usage.
--------------------	-------------------------------

<code>-continue_on_error</code>	Continues to read the file if an error occurs. By default, when an error occurs, the <code>read_sdf</code> command stops reading the file. Use this parameter to attempt to continue reading the file, if possible. Note: Use caution if you continue, because some annotations could be missing due to the errors. Use the <code>report_annotated_parasitics</code> command to find missing annotations. If the file is corrupted, obtain a new SDF file.
<code>-early_sdf_field {min typ max}</code>	Annotates the early delay from the specified SDF field. In single-corner mode, all setup and hold analyses use the max delays. The software ignores any min/early delay information. In best-case worst-case mode, use this parameter to specify which SDF field to use for hold analysis. In on-chip variation mode, the software uses the early delay to calculate fast paths.
<code>-ilm_filter</code>	Ignores ILM-related errors. By default, the <code>read_sdf</code> command does not ignore errors reported due to missing arcs in the core ILM.
<code>-increment</code>	Increments the total delay without affecting the delta delay. By default, when the SDF file contains the INCREMENT delay type, the <code>read_sdf</code> command adjusts the delta delay slot to model SI cross-talk push-out/pull-in effects. When this option is used, the INCREMENT delays will be applied against the static base delay values instead.
<code>-late_sdf_field {min typ max}</code>	Annotates the late delay from the specified SDF field. In single-corner mode, use this parameter to specify which of the SDF triplet values to annotate for both setup and hold checks. In best-case worst-case mode, use this parameter to specify which SDF field to use for setup analysis. In on-chip variation mode, the software uses the late delay for the calculation of slow paths.

<code>- overwrite_incremental_delay</code>	When annotating an incremental SDF file, it replaces any pre-existing delta delays with the new SDF values. The total delay will be equivalent to the original base delay, plus the new delta delay. If the delays in the incremental SDF file represent crosstalk delay push-outs from a signal integrity analysis, use the <code>-overwrite_incremental_delay</code> parameter to ensure that any stale data from a previous analysis is removed. By default, when an incremental SDF file (SDF has an internal <code>INCREMENT</code> keyword) is annotated, the delays are added to the base delay and also stored separately as delta delays. If an additional incremental SDF file is annotated, the software adds the existing base and delta delays with the new values.
<code>-path instanceName</code>	Specifies the instance in the design to which the specified SDF file should be applied.
<code>-persistent</code>	Keeps the delays from certain SDF files persistently in use during various steps of the timing optimization. Use this parameter to support pre-timed structures, such as clock meshes.
<code>-print_summary</code>	Outputs the summary information of annotated delays.
<code>-scale float</code>	Overrides the automatic scaling of SDF delays to library units with the given value. To disable all scaling, use a value of 1.0.
<code>-sdf_field {min typ max}</code>	Annotates the delay from the specified SDF slot. In single-corner and best-case worst-case modes, the software uses this value for both setup and hold analysis. In on-chip variation mode, the software uses the same value from the specified SDF slot for both the early and late delays of the specified arc.
<code>sdf_filename</code>	Specifies the name of the SDF file that has the back annotation data.
<code>-strict_cond_matching</code>	Annotates only the conditions that are an exact syntax-to-syntax match between the SDF file and the <code>.lib</code> file. <i>Default:</i> Uses looser conditional matching rules between the <code>.lib</code> and SDF arc specifications.
<code>-view viewName</code>	Annotates the specified analysis view(s) with the delays from the SDF file. You can use this parameter only when you are in multi-mode multi-corner analysis mode.

Examples

Here are some examples.

- The following command annotates delay information from the SDF min field into the min/early slots, and maps the SDF max field data to the max/late delay slots:

```
tempus> read_sdf sdf_filename
```

Given the following SDF, the default annotation would be:

```
(INTERCONNECT C1/Y RS/CK (0.123:0.456:0.789) (0.123:0.456:0.789))
```

- In single-corner analysis mode, a delay of 0.789 ns is annotated to the max delay slot.
 - In best-case worst-case mode, a delay of 0.123 ns is annotated to the min delay slot, and a value of 0.789 ns is placed in the max delay slot.
 - In on-chip variation mode, a delay of 0.123 is used for early delays, and a value of 0.789 ns is used for late delays.
- The following command annotates the SDF typ field to both the early and late delay slots:

```
tempus> read_sdf -sdf_field typ sdf_filename
```

- The following command annotates the SDF typ field to the early delay slot, and the SDF max field to the late delay slot:

```
tempus> read_sdf -early_sdf_field typ -late_sdf_field max sdf_filename
```

- The following command annotates the SDF typ field to the late delay slot. The early delay slot is annotated with the default min SDF field:

```
tempus> read_sdf -late_sdf_field typ sdf_filename
```

Related Commands

- `write_sdf`
- `set_analysis_mode`
- `set_annotated_delay`
- `report_timing`
- `report_annotated_parasitics`

read_spdf

```
read_spdf  
[-help]  
SPDF_file_name
```

Reads the statistical parameter distribution format (SPDF) file. An SPDF file contains statistics of the variation of process parameters. The SPDF file is required for statistical static timing analysis (SSTA).

Parameters

-help	Prints out the command usage.
<i>SPDF_file_name</i>	Specifies the name of the SPDF file.

read_spef

```
read_spef
[-help]
fileName fileName
[-decoupled]
[-extended]
[-idx]
[-rc_corner rcCornerName(s)]
[-rc_scale]
[-spef_field integer list]
[-starN | -noStarN]
[-via_variation_file via_variation_file]
[-wire_variation_file wire_variation_file]
```

Loads resistors and capacitors for the interconnects in SPEF into the software to calculate delays.

Note: This command supports multi-threaded SPEF reading, scalable up to a maximum of eight CPUs. The number of CPUs can be specified using Tempus's standard usage model.

Note: The Tempus software builds the RC network by using the information only from SPEF. If the SPEF is missing any connectivity information, then Tempus will issue a warning (SPEF-1011), and the internal connection between nodes will be made. The Tempus software ensures that during each run, the same node connections (which may or may not be correct) are used, resulting in consistent results across multiple runs.

Parameters

-help	Prints out the command usage.
-decoupled	Grounds the coupling capacitance in the SPEF file and merges the capacitance on the same node. This parameter can be used only for static timing analysis and cannot be used for SI analysis.
-extended	Enables the reading of “\$llx=<value1> \$lly=<value2> \$urx=<value3> \$ury=<value4> \$LAYER=<number>” to set the layer number for nodes of a resistance from a Quantus extended SPEF file. It also reads in the extended *LAYER_MAP section, if it is present. This parameter can also be read in “\$l=<length> \$w=<width> \$si_w=<swidth> \$lvl=<number>” to set the width, silicon width, or layer for nodes of a resistance from StarRC extended SPEF file.

<code>-idx</code>	Enables Inter-Die Extraction (IDX) SPEF reading. IDX SPEF models the Inter-Die capacitive impact in the form of coupling and ground capacitances. Individual die and IDX SPEFs need to be extracted by Quantus and in the extended SPEF (xSPEF) format. The <code>-extended</code> parameter must also be specified to read the xSPEF format. The software also supports multiple IDX SPEF readings. Flat die-level SPEFs need to be specified along with IDX SPEFs. With the <code>-idx</code> parameter, die-level and IDX SPEFs can be specified in any order.
<code>fileName fileName</code>	Specifies the SPEF file. You can specify multiple files in a single command.
<code>-rc_corner</code> <code>rcCornerName(s)</code>	Annotates the parasitics to the predefined RC corner for multi-corner analysis. If you want to use SPEF information when calculating delays in multi-mode multi-corner analysis mode, you must annotate the parasitics for each RC corner in the design. For example, if you have four active RC corners in the design, you must annotate the parasitics for each RC corner. <pre>read_spef -rc_corner corner1 rc1.spef read_spef -rc_corner corner2 rc2.spef read_spef -rc_corner corner3 rc3.spef read_spef -rc_corner corner4 rc4.spef</pre> Note: SPEF reading will not be triggered until SPEF files for all active RC corners are specified.
<code>-rc_scale</code>	Scales the SPEF read in for the specified corner as per the <code>postRoute_cap</code>, <code>postRoute_xcap</code>, and <code>postRoute_res</code> scale factors defined using the <code>create_rc_corner</code> and <code>update_rc_corner</code> commands.
<code>-spef_field integer</code> <code>list</code>	Specifies which field of the triplet values in the SPEF file is to be loaded. The <code>-spef_field</code> parameter specifies the list of RC corner indices {1 2 3...N} in the SPEF file to be loaded instead of triplet values. This parameter provides the flexibility to perform order-independent and field-independent mapping to RC corners.
<code>-starN -nostarN</code>	Specifies whether or not to read the *N statements in the SPEF file. The *N statements define the location of the RC nodes. The <code>-starN</code> parameter reads the *N statements, and the <code>-noStarN</code> parameter ignores the *N statements. Default: <code>-noStarN</code> (By default, the software ignores the *N statements in the SPEF file.)
<code>-via_variation_file</code> <code>via_variation_file</code>	Specifies the name of the file for modeling via variation. The contents of the file include all the via layer names with corresponding MEAN_SHIFT and STDDEV, so that all the VIA resistance in SPEF can be mapped to their own

variation multiplier. This option is used to specify the via_variation file only when the design has a single corner.

For example,

```
read_spef
    -rc_corner rcCornerName \
-via_variation_file viaVariationFileName
```

A sample via_variation file is provided below:

```
// corner_name : rcMax

// wildcard to cover default
object_type : design
delay_type : via

layer_name : *

area : 0.20 0.40 0.80

// no nominal for wildcard default
table_min_sigma : 0.80 0.90 0.95
table_max_sigma : 1.60 1.30 1.15
table_meanshift : 0.05 0.03 0.02
table_stdev : 0.10 0.05 0.02

// per via layer data
object_type : design
delay_type : via
layer_name : VIA1

Area : 0.0004 0.0008 0.0016

table_min_sigma : 0.80 0.90 0.95
table_max_sigma : 1.60 1.30 1.15
table_meanshift : 0.015 0.015 0.012
table_stdev : 0.214 0.150 0.117

object_type : design
delay_type : via
layer_name : VIA2

area : 0.0004 0.0008 0.0016

table_min_sigma : 0.80 0.90 0.95
table_max_sigma : 1.60 1.30 1.15
table_meanshift : 0.015 0.012 0.014
```

<pre>- wire_variation_file wire_variation_file</pre>	<pre>.... table_stdev : 0.223 0.183 0.140</pre> Note: Each line in the file lists the basic components of the statistical variation approach. The values in the "area" line list all possible areas. In the above example, the area line lists three cases that can happen in the design. It can list different number of cases, for example one or four. The subsequent lines list the corresponding values matching to the areas defined in the area line.
<pre>- wire_variation_file wire_variation_file</pre>	When specified, the software loads the wire variation side file.

Examples

- The following command loads the SPEF file `TOPCHIP_SP.spef`:

```
read_spef TOPCHIP_SP.spef
```

- The following command loads multiple SPEF files for hierarchical SPEF stitching:

```
read_spef { PTN/results_conv/results_conv.spef tdsp_core/tdsp_core/
tdsp_core.spef PTN/TOPCHIP_SP/TOPCHIP_SP.spef }
```

- The following commands a) setup the appropriate scale factors for the corners `RCmax` and `RCmin`, b) scale the SPEF read for the defined corners `RCmax` and `RCmin`:

```
=== MMMC setup ===

create_rc_corner -name RCmax -postRoute_cap 1.4 -postRoute_xcap 1.4 -postRoute_res 1.4
create_rc_corner -name RCmin -postRoute_cap 0.8 -postRoute_xcap 0.8 -postRoute_res 0.8

..

=== run script===

...

read_spef -rc_corner RCmax typSPEF.spef -rc_scale
read_spef -rc_corner RCmin typSPEF.spef -rc_scale
```

- The following command loads a SPEF file `pnta-spef.gz` that has RC fields from 5 RC corners:

```
read_spef -rc_corner {rcCorner1 rcCorner2 rcCorner3 rcCorner4 rcCorner5} -spef_field {1
2 3 4 5} pnta-spef.gz
```

The above command will be interpreted as follows:

1st field in SPEF maps to RC corner rcCorner1
2nd field in SPEF maps to RC corner rcCorner2
3rd field in SPEF maps to RC corner rcCorner3
4th field in SPEF maps to RC corner rcCorner4
5th field in SPEF maps to RC corner rcCorner5

- The following command loads a SPEF file `pnta-spef.gz` that has RC fields from N RC corners:

```
read_spef -rc_corner {rc_corner1 rc_corner2} -spef_field {2 5} pnta-spef.gz
```

The above command will read SPEF data for 2 RC corners only and will be interpreted as follows:

2nd field in SPEF maps to RC corner rc_corner1
5th field in SPEF maps to RC corner rc_corner2

- The following command loads a SPEF file `pnta-spef.gz` that has RC fields from N RC corners:

```
read_spef -rc_corner {rcCorner1 rcCorner2 rcCorner3} -spef_field {1 1 3} pnta-spef.gz
```

The above command will read SPEF data for 3 RC corners only and will be interpreted as follows:

1st field in SPEF maps to RC corner rcCorner1
1st field in SPEF maps to RC corner rcCorner2 as well.
3rd field in SPEF maps to RC corner rcCorner3

- The following command loads hierarchical design SPEF files. The software reads the top level multi-field SPEF file, followed by the multi-field SPEF file from `block1` and `block2`:

```
read_spef -rc_corner {rcCorner1 rcCorner2 rcCorner3 rcCorner4 rcCorner5} -spef_field {1  
2 3 4 5} {top.spef.gz block1.spef.gz block2.spef.gz}
```

Related Commands

- `create_parasitic_coordinate_transform`
- `create_rc_corner`
- `update_rc_corner`
- `report_annotated_parasitics`
- `set_multi_cpu_usage`

read_timing_context

```
read_timing_context
[-help]
[-clock_mapping_file clock_mapping_file]
[-instances instance_list]
[-skip_interface_paths]
[-view_map_file view_map_file]
[dir_name]
```

Reads the timing context of a block specified in the input directory.

Note: The `read_timing_context` command is not supported in DSTA. The context written from top level DSTA can be read in lower-level STA runs.

Parameters

<code>-help</code>	Prints out the command usage.
<code><i>dir_name</i></code>	Specifies the name of the input directory to load context data.
<code>-clock_mapping_file</code> <code><i>clock_mapping_file</i></code>	Specifies the clock mapping file. This file contains user-provided information to guide clock mapping. The software-inferred clock mapping will be overwritten by the user-specified clock mapping defined in the <code>clock_mapping_file</code> file.
<code>-instances</code> <code><i>instance_list</i></code>	Specifies the list of instances for which analysis needs to be performed. The instances provided should belong to a single cluster file; otherwise, context reading will error out. When this parameter is not specified, the software performs context analysis of all the instances (belonging to a cluster) of a design.
<code>-</code> <code>skip_interface_paths</code>	Skips the interface paths.
<code>-view_map_file</code> <code><i>view_map_file</i></code>	Specifies the view mapping file. This file contains user-specified information to guide view mapping using the <code>set_timing_context_view_mapping</code> command. In the case of a MMMC flow with different view names in the block and top STA, the equivalent block and top view names can be specified in <code>view_map_file</code> .

Examples

- The following command will read context models of all the instances of the `tdsp_core` design that are part of one cluster:

```
read_timing_context context
```
- The following command will read context models corresponding to the `tdsp_core_inst1` instance of the `tdsp_core` design:

```
read_timing_context context -instance {tdsp_core_inst1}
```
- The following command will read merged context model corresponding to the `tdsp_core_inst1` and `tdsp_core_inst2` instances of `tdsp_core` design - provided that both the instances are part of a single cluster:

```
read_timing_context context -instance {tdsp_core_inst1 tdsp_core_inst2}
```

The command will error out if these instances are part of different clusters.
- The following command will read context model of `tdsp_core` design with user-defined clock mapping and view mapping files:

```
read_timing_context context -clock_mapping_file clkMap.txt -view_map_file viewMap.txt
```

Related Commands

- `write_timing_context`
- `set_timing_context_view_mapping`
- `report_timing_context_clock_mapping`
- `compare_timing_context`

read_twf

```
read_twf  
[-help]  
[ filenames]  
[-cell string]  
[-prefix string]  
[-quiet]  
[-reset]  
[-scope string]  
[-skipconst]  
[-skiptw]  
[-strip_prefix string ]  
[-verbose]  
[-view]
```

Reads the external timing window file (TWF) for noise (Signal Integrity) and power calculation (Common Power Engine - CPE). This file is used to obtain clocks, constants, external load, slews, timing windows, and slack information. The external TWF specification is optional. In the absence of a TWF, this information is automatically derived using the Common Timing Engine (CTE) if timing constraints are read.

Note: If the `VOLTAGE_THRESHOLD` construct is missing from the TWF file header, the values 10 and 90 are automatically assigned as the minimum and maximum threshold values, respectively. For example:

```
VOLTAGE_THRESHOLD (10 90)
```

Parameters

<code>-help</code>	Prints out the command usage.
<code>-cell <i>string</i></code>	Specifies the master cell for the timing window file.
<code><i>filenames</i></code>	Specifies the names of the timing window files.
<code>-prefix <i>string</i></code>	Adds a prefix to the net or pin name.
<code>-quiet</code>	Skips information and warning messages.
<code>-reset</code>	Clears the previously read TWF files.
<code>-scope <i>string</i></code>	Specifies the scope for the given hierarchical timing window file.
<code>-skipconst</code>	Skips constant processing.
<code>-skiptw</code>	Skips timing windows processing.
<code>-strip_prefix <i>string</i></code>	Removes hierarchical path name.
<code>-verbose</code>	Prints the command progress details.
<code>-view <i>string</i></code>	Specifies the view for the timing window file.

Examples

- The following command reads the timing window file TW.in for the view 'view01', while skipping the constant processing from the timing window file:

```
tempus> read_twf TW.in -skipconst -view view01
```

- The following command clears the previously read TWF file:

```
tempus> read_twf -reset
```

Related Commands

- `write_twf`

report_analysis_coverage

```
report_analysis_coverage
[-help]
[pin_port_list]
[-check_type check_type_list]
[-exclude_untested reason_list]
[-include_dyn_checks]
[-include_parallel_check_arcs]
[-include_when_cond]
[-max_paths {type integer}]
[-nworst <integer>]
[-retime {aocv | ssta | path_slew_propagation | aocv_path_slew_propagation}]
[-sort {pin | refpin | checktype | slack | reason}]
[-tcl_list]
[-unsorted]
[-verbose check_status_list]
[-view view_name]
[> <filename[.gz]>]
[>> <filename[.gz]>]
```

Provides information about timing checks in the design.

For each type of timing check, the command reports the number of checks that meet constraints, violated constraints, or that are untested in the `SUMMARY` section of the report. The `DETAILS` section includes information about the signal pin, reference pin, check type, slack, and the reason for being untested.

When the `timing_report_group_based_mode` global variable is set to `true`, this command will report a check for endpoints for all clocks. In verbose mode, the "clock" column will be displayed.

✔ Use `check_timing` to look for missing constraints.

Parameters

-help	Prints out the command usage.
-------	-------------------------------

{> | >>} *file_name*

When specified with >, writes the report to the specified file name. If the file already exists, the software overwrites it.

When specified with >>, writes the report to the specified file name. If the file already exists, the software concatenates the report to the end of the file.

The *filename* parameter must be the last argument in the list. The *filename* and `-tcl_list` parameters are mutually exclusive; you cannot specify them together.

Note: To write a compressed report, add the `.gz` extension to the file name.

Default: Report is displayed on standard output without being saved.

`-check_type check_type_list`

Reports checks that are specified in the `check_type_list`.

Valid check types are:

- `clock_gating_hold`
- `clock_gating_pulse_width`
- `clock_gating_setup`
- `clock_period`
- `clock_separation`
- `data_hold`
- `data_setup`
- `external_delay`:
- `hold`
- `no_change_hold`
- `no_change_setup`
- `path_delay`
- `pulse_width`
- `recovery`
- `removal`
- `setup`
- `skew`
- `time_borrow`

Note: If there are no specified timing checks in the design, the report shows "No timing checks found."

Default: All (all types of check(s) are reported)

<code>-exclude_untested reason_list</code>	Excludes from the report endpoints that are unconstrained due to a specific reason. Valid reasons are: • <code>const</code>: Constant reaching signal/reference pin of any timing check. • <code>false_path</code>: Path disabled due to <code>set_false_path</code>. • <code>no_endpoint_clock</code>: Missing clock at the reference pin of the timing check. • <code>no_startpoint_clock</code>: Missing clock at the signal pin of the timing check. This may happen if the clock signal does not reach the launching register. • <code>unknown</code>: None of the known reasons are identified for an untested path. • <code>user_disable</code>: Checks are disabled due to <code>set_disable_timing</code>.
<code>-include_dyn_checks</code>	Reports the dynamic checks (<code>data_to_data</code>, <code>path_delay</code>) on pins that are constrained by <code>set_data_check</code>, <code>set_max_delay</code>, or <code>set_min_delay</code> constraints. By default, these checks will not be reported, and the command only reports the pins constrained by all library checks, <code>set_output_delay</code>, or inferred clock gating checks. The dynamic checks, such as <code>set_data_check</code> or <code>set_max_delay</code> are not reported by default, when <code>-include_dyn_checks</code> parameter is not specified.
<code>-include_parallel_check_arcs</code>	Reports all the parallel check arcs that have been merged.
<code>-include_when_cond</code>	Reports the when condition of check arcs.
<code>-max_paths {type integer}</code>	Enumerates the specified number of worst paths in the design, regardless of the endpoint.
<code>-nworst integer</code>	Enumerates the specified number of paths for each endpoint.
<code>-retime {aocv ssta path_slew_propagation aocv_path_slew_propagation}</code>	Reanalyzes the set of paths using the specified method.

<code>pin_port_list</code>	Limits the report to timing checks on pins in the <code>pin_port_list</code>. Note: This parameter is obsolete and will be removed in a future release. Cadence recommends using <code>pin_port_list</code> instead.
<code>-sort {pin reffin checktype slack reason}</code>	Sorts checks by the specified option when reporting detailed information with the <code>-verbose</code> option. The <code>-sort</code> parameter does not change the column order. The valid sort options are: <code>pin</code>, <code>reffin</code>, <code>checktype</code>, <code>slack</code>, <code>reason</code> When sorting by pin name, if the pin names are the same, the software then attempts to sort the checks by slack value. If the slack values are the same, the software attempts to sort by timing check type. When sorting by timing check type, if the check types are the same, the software then attempts to sort by slack value. If the slack values are the same, the software attempts to sort by pin name. When sorting by Slack, if the Slack values are the same, the software then attempts to sort the checks by pin name. If the pin names are the same, the software attempts to sort by timing check type. When you sort by <code>slack</code>, the Slack value is reported in order of least negative to most negative (worst Slack last). You can use the following commands for debugging purposes and to improve the readability of the report: <pre>report_analysis_coverage -verbose untested -sort reason report_analysis_coverage -verbose violated -sort slack</pre> Default: By default, checks are sorted by slack value (<code>slack</code>).
<code>-tcl_list</code>	Produces the report in Tcl list format instead of a tabular format. This is useful for integrating timing with custom Tcl functions, and also for customizing report generation. The <code>-tcl_list</code> and <code>filename</code> parameters are mutually exclusive; you cannot specify them together.
<code>-unsorted</code>	Reports unsorted slacks with untested endpoints at the end of the report.

<code>-verbose check_status_list</code>	Limits detailed information to checks with the specified status. The valid status types are: <code>met</code>, <code>violated</code>, and <code>untested</code>. • <code>met</code>: Shows as a positive value in the Slack column. • <code>violated</code>: Shows as a negative value in the Slack column. • <code>untested</code>: Shows as UNTESTED in the Slack column, and the reason is given in the Reason column. The Reason column will be blank when a Slack number is displayed - this indicates the check has been tested. You can reduce the number of lines in the DETAILS section by filtering out the met checks, as shown in the example below: <pre>report_analysis_coverage -verbose {violated untested}</pre> Note: The <code>-verbose</code> parameter supports multi-threading.
<code>-view view_name</code>	Generates a timing check report for the specified analysis view only. You can specify this parameter only when the software is in multi-mode multi-corner analysis mode.

Examples

- The following command filters from the report all endpoints that are unconstrained because of the reason `const`:

```
tempus> report_analysis_coverage -exclude_untested const -verbose untested
```

----- TIMING CHECK COVERAGE SUMMARY

Check Type	No. of Checks	Met	Violated	Untested
Clock Gating Hold	1	0 (0%)	1 (100%)	0 (0%)
Clock Gating Setup	1	1 (100%)	0 (0%)	0 (0%)
ClockPeriod	28	24 (85%)	0 (0%)	4 (14%)
ExternalDelay (Early)	1	0 (0%)	0 (0%)	1 (100%)
ExternalDelay (Late)	1	0 (0%)	0 (0%)	1 (100%)
Hold	13	8 (61%)	1 (7%)	4 (30%)
PulseWidth	28	24 (85%)	0 (0%)	4 (14%)
Setup	13	8 (61%)	1 (7%)	4 (30%)

----- TIMING CHECK COVERAGE DETAILS

Tempus Text Command Reference

Timing Analysis Commands

Pin	Reference Pin	Check Type	Slack	Reason

out1	-	ExternalDelay (Early)	UNTESTED	No data signal
out1	-	ExternalDelay (Late)	UNTESTED	No data signal
u99/CK ^	u99/CK ^	ClockPeriod	UNTESTED	No endpoint clock
u99/CK ^	u99/CK v	PulseWidth	UNTESTED	No endpoint clock
u99/CK v	u99/CK ^	PulseWidth	UNTESTED	No endpoint clock
u99/CK v	u99/CK v	ClockPeriod	UNTESTED	No endpoint clock
u99/D	u99/CK ^	Hold	UNTESTED	No endpoint clock
u99/D	u99/CK ^	Setup	UNTESTED	No endpoint clock
u8/CK ^	u8/CK ^	ClockPeriod	UNTESTED	No endpoint clock
u8/CK ^	u8/CK v	PulseWidth	UNTESTED	No endpoint clock
u8/CK v	u8/CK ^	PulseWidth	UNTESTED	No endpoint clock
u8/CK v	u8/CK v	ClockPeriod	UNTESTED	No endpoint clock
seg1/u9/D	seg1/u9/CK ^	Hold	UNTESTED	No data signal
seg1/u9/D	seg1/u9/CK ^	Setup	UNTESTED	No data signal
seg2/u9/D	seg2/u9/CK ^	Hold	UNTESTED	No data signal
seg2/u9/D	seg2/u9/CK ^	Setup	UNTESTED	No data signal

When the software is in multi-mode multi-corner timing analysis mode, You can specify the -view parameter to generate a timing check report for the specified analysis view only:

```
tempus 150> report_analysis_coverage -view view1
```

```
-----
```

```
TIMING CHECK COVERAGE SUMMARY (Analysis View: view1)
```

```
-----
```

Check Type	No. of Checks	Met	Violated	Untested

Clock Gating Hold	1	0 (0%)	1 (100%)	0 (0%)
Clock Gating Setup	1	1 (100%)	0 (0%)	0 (0%)
ClockPeriod	28	24 (85%)	0 (0%)	4 (14%)
ExternalDelay (Early)	1	0 (0%)	0 (0%)	1 (100%)
ExternalDelay (Late)	1	0 (0%)	0 (0%)	1 (100%)
Hold	14	8 (57%)	1 (7%)	5 (35%)
PulseWidth	28	24 (85%)	0 (0%)	4 (14%)
Setup	14	8 (57%)	1 (7%)	5 (35%)

Related Commands

- [check_timing](#)
- [report_timing](#)

report_analysis_mode

report_analysis_mode

Reports the timing analysis mode settings for the current Tempus session.

Parameters

None

Examples

- The following `report_analysis_mode` command example displays the current `set_analysis_mode` settings:

```
tempus> report_analysis_mode
set_analysis_mode -domain allClockDomain -checkType setup -skew true -log true -warn
true -sequentialConstProp true -clockPropagation sdcControl -clkSrcPath true -
timingSelfLoopsNoSkew false -asyncChecks async -useOutputPinCap true -timeBorrowing true
-clockGatingCheck true -enableMultipleDriveNet true -analysisType onChipVariation -cpvr
both -timingEngine statistical
```

Related Commands

- `set_analysis_mode`
- `set_input_delay`
- `set_output_delay`

report_analysis_summary

```
report_analysis_summary  
[-help]  
[-csv string]  
[-merged_views]  
[-retime {path_slew_propagation | aocv_path_slew_propagation}]  
[-retime_mode {exhaustive}]  
[-view viewName]  
[> <filename[.gz]>]  
[>> <filename[.gz]>]  
[-early | -late ]  
[-expanded_summary_clocks | -merged_groups]
```

Generates timing slack summary reports for specified analysis view(s). This command categorizes the slack summary in terms of groups: in2reg, reg2reg, reg2Out, and in2out. A column "ALL" is displayed that regroups in2reg, in2out, reg2reg, and reg2out details.

By default, slack is categorized into worst negative slack (WNS), total negative slack (TNS), and the total number of violations in group-based mode.

The `report_analysis_summary` command is supported in path-based analysis (PBA) mode.

Parameters

<code>-help</code>	Prints out the command usage.
<code>{> >>} filename</code>	When specified with <code>></code>, writes the report to the specified file name. If the file already exists, the software overwrites it. When specified with <code>>></code>, writes the report to the specified file name. If the file already exists, the software concatenates the report to the end of the file. Note: To write a compressed report, add the <code>.gz</code> extension to the file name. <i>Default:</i> Report is displayed on standard output without being saved.
<code>-csv string</code>	Specifies the name of the CSV output file.
<code>-early</code>	Generates a report for hold analysis mode for early paths.
<code>-expanded_summary_clocks</code>	Generates a report in terms of each possible clock domain pair separately.
<code>-late</code>	Generates a report for the setup analysis mode for late paths.
<code>-merged_groups</code>	Merges all the clock groups for a consolidated timing summary.
<code>-merged_views</code>	Generates a merged report across all the views.
<code>-retime</code> <code>{path_slew_propagation </code> <code>aocv_path_slew_propagation}</code>	Reanalyzes the defined set of paths using the specified method. By default, the timing report shows up to 2 million paths for retime PBA analysis.
<code>-retime_mode {exhaustive}</code>	Reanalyzes the specified set of paths using the specified retime mode.
<code>-view viewName</code>	Generates a report for the specified analysis views only. By default, all the active views are analyzed.

Examples

- The `report_analysis_summary` command shows the following report output:

```
tempus> report_analysis_summary

REPORT ANALYSIS SUMMARY (View:default_emulate_view) (EL:Setup)
=====

REPORT ANALYSIS SUMMARY (View:default_emulate_view) (EL:Hold)
=====

GroupType:ClockGate
=====

All           IN2REG           IN2OUT           REG2REG           REG2OUT
-----
WNS:TNS:#VIO  WNS:TNS:#VIO  WNS:TNS:#VIO  WNS:TNS:#VIO  WNS:TNS:#VIO  FromClk-
>ToClk
-----
-5.614:-5.614:1  0.000:0.000:0  0.000:0.000:0  -5.614:-5.614:1  0.000:0.000:0  *->*
-----

GroupType:Standard
=====

All           IN2REG           IN2OUT           REG2REG           REG2OUT
-----
WNS:TNS:#VIO  WNS:TNS:#VIO  WNS:TNS:#VIO  WNS:TNS:#VIO  WNS:TNS:#VIO  FromClk-
>ToClk
-----
-8.897:-8.897:1  0.000:0.000:0  0.000:0.000:0  -8.897:-8.897:1  0.000:0.000:0  *->
>CLK_W_1
-----
-0.571:-0.571:1  -0.571:-0.571:1  0.000:0.000:0  0.000:0.000:0  0.000:0.000:0  *->
>CLK_W_2
-----
```

Related Commands

- `set_default_view`
- `set_analysis_view`

report_analysis_views

```
report_analysis_views
-help
[-type {all | setup | hold | active}]
```

Generates a hierarchical report of the current multi-mode multi-corner configuration. By default, the `report_analysis_views` command generates a report of all defined views in the design. Use the `-type` parameter to generate a report on a specific type of analysis view.

Use this command after creating analysis views ([create_analysis_view](#)) and setting active setup and hold views ([set_analysis_view](#)) for the design.

Parameters

<code>-help</code>	Outputs a brief description that includes type and default information for each <code>report_analysis_views</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man report_analysis_views</code> .
<code>-type</code> {all setup hold active}	Specifies the type of report to generate. <i>Default:</i> all • active: Generates a report of all active setup and hold views in the design. • all: Generates a report of all defined views in the design, including those that are currently inactive. • hold: Generates a report of all active hold views in the design. • setup: Generates a report of all active setup views in the design.

Examples

- The following command generates a hierarchical report of the active setup analysis views in the design:

```
tempus> report_analysis_views -type setup
```

The software reports the following results:

```
Multi-Mode Analysis View Report
-----
+ SETUP Views
|
+ Analysis View: default_emulate_view
```

```

|
+ Delay Calc Corner: default_emulate_delay_corner
| |
| |
| |
| + rc_corner: default_emulate_rc_corner
| | |
| | + preRoute_res: 1
| | |
| | + postRoute_res: 1
| | |
| | + preRoute_cap: 1
| | |
| | + postRoute_cap: 1
| | |
| | + postRoute_xcap: 1
| | |
| | + preRoute_clkcap: 0
| | |
| | + postRoute_clkcap: 0 0 0
| | |
| | + preRoute_clkres: 0
| | |
| | + postRoute_clkres: 0 0 0
| |
| + late_library_set: default_emulate_libset_max
| | |
| | + timing: /icd/ssv_pe_t2b/timing/pawang/unit_test_cases/unit_2/data/cell_w.lib
| | |
| | + si: /icd/ssv_pe_t2b/timing/pawang/unit_test_cases/unit_2/data/celtic-NDC.cdB
| |
| + early_library_set: default_emulate_libset_min
| | |
| | + timing: /icd/ssv_pe_t2b/timing/pawang/unit_test_cases/unit_2/data/cell_w.lib
| | |
| | + si: /icd/ssv_pe_t2b/timing/pawang/unit_test_cases/unit_2/data/celtic-NDC.cdB
| |
| + si_enabled: true
|
+ Constraint Mode: default_emulate_constraint_mode
| |
| + SDC Constraint Files: /dev/null

```

Related Commands

- `create_analysis_view`
- `set_analysis_view`

report_annotated_assertions

```
report_annotated_assertions
[-help]
-type {load | transition}
[-list_annotated]
[-list_non_annotated]
[-view viewName ]
[-max_lines number ]
[-min | -max | -min_max]
object_list
[{> | >>} file_name ]
```

Reports information for annotated assertions, set using the `set_load` or `set_annotated_transition` command, according to the specified assertion type. You can select load or transition reporting.

Note: The `report_annotated_assertions` command is now obsolete. This command will continue to be supported in the current release, but will be removed in the next major release of the software.

Parameters

<code>{> >>} file_name</code>	Writes the report to the specified file name. When specified with <code>></code>, writes the report to the specified file name. If the file already exists, the software overwrites it. When specified with <code>>></code>, writes the report to the specified filename. If the file already exists, the software concatenates the report to the end of the file.
<code>-list_annotated</code>	Lists the detailed information for the specified assertion types.
<code>-list_non_annotated</code>	Lists information for non-annotated assertions.
<code>-min -max -min_max</code>	Specifies to report the minimum, maximum, or both values of the specified assertion type. When input assertions are applied using both <code>-min</code> and <code>-max</code> parameters, you can specify the <code>-min_max</code> parameter to report both the assertion values. In this case, you do not need to specify this command twice, that is, separately with <code>-min</code> and <code>-max</code> parameters. <i>Default:</i> max
<code>-max_lines number</code>	Controls the number of output lines reported by the <code>-list_annotated</code> or <code>-list_non_annotated</code> parameter. By default, the specified net, pin, or port names are reported.
<code>object_list</code>	Specifies the list of nets, pins, or ports to be reported.
<code>-type {load transition}</code>	Specifies the assertion type (load or transition) to be reported.
<code>-view viewName</code>	Reports the annotations for the specified analysis view. In multi-mode multi-corner timing analysis mode, annotations are reported for the specified mode or corner.

Examples

- The following command reports all the load annotated assertions:

```
tempus> report_annotated_assertions -type load -list_annotated -min_max
```

Related Commands

- `set_load`
- `set_annotated_transition`

report_annotated_check

```
report_annotated_check
[-help]
[-check_type check_type_list]
[-max_line number]
[instance_list]
[-list_annotated]
[-list_non_annotated]
[-view viewName]
[> | >> filename[.gz] | -tcl_list]
```

Reports coverage of annotated timing checks.

Parameters

-help	Prints out the command usage.
-check_type <i>check_type_list</i>	Allows you to specify the check type. The valid check types are <code>setup</code> , <code>hold</code> , <code>pulse_width</code> , <code>clock_period</code> , <code>clock_gating_setup</code> , <code>clock_gating_hold</code> , <code>clock_gating_pulse_width</code> , <code>data_setup</code> , <code>data_hold</code> , <code>recovery</code> , <code>removal</code> , <code>clock_separation</code> , <code>no_change_setup</code> , <code>no_change_hold</code> , and <code>skew</code> .
> <i>filename</i>	When specified with <code>></code> , writes the report to the specified file name. If the file already exists, the software overwrites it. When specified with <code>>></code> , writes the report to the specified file name. If the file already exists, the software concatenates the report to the end of the file. The <i>filename</i> parameter must be the last argument in the list. The <i>filename</i> and <code>-tcl_list</code> parameters are mutually exclusive; you cannot specify them together. Note: To write a compressed report, add the <code>.gz</code> extension to the file name. <i>Default:</i> Report is displayed on standard output without being saved.
instance_list	Specifies instance names for which annotations of check arcs are to be reported.

<code>-list_annotated</code>	Lists the detailed annotation information for the following timing checks: • setup • hold • recovery • removal • pulse-width • nochange-setup • clock-period • skew Note: If the software is in multi-mode multi-corner timing analysis mode, you need to specify the <code>-view</code> parameter.
<code>-list_non_annotated</code>	Lists the timing checks that do not have annotated delay information. The software checks for the following timing checks: • setup • hold • recovery • removal • pulse-width • nochange-setup • clock-period • skew Note: If the software is in multi-mode multi-corner timing analysis mode, you need to specify the <code>-view</code> parameter.
<code>-max_line <i>number</i></code>	Controls the number of output lines reported by the <code>-list_annotated</code> or <code>-list_non_annotated</code> parameter.
<code>-tcl_list</code>	Produces the report in Tcl list format instead of a tabular format. This is useful for integrating timing with custom Tcl functions, and also for customizing report generation. The <code>-tcl_list</code> and <code>filename</code> parameters are mutually exclusive; you cannot specify them together.

`-view viewName`

Reports coverage of annotated timing checks for the specified analysis view. You can specify this parameter only when the software is in multi-mode multi-corner timing analysis mode.

Default: Reports annotated timing checks for the default analysis view.

Examples

- The `report_annotated_check` command shows the following report output in simultaneous setup and hold analysis mode:

```
tempus> report_annotated_check
```

```
-----
```

Check arc type	Total	Annotated	Not Annotated	Percentage
				Arcs
				Annotated
-----+-----+-----+-----+-----				
Setup arcs	1642	2	1640	0.121803
Hold arcs	1642	1	1641	0.060901
Recovery arcs	368	0	368	0.000000
Removal arcs	367	1	366	0.272480
Min pulse width arcs	1403	0	1403	0.000000
Period arcs	6	0	6	0.000000
Clock gating hold arcs	2	0	2	0.000000
+-----+-----+-----+-----+-----				
All arcs	5430	4	5426	0.073665
+-----+-----+-----+-----+-----				

- The following command lists down all the annotated checks in the `f_slow` view:

```
tempus> report_annotated_check -list_annotated -view f_slow
```

```
-----
```

Analysis View: f_slow

```
+-----+
-----+
| Setup Checks |
```

SourceSignal	RefSignal	Delays	Type	Cond
View				
-----+-----+-----+-----+-----				

DTMF_IN/TDSP_CR/PROG	DTMF_IN/TDSP_CR/PROG	0.500:0.500:0.500		

Tempus Text Command Reference

Timing Analysis Commands

```

setup_rising_clk_rise|RN_EQ_1_AN|f_slow |
| BUS_MC_IN/dt_reg_2 | BUS_MC_IN/dt_reg_2 | | |
|
| D ^ | CK ^ | | |
|
| DTMF_IN/TDSP_CR/PROG| DTMF_IN/TDSP_CR/PROG|
0.500:0.500:0.500|setup_falling_clk_rise|RN_EQ_1_AN|f_slow |
| BUS_MC_IN/dt_reg_2 | BUS_MC_IN/dt_reg_2 | | |
|
| D v | CK ^ | | |
|
+-----+
-----

```

Check arc type	Total	Annotated	Not Annotated	Percentage
Setup arcs	1642	2	1640	0.121803
Recovery arcs	368	0	368	0.000000
Min pulse width arcs	1403	0	1403	0.000000
Period arcs	6	0	6	0.000000
All arcs	3419	2	3417	0.058497

- The following command lists down all the annotated checks in the f_fast view:

```
tempus > report_annotated_check -list_annotated -view f_fast
```

Analysis View: f_fast

SourceSignal	RefSignal	Delays	Type	Cond
View				
DTMF_INST/TEST_INST	DTMF_INST/TEST_INST	0.400:0.400:0.400	hold_rising_clk_rise	
f_fast				
RAM_INST/A[5]	RAM_INST/CLK			
DTMF_INST/TEST_INST	DTMF_INST/TEST_INST	0.400:0.400:0.400	hold_falling_clk_rise	
f_fast				

Tempus Text Command Reference

Timing Analysis Commands

```

| RAM_INST/A[5] v      | RAM_INST/CLK ^      |
|      |      |
+-----+
-----+
+-----+
-----+
| Removal Checks |
|-----+-----+-----+
-----|
| SourceSignal    | RefSignal          | Delays            | Type              | Cond
|View      |
|-----+-----+-----+-----+
-----|

| DTMF_IN/SPI_IN/dflag| DTMF_IN/SPI_IN/dflag|0.300:0.300:0.300|removal_rising_clk_rise|
| f_fast |
| N ^          | K ^          |
|      |      |
+-----+
-----+
+-----+

| Check arc type      | Total | Annotated | Not      | Percentage|
|      |      |      | Annotated | Arcs      |
|      |      |      |      | Annotated |
|-----+-----+-----+-----+
| Hold arcs          | 1642  | 1        | 1641     | 0.060901 |
| Removal arcs       | 367   | 1        | 366      | 0.272480 |
| Clock gating hold arcs | 2     | 0        | 2         | 0.000000 |
+-----+-----+-----+-----+
| All arcs           | 2011  | 2        | 2009     | 0.099453 |
+-----+-----+-----+-----+

```

Related Commands

- [set_annotated_check](#)
- [reset_annotated_check](#)

report_annotated_delay

```
report_annotated_delay
[-help]
[pin_port_list]
[-ignore_tied_low_high_net_arcs]
[-list_annotated]
[-list_non_annotated]
[-max_line number]
[-tcl_list]
[-view viewName]
[> file_name [.gz]]
[>> file_name [.gz]]
```

Reports coverage of annotations on a design. These annotations include SDF delay arc annotations. In addition to SDF annotations, the command also reports annotations set using the [set_annotated_delay](#) command.

Parameters

-help	Prints out the command usage.
> >> <i>filename</i>	When specified with >, writes the report to the specified file name. If the file already exists, the software overwrites it. When specified with >>, writes the report to the specified file name. If the file already exists, the software concatenates the report to the end of the file. The <i>filename</i> parameter must be the last argument in the list. The <i>filename</i> and -tcl_list parameters are mutually exclusive; you cannot specify them together. Note: To write a compressed report, add the .gz extension to the file name. <i>Default:</i> Report is displayed on standard output without being saved.
- ignore_tied_low_high_net_arcs	Ignores timing arcs of power and ground nets.

<code>-list_annotated</code>	Lists detailed annotation information for the following annotated arcs: • annotated internal net arcs • annotated input port arcs • annotated output port arcs • annotated cell arcs
<code>-list_non_annotated</code>	Lists the arc information for the non-annotated arcs corresponding to the following types of arcs: • non-annotated internal net arcs • non-annotated input port arcs • non-annotated output port arcs • non-annotated cell arcs
<code>-max_line <i>number</i></code>	Controls the number of output lines reported by the <code>-list_annotated</code> or <code>-list_non_annotated</code> parameter.
<code>pin_port_list</code>	Specifies pins or ports for which annotations of outgoing delay arcs are to be reported.
<code>-tcl_list</code>	Produces the report in Tcl list format instead of a tabular format. This is useful for integrating timing with custom Tcl functions, and also for customizing report generation. The <code>-tcl_list</code> and <code>filename</code> parameters are mutually exclusive; you cannot specify them together.
<code>-view <i>viewName</i></code>	Generates the report for the specified analysis view only. You can specify this parameter only when the software is in multi-mode multi-corner timing analysis mode.

Examples

Here are some examples.

- If your SDF has only cell arcs, and net delays are included in the cell delays, you can report the status of annotation for cell arcs, internal net arcs, net arcs from primary inputs, and net arcs to primary outputs, and all timing arcs separately.

```
tempus> report_annotated_delay
```

```
-----
```

Tempus Text Command Reference
Timing Analysis Commands

Delay Arc Type	Total	Annotated	Not Annotated	Percentage Annotated

Cell arcs	128	0	128	0.000000
Input net arcs	14	0	14	0.000000
Internal net arcs	107	0	107	0.000000
Output net arcs	7	0	7	0.000000

All timing arcs	256	0	256	0.000000

Related Commands

- [set_annotated_delay](#)
- [report_net](#)

report_annotated_parasitics

```
report_annotated_parasitics
[-help]
[-net_file string]
[-nets string]
[-view {viewName}]
[> {write_fileName}]
[>> {append_fileName}]
[{-list_annotated}
[-list_broken_net]
[-list_extra_pin_net]
[-list_float_net]
[-list_nodriver_net]
[-list_noload_net]
[-list_not_annotated]
[-list_real_net]
[-list_supply_net]
[-list_zero_cap_net]
[-max_missing {max_missing_int}}}]
```

Reports the back-annotated parasitics of the design. By default, only the summary report is generated. The software groups all the nets by net type and annotation status (annotated or not annotated).

Parameters

<code>-help</code>	Outputs a brief description that includes type and default information for the <code>report_annotated_parasitics</code> parameters. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man report_annotated_parasitics</code>.
<code>>></code> <code>{append_fileName}</code>	Appends the report to the specified file name.
<code>-list_annotated</code>	Generates a list of annotated nets.
<code>-list_broken_net</code>	Reports broken nets in the design.
<code>-list_extra_pin_net</code>	Generates a list of nets that have extra pins in the SPEF but not in the Verilog.
<code>-list_float_net</code>	Generates a list of floating nets.
<code>-list_nodriver_net</code>	Lists all terminal nets without a driver.
<code>-list_noload_net</code>	Lists all terminal nets with no load.
<code>-list_not_annotated</code>	Generates a list of not annotated nets.
<code>-list_real_net</code>	Lists all real nets.
<code>-list_supply_net</code>	Lists all supply nets.
<code>-list_zero_cap_net</code>	Generates a list of zero capacitance nets.
<code>-max_missing</code> <code>{max_missing_int}</code>	Limits the number of missing annotations to the specified number of annotations. <i>Default:</i> 1000
<code>-net_file string</code>	Accepts a file name, which contains a list of net names, as input. When this parameter is specified, the parasitic annotation summary is provided only for the listed nets. The net names in the file can be specified as one net per line.
<code>-nets string</code>	Specifies the list of net names for which an annotation summary is required.
<code>-view viewname</code>	Reports the back-annotated parasitics for the specified analysis view. You can specify this parameter only when the software is in multi-mode multi-corner timing analysis mode. <i>Default:</i> Reports against the default analysis view
<code>> {write_fileName}</code>	Writes the report to the specified file name.

Examples

- The following command generates a list of annotated and not-annotated nets:

```
report_annotated_parasitics -list_annotated -list_not_annotated -list_real_net -
list_float_net -list_nodriver_net \ + -list_noload_net -list_supply_net -max_missing
1000

# list syntax: type: net_name [nrCap Cap nrXCap XCap nrRes Res]
# unit: pF, Ohm
# List the first 82 of total 82 real nets:
Annotated Real Net: [seg3/w7] [ 10 0.0690723 25 0.0512442 9 219.251 ]
Annotated Real Net: [seg3/w5] [ 14 0.0373353 24 0.0265601 13 161.843 ]
:
:
Annotated Real Net: [CLK3] [ 2 0.00046867 0 0 1 18.872 ]
Annotated Real Net (broken): [CLK4] [ 1 0.0002379 0 0 2 0.0002 ]
# List the first 20 of total 20 not annotated real nets:
Not Annotated Real Net: [seg3/wout]
Not Annotated Real Net: [CLK44]
:
:
Not Annotated Real Net: [out1]
Not Annotated Real Net: [in1]
# No annotated supply net.
# List the first 4 of total 4 not annotated supply nets:
Not Annotated Supply Net: [gnd]
Not Annotated Supply Net: [GND]
Not Annotated Supply Net: [VDD]
Not Annotated Supply Net: [vdd]
# No annotated floating net.
# No not annotated floating net.
# No annotated no driver net.
# List the first 9 of total 9 not annotated no driver dangling nets:
Not Annotated Dangling Net (no driver): [seg3/n8]
Not Annotated Dangling Net (no driver): [seg3/n111111]
:
:
Not Annotated Dangling Net (no driver): [in]
Not Annotated Dangling Net (no driver): [out5]
# List the first 8 of total 8 annotated no load dangling nets:
Annotated Dangling Net (no load): [seg3/n14] [ 1 0.00520893 0 0 0 0 ]
Annotated Dangling Net (no load): [seg2/n14] [ 1 0.00636874 0 0 0 0 ]
:
:
Annotated Dangling Net (no load): [n6] [ 2 0.1086 0 0 1 0.0001 ]
Annotated Dangling Net (no load): [n5] [ 2 0.1065 0 0 1 0.0001 ]
```

List the first 16 of total 16 not annotated no load dangling nets:

Not Annotated Dangling Net (no load): [seg3/u14:QN]

Not Annotated Dangling Net (no load): [seg3/u9:QN]

:

:

Not Annotated Dangling Net (no load): [u2:QN]

Not Annotated Dangling Net (no load): [in4]

Summary of Annotated Parasitics:

+-----+				
Net Type	Count	Annotated (%)	Not Annotated (%)	
+-----+				
total	135	90 66.7%	45 33.3%	
+-----+				
no drive (*)	9	0 0.0%	9 100.0%	
1-term:no load	24	8 33.3%	16 66.7%	
real net (complete)	81	61 75.3%	20 24.7%	
real net (broken)	21	21 100.0%	0 0.0%	
+-----+				

Note for Net Types marked "(*)": Such nets are never timed, but reported here for informational purpose.

+-----+				
Annotated	Res (KOhm)	Cap (pF)	XCap (pF)	
+-----+				
Count	982	1011	1152	
Value	20.0869	7.7289	5.4598	
+-----+				

Related Commands

- [read_sdf](#)
- [read_spef](#)

report_aocv_derate

```
report_aocv_derate  
[-help]  
[-check_global_derating]  
[-clock]  
[-data]  
-file string  
[-list_cell_not_annotated]  
[-mesh]  
[-voltage float]  
[-design | -cell string ]  
[{-delay_corner string [-power_domain string] } | -max | -min]  
[-early | -late]  
[-rise | -fall]
```

Reports the AOCV derating tables from AOCV libraries. The command generates the output report in AOCV library format.

Parameters

<code>-help</code>	Prints out command usage.
<code>-cell <i>string</i></code>	Reports the table for the specified library cell name. You can specify a regular expression corresponding to library cell names.
<code>-check_global_derating</code>	Reports the list of cells for which the derating table or global derating table was not specified.
<code>-clock</code>	Reports derate tables read for clock paths.
<code>-delay_corner <i>string</i></code>	Reports the derating tables from the specified delay corner. For a MMMC design, if a delay corner-specific table is read using the <code>create_library_set -aocv</code> command, that derating table is reported. Otherwise, the default table is reported. For a non-MMMC design, no details are reported. The <code>-delay_corner</code> and <code>-min/-max</code> parameters are mutually exclusive.
<code>-data</code>	Reports derate tables read for data paths.
<code>-design</code>	Reports the default cell or net derating tables.
<code>-early -late</code>	Reports the early/late derating table.
<code>-file <i>string</i></code>	Reports all the derating tables. This parameter is mandatory.
<code>-list_cell_not_annotated</code>	Reports the list of cells for which the derating table is not specified. This parameter reports an error when specified with <code>-voltage</code> , <code>-design</code> , or <code>-cell</code> parameters.
<code>-max</code>	Reports derate tables from the max AOCV libraries.
<code>-mesh</code>	Reports derate tables read for mesh.
<code>-min</code>	Reports derate tables from the min AOCV libraries.
<code>-power_domain <i>string</i></code>	Reports derate tables from the specified power domain. This option should always be specified with a delay corner.
<code>-rise -fall</code>	Reports the rise/fall table for the specified transition.
<code>-voltage <i>float</i></code>	Reports the derating tables for the specified voltage value.

Examples

Here are some examples.

- The following command reports derating tables for the specified cells:

```
tempus> report_aocv_derate -cell "BUF" -delay_corner default_emulate_delay_corner -file
report.lib
```

```
#content of file "report.lib" :
```

```
Cell BUF
```

```
LATE-DATA-RISE
```

```
Stage 1.000000 2.000000 3.000000 4.000000 5.000000 8.000000 9.000000 18.000000 19.000000
1.180000 1.160000 1.140000 1.120000 1.100000 1.080000 1.060000 1.040000 1.020000
```

```
Cell BUF
```

```
LATE-DATA-FALL
```

```
Stage 1.000000 2.000000 3.000000 4.000000 5.000000 8.000000 9.000000 18.000000 19.000000
1.180000 1.160000 1.140000 1.120000 1.100000 1.080000 1.060000 1.040000 1.020000
```

```
Cell BUF
```

```
LATE-CLOCK-RISE
```

```
Stage 1.000000 2.000000 3.000000 4.000000 5.000000 8.000000 9.000000 18.000000 19.000000
1.180000 1.160000 1.140000 1.120000 1.100000 1.080000 1.060000 1.040000 1.020000
```

```
Cell BUF
```

```
LATE-CLOCK-FALL
```

```
Stage 1.000000 2.000000 3.000000 4.000000 5.000000 8.000000 9.000000 18.000000 19.000000
1.180000 1.160000 1.140000 1.120000 1.100000 1.080000 1.060000 1.040000 1.020000
```

- The following command reports derating tables for the cells:

```
tempus > report_aocv_derate -clock -file aocv_clock_derate.rpt
```

```
Cell AN2D1
```

```
EARLY-CLOCK-RISE
```

```
Stage 1.000000 2.000000 3.000000 4.000000 5.000000 6.000000 7.000000 8.000000 9.000000
10.000000
0.70000 0.850000 0.880000 0.890000 0.900000 0.910000 0.920000 0.920000 0.930000
0.935000
```

```
Cell AN2D1
```

```
EARLY-CLOCK-FALL
```

```
Stage 1.000000 2.000000 3.000000 4.000000 5.000000 6.000000 7.000000 8.000000 9.000000
10.000000
0.760090 0.830010 0.860060 0.880300 0.890200 0.901000 0.902000 0.910000 0.920000
0.9202594
```

```
Cell BUFF
```

```
EARLY-CLOCK-RISE
```

```
Stage 1.000000 2.000000 3.000000 4.000000 5.000000 6.000000 7.000000 8.000000 9.000000
10.000000
```

```
0.842000 0.888258 0.908854 0.921954 0.929953 0.935137 0.940946 0.944305 0.942031
0.959500
```

```
Cell BUFF
```

```
EARLY-CLOCK-FALL
```

```
Stage 1.000000 2.000000 3.000000 4.000000 5.000000 6.000000 7.000000 8.000000 9.000000
10.000000
```

```
0.739532 0.810305 0.840208 0.861346 0.880605 0.899462 0.903028 0.900408 0.912154
0.916708
```

```
* The following command reports derating tables for the cells in delay corner in mesh
report_aocv_derate -mesh -delay_corner dc_125c_rcworst -file
aocv_clock_delay_corner_dc_125c_rcworst.rpt
```

```
Cell AN2D1
```

```
EARLY-MESH-RISE
```

```
Cell AN2D1
```

```
EARLY-MESH-FALL
```

```
Stage 1.000000 2.000000 3.000000 4.000000 5.000000 6.000000 7.000000 8.000000 9.000000
10.000000
```

```
0.739532 0.810305 0.840208 0.861346 0.880605 0.899462 0.903028 0.900408 0.912154
0.916708
```

```
Cell AN2D1
```

```
LATE-MESH-RISE
```

```
Stage 1.000000 2.000000 3.000000 4.000000 5.000000 6.000000 7.000000 8.000000 9.000000
10.000000
```

```
0.842000 0.888258 0.908854 0.921954 0.929953 0.935137 0.940946 0.944305 0.942031
0.959500
```

```
Cell AN2D1
```

```
LATE-MESH-FALL
```

```
Stage 1.000000 2.000000 3.000000 4.000000 5.000000 6.000000 7.000000 8.000000 9.000000
10.000000
```

```
0.842000 0.888258 0.908854 0.921954 0.929953 0.935137 0.940946 0.944305 0.942031
0.959500
```

Related Commands

- [report_pba_aocv_derate](#)

report_boundary_model_constraints

report_boundary_model_constraints
[-help]

Reports the boundary model constraints that are applied using the `select_boundary_model_constraints` command.

Parameters

-help	Prints out the command usage.
-------	-------------------------------

report_case_analysis

```
report_case_analysis
[-help]
[<pins_and_ports_list>]
[-all]
[-dir]
[-include_ref_name]
[-nosplit]
[-tcl_list]
[-verbose]
[-view <view_name>]
[> <filename[.gz]>]
[>> <filename[.gz]>]
[-propagated [-include_function ]]
```

Generates a User Case Analysis table that reports ports and pins with a `set_case_analysis` constraint. These ports and pins are constant only for timing analysis. You can optionally generate a Logic Constant Value table, which reports ports and pins whose constant status is part of the design. Information from this table includes ports or pins with `set_logic` constraints (`set_logic_zero` and `set_logic_one`), netlist explicit constants, and output pins with constant output functions.

Parameters

-help	Prints out the command usage.
{> >>} <i>filename</i>	When specified with <code>></code>, writes the report to the specified file name. If the file already exists, the software overwrites it. When specified with <code>>></code>, writes the report to the specified file name. If the file already exists, the software concatenates the report to the end of the file. The <i>filename</i> parameter must be the last argument in the list. The <i>filename</i> and <code>-tcl_list</code> parameters are mutually exclusive; you cannot specify them together. Note: To write a compressed report, add the <code>.gz</code> extension to the file name. <i>Default:</i> Report is displayed on standard output without being saved.
-all	Specifies that the Logic Constant Value table report pins or ports with <code>set_logic</code> constants, constants in the netlist, and output pins with a constant function. These output pins are from library cells or netlist modules.

<code>-include_function</code>	Displays the pin function of a pin. This must be specified with the <code>-propagated</code> parameter.
<code>-include_ref_name</code>	Displays the reference name of an instance in the report.
<code>-nosplit</code>	Specifies that reports with long names will not be split into multiple lines. Note: The <code>-nosplit</code> parameter works if the <code>set_table_style -no_frame_fix_width</code> command is specified.
<code>-dir</code>	Specifies that the report include the direction of the pin or port, which is reported as input, output, or inout (internal pins may not have constant constraints).
<code>pins_and_ports_list</code>	Generates a report only for the pins and ports in the specified list. <i>Default:</i> Generates a report for all pins and ports in the design.
<code>-propagated</code>	Generates a report that includes all the constant pins, including pins with propagated constants. When specified, the report column header shows the constant values - the column header for constant values is <code>User case analysis value</code> . <i>Default:</i> Report shows pins specified with the <code>set_case_analysis</code> command.
<code>-tcl_list</code>	Produces the report in Tcl list format instead of a tabular format. This is useful for integrating timing with custom Tcl functions, and also for customizing report generation. The <code>-tcl_list</code> and <code>filename</code> parameters are mutually exclusive; you cannot specify them together.

Examples

Here are some examples.

- The following command uses the `-dir` option, which shows that 1 is on the input part of the `timing_const_on_Z/Z` pin:
tempus> report_case_analysis -dir

The following report is generated:

```

-----
Pin name          Dir          User case
                  Analysis value
-----
gscanmode         input         0

```

```
gscanenable      input      0
-----
```

- The following command generates a report that includes all constant pins, including pins with propagated constants. When specified, the report column header for constant values changes to Constant value.

```
tempus> report_case_analysis -propagated
```

The following report is generated:

```
-----
Pin name                                Constant
                                      value
-----
clockandlatch/U8/TE                    0
clockandlatch_dup1/U8/TE                1
clockandlatch_dup2/U8/TE                1
clockandlatch_dup3/U8/TE                0
-----
```

- The following command generates a report in which each pin is listed on a separate line. The report includes all constant pins, including pins with propagated constants. The pin `clockandlatch_dup3/U8/TE` is reported because of a `set_case_analysis 0` assertion on pin `gscanenable`:

```
tempus> report_case_analysis -propagated -verbose clockandlatch_dup3/U8/TE
```

```
-----
Pin name                                Constant
                                      Value
-----
gscanenable                            0
FE_OFC306_gscanenable/I                 0
FE_OFC306_gscanenable/ZN                 1
FE_OFC307_gscanenable/I                 1
FE_OFC307_gscanenable/ZN                 0
FE_OFC315_gscanenable/I                 0
FE_OFC315_gscanenable/Z                 0
coreclockandlatch_dup3/U8/TE            0
-----
```

- The following report shows that there are no `set_case_analysis` constraints but there are logical constraints:

```
tempus> report_case_analysis -propagated -verbose core_q_reg_1_/SI
```

```
-----
Pin name                                Logic
                                      constant
-----
```

	value

ram256x10/TEST1	0
ram1024x32/TEST1	0
core_q_reg_2_/SI	1
core_q_reg_11_/SI	0
core_q_reg_1_/SI	1

Related Commands

- [set_case_analysis](#)

report_cell_instance_timing

```
report_cell_instance_timing
[-help]
list_of_cell_instances
[-early | -late]
[-clock_timing]
[-tcl_list]
[-view view_name]
[{> | >>} filename]
```

Reports the following information about the timing context for all the pin and delay arcs of the specified instances:

- Arrival time
- Required time
- Slack
- Delays

Note: Internal pins (pins inside the instance boundary) are also reported.

This command uses the results of earlier timing analysis runs; therefore, you must run the `report_timing` command before using `report_cell_instance_timing`.

To analyze the setup (maximum) paths, you must use the `set_analysis_mode -checkType setup` command and run the `report_timing` command before using the `-late` parameter. Similarly, to analyze the hold (minimum) paths, you must use the `set_analysis_mode -checkType hold` command and run the `report_timing` command before using the `-early` parameter.

The `report_cell_instance_timing` timing report contains the following tables:

- The first table in the report contains information about the pins of the instance.
The propagated slew is the slew applied to the pin, depending on the slew propagation mode.
- The second table contains information about the arcs through the instance.
The slew out is the slew calculated at the arc output from the arc input slew, using the timing model.
- The third table, generated for sequential devices, contains information about the timing checks, if present, on the cell.
Only checks that are from the library are shown in this table. Automatic checks, such as clock gating setup and hold, are not reported.

The *Phase* column of each table can contain the following symbols:

- `D` indicates it is a data signal in the clock domain.
- `C` indicates it is a clock signal in the clock domain.

- P indicates it is a positive phase.
- N indicates it is a negative phase.
- L indicates it is a latency phase.
- Asterisk (*) indicates there is a timing exception associated with the signal.

Note: Currently, it is not possible to expand the asterisk to determine which exception is associated with the reported timing phase.

When the software is in multi-mode multi-corner timing analysis mode, the timing context information is reported for each active analysis view, with the view name listed in the *Phase* column.

Parameters

<code>-help</code>	Prints out the command usage.
<code>{> >>} filename</code>	When specified with <code>></code>, writes the report to the specified file name. If the file already exists, the software overwrites it. When specified with <code>>></code>, writes the report to the specified file name. If the file already exists, the software concatenates the report to the end of the file. The <code>filename</code> parameter must be the last argument in the list. The <code>filename</code> and <code>-tcl_list</code> parameters are mutually exclusive; you cannot specify them together. Note: To write a compressed report, add the <code>.gz</code> extension to the file name. <i>Default:</i> Report is displayed on standard output without being saved.
<code>-clock_timing</code>	Specifies that slack is to be reported for clock style checks.
<code>-early -late</code>	Reports the information for early (maximum) path (data hold/clock setup) or late (minimum) path (data setup/clock hold). You must use the <code>set_analysis_mode -checkType hold</code> command and run the <code>report_timing</code> command before using the <code>-early</code> parameter. Similarly, you must use the <code>set_analysis_mode-checkType setup</code> command and run the <code>report_timing</code> command before using the <code>-late</code> parameter. <i>Default:</i> If you do not specify any of the parameters, the software uses the <code>-late</code> parameter.
<code>list_of_cell_instances</code>	Specifies a flat (non-hierarchical) cell instance or list of cell instances. Use curly braces (<code>{}</code>) to enclose the list and separate the instance names with white space.
<code>-tcl_list</code>	Produces the report in Tcl list format instead of a tabular format. This is useful for integrating timing with custom Tcl functions, and also for customizing report generation. The <code>-tcl_list</code> and <code>filename</code> parameters are mutually exclusive; you cannot specify them together.
<code>-view view_name</code>	Specifies the analysis view in which to query the object attribute.

Examples

- The following example reports information about the timing context of instance seg1/u9:

```
tempus > report_cell_instance_timing seg1/u9
```

```
-----
Instance seg3/u14 of DFF
-----
```

Pin	Dir	Propagated Slew	Arrival	Required	Slack	Phase
CK ^	IN	0.062	0.621			CLK_W_3 (C) (P) *
CK v	IN	0.065				CLK_W_3 (D) (P) *
D v	IN	0.003	1.815	2.465	0.650	CLK_W_3 (D) (P)
CK ^	IN	0.062				CLK_W_3 (D) (N) *

```
-----
Instance seg3/u14 of DFF
-----
```

```
Arc Slew
-----
```

From	To	In	Out	Load	Delay	Phase
CK ^	QN ^	0.062	0.058	0.000	0.196	CLK_W_3 (C) (P) *
CK ^	QN v	0.062	0.041	0.000	0.236	CLK_W_3 (D) (P) *
CK ^	QN ^	0.062	0.000			CLK_W_3 (D) (N) *

```
-----
Instance seg3/u14 of DFF
-----
```

```
Pins Phase
-----
```

Check	Sig	Ref	Slack	Delay	Adjust	Sig	Ref
HOLD	D ^	CK ^	1.570	-0.129	0.000	CLK_W_3 (D) (P)	CLK_W_3 (C) (P) *
SETUP	D ^	CK ^	0.372	0.186	-2.000	CLK_W_3 (D) (P)	CLK_W_3 (C) (P) *

- If a constant is asserted or propagated to the instance pin, it is reported in the Phase column of the pin table, as shown in the following example:

```
tempus > report_cell_instance_timing p5042A
```

```
-----
Instance p5042A of MUX2HVTD1
-----
```

Pin	Dir	Propagated Slew	Arrival	Required	Slack	Phase
I0 ^	IN	0.032	0.127	8.583	8.455	EJ_TCK (D) (P)
I0 v	IN	0.021	0.127	8.568	8.441	EJ_TCK (D) (P)
S	IN					Constant 0

```
Z ^      OUT      0.020      0.168      8.624      8.455  EJ_TCK(D) (P)
```

```
-----
Instance p5042A of MUX2HVTd1
-----
```

```
Arc Slew
```

```
-----
From    To    In      Out      Load    Delay Phase
-----
```

```
I0 ^    Z ^    0.032    0.020    0.002    0.041  EJ_TCK(D) (P)
```

```
I1 ^    Z ^    0.060    0.002                      EJ_TCK(D) (P)
-----
```

- When the software is in multi-mode multi-corner analysis mode, the timing context information is reported for each active analysis view. The view name is reported in the Phase column, as shown in the following example:

```
tempus > report_cell_instance_timing FE_OFC304_EJ_TDO
```

```
-----
Instance FE_OFC304_EJ_TDO of BUFFHVTd8
-----
```

```
Pin      Dir      Propagated  Arrival Required  Slack  Phase
Slew
```

```
-----
I ^      IN      0.052      8.812      16.260      7.448  EJ_TCK(D) (N) (setup_view1) *
```

```
I v      IN      0.033      8.799      16.319      7.520  EJ_TCK(D) (N) (setup_view1) *
-----
```

Related Commands

- [report_net](#)
- [report_timing](#)

report_clock_gating_check

```
report_clock_gating_check  
[-help]  
[ object_list ]  
[-view view_name ]  
[{> | >>} file_name ]
```

Reports information of all or specific clock gating checks of the current design. If you specify a list of pins, cells, or clocks, the clock gating check information is reported only for the specified objects. If you use the `report_clock_gating_check` command without an object list, all clock gating check information is reported.

The `Type` column of the generated report can contain one of the following abbreviations:

- `I` indicates that the clock gating check is inferred.
- `L` indicates that the clock gating check is from the library (integrated clock gating check).

Set the `report_precision` global variable using the `set_global` command to specify the accuracy of the result in terms of the number of significant digits.

Parameters

<code>-help</code>	Prints out the command usage.
<code>object_list</code>	Specifies a list of pins, instances, or clocks. A collection of these can also be specified. - Pin: Any pin of a clock gating cell can be specified, and the check information will be reported for that pin. - Instance: Any instance name of a clock gating cell can be specified, and the check information will be reported for that instance. - Clock: Only a clock at a given clock (reference) pin of a cell can be specified for reporting the check information. Check information about the clocks at the enable (signal) pin cannot be reported. If the <code>object_list</code> parameter is not specified, all the clock gating checkpoints (inferred and library) of the current design are reported.
<code>-view</code> <code>view_name</code>	Specifies a view for which clock gating check is to be reported. The default is the worst of all the views.
<code>{> >>}</code> <code>filename</code>	When specified with <code>></code>, writes the report to the specified file name. If the file already exists, the software overwrites it. When specified with <code>>></code>, writes the report to the specified file name. If the file already exists, the software concatenates the report to the end of the file.

Examples

The following command returns a clock gating check report for the current design:

```
tempus> report_clock_gating_check
```

The clock gating check report returned by this command is as follows:

```
-----
Clock Gating Check Report
-----
Instance  Enable  Clock  Type  Level  Setup  Setup  Hold  Hold  View
          Rise   Fall   Rise   Fall
-----
C_and_1   A       B      I      H      0.000  0.000  -     -     view1
-----
```

Related Commands

- `report_precision`
- `set_global`

report_clock_network_transitions

```
report_clock_network_transitions  
[-help]  
[-all]  
[-clock string]  
[-exclude_pins string]  
[-threshold float]  
[-verbose]  
[-view string]
```

Reports all the clock tree pins that have valid rise/fall maximum transition values.

This command generates reports on a per clock basis. For virtual clocks, no pins will be reported.

Parameters

-help	Prints out the command usage.
-all	Reports all the pins having a valid transition value.
-clock <i>string</i>	Specifies a list of clocks.
-exclude_pins <i>string</i>	Does not report the specified pins.
-threshold <i>float</i>	Does not report pins that have at least one of the transition values (rise or fall) greater than the specified threshold value.
-verbose	Specifies to include the library cell information.
-view <i>string</i>	Specifies the name of the view.

Examples

- The following command reports all the pins along with the rise/fall transition values for the specified clock:

```
report_clock_network_transitions -all -clock clk_48
```

```
Clock: clk_48
```

```
Clock Network:
```

Pin	Rise	Fall
chip_core_inst/sb_inst/u_usb20_phy_top/output_buffer_ring_5/o	0.030	0.036
chip_core_inst/sb_inst/clk48_cts19_iso0/i	0.030	0.037
chip_core_inst/sb_inst/clk48_cts19_iso0/o	0.060	0.079
chip_core_inst/sb_inst/scan_pll_mux_clk_48/clock_bist_mux_ch/i1	0.060	0.078
chip_core_inst/sb_inst/scan_clk_bypas_clk_48/scan_clock_mux_ch/i0	0.060	0.078
chip_core_inst/sb_inst/scan_clk_bypas_clk_48/scan_clock_mux_ch/o	0.076	0.105
:		
:		

report_clock_propagation

```
report_clock_propagation
[-help]
-clock clock_list
[-max_paths number of paths]
-to pin_port_list
[-verbose]
[-view viewName]
[> filename[.gz]]
[>> filename[.gz]]
```

This command allows you to determine clocks that do not propagate to a particular node in the network and report the causes of such occurrences.

Parameters

-help	Prints out the command usage
-clock <i>clock_list</i>	List or collection of clocks that are to be checked.
-max_paths <i>number of paths</i>	Reports the number of maximum clock paths per clock.
-to <i>pin_port_list</i>	Specifies the pin/port list.
-verbose	Generates a verbose report for the clock propagation stop reason.
-view <i>viewName</i>	Generates a report for the specified analysis view.
> <i>filename[.gz]</i> / >> <i>filename[.gz]</i>	When specified with >, writes the report to the specified file name. If the file already exists, the software overwrites it. When specified with >>, writes the report to the specified file name. If the file already exists, the software concatenates the report to the end of the file. Note: To write a compressed report, add the .gz extension to the file name.

Examples

- The following command shows that clock propagation stops due to an inactive timing arc from b12/A to b12/Y

```
tempus> set_disable_timing -from A -to Y b12
tempus> report_clock_propagation -to u99/CK -clock CLK_W_4

Clock: CLK_W_4
Point: u99/CK
Reason 1: The clock 'CLK_W_4' propagation stops due to the inactive timing arc:
(b12/A->b12/Y)
```

The partial clock path:

Source	Sink	ArcType	ArcSense
CLK4	b11/A	(net)	(net)
b11/A	b11/Y	combinational	positive_unate
b11/Y	b12/A	(net)	(net)
b12/A	b12/Y	combinational	positive_unate

- The following command shows that clock propagation stops due to no clock source found:

```
tempus> report_clock_propagation -to launchff/Q -clock [get_clocks gen_clk]
tempus> report_clock_propagation -to seg3/u15/Q -clock CLK_W_5_gen
```

```
Clock: CLK_W_5_gen
Point: seg3/u15/Q
Reason 1: The generated clock 'CLK_W_5_gen' could not be created on its root pin
'seg3/u15/Q' because no clock source found for it
```

- The following command shows that clock propagation stops at the register clock pin as all the arcs from the clock pin of the register have been disabled:

```
tempus> set_disable_timing -from A -to Y seg3/u17_a
tempus> report_clock_propagation -to seg3/u14/CK -clock CLK_W_3
```

```
Clock: CLK_W_3
Point: seg3/u14/CK
Reason 1: The clock 'CLK_W_3' propagation stops due to the inactive timing arc:
(seg3/u17_a/A->seg3/u17_a/Y)
```

The partial clock path:

Source	Sink	ArcType	ArcSense
CLK3	seg3/u16/A	(net)	(net)

seg3/u16/A	seg3/u16/Y	combinational	positive_unate
seg3/u16/Y	seg3/u17/A	(net)	(net)
seg3/u17/A	seg3/u17/Y	combinational	positive_unate
seg3/u17/Y	seg3/u17_a/A	(net)	(net)
seg3/u17_a/A	seg3/u17_a/Y	combinational	positive_unate

Related Commands

- [set_disable_timing](#)

report_clock_sense

report_clock_sense
[-help]

Reports all clock sense settings in the design.

Parameters

-help	Prints out the command usage.
-------	-------------------------------

Related Commands

- `set_clock_sense`
- `reset_clock_sense`
- `get_pins`
- `get_ports`

report_clock_timing

```
report_clock_timing
[-help]
[-absolute_compare]
[-cpr_relative]
[-latency_greater_than <lower_limit>]
-type {skew interclock_skew jitter summary latency cpr_stage_count}
    [ [-launch | -capture]
      [-rise | -fall]
      | [-histogram [-histogram_range interval size]]
      [-logic_level [-source {clock_root | generated_clock}]]
    ]
]
[-early | -late]
[-clock clock_list]
[-from_clock from_clock_list]
[-to_clock to_clock_list]
[-from from_list]
[-to to_list]
[-nworst worst_entries]
[-greater_than lower_limit]
[-view view_name]
[-verbose]
[-format column_list]
[-tcl_list]
[ {> | >>} filename ]
```

Generates a clock analysis report for the current design. The software generates a separate report for each specified clock or pair of clocks. You specify the clocks using the `-clock` parameter, and clock pairs using the `-from_clock` and `-to_clock` parameters. You can use the `-nworst` and `-greater_than` parameters with the `-from_clock` and `-to_clock` parameters to generate reports with skew to the specified sink pins. You can use the `-early` and `-late` parameters to specify the type of skew to be reported.

Note: To remove common path pessimism from the reported skew, use the `set_analysis_mode -cpr both` command before using the `report_clock_timing` command. The report contains a separate column for common path adjustment values.

You can use the `report_clock_timing` command to perform MMMC-based SSTA analysis. In MMMC mode, the command reports the latency and clock skew for all individual views. You can use the `report_clock_timing -view` parameter to report clock timing for a user-specified view.

Parameters

<code>-help</code>	Prints out the command usage.
<code>{> >>} filename</code>	When specified with <code>></code>, writes the report to the specified file name. If the file already exists, the software overwrites it. When specified with <code>>></code>, writes the report to the specified file name. If the file already exists, the software concatenates the report to the end of the file. The <code>filename</code> parameter must be the last argument in the list. The <code>filename</code> and <code>-tcl_list</code> parameters are mutually exclusive; you cannot specify them together. Note: To write a compressed report, add the <code>.gz</code> extension to the file name. <i>Default:</i> Report is displayed on standard output without being saved.
<code>-absolute_compare</code>	Reports skews with values compared in absolute form.
<code>-clock clock_list</code>	Specifies the list of clocks to be reported. The software generates a separate report for each clock specified in the <code>clock_list</code>. <i>Default:</i> Skew report is generated for all clocks in the design.
<code>-cppr_relative</code>	Reports the latency in skew reports from the CPPR common pin. When specified, the reports are printed with latency measured from the branch point to the register's CK pin. The CPPR value is no longer required in reports.
<code>-early</code>	Uses hold skew to generate the report. Hold skew is the difference between the minimum latency at the start point flip-flop and the maximum latency at the end point flip-flop. <i>Default:</i> Uses the setup skew.
<code>-format column_list</code>	Specifies which columns to display for path reports, and the order in which they should appear. For example: <pre>-format {instance arc pin cell slew load}</pre> The <code>-format</code> parameter can be used only for latency or skew reports (<code>-type skew</code>, <code>-type interclock_skew</code>, or <code>-type latency</code>), and if the <code>-verbose</code> parameter is specified. See Report Timing--Column List Options for a list of valid arguments.

<code>-from from_list</code>	Specifies the list of sequential device clock pins or ports to consider as from-pins in the generated report. If any pin in the <i>from_list</i> is not a sequential device clock pin, it is replaced with all the sequential device clock pins in the transitive fanout of the named pin. If there are no sequential pins in the transitive fanout of the pin, the software drops the pin from the list and generates a warning message.
<code>- from_clock from_clock_list</code>	Specifies the list of from-clocks to be used in generating the current interclock skew report. You can use this option only in an interclock skew report. <i>Default:</i> Report is generated considering all clocks as from-clocks.
<code>-greater_than lower_limit</code>	Reports skews with values greater than the <i>lower_limit</i> value.
<code>-histogram</code>	Generates a textual histogram of the latency report. The <code>-histogram</code> parameter can be used only when <code>-type latency</code> is specified. When specified in conjunction with the <code>-clock</code> parameter, the software limits latency reporting to the specified clocks only. When specified with the <code>-greater_than</code> parameter, the software limits latency reporting to clocks with a latency value greater than the specified value. Note: The <code>-histogram</code> parameter cannot be specified in conjunction with the following parameters: <code>-launch</code> , <code>-capture</code> , <code>-from</code> , <code>-to</code> , <code>-from_clock</code> , <code>-to_clock</code> , <code>-verbose</code> , <code>-nworst</code> .
<code>-histogram_range interval_size</code>	Generates the histogram report at the specified intervals. The interval size specified must be a positive integer or a decimal value. For example, if the maximum latency value is 437 and you specify <code>-histogram_range 50</code> , the histogram would be generated with intervals of: 0-50, 50-100, 100-150, ... 400-450. Note: The unit of <i>interval_size</i> will be the same as that used for generating the timing report. This parameter can be used only when the <code>-histogram</code> parameter is specified.
<code>-late</code>	Uses setup skew to generate the report. Setup skew is the difference between the maximum latency at the start point flip-flop and the minimum latency at the end point flip-flop.

<code>-latency_greater_than <lower_limit></code>	Reports are filtered based on the specified <code>-type</code> option. When specified with: • <code>-type skew</code> or <code>interclock_skew</code>: Reports are filtered for latency greater than the provided value or depth. • <code>-type cprpr_stage_count</code>: Reports are filtered for depth greater than the provided value.
<code>-launch -capture</code>	When specified with <code>-type latency</code>, it generates a report of either the launch latency values (<code>-launch</code>), or the capture latency values (<code>-capture</code>). Note: You can specify these parameters only with <code>-type latency</code>. <i>Default:</i> <code>-launch</code>
<code>-logic_level</code>	Reports the levels (the stage count of gates) between the clock root, or generated clock root, and the capturing register. The <code>-logic_level</code> parameter can be used only when <code>-type latency</code> is specified. The <code>-logic_level</code> parameter cannot be specified in conjunction with the following parameters: <code>-launch</code>, <code>-capture</code>, <code>-from</code>, <code>-to</code>, <code>-from_clock</code>, <code>-to_clock</code>, <code>-verbose</code>, <code>-nworst</code>.
<code>-nworst worst_entries</code>	Specifies the number of entries to be reported per clock domain. <i>Default:</i> 1
<code>-rise -fall</code>	Specifies active transition at the sequential devices clock pins. The <code>-rise</code> and <code>-fall</code> parameters are mutually exclusive and cannot be specified together. Note: You can specify these parameters only with the <code>-type latency</code> parameter.

<pre>-source {clock_root generated_clock}</pre>	Reports the latency from either the master clock, or the generated clock that was derived from it. The <code>-source</code> parameter can be used only when <code>-logic_level</code> is specified. The <code>-source</code> parameter cannot be specified in conjunction with the following parameters: <code>-launch</code>, <code>-capture</code>, <code>-from</code>, <code>-to</code>, <code>-from_clock</code>, <code>-to_clock</code>, <code>-verbose</code>, <code>-nworst</code>. <i>Default:</i> Reports the latency from the master clock (<code>-source clock_root</code>)
<pre>-tcl_list</pre>	Produces the report in Tcl list format instead of a tabular format. This is useful for integrating timing with custom Tcl functions, and also for customizing report generation. The <code>-tcl_list</code> and <code>filename</code> parameters are mutually exclusive; you cannot specify them together.
<pre>-to to_list</pre>	Specifies the list or collection of sequential device clock pins or ports to be considered for reporting. If any pin in the <code>to_list</code> is not a sequential device clock pin, it is replaced with all the sequential device clock pins in the transitive fanout of the named pin. If there are no sequential pins in the transitive fanout of the pin, the software drops the pin from the list and generates a warning message.
<pre>-to_clock to_clock_list</pre>	Specifies the list of to-clocks to be used in generating the current interclock skew report. You can use this option only in an interclock skew report. <i>Default:</i> Report is generated considering all clocks as to-clocks.
<pre>-type {skew interclock_skew jitter summary latency cprpr_stage_count}</pre>	Specifies the type of report to generate. The <code>-type</code> options are described below. • <code>skew</code>: Reports the skew for all clocks, or for clocks that you specify using the <code>-clock</code> parameter. • <code>interclock_skew</code>: Reports the skew between all clocks (excluding the same clock pair) in the design, or between clocks that you specify using the <code>-from_clock</code> and <code>-to_clock</code> parameters. Note: The <code>skew</code> option can be used to report the previous results that show the missed out same clock skew values (reported using the <code>interclock_skew</code> option).

- **jitter:** Reports the difference in late and early arrival times of a clock phase at a clock end point. When specified, the software generates a clock jitter report for one clock end point, for each clock in the design.

You can generate a jitter report for all clocks, or clocks specified with the `-clock` parameter. You can limit the report to a specific clock end point using the `-to` parameter.

Note: The `report_clock_timing -type jitter` option can be used only when the software is in On-Chip Variation (OCV) timing analysis mode. Jitter analysis does not remove any common path pessimism.

- **summary:** Generates a summary report for each analyzed clock that includes the following information:
 - When setup skew is used (`-late`):
 - maximum launch latency
 - minimum capture latency
 - maximum setup skew
 - When hold skew is used (`-early`):
 - minimum launch latency
 - maximum capture latency
 - maximum hold skew
- **latency:** Reports the network and source latency values for all the clock pins in the design, or for clock pins that you specify using the `-clock` parameter.
- **cppr_stage_count:** The generated report is arranged in terms of difference of stages from common pin to CK pin. The reports are generated on the basis of logical CPPR points derived between each pair of phases reaching the launch/capture CK pins. Only same clock pairs (similar to type skew) are picked. For example,

```
Branch Point Difference Depth Clock Pin
-----
inv2/Y
5 r ffc1/CK
4 1 f ffc3/CKN
```

<code>-verbose</code>	Generates a report that contains the full clock path from the clock source to the sink pin (the clock pin of the flop or latch). You can specify the <code>-verbose</code> parameter only for skew or latency reports (<code>-type skew</code>, <code>-type interclock_skew</code>, or <code>-type latency</code>). When this parameter is specified in SOCV mode, the intermediate arrival times of pins are shown as "-" in the report output.
<code>-view view_name</code>	Generates a report for the specified analysis view only. You can specify this parameter only when the software is in multi-mode multi-corner timing analysis mode.

Examples

- The following command reports worst case skew for all clocks in a design:

```
tempus > report_clock_timing -type skew -nworst 4
```

```
Clock: CLK_W_4
```

```
Skew      Latency   Clock Pin
```

```
-----
```

```
          0.150 r    u99/CK
```

```
0.000     0.150 r    u55/CK
```

```
Clock: CLK_W_3
```

```
Skew      Latency   Clock Pin
```

```
-----
```

```
          0.682 r seg3/u9/CK
```

```
0.061     0.621 r seg3/u14/CK
```

```
Clock: CLK_W_2
```

```
Skew      Latency   Clock Pin
```

```
-----
```

```
          0.682 r seg2/u9/CK
```

```
0.061     0.621 r seg2/u14/CK
```

```
Clock: CLK_W_1
```

```
Skew      Latency   Clock Pin
```

```
-----
```

```
          0.682 r seg1/u9/CK
```

```
0.061     0.621 r seg1/u14/CK
```

Here 'r' indicates the transition (r - rise, f - fall) at the reported register clock pin.

- The following command reports interclock skew with -from_clock, -to_clock, -from, and -to parameters defined:

```
tempus > report_clock_timing -type interclock_skew -nworst 10 -from CLK4 -to u5/CK -
from_clock CLK_W_4 -to_clock CLK_W_4_gen2
```

```
Clocks: CLK_W_4 -> CLK_W_4_gen2
```

```
Skew      Latency   Clock Pin
```

```
-----
```

CLK4

```

0.180 r    u99/CK
          u2/Q
          u8/Q
-0.205    0.386 ri  u5/CK

```

Here 'r' indicates the transition (r - rise, f - fall) at the reported register clock pin. The 'i' refers to the 'inverted' polarity of the clock signal at the reported register clock pin.

- The following command reports skew for all clocks in the design including common path pessimism removal:

```

tempus > set_analysis_mode -cpr both
tempus > report_clock_timing -type skew -nworst 10

```

```

Clock: CLK_W_4
Skew    Latency    CPR    Clock Pin
-----
          0.180          r u99/CK
0.000    0.120    -0.060 r u55/CK

Clock: CLK_W_3
Skew    Latency    CPR    Clock Pin
-----
          0.818          r seg3/u9/CK
0.139    0.496 -0.183  r seg3/u14/CK

Clock: CLK_W_2
Skew    Latency    CPR    Clock Pin
-----
          0.818          r seg2/u9/CK
0.139    0.496    -0.183  r seg2/u14/CK

Clock: CLK_W_1
Skew    Latency    CPR    Clock Pin
-----
          0.818          r seg1/u9/CK
0.139    0.496    -0.183  r seg1/u14/CK

```

- The following command generates a summary report for the clock:

```

tempus > report_clock_timing -type summary -nworst 10 -clock CLK_W_1

```

```

Clock: CLK_W_1

          Skew    Latency    Clock Pin
-----
Maximum launch latency:  0.818 r    seg1/u9/CK
Minimum capture latency: 0.000 r    u2/CK

```

```
Maximum skew:          0.818 r    seg1/u9/CK
                     0.322  0.496 r    seg1/u14/CK
```

- The following command reports the Network and source latency values for the clock CLK_W_1:

```
tempus > report_clock_timing -type latency -nworst 10 -clock CLK_W_1
```

```
Clock: CLK_W_1
```

```
---Latency---
```

```
Source Network Total Clock Pin
```

```
0.000 0.818 0.818 r seg1/u9/CK
```

```
0.000 0.000 0.000 r u2/CK
```

- The following command reports the Network and source latency values for the clock GCLK1, as well as the full clock paths:

```
tempus > report_clock_timing -type latency -verbose -clock CLK_W_1
```

```
Clock: CLK_W_1
```

```
---Latency---
```

```
Source Network Total Clock Pin
```

```
0.000 0.818 0.818 r seg1/u9/CK
```

```
Clock latency Path
```

```
Clock Rise Edge 0.000
```

```
= Beginpoint Arrival Time 0.000
```

```
Other End Path:
```

Instance	Arc	Cell	Delay	Clock Latency
----------	-----	------	-------	---------------

-	CLK1 ^	-	-	0.000
---	--------	---	---	-------

seg1/u16	A ^ -> Y ^	CLKBUF	0.111	0.111
----------	------------	--------	-------	-------

seg1/u17	A ^ -> Y ^	CLKBUF	0.126	0.236
----------	------------	--------	-------	-------

seg1/u17_a	A ^ -> Y ^	CLKBUF	0.107	0.343
------------	------------	--------	-------	-------

seg1/and1	B ^ -> Y ^	AND	0.071	0.818
-----------	------------	-----	-------	-------

seg1/u9	CK ^	DFF	0.000	0.818
---------	------	-----	-------	-------

- When the software is in multi-mode multi-corner timing analysis mode, the software generates a report for every active analysis view in the design:

```
tempus > report_clock_timing -type latency
```

```
Clock: EJ_TCK
```

```
Analysis View: hold_view1
```

```
---Latency---
```



```
Source Network Total Clock Pin
-----
0.000 0.000 0.000 r cregister_q_reg_3_/CP

Clock: EJ_TCK
Analysis View: hold_view2
---Latency---
Source Network Total Clock Pin
-----
0.000 0.000 0.000 r cregister_q_reg_3_/CP

Clock: EJ_TCK
Analysis View: setup_view1
---Latency---
Source Network Total Clock Pin
-----
0.000 0.000 0.000 r cregister_q_reg_3_/CP

Clock: EJ_TCK
Analysis View: setup_view2
---Latency---
Source Network Total Clock Pin
-----
0.000 0.000 0.000 r cregister_q_reg_3_/CP
```

- Specify the **-view** parameter to generate a report only for a specific analysis view

```
tempus > report_clock_timing -view setup_view1 -type latency
```

```
Clock: EJ_TCK
Analysis View: setup_view1
---Latency---
Source      Network      Total      Clock      Pin
-----
0.000      0.000      0.000      r      cregister_q_reg_3_/CP

Clock: SI_ClkIn
Analysis View: setup_view1
---Latency---
Source      Network      Total      Clock      Pin
-----
0.000      0.000      0.000      r      cregister_q_reg_13_/CP
```

- The following command generates a textual histogram of the latency report:

```
tempus > report_clock_timing -type latency -histogram
```

```
[Delay Chart: CLK4(CLK_W_4_gen2)]
*****
Min Delay : 0.778
Max Delay : 0.778
Count : 1
```

```
*****
-----
Delay      Range
Min      Max      Count      1
-----
0.000    100.000  1          |o
```

- The following command generates the histogram report at intervals of 0.5:

```
tempus > report_clock_timing -type latency -histogram -histogram_range 0.5
```

```
[Delay Chart: CLK4 (CLK_W_4_gen2)]
*****
Min Delay : 0.778
Max Delay : 0.778
Count : 1
*****
-----
Delay Range
Min      Max      Count      1
-----
0.500    1.000  1          |o
0.000    0.500  0
```

- The following command generates a latency report that includes the stage gate counts between the clock root or generated clock root, and the capturing register:

```
tempus > report_clock_timing -type latency -logic_level
```

```
*****
```

```
Start point:CLK2 (CLK_W_4_gen1 (CLK_W_3))
```

```
-----
```

Delay	Count	End Point
1.018	8	seg2/u9/CK r
1.018	8	seg2/u3/CK r
0.945	7	seg2/u14/CK r

```
*****
```

```
Start point:CLK3 (CLK_W_4_gen2 (CLK_W_4))
```

```
-----
```

Delay	Count	End Point
1.345	8	seg3/u6/CK r
1.345	8	seg3/u1/CK r
1.218	7	seg3/u12/CK r

- The following command generates a latency report starting from the generated clock, and includes the stage gate counts between the generated clock and the capturing register:

```
tempus > report_clock_timing -type latency -logic_level -source generated_clock
```

```
*****
```

```
Start point:CLK2 (CLK_W_4_gen1 (CLK_W_3))
```

```
-----
```

Delay	Count	End Point
0.818	8	seg2/u9/CK r
0.818	8	seg2/u3/CK r
0.745	7	seg2/u14/CK r

```
*****
```

```
Start point:CLK3 (CLK_W_4_gen2 (CLK_W_4))
```

```
-----
```

Delay	Count	End Point
0.845	8	seg3/u6/CK r
0.845	8	seg3/u1/CK r
0.718	7	seg3/u12/CK r

- The following command generates a latency report starting from the generated clock CLK_W_4_gen2, and includes the stage gate counts between the generated clock and the capturing register:

```
tempus > report_clock_timing -type latency -logic_level -source generated_clock -clock CLK_W_4_gen2
```

```
*****
```

```
Start point:CLK3 (CLK_W_4_gen2 (CLK_W_4))
```

```
-----
```

Delay	Count	End Point
0.845	8	seg3/u6/CK r
0.845	8	seg3/u1/CK r
0.718	7	seg3/u12/CK r

- The following command uses the -format parameter to customize the generated latency report:

```
tempus > report_clock_timing -type latency -verbose -format {instance arc pin cell slew load user_derate}
```

```
Clock: CLK_W_4
```

---Latency---

Source	Network	Total	Clock Pin
--------	---------	-------	-----------

0.000	0.152	0.152	r u55/CK
-------	-------	-------	----------

Clock latency Path

Clock Rise Edge 0.000

= Beginpoint Arrival Time 0.000

Other End Path:

Instance	Arc	Pin	Cell	Slew	Load	User Derate
-	CLK4 ^	CLK4	-	0.003	0.025	-
b11	A ^ -> Y ^	Y	BUF	0.131	0.011	1.000
b12	A ^ -> Y ^	Y	BUF	0.117	0.015	1.000
u55	CK ^	CK	DFF	0.117	0.015	1.000

- The following command reports the worst case jitter for the clock CLK_W_1:

```
tempus > report_clock_timing -type jitter -clock CLK_W_1 -nworst 2
```

Clock: CLK_W_1

Jitter	Latency (Late)	Latency (Early)	Clock Pin
0.273	0.818	r	0.545 r seg1/u9/CK
0.273	0.818	r	0.545 r seg1/u3/CK

- The following command reports the worst case jitter for the specified clock end point:

```
tempus > report_clock_timing -type jitter -to seg1/u3/CK
```

Clock: CLK_W_1

Jitter	Latency (Late)	Latency (Early)	Clock Pin
0.273	0.818	r	0.545 r seg1/u3/CK

Related Commands

- [set_analysis_mode](#)
- [report_timing](#)

report_clocks

```
report_clocks
[-help]
[-description]
[-arrival_points]
[-phase_shift_table]
[-total_shift_table]
[-uncertainty_table]
[-adjustment_table]
[-delay_adjustment_table]
[-source_insertion]
[-insertion]
[-generated]
[-groups]
[-hierarchy]
[-clocks clk_signame | clk_signame_list]
[-view view_name]
[{> | >>} filename | -tcl_list]
```

Reports information on clocks created with the [create_clock](#) and [create_generated_clock](#) commands. The report includes information about all clock waveforms, clock arrival points, and clock uncertainties.

Use the `-description`, `-arrival_points`, `-uncertainty_table`, and the `*_table` parameters to specify different parts of the report. If one of these options is provided, only the section it corresponds to is reported. If other options are supplied, those sections are also reported.

Based on the current setup or hold analysis mode, the following parameters report only one of the late or early tables:

- `-phase_shift_table`
- `-total_shift_table`
- `-uncertainty_table`
- `-adjustment_table`
- `-delay_adjustment_table`

However, in simultaneous (setup and hold) mode, both tables are reported.



The `report_clocks` command reports only constraints related to clock waveforms. Use the `report_ports -type` command for constraints applied to clock pins.

Parameters

-help	Prints out the command usage.
{> >>} <i>filename</i>	When specified with >, writes the report to the specified file name. If the file already exists, the software overwrites it. When specified with >>, writes the report to the specified file name. If the file already exists, the software concatenates the report to the end of the file. The <i>filename</i> parameter must be the last argument in the list. The <i>filename</i> and -tcl_list parameters are mutually exclusive; you cannot specify them together. Note: To write a compressed report, add the .gz extension to the file name. <i>Default:</i> Report is displayed on standard output without being saved.
-adjustment_table	Reports two tables that detail the hold and setup cycle adjustments between clock waveforms as set by the set_multicycle_path command. Note: If specific ideal (or generated) clocks are requested with the -clocks <i>clock_name_list</i> option, then only the relationships between those clock signals are reported.
-arrival_points	Reports only the clock arrival points. A clock arrival point is the pin where the create_clock or create_generated_clock command has been asserted. Note: This parameter is obsolete and will be removed in a future release.
-clocks <i>clk_signame</i> <i>clk_signame_list</i>	Requests information on a specific ideal (or generated) clock. If a list is given, only information about those clocks is reported. You can use wildcards in the names. <i>Default:</i> All clocks, except generated clocks, are reported. Use the -generated option to report generated clocks. Note: This parameter is obsolete and will be removed in a future release. Cadence recommends that you specify clock names without the -clocks parameter.
- delay_adjustment_table	Reports two tables that detail the early and late path delay adjustments between clock waveforms as set by the set_max_delay and set_min_delay commands. Note: If specific ideal (or generated) clocks are requested with the -clocks <i>clock_name_list</i> option, then only the relationships between those clock signals are reported.
-description	Reports only the description of the specified clock waveform(s).

-generated	Includes information on generated clocks in the report, such as the source and master clock of the generated clock, and other information specified in the <code>create_generated_clock</code> constraints. The report uses the internal clock names created by the system if you did not specify new clock names while creating these clocks. Note: This parameter is obsolete and will be removed in a future release. The software now reports generated clocks by default.
-groups	Reports the existing clock groups.
-hierarchy	Reports the inter-clock relationship in tree style format.
-insertion	Reports the network insertion delays specified on clock waveforms. The values in the table are in <code>min:typ:max</code> format.
-phase_shift_table	Reports the table that details the phase shift between clock waveforms. Based on the current analysis mode, the table shows the phase shifts either in late or early mode. In the case of simultaneous setup hold mode, both tables are reported for phase shift in late and early modes. Note: If specific ideal (or generated) clocks are requested with the <code>-clocks clock_name_list</code> option, then only the relationships between those clock signals are reported.
-source_insertion	Reports the source insertion delays specified on clock waveforms.
-tcl_list	Produces the report in Tcl list format instead of a tabular format. This is useful for integrating timing with custom Tcl functions, and also for customizing report generation. The <code>-tcl_list</code> and <code>filename</code> parameters are mutually exclusive; you cannot specify them together.
-total_shift_table	Based on the current analysis mode, it reports one of the tables (late or early mode) that details the total relationship between clock waveforms. In the case of a simultaneous setup hold mode, both tables are reported. Note: If specific ideal (or generated) clocks are requested with the <code>-clocks clock_name_list</code> option, then only the relationships between those clock signals are reported.

<code>-uncertainty_table</code>	Based on the current analysis mode, reports one of the tables (late mode or early mode) that detail the uncertainty between clock waveforms as specified using the <code>set_clock_uncertainty</code> command. In the case of a simultaneous setup hold mode, both tables are reported. The tables report only those clock waveforms for which uncertainty has been defined using the <code>set_clock_uncertainty</code> command. Note: If specific ideal (or generated) clocks are requested with the <code>-clocks clock_name_list</code> option, then only the relationships between those clock signals are reported.
<code>-view view_name</code>	Generates a clock report for the specified analysis view only. You can specify this parameter only when the software is in multi-mode multi-corner timing analysis mode.

Examples

Here are some examples.

- `tempus > report_clocks`

```
-----
Clock Descriptions
-----
Attributes
-----
Clock Name  Source  Period  Lead   Trail  Generated Propagated
-----
CLK1        CLK1    20.000  0.000  10.000  n         y
CLK2        CLK1    10.000  0.000   5.000  n         y
-----
Generated-Clock Descriptions
-----
-----
Name          Generated  Master    Master-clock Invert Freq.    Duty-Cycle Edges
Edge-Shift
              Source(pin) Source(pin)                               Multiplier
-----
-----
GCLK         u8/Q      CK1       CLK1        n        1/2      -        -        -
-----
```

- The following command displays only the uncertainty information:

```
tempus > set_clock_uncertainty 5 CLK_W_1
tempus > set_clock_uncertainty 4 -from CLK_W_1 -to CLK_W_4_gen1
tempus > set_clock_uncertainty 6 -from CLK_W_2 -to CLK_W_4_gen2
tempus > report_clocks -uncertainty_table
```

Clock To Clock Uncertainty (Late)

From	To	L->L	L->T	T->L	T->T
*	CLK_W_1	5.000	5.000	5.000	5.000
CLK_W_1	CLK_W_4_gen1	4.000	4.000	4.000	4.000
CLK_W_2	CLK_W_4_gen2	6.000	6.000	6.000	6.000

Clock To Clock Uncertainty (Early)

From	To	L->L	L->T	T->L	T->T
*	CLK_W_1	-5.000	-5.000	-5.000	-5.000
CLK_W_1	CLK_W_4_gen1	-4.000	-4.000	-4.000	-4.000
CLK_W_2	CLK_W_4_gen2	-6.000	-6.000	-6.000	-6.000

- Consider the following clock definition:

```
tempus > create_clock -name "CLK_W_1" -add -period 60 -waveform {0 30}
tempus > report_clocks CLK_W_1 -phase_shift_table
```

Clock To Clock Phase Shift (Late)

From	To	L->L	L->T	T->L	T->T
CLK_W_1	CLK_W_1	60.000	0.000	60.000	60.000

Clock To Clock Phase Shift (Early)

From	To	L->L	L->T	T->L	T->T
CLK_W_1	CLK_W_1	0.000	-60.000	0.000	0.000

- You can use the following command to add uncertainty:

```
tempus > set_clock_uncertainty 2 CLK_W_1
```

The phase shift and total shift table values will be different. The following command displays the changed shift values as given below:

```
tempus > report_clocks -total_shift_table CLK_W_1
```

```
-----
Total Clock To Clock Phase Shift Required (Late)
-----
```

```
From      To      L->L      L->T      T->L      T->T
-----
```

```
CLK_W_1 CLK_W_1 58.000 -2.000 58.000 58.000
-----
```

```
-----
Total Clock To Clock Phase Shift Required (Early)
-----
```

```
From      To      L->L      L->T      T->L      T->T
-----
```

```
CLK_W_1 CLK_W_1 2.000 -58.000 2.000 2.000
-----
```

You can see that the uncertainty is subtracted from the phase shift.
Consider the following example path:

Path 1: MET Setup Check with Pin u2/CK

Endpoint: u2/D (v) checked with leading edge of 'CLK_W_1'

Beginpoint: CLK2 (v) triggered by trailing edge of 'CLK_W_2'

Path Groups: {CLK_W_1}

Other End Arrival Time 0.000

- Setup 0.188

+ Phase Shift 15.000

- Uncertainty 2.000

= Required Time 12.812

- Arrival Time 12.463

= Slack Time 0.349

Clock Fall Edge 12.500

= Beginpoint Arrival Time 12.500

```
-----
Load  Slew    Delay  Arrival  Cell  Arc          Pin
      Time
-----
0.034  0.003  12.500                CLK2 v      CLK2
0.003  0.003  -0.037  12.463    BUF   A v -> Y v  u10999/Y
0.003  0.003  0.000  12.463    DFF   D v          u2/D
-----
```

Here the begin point CLK_W_2 to end point CLK_W_1 shows a phase shift value of 15.00. This

means that if a path starts from the falling edge of CLK_W_2 , and ends at the rising edge of CLK_W_1, the phase shift is 15.00. In other words, when the timing (setup time) is reported, the required time at the end point begins with 15.00 (in addition to adjustments like setup time, time borrowing, uncertainty, etc). However, the arrival time of this path starts at 12.500.

- When the software is in multi-mode multi-corner timing analysis mode, the software reports the clock information for each active analysis view:

```
tempus > report_clocks -total_shift_table -description
```

```
-----
Clock Descriptions
-----
```

```
Attributes
-----
```

Clock Name	Source	View	Period	Lead	Trail	Generated	Propagated
EJ_TCK	EJ_TCK	hold_view2	17.400	0.000	8.700	n	n
EJ_TCK	EJ_TCK	hold_view1	17.400	0.000	8.700	n	n
EJ_TCK	EJ_TCK	setup_view1	17.400	0.000	8.700	n	n

```
-----
Total Clock To Clock Phase Shift Required (Late)
-----
```

From	To	View	L->L	L->T	T->L	T->T
EJ_TCK	EJ_TCK	hold_view2	17.400	0.000	17.400	17.400
EJ_TCK	EJ_TCK	hold_view1	17.400	0.000	17.400	17.400
EJ_TCK	EJ_TCK	setup_view2	17.400	0.000	17.400	17.400

- Specify the `-view` parameter to generate clock information for a specific analysis view only:

```
tempus > report_clocks -total_shift_table -description -view setup_view1 > clk_sv1
```

```
-----
Clock Descriptions
-----
```

```
Attributes
-----
```

Clock Name	Source	View	Period	Lead	Trail	Generated	Propagated
EJ_TCK	EJ_TCK	setup_view1	17.400	0.000	8.700	n	n
virt_in_EJ_TCK	-	setup_view1	17.400	0.000	8.700	n	n

```
-----
Total Clock To Clock Phase Shift Required (Late)
-----
```

From	To	View	L->L	L->T	T->L	T->T
------	----	------	------	------	------	------

```
EJ_TCK          EJ_TCK setup_view1 17.400    0.000   17.400   17.400
virt_in_EJ_TCK  EJ_TCK setup_view1 17.400    0.000   17.400   17.400
-----
```

- The `report_clocks` command output displays a column showing active or inactive clocks in the design:

```
-----
Clock Descriptions
-----
```

```
Attributes
-----
```

Clock Name	Source	Period	Lead	Trail	Generated	Propagated	Active
CLK_W_1	CLK1	20.000	0.000	10.000	n	y	y
CLK_W_2	CLK2	25.000	0.000	12.500	n	y	y
CLK_W_4_gen2	u8/Q	20.000	0.000	10.000	y	y	y

```
-----
```

Related Commands

- `create_clock`
- `create_generated_clock`
- `set_clock_uncertainty`

report_constraint

```
report_constraint
[-help]
[-all_violators]
[-verbose]
[-late]
[-early]
[-em_type {peak | rms | avg | all}]
[-check_type { pulse_width | clock_period | recovery | removal | clock_gating_setup |
clock_gating_hold | skew | pulse_clock_max_width | pulse_clock_min_width | electromigration |
rail_swing}]
[-clock]
[-connection_class]
[-data]
[-drv_abs_margin float]
[-drv_fields {field_list}]
[-drv_output_format {txt|csv}]
[-drv_ratio_margin float]
[-drv_violation_type { max_capacitance | max_transition | max_fanout | min_capacitance |
min_transition | min_fanout | pulse_clock_max_transition | pulse_clock_min_transition}]
[-max_pins_per_drv <integer>]
[-retime {aocv | path_slew_propagation | aocv_path_slew_propagation}]
[-retime_mode {path | exhaustive}]
[-view viewName ]
[-worst_drv_per_net]
[ pin_port_list ]
[{> | >>} filename[.gz]]
```

Reports constraint information of the current design. The constraints information includes information such as whether or not a constraint is violated, by how much, and the worst violator object.

The following timing checks are reported when you use the `report_constraint` command:

- All setup and hold checks
- DRV
- Clock and asynchronous checks including `pulse_width`, `min_period`, `recovery`, `removal`, `clock_gating_setup`, `clock_gating_hold`, `skew`

You can specify the design rule constraints on pins, ports, cells, or a design using the commands: `set_min_fanout`, `set_max_fanout`, `set_min_transition`, `set_max_transition`, `set_min_capacitance`, and `set_max_capacitance`.

The `report_constraint` command checks against the worst design rule constraint between the library constraint limit and the user-specified constraint value when both are applied to the violating pin or port.

The `report_constraint` command supports multi-threaded reporting. You can make the following

settings to enable multi-threaded reporting:

```
set_multi_cpu_usage -localCpu number_of_CPU  
set_table_style -no_frame_fix_width [-nosplit]
```

By default, the `report_constraint` command reports only the mean slews in transition DRV checks. You can enable DRV checking to use mean as well as the n-sigma slew values (in SOCV mode), with the following setting:

```
set timing_report_max_transition_check_using_nsigma_slew true
```

The `report_constraint` command allows merging of reports. The following check type report outputs from various MMMC analysis views can be merged into a single report using the `merge_timing_reports` command:

- min/max delay reports
- recovery/removal reports
- DRV reports
- pulse width reports
- skew/period checks

The input reports must be created with the reporting table format conforming to the `set_table_style -no_frame_fix_width -nosplit` format, which can be any of the following:

- group or non-group
- verbose or non-verbose
- compressed or uncompressed

The merged report will have endpoints from various views sorted in order of slack. For a pin reported from multiple analysis views, slack from all the views is available.

You can use a one-step method for generating merged MMMC view reports from the DMMMC master shell. After running the `distribute_views` command, you can issue the `report_constraint` command, which generates reports collectively from all active distributed views (that is, views available for interactive DMMMC analysis). The merged output report is arranged according to slack/criticality order across active views.

All the options of the `report_constraint` command are supported in this mode.

Configurable DRV Reports

Tempus allows you to perform design rule violation (DRV) checks to ensure that the netlist meets the design rule constraints specified in the design library. To enable configurable DRV reporting, you can use the `timing_report_constraint_format` global variable. For example,

```
set timing_report_constraint_format {list of DRV fields}  
report_constraint -drv_violation_type {list of DRV checks} \  
    -all_violators | pin_or_port_list
```

You can also use the following command to specify DRV fields:

```
report_constraint -drv_violation_type {list of DRV checks} -drv_fields field_list
```

The field list specified with the `-drv_fields` parameter has higher precedence and will override the field list specified with the `timing_report_constraint_format` global variable. If DRV fields are not specified with this global variable (or with the `report_constraint -drv_fields` command), then the default set of DRV fields will be displayed. For a complete list of DRV fields, refer to the [DRV Fields](#) table.

For more information, refer to the [Design Rule Violation \(DRV\) Reporting](#) section in the *Tempus User Guide*.

Parameters

<code>-help</code>	Prints out the command usage.
<code>{> >>} filename</code>	When specified with <code>></code>, writes the report to the specified file name. If the file already exists, the software overwrites it. When specified with <code>>></code>, writes the report to the specified file name. If the file already exists, the software concatenates the report to the end of the file. The <code>filename</code> parameter must be the last argument in the list. Note: To write a compressed report, add the <code>.gz</code> extension to the file name. <i>Default:</i> Report is displayed on standard output without being saved.

<code>-all_violators</code>	Reports the worst negative slack per violating pin against all the timing and design rule constraint checks. For timing checks, the violations are reported at timing endpoints. The DRV is checked and reported on leaf pins and ports. The supported check types are: • setup • hold • clock_gating_setup • clock_gating_hold • recovery • removal • clock_period • pulse_width • max_capacitance • max_transition • max_fanout • min_capacitance • min_transition • min_fanout • skew • pulse_clock_max_width • pulse_clock_min_width • pulse_clock_max_transition • pulse_clock_min_transition You can use the <code>-all_violators</code> parameter for reporting specific check types to generate a report for that particular check type only. For example, you can use <code>-all_violators</code> with <code>-late</code> to report setup checks and path delays from <code>set_max_delay</code>.
<code>-check_type</code>	Reports setup or hold checks. The supported check types are: • <code>clock_gating_setup</code>: Reports only the paths that end at the <code>clock_gating_setup</code> timing check. • <code>clock_gating_hold</code>: Reports only the paths that end at the

clock_gating_hold timing check.

- `electromigration`: Reports signal EM analysis related violations. When high-density current passes through a thin metal wire, the high-energy electrons exert forces on atoms, causing them to migrate. Electromigration (EM) can drastically reduce the lifetime of a chip by increasing the resistivity of a metal wire or creating a short circuit between adjacent lines. When this option is specified, the software reports standard cell-level electromigration violations.
- `recovery`: Reports paths that end at the recovery timing check.
- `removal`: Reports paths that end at the removal timing check.
- `pulse_width`: Reports only the paths that end at the pulse_width timing check.
If you do not specify the `-all_violators` parameter, then the `pulse_width` option reports only one entry of the most critical pulse_width check. The DRV checks also report only the most critical entry.

To report all the violated pulse_width checks, you need to specify the `report_constraint -check_type pulse_width -all_violators` command (which is consistent with other check types of the `report_constraint` command).

To report both violating and met entries, you can use the `report_min_pulse_width` command.

- `clock_period`: Reports only the paths that end at the clock_period timing check.
- `skew`: Reports skew checks.
The `report_constraint` command performs skew checks by default.
Skew checks are implemented as setup checks that are reported in the late analysis or simultaneous mode. These checks are not reported in the early mode. Skew checks are performed assuming the signal and reference clocks have the same frequency. If the frequencies of the signal and reference clocks differ, a warning message is displayed.
- `pulse_clock_max_width`: Reports the maximum pulse clock width checks set using the `set_pulse_clock_max_width` command.

	• <code>pulse_clock_min_width</code>: Reports the minimum pulse clock width checks. • <code>rail_swing</code>: Reports details related to the rail swing checks. For rail swing check settings, you can use the following global variables: ◦ <code>timing_rail_swing_checks_low_voltage_threshold</code> ◦ <code>timing_rail_swing_checks_high_voltage_threshold</code>
<code>-clock</code>	Specifies that the DRV checks will be restricted to the clock network. If any pin is part of both the clock and data network, it will be reported as a clock.
<code>-connection_class</code>	Reports violations if the connection class attributes of the driving and receiving pins of a net are different.
<code>-data</code>	Specifies that the DRV check will be restricted to the data network. If any pin is part of both the clock and data network, it will be reported as a clock network pin. By default, the DRV checks are applied to both clock and data networks.
<code>-drv_abs_margin float</code>	Specifies the absolute DRV margin value applied to obtain the threshold transition.
<code>-drv_fields {field_list}</code>	Allows only the specified fields to be included in the DRV report - in the same order as defined. For a complete list of DRV fields, refer to the DRV Fields table.
<code>-drv_output_format {txt csv}</code>	Specifies the DRV output format – txt or CSV. The default value is “txt”.
<code>-drv_ratio_margin float</code>	Specifies the ratio of the DRV margin value applied to obtain the threshold transition.
<code>-drv_violation_type</code>	Reports DRV violations. The supported DRV check types are: • <code>max_capacitance</code>: Reports pins having the worst <code>max_capacitance</code> constraint. If you specify the <code>-all_violators</code> parameter, you can see all the pins whose actual capacitance is greater than the <code>max_capacitance</code> value specified in the library and timing constraints file. • <code>min_capacitance</code>: Reports pins having the worst <code>min_capacitance</code> constraint. If you specify the <code>-all_violators</code> parameter, you can see all the pins whose actual capacitance is

	less than the <code>min_capacitance</code> value specified in the library and timing constraints file. • <code>max_transition</code>: Reports pins having the worst <code>max_transition</code> constraint. If you specify the <code>-all_violators</code> parameter, you can see all the pins whose actual transition is greater than the <code>max_transition</code> value specified in the library and timing constraint file. Note: These checks are not reported in hold analysis mode. The <code>set_si_mode -enable_drv_with_delta_slew true</code> setting is required for <code>max_transition</code> DRV violations with SI slew. • <code>min_transition</code>: Reports pins having the worst <code>min_transition</code> constraint. If you specify the <code>-all_violators</code> parameter, you can see all the pins whose actual transition is less than the <code>min_transition</code> value specified in the library and timing constraints file. • <code>max_fanout</code>: Reports pins having the worst <code>max_fanout</code> constraint. If you specify the <code>-all_violators</code> parameter, you can see all the pins whose actual fanout is greater than the <code>max_fanout</code> value specified in the library and timing constraints file. • <code>min_fanout</code>: Reports pins having the worst <code>min_fanout</code> constraint. If you specify the <code>-all_violators</code> parameter, you can see all the pins whose actual fanout is less than the <code>min_fanout</code> value specified in the library and timing constraints file. • <code>pulse_clock_max_transition</code>: Reports maximum pulse clock transition checks. • <code>pulse_clock_min_transition</code>: Reports minimum pulse clock transition checks.
<code>-early</code>	Reports all of the hold checks, including regular, clock_gating_hold, and data-to-data hold checks. Also reports path delays from <code>set_min_delay</code>. Note: This parameter does not report removal checks.
<code>-em_type {peak rms avg all}</code>	Reports DRV violations as per the specified EM type table lookup. When this parameter is specified, the software reads the <code>em_max_toggle_rate</code> table in the Liberty library for the specified EM current type (RMS, peak, or average).

<code>-late</code>	Reports all of the setup checks, including regular, clock_gating_setup, and data-to-data setup checks. Also reports path delays from set_max_delay. Note: This parameter does not report recovery checks.
<code>-max_pins_per_drv</code>	Reports the specified number of top worst violations per DRV category, such as max_transition, max_capacitance. If this parameter is not specified, 1000 entries per DRV category will be reported by default.
<code>pin_port_list</code>	Controls the output of DRV constraints. When specified, all DRV violations are reported only for the specified pins or ports.
<code>-retime {aocv path_slew_propagation aocv_path_slew_propagation}</code>	Reanalyzes the specified set of paths using the specified method. Note: The parameter supports only setup/hold check types. <code>aocv</code>: Re-evaluates the given set of graph-based analysis (GBA) paths by reverting back graph-based AOCV derate factors and applying path-based AOCV derates instead of less pessimistic derates. In this case, the slew (and respective delays) values are worst that are propagated during graph-based analysis. <code>aocv_path_slew_propagation</code>: Re-evaluates the given set of graph-based analysis (GBA) paths by using path-based AOCV derates and delays using actual path-based propagated slew. <code>path_slew_propagation</code>: Re-evaluates the given set of graph-based analysis (GBA) paths by recalculating the delay values based upon actual propagated slew across the path. In this case, the AOCV derates are the ones that are reported during graph-based analysis.

<pre>-retime_mode {path exhaustive}</pre>	Allows the <code>report_constraint</code> command to be run in exhaustive path-based analysis (PBA) mode. In this mode, the software will exhaustively examine <code>nworst</code> paths of the selected endpoints and then select the true worst path to be reported - based on retimed slacks of all the examined paths. To enable exhaustive mode, the use model is given below: <pre>report_constraint -all_violators -retime path_slew_propagation -retime_mode exhaustive</pre> When you use this parameter, reporting can be run time intensive as the software needs to do an exhaustive search for the true worst paths. Therefore, it is recommended to use this option in the later phases of the sign-off stage when there are very few violations left for analysis. <i>Default:</i> <code>path</code>
<pre>-verbose</pre>	Generates a report that displays the full data path, along with the accompanying required time and slack calculations. This report is equivalent to the default <code>report_timing</code> report. When you specify the <code>-verbose</code> parameter, you can set the <code>timing_report_constraint_use_report_timing_format</code> global variable to honor the <code>report_timing_format</code> settings. The DRV reports, with or without the <code>-verbose</code> parameter, are the same.
<pre>-view viewName</pre>	Generates a report containing endpoint and other constraint information for the specified analysis view. You can only specify this parameter when the software is in multi-mode multi-corner analysis mode. <i>Default:</i> Generates a report containing the worst endpoint and constraint information based on all of the analysis views.
<pre>-worst_drv_per_net</pre>	Reports only the worst DRV violation per net.

DRV Fields

Field Name	Report Column	Description
------------	---------------	-------------

capacitance_actual	Actual Capacitance	Specifies the actual minimum or maximum capacitance at a pin or port. This field is used to determine the capacitance value.
capacitance_slack	Capacitance Slack	Specifies the max_cap/min_cap slack at a pin or port.
direction	Direction	Specifies the pin or port direction.
driver_max_capacitance	Driver Required Maximum Capacitance	Specifies the max_capacitance limit at the driver pin or port.
driver_max_fanout	Driver Required Maximum Fanout	Specifies the worst max_fanout limit on the driver pin or port.
driver_max_transition	Driver Required Maximum Transition	Specifies the max_transition limit at the driver pin or port.
driver_min_capacitance	Driver Required Minimum Capacitance	Specifies the min_capacitance limit at the driver pin or port.
driver_min_transition	Driver Required Minimum Transition	Specifies the min_transition limit at the driver pin or port.
driver_pin	Driver	Specifies the leaf driver pin or port name.
driver_ref_cell	Driver Libcell	Specifies the leaf driver reference cell or port.
driver_transition	Driver Actual Transition	Specifies the actual transition at the driver pin or port.
drv_source	Source of DRV	Specifies the source of the DRV constraint, which is used for the DRV check. Reports 'library_constraint' or 'user_constraint' depending on which DRV violation check is used.
edge	Edge	The signal transition direction on the net. Returns "rise" or "fall" if the DRV report corresponds to a rising or falling transition on the net.
fanout_actual	Actual Fanout	Specifies the actual minimum or maximum fanout at a pin or port. This field is used to determine the fanout slack value.
fanout_slack	Fanout Slack	Specifies the max_fanout/min_fanout slack at a pin or port.

hnet	Hierarchical Nets	Reports the hierarchical nets associated with violating
lib_constraint	Library Constraint	Specifies the value of the library constraint on the viola not defined, reports as "NA".
max_capacitance	Required Capacitance	Specifies the maximum capacitance limit for a pin.
max_clock_frequency	Frequency	Specifies the frequency of the clock reported under the max_frequency_clock field.
max_fanout	Required Fanout	Specifies the maximum fanout limit for a pin.
max_frequency_clock	Maximum Frequency Clock	Specifies the name of the highest frequency clock cons the pin. For clock network pins, the fastest clock reachi will be reported. For data network pins, the fastest laun reaching the pin will be reported.
max_slack	Worst Setup Slack	Specifies the worst setup slack through a pin or port.
max_transition	Required Transition	Specifies the maximum transition limit for a pin or port.
min_capacitance	Required Capacitance	Specifies the minimum capacitance limit for a pin.
min_fanout	Required Fanout	Specifies the minimum fanout limit for a pin.
min_slack	Worst Hold Slack	Specifies the worst hold slack through at a pin or port.
min_transition	Required Transition	Specifies the minimum transition limit for a pin or port.
net	Net	Specifies the top flat net connected to the violating pin (at topmost hierarchy) flat net.
hnet	Hierarchical Nets	Reports hierarchical nets. When the <code>timing_enable_hierarchical_get_nets_support</code> global is set to <code>true</code> , the software enables support for displayi hierarchical net names.
net_fanout	Net Fanout	Specifies the number of leaf level load pins of the net c to a pin or port.
net_resistance	NetResistance	Resistance of the net connected to a pin or port.

net_type	Net Type	Specifies one or more of the following identifiers to indicate a special net: • ideal: the net is marked ideal using the set_ideal_net command • clock: the net is part of a clock network • constant: the net is tied to logic 1/0, respectively • multidriven: the net has multiple driver • dont_touch: the net is marked as dont_touch using set_dont_touch command • "-": None of the above
partition	Partition	Specifies the name of the SPEF hierarchy in which the pin appears.
pin	Pin	Reports the pin or port on which the DRV check is performed.
pin_type	Type	Specifies whether the pin/port is a data or a clock pin/port. Any pin/port that belongs to only a data network will be reported as DATA. Any pin/port that belongs to both data and clock networks, it will be reported as CLOCK.
receiver_max_transition	Receiver Required Maximum Transition	Specifies the max_transition limit at a receiver pin or port.
receiver_pin	Receiver	Specifies the leaf receiver pin name.
receiver_ref_cell	Receiver Libcell	Specifies the leaf reference cell name of a receiver pin.
receiver_transition	Receiver Actual Transition	Specifies the actual transition at a receiver pin or port.
ref_cell	Violating Libcell	Specifies the reference library cell to which the violation is attached.
sequential	Sequential	Specifies if the pin belongs to a sequential cell or a combinational cell. For ports, the report will show "false".
total_capacitance	TotalCapacitance	Total capacitance of the net connected to a pin or port.

transition_actual	Actual Transition	Specifies the actual minimum/maximum transition at a pin or port. This may be a base transition or a base plus crosstalk-transition, depending on the <code>timing_max_transition_use_si_transition</code> setting to false (default) or true, respectively.
transition_si	SI Transition	Specifies only the delta SI transition time at a pin or port. The total transition time comprises the base transition time (transition value without SI crosstalk analysis) as well as the SI transition time.
transition_sigma	Transition N-Sigma	Specifies the N-Sigma multiplier defined for transition violation. <code>set_socv_reporting_nsigma_multiplier -transition_violation</code> command. This field will be reported for SOCV STA, under <code>timing_report_max_transition_check_using_nsigma</code> to the <code>true</code> setting. This is sensitive to views and <code>-data</code> and <code>-clock</code> specific. Note: This field will impact transition_actual and transition_si computation for max_transition, min_transition, pulse_clock_max_transition, and pulse_clock_min_transition checks.
transition_slack	Transition Slack	Specifies the max_transition/min_transition slack at a pin/port.
user_constraint	User Constraint	Specifies user defined tran/cap/fanout limit value set by constraints as a command or through SDC on the violation. If not defined, reports as "NA".
view	View	Name of the pertaining analysis view.

Examples

- Consider a design having 3 sequential endpoints:
 - setup_pin: Having a setup check with slack equal to -9.000
 - recovery_pin: Having a recovery check with slack equal to -8.000
 - both_setup_recovery_pin: Having both setup check (slack = -6.000) and recovery check (slack = -7.000)

In this case, the `report_constraint -late -all_violators` command will display the following output:

```
max_delay/setup
-----
End Point                Slack                Cause
-----
setup_pin f              -9.000          VIOLATED
both_setup_recovery_pin  -6.000          VIOLATED
-----
```

- The `report_constraint -check_type recovery -all_violators` command will display the following output:

```
Check type : recovery
-----
End Point                Slack                Cause
-----
recovery_pin r           -8.000          VIOLATED
both_setup_recovery_pin  -7.000          VIOLATED
-----
```

- The `report_constraint -late -check_type recovery -all_violators` command displays the following output:

```
max_delay/setup
-----
End Point                Slack                Cause
-----
setup_pin f              -9.000          VIOLATED
recovery_pin r           -8.000          VIOLATED
both_setup_recovery_pin  -7.000          VIOLATED
-----
```

- The `report_constraint` command displays the following output:

```
max_delay/setup
-----
End Point                Slack                Cause
-----
out2 r                   -4111.834        VIOLATED
-----

Check type : clock_period
-----

No paths found
```

Check type : pulse_width

SEQUENTIAL CLOCK PULSE WIDTH			
Pin	Required Pulse Width	Actual Pulse Width	Slack
dff2/CK (low)	0.260	-670.800	-671.060

Check type : max_transition

Pin : sub1/dff1/D
max_transition : 4.500
Transition Time: 5114.060

slack: -5109.560 (VIOLATED)

Check type : max_capacitance

Pin : buf133/Y
max_capacitance: 0.000
-Capacitance : 0.423

slack : -0.423 (VIOLATED)

Check type : max_fanout

Pin : specialClkGate/o
max_fanout : 0.557
fanout : 1.000

slack : -0.443 (VIOLATED)

- The `report_constraint -all_violators` command displays the following output:

max_delay/setup

End Point	Slack	Cause
out2 r	-4111.834	VIOLATED
dff3/D r	-4110.260	VIOLATED
lat2/D r	-3774.174	VIOLATED

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dff2/D r	-3770.285	VIOLATED
out r	-3161.115	VIOLATED
and1/B r	-2464.880	VIOLATED
dff1/D r	-2098.868	VIOLATED
sub2/bufH1/dff1/D r	-2098.821	VIOLATED
sub2/bufH2/dff1/D r	-2028.843	VIOLATED
sub1/dff1/D r	-2028.843	VIOLATED
dff5/D r	-1279.155	VIOLATED

Check type : clock_period

No paths found

Check type : skew

No paths found

Check type : pulse_width

SEQUENTIAL CLOCK PULSE WIDTH

Pin	Required Pulse Width	Actual Pulse Width	Slack
dff2/CK (low)	0.260	-670.800	-671.060
lat2/GN (low)	0.201	-670.800	-671.001
sub2/bufH1/dff1/CK (high)	0.167	-670.800	-670.968
dff1/CK (high)	0.167	-670.800	-670.968
sub2/bufH2/dff1/CK (high)	0.167	-670.800	-670.968
sub1/dff1/CK (high)	0.167	-670.800	-670.968
dff11/CK (high)	0.167	-670.800	-670.968

Check type : max_transition

Pin Name	Required	Actual	Slack
and_parallel_1/A	4.500	5114.060	-5109.560
buf4/A	4.500	5114.060	-5109.560
sub1/dff1/D	4.500	5114.060	-5109.560

Check type : max_capacitance

```
-----
```

Pin Name	Required	Actual	Slack

buf133/Y	0.000	0.423	-0.423
buf11/Y	0.000	0.105	-0.105
buf5/Y	0.000	0.105	-0.105
dff11/Q	0.000	0.105	-0.105
Check type : max_fanout			

```
-----
```

```
-----
```

Pin Name	Required	Actual	Slack

specialClkGate/o	0.557	1.000	-0.443

- The `report_constraint -all_violators -drv_violation_type min_capacitance` command reports the following:

```
-----
```

Pin Name	Required	Actual	Slack

DIN	0.0400	0.0013	-0.0387
F2/Q	0.0400	0.0013	-0.0387
CLK1	0.0400	0.0051	-0.0349

- The `report_constraint -drv_violation_type min_capacitance` command outputs the following:
Check type : min_capacitance

```
-----
```

Pin : F2/Q
Capacitance: 0.0013
- min_capacitance: 0.0400

```
-----
```

slack: -0.0387 (VIOLATED)

- The `report_constraint -all_violators -drv_violation_type min_transition` command outputs the following:

```
-----
```

Pin Name	Required	Actual	Slack

I0/A	0.0300	0.0001	-0.0299
I1/A	0.0300	0.0001	-0.0299
CLK0	0.0300	0.0001	-0.0299
CLK1	0.0300	0.0001	-0.0299

```
DIN      0.0300      0.0001      -0.0299
I4/S0    0.0300      0.0001      -0.0299
F1/DATA  0.0300      0.0001      -0.0299
SEL      0.0300      0.0001      -0.0299
```

- The `report_constraint -drv_violation_type min_transition` command outputs the following:

```
Check type : min_transition
```

```
-----
```

```
Pin : F1/DATA
```

```
Transition Time:0.0001
```

```
- min_transition :0.0300
```

```
-----
```

```
slack : -0.0299 ( VIOLATED )
```

- The `report_constraint -all_violators -drv_violation_type min_fanout` command outputs the following:

```
-----
```

Pin Name	Required	Actual	Slack
----------	----------	--------	-------

```
-----
```

F5/Q	2.0000	0.0000	-2.0000
------	--------	--------	---------

CLK0	2.0000	1.0000	-1.0000
------	--------	--------	---------

F3/Q	2.0000	1.0000	-1.0000
------	--------	--------	---------

F2/Q	2.0000	1.0000	-1.0000
------	--------	--------	---------

F1/Q	2.0000	1.0000	-1.0000
------	--------	--------	---------

SEL	2.0000	1.0000	-1.0000
-----	--------	--------	---------

CLK1	2.0000	1.0000	-1.0000
------	--------	--------	---------

- The `report_constraint -drv_violation_type min_fanout` command outputs the following:

```
Check type : min_fanout
```

```
-----
```

```
Pin : F5/Q
```

```
Fanout: 0.0000
```

```
- min_fanout: 2.0000
```

```
-----
```

```
slack : -2.0000 ( VIOLATED )
```

- The `report_constraint -check_type clock_period` command reports the following:

```
Check type : clock_period
```

```
-----
```

End Point	Slack	Cause
-----------	-------	-------

```
-----
ffo_ram/mm0/mm1/CLKB f    -0.027      VIOLATED
-----
```

- The following example illustrates how to restrict the number of DRV violations to display for each of the specified DRV checks:

```
report_constraint -drv_violation_type {max_transition max_capacitance} -drv_fields
{max_transition transition_actual transition_slack max_capacitance capacitance_actual
capacitance_slack max_slack net  max_frequency_clock max_clock_frequency pin_type
ref_cell driver_pin pin} -all_violators -max_pins_per_drv 6
#####
# format : frame 0 : split 0
# Units: Time (1ns), Capacitance (1pf), Frequency (1MHz).
Check type : max_transition
-----
```

```
-----
Required      Actual      Transition   Worst      Net      Maximum      Frequency      Pin
Violating     Driver      Pin
Transition    Transition  Slack       Setup      Frequency      Type
Libcell

Slack       Clock

0.002      0.120      -0.118      NA      clk1      CK1      100.000      CLOCK
DFFRX2     clk1      ff1/CK
0.002      0.120      -0.118      NA      clk2      CK2      200.000      CLOCK
DFFRX2     clk2      ff2/CK
0.002      0.059      -0.057      0.894      n4      CK1      100.000      DATA
  BUFX1     buf2/Y     buf2/Y
0.002      0.059      -0.057      4.362      n4      CK1      100.000      DATA
  DFFRX2     buf2/Y     ff2/D
0.002      0.059      -0.057      0.894      n4      CK1      100.000      DATA
  BUFX1     buf2/Y     buf3/A
0.002      0.045      -0.043      3.840      n1      CK2      200.000      DATA
  BUFX1     buf0/Y     buf1/A
```

```
Check type : max_capacitance
-----
```

```
-----
Required      Actual      Capacitance   Worst      Net      Maximum      Frequency      Pin
Violating     Driver      Pin
Capacitance   Capacitance  Slack       Setup      Frequency      Type
Libcell

Slack       Clock
```

```
-----
0.002      0.004      -0.002      0.894      n4      CK1      100.000      DATA
  BUFX1      buf2/Y      buf2/Y
0.002      0.002      -0.000      3.840      n1      CK2      200.000      DATA
  BUFX1      buf0/Y      buf0/Y
0.002      0.002      -0.000      3.840      n2      CK2      200.000      DATA
  BUFX1      buf1/Y      buf1/Y
0.002      0.002      -0.000      0.894      n3      CK1      100.000      DATA
  DFFRX2      ff1/Q      ff1/Q
```

- The following examples shows the pin-wise DRV report without the worst DRV per net:

```
set timing_report_constraint_format "max_transition transition_actual transition_slack
max_capacitance capacitance_actual capacitance_slack max_slack net max_frequency_clock
max_clock_frequency pin_type ref_cell driver_pin pin"
```

```
report_constraint -drv_violation_type {max_transition max_capacitance} -all_violators
#####
# format : frame 0 : split 0
# Units: Time (1ns), Capacitance (1pf), Frequency (1MHz).
Check type : max_transition
```

```
-----
-----
Required      Actual      Transition      Worst      Net      Maximum      Frequency      Pin
Violating      Driver      Pin
Transition      Transition      Slack      Setup      Frequency      Type
Libcell
Slack      Clock
-----
-----
0.002      0.120      -0.118      NA      clk1      CK1      100.000      CLOCK
DFFRX2      clk1      ff1/CK
0.002      0.120      -0.118      NA      clk2      CK2      200.000      CLOCK
DFFRX2      clk2      ff2/CK
0.002      0.059      -0.057      0.894      n4      CK1      100.000      DATA
  BUFX1      buf2/Y      buf2/Y
0.002      0.059      -0.057      4.362      n4      CK1      100.000      DATA
  DFFRX2      buf2/Y      ff2/D
0.002      0.059      -0.057      0.894      n4      CK1      100.000      DATA
  BUFX1      buf2/Y      buf3/A
0.002      0.045      -0.043      3.840      n1      CK2      200.000      DATA
  BUFX1      buf0/Y      buf1/A
0.002      0.045      -0.043      3.840      n1      CK2      200.000      DATA
  BUFX1      buf0/Y      buf0/Y
0.002      0.044      -0.042      3.840      n2      CK2      200.000      DATA
```


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```

    BUFX1      buf1/Y      buf1/Y
0.002         0.044        -0.042      3.840      n2      CK2      200.000      DATA
    DFFRX2     buf1/Y      ff1/D
0.002         0.038        -0.036      0.894      n3      CK1      100.000      DATA
    BUFX1      ff1/Q       buf2/A
0.002         0.038        -0.036      0.894      n3      CK1      100.000      DATA
    DFFRX2     ff1/Q       ff1/Q
0.002         0.031        -0.029      1.208      out2     CK2      200.000      DATA
    DFFRX2     ff2/Q       ff2/Q
0.002         0.030        -0.028      0.894      out1     CK1      100.000      DATA
    BUFX1      buf3/Y      buf3/Y

```

Check type : max_capacitance

Required Violating Capacitance	Actual Driver Capacitance	Capacitance Pin Slack	Worst Setup Slack	Net	Maximum Frequency Clock	Frequency	Pin Type	Libcell
--------------------------------------	---------------------------------	-----------------------------	-------------------------	-----	-------------------------------	-----------	-------------	---------

0.002	0.004	-0.002	0.894	n4	CK1	100.000	DATA	BUFX1
buf2/Y	buf2/Y							
0.002	0.002	-0.000	3.840	n1	CK2	200.000	DATA	BUFX1
buf0/Y	buf0/Y							
0.002	0.002	-0.000	3.840	n2	CK2	200.000	DATA	BUFX1
buf1/Y	buf1/Y							
0.002	0.002	-0.000	0.894	n3	CK1	100.000	DATA	DFFRX2
ff1/Q	ff1/Q							

- The following command example shows pins per net with the worst DRV slack:

```

report_constraint -drv_violation_type {max_transition max_capacitance} -all_violators -
worst_drv_per_net -max_pins_per_drv 5

```

#####

format : frame 0 : split 0

Units: Time (1ns), Capacitance (1pf), Frequency (1MHz).

Check type : max_transition

Required Violating Transition Libcell	Actual Driver Transition	Transition Pin Slack	Worst Setup Slack	Net	Maximum Frequency Clock	Frequency	Pin Type
--	--------------------------------	----------------------------	-------------------------	-----	-------------------------------	-----------	-------------

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```

-----
-----
0.002      0.120      -0.118      NA      clk1  CK1      100.000      CLOCK
DFFRX2    clk1      ff1/CK
0.002      0.120      -0.118      NA      clk2  CK2      200.000      CLOCK
DFFRX2    clk2      ff2/CK
0.002      0.059      -0.057      0.894   n4      CK1      100.000      DATA
  BUFX1    buf2/Y    buf2/Y
0.002      0.045      -0.043      3.840   n1      CK2      200.000      DATA
  BUFX1    buf0/Y    buf1/A
0.002      0.044      -0.042      3.840   n2      CK2      200.000      DATA
  BUFX1    buf1/Y    buf1/Y

```

Check type : max_capacitance

```

-----
-----
Required      Actual      Capacitance  Worst   Net   Maximum   Frequency   Pin
Violating     Driver   Pin
Capacitance   Capacitance Slack      Setup      Frequency      Type
Libcell
              Slack      Clock
-----
-----
0.002      0.004      -0.002      0.894   n4      CK1      100.000      DATA
  BUFX1    buf2/Y    buf2/Y
0.002      0.002      -0.000      3.840   n1      CK2      200.000      DATA
  BUFX1    buf0/Y    buf0/Y
0.002      0.002      -0.000      3.840   n2      CK2      200.000      DATA
  BUFX1    buf1/Y    buf1/Y

```

- The following command reports the hierarchical nets associated with the violating pins:

```
tempus > report_constraint -drv_violation_type max_transition -drv_fields {pin net hnet}
```

Check type : max_transition

```

-----
+-----+
| Pin      | Net      | Hierarchical |
|          |          | Net          |
|-----+-----+-----|
| data_bidi2 | data_bidi2 | data_bidi2   |
+-----+

```

Related Commands

- `set_min_fanout`
- `set_max_fanout`
- `set_min_transition`
- `set_max_transition`
- `set_min_capacitance`
- `set_max_capacitance`
- `report_timing_format`
- `report_timing`

report_cppr

```
report_cppr
[-help]
[-check_type {setup | hold | clock_gating_setup | clock_gating_hold |
clock_gating_pulse_width | data_setup | data_hold | recovery | removal | pulse_width |
clock_period | clock_separation | skew | no_change_setup | no_change_hold | time_borrow}]
-from pin_or_port
-to pin_or_port
[-from_clock clkname ]
[-to_clock clkname ]
[-early | -late]
[-view view_name ]
[> filename ]
```

Reports the clock reconvergence pessimism value at the common branching point of the early and late paths in the clock network of the specified data path.

The `report_cppr` command reports separate values for both opening and closing edges for paths with latches as destinations. If both source and destination flops are triggered by multiple clocks, then one report is generated for each separate clock.

When the software is in multi-mode multi-corner analysis mode, `report_cppr` reports the results for each active analysis view.

To use the `report_cppr` command, you must first set the following global variables:

- `timing_cppr_transition_sense`
- `timing_cppr_threshold_ps`

Parameters

-help	Prints out the command usage.
-------	-------------------------------

<pre>-check_type {setup hold clock_gating_setup clock_gating_hold clock_gating_pulse_width data_setup data_hold recovery removal pulse_width clock_period clock_separation skew no_change_setup no_change_hold time_borrow}</pre>	Reports the CPPR for the paths that end at the specified timing check. The following check types are reported when the software is in setup analysis mode: <code>setup</code>, <code>pulse_width</code>, <code>clock_period</code>, <code>clock_gating_setup</code>, <code>clock_gating_hold</code>, <code>clock_gating_pulse_width</code>, <code>clock_separation</code>, <code>time_borrow</code>, and <code>no_change_setup</code>. The following check types are reported when the software is in hold analysis mode: <code>hold</code>, <code>clock_gating_hold</code>, <code>data_hold</code>, <code>removal</code>, and <code>no_change_hold</code>. If the software is in simultaneous setup and hold analysis mode, all of the
<pre>> filename</pre>	Writes the report to the specified file name. If the file already exists, the software will overwrite the file. The <code>filename</code> parameter must be the last argument in the list. Note: To write a compressed report, add the <code>.gz</code> extension to the file name. Default: Report is displayed on standard output without being saved.
<pre>-early</pre>	Reports the hold analysis clock reconvergence pessimism value. You can specify this parameter when the software is in hold analysis mode or simultaneous setup and hold analysis mode (<code>set_global timing_enable_simultaneous_setup_hold_mode</code>).
<pre>-from pin_or_port</pre>	Reports the clock reconvergence pessimism value for paths starting from the specified port.
<pre>-from_clock clkname</pre>	Reports the clock reconvergence pessimism value for paths with the specified clock when more than one clock triggers the source flop.
<pre>-late</pre>	Reports the setup analysis clock reconvergence pessimism value. You can specify this parameter when the software is in setup analysis mode or simultaneous setup and hold analysis mode (<code>set_global timing_enable_simultaneous_setup_hold_mode</code>).
<pre>-to pin_or_port</pre>	Reports the clock reconvergence pessimism value for paths ending at the specified port.
<pre>-to_clock clkname</pre>	Reports the clock reconvergence pessimism value for paths with the specified clock when more than one clock captures the destination flop.
<pre>-view view_name</pre>	Reports the clock reconvergence pessimism values for paths in the specified view. The <code>view_name</code> parameter only when the software is in multi-mode multi-corner timing analysis.

Examples

Here are some examples.

- When the software is in multi-mode multi-corner timing analysis mode, you can specify the `-view` parameter to report the CPPR values for a specific analysis view:

```
tempus > report_cppr -from CLK3 -to seg3/u14/CK
```

```
Start Clock Pin: CLK3 (Triggered by rising edge of clock CLK_W_3)
End Clock Pin: seg3/u14/CK (Triggered by rising edge of clock CLK_W_3)
Analysis Type: Setup
Analysis View: setup_view1
Common Branch Point : seg3/u17_c/Y
Launching Edge Type At Common Point : Rising
Capturing Edge Type At Common Point : Rising
Pessimism Threshold Value : 0.020
```

```
-----
Arrival    Late    Early    Pessimism
Times
-----
Rise       0.549    0.366    0.183
Fall       0.400    0.267    0.133
-----
```

```
Edge Selection Mode : Normal
Edge Selection : Match, Using Rise
CPPR Adjustment : 0.183
```

- The following report shows sigma and mean values:

```
tempus > report_cppr -from FF01/CP -to FF02/CP
```

```
Start Clock Pin: FF01/CP (Triggered by rising edge of clock CK)
End Clock Pin: FF02/CP (Triggered by rising edge of clock CK)
Analysis Type: Setup
Analysis View: view1
Common Branch Point : CK
Launching Edge Type At Common Point : Rising
Capturing Edge Type At Common Point : Rising
Pessimism Threshold Value : 0.020
```

```
+-----+
| Arrival | Late  | Late  | Early | Early | Pessimism | Pessimism | Pessimism |
| Times   | Mean  | Sigma | Mean  | Sigma | Mean      | Sigma     |           |
+-----+-----+-----+-----+-----+-----+-----+-----+
| Rise    | 0.000 | 0.000 | 0.000 | 0.000 | 0.000     | 0.000     | 0.000     |
```

Tempus Text Command Reference
Timing Analysis Commands

```
| Fall      | 0.000 | 0.000 | 0.000 | 0.000 | 0.000      | 0.000      | 0.000      |
+-----+
```

Edge Selection Mode : Normal

Edge Selection : Match, Using Rise

CPPR Adjustment : 0.000 (+3.00S) | 0.000 | 0.00

report_critical_instance

```
report_critical_instance
[-help]
[-max_slack slack]
[-max_insts value]
[-estimate]
[-sequential_insts]
[-name_length length]
[-outfile <filename>]
[-cost_type {worst_slack | total_slack | path_count | start_end_point_count}]
[ > filename]
```

Reports the timing critical instance (bottleneck) information, including total negative slack for each instance, total count of connected start and end points and total count of paths through an instance.

Parameters

-help	Prints out the command usage.
> <i>filename</i>	Writes the report to the specified file name. If the file already exists, the software overwrites it. Note: The file name must be the last argument in the list. <i>Default:</i> Displays the results in the Tempus console window without being saved.
-cost_type	Specifies the criteria for choosing the critical instance. You can use the -max_slack and -cost_type parameters together to filter the paths to only those with slack worse than the specified <i>slack</i> value, then report the filtered set of paths based on the specified cost. <i>Default:</i> worst_slack Select one of the following options:
	• worst_slack: Sets the criteria as the worst slack of all points that pass through an instance.
	• total_slack: Sets the criteria as a sum of the slack of the end points.
	• path_count: Sets the criteria as the sum of the total paths through the instance.

	<code>start_end_point_count</code>: Sets the criteria as the sum of the total start and end points connected to the instance.
<code>-estimate</code>	Generates a critical instance report based on the fast algorithm (linear-complexity) to give a slightly pessimistic estimation that may include some false paths. Use this parameter in the early design phase. <i>Default:</i> Uses <code>report_timing</code> to find the accurate critical instance information.
<code>-max_insts value</code>	Specifies the number of worst cost instances to report. <i>Default:</i> 20
<code>-max_slack slack</code>	Specifies the lower bound of the slack range. Only paths with slack worse than the specified <code>slack</code> value are reported. You can use the <code>-max_slack</code> and <code>-cost_type</code> parameters together to filter the paths to only those with slack worse than the specified <code>slack</code> value, then report the filtered set of paths based on the specified cost. <i>Default:</i> 0
<code>-name_length length</code>	Specifies the maximum length for instance names in the final output report. <i>Default:</i> 48
<code>-outfile filename</code>	Writes out the report into the output file.
<code>-sequential_insts</code>	Includes the sequential instances group in the output. <i>Default:</i> Reports combinatorial instances only.

Examples

Here are some examples.

- The following command reports the critical instance information, including the sequential instances group for the 5 worst cost instances:

```
tempus> report_critical_instance -sequential_insts -name_length 42 -max_insts 5
```

The following report is generated:

```
# -----
# Inst:combinatorial   Cell           Cost
# -----
p0042A82620           CKNHVT4    -0.124
p0009A84488           CKND2HVT2  -0.124
```

```
p0009A84792      IND2D1      -0.124
p0012A85164      CKNXHVTD1  -0.124
# -----
# Inst:sequential  Cell          Cost
# -----
dcache_/ram256    v11rbw_wsz10_hz8 -0.124
dcache_data3/ram10 v11rbw_wsz32_hz8 -0.120
dcache_tagram2/ram256 v11rbw_wsz24_hd8 -0.120
# -----
```

- The following command reports the critical instance information based on the worst slack of all points that pass through an instance:

```
tempus> report_critical_instance -cost_type worst_slack
# -----
# Inst:combinatorial  Cell          Cost
# -----
I3                    BUFX2          -0.336
I10                   AND2X1          -0.336
I1                    BUFX2          -0.117
I2                    BUFX2          -0.117
# -----
```

- The following command reports the critical instance information based on the worst slack of all points that pass through an instance, for paths with slack worse than -0.2:

```
tempus> report_critical_instance -cost_type worst_slack -max_slack -0.2
# -----
# Inst:combinatorial  Cell          Cost
# -----
I3                    BUFX2          -0.336
I10                   AND2X1          -0.336
# -----
```

- The following command reports the critical instance information based on the sum of the slack of the end points:

```
tempus> report_critical_instance -cost_type total_slack
# -----
# Inst:combinatorial  Cell          Cost    Start    End
# -----
I3                    BUFX2          -0.336     1         1
I10                   AND2X1          -0.336     2         1
I1                    BUFX2          -0.117     0         1
I2                    BUFX2          -0.117     0         1
```

- The following command reports the critical instance information based on the sum of the total paths through the instance:

```
tempus> report_critical_instance -cost_type path_count
```

```
# -----
# Inst:combinatorial   Cell           Cost      Start   End
# -----
I10                    AND2X1         4          2       1
I3                    BUFX2         2          1       1
I1                    BUFX2         2          0       1
I2                    BUFX2         2          0       1
# -----
```

- The following command reports the critical instance information based on the sum of the total start and end points connected to the instance:

```
tempus> report_critical_instance -cost_type start_end_point_count
```

```
# -----
# Inst:combinatorial   Cell           Cost      Start   End
# -----
I10                    AND2X1         2          1       1
I3                    BUFX2         2          1       1
I1                    BUFX2         2          1       1
I2                    BUFX2         2          1       1
# -----
```

- The following command reports the critical instance information based on the sum of the total start and end points connected to the instance, for paths with slack worse than -0.2:

```
tempus> report_critical_instance -cost_type start_end_point_count -max_slack -0.2
```

```
# -----
# Inst:combinatorial   Cell           Cost      Start   End
# -----
I10                    AND2X1         2          1       1
I3                    BUFX2         2          1       1
# -----
```

Related Commands

- [report_timing](#)

report_delay_calculation

```
report_delay_calculation
[-help]
[-active_arcs]
-from port_or_pin_name
-to port_or_pin_name
[-outfile filename]
[-min]
[-max]
[-mis]
[-show_all_attackers]
[-si]
[-thresholds]
[-view viewname]
[-voltage]
[-waveform]
```

Reports the delay calculation information for a cell or net timing arc.

You can run the [report_timing](#) command before running `report_delay_calculation` to have the correct timing window and aggressor slew.

The `report_delay_calculation` output shows a 3-sigma multiplier, instead of a single sigma. This ensures that the sigma representation for both the delay and transition time is the same.

Parameters

-help	Outputs a brief description that includes type and default information for each <code>report_delay_calculation</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man report_delay_calculation</code> .
-active_arcs	Disables reporting of all the non-active parallel delay arcs and reports only the valid check arcs.
-from port_or_pin_name	Specifies the name of the From port or pin. If the From port or pin is the output of a cell and the To port or pin is the input of a cell, delay is calculated for the net, and the delay type in the report is 'net delay.' If the From port or pin is the input of a cell and the To port or pin is the output of a cell, delay is calculated for the cell, and the delay type in the report is 'cell delay.'

<code>-max</code>	Indicates that the delay calculation report is for the worst-case operating conditions in min-max mode. If neither <code>min</code> nor <code>max</code> arguments are specified, the default mode is set to <code>max</code> .
<code>-min</code>	Indicates that the delay calculation report is for the best-case operating conditions in min-max mode. If neither <code>min</code> nor <code>max</code> arguments are specified, the default mode is set to <code>max</code> .
<code>-mis</code>	Reports advanced multi-input switching (AMIS) details.
<code>-outfile filename</code>	Specifies the name of the output file in which to save the delay calculation report.
<code>-si</code>	Additionally, reports the SI components, including the incremental delay if <code>set_si_mode -separate_delta_delay_on_data</code> is used.
<code>-show_all_attackers</code>	Allows you to report all the small attackers.
<code>-thresholds</code>	Displays the rise delay, rise slew, fall delay, fall slew, and slew delay threshold values for the From and To pins.
<code>-to port_or_pin_name</code>	Specifies the name of the To port or pin. If the From port or pin is the output of a cell and the To port or pin is the input of a cell, delay is calculated for the net, and the delay type in the report is 'net delay.' If the From port or pin is the input of a cell and the To port or pin is the output of a cell, delay is calculated for the cell, and the delay type in the report is 'cell delay.'
<code>-view viewname</code>	Specifies the view name for which to print the delay calculation report when using the multi-mode multi-corner (MMMC) mode.
<code>-voltage</code>	Displays the voltage of the cell or net timing arc.
<code>-waveform</code>	Reports waveforms of the specified pins.

Examples

- The following command shows the cell delay calculation report for the From pin I1/A and To pin I1/Y:

```
tempus> report_delay_calculation -from I1/A -to I1/Y
```

```
From pin : I1/A
```

```
To Pin : I1/Y
```

Cell : S2INVV1D1

Library : test

Arc sense : negative unate

Delay type : cell delay

RC Summary for net N2

Number of capacitance : 7

Net capacitance : 0.026057 pF

Total rise capacitance: 0.027172 pF

Total fall capacitance: 0.027169 pF

Number of resistance : 7

Total resistance : 1094.135127 Ohm

- The following command shows the net delay calculation report for the from pin ff1/Q and to pin A1/buf1/A with waveforms:

```
tempus> report_delay_calculation -from ff1/Q -to A1/buf1/A -waveform
```

From pin : ff1/Q

Cell : DFFX2

Library : gpdk045bc

To pin : A1/buf1/A

Cell : BUF2

Library : gpdk045bc

Delay type: net

From pin waveforms

```
Rise Waveform : { 0.00e+00 0.00000 5.35e-10 0.05400 1.07e-09 0.10800 1.61e-09 0.16200
2.14e-09 0.21600 2.68e-09 0.27000 3.21e-09 0.32400 3.75e-09 0.37800 4.28e-09 0.43200
4.82e-09 0.48600 5.35e-09 0.54000 5.89e-09 0.59400 6.42e-09 0.64800 6.96e-09 0.70200
7.50e-09 0.75600 8.03e-09 0.81000 8.57e-09 0.86400 9.10e-09 0.91800 9.64e-09 0.97200
1.02e-08 1.02600 1.07e-08 1.08000 }
```

```
Fall Waveform : { 0.00e+00 1.08000 3.69e-10 1.02600 7.39e-10 0.97200 1.11e-09 0.91800
1.48e-09 0.86400 1.85e-09 0.81000 2.22e-09 0.75600 2.59e-09 0.70200 2.96e-09 0.64800
3.32e-09 0.59400 3.69e-09 0.54000 4.06e-09 0.48600 4.43e-09 0.43200 4.80e-09 0.37800
5.17e-09 0.32400 5.54e-09 0.27000 5.91e-09 0.21600 6.28e-09 0.16200 6.65e-09 0.10800
7.02e-09 0.05400 7.39e-09 -0.00000 }
```

To pin waveforms

```
Rise Waveform : { 2.73e-14 0.00000 5.35e-10 0.05318 1.07e-09 0.10793 1.61e-09 0.16125
2.14e-09 0.21587 2.68e-09 0.26930 3.21e-09 0.32383 3.75e-09 0.37734 4.28e-09 0.43179
4.82e-09 0.48538 5.35e-09 0.53975 5.89e-09 0.59341 6.42e-09 0.64772 6.96e-09 0.70144
7.50e-09 0.75570 8.03e-09 0.80946 8.57e-09 0.86368 9.10e-09 0.91748 9.64e-09 0.97166
1.02e-08 1.02549 1.07e-08 1.07965 }
```

```
Fall Waveform : { 2.38e-14 1.08000 3.69e-10 1.02719 7.39e-10 0.97209 1.11e-09 0.91911
1.48e-09 0.86416 1.85e-09 0.81104 2.22e-09 0.75623 2.59e-09 0.70298 2.96e-09 0.64828
3.32e-09 0.59493 3.69e-09 0.54033 4.06e-09 0.48688 4.43e-09 0.43237 4.80e-09 0.37885
5.17e-09 0.32440 5.54e-09 0.27081 5.91e-09 0.21643 6.28e-09 0.16279 6.65e-09 0.10846
7.02e-09 0.05476 7.39e-09 0.00048 }
```

RC Summary for net n2

```
Number of capacitance : 19
Net capacitance : 0.714975 pF
Total rise capacitance: 0.716177 pF
Total fall capacitance: 0.716179 pF
Number of resistance : 24
Total resistance : 101.624695 Ohm
```

Rise Fall

```
Net delay : 0.002500 ns 0.002300 ns
From pin transition time: 8.565800 ns 5.910500 ns
To pin transition time : 8.568400 ns 5.913000 ns
```

- The following command shows the attacker status in the SI delay calculation report output (when `si_delay_enable_report` attribute is set to `true`):

```
tempus> report_delay_calculation -si -from seg3/u5/Y -to seg3/u6/A
```

```
From pin : seg3/u5/Y
Cell : INV
Library : cell_w
To pin : seg3/u6/A
Cell : INV
Library : cell_w
Base or SI: SI delay
Delay type: net
```

```
-----
RC Summary for net seg3/n5
-----
```

```
Number of capacitance : 17
Net capacitance : 0.293534 pF
Total rise capacitance: 0.320849 pF
Total fall capacitance: 0.320780 pF
Number of resistance : 17
Total resistance : 567.671387 Ohm
-----
```

```
SI information for rise
-----
```

```
Victim timing window : 1.0743 ns 2.7024 ns CLK_W_3
-----
```

```
Attacker Status Direction Slew Coupling cap Fanout Voltage Bump Timing window Clock
-----
```

```
seg2/n5 INF Fall 0.303667 ns 0.138759 pF 1 1.080 v 0.198396 v 1.1917 ns 2.8626 ns
CLK_W_2
seg1/n5 INF Fall 0.314333 ns 0.138787 pF 1 1.080 v 0.182412 v 1.3881 ns 3.0723 ns
CLK_W_1
-----
```

```
SI information for fall
-----
```

```
Victim timing window : 1.2016 ns 5.0700 ns CLK_W_3
-----
```

```
Attacker Status Direction Slew Coupling cap Fanout Voltage Bump Timing window Clock
-----
```

```
seg2/n5 INF Rise 0.550667 ns 0.138759 pF 1 1.080 v 0.138780 v 1.0887 ns 2.0217 ns
CLK_W_2
seg1/n5 INF Rise 0.800167 ns 0.138787 pF 1 1.080 v 0.107892 v 1.2751 ns 2.3031 ns
CLK_W_1
-----
```

```
Model types
-----
```


Net Term Cell Model

```
-----  
seg3/n5 seg3/u5/Y INV NLDM  
seg3/n5 seg3/u6/A INV NLDM  
seg2/n5 Y INV NLDM  
seg1/n5 Y INV NLDM  
-----
```

Delays Rise Fall

```
-----  
Incr Net Delay : 0.565900 ns 0.093700 ns  
Net delay (mean) : 0.696000 ns 0.188900 ns  
(3-sigma) : 0.000000 ns 0.000000 ns  
Timing Derate (mean) : 1.000000 1.000000  
(sigma) : 1.000000 1.000000  
Derated Net delay (mean) : 0.696000 ns 0.188900 ns  
(3-sigma) : 0.000000 ns 0.000000 ns  
-----
```

Transitions Rise Fall

```
-----  
Input transition (mean) : 0.482600 ns 0.234800 ns  
(sigma) : 0.000000 ns 0.000000 ns  
Output transition (mean) : 0.622200 ns 0.274100 ns  
(sigma) : 0.000000 ns 0.000000 ns  
-----
```

Abbreviative attacker status and filtered reason:

```
-----  
SY : Synchronous  
INF: Attacker with infinite timing window  
PE : Filtered due to physical exclusion  
UF : Filtered by user  
CN : Filtered because it is a constant net  
SB : Filtered due to small bump  
TA : Filtered because of timing window does not overlap  
LC : Filtered due to logical correlation  
ACT: Attacker is Active.  
-----
```

Related Commands

- [report_timing](#)
- [set_delay_cal_mode](#)
- [set_si_mode](#)

report_design

```
report_design  
[-help]  
[-view viewName]  
[{> | >>} filename | -tcl_list]  
[-early | -late]
```

Reports the minimum and/or maximum operating conditions and active design rules for the current design.

When `set_analysis_mode -analysisType single` option is specified, the `report_design` command reports only the maximum operating conditions.

When the `set_analysis_mode -analysisType bcWc` option is specified, the `report_design` command reports the minimum operating conditions, if the `set_analysis_mode -checkType hold` option is specified.

Both minimum and maximum operating conditions are available for on-chip variation analysis mode.

When simultaneous setup hold analysis mode is set to `on` (this is the default in Tempus), the minimum or maximum conditions can be reported using the `-early/-late` parameters.

Parameters

-help	Prints out the command usage.
{> >>} <i>filename</i>	When specified with >, writes the report to the specified file name. If the file already exists, the software overwrites it. When specified with >>, writes the report to the specified file name. If the file already exists, the software concatenates the report to the end of the file. The <i>filename</i> parameter must be the last argument in the list. The <i>filename</i> and -tcl_list parameters are mutually exclusive; you cannot specify them together. Note: To write a compressed report, add the .gz extension to the file name. Default: Report is displayed on standard output without being saved.
-early -late	The -early parameter reports the minimum operating condition in bcwc or onChipVariation analysis mode. The -late parameter reports the maximum operating condition. The -late and -early parameters are mutually exclusive.
- tcl_list	Produces the report in Tcl list format instead of a tabular format. This is useful for integrating timing with custom Tcl functions, and also for customizing report generation. The -tcl_list and <i>filename</i> parameters are mutually exclusive; you cannot specify them together.
-view <i>viewName</i>	Reports view specific operating conditions.

Examples

- The following is a sample report generated using the `report_design` command:

```
tempus> report_design
```

Design Property	Value
Design Name	dsp_core
Operating Condition Name	slow
Process	1.000
Voltage	1.620
Temperature	125.000
Tree Type	balanced

Design Rule	Value
Max Transition	NA
Min Transition	NA
Max Capacitance	NA
Min Capacitance	NA
Max Fanout	24.000

- The following sample report shows both the minimum and maximum operating condition details in simultaneous mode:

```
tempus> report_design
```

Design Property	Value	
Design Name	dsp_test	
	MAX	MIN
Operating Condition Name	tdsp_core/%NOM_PVT	tdsp_core/%NOM_PVT
Process	1.000	1.000
Voltage	1.800	1.800
Temperature	25.000	25.000

Tree Type	worst_case	worst_case
-----------	------------	------------

Design Rule	Value
Max Transition	NA
Min Transition	NA
Max Capacitance	NA
Min Capacitance	NA
Max Fanout	NA
Min Fanout	NA

Related Commands

- `report_timing`
- `set_analysis_mode`

report_design_rule

```
report_design_rule  
[-help]  
[-byDriver]  
[-byNet]  
[-byPin]  
[-cap]  
[-excNetFile filename]  
[-fanout]  
[-max]  
[-min]  
[-outfile filename]  
[-selNetFile filename]  
[-tran]  
[> filename]  
[>> filename]
```

Reports the following design rule violations:

- Nets that exceed the maximum capacitance constraints in the timing library and timing constraints file.
- Nets that exceed the transition constraints in the timing constraints file.
- Pins that exceed the maximum fanout constraints in the timing library and timing constraints file.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-byDriver</code>	Reports violations by each driver.
<code>-byNet</code>	Reports violations by each net. <i>Default:</i> Violations are reported by net.
<code>-byPin</code>	Reports violations by each pin.
<code>-cap</code>	Reports violations of capacitance constraints.
<code>-excNetFile filename</code>	Specifies the file that contains the hierarchical net names that are excluded from the capacitance violation report.
<code>-fanout</code>	Reports violations of fanout constraints.
<code>-max</code>	Reports only maximum capacitance violations. <i>Default:</i> <code>-max</code>
<code>-min</code>	Reports only minimum capacitance violations.
<code>-outfile filename</code>	Specifies the output filename to which the report violations are saved.
<code>-selNetFile filename</code>	Specifies the file that contains the hierarchical net names selected for the report. Only these net names are considered for inclusion in the capacitance violation report.
<code>-tran</code>	Reports violations of transition constraints. <i>Default:</i> Reports all design rule violations.
<code>>filename</code>	Redirects report into the outfile. <i>Default:</i> <code>stdout</code>
<code>>> filename</code>	Redirects report into the outfile. <i>Default:</i> <code>stdout</code>

Command Order

Use this command after running the timing analysis.

Examples

Here are some examples.

- The following command reports the design rule violations in the report file `maxcap.report`:

```
tempus> report_design_rule -outfile maxCap.report
```

The following report output is displayed.

```
# Net                MaxTranTime   TranTime      TranSlack      CellPort      Remark
#
DTMF_INST/lpcm[15]  2.700r/2.700f 2.783r/1.211f -0.083r/1.490f INVXL/A      #12,0.000
DTMF_INST/lpcm[15]  2.700r/2.700f 2.783r/1.211f -0.083r/1.490f AOI21X1/Y    #12,0.000
DTMF_INST/lpcm[15]  2.700r/2.700f 2.783r/1.211f -0.083r/1.490f NAND2X1/A    #12,0.000
DTMF_INST/lpcm[15]  2.700r/2.700f 2.783r/1.211f -0.083r/1.490f OAI21XL/B0   #12,0.000

# Net MaxCap Cap CapSlack CellPort Remark
#

# Net                MaxFanLoad   FanLoad      FanLoadSlk     CellPort      Remark
#
DTMF_INST/m_rcc_clk__L2_N0  15.000      129.000     -114.000      CLKINVX20/Y   #129,0.000,C
DTMF_INST/m_clk__L2_N7     15.000      51.000      -36.000       CLKINVX20/Y   #51,0.000,C
DTMF_INST/m_clk__L2_N4     15.000      50.000      -35.000       CLKINVX20/Y   #50,0.000,C
DTMF_INST/m_clk__L2_N3     15.000      49.000      -34.000       CLKINVX20/Y   #49,0.000,C

*info: there are 56 nets with violation reported.
*info: remark format #FanOut,Cap,NetType(I,O,C,T).
*info: 10 clock nets have violation ( remark C ).
```

Related Commands

- `set_min_fanout`
- `set_max_fanout`
- `set_min_transition`
- `set_max_transition`
- `set_min_capacitance`
- `set_max_capacitance`

report_fanin

```
report_fanin  
[-help]  
-to pin_port_net_list  
[-trace_through {case_disable | user_disable | all}]  
[-pin_levels numOfPinLevels]  
[-view viewName]  
[-nosplit]  
[{> | >> } filename]
```

Reports a cone traversal that is not tied to the timing graph, that is, not blocked by `set_disable_timing` and case analysis. By default, the trace matches the timing view and the reports generated using the `all_fanin` and `all_fanout` commands. The reports indicate whether disabled or case analysis arcs were traversed.

Parameters

<code>-help</code>	Prints out the command usage.
<code>> >> filename</code>	When specified with <code>></code>, writes the report to the specified file name. If the file already exists, the software overwrites it. When specified with <code>>></code>, writes the report to the specified file name. If the file already exists, the software concatenates the report to the end of the file.
<code>-nosplit</code>	Specifies that reports with long names will not be split into multiple lines.
<code>-pin_levels numOfPinLevels</code>	Specifies the desired depth of the output.
<code>-trace_through {case_disable user_disable all}</code>	Allows traversal through disabled arcs. You can specify one of the following options: • <code>case_disable</code>: Arcs which are disabled because of <code>set_case_analysis</code> are traversed but because of <code>set_disable_timing</code> are not traversed • <code>user_disable</code>: Arcs disabled because of <code>set_disable_timing</code> are traversed but because of <code>set_case_analysis</code> are not traversed • <code>all</code>: Trace through all arcs
<code>-to pin_port_net_list</code>	Specifies a list or collection of pins, ports, or nets. A mixed list or heterogeneous collection is also allowed.
<code>-view viewName</code>	Specifies disabled arcs in the specified view.

Examples

- The `report_fanin -to h0/A` command shows the following output:

Fanin Network for Pin: h0/A

Source Pin	Sink Pin	Cell Type	Timing Sense	Arc	Case
				Disabled	Analysis
u1/Y	h0/A	(net)	-	-	-
u1/A	u1/Y	BUFX2	positive_unate	-	-
in	u1/A	(net)	-	-	-

Related Commands

- `set_disable_timing`
- `set_case_analysis`
- `all_fanin`
- `all_fanout`

report_fanout

```
report_fanout  
[-help]  
{-from pin_port_net_list | -clock_tree}  
[-trace_through {case_disable | user_disable | all}]  
[-pin_levels numOfPinLevels ]  
[-view viewName ]  
[-nosplit]  
[ {> | >> } filename ]
```

Allows a cone traversal that is not tied to the timing graph, that is, not blocked by `set_disable_timing` and case analysis. By default, the trace matches the timing view and the reports generated using the `all_fanin` and `all_fanout` commands. The reports indicate whether disabled or case analysis arcs were traversed.

Parameters

<code>-help</code>	Prints out the command usage.
<code>> >> filename</code>	When specified with <code>></code>, writes the report to the specified file name. If the file already exists, the software overwrites it. When specified with <code>>></code>, writes the report to the specified file name. If the file already exists, the software concatenates the report to the end of the file.
<code>-clock_tree</code>	Reports the fanout of the source pin, one after the other, for all the clocks in the design.
<code>-nosplit</code>	Specifies that reports with long names will not be split into multiple lines.
<code>-pin_levels numOfPinLevels</code>	Specifies the desired depth of the output.
<code>-trace_through {case_disable user_disable all}</code>	Allows traversal through disabled arcs. You can specify one of the following options: • <code>case_disable</code>: Arcs which are disabled because of <code>set_case_analysis</code> are traversed but because of <code>set_disable_timing</code> are not traversed • <code>user_disable</code>: Arcs disabled because of <code>set_disable_timing</code> are traversed but because of <code>set_case_analysis</code> are not traversed • <code>all</code>: Trace through all arcs
<code>-from pin_port_net_list</code>	Specifies a list or collection of pins, ports, or nets. A mixed list or heterogeneous collection is also allowed.
<code>-view viewName</code>	Specifies disabled arcs in the specified view.

Examples

- Suppose arc seg1/u1 is disabled due to the following constraint:

```
tempus> set_disable_timing -from A -to Y seg1/u1
```

```
tempus> report_fanout -from IN1 (command will show the following output):
```

```
Fanout Network for Port: IN1
```

Source Pin	Sink Pin	Cell Type	Timing Sense	Arc Disabled	Case Analysis
IN1	seg1/u1/A	(net)	-	-	-

- The following command will show the output below:

```
tempus> report_fanout -from IN1 -trace_through user_disable
```

```
Fanout Network for Port: IN1
```

Source Pin	Sink Pin	Cell Type	Timing Sense	Arc Disabled	Case Analysis
IN1	seg1/u1/A	(net)	-	-	-
seg1/u1/A	seg1/u1/Y	CLKBUF	positive_unate	Y	-
seg1/u1/Y	seg1/u2/A	(net)	-	-	-

Related Commands

- `set_disable_timing`
- `set_case_analysis`
- `all_fanin`
- `all_fanout`

report_freq_violation

```
report_freq_violation
[-help]
[-detailed]
[-em_type {avg | peak | rms | all}]
[-excInstFile string]
[-outfile string]
[-reportFormat integer]
[-selInstFile string]
[-slew {min | max}]
[-sort {name | freq | minDiff | maxDiff | minFreq | maxFreq}]
[-tableScale float]
[-view string]
```

Runs electromigration (EM) analysis based on the frequency table defined in the Liberty library and generates a frequency violation report.

You must run the following commands prior to setting the `report_freq_violation` command:

- `write_tcf activityFile`
- `read_activity_file -format TCF activityFile`
- `propagate_activity`
- `read_activity_file -set_net_freq true -format TCF activityFile`
- `set_analysis_view -setup default_view_setup -hold default_view_hold`
- `set_analysis_mode -checkType hold`

Note: If activity files are not defined, the frequency can also be calculated on a pin by the clock period. There will also be a default frequency of 500M (toggle count 1e+9) if there is no clock phase on this pin.

Parameters

-help	Prints a brief description that includes type and default information for each <code>report_freq_violation</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man report_freq_violation.</code>
-------	--

<code>-detailed</code>	Generates a detailed report containing information about instances with frequency violations. The report includes per-instance details rather than just the summary statistics. It is particularly useful for debugging and root cause analysis as it provides instance-level information, expanded metrics, and sorting and filtering compatibility. The report generated by this parameter includes the following frequency violation values: • [V]: Represents the operating voltage applied to the instance during EM analysis. • [F]: Indicates the switching frequency or activity rate of the instance. • [P]: Represents the power consumption for the instance derived from the dynamic and leakage components.
<code>-em_type {avg peak rms all}</code>	Specifies the electromigration analysis type to apply when reporting a frequency violation. This parameter determines the current stress model used for evaluating EM reliability. Use one of the following values with this parameter: • <code>avg</code>: Uses the average current over time to estimate the EM stress. Use this value for the nets with relatively stable current flow. • <code>peak</code>: Uses the maximum instantaneous current observed while reporting a frequency violation. This is the most appropriate value to be used for worst case analysis. • <code>rms</code>: Uses the root mean square (RMS) current, which accounts for variations and switching activity. • <code>all</code>: Reports results for all the above three values (<code>avg</code>, <code>peak</code>, and <code>rms</code>) in the same run for comprehensive analysis. <i>Default:</i> <code>avg</code>
<code>-excInstFile excInstFileName</code>	Specifies an instance file that lists the names of instances to be excluded from the frequency violation report. Note: The instance file is a text file that contains a list of instances. Each instance name appears on a single line in the file. The comments begin with "#". Leading or ending spaces in instance names are ignored.
<code>-outfile fileName</code>	Specifies the report file for the violation data.
<code>-reportFormat integer</code>	Specifies a format for the violation report. The report details vary according to the selected report format. You can specify report formats 0, 1, or 2. For more information on the report formats, see the <i>Report Format</i> section below. <i>Default:</i> 0

<code>-selInstFile selInstFileName</code>	Specifies a file that lists the names of instances to be included in the frequency violation report. Note: The instance file is a text file that contains a list of instances. Each instance name appears on a single line in the file. The comments begin with "#". Leading or ending spaces in instance names are ignored.
<code>-slew {min max}</code>	Specifies the minimum or maximum slew value. <i>Default: min</i>
<code>-sort {name freq minDiff maxDiff minFreq maxFreq}</code>	Sorts the instance-based frequency data by name, freq, minDiff, maxDiff, minFreq, and maxFreq. By default, the violation report is not sorted. You can use one of the following options: ◦ <code>name</code>: Name of a pin or arc. ◦ <code>freq</code>: Frequency value of a pin or arc. ◦ <code>minDiff</code>: Difference between freq and minFreq of a pin or arc. ◦ <code>maxDiff</code>: Difference between freq and maxFreq of a pin or arc. ◦ <code>minFreq</code>: Minimum frequency value of a pin or arc as defined in the frequency table. ◦ <code>maxFreq</code>: Maximum frequency value of a pin or arc as defined in the frequency table.
<code>-tableScale float</code>	Specifies the scale factor for the EM table value. <i>Default: 1.0</i>
<code>-view viewName</code>	Specifies the view for which to generate the frequency violation report. Use this parameter when working with multi-mode multi-corner (MMMC) designs.

Example

- The following command reports instances that violate frequency constraints and lists them in the `freqViol.rpt` report file:

```
report_freq_violation -outfile freqViol.rpt
```

Report Format

The report formats are described below.

- **Format 0**

The following command generates a report providing details of minFreq, Freq, Tran, Cap, Remark, and Arc/Pin and directs the output to the report file summaryViol.rpt:

```
report_freq_violation -outfile summaryViol.rpt -reportFormat 0
```

For example:

minFreq	Freq	Tran	Cap	Remark	Arc/Pin
5.985	1.009	0.0356	0.003103		tt0/data/r3/add_27/U1_1_6/Y

- **Format 1**

When `report_freq_violation -reportFormat` is set to 1, the report provides details as shown in the following example:

Status	Remark	freq	minFreq	maxFreq	Arc/Pin
[P]	*	1.009	5.985	5.995	tt0/data/r3/add_27/U1_1_6/Y

- **Format 2**

A detailed report is generated when `report_freq_violation-reportFormat` is set to 2, as shown in the following example:

Status	Remark	freq	minFreq	maxFreq	minTran	maxTran	minCap	maxCap
Arc/Pin								
[P]	*	1.009	5.985	5.995	3.560e-02	3.790e-02	3.103e-03	3.103e-03
03 tt0/data/r3/add_27/U1_1_6/Y								

Where:

Status	Arc or pin status - F (fail), V (violation), P (pass), or ? (unknown)
Remark	Indicates whether the pin is a clock
freq	Working frequency of the arc or pin
minFreq	Minimum frequency value taken from the frequency table
maxFreq	Maximum frequency value taken from the frequency table
minTran	Minimum slew value
maxTran	Maximum slew value

minCap	Minimum capacitance load
maxCap	Maximum capacitance load
Arc/Pin	Name of the arc or pin

Related Commands

- `write_tcf`
- `read_activity_file`
- `propagate_activity`
- `set_analysis_view`
- `set_analysis_mode`

report_inactive_arcs

```
report_inactive_arcs
[-help]
[instance_or_port_list]
[-exclude_type {const snipped disable library_disable global_disable conditional_disable
Library_Mode power_mode_disable}]
[-include_net_arcs]
[-include_when_cond]
[-type {const snipped disable library_disable global_disable conditional_disable Library_Mode
power_mode_disable}]
[-view view_name]
[> filename[.gz]]
[>> filename[.gz]]
[-delay_arcs_only | -check_arcs_only]
```

Reports information about disabled timing and timing check arcs in the design. Reports all arcs disabled due to user-specified exceptions, such as [set_disable_timing](#) or [set_case_analysis](#), as well as information about arcs disabled by constant propagation during timing analysis, snipped loop arcs, and arcs disabled in the timing library.

The report output prints only the 'from' and 'to' information of arcs, and not the full instance name for each arc.

Parameters

-help	Prints out the command usage.
<i>instance_or_port_list</i>	Reports disabled timing arcs for the specified list of instances or ports. If you specify a list of instances, the software reports all disabled arcs of the specified cells. If you specify a list of ports, the software reports all disabled arcs directly connected to the ports. <i>Default:</i> Reports all disabled net and cell arcs.
-check_arcs_only	Lists only disabled check arcs. The <code>check_arcs</code> are disabled by using the <code>set_disable_timing</code> command. <i>Default:</i> Both disabled timing delay arcs and check arcs are listed.
-delay_arcs_only	Lists only disabled timing delay arcs.

<code>-exclude_type {const snipped disable library_disable global_disable conditional_disable Library_Mode power_mode_disable}</code>	Excludes the disabled arcs that belong to the specified category of disabled arcs. The valid options are: const, snipped, disable, library_disable, global_disable, conditional_disable, Library_Mode, and power_mode_disable.
<code>-include_net_arcs</code>	Includes disabled net arcs in the report.
<code>-include_when_cond</code>	Shows the when_cond of arcs in the output report.

<pre>-type {const snipped disable library_disable global_disable conditional_disable Library_Mode power_mode_disable}</pre>	Lists only the disabled arcs that belong to the specified category of disabled arcs. The <code>-type</code> options restrict the report to only those arcs which are disabled due to a particular timing analysis step, such as constant propagation, specified disables, or loop arc disables. The valid type options are: • <code>const</code>: Propagated constant -- arcs disabled due to a constant value. The constant value can apply to one of the arc nets/pins, to side inputs that disable a condition for the arc, to nets that disable an active mode for the arc, or to disable a default conditional arc. • <code>snipped</code>: Loop snipped arcs -- arcs automatically disabled by the software as part of a feedback loop. • <code>disable</code>: Defined by using the <code>set_disable_timing</code> command. • <code>library_disable</code>: Delay arc disabled due to library disables. These disables may be caused by the following conditions: ◦ Disables specified using the <code>set_disable_timing</code> command. ◦ The arc is an asynchronous arc, and the <code>timing_enable_preset_clear_arcs</code> global variable is set to <code>false</code>. • <code>global_disable</code>: Lists the arcs that are disabled through user-specified global variable settings. • <code>conditional_disable</code>: Arcs disabled if a <code>when</code> condition is specified. There is no constant on the sink or source pin, and the arc is disabled due to the <code>when</code> condition. • <code>Library_Mode</code>: Displays the library mode of disabled arcs. • <code>power_mode_disable</code>: Reports the arcs for which power mode is disabled.
<pre>-view view_name</pre>	Generates the report for the specified analysis view only. You can specify this parameter only when the software is in multi-mode multi-corner timing analysis mode.

```
{> | >>} filename
```

When specified with >, writes the report to the specified file name. If the file already exists, the software overwrites it.

When specified with >>, writes the report to the specified file name. If the file already exists, the software concatenates the report to the end of the file.

The *filename* parameter must be the last argument in the list.

Note: To write a compressed report, add the .gz extension to the file name.

Default: Report is displayed on standard output without being saved.

Examples

Here are some examples.

- The following command reports all disabled arcs in the design:

```
tempus> report_inactive_arcs > Report
```

The following example shows the report generated:

```
-----
Flags :   const                propagated constant
         snipped              loop snipped arcs
         disable              set_disable_timing
         disable_clock_gating set_disable_clock_gating
         library_disable      set_disable_cell_timing
         Missing_Phase        No Arc Phase Data
From      To      DisableType ArcType      Reason
-----
u8/CK ^    u8/D ^    const      hold_rising_clk_rise  u8/D = 1
u8/CK ^    u8/D v    const      hold_falling_clk_rise u8/D = 1
seg3/u12/A ^ seg3/u12/Y v disable    combinational      User Disable
-----
```

- When the software is in multi-mode multi-corner analysis mode, the report generated lists the disabled arcs by active analysis view, as shown in the following example:

```
-----
Flags : const propagated constant
         snipped loop snipped arcs
         disable set_disable_timing
         disable_clock_gating set_disable_clock_gating
         library_disable set_disable_cell_timing
```

Missing_Phase No Arc Phase Data

From	To	DisableType	ArcType	Reason	View

gscanmode ^	FE_0305_gmode/I ^	const	net_arc	gscanmode = 0	setup_view2
gscanmode v	FE_0305_gmode/I v	const	net_arc	gscanmode = 0	setup_view2
gscanmode ^	FE_0305_gmode/I ^	const	net_arc	gscanmode = 0	hold_view1
gscanmode v	FE_0305_gmode/I v	const	net_arc	gscanmode = 0	hold_view1
gscanmode ^	FE_0305_gmode/I ^	const	net_arc	gscanmode = 0	hold_view2

Related Commands

- [set_disable_timing](#)
- [set_case_analysis](#)

report_instance_library

```
report_instance_library  
[-help]  
[-file fileName]  
[-instance instanceName ]  
[-view viewName | -delay_corner delayCornerName ]  
[-power_domain domainName ]  
[-early | -late]
```

Reports library related information for an instance(s). The report prints the following information about an instance:

- Instance name
- Cell type
- View name or delay corner
- Power domain name
- Library/Libset (early/late)
- Library file (early/late)
- Operating conditions
- Binding Status : Prints out the status of the available voltage library. If the user setting is `True`, then the binding status is not displayed.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-delay_corner</code> <i>delayCornerName</i>	Specifies the delay corner from which an instance library is to be selected. If no delay corner is specified, then all the instance libraries will be printed.
<code>-early -late</code>	Specifies the library set (early/late) from which a library instance is to be printed. If either option is not specified, both the early and late library sets are used.
<code>-file</code> <i>fileName</i>	Specifies the name of the output file. If no file is specified, the report will be printed to the standard output file.
<code>-instance</code> <i>instanceName</i>	Specifies the instance(s) for which the report is to be printed. If no instance is provided, the report will print information about all the instances. The value of instance can be a list of regular expressions.
<code>-power_domain</code> <i>domainName</i>	Specifies the power domain from which instance(s) are to be selected. This option should be specified with the <code>-delay_corner</code> or <code>-view</code> options.
<code>-view</code> <i>viewName</i>	Specifies the view from which the instance library is to be selected.

Examples

- The following command displays the information below about instance `LIN_CONV_INST/F6A` for the specified view:

```
report_instance_library -instance LIN_CONV_INST/F6A -early -view scan_slow_RCMAX
```

```
Instance           : LIN_CONV_INST/F6A
Cell               : INV_X2
Analysis View      : scan_slow_RCMAX
Power Domain       : Default
Library/Libset(early) : F_lib_v1.0/libset_slow
Library File(early)  : /libs/F_lib_v1.0_worst.lib
Binding Status(early) : All PG Pins Voltage Match
Op Cond(early)      : P->1.000000, V->0.800000, T->25.000000 (TypTyp_0.8V_25C)
Cell Type          : gateCell
```

- The following command displays the information below about instance `FE_C674` for the specified delay corner:

```
report_instance_library -delay_corner delay_corner_slow_RCMAX -late
```

```
Instance           : FE_C674
Cell               : BUF_X32
Delay Corner        : delay_corner_slow_RCMAX
PowerDomain         : Default
```

```
Library/Libset(late) : F_lib_v1.0/libset_fast
Library File(late)   : /libs/F_lib_v1.0_typic
Binding Status(late) : All PG Pins Voltage Match
Op Cond(late)        : P->1.000000, V->0.800000, T->25.000000 (TypTyp_0.8V_25C)
Cell Type            : gateCell
```

Related Commands

- [check_instance_library_in_views](#)

report_ipd_configuration

```
report_ipd_configuration
```

```
[-help]
```

```
[-all]
```

Reports the rail sweep information and the software created voltage combinations, that is, the IPD views.

This command reports and indicates the valid voltage combinations along with the excluded voltage combinations.

Additionally:

- IPD views are named automatically as “ipd_view<count>”.
- “Excluded” column reports “yes” against the excluded voltage combinations - based on `set_ipd_rail_voltage -exclude_voltage_signatures settings`.

Parameters

-help	Prints out the command usage.
-all	Reports all the valid and excluded configurations. By default, the view exclusion status is not printed.

Examples

- A sample report is shown below, when the following command is used:

```
report_ipd_configuration
```

```
POWER RAIL/GROUP MAPPING
```

```
-----
```

```
Mapping Power rail/group
```

```
-----
```

```
PG1 VDD
```

```
PG2 VDDR
```

```
PG3 VDDE
```

```
VOLTAGE SWEEP
```

```
-----
```

```
Power Sweep
```

```
Rail
```

```

-----
PG1 { 0.9 }
PG2 { 1.1 1.2 }
PG3 { 0.8 0.9 }
VOLTAGE SIGNATURES
-----
PG1      PG2      PG3      View      Excluded
-----
0.9      1.1      0.8      ipd_view1  no
0.9      1.2      0.8      ipd_view2  no
0.9      1.1      0.9      ipd_view3  no
0.9      1.2      0.9      ipd_view4  yes

```

- **Sample report when -all parameter is used:**

```
tempus > report_ipd_configuration -all
```

```

POWER RAIL/GROUP MAPPING
-----
Mapping Power rail/group
-----

PG1 VDD
PG2 VDDR
PG3 VDDE
VOLTAGE SWEEP
-----

Power Sweep
Rail
-----

PG1 { 0.9 }
PG2 { 1.1 1.2 }
PG3 { 0.8 0.9 }
VOLTAGE SIGNATURES
-----
PG1      PG2      PG3      View      Excluded
-----
0.9      1.1      0.8      ipd_view1  no
0.9      1.2      0.8      ipd_view2  no
0.9      1.1      0.9      ipd_view3  no
0.9      1.2      0.9      ipd_view4  yes

```

Related Commands

- `set_ipd_rail_voltage`
- `create_pg_net_group`
- `report_pg_net_group`
- `create_ipd_configuration`
- `write_ipd_scope`
- `read_ipd_scope`
- `report_ipd_crossings`
- `check_ipd_logic`

Related Information

- For more information, refer to the [Inter Power Domain \(IPD\) Analysis](#) chapter in the *Tempus User Guide*.

report_ipd_crossings

```
report_ipd_crossings
[-help]
[-all]
[-cprpr_aware]
[-from <pin_list>]
[-from_power_domain <string>]
[-out_file <file_name>]
[-to <pin_list>]
[-to_power_domain <string>]
```

Reports the IPD crossing information summary on the design. This command also provides details on the pins and instances that are part of these crossings.

Parameters

-help	Prints out the command usage.
-all	Reports detailed crossing information.
-cprpr_aware	Reports all the IPD crossings with CPPR awareness.
-from <pin_list>	Specifies the source pin for which crossing is to be reported.
-from_power_domain <string>	Specifies the source power domain for which crossing is to be reported.
-out_file <file_name>	Specifies the output file name - in case the output is to be redirected to a file.
-to <pin_list>	Specifies the sink pin for which crossing is to be reported.
-to_power_domain <string>	Specifies the sink power domain for which crossing is to be reported.

Examples

- The following output is generated by the `report_ipd_crossings` command:

```
tempus > report_ipd_crossings
-----
----
IPD Crossing              Inferred   Inferred   User       Excluded   Excluded
```

Tempus Text Command Reference

Timing Analysis Commands

Active						
	IPD	IPD	Excluded	due to	due to	IPD
	Crossings	Crossings		Same	Same	
Crossings						
	(Clock)	(Data)		Voltage	Supply	
					Net	

PD_A0VCC_A0VSS -> PD_VDD_VSS	0	2	0	0	0	2
PD_A1VCC_A1VSS -> PD_VDD_VSS	2	0	0	0	0	2
PD_A2VCC_A2VSS -> PD_VDD_VSS	0	1	0	0	0	1
PD_A3VCC_A3VSS -> PD_VDD_VSS	0	1	0	0	0	1

- The following report output shows the crossing instances for all the crossings:

```
tempus > report_ipd_crossings -all
```

Source Power	Sink Power	Object	IPD	IPD	Source Crossing	Sink
Crossing						
Domain	Domain	Type	Crossing	Crossing	Pin	Pin
			Type	Element		

PD_A0VCC_A0VSS	PD_VDD_VSS	Net	Inferred	Data	inst1/n[3]	inst5/A
PD_A0VCC_A0VSS	PD_VDD_VSS	Net	Inferred	Data	inst1/n[4]	inst5/A
PD_A1VCC_A0VSS	PD_VDD_VSS	Net	Inferred	Clock	inst2/n[3]	inst6/A
PD_A1VCC_A0VSS	PD_VDD_VSS	Net	Inferred	Clock	inst2/n[4]	inst6/A
PD_A2VCC_A0VSS	PD_VDD_VSS	Cell	Inferred	Data	inst3/A	inst3/Y
PD_A3VCC_A0VSS	PD_VDD_VSS	Cell	Inferred	Data	inst4/A	inst4/Y

- The following command reports crossings between specific pins:

```
tempus > report_ipd_crossings -from inst3/A -to inst3/Y
```

Source Power	Sink Power	Object	IPD	IPD	Source Crossing	Sink
--------------	------------	--------	-----	-----	-----------------	------

Crossing						
Domain	Domain	Type	Crossing Type	Crossing Element	Pin	Pin

-						
PD_A2VCC_A0VSS	PD_VDD_VSS	Cell	Inferred	Data	inst3/A	inst3/Y

-						

- The following command reports crossings between specific power domains:

```
tempus > report_ipd_crossings -from_to_power_domain PD_A0VCC_A0_VSS -to_power_domain PD_VDD_VSS
```

Source Power Domain	Sink Power Domain	Object Type	IPD Crossing Type	IPD Crossing Element	Source Crossing Pin	Sink Crossing Pin

PD_A0VCC_A0VSS	PD_VDD_VSS	Net	Inferred	Data	inst1/n[3]	inst5/A
PD_A0VCC_A0VSS	PD_VDD_VSS	Net	Inferred	Data	inst1/n[4]	inst5/A

Related Commands

- `set_ipd_rail_voltage`
- `create_pg_net_group`
- `report_pg_net_group`
- `create_ipd_configuration`
- `report_ipd_configuration`
- `write_ipd_scope`
- `read_ipd_scope`
- `check_ipd_logic`

Related Information

- For more information, refer to the [Inter Power Domain \(IPD\) Analysis](#) chapter in the *Tempus User Guide*.

report_lib_arcs

```
report_lib_arcs
[-help]
-arc <string>
[-fall]
[-nldm]
[-quiet]
[-rise]
[-lvf [-early]]
[-lvf [-late]]
[-moments [-mean_shift]]
[-moments [-std_dev]]
[-moments [-skewness]]
[-input_net_transition <float> | -constrained_pin_transition <float>]
[-input_net_transition <float> | -related_pin_transition <float>]
[-total_output_net_capacitance <float> | -constrained_pin_transition <float>]
[-total_output_net_capacitance <float> | -related_pin_transition <float>]
[-constrained_pin_transition <float> | -delay ] [-constrained_pin_transition <float> | -
transition ]
[-related_pin_transition <float> | -delay ]
[-related_pin_transition <float> | -transition ]
```

Reports the library arc data. The command also returns the data in a Tcl object, which can be used in scripts.

This command uses a collection of arcs as input, which can be queried using the `get_lib_arcs` command.

The units for the input and output data can be obtained using the `get_time_unit` and `get_capacitance_unit` commands.

Parameters

-help	Prints out the command usage.
-arc <string>	Specifies the arc(s) for which the report is to be printed.
-constrained_pin_transition <float>	Specifies the constrained pin slew (valid for constraint arcs) at which the parameters need to be reported. The slew units must be specified using the <code>get_time_unit</code> command.
-delay	Specifies if the delay is to be reported.

<code>-early</code>	Specifies if early LVF data is to be reported. This parameter can be used only in conjunction with the <code>-lvf</code> .
<code>-fall</code>	Specifies if details of fall tables are to be reported.
<code>-input_net_transition</code> <float>	Specifies the slew at which the parameters need to be reported. The slew units must be specified using the <code>get_time_unit</code> command.
<code>-late</code>	Specifies if late LVF data is to be reported. This parameter can be used only in conjunction with the <code>-lvf</code> .
<code>-lvf</code>	Specifies if LVF data is to be reported.
<code>-mean_shift</code>	Specifies if the mean shift data is to be reported. This parameter can be used only in conjunction with the <code>-moments</code> .
<code>-moments</code>	Specifies if the moments data is to be reported.
<code>-nldm</code>	Specifies if NLDM data is to be reported.
<code>-quiet</code>	Disables output to be displayed on the screen.
<code>-related_pin_transition</code> <float>	Specifies the related pin slew (valid for constraint arcs) at which the parameters need to be reported. The slew units must be specified using the <code>get_time_unit</code> command.
<code>-rise</code>	Specifies if details of rise tables are to be reported.
<code>-skewness</code>	Specifies whether skewness data should be reported. This parameter can be used only in conjunction with the <code>-moments</code> .
<code>-std_dev</code>	Specifies if standard deviation data is to be reported. This parameter can be used only in conjunction with the <code>-moments</code> .
<code>-</code> <code>total_output_net_capacitance</code> <float>	Specifies the load at which the parameters need to be reported. The load must be specified using the <code>get_capacitance_unit</code> command.
<code>-transition</code>	Specifies if the output slew is to be reported.

— .

Examples

- The following command reports the library arc data containing mean shift rise tables:

```
tempus > report_lib_arcs -arc [get_lib_arcs -from cell_w/DFF/CK -to cell_w/DFF/Q] -
input_net_transition 0.08 /
      -total_output_net_capacitance 0.055 -mean_shift -rise -delay -moments -nldm
```

Units of Report:- Time: '1000ps', Capacitance: '1000fF'.

input_net_transition:0.080000, total_output_net_capacitance:0.055000

```
-----
| Index | Timing Arc | Value(s) | | | |
|---|---|---|---|---|---|
|      | Cell      | Source/Sink Pin | Timing Sense | Timing Type | cell_rise |
|-----|-----|-----|-----|-----|-----|
| 1     | cell_w/DFF | CK -> Q      | non_unate   | rising_edge | 0.258444 |
|-----|-----|-----|-----|-----|-----|
{{cell_rise 0.258444279432} {ocv_mean_shift_cell_rise NA}}
```

- The following command reports the library arc data containing fall tables for moments data:

```
tempus > report_lib_arcs -arc [get_lib_arcs -from cell_w/BUF/A -to cell_w/BUF/Y] /
      -input_net_transition 0.055 -total_output_net_capacitance 0.15 -fall -delay -moments
      -nldm
```

Units of Report:- Time: '1000ps', Capacitance: '1000fF'.

input_net_transition:0.055000, total_output_net_capacitance:0.150000

```
-----
| Index | Timing Arc | Value(s) | | | |
|---|---|---|---|---|---|
|      | Cell      | Source/Sink Pin | Timing Sense | Timing Type | cell_fall |
|-----|-----|-----|-----|-----|-----|
| 1     | cell_w/BUF | A -> Y      | positive_unate | combinational| 0.129744 |
|-----|-----|-----|-----|-----|-----|
{{cell_fall 0.129} {ocv_std_dev_cell_fall NA} {ocv_mean_shift_cell_fall NA}
{ocv_skewness_cell_fall NA}}
```

- The following command reports the library arc data containing rise tables data with output slew:

```
tempus > report_lib_arcs -arc [get_lib_arcs -from cell_w/DFF/CK -to cell_w/DFF/Q] -
input_net_transition 0.08 /
      -total_output_net_capacitance 0.055 -mean_shift -rise -delay -moments -nldm -
transition
```

Units of Report:- Time: '1000ps', Capacitance: '1000fF'.

```
input_net_transition:0.080000, total_output_net_capacitance:0.055000
```

```
-----
| Index | Timing Arc |
| Value(s) |
| -----
|
| Cell | Source/Sink Pin | Timing Sense | Timing Type | rise_transition |
cell_rise |
|-----
| 1 | cell_w/DFF | CK -> Q | non_unate | rising_edge | 0.21692 |
0.258444 |
|-----
{{cell_rise 0.258} {rise_transition 0.216} {ocv_mean_shift_cell_rise NA}
{ocv_mean_shift_rise_transition NA}}
```

This command also returns the data in a Tcl object that can be parsed and used in scripts. A 3-Dimensional Tcl List will return the following information:

```
{{TableName Value} {TableName Value}} {{TableName Value} {TableName Value}}
```

Here, the first list (left column) indicates the Table-Value pair in one arc. The second list indicates the various tables present per arc. And the third list (last column) indicates the list of total arcs in the collection.

- The following command reports the slew and capacitance load:

```
tempus > set items [report_lib_arcs -arc [ get_lib_arcs -from sc_cmoslp/A022X/clk -to
sc_cmoslp/A022X /out1] /
-input_net_transition 0.003 -total_output_net_capacitance 0.021]
```

Units of Report:- Time: '1000ps', Capacitance: '1000fF'.

```
input_net_transition:0.003, total_output_net_capacitance:0.021
```

```
-----
| Index | Timing Arc |
| Value(s) |
|-----
|
| Cell | Source/Sink Pin | Timing Sense | Timing Type |
rise_transition | fall_transition | cell_rise | cell_fall |
|-----
| 1 | sc_cmoslp/A022X | clk -> out1 | positive_unate | combinational | 0.20**
| 0.30** | 1.00** | 1.20** |
|-----
```

```
-----  
| 2      | sc_cmoslp/A022X | clk -> out1      | negative_unate | combinational | 0.20**  
| 0.30**          | 1.00**          | 1.20**          |  
-----
```

```
-----  
** --> Scalar Value{{cell_rise 0.999} {rise_transition 0.199} {cell_fall 1.199}  
{fall_transition 0.299}} {{cell_rise 0.999} {rise_transition 0.199} {cell_fall 1.199}  
{fall_transition 0.299}}
```

```
tempus 5 > puts $items  
{{cell_rise 0.999} {rise_transition 0.199} {cell_fall 1.199} {fall_transition 0.299}}  
{{cell_rise 0.999} {rise_transition 0.199} {cell_fall 1.199} {fall_transition 0.299}}
```

Related Commands

- [get_lib_arcs](#)
- [get_time_unit](#)
- [get_capacitance_unit](#)

report_min_pulse_width

```
report_min_pulse_width
[-help]
[-clock <object_list>]
[-net]
[-path_type {summary | short | full_clock}]
[-pins <object_list>]
[-tcl_list]
[-verbose]
[-view <view_name>]
[-violation_only]
[-waveform_aware_type {regular | waveform_aware | both}]
[> <filename[.gz]>]
[>> <filename[.gz]]
```

Reports a set of minimum pulse width violations on pins and ports on the clock network. The report shows the required pulse width, the actual pulse width, and by how much the constraint is violated or met. This is reported separately for sequential elements and instances on the clock tree.

You can customize pulse width reporting format by using the `report_timing_format` global variable.

The report header shows clock edge information to make it more intuitive for clock style checks.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-clock <object_list></code>	Specifies a set of clocks to for which to report the minimum pulse width.
<code>{> >>} filename</code>	When specified with <code>></code>, writes the report to the specified file name. If the file already exists, the software overwrites it. When specified with <code>>></code>, writes the report to the specified file name. If the file already exists, the software concatenates the report to the end of the file. The <i>filename</i> parameter must be the last argument in the list. The <i>filename</i> and <code>-tcl_list</code> parameters are mutually exclusive; you cannot specify them together. Note: To write a compressed report, add the <code>.gz</code> extension to the file name. Default: Report is displayed on standard output without being saved.

<code>-net</code>	Reports the net arcs. This parameter can be specified with the <code>-path_type full_clock</code> option only.
<code>-path_type {summary short full_clock}</code>	Specifies the type of report to generate. <i>Default:</i> <code>summary</code>
	summary: Generates a summary report for each pulse width check that shows the required width, the actual width, and the resulting slack.
	short: Generates a path report for each pulse width check that lists the latency of each edge of the pulse as a single lumped value, instead of reporting a detailed path report.
	full_clock: Generates a path report for each pulse width check that shows the detailed pin-to-pin path for each edge of the pulse. Note: The <code>-path_type full_clock</code> parameter may cause an increased runtime. In that case, you can use the <code>-path_type short</code> parameter.
<code>-pins object_list</code>	Specifies a set of ports and pins on the clock network. If this list is not specified, then the report contains all the pins and ports in the current design.
<code>-tcl_list</code>	Produces the report in Tcl list format instead of a tabular format. This is useful for integrating timing with custom Tcl functions, and also for customizing report generation. The <code>-tcl_list</code> and <code>filename</code> parameters are mutually exclusive; you cannot specify them together.
<code>-verbose</code>	Provides detailed information about minimum pulse width calculations. Note: This parameter is obsolete. The parameter still works in this release, but to ensure compatibility with future releases, use the <code>-path_type</code> parameter instead.
<code>-view view_name</code>	Generates the report for the specified analysis view only. You can specify this parameter only when the software is in multi-mode multi-corner timing analysis mode.
<code>-violation_only</code>	Reports only the negative slack minimum pulse width violations.


```
-waveform_aware_type
{regular |
 waveform_aware | both}
```

Reports the waveform-aware pulse width checks. You can specify the following options:

- `waveform_aware`: Reports only the waveform-aware pulse width checks.
- `both`: Reports waveform-aware as well as regular pulse width checks.
- `regular`: Reports regular pulse width checks. This is the default setting.

Examples

Here are some examples.

- The `report_min_pulse_width` will generate a summary report as shown below:

```
tempus> report_min_pulse_width
```

```
-----
SEQUENTIAL CLOCK PULSE WIDTH
-----
```

Pin	Required Pulse Width	Actual Pulse Width	Slack
-----	-------------------------	-----------------------	-------

seg3/u14/CK (high)	0.169	0.860	0.691
seg3/u3/CK (high)	0.169	0.875	0.706
seg3/u9/CK (high)	0.169	0.875	0.706
seg3/u3/CK (low)	0.195	1.083	0.888
seg3/u9/CK (low)	0.195	1.083	0.888

- A short report can be generated at any clock pin having pulse width check, as shown below. The pulse width is shown as sequential clock pulse width in case it is reported on sequential pins:

```
tempus> report_min_pulse_width -path_type short
```

```
SEQUENTIAL CLOCK PULSE WIDTH
-----
```

```
Path 1: MET PulseWidth Check with Pin seg3/u14/CK
```

```
Ending Clock Edge: seg3/u14/CK (v) checked with trailing edge of 'CLK_W_3'
```

```
Beginning Clock Edge: seg3/u14/CK (^) triggered by leading edge of 'CLK_W_3'
```

```
Other End Arrival Time 1.503
```

```
- PulseWidth 0.169
```

```
+ Phase Shift 0.000
```

```
= Required Time 1.334
```

```
- Arrival Time 0.621
```

```
= Slack Time 0.714
```

```
Clock Rise Edge 0.000
```

```
+ Clock Network Latency (Prop) 0.621
```

```
= Beginpoint Arrival Time 0.621
```

```

-----
Pin           Cell      Delay  Arrival  Edge
                        Time
-----
seg3/u14/CK DFF              0.621   ^
-----

Path 2: MET PulseWidth Check with Pin seg3/u3/CK
Ending Clock Edge: seg3/u3/CK (v) checked with trailing edge of 'CLK_W_3'
Beginning Clock Edge: seg3/u3/CK (^) triggered by leading edge of 'CLK_W_3'
Other End Arrival Time 1.578
- PulseWidth 0.169
+ Phase Shift 0.000
= Required Time 1.409
- Arrival Time 0.682
= Slack Time 0.727
Clock Rise Edge 0.000
+ Clock Network Latency (Prop) 0.682
= Beginpoint Arrival Time 0.682
-----
Pin           Cell      Delay  Arrival  Edge
                        Time
-----
seg3/u3/CK DFF              0.682   ^
-----

```

Related Commands

- `report_analysis_coverage`
- `report_timing`
- `set_min_pulse_width`
- `reset_min_pulse_width`

report_mode

```
report_mode
[-help]
instance_list
[-view viewName]
```

Reports the status of library modes associated with a specified instance. The report includes the status and condition for each mode in each mode group defined in the timing library.

Parameters

-help	Prints out the command usage.
instance_list	Generates a report for the specified instances.
-view viewName	Reports the status of library modes in the specified active analysis view. <i>Default:</i> Reports the status of library modes in all active analysis views.

Examples

Here are some examples.

- The following command activates the `tst_mode_disable` library mode for instances of `i_ram`:

```
tempus> set_mode tst_mode_disable [get_cells i_ram]
```

- The following command sets the active setup and hold analysis views:

```
tempus> set_analysis_views -setup {v1 v2} -hold {v3}
```

- The following command reports the status of all modes associated with `m4k_top`:

```
tempus> report_mode m4k_top
```

```
-----
Mode Group tst_mode of m4k_top
-----
Mode Name      Status    Condition  View Name
-----
test_mode_enable  ACTIVE    TST_BYPASS  View1
test_mode_disable INACTIVE  !(TST_BYPASS) View2
-----
```

- The following command reports the status of all modes in the analysis view `View1`:

```
tempus> report_mode m4k_top -view View1
```

```
-----
```

Mode Group tst_mode of m4k_top

```
-----  
Mode Name      Status  Condition  View Name  
-----  
test_mode_enable ACTIVE  TST_BYPASS View1  
-----
```

Related Commands

- [set_analysis_view](#)
- [set_mode](#)

report_net

```
report_net  
[-help]  
[-min_fanout int]  
[-max_fanout int]  
{-net list_of_net_name_or_id | -pin list_of_pin_name_or_id}  
[-tcl_list]  
[-hier]  
[-output filename]  
[-tcl_list]  
[> | >>} <filename[.gz]>]
```

Reports the net information on the current module. The information includes the net name, the number of source pins, sink pins, and bidirectional pins on the net, wire capacitance, and total capacitance.

Electrical information, such as pin capacitance and wire capacitance with which the net is associated are also included.

Use the `-pin` option to report the net that is connected to the specified pin (or port). Both `-net` and `-pin` cannot be specified at the same time. The net (or pin) names can be hierarchical names that are unique relative to the top cell.

This command only searches for the defined nets and does not modify any objects in the database.

Parameters

<code>-help</code>	Prints out the command usage.
<code>{> >>} filename</code>	When specified with <code>></code>, writes the report to the specified file name. If the file already exists, the software overwrites it. When specified with <code>>></code>, writes the report to the specified file name. If the file already exists, the software concatenates the report to the end of the file. The <i>filename</i> parameter must be the last argument in the list. The <i>filename</i>, <code>-output</code>, and <code>-tcl_list</code> parameters are mutually exclusive; you cannot specify them together. Note: To write a compressed report, add the <code>.gz</code> extension to the file name. <i>Default:</i> Report is displayed on standard output without being saved.
<code>-hier</code>	Reports all of the nets in the design hierarchy. Without this argument, only the nets in the current level of hierarchy are reported.
<code>-max_fanout int</code>	Reports the net whose number of fanouts is less than or equal to the maximum number specified by <i>integer</i> .
<code>-min_fanout int</code>	Reports the net whose number of fanouts is greater than or equal to the minimum number specified by <i>integer</i> .
<code>-net list_of_net_name_or_id</code>	Reports the set of nets on the list of net names. This argument cannot be specified at the same time as <code>-pin</code>. If neither <code>-net</code> or <code>-pin</code> is specified, all nets are listed by default. Note: This argument also reports the capacitance of the net and the input pins.
<code>-output filename</code>	Dumps the report to the specified file instead of to the standard output. The <i>filename</i> argument must be the last argument in the list. This parameter cannot be used simultaneously with the <code>-tcl_list</code> option.
<code>-pin list_of_pin_name_or_id</code>	Reports the set of nets to which the pins on the list <i>list_of_pin_name_or_id</i> are connected. This argument cannot be specified at the same time as <code>-net</code>.
<code>-tcl_list</code>	Produces the report in Tcl list format instead of a tabular format. This is useful for extracting information from the report in a Tcl program. The <code>-tcl_list</code>, <i>filename</i>, and <code>-output</code> parameters are mutually exclusive; you cannot specify them together.

You can control the `report_net` table fields, such as setting the minimum and maximum column width using the `set_table_style` command. The table names are `report_net_source_table`, `report_net_sink_table`, `report_net_summary_table`, `report_net`, and `report_net_header`.

For example, the following command sets an indentation of six columns for the table and disables printing of the first column:

```
tempus > set_table_style -name report_net_source_table -indent 6 -max_widths {0,}
tempus > report_net
```

Examples

Here are some examples.

- The format of the `report_net` report is the same for all the examples, as shown in the first example. The following command reports information about net named `in`, as shown in the following example:

```
tempus > report_net -net n5
```

```
Net Name : n5
```

```
Number of Sources : 1
```

```
Number of Sinks : 1
```

```
Number of Bidis : 0
```

```
Source of parasitics : Wireload model(.lib None Generated)
```

```
Net Capacitance : 0.000
```

```
Total Capacitance : 0.011
```

```
-----
Source
-----
```

```
Pin R Driver Load Slew
-----
```

Name	Dir	Cell	Cap	Model	Model	Ctotal	Rise	Fall	Phase
------	-----	------	-----	-------	-------	--------	------	------	-------

u5/Q	OUT	DFE	0.000	Unknown	CTOTAL	1.000	0.003	0.003	CLK_W_4_gen2 (D) (N)
------	-----	-----	-------	---------	--------	-------	-------	-------	----------------------

u5/Q	OUT	DFE	0.000	Unknown	CTOTAL	1.000	0.003	0.003	CLK_W_4_gen1_inv (D) (P)
------	-----	-----	-------	---------	--------	-------	-------	-------	--------------------------

```
-----
Sinks for pin u5/Q
-----
```

```
Pin R Delay Slew
-----
```

Name	Dir	Cell	Cap	Parameters	Rise	Fall	Rise	Fall	Phase
------	-----	------	-----	------------	------	------	------	------	-------

u7/A	IN	BUF	0.011	-	0.000	0.000	0.003	0.003	CLK_W_4_gen2 (D) (N)
------	----	-----	-------	---	-------	-------	-------	-------	----------------------

u7/A	IN	BUF	0.011	-	0.000	0.000	0.003	0.003	CLK_W_4_gen1_inv (D) (P)
------	----	-----	-------	---	-------	-------	-------	-------	--------------------------

- The following command reports the net to which the pin `interesting_pin` is connected:

```
tempus > report_net -pin interesting_pin
```

- The following command reports all the nets in the current module:

```
tempus > report_net -net *
```

- The following command reports all nets with a fanout greater than 16:

```
tempus > report_net -net * -min_fanout 16
```

Related Commands

- [report_path_exceptions](#)
- [report_timing](#)

report_parasitic_coordinate_transforms

report_parasitic_coordinate_transforms

[-help]

[-user]

Reports the transformations applied to the hierarchical SPEF instances to translate the hierarchical locations given in the SPEF files to the flat locations for the design.

Parameters

-help	Prints a brief description that includes type and default information for each report_parasitic_coordinate_transforms parameter. For a detailed description of the command and all of its parameters, use the man command: man report_parasitic_coordinate_transforms.
-user	Reports user-specified transforms.

Examples

```
#Final origin and orientation in top-level from merging of hierarchical parasitic data
#
#cell      x_offset  y_offset  orient  x_block_origin  y_block_origin  source  hinst
BLOCK2     66.64     595.69    R0       0                0              Auto    hblock1
BLOCK1     186.85     244.62    R90       0                0              User    hblock2
BLOCK2     559.55     364.455   R270      0                0              Auto    hblock3
BLOCK3     857.2      879.235   MY90      4.24             9.45           Auto    hblock4
BLOCK1     91.77      629.79    MX90      0                3              Auto    hblock1/block1
BLOCK1     62.965     808.145   R180      2.35             0.35           Auto    hblock1/block2
```

Related Commands

- [create_parasitic_coordinate_transform](#)
- [read_parasitics](#)
- [read_spef](#)

report_path_exceptions

```
report_path_exceptions  
[-help]  
[-all]  
[-early | -late]  
[ -ignored]  
[-view view_name ]  
[{> | >>} filename | -tcl_list]
```

Generates a report about path exceptions specified using the `set_false_path`, `set_min_delay`, `set_max_delay`, and `set_multicycle_path` commands.

By default, the `report_path_exceptions` command only reports activated path exceptions. To report only the path exceptions that have been ignored by timing analysis, specify the `-ignored` parameter.

If more than one path exception is applied to a timing path, multiple path exceptions that match a given path are prioritized. The adjustment on the path honors the path exception with the highest priority.

For more information on path exception priorities, refer to the "Path Exception Priorities" section in the Timing Analysis Commands chapter.

When the software is in multi-mode multi-corner timing analysis mode, the report generates path exception information for every active analysis view.

The `report_path_exceptions` command output also includes 2 columns that show the file name and line number. The report output shows the exact file name and line number of the applied/ignored constraints. The compressed constraints will be decompressed first and then shown as individual constraints as written in the SDC file.

Parameters

<code>-help</code>	Prints out command usage.
<code>-all</code>	Reports all the applied and ignored path exceptions.
<code>{> >>} filename</code>	When specified with <code>></code>, writes the report to the specified file name. If the file already exists, the software overwrites it. When specified with <code>>></code>, writes the report to the specified file name. If the file already exists, the software concatenates the report to the end of the file. The <code>filename</code> parameter must be the last argument in the list. The <code>filename</code> and <code>-tcl_list</code> parameters are mutually exclusive; you cannot specify them together. Note: To write a compressed report, add the <code>.gz</code> extension to the file name. <i>Default:</i> Report is displayed on standard output without being saved.
<code>-early -late</code>	Generates the path exception report for either early paths or late paths. <i>Default:</i> Generates a report based on the current timing analysis mode.
<code>-ignored</code>	Reports only the path exceptions that have been ignored by timing analysis. This parameter can be used for debugging constraints that may be ineffective or incorrectly entered. A path exception may be ineffective if it is covered by, or overridden by, another path exception. For example: <pre>set_false_path -to OUT set_multicycle_path -to OUT 2</pre> In the above scenario, the <code>set_multicycle_path</code> command will be ignored as false paths have a higher priority over cycle addition on the same timing paths.
<code>-tcl_list</code>	Produces the report in Tcl list format instead of a tabular format. This is useful for integrating timing with custom Tcl functions, and also for customizing report generation. The <code>-tcl_list</code> and <code>filename</code> parameters are mutually exclusive; you cannot specify them together.
<code>-view view_name</code>	Generates a path exception report for the specified analysis view only. You can specify this parameter only when the software is in multi-mode multi-corner timing analysis mode.

Examples

Here are some examples.

- The following commands show that a false path has priority over cycle addition on the same path:

```
tempus > set_false_path -from seg1/u3/q -to seg1/u9/d
tempus > set_multicycle_path -from seg1/u3/Q -to seg1/u9/D 2
```

```
tempus > report_path_exceptions -ignored
```

```
-----
From          To          Early          Late
-----
seg1/u3/Q     seg1/u9/D     false ignored  false ignored
seg1/u3/Q     seg1/u9/D     -              add 1 ignored
seg1/u3/Q     seg1/u9/D     add 0 ignored  -
-----
```

The following command shows how edges are handled:

```
tempus > set_false_path -rise_from {IN1} -through {OUT1}
tempus > set_false_path -rise_from {IN1 IN2} -fall_to {OUT1}
tempus > set_false_path -rise_from {IN1}
tempus > report_path_exceptions -ignored
```

```
-----
From          Through      To          Early          Late
-----
^ IN1 IN2     -              v OUT1     false ignored  false ignored
^ IN1         OUT1      -          false ignored  false ignored
-----
```

```
tempus > report_path_exceptions -early
```

```
-----
From          Through      To          Early
-----
^ IN1         -              -          false
-----
```

```
tempus > report_path_exceptions -late
```

```
-----
From          Through      To          Late
-----
^ IN1         -              -          false
-----
```

When the software is in multi-mode multi-corner timing analysis mode, the software generates path exception information for every active analysis view:

```
tempus > set_analysis_view -setup {setup1 setup2} -hold {hold1 hold2}
tempus > report_path_exceptions
```

```
-----
From          Early    Late    View
              Name
-----
```

```
test_mode reset    false      setup2
test_mode reset    false      setup1
test_mode reset           false hold1
test_mode reset           false hold2
-----
```

Specify the `-view` parameter to generate a report only for a specific active analysis view:

```
tempus > report_path_exceptions -view setup1
```

```
-----
From      Early    Late      View
              Name
-----
test_mode  reset    false    setup1
-----
```

Related Commands

- [report_timing](#)

report_path_groups

```
report_path_groups
[-help]
[-name group_name]
[> | >>] filename | -tcl_list]
```

Lists the names of all path groups with the corresponding paths.

Parameters

-help	Prints out the command usage.
{> >>} filename	When specified with >, writes the report to the specified file name. If the file already exists, the software overwrites it. When specified with >>, writes the report to the specified file name. If the file already exists, the software concatenates the report to the end of the file. The <i>filename</i> parameter must be the last argument in the list. The <i>filename</i> and -tcl_list parameters are mutually exclusive; you cannot specify them together. Note: To write a compressed report, add the .gz extension to the file name. <i>Default:</i> Report is displayed on standard output without being saved.
-name group_name	Limits the report to the specified path group. <i>Default:</i> All path groups are reported.
-tcl_list	Produces the report in Tcl list format instead of a tabular format. This is useful for integrating timing with custom Tcl functions, and also for customizing report generation. The -tcl_list and <i>filename</i> parameters are mutually exclusive; you cannot specify them together.

Examples

The following command shows a sample report:

```
tempus> report_path_groups
-----
From      To          Group Name
-----
CLK1      seg1/u3/D    path1
-----
```

Related Commands

- `set_clock_groups`
- `reset_clock_groups`
- `reset_path_group`
- `get_path_groups`
- `group_path`

report_pba_aocv_derate

```
report_pba_aocv_derate
[-help]
[-aocv_derate_only]
[-delay_corner delayCornerName]
[-derate_mode {worst_slack | worst_derate}]
[-early]
[-include_clock_derate]
[-late]
[-max_slack val]
[-max_paths integer | -max_points integer]
[-nworst worst_entries]
[-retime {aocv | aocv_path_slew_propagation}]
[-retime_mode {path | exhaustive}]
[> filename [.gz]]
[>> filename [.gz]]
```

Reports path-based AOCV derates.

Note: The `report_pba_aocv_derate` command is supported only in AOCV mode. You must use the following setting to enable AOCV mode:

```
set_analysis_mode -aocv true
```


Parameters

<code>-help</code>	Prints out the command usage.
<code>-aocv_derate_only</code>	Allows AOCV reporting only.
<code>-delay_corner</code> <code>delayCornerName</code>	Specifies the name of the delay corner.
<code>-derate_mode {worst_slack worst_derate}</code>	Specifies the derating mode. <i>Default:</i> <code>worst_derate</code>
<code>-early</code>	Specifies derates for early paths.
<code>-include_clock_derate</code>	Includes clock derate.
<code>-late</code>	Specifies the derates for late paths.
<code>-max_paths integer</code>	Reports derates based on the specified number of paths.
<code>-max_points integer</code>	Reports derates based on the specified number of endpoints.
<code>-max_slack val</code>	Reports the upper bound value of the reported slack.
<code>-nworst worst_entries</code>	Reports the derates based on the specified number of worst paths per end point.
<code>-retime {aocv aocv_path_slew_propagation}</code>	Reanalyzes the specified set of paths using the specified method. <i>Default:</i> <code>aocv</code>
<code>-retime_mode {path exhaustive}</code>	Specifies the retime mode to be either <code>path</code> or <code>exhaustive</code> .
<code>> filename[.gz]</code>	Redirects output to the specified file.
<code>>> filename[.gz]</code>	Redirects output to the specified file.

Related Commands

- `set_analysis_mode`
- `report_aocv_derate`

report_pg_net_group

```
report_pg_net_group  
[-help]  
[-group <name>]
```

Reports the output of the `create_pg_net_group` command settings. The output shows the PG net groups created in the design and the PG nets associated with them.

Parameters

<code>-help</code>	Prints out the command usage
<code>-group <name></code>	Specifies a list of power nets associated with the specified group.

Examples

- The following command shows the report output of the previously created group PG1:

```
tempus > report_pg_net_group
```

```
create_pg_net_group -name PG1 -pg_nets {pgn1 pgn2 pgn3}
```

- The following command shows the PG nets associated with the PG net group PG1:

```
tempus > report_pg_net_group -group PG1
```

```
pgn1 pgn2 pgn3
```

Related Commands

- `set_ipd_rail_voltage`
- `create_pg_net_group`
- `create_ipd_configuration`
- `report_ipd_configuration`
- `write_ipd_scope`
- `read_ipd_scope`
- `report_ipd_crossings`

Related Information

- For more information, refer to the [Inter Power Domain \(IPD\) Analysis](#) chapter in the *Tempus User Guide*.

report_ports

```
report_ports
[-help]
[-type {[input] | [source_insertion] | [insertion] | [clock_root]
| [uncertainty] | [arrival] | [required] | [external] | [clk_arrival]
| [port_cap] | [fanout_load] | [fanout_load_limit] | [drive_resistance]
| [drive_cell] | [slew_time] | [slew_limit] | [constant] | external_detail
| drive_resistance_detail}]
[-include_pins]
[-pins port_name_list]
[-view viewName]
[> filename | -tcl_list]
```

Reports timing constraints on ports. By default, the command reports all timing constraints on all ports of the current module.

The direction (*DIR*) column of the generated report can contain one of the following abbreviations:

- **IN** indicates that the pin is an input pin.
- **OUT** indicates that the pin is an output pin.
- **INTL** indicates that the pin is an internal pin (a pin inside the instance boundary).

The *Clock Name* column of the report can contain the following symbols:

- **D** indicates it is a data signal in the clock domain.
- **C** indicates it is a clock signal in the clock domain.
- **P** indicates it is a positive phase.
- **N** indicates it is a negative phase.
- **Asterisk (*)** indicates there is a timing exception associated with the signal.

Note: Currently, it is not possible to expand the asterisk to determine which exception is associated with the reported timing phase.

Parameters

-help	Prints out the command usage.
-------	-------------------------------

<code>> filename</code>	Writes the report to the specified file name. If the file already exists, the software overwrites it. The <code>filename</code> parameter must be the last argument in the list. The <code>filename</code> and <code>-tcl_list</code> parameters are mutually exclusive; you cannot specify them together. Note: To write a compressed report, add the <code>.gz</code> extension to the file name. <i>Default:</i> Report is displayed on standard output without being saved.
<code>-include_pins</code>	Reports timing constraints on instance pins, as well as on ports. <i>Default:</i> Reports timing constraints on ports only.
<code>-pins port_name_list</code>	Specifies the pins or ports to be reported. Pins can be specified by name or by object ID in a Tcl list.
<code>-tcl_list</code>	Produces the report in Tcl list format instead of a tabular format. This is useful for integrating timing with custom Tcl functions, and also for customizing report generation. The <code>-tcl_list</code> and <code>filename</code> parameters are mutually exclusive; you cannot specify them together.
<code>-type {assertion_type assertion_type_list}</code>	Specifies the constraints to be reported. <i>Default:</i> All constraints are reported. The following are valid <code>assertion_type</code> values: • <code>arrival</code> Reports arrival time constraints as set by the <code>set_input_delay</code> command. • <code>clk_arrival</code> Reports clock arrival time constraints as set by the <code>set_clock_latency</code> command. • <code>clock_root</code> Reports the clock root as set by the <code>create_clock</code> command. • <code>constant</code> Reports constant constraints as set by the <code>set_case_analysis</code> command. • <code>drive_cell</code> Reports drive cell constraints as set by the <code>set_driving_cell</code> command.

- `drive_resistance`
Reports drive resistance constraints as set by the `set_drive` command.
- `drive_resistance_detail`
Reports detail about the input ports with drive resistance constraints.
- `external`
Reports external delay constraints as set by the `set_output_delay` command.
- `external_detail`
Reports detail about the output ports with external delay constraints.
- `fanout_load`
Reports fanout load constraints as set by the `set_fanout_load` command.
- `fanout_load_limit`
Reports fanout load limit constraints as set by the `set_max_fanout` command.
- `input`
Reports the input delay constraints as set by the `set_input_delay` command.
- `insertion`
Reports the network insertion delay as set by the `set_clock_latency` command.

	• <code>port_cap</code> Reports port capacitance constraints as set by the <code>set_load</code> command. • <code>required</code> Reports required time constraints as set by the <code>set_output_delay</code> command. • <code>slew_limit</code> Reports slew time limit constraints as set by the <code>set_max_transition</code> command. • <code>slew_time</code> Reports slew time constraints as set by the <code>set_input_transition</code> command. • <code>source_insertion</code> Reports the source insertion delay as set by the <code>set_clock_latency - source</code> command. • <code>uncertainty</code> Reports the clock uncertainty as set by the <code>set_clock_uncertainty</code> command.
<code>-view viewName</code>	Specifies the view name.

Examples

Here are some examples.

- The following command shows a port report, as shown in the following example:

```
tempus > report_ports
```

```
-----
Early Late
-----
Pin      Dir      Assertion      Clock Name      Rise      Fall      Rise      Fall
Name
-----
CLK1     IN       clock_root      CK2 (C) (P)
CLK2     IN       clock_root      CLK2 (C) (P)
CLK3     IN       clock_root      CLK3 (C) (P)
IN1      IN       drive_resistance * (D) (P)      2.000     2.000     2.000     2.000
IN2      IN       drive_resistance * (D) (P)      2.000     2.000     2.000     2.000
IN3      IN       drive_cell      * (D) (P)      INV       INV       INV       INV
```

Tempus Text Command Reference

Timing Analysis Commands

```
OUT1  OUT      external          GCLK1 (C) (P)  1.000  1.000  1.000  1.000
OUT2  OUT
OUT3  OUT derived_clock GCLK4 (C) (P)
```

```
-----
Pin   Dir   Assertion                      Value
Name
-----
CLK3  IN    slew_limit                        ( 0.900 : )
CLK3  IN    port_cap_limit                   ( 0.020 : 0.900 )
CLK3  IN    fanout_load_limit              5.000
CLK2  IN    slew_limit                        ( 0.900 : )
CLK2  IN    port_cap_limit                   ( 0.020 : 0.900 )
CLK2  IN    fanout_load_limit              5.000
CLK1  IN    slew_limit                        ( 0.900 : )
CLK1  IN    port_cap_limit                   ( 0.020 : 0.900 )
CLK1  IN    fanout_load_limit              5.000
IN3   IN    slew_limit                        ( 0.900 : )
IN3   IN    port_cap                      ( 1.500 : : 2.600 )
IN3   IN    port_wire_cap                 ( 1.200 : : 3.000 )
IN3   IN    port_cap_limit                   ( 0.020 : 0.900 )
IN3   IN    fanout_load_limit              5.000

IN2   IN    slew_limit                        ( 0.900 : )
IN2   IN    port_cap                      ( 1.500 : : 2.600 )
IN2   IN    port_wire_cap                 ( 1.200 : : 3.000 )
IN2   IN    port_cap_limit                   ( 0.020 : 0.900 )
IN2   IN    fanout_load_limit              5.000
IN1   IN    slew_limit                        ( 0.900 : )
IN1   IN    port_cap                      ( 1.500 : : 2.600 )
IN1   IN    port_wire_cap                 ( 1.200 : : 3.000 )
IN1   IN    port_cap_limit                   ( 0.020 : 0.900 )
IN1   IN    fanout_load_limit              5.000
OUT3  OUT    slew_limit                        ( 0.900 : )
OUT3  OUT    port_cap_limit                   ( 0.020 : 0.900 )
OUT2  OUT    slew_limit                        ( 0.900 : )
OUT2  OUT    port_cap_limit                   ( 0.020 : 0.900 )
OUT1  OUT    slew_limit                        ( 0.900 : )
OUT1  OUT    port_cap_limit                   ( 0.020 : 0.900 )
```

- The following command shows `source_insertion` and `uncertainty` assertion type options as shown in the following example:

```
tempus > report_ports -type clock_root source_insertion input -pins clk in
tempus > report_ports -type {uncertainty source_insertion} -pins CLK1
```

```
-----
Early Late
-----
```


Tempus Text Command Reference

Timing Analysis Commands

Pin Name	Dir	Assertion	Clock Name	Rise	Fall	Rise	Fall
CLK1	IN	source_insertion	*(C) (P)	-0.500	-0.500	-0.500	-0.500

Pin Name	Dir	Assertion	Value
CLK1	IN	uncertainty	0.400 (T) 0.400 (T) 0.400 (T) 0.400 (T)

- The following command shows `clock_root` and `uncertainty` assertion type options as shown in the following example:

```
tempus > report_ports -type {input uncertainty clock_root}
```

Early Late							

Pin	Dir	Assertion	Clock Name	Rise	Fall	Rise	Fall
Name							

CLK1	IN	clock_root	CLK_W_1 (C) (P)				
in1	IN	input	@ (D) (P)	1.000	1.000	1.000	1.000

Pin	Dir	Assertion	Value				
Name							

CLK1	IN	uncertainty	0.400 (T) 0.400 (T) 0.400 (T) 0.400 (T)				

Related Commands

- [report_clocks](#)
- [report_net](#)
- [report_timing](#)

report_reference

report_reference

[-help]

[-flat]

[-nosplit]

[-view]

[> *file_name*]

[>> *file_name*]

Reports reference cell usage statistics.

The command reports the unit area, library name, and total count of the reference cells. By default, the report is sorted in order of number of instances.

Parameters

-help	Prints out the command usage.
-flat	Disables printing of hierarchical instances.
-nosplit	Generates the report in no split mode.
-view	Generates the report for the specified analysis view.
> <i>file_name</i>	Specifies the name of the write mode file name.
>> <i>file_name</i>	Specifies the name of the append mode file name.

Examples

- The following report output provides reference cell details:

```
tempus> report_reference
```

```
Design : top
```

```
Attributes
```

```
s - standard cell
```

```
h - hierarchical
```

```
m - macro
```

```
b - blackbox
```

Reference	Library	Unit Area	Count	Total Area	Attributes
PCORNERDG	cell_w	55225.000	4	220900.000	s
PCORNERDG	cell_w	55225.000	4	220900.000	s
PDIDGZ	cell_w	9400.000	28	263200.000	s
PDIDGZ	cell_w	9400.000	28	263200.000	s

report_settings

report_settings
-help
pattern

Reports all the timing and signal integrity global variables and mode settings based on the specified pattern. The command accepts wildcard characters.

Parameters

-help	Prints out the command usage.
pattern	Specifies the pattern to which the global variables and mode settings are to be matched.

Examples

- The following command reports all the mode settings, such as design mode, analysis mode:

```
tempus> report_settings *cpr*
```

# Application Globals	User Setting	Default Value

timing_cpr_remove_clock_to_data_crp	false	false
timing_cpr_self_loop_mode	true	true
timing_cpr_skip_clock_reconvergence	false	false
timing_cpr_threshold_ps	20	20
timing_cpr_transition_sense	normal	normal
timing_enable_pessimistic_cpr_for_reconvergent_clock_paths	false	false
timing_enable_si_cpr	true	true

report_slack_histogram

```
report_slack_histogram
[-help]
[-compact]
[-early]
[-infile string]
[-late]
[-max_slack float]
[-min_slack float]
[-outfile string]
[-plus_non_violating]
[-retime {aocv | ssta | path_slew_propagation | aocv_path_slew_propagation}]
[-retime_mode {path | exhaustive}]
[-step float]
[-target_slack float]
[-view string]
```

Generates a histogram of end-point slacks.

The `report_slack_histogram` command can be used in path-based analysis (PBA) mode.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-compact</code>	Specifies the compact column report format. This format removes any histogram rows that have a count of 0.
<code>-infile string</code>	Specifies the input filename to be used for generating the histogram. When the <code>-infile</code> parameter is specified, the <code>report_slack_histogram</code> command generates reports using data from the specified file. <i>Default:</i> The software will generate this data with the default <code>max_paths</code> limit.
<code>-early</code>	Specifies early mode analysis for computing the histogram.
<code>-late</code>	Specifies late mode analysis for computing the histogram.
<code>-max_slack float</code>	Specifies the upper slack limit for the generated report. <i>Default:</i> 0

<code>-min_slack float</code>	Specifies the lower slack limit for the generated report. <i>Default:</i> Worst negative slack
<code>-outfile string</code>	Specifies the name of the output file. <i>Default:</i> Displays the report on the Tempus console.
<code>-plus_non_violating</code>	Specifies to include the non-violating slack values.
<code>-retime {aocv ssta path_slew_propagation aocv_path_slew_propagation}</code>	Re-analyzes the specified set of paths using the specified method.
<code>-retime_mode {path exhaustive}</code>	Re-analyzes the specified set of paths using the specified method.
<code>-step float</code>	Specifies the precision for the slack histogram. <i>Default:</i> 0.5ns
<code>-target_slack float</code>	When specified, the slack histogram is centered around the defined target slack value.
<code>-view string</code>	Writes out slack histograms for the specified view. If you do not specify this option, the command outputs details for all the views.

Examples

- The following command reports the lower slack limits along with the non-violating slack values:

```
report_slack_histogram -compact -min_slack -10 -plus_non_violating
```

```
-----
```

Slack Range (ns)	Count	Sum
+-----+-----+-----+		
(-9.949 ~ -9.449]	15	15
(-9.449 ~ -8.949]	12	27
(-8.949 ~ -8.449]	18	45
(-8.449 ~ -7.949]	16	61

```
-----
```

- The following command reports the upper slack limits for the specified view:

```
report_slack_histogram -max_slack -20 -step 0.5 -view func_fast_RCMAx
```

```
-----
```

Slack Range (ns)	Count	Sum
+-----+-----+-----+		
(-21.182 ~ -21.000]	7	7
(-21.000 ~ -20.500]	2	9
(-20.500 ~ -20.000]	5	14

```
-----
```

- The following command shows a report for the specified target slack value:

```
report_slack_histogram -target_slack -21
```

```
-----
```

Slack Range (ns)	Count	Sum
+-----+-----+-----+		
(-22.239 ~ -22.000]	7	7
(-22.000 ~ -21.500]	2	9
(-21.500 ~ -21.000]	5	14
(-21.000 ~ -20.500]	2	16
(-20.500 ~ -20.000]	3	19

```
-----
```

Related Commands

- `write_global_slack_report`

report_socv_interconnect_variation

```
report_socv_interconnect_variation  
[-help]  
[-delay_corner string]  
[-power_domain string]
```

Reports the RC variation multiplier data provided in the SOCV files.

Parameters

-help	Prints out the command usage.
-delay_corner <i>string</i>	Specifies the name of the delay corner.
-power_domain <i>string</i>	Specifies the name of the power domain.

report_socv_library

```
report_socv_library
[-help]
[-categories string]
[-coverage_verbose]
[-list_not_annotated]
[ -libset string | { -delay_corner string [ -power_domain string] } | { -view string [ -power_domain string] } ] [ -libs string | -lib_cells string]
[-coverage]
```

Reports the coverage of the SOCV library(s) loaded in the design.

Parameters

-help	Prints out the command usage.
-categories <i>string</i>	Allows you to control the types of table data to be used to check delay arcs, transition arcs, and check arcs. <i>Default:</i> all
-coverage	Reports coverage library data.
-coverage_verbose	Generates a detailed report table for each cell.
-delay_corner <i>string</i>	Checks all the libraries defined in the specified delay corner.
-lib_cells <i>string</i>	Specifies library cells. The allowed format is \u201clibrary/cell\u201d name or pattern or collection. This is typically used for debugging purposes.
-libs <i>string</i>	Specifies the name of the library. If the -library_set or -delay_corner parameter is not specified, then all libraries matching the library name across the MMMC configuration will get checked.
-libset <i>string</i>	Checks all the libraries defined in the specified library set.
-list_not_annotated	Control the types of table data to be used to check delay arcs, transition arcs, and check arcs. <i>Default:</i> all
-power_domain <i>string</i>	Checks and reports the specified power domain.

-view *string*

Checks all the delay corners associated with a view.

report_statistical_timing_derate_factors

```
report_statistical_timing_derate_factors
[-help]
[-derate_cell]
[-derate_cell_check]
[-derate_net]
[-max_slack <float>]
[-path_based_ssta]
[-path_group <groupname_list>]
[-ssta_view <ssta_view_name>]
[-sta_view <sta_view_name>]
[-type {arrival_time_based | slack_time_based}]
[-early | -late]
[-clock_factor | -data_factor]
[-max_paths <integer> | -max_points <integer>]
```

Generates design specific OCV factor. The scaling factor (hold or setup) is computed either by using the arrival time or the slack time approach.

Parameters

-help	Prints out the command usage.
-clock_factor -data_factor	Specifies if the user-defined factor value is for clock or data. If you specify the -clock_factor parameter, the software calculates the -data_factor parameter, and vice versa. These parameters are applicable to the type, slack_time_based. Note: The -clock_factor parameter and the -data_factor parameter are mutually exclusive. You cannot specify them together.
-derate_cell	Applies user-specified scaling factors on cell delays.
-derate_cell_check	Applies user-specified scaling factors on timing checks.
-derate_net	Applies user-specified scaling factors on net delays.
-early -late	Specifies early or late mode analysis. <i>Default:</i> late.
-max_paths integer	Reports the specified number of worst paths in the design, regardless of the endpoint.

<code>-max_points</code> <i>integer</i>	Reports the worst path that is found to each endpoint up to the number specified by the <code>-max_points</code> option.
<code>-max_slack</code> <i>float</i>	Reports only those paths with slack equal to or less than the specified float value.
<code>-path_based_ssta</code>	Enables the generation of the statistical OCV factor using path-based SSTA analysis.
<code>-path_group</code> <i>groupname_list</i>	Considers only the paths contained in the groups in the specified <i>groupname_list</i> for computing the statistical timing derate factor.
<code>-ssta_view</code> <i>ssta_view_name</i>	Specifies the view for the SSTA run.
<code>-sta_view</code> <i>sta_view_name</i>	Specifies the view for the STA run.
<code>-type</code> <i>arrival_time_based</i> <i>slack_time_based</i>	Specifies if the OCV factor needs to be generated using the arrival time or the slack time approach. In the arrival time approach, the SSTA arrival time values are used to compute the statistical OCV factor for clock and data. In the slack time approach, the STA and SSTA slacks are used to generate the statistical OCV factor for either clock or data, depending on whether you specify the factor for data or clock. Default: <code>arrival_time_based</code> You can use the <code>-type arrival_time_based</code> parameter with the following parameters only: • <code>-late</code> • <code>-early</code> • <code>-path_based_ssta</code>

You can use the `-type slack_time_based` parameter with the following parameters only:

- `-ssta_view`
- `-sta_view`
- `-late`
- `-early`
- `-clock_factor`
- `-data_factor`
- `-derate_cell`
- `-derate_net`
- `-derate_cell_check`
- `-max_paths`
- `-max_slack`

Examples

Here are some examples.

- The following example computes the factors for setup analysis using the arrival time-based approach:

```
tempus> report_statistical_timing_derate_factors -type arrival_time_based -late
```

- The following example computes the data factor for setup analysis using the slack time-based approach, applying a user defined clock factor of 0.95:

```
tempus> report_statistical_timing_derate_factors -type slack_time_based -ssta_view  
viewname -sta_view viewname -clock_factor 0.95
```

- The following example computes the clock factor for hold analysis using the slack time-based approach, applying a user defined data factor of 1.05:

```
tempus> report_statistical_timing_derate_factors -type slack_time_based -ssta_view  
viewname -sta_view viewname -data_factor 1.05 -early
```

report_timing

```
report_timing
[-help]
[-analysis_summary_csv_extended_file <string>]
[-analysis_summary_csv_file <string>]
[-analysis_summary_file <string>]
[-check_clocks]
[-check_type {setup | hold | clock_gating_setup | clock_gating_hold |
clock_gating_pulse_width | data_setup data_hold | recovery | removal | pulse_width |
clock_period clock_separation | skew | no_change_setup | no_change_hold | max_delay |
min_delay}]
[-debug {unconstrained | time_borrow | cpr_point | constraint_arc | context | wire_variation
| vt_skew}]
[-delay_limit float]
[-derate_summary]
[-gui]
[-format column_list]
[-hpin]
[-max_slack float]
[-min_slack float]
[-net]
[-output_format {text | csv | gtd | binary}]
[-path_exceptions {applied | ignored | all}]
[-path_group groupname_list]
[-path_type {end | summary | full | full_clock | end_slack_only | summary_slack_only}]
[-retime {aocv | path_slew_propagation | aocv_path_slew_propagation}]
[-retime_delaycal_pins <pin_name>]
[-retime_format {manual | retime_compare | retime_replace}]
[-retime_mode {path | exhaustive}]
[-skip_io_paths]
[-unique_pins]
[-view viewName]
[-worst_rc_corner]
[> <filename[.gz]>]
[>> <filename[.gz]>]
[-late | -early]
[-rise | -fall]
[-begin_end_pair | [[ -max_paths integer]
[-nworst integer]]]
[-unconstrained | -point_to_point]
[-collection | -machine_readable | -tcl_list]
[-not_through object_list | -not_rise_through object_list | -not_fall_through object_list]
[[-from object_list | -from_rise object_list | -from_fall object_list]
[-clock_from clk_signame_list [-edge_from {lead trail}]]]
```

```
[-through object_list | -through_rise object_list | -through_fall object_list]  
[-to object_list | -to_rise object_list | -to_fall object_list]  
[-clock_to clk_signame_list [-edge_to {lead trail}]]]
```

Generates a timing report that provides information about various paths in a design.

The reports contain data on delays along the entire path. The start and end nodes of each path are identified.

The timing report contains the following information:

- The slack times of the arriving signal at the end node
- The start node
- The associated transitions
- The signal required times and the actual signal arrival times
- Summary of propagated vs ideal status of launching and capturing clocks
- Any phase shifts applied when evaluating timing checks
- Any CPPR values applied to timing check evaluation

The `report_timing` command output accommodates both the instance pin names and net names in the same column. Since both instance pin names and net names can be long, to save screen space, alternating instance pin and net names (depending on whether the row displays gate delay or net delay) are reported in a single column.

When the `timing_report_timing_header_detail_info` timing global variable is set to `extended`, the software generates reports with the following header details:

- Analysis mode - best-case worst-case (BCWC) or on-chip variation (OCV)
- Active views (displayed only in MMMC mode)
- Clock reconvergence pessimism (CPPR)
- Latch slack mode
- Specified clock propagation value
- Delay calculation
- Specified delay calculation engine

When the `timing_report_group_based_mode` global variable is set to `true`, the paths are grouped by clock domain. In this mode, the `report_timing` command will show the following four types of groups:

- Clock gating group: This is an internal group that reports the clock gating check type paths.
- Asynchronous group: This is also an internal group that reports the asynchronous check type paths.
- Groups for all clocks in a design: These are internal groups that are classified on the basis of each

reference clock. The paths belonging to each reference clock are reported separately under each clock name's group. These groups do not honor the clock gating/asynchronous check types. Suppose there are three clocks, C1, C2, and C3, in a design, then three groups will be reported for each of these clocks. The endpoint sharing the same reference clock will be part of the same group. In case the endpoint belongs to multiple groups, then it will be reported separately in each group.

- Default group: This is an internal group that reports combinational maximum or minimum delay paths.
- User-defined groups: These are user-defined path groups that can be created by users with single or multiple constraints. These groups honor all the check types and are reported by default.

By default, the `timing_report_group_based_mode` global variable is set to `false`.

Note: This global variable does not honor user-defined path groups, so the paths will still be grouped by different clocks.

In this mode, the following commands will have the same behavior and can be used interchangeably:

```
report_timing -path_group Clock_Name
```

```
report_timing -clock_to Clock_Name
```

The `report_timing` command supports multi-threaded reporting. To enable multi-threaded reporting, you can use the following commands:

```
set_multi_cpu_usage -localCpu number_of_CPU
```

```
set_table_style -no_frame_fix_width [-nosplit]
```

Note: When the `timing_allow_input_delay_on_clock_source` global variable is set to `true` and the `set_input_delay` command is applied to a clock input, the specified delay value will be considered as the source clock latency for clock paths and as regular data path arrival time for clock source paths.

You can use a one-step method for generating merged MMMC view reports from a DMMMC master shell. After running the `distribute_views` command, you can use the `report_timing` command to generate reports collectively from all the active distributed views (that is, views available for interactive DMMMC analysis). The merged output report is arranged as per slack/criticality order across active views.

When the `timing_enable_wire_variation` global variable is set to `true`, the timing reports show wire variation details in the below columns:

- `retime_slew_wire_stddev`
- `retime_delay_wire_stddev`
- `arrival_wire_stddev`

For more details on wire variation, refer to the "Wire Variation Analysis" chapter in the *Tempus User Guide*.

- [Parameters](#)
- [Group Path Reporting for User-Defined Groups](#)
- [Use of Informational Flags](#)
- [Retime Analysis Report Header Flags](#)
- [Report Timing Column List Options](#)
- [Reporting Timing Checks Between Various Types of Clocks](#)
- [report_timing -from latch/D Settings](#)
- [Examples](#)
- [Related Commands](#)

Parameters

<code>-help</code>	Outputs a brief description that includes type and default information for each <code>report_timing</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man report_timing</code> .
<code>{> >>} filename</code>	When specified with <code>></code>, writes the report to the specified file name. If the file already exists, the software will overwrite it. When specified with <code>>></code>, writes the report to the specified file name. If the file already exists, the software will concatenate the report at the end of the file. The <code>filename</code> parameter must be the last argument in the list. The <code>filename</code> and <code>-tcl_list</code> parameters are mutually exclusive; you cannot specify them together. Note: To write a compressed report, add the <code>.gz</code> extension to the file name. Note: The last argument specified with the <code>report_timing</code> command is not considered a filename. <i>Default:</i> Report is displayed on standard output (not saved).
<code>-analysis_summary_csv_extended_file <string></code>	Generates the CSV file to be used for merging. <i>Data_type:</i> string, optional
<code>-analysis_summary_csv_file <string></code>	Generates the analysis summary CSV file. <i>Data_type:</i> string, optional

<code>-analysis_summary_file <string></code>	Generates the analysis summary file. <i>Data_type</i>: string, optional
<code>-begin_end_pair</code>	Reports all the violating paths between unique source and target register pairs. To report other paths, you can specify this parameter with the <code>-max_slack</code> parameter. The paths are not sorted by slack - paths with the same source pins are reported adjacent to each other. When specified with the <code>-from</code>, <code>-through</code>, or <code>-to</code> parameter, the software only reports the unique register pair paths that match the path description. This parameter reports the paths that are grouped by <code>path_groups</code>, irrespective of the <code>timing_report_group_based_mode</code> global variable setting. You can use the <code>timing_report_begin_end_pair_max_path_limit</code> global variable to specify the maximum path limit. When the number of maximum paths to be reported by the <code>-begin_end_pair</code> parameter exceeds the specified limit (default is 200000), a warning message will be displayed. Also, some of the paths may not be reported. Note: You cannot specify this parameter with the <code>-max_paths</code>, <code>-nworst</code>, <code>-unique_pins</code>, and <code>-min_slack</code> parameters. <i>Data_type</i>: bool, optional
<code>-check_clocks</code>	Generates reports based on timing paths on the clock network instead of the standard timing to data endpoints. This report includes clock paths that end at the reference end of a check or a clock gating end point. <i>Default:</i> Data paths and all the other clock paths (for example, a clock path ending at a D pin of a register) are reported. <i>Data_type</i>: bool, optional
<code>-check_type check_type</code>	Reports only the paths that end at the specified timing check. You can specify the following options: <code>setup</code>, <code>hold</code>, <code>clock_gating_setup</code>, <code>clock_gating_hold</code>, <code>clock_gating_pulse_width</code>, <code>data_setup</code>, <code>data_hold</code>, <code>recovery</code>, <code>removal</code>, <code>pulse_width</code>, <code>clock_period</code>, <code>clock_separation</code>, <code>skew</code>,

`no_change_setup`, `no_change_hold`, `max_delay`, and `min_delay`.

- The clock style checks are: `pulse_width`, `clock_gating_pulse_width`, `clock_period`, `skew`, and `clock_separation`.
- In setup analysis mode, the reported check types will be: `setup`, `skew`, `pulse_width`, `clock_period`, `clock_gating_setup`, `clock_gating_pulse_width`, `data_setup`, `recovery`, `clock_separation`, and `no_change_setup`.
- In hold analysis mode, the reported check types will be: `hold`, `clock_gating_hold`, `data_hold`, `removal`, and `no_change_hold`.

If the software is in simultaneous setup and hold analysis mode, all of the check types listed above will be reported.

When the `max_delay` and `min_delay` options are specified, the timing report will display details of timing paths for which the `set_max_delay` or `set_min_delay` constraints are defined.

By default, the following checks are reported:

- `setup`
- `hold`
- `clock_gating_setup`
- `clock_gating_hold`
- `data_setup`
- `data_hold`
- `recovery`
- `removal`
- `no_change_setup`
- `no_change_hold`

Do not use this parameter with the `-unconstrained`, `-early`, or `-late` parameters.

Note: The `report_timing` command does not perform skew checks by default.

	<i>Data_type</i> : enum, optional
<code>-clock_from clk_signame_list</code>	Generates reports based on the source clock waveform(s). This parameter reports only those paths whose source clocks are the clock signals in <i>clk_signame_list</i>. <i>Data_type</i>: bool, optional
<code>-clock_to clk_signame_list</code>	Generates reports based on target clock waveform(s). This parameter reports only those paths whose target clocks are the clock signals in <i>clk_signame_list</i>.
<code>-collection</code>	Returns a collection of timing paths, which is useful for performing Tcl queries on selected timing reports. To generate a collection of paths analyzed by Path-Based Analysis (PBA), you can use the <code>-retime</code> parameter. Note: This parameter is not supported in DMMMC mode. <i>Data_type</i>: bool, optional

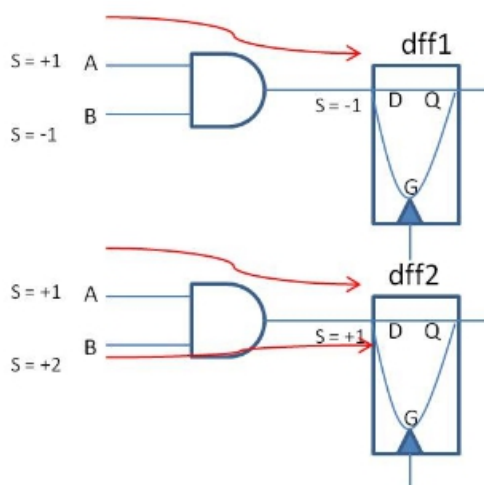
<pre>-debug {unconstrained time_borrow cppr_point constraint_arc context wire_variation vt_skew}</pre>	You can specify the following options for debugging: • unconstrained: Determines the reasons for a path to be unconstrained. The report output will list all the applicable arcs/constraints that could not constrain the paths, along with the reasons. • time_borrow: Reports time borrowing calculations. Time borrowing is the amount of time borrowed by a previous logic. • cppr_point: Reports the CPPR point information for the reported paths. • constraint_arc: Reports check arc details, such as constraint arc delay, constraint pins (along with the related pin slews), skewness, user derate, annotation, and when_cond. • context: Reports the launch and capture path interface timing for each instance and indicates the block port that shows latency in the MIM flow. • wire_variation: Reports the corresponding slack shift details at the end of each timing path report. • vt_skew: Reports voltage threshold (VT) skew analysis details. <i>Data_type:</i> string, optional
<pre>-delay_limit float</pre>	Specifies the path delay limit for unconstrained paths (when the <code>-unconstrained</code> parameter is specified). For early paths (<code>-early</code> parameter), the software reports only those paths with a path delay less than the delay limit. For late paths (<code>-late</code> parameter), only those paths with a path delay more than the delay limit will be reported. Note: The <code>-delay_limit</code> can only be used in conjunction with the <code>-unconstrained</code> parameter. <i>Data_type:</i> float, optional
<pre>-derate_summary</pre>	Generates a summary table for AOCV derating for launch and capture paths. <i>Data_type:</i> bool, optional

<code>-early -late</code>	Reports for early paths (hold check) or late paths (setup check). If you specify a check type, setup, or hold with the <code>set_analysis_mode</code> command, the <code>report_timing</code> command reports the specified check by default. Note: The <code>-early</code> and <code>-late</code> parameters are not mutually exclusive; you can specify them together. <i>Default:</i> <code>-late</code> <i>Data_type:</i> bool, optional
<code>-edge_from {lead trail}</code>	Generates reports based on the source clock edge - leading or trailing. Note: The <code>-edge_from</code> parameter can only be used in conjunction with the <code>-clock_from</code> parameter. <i>Default:</i> Reports both source clock edges <i>Data_type:</i> bool, optional
<code>-edge_to {lead trail}</code>	Generates reports based on the target clock edge, either leading or trailing. Note: The <code>-edge_to</code> parameter can only be used in conjunction with the <code>-clock_to</code> parameter. <i>Default:</i> Reports both the target clock edges <i>Data_type:</i> bool, optional
<code>-format column_list</code>	Formats the report according to the specified <code>column_list</code>. The <code>column_list</code> specifies the columns to be displayed in the timing report and the order in which they appear. The valid format columns are: adjustment, abs_delay_err, abs_slew_err, annotation, aocv_adj_stages, aocv_derate, aocv_weight, arc, arrival, arrival_mean, arrival_sigma, cell, delay, delay_mean, delay_sigma, direction, edge, fanin, fanout, flags, hpin, incr_delay, instance, instance_location, latch_window, load, module, net, pct_delay_err, pct_slew_err, phase, phys_info, pin, pin_load, pin_location, power_domain, required, retime_delay, retime_delay_mean, retime_delay_sigma, retime_mis_delta, retime_mis_derate, retime_slew, retime_slew_mean, retime_slew_sigma, stage_count, slew, slew_mean, slew_sigma, socv_derate, spice_arrival, spice_delay, spice_slew, stolen, timing_point, total_derate,

	user_derate, voltage, vt_skew_derate, when_cond, wire_load, wlmmodel For example: <pre>-format {hpin cell delay required arrival required edge}</pre> Note: • The default net format (with the <code>-net</code> parameter) for the full path will be: <code>{hpin edge net cell delay arrival required}</code> • The <code>required</code> column will not be displayed when the <code>-unconstrained</code> parameter is specified. • The <code>adjustment</code> column is automatically added to the timing report. To disable this feature, you can set the <code>timing_report_generated_clock_info</code> global variable to <code>false</code>. By default, this global variable is set to <code>true</code>. • You can use a combination of the <code>-format</code> and <code>-tcl_list</code> parameters to integrate the timing reports into your Tcl scripts. You can use the <code>-from</code> parameter to limit the number of paths reported and to find specific paths in the design. • The <code>-format</code> option cannot be used with the <code>-path_type end</code> or <code>-path_type summary</code> options. • You can manage the retiming columns with the <code>timing_report_retime_formatting_mode</code> global variable. • Informational tags are displayed for reporting additional details with the <code>flags</code> option. For details on the 'Flags' column, refer to the section on Use of Informational Flags. For more details on the report format options, see the Report Timing Column List Options table. <i>Data_type</i>: string, optional
<code>-gui</code>	Displays the timing report in a GUI window. <i>Data_type</i>: bool, optional

<pre>{-from -from_rise -from_fall} object_list</pre>	Reports paths starting from the object(s) specified in the <i>object_list</i>. The supported objects are pins and ports. Using <i>-from_rise</i> (or <i>-from_fall</i>) specifies that the rising (or falling) edge of the signals on the pins/ports in the <i>object_list</i> are the start of the paths. You can use this option with the <i>-through</i> and <i>-to</i> parameters for specifying particular paths in the design. Note: These parameters do not allow clock objects. You can use the <i>-clock_from</i> and <i>-clock_to</i> parameters to reference clock waveforms. <i>Data_type</i>: string, optional
<pre>-hpin</pre>	Prints all the hierarchical crossings of an arc under the <i>hpin</i> column. <i>Data_type</i>: bool, optional
<pre>-machine_readable</pre>	Generates a detailed timing report in machine-readable format. This report can be used for debugging timing results using the timing debug feature. <i>Data_type</i>: bool, optional
<pre>-max_paths integer</pre>	Reports the specified number of worst paths in the design, with a maximum of <i>nworst</i> (<i>-nworst</i> parameter) paths to any single endpoint (default 1). In case of group-based mode (when <i>timing_report_group_based_mode</i> is set to <i>true</i>), the <i>-max_paths</i> parameter specifies the number of worst paths reported per clock group. <i>Default</i>: 1 <i>Data_type</i>: int, optional

<code>-max_slack float</code>	Reports only those paths with slack less than the specified maximum slack value. The <code>-max_slack</code> parameter reports paths that fall within the specified range. A positive slack value indicates that timing was met, and a negative slack value indicates a timing violation. When the <code>-retime_mode exhaustive</code> option is specified, the <code>-max_slack</code> value is set to 0. The <code>-max_slack</code> parameter cannot be used with the <code>-unconstrained</code> option. <i>Data_type</i>: string, optional
<code>-min_slack float</code>	Reports only those paths whose slack is greater than the specified <code>-min_slack</code> value. You can generate a timing report showing paths with slack greater than the specified slack using the <code>-max_paths</code> parameter. For example, to report all paths with a slack greater than 2ns, you can use the following command: <pre>min_slack 2.0 -max_paths 1000</pre> In this case, the maximum number of paths reported will be 1000. When the <code>-min_slack</code> parameter is specified, the total number of paths processed will be as determined by the <code>-max_paths/-nworst</code> settings, and only those paths satisfying the <code>-min_slack</code> criteria will be reported. The software will not process paths beyond the specified <code>max_paths/nworst</code> limit, even if the <code>min_slack</code> criteria are satisfied. In the diagram below, if <code>-min_slack</code> is 0 and <code>-max_paths 4 -nworst 2</code> is specified, then only 3 paths will be reported; the path with a slack value of -1 will be ignored. Here, even if there is another path with slack that is greater than or equal to 2, then it will not be reported because the worst <code>max_paths</code> have been processed.



The `-min_slack` option cannot be used with the `-unconstrained` option.

Data_type: string, optional

`-net`

Adds a row for the net arc. This parameter also separates the cell delay from the wire delay.

The `report_timing -net` command displays the net delays separate from the cell delays. However, the net delay may sometimes be shown as 0.0. To get the net delays to show up, you can increase the report precision using the `report_precision` global variable.

By default, the net arc is not shown, and the net delay is added to the following delay.

Data_type: bool, optional

<pre>{-not_through -not_rise_through -not_fall_through} object_list</pre>	You can specify one of the following options: • -not_through: Reports paths that do not traverse through the specified nets, ports, or pins of a cell. By default, this parameter will be ignored if an object has already been specified. In this case, the parameter shows the following behavior: ◦ If a collection of all the input ports has been specified (using the <code>all_inputs</code> command), then the report will show paths to the inout part, and paths starting from the internal part of inout will be excluded. ◦ If a collection of all the output ports has been specified (using the <code>all_outputs</code> command), then the report will show paths starting from the internal part of the inout, and paths ending at the inout part will be excluded. ◦ If any other collection (not specified with the <code>all_inputs</code> or <code>all_outputs</code> command) is specified, then paths to both the ports will not be reported. • -not_rise_through: Does not report paths with rise transition. • -not_fall_through: Does not report paths with fall transition. <i>Data_type:</i> string, optional
<pre>-nworst integer</pre>	Specifies the maximum number of paths which can be reported for a particular endpoint (the total number of paths reported is specified by the <code>-max_paths</code> parameter). When the <code>timing_report_group_based_mode</code> global variable is set to <code>true</code> (group-based mode), the <code>-nworst</code> parameter specifies the maximum number of paths for a particular endpoint per clock group. You can use the <code>-nworst</code> parameter to report all the checks at an endpoint (or use <code>-check_type</code>) to report a specific check. <i>Default:</i> 1 <i>Data_type:</i> int, optional

<pre>-output_format {text csv gtd binary}</pre>	Generates a detailed timing report in the specified format - text, CSV, GTD, or binary. <i>Data_type</i>: enum, optional
<pre>-path_exceptions {applied ignored all}</pre>	Includes information on path exceptions applied and considered for the reported path. You can specify the following options: • applied: Reports the honored exception on the reported path. • ignored: Reports the ignored exceptions on the reported path. • all: Reports both applied and ignored sets of exceptions on the reported. If multiple from, to, or through pins are specified, the exception report for that path will show only those pins that are part of the current path. Note: If a path is already unconstrained, then all further path exceptions applied to that particular path will be reported as ignored exceptions. The <code>-path_exceptions</code> parameter also reports <code>-start/-end</code> options with multi-cycle paths. Without the start or end specification, by default, the setup check is relative to the end clock, and the hold check is relative to the start clock. The <code>report_timing -path_exceptions all</code> command will report all the non-default <code>-start/-end</code> specifications. You cannot use this parameter with the following parameters: • <code>-point_to_point</code> • <code>-collection</code> • <code>-path_type</code> • <code>-tcl_list</code> Also, you cannot use this parameter with block-based SSTA. Note: Analyzing all the exceptions for a path will increase the runtime. <i>Data_type</i>: enum, optional

<pre>-path_group groupname_list</pre>	Reports only the paths that are contained in the groups specified in the <i>groupname_list</i>. This parameter reports the paths belonging to the default path group by using <code>-path_group default</code>. The paths that do not belong to any user-specified path group belong to the default path group named <code>default</code>. In group-based mode, the internal groups, such as, clock gating, clock names, asynchronous, and user-defined path groups can be selected with the <code>-path_group</code> parameter. <i>Data_type</i>: string, optional
<pre>-path_type {end summary full full_clock end_slack_only summary_slack_only}</pre>	The <code>-path_type</code> parameter allows to you choose the report format by path type. You can specify one of the following options: • <code>end</code>: Generates an end point report for each path consisting of an endpoint, cause, slack, arrival time, required time, and phase. This option generates a very fast report. The <code>-format</code> option does not have any impact on this report format. • <code>summary</code>: Generates a summary report for each path consisting of a start point, endpoint, cause, slack, arrival time, required time, and phase. The <code>-format</code> option does not have any impact on this report format. • <code>full</code>: Generates a report that displays the full path with accompanying required time and slack calculation. This is the default path type. You can control the report format using the <code>-format</code> option. • <code>full_clock</code>: If the reported path ends at a timing check, the software reports the full clock path (Other End Path), in addition to the full data path (Timing Path). • <code>end_slack_only</code>: Generates an end point report (similar to <code>-path_type end</code> report), for each path consisting of an endpoint, slack, and cause (as Violated/Met). When this option is used, the path collections will not be reported. This option generates a very fast report. The <code>-format</code> option does not have any impact on this report format. • <code>summary_slack_only</code>: Generates a summary report

(similar to `-path_type summary`) for each path, consisting of a start point, endpoint, slack, and cause (as Violated/Met). The `-format` parameter does not have any impact on this report format.

When the `-path_type` parameter is not specified, the default report format (`full`) is used.

In MMC mode, when you specify the `report_timing -path_type` parameter (with `end`, `end_slack_only`, `summary`, or `summary_slack_only` options), the output report displays a view column for all the reported paths.

Note: The paths contained in a collection are not reported with the upgraded path type.

Data_type: enum, optional

`-point_to_point`

Traces the worst delay path between a pair of from-to pins. The paths reported with this parameter include only the cumulative delay of a path.

You must specify either the `-from` or `-to` parameter to define the begin and end points of the path.

If you do not specify a `-from` pin, the standard definition of the begin point of a path applies. If you specify a `-from` pin, it is treated as the begin point of the path. If you specify a `-from` pin that is not a valid begin point, the arrival time of the first pin starts from 0.

If you do not specify a `-to` pin, the standard definition of the end point applies. If you specify a `-to` pin, it is treated as the end point, and only those paths ending in `-to` pins are reported. If you specify more than one `-to` pin, the worst path to each pin-to-pin is reported.

When from-to pins are specified, the software will trace across the trigger arcs and report a path if there is a clock reaching the CK pin of the trigger arc. However, if there is a begin or end point present between the `-from` and `-to` pins, the software will not trace beyond it.

If there is no clock reaching the CK pin of the trigger arc, the software cannot trace across the trigger arc, and no path is reported. Also, a clock path delay is included only when the clock is in propagated mode.

The `-point_to_point` parameter can be used with the `-max_paths` and `-nworst` parameters.

The `-point_to_point` and `-through` parameters are mutually exclusive; you cannot specify them together.

Note: The `-point_to_point` parameter supports a single `-from` and `-to` pin; a list of pins is not supported.

Data_type: bool, optional

<pre>-retime_delaycal_pins <pin_name></pre>	Reports the PBA delay calculation details for timing arcs that fanin to the user-specified to-pin list. You can specify multiple pins in the pin list. The delay calculation report will be printed at the end of each timing path for the arcs, which are retimed in that path. The PBA delay calculation report format is similar to the <code>report_delay_calculation</code> output in GBA mode; however, the results may vary because this output is based on path-based analysis. When a cell output pin list is provided, the generated report is for the cell arcs; the cell input pin list generates the net arc report. Note: This parameter is supported for PBA reporting only. <i>Data_type:</i> string, optional
<pre>-retime_format {manual retime_compare retime_replace}</pre>	Allows you to set the format of timing reports for retime-related fields. You can specify one of the following options: • <code>manual</code>: Does not change the <code>report_timing</code> format. • <code>retime_compare</code>: If GBA columns are present in the report format, then the corresponding retime options are added. If PBA columns are already present, then there is no change in the format. • <code>retime_replace</code>: If GBA columns are present in the report format, then the GBA columns are replaced by the corresponding PBA columns. If PBA columns are already in the report, then there is no change in the format. Note: The timing report format can also be changed with the <code>timing_report_retime_formatting_mode</code> global variable. The <code>report_timing</code> command setting has precedence over this global variable. <i>Data_type:</i> enum, optional

<pre>-retime_mode {path exhaustive}</pre>	Reanalyzes the specified set of paths using the specified method. You can use one of the following options: • path: Reanalyzes the specified set of paths. By default, the software uses <code>-retime_mode path</code> setting, when the <code>-retime path_slew_propagation</code> option is specified. • exhaustive: Enables exhaustive path-based analysis mode. In this mode, the software will exhaustively examine <code>nworst</code> paths of the selected endpoints and then select the true worst path to be reported - based on retimed slacks of all the examined paths. When you use this option, reporting can be runtime intensive as the software needs to do an exhaustive search for true worst paths. Therefore, it is recommended to use this option in the later phases of the signoff stage, when there are very few violations left for analysis. <i>Default:</i> <code>path</code> <i>Data_type:</i> enum, optional
<pre>-retime {aocv path_slew_propagation aocv_path_slew_propagation}</pre>	Reanalyzes the specified set of paths using path-based analysis (1 path per endpoint). Note: This feature is not meant to be run on all the paths of a design. You can specify the following options: • aocv: Computes AOCV stage count based on specific paths and applies the AOCV derate values to GBA (graph-based analysis) delays. The <code>-retime aocv</code> option can be used with the following parameters only: ◦ <code>-clock_from</code> ◦ <code>-edge_from</code> ◦ <code>-clock_to</code> ◦ <code>-edge_to</code> ◦ <code>-rise/-fall</code> ◦ <code>-early/-late</code> ◦ <code>-check_type</code> ◦ <code>-max_paths</code>

- `-begin_end_pair`
- `-from/-from_rise/-from_fall`
- `-through/-through_rise/ -through_fall`
- `-to/-to_rise/-to_fall`
- `-path_group`
- `-path_exceptions`
- `-net`
- `-path_type`
- `-max_slack`
- `-min_slack`
- `-view`
- `-format column_list`
- `-derate_summary`
- `-machine_readable`
- `-nworst`
- **path_slew_propagation:** Re-evaluates the given set of graph-based analysis (GBA) paths by recalculating the delay values based upon actual propagated slew across the path. In this case, the AOCV derates are reported during graph-based analysis.

The default timing analysis uses worst-case slew propagation to calculate the delays at each stage. As a result, timing analysis results can be pessimistic.

Note: This feature should be run only on paths that do not meet the slack margin criteria, to help identify actual failing paths from those that are simply failing due to the conservative delay calculation.

The signal integrity incremental delays can be recomputed for the entire path. You can use the following command to perform path-based signal integrity analysis:

```
report_timing -retime path_slew_propagation
```

In the `report_timing` command output, the Retime Incr Delay column reports recomputed signal integrity delays. To

	view this column, specify the <code>report_timing -format retime_incr_delay</code> option or use the <code>report_timing_format</code> global variable. The <code>-retime</code> parameter also supports re-timing a collection of paths. This option does not retime an already retimed path collection. This feature works in graph-based analysis (GBA) mode for path collections. You can use the following model to retime collections: <pre>report_timing \$collection -retime path_slew_propagation</pre> A path collection stores context-dependent data, that is, a delay property in GBA will reflect the delay data of the timing path report generated in GBA mode. The same property in PBA will reflect the retime delay data of the timing path report in PBA mode. The delay column in timing reports is also context dependent, under the <code>timing_report_retime_formatting_mode retime_replace</code> global variable setting. Note: For memory optimization purposes, both delays will not be stored in the same path collection. You can use this setting for comparison purposes. • <code>aocv_path_slew_propagation</code>: Re-evaluates the given set of graph-based analysis (GBA) paths by using path-based AOCV derates and delays using actual path-based propagated slew. This option is a combination of options <code>-aocv</code> and <code>-path_slew_propagation</code>. For retime report header details, see Retime Analysis Report Header Flags. <i>Data_type</i>: enum, optional
<code>-skip_io_paths</code>	Specifies whether the input/output paths are to be ignored. <i>Data_type</i>: bool, optional
<code>-rise -fall</code>	Reports the path with the specified edge on the endpoint. If an endpoint is specified using the <code>-to_rise</code> (or <code>-to_fall</code>) parameter, the <code>-rise</code> (or <code>-fall</code>) parameter is ignored, and paths with an edge specified by <code>-to_rise</code> (or <code>-to_fall</code>) are reported. The <code>-rise</code> option applies to end points only. <i>Data_type</i>: bool, optional

<code>-tcl_list</code>	Produces the report in Tcl list format instead of a tabular format. This is useful for integrating timing with custom Tcl functions, and also for customizing report generation. The <code>-tcl_list</code> and <code>filename</code> parameters are mutually exclusive; you cannot specify them together. <i>Data_type</i>: bool, optional
<code>{-through -through_rise -through_fall} object_list</code>	Reports paths that pass through pin(s) or net(s) specified by the <code>object_list</code>. Any number of <code>-through</code> pins or nets can be specified. When you specify the <code>-through_rise</code> (or <code>-through_fall</code>) parameter, the paths go through the rising (or falling) edge of the signals on the pins/nets in the <code>object_list</code>. The <code>object_list</code> is a logical OR function. The resulting path may pass through any of the pins or nets in the <code>-through object_list</code>. To force the report to pass through multiple pins or nets, separate <code>-through</code> statements might be required. You can use these parameters with the <code>-net</code> option. <i>Data_type</i>: string, optional
<code>{-to -to_fall -to_rise} object_list</code>	Reports paths leading to the pin(s) specified in the <code>object_list</code>. The pins in the <code>object_list</code> can be either pins on the design boundary (ports) or pins on an instance. Only one list of <code>-to</code> pins can be specified per report. Using <code>-to_rise</code> (or <code>-to_fall</code>) specifies that the rising (or falling) edge of the signals on the pins in <code>object_list</code> are at the end of the paths. The <code>-to_rise</code> option applies to end point as well as intermediate pins. <i>Data_type</i>: string, optional

<code>-unconstrained</code>	Reports only the unconstrained paths (paths with no slack). Each signal arriving at the path end node, which does not have a matching required time, results in an unconstrained path. The <code>report_timing</code> command without the <code>-unconstrained</code> option reports only constrained paths. If no constrained path is found, there may be unconstrained paths or the path may not exist. Note: The <code>-min_slack</code> and <code>-max_slack</code> cannot be specified with the <code>-unconstrained</code> parameter. See Examples for the conditions under which a path is reported as constrained or unconstrained. <i>Data_type:</i> bool, optional
<code>-unique_pins</code>	Reports paths through a unique set of pins. When this parameter is specified in group-based mode for a given begin-end point pair, the software will consider all different launch clock paths for reporting unique paths. If, for a given begin-end pair and a given capture clock, there is a possibility of different launch clock paths, then each such path will be reported. Note: Using this parameter may significantly affect runtime and memory usage if the number of paths to be reported is high. <i>Data_type:</i> bool, optional
<code>-view {viewName}</code>	Generates a report for the specified analysis view. By default, all the active views are analyzed. You must be in multi-mode multi-corner (MMMC) analysis mode in order to use this parameter. To create a view, you use the <code>create_analysis_view</code> command. To set an active view for analysis, you use the <code>set_analysis_view</code> command. The <code>report_timing</code> command performs block-based analysis in MMMC mode. By default, the command reports the worst endpoint(s) across all views. You can use the <code>-view</code> parameter to report for individual views. <i>Default:</i> Reports the worst path to each endpoint across all the analysis views. <i>Data_type:</i> bool, optional

<code>-worst_rc_corner</code>	Identifies the worst interconnect parameter combination that leads to minimum slack (desired) and generates a report for the corner. <i>Data_type</i> : bool, optional
-------------------------------	---

Group Path Reporting for User-Defined Groups

- A user-defined group is created with the same name as any of the defined reference clocks and with the same constraint as for the "to" clock. For example, `group_path -name CLK -to CLK`. In this case only one group named `CLK` will be displayed in the timing report. This will be the same default group with reference clock `CLK`. Note that default groups - clock gating and asynchronous checks - will not be honored for this group.
- A user-defined group is created with the same name as any of the reference clocks but with different constraint as specified for the "to" clock. For example, `group_path -name CLK -from pin1 -to CLK`. In this case the two groups named `CLK` will be displayed in the timing report. One group will be the same as the default group with reference clock `CLK` (which will not honor clock gating or asynchronous checks) and the other will be the user-defined path group with name `CLK`, which will honor all the checks.

Use of Informational Flags

Informational tags are used providing additional details. The following informational tags are supported:

Flags	Description
(i)	Net is ideal
(P)	Instance is preserved / don't_touch
(p)	Instance is preserved / don't_touch but may be resized

The output format uses the flag keys on the timing report and adds a description at the end of the report for reference. A sample output is as shown below:

```
#-----
# Timing Point   Delay (ns) Latch   Window   Arrival   Arc   Flags   Adjustment
# (ns)
#-----
f0/CP           -           -       1.000000           CP           -           -
```

```
f0/Q      0.254100  -   1.254100      CP->Q  -   -
b32/Z     0.598446  -   1.852546      I->Z   -   -
...
b4/Z      0.086940  -   2.502181      I->Z   -   -
f1/D      0.000000  -   2.502181      D      <<<  -
#-----
```

(P) : Instance is preserved
 (p) : Instance is preserved but may be resized
 (i) : Net is ideal.
 (b) : Timing paths are broken.

Retime Analysis Report Header Flags

AOCV(default)	Indicates that the selected retime analysis includes recalculation of path depths and AOCV derates. The AOCV retime is enabled with the <code>-retime aocv</code> or <code>aocv_path_slew_propagation</code> parameter. The AOCV stage counting mode (set using the <code>timing_aocv_analysis_mode</code> global variable) is indicated in the parenthesis. The stage counting modes are given below: ● <code>launch_capture (default)</code> (Default) ● <code>combine_launch_capture</code> (Combine) ● <code>separate_clock_data</code> (Separate) ● <code>clock_only</code> (Clock)
Corr	Indicates that the current SOCV accuracy mode considers the correlation of variation between stages. Starting with SOCV accuracy mode <code>medium</code> (specified using the <code>set_delay_cal_mode -socv_accuracy_mode</code> parameter), it is assumed that some portion of the variation is not completely uncorrelated random variation, so the software tracks a correlated and uncorrelated component for each of the applicable moments.

Data Clock	Indicates the portion of the path that has been retimed. By default, only the data path is retimed. If the <code>timing_disable_retime_clock_path_slew_propagation</code> global variable is set to <code>false</code>, then the clock path will also be retimed, and “Clock” will be added to the retime status.
EWM	Indicates that the equivalent waveform model is used (<code>set_delay_cal_mode -equivalent_waveform_model</code> parameter).
GBA COLL	Shows a collection of paths for which retime analysis is not performed. This can be controlled with the When the <code>report_timing -path_type exhaustive -collection</code> command. By default, the <code>report_timing -collection</code> parameter does not show GBA and PBA slack values separately.
GP	Indicates that graph-based PBA mode settings are enabled.
GS	Indicates that GBA skewness mode settings are enabled.
PBA COLL	Shows a collection of paths for which retime analysis is performed while creating the collection. This can be controlled with the When the <code>report_timing -path_type exhaustive -collection</code> command. By default, the <code>report_timing -collection</code> parameter does not show GBA and PBA slack values separately.
Path-Slew	Indicates the selected retime analysis is set to use path correct slew rates to reduce the pessimism from slew merging. This can be controlled with the <code>report_timing -retime</code> parameter. The path slews are recalculated in <code>path_slew_propagation</code> and <code>aocv_path_slew_propagation</code> modes.
PG	Indicates that group-based mode reporting is enabled when N paths are reported for every group.

RC	Indicates that RC variation is enabled with the <code>timing_socv_rc_variation_mode</code> setting, and an SOCV format RC variation side file library is available.
SI	Indicates that SI analysis is enabled during the retime analysis. The path-based analysis will recompute SI effects using path-correct victim timing windows.
SOCV(Ultra)	Indicates that SOCV analysis is enabled and has been incorporated into the retiming. The SOCV analysis mode (specified with the <code>set_delay_cal_mode -socv_accuracy_mode</code> parameter) is shown in parenthesis. The accuracy modes are mapped as follows: • low <not shown> • medium Medium • high High • ultra Ultra
Spatial (CS)	Indicates that spatial derating side files have been loaded, and an explicit spatial derate factor will be calculated. This is not shown for AOCV 2D derate tables, where the spatial and depth derates are combined. The GBA spatial derate mode (specified with the <code>timing_spatial_derate_distance_mode</code> setting) is shown in parenthesis with the possible values given below: • chip_size (CS) • arc_bounding_box (ARC) The GBA spatial derate mode does not affect the path-based analysis bounding box calculation.
WP	Indicates that waveform propagation is enabled (using the <code>set_delay_cal_mode -equivalent_waveform_model propagation</code> setting). The path-based retime will also use this delay calculation accuracy setting when recomputing delays and transitions for the path.

Examples of GBA/PBA Collection Reports

- When the `report_timing -path_type exhaustive -collection` command is specified, the following command generates a GBA collection report as shown below:

```
tempus > report_timing $tim_coll ; set tim_coll [report_timing -path_type exhaustive -collection]
```

```
**WARN: (TCLCMD-1248): Since no '-max_slack' has been specified with exhaustive '-retime_mode', tool will assume it's default value in this case which is '0'
**WARN: (TCLCMD-1251): The PBA exhaustive mode '-nworst' limit has been exhausted for some of the endpoints. Refer to verbose log file of current session for more details.
INFO: Path Based Analysis (PBA) performed on total '3' paths
INFO: Max depth 3 reached for end point 'htb/CG_REBAL_HIER_IN21/RC_IC_IN_1/E' during Path Based Analysis
```

```
Path 1: VIOLATED Clock Gating Setup Check with Pin htb/CG_REBAL_HIER_IN21/RC_IC_IN_1/CK
Endpoint: htb/CG_REBAL_HIER_IN21/RC_IC_IN_1/E (v) checked with leading edge of 'clk'
Beginpoint: htb/mc_vl_reg_2/Q (v) triggered by leading edge of 'clk'
Common point: SPINE_CELL_BBF1/Y
Path Groups: {clock_gating_default}
Analysis View: sg_05v_15c_sigmax
Retime Analysis {GBA COLL}
```

- When the `report_timing -path_type exhaustive -collection` command is specified, the following command shows a PBA collection report:

```
tempus > report_timing $tim_coll ; set tim_coll [report_timing -path_type exhaustive -collection]
```

```
**WARN: (TCLCMD-1248): Since no '-max_slack' has been specified with exhaustive '-retime_mode', tool will assume it's default value in this case which is '0'
INFO: Path Based Analysis (PBA) performed on total '530' paths
INFO: Max depth 131 reached for end point 'htb/CG_REBAL_HIER_IN21/RC_IC_IN_1/E' during Path Based Analysis
```

```
Path 1: VIOLATED Clock Gating Setup Check with Pin
htb/RC_CG_REBALANCE_HIER_INST2989/RC_IC_IN_1/CK
Endpoint: htb/RC_CG_REBALANCE_HIER_INST2989/RC_IC_IN_1/E (v) checked with leading edge of 'clk'
```

```

Beginpoint: htb/mc_vl_reg_2/Q                                (^) triggered by leading
edge of 'clk'
Common point: SPINE_CELL_BBF1/Y
Closest CPPR Driver Pin: { Launch: SPINE_CELL_BBF1/Y Capture: SPINE_CELL_BBF1/Y }
Path Groups: {clock_gating_default}
Analysis View: sg_05v_15c_sigmax
Retime Analysis {PBA COLL}

```

VT Skew Reports

The following informational tags are supported for VT skew reports:

Flags	Description	Command Settings
VT-Skew	Shows VT skew details.	set timing_enable_vtskew_derate_mode true
PBA	Shows that the current path has undergone VT skew analysis. However, all the paths might not be considered VT skew candidates - even if VT skew analysis mode is enabled.	set_vt_skew_config - unique_vt_mode {none slow fast nominal}
PU	Indicates unique VT skew for PBA. Displayed when unique VT mode is enabled.	set_vt_skew_config - unique_vt_mode {slow fast nominal}
GU	Indicates unique VT skew for GBA. Displayed when unique VT mode is enabled.	

The following report shows report header flags for retime analysis:

```

Path 1: MET Hold Check with Pin TDSP_CORE_INST/EXECUTE_INST/top_reg[11]/CP
Endpoint:   TDSP_CORE_INST/EXECUTE_INST/top_reg[11]/D                (v) checked
with leading edge of 'm_clk'
Beginpoint: TDSP_CORE_INST/DATA_BUS_MACH_INST/data_out_reg[11]/Q (v) triggered
by leading edge of 'm_clk'
Path Groups: {m_clk}
Analysis View: view2
Retime Analysis {Data Path-Slew SOCV(High)-Corr SI Vt-Skew (PBA GU PU) }
Retime Vt-Skew Analysis {VT_group_1@fast VT_group_2@fast VT_group_3@fast }
Retime Vt-Skew Stats { Dominant_VT_group: NA Launch_Clock_VTGroups: 2 Capture_
Clock_VTGroups: 1 Total_Clock_VTGroups: 2}

```

```

Other End Arrival Time      0.569769 (+3.00S) | 0.531640 | 0.012710
+ Hold                      -0.035400 (+3.00S) | -0.035400 | 0.000000
+ Phase Shift              0.000000
- CPPR Adjustment          0.051262 (+3.00S) | 0.000100 | 0.017054
= Required Time            0.462027 (+3.00S) | 0.496140 | 0.011371 (-)
  Arrival Time             0.769317 (-3.00S) | 0.803430 | 0.011371
  Slack Time               0.307290 (-3.00S) | 0.307290 | 0.000000
= Slack Time(original)     0.356572 (-3.00S) | 0.356572 | 0.000000
  Clock Rise Edge          0.000000
  = Beginpoint Arrival Time 0.000000 (-3.00S) | 0.000000 | 0.000000

```

Report Timing Column List Options

Option	Description
adjustment	Reports generated clock adjustment values. When reporting the detailed path for a generated clock, there are circumstances where the arrival time needs to be adjusted when a <code>create_generated_clock</code> assertion is encountered along the path. The adjustment argument can be used to provide a clearer indication of why a "jump" in the arrival time is seen in the path report. When the <code>create_generated_clock -edge_shift</code> parameter is specified, the adjustment column will show the amount of shift specified.
abs_delay_err	Reports the absolute delay difference of Tempus from Spice.
abs_slew_err	Reports the absolute slew difference of Tempus from Spice.
annotation	Reports RC parasitic annotations for net arcs, and delay annotations for either gate or net arcs. The following annotations are reported: • SDF: SDF back-annotation. The annotation status reflects the gate arc, not the interconnect arc. • SPEF: This entry appears if parasitic extraction has been done or annotated using SPEF. • LumpedRC: Lumped RC annotation • DlyAssert: Assertion set with <code>set_annotated_delay</code> command • WLM: Wire load model back-annotation • <none>: All others

aocv_adj_stages	Shows up "Aocv Adjust Stages" column. This column reports the AOCV adjustment stages that indicate the total effective stage count.
aocv_derate	Reports the graph-based AOCV derating factor used for cells/nets in AOCV mode. In retime mode, this column will report graph-based derating for cells/nets before the AOCV branch point and path-based derating.
aocv_weight	Reports AOCV weight values for the path elements.
arc	Reports the arc as described by the from pin, from pin edge, to pin, and to pin edge. For example, the arc from the rising edge of pin A to the falling edge of pin Z is reported as A -> Z.
arrival	Reports the arrival time on the pin.
arrival_mean	Reports mean of the arrival time on a pin.
arrival_sigma	Reports the standard deviation of the arrival time on a pin.
cell	Reports the cell name of the given pin's instance.
delay	Reports the arc delay. If the stolen slack at the arc output pin (output of transparent latches) is not zero, the stolen column is also displayed along with this column.
delay_mean	Reports the mean of each arc's delay.
delay_sigma	Reports the standard deviation of each arc's delay. Note: In the absence of correlated and uncorrelated components of variation, the individual delay sigma values will not be summed up.
direction	Reports the pin direction (IN, OUT).
edge	Reports the edge of a pin (^=R, v=F).
fanin	Reports the number of source nodes of a net connected to a timing pin.
fanout	Reports the number of sinks of a net connected to a timing pin.
flags	Indicates the size_only, dont_use, dont_touch, and ideal attributes. When the 'flags' option is specified, a capacitance group entry (if it exists) will be displayed under the "Flags" column. This value indicates the input capacitance group being used for delay calculations – C1C2 (2-piece model) or C1CN (n-piece model).

hpin	Reports the hierarchical name for a given pin. Note: The hpin, pin, and timing_point will print the same values in the report output.
incr_delay	Reports the incremental delay of a pertaining arc. The software reads the incremental delay from an incremental SDF file using the <code>read_sdf</code> command.
instance	Reports the hierarchical name of a given pin's instance.
instance_location	Reports the location (x,y) of an instance. If a design is not placed, this column will be blank. This option displays the <code>instance</code> column automatically.
latch_window	Reports the available window for the latch to sample the data.
load	Reports the total capacitance load on a given pin.
module	Reports the module name of the corresponding block hierarchy. This hierarchical module name will be reported only when the <code>-hpin</code> parameter of the <code>report_timing</code> command is specified.
net	Reports the hierarchical name of a net connected to a given pin.
pct_delay_err	Reports the percent delay difference of Tempus from Spice.
pct_slew_err	Reports the percent slew difference of Tempus from Spice.
phase	Reports the phase name of a pin.
phys_info	Shows minimum/maximum assigned layer, and non-default rule name information.
pin	Reports the reference name for the given pin.
pin_load	Reports the pin load data. The pin load refers to the sum of the pin capacitance of all the pins connected to the current net.
pin_location	Reports the location (x,y) of a pin. If a design is not placed, this column will be blank.
power_domain	Reports the power domain details of the specified path instance.
required	Reports the required time on a pin.
retime_delay	Reports the recalculated arc delay based on the path being reanalyzed with the specified retime method.
retime_delay_mean	Reports the mean of the recalculated arc delay on the path being reanalyzed with the specified retime method.

retime_delay_sigma	Reports the standard deviation of each parameter of an arc's retimed delay.
retime_mis_delta	Reports MIS delta delay in PBA mode for MIS-aware timing arcs in the path.
retime_mis_derate	Reports MIS derate values (in PBA mode) when derate-based advanced multi-input switching (AMIS-Derate) analysis is enabled.
retime_slew	Reports the recalculated slew based on the path being reanalyzed with the specified retime method.
retime_slew_mean	Reports the mean of each parameter of an arc sink pin's retimed slew.
retime_slew_sigma	Reports the standard deviation of each parameter of an arc sink pin's retimed slew.
stage_count	Displays the aggregate stage count of a specific instance in graph-based analysis (GBA) mode, and the actual stage count in path-based analysis (PBA) mode.
slew	Reports the propagated slew at the given pin.
slew_mean	Reports the mean of each parameter of an arc sink pin's slew.
slew_sigma	Reports the standard deviation of each parameter of an arc sink pin's slew.
socv_derate	Reports the SOCV or the spatial derating factor applied to cells or nets in SOCV analysis mode.
spice_arrival	Reports the Spice arrival time at the timing point.
spice_delay	Reports the Spice delay of a timing arc.
spice_slew	Reports the Spice slew of a timing point.
stolen	Reports the slack stolen (or the time given to the previous stage) at the given pin. If the slack is not zero and the <code>delay</code> column is specified, then this column is displayed by default (but only visible on the output of transparent latches).
timing_point	Reports the hierarchical names for the given points.

total_derate	Reports the total derating factor applied on the corresponding cell or nets that reflect the combined impact of all the derates (OCV/AOCV/spatial) applied on each stage of the timing path. Note: By default, the software will take into account the negative delays and keep the SOCV derate column in sync with the user-derate column. The software multiplies the negative delay by (2.0 - derating factor) to make it more pessimistic. However, reporting will always show derates based on the positive delay. The timing report output will show similar values for the SOCV Derate and Total Derate columns, but there will be no changes in the slack values.
user_derate	Reports the derating scale factors set with the <code>set_timing_derate</code> command. Note: The timing system cannot provide an accurate derating factor for the lumped gate and wire delay when the <code>set_timing_derate</code> factors for <code>-cell_delay</code> and <code>-net_delay</code> are not equivalent. In this case, an asterisk (*) is reported in the SSI Derate column.
voltage	Returns the voltage that is used during delay calculations.
vt_skew_derate	Returns voltage threshold (VT) skew analysis details.
when_cond	Reports the "when" condition of the arc specified in the library.
wire_load	Reports the wire load.
wlmodel	Reports the kind of parasitic used by a pin or net, such as SPEF. If the wire load model is derived from a Liberty file, the information is reported in the following format: <pre>Wireload model (.lib model_name library_name)</pre>

Reporting Timing Checks Between Various Types of Clocks

The following table summarizes the timing checks and the min/max delay checks that are performed based on the types of launching and capturing clocks:

- NZ (Non Zero Period Clock - Regular Clock)
- Z (Zero Period Clock - Regular clock whose period is equal to zero)
- @ (Asynchronous Clock - Clock which triggers the default input delay and the input delay without a clock)
- CPD (Combinational Path Delay)

Launching Clock	Capturing Clock	Timing Checks	Path Delay Checks
@	@	NO	CPD
Z	@	NO	CPD
NZ	@	NO	CPD
@	Z	YES	SPD
@	NZ	YES	SPD
NZ	Z	NO	SPD
Z	NZ	YES	SPD
Z	Z	YES	SPD
NZ	NZ	YES	SPD

report_timing -from latch/D Settings

The `report_timing -from [register/latch]/CK` command settings can be defined in line with the path reported in the `full_clock` path report (as the reported path might be `clock->lat/CK->lat/Q->ff2/D`). So the `report_timing -from CK-pin` can be very intuitive.

However, in latch case scenarios where there are multiple D pins on a latch, Tempus supports `-from latch/D pin`, which is generally used for debug purposes. When propagating across the `CK->Q` arc, the software will perform worst-case across all the `D->Q` arcs and report the worst path possible. The `report_timing -from latch/D2` command can be used to propagate arrival only across the `D2->Q` arc, so that no worst case occurs. And the path report or slack with respect to a particular D pin of the latch can be attained.

Examples,

- The following command uses `lat/G` (clock pin) , `lat/D` and `lat/D2` (2 data pins), `lat/Q` (output/begin pin), and shows the report below:

```
report_timing -to ff2/D -path_type full -format {hpin edge delay arrival arc} -net -from lat/G
```

```
-----

Path 1: MET Setup Check with Pin ff2/CK
Endpoint:  ff2/D (v) checked with trailing edge of 'CK1'
Beginpoint: lat/Q (v) triggered by trailing edge of 'CK1'
```

```

Path Groups: {CK1}
Other End Arrival Time          1.400
- Setup                        0.110
+ Phase Shift                  2.500
+ CPPR Adjustment              0.000
= Required Time                3.790
- Arrival Time                 2.073
= Slack Time                   1.717

    Clock Fall Edge            1.250
    + Clock Network Latency (Prop) 0.120
    = Beginpoint Arrival Time    1.370
    -----
    Pin      Edge  Delay  Time  Arrival  Arc
                   Given  Time
                   To
                   Start
                   Point
    -----
    lat/G ->  v    -      -      1.370    G v
    lat/Q     v    0.150  0.493  2.013    D v -> Q v
    buf9/A    v    0.000  -      2.013    -
    buf9/Y    v    0.030  -      2.043    A v -> Y v
    buf10/A   v    0.000  -      2.043    -
    buf10/Y   v    0.030  -      2.073    A v -> Y v
    ff2/D ->  v    0.000  -      2.073    -
    -----

```

- The following command shows the report below:

```

report_timing -to ff2/D -path_type full -format {hpin edge delay arrival arc} -net -from
lat/D2

```

```

-----
Path 1: MET Setup Check with Pin ff2/CK
Endpoint:  ff2/D (v) checked with trailing edge of 'CK1'
Beginpoint: lat/Q (v) triggered by trailing edge of 'CK1'
Path Groups: {CK1}
Other End Arrival Time          1.400
- Setup                        0.110
+ Phase Shift                  2.500
+ CPPR Adjustment              0.000

```

```

= Required Time                3.790
- Arrival Time                1.773
= Slack Time                  2.017

  Clock Fall Edge              1.250
+ Clock Network Latency (Prop) 0.120
= Beginpoint Arrival Time      1.370
-----
Pin      Edge  Delay  Time  Arrival  Arc
              Given  Time
              To
              Start
              Point
-----
lat/G     v    -      -      1.370    G v
lat/Q     v    0.150  0.193  1.713    D2 v -> Q v
buf9/A    v    0.000  -      1.713    -
buf9/Y    v    0.030  -      1.743    A v -> Y v
buf10/A   v    0.000  -      1.743    -
buf10/Y   v    0.030  -      1.773    A v -> Y v
ff2/D -> v    0.000  -      1.773    -
-----

```

Examples

SPD (Sequential Path Delay) Examples

- The following command reports paths leading to the pin(s) specified by the *object_list*:

```
tempus > report_timing -to u55/D
```

```
Path 1: MET Setup Check with Pin u55/CK
```

```
Endpoint: u55/D (^) checked with leading edge of 'CLK_W_4'
```

```
Beginpoint: u55/QN (^) triggered by leading edge of 'CLK_W_4'
```

```
Path Groups: {CLK_W_4}
```

```
Other End Arrival Time 0.152
```

```
- Setup 0.306
```

```
+ Phase Shift 10.000
```

```
= Required Time 9.845
```

```
- Arrival Time 1.902
```

```
= Slack Time 7.943
```

```
Clock Rise Edge 0.000
+ Clock Network Latency (Prop) 0.152
= Beginpoint Arrival Time 0.152
```

Pin	Arc	Cell	Delay	Arrival Time	Required Time	Instance

-						
u55/CK	CK ^	-	- 0.151	8.094		u55
u55/QN	CK ^ -> QN ^	DFF	1.750	1.901	9.844	u55
u55/D ->	D ^	DFF	0.001	1.902	9.845	u55

- The following command shows a report similar to the first with the addition of net arc information:

```
tempus > report_timing -to u55/D -net
```

```
Path 1: MET Setup Check with Pin u55/CK
Endpoint: u55/D (^) checked with leading edge of 'CLK_W_4'
Beginpoint: u55/QN (^) triggered by leading edge of 'CLK_W_4'
Path Groups: {CLK_W_4}
Other End Arrival Time 0.152
- Setup 0.306
+ Phase Shift 10.000
= Required Time 9.845
- Arrival Time 1.902
= Slack Time 7.943
Clock Rise Edge 0.000
+ Clock Network Latency (Prop) 0.152
= Beginpoint Arrival Time 0.152
```

Pin	Arc	Cell	Delay Time	Arrival Time	Required Time	Instance

u55/CK	CK ^	-	-	0.151	8.094	u55
u55/QN	CK ^ -> QN ^	DFF	1.750	1.901	9.844	u55
u55/D ->	-	DFF	0.001	1.902	9.845	u55

- The `report_timing` command also accepts `-collection` as an argument. For example,

```
tempus > report_timing [report_timing -collection]
tempus > set rpt [report_timing -collection]
tempus > report_timing $rpt
```

- The following command displays the worst late path in the design. The format of the report is similar to the first example:

```
tempus > report_timing
```

- The following command displays the worst late path to each violating endpoint that has a slack less than -1.0:

```
tempus > report_timing -max_slack -1.0
```
- The following command displays all the late paths that end at port `out[2]` and that have negative slack up to a maximum of 1000 worst paths. If there are more than 1000 paths, only the 1000 worst paths are reported:

```
tempus > report_timing -to out[2] -max_paths 1000 -max_slack 0.0
```
- The following command reports the worst late path that starts at `in[0]` and ends at `out[1]`:

```
tempus > report_timing -from in[0] -to out[1]
```
- The following command displays the three worst paths that start at `in[1]` and end at `out[3]`. With reconvergent fanout, more than one path may exist:

```
tempus > report_timing -from in[1] -to out[3] -max_paths 3
```
- The following command displays the ten worst paths. Only the paths between the specified pins are enumerated. With reconvergent fanout, more than one path may exist:

```
tempus > report_timing -from i102/Z -to i123/A -max_paths 10
```
- The following command enumerates the worst ten paths through the given two pins, starting at the beginning points in the design and ending at the endpoints, similar to using the `-from` option:

```
tempus > report_timing -through i102/Z -through i123/A -max_paths 10
```
- The following command reports the ten worst paths through blocks A and C, or blocks B and C. The path only has to satisfy one element in a through list, but all through lists must be satisfied. The example can be thought of as ((A or B) and C):

```
tempus > report_timing -through {A/*B/*} -through {C/*} -max_paths 10
```
- The following command forces a path through pin A and pin B. Use the syntax shown in the example. Do not use `-through {A B}`:

```
tempus > report_timing -through A -through B
```
- The following command displays the clock path through the clock root to Launch flop:

```
tempus > report_timing -through aBC_bs/bs_mex1/r2_00_q1_reg_10_/CK -to .... -  
path_type full_clock
```
- You will not see the path for the launch clock if you specify the following command because the `report_timing` command starts reporting from the CK pin:

```
tempus > report_timing -from aBC_bs/bs_mex1/r2_00_q1_reg_10_/CK
```

The first command displays the path leading to the CK pin.
- The following command shows the `-unconstrained` option:

```
tempus > report_timing -unconstrained
```

```

Path 1:Endpoint: OUT4 (^) (unconstrained output)
Beginpoint: u55/Q (^) triggered by leading edge of 'CLK_W_4'
Arrival Time 0.365
+ Clock Network Latency (Prop) 0.150
= Beginpoint Arrival Time 0.150

-----

Load Slew Delay Arrival Cell Arc Pin
Time

-----

0.013 0.116 - 0.150 - CK ^ u55/CK
0.011 0.072 0.215 0.365 DFF CK ^ -> Q ^ u55/Q
0.011 0.131 0.000 0.365 - OUT4 ^ OUT4 ->

-----

```

- The following command shows the `-unconstrained -path_type end` report:

```
tempus > report_timing -nworst 5 -unconstrained -path_type end -format {pin arrival phase}
```

```

-----

```

Path No.	Pin	Cause	Arrival	Phase

1	seg3/u9/CK ^	Unconstrained Path	0.682	CLK_W_3 (D) (P) *
2	seg3/u9/CK ^	Unconstrained Path	0.682	CLK_W_3 (D) (P) *
3	seg3/u3/CK ^	Unconstrained Path	0.682	CLK_W_3 (D) (P) *
4	seg3/u3/CK ^	Unconstrained Path	0.682	CLK_W_3 (D) (P) *
5	seg2/u9/CK ^	Unconstrained Path	0.682	CLK_W_2 (D) (P) *

- The following command sets a false path from u99/Q to u55/D:

```
tempus > set_false_path -from u99/Q -to u55/D
tempus > set_false_path -through u4/Y
tempus > report_timing -from u99/Q -to u55/D -unconstrained -path_exceptions all
```

```

Path 1:Endpoint: u55/D (^) (unconstrained output)
Beginpoint: u99/Q (v) triggered by leading edge of 'CLK_W_4'
Arrival Time 0.827
+ Clock Network Latency (Prop) 0.150
= Beginpoint Arrival Time 0.150

-----

Load      Slew      Delay      Arrival      Cell      Arc      Pin
Time

-----

```

```
0.013    0.116    -          0.150    -      CK ^          u99/CK
0.027    0.086    0.195     0.345    DFF    CK ^ -> Q v u99/Q ->
1.006    1.140    0.482     0.827    INV     A v -> Y ^  u4/Y
1.006    1.140    0.000     0.827    DFF     D ^          u55/D ->
```

Applied exceptions:

Through Late

u4/Y false

Ignored exceptions:

From To Late

u99/Q u55/D false

- For hold paths, if the path has a setup multi-cycle path then those exceptions will also be reported because hold slacks get impacted by inferred path exceptions. For example:

```
tempus > set_multicycle_path 2 -setup -from IN1
```

Path 1: VIOLATED Hold Check with Pin seg1/u3/CK

Endpoint: seg1/u3/D (v) checked with leading edge of 'CLK_W_1'

Beginpoint: IN1 (v) triggered by leading edge of '@'

Path Groups: {CLK_W_1}

Other End Arrival Time 0.682

+ Hold -0.034

+ Phase Shift 0.000

- Cycle Adjustment -20.000

= Required Time 20.648

Arrival Time 0.117

Slack Time -20.531

Clock Rise Edge 0.000

+ Input Delay 0.000

+ Drive Adjustment -0.014

= Beginpoint Arrival Time -0.014

Load	Slew	Delay	Arrival	Cell	Arc	Pin
------	------	-------	---------	------	-----	-----


```

                                Time
-----
0.013    0.029    -      -0.014    -      IN1 v      IN1
0.013    0.057    0.054  0.040    CLKBUF  A v -> Y v  seg1/u1/Y
0.003    0.050    0.077  0.117    CLKBUF  A v -> Y v  seg1/u2/Y
0.003    0.050    0.000  0.117    DFF     D v      seg1/u3/D
-----

```

Applied exceptions:

```

-----
From Early Late
-----

```

```

IN1 - cycles 2
-----

```

- The following command displays the timing report for analysis view `setup_view1`:

```
tempus > report_timing -view setup_view1
```

```

Path 1: MET Late External Delay Assertion
Endpoint: EJ_TDOzstate (^) checked with leading edge of 'virt_out_EJ_TCK'
Beginpoint: core_ejt_ejt_tap_ejt_tck__EJ_TDOzstate_q_reg_0/Q (^) triggered by trailing
edge of 'EJ_TCK'
Path Groups: {virt_out_EJ_TCK}
Analysis View: setup_view1
Other End Arrival Time 0.000
+ Source Insertion Delay 0.300
- External Delay 1.250
+ Phase Shift 17.400
+ CPPR Adjustment 0.000
= Required Time 16.450
- Arrival Time 9.318
= Slack Time 7.132
Clock Fall Edge 8.700
+ Clock Network Latency (Ideal) 0.000
= Beginpoint Arrival Time 8.700

```

```

-----
Load    Slew    Delay    Arrival    Cell        Arc          Pin
                                Time
-----
0.010   0.000                8.700                CP ^      core_ejt_ejt

```

```
0.118  0.145  0.127  8.828      DFSVTD4  CP ^ -> Q ^      core_reg_0_/Q
0.525  0.438  0.411  9.238      BUFVTD6  I ^ -> Z ^      FE_OFC260_/Z
0.525  0.474  0.080  9.318              EJ_TDOzstate ^ EJ_TDOzstate
```

- **When -from is intermediate begin point:**

```
tempus > report_timing -to u55/D -from u99/Q -point_to_point
```

```
Path 1:Endpoint: u55/D (^) (unconstrained output)
Beginpoint: u99/Q (v) triggered by leading edge of 'CLK_W_4'
Arrival Time 0.482
```

```
-----
Load      Slew      Delay      Arrival      Cell      Arc      Pin
              Time
-----
0.027  0.086              0.000              Q v      u99/Q ->
1.006  1.140  0.482  0.482      INV      A v -> Y ^ u4/Y
1.006  1.140  0.000  0.482      DFF      D ^      u55/D ->
-----
```

- **The following command recalculates the specified critical paths using the AOCV factor and generates the following timing report:**

```
tempus > report_timing -from in3 -to Top_I9/D -retime aocv -path_type full_clock
```

```
Path 1: MET Setup Check with Pin Top_I9/CK
Endpoint: Top_I9/D (v) checked with leading edge of 'clk2'
Beginpoint: in3 (^) triggered by leading edge of '@'
Path Groups: {clk2}
Retime Analysis { AOCV(Default) }
Other End Arrival Time 0.000
- Setup 0.248
+ Phase Shift 1.000
= Required Time 0.752
- Arrival Time 0.061
= Slack Time 0.691
Clock Rise Edge 0.000
+ Input Delay 0.000
= Beginpoint Arrival Time 0.000
Timing Path:
```

```
-----
Load      Slew      Delay      Aocv      Cell      Arc      Pin
              Derate
-----
```

```
-----
0.011  0.004                      in3 ^      in3 ->
0.011  0.004  0.001  1.173  INVX1                      Top_I8/A
0.007  0.076  0.060  1.175  INVX1  A ^ -> Y v  Top_I8/Y
0.007  0.076  0.000  1.173  DFFX1                      Top_I9/D ->
-----
```

```
Clock Rise Edge 0.000
= Beginpoint Arrival Time 0.000
Other End Path:
```

```
-----
Load      Slew      Delay  Aocv  Cell   Arc      Pin
          Derate
-----
0.012     0.000                      clk2 ^    clk2
0.012     0.000  0.000   0.825 DFFX1          Top_I9/CK
-----
```

- The following command recalculates the specified critical paths using the AOCV factor and generates an AOCV derating summary table for the launch and capture paths, in addition to the timing report:

```
tempus > report_timing -from in3 -to Top_I9/D -retime aocv -path_type full_clock -
derate_summary
```

```
Path 1: MET Setup Check with Pin Top_I9/CK
Endpoint: Top_I9/D (v) checked with leading edge of 'clk2'
Beginpoint: in3 (^) triggered by leading edge of '@'
Path Groups: {clk2}
Retime Analysis { AOCV(Default) }
Other End Arrival Time 0.000
- Setup 0.248
+ Phase Shift 1.000
= Required Time 0.752
- Arrival Time 0.061
= Slack Time 0.691
Clock Rise Edge 0.000
+ Input Delay 0.000
= Beginpoint Arrival Time 0.000
Timing Path:
```

```
-----
Load      Slew      Delay  Aocv  Cell   Arc      Pin
          Derate
-----
0.011     0.004                      in3 ^      in3 ->
```

```
0.011  0.004  0.001  1.173  INVX1          Top_I8/A
0.007  0.076  0.060  1.175  INVX1  A ^ -> Y v Top_I8/Y
0.007  0.076  0.000  1.173  DFFX1          Top_I9/D ->
```

```
-----
Clock Rise Edge 0.000
= Beginpoint Arrival Time 0.000
Other End Path:
```

```
-----
Load      Slew      Delay  Aocv    Cell    Arc      Pin
          Derate
-----
0.012  0.000          clk2 ^  clk2
0.012  0.000  0.000   0.825  DFFX1    Top_I9/CK
-----
```

```
DUAL LOCV Derate Summary
Clock Branch Point None
Distance 89.944
```

- The following command reports pin names and net names in the same column alternatively:

```
tempus > report_timing -format { load slew delay arrival cell pin_load fanout
timing_point} -net
```

```
Path 1: MET Setup Check with Pin u55/CK
Endpoint: u55/D (^) checked with leading edge of 'CLK_W_4'
Beginpoint: u99/Q (v) triggered by leading edge of 'CLK_W_4'
Path Groups: {CLK_W_4}
Other End Arrival Time 0.150
- Setup 0.176
+ Phase Shift 10.000
= Required Time 9.974
- Arrival Time 0.827
= Slack Time 9.148
Clock Rise Edge 0.000
+ Clock Network Latency (Prop) 0.150
= Beginpoint Arrival Time 0.150
```

```
-----
Load      Slew      Delay  Arrival  Cell    Pin Load Fanout Timing Point
          Time
-----
          0.116          0.150          u99/CK
          0.086  0.195  0.345    DFF          u99/Q
0.027          0.086  0.000  0.345    INV          u4/A
          1.140  0.482  0.827    INV          u4/Y
1.006          0.086  0.000  0.345    INV          u4/Y
          1.140  0.482  0.827    INV          u4/Y
          1.140  0.000  0.827    DFF          u55/D ->
-----
```

- The following command reports power domain details of path instances:

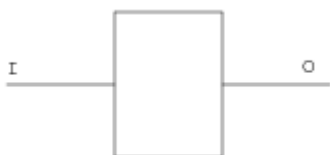
```
tempus > report_timing -format { load slew delay arrival cell arc hpin power_domain}
```

```
Path 1: MET Setup Check with Pin u55/CK
Endpoint: u55/D (^) checked with leading edge of 'CLK_W_4'
Beginpoint: u99/Q (v) triggered by leading edge of 'CLK_W_4'
Path Groups: {CLK_W_4}
Other End Arrival Time 0.150
- Setup 0.176
+ Phase Shift 10.000
= Required Time 9.974
- Arrival Time 0.827
= Slack Time 9.148
Clock Rise Edge 0.000
+ Clock Network Latency (Prop) 0.150
= Beginpoint Arrival Time 0.150
```

Load	Slew	Delay	Arrival Time	Cell	Arc	Pin	Power Domain
0.013	0.116		0.150		CK ^	u99/CK	
0.027	0.086	0.195	0.345	DFF	CK ^ -> Q v	u99/Q	PD1
1.006	1.140	0.482	0.827	INV	A v -> Y ^	u4/Y	PD2
1.006	1.140	0.000	0.827	DFF	D ^	u55/D ->	PD1

- The following examples refer to the figure shown below:

Conditions for Unconstrained and Constrained Paths



- Shows the path from I to O is reported as an unconstrained path under the following conditions (constraints on I and O):
- No constraints on either I or O
- set_input_delay -clock ck1 1.0 I
No set_output_delay constraint on O
- set_input_delay 1.0 I
set_output_delay 1.1 O

- Shows the path from I to O is reported as a constrained path under the following conditions (constraints on I and O):

- set_input_delay -clock ck1 1.0 I
set_output_delay -clock ck1 1.0 O

- set_input_delay -clock ck1 1.0 I
set_output_delay -clock ck2 1.0 O

- set_output_delay -clock ck1 1.0 O

No set_input_delay constraint on I. It uses default delay of zero.

Default: -late

- The following command reports AOCV weight details of path elements:

```
tempus > report_timing -from in3 -to Top_I9/D -format {instance cell arc delay
aocv_derate stage_count arrival aocv_weight} -path_type full_clock
```

Path 1: MET Setup Check with Pin Top_I9/CK

Endpoint: Top_I9/D (v) checked with leading edge of 'clk2'

Beginpoint: in3 (^) triggered by leading edge of '@'

Path Groups: {clk2}

Other End Arrival Time 0.000

- Setup 0.248

+ Phase Shift 1.000

= Required Time 0.752

- Arrival Time 0.061

= Slack Time 0.691

Clock Rise Edge 0.000

+ Input Delay 0.000

= Beginpoint Arrival Time 0.000

Timing Path:

Instance	Cell	Arc	Delay	Aocv Derate	Aocv Stage Count	Arrival Time	Aocv Weight

		in3 ^				0.000	
Top_I8	INVX1	A ^ -> Y v	0.061	1.175	1.000	0.061	1
Top_I9	DFFX1	D v	0.000	1.173	2.000	0.061	1

Clock Rise Edge 0.000

= Beginpoint Arrival Time 0.000

Other End Path:

Instance	Cell	Arc	Delay	Aocv	Aocv	Arrival	Aocv
				Derate	Stage	Time	Weight
					Count		

		clk2 ^				0.000	
Top_I9	DFFX1	CK ^	0.000	0.825	1.000	0.000	1

- The following command reports paths through unique set of pins:

```
tempus > report_timing -nworst 2 -unique_pins
```

Path 1: MET Setup Check with Pin seg3/u14/CK

Endpoint: seg3/u14/D (^) checked with leading edge of 'CLK_W_3'

Beginpoint: seg3/u9/Q (^) triggered by leading edge of 'CLK_W_3'

Path Groups: {CLK_W_3}

Other End Arrival Time 0.621

- Setup 0.186

+ Phase Shift 2.000

= Required Time 2.434

- Arrival Time 2.062

= Slack Time 0.372

Clock Rise Edge 0.000

+ Clock Network Latency (Prop) 0.682

= Beginpoint Arrival Time 0.682

Load	Slew	Delay	Arrival Time	Cell	Arc	Pin
0.008	0.049	0.682			CK ^	seg3/u9/CK
0.011	0.075	0.185	0.867	DFF	CK ^ -> Q ^	seg3/u9/Q
0.027	0.077	0.073	0.940	BUF	A ^ -> Y ^	seg3/u10/Y
1.027	0.644	0.317	1.257	INV	A ^ -> Y v	seg3/u11/Y
1.011	1.087	0.402	1.659	INV	A v -> Y ^	seg3/u12/Y
0.003	1.077	0.403	2.062	BUF	A ^ -> Y ^	seg3/u13/Y
0.003	1.077	0.000	2.062	DFF	D ^	seg3/u14/D

```
Path 2: MET Setup Check with Pin u2/CK
Endpoint: u2/D (v) checked with leading edge of 'CLK_W_1'
Beginpoint: CLK2 (v) triggered by trailing edge of 'CLK_W_2'
Path Groups: {CLK_W_1}
Other End Arrival Time 0.000
- Setup 0.188
+ Phase Shift 15.000
= Required Time 14.812
- Arrival Time 12.463
= Slack Time 2.349
Clock Fall Edge 12.500
= Beginpoint Arrival Time 12.500
```

```
-----
Load      Slew      Delay      Arrival      Cell      Arc      Pin
              Time
-----
0.034  0.003   12.500                      CLK2 v      CLK2
0.003  0.003   -0.037   12.463      BUF      A v -> Y v  u10999/Y
0.003  0.003    0.000   12.463      DFF      D v          u2/D
-----
```

- o The following command reports all the violating paths (or having slack greater than the value specified with '-max_slack') between unique source and target register pairs:

```
tempus > report_timing -to u55/D -begin_end_pair -max_slack 10
```

```
Path 1: MET Setup Check with Pin u55/CK
Endpoint: u55/D (^) checked with leading edge of 'CLK_W_4'
Beginpoint: u99/Q (v) triggered by leading edge of 'CLK_W_4'
Path Groups: {CLK_W_4}
Other End Arrival Time 0.150
- Setup 0.176
+ Phase Shift 10.000
= Required Time 9.974
- Arrival Time 0.827
= Slack Time 9.148
Clock Rise Edge 0.000
+ Clock Network Latency (Prop) 0.150
= Beginpoint Arrival Time 0.150
-----
```


Tempus Text Command Reference

Timing Analysis Commands

Load	Slew	Delay	Arrival Time	Cell	Arc	Pin
0.013	0.116	0.150			CK ^	u99/CK
0.027	0.086	0.195	0.345	DFF	CK ^ -> Q v	u99/Q
1.006	1.140	0.482	0.827	INV	A v -> Y ^	u4/Y
1.006	1.140	0.000	0.827	DFF	D ^	u55/D ->

- The following command generates a group based report in non-group mode:

```
tempus > report_timing -begin_end_pair -max_slack 10 -path_type end_slack_only
```

```
Path Groups: max_delay/setup {CLK_W_1}
```

```
End Point Slack Cause
```

```
u2/D f 2.349 MET
```

```
Path Groups: max_delay/setup {CLK_W_3}
```

```
End Point Slack Cause
```

```
seg3/u14/D r 0.372 MET
```

```
seg3/u3/D f 2.397 MET
```

```
report_timing -path_type full_clock -net -debug time_borrow
```

```
Path 1: MET Latch Borrowed Time Check with Pin latch2/GN
```

```
Endpoint: latch2/D (^) checked with trailing edge of 'CK1'
```

```
Beginpoint: ff1/Q (v) triggered by leading edge of 'CK1'
```

```
Path Groups: {CK1}
```

```
Other End Arrival Time 2.558
```

```
+ Borrow Edge Adjust 0.043
```

```
+ Phase Shift 0.000
```

```
+ CPPR Adjustment 0.000
```

```
= Required Time 2.601
```

```
- Arrival Time 0.544
```

```
= Slack Time 2.057
```

```
Clock Rise Edge 0.000
```

```
+ Clock Network Latency (Prop) 0.313
```

```
= Beginpoint Arrival Time 0.313
```

Load	Slew	Delay	Incr	Arc	Fanout	Cell	Arc	User	Arrival
Instance									

Tempus Text Command Reference

Timing Analysis Commands

Delay	Annotation					Derate	Time		

--									
0.005	0.055			WLM	2		CK ^		0.313
ff1									
0.004	0.042	0.123	0.000	WLM	1	DDFHQX1	CK ^ -> Q v	1.000	0.436
ff1									
0.004	0.044	0.035	0.000	WLM	1	INVX1	A v -> Y ^	1.000	0.471
invd4									
0.004	0.026	0.019	0.000	WLM	1	INVX1	A ^ -> Y v	1.000	0.491
invd5									
0.010	0.082	0.053	0.000	WLM	1	INVX1	A v -> Y ^	1.000	0.544
invd6									
0.010	0.082	0.000	0.000	WLM		TLATNRX4	D ^	1.000	0.544
latch2									

Time Borrow Calculations									

Latch Early Close Edge 4.482									
- Library Setup 0.057									
- Latch Early Open Edge 2.558									
= Max Allowed Borrow Time 1.866									
- Time Borrowed 0.043									
= Available Time Borrow 1.824									

- When `timing_report_retime_formatting_mode` global variable is set to `retime_compare`, then the following report is displayed:

```
tempus > report_timing -retime path_slew_propagation -format { delay retime_delay slew
}
+-----+
| Delay  | Retime | Slew  | Retime |
|        | Delay  |       | Slew   |
|-----+-----+-----+-----|
|        |        | 0.005 | 0.005 |
| 0.000  | 0.000 | 0.005 | 0.005 |
| 1.412  | 1.553 | 0.880 | 0.880 |
| 0.000  | 0.000 | 0.913 | 0.880 |
+-----+
```

When the `timing_report_retime_formatting_mode` global variable is set to `retime_replace`, then the GBA columns are replaced by the corresponding PBA columns, as shown in the example below:

```
tempus > report_timing -retime path_slew_propagation -format { delay retime_delay slew
}
```

```
+-----+
| Delay | Retime | Retime |
|       | Delay  | Slew   |
|-----+-----+-----|
|       |       | 0.005  |
| 0.000 | 0.000 | 0.005  |
| 1.412 | 1.553 | 0.880  |
| 0.000 | 0.000 | 0.880  |
+-----+
```

- The following command generates the PBA delay calculation report for the net arc terminating at ff2/D pin.

```
tempus > report_timing -to ff2/D -retime path_slew_propagation -retime_delaycal_pins
ff2/D
```

```
-----
report_delay_calculation PBA Report
-----
```

```
From pin : inv_inst/ZN
Voltage : 0.765 V
Cell : INV
Library : ccs_timing_si
To pin : ff2/D
Voltage : 0.765 V
Cell : DFF
Library : ccs_timing_si
Base or SI: SI delay
Delay type: net
```

```
Rise Rise Fall Fall Slew
Thresholds: Delay Slew Delay Slew Derate
From pin : 50 30->70 50 70->30 0.5
To pin : 50 30->70 50 70->30 0.5
```

```
-----
From pin waveforms
-----
```

```
Rise Waveform : { 0 0 6.88001e-12 -0.000345826 1.376e-11 -0.000759363 2.064e-11 -
0.00122762 2.752e-11 -0.00172836 3.44001e-11 -0.00223374 4.81601e-11 -0.00319517
6.19201e-11 -0.00401515 7.56801e-11 -0.00442153 1.032e-10 -0.00183165 1.1696e-10
0.0012542 1.2384e-10 0.00306022 1.3072e-10 0.00495619 1.4448e-10 0.00885701 1.5824e-10
0.0128341 1.8576e-10 0.0208058 2.408e-10 0.0366083 3.50881e-10 0.067488 5.71041e-10
0.126359 6.26081e-10 0.140509 6.53601e-10 0.147502 7.08641e-10 0.161279 7.36161e-10
0.164503 7.49921e-10 0.164456 7.63681e-10 0.164333 7.77441e-10 0.164129 8.04961e-10
0.161906 8.32481e-10 0.158058 8.60001e-10 0.154065 8.87521e-10 0.149911 9.42561e-10
```

Tempus Text Command Reference

Timing Analysis Commands

```
0.141834 9.97601e-10 0.134669 1.05264e-09 0.129404 1.10768e-09 0.126477 1.12144e-09
0.123602 1.12832e-09 0.119572 1.1352e-09 0.114331 1.14208e-09 0.108821 1.14896e-09
0.10317 1.16272e-09 0.0926769 1.17648e-09 0.0833663 1.204e-09 0.0657246 1.21776e-09
0.0578875 1.23152e-09 0.0515706 1.24528e-09 0.0458775 1.25904e-09 0.0406301 1.28656e-09
0.0323652 1.31408e-09 0.0270132 1.36912e-09 0.0280973 1.39664e-09 0.0336656 1.42416e-09
0.0379847 1.43792e-09 0.0378836 1.45168e-09 0.0365497 1.4792e-09 0.0332861 1.49296e-09
0.0314441 1.52048e-09 0.0278608 1.548e-09 0.0244482 1.60304e-09 0.0191201 1.65808e-09
0.0162272 1.71312e-09 0.0171329 1.76816e-09 0.0205911 1.8232e-09 0.0256714 1.87824e-09
0.0351823 1.90576e-09 0.0413911 1.91952e-09 0.0445492 1.9264e-09 0.0461416 1.94016e-09
0.048697 1.9436e-09 0.046998 1.94704e-09 0.0443201 1.95392e-09 0.038436 1.95736e-09
0.0352619 1.96424e-09 0.0289844 1.97112e-09 0.022866 1.98488e-09 0.0125791 1.99864e-09
0.00580478 2.0124e-09 0.000710309 2.02616e-09 -0.00178897 2.03992e-09 -0.00114673
2.05368e-09 0.000872076 2.06744e-09 0.00382227 2.0812e-09 0.00744581 2.10872e-09
0.0151135 2.13624e-09 0.0227224 2.16376e-09 0.0302613 2.2188e-09 0.0451577 2.32888e-09
0.0742571 2.54904e-09 0.129727 2.60408e-09 0.143045 2.6316e-09 0.143044 2.64536e-09
0.140082 2.65912e-09 0.137093 2.67288e-09 0.134064 2.7004e-09 0.12951 2.72792e-09
0.126729 2.78296e-09 0.1221 2.81048e-09 0.120269 2.86553e-09 0.118138 2.92057e-09
0.118162 3.03065e-09 0.12258 3.14073e-09 0.13482 3.25081e-09 0.155185 3.30585e-09
0.166687 3.36089e-09 0.178314 3.41593e-09 0.190705 3.47097e-09 0.203779 3.58105e-09
0.229323 3.63609e-09 0.241791 3.74617e-09 0.266165 3.96633e-09 0.312679 4.18649e-09
0.35642 4.29657e-09 0.377296 4.40665e-09 0.397506 4.51673e-09 0.417121 4.62681e-09
0.436119 4.84697e-09 0.472392 5.06713e-09 0.506472 5.28729e-09 0.538488 5.50745e-09
0.568534 5.72761e-09 0.596685 5.94777e-09 0.62296 6.16793e-09 0.647336 6.38809e-09
0.669575 6.60825e-09 0.689497 6.82841e-09 0.706756 7.04857e-09 0.721042 7.26873e-09
0.732528 7.48889e-09 0.741269 7.70905e-09 0.747704 7.92921e-09 0.752418 }
```

To pin waveforms

```
-----  
Rise Waveform : { 0 0 6.88001e-12 -0.00031215 1.376e-11 -0.000717461 2.064e-11 -  
0.00118119 2.752e-11 -0.00168008 3.44001e-11 -0.00218582 4.81601e-11 -0.00315118  
6.19201e-11 -0.0039801 7.56801e-11 -0.00441444 1.032e-10 -0.00195223 1.1696e-10  
0.00108838 1.2384e-10 0.00288516 1.3072e-10 0.00477386 1.4448e-10 0.00866932 1.5824e-10  
0.0126446 1.8576e-10 0.0206168 2.408e-10 0.0364222 3.50881e-10 0.0673075 5.71041e-10  
0.12619 6.26081e-10 0.140342 6.53601e-10 0.147336 7.08641e-10 0.161117 7.36161e-10  
0.164325 7.49921e-10 0.16428 7.63681e-10 0.164156 7.77441e-10 0.163952 8.04961e-10  
0.161721 8.32481e-10 0.157871 8.60001e-10 0.153877 8.87521e-10 0.149721 9.42561e-10  
0.141646 9.97601e-10 0.134476 1.05264e-09 0.129217 1.10768e-09 0.126291 1.12144e-09  
0.123396 1.12832e-09 0.11934 1.1352e-09 0.114094 1.14208e-09 0.108576 1.14896e-09  
0.102955 1.16272e-09 0.0924557 1.17648e-09 0.0831177 1.204e-09 0.0655087 1.21776e-09  
0.05769 1.23152e-09 0.0513455 1.24528e-09 0.0456527 1.25904e-09 0.0404119 1.28656e-09  
0.0321485 1.31408e-09 0.0268007 1.36912e-09 0.0278778 1.39664e-09 0.0334662 1.42416e-09  
0.0377678 1.43792e-09 0.0376836 1.45168e-09 0.0363449 1.4792e-09 0.0330793 1.49296e-09  
0.0312404 1.52048e-09 0.027657 1.548e-09 0.0242445 1.60304e-09 0.0189107 1.65808e-09  
0.0160238 1.71312e-09 0.0169355 1.76816e-09 0.020394 1.8232e-09 0.0254886 1.87824e-09  
0.034983 1.90576e-09 0.0412062 1.91952e-09 0.0443699 1.9264e-09 0.0459651 1.94016e-09  
0.0485097 1.9436e-09 0.0467553 1.94704e-09 0.0440755 1.95392e-09 0.0381743 1.95736e-09
```

```
0.0350146 1.96424e-09 0.0287352 1.97112e-09 0.0226158 1.98488e-09 0.0123552 1.99864e-09
0.00553888 2.0124e-09 0.000528693 2.02616e-09 -0.00200176 2.03992e-09 -0.00134081
2.05368e-09 0.000687182 2.06744e-09 0.00364017 2.0812e-09 0.00726146 2.10872e-09
0.0149298 2.13624e-09 0.0225402 2.16376e-09 0.0300808 2.2188e-09 0.0449804 2.32888e-09
0.0740856 2.54904e-09 0.129567 2.60408e-09 0.142888 2.6316e-09 0.142859 2.64536e-09
0.139899 2.65912e-09 0.136908 2.67288e-09 0.133879 2.7004e-09 0.129349 2.72792e-09
0.126549 2.78296e-09 0.121913 2.81048e-09 0.120103 2.86553e-09 0.117961 2.92057e-09
0.117997 3.03065e-09 0.122409 3.14073e-09 0.134658 3.25081e-09 0.155035 3.30585e-09
0.166534 3.36089e-09 0.178163 3.41593e-09 0.190548 3.47097e-09 0.203625 3.58105e-09
0.229173 3.63609e-09 0.241644 3.74617e-09 0.266022 3.96633e-09 0.312545 4.18649e-09
0.356294 4.29657e-09 0.377174 4.40665e-09 0.397388 4.51673e-09 0.417006 4.62681e-09
0.436007 4.84697e-09 0.472288 5.06713e-09 0.506374 5.28729e-09 0.538396 5.50745e-09
0.568448 5.72761e-09 0.596605 5.94777e-09 0.622885 6.16793e-09 0.647267 6.38809e-09
0.669513 6.60825e-09 0.689441 6.82841e-09 0.70671 7.04857e-09 0.721003 7.26873e-09
0.732498 7.48889e-09 0.741247 7.70905e-09 0.747687 7.92921e-09 0.752406 }
```

```
-----
RC Summary for net vic_net
-----
```

```
Number of capacitance : 3
Net capacitance : 0.0568429 pF
Total rise capacitance: 0.0572236 pF
Total fall capacitance: 0.0572328 pF
Number of resistance : 5
Total resistance : 15.6659 Ohm
-----
```

```
SI information for rise
-----
```

```
Victim clock : myclk
-----
```

```
Attacker Status Direction Slew Coupling cap Fanout Voltage Bump Timing window Clock
-----
```

```
att1 SY Fall 0.404581 ns 0.012500 pF 1 0.765 v 0.135482 v 1.123200 ns 1.123300 ns myclk
att2 SY Fall 0.205181 ns 0.010500 pF 1 0.765 v 0.125344 v 1.390900 ns 1.419600 ns myclk
-----
```

```
Model types
-----
```

```
Net Term Cell Model
-----
```

```
vic_net inv_inst/ZN INV CCS
vic_net ff2/D DFF CCS
att1 ZN INV CCS
att2 Q DFF CCS
-----
```

```
Underated Delays Rise
```

```
-----
Incr Net delay : 2.586069 ns
Net delay : 2.586721 ns
-----
```

```
Transitions Rise
-----
```

```
Input transition : 3.370655 ns
Output transition : 3.370683 ns
-----
```

```
Abbreviative attacker status and filtered reason:
-----
```

```
SY : Synchronous
INF: Attacker with infinite timing window
PE : Filtered due to physical exclusion
UF : Filtered by user
CN : Filtered because it is a constant net
SB : Filtered due to small bump
TA : Filtered because of timing window does not overlap
LC : Filtered due to logical correlation
ACT: Attacker is Active.
-----
```

- The following command generates a report that shows timing paths for which `set_max_delay` constraints are defined:

```
report_timing -check_type max_delay -max_paths 10 > 11
```

```
Path 1: MET (0.721 ns) Path Delay Check with Pin ff2/D
```

```
    Group: CK1
    Startpoint: (R) ff1/Q
      Clock: (R) CK1
    Endpoint: (R) ff2/D
      Clock: (R) CK1
```

	Capture	Launch
Path Delay:+	1.000	-
Src Latency:+	0.000	0.000
Net Latency:+	0.010 (P)	0.010 (P)
Arrival:=	1.010	0.010

```
Setup:-      0.100
Required Time:= 0.910
Launch Clock:= 0.010
Data Path:++ 0.179
Slack:=      0.721
```

```
-----
Pin      Edge  Direction  Phase      Arrival  Delay
                               Time
-----
```

Tempus Text Command Reference

Timing Analysis Commands

```

ff1/CK ^ IN CK1 (C) (P) * 0.010 -
ff1/Q ^ OUT CK1 (D) (P) * 0.013 0.003
buf4/Y ^ OUT CK1 (D) (P) * 0.023 0.010
buf5/Y ^ OUT CK1 (D) (P) * 0.033 0.010
and1/Y ^ OUT CK1 (D) (P) * 0.179 0.146
buf6/Y ^ OUT CK1 (D) (P) * 0.189 0.010
ff2/D ^ IN CK1 (D) (P) * 0.189 0.000
-----

```

Path 2: MET (0.987 ns) Path Delay Check with Pin out1

Group: default

Startpoint: (R) ff2/Q

Clock: (R) CK1

Endpoint: (R) out1

Clock:

	Capture	Launch
Path Delay:+	1.000	-
Src Latency:+	0.000	0.000
Net Latency:+	0.000 (I)	0.010 (P)
Arrival:=	1.000	0.010

Output Delay:- 0.000

Required Time:= 1.000

Launch Clock:= 0.010

Data Path:+ 0.003

Slack:= 0.987

```

-----
Pin      Edge  Direction  Phase          Arrival  Delay
              Time
-----
ff2/CK ^ IN CK1 (C) (P) * 0.010 -
ff2/Q ^ OUT CK1 (D) (P) 0.013 0.003
out1 ^ OUT CK1 (D) (P) * 0.013 0.000
-----

```

- In PBA, EPBA, and IPBA mode, the timing report shows PBA header for columns (by default):

```
report_timing -retime path_slew_propagation -path_type full_clock
```

Path 1: VIOLATED Late External Delay Assertion

Endpoint: o5 (^) checked with leading edge of 'CLK'

Beginpoint: ff3/Q (^) triggered by leading edge of 'CLK'

Analysis View: an_view_setup

Retime Analysis {Data Clock Path-Slew Spatial(CS) SOCV}

Retime SOCV Analysis { Distance(um) : 27.653 }

NSigma		Mean		Sigma		Stddev
--------	--	------	--	-------	--	--------

Tempus Text Command Reference
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```

Other End Arrival Time      0.0000 (-3.00S) | 0.0000 | 0.0000 | 0.0000
- External Delay            2.5000
+ Phase Shift              0.5000
+ CPPR Adjustment          0.0000 (+3.00S) | 0.0000 | 0.0000 | 0.0000
- Uncertainty              0.5000
= Required Time            -2.5000 (-3.00S) | -2.5000 | 0.0000 | 0.0000
- Arrival Time             18.4496 (+3.00S) | 18.4486 | 0.0004 | 0.0004
= Slack Time               -20.9496 (-3.00S) | -20.9486 | 0.0004 | 0.0004
= Slack Time(original)     -20.9484 (-3.00S) | -20.9484 | 0.0000 | 0.0000

    Clock Rise Edge                0.0000
    + Source Insertion Delay        1.0000 (+3.00S) | 1.0000 | 0.0000 | 0.0000
    = Beginpoint Arrival Time       1.0000 (+3.00S) | 1.0000 | 0.0000 | 0.0000

```

```

-----
Pin          Phase          Edge Delay      Delay Mean  Delay Stddev
Retime Delay Retime Delay Retime Delay  Arrival Time

    Mean          Stddev
-----
clk          CLK(C) (P) (an_view_setup)* ^ - - -
-            -            1.0000

clk_L1/A     CLK(C) (P) (an_view_setup)* ^ 0.0004 0.0004 0.0000
0.0014       0.0005       0.0003       1.0014

clk_L1/Y     CLK(C) (P) (an_view_setup)* ^ 0.0600 0.0600 0.0000
0.0600       0.0600       0.0000       1.0613

clk_L2/A     CLK(C) (P) (an_view_setup)* ^ 0.0001 0.0001 0.0000
0.0004       0.0002       0.0001       1.0613

clk_L2/Y     CLK(C) (P) (an_view_setup)* ^ 0.0604 0.0604 0.0000
0.0604       0.0604       0.0000       1.1217

ff3/CLK      CLK(C) (P) (an_view_setup)* ^ 0.0001 0.0001 0.0000
0.0011       0.0001       0.0003       1.1215

ff3/Q        CLK(D) (P) (an_view_setup)* ^ 17.3092 17.3092 0.0000
17.3091      17.3091      0.0000      18.4306

o5           CLK(D) (P) (an_view_setup)   ^ 0.0182 0.0182 0.0000
0.0197       0.0183       0.0005      18.4496

```


Related Commands

- `report_net`
- `report_ports`
- `set_multi_cpu_usage`
- `set_table_style`
- `set_input_delay`
- `set_max_delay`
- `set_min_delay`
- `set_analysis_mode`
- `create_analysis_view`

report_timing_context_clock_mapping

```
report_timing_context_clock_mapping  
[-help]  
[-format <text | csv>]  
[> <fileName>]  
[>> <fileName>]
```

Generates clock mapping report.

By default, Tempus generates the clock mapping report when the `read_timing_context` command is used. The `report_timing_context_clock_mapping` command can be used to generate a clock mapping report after `read_timing_context`. This command can generate clock mapping reports in either text format or CSV (which helps in easier post-processing of a report).

Parameters

<code>-help</code>	Prints out the command usage.
<code>-format <text csv></code>	Writes the report in text or csv formats, default is <code>text</code> .
<code>> <fileName></code>	Writes the report to the specified file name.
<code>>> <fileName></code>	Writes the report to the specified filename.

Examples

- The example below shows the `report_timing_context_clock_mapping` command output sample CSV report:

```
# View V1
# Inst I1
# Block Clock, Block Clock Period, Block Clock Waveform, Top Clock, Top Clock Period,
Top Clock Waveform, Port, Comment
# Block clocks mapped to top clocks on clock pins/ports:
BCLK1, 10, {0 5}, TCLK1, 10, {0 5}, CPORT1, auto mapped
BCLK2, 10, {0 5}, TCLK2, 20, {0 10}, CPORT2, user mapped
# Block Clocks NOT Mapped to top clocks on clock pin/ports:
BCLK3, 10, {0 5}, NA, NA, NA, CPORT3, No possible match by parameter
# Top clocks NOT mapped to block clocks on clock pins/ports:
NA, NA, NA, TCLK3, 10, {0 5}, CPORT4, No possible match by parameter
# Virtual block clocks NOT mapped to top clocks:
VCLK1, 10, {0 5}, NA, NA, NA, NA, NA
# Block clocks mapped to top clocks on data ports:
VCLK2, 10, {0 5}, TCLK4, 10, {0 5}, DPORT1, auto mapped
# Block clocks NOT mapped to top clocks on data ports:
VCLK2, 10, {0 5}, NA, NA, NA, DPORT2, No possible match by parameter
# Top clocks NOT mapped to block clocks on data ports:
NA, NA, NA, TCLK5, 10, {0 5}, DPORT3, No possible match by parameter
```

Related Commands

- `read_timing_context`
- `write_timing_context`
- `set_timing_context_view_mapping`
- `compare_timing_context`

report_timing_derate

```
report_timing_derate  
[-help]  
[-delay_corner delayCornerName ]  
[-include_inherited]  
[ object_list ]  
[{> | >>} filename [.gz]]
```

Reports delay scaling or derating factors for early and late clock and data paths in the current design.

By default, the `report_timing_derate` output shows `pulse_width_check` and `clock_period_check` details for instance objects (see `set_timing_derate`).

If you specify this command without any parameters, reports are generated for global derating factors for each instance, net, or cell for which derating is asserted.

The following reports are also generated:

- In the best case, worst case, maximum, and minimum corner
- Early values from the minimum corner and late values from the maximum corner in on-chip variation mode
- In MMMC mode, specified corners, or for all corners when a corner is not specified
- Power domains (if present in the design)
- Derates for pulse width and clock period checks. Also, the report will show mean/sigma values when the `set_timing_derate -mean/-sigma` parameters are specified.

Parameters

<code>-help</code>	Prints out the command usage.
<code>> >> filename</code>	When specified with <code>></code>, writes the report to the specified file name. If the file already exists, the software overwrites it. When specified with <code>>></code>, writes the report to the specified file name. If the file already exists, the software concatenates the report to the end of the file. Note: To write a compressed report, add the <code>.gz</code> extension to the file name.
<code>-delay_corner delayCornerName</code>	Reports the derating factors for the specified delay calculation corner only. <i>Default:</i> Reports derating factors for all delay calculation corners.
<code>- include_inherited</code>	Reports the object names from which the derating factor values are inherited. When specified, the report displays the name of the cell for each instance from which derate values are inherited.
<code>object_list</code>	Specifies the list of cells, nets, or library cells.

Examples

Here are some examples.

- The following commands display global derating factors:

```
tempus > set_timing_derate 1.2
tempus > set_timing_derate -clock -early 1.1
tempus > set_timing_derate -data -late 1.2
tempus > report_timing_derate
```

	-----Clock-----		-----Data-----	
	Early	Late	Early	Late
Cell Delay	1.100000	1.200000	1.200000	1.200000
Net Delay Static	1.100000	1.200000	1.200000	1.200000
Net Delay Dynamic	1.100000	1.200000	1.200000	1.200000
Cell Check	--	--	--	--
Retain delay	--	--	--	--

- The following commands display instance derating factors:

```
tempus > set_timing_derate -cell_delay -clock -late 1.3 [get_cells {u55}]
```

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```
tempus > set_timing_derate -cell_check -late 1.4 [get_cells {u55}]
tempus > report_timing_derate

---- Report Timing Derate for DC Corner: default_emulate_delay_corner
```

```
-----Clock-----Data-----
                Rise      Fall      Rise      Fall
                Early Late Early Late Early Late Early Late
-----
Cell (leaf): u55
Cell Delay      -      1.300  --      1.300  --      --      --      --
Cell Check      --      1.400  --      1.400  --      1.400  --      1.400
```

- The following commands display hierarchical derating factors:

```
tempus > set_timing_derate 1.2
tempus > set_timing_derate 1.1 -late h4
tempus > set_timing_derate 1.2 -early h4/h2
tempus > set_timing_derate -cell_check 1.2 h4
tempus > report_timing_derate -include_inherited
```

```
-----Clock-----Data-----
                Early      Late      Early      Late
-----
Cell Delay      1.200000      1.200000      1.200000      1.200000
Net Delay Static 1.200000      1.200000      1.200000      1.200000
Net Delay Dynamic 1.200000      1.200000      1.200000      1.200000
Cell Check      --              --              --              --
Retain delay    --              --              --              --

Cell (hier): h4
Cell Delay      1.200000      1.100000      1.200000      1.100000
inherited      top              --              top              --
Net Delay Static 1.200000      1.100000      1.200000      1.100000
inherited      top              --              top              --
Net Delay Dynamic 1.200000      1.100000      1.200000      1.100000
inherited      top              --              top              --
Cell Check      1.200000      1.200000      1.200000      1.200000
```

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Timing Analysis Commands

```

inherited      --              --              --              --

Cell (hier): h4/h2

Cell Delay      1.200000      1.100000      1.200000      1.100000
inherited      --              h4              --              h4
Net Delay Static 1.200000      1.100000      1.200000      1.100000
inherited      --              h4              --              h4
Net Delay Dynamic 1.200000      1.100000      1.200000      1.100000
inherited      --              h4              --              h4

```

- The following commands display best case worst case derating factors:

```

tempus > set_timing_derate -cell_delay -clock -late 1.2 [get_cells {f2}] -min
tempus > set_timing_derate -cell_delay -late 1.3 [get_cells {f2}] -max
tempus > report_timing_derate

```

```

-----Report Timing Derate for MIN corner-----
                -----Clock-----                -----Data-----
                Early          Late          Early          Late
                -----
Cell Delay      --              --              --              --
Net Delay Static --              --              --              --
Net Delay Dynamic --              --              --              --
Cell Check      --              --              --              --
Retain delay    --              --              --              --

```

```

Cell (leaf): f2
Cell Delay      --              1.200000      --              --

```

```

-----Report Timing Derate for MAX corner-----
                -----Clock-----                -----Data-----
                Early          Late          Early          Late
                -----
Cell Delay      --              --              --              --
Net Delay Static --              --              --              --
Net Delay Dynamic --              --              --              --
Cell Check      --              --              --              --
Retain delay    --              --              --              --

```

```

Cell (leaf): f2

```

```
Cell Delay      --                1.300000      --                1.300000
-----
```

- The following example shows a report that is displayed when sigma/mean derates are applied in the SOCV mode:

```
tempus > set_timing_derate 1.11 -mean -delay_corner [all_delay_corners ] [get_lib_cells
XOR3_X4M_A12TS_C35] -> derate set for mean on the lib cell XOR3_X4M_A12TS_C35
tempus > set_timing_derate 1.22 -sigma -delay_corner [all_delay_corners ] [get_lib_cells
XOR3_X4M_A12TS_C35] -> derate set for sigma on the lib cell XOR3_X4M_A12TS_C35
tempus > report_timing_derate [get_lib_cells XOR3_X4M_A12TS_C35] -> report_timing_derate
shows derate split in the format mean_derate:sigma_derate
```

```
---- Report Timing Derate for DC Corner: max_delay_corner
```

```
-----Clock-----
```

```
-----Data-----
```

		Rise				Fall			
Rise				Fall					
	Early			Late		Early		Late	Early
Late		Early			Late				

```
-----
LibraryCell: base_svt_c35Fsc12mc_cln28hpmPbcs30V1100T125/XOR3_X4M_A12TS_C35
```

```
Cell_delay      1.11000:1.22000 1.11000:1.22000 1.11000:1.22000 1.11000:1.22000
1.11000:1.22000 1.11000:1.22000 1.11000:1.22000 1.11000:1.22000
```

```
Retain_delay    --                --                --                --                --
--                --                --                --                --
```

```
Cell Check      --                --                --                --                --
--                --                --                --                --
```

Related Commands

- [set_timing_derate](#)
- [reset_timing_derate](#)

report_timing_lib

```
report_timing_lib  
[-help]  
-outfile fileName  
[-byLib]  
[-all]  
[-pin]  
[-threshold]  
[-sibling]  
[-lib library_name]  
[cellName]
```

Reports the contents of the timing library.

Parameters

-help	Prints out the command usage.
-all	Prints all the pin properties, siblings of a cell, the libraries where they are defined, and the voltage.
-byLib	Organizes the generated report by libraries. If you do not specify this parameter, the <code>report_timing_lib</code> command prints out cells with the same name together. Use this parameter to support the multi-supply multi-voltage feature of the software, where you can define the same cell names in more than one library with different voltage values and timing data.
<i>cellName</i>	Specifies the individual cell name to be reported.
-lib <i>library_name</i>	Reports a specified library. If you do not specify this parameter, all the library names are reported.
-outfile <i>fileName</i>	Specifies the name of the file to which the results are written.
-pin	Reports the pin properties, such as the pin name, direction, capacitance, max load, max fanout, and fanout load.
-sibling	Reports sibling cells, the libraries where they are defined, and the voltage.
-threshold	Reports the input thresholds, output thresholds, and slew thresholds of a cell.

Command Order

Use this command after importing the design when the timing library is also loaded.

Examples

- The following command reports the cell information in the timing library, and writes the results to the file `xyz.dump.lib`:

```
tempus > report_timing_lib -outfile xyz.dump.lib
```

Related Commands

- `read_lib`

report_timing_robustness

```
report_timing_robustness
[-help]
-check_type {setup | hold | setup_CGS | hold_CGH}
[-opt_list <string>]
[-path_group <string>]
-run_time_limit <integer>
-target_PTR <float>
[-view <string>]
```

Reports the PTR score of a design.

The Parametric Timing Robustness (PTR) analysis calculates the probability of an entire chip meeting timing requirements, instead of relying on the probabilities of individual paths to compute the robustness of a chip. This command reports the percentage of parts (to be manufactured) that meet timing at all the endpoints.

The PTR score of a design, which is a statistical measure of the timing slack for the whole design, is based on the following steps:

- Enumerates all the instances in a design
- Identifies paths that are critical for PTR of a design
- Computes the PTR based on Monte-Carlo analysis

Parameters

-help	Prints out the command usage.
-check_type {setup hold setup_CGS hold_CGH}	Specifies the check type.
-opt_list <string>	Specifies the PTR options list.
-path_group <string>	Specifies the user-defined path groups created using the group_path command.
-run_time_limit <integer>	Specifies the runtime limit. Unit in minutes.
-target_PTR <float>	Specifies the target PTR score.
-view <string>	Specifies the single view corresponding to the specified check type.

Examples

- The following examples show the calculated PTR value and the report output:

```
report_timing_robustness -check_type setup -path_group R2R -target_PTR 0.8 -  
run_time_limit 15  
collecting paths and reporting PTR .....  
Final PTR 0.89735
```

```
report_timing_robustness -check_type setup -path_group R2R -target_PTR 0.95 -  
run_time_limit 15  
collecting paths and reporting PTR .....  
Final PTR 0.91295
```

```
set P1 [report_timing_robustness -check_type setup -path_group R2R -target_PTR 0.95 -  
run_time_limit 15]  
collecting paths and reporting PTR .....  
Final PTR 0.91295
```

```
report_property $P1  
-----  
property                | value  
-----  
slack_mean              | 0.961705  
slack_ptr               | 0.912950  
slack_quantile          | -2.085161  
slack_skewness          | -0.668439  
slack_stddev            | 0.682927  
-----
```

Related Commands

- [report_timing](#)
- [group_path](#)

Related Information

- [Parametric Timing Robustness Analysis](#) section in the *Tempus User Guide*.

report_wire_load

```
report_wire_load  
[-help]  
[-inst name_or_collection_of_hierarchical_instances ]  
[-ports name_or_collection_of_ports ]  
[-late | -early]  
[> | >> filename ]
```

Reports the wireload name, library, wire load mode, and selection type for all hierarchical instances in the design.

Note: To use the wireload model functionality, you must enable the `enable_wire_load_model_support` global variable using the following command:

```
set_var enable_wire_load_model_support 1
```

Parameters

<code>> >> filename</code>	When specified with <code>></code>, writes the report to the specified file name. If the file already exists, the software overwrites it. When specified with <code>>></code>, writes the report to the specified file name. If the file already exists, the software concatenates the report to the end of the file. The <i>filename</i> parameter must be the last argument in the list. Note: To write a compressed report, add the <code>.gz</code> extension to the file name. <i>Default:</i> Report is displayed on standard output without being saved.
<code>-early</code>	Reports the wireload model information for the best-case condition.
<code>-inst</code> <code>name_or_collection_of_hierarchical_instances</code>	Specifies the name of an instance or collection for which the <code>report_wire_load</code> command reports the wireload name, library, wire load mode, and selection type. The <code>report_wire_load</code> command also supports wildcard names such as <code>U*</code>.
<code>-late</code>	Reports the wireload model information in the worst-case condition. <i>Data_type:</i> bool, optional
<code>-ports name_or_collection_of_ports</code>	Reports wireload information for the specified ports.

Examples

- The following example shows the default report when wireload mode is set to top:

```
tempus> report_wire_load

-----

Design Name top_chip
Wire Load Mode top

Instance Module WireLoadModel Library SelectionType
-----

- top_chip model4 fast library-default
-----
```

- The following example shows the report when wireload mode is set to enclosed to report lower hierarchy:

```
tempus> set_wire_load_mode enclosed
tempus> report_wire_load

-----

Design Name top_chip
Wire Load Mode enclosed
Instance Module WireLoadModel Library SelctionType
-----

- top_chip model4 fast library-default
U1 randr model4 fast library-default
U2 randr model4 fast library-default
-----
```

- The following example shows model assertions using the set_wire_load_model command the corresponding reports using the report_wire_load command:

```
tempus> set_wire_load_model -name model3
tempus> report_wire_load

-----

Design Name top_chip
Wire Load Mode enclosed
Instance Module WireLoadModel Library SelctionType
-----

- top_chip model3 fast user-asserted
U1 randr model3 fast user-asserted
U2 randr model3 fast user-asserted
-----
```

```
tempus> set_wire_load_model -name model2 [get_cells U1]
tempus> set_wire_load_model -name model4 [get_cells U2]
tempus> report_wire_load
```

```
-----
Design Name top_chip
Wire Load Mode enclosed
Instance Module WireLoadModel Library SelctionType
-----
- top_chip model3 fast user-asserted
U1 randr model2 fast user-asserted
U2 randr model4 fast user-asserted
-----
```

- The following example shows the report including port information:

```
tempus> report_wire_load -ports [get_ports XYZ/A]
-----
Design Name    randr
Wire Load Mode enclosed
Port      Module WireLoadModel Library      SelctionType
-----
P1        randr model2                slow    user-asserted
P2        randr model4                slow    user-asserted
-----
```

Related Commands

- [set_wire_load_model](#)
- [enable_wire_load_model_support](#)
- [set_analysis_mode](#)

reset_instance_library

```
reset_instance_library  
[-help]
```

Resets the settings/bindings done using the `set_instance_library` command.

Parameters

<code>-help</code>	Prints out the command usage.
--------------------	-------------------------------

Related Commands

- `set_instance_library`

reset_path_adjust

```
reset_path_adjust  
[-help]  
-path_adjust_group path_adjust_group_name  
-view string  
[-setup | -hold]
```

Removes the specified slack adjustment values for timing paths that are set using the `set_path_adjust` command.

Parameters

-help	Prints out the command usage.
-hold	Resets the adjust value for early paths.
-path_adjust_group <i>path_adjust_group_name</i>	Specifies the name of the path adjust group that defines the path for which the adjustment will be applied.
-setup	Resets the adjust value for late paths.
-view <i>string</i>	Specifies the view for which the adjustment will be reset.

Related Commands

- `set_path_adjust`

reset_path_adjust_group

```
reset_path_adjust_group  
[-help]  
[-name group_name | -all ]
```

Removes all path adjust groups that are set using the [set_path_adjust_group](#) command for setting path slack adjustment values.

Parameters

-help	Prints out the command usage.
-all	Removes all the path adjust groups.
-name <i>group_name</i>	Removes the named path adjust group completely.

Related Commands

- [set_path_adjust_group](#)
- [set_path_adjust](#)

reset_sdf_assertions

```
reset_sdf_assertions
```

```
[-help]
```

Resets the SDF delays to the values defined in the library. When you perform timing analysis, the software uses the SDF values defined with the `read_sdf` command.

You can use this command to discard the delay values specified with the SDF files and use the delays defined in the library.

Note: You do not need to read in the SDF delays every time you perform delay calculations. The software uses the SDF files specified earlier in the flow.

Parameters

-help	Prints out the command usage.
-------	-------------------------------

Related Commands

- `read_sdf`

reset_timing_derate

```
reset_timing_derate
[-help]
[-delay_corner delayCornerName]
[object_list]
[-increment]
[-transition]
```

Resets the derate factors specified on a design or a list of instances (cells, net, or library cells). All the derating factors that are set globally or on specific instances are reset to a default value of 1.0. The derating factors for early or late paths in the design are set using the `set_timing_derate` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-delay_corner</code> <i>delayCornerName</i>	Specifies the delay corner or a list of delay corners on which reset derating is to be applied.
<code>-increment</code>	Resets the incremental derate values that are set using the <code>set_timing_derate</code> command. The <code>set_timing_derate</code> command saves two different values – the “base” derate and the “incremental” derate. When the <code>-incremental</code> parameter is used, the software populates the incremental derates. The <code>reset_timing_derate</code> command will reset the incremental derate values only.
<i>object_list</i>	Specifies the list of cells, nets, or library cells to reset. If the <i>object_list</i> contains hierarchical cells, then all cells within the hierarchical cells and across lower hierarchies have reset derate factors.
<code>-transition</code>	Resets the transition derate values.

Examples

- The following example resets all the object-specific and global derating factors:
tempus> `reset_timing_derate *`
- The following example resets only global derating factors:
tempus> `reset_timing_derate [current_design]`
- The following example resets the derating factor values for the specified delay corner:

```
tempus> reset_timing_derate -delay_corner delayCornerName
```

- The following example resets the global derating values for the specified delay corner:

```
tempus> reset_timing_derate -delay_corner delayCornerName [current_design]
```

- The following example resets the timing derate on instance `u1`:

```
tempus> reset_timing_derate [get_cells u1]
```

Related Commands

- [set_timing_derate](#)

reset_wire_load_mode

```
reset_wire_load_mode  
[-help]
```

Resets the wire-load mode to the default mode of `top`.

Note: To use the wireload model functionality, you must enable the `enable_wire_load_model_support` global variable using the following command:

```
set_var enable_wire_load_model_support 1
```

Related Commands

- `set_wire_load_mode`

reset_wire_load_model

```
reset_wire_load_model  
[-help]  
[list_of_instances_or_ports]
```

Resets wireload models for the specified list of objects. The list of objects can include designs, hierarchical cells, or ports.

When objects are not specified, the software removes the wireload model specified using the `set_wire_load_model` command from the current instance only.

Note: To use the wireload model functionality, you must enable the `enable_wire_load_model_support` global variable using the following command:

```
set_var enable_wire_load_model_support 1
```

Parameters

<code>-help</code>	Prints out the command usage.
<code>list_of_instances_or_ports</code>	Specifies the list of instances or ports for which wireload models will be reset.

Examples

- The following command removes the wireload model for `clk1`:

```
tempus> reset_wire_load_model clk1
```

Related Commands

- `set_wire_load_model`

reset_wire_load_selection_group

```
reset_wire_load_selection_group  
[-help]  
[{ list_of_objects }]
```

Removes the wireload selection group for the specified list of objects. The list of objects can include designs, hierarchical cells, or ports. When objects are not specified, wireload selection groups are removed for all objects.

Note: To use the wireload model functionality, you must enable the `enable_wire_load_model_support` global variable using the following command:

```
set_var enable_wire_load_model_support 1
```

Parameters

<code>-help</code>	Prints out the command usage.
<code>list_of_objects</code>	Specifies the list of object names for which to remove the wireload selection groups.

Example

- The following command removes the wireload selection group for `clk1`:

```
tempus> reset_wire_load_selection_group clk1
```

Related Commands

- `set_wire_load_selection_group`

select_boundary_model_constraints

```
select_boundary_model_constraints  
[-help]  
[-block string]  
[-define_bundled_bus string]  
[-instance string]  
[-set_instance_library <string>]  
[-set_quiet_attacker string]
```

Applies boundary model constraints to specific blocks/instances.

Parameters

-help	Prints out the command usage.
-block <i>string</i>	Specifies the block for which the constraints will be applied.
-define_bundled_bus <i>string</i>	Applies the <code>define_bundled_bus</code> constraints.
-instance <i>string</i>	Specifies the instance for which the constraints will be applied.
-set_instance_library <string>	Applies the instance library constraints defined with the <code>set_instance_library</code> command.
-set_quiet_attacker <i>string</i>	Applies the <code>set_quiet_attacker</code> constraints.

Related Commands

- `report_boundary_model_constraints`
- `define_bundled_bus`
- `set_quiet_attacker`
- `set_instance_library`

set_analysis_mode

```
set_analysis_mode
[-help]
[-aging {true | false}]
[-aocv {true | false}]
[-enable_ipd {true | false}]
[-reset]
[-skew {true | false}]
[-checkType {setup | hold}]
[-warn {true | false}]
[-clockPropagation {sdcControl | forcedIdeal | autoDetectClockTree}]
[-asyncChecks {async | noAsync | asyncOnly}]
[-clkSrcPath {true | false}]
[-useOutputPinCap {true | false}]
[-timing_robustness {true | false}]
[-timeBorrowing {true | false}]
{-graph_pba {true | false}}
[-honorClockDomains {true | false}]
[-log {true | false}]
[-multi_input_switching_mode {0 | 1 | 2 | 3}]
[-propSlew {true | false}]
[-clockGatingCheck {true | false}]
[-sequentialConstProp {true | false}]
[-timingSelfLoopsNoSkew {true | false}]
[-analysisType {single | bcwc | onChipVariation}]
[-socv {true | false}]
[-cpr {none | both | setup | hold}]
[-timingEngine {statistical | static}]
```

Sets global analysis modes for timing analysis. The software uses these modes for all timing analysis commands unless you specify specific modes in the reporting commands. The modes that you specify in the reporting commands override these modes.

When you use the `set_analysis_mode` command, the software sets the following defaults:

- Sets the constants to be propagated while building the timing graph. The software reads the constants from the timing constraints file or the netlist.
- Sets the analysis mode to accept the module-level timing constraints of the internally specified `set_input_delay` and `set_output_delay` variables for the I/Os of the module for running timing analysis and timing optimization.
The software applies the `set_input_delay` command and the `set_output_delay` command on internal pins during timing analysis.

- Treats the latches as latches.
- Reports the end points as many times as the number of clocks that control that end point.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-aging {true false}</code>	Enables timing-aware aging analysis. The aging feature must be enabled before the timing updates are done. Note: You cannot disable this feature in the same session later. <i>Default:</i> <code>false</code>
<code>-analysisType {single bcwc onChipVariation}</code>	Sets the timing analysis type to single, best case, worst case, or on-chip variation. • <code>single</code>: Scales the delay values based on one operating condition. • <code>bcwc</code>: Checks the design for two extreme operating conditions. The software uses the maximum delays for all paths during setup checks and the minimum delays for all paths during hold checks. • <code>onChipVariation</code>: Calculates the delay for one path based on the maximum operating condition, while calculating the delay for another path based on the minimum operating condition for setup or hold checks. <i>Default:</i> <code>onChipVariation</code>
<code>-aocv {true false}</code>	Sets the analysis mode to advanced on-chip variation (AOCV) analysis.

<pre>-asyncChecks {async noAsync asyncOnly}</pre>	Reports timing violations for asynchronous timing checks, including recovery and removal, and other checks on pins defined as preset or clear in the Liberty file. • <code>async</code>: Reports timing checks on pins that are defined as preset or clear in the Liberty file. • <code>noAsync</code>: Reports timing violations while ignoring the two time arc attributes. • <code>asyncOnly</code>: Reports only recovery and removal checks. <i>Default:</i> <code>async</code>
<pre>-checkType {setup hold}</pre>	Checks the design for setup or hold violations in the current analysis. <i>Default:</i> <code>setup</code> Note: When simultaneous setup/hold mode analysis is on, this parameter does not have any impact. You can use the <code>report_timing -late/-early</code> parameter to report setup/hold timing reports.
<pre>-clkSrcPath {true false}</pre>	Reports combinational paths from the clock source to the timing checks on data pins. Note: If <code>-clkSrcPath false</code> is specified and the clock portion of the path is considered as ideal (due to the SDC constraints or the <code>-clockPropagation forcedIdeal</code> parameter setting), the delay calculation software treats the clock nets as ideal nets when calculating the delays and slews along the data path. For all pre-clock tree synthesis (CTS) steps, the software uses <code>-clkSrcPath false</code> in combination with <code>-clockPropagation forcedIdeal</code> to calculate the delays and slews along the data path. For all post-CTS flow steps, the software uses the default <code>-clkSrcPath true</code> in combination with <code>-clockPropagation sdcControl</code> to calculate the delays and slews along the data path. <i>Default:</i> <code>true</code>
<pre>-clockGatingCheck {true false}</pre>	Reports the gating checks. <i>Default:</i> <code>true</code>
<pre>-clockPropagation {sdcControl forcedIdeal autoDetectClockTree}</pre>	Specifies how to determine whether clock end points are interpreted as having either ideal or propagated clock timing. <i>Default:</i> <code>sdcControl</code>

	• <code>sdcControl</code>: Reports the timing violations considering clock tree and enables propagating clocks to report clock latency. You must set the <code>set_propagated_clock</code> constraint to propagate the clocks.
	• <code>forcedIdeal</code>: Does not report the timing violations considering the clock tree.
	• <code>autoDetectClockTree</code>: This parameter is obsolete. The timing system will revert to <code>sdcControl</code> if this parameter is specified.
<code>-cprr {none both setup hold}</code>	Removes pessimism from clock paths that have a portion of the clock network in common between the clock source and clock destination paths. The pessimism is introduced when the timing analysis tools assume that the common path has different delay values for two different paths in the case of on-chip variation. The pessimism can also be because of clock reconvergence in the clock network. The pessimism affects all analysis modes, including single, best case-worst case (bcwc), and on-chip variation. <i>Default:</i> none
	• <code>none</code>: Disables removal of clock reconvergence pessimism.
	• <code>both</code>: Enables removal of clock reconvergence pessimism for both setup and hold modes. When <code>set_analysis_mode -cprr</code> is specified (without any arguments), the value is set to <code>both</code>.
	• <code>setup</code>: Enables removal of clock reconvergence pessimism in setup mode only.
	• <code>hold</code>: Enables removal of clock reconvergence pessimism in hold mode only.
<code>-enable_ipd {true false}</code>	Enables IPD analysis. For more information, refer to the Inter Power Domain (IPD) Analysis chapter in the <i>Tempus User Guide</i>.
<code>-graph_pba {true false}</code>	When specified, the software performs partial PBA analysis during GBA to reduce GBA pessimism.

<code>-honorClockDomains {true false}</code>	Honors clock domains. <i>Default:</i> false
<code>-log {true false}</code>	Writes warning messages to the log file. <i>Default:</i> true
<code>-multi_input_switching_mode {0 1 2 3}</code>	Sets the multiple input switching (MIS) mode. You can specify one of the following options. When set to: • 0: Simultaneous input switching is not considered. • 1: MIS analysis considers cells for which multiple inputs of a cell are tied together (tied-inputs). • 2: MIS analysis considers both the tied-inputs and the inputs of cells switching independent to each other. • 3: Enables derate-based MIS. When MIS-derate is enabled, Tempus will automatically disable AMIS-simulation. <i>Default:</i> 0
<code>-propSlew {true false}</code>	Controls whether slews in early path delay calculation are degraded as they pass across the interconnect from driver to receiver. You can specify a value of false to turn off interconnect slew degradation for early paths and have the driver slews copied to the receivers. Early path delay calculation affects all Hold paths in best-case/worst-case analysis mode, captures clock paths in on-chip variation Setup analysis, and clock launch and data paths in on-chip variation Hold analysis. <i>Default:</i> true
<code>-reset</code>	Resets parameters to their default values. The <code>-reset</code> parameter <i>must</i> be the first parameter specified. If you specify <code>-reset</code> by itself, the software resets all <code>set_analysis_mode</code> parameters to their default values. If you specify parameters after <code>-reset</code>, the software resets only those parameters to their default values.
<code>-sequentialConstProp {true false}</code>	Propagates constants through sequential elements. If you specify <code>-sequentialConstProp false</code>, constants will stop at sequential elements. <i>Default:</i> false

<pre>-skew {true false}</pre>	Reports clock skew. The <code>-skew false</code> setting treats all clock latencies as 0. <i>Default:</i> <code>true</code> With the <code>-skew true</code> setting, you can specify the <code>-clockPropagation</code> parameter. • Using the default <code>-skew true -clockPropagation sdcControl</code> setting accounts for the clock latency during timing analysis and proceeds as follows: ◦ If the <code>set_propagated_clock</code> command is specified in the timing constraint file, the clock latency is computed from the netlist (propagated mode). ◦ If the <code>set_clock_latency</code> command is specified in the timing constraint file, the clock latency specified in the constraint is used. Otherwise, no latency is reported (ideal mode)
<pre>-socv {true false}</pre>	Enables SOCV analysis. <i>Default:</i> <code>false</code>
<pre>-timeBorrowing {true false}</pre>	Considers time borrowing during timing analysis. Time borrowing is the amount of time borrowed by a previous logic. <i>Default:</i> <code>true</code>
<pre>-timingEngine {statistical static}</pre>	Specifies the type of timing analysis to perform. • <code>statistical</code>: Sets to analysis mode to statistical static timing analysis (SSTA). • <code>static</code>: Performs static timing analysis. <i>Default:</i> <code>static</code>
<pre>-timing_robustness {true false}</pre>	Enables PTR analysis. The Parametric Timing Robustness (PTR) analysis calculates the probability of an entire chip meeting timing requirements, instead of relying on the probabilities of individual paths to compute the robustness of a chip. <i>Default:</i> <code>false</code>

<code>-timingSelfLoopsNoSkew {true false}</code>	Eliminates clock skew due to clock uncertainty for a path starting and ending at the same register. If the clock skew is not eliminated, the timing for such paths is pessimistic. Note: Use this parameter before building the timing graph. <i>Default:</i> false
<code>-useOutputPinCap {true false}</code>	Includes the output pin capacitance when calculating delay values. When you specify <code>-useOutputPinCap false</code>, the software excludes only the pin capacitance of the driver for which you are calculating the delay. <i>Default:</i> true
<code>-warn {true false}</code>	Displays warning messages as the timing constraints file is read. <i>Default:</i> true

Examples

- The following command sets timing analysis to report clock skew, suppresses warning messages, and checks for setup timing:

```
tempus> set_analysis_mode -checkType setup -skew true -warn false
```
- The following command sets the timing analysis type to onChipvariation, with cpr enabled for both setup as well as hold checks:

```
tempus> set_analysis_mode -analysisType onChipVariation -cpr both
```
- The following command enables the advanced on-chip variation mode in the design:

```
tempus> set_analysis_mode -aocv true
```
- The following command disables the clock-gating checks:

```
tempus> set_analysis_mode -clockGatingCheck false
```
- The following command forces the tool to treat all the clocks as ideal networks, even if the clocks are defined as propagated in the constraints:

```
tempus> set_analysis_mode -clockPropagation forcedIdeal
```
- The following command results in the propagation of constants through the sequential elements:

```
tempus> set_analysis_mode -sequentialConstProp true
```
- The following command disables the time borrowing in the design:

```
tempus> set_analysis_mode -timeBorrowing false
```

Related Commands

- `get_analysis_mode`
- `report_analysis_mode`
- `report_timing`
- `set_aging_analysis_mode`
- `set_aging_activity`
- `create_aging_stress_profile`
- `create_spice_deck`
- `set_ipd_rail_voltage`
- `create_pg_net_group`
- `create_ipd_configuration`
- `report_ipd_configuration`
- `write_ipd_scope`
- `read_ipd_scope`

set_analysis_view

```
set_analysis_view
[-help]
[-drv string]
[-dynamic string]
[-ir string]
[-leakage string]
[-update_timing]
[-setup string | -hold string]
```

Defines the analysis views to use for setup and hold analysis and optimization. You must define at least one setup and one hold analysis view. The "active" views represent the different design variations that will be analyzed. These views can be changed throughout the flow to utilize different subsets of views. The libraries and data are loaded into the system that are required to support the selected set of active views.

When the `set_analysis_view` command is issued, it will cause a full timing reset. During a timing reset, all the interconnect parasitics, delays, and timing slacks will be recalculated.

The order in which you specify the views with the `-setup` and `-hold` parameters is important. By default, the first views defined in the `-setup` and `-hold` lists are the default views. The default views serve two purposes:

- Provide the library data that is used during the design initialization process.
- Indicate to parts of the application that are not MMMC-aware, which view's information to use for a particular analysis step (in most cases, the default view is the first setup or hold view). You can use the `set_default_view` command to change the default view to any currently active view without incurring a timing reset. This control is not frequently used since most applications either support MMMC or provide a direct control for selecting the appropriate view to use for that specific analysis.

Using the `set_default_view` command does not affect software performance because it only uses views that are already active in the design. If you use the `set_analysis_view` command to change the default views, the existing timing and delay calculation data is reset.

Use this command after creating multiple analysis views (`create_analysis_view`).

Note: The `set_analysis_view` command does not work in an incremental manner for setting parameters. Each invocation of this command must include the full specification of all the active views required for analysis.

Parameters

-help	Prints out the command usage.
-------	-------------------------------

<code>-drv string</code>	Selects specific views for DRV analysis, e.g., <code>max_transition</code> and <code>max_capacitance</code> checking. When this parameter is specified, additional analysis views are specified for DRV analysis, so the timing system efficiently computes the required data for these checks. This methodology is efficient with respect to memory and performance, because not all the views, as full Setup views, are included. The following guidelines should be followed when selecting DRV views: • Any setup views that require DRV analysis should not be specified as DRV views. The setup views are automatically considered for DRV. • If a subset of hold-only views requires DRV analysis, they should be specified as DRV views as well. The timing system will augment the timing information normally computed for Hold views with additional data required for the DRV analysis. • Additional views that are not active setup or hold views can be specified as DRV only views. The system will efficiently compute the required data for performing accurate DRV checks.
<code>-dynamic string</code>	Sets a view as a dynamic view. Delay calculation is required to provide transition times. Delay calculation should not perform SI analysis if the delay corner is only required for the dynamic power. The timer is not required to propagate required times or perform slack computations.
<code>-hold string</code>	Sets the active views for hold analysis.
<code>-ir string</code>	Specifies the IR view.
<code>-leakage string</code>	Sets a view as a leakage view. The timing libraries must be loaded, if not already in use. The Timer constant propagation is required, whereas delay calculation is not required.

<code>-setup <i>string</i></code>	Sets the active views for setup analysis.
<code>-update_timing</code>	Reloads the configuration described by the MMMC view and mode objects. When the timing global <code>timing_defer_mmmc_object_updates</code> is set to <code>true</code>, this parameter is required to update the MMMC timing environment. The <code>-update_timing</code> parameter may be specified without any <code>-setup</code> or <code>-hold</code> arguments. In this case, the data associated with the current set of setup and/or hold views will be loaded. If no active setup or hold views exist, then using this parameter alone may display an error. The <code>-update_timing</code> parameter may also be used in conjunction with the <code>-setup</code> and <code>-hold</code> options. In this case, the newly specified setup/hold view configuration will be loaded with the updated MMMC object information.

When you issue the `set_analysis_view` command, the software will perform a timing reset, so that only the new active view information is loaded. The software requires the `-setup` and `-hold` view parameters to be explicitly specified for each `set_analysis_view` command run. If these parameters are not specified, the software issues an error message. Any optional settings, such as `-leakage` or `-dynamic` views, will be reset if they are not specified with the new `set_analysis_view` command.

Note: It is recommended to use the same power view for both leakage and dynamic, that is, both the `-dynamic` and `-leakage` parameters should have the same user-specified view settings.

Examples

- The following command sets missionWCCOM and mission2WCCOM as the active views for setup analysis, and missionBCCOM and testBCCOM as the active views for hold analysis:

```
tempus> set_analysis_view -setup {missionWCCOM mission2WCCOM} -hold {missionBCCOM
testBCCOM}
```

Related Commands

- `create_analysis_view`
- `update_analysis_view`
- `set_default_view`

set_aocv_thresholds

```
set_aocv_thresholds  
[-help]  
[-min_stage_count_setup int]  
[-min_stage_count_hold int]  
[-reset]  
[-slack_pruning_threshold float]  
-view list_of_views
```

Limits the minimum AOCV stage depth in graph-based analysis (GBA) mode.

Parameters

-help	Prints out the command usage
-min_stage_count_setup <i>int</i>	Applies the AOCV stage count floor during Setup analysis, so that all the instances are guaranteed to have a stage depth no less than the specified value. A value of '0' will remove the assertion.
-min_stage_count_hold <i>int</i>	Applies the AOCV stage count floor during Hold analysis, so that all the instances are guaranteed to have a stage depth no less than the specified value. A value of '0' will remove the assertion.
-reset	Removes/sets to default all of the AOCV threshold assertions for the specified views.
- slack_pruning_threshold <i>float</i>	Controls the slack threshold (in two-pass/slack-based AOCV analysis mode), at which paths are removed from consideration for the next phase of AOCV stage counting.
-view <i>list_of_views</i>	Applies the specified thresholds to the list of analysis views.

set_boundary_model_clock_mapping

```
set_boundary_model_clock_mapping  
[-help]  
-block_clock <string>  
-instance <string>  
-top_clock <string>  
[-view <string>]
```

Provides the boundary model view mapping data.

Parameters

-help	Prints out the command usage.
-block_clock <string>	Specifies the block view.
-instance <string>	Specifies the name of the instance.
-top_clock <string>	Specifies the name of the top clock.
-view <string>	Specifies the name of the view.

Related Commands

- [read_boundary_model](#)
- [set_boundary_model_view_mapping](#)

set_boundary_model_view_mapping

```
set_boundary_model_view_mapping  
[-help]  
-block_view <string>  
-top_view <string>
```

Provides the boundary model view mapping data.

Parameters

-help	Prints out the command usage.
-block_view <string>	Specifies the block view.
-top_view <string>	Specifies the top view.

Examples

- The following command specifies the block view name as `bview`, and the corresponding view name at the top level is `tview`:

```
set_boundary_model_view_mapping -block_view bview -top_view tview
```

Related Commands

- `read_boundary_model`
- `set_boundary_model_clock_mapping`

set_default_view

```
set_default_view
[-help]
{[-hold newDefaultHoldView] | [-setup newDefaultSetupView]}
```

Temporarily changes the default active analysis view to a different active view.

Some applications can function only on a single analysis view at a time. By default, these single-view applications use the default analysis view. If an application or flow step does not provide a native option for specifying which view to use, the `set_default_view` command can be used to temporarily change the default view to a different active view that is better suited to that flow step.

Note: Using the `set_default_view` command does not affect software performance because it only uses views that are already active in the design. If you use the `set_analysis_view` command to change the default views, the existing timing, delay calculation, and RC data is reset.

Use this command after defining active views for the design (`set_analysis_view`).

Parameters

<code>-help</code>	Prints out the command usage.
<code>-hold newDefaultHoldView</code>	Changes the default active hold analysis view to the specified analysis view.
<code>-setup newDefaultSetupView</code>	Changes the default setup analysis view to the specified analysis view.

Examples

- The following command sets `missionWCCOM`, `mission2WCCOM`, and `missionSlow` as the active views for setup analysis, and `missionBCCOM` and `testBCCOM` as the active views for hold analysis. The default active setup view is `missionWCCOM`, and the default active hold view is `missionBCCOM`.

```
tempus> set_analysis_view -setup {missionWCCOM mission2WCCOM missionSlow} -hold
{missionBCCOM testBCCOM}
```

- The following command changes the default active setup view to `missionSlow`:

```
tempus> set_default_view -setup missionSlow
```

- The following command changes the default active hold view to `testBCCOM`:

```
tempus> set_default_view -hold testBCCOM
```

Related Commands

- `create_analysis_view`
- `set_analysis_view`

set_exclude_net

```
set_exclude_net  
[-help]  
{-file filename | -template}
```

Specifies an exclude net file to use during delay calculation. Normally, nets that exceed the default value of 1,000 pins are excluded from delay calculation, and default delay values are used. If you specify an exclude net file, nets (including hierarchical nets) listed in the file are excluded from calculation, and their corresponding delay values specified in the file are used. Nets usually specified in the file include high-fanout nets that increase the run time of the program.

The exclude net file can have one of two formats:

- *netName netDelay netOutputTran netLoad*
- *netName netDelay*

Parameters

-help	Prints out the command usage.
-file <i>filename</i>	Specifies the name of the excluded net file.
-template	Generates a sample file that contains the net and delay formats to use to specify the file.

Examples

- The following command sets an exclude net file named `myfile.exc`:

```
tempus> set_exclude_net -file myfile.exc
```

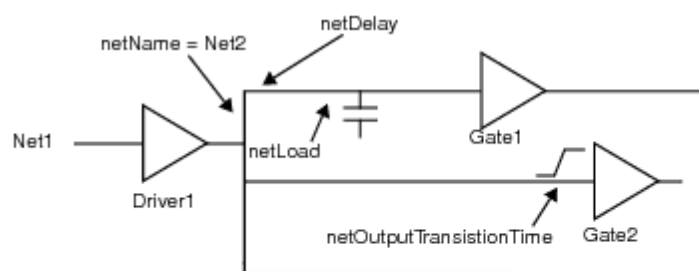
Each of the excluded nets uses the corresponding delay value specified in the `myfile.exc` file.

- The following command creates a sample file that contains net and delay formats to use to create an exclude file:

```
tempus> set_exclude_net -template
```

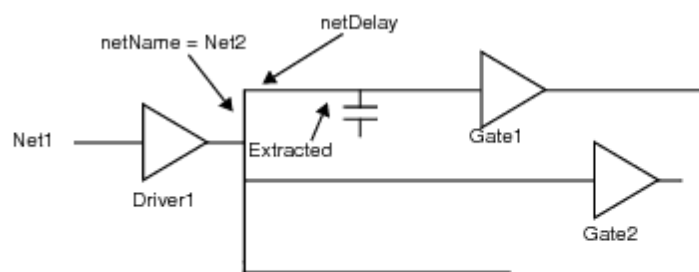
The following figures show the usage of the exclude net file format:

Format 1: <netName> <netDelay> <netOutputTran> <netLoad>



- ☐ Driver delay calculated using `netLoad` value.
- ☐ Gate delay calculated using `netOutputTran` value.
- ☐ Default delay values are used for remaining nets with a terminal count greater or equal to 1000.

Format 2: <netName> <netDelay>



- ☐ Driver delay calculated using extracted load value.
- ☐ Gate delay calculated using extracted load value.
- ☐ Default delay values are used for remaining nets with a terminal count greater or equal to 1000.

The timing units for the file are:

Symbol	Unit
fs, Fs	10^{-15} Second
ps, Ps	10^{-12} Second
ns, Ns	10^{-9} Second
us, Us	10^{-6} Second
ms	10^{-3} Second

set_instance_library

```
set_instance_library  
[-help]  
-library string  
-instance string  
[-early | -late]
```

Binds a cell library to an instance. When specified, the software overrides the default binding mechanism for a cell from the library set.

Note: The `set_instance_library` command cannot be used at the command prompt - this command can be specified in the ViewDefinition file. For more details, refer to the *Tempus User Guide*.

Parameters

-help	Prints out the command usage.
-early	Uses the library binding for early timing analysis.
-instance <string>	Specifies the instance for which the library binding needs to be done.
-late	Uses the library binding for late timing analysis.
-library <string>	Specifies the library file path.

Examples

The following commands specify the instance for which the library binding will be done:

- `set_instance_library -library newcell.lib -instance {i1}`
- `set_instance_library -library mysil.lib -instance {i1 i2}`
- `set_instance_library -library {new.lib mysil.lib} -instance {i1} -early`
- `set_instance_library -library {mynew.lib mysil.lib} -instance {i1 i2} -late`

Related Topics

For more information, refer to the [Set Instance Library Flow](#) section in the *Tempus User Guide*.

Related Commands

- `reset_instance_library`

set_interactive_constraint_modes

```
eset_interactive_constraint_modes  
[-help]  
{list_of_constraint_modes}
```

Puts the software into interactive constraint entry mode for the specified multi-mode multi-corner constraint mode objects. Any timing constraints that you specify after this command take effect immediately on all active analysis views that are associated with the specified constraint modes. By default, no constraint modes are considered active.

All new assertions are saved in the SDC files of the specified constraint modes when you save the design.

The `all_constraint_modes` command can be used to generate a list of constraint modes as the argument for this command.

Use the `get_interactive_constraint_modes` command to return a list of the constraint mode objects in constraint entry mode.

Note: The interactive constraint mode works only when the software is in a regular multi-mode multi-corner timing-analysis mode. In distributed multi-mode multi-corner analysis, the same can be used in interactive mode as follows:

```
set_distribute_mmmc_mode -interactive true  
set_interactive_constraint_modes [get_analysis_view \  
[all_setup_analysis_view] -constraint_mode]  
## apply constraints ###  
set_input_delay -clock CK1 0.2 {EN}  
set_case_analysis 0 SEL  
## end of constraints ###  
set_distribute_mmmc_mode -interactive false
```

When interactive constraints are run using scripts in the DMMMC batch mode, the following can be used:

```
distribute_command { set_interactive_constraint_modes \  
[get_analysis_view [all_setup_analysis_view] -constraint_mode]  
## apply constraints ###  
set_input_delay -clock CK1 0.2 {EN}  
set_case_analysis 0 SEL  
## end of constraints ###  
}
```


Parameters

<code>-help</code>	Prints out the command usage.
<code>{list_of_constraint_modes}</code>	Specifies the constraint mode objects to update with the new assertions.

Examples

- The following commands put the software into interactive constraint entry mode, and apply the `set_propagated_clock` assertion on all views in the current session that are associated with the constraint mode `func1`:

```
tempus> set_interactive_constraint_modes func1
tempus> set_propagated_clock [all_clocks]
```

When you specify `saveDesign`, the updated constraints file for the `func1` constraint mode will include the `set_propagated_clock`.

- The following commands put the software into interactive constraint entry mode and apply the `set_propagated_clock` assertion on all active analysis views in the current session.

```
tempus> set_interactive_constraint_modes [all_constraint_modes -active]
tempus> set_propagated_clock [all_clocks]
```

When you specify the `save_design` command, the `set_propagated_clock` assertion is saved persistently in a new `.sdc` file.

Related Commands

- `get_interactive_constraint_modes`
- `all_constraint_modes`

set_io_thresholds

```
set_io_thresholds
[-help]
[-slew_lower_threshold_pct pctValue]
[-slew_upper_threshold_pct pctValue]
[-input_threshold_pct pctValue]
[-output_threshold_pct pctValue]
[-slew_lower_threshold_pct_rise pctValue]
[-slew_lower_threshold_pct_fall pctValue]
[-slew_upper_threshold_pct_rise pctValue]
[-slew_upper_threshold_pct_fall pctValue]
[-input_threshold_pct_rise pctValue]
[-input_threshold_pct_fall pctValue]
[-output_threshold_pct_rise pctValue]
[-output_threshold_pct_fall pctValue]
```

Sets the threshold trip-points used for delay calculation to and from the primary I/Os. The `set_io_thresholds` command does not affect the interpretation of threshold settings in the Liberty timing libraries; the effect is strictly limited to the I/O ports.

If you do not specify I/O thresholds with the `set_io_thresholds` command, delay calculation assumes that the I/O thresholds are equivalent to those of the logic gates to which they are connected.

The `set_io_thresholds` command supports separate rise and fall slew thresholds as normally found in Liberty timing libraries.

Note: Some third-party applications derive the I/O threshold values from the first Liberty timing library in the library search path. The software currently does not support this methodology. Explicit setting of I/O thresholds with the `set_io_thresholds` command should be used to complete a compatible analysis environment.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-input_threshold_pct <i>pctValue</i></code>	Specifies the slew input threshold value.
<code>-input_threshold_pct_fall <i>pctValue</i></code>	Specifies the measurement point for a falling signal that will be used for calculating the interconnect delay from a primary input to internal logic. <i>Value:</i> 0 - 100 <i>Default:</i> 50

<code>-input_threshold_pct_rise pctValue</code>	Specifies the measurement point for a rising signal that will be used for calculating the interconnect delay from a primary input to internal logic. <i>Value:</i> 0 - 100 <i>Default:</i> 50
<code>-output_threshold_pct pctValue</code>	Specifies the slew output threshold value.
<code>-output_threshold_pct_fall pctValue</code>	Specifies the measurement point for a falling signal that will be used for calculating the interconnect delay from internal logic to a primary output. <i>Value:</i> 0 - 100 <i>Default:</i> 50
<code>-output_threshold_pct_rise pctValue</code>	Specifies the measurement point for a rising signal that will be used for calculating the interconnect delay from internal logic to a primary output. <i>Value:</i> 0 - 100 <i>Default:</i> 50
<code>-slew_lower_threshold_pct pctValue</code>	Specifies the slew lower threshold value.
<code>-slew_lower_threshold_pct_fall pctValue</code>	Specifies the lower trip-point for a falling signal that will be used to measure transition times. <i>Value:</i> 0 - 100 <i>Default:</i> 20
<code>-slew_lower_threshold_pct_rise pctValue</code>	Specifies the lower trip-point for a rising signal used to measure transition times. <i>Value:</i> 0 - 100 <i>Default:</i> 20
<code>-slew_upper_threshold_pct_fall pctValue</code>	Specifies the upper trip-point for a falling signal that will be used to measure transition times. <i>Value:</i> 0 - 100 <i>Default:</i> 80

<code>-slew_upper_threshold_pct_rise pctValue</code>	Specifies the upper trip-point for a rising signal that will be used to measure transition times. <i>Value:</i> 0 - 100 <i>Default:</i> 80
<code>- slew_upper_threshold_pct pctValue</code>	Specifies the slew upper threshold value.

Examples

- The following command specifies trip-points for measuring delay and transition times:

```
tempus> set_io_thresholds -slew_lower_threshold_pct_rise 10
                        -slew_upper_threshold_pct_fall 80
                        -slew_upper_threshold_pct_rise 90
                        -slew_lower_threshold_pct_fall 20
                        -input_threshold_pct_rise 40
                        -input_threshold_pct_fall 70
                        -output_threshold_pct_rise 60
                        -output_threshold_pct_fall 30
```

set_ipd_rail_voltage

```
set_ipd_rail_voltage
[-help]
[-exclude_voltage_signatures <List of combinations to be excluded>]
[-add | -sweep <List of power rails and corresponding voltage sweep list>]
```

Specifies the PG net voltage and excludes those voltage combinations that do not require analysis. When power intent is specified through CPF/IEEE-1801, the `set_ipd_rail_voltage` specification refers to the primary power net of the power domain.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-add</code>	Specifies with the <code>-exclude_voltage_signatures</code> parameter to exclude voltage combinations incrementally.
<code>-exclude_voltage_signatures <List of combinations to be excluded></code>	Specifies the list of voltage combinations to be excluded from all the possible combinations of voltage.
<code>-sweep <List of power rails and corresponding voltage sweep list></code>	Specifies the list of power rails and their corresponding voltage values.

Examples

- The following command is used to specify the PG nets in the design and the corresponding voltages (in volts):

```
tempus > set_ipd_rail_voltage -sweep { PGN1 {0.7 0.8} PGN2 {0.9 1.0} PGN3 {1.1 1.2} PGN4 {0.6 0.7} }
```
- The following command removes a particular combination from the analysis:

```
tempus > set_ipd_rail_voltage -exclude_voltage_signatures { {PGN1 0.7 PGN2 0.9 PGN3 1.2 PGN4 0.6} }
```
- The following command excludes more combinations:

```
tempus > set_ipd_rail_voltage -exclude_voltage_signatures { {PGN1 0.7 PGN2 0.9 PGN3 1.2 PGN4 0.7} } -add
```

Related Commands

- `create_pg_net_group`
- `report_pg_net_group`
- `create_ipd_configuration`
- `report_ipd_configuration`
- `write_ipd_scope`
- `read_ipd_scope`
- `report_ipd_crossings`
- `check_ipd_logic`

Related Information

- For more information, refer to the [Inter Power Domain \(IPD\) Analysis](#) chapter in the *Tempus User Guide*.

set_library_unit

```
set_library_unit
[-help]
[-cap capUnit]
[-time timeUnit]
```

Sets the time and capacitance unit values to be used for constraints and the generation of reports.

The timing constraints (SDC) files do not contain any unit specification for capacitance and time values. Hence, the units for these values are considered to be the same as those specified in the timing libraries. The commands, e.g., `report_timing`, require consistent units to generate reports. If an inconsistency occurs in the timing libraries, then the default unit of time (as ns) and capacitance (as pf) is used.

You can use this command when you need to override the library units or force the setting of units when libraries have different units.

Note: You must set the `set_library_unit` command before the design is read and the constraints are loaded.

Parameters

-help	Prints out the command usage.
-cap <i>capUnit</i>	Specifies the capacitance load unit in the timing library. The valid values are 1pf (pico farad) and 1ff (femto farad). <i>Default:</i> 1.0 pf
-time <i>timeUnit</i>	Specifies the time unit to be used for timing constraints. The valid values are 1ns, 1ps, 10ps, and 100ps. <i>Default:</i> 1.0 ns

Examples

- The following command sets the time unit for the design to 10ps:

```
tempus> set_library_unit -time 10ps
```

set_max_cap_per_freq

```
set_max_cap_per_freq  
[-help]  
[-lib libName]  
[-force]  
-freq { freq1 freq2 ... }  
-cap { cap1 cap2 ... }  
-pins { cell pin }
```

Adds a maximum capacitance per frequency table to the existing `max_capacitance` statement in the timing library. Timing analysis uses this table to detect violations.

The timing engine derives the frequency of an instance from the clock domain to which it is connected (the fastest clock domain, if there is more than one). If the instance is a clock instance (that is, part of the clock tree), the actual clock frequency is used. If the instance is part of the combinational logic (that is, a flip-flop, or the logic connected to the output of a flip-flop), the frequency used equals one-half of the associated clock domain frequency because a flip-flop output can only toggle at one-half the rate of the input clock. For example, a clock of 1.0 Mhz has 1,000,000 rise and fall transitions per second; however, the output of a flip-flop connected to a 1.0 Mhz clock can only have 500,000 rise and fall transitions per second.

Timing analysis computes frequency directly from the toggle count format (TCF) file.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-cap { cap1 cap2 ... }</code>	Specifies a list of capacitance values that must be greater than or equal to 0. You must specify the same number of <code>-freq</code> and <code>-cap</code> values. The software derives the maximum capacitance from a linear interpolation (or extrapolation) for a given frequency value using two frequency values in the table. For interpolation, the software uses the closest value above the frequency and the closest value below the frequency. For extrapolation, the software uses the two closest values to the frequency. <i>Type:</i> Float, specified in units of pF
<code>-force</code>	Computes the maximum capacitance from the interpolated <code>-freq</code> and <code>-cap</code> values only, and ignores any previously defined <code>max_capacitance</code> values for these pins in the library. <i>Default:</i> Computes the maximum capacitance using the worst case (minimum value) between the library <code>max_capacitance</code> values and the interpolated <code>-cap</code> values.
<code>-freq { freq1 freq2 ... }</code>	Specifies a list of at least two frequency values that must be greater than or equal to 0. You must specify the same number of <code>-freq</code> and <code>-cap</code> values, and the frequency values must be specified in ascending order. <i>Type:</i> Float, specified in units of Hz
<code>-lib libName</code>	Modifies only the matching cell-pins in the specified library. <i>Default:</i> Modifies the matching cell-pins in all libraries
<code>-pins {cell pin}</code>	Adds the new capacitance constraints to the specified set of cells and pins. You can specify output or input pins. The asterisk (*) wildcard character can be used in a cell or pin name. If a table associated with the pin already exists, it is replaced with the new table. For analysis and optimization, the software uses the most restrictive (lowest capacitance) value from either the table or the <code>.lib max_capacitance</code>. If you also specify the <code>-force</code> parameter, the <code>max_capacitance</code> value is ignored. The final value can be further modified using <code>.sdc set_max_capacitance</code> statements.

Command Order

Use this command any time after loading the timing library files (that is, after importing or restoring the design).

Example

Assume all output pins of a library have a `max_capacitance` of 3.5 pF from the library file.

- The following command sets the frequency dependent `max_capacitance` to 3.0 pF at a frequency of 100 Mhz, and to 1.5 pF at a frequency of 200 Mhz, for all output or inout pins for all cells in the library named `fast`:

```
tempus> set_max_cap_per_freq -lib fast -freq {100e6 200e6} -cap {3.0 1.5} -pins {* *}
```

As a result, for a net at 300 Mhz, an and2 cell output pin has an extrapolated `max_capacitance` value of 0 pF (which means the cell cannot be used at that frequency). An extrapolation to 50 Mhz results in a `max_capacitance` value of 3.75 pF. However, the 3.75 pF value is greater than the existing `.lib max_capacitance` limit of 3.5 pF; therefore, timing analysis uses the 3.5 pF value.

- The following command applies the same constraints to all libraries with cell names matching `dff*` that have a pin named `Qbar`:

```
tempus> set_max_cap_per_freq -force -freq {100e6 200e6} -cap {3.0 1.5} -pins {dff*  
Qbar}
```

Note: The original `.lib max_capacitance` limit is ignored because the `-force` parameter is specified; therefore, extrapolation to 50 Mhz results in a final `max_capacitance` constraint value of 3.75 pF.

Related Commands

- `set_max_trans_per_freq`
- `report_constraint`

set_max_tran_per_freq

```
set_max_tran_per_freq  
[-help]  
[-force]  
-freq <string>  
[-lib <string>]  
-pins <string>  
-tran <string>
```

Adds a maximum transition per frequency table to the existing `max_transition` statement in the timing library. Timing analysis uses this table to detect violations.

The timing engine derives the frequency of an instance from the clock domain to which it is connected (the fastest clock domain, if there is more than one). If the instance is a clock instance (that is, part of the clock tree), the actual clock frequency is used. If the instance is part of the combinational logic (that is, a flip-flop, or the logic connected to the output of a flip-flop), the frequency used equals one-half of the associated clock domain frequency because a flip-flop output can only toggle at one-half the rate of the input clock. For example, a clock of 1.0 Mhz has 1,000,000 rise and fall transitions per second; however, the output of a flip-flop connected to a 1.0 Mhz clock can only have 500,000 rise and fall transitions per second.

Timing analysis computes frequency directly from the toggle count format (TCF) file.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-force</code>	Computes the maximum transition from the interpolated <code>-freq</code> and <code>-tran</code> values only and ignores any previously defined <code>max_transition</code> values for these pins in the library. <i>Default:</i> Uses the worst case (minimum value) between the <code>.lib max_transition</code> values and the interpolated <code>-tran</code> values.
<code>-freq <string></code>	Specifies a list of at least two frequency values that must be greater than or equal to 0. You must specify the same number of <code>-freq</code> and <code>-tran</code> values, and the frequency values must be specified in ascending order. <i>Type:</i> Float, specified in units of Hz
<code>-lib <string></code>	Modifies only the matching cell-pins in the specified library. <i>Default:</i> Modifies the matching cell-pins in all libraries.
<code>-pins <string></code>	Adds the new transition constraints to the specified set of cells and pins. You can specify input, output, or inout pins. The asterisk (*) wildcard character can be used in a cell or pin name. If a table associated with the pin already exists, it is replaced with the new table. For analysis and optimization, the software uses the most restrictive (the lowest transition) value from either the table or the <code>.lib max_transition</code>. If you also specify the <code>-force</code> parameter, the <code>max_transition</code> value is ignored. The final value can be further modified using <code>.sdc set_max_transition</code> statements.
<code>-tran <string></code>	Specifies a list of transition values that must be greater than or equal to 0. You must specify the same number of <code>-freq</code> and <code>-tran</code> values. The software derives the maximum transition limit from a linear interpolation (or extrapolation) for a given frequency value using two frequency values in the table. For interpolation, the software uses the closest value above the frequency and the closest value below the frequency. For extrapolation, the software uses the two closest values to the frequency. <i>Type:</i> Float, specified in units of ns

Command Order

Use this command any time after loading the timing library files (that is, after importing or restoring the design).

Examples

Assume all output pins of a library have a `max_transition` value of 0.9 ns from the library file.

- The following command sets the frequency dependent `max_transition` to 1.0 ns at a frequency of 100 Mhz, and to 0.7 ns at a frequency of 200 Mhz, for pins matching `in*` of cells matching `*` (that is, all cells) in the library named `slow`:

```
tempus> set_max_tran_per_freq -lib slow -freq {100e6 200e6} -tran {1.0 0.7} -pins {*  
in*}
```

As a result, at 300 Mhz, an `and2` cell input pin named `in1` has an extrapolated `max_transition` value of 0.4ns. An extrapolation to 50 Mhz results in a `max_transition` value of 1.15 ns. However, the 1.15 ns value is greater than the existing `.lib max_transition` value of 0.9 ns; therefore, timing analysis uses the 0.9 value.

- The following command applies the same constraints to all libraries with cell names matching `dff*` that have a pin named `clk`:

```
tempus> set_max_tran_per_freq -force -freq {100e6 200e6} -tran {1.0 0.7} -pins {dff*  
clk}
```

Note: The original `.lib max_transition` value is ignored because the `-force` parameter is specified; therefore, an extrapolation to 50 Mhz results in a final `max_transition` limit of 1.15 ns.

Related Commands

- `set_max_cap_per_freq`
- `report_constraint`

set_multi_input_switching_mode

```
set_multi_input_switching_mode  
[-help]  
[-disable_lib_cells <string>]  
[-mis_alignment_mode {0 | 1}]  
[-switching_alignment_factor <float>]  
[-view <string>]
```

Configures multi-input switching (MIS) mode settings.

You can enable the MIS feature using the `set_analysis_mode -multi_input_switching_mode` command parameter.

For reporting advanced MIS in delay calculation and delta delay in timing reports, you can use the following options:

- `report_delay_calculation -mis`
- `report_timing -format retime_mis_delta`
- `report_timing_format {retime_mis_delta}`

Parameters

<code>-help</code>	Prints out the command usage.
<code>-disable_lib_cells</code> <code><string></code>	Enables or disables the multi-input switching (MIS) mode support for the specified library cells. <i>Default:</i> " "
<code>-mis_alignment_mode {0 1}</code>	Controls the MIS alignment mode. <i>Default:</i> 0
<code>-switching_alignment_factor</code> <code><float></code>	Allows to configure the relative alignment of side inputs switching with respect to the primary inputs. The switching alignment factor is applied for all the instances for which MIS analysis is to be performed. This parameter accepts values between -1 and 1. These are described below: • 1: Side inputs switch at the earliest arrival. • 0: Side inputs switch at the same time as the actual input. • -1: All the side inputs switch at the same time and align at the completion of the output pin transition. <i>Default:</i> 0 Note: Setting the <code>-switching_alignment_factor</code> to 1 can be pessimistic if a side input switches much earlier than the main input. You can use the <code>-mis_alignment_mode</code> parameter to reduce pessimism in this mode.
<code>-view <string></code>	Specifies the name of the view.

Examples

- The following commands disable MIS support for the specified library cells:

```
set_multi_input_switching_mode -disable_lib_cells [get_libs_cell aoi*]
```

```
set_multi_input_switching_mode -disable_lib_cells [get_libs_cell lib_0.510v/aoi*]
```

Related Commands

- `get_multi_input_switching_mode`
- `set_analysis_mode`
- `report_timing`
- `report_delay_calculation`
- `report_timing_format`
- `create_spice_deck`

Related Information

- [Advanced Multiple-Input Switching Analysis](#) section in the *Tempus User Guide*.

set_op_cond

```
set_op_cond  
[-help]  
[<opCond>]  
[-forceLibrary <string>]  
[-library <string>]  
[-powerDomain <string>]  
[-min <string> -minLibrary <string>]  
[-max <string> -maxLibrary <string>]
```

Sets the operating temperature, process, or voltage conditions for the design. The operating conditions are contained in the timing library, which is normally read in during design import.

The `set_op_cond` command can use a virtual operating condition that was created with the multi-mode multi-corner `create_op_cond` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-forceLibrary</code> <i>string</i>	Specifies the library in which the <i>opCond</i> will be set. Do not use this parameter with the <code>-min</code> , <code>-max</code> , or <code>-powerDomain</code> parameters. Use this parameter to specify different operating conditions for different libraries in a single session.
<code>-library</code> <i>string</i>	Specifies the library in which the operating condition (<i>opCond</i>) is defined. You can define a single operating condition from a library, and the software uses the PVT characteristics associated with that library for every library in the design. When you use this parameter in multiple <code>set_op_cond</code> commands, the software uses the value specified in the last command. If the <code>-library</code> parameter is not specified, the <i>opCond</i> in the first library read is used. For example, consider the following commands: <pre>set_op_cond WCCOM -library wcTestLib set_op_cond BCCOM -library bcTestLib set_op_cond WCCOM -library wcTestLib1 set_op_cond BCCOM -library bctestLib1 set_op_cond -library wcTestLib2 set_op_cond -library bcTestLib2</pre> In this case, the software uses the last value defined (<i>bcTestLib2</i>). Therefore, the operating conditions in the <i>bcTestLib2</i> library are used for both setup and hold analysis.
<code>-max</code> <i>string</i>	Specifies the operating condition normally used for setup analysis.
<code>-maxLibrary</code> <i>string</i>	Specifies the library in which the <i>maxOpCond</i> is defined. If the <code>-maxLibrary</code> parameter is not specified, and <code>-max</code> is used, the <i>maxOpCond</i> from the first library listed in the maximum libraries is used.
<code>-min</code> <i>string</i>	Specifies the operating condition normally used for hold analysis.
<code>-minLibrary</code> <i>string</i>	Specifies the library in which the <i>minOpCond</i> is defined. If the <code>-minLibrary</code> parameter is not specified, and <code>-min</code> is used; the <i>minOpCond</i> from the first library listed in the minimum libraries is used.
<i>opCond</i>	Specifies the operating condition used for setup and hold analysis. You can specify a virtual operating condition created with the multi-mode multi-corner <code>create_op_cond</code> command.
<code>-powerDomain</code> <i>string</i>	Specifies that the <i>opCond</i> is used for the power domain.

Command Order

Use this command after importing the design and reading the libraries.

Examples

- The following command sets the operating condition of the design to `WCCOM`, defined in the `slowLib` timing library. The worst case (WCCOM) operation condition will be used for both setup and hold analysis:

```
tempus> set_op_cond WCCOM -library slowLib
```

- The following Tcl command sets the operating condition to `WCCOM`, defined in the `slowlib` library, for setup analysis and sets the operating condition to `BCCOM`, defined in the `fastlib` library, for hold analysis:

```
tempus> set_op_cond -max WCCOM -maxLibrary slowlib -min BCCOM -minLibrary fastlib
```

- The following example specifies the `slow` operating condition defined in the `slow` library for setup analysis and the `fast` operating condition defined in the `fast` library for hold analysis:

```
tempus> set_op_cond -maxLibrary slow -max slow -minLibrary fast -min fast
```

- The following example sets the timing libraries to associate the fast library (Min Libraries) to setup and the slow library (Max Libraries) to hold:

```
tempus> set_timing_library -max Min -min Max
```

However, this means that setup analysis uses the `fast` library (defined in the Min Libraries list) with a `slow` operating condition defined in the `slow` library, and hold analyses uses the `slow` library (defined in the Max Libraries list) but with a `fast` operating condition defined in the `fast` library.

To get a consistent result, reset the `set_op_cond` command to:

```
tempus> set_op_cond -maxLibrary fast -max fast -minLibrary slow -min slow
```

Now, setup analysis uses the `fast` library (defined in the Min Libraries list) with a `fast` operating condition defined in the `fast` library, and hold analyses uses the `slow` library (defined in the Max Libraries list) but with a `slow` operating condition defined in the `slow` library.

Related Commands

- `create_op_cond`
- `all_op_conds`

set_path_adjust

```
set_path_adjust
[-help]
adjust_value
-path_adjust_group path_adjust_group_name
-view string
[-setup | -hold]
```

Defines the slack adjustment value for timing paths. The defined adjustment value is added to the slack of a path specified by the path adjust group, referred to by the `-path_adjust_group` parameter. You can specify a path adjust group by using the `set_path_adjust_group` command.

If multiple path adjustment values are applicable on the same pin, which is on the path specified by multiple `set_path_adjust_group` constraints, then the smallest adjustment value will be applied.

Note: The `set_path_adjust` command is not allowed to be used inside the SDC constraints file. It is only allowed in the software run script.

Parameters

<code>-help</code>	Prints out the command usage.
<code>adjust_value</code>	Specifies the value of the slack adjustment.
<code>-hold</code>	Specifies to apply the adjustment value to early paths.
<code>-path_adjust_group</code> <code>path_adjust_group_name</code>	Specifies the name of the path adjust group that defines the path for which the adjustment will be applied.
<code>-setup</code>	Specifies to apply the adjustment value to late paths. Note: If you do not specify the <code>-setup</code> or <code>-hold</code> parameter, the software sets the adjust value to both setup and hold analysis.
<code>-view string</code>	Specifies a list of analysis views for which the adjustment will be applied. This parameter accepts multiple views as arguments.

Examples

- The following commands define the slack adjustment values for the specified views:

```
tempus 1> set_path_adjust 0.44 -path_adjust_group AG1 -view view1 -hold  
tempus 1> set_path_adjust 0.44 -path_adjust_group AG2 -view view2 -setup  
tempus 1> set_path_adjust 0.44 -path_adjust_group AG3 -view view3
```

Related Commands

- `set_path_adjust_group`
- `reset_path_adjust_group`

set_path_adjust_group

```
set_path_adjust_group  
[-help]  
[-fall]  
[-fall_through through_list]  
-name path_adjust_group_name  
[-rise]  
[-rise_through through_list]  
[-through through_list]  
[-from from_list | -rise_from from_list | -fall_from from_list]  
[-to to_list | -rise_to to_list | -fall_to to_list]
```

Allows you to specify the path that is used by the `set_path_adjust` command for setting path slack adjustment values.

The `set_path_adjust_group` command is not allowed to be used inside the SDC constraints file. It is only allowed in the tool run script.

If path group definitions are matching/overlapping (when the `-from/-through/-to` parameters are used), then path group resetting is applied. The path group overwriting and prioritization rules are similar to path exceptions.

For more details, refer to [group_path](#).

Parameters

<code>-help</code>	Prints out the command usage.
<code>-fall</code>	Applies <code>set_path_adjust_group</code> at the falling edge of the signal on the path endpoints.
<code>-fall_from from_list</code>	Applies <code>set_path_adjust_group</code> at the falling edge of the signal on the from pins.
<code>-fall_through through_list</code>	Applies <code>set_path_adjust_group</code> at the falling edge of the signal on the through pins.
<code>-fall_to to_list]</code>	Applies <code>set_path_adjust_group</code> at the falling edge of the signal on the to pins.
<code>-from from_list</code>	Specifies the list of pins, instances, or clocks that are the start of the paths.
<code>-name path_adjust_group_name</code>	Specifies the name of the path adjust group.
<code>-rise</code>	Applies <code>set_path_adjust_group</code> at the rising edge of the signal on the path endpoints.
<code>-rise_from from_list</code>	Applies <code>set_path_adjust_group</code> at the rising edge of the signal on the from pins.
<code>-rise_through through_list</code>	Applies <code>set_path_adjust_group</code> at the rising edge of the signal on the through pins.
<code>-rise_to to_list</code>	Applies <code>set_path_adjust_group</code> at the rising edge of the signal on the to pins.
<code>-through through_list</code>	Specifies the list of pins or instances where the paths go through.
<code>-to to_list</code>	Specifies the list of pins, instances, or clocks where the path ends.

Examples

The following commands specify the adjusted path groups:

- `set_path_adjust_group -name AG1 -from in1 -through buf1/A -to out1`
- `set_path_adjust_group -name AG2 -from [get_clocks CK1] -through buf1/A -to [get_clocks CK2]`

- `set_path_adjust_group -name AG3 -from [get_clocks {CK1 CK2}] -to [get_clocks {CK1 CK2}]`

Related Commands

- `reset_path_adjust_group`
- `set_path_adjust`
- `group_path`

set_pba_mode

```
set_pba_mode
[-help]
[-reset]
[-epba_bounded_mode {true | false}]
[-ipba_endpoints_max_limit {100000}]
```

Enables IPBA infinite depth path-based analysis (IPBA).

This feature runs faster analysis (vs full depth exhaustive PBA (EPBA)) and attains PBA design coverage with no depth limit per endpoint.

Parameters

-help	Prints out the command usage.
-epba_bounded_mode {true false}	Enables IPBA mode with exhaustive PBA analysis (run using the <code>report_timing -retime_mode exhaustive</code> command). In this mode, the software ensures that all the paths to the violating endpoints are covered for PBA, and the slack is the worst PBA slack reported across all the paths to the endpoint. For any endpoint that is considered for IPBA analysis, the complete fanin cone (all timing paths ending at that endpoint) is evaluated for any possible PBA violation. <i>Default:</i> false
-ipba_endpoints_max_limit {100000}	Specifies the maximum number of worst critical GBA endpoints, which will be picked for IPBA analysis. The rest of the GBA endpoints will report GBA slack. This parameter should be set to a number higher than the total number of GBA violating endpoints, in order to cover all the violating endpoints for IPBA. To ensure that all the GBA violations are considered for IPBA analysis, the <code>report_timing -max_paths</code> parameter should be set equal to or greater than the total number of GBA violating endpoints. <i>Default:</i> 100000 <i>Min:</i> 1 <i>Max:</i> 1000000
-reset	Resets the specified or all the options to their default value.

Related Commands

- [get_pba_mode](#)
- [report_timing](#)

set_socv_constraint_config

```
set_socv_constraint_config  
[-help]  
[-hold_folding float]  
{-libs <string> }  
[-nochange_hold_folding float]  
[-nochange_setup_folding float]  
[-setup_folding float]
```

Enhances the built-in sigma pessimism of check values for setup and hold constraints in the library. The folding factor attribute records the amount of on-chip-variation pessimism added during cell characterization to the nominal setup/hold tables in the libraries. The folding factor attributes are library-level attributes.

Notes

- The `set_socv_constraint_config` command does not reset the timing, so it should be used before the `report_timing` command.
- To change the setting later in the flow, you need to use the `update_timing -full` command.
- You cannot use this command if any of the following global variables are specified:

```
timing_library_setup_sigma_multiplier  
timing_library_hold_sigma_multiplier
```

Parameters

<code>-help</code>	Prints out the command usage.
<code>-hold_folding float</code>	Specifies a number of additional sigma units added on top of the existing nominal hold value. <i>Min</i> = -100 <i>Max</i> = 100
<code>-libs string</code>	Specifies a list of library names (not library paths) or collection objects.
<code>-nochange_hold_folding float</code>	Specifies a number of additional sigma units added on top of the existing nominal hold value for no-change timing arcs. <i>Min</i> = -100 <i>Max</i> = 100
<code>-nochange_setup_folding float</code>	Specifies a number of additional sigma units added on top of the existing nominal setup value for no-change timing arcs. <i>Min</i> = -100 <i>Max</i> = 100
<code>-setup_folding float</code>	Specifies a number of additional sigma units added on top of the existing nominal setup value. <i>Min</i> = -100 <i>Max</i> = 100

Examples

- The following command is used when an additional margin of 10 sigma units is applied to the nominal hold tables in the library collection `test.lib`:

```
set_socv_constraint_config -hold_folding 10 -libs [get_libs test.lib]
```

Related Commands

- [report_timing](#)
- [update_timing](#)

set_socv_rc_variation_factor

```
set_socv_rc_variation_factor  
[-help]  
variation_factor  
[-early]  
[-late]  
[-view list_of_views]
```

Defines the value of the interconnect delay variation factor.

This command applies a global variation as a percentage of the nominal delay.

To enable this feature, you can set the `timing_socv_rc_variation_mode` global variable to `true`.

Parameters

<code>-help</code>	Prints out the command usage
<i>variation_factor</i>	Specifies the value of the interconnect delay variation factor to be used.
<code>-early</code>	Specifies the variation factor for early interconnect delays.
<code>-late</code>	Specifies the variation factor for late interconnect delays.
<code>-view</code> <i>list_of_views</i>	Apply the SOCV interconnect variation factor to the specified list of analysis views.

Related Commands

- `get_socv_rc_variation_factor`

set_socv_reporting_nsigma_multiplier

```
set_socv_reporting_nsigma_multiplier
[-help]
[-view list_of_views]
{[-setup float] [-hold float]}
[-transition float]
```

Controls n-sigma value for reporting purposes.

This command supports n-sigma specification per view per setup/hold combination.

Parameters

-help	Prints out the command usage.
-hold <i>float</i>	Specifies the sigma multiplier for hold mode.
-setup <i>float</i>	Specifies the sigma multiplier for setup mode.
-transition <i>float</i>	Specifies the sigma multiplier to use for transition related DRV checks – including max_transition, min_transition, pulse_clock_max_transition, and pulse_clock_min_transition. When this parameter is not specified, the sigma multiplier specified for setup will be used for transitions. When the -transition parameter is specified, you can set the <code>timing_report_max_transition_check_using_nsigma_slew</code> global variable to <code>true</code> to report transition related DRV checks, such as max_transition, min_transition, pulse_clock_max_transition, and pulse_clock_min_transition. This setting enables the <code>report_constraint</code> command to use mean and sigma slew values for reporting transition values in DRV checks.
-view <i>list_of_views</i>	Applies the sigma multiplier factor to the specified list of analysis views.

Examples

- The following command applies the sigma multiplier to the specified view in setup mode :

```
set_socv_reporting_nsigma_multiplier -setup 2.5 -view std_max_setup
```
- The following command applies the sigma multiplier to the specified view in hold mode:

```
set_socv_reporting_nsigma_multiplier -hold 3.0 -view std_min_hold
```

Related Commands

- `get_socv_reporting_nsigma_multiplier`

set_table_style

```
set_table_style
[-help]
[-indent integer]
[-major_sort integer]
[-max_widths integer]
[-min_widths integer]
[-minor_sort integer]
[-name string]
[-nosplit]
[-reverse_rows] | -no_frame_fix_width
```

Disables printing specific columns and allows specifying the minimum and maximum sizes of each column in a report. It allows reversing the data, controlling the left indent, and controlling column sorting.

Note: Do *not* include a blank space between commas ", " when specifying widths. If you are specifying a value for the width(s), make sure there is *no* leading or trailing blank space.

The `-frame` and `-no_frame_fix_width` parameters work in a global manner. If you specify one of these parameters, it will affect all reports that can be formatted with this command. You cannot specify them with the `-name` parameter to change the format of a specific report.

Parameters

-help	Outputs a brief description that includes type and default information for each <code>set_table_style</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man set_table_style</code> .
-indent integer	Specifies the number of spaces to leave on the left for that particular table.
-major_sort integer	Specifies the column to use for the major sort. If the column number is negative, the sort reverses.

`-max_widths`
list_of_integers

Specifies the maximum width for each table column in the report specified by the `-name` parameter.

Note: You cannot specify the `-max_widths` parameter without `-name`.

You can separate each specified width value with a comma (,). For example, if you specify:

```
-max_widths {10,20,30}
```

The first column will have a maximum width of 10, the second column will have a maximum width of 20, and the third column will have a maximum width of 30. The fourth and fifth columns will remain the default width spacing.

Commas are also specified as column "placeholders", when you want to change the spacing of a column without changing the spacing of the preceding column. For example, if you specify:

```
-max_widths {,,20,30}
```

The third column will have a maximum width of 20, the fourth column will have a maximum width of 30, and the first, second, and fifth columns will remain the default width spacing.

If you specify:

```
-max_widths {20,,30}
```

The first column will have a maximum width of 20, the third column will have a maximum width of 30, and the second, fourth, and fifth columns will remain the default width spacing.

Each column resizes itself to be between the range of the maximum and minimum widths specified for that column, based on the data in that column. Overruns are wrapped within the same column, unless otherwise specified.

Default: {50, 50, 50, 50, 50}

<code>-min_widths</code> <i>list_of_integers</i>	Specifies the minimum width for each table column in the report specified by the <code>-name</code> parameter. Note: You cannot specify the <code>-min_widths</code> parameter without <code>-name</code>. You can separate each specified width value with a comma (,). For example, • If you specify <code>-min_widths {7,10,15}</code>, the first column will have a minimum width of 7, the second column will have a minimum width of 10, and the third column will have a minimum width of 15. The fourth and fifth columns will remain the default width spacing. • The commas are also specified as column "placeholders" when you want to change the spacing of a column without changing the spacing of the preceding column. If you specify <code>-min_width {,,10,15}</code>, the third column will have a minimum width of 10, the fourth column will have a minimum width of 15, and the first, second, and fifth columns will remain the default width spacing. • If you specify <code>-min_width {7,,10}</code>, the first column will have a minimum width of 7, the third column will have a width of 10, and the second, fourth, and fifth columns will remain the default width spacing. Each column resizes itself to be between the range of the maximum and minimum widths specified for that column, based on the data in that column. Overruns are wrapped within the same column, unless otherwise specified. <i>Default:</i> {5,5,5,5,5}
<code>-minor_sort</code> <i>integer</i>	Specifies the column to be used in the minor sort. It applies to <i>all</i> rows that have the same values in the <code>major_sort</code> column. If the column number is negative, the software reverses the sort.
<code>-name</code> <i>string</i>	Changes the formatting for the specified table. See List of Table Names for a list of table names you can use, and the commands with which they work. This parameter should only be used for <code>-max_width</code> or <code>-min_width</code> options.

<code>-no_frame_fix_width</code>	Specifies that the table columns have a fixed minimum width, but no maximum width (the data does not wrap). Column data that fits within the column's minimum width will be left-justified and pad its location to fill the specified column space. Column data that is longer than the column's minimum width will take as much space in the table as needed. If a column is overwritten by the data from the preceding column, its data will begin on a new line at the correct column location. This is the default format for tables.
<code>-nosplit</code>	Specifies that each column follows the preceding column without splitting to a new line if the data in one column overruns the next column. The wrapping behavior for data is controlled by the size of the display window. This format is useful for machine parsing the report. Note: You must specify <code>-no_frame_fix_width</code> in order to use this parameter.
<code>-reverse_rows</code>	Reverses the data rows. This is useful, for example, if you want to trace back from an end point and the default report traces forward from the end point.

The timing report formatting depends on the specified `set_table_style` parameters. These are described below:

- When the `set_table_style -no_frame_fix_width` parameter is specified, the timing report displays long inst/hpin names, and the rest of the column values are shown on the next line.
- When `set_table_style -no_frame_fix_width -nosplit` is specified, the timing report displays long inst/hpin names and the rest of the column values are shown on the same line, because the `-nosplit` table style is specified. In this case, other column headers (beyond inst/pin) may not be aligned properly.

- The following settings are specified:

```
tempus> set_table_style -no_frame_fix_width -nosplit
tempus> set_table_style -name report_timing -max_width {50,50,,,,,}
```

The generated report will show properly aligned column headers, and end-points/begin-points are not split.

- The following settings generate a report displaying longer (than the above reports) inst/pin names:

```
tempus> set_table_style -no_frame_fix_width -nosplit
tempus> set_table_style -name report_timing -max_width {120,120,,,,,}
```

In this report output, the column headers are aligned properly and are not split.

List of Table Names

Table Name	Formats Reports Generated by
report_cell_instance_delays_arcs	report_cell_instance_timing
report_cell_instance_delays_checks	report_cell_instance_timing
report_cell_instance_delays_pins	report_cell_instance_timing
report_inactive_arcs	report_inactive_arcs
report_net	report_net
report_net_header	report_net
report_net_sink_table	report_net
report_net_source_table	report_net
report_net_summary_table	report_net
report_path_exceptions	report_path_exceptions
report_port	report_ports
report_timing	report_timing
report_timing_summary	report_timing

Examples

- The following command resizes the first column to a maximum range of 70 and specifies a default width of 50 for the second column. The default for the maximum width is 50, and the default for the minimum width is 5:

```
tempus> set_table_style -name report_timing -max_widths {70,}
```

Note: Do *not* include a blank space between commas ", " when specifying widths. If you are specifying a value for the width(s), make sure there is *no* leading or trailing blank space.

- The following command specifies a default width of 50 for the first and second columns:

```
tempus> set_table_style -name report_timing -max_widths {,}
```

- The following timing report shows the table style generated with the `-no_frame_fix_width` parameter:

```
tempus> report_timing -format {cell arc delay arrival required instance}
```

```
Path 1: MET Setup Check with Pin seg3/u14/CK
```

```
Endpoint: seg3/u14/D (^) checked with leading edge of 'CLK3'
```

```

Beginpoint: seg3/u9/Q (^) triggered by leading edge of 'CLK3'
Path Groups: {CLK3}
Other End Arrival Time 0.621
- Setup 0.186
+ Phase Shift 2.000
= Required Time 2.434
- Arrival Time 2.062
= Slack Time 0.372
Clock Rise Edge 0.000
+ Clock Network Latency (Prop) 0.682
= Beginpoint Arrival Time 0.682

```

Cell	Arc	Delay	Arrival Time	Required Time	Instance
	CK ^		0.682	1.054	seg3/u9
DFF	CK ^ -> Q ^	0.185	0.867	1.240	seg3/u9
BUF	A ^ -> Y ^	0.073	0.940	1.312	seg3/u10
INV	A ^ -> Y v	0.317	1.257	1.629	seg3/u11
INV	A v -> Y ^	0.402	1.659	2.031	seg3/u12
BUF	A ^ -> Y ^	0.403	2.062	2.434	seg3/u13
DFF	D ^	0.000	2.062	2.434	seg3/u14

- The following sample timing report shows the table style generated with the `-no_frame_fix_width` and `-nosplit` parameters:

```
tempus> report_timing -format {cell arc delay arrival required instance}
```

```

Path 1: MET Setup Check with Pin seg3/u14/CK
Endpoint: seg3/u14/D (^) checked with leading edge of 'CLK3'
Beginpoint: seg3/u9/Q (^) triggered by leading edge of 'CLK3'
Path Groups: {CLK3}
Other End Arrival Time 0.621
- Setup 0.186
+ Phase Shift 2.000
= Required Time 2.434
- Arrival Time 2.062
= Slack Time 0.372
Clock Rise Edge 0.000
+ Clock Network Latency (Prop) 0.682
= Beginpoint Arrival Time 0.682

```

Cell	Arc	Delay	Arrival Time	Required Time	Instance
CK ^			0.682	1.054	seg3/u9

Tempus Text Command Reference

Timing Analysis Commands

DFF	CK ^ -> Q ^	0.185	0.867	1.240	seg3/u9
BUF	A ^ -> Y ^	0.073	0.940	1.312	seg3/u10
INV	A ^ -> Y v	0.317	1.257	1.629	seg3/u11
INV	A v -> Y ^	0.402	1.659	2.031	seg3/u12
BUF	A ^ -> Y ^	0.403	2.062	2.434	seg3/u13
DFF	D ^	0.000	2.062	2.434	seg3/u14

Related Commands

- [report_cell_instance_timing](#)
- [report_inactive_arcs](#)
- [report_net](#)
- [report_path_exceptions](#)
- [report_ports](#)
- [report_timing](#)

set_threshold_voltage_group

```
set_threshold_voltage_group  
[-help]  
[-hinst <string>]  
-name <group_name>  
[-override]  
{-libraries <library list> | -lib_cells <lib cell list>}
```

Groups timing libraries into different voltage threshold (Vth) classes (e.g., HVT, LVT, ULVT).

This command can be used to define (or redefine) the threshold voltage groups associated with the specified timing library cells. You can group entire libraries into a single group, or group by per library cell specification.

Parameters

-help	Prints out the command usage.
-hinst <string>	Specifies the hierarchical instances.
-lib_cells <lib cell list>	Specifies a list of library cells to be defined in the VT group.
-libraries <library list>	Specifies a list of libraries to be defined in the VT group.
-name <group_name>	Specifies the name of the threshold voltage group being defined.
-override	Allows you to override Liberty-defined VT groups. If the Liberty timing library already has native <code>threshold_voltage_group</code> attributes at the library or library cell level, you can use the -override parameter to customize it with the specified threshold voltage group.

Examples

- The following commands define library threshold voltage groups if they do not preexist in the .lib:

```
set_threshold_voltage_group -name SVT -libraries stdcell_vt_1p62V_cworst
set_threshold_voltage_group -name LVT -libraries stdcell_2vt_1p62V_cworst
set_threshold_voltage_group -name ULVT -libraries stdcell_vlvt_1p62V_cworst
set_threshold_voltage_group -name ELVT -libraries stdcell_flvt_1p62V_cworst
```


Related Commands

- `set_vt_skew_derate`
- `set_vt_skew_config`

Related Topics

For more information, refer to the "Voltage Threshold Skew Modeling and Analysis" section of the [Analysis and Reporting](#) chapter in the *Tempus User Guide*.

set_timing_context_clock_mapping

```
set_timing_context_clock_mapping
[-help]
[-add]
-block_clock block_clock_name
[-instance <string>]
-top_clock top_clock_name
[-top_view view_name]
[-view view_name]
```

Specifies user-defined clock mapping in context analysis. The auto-inferred clock mappings will be overridden by user-specified mappings. This command should not be specified directly in the Tempus shell.

Note: To ensure correct behavior, this command must be specified in the clock mapping file and subsequently loaded into the Tempus environment using the following command:

```
read_timing_context -clock_mapping_file clock_mapping_file
```

This command should not be run as a standalone command; it must be enclosed in the mapping file.

Parameters

-help	Prints out the command usage.
-add	Specifies the multiple top clocks to a single block clock mapping. When -add is not specified, the last specified top to block clock mapping will override the previously specified mapping for that block clock.
-block_clock <i>block_clock_name</i>	Specifies the name of the clock from the block-level analysis. The software maps the specified clock with the clock specified using the -top_clock parameter.
-instance <i>string</i>	Specifies the list of block instances for which the clock mapping is to be applied. When this parameter is not specified, the clock mapping will be applied to all the instances for which context analysis is to be performed.
-top_clock <i>top_clock_name</i>	Specifies the name of the clock from top-level analysis. The software maps the specified clock with the clock specified using the -block_clock parameter.
-top_view <i>view_name</i>	Specifies the top analysis view for top or block level clock matching.
-view <i>view_name</i>	Specifies the MMMC analysis view for which the top-block clock mapping is to be applied.

Examples

- In the following example, the block clock BCLK is mapped to the top clock TCLK in block view B_view:

```
set_timing_context_clock_mapping -top_clock TCLK -block_clock BCLK -view B_view
```

- In the following example, the block clock BCLK is mapped to both top clocks TCLK1 and TCLK2 in block view B_view:

```
set_timing_context_clock_mapping -block_clock BCLK -top_clock TCLK1 -view B_view
```

```
set_timing_context_clock_mapping -block_clock BCLK -top_clock TCLK2 -view B_view -add
```

- In the following example, the block clock BCLK is mapped only to TCLK2 in block view B_view since `-add` is not specified:

```
set_timing_context_clock_mapping -block_clock BCLK -top_clock TCLK1 -view B_view
```

```
set_timing_context_clock_mapping -block_clock BCLK -top_clock TCLK2 -view B_view
```

Related Commands

- `write_timing_context`
- `read_timing_context`
- `set_timing_context_view_mapping`

set_timing_context_view_mapping

```
set_timing_context_view_mapping  
[-help]  
[-block_clock_suffix block_clock_suffix]  
-block_view block_view_name  
-top_view top_view_name
```

Provides mapping between the top and block STA views. This command can be used only if the top and block STA view names differ.

A text file containing a set of `set_timing_context_view_mapping` commands can be provided to the `read_timing_context -view_map_file` parameter.

Parameters

<code>-help</code>	Prints out the command usage.
<code>- block_clock_suffix block_clock_suffix</code>	Specifies the suffix for block clock names. The block clocks with the specified suffix in the <code>block_view</code> name will only be mapped to the top clocks from the <code>top_view</code> .
<code>-block_view block_view_name</code>	Specifies the name of the view in the block-level analysis. The software maps the specified view with the <code>-top_view</code> view.
<code>-top_view top_view_name</code>	Specifies the name of the view in the top-level analysis. The software maps the specified view with the <code>-block_view</code> view.

Examples

- The following example shows how a view mapping file can be set up:

```
set_timing_context_view_mapping -top_view T1_view -block_view B1_view  
set_timing_context_view_mapping -top_view T2_view -block_view B2_view
```

Related Commands

- `write_timing_context`
- `read_timing_context`
- `set_timing_context_clock_mapping`

set_timing_derate

```
set_timing_derate
[-help]
derate_value
[object_list]
[-add]
[-cell_check]
[-cell_delay]
[-clock]
[-clock_period_check]
[-corner]
[-data]
[-dynamic]
[-early float]
[-fall]
[-increment]
[-input_switching]
[-late float]
[-mean]
[-multiply]
[-net_delay]
[-power_domain string]
[-pulse_width_check]
[-retain_delay]
[-rise]
[-sigma]
[-static]
[-statistical]
[-transition]
[-min | -max | -delay_corner delayCornerName]
```

Sets delay scaling or derating factors for early and late paths in the current design. You use this command to perform scaled on-chip variation around best-case and worst-case corners. The timing scaling factors affect the delay values generated in the timing reports. You can set scaling factors for data paths, clock paths, and minimum and maximum operating conditions.

If you do not specify the `-clock` or `-data` parameter, the software applies the scaling factors to both clock and data paths. If you specify only one of the two parameters, `-data` or `-clock`, the software uses the last defined value for the other.

Similarly, if you do not specify the `-min` or `-max` parameter, the software applies the scaling factor to both minimum and maximum operating conditions. You should only use the `-min` or `-max` parameter for best-case worst-case analysis. If you specify them in on-chip variation (OCV) mode, the software will use the `-min -early` derating factor for early path scaling, and the `-max -late` derating factor for late path adjustments.

When there are separate max and min libraries specified, only those `set_timing_derate` constraints are honored that are specified with the max library. The constraints specified with the min library are ignored.

In MMMC mode, for delay corners having different late and early library sets, only those `set_timing_derate` constraints are honored that are specified with the late library set of the delay corner. The constraints specified with the early library set are ignored.

Note:

- The `set_timing_derate` command can only be used on the command-line. You cannot import it as part of an SDC file.
- If the `set_timing_derate` constraint is added incrementally in SI mode, an explicit `update_timing -full` command is required (before `report_timing` or `report_constraint`) for SI full accuracy.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-add</code>	Incrementally adds or subtracts derates. You can also apply derates to specific leaf instances. This parameter also supports net objects.
<code>-cell_check</code>	Specifies scaling on timing checks. Use the <code>-cell_check</code> option with the <code>-early</code> option to specify scaling on hold arcs and use with the <code>-late</code> option to specify scaling on setup arcs. When you specify this parameter, derating factors are also applied to user constraints such as <code>set_output_delay</code> , <code>set_data_check</code> , and <code>set_clock_gating_check</code> . The <code>-cell_check</code> parameter, by default does not apply check derate to output external delay (e.g., <code>set_output_delay</code>). To enable this feature, you can set the <code>timing_apply_check_derate_to_external_output_delay</code> timing global variable to <code>true</code> . When set to <code>true</code> , this global applies cell derate to the top boundary port.
<code>-cell_delay</code>	Specifies scaling on cell delays. If neither the <code>-cell_delay</code> nor the <code>-net_delay</code> option is specified, the specified scaling factor applies to both types of delays.
<code>-clock</code>	Applies the derating factors to clock paths only.

<code>-clock_period_check</code>	Specifies derate values for clock period checks. Note: This parameter cannot be specified with the <code>-clock</code>, <code>-data</code>, <code>-cell_check</code>, <code>-cell_delay</code>, <code>-early</code>, <code>-retain_delay</code>, or <code>-net_delay</code> parameter.
<code>-corner</code>	Applies derating on cells or nets that are modeled as corner data elements. The specified derating factor is applied to both the mean and sensitivities of the delays (corner objects can have propagated sensitivities). A cell with no timing arc data in a library that has any sensitivity information with respect to any parameter defined in SPDF is considered a corner object. All cells that do not qualify are considered to be statistical objects. A net with parasitic data that has no sensitivity information with respect to any parameter defined in SPDF is considered a corner object. All nets that do not qualify are considered to be statistical objects.
<code>-data</code>	Applies the derating factors to data paths only.
<code>-delay_corner</code> <i>delayCornerName</i>	Applies the derating factor to the specified delay calculation corner or a list of delay corners. Note: The timing derate can be applied to corners associated with active views only.
<i>derate_value</i>	Specifies the derating value to be used.

<code>-dynamic</code>	Applies the specified derating factor to the delta portion of the net and cell delays. The delay typically models SI crosstalk adjustments from detailed RC analysis or annotation. You can use this parameter in conjunction with the <code>-net_delay</code> or <code>-cell_delay</code> parameter to target the derate to desired components of the overall stage delay. The delay calculation controls allow for the SI crosstalk impact to be apportioned to the cell delay and/or net delay. You should coordinate the use of <code>-dynamic</code> derate with your delay calculation configuration. For more information, you can refer to the following <code>set_si_mode</code> command parameters: • <code>-delta_delay_annotation_mode</code> • <code>-separate_delta_delay_on_data</code> Notes • To apply the derate with <code>-dynamic</code>, you must specify <code>set_si_mode -separate_delta_delay_on_data true</code> setting. • When <code>-dynamic</code> is specified with: ◦ <code>set_si_mode -delta_delay_annotation_mode lumpedOnNet</code>, derate applies only to the SI component of net delays. ◦ <code>set_si_mode -delta_delay_annotation_mode arc</code>, derate applies to SI components of both cell and net delays. ◦ <code>-cell_delay</code> parameter and <code>set_si_mode -delta_delay_annotation_mode arc</code>, derate applies only to the SI component of cell delays. ◦ <code>-net_delay</code>, the derate applies to the SI component of net delays. • You cannot specify <code>-dynamic</code> with <code>-mean</code> or <code>-sigma</code> parameter.
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<code>-early float</code>	Specifies that the derating factor applies to early paths. Early paths are launching the clock during hold analysis, receiving the clock during setup analysis, and data path during hold analysis. Note: To use this parameter, you must specify <code>-delay_corner</code>.
<code>-fall</code>	Applies the derating factor to falling delay and checks arcs based on the transition direction at the arc's sink pin. This parameter can be used to selectively apply timing derates for the fall arcs (cell, net, or check arcs).

<code>-increment</code>	Incrementally adds the derate value to the total derate value in all the modes - OCV, AOCV, and SOCV modes. The <code>set_timing_derate</code> command saves two different values – the “base” derate and the “incremental” derate. When the <code>-incremental</code> parameter is specified, the software populates the incremental derates. Notes: • This parameter is used to specify incremental derates for leaf cell instances or library cell objects only. • <code>-increment</code> and <code>-multiply</code> are mutually exclusive parameters. • <code>-increment/-multiply</code> are supported with existing parameters, such as, <code>-late/-early</code>, <code>-rise/-fall</code>, <code>-mean/-sigma</code>. • Unlike base transition derate, the derate values with <code>-increment/-multiply</code> can be greater or smaller than 1. • Base transition derate values will be limited to: Late ≥ 1.0 and < 100.0, Early ≤ 1.0 and > 0.0. By default, a limit check on the total derate value will be performed during transition for delay calculation. If the total transition derate is out of range, a warning message will be issued. By default, the incremental derates are overwritten. To disable this behavior, you can use the following global variables: • <code>timing_derate_incremental_adjust_additive_mode</code>: Incremental derates are added to each other to obtain the final incremental derate value. • <code>timing_derate_incremental_multiply_accumulative_mode</code>: Incremental derates are multiplied with each other to obtain the final incremental derate value. Note: The <code>-increment</code> parameter is mutually exclusive with <code>-pulse_width_check</code>, <code>-clock_period_check</code>, and <code>-input_switching</code> parameters.
<code>-input_switching</code>	Considers the given derate as a simultaneous switching input (SSI) derate that is applicable to early paths only.

<code>-late float</code>	Specifies that the derating factor applies to late paths. The late paths are receiving clock during hold analysis, launching clock during setup analysis, and the data path during setup analysis. Note: To use this parameter, you must specify <code>-delay_corner</code>.
<code>-max</code>	Applies the derating factor to the maximum operating condition during best-case worst-case analysis. This parameter is ignored during on-chip variation analysis. Note: You cannot use the <code>-max</code> parameter when the software is in multi-mode multi-corner (MMMC) mode. You must use the <code>-delay_corner</code> parameter instead.
<code>-min</code>	Applies the derating factor to the minimum operating condition during best-case worst-case analysis. This parameter is ignored during on-chip variation analysis. Note: You cannot use the <code>-min</code> parameter when the software is in multi-mode multi-corner (MMMC) mode. You must use the <code>-delay_corner</code> parameter instead.
<code>-mean</code>	Controls nominal derating. You can apply separate derating to the mean components of the delay.
<code>-multiply</code>	Incrementally multiplies derates. You can also apply derates to specific leaf instances. This parameter also support nets objects.
<code>-net_delay</code>	Specifies scaling on net delays.
<code>object_list</code>	Specifies the list of instances, library cells, or nets to which the derating will be applied.
<code>-power_domain string</code>	Allows you to specify a specific power domain for the timing derate factors to be applied.
<code>-pulse_width_check</code>	Specifies derate values for pulse width checks. Note: The <code>-pulse_width_check</code> and <code>-clock_period_check</code> parameters are mutually exclusive. To apply check derating to pulse width checks, set the <code>timing_enable_derating_for_pulsewidth_checks</code> global variable to <code>true</code>. By default, the global is set to <code>false</code>.

<code>-retain_delay</code>	Specifies scaling on retain constructs. If you specify both <code>-retain_delay</code> and <code>-cell_delay</code> parameters, the software uses the <code>retain_delay</code> value for retain construct scaling. If you do not specify the <code>-retain_delay</code> parameter, retain constructs are scaled in the same manner as cell delays. You cannot specify <code>-retain_delay</code> with <code>-mean</code> or <code>-sigma</code> parameters.
<code>-rise</code>	Applies the derating factor to rising delay and checks arcs based on the transition direction at the arc's sink pin. This parameter can be used to selectively apply timing derates for the rise arcs (cell, net, or check arcs).
<code>-sigma</code>	Controls derating on the variation component. You can apply a separate derating to the sigma components of the delay.
<code>-static</code>	Applies the specified derating factor only to the non-delta portion of net delays. You can specify the <code>-net_delay</code> or <code>-cell_delay</code> parameter with the <code>-static</code> parameter. You cannot specify <code>-static</code> with <code>-mean</code> or <code>-sigma</code> parameters.
<code>-statistical</code>	Applies derating on cells or nets that are modeled as statistical data elements. The specified derating factor is applied to both the mean and the sigma of delays. If both the <code>set_timing_derate -corner</code> and <code>-statistical</code> options are not specified, the derating factor is equally applied to variation and corner objects. This is equivalent to issuing two separate <code>set_timing_derate</code> commands, one with the <code>-statistical</code> option and the other with the <code>-corner</code> option.

<code>-transition</code>	Specifies scaling on transitions. When specified, the software allows derate transition and stretch waveforms. For incremental derates applied to the base derate (without the <code>-increment</code> parameter), you can use the following parameters with the <code>-transition</code> parameter: • <code>-increment</code> parameter adds to the base derate to obtain the final derate. • <code>-multiply</code> parameter multiplies the incremental derate by the base derate to obtain the final derate value (<code>-increment</code> parameter is not required). • Validity of final derate value (same limits as base transition derate) is applied only during timing analysis, and not at the time of constraint processing. Note: You can use the <code>-transition</code> parameter with the following parameters: • <code>-delay_corner</code> • <code>-early/-late</code> • <code>-rise/-fall</code> • <code>-mean/-sigma</code> • <code>-increment</code> • <code>-multiply</code>
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Examples

Here are some examples.

- The following command (1) can be used when the software is in on-chip variation analysis mode to add pessimism to both the setup and hold analyses by making the early paths 20 percent faster:

```
tempus > set_analysis_mode -analysisType onChipVariation
```

...

```
(1) tempus > set_timing_derate -early 0.8 -late 1.0
```

- The following example uses two `set_timing_derate` commands to set the scaling factors independently for the min and max corners. When the software is in best-case worst-case analysis mode:
 - Command (1) adds pessimism to the setup analysis by making the latching clock path 20

percent faster.

- Command (2) adds pessimism to the hold analysis by making the latching clock path 10 percent slower.

```
(1) tempus > set_timing_derate -max -early 0.8 -late 1.0
```

```
(2) tempus > set_timing_derate -min -early 1.0 -late 1.1
```

- When the software is in on-chip variation analysis mode, there is only one active delay corner; therefore, the `-min` and `-max` parameters should not be used. The following command sequence will generate a warning:

```
tempus > set_analysis_mode -analysisType onChipVariation
```

```
...
```

```
tempus > set_timing_derate -min -early 0.8 -late 0.9
```

```
tempus > set_timing_derate -max -early 1.1 -late 1.2
```

In this situation, the software uses 0.8 for derating early paths and 1.2 for derating late paths.

- The following command sets the early path scaling factor of 0.85, for instance, `MULT_9` cell delays:

```
tempus > set_timing_derate -early -cell_delay 0.85 [get_cells {MULT_9}]
```

- The following command sets the early path scaling factor of 0.75 for the net named `MUX2/Z`:

```
tempus > set_timing_derate -net_delay -early 0.75 [get_nets {MUX2/Z}]
```

- The following command specifies power domain `PD1` for derating instances:

```
tempus > set_timing_derate -power_domain PD1 -cell_delay 1.1
```

- In the following example, a design having max library as 'slow' and min library as 'fast', only the first two `set_timing_derate` constraints are honored:

```
tempus > set_timing_derate -late 1.2 [get_lib_cells {slow/BUFX1}] -cell_delay
```

```
tempus > set_timing_derate -early 0.8 [get_lib_cells {slow/BUFX1}] -cell_delay
```

```
tempus > set_timing_derate -late 1.1 [get_lib_cells {fast/BUFX1}] -cell_delay
```

```
tempus > set_timing_derate -early 0.9 [get_lib_cells {fast/BUFX1}] -cell_delay
```

- Consider the following library sets, in MMMC mode:

```
tempus > create_library_set -name libset1 -timing slow.lib
```

```
tempus > create_library_set -name libset2 -timing fast.lib
```

```
tempus > create_delay_corner -name corner1 -late_library_set libset1 -early_library_set libset2
```

From the following `set_timing_derate` constraints, the first and last are honored:

```
tempus > set_timing_derate -late 1.2 -cell_delay [get_lib_cells slow/*] -delay_corner corner1
```

```
tempus > set_timing_derate -early 2.3 -cell_delay [get_lib_cells slow/*] -delay_corner corner1
```

```
tempus > set_timing_derate -late 3.4 -cell_delay [get_lib_cells fast/*] -delay_corner
corner1

tempus > set_timing_derate -early 4.5 -cell_delay [get_lib_cells fast/*] -delay_corner
corner1
```

- The following command can be used when the software is in on-chip variation analysis mode to add pessimism to both the setup and hold analyses by making the early paths 20 percent faster:

```
set_analysis_mode -analysisType onChipVariation
...
set_timing_derate -early 0.8 -late 1.0
```

This command adds pessimism to the setup analysis by making the latching clock path 20 percent faster.

- The following example uses two `set_timing_derate` commands to set the scaling factors independently for the min and max corners. When the software is in best-case worst-case analysis mode:

- The following command sets the early path scaling factor of 0.85, for instance, MULT_9 cell delays:

```
set_timing_derate -early -cell_delay 0.85 [get_cells {MULT_9}]
```

- The following command sets the early path scaling factor of 0.75 for the net named MUX2/Z:

```
set_timing_derate -net_delay -early 0.75 [get_nets {MUX2/Z}]
```

- The following command specifies power domain PD1 for derating instances:

```
set_timing_derate -power_domain PD1 -cell_delay 1.1
```

- The following command shows library object level support for clock (and pulse width) check options:

```
set_timing_derate -delay_corner [all_delay_corners] -clock_period_check 1.05
[object_list]
```

- The following command sets an increment derate of 0.5 on the mean and 0.8 on the sigma of the cell:

```
set_timing_derate -increment 0.8 -sigma MP_32_INST/p11A
set_timing_derate -increment 0.5 -mean MP_32_INST/p11A
```

To increment the sigma derate further by 0.9, you can use the following command:

```
set_timing_derate -increment 0.9 -sigma MP_32_INST/p11A
```

The resultant sigma derate can be calculated as $1+0.9=1.9$.

When the `timing_derate_incremental_adjust_additive_mode` global variable is set to `true`, the resultant sigma derate will be $1+0.8+0.9=2.7$.

Related Commands

- `report_timing_derate`
- `reset_timing_derate`

set_timing_library

```
set_timing_library
[-help]
[-max {Max | Min}]
[-min {Max | Min}]
```

Sets the timing libraries to be used in setup analysis (late-path delay calculation) or hold analysis (early-path delay calculation).

The default setting, if this command is not used, is that only maximum libraries will be used for setup analysis (late-path delay calculation) and only minimum libraries will be used for hold analysis (early-path delay calculation).

Use this command only after loading all the timing libraries.

Note: This command will become obsolete in the next major release as the software moves to using the Multi-Mode/Multi-Corner (MMMC) architecture for design import.

Parameters

-help	Prints out the command usage.
-max {Max Min}	Specifies the Max or Min libraries to be used in setup analysis.
-min {Max Min}	Specifies the Max or Min libraries to be used in hold analysis.

Examples

- The following command sets the timing library to `Max` used for both setup and hold analysis (early-path delay calculation):

```
tempus> set_timing_library -max Max -min Max
```

- The following script shows how to set a Min library in setup analysis and a Max library in hold analysis for reporting the detailed data paths that do not meet timing in a report file:

```
tempus> set_timing_library -max Min -min Max
tempus> set_analysis_mode -checkType setup
tempus> report_timing -max_points 10 > setupTiming.rpt
tempus> set_analysis_mode -checkType hold
tempus> report_timing -max_points 10 > holdTiming.rpt
```

- The `set_timing_library -max Min -min Max` command sets the Min libraries for setup analysis and the Max libraries for hold analysis.

- The `set_analysis_mode -checkType setup` command sets the timing analysis mode to `setup`.
- The `report_timing -max_points 10 > setupTiming.rpt` command runs a setup analysis using the `Min Libraries` and writes the result to the `setupTiming.rpt` file.
- The `set_analysis_mode -checkType hold` command sets the timing analysis mode to `hold`.
- The `report_timing -max_points 10 > holdTiming.rpt` command runs a hold analysis using `Max Libraries` and writes the output to the `holdTiming.rpt` file.

set_vt_skew_config

```
set_vt_skew_config
[-help]
[-analysis_mode {setup | hold | both}]
[-clocks_only]
[-correlated_vt_groups <corr_group>]
[-exclude <exclude>]
[-exclude_nominal]
[-include_only <include_only>]
[-unique_vt_mode {none | slow | fast | nominal}]
```

Allows customization of the combinations of Vth classes to be analyzed. By default, the software will analyze all the possible combinations.

This command is not cumulative, as each invocation will reset any options previously set by its default value unless that option is re-specified.

Parameters

-help	Prints out the command usage.
-analysis_mode {setup hold both}	Specifies the analysis mode for which the configuration is to be used.
-clocks_only	Allows the software to consider only launch and capture clock VTs post CPPR point when enumerating Vth combinations. By default, the software will consider Vth classes on the data path as well.
-correlated_vt_groups <corr_group>	Specifies the groups for correlation. By default, all Vth groups are considered uncorrelated. If a set of Vth groups is specified as correlated, then they will be considered as one entity when determining the combinations of Vth groups. All the correlated VTs will use the same (fast, slow, nominal) derate factor.
-exclude <exclude>	Specifies the configurations to be excluded. When this parameter is specified, the software excludes some combinations of Vth classes from the full elaborated set of combinations. The argument to this parameter is a Tcl list-of-lists as described for <code>-include_only</code> . The <code>-exclude</code> expression is a secondary filter which acts against the current included set of enumerations – either all by default, or a subset configured by <code>-include_only</code> .

<code>-exclude_nominal</code>	Determines whether to enumerate fast, slow, nominal combinations, or just fast, slow.
<code>-include_only</code> <code><include_only></code>	Specifies the configurations to be included. You can specify only the Vth combinations of interest. By default, all the possible combinations are analyzed. The argument to this parameter is a Tcl list-of-lists structure, which associates a Vth group with the set of analyses to consider for that group. Note: The options for <code>-include_only</code> and <code>-exclude</code> parameters are a Tcl list-of-lists that provides a way to customize the Vth enumerations that are analyzed. The format is a Vth group name followed by an enum list of combinations of fast, slow, or nominal. If a Vth group is missing from the specification, a value of 'all' will be assumed.
<code>-unique_vt_mode</code> {none slow fast nominal}	Allows the software to consider only the launch and capture clock VTs post CPPR point when enumerating Vth combinations. By default, the software will consider Vth classes on the data path as well. The unique Vt analysis mode is applicable only during the path-based analysis (PBA) step.

Examples

- The following command defines the derate mode to use in both GBA and PBA when the path has a unique VT:

```
set_vt_skew_config -unique_vt_mode slow
```

Setting a unique Vt mode is required for the GBA unique Vt flow.

Related Commands

- [set_threshold_voltage_group](#)
- [set_vt_skew_derate](#)

Related Topics

For more information, refer to the "Voltage Threshold Skew Modeling and Analysis" section of the [Analysis and Reporting](#) chapter in the *Tempus User Guide*.

set_vt_skew_derate

```
set_vt_skew_derate  
[-help]  
[-delay_corner <string>]  
[-fast <float>]  
[-hold]  
[-mean]  
[-setup]  
[-sigma]  
[-slow <float>]  
[-threshold_voltage_group <group name>]
```

Assigns fast and slow derate factors to specific threshold voltage groups.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-delay_corner <string></code>	Specifies the list of delay corners.
<code>-fast</code> <code><float></code>	Specifies the derate factor to use when evaluating both the early and late paths for a speed-up due to Vth skew shift. This derate value is expected to be less than 1.0.
<code>-hold</code>	Specifies that the derate is applied to the hold component.
<code>-mean</code>	Specifies that the derate is applied to the mean component. The specified derate only applies to mean delays in SOCV analysis. If the <code>-mean</code> or <code>-sigma</code> parameter is not specified, the derate applies to both the delay components.
<code>-setup</code>	Specifies that the derate is applied to the setup component.
<code>-</code> <code>sigma</code>	Specifies that the derate is applied to the sigma component. The specified derate applies only to the delay-variation component in SOCV analysis.
<code>-slow <float></code>	Specifies the derate factor to use when evaluating both the early and late paths for a slowdown due to Vth skew shift. This derate value is expected to be greater than 1.0.
<code>-</code> <code>threshold_voltage_group</code> <code><group name></code>	Specifies the threshold voltage group that is either predefined in Liberty or defined using the <code>set_threshold_voltage_group</code> command.

Examples

- The following commands define the VT skew derate factors and customize the different path combinations to be analyzed in PBA mode:

```
set_vt_skew_derate -threshold_voltage_group SVT -fast 0.90 -slow 1.10
set_vt_skew_derate -threshold_voltage_group LVT -fast 0.91 -slow 1.11
set_vt_skew_derate -threshold_voltage_group ULVT -fast 0.92 -slow 1.12
set_vt_skew_derate -threshold_voltage_group ELVT -fast 0.93 -slow 1.13
```

Related Commands

- `set_threshold_voltage_group`
- `set_vt_skew_config`

Related Topics

For more information, refer to the "Voltage Threshold Skew Modeling and Analysis" section of the [Analysis and Reporting](#) chapter in the *Tempus User Guide*.

set_wire_load_mode

```
set_wire_load_mode
[-help]
mode
```

Controls how the software searches through the design hierarchy to find the appropriate wire load model for a net, or a hierarchical part of a net.

Note: To disable wire load model functionality, you can make the following setting:

```
set enable_wire_load_model_support 0
```

Parameters

-help	Outputs a brief description that includes type and default information for each <code>set_wire_load_mode</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man set_wire_load_mode</code> .
mode	Specifies the wire-load model. You can specify one of the following options. - enclosed - segmented - top

	enclosed: Selects the wire load model using an area lookup table for the module that is at the lowest level in the design hierarchy and contains the entire net. When you set the mode to <code>enclosed</code>, the software uses the area of the smallest block enclosing the net to select the wire load model. In the <code>enclosed</code> mode, the software uses the following order of precedence to search the wire load model in: • The inner-most block enclosing the net on which you set the wire load selection group using the <code>set_wire_load_selection_group</code> command, if the <code>auto_wire_load_selection</code> global variable is enabled. Or: The inner-most block enclosing the net on which you set the wire load model using the <code>set_wire_load_model</code> command. • Each higher-level hierarchical block for a <code>set_wire_load_selection_group</code> assertion, all the way up to the top-level block, if the <code>auto_wire_load_selection</code> global variable is enabled. Or: Each higher-level hierarchical block for a <code>set_wire_load_model</code> assertion, all the way up to the top-level block. • The <code>default_wire_load_selection</code> field in the first timing library. Or, The <code>default_wire_load</code> field is in the first timing library. Note: If even the default values are not present in the timing library, you will not be able to use the wire load model functionality.
	segmented: Selects the wire load model in a way similar to the <code>enclosed</code> mode, except that each hierarchical segment of the net is treated separately. In the <code>segmented</code> mode: • A segment is defined by pins (including hierarchical pins) connected to a particular net. • When counting the drivers and receivers, only the pins connected to the segment are considered. This information, along with the wire load models determine the parasitic information for the segment. • The parasitics of each segment sum up to form the parasitics for the entire net. Note: Each segment may have different wire load models.

`top`: Selects the wire load model in a way similar to `enclosed` and `segmented`, except that all nets use the wire load selection or wire load models defined by the top-level (largest) block enclosing the net.

Note: If the first library read in does not define the default wire load mode, the software uses `top` as the default mode.

Examples

- The following command sets the wire load mode to `enclosed`:

```
tempus > set_wire_load_mode enclosed
```

Related Commands

- `reset_wire_load_mode`

set_wire_load_model

```
set_wire_load_model
[-help]
[list_of_instances_or_ports]
[-library library_name]
[-max]
[-min]
-name wireload_model
```

Specifies the wire load model to be used from the technology library, and sets the wire load model on the design or on a hierarchical object (instance). If the list of instances is not specified, the wire load model is set on the current instance. Otherwise, the wire load model is applied to all specified instances or ports.

Use wire-load models to estimate delays before the actual wire loads are back-annotated. A technology library contains different wire load models, previously computed based on the analysis of several designs differing in area.

To disable the wire load model functionality, you can make the following setting:

```
set enable_wire_load_model_support 0
```

Parameters

-help	Prints out the command usage.
-library library_name	Specifies the technology library to use when searching for the name of the wire load model. You can specify a library name or a collection of libraries. <i>Default:</i> All technology libraries are searched.
list_of_instances_or_ports	Specifies the list of objects - designs, ports, or hierarchical instances.
-max	Sets the wire load model only for the maximum PVT (process, voltage, temperature) corner. <i>Default:</i> The specified wire load model applies to all three PVT corners if neither the -min nor the -max parameter is specified.
-min	Sets the wire load model only for the minimum PVT (process, voltage, temperature) corner. <i>Default:</i> The specified wire load model applies to all three PVT corners if neither the -min nor the -max parameter is specified.
-name wireload_model	Specifies the name of the wire load model in the library.

Examples

- The following command sets the `COM3_LV_WLM_0.0` wire load model from the `COM3_LV_WIRE` library on the current module for all PVT corners:

```
tempus> set_wire_load_model -library COM3_LV_WIRE -name COM3_LV_WLM_0.0
```

- The following command sets the `B1X1` wire load from the `lca500kv` library on the `clk1` port for the minimum PVT corner:

```
tempus> set_wire_load_model -library lca500kv -min -name B1X1 clk1
```

- The following command sets the `3M_TO_4500K` wire load from the `c13tsmc_slow2s108v125c` library on the current module for the minimum PVT corner:

```
tempus> set_wire_load_model -library c13tsmc_slow2s108v125c -min -name 3M_TO_4500K
```

Related Commands

- `report_wire_load`
- `reset_wire_load_model`

set_wire_load_selection_group

```
set_wire_load_selection_group
[-help]
wireload_selection_group_name
[list_of_instances]
[-library library_name]
[-max]
[-min]
```

Overrides the wire load selection table in the specified library.

To disable the wire load model functionality, you can make the following setting:

```
set enable_wire_load_model_support 0
```

Parameters

-help	Outputs a brief description that includes type and default information for each <code>set_wire_load_selection_group</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man set_wire_load_selection_group</code>
-library <i>library_name</i>	Specifies the name of the technology library to search for the wire load selection table. When this parameter is not specified, the software searches all the libraries.
<i>list_of_instances</i>	Specifies the list of instances.
-max	Controls the choice of a wire load selection table only for the PVT max corner. <i>Default:</i> Selects the table for all PVT corners if neither the <code>-min</code> nor the <code>-max</code> parameter is specified.
-min	Controls the choice of a wire load selection table only for the minimum PVT corner. <i>Default:</i> Selects the table for all PVT corners if neither the <code>-min</code> nor the <code>-max</code> parameter is specified.
<i>wireload_selection_group_name</i>	Specifies the name of the wire load selection group.

Examples

- The following command selects `TLM` as the selection table used by the library:

```
tempus> set_wire_load_selection_group TLM
```
- The following command selects `ASPEC1` as the selection table used by the `sm333s` library; only the minimum PVT corner is affected:

```
tempus> set_wire_load_selection_group -min -library sm333s ASPEC1
```

Related Commands

- `reset_wire_load_selection_group`

timing_enable_boundary_model_eco

```
timing_enable_boundary_model_eco  
[-help]
```

Enables all boundary model settings required for the proper working of the ECO flow.

This command should be used after all the STA settings have been made. You can use this command in the `create_boundary_model` and `read_boundary_model` command sessions.

Parameters

-help	Prints out the command usage.
-------	-------------------------------

Related Commands

- `create_boundary_model`
- `read_boundary_model`

update_analysis_view

```
update_analysis_view  
[-help]  
-name <existingAnalysisViewName>  
{[-constraint_mode <newModeName>]  
[-delay_corner <newDelayCornerObj>]  
[-latency_file <string>]}
```

Changes attribute values for the specified existing analysis view.

Note: You can update an analysis view before or after the multi-mode multi-corner definition file is loaded into the design. However, after the software processes the file, any change to an existing object

when the software is in multi-mode multi-corner mode results in the timing, delay calculation, and RC data being reset.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-constraint_mode</code> <i>newModeName</i>	Specifies the name of the new constraint mode to associate with this analysis view.
<code>-delay_corner</code> <i>newDelayCornerObj</i>	Specifies the name of the new delay calculation corner to associate with this analysis view.
<code>-latency_file</code> <i>string</i>	Specifies the latency file name.
<code>-name</code> <i>existingAnalysisViewName</i>	Specifies the name of the existing analysis view to update.

Examples

- The following command creates an analysis view called `missionSlow` using the constraint mode `missionSetup` and the delay calculation corner `dcWCCOM`:

```
tempus> create_analysis_view  
-name missionSlow  
-constraint_mode missionSetup  
-delay_corner dcWCCOM
```

The following command changes the delay calculation corner associated with the analysis view `missionSlow` to `dcMax`:

```
tempus> update_analysis_view -name missionSlow -delay_corner dcMax
```

Related Commands

- `create_analysis_view`
- `set_analysis_view`
- `get_analysis_view`

update_constraint_mode

```
update_constraint_mode  
[-help]  
[-name modeName]  
{[[-sdc_files {file1.sdc file2.sdc ...}]  
[-ilm_sdc_files {file1.sdc file2.sdc ...}]  
[-tcl_vars {{var_name1 value1} {var_name2 value2} {var_name3 value3} ...}] [-  
reset_latency_files ]] }
```

Updates the SDC constraint file information for the specified existing constraint mode object.

Note: The `update_constraint_mode` command overwrites the previous constraint mode settings. However, by default, this command updates a constraint mode without affecting other constraint modes, including interactive constraints, such as `set_dont_use` and `set_dont_touch`.

You can update a constraint mode object before or after the multi-mode multi-corner definition file is loaded into the design. However, after the software processes the file, any change to an existing object when the software is in multi-mode multi-corner mode results in the timing, delay calculation, and RC data being reset.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-ilm_sdc_files</code> { <i>file1.sdc file2.sdc</i> ...}	Specifies the new full chip SDC constraint files for the ILM flow to associate with this constraint mode.
<code>-name modeName</code>	Specifies the name of the existing constraint mode object to update.
<code>-reset_latency_files</code>	Cleans up the latency files of all the related views. The timing system maintains view-specific latency files to provide <code>set_clock_latency</code> assertions for useful-skew modeling and for providing post-CTS balancing of IO timing. When the <code>update_constraint_mode</code> is asserted to provide new constraints for a mode, you can decide whether to remove the related constraints in the latency files.
<code>-sdc_files</code> { <i>file1.sdc file2.sdc</i> ...}	Specifies the new SDC constraint files to associate with this constraint mode.
<code>-</code> <code>tcl_vars {{var_name1</code> <code>value1} {var_name2</code> <code>value2} {var_name3</code> <code>value3} ...}</code>	Specifies a list of TCL variables and values to be used in the constraint mode.

Examples

- The following command groups the SDC files `io.sdc`, `mission1-clks.sdc`, and `mission1-except.sdc` to create a mode object named `missionSetup`:

```
tempus> create_constraint_mode -name missionSetup
-sdc_files [list io.sdc mission1-clks.sdc mission1-except.sdc]
```

The following command changes the SDC files associated with the constraint mode `missionSetup` to `io.sdc`, `mission3-clks.sdc`, and `mission3-except.sdc`:

```
tempus> update_constraint_mode -name missionSetup
-sdc_files {io.sdc mission3-clks.sdc mission3-except.sdc}
```

Related Commands

- `create_constraint_mode`
- `get_constraint_mode`
- `all_constraint_modes`

update_delay_corner

```
update_delay_corner
[-help]
[-early_estimated_worst_irDrop_factor percent]
[-early_irdrop_data files_or_directories]
[-early_irdrop_file list_of_files]
[-early_library_set libSetName]
[-early_opcond opcondName]
[-early_opcond_library libName]
[-early_rc_corner rcCornerObj]
[-early_temp_file tempName]
[-irdrop_data files_or_directories]
[-irdrop_file list_of_files]
[-late_estimated_worst_irDrop_factor percent]
[-late_irdrop_data files_or_directories]
[-late_irdrop_file list_of_files]
[-late_library_set libSetName]
[-late_opcond opcondName]
[-late_opcond_library libName]
[-late_rc_corner rcCornerObj]
[-late_temp_file tempName]
[-library_set libSetObj]
[-name delayCornerObj]
[-opcond opcondName]
[-opcond_library libName]
[-pg_net_voltages voltage_pairs]
[-power_domain powerDomainName]
[-rc_corner rcCornerObj]
[-si_enabled {true | false}]
[-supply_set supplySetName]
[-temp_file tempName]
```

Modifies the parameters of an existing delay calculation corner object. You can use the `update_delay_corner` command to add or change attribute values for an existing delay calculation corner object, or for an existing power domain definition.

You can update a delay calculation corner object before the multi-mode multi-corner definition file is loaded into the design, or after. However, after the software processes the file, any change to an existing object when the software is in multi-mode multi-corner mode results in the timing, delay calculation, and RC data being reset.

You can also use the `update_delay_corner` command to add a power domain definition to the specified delay calculation corner. Instances contained within an MSMV power domain often require a different library or operating condition that used for the default power domain. Power domain definitions within a delay calculation corner object can be used to assign specific libraries and PVT settings per power domain.

Note:

- You must use the same power domain names that you intend to specify in the power domain file to define the physical aspects of the domain.
- Use separate delay calculation corners to define major PVT corner differences. Use the `-early_*` and `-late_*` parameters within a single delay calculation corner to control on-chip-variation.

The `-early_*` and `-late_*` parameters can be specified separately, to update a delay corner object with early or late information. If you only specify one of the `-early_*` or `-late_*` parameters, the software uses the original information pointer in place of the missing early/late parameter.

For example, if you initially created a delay corner object using:

```
create_delay_corner
-library_set libsNominal
...
```

If you specify `update_delay_corner -early_library_set libsEarly`, the software actually uses:

```
update_delay_corner
-early_library_set libsEarly
-late_library_set libsNominal
```

Use this command after creating at least one delay calculation corner (`create_delay_corner`).

Parameters

<code>-help</code>	Prints out the command usage.
<code>-early_estimated_worst_irDrop_factor percent</code>	Specifies the early voltage derating factor. When this parameter is specified, the software performs delay calculations using the voltage = $\text{nomVolt} * (1 - \text{value})$. <i>Default:</i> 0.0
<code>-early_irdrop_data files_or_directories</code>	Specifies the IR drop files or directories to apply when calculating the early arrival times at a single delay corner.

<code>-early_irdrop_file list_of_files</code>	Specifies the list of IR drop files to apply when calculating early arrival times at a single power domain. This parameter is used primarily when configuring the multi-mode multi-corner environment for on-chip-variation (OCV) analysis. Note: This parameter is obsolete and will be removed in a future release. To ensure compatibility with future releases, you should update your scripts.
<code>-early_library_set libSetName</code>	Specifies the library set to associate with the power domain for calculating early arrival times for this power domain. This parameter is used primarily when configuring the multi-mode multi-corner environment for on-chip-variation (OCV) analysis.
<code>-early_opcond opcondName</code>	Specifies the operating condition to use for calculating early arrival times for this power domain. This parameter is used primarily when configuring the multi-mode multi-corner environment for on-chip-variation (OCV) analysis. <i>Default:</i> Each library in the early library set uses its own default operating condition.
<code>-early_opcond_library libName</code>	Specifies the internal library name for the library in which the early operating condition is defined. This is <i>not</i> the library file name. This parameter is used primarily when configuring the multi-mode multi-corner environment for on-chip-variation (OCV) analysis. <i>Default:</i> The software searches the early library set for the specified operating condition (<code>-early_opcond</code>), starting with the master library.
<code>-early_rc_corner rcCornerObj</code>	Specifies the RC corner object to associate with the early corner object.

<code>-early_temp_file tempName</code>	Specifies the name of the temperature file for temperature-aware delay calculation for the early corner.
<code>-irdrop_data files_or_directories</code>	Specifies the IR drop files or directories to apply to both the early and late delay calculation for the specified delay corner object.
<code>-irdrop_file list_of_files</code>	Specifies the list of IR drop files to apply to both early and late delay calculation for this power domain object. This parameter is used primarily to configure single-corner or Best-Case Worst-Case (BC-WC) analysis modes. Note: This parameter is obsolete and will be removed in a future release. To ensure compatibility with future releases, you should update your scripts.
<code>-late_estimated_worst_irDrop_factor percent</code>	Specifies the late voltage derating factor. When this parameter is specified, the software performs delay calculations using the voltage = $\text{nomVolt} * (1 - \text{value})$. <i>Default:</i> 0.0
<code>-late_irdrop_data files_or_directories</code>	Specifies the IR drop files or directories to apply when calculating the late arrival times at a single delay corner.
<code>-late_irdrop_file list_of_files</code>	Specifies the list of IR drop files to apply when calculating late arrival times at a single power domain. This parameter is used primarily when configuring the multi-mode multi-corner environment for on-chip-variation (OCV) analysis. Note: This parameter is obsolete and will be removed in a future release. To ensure compatibility with future releases, you should update your scripts.

<code>-late_library_set libSetName</code>	Specifies the library set to associate with this delay corner object for calculating late arrival times for this power domain. This parameter is used primarily when configuring the multi-mode multi-corner environment for on-chip-variation (OCV) analysis.
<code>-late_opcond opcondName</code>	Specifies the operating condition to use for calculating late arrival times for this power domain. This parameter is used primarily when configuring the multi-mode multi-corner environment for on-chip-variation (OCV) analysis. <i>Default:</i> Each library in the late library set uses its own default operating condition.
<code>-late_opcond_library libName</code>	Specifies the internal library name for the library in which the late operating condition is defined. This is <i>not</i> the library file name. This parameter is used primarily when configuring the multi-mode multi-corner environment for on-chip-variation (OCV) analysis. <i>Default:</i> The software searches the late library set for the specified operating condition (<code>-late_opcond</code>), starting with the master library.
<code>-late_temp_file tempName</code>	Specifies the name of the temperature file for temperature aware delay calculation for late corner.
<code>-late_rc_corner rcCornerObj</code>	Specifies the RC corner object to associate with the late corner object.
<code>-library_set libSetName</code>	Specifies the library set to associate with the power domain. All of the cells and macros in the power domain get their timing and power information from these libraries. This parameter is used primarily to configure single-corner or Best-Case Worst-Case (BC-WC) analysis modes.

<code>-name dcCornerObj</code>	Specifies the delay calculation corner to which to add the power domain definition, or to which to update the delay corner or power domain attribute values.
<code>-opcond OpcondName</code>	Specifies the operating condition to use for the power domain. This parameter is used primarily to configure single-corner or Best-Case Worst-Case (BC-WC) analysis modes. <i>Default:</i> Each library in the library set uses its own default operating condition.
<code>-opcond_library libName</code>	Specifies the internal library name for the library in which the operating condition is defined. This is <i>not</i> the library file name. This parameter is used primarily to configure single-corner or Best-Case Worst-Case (BC-WC) analysis modes. <i>Default:</i> The software searches the library set for the specified operating condition (<code>-opcond</code>), starting with the master library.
<code>-pg_net_voltages voltage_pairs</code>	Specifies the power/ground net voltages to be used for both early/late timing analysis.
<code>-power_domain powerDomainName</code>	Specifies the name of the power domain to add to the delay calculation corner. This name must match the name of a power domain specified in the power domain file. Note: This parameter is required for all the designs that use power domains and should be specified for each delay corner in the MMMC file.
<code>-rc_corner rcCornerObj</code>	Specifies the new RC corner object to associate with this delay corner.

<code>-si_enabled {true false}</code>	Controls signal integrity (SI) analysis. When set to <code>true</code>, the analysis of all MMMC views will include SI analysis by default. You can use this parameter to further determine which analysis views receive SI analysis and which do not. When this parameter is set to <code>false</code> on a delay corner, the software will disable SI analysis for all the views that reference that corner. By default (or when this parameter is set to <code>true</code>), SI analysis is computed for all the views related to that corner - whenever SI is globally enabled. (The global control of signal integrity (SI) analysis is determined by the <code>set_delay_cal_mode -SIAware</code> setting.)
<code>-supply_set <i>supplySetName</i></code>	Specifies the name of the supply set to add to the delay calculation corner.
<code>-temp_file <i>tempName</i></code>	Specifies the name of the temperature file for temperature-aware delay calculation for a single corner.

Examples

- The following command creates the delay calculation corner `dc1`:

```
tempus> create_delay_corner -name dc1
-library_set libs-2volt
-opcond_library delayvolt_2V
-opcond slow_2V
-rc_corner rc-cworst
```

The following commands add definitions for the power domains `domain-3V` and `domain-5V` to the `dc1` delay calculation corner:

```
tempus> update_delay_corner -name dc1
-power_domain domain-3V
-library_set libs-3volt
-opcond_library delayvolt_3V
-opcond slow_3V
update_delay_corner -name dc1
-power_domain domain-5V
-library_set libs-5volt
-opcond_library delayvolt_5V
-opcond slow_5V
```

- The following command changes the libraries used by the delay calculation corner `dc1` to those in the library set `IsCOM-2v`, and changes the RC corner associated with `dc1` to `rc-ctyp`:

```
tempus> update_delay_corner -name dc1
-library_set IsCOM-2V
-rc_corner rc-ctyp
```

Related Commands

- [create_delay_corner](#)
- [get_delay_corner](#)

update_io_latency

```
update_io_latency  
[-help]  
[-source]  
-verbose
```

Provides an automated means of balancing IO and Core clock latencies after the clock network has been added to the design so that the pre and post clock tree skew between the clocks is maintained as closely as possible. This results in the minimum change of inbound and outbound IO slacks. The purpose of the `update_io_latency` command is to match the skew before and after CTS. The command must be run once with the clock in the ideal mode.

This command performs the following actions:

- Measures and calculates the ideal average latency for each Core clock.
- Calculates preCTS skew between IO and Core clocks.
- Asserts `set_propagated_clock` on all clock root pins or ports to put clock trees in propagated mode.
- Measures and calculates propagated mode average latency for each Core clock.
- Adjusts `set_clock_latency` on the IO clocks network latency to minimize pre to post CTS IO-to-Core clock skew.

In case a single IO clock interacts with multiple Core clocks, the IO clock will be adjusted by the overall average latency of all the related Core clocks combined.

For example, let us consider that in the ideal mode, the IO latency is 500ps and the core latency is 600ps, resulting in a skew of 100ps. After CTS, we need to match the skew. Therefore, if the core comes in at 650ps, then the IO should be adjusted to 550ps.

To be effective, this command should only be used after clock trees have been implemented. This allows the command to see both the ideal and propagated state of the clock trees.

Note: The `set_propagated_clock` command should not be issued after the `update_io_latency` command.

Parameters

-help	Prints out the command usage.
-source	Adjusts the source latency of the core clock.
-verbose	Provides additional information to the console as adjustments are made to each IO clock.

Related Commands

- `set_propagated_clock`

update_library_set

```
update_library_set
[-help]
[-aocv string]
[-library_side_file string]
-name libSetName
[-si string]
[-socv string]
[-timing string]
```

Updates an existing specified library set.

Note: You can edit a library set before (or after) the multi-mode multi-corner view definition file is loaded into the design. However, after the software processes the file, any change to an existing object when the software is in multi-mode multi-corner mode resets the timing, delay calculation, and RC data.

Parameters

-help	Prints out the command usage.
-aocv <i>string</i>	Specifies a list of new AOCV libraries to include in the specified library set.
- library_side_file <i>string</i>	Specifies the library side file.
-name <i>libSetName</i>	Specifies the name of an existing library set to be updated. The library set must have been defined using the create_library_set command.
-si <i>string</i>	Specifies a list of new cdB libraries and user-defined noise (UDN) models to include in the library set.
-socv <i>string</i>	Specifies a list of new SOCV libraries to be included in the library set.
-timing <i>string</i>	Specifies a list of new timing libraries to include in the specified library set. Note: The ECSM and CCS libraries can be very large, and loading them may take a significant amount of time. You can convert the ASCII NLDM (Liberty .lib), ECSM, and CCS libraries to a binary version using the write_ldb command. For converting libraries to a binary format, refer to write_ldb. Instead of the original ASCII libraries, you can specify the following binary libraries in the configuration file: • <code>create_library_set -name libName -timing "AAA.lib BBB.ecsm CCC.ccs"</code>

Examples

- The following command creates a library set that associates timing libraries and cdB libraries with a nominal voltage of 1 volt, with the library name IsCOM-1V:

```
tempus> create_library_set -name IsCOM-1V -timing [list stdcell_F_1V.lib ram_F.lib  
pad.lib]
```

The following command changes the timing libraries associated with the library set IsCOM-1V:

```
tempus> update_library_set -name IsCOM-1V -timing [list stdcell_CVF_1V.lib ram_CVF.lib  
pad.lib ]
```

Related Commands

- `create_library_set`
- `get_library_set`
- `all_library_sets`

update_model_context

```
update_model_context  
[-help]  
-dir string  
[-instances string]
```

Allows the software to read in the context-only boundary models and perform timing analysis without the need to set up a new STA run with a different boundary model.

Parameters

-help	Prints out the command usage.
-dir <i>string</i>	Specifies the directory for the model context.
-instances <i>string</i>	Specifies the instances for context update.

Examples

- The following example shows that commands that are used to update the timing context of a boundary model:

```
read_boundary_model -dir dir_derate_nominal  
...  
set_timing_derate -early 1  
set_timing_derate -late 1  
update_timing  
update_model_context -dir dir_derate_extreme  
set_timing_derate -early 0.8  
set_timing_derate -late 1.2  
update_timing
```

Related Commands

- [write_model_context](#)
- [create_boundary_model](#)
- [read_boundary_model](#)

update_rc_corner

```
update_rc_corner
[-help]
[-T float]
-name string
[-postroute_cap string]
[-postroute_res string]
[-postroute_xcap string]
[-qx_tech_file string]
[-via_variation_file string]
```

Adds or changes attribute values for the specified existing RC corner object.

Note: You can edit an RC corner object before the multi-mode multi-corner definition file is loaded into the design or after. However, after the software processes the file, any change to an existing object when the software is in multi-mode multi-corner mode results in the timing, delay calculation, and RC data being reset.

Note: If the `read_spef` command is specified without the `-rc_scale` parameter, for example, "`read_spef -rc_corner early_spef_${rc_corner}`", then the software does not apply the capacitance scale factor specified in `-postroute_cap` and `-postroute_xcap` parameters. If the `read_spef` command is specified with the `-rc_scale` parameter, for example, "`read_spef -rc_corner early_spef_${rc_corner} -rc_scale` ", then the software applies the capacitance scale factor. Here, no factor is specified in the `-rc_scale` option.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-name <i>string</i></code>	Specifies the name for the RC corner object being created. You can choose any name for an RC corner object, but it is recommended that you use a name that can be easily recognized as referring to a specific RC corner (for example, rc-min, rc-max, c-min, c-max, and so on).
<code>-postroute_cap <i>string</i></code>	Specifies the capacitance scale factor for RC extraction in post-route mode.
<code>-postroute_res <i>string</i></code>	Specifies the resistance scale factor for RC extraction in post-route mode.
<code>-postroute_xcap <i>string</i></code>	Specifies the coupling capacitance scale factor for RC extraction in post-route mode.

<code>-qx_tech_file string</code>	Specifies the name of a Quantus QRC technology (.tch) file used by Quantus QRC sign-off RC extraction. The technology file is generated by TechGen, a built-in functionality of Quantus QRC. For most technologies, you can obtain this file from the technology vendor. For technologies not available from the technology vendors, you can enter fabrication process information into an ASCII-format interconnect technology (ICT) input file. The ICT file is then used as input to IceCaps, which in turn generates the technology file.
<code>-T float</code>	Specifies the temperature, in units of Celsius, to use for derating resistance values. Use this parameter to override the default temperature specified in the Quantus QRC technology file. Note: The software does not require that the temperature specified for the PVT operating condition at the delay calculation corner level match the RC nominal or user-specified temperature value.
<code>-via_variation_file string</code>	Specifies a file for modeling via variation. The VIA variation file lists all VIA layer names and their corresponding multipliers. Each RC corner can have its own VIA variation file. For example: <pre>update_rc_corner -name rcCornerName1 \ -via_variation_file viaVariationFileName1 update_rc_corner -name rcCornerName2 \ -via_variation_file viaVariationFileName2</pre>

Examples

- The following command creates an RC corner called `rc-cbest` that uses the capacitance table `myTech_bc.CapTbl`, and derates the resistance values based on the temperature of 50 Celsius:
`create_rc_corner -name rc-cbest -cap_table myTech.4_bc.CapTbl -temperature 50`
- The following command changes the capacitance table used by RC corner `rc-cbest` to `myCap.2_bc.capTbl`, and the temperature by which resistance values are derated to 25 Celsius:
`update_rc_corner -name rc-cbest -cap_table myCap.2_bc.capTbl -temperature 25`

Related Commands

- `create_rc_corner`
- `all_rc_corners`

update_timing

update_timing

[-help]

[-full]

Updates timing for the current design.

Parameter

-help	Prints out the command usage.
-full	Runs a full timing update for the current design, overriding the default behavior of running incremental timing updates, where only a minimal amount of computation is necessary to update the timing analysis information. Note that for full SI accuracy during reporting, an explicit <code>update_timing -full</code> command is required after incrementally applying constraints such as <code>set_timing_derate</code>, <code>set_input_delay</code>, <code>set_max_delay</code>, <code>set_clock_groups</code>, or <code>create_clock</code> (these constraints only affect SI delay calculation). Note: It is not required to run the <code>update_timing</code> command without the <code>-full</code> parameter because timing is automatically updated by the <code>report_timing</code> and <code>report_constraint</code> commands - except after incrementally applying constraints that only affect SI delay calculation.

Related Commands

- `check_timing`
- `report_timing`
- `report_constraint`

write_global_slack_report

```
write_global_slack_report
```

```
[-help]
```

```
[-early]
```

```
[-late]
```

```
[-rise]
```

```
[-fall]
```

```
[-include_ports]
```

```
[-max_slack val ]
```

```
[-min_slack val ]
```

```
[-view view_name ]
```

```
[> | >> file_name ]
```

```
[ pin_port_list ]
```

Reports early and late slack values on every instance pin in the design.

This command computes and reports the required time and slack values of the specified pins.

Note: In DSTA mode, the `write_global_slack_report` command does not print information on pins that do not have any valid slack values.

Parameters

<code>-help</code>	Prints out the command usage.
<code>> </code> <code>>> filename</code>	Specifies the output file name. When specified with <code>></code>, writes the report to the specified file name. If the file already exists, the software overwrites it. When specified with <code>>></code>, writes the report to the specified file name. If the file already exists, the software concatenates the report to the end of the file. Note: To write a compressed report, add the <code>.gz</code> extension to the file name.
<code>-early/-late</code>	Specifies the mode for which slack is to be reported.
<code>-fall/-rise</code>	Specifies the transition for which slack is to be reported.
<code>-include_ports</code>	Reports slack information for ports. By default, the slack information is reported for pins only.
<code>-max_slack</code> <code>val</code>	Reports pins with upper-bound slack values less than the specified value.
<code>-min_slack</code> <code>val</code>	Reports pins with lower-bound slack values less than the specified value. Note: This parameter is obsolete and will be removed in a future release. Cadence recommends using the <code>-max_slack</code> parameter.
<code>pin_port_list</code>	Specifies the list of pins or ports for which slack information is to be reported.
<code>-view</code> <code>view_name</code>	Reports slack information for the specified view.

Examples

- The following command reports the early and late slacks on every instance pin in the design:

```
tempus > write_global_slack_report > out.rpt

# Unit: 1ns
# <late_rise> <late_fall> <early_rise> <early_fall> <view_name> <hpin_name>
# -----
*          *          *          *          (null)      u2/Q
*          *          *          *          (null)      u8/Q
*          *          *          *          (null)      b11/A
*          *          *          *          (null)      b11/Y
0.823      0.347      0.115      0.501      (null)      u10999/A
0.823      0.347      0.115      0.501      (null)      u10999/Y
# -----
```

- The following command reports slack for a pin collection:

```
tempus> set x [get_pins -hier */A]

0x14 <---- 'x' stores Pointer for pin collection
```

- The following command will generate a global slack report for pin collection as defined in "x":

```
tempus> write_global_slack_report $x

# Unit: 1ns
# <late_rise> <late_fall> <early_rise> <early_fall> <view_name> <hpin_name>
# -----
19.622      19.742      0.321      0.106      (null)      seg1/u1/A
19.622      19.742      0.321      0.106      (null)      seg1/u2/A
18.018      18.362      1.876      1.486      (null)      seg1/u10/A
18.018      18.362      1.876      1.486      (null)      seg1/u11/A
18.362      18.018      1.486      1.876      (null)      seg1/u12/A
18.018      18.362      1.876      1.486      (null)      seg1/u13/A
25.845      25.883      -0.934     -1.053     (null)      seg2/u1/A
25.845      25.883      -0.934     -1.053     (null)      seg2/u2/A
# -----
```

- The following example shows slack calculations, when you run the [report_timing](#) command for the same hpin.

```
tempus > write_global_slack_report DT_IN/RE_CO/r697_reg_2/D

# Unit: 1ns
# <late_rise> <view_name> <late_fall> <view_name> <early_rise>
<view_name> <early_fall> <view_name> <hpin_name>
# -----
```

```
12.641 dtmf_view_setup 12.127 dtmf_view_setup * *
DT_IN/RE_CO/r697_reg_2/D
```

```
tempus > report_timing -to DT_IN/RE_CO/r697_reg_2/D -late -fall
```

```
Path 1: MET Setup Check with Pin DT_IN/RE_CO/r697_reg_2/CKN
Endpoint: DT_IN/RE_CO/r697_reg_2/D (v) checked with trailing
edge of 'vclk2'
Beginpoint: DT_IN/RE_CO/r697_reg_2/Q (v) triggered by trailing
edge of 'vclk2'
Path Groups: {vclk2}
Analysis View: dtmf_view_setup
Other End Arrival Time 7.015
- Setup 0.600
+ Phase Shift 14.000
- Uncertainty 0.250
= Required Time 20.164
- Arrival Time 8.037
= Slack Time 12.127
Clock Fall Edge 7.000
+ Clock Network Latency (Prop) 0.015
= Beginpoint Arrival Time 7.015
```

+-----+-----+-----+-----+-----+-----+						
Instance	Arc	Cell	Delay	Arrival	Required	
				Time	Time	
+-----+-----+-----+-----+-----+-----+						
DT_IN/RE_CO/r697_reg_2	CKN v			7.015	19.142	
DT_IN/RE_CO/r697_reg_2	CKN v -> Q v	SDDFFNX1	0.710	7.725	19.852	
DT_IN/RE_CO/i_6594	B v -> Y v	MX2X1	0.312	8.037	20.164	
DT_IN/RE_CO/r697_reg_2	D v	SDDFFNX1	0.000	8.037	20.164	
+-----+-----+-----+-----+-----+-----+						

```
tempus > report_timing -to DT_IN/RE_CO/r697_reg_2/D -late -rise
```

```
Path 1: MET Setup Check with Pin DT_IN/RE_CO/r697_reg_2/CKN
Endpoint: DT_IN/RE_CO/r697_reg_2/D (^) checked with trailing
edge of 'vclk2'
Beginpoint: DT_IN/RE_CO/r697_reg_2/Q (^) triggered by trailing
edge of 'vclk2'
Path Groups: {vclk2}
Analysis View: dtmf_view_setup
Other End Arrival Time 7.015
- Setup 0.204
+ Phase Shift 14.000
- Uncertainty 0.250
= Required Time 20.560
- Arrival Time 7.919
```



```
= Slack Time 12.641
Clock Fall Edge 7.000
+ Clock Network Latency (Prop) 0.015
= Beginpoint Arrival Time 7.015
```

+-----+					
Instance	Arc	Cell	Delay	Arrival Time	Required Time
+-----+					
DT_IN/RE_CO/r697_reg_2	CKN v			7.015	19.656
DT_IN/RE_CO/r697_reg_2	CKN v -> Q ^	SDDFNX1	0.628	7.643	20.284
DT_IN/RE_CO/i_6594	B ^ -> Y ^	MX2X1	0.276	7.919	20.560
DT_IN/RE_CO/r697_reg_2	D ^	SDDFNX1	0.000	7.919	20.560
+-----+					

Related Commands

- `write_global_slack_worst_trigger_path_on_clocks`

write_ipd_scope

```
write_ipd_scope
[-help]
[-context_only]
[-exclude <string>]
[-include <string>]
[-scope_data <string>]
[[[-bounded_corners ] [-include_derates ] [-include_constraints] [-overwrite ] [<dir_name>]]
| [[[-early_structural_checks <string>] [-out_file_prefix <string>] [-
power_domain_count_threshold <integer>]] | [-write_all_paths <string>]]
```

Writes out the IPD scope in the scope creation run. The scope is generated in the directory location specified as the command argument.

This command is used to write out the IPD Scope in the scope creation run. The scope is generated in the directory location specified as a command argument, which provides options to control whether constraints and derates are to be written out in the scope or not.

Parameters

-help	Prints out the command usage.
-context_only	Re-generates constants that will be a part of context data.
-early_structural_checks <string>	Specifies the early check types to be performed. The valid values are: • clock_divergence • multiple_pd_data_paths • multiple_pd_clock_paths • multiple_pd_paths
-exclude <string>	Specifies the netlist exclusions. The supported options are clock, port, async, scanchain, and <\$list_of_PD_combinations>.
-include <string>	Specifies the netlist filter inclusions - the supported options are: • equivalent_power_domains • internal_power_domains • timing_disabled_network • equivalent_supply_voltage

<code>-include_constraints</code>	Allows the IPD scope to be written with victim constraints.
<code>-include_derates</code>	Allows the IPD scope to be written with timing derates.
<code><dir_name></code>	Specifies the output directory containing partial Verilog and partial SPEF, or marking data, and context data.
<code>-bounded_corners</code>	Generates a bounded corner scope based on the minimum/maximum voltage combination analyzed in the scope creation run. By default, the software writes out the maximum voltage combination details in the scope if this parameter is not specified in the bounded corner run.
<code>-scope_data <string></code>	Specifies the scope data to be written from the minimum or maximum voltage combination.
<code>-out_file_prefix <string></code>	Specifies the outfile for structural checks.
<code>-overwrite</code>	Overwrites the already present directory's content with the new content.
<code>- power_domain_count_threshold <integer></code>	Specifies the threshold for the power domain count reported using the <code>multiple_pd_paths</code> option.
<code>-write_all_paths <string></code>	Writes all the IPD paths and uses the name of the path descriptions' file.

Examples

- The following command generates a bounded corner scope based on the minimum/maximum voltage combination analyzed in the scope creation run:

```
tempus > write_ipd_scope -bounded_corner -overwrite bounded_corner_scope
```

This command writes out the derates and constraints in the IPD scope, and overwrites an existing scope generated with the same directory name argument.

Related Commands

- `set_ipd_rail_voltage`
- `create_pg_net_group`
- `report_pg_net_group`
- `create_ipd_configuration`
- `report_ipd_configuration`
- `read_ipd_scope`
- `report_ipd_crossings`
- `check_ipd_logic`

Related Information

- For more information, refer to the [Inter Power Domain \(IPD\) Analysis](#) chapter in the *Tempus User Guide*.

write_ldb

```
write_ldb  
[-help]  
-library string  
-outfile string
```

Converts the text library to a Cadence binary format library called LDB.

You can use LDB to run Tempus analysis. The ECSM and CCS libraries can be very large, and loading them may take significant time. You can convert the ASCII NLDM (Liberty .lib), ECSM, and CCS libraries to LDB by using the `write_ldb` command.

When loading the design, the timing libraries will be read faster because the software reads the binary versions.

Instead of the original ASCII libraries, you can specify LDB in the `read_lib` command or the `create_library_set` command in the MMMC configuration file.

Notes:

- Tempus-generated LDB files can be used in Tempus, Innovus, Voltus, Genus, and Conformal software.
- The LDB files are not forward compatible; that is, the LDB generated in one version cannot be used with the prior versions of the software.
- The software does not support concatenated Liberty library (multiple libraries in the same dotlib file) files during LDB conversion. To use these files, you are required to separate individual libraries before converting to individual LDB files.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-library <i>string</i></code>	Specifies the name of the input library file.
<code>-outfile <i>string</i></code>	Specifies the name of the generated LDB file.

Examples

- The following example illustrates the conversion of libraries to LDB:

```
write_ldb -library A.lib -outfile A.ldb
```

Related Commands

- `check_ldb_version`

write_loop_break_sdc

```
write_loop_break_sdc
[-help]
-input_file fileName
-outfile fileName
```

Generates an SDC file containing constraints for combinational loop breaking. The command selects the disable arc(s) with flag set to 1 from the third-party report file, and generates the equivalent SDC commands in an output file.

You can load the output SDC file at any time during a Tempus session. To load the file, use the [read_sdc](#) command.

Parameters

-help	Prints out the command usage.
-input_file fileName	Specifies the disabled time arcs report that you generate using a third-party software. To generate the report, use the following command: report_disable_timing -nosplit > disable_arcs.rpt
-outfile fileName	Specifies the name of the output SDC file.

Command Order

Use this command after generating the disable arcs report.

Examples

The following command generates an `etsCombLoopBreak.sdc` file when the input is a `disable_arcs.rpt` file:

```
tempus> write_loop_break_sdc -input_file disable_arcs.rpt -outfile etsCombLoopBreak.sdc
```

The contents of `disable_arcs.rpt` file are as follows:

Cell or Port	From	To	Sense	Flag Reason

:				

Tempus Text Command Reference
Timing Analysis Commands

uchip_core/u_s5_power	oLpcSd_4_	orig_oSmAcpiSd_4_	*	1
:				

The contents of output file `etsCombLoopBreak.sdc` are as follows:

```
:  
set_disable_timing -from {oLpcSd_4_} -to {orig_oSmAcpiSd_4_}  
    [get_cells {uchip_core/u_s5_power}]  
:
```


write_model_context

```
write_model_context
[-help]
-dir string
[-overwrite]
-reference_dir string
```

Allows the software to create a single complete boundary model and a timing context-only boundary model when several boundary models for a block need to be created with different timing settings (such as derates, timing global variables).

The timing context-only boundary models do not have any netlist or parasitic information and provide only timing context data to perform timing analysis using the boundary models. Thus, these can be created with a much smaller run time using the `write_model_context` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-dir <i>string</i></code>	Specifies the directory for the model context.
<code>-overwrite</code>	Allows the software to overwrite the existing directory.
<code>-reference_dir <i>string</i></code>	Specifies the directory path of the original model.

Examples

- The following example shows the commands that are used to create two boundary models. The first boundary model, `dir_derate_nominal`, is created using the `create_boundary_model` command and is a complete boundary model.

```
set_timing_derate -early 1
set_timing_derate -late 1
update_timing
create_boundary_model -dir dir_derate_nominal -overwrite
```

The second boundary model, `dir_derate_extrem`, is a context-only boundary model that is created using the `write_model_context` command. This model provides the timing context for the `dir_derate_nominal` boundary model:

```
set_timing_derate -early 0.8
set_timing_derate -late 1.2
update_timing
write_model_context -dir dir_derate_extreme -overwrite -reference_dir dir_derate_nominal
```

- The following example shows the commands that are used to read the complete boundary model. The context can be updated using the `update_model_context` command to update the timing context of the boundary model:

```
read_boundary_model -dir dir_derate_nominal
...
set_timing_derate -early 1
set_timing_derate -late 1
update_timing
update_model_context -dir dir_derate_extreme
set_timing_derate -early 0.8
set_timing_derate -late 1.2
update_timing
```

Related Commands

- `update_model_context`
- `create_boundary_model`
- `read_boundary_model`

write_path_descriptions

```
write_path_descriptions  
[-help]  
var_name  
-out_file string
```

Writes out path descriptions of all the timing paths specified in the variable (path collection) to an ASCII file.

The output file contains a set of [report_timing](#) commands for each timing path in the given path collection. The file also includes a complete description of each timing path captured through timing points, along with the clock and edge information.

The path description of a given collection of paths can be read back, and the same path configuration can be recreated in another Tempus session. This allows for processing and comparing the timing, attributes, and other characteristics of the specified paths.

Parameters

<code>-help</code>	Prints out the command usage.
<code>var_name</code>	Specifies the object collection of timing paths. The input must be a single collection. If you need to write out path descriptions of collections generated with different analysis types or path collections, a single collection can be created by appending all the collections using the append_to_collection command.
<code>-out_file string</code>	Redirects output to the specified file.

Examples

- The following command writes out descriptions of late GBA paths when the `-path_type` parameter is not specified:

```
set path_coll [report_timing -collection]
write_path_descriptions $path_coll > PathSpecFile.txt
```

The output file `PathSpecFile.txt` contains:

```
report_timing -net -path_type full_clock -late -from_rise A -through_rise B -
through_rise C \
    -to_rise D -view slow_v -clock_from clk1 -edge_from lead -clock_to clk2 -edge_to
lead
```

- The following command writes out descriptions of late PBA paths when the `-path_type full` option is specified:

```
set path_coll [report_timing -collection -retime path_slew_propagation -path_type full]

write_path_descriptions $path_coll > PathSpecFile.txt
```

The output file `PathSpecFile.txt` contains:

```
report_timing -net -path_type full_clock -late -from_rise A -through_rise B -
through_rise C \
    -to_rise D -view slow_v -clock_from clk1 -edge_from lead -clock_to clk2 -edge_to
lead
```

Note: The output does not show retime details. You should use the `read_path_descriptions -retime` parameter to analyze retime collections that are defined with the `write_path_descriptions` command.

- The following command writes out descriptions of both the early and late GBA paths:

```
set path_coll [report_timing -collection]
set path_coll_early [report_timing -collection -early]
append_to_collection path_coll $path_coll_early
write_path_descriptions $path_coll > PathSpecFile.txt
```

The output file `PathSpecFile.txt` contains:

```
report_timing -net -path_type full_clock -late -from_rise A -through_rise B -
through_rise C \
    -to_rise D -view slow_v -clock_from clk1 -edge_from lead -clock_to clk2 -edge_to
lead
```

```
report_timing -net -path_type full_clock -early -from_rise A through_rise B -to_rise D -  
view \  
    slow_v -clock_from clk1 -edge_from lead -clock_to clk2 -edge_to lead
```

Related Commands

- [read_path_descriptions](#)
- [report_timing](#)
- [append_to_collection](#)

write_sdc

```
write_sdc  
[-help]  
[-clocks <string>]  
[-commands <string>]  
filename  
[-exclude_annotated <string>]  
[-path_adjust_only]  
[-path_group_only]  
[-pins <string>]  
[-sdc]  
[-view <string>]  
[-vt_skew_derate_only]
```

Generates the timing constraints file in a third-party compiler format.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-clocks <string></code>	Specifies the name of the clocks to be included in the SDC file.
<code>-commands <string></code>	Specifies the name of the commands to be included in the SDC file.
<code>-exclude_annotated string</code>	Disables listing of annotated constraints. You can specify one of the following options: • <code>delays</code>: Suppresses dumping of <code>set_annotated_delay</code> constraints. • <code>checks</code>: Suppresses dumping of <code>set_annotated_check</code> constraints. • <code>transitions</code>: Suppresses dumping of <code>set_annotated_transition</code> constraints.
<code>filename</code>	Specifies the name of the constraints file. By default, the constraint file is generated in the current directory. To save to a different directory, you need to specify the complete path.
<code>-path_adjust_only</code>	Exports <code>set_path_adjust_group</code> and <code>set_path_adjust</code> constraints.
<code>-path_group_only</code>	Exports <code>group_path</code> constraints.
<code>-pins <string></code>	Specifies the name of the pins to be included in the SDC file.
<code>-sdc</code>	Produces an SDC file in which functions for manipulating collections are expanded. For example, the <code>remove_from_collection</code> <code>all_inputs</code>, <code>all_outputs</code>, <code>all_clocks</code>, and <code>filter</code> commands are replaced by the resulting list of objects. Note: When you specify the <code>-sdc</code> parameter, a warning is issued stating that using the parameter produces a constraint file containing SDC standard constraints only. This may cause some design constraints to be dropped.
<code>-view string</code>	Specifies a view name to obtain the desired SDC.
<code>- vt_skew_derate_only</code>	When specified, the software will check the voltage threshold (VT) skew configuration and report the VT skew derate details as per the voltage groups defined.

Command Order

Use this command after reading in a timing constraints file.

Examples

- The following command generates a constraints file, `newc`, and saves the generated file in the directory `new`:

```
tempus> write_sdc new/newc
```

- Specifying the following constraints will have new entries in the output constraints file:

```
tempus> set_annotated_delay -from u1/A -to u1/Y -cell 2
tempus> set_annotated_check 2 -setup -from f0/CK -to f0/D
```

The `write_sdc` output will be as follows:

```
set_annotated_delay -cell 2 -from [get_pins {u1/A}] -to [get_pins {u1/Y}]
set_annotated_check -setup -from [get_pins {f0/CK}] -to [get_pins {f0/D}] 2
```

- For the following constraints in the design :

```
tempus> set_annotated_delay -from u1/A -to u1/Y -cell 2
tempus> set_annotated_transition 2 u1/A
tempus> set_annotated_check 2 -setup -from f0/CK -to f0/D
```

The `write_sdc -exclude_annotated delays` output will be as follows:

```
set_annotated_transition 2 [get_pins {u1/A}]
set_annotated_check -setup -from [get_pins {f0/CK}] -to [get_pins {f0/D}] 2
```

The `write_sdc -exclude_annotated {delays transitions}` command outputs the following:

```
set_annotated_check -setup -from [get_pins {f0/CK}] to [get_pins {f0/D}] 2
```

Related Commands

- [read_sdc](#)
- [reset_sdc](#)
- [write_loop_break_sdc](#)

write_sdf

```
write_sdf
[-help]
[filename]
[-abstracted_model]
[-adjust_setuphold_for_zero_hold_slack]
[-base_delay]
[-celltiming {all | none | nochecks}]
[-collapse_internal_pins]
[-condelse]
[-delimiter char]
[-delta_delay]
[-edges {edged | library | noedge | check_edge}]
[-exclude_cells]
[-exclude_cells_keep_io_nets]
[-exclude_cells_keep_top_level_nets]
[-exclude_disabled_arcs]
[-exclude_disabled_modes]
[-exclude_whatif_arcs]
[-filter]
[-ideal_clock_network]
[-include_variation]
[-interconn {all | none | noport | noinput | nooutput}]
[-map_file {file1 file2 ...}]
[-max_view viewName]
[-min_clock_threshold number]
[-min_data_threshold number]
[-min_period_edges {posedge | negedge | both | none}]
[-min_view viewName]
[-multi_drive_net_syntax string]
[-no_derate]
[-no_escape]
[-nonegchecks]
[-path name]
[-precision non_neg_integer]
[-process]
[-recompute_delay_calc]
[-recompute_parallel_arcs]
[-recrem [merge_always | split | merge_when_paired]]
[-remashold]
[-scale float]
[-setuphold [merge_always | split | merge_when_paired]]
```

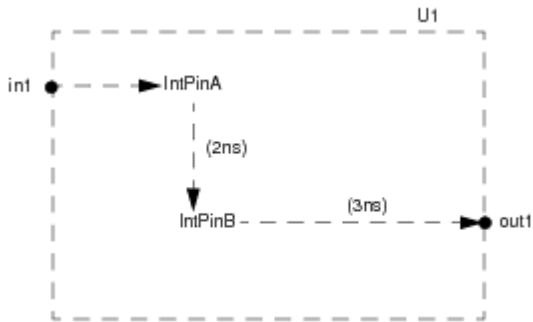
```
[-splitrecrem]
[-splitsetuphold]
[-target_application {sta | verilog}]
[-gate_level_sim_model]
[-temperature]
[-timescale string]
[-transform_out_to_out_arcs]
[-typ_view viewName]
[-version { 2.1 | 3.0 }]
[-view viewName]
[-voltage]
[-worst_case_illegal_cond_arcs string]
```

Writes delays to a Standard Delay Format (SDF) file. The `write_sdf` command can generate both plain-text and GNU zipped (.gz) SDF files.

By default, the `write_sdf` command uses the delays already cached by the system and writes only the values of the triplets.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-abstracted_model</code>	Writes out the gate-level simulation (GLS) SDF file.
<code>-adjust_setuphold_for_zero_hold_slack</code>	Adjusts the hold timing check to meet non-negative slack values.
<code>-base_delay</code>	Writes out base delay information in the SDF file.
<code>-celltiming {all none nochecks}</code>	Specifies which cell delays and timing checks to write out. <i>Default:</i> <code>all</code> <code>all</code> - Writes all cell delays and timing checks to the SDF file. This is the default. <code>nochecks</code> - Disables the writing of timing checks only. <code>none</code> - Prevents cell delays and timing checks from being written into the SDF file.

<code>-collapse_internal_pins</code>	Collapses arcs across internal pins to a single primary input to primary output arc. Use the <code>-collapse_internal_pins</code> parameter when the SDF file being generated is intended for use by an application that does not support the same internal pin structure in its leaf cell libraries (for example, Cadence VXL and NCsim Verilog simulation products). Do <i>not</i> use this parameter to generate an SDF file intended for use within the Tempus flow, or by third party tools that utilize the same Liberty timing models. <i>Default:</i> Arcs between internal pins in a timing model are written to the SDF file as explicit annotations to and from the internal pins.  For example, assume the following arc between internal pins: The following SDF would be generated by default: <pre>(U1/in1 U1/IntPinA (1::1) (1::1)) (U1/IntPinA U1/IntPinB (2::2) (2::2)) (U1/IntPinB U1/out1 (3::3) (3::3))</pre> If you specify <code>-collapse_internal_pins</code>, the resulting SDF would show a single arc with the merged delays: <pre>(U1/in1 U1/out1 (6::6) (6::6))</pre>
<code>-condelse</code>	Writes <code>CONDELSE</code> constructs with the default value when a <code>COND</code> construct is written. Note: Use this argument with the <code>-version 3.0</code> option.

<code>-delimiter char</code>	Specifies the hierarchy divider character to use in the output file. This option sets the hierarchy divider of only the <code>write_sdf</code> command. If omitted, the default hierarchy divider character (/) is used as the <code>write_sdf</code> default.
<code>-delta_delay</code>	Writes out delta delay values to the SDF file.
<code>-edges {edged library noedge check_edge}</code>	Specifies the edge values. Refer to the Table for the full qualifications of the possible values for the edges option. <i>Default:</i> edged • <code>check_edge</code> - Keeps edge specifiers on timing check arcs but does not add edge specifiers on delay arcs. • <code>edged</code> - Keeps edge specifiers on timing check arcs and delay arcs. • <code>library</code> - Assumes that the <code>sdf_edges</code> attributes are specified in the <code>.lib</code> file. The default <code>sdf_edges</code> value of a timing arc is <code>noedge</code>. • <code>noedge</code> - Does not keep edge specifiers.
<code>-exclude_cells</code>	Specifies a list of leaf-cell or hierarchical instances whose timing data will be excluded from the SDF (Standard Delay Format) file. If you specify a hierarchical instance, the timing data for all leaf-cells below the level of hierarchy will be omitted from the SDF file.
<code>-exclude_cells_keep_io_nets</code>	Restricts the filtering to just the annotations that are completely enclosed by the cell. Annotations to or from the cell will be preserved. By default, when the <code>-exclude_cells</code> parameter is used, any annotations within the excluded cell and any annotations to or from the excluded cell are filtered from the SDF.

<code>-exclude_cells_keep_top_level_nets</code>	Allows you to configure the handling of top-level interconnect annotations to an instance that is excluded from the SDF using the <code>write_sdf -exclude_cells</code> parameter. When this parameter is specified, the interconnect arcs that span to the ports of the instance will be included in the SDF. By default, no interconnect arcs to the instance are included. The delay annotated will be the worst-case delays of the top-level nets to/from the first-level sink/source pins within the block that traverse a given input/output port of the instance, respectively.
<code>-exclude_disabled_arcs</code>	Excludes user-set disabled arcs in the Standard Delay Format (SDF) file. If the user sets this option, the disabled arcs set by commands <code>set_mode</code> , <code>set_disable_timing</code> , and <code>set_case_analysis</code> are excluded.
<code>-exclude_disabled_modes</code>	Excludes timing arcs that have been disabled due to an active library mode set by the <code>set_mode</code> command from appearing in the SDF file.
<code>-exclude_whatif_arcs</code>	Excludes what-if arcs in the SDF file.
<code>filename</code>	Specifies the name of the SDF output file.
<code>-filter</code>	Converts all negative interconnect and <code>IOPATH</code> delays to zero.
<code>-ideal_clock_network</code>	Marks delays of delay arcs on ideal clock networks, as 0 in the SDF file.
<code>-include_variation</code>	When the selected analysis type is SOCV, the <code>-include_variation</code> parameter can be used to export SDF delays and check values, including the variation component. By default, the SDF delays and check values will not have the mean + N*sigma value.

<pre>-interconn {all none noport noinport nooutport}</pre>	Specifies which interconnect delays to write. • <code>all</code> - Writes all <code>INTERCONN</code> delays into the SDF file. This is the default if the <code>-interconn</code> option is not specified. • <code>none</code> - No <code>INTERCONN</code> delays are written into the SDF file. • <code>noport</code> - No top level port <code>INTERCONN</code> delays are written into the SDF file. • <code>nooutport</code> - Disables only the output port <code>INTERCONN</code> delays. <i>Default:</i> <code>all</code>
<pre>-map_file {file1 file2 ...}</pre>	Specifies a list of map files. The map files can be used to determine what is written into the SDF files. For example, a map file can provide delay information on internal pins that are not defined in the timing library. Syntax for a map file (for a cell) is as follows: ◦ A template section (the text to output) that references variables of the form <code>\$<number></code> ◦ A variable definition section that tells the expression to evaluate to obtain the value
<pre>-max_view viewName</pre>	Uses the late delay from the specified analysis view to populate the SDF max slot when in multi-mode multi-corner analysis mode.
<pre>-merge_arcs</pre>	Merges two arcs that are related to the same internal pin and forms one timing arc. For example, consider a cell's <code>IOPATH</code> that forms an arc from pin <code>I</code> to internal pin <code>A</code> and another from pin <code>A</code> to output pin <code>O</code>. In this case, if you use the <code>-merge_arcs</code> option, the software will merge both the arcs related to internal pin <code>A</code> and form one timing arc from pin <code>I</code> to output pin <code>O</code>. Note: The <code>-merge_arcs</code> parameter is obsolete and has been replaced by the <code>-collapse_internal_pins</code> parameter. This parameter still works in this release, but to ensure compatibility with future releases, update your script to use the <code>-collapse_internal_pins</code> parameter.

<code>-min_clock_threshold number</code>	Specifies the absolute minimum delta delay change threshold on the clock path to be written into the SDF file. Time unit is in seconds. <i>Default: 1e-20</i>
<code>-min_data_threshold number</code>	Specifies the absolute minimum delta delay change threshold on the data path to be written into the SDF file. Time unit is in seconds. <i>Default: 1e-20</i>
<code>-min_period_edges {posedge negedge both none}</code>	Specifies how to write out minimum <code>PERIOD</code> checks. Two check arcs are always created for the <code>min_period</code> check when you read in a <code>.lib</code> file. The following options let you keep or remove the positive edge, the negative edge, both edges, or no edges from the <code>PERIOD</code> statement in the <code>.sdf</code> file. • <code>posedge</code> - Writes out one positive edge <code>PERIOD</code> check. • <code>negedge</code> - Writes out one negative edge <code>PERIOD</code> check. • <code>both</code> - Writes out both the positive edge and the negative edge <code>PERIOD</code> checks. This is the default. • <code>none</code> - Writes out one <code>PERIOD</code> check without any edge specifier.
<code>-min_view viewName</code>	Uses the early delay from the specified analysis view to populate the SDF <code>min</code> slot when in multi-mode multi-corner analysis mode.
<code>-multi_drive_net_syntax string</code>	Specifies the multi-drive nets as ports or interconnects. You can use the <code>interconnect</code> and <code>port</code> options if you need to define the net delays of multi-driven nets (clock mesh) as interconnects or ports respectively, in the SDF.
<code>-no_derate</code>	Writes non-derated delays to the Standard Delay Format (SDF) file. By default, the <code>set_timing_derate</code> command writes scaled delays to the SDF file. When the <code>-no_derate</code> parameter is specified, the <code>write_sdf</code> command removes all the applied derates.

<code>-no_escape</code>	Writes out object names without escaping hierarchical dividers. Normally, when writing out an SDF file, the hierarchical divider (/) is escaped because it is a special character.
<code>-nonegchecks</code>	Converts all negative timing check values to 0.0. The <code>-nonegchecks</code> parameter has an effect only when the negative checks are legal. In SDF v3.0, the negative checks are only legal in the context of <code>RECREM</code> or <code>SETUPHOLD</code> checks – they are not legal in the case of split checks. In SDC v2.1, the negative values are permitted in <code>SETUP</code>, <code>HOLD</code>, and <code>RECOVERY</code> checks. Therefore, for these checks, the <code>-nonegchecks</code> parameter is required to convert negative values to zero. The negative checks are legal in the split checks, so the <code>-nonegchecks</code> parameter can have an effect. The SDF v2.1, however, does not support <code>REMOVAL</code> checks.
<code>-path <i>name</i></code>	Specifies the sub-block name.
<code>-precision <i>non_neg_integer</i></code>	Controls the number of digits appearing after the decimal point in the output SDF file. The default precision is the value of the <code>report_precision</code> .
<code>-process</code>	Specifies one or a combination of the following process names that will appear in the header of the generated SDF file: • <code>best</code> • <code>typical</code> • <code>worst</code> Use the colon delimiter to specify multiple process names. For example: <pre>-process best:typical:worst</pre> In the above example, the header of the generated SDF file will contain: <pre>(PROCESS "best:typical:worst")</pre>

<code>-recompute_delay_calc</code>	Instructs the software to recompute any necessary data required for a complete SDF file before exporting. By default, only the existing data that has been computed for STA is exported to the SDF. You must use this parameter to ensure that the SDF file is complete for general use in other applications.
<code>-recompute_parallel_arcs</code>	Instructs the software to temporarily recompute delays with separate delay calculation for parallel arcs. The software normally operates in a high performance mode, which optimizes the handling of multiple timing arcs between input and output pin pairs. By default, the SDF file generated will contain the same delay information for all of these arcs. To have the SDF recalculated with explicit pin pair data, you should use the option <code>-recompute_parallel_arcs</code> to cause delays for each parallel arc to be explicitly calculated. It is recommended to use this option for generating SDF for functional simulation applications. Note: When you use the <code>write_sdf -recompute_parallel_arcs</code> option, timing analysis is performed again. You need to obtain a new delay calculation to generate the SDF, and another delay calculation to put the system back to where it was before <code>write_sdf</code> was specified. Since the delay calculation is run again, the timing analysis will be reset. Note: The <code>-recompute_parallel_arcs</code> parameter is obsolete. This parameter still works in this release, but to ensure compatibility with future releases, update your scripts to use the <code>-recompute_delay_calc</code> parameter instead.

```
-recrem [merge_always | split |  
merge_when_paired]
```

Determines how RECOVERY and REMOVAL timing checks are written.

- `merge_always` - Always writes combined RECREM checks, even if either a RECOVERY, or a REMOVAL, check does not exist.

For example, if a RECOVERY check exists, but a REMOVAL check does not, the software writes a combined RECREM check in the SDF file as follows:

```
RECREM ( value ) ( value ) ( )
```

- `split` - Splits the RECREM checks into a RECOVERY check and a REMOVAL check.
- `merge_when_paired` - Writes combined RECREM checks only when both a RECOVERY and a REMOVAL check exist. For example, if both a RECOVERY and a REMOVAL check exist, the software writes a combined RECREM check in the SDF file as follows:

```
RECREM ( value ) ( value ) ( value ) ( value )
```

However, if a RECOVERY check exists, but a REMOVAL check does not, the software only writes the RECOVERY check in the SDF file:

```
RECOVERY ( value ) ( value )
```

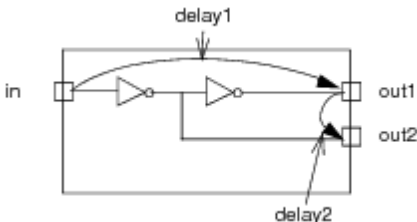
Default: `merge_always`

Note: The RECREM sum should be either zero or greater than zero. A negative value will be adjusted to enforce this condition. This is an SDF v3.0 requirement (the software may issue a warning).

<code>-remashold</code>	Converts the SDF v2.1 unsupported <code>REMOVAL</code> (and the <code>REMOVAL</code> portion of the <code>RECREM</code> constructs) to <code>HOLD</code> checks when used with the <code>-version 2.1</code> option. <i>Default:</i> Not to write out the unsupported checks. Converts the <code>REMOVAL</code> checks to <code>HOLD</code> checks when combined with the <code>-version 3.0</code> option. The <code>RECREM</code> checks are split into a <code>RECOVERY</code> check and a <code>HOLD</code> check. This option automatically splits the <code>RECREM</code> checks. <i>Default:</i> To maintain the version 3.0 checks.
<code>-scale float</code>	Specifies a multiplier to the delay values used in the SDF file. This option does not change the <code>timescale</code> setting of the SDF file. The delay values are scaled, but the units are the same. <i>Default:</i> 1

<pre>-setuphold [merge_always split merge_when_paired]</pre>	Determines how <code>SETUP</code> and <code>HOLD</code> timing checks are written. • <code>merge_always</code> - Always writes combined <code>SETUPHOLD</code> checks, even if either a <code>SETUP</code> or a <code>HOLD</code>, check does not exist. For example, if a <code>SETUP</code> check exists, but a <code>HOLD</code> check does not, the software writes a combined <code>SETUPHOLD</code> check in the SDF file as follows: <pre>SETUPHOLD (value) (value) () ()</pre> • <code>split</code> - Splits the <code>SETUPHOLD</code> timing checks into separate <code>SETUP</code> and <code>HOLD</code> checks. • <code>merge_when_paired</code> - Writes combined <code>SETUPHOLD</code> checks only when both a <code>SETUP</code> and a <code>HOLD</code> check exist. For example, if both a <code>SETUP</code> and a <code>HOLD</code> check exist, the software writes a combined <code>SETUPHOLD</code> check in the SDF file as follows: <pre>SETUPHOLD (value) (value) (value) (value)</pre> However, if a <code>SETUP</code> check exists, but a <code>HOLD</code> check does not, the software only writes the <code>SETUP</code> check in the SDF file: <pre>SETUP (value) (value)</pre> <i>Default:</i> <code>merge_always</code>
<pre>-splitrecrem</pre>	Splits the <code>RECREM</code> checks into a <code>RECOVERY</code> check and a <code>REMOVAL</code> check. <i>Default:</i> To write the combined <code>RECREM</code> check. Note: Use this option only with the <code>version 3.0</code> option.

<code>-splitsetuphold</code>	Splits the <code>SETUPHOLD</code> timing checks into separate <code>SETUP</code> and <code>HOLD</code> checks. <i>Default:</i> To write the combined <code>SETUPHOLD</code> checks. Note: In SDC v3.0, the negative values are illegal in <code>SETUP</code>, <code>HOLD</code>, <code>RECOVERY</code>, and <code>REMOVAL</code> checks. Therefore, negative values are automatically converted to zero with the <code>-splitsetuphold</code> and <code>-splitrecrem</code> parameters.
<code>-target_application { sta verilog }</code>	Allows customization of the SDF output for use with STA or Verilog applications. You can select the following options: • <code>sta</code>: You can use the <code>sta</code> option to configure the SDF output for use with static-timing analysis (STA) applications. This includes sign-off timing analysis with products, such as Cadence Tempus, as well as working with implementation flows that use STA for internal timing analysis. The <code>sta</code> option preserves maximum accuracy and generates SDF output that matches the structure of the Liberty timing libraries. This option is equivalent to the following options: <pre>write_sdf -interconn all, -celltiming all, -edges edged, -setuphold merge_when_paired, and -recrem merge_when_paired.</pre> • <code>verilog</code>: You can use the <code>verilog</code> option to generate SDF output that is compatible with Cadence Verilog-XL or Verilog-NC simulators. This option is equivalent to the following options: <pre>write_sdf -filter -edges check_edge, -setuphold merge_when_paired, -recrem merge_when_paired, - min_period_edges -none, -collapse_internal_pins, and -recompute_parallel_arcs.</pre> When the <code>-target_application</code> option is not specified, then by default SDF output is configured for use with STA applications.
<code>-gate_level_sim_model</code>	Writes out the GLS SDF file.

<code>-temperature</code>	Defines the temperature values that will appear in the header of the generated SDF file. Use the colon delimiter to specify multiple temperature values. For example: <pre>-temperature -40.0:25.0:125</pre> In the above example, the header of the generated SDF file will contain: <pre>(TEMPERATURE -40.0:25.0:125)</pre>
<code>-timescale <i>string</i></code>	Specifies the units of time in ns (nanoseconds) or ps (picoseconds). When <code>-si</code> option is used, the default <code>-timescale</code> value is ps.
<code>-transform_out_to_out_arcs</code>	Writes out all output-to-output timing arcs as input-to-output arcs. Input-to-output arc delays are computed by adding the delay from the input pin to one of the output pins plus the delay from that output pin to the other output pin. For instance, in the following figure, the delay between <code>in</code> and <code>out2</code> would be the sum of <code>delay1</code> and <code>delay2</code>. 
<code>-typ_view <i>viewName</i></code>	Uses the late delay from the specified analysis view to populate the SDF <code>typ</code> slot when in multi-mode multi-corner analysis mode.
<code>-version {2.1 3.0}</code>	Specifies whether to generate the SDF V2.1 or V3.0. The default SDF version is 3.0. Used in combination with the <code>-remashold</code> option.

<code>-view viewName</code>	Uses the early and late delays for the specified analysis view to populate the <code>min</code> and <code>max</code> SDF triplet slots when in multi-mode multi-corner analysis mode. The SDF <code>typ</code> slot remains empty (<code>earlyVal :: lateVal</code>). In multi-mode multi-corner analysis mode, if you want to use SPEF information when calculating the delays, you must use <code>read_spef -rc_corner</code> to import the SPEF files for each RC corner in the design.
<code>-voltage</code>	Defines the voltage values that will appear in the header of the generated SDF file. Use the colon delimiter to specify multiple voltage values. For example: <pre>-voltage 1.980:1.800:1.620</pre> In the above example, the header of the generated SDF file will contain: <pre>(VOLTAGE 1.980:1.800:1.620)</pre>
<code>-worst_case_illegal_cond_arcs string</code>	Specifies the <code>sdf_cond</code> - all, none, delays, or checks - for the arcs. When this parameter is specified, the <code>write_sdf</code> command detects illegal SDF condition syntax coming from the Liberty library and generates an unconditional annotation in the SDF file that worst-cases the delays of the related timing arcs. This prevents downstream tools (such as Verilog simulators) from complaining about illegal syntax in the SDF file.

Definition of -edges Option Values

	edged Value	lib rary Val ue	no edg e Val ue

IOPATH from a unate input pin to output pin	No edge specifier	No edge specifier	No edge specifier
IOPATH from a non-unate input pin to output pin	Edge specifier at the input pin	Edge specifier at the input pin when merging edges, —edge	No edge specifier

is either "b" or "s" for MAX and TYP fields; the minimum value

			s a r e u s e d f o r M I N f i e l d s.
IOPATH of a clock/enable pin to data out pin	Edge specifier at the input pin	E d g e s p e c i f i e r a t t h e i n p u t p i n w h e n s e e d f o r e d g e s p e c i f i e r	N o e d g e s p e c i f i e r

		g e s is ei th e r "b o t h — e d g e s" O r "s t a r t — e d g e"	
Timing Checks	Edge specifiers match the check arcs specified in the timing library	O v e r r i d d e n b y th e v al u	O n l y th e e d g e s p e ci fi er

e
of
th
e
s
d
f
—
e
d
g
e
s
at
tri
b
ut
e

s
of
th
e
re
fe
re
n
c
e
pi
n
ar
e
w
rit
te
n

F
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—
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e
ti
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h
e
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g
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d
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(f
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rHOLD style check) or the starting pin (for SETUP type check) is edge -

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e
c
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f
i
e
d

T
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t
i
m
i
n
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—
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a
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—
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r

stand second events, respectively, of the Verilog-XL's \$hold() and \$s

Examples

- The following command writes out an SDF 2.1 compliant output file named *my.sdf* with 4 digits after the decimal point:

```
tempus> write_sdf -version 2.1 -precision 4 my.sdf
```

The following SDF file output shows a comparison between the output of `-edges edged` with `-edges noedge`:

```
(HOLD (posedge D) (posedge CK) (-0.14:-0.14:-0.14))  
(HOLD (negedge D) (posedge CK) (-0.06:-0.06:-0.06))  
(SETUP (posedge D) (posedge CK) (0.22:0.22:0.22))  
(SETUP (negedge D) (posedge CK) (0.41:0.41:0.41))
```

When the `-edges` parameter is set to `noedge`, only the edge specifiers of the clock pin will be written, as shown below:

```
(HOLD D (posedge CK) (-0.14:-0.06:-0.06))  
(SETUP D (posedge CK) (0.22:0.41:0.41))
```

- The following command writes out the delta delay values for all the analysis views to the specified SDF file:

```
write_sdf -delta_delay -view [all_analysis_views -type active] tst.sdf
```

- The following command writes non-derated delays to the specified SDF file:

```
write_sdf tst.sdf -no_derate
```

Related Commands

- `read_sdf`

write_timing_context

```
write_timing_context  
[-help]  
[dir_name]  
[-instances instance_list]  
-module module_name  
[-overwrite]  
[-skip_interface_paths]
```

Writes the timing context model of the specified lower-level modules/instances in the given output directory. Within this directory, Tempus creates sub-directories for each of the specified modules and writes the context model of each module within the corresponding sub-directory.

Parameters

-help	Prints out the command usage.
dir_name	Specifies the name of the output directory.
-instances instance_list	Specifies the instance list. If not specified, then the context of all the instances of modules (specified with the -module parameter) will be generated.
-module module_name	Specifies the name of the module. You can use this parameter to write timing context information or the required lower-level block modules. The timing context information for multiple modules can be written by specifying more than one module name.
-overwrite	Allows the software to overwrite any existing context directory data. If the context output directory already exists, the software will issue an error.
- skip_interface_paths	Skips the interface paths. However, the context data impacting only the internal reg-reg paths is preserved. During context analysis, the interface paths will be unconstrained.

Examples

- The following command will write the context model of block module `tdsp_core` and `results_conv` in respective sub-directories under the context directory `context/tdsp_core`

`context/results_conv:`

```
write_timing_context -module {tdsp_core results_conv} context
```

Since the `-instance` parameter is not specified with the `write_timing_context` command, the output directory will contain context models for all instances of `tdsp_core` and `results_conv` modules.

- The following command will write context for the `tdsp_core_inst1` instance of the `tdsp_core` module and all the instances of the `results_conv` module.

```
write_timing_context -module {tdsp_core results_conv} -instances {tdsp_core_inst1}  
context
```

Related Commands

- `set_analysis_mode`
- `read_timing_context`
- `set_timing_context_view_mapping`
- `report_timing_context_clock_mapping`
- `compare_timing_context`

write_timing_windows

```
write_timing_windows  
[-help]  
out_file  
[-pin]  
[-power_compatible]  
[-ssta_sigma_multiplier value]  
[-view viewName]  
[-voltage_threshold {lower_threshold upper_threshold}]
```

Generates a Timing Window File (TWF). The TWF file primarily contains the earliest and latest possible arrival times for a signal on a net or a pin.

The format of a timing window file consists of eight pairs of data as given below:

```
(NET <netname>
```

```
minRTW:maxRTW minRt:maxRt minRDr:maxRDr minRSlk:maxRSlk minFTW:maxFTW minFt:maxFt  
minFDr:maxFDr minFSlk:maxFSlk)
```

Where:

- minRTW: Minimum rise arrival time
- maxRTW: Maximum rise arrival time
- minRt: Minimum rise transition time
- maxRt: Maximum rise transition time
- minRDr: Minimum rise source resistance
- maxRDr: Maximum rise source resistance
- minRSlk: Minimum rise slack time
- maxRSlk: Maximum rise slack time
- minFTW: Minimum fall timing window
- maxFTW: Maximum fall timing window
- minFt: Minimum fall transition time
- maxFt: Maximum fall transition time
- minFDr: Minimum fall source resistance
- maxFDr: Maximum fall source resistance
- minFSlk: Minimum fall slack time
- maxFSlk: Maximum fall slack time

Parameters

<code>-help</code>	Prints out the command usage.
<code>out_file</code>	Writes the TWF file to the specified file name. The <i>filename</i> parameter must be the last argument in the list. Note: To write a compressed file, add the <code>.gz</code> extension to the file name.
<code>-pin</code>	Generates a TWF for each pin instead of each net. <i>Default:</i> Generates a TWF for the crosstalk analysis tool, which contains timing windows on each net.
<code>-power_compatible</code>	Generates a timing windows file that is suitable for power analysis.
<code>-ssta_sigma_multiplier value</code>	Specifies a sigma multiplier between 0 and 3. The value that you specify increases the arrival window by that amount, and the worst slack value is generated. The transition and impedance values are not impacted.
<code>-view viewName</code>	Generates a TWF for the specified multi-mode multi-corner view only. You must be in multi-mode multi-corner analysis mode to use this parameter. If the <code>-view</code> parameter is not specified, then by default the first active view will be used to generate timing windows.
<code>-voltage_threshold</code> <code>{lower_thresholdupper_threshold}</code>	Specifies voltage threshold values for generating a Timing Window File (TWF). All the slews in the TWF file will be scaled with these values. The lower threshold value must be less than the higher threshold value. <i>Value range:</i> 0 to 100

Examples

- The `write_timing_windows -voltage_threshold {20 80}` command outputs the following in the TWF file:
(VOLTAGE_THRESHOLD 20 80)
- When active clocks are set using the `set_active_clocks` command, the `write_timing_windows` command will not display windows for inactive clocks in the output. The example report is shown below:

```
(TIMING_WINDOWS
(HEADER
(VERSION "0.1")
(DATE "Tue Apr 2 23:07:58 2013")
(DESIGN "top")
(DELIMITERS "/"[""])
(VOLTAGE_THRESHOLD 20 80)
(PROCESS 1.000 1.000 1.000)
(VOLTAGE 1.012 1.012 1.012)
(TEMPERATURE 125.000 125.000 125.000)
(OPERATING_CONDITIONS DEFAULT DEFAULT DEFAULT)
(TIME_SCALE 1.000E-12)
(CAP_SCALE 1.000E-12)
(RESISTANCE_SCALE 1.000E+00)
(DEFAULT_INPUT_SLEW 0.000000 0.000000 0.000000 0.000000)
(PROGRAM "novus" "v13.10-dev.festa ((64bit) 04/02/2013 23:24 (Linux 2.6))" "Cadence")
)
(WAVEFORM "Iclk1" 10000.000000 (POSEDGE 0.000000) (NEGEDGE 5000.000000))
(WAVEFORM "Iclk2" 15000.000000 (POSEDGE 0.000000) (NEGEDGE 7500.000000))
(WAVEFORM "Iclk3" 15000.000000 (POSEDGE 0.000000) (NEGEDGE 7500.000000))
(WAVEFORM "Iclk4" 10000.000000 (POSEDGE 0.000000) (NEGEDGE 5000.000000))
(CAUSED_BY "Iclk1" RISE
(NET "in1" 0:200 120:120 * 9200 0:200 120:120 * 9200 D)
)
(CAUSED_BY NULL
(NET "sel" 0:200 120:120 * * 0:200 120:120 * * D)
)
(CAUSED_BY "Iclk4" RISE
(NET "clk1" 0:200 120:120 * * * 120:120 * * D)
)
(CAUSED_BY "Iclk4" FALL
(NET "clk1" * 120:120 * * 5000:5200 120:120 * * D)
)
(CAUSED_BY "Iclk1" RISE
(NET "clk1" 0:200 120:120 * * * 120:120 * * D)
)
)
```

```
(CAUSED_BY "Iclk1" FALL
(NET "clk1" * 120:120 * * 5000:5200 120:120 * * D)
)
(CAUSED_BY "Iclk1" RISE
(NET "clk1" 0:200 120:120 * * 5000:5200 120:120 * * C)
)
(CAUSED_BY "Iclk4" RISE
(NET "clk1" 0:0 120:120 * * 5000:5000 120:120 * * C)
)
(CAUSED_BY NULL
(NET "clk2" 0:200 120:120 * * 0:200 120:120 * * D)
)
(CAUSED_BY "Iclk1" RISE
(NET "out1" 1200:2600 12.7:12.7 * 1300:7400 1200:2600 10.8:10.8 * 1300:7400 D)
)
(CAUSED_BY "Iclk4" RISE
(NET "out1" 300:800 12.7:12.7 * 400:9200 300:800 10.8:10.8 * 400:9200 D)
```

Related Commands

- `write_twf`
- `read_twf`

write_twf

```
write_twf
-help
outfile
[-exclude_instances string]
[-format string]
[-include_instances string]
[-include_lib_cells string]
[-pin]
[-power_compatible]
[-ssta_sigma_multiplier float]
[-third_party_file_format string]
[-view string]
[-voltage_threshold string]
```

Generates a timing window file (TWF). The TWF file mainly contains the earliest and the latest possible arrival times that a signal may arrive on a net or a pin.

The format of a timing window file consists of eight pairs of data as given below:

```
(NET <netname>
```

```
minRTW:maxRTW minRt:maxRt minRDr:maxRDr minRSlk:maxRSlk minFTW:maxFTW minFt:maxFt
minFDr:maxFDr minFSlk:maxFSlk)
```

Where:

- minRTW: Minimum rise arrival time
- maxRTW: Maximum rise arrival time
- minRt: Minimum rise transition time
- maxRt: Maximum rise transition time
- minRDr: Minimum rise source resistance
- maxRDr: Maximum rise source resistance
- minRSlk: Minimum rise slack time
- maxRSlk: Maximum rise slack time
- minFTW: Minimum fall timing window
- maxFTW: Maximum fall timing window
- minFt: Minimum fall transition time
- maxFt: Minimum fall transition time
- minFDr: Minimum fall source resistance

- `maxFDr`: Maximum fall source resistance
- `minFSlk`: Minimum fall slack time
- `maxFSlk`: Maximum fall slack time

Parameters

<code>-help</code>	Prints out the command usage.
<code>-exclude_instances string></code>	Specifies a list of hierarchical instances to be excluded from the TWF file.
<code>-format string</code>	Specifies the name of the format to be used for TWF generation. Note: This parameter supports third-party TWF generation.
<code>-include_instances string</code>	Specifies a list of hierarchical instances to be included in the TWF file.
<code>-include_lib_cells string</code>	Specifies the library cells that require the input timing window.
<code>outfile</code>	Specifies the name of the output file. The <code>out_file</code> parameter must be the last argument in the list. To write a compressed file, add the <code>.gz</code> extension to the file name. Note: This parameter supports third-party TWF generation.
<code>-pin</code>	Generates a TWF for each pin instead of each net. <i>Default:</i> Generates a TWF for the crosstalk analysis tool, which contains timing windows on each net.
<code>-power_compatible</code>	Generates a timing window file (TWF) that can be used during power analysis.
<code>-ssta_sigma_multiplier float</code>	Specifies a sigma multiplier between 0 and 3. The value that you specify increases the arrival window by that amount, and the worst slack value is generated. The transition and impedance values are not impacted.
<code>- third_party_file_format string</code>	Specifies the name of the third-party file format to be used during TWF generation. You can specify the <code><ir_drop></code> or <code><jitter></code> format option. Note: This parameter supports third-party TWF generation.

<code>-view <i>string</i></code>	Specifies the name of the analysis view. Note: This parameter supports third-party TWF generation.
<code>-voltage_threshold <i>string</i></code>	Specifies voltage threshold values for generating a Timing Window File (TWF). All the slews in the TWF file will be scaled to these values. The lower threshold must be less than the higher threshold. <i>Value range:</i> 0 to 100

Examples

- The `write_twf -voltage_threshold {20 80}` command writes the following in the TWF file:
- The following example shows third-party TWF report with IR drop option:

```

I m_clk_I10
S A 0.508 0.509 0.175 0.195
S Z 0.077 0.064 0.054 0.054
K Z -- -- -- --
T Z 1 -- -- 2.353 2.280 3 f
T Z 1 0.374 0.268 2.353 2.280 3 r

```

Third-party TWF report with Jitter option is shown below:

```

I m_clk_I10
T Z 1 -- -- 2.353 2.280 3 f
T Z 1 0.374 0.268 2.353 2.280 3 r

```

Where I - Instance, S - slew data, and K - slack data.

Related Commands

- `read_twf`
- `write_timing_windows`

Timing Constraint Commands

- [create_clock](#)
- [create_generated_clock](#)
- [current_design](#)
- [current_instance](#)
- [get_sdc_mode](#)
- [group_path](#)
- [reset_annotated_check](#)
- [reset_annotated_delay](#)
- [reset_annotated_transition](#)
- [reset_aocv_stage_weight](#)
- [reset_case_analysis](#)
- [reset_clock](#)
- [reset_clock_exclusivity](#)
- [reset_clock_gating_check](#)
- [reset_clock_groups](#)
- [reset_clock_latency](#)
- [reset_clock_sense](#)
- [reset_clock_transition](#)
- [reset_clock_tree_latency](#)
- [reset_clock_uncertainty](#)
- [reset_data_check](#)
- [reset_disable_clock_gating_check](#)
- [reset_disable_timing](#)
- [reset_drive](#)
- [reset_driving_cell](#)

- `reset_generated_clock`
- `reset_ideal_latency`
- `reset_ideal_network`
- `reset_ideal_transition`
- `reset_input_delay`
- `reset_load`
- `reset_max_capacitance`
- `reset_max_fanout`
- `reset_max_time_borrow`
- `reset_max_transition`
- `reset_min_capacitance`
- `reset_min_fanout`
- `reset_min_pulse_width`
- `reset_min_transition`
- `reset_mode`
- `reset_output_delay`
- `reset_path_exception`
- `reset_path_group`
- `reset_pll_timing`
- `reset_propagated_clock`
- `reset_property`
- `reset_pulse_clock_max_transition`
- `reset_pulse_clock_max_width`
- `reset_pulse_clock_min_transition`
- `reset_pulse_clock_min_width`
- `reset_resistance`
- `reset_sense`
- `set_active_clocks`
- `set_annotated_check`
- `set_annotated_delay`

- `set_annotated_transition`
- `set_aocv_interface_path_offset`
- `set_aocv_stage_weight`
- `set_case_analysis`
- `set_clock_exclusivity`
- `set_clock_gating_check`
- `set_clock_groups`
- `set_clock_latency`
- `set_clock_sense`
- `set_clock_transition`
- `set_clock_uncertainty`
- `set_data_check`
- `set_disable_clock_gating_check`
- `set_disable_timing`
- `set_dont_touch`
- `set_dont_touch_network`
- `set_dont_use`
- `set_drive`
- `set_driving_cell`
- `set_false_path`
- `set_fanout_load`
- `set_ideal_latency`
- `set_ideal_network`
- `set_ideal_transition`
- `set_input_delay`
- `set_input_transition`
- `set_load`
- `set_logic_one`
- `set_logic_zero`
- `set_max_capacitance`

- [set_max_delay](#)
- [set_max_fanout](#)
- [set_max_time_borrow](#)
- [set_max_transition](#)
- [set_min_capacitance](#)
- [set_min_delay](#)
- [set_min_fanout](#)
- [set_min_pulse_width](#)
- [set_min_transition](#)
- [set_mode](#)
- [set_multicycle_path](#)
- [set_normalized_driver_waveform_lib](#)
- [set_output_delay](#)
- [set_pll_timing](#)
- [set_propagated_clock](#)
- [set_property](#)
- [set_pulse_clock_max_transition](#)
- [set_pulse_clock_max_width](#)
- [set_pulse_clock_min_transition](#)
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Path Exception Priorities - Overview

This section describes the path exception priority rules, which are followed by Tempus to determine the effective path exceptions to be applied on a particular path if multiple path exceptions are applied to it. The path exception priority rules are explained as per the path exception command type and various command parameters. The path exception commands are given below:

- `set_false_path`
- `set_max_delay`
- `set_min_delay`
- `set_multicycle_path`

Note: The priority rules (as explained in the following sections) are applicable to `set_path_adjust_group` and `group_path` commands also.

Priority Rules on the Basis of Path Exception Type

If the types of path exception commands are different, the priority order (in decreasing priority) is as follows:

1. `set_false_path`
2. `set_min_delay/set_max_delay`
3. `set_multicycle_path`

For example, `set_false_path -from A` is given priority over `set_multicycle_path -from A 2`.

Priority Rules on the Basis of Path Exception Command Reference Points

The path exception commands use parameters to specify reference points, such as, `-from/-through/-to/-clock_from/-clock_to`.

If there is more than one exception of a given type, for example the `set_multicycle_path` command, the path exception that is more specific has higher priority. A path exception has higher priority if it specifies more of the `-from/-through/-to` parameters. For example, the `-from -to` options will have priority over the `-from` option.

Here is an example of path exception priorities for `-from/-through/-to` parameters:

Priority	Path Exception
1 (highest)	<code>set_multicycle_path -from pins -through pins -to pins</code>
2	<code>set_multicycle_path -from pins -to pins</code>
3	<code>set_multicycle_path -from pins -through pins -to clocks</code>

If the path has the same number of reference points:

- `-from` and `-to` option has priority over the `-from` option
- `-from` option has priority over the `-to` option
- `-to` option has priority over the `-through` option

For example:

- i. `set_multicycle_path -from A 2` is given priority over `set_multicycle_path -through B 4`.
- ii. `set_false_path -from A` is given priority over `set_false_path -through B`.

2. If there are more than one exceptions of a given type with the same reference point, the path exception that is more specific has a higher priority. A path exception is more specific if it specifies a longer path than the other. For example, the '`-from -to`' parameters will have a higher priority over `-from`.

3. If a path exception differs only in terms of number of `-through` commands specified, the exception with a more constraining value will be honored.

Priority Rules on the Basis of `-start/-end` (only for `set_multicycle_path`)

For the `set_multicycle_path` constraint, if the reference points are the same and both the constraints differ only in `-start/-end`, the precedence rule is as follows:

- Command with `-start` will be higher priority, whereas `-end` will be lower priority.

Path Exception Priority Rules Table

The following table shows the priorities for path exceptions applied to the same path:

Path Exception Priorities

Priority	Path Exception
1. (Highest)	<code>set_false_path</code> <code>set_false_path -from pin_list</code> <code>set_false_path -to pin_list</code> <code>set_false_path -through pin_list</code>
2.	<code>set_max_delay -from pin_list</code>
3.	<code>set_max_delay -to pin_list</code>
4.	<code>set_max_delay -through pin_list</code>
5.	<code>set_max_delay -from clkwave_name</code>
6.	<code>set_max_delay -to clkwave_name</code>
7.	<code>set_max_delay</code> (The most constraining adjustment has the higher priority over less constraining adjustments.)
8.	<code>set_multicycle_path -from pin_list -to pin_list</code>
9.	<code>set_multicycle_path -from pin_list</code>
10.	<code>set_multicycle_path -to pin_list</code>
11.	<code>set_multicycle_path -through pin_list</code>
12.	<code>set_multicycle_path -from clkwave_name</code>
13.	<code>set_multicycle_path -to clkwave_name</code>
14. (Lowest)	<code>set_multicycle_path</code> (The most constraining adjustment has the higher priority over less constraining adjustments.)

Note: To check for applied/ignored path exceptions, you can use the following command:

```
report_timing -path_exceptions <applied | ignored | all>
```

To report all the path exceptions that have been ignored by timing analysis, you can use the following command:

```
report_path_exceptions [-ignored]
```

Examples of Path Exception Priorities

The following are examples of the path exception priorities for a path starting at **A**, going through **B**, and ending with **C**. The pairs of examples are ordered such that the highest priority constraint in one pair has precedence over the highest priority constraint in the next pair.

- In the following pair of constraints, the false path overrides the cycle addition for the path from **A** through **B** to **C**. For all other paths from **A**, the cycle addition applies.

```
set_false_path -from A -through B -to C
set_multicycle_path -from A 2
```

- In the following pair, the first constraint is applied (two cycles added) because the `-from` option has priority over the `-to` option:

```
set_multicycle_path -from A 3
set_multicycle_path -to C 2
```

- In the following pair, the first constraint is applied (two cycles added) because it specifies a `-through` pin as well as a `-to` pin:

```
set_multicycle_path -through A -to B 3
set_multicycle_path -to B 2
```

- In the following pair, the first constraint is applied (two cycles added) because the `-from` option has precedence over the `-to` option:

```
set_multicycle_path -from CLKA 3
set_multicycle_path -to CLKB 2
```

- In the following pair, the second constraint is applied because the second constraint overwrites the first one:

```
set_multicycle_path -from A 2
set_multicycle_path -from A 3
```

Bidirectional Pin Defaults

Bidirectional pins have both an input pin and an output pin type. This section explains the default pin type that the system uses for bidirectional pins.

Defaults for Pin (or Port) Constraints

The following defines the pin and port types that are used in the next table.

Bidirectional Pin (or Port) Type Definitions

Pin Type	Type 1	Type 2
----------	--------	--------

Blackbox pins	out	in
Instance pins	out	in
Netlist (primary) ports	in	out
Hierarchical ports	in	out

Bidirectional Pin Type Defaults

Pin or Port Constraint	Type	Hierarchical Port
<code>set_input_delay</code>	1	ignored
<code>set_output_delay</code>	2	ignored
<code>set_clock_latency</code>	1	ignored
<code>create_generated_clock</code>	1	used
<code>set_input_transition</code>	1	ignored
<code>set_driving_cell</code>	1	ignored
<code>set_drive</code>	1	ignored

Defaults for Port Constraints

The following table defines the port types that are used in the next table.

Bidirectional Port Type Definitions

Pin Type	Type 1	Type 2
Netlist (primary) ports	in	out
Hierarchical ports	in	out

Bidirectional Port Type Defaults

Port Constraint	Type	Hierarchical Port
<code>set_fanout_load</code>	2	ignored
<code>set_load</code>	2	ignored
<code>set_max_transition</code>	2	ignored

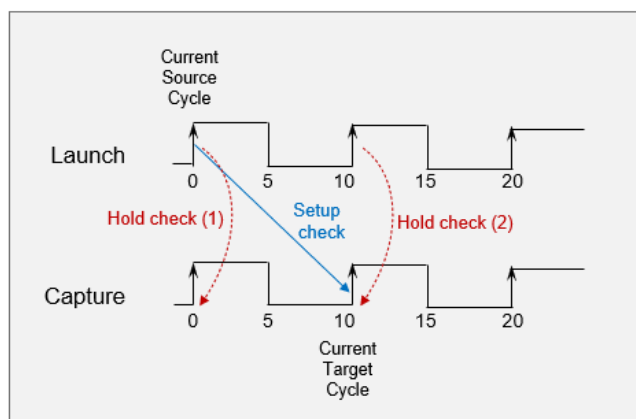
Cycle Accounting Rules - Data Checks

The following cycle accounting rules apply in the given below scenarios.

General Single-Cycle Timing

- When the launch and capture root clock edges align, the Setup capture edge is adjusted so that a full clock cycle is allowed for the path.
- When launch and capture Setup edges do not align, then the most constraining pair of root edges defines the cycle time. The launch and capture Hold edges are derived from the adjusted Setup edges:
 - Any multi-cycle adjustment to Setup is done before deriving the Hold edges.
 - Two Hold rules are checked and the most constraining edge pair is used (often they are the same).
 - Data launched from the current cycle at the source cannot be captured by the previous cycle at the target.
 - Data from the next cycle at the source cannot be captured by the current cycle at the target.
- Hold edges typically align - allowing full CPPR credit - including SI to be given.

The following diagram illustrates this behavior:



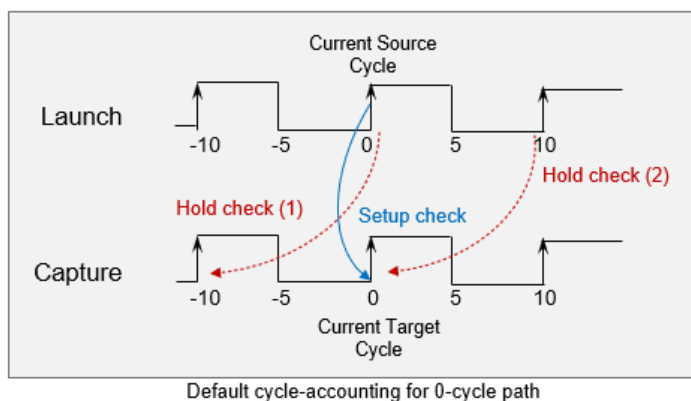
Zero-Cycle Timing

The zero-cycle Setup paths launch and capture data off of the same cycle. The zero-cycle Setup paths are obtained from the `set_data_check` constraint and non-sequential type library checks.

The following report outputs shows setup and hold data check output comparison:

Path 1: MET Data To Data Setup Check with Pin ref					Path 1: MET Data To Data Hold Check with Pin ref				
Endpoint: data (v) checked with leading edge of 'PH10'					Endpoint: data (v) checked with leading edge of 'PH10'				
Beginpoint: RB/Q (v) triggered by leading edge of 'PH10'					Beginpoint: RB/Q (v) triggered by leading edge of 'PH10'				
Other End Arrival Time					Other End Arrival Time				
- Data Check Setup					+ Data Check Hold				
+ Phase Shift					+ Phase Shift				
+ CPPR Adjustment					- CPPR Adjustment				
= Required Time					= Required Time				
- Arrival Time					Arrival Time				
= Slack Time					Slack Time				
Timing Point	Edge	Cell	Delay	Arrival Time	Timing Point	Edge	Cell	Delay	Arrival Time
clk	^	-	-	0.000	clk	^	-	-	0.000
Cl/Y	^	BUF1	0.260	0.260	Cl/Y	^	BUF1	0.110	0.110
...					...				
data ->	v	-	0.000	3.260	data ->	v	-	0.000	3.110
Timing Point	Edge	Cell	Delay	Arrival Time	Timing Point	Edge	Cell	Delay	Arrival Time
clk	^	-	-	0.000	clk	^	-	-	0.000
Cl/Y	^	BUF1	0.110	0.110	Cl/Y	^	BUF1	0.260	0.260
...					...				
ref	^	-	0.000	6.110	ref	^	-	0.000	6.260

The following diagram shows the default cycle accounting for a zero-cycle path:



In this case:

- The adjusted Hold edges mostly yield positive slack.
- The SI CPPR (50ps) is credited to Setup, but not Hold due to the Hold launch and capture edges not aligning.

Zero-Cycle Timing with Multi-Cycle Adjust

The following command will move the Hold launch back one cycle - making the Hold checks zero-cycle as well:

```
set_multicycle_path -1 -hold -to [get_ports data]
```

Path 1: MET Data To Data Setup Check with Pin ref
Endpoint: data (v) checked with leading edge of 'PH10'
Beginpoint: RB/Q (v) triggered by leading edge of 'PH10'
Other End Arrival Time 6.110
- Data Check Setup 3.000
+ Phase Shift 0.000
+ CPPR Adjustment 0.150
= Required Time 3.260
- Arrival Time 3.260
= Slack Time 0.000

Timing Point	Edge	Cell	Delay	Arrival Time
clk	^	-	-	0.000
C1/Y	^	BUFX1	0.260	0.260
...				
data ->	v	-	0.000	3.260

Timing Point	Edge	Cell	Delay	Arrival Time
clk	^	-	-	0.000
C1/Y	^	BUFX1	0.110	0.110
...				
ref	^	-	0.000	6.110

Path 1: VIOLATED Data To Data Hold Check with Pin ref
Endpoint: data (v) checked with leading edge of 'PH10'
Beginpoint: RB/Q (v) triggered by leading edge of 'PH10'
Other End Arrival Time 6.260
+ Data Check Hold 3.000
+ Phase Shift -10.000
- CPPR Adjustment 0.150
- Cycle Adjustment -10.000
= Required Time 9.110
Arrival Time 3.110
Slack Time -6.000

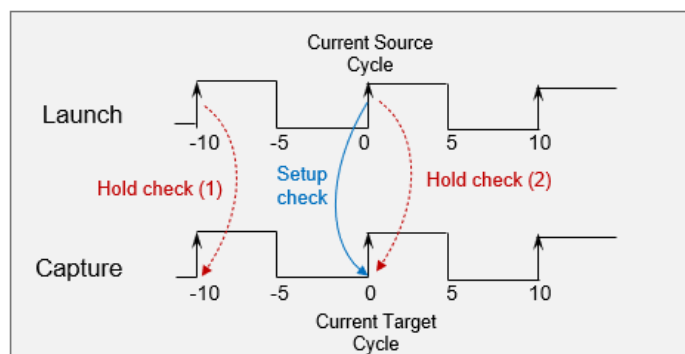
Timing Point	Edge	Cell	Delay	Arrival Time
clk	^	-	-	0.000
C1/Y	^	BUFX1	0.110	0.110
...				
data ->	v	-	0.000	3.110

Timing Point	Edge	Cell	Delay	Arrival Time
clk	^	-	-	0.000
C1/Y	^	BUFX1	0.260	0.260
...				
ref	^	-	0.000	6.260

The current syntax does not restrict the multi-cycle application to only data-to-data checks.

Since both the Setup and Hold are zero-cycle, both receive the full CPPR credit - including SI.

The following diagram illustrates accounting for zero-cycle path when the above mentioned command is specified.



Accounting for 0-cycle path with: `set_multicycle_path -1 -hold`

set_multicycle_path -start

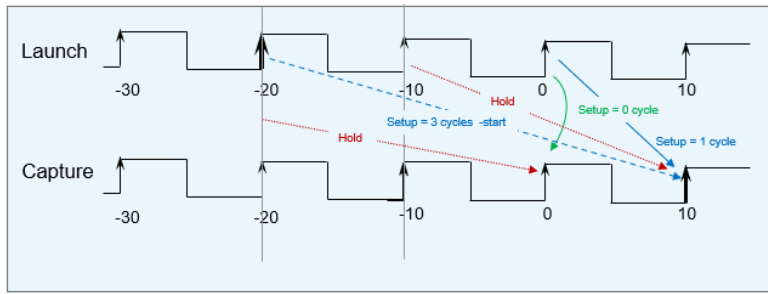
When the `set_multicycle_path -start` parameter is used to adjust data-to-data paths, there might be some compatibility issues with third-party software.

You can set the `timing_enable_sdc_compatible_data_check_mcp` global variable to `true` to enable checking for SDC compatible data checks. When the `-start` parameter is specified, the parameter reverses the reference clock by one clock cycle.

The following diagram shows Tempus cycle accounting when `set_multicycle_path -start 3` command is specified.

Tempus Text Command Reference

Timing Constraint Commands



create_clock

```
create_clock  
[-help]  
-period period_value  
[-name clock_name]  
[-waveform edge_list]  
[-add]  
[-comment string]  
[sources]
```

Creates a clock object and defines its waveform in the current design. If you do not specify any sources, but you specify the `-name clock_name` argument, a virtual clock is created. The new clock has an ideal clock latency, and the software does not assume any propagation delay through the clock network.

The clock that you create using the `create_clock` command cannot have multiple pulses per period. You can specify multiple constraints on the same pin using the `-add` parameter.

If one of the sources is already the source of a clock, the source is removed from that clock. If you need to specify multiple clocks on the same source for simultaneous analysis, you must specify the `-add` parameter, to add this clock to the existing clock.

For example, if you specify the following command sequence, the PH2 clock waveform overwrites the PH1 waveform.:

```
create_clock -name PH1 ... [get_ports sysclk]  
create_clock -name PH2 ... [get_ports sysclk]
```

To have both clocks applied, you must specify the following command sequence:

```
create_clock -name PH1 ... [get_ports sysclk]  
create_clock -name PH2 -add ... [get_ports sysclk]
```

You can create a collection of clocks, which is a group of objects referenced by a string identifier using the `get_clocks` command.

Note: If the `create_clock` constraint is added incrementally in SI mode, an explicit `update_timing -full` command is required (before `report_timing` or `report_constraint`) for SI full accuracy.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-add</code>	Specifies multiple clocks on the same source for simultaneous analysis with different clock waveforms. You must specify the <code>-name</code> parameter with this parameter.
<code>-comment <i>string</i></code>	Allows you to insert comments. The comments are read but are ignored with a one-time warning message. They are not used during timing analysis reporting nor are they written out during saving of constraints.
<code>-name <i>clock_name</i></code>	Specifies the string name of the ideal clock being created. If you do not specify this option, the clock gets the same name as the first clock source that you specify using the <code>-sources</code> option. If you do not specify any sources, you must specify the <code>-name <i>clock_name</i></code> option, which creates a virtual clock not associated with a port or pin.
<code>-period <i>period_value</i></code>	Specifies the value of the ideal clock in library time units.
<code><i>sources</i></code>	Specifies the list of pins or ports associated with an ideal clock waveform of the clock. If you do not specify this option, you must specify the <code>-name <i>clock_name</i></code> option, which creates a virtual clock not associated with a port or pin. If you specify a clock on a pin that already has a clock, the new clock replaces the old one.
<code>-waveform <i>edge_list</i></code>	Specifies the rise and fall edge times of the clock waveforms in library time units over an entire clock period. The first value in the edge list is the time at which the first transition on the clock occurs, which is the first rising transition after time zero, and edges are monotonically increasing. The numbers should represent one full clock period. If you do not specify the <code>-waveform <i>edge_list</i></code> option, a default waveform is assumed, and the leading edge is placed at 0 and the trailing edge is placed at the midpoint of the period, such that a symmetric clock is generated.

Examples

- The following command creates an `ideal_clk1` clock on port `clkA` with a period of 5, a rise at 0, and a fall at 2.5:

```
tempus > create_clock [get_ports {clkA}] -name ideal_clk1 -period 5 -waveform {0 2.5}
```
- The following command creates an `ideal_clk2` clock on port `clkB` with a period of 10, a rise at 1, and a fall at 9:

```
tempus > create_clock [get_ports {clkB}] -name ideal_clk2 -period 10 -waveform {1 9}
```

Related Commands

- `all_clocks`
- `get_clocks`
- `get_pins`
- `get_ports`
- `set_clock_latency`
- `report_clocks`
- `reset_clock`
- `set_clock_uncertainty`
- `set_input_delay`
- `set_output_delay`

create_generated_clock

```
create_generated_clock
[-help]
target_pin_list
[-add]
[-comment string]
[-duty_cycle percent]
[-edge_shift edge_shift_list]
[-invert]
[-master_clock source_clock_name]
[-name clock_name]
[-preinvert]
-source source_pin
{[-multiply_by factor]
[-divide_by factor]
[-edges edge_list]
[-combinational]}
```

Creates a new clock signal from the clock waveform of a given pin in the design, and binds it with the pins or hierarchical pins in the *target_pin_list* argument. Whenever the source clock changes, the derived clock(s) change automatically.

You can generate a new clock waveform using one of the three ways:

- Multiply the frequency of the source clock
- Divide the frequency of the source clock
- Select the edges of the source clock to be mapped to the edges of the new clock

Given below are the SDC semantics to be used when the `create_generated_clock` command is defined:

- The edges are interpreted as alternating rise and falling edges.
- Each edge must be greater than and equal to its previous edge.
- Less than 3 edges are not allowed.
- Legal edge numbers start at “1”, edge 0 is undefined.
- Legal assertion of period 0 will be -edges {1 1 1 }.

If incorrect clock definitions are specified, an error will occur and the clock will not be generated.

When the `create_generated_clock` command is specified without the `-master_clock` parameter, and the software finds multiple clocks reaching the specified source pin, then the regular clocks are picked ahead of the generated clocks. In case if multiple regular clocks are present, the master clock is picked based on alphabetical sorting order of the clock names.

When this constraint is set on a pin, the software will stop the propagation of clock phases from all upstream clocks, including the generated clock's own master clock. The special latency-only phases of

the master clock may still be propagated to fulfill the requirements of other downstream generated clocks, which are used only for latency path calculations.

Note: A `create_generated_clock` assertion on a pin overrides any existing `create_generated_clock` assertion if both the constraints have the same source pin. In the following command sequence, the second assertion overrides the first:

```
create_generated_clock -name GCK1 -source CK -divide_by 2 U_FD1/Q
create_generated_clock -name GCK2 -source CK -multiply_by 2 U_FD1/Q
```

To have both assertions applied, you must specify the `-add` parameter. For example:

```
create_generated_clock -name GCK1 -source CK -divide_by 2 U_FD1/Q
create_generated_clock -name GCK2 -add -master_clock CLK -source CK -multiply_by 2      U_FD1/Q
```

Parameters

<code>-help</code>	Prints out the command usage.
<code>-add</code>	Models multiple generated clocks on the same source when multiple clocks must fan into the source pin. Ideally, one generated clock must be specified for each clock that fans into the master pin. Specify this parameter with the <code>-name</code> and <code>-master_clock</code> parameters. You can use the <code>-add</code> parameter to specify a different clock name for each generated clock, when used with the <code>-master_clock</code> option. Subsequently, you can use this clock name for setting other constraints, such as, <code>set_false_path</code> or <code>set_input_delay</code>.
<code>-combinational</code>	Restricts the path search for latency to only paths which do not cross sequential devices, or the source pins of other generated clocks. When you specify <code>-combinational</code>, if there are both sequential and combinational latency paths, the software only considers the combinational paths for the generated clock. If there are no combinational latency paths for the generated clock, the generated clock does not inherit latency from its parent clock hierarchy. By default, when calculating the latency of a generated clock, the software searches for the longest (and shortest) paths from the parent waveform to the generated clock point. All paths are considered, including those which traverse sequential elements such as flip-flops, registers, and latches. The default behavior is intended for cases where there is a cascade of generated clocks, and some of the clocks are not explicitly defined. The <code>-combinational</code> option can be used in conjunction with the <code>-multiply_by</code>, <code>-divide_by</code>, or <code>-edges</code> options.

<code>-comment <i>string</i></code>	Allows you to insert comments. The comments are read but are ignored with a one-time warning message. They are not used during timing analysis reporting nor are they written out during saving of constraints.
<code>-divide_by <i>integer</i></code>	Determines the frequency of the new clock by dividing the frequency of the source clock by <i>dfactor</i> . For example, if <i>dfactor</i> =5, the new frequency is 1/5 of the source frequency, and the new clock period is 5 times the source clock period.
<code>-duty_cycle <i>percent</i></code>	Sets the duty cycle (high pulse width or clock period). The number (<i>percent</i>) is between 0 and 100. For example, if <i>percent</i> =50, the high pulse width of the new clock waveform is the same as its low pulse width. Note: The <code>-duty_cycle</code> option can only be used with the <code>-multiply_by</code> option, otherwise it will be ignored with a warning.
<code>-edges <i>edge_list</i></code>	Selects a list of edges from the source clock that form the edges of the derived clock. See the frequency division example below.
<code>-edge_shift <i>edge_shift_list</i></code>	Specifies the amount of shift for each edge in the <i>edge_list</i> option. The amount of shift will be in library time units.
<code>-invert</code>	Inverts the source clock. Use with the <code>-source</code> option and the <code>-multiply_by</code> or <code>-divide_by</code> options.
<code>-master_clock <i>source_clock_name</i></code>	Generates a target clock from the specified clock name. If you specify the <code>-master_clock</code> option and multiple signals arrive at the source pin, the clock specified by this option is used to generate the target clock. Use the <code>-add</code> parameter along with <code>-master_clock</code> parameter to add a clock to the existing clock when you want to specify multiple clocks on the same source for simultaneous analysis.
<code>-multiply_by <i>integer</i></code>	Determines the frequency of the new clock by multiplying the frequency of the source clock by <i>mfactor</i> . For example, if <i>mfactor</i> =4, the new frequency is 4 times the source frequency, and the new clock period is 1/4 of the source clock period.
<code>-name <i>clock_name</i></code>	Specifies the name for the generated clock. <i>Default:</i> The system creates a name for you. Always specify a name for the generated clock. Some commands issue an error if you use the system-generated name.
<code>-preinvert</code>	Inverts the polarity of a master clock at the specified source pin.

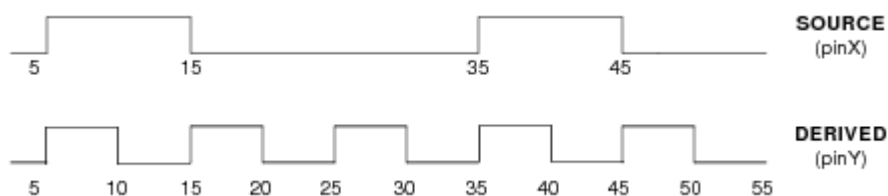
<code>-source source_pin</code>	Specifies the name of the source pin from which the generated clock is to be derived - with respect to the master clock containing the master clock as well as the generated clock definition.
<code>target_pin_list</code>	Specifies the list of pins where the generated clock assertion is applied. The clock signal propagating downstream from each of these pins becomes associated with the generated clock.

Examples

- The following command generates the derived clock waveform, as shown in the figure below:

```
tempus > create_generated_clock -name mult_3 -source pinX -multiply_by 3 -duty_cycle 50 pinY
```

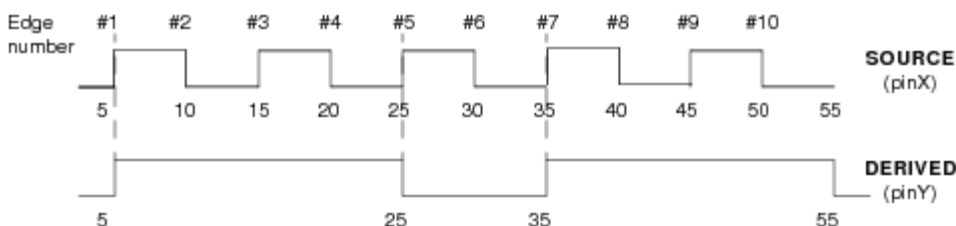
Figure: Derived Clock (Frequency Multiplication)



- The following command shows one way to create a divide-by-3 clock by using edge numbers of the source waveform to specify the new waveform, as shown in the figure below:

```
tempus > create_generated_clock -name genclk -source pinX -edges {1 5 7} pinY
```

Figure: Derived Clock (Frequency Division)

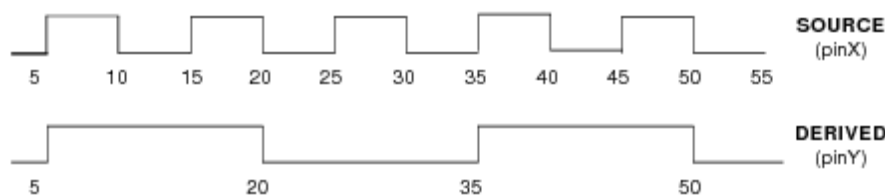


- The following command uses the `-divide_by` option to get the same clock period; the difference is in the duty cycle:

```
tempus > create_generated_clock -name div_3 -source pinX -divide_by 3 pinY
```

With the `-edges` option, the above example creates a clock with a duty cycle of 67 percent (20/30). However, using the `-divide_by 3` option preserves the 50 percent duty cycle of the source clock as shown in the figure below:

Figure: Derived Clock (Frequency Division, 50 Percent Duty-cycle)



- Using the following commands, when there is more than 1 source clock, the tool creates the generated clock based on the source clock specified with the `-master` option: `create_clock -name c1 -period 10 [get_ports {ck} ] -add` `create_clock -name c2 -period 10 [get_ports {ck} ] -add` `create_generated_clock -name newclk -source ck -divide_by 2 -master_clock c1 reg/Q -add`

Using the `report_clocks` command, the output waveform of the `newclk` clock is a divide-by-2 of the `c1` clock and is reported as shown in the example below.

Example: `report_clocks` after using the `-master` option

```
tempus > create_clock -name CK1 -period 20 [get_ports {CLK1} ] -add
tempus > create_clock -name CK2 -period 10 [get_ports {CLK1} ] -add
tempus > create_generated_clock -name GCLK -source CLK1 -divide_by 2 u8/Q -add -master_clock CK2
```

```
tempus > report_clocks
```

Clock Descriptions

Attributes

Clock Name	Source	Period	Lead	Trail	Generated	Propagated
CK1	CLK1	20.000	0.000	10.000	n	n
CK2	CLK1	10.000	0.000	5.000	n	n

Generated-Clock Descriptions

Name	Generated Source (pin)	Master Source (pin)	Master-clock	Invert	Freq. Multiplier	Duty-Cycle	Edges	Edge-Shift
GCLK	u8/Q	CLK1	CK2	n	1/2	-	-	-

Related Commands

- `create_clock`
- `get_clocks`
- `report_clocks`
- `reset_generated_clock`

current_design

```
current_design  
[-help]  
[design_name]
```

Changes the current directory in the design hierarchy to the specified design and returns the path to the design.

If no design is specified, the command returns the top-level design.

Parameters

<code>-help</code>	Prints out the command usage.
<code><i>design_name</i></code>	Specifies the name of the top design.

Related Commands

- `set_max_transition`
- `set_min_capacitance`
- `set_min_transition`
- `list_libraries`

current_instance

```
current_instance  
[-help]  
[hierarchical_instance]
```

Sets the specified instance to be the current instance

The design objects referenced by the subsequent commands will be relative to this instance and all the searches for design objects start from this point. If the design objects are referenced hierarchically, the specified instance is used as the root (top) of the hierarchy.

Parameters

<code>-help</code>	Prints out the command usage.
<code>hierarchical_instance</code>	Specifies the name of the instance to be set as the current instance. When this parameter is not specified, the current instance is set to the top instance of the hierarchy.

Examples

- The following command sets the current instance to MAC/MULTI, then sets a constant on a register pin inside the hierarchical instance and later sets the current instance back to top:

```
current_instance MAC/MULTI  
set_case_analysis 0 [get_pins {reg6/SE}]  
...  
current_instance
```

Related Commands

- `set_case_analysis`

get_sdc_mode

get_sdc_mode
[-help]

Determines the constraint mode that is being currently read.

This command can be used inside SDC constraint files which are shared across multiple constraint modes to conditionalize how the constraints are interpreted for specific modes.

Parameters

None

group_path

```
group_path  
[-help]  
[-comment string]  
-name path_group_name  
[[-from from_list]  
  [-to to_list]  
  [-through through_list]]
```

Groups paths in a design, and identifies them with a path group name. You can use the path group names for reporting.

You cannot use `group_path` statements in the SDC file when the software is in MMMC mode. In this case a warning is issued. However, the software supports the `group_path` command in MMMC mode as a command outside of the SDC file. The path groups are considered to be global in nature and apply to all the views, and are not supported on MMMC mode-specific basis. The path groups are persistent through a `saveDesign` to `restoreDesign` sequence. They are saved in the `<saveDir>/mmmc/pathgroup.sdc` file of a MMMC save directory.

Note that if a path can be included in multiple path groups, it will only be included in the path group with the most specific definition (`-from ... -through ... -to ...` is more specific than `-from ... -to ...` is more specific than `-from ...`).

Parameters

<code>-help</code>	Prints out the command usage.
<code>-comment <i>string</i></code>	Allows you to insert comments. The comments are read but are ignored with a one-time warning message. They are not used during timing analysis reporting nor are they written out during saving of constraints.
<code>-from <i>from_list</i></code>	Specifies a list of pins, instances, or clocks that are the start of the paths. When you specify a clock object, all paths that are triggered by the clock are included in the path group.
<code>-name <i>path_group_name</i></code>	Specifies the name of the path group. If the <code>path_group_name</code> already exists, the specified paths are added to the existing path group.
<code>-through <i>through_list</i></code>	Specifies the pins or instances where the paths go through. When you specify multiple through list, the paths that go through the objects in the specified order are added to the path group.
<code>-to <i>to_list</i></code>	Specifies a list of pins, instances, or clocks where the path ends. You must specify valid endpoints in the <code>to_list</code> or else the software ignores the constraint. When you specify a clock object, all paths that are checked by the clock are included in the path group. If you specify an instance name in the <code>to_list</code> , all the data input pins of the instance are added to the path group.

Examples

Here are some examples.

- The following example groups paths that start from specific pins and end at the D pin of any instances whose name start with the string ff. The example also reports the specified path groups:

```
tempus > group_path -name path1 -from {ff1/CLK ff2/CLK} -to {ff*/D}
tempus > report_timing -path_group path1
```

- The following command groups paths that go through specific points.
- If path group definitions are matching/overlapping (when the `-from/-through/-to` parameters are used), then path group resetting is applied. The path group overwriting and prioritization rules are similar to path exceptions.

The following examples show how path group resetting is applied.

If the timing path from u1/CK to u2/CK is part of both the path groups, then the path will be reported in the second path group only (that is, the most recent matching path group).

```
group_path -name g2 -from u1/CK,u4/Y -to u2/CK
```

```
group_path -name g1 -from u1/CK -to u2/CK
```

In the following example, the path groups are not overlapping and have different `-from/-thru/-to` and map to the same path, then the path will be reported only in the first path group (that is, the most specific matching path group):

```
group_path -name g2 -from u1/CK -to u2/CK  
group_path -name g1 -to u2/CK
```

In this case, the path will be reported only in the first path group (that is, the most specific matching path group).

Related Commands

- `report_timing`
- `get_path_groups`
- `reset_path_group`
- `report_path_groups`

reset_annotated_check

```
reset_annotated_check
[-help]
[-clock [rise | fall]]
[-cond sdf_condition]
[-setup | -recovery]
[-hold | -removal]
[-nochange_high | -nochange_low]
[-rise] [-fall]
{[[-from from_pins][-to to_pins]] | -all}
[object_list]
```

Removes annotated timing check values that you had set by using the `set_annotated_check` command or by SDF annotation.

Note: The `reset_annotated_check` command does not reset the timing check values set by SDF annotations.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-all</code>	Resets all annotated timing check values in the design.
<code>-clock [rise fall]</code>	Specifies the direction of the related pin of the check to reset. <i>Default:</i> Resets checks against both edges of the clock
<code>-cond sdf_condition</code>	Resets all annotated timing check values for which a condition is specified.
<code>-fall</code>	Restricts the reset to only affect the falling edge of the constrained pins. <i>Default:</i> Checks on both rising and falling signals are reset
<code>-from from_pins</code>	Specifies the beginning instance pin of the timing arcs for which the annotated check is reset. This is usually the reference (for example, clock pin) of the constraint. The selected timing arcs are those that have their source pins specified using this option and the sink pins specified using the <code>-to</code> option. If this option is not specified, only timing arcs whose sink pins are specified using the <code>-to</code> option are selected. You cannot use this parameter with <code>object_list</code> parameter.
<code>-hold -removal</code>	Resets annotations on the hold or removal check type.

<code>-nochange_high -nochange_low</code>	Resets the annotations on the specified timing check type.
<code>object_list</code>	Resets the annotated checks on specified list or collection of cell instances.
<code>-rise</code>	Restricts the reset to only affect the rising edge of the constrained pins. <i>Default:</i> Checks on both rising and falling signals are reset
<code>-setup -recovery</code>	Resets annotations on the setup or recovery check type.
<code>-to to_pins</code>	Specifies the terminating instance pin of the timing arcs for which the annotated check is reset. This is usually the constrained pin (data pin) of the check. The selected timing arcs are those that have their sink pins specified using this option and the source pins specified using the <code>-from</code> option. If this option is not specified, only timing arcs whose source pins are specified using the <code>-from</code> option are selected. You cannot use this parameter with <code>object_list</code> parameter.

Examples

- The following command resets all the timing check annotations in the design:
`tempus > reset_annotated_check -all`
- The following command resets all timing check annotations on a particular cell:
`tempus > reset_annotated_check [get_cells BLK/BR1]`
- The following command resets specific checks on a specific instance:
`tempus > reset_annotated_check -setup -hold [get_cells BLK/BR2]`
- The following command resets checks on a specific timing check arc:
`tempus > reset_annotated_check -from BLK/BR1/CK -to BLK/BR1/D`

Related Commands

- [set_annotated_check](#)

reset_annotated_delay

```
reset_annotated_delay
[-help]
{[-all] | [[-from from_pins] [-to to_pins]] | [object_spec]}
```

Resets annotated delay assertions that have been previously specified using the `set_annotated_delay` command. When you reset an annotated delay, the system uses the delays which are calculated from the original library for the arc.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-all</code>	Resets all annotated delay assertions in the design.
<code>-from</code> <i>from_pins</i>	Specifies timing arcs for which the delay is reset. The selected timing arcs are those that have their source pins specified using this option and the sink pins specified using the <code>-to</code> option. If this option is not specified, only timing arcs whose sink pins are specified using the <code>-to</code> option are selected.
<i>object_spec</i>	Resets the annotated delays on specified list or collection of cell instances.
<code>-to</code> <i>to_pins</i>	Specifies timing arcs for which the delay is reset. The selected timing arcs are those that have their sink pins specified using this option and the source pins specified using the <code>-from</code> option. If this option is not specified, only timing arcs whose source pins are specified using the <code>-from</code> option are selected.

Examples

- The following command resets all the annotated delay assertions in the design:
tempus > reset_annotated_delay -all
- The following command resets all the `set_annotated_delay` assertions applied on pins (either `from` or `to`) of instance `f0`:
tempus > reset_annotated_delay f0
- The following command resets the `set_annotated_delay` assertions with the `from` and `to` pin pair combination:
tempus > reset_annotated_delay -from u1/Y -to u2/A

Related Commands

- `set_annotated_delay`

reset_annotated_transition

```
reset_annotated_transition  
[-help]  
{-all | port_or_pin_list}
```

Resets annotated transitions previously asserted by the `set_annotated_transition` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-all</code>	Resets all annotated transitions in the design.
<code>port_or_pin_list</code>	Specifies a list of specific pins or primary ports to reset the transition.

Examples

Here are some examples.

- The following commands reset the `set_annotated_transition` constraint on an internal pin `pad/Y0`.

```
tempus > reset_annotated_transition pad/Y0
```

or

```
tempus > reset_annotated_transition [get_pins {pad/Y0}]
```

You can also specify a collection as the pin argument. For example:

```
tempus > reset_annotated_transition [get_pins pad/*]
```

- The following commands reset the `set_annotated_transition` constraint set on a port `in`:

```
reset_annotated_transition in
```

or

```
tempus > reset_annotated_transition [get_ports {in}]
```

Any `set_input_transition` constraint applied on the same port is not removed. This is applicable only for input ports, as `set_input_transition` cannot be applied on output ports.

- The following command resets all the `set_annotated_transition` constraints:

```
tempus > reset_annotated_transition -all
```

Related Commands

- `set_annotated_transition`

reset_aocv_stage_weight

```
reset_aocv_stage_weight
[-help]
port_list
[-early]
[-late]
```

Resets the existing stage weight values set for the specified ports.

Parameters

-help	Prints out the command usage.
-early	Resets weight for early paths.
-late	Resets weight for late paths.
port_list	Specifies a list or collection of input, output, and bidi ports for which stage weight values are to be reset.

Examples

- The following example resets the stage weight values of ports `clock1` and `out0`:
tempus > reset_aocv_stage_weight [get_ports {clock1 out0}]

Related Commands

- `set_aocv_stage_weight`

reset_case_analysis

```
reset_case_analysis  
[-help]  
list_of_ports_or_pins
```

Removes the assertions set by the `set_case_analysis` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>list_of_ports_or_pins</code>	Specifies the list of ports or pins for which the assertions are to be removed.

Examples

- The following commands sets the assertion to 0 and then resets them:

```
tempus > set_case_analysis 0 CM1/S0  
tempus > get_property [get_pins CM1/S0] constant_value
```

Returns a value of 0

```
tempus > reset_case_analysis [get_pins CM1/S0]  
tempus > get_property [get_pins CM1/S0] constant_value
```

Returns a value of NA

Related Commands

- `set_case_analysis`

reset_clock

```
reset_clock
[-help]
{-all | clock_list}
```

Removes previously created clock assertions.

Parameters

-help	Prints out the command usage.
-all	Removes all clocks in the design.
clock_list	Specifies the list of clocks for which the clock assertion is removed.

Examples

- The following commands reset a list of clocks and all clocks, and display the results using the `report_clocks` command:

```
tempus > report_clocks
```

Generates the following report:

```
-----
Clock Descriptions
-----
Attributes
-----
Clock Name    Source    Period    Lead    Trail    Generated    Propagated
-----
CK1           CLK1      20.000    0.000   10.000    n            n
CK2           CLK1      10.000    0.000   5.000     n            n
-----
Generated-Clock Descriptions
-----
--
Name          Generated  Master    Master-clock Invert Freq.    Duty-Cycle  Edges  Edge-
Shift
              Source(pin) Source(pin)                               Multiplier
-----
--
GCLK4        u8/Q      CLK4      CLK_W_4      n          1/2        -      -      -
tempus> reset_clock [get_clocks CK1]
```

tempus> report_clocks

Clock Descriptions

Attributes

Clock Name	Source	Period	Lead	Trail	Generated	Propagated
CK2	CLK1	10.000	0.000	5.000	n	n

Generated-Clock Descriptions

Name	Generated	Master	Master-clock	Invert	Freq.	Duty-Cycle	Edges
Edge-Shift	Source (pin)	Source (pin)			Multiplier		
GCLK4	u8/Q	CLK4	CLK_W_4	n	1/2	-	-

Related Commands

- `create_clock`
- `report_clocks`

reset_clock_exclusivity

```
reset_clock_exclusivity  
[-help]  
<pin_name>
```

Allows to reset the `set_clock_exclusivity` constraints applied on pins.

Parameters

<code>-help</code>	Prints out the command usage.
<code><pin_name></code>	Specifies the name of pins for which clock exclusivity constraints need to be reset.

Related Commands

- `set_clock_exclusivity`

reset_clock_gating_check

```
reset_clock_gating_check  
[-help]  
[-setup]  
[-hold]  
[-rise]  
[-fall]  
[-high | -low]  
[object_list]
```

Resets the specified setup and hold values for clock gating checks to the default values.

Parameters

-help	Prints out the command usage.
-high -low	Resets the non-controlling value (high or low) of the clock.
-hold	Resets clock gating hold values.
-rise -fall	Resets the specified setup (or hold) value to the rising (or falling) signals on the specified pins.
-setup	Resets clock gating setup values.
object_list	Specifies a list of clocks, instances, or pins for which clock gating checks are to be reset to default values. If the <i>object_list</i> parameter is not specified, the clock gating check is reset on the current design.

Related Commands

- [set_clock_gating_check](#)

reset_clock_groups

```
reset_clock_groups  
[-help]  
{-physically_exclusive | -logically_exclusive | -asynchronous}  
{-name name_list | -all}
```

Resets clock groups that have been set using the `set_clock_groups` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-all</code>	Removes all the physically exclusive, logically exclusive, or asynchronous clock groups in the design. This parameter should be used with one of the parameters: <code>-logically_exclusive</code> , <code>-physically_exclusive</code> , or <code>-asynchronous</code> at one time, and should not be used with the <code>-name</code> parameter.
<code>-asynchronous</code>	Specifies the asynchronous clock groups to be removed. This parameter must be used with either <code>-all</code> or <code>-name</code> parameter.
<code>-logically_exclusive</code>	Specifies the logically exclusive clock groups to be removed. This parameter must be used with either <code>-all</code> or <code>-name</code> parameter.
<code>-name <i>name_list</i></code>	Specifies the previously defined names of clock groups to be removed. This parameter should be used with one of the parameters: <code>-logically_exclusive</code> , <code>-physically_exclusive</code> , or <code>-asynchronous</code> at one time.
<code>-physically_exclusive</code>	Specifies all the physically exclusive clock groups to be removed. This parameter must be used with either <code>-all</code> or <code>-name</code> parameter.

Examples

- The following command resets all physically exclusive clock groups in the design:

```
tempus > reset_clock_groups -physically_exclusive -all
```

- The following command resets physically exclusive clock group `CLKGRP0`:
`tempus > reset_clock_groups -physically_exclusive -name CLKGRP0`

Related Commands

- `set_clock_groups`

reset_clock_latency

```
reset_clock_latency
[-help]
[-source]
[-clock clock_list]
pin_or_clock_list
```

Resets assertions made by previous `set_clock_latency` commands.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-clock <i>clock_list</i></code>	Specifies the list of clocks with which associated <code>set_clock_latency</code> constraints, at the pin level, are to be removed. Note: This parameter cannot be used with clocks specified using the <code>pin_or_clock_list</code> parameter.
<code><i>pin_or_clock_list</i></code>	Specifies a list of pins, ports, or clock objects for which the assertions are to be removed.
<code>-source</code>	Removes the source latency.

Examples

- The following commands show the use of the `reset_clock_latency` command to remove a leaf-level `set_clock_latency` assertion, after the clock tree has been put into propagated mode:

```
tempus > set_clock_latency 3.141 [get_pins BLK/BR2/CK]
tempus > report_timing -to BLK/BR2/D -path_type full_clock
```

The following report is generated:

```
Path 1: MET Setup Check with Pin BLK/BR2/CK
Endpoint: BLK/BR2/D (v) checked with leading edge of 'WAVE'
Beginpoint: BLK/BR1/Q (v) triggered by leading edge of 'WAVE'
Other End Arrival Time 3.141 <---
- Setup 0.191
+ Phase Shift 10.000
= Required Time 12.950
```

```
- Arrival Time 0.249
= Slack Time 12.701
...
```

```
tempus > set_propagated_clock [get_ports tclk]
tempus > report_timing -path_type full_clock -net -to BLK/BR2/D
```

The following report is generated:

```
Path 1: MET Setup Check with Pin BLK/BR2/CK
Endpoint: BLK/BR2/D (v) checked with leading edge of 'WAVE'
Beginpoint: BLK/BR1/Q (v) triggered by leading edge of 'WAVE'
Other End Arrival Time 3.141 <-----
- Setup 0.191
+ Phase Shift 10.000
= Required Time 12.950
- Arrival Time 0.594
= Slack Time 12.356
...
```

In this example, setting the root of the clock tree to propagated mode is not sufficient to override the `set_clock_latency` assertion at the clock endpoint. You must use the `reset_clock_latency` command to remove the assertion, so that the register sees propagated clock timing.

```
tempus > reset_clock_latency [get_pins BLK/BR2/CK]
tempus > report_timing -path_type full_clock -net -to BLK/BR2/D
```

The following report is generated:

```
Path 1: MET Setup Check with Pin BLK/BR2/CK
Endpoint: BLK/BR2/D (v) checked with leading edge of 'WAVE'
Beginpoint: BLK/BR1/Q (v) triggered by leading edge of 'WAVE'
Other End Arrival Time 0.345 <-----
- Setup 0.186
+ Phase Shift 10.000
= Required Time 10.159
- Arrival Time 0.588
= Slack Time 9.571
```

- The following commands show that `reset_clock_latency` is only effective when it is applied on the

same object for which you had set the assertion using the `set_clock_latency` command:
Latency here is specified on a pin object:

```
tempus > set_clock_latency 3.141 [get_pins BLK/BR2/CK]
tempus > report_timing -path_type full -net -to BLK/BR2/D
```

The following report is generated:

```
Path 1: MET Setup Check with Pin BLK/BR2/CK
Endpoint: BLK/BR2/D (v) checked with leading edge of 'WAVE'
Beginpoint: BLK/BR1/Q (v) triggered by leading edge of 'WAVE'
Other End Arrival Time 3.141 <-
- Setup 0.191
+ Phase Shift 10.000
= Required Time 12.950
- Arrival Time 0.249
= Slack Time 12.701
```

Resetting of the latency is done on a clock object and has no effect on downstream latencies specified on pin objects.

In this example, a `reset_clock_latency` on the clock waveform that is driving BLK/BR2/CK will have no effect.

```
tempus > reset_clock_latency [get_clocks WAVE]
tempus > report_timing -path_type full -net -to BLK/BR2/D
```

The following report is generated:

```
Path 1: MET Setup Check with Pin BLK/BR2/CK
Endpoint: BLK/BR2/D (v) checked with leading edge of 'WAVE'
Beginpoint: BLK/BR1/Q (v) triggered by leading edge of 'WAVE'
Other End Arrival Time 3.141 <-
- Setup 0.191
+ Phase Shift 10.000
= Required Time 12.950
- Arrival Time 0.249
= Slack Time 12.701
```

Related Commands

- `set_clock_latency`

reset_clock_sense

```
reset_clock_sense
[-help]
[-all]
| [[-clocks clock_list] pin_or_port_list]]
```

Resets clock sense constraint(s) specified using the `set_clock_sense` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-all</code>	Resets all the specified <code>clock_sense</code> constraints.
<code>-clocks</code> <i>clock_list</i>	Specifies the list of clocks for which the clock sense constraints are to be removed. The clocks must have previously been defined using the <code>set_clock_sense -clock</code> parameter.
<i>pin_or_port_list</i>	Specifies the list or collection of cell output pins or ports for which the clock sense constraints will be removed.

Examples

- The following command resets all the previously defined `clock_sense` constraints from a design:
`tempus > reset_clock_sense -all`

Related Commands

- `set_clock_sense`

reset_clock_transition

```
reset_clock_transition
[-help]
clock_list
```

Removes the clock transition assertions you had set on the specified list of clock waveforms.

Parameters

-help	Prints out the command usage.
clock_list	Specifies a list of clock waveforms for which the clock transition assertions are removed. The <i>clock_list</i> variable can be a Tcl list or collection of clock waveforms.

Examples

- The following commands set the clock transition assertions, and resets them and displays the results using the `report_timing` command:

```
tempus > set_clock_transition 0.123 [get_clocks WAVE]
tempus > report_timing -net -to BLK/BR2/D
```

Generates the following report:

```
Path 1: MET Setup Check with Pin BLK/BR2/CK
Endpoint: BLK/BR2/D (v) checked with leading edge of 'WAVE'
Beginpoint: BLK/BR1/Q (v) triggered by leading edge of 'WAVE'
Other End Arrival Time 0.000
- Setup 0.192
+ Phase Shift 10.000
= Required Time 9.808
- Arrival Time 0.249
= Slack Time 9.559
Clock Rise Edge 0.000
= Beginpoint Arrival Time 0.000
+-----+
| Instance | Cell | Arc | Pin | Slew | Delay | Arrival |
|          |     |    |     |      |      | Time    |
|-----+-----+-----+-----+-----+-----+
|          |     |    |     |      |      |         |
```

Tempus Text Command Reference

Timing Constraint Commands

		tclk ^	tclk	0.123		0.000
CG/BC1	BUFX2		A	0.123	0.000	0.000
CG/BC1	BUFX2	A ^ -> Y ^	Y	0.123	0.000	0.000
TC1	BUFX2		A	0.123	0.000	0.000
TC1	BUFX2	A ^ -> Y ^	Y	0.123	0.000	0.000
CG1	AND2X4		B	0.123	0.000	0.000
CG1	AND2X4	B ^ -> Y ^	Y	0.123	0.000	0.000
BLK/blkint	BUFX2		A	0.123	0.000	0.000
BLK/blkint	BUFX2	A ^ -> Y ^	Y	0.123	0.000	0.000
BLK/BR1	DFFHQX1		CK	0.123	0.000	0.000

```
tempus > reset_clock_transition [get_clocks WAVE]
```

```
tempus > report_timing -net -to BLK/BR2/D
```

Generates the following report:

Path 1: MET Setup Check with Pin BLK/BR2/CK

Endpoint: BLK/BR2/D (v) checked with leading edge of 'WAVE'

Beginpoint: BLK/BR1/Q (v) triggered by leading edge of 'WAVE'

Other End Arrival Time 0.000

- Setup 0.191

+ Phase Shift 10.000

= Required Time 9.809

- Arrival Time 0.249

= Slack Time 9.560

Clock Rise Edge 0.000

= Beginpoint Arrival Time 0.000

Instance	Cell	Arc	Pin	Slew	Delay	Arrival Time
		tclk ^	tclk	0.000		0.000
CG/BC1	BUFX2		A	0.000	0.000	0.000
CG/BC1	BUFX2	A ^ -> Y ^	Y	0.000	0.000	0.000
TC1	BUFX2		A	0.000	0.000	0.000
TC1	BUFX2	A ^ -> Y ^	Y	0.000	0.000	0.000
CG1	AND2X4		B	0.000	0.000	0.000
CG1	AND2X4	B ^ -> Y ^	Y	0.000	0.000	0.000

```
| BLK/blkint | BUFX2 | | A | 0.000 | 0.000 | 0.000 |
| BLK/blkint | BUFX2 | A ^ -> Y ^ | Y | 0.000 | 0.000 | 0.000 |
| BLK/BR1 | DFFHQX1 | | CK | 0.000 | 0.000 | 0.000 |
```

- The following commands set the clock transition assertions, and resets them and displays the results using the `report_timing` command:

```
tempus > set_clock_transition 0.123 [get_clocks CLK_W_2]
tempus > report_timing -net -format {pin arc cell slew delay arrival} -to seg3/u14/D
```

```
Path 1: MET Setup Check with Pin seg3/u14/CK
Endpoint: seg3/u14/D (^) checked with leading edge of 'CLK_W_3'
Beginpoint: seg3/u9/Q (^) triggered by leading edge of 'CLK_W_3'
Path Groups: {CLK_W_3}
Other End Arrival Time 0.000
- Setup 0.171
+ Phase Shift 2.000
= Required Time 1.829
- Arrival Time 1.409
= Slack Time 0.420
Clock Rise Edge 0.000
+ Clock Network Latency (Ideal) 0.000
= Beginpoint Arrival Time 0.000
```

Pin	Edge	Net	Cell	Slew	Delay Time	Arrival Time	Required
seg3/u9/CK	^	seg3/wout	-	0.123	-	0.000	0.420
seg3/u9/Q	^	seg3/n9	DFF	0.071	0.220	0.220	0.640
seg3/u10/Y	^	seg3/n10	BUF	0.074	0.071	0.291	0.711
seg3/u11/Y	v	seg3/n11	INV	0.638	0.312	0.603	1.023
seg3/u13/Y	^	seg3/n13	BUF	1.078	0.403	1.409	1.829
seg3/u14/D	^	-	DFF	1.078	0.000	1.409	1.829

Related Commands

- `set_clock_transition`

reset_clock_tree_latency

reset_clock_tree_latency

[-help]

[object_list]

Resets all network clock latency set using the [set_clock_latency](#) command in the fanout of the specified clock or pins and ports. The `reset_clock_tree_latency` command resets the latency until the root pins of clocks and generated clocks.

Note: You can use the `reset_clock_tree_latency` command only on command line and command script. You cannot specify it in an SDC file.

Parameters

-help	Prints out the command usage.
<i>object_list</i>	Specifies list of objects - pins, ports, or clocks for which latency will be reset.

Related Commands

- [set_clock_latency](#)
- [set_propagated_clock](#)

reset_clock_uncertainty

```
reset_clock_uncertainty
[-help]
[-setup]
[-hold]
[-half_cycle_jitter | -full_cycle_jitter]
{ {-from clk_sig_from_list | -rise_from clk_sig_from_list | -fall_from clk_sig_from_list}
  {-to clk_sig_to_list | -rise_to clk_sig_to_list | -fall_to clk_sig_to_list}
  | pin_or_clock_list
}
```

Removes the assertions that were made by previous `set_clock_uncertainty` commands.

Parameters

-help	Prints out the command usage.
{-from -rise_from -fall_from} <i>clk_sig_from_list</i>	Removes the assertions for uncertainty value for the specified list of launching clocks. Note: You must specify one of these parameters with one of the -to, -rise_to, or -fall_to parameters. You cannot use these parameters when pins are specified.
	-fall_from: Removes the uncertainty value only from the falling edge of the launching clocks.
	-from: Removes the uncertainty value from both the rising and falling edges of the launching clocks.
	-rise_from: Removes the uncertainty value only from the rising edge of the launching clocks.
-half_cycle_jitter -full_cycle_jitter	Removes or resets the clock source jitter assertion(s) of the specified type.

<code>pin_or_clock_list</code>	Removes the target uncertainty assertions from the specified list of clocks or pins. The uncertainty value is removed from the registers clocked by the target waveform, or register pins in the fanout of the specified pin. You can specify a list of clock names, or clock root pin or port names relative to the current module. You also can specify a collection of clocks, or clock root pins or ports. Note: You cannot use this parameter when any of the <code>-from</code> and <code>-to</code> parameter pairs are specified.
<code>-setup/-hold</code>	Specifies whether the clock uncertainty is removed for setup or hold checks. <i>Default:</i> Removes clock uncertainty for both setup and hold check.
<code>{-to -rise_to -fall_to} clksig_to_list</code>	Removes the uncertainty value from the specified list of capture clocks. Note: You must specify one of these parameters with one of the <code>-from</code>, <code>-rise_from</code>, or <code>-fall_from</code> parameters. You cannot use these parameters when pins are specified.
	• <code>-fall_to</code>: Removes the uncertainty value only from the falling edge of the capture clocks.
	• <code>-rise_to</code>: Removes the uncertainty value only from the rising edge of the capture clocks.
	• <code>-to</code>: Removes the uncertainty value from both the rising and falling edges of the capture clocks.

Examples

- The following commands sets and resets the clock uncertainty assertions and displays the results using the `report_timing` command:

```
tempus > set_clock_uncertainty 0.345 -from [get_clocks VCLK] -to [get_clocks WAVE]
tempus > report_timing -net -from tin
```

Generates the following report:

```
Path 1: MET Setup Check with Pin TR1/BR1/CK
Endpoint: TR1/BR1/D (v) checked with leading edge of 'WAVE'
Beginpoint: tin (v) triggered by trailing edge of 'VCLK'
Other End Arrival Time 0.000
- Setup 0.192
+ Phase Shift 10.000
- Uncertainty 0.345 <---
= Required Time 9.463
- Arrival Time 5.877
= Slack Time 3.586   reset_clock_uncertainty -from [get_clocks VCLK] -to [get_clocks
WAVE]

tempus > report_timing -net -from tin
```

Generates the following report:

```
Path 1: MET Setup Check with Pin TR1/BR1/CK
Endpoint: TR1/BR1/D (v) checked with leading edge of 'WAVE'
Beginpoint: tin (v) triggered by trailing edge of 'VCLK'
Other End Arrival Time 0.000
- Setup 0.192
+ Phase Shift 10.000
= Required Time 9.808
- Arrival Time 5.877
= Slack Time 3.931
```

Related Commands

- [set_clock_uncertainty](#)

reset_data_check

```
reset_data_check  
[-help]  
[-setup]  
[-hold]  
{-from pin_or_port_list | -rise_from pin_or_port_list | -fall_from pin_or_port_list}  
{-to pin_or_port_list | -rise_to pin_or_port_list | -fall_to pin_or_port_list}  
[-clock clock_object]
```

Removes specified data-to-data checks that you have specified using the [set_data_check](#) command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-clock <i>clock_object</i></code>	Removes the data check with a related (<code>-from</code>) pin event that is triggered by the specified clock. Use this parameter only if the original <code>set_data_check</code> constraint also used the <code>-clock</code> parameter to specify an explicit clock relationship.
<code>{-from -rise_from -fall_from}</code> <code><i>pin_or_port_list</i></code>	Specifies the related pins of the data-to-data checks to be removed. The <code><i>pin_or_port_list</i></code> list can contain either object IDs or hierarchical names relative to the current module. Leaf design cells (instances) can also be specified. Use only one <code>-from</code>, <code>-rise_from</code>, or <code>-fall_from</code> parameter per command.
	• By default, the <code>-from</code> parameter removes checks for both the rising and the falling edges. • Using <code>-rise_from</code> parameter removes checks at the rising edge of the signal on the from pins. • Using <code>-fall_from</code> parameter removes checks at the falling edge of the signal on the from pins.
<code>-setup/-hold</code>	Specifies whether the data check to be removed is for setup or hold check. If you do not specify either <code>-setup</code> or <code>-hold</code> option, both setup and hold checks are removed.
<code>{-to -rise_to -fall_to}</code> <code><i>pin_or_port_list</i></code>	Specifies the constrained pins of data-to-data checks to be removed. The <code><i>pin_or_port_list</i></code> list can contain either object IDs or hierarchical names relative to the current module. Leaf design cells (instances) can also be specified. Use only one <code>-to</code>, <code>-rise_to</code>, or <code>-fall_to</code> parameter per command. • By default, the <code>-to</code> parameters removes both the rising and the falling edges. • Using <code>-rise_to</code> parameter removes the rising edge of the signal on the to pins. • Using <code>-fall_to</code> parameter removes the falling edge of the signal on the to pins.

Examples

Here are some examples.

- The following commands set and reset the data checks from port30 to port28:

```
tempus > set_data_check 0.345 -rise_from port30 -to port28
tempus > report_timing -rise_from port30 -to port28
```

The following report is displayed:

```
Path 1: VIOLATED Data To Data Setup Check with Pin port30
Endpoint: port28 (v) checked with leading edge of 'PH1'
Beginpoint: U2/Q (v) triggered by leading edge of 'PH1'
Other End Arrival Time 0.465
- Data Check Setup 0.345
+ Phase Shift 0.000
= Required Time 0.120
- Arrival Time 0.309
= Slack Time -0.189
```

```
tempus > reset_data_check -rise_from port30 -to port28
tempus > report_timing -rise_from port30 -to port28
```

The following information is displayed:

```
No constrained timing paths with given description found.
Paths may be unconstrained (try '-unconstrained' option) or may not exist.
```

Related Commands

- [set_data_check](#)
- [report_timing](#)

reset_disable_clock_gating_check

```
reset_disable_clock_gating_check
[-help]
object_list
```

Enables clock gating checks for the specified instances or pins.

Parameters

-help	Prints out the command usage.
object_list	Specifies the list of disabled pins or instances that are to be enabled.

Related Commands

- `set_disable_clock_gating_check`

reset_disable_timing

```
reset_disable_timing  
[-help]  
[-from pin_name -to pin_name]  
object_list
```

Restores timing arcs that were disabled using the `set_disable_timing` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-from</code> <code><i>pin_name</i></code> - <code>to <i>pin_name</i></code>	Specifies the arc(s) that are to be restored. Specify the <code>-from</code> and <code>-to</code> options together. You can specify the <code>-from</code> and <code>-to</code> parameters when the <code>object_list</code> contains instance or library cell objects. Do not specify the <code>-from</code> and <code>-to</code> parameters when the <code>object_list</code> includes instance pin or library cell pin objects. The <code>-from</code> and <code>-to</code> options are optional. If not specified, all cell arcs found in the instance or library cell objects are restored.
<code><i>object_list</i></code>	Specifies the instance, instance pin, library arc, library cell, or library cell pin objects. If the <code>object_list</code> includes instance pin or library cell pin objects, all cell arcs attached to the specified pin objects are restored. If the <code>object_list</code> contains instance or library cell objects, and the <code>-from</code> and <code>-to</code> parameters are not specified, all arcs found in the specified instance or library cell objects are restored.

Examples

- The following commands set and reset the timing arcs and display the resulting reports:

```
tempus > set_disable_timing -from A -to Y [get_cells b1]
tempus > report_inactive_arcs
```

```
-----
Flags :                const                propagated constant
                snipped                loop snipped arcs
                disable                set_disable_timing
                disable_clock_gating    set_disable_clock_gating
                library_disable        set_disable_cell_timing
                Missing_Phase          No Arc Phase Data

From      To      DisableType  ArcType      Reason
-----
b1/A ^    b1/Y ^    disable      combinational  User Disable
b1/A v    b1/Y v    disable      combinational  User Disable
-----
```

- The following commands disable the timing report header:

```
tempus > set_global timing_disable_report_header_info true
tempus > report_inactive_arcs
```

```
-----
Flags :                const                propagated constant
                snipped                loop snipped arcs
                disable                set_disable_timing
                disable_clock_gating    set_disable_clock_gating
                library_disable        set_disable_cell_timing
                Missing_Phase          No Arc Phase Data

From      To      DisableType  ArcType      Reason
-----
u8/CK ^    u8/D ^    const        hold_rising_clk_rise  u8/D = 1
u8/CK ^    u8/D v    const        hold_falling_clk_rise u8/D = 1
u8/CK ^    u8/D ^    const        setup_rising_clk_rise u8/D = 1
u8/CK ^    u8/D v    const        setup_falling_clk_rise u8/D = 1
-----
```

- The following commands disable the timing arcs from CK to Q for instance U8, then reset the assertions using the instance reference:

```
tempus > set_disable_timing -from CK -to Q [get_cells u8]
tempus > report_inactive_arcs
```

```
-----
Flags :          const          propagated constant
          snipped              loop snipped arcs
          disable              set_disable_timing
          disable_clock_gating set_disable_clock_gating
          library_disable      set_disable_cell_timing
          Missing_Phase        No Arc Phase Data
From      To      DisableType  ArcType      Reason
-----
u8/CK ^    u8/Q ^    disable      rising_edge  User Disable
u8/CK ^    u8/Q v    disable      rising_edge  User Disable
-----
```

- The following commands disable the timing arcs from CK to Q for instance U8, then reset the assertions using the instance reference:

```
tempus > reset_disable_timing u8
tempus > report_inactive_arcs
```

```
-----
Flags :          const          propagated constant
          snipped              loop snipped arcs
          disable              set_disable_timing
          disable_clock_gating set_disable_clock_gating
          library_disable      set_disable_cell_timing
          Missing_Phase        No Arc Phase Data
From      To      DisableType  ArcType      Reason
-----
u8/CK ^    u8/D ^    const        hold_rising_clk_rise  u8/D = 1
u8/CK ^    u8/D v    const        hold_falling_clk_rise u8/D = 1
u8/CK ^    u8/D ^    const        setup_rising_clk_rise  u8/D = 1
u8/CK ^    u8/D v    const        setup_falling_clk_rise u8/D = 1
-----
```

Related Commands

- `set_disable_timing`
- `report_inactive_arcs`

reset_drive

```
reset_drive  
[-help]  
port_list
```

Resets the drive previously asserted with the `set_drive` command on input ports. The `reset_drive` command removes assertions for all rise/fall, minimum/maximum combinations for which the assertion exists.

Parameters

<code>-help</code>	Prints out the command usage.
<code>port_list</code>	Specifies a list of input or bidirectional ports for which the drive resistance is reset.

Related Commands

- `set_drive`

reset_driving_cell

```
reset_driving_cell  
[-help]  
[-rise]  
[-fall]  
[-min]  
[-max]  
port_list
```

Resets the driving cell assertions applied with the `set_driving_cell` command for the given rise/fall, minimum/maximum combination on ports.

If `-rise`, `-fall`, `-min`, or `-max` parameters are not specified, the `reset_driving_cell` command resets the assertion for all rise/fall, minimum/maximum combinations for which the assertion exists.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-rise</code>	Specifies the rising drive assertions.
<code>-fall</code>	Specifies the falling drive assertions.
<code>-min</code>	Specifies the early drive assertion.
<code>-max</code>	Specifies the late drive assertion.
<i>port_list</i>	Specifies a list of ports for which the driving cell assertions are reset.

Related Commands

- `set_driving_cell`

reset_generated_clock

```
reset_generated_clock
[-help]
{-all | clock_list}
```

Removes generated clock assertions previously created using the `create_generated_clock` command.

To use the `reset_generated_clock` command you must specify the `set_interactive_constraint_modes` constraints for the applicable modes. You can also include these commands in the SDC constraint file - applicable only for the views that include that SDC.

Note: The generated clocks may originate from the library files dotlib - you can use `reset_generated_clock` from the command line, or add it to all the SDC files in use.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-all</code>	Removes all the generated clocks in the design.
<code><i>clock_list</i></code>	Specifies the list of clocks (in collection format) for which the generated clock assertions are to be removed.

Examples

Here are some examples.

- The following command removes the generated clock assertions on all the generated clocks `RC/Q`:
tempus > report_clocks

The following report is displayed:

```
-----
                                Clock Descriptions
-----
                                Attributes
-----
```

Clock Name	Source	Period	Lead	Trail	Generated	Propagated
CK1	CLK1	20.000	0.000	10.000	n	n
CK2	CLK1	10.000	0.000	5.000	n	n
GCLK	u8/Q	20.000	0.000	10.000	y	y

Generated-Clock Descriptions

Name	Generated	Master	Master-clock	Invert	Freq.	Duty-Cycle	Edges	Edge-Shift
	Source (pin)	Source (pin)			Multiplier			
GCLK	u8/Q	CK1	CLK1	n	1/2	-	-	-

```
tempus > set_interactive_constraint_modes [all_constraint_modes -active]
tempus > reset_generated_clock [get_clocks GCLK]
tempus > report_clocks
```

The following report is displayed:

Clock Descriptions

Attributes						
Clock Name	Source	Period	Lead	Trail	Generated	Propagated
CK1	CLK1	20.000	0.000	10.000	n	n
CK2	CLK1	10.000	0.000	5.000	n	n
GCLK	u8/Q	20.000	0.000	10.000	y	y

Generated-Clock Descriptions

Name	Generated	Master	Master-clock	Invert	Freq.	Duty-Cycle	Edges	Edge-Shift
	Source (pin)	Source (pin)			Multiplier			

Related Commands

- [create_generated_clock](#)
- [report_clocks](#)
- [set_interactive_constraint_modes](#)

reset_ideal_latency

```
reset_ideal_latency  
[-help]  
[-min]  
[-max]  
[-rise]  
[-fall]  
object_list
```

Resets the ideal latency constraints, specified using the `set_ideal_latency` command, on instance pins or ports.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-fall</code>	Resets the ideal latency for the falling edge.
<code>-max</code>	Resets the ideal latency for the maximum corner.
<code>-min</code>	Resets the ideal latency for the minimum corner.
<i>object_list</i>	Resets a list of objects - instance pins, ports, or nets.
<code>-rise</code>	Resets the ideal latency for the rising edge.

Related Commands

- `set_ideal_latency`

reset_ideal_network

```
reset_ideal_network  
[-help]  
object_list
```

Resets the ideal network constraints, specified using the `set_ideal_network` command, on pins, ports, or nets.

Note: The `reset_ideal_network` command applies to all the active views and is not view/mode-based.

Parameters

<code>-help</code>	Prints out the command usage.
<code>object_list</code>	Resets a list of objects - instance pins, ports, or nets.

Related Commands

- `set_ideal_network`

reset_ideal_transition

```
reset_ideal_transition  
[-help]  
[-min]  
[-max]  
[-rise]  
[-fall]  
object_list
```

Resets the ideal transition constraint, specified using the `set_ideal_transition` command, on instance pins or ports.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-fall</code>	Resets the ideal transition for the falling edge.
<code>-max</code>	Resets the ideal transition for the maximum corner.
<code>-min</code>	Resets the ideal transition for the minimum corner.
<i>object_list</i>	Resets a list of objects - instance pins, ports, or nets.
<code>-rise</code>	Resets the ideal transition for the rising edge.

Related Commands

- `set_ideal_transition`

reset_input_delay

```
reset_input_delay  
[-help]  
[-clock clock_name]  
[-clock_fall]  
[-rise]  
[-fall]  
[-max]  
[-min]  
[-level_sensitive]  
pin_or_port_list
```

Resets previously specified input delay assertions.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-clock</code> <i>clock_name</i>	Specifies the name of the clock. <i>Default:</i> The asynchronous (@) clock. If you specify multiple clocks, an error occurs and the constraint is ignored. When you specify the <code>-clock</code> parameter, the constraints specified for a pin with matching <code>-clock</code> parameter are reset. If you do not specify the <code>-clock</code> parameter, all the constraints applied on the specified pin, with or without the <code>-clock</code> parameter specification, are reset.
<code>-clock_fall</code>	Removes the assertions for the delay that is relative to the falling edge of the clock. You must specify the <code>-clock</code> parameter with this parameter. <i>Default:</i> Removes the delay assertions relative to the falling edge.
<code>-fall</code>	Resets the input delay for the falling edge at the input port. If both the <code>-rise</code> and <code>-fall</code> options are not specified, then the input delay is reset on both the edges.
<code>-level_sensitive</code>	Resets input delay that is launched by a positive level sensitive latch. When you specify the <code>-level_sensitive</code> parameter, the command resets all the <code>set_input_delay</code> constraints specified with the <code>-level_sensitive</code> parameter. If <code>-level_sensitive</code> parameter is not specified, the constraints specified for the same pin, with or without the <code>-level_sensitive</code> parameter, are reset. Note: You must specify the <code>-clock</code> parameter with this parameter.
<code>-max</code>	Removes the delay assertions that refer to the setup analysis. If you do not specify the <code>-max</code> parameter, the input delay assertions for both setup and hold analysis are removed.
<code>-min</code>	Removes the delay assertions that refer to the hold analysis. If you do not specify the <code>-min</code> parameter, the input delay assertions for both setup and hold analysis are removed.
<i>pin_or_port_list</i>	Specifies a single or multiple pins or ports for which the input delay is reset. To specify multiple pins, enclose the list with curly braces ({}) and separate the pin information with white space.
<code>-rise</code>	Resets the input delay for the rising edge at the input port. If both <code>-rise</code> and <code>-fall</code> options are omitted, the input delay is reset on both the edges.

Related Commands

- `set_input_delay`

reset_load

```
reset_load  
[-help]  
object_list
```

Resets the existing `set_load` assertions specified on nets or ports.

Parameters

<code>-help</code>	Prints out the command usage.
<code>object_list</code>	Specifies a list of nets or ports in the top cell for which assertions are reset.

Related Commands

- `set_load`

reset_max_capacitance

reset_max_capacitance

[-help]

object_list

Resets maximum capacitance limit specified (using the the `set_max_capacitance` command) on the objects in the object list. The valid objects in the object list are primary ports, clocks, and current design.

Parameters

<code>-help</code>	Prints out the command usage.
<i>object_list</i>	Specifies the list of ports, modules, and/or clock waveforms for which assertions are reset.

Related Commands

- `set_max_capacitance`

reset_max_fanout

reset_max_fanout

[-help]

object_list

Resets maximum fanout limit specified (using the `set_max_fanout` command) on the objects in the object list. The valid objects in the object list are input and bidirectional ports and current design.

Parameters

<code>-help</code>	Prints out the command usage.
<i>object_list</i>	Specifies the list of ports or modules for which assertions are reset.

Related Commands

- `set_max_fanout`

reset_max_time_borrow

```
reset_max_time_borrow  
[-help]  
object_list
```

Resets the specified maximum time borrow limit of the specified objects.

Parameters

<code>-help</code>	Prints out the command usage.
<code>object_list</code>	Specifies the list of pins, instances, or clocks for which the time borrow limit is reset.

Related Commands

- `set_max_time_borrow`

reset_max_transition

```
reset_max_transition  
[-help]  
object_list
```

Resets the maximum slew (transition) limit specified using the `set_max_transition` command (from the SDC or the Tempus command line) on the objects in the given object list. The valid objects in the object list are primary ports, clocks, and current design.

The maximum transition limit that is set by the user overrides the dotlib value so that sharper slews are implemented on the timing sensitive nets, such as clocks. A violation due to the `set_max_transition` setting does not impact the timing accuracy.

Note: For command-line usage of this command, you need to use `set_interactive_constraint_modes`, for example,

```
set_interactive_constraint_modes [all_constraint_modes -active]
```

Parameters

<code>-help</code>	Prints out the command usage.
<code>object_list</code>	Specifies the list (or collection) of ports, modules, and/or clock waveforms for which assertions are to be reset. For example, <code>reset_max_transition [get_ports *]</code>

Related Commands

- `set_max_transition`

reset_min_capacitance

reset_min_capacitance

[-help]

object_list

Resets minimum capacitance limit specified (using the the `set_min_capacitance` command) on objects in the object list. The valid objects in the object list are primary ports, clocks, and the current design.

Parameters

<code>-help</code>	Prints out the command usage.
<i>object_list</i>	Specifies the list of ports, modules, and/or clock waveforms for which assertions are reset.

Related Commands

- `set_min_capacitance`

reset_min_fanout

reset_min_fanout

[-help]

object_list

Resets minimum fanout limit specified (using the `set_min_fanout` command) on the objects in the object list. The valid objects in the object list are input and bidirectional ports and the current design.

Parameters

<code>-help</code>	Prints out the command usage.
<i>object_list</i>	Specifies the list of ports or modules for which assertions are reset.

Related Commands

- `set_min_fanout`

reset_min_pulse_width

```
reset_min_pulse_width
[-help]
object_list
[-high]
[-low]
[-waveform_aware_type {absolute | source_width_ratio | all}]
```

Resets minimum pulse width constraint on the specified input ports of the top cell and the specified modules.

Parameters

-help	Prints out the command usage.
-low	Resets the minimum pulse width checks that are applied to low signal levels only.
-high	Resets the minimum pulse width checks that are applied to high signal levels only.
object_list	Specifies a list of pins or clock waveforms for which the minimum pulse width is to be reset. If the list is not specified, the minimum pulse width check applies to all the pins in the clock tree of the current design. If a clock waveform is specified, all the pins driven by the clock are affected.
- waveform_aware_type {absolute source_width_ratio all}	Resets the regular and waveform-aware pulse width assertions.

Related Commands

- [set_min_pulse_width](#)

reset_min_transition

```
reset_min_transition
```

```
[-help]
```

```
object_list
```

Resets minimum slew (transition) limit specified (using the `set_min_transition` command) on the objects in the object list. The valid objects in the object list are primary ports, clocks, and current design.

Parameters

<code>-help</code>	Prints out the command usage.
<code>object_list</code>	Specifies the list of ports, modules, and/or clock waveforms for which assertions are reset.

Related Commands

- `set_min_transition`

reset_mode

```
reset_mode
[-help]
[-type {cell}]
list_of_modes
object_list
```

Resets the Liberty timing library modes for a specified instance.

Library modes are set using the `set_mode` command. By default, all library modes in the timing library are active. If you specify one or more library modes using the `set_mode` command, only the timing arcs consistent with the specified mode(s) are active for the instance. All other library modes become inactive. When you reset the library mode for the instance, all library modes become active again.

Parameters

-help	Prints out the command usage.
object_list	Specifies the instances to which to reset the library modes.
list_of_modes	Specifies the name of the library modes to reset.
-type {cell}	Resets the library mode at the instance level. Note: Setting the library mode at the instance level is the only form of <code>set_mode</code> that is currently supported. This parameter exists to support constraints coming from external sources.

Examples

- The following command specifies that only the timing arcs consistent with the `test_A` library mode are active for instances of `b1`:

The `test_A` library mode is made inactive.

```
tempus > set_mode test_mode [get_cells BUF]
```

The following command reports the status of all library modes associated with BUF 'b1'. It shows that `test_A` is active and `test_B` is inactive.

```
tempus > report_mode b1
```

```
-----  
Mode Group test_mode of b1  
-----
```

Mode Name	Status	Condition
--------------	--------	-----------

test_B	Inactive	!(Y)
--------	----------	------

test_A	ACTIVE	Y
--------	--------	---

```
-----
```

The following command resets the library mode for instances of `b1` :

```
tempus > reset_mode test_A b1
```

If you generate a report of the library mode status for `b1`, it shows that both modes are now active:

```
tempus > report_mode b1
```

```
-----  
Mode Group test_mode of b1  
-----
```

Mode Name	Status	Condition
--------------	--------	-----------

test_B	ACTIVE	!(Y)
--------	--------	------

test_A	ACTIVE	Y
--------	--------	---

```
-----
```

Related Commands

- [set_mode](#)
- [report_mode](#)

reset_output_delay

```
reset_output_delay  
[-help]  
port_or_pin_list  
[-clock clock_name]  
[-clock_fall]  
[-fall]  
[-level_sensitive]  
[-max]  
[-min]  
[-rise]
```

Resets previously specified output delay assertions.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-clock</code> <i>clock_name</i>	Specifies the name of the clock. <i>Default:</i> The asynchronous (@) clock. If you specify multiple clocks, an error occurs and the constraint is ignored. When you specify the <code>-clock</code> parameter, the constraints specified for a pin with matching <code>-clock</code> parameter are reset. If you do not specify the <code>-clock</code> parameter, all the constraints applied on the specified pin, with or without the <code>-clock</code> parameter specification, are reset.
<code>-clock_fall</code>	Removes the assertions for the delay that is relative to the falling edge of the clock. You must specify the <code>-clock</code> parameter with this parameter. <i>Default:</i> Removes the delay assertions relative to the rising edge.
<code>-fall</code>	Resets the output delay for the falling edge at the output port. If both <code>-rise</code> and <code>-fall</code> options are omitted, the output delay is reset on both the edges.
<code>-level_sensitive</code>	Resets output delay that is launched by a positive level sensitive latch. When you specify the <code>-level_sensitive</code> parameter, the command resets all the <code>set_output_delay</code> constraints specified with the <code>-level_sensitive</code> parameter. If <code>-level_sensitive</code> parameter is not specified, the constraints specified for the same pin, with or without the <code>-level_sensitive</code> parameter, are reset. Note: You must specify the <code>-clock</code> parameter with this parameter.
<code>-max</code>	Removes the delay assertions that refer to the setup analysis. If you do not specify the <code>-max</code> parameter, the output delay assertions for both setup and hold analysis are removed.
<code>-min</code>	Removes the delay assertions that refer to the hold analysis. If you do not specify the <code>-min</code> parameter, the output delay assertions for both setup and hold analysis are removed.
<i>pin_or_port_list</i>	Specifies a single or multiple pins or ports for which the output delay is reset. To specify multiple pins, enclose the list with curly braces ({}) and separate the pin information with white space.
<code>-rise</code>	Resets the output delay for the rising edge at the output port. If both <code>-rise</code> and <code>-fall</code> options are omitted, the output delay is reset on both the edges.

Examples

- The following example sets the output delays for the output port Z. The output signal is required at Z at 2 units before the rising edge of clock Clk1, and at 2.5 units before falling edge of clock Clk2:

```
tempus > set_output_delay 2 Z -clock Clk1
tempus > set_output_delay 2.5 Z -clock Clk2 -clock_fall -add_delay
```

- The following command will reset the output delay for the rising edge of clock Clk2:

```
tempus > reset_output_delay Z -clock Clk2
```

```
report_timing -to_rise Z -clock_to Clk2
#####
# Generated by: Cadence Tempus 18.11-s094_1
# OS: Linux x86_64 (Host ID noi-rakeshp)
# Generated on: Thu Apr 18 09:50:52 2019
# Design: top
# Command: report_timing -to_rise Z -clock_to Clk2
#####
No constrained timing paths with given description found.
Paths may be unconstrained (try '-unconstrained' option) or may not exist.
```

```
report_timing -to_fall Z -clock_to Clk2
#####
Path 1: VIOLATED Late External Delay Assertion
Endpoint: Z (v) checked with trailing edge of 'Clk2'
Beginpoint: dff3/Q (v) triggered by leading edge of 'Clk'
Path Groups: {vclk2}
Other End Arrival Time 2.000
- External Delay 2.500
+ Phase Shift 0.000
= Required Time -0.500
- Arrival Time 0.165
= Slack Time -0.665
Clock Rise Edge 0.000
+ Clock Network Latency (Prop) 0.052
= Beginpoint Arrival Time 0.052
-----
Instance Arc Cell Delay Arrival Required
Time Time
-----
```

```
dff3 CK ^ - - 0.052 -0.613
dff3 CK ^ -> Q v DFFHQX1 0.113 0.165 -0.500
- Z v - 0.000 0.165 -0.500
-----
```

- The following command resets the output delay for clock Clk2 with respect to the falling edge:
tempus > reset_output_delay Z -clock Clk2 -clock_fall
- The following command will reset all the output delays with respect to all the edges for both the clocks - Clk1 and Clk2:

```
tempus > reset_output_delay Z
```

Related Commands

- `set_output_delay`

reset_path_exception

```
reset_path_exception  
[-help]  
[-exact]  
[-type exception_type_list]  
[-rise]  
[-fall]  
[-from from_list | -rise_from from_list | -fall_from from_list]  
[-ignore_clock_latency]  
[-through through_list | -rise_through through_list | -fall_through through_list]  
[-to to_list | -rise_to to_list | -fall_to to_list]  
[-setup | -hold]  
[-all]
```

Removes any previously set path exceptions (or path exception of the specified type) for the given paths. To remove all path exceptions, use the `-all` option with the `reset_path_exception` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-all</code>	Removes all the path exceptions. This parameter cannot be used with the <code>-from</code> , <code>-through</code> and <code>-to</code> .
<code>-exact</code>	Forces exact matching of the constraints. When this parameter is specified, the reset settings (<code>-from/-through/-to</code>) match with the original constrained path settings.
<code>-fall</code>	Applies the assertion at the falling edge of the signal on the path endpoints.
<code>{-from -rise_from -fall_from} from_list</code>	Specifies pins at the start of the path exceptions that are to be reset.
<code>-ignore_clock_latency</code>	Resets the exceptions specified with the <code>set_max_delay/set_min_delay -ignore_clock_latency</code> parameter.
<code>-rise</code>	Applies the assertion at the rising edge of the signal on the path endpoints.
<code>{-through -rise_through -fall_through} through_list</code>	Specifies the pins that the path to be reset goes through.
<code>{-to -rise_to -fall_to} to_list</code>	Specifies pins at the end of the path exceptions that are to be reset.
<code>-setup -hold</code>	Resets only the setup (or hold) paths. The <code>-setup/-hold</code> parameters should be used to reset maximum or minimum delay that is set using the <code>set_max_delay</code> and <code>set_min_delay</code> commands. <i>Default:</i> Both <code>-setup</code> and <code>-hold</code> are reset.
<code>-type exception_type_list</code>	Removes only path exceptions of the specified type. Valid types are: <code>false_path</code> , <code>multicycle</code> , and <code>path_delay</code> . <i>Default:</i> All path exception assertions are reset.

Examples

- In the example below the rising path from R1 to R2 has three different path exceptions asserted on it - they are listed in priority from lowest to highest. As shown in the reports below, the false path is

the highest precedence and so no path is reported.

```
tempus > set_multicycle_path -from R1/CK -rise_to R2/D 2
tempus > set_max_delay 3 -from R1/CK -rise_to R2/D
tempus > set_false_path -from R1/CK -rise_to R2/D
tempus > report_timing -from R1/CK -rise_to R2/D
```

The following information is displayed:

No constrained timing paths with given description found.
Paths may be unconstrained (try '-unconstrained' option) or may not exist.

```
tempus > report_path_exceptions
```

The following report is displayed:

```
-----
From      To      Early      Late
-----
R1/CK     ^ R2/D     false      false
-----
```

To report ignored path exceptions, you can use the following command:

```
tempus > report_path_exception -ignored
```

```
-----
From      To      Early      Late
-----
R1/CK     ^ R2/D                      delay 3 ignored
R1/CK     ^ R2/D      add 0 ignored
R1/CK     ^ R2/D                      add 1 ignored
-----
```

- In this example, the false path exception is removed from the path. This causes the max path delay assertion to become active - as it is the next highest precedence.

```
tempus > reset_path_exception -from R1/CK -rise_to R2/D -type false_path
tempus > report_path_exceptions
```

The following report is displayed:

```
-----
From      To      Early      Late
-----
R1/CK     ^ R2/D                      delay 3
R1/CK     ^ R2/D      add 0
-----
```

```
tempus > report_timing -from R1/CK -rise_to R2/D
```


The following report is displayed:

```
Path 1: MET Setup Check with Pin R2/CK
Endpoint: R2/D (^) checked with leading edge of 'CLK_W_1'
Beginpoint: R1/Q (^) triggered by leading edge of 'CLK_W_1'
Path Groups: {CLK_W_1}
Other End Arrival Time 0.000
- Setup 0.097
+ Path Delay 3.000
= Required Time 2.903
- Arrival Time 0.243
= Slack Time 2.660
Clock Rise Edge 0.000
+ Clock Network Latency (Prop) 0.000
= Beginpoint Arrival Time 0.000
-----
```

Load	Slew	Delay	Arrival Time	Cell	Arc	Pin
0.026	0.003		0.000		CK ^	R1/CK ->
0.011	0.063	0.187	0.187	DFF	CK ^ -> Q ^	R1/Q
0.003	0.042	0.056	0.243	BUF	A ^ -> Y ^	B1/Y
0.003	0.042	0.000	0.243	DFF	D ^	R2/D ->

```
-----
```

- In this example, the `set_max_delay` exception is removed, leaving only the `set_multicycle_path` exception as active.

```
tempus > reset_path_exception -from R1/CK -rise_to R2/D -type path_delay
tempus > report_timing -from R1/CK -rise_to R2/D
```

The following report is displayed:

```
Path 1: MET Setup Check with Pin R2/CK
Endpoint: R2/D (^) checked with leading edge of 'CLK_W_1'
Beginpoint: R1/Q (^) triggered by leading edge of 'CLK_W_1'
Path Groups: {CLK_W_1}
Other End Arrival Time 0.000
- Setup 0.097
+ Phase Shift 20.000
+ Cycle Adjustment 20.000 <----- Multi-Cycle timing
= Required Time 39.903
- Arrival Time 0.243
= Slack Time 39.660
Clock Rise Edge 0.000
+ Clock Network Latency (Prop) 0.000
= Beginpoint Arrival Time 0.000
-----
```

Load	Slew	Delay	Arrival Time	Cell	Arc	Pin

0.026	0.003		0.000		CK ^	R1/CK ->
0.011	0.063	0.187	0.187	DFF	CK ^ -> Q ^	R1/Q
0.003	0.042	0.056	0.243	BUF	A ^ -> Y ^	B1/Y
0.003	0.042	0.000	0.243	DFF	D ^	R2/D ->

- In this example, the multi-cycle path assertion is removed, and the analysis reverts back to single-cycle timing

```
tempus > reset_path_exception -rise_to R2/D -type multicycle
tempus > report_path_exceptions
```

None reported

```
tempus > report_timing -from R1 -rise_to R2/D
```

The following report is displayed:

```
Path 1: MET Setup Check with Pin R2/CK
Endpoint: R2/D (^) checked with leading edge of 'CLK_W_1'
Beginpoint: R1/Q (^) triggered by leading edge of 'CLK_W_1'
Path Groups: {CLK_W_1}
Other End Arrival Time 0.000
- Setup 0.097
+ Phase Shift 20.000 <--- Single-cycle timing
= Required Time 19.903
- Arrival Time 0.243
= Slack Time 19.660
Clock Rise Edge 0.000
+ Clock Network Latency (Prop) 0.000
= Beginpoint Arrival Time 0.000
```

Load	Slew	Delay	Arrival Time	Cell	Arc	Pin

0.026	0.003	0.000			CK ^	R1/CK ->
0.011	0.063	0.187	0.187	DFF	CK ^ -> Q ^	R1/Q
0.003	0.042	0.056	0.243	BUF	A ^ -> Y ^	B1/Y
0.003	0.042	0.000	0.243	DFF	D ^	R2/D ->

- The example below shows how the `-hold` and `-setup` options can be used to control the effect of issuing `reset_path_exception`. The example will issue multi-cycle path constraints against both setup and hold modes, and then reset only the setup side.

```
tempus > set_multicycle_path 3 -setup -rise_from [get_clocks PH1] -rise_through U2/A
tempus > set_multicycle_path 2 -hold -rise_from [get_clocks PH1] -rise_through U2/A
tempus > report_timing -late -from R1 -rise_to R2/D
```

The following report is displayed:

```
Path 1: MET Setup Check with Pin R2/CK
Endpoint: R2/D (^) checked with leading edge of 'CLK_W_1'
Beginpoint: R1/Q (^) triggered by leading edge of 'CLK_W_1'
Path Groups: {CLK_W_1}
Other End Arrival Time 0.000
- Setup 0.097
+ Phase Shift 20.000
+ Cycle Adjustment 40.000 <-- 3 cycles
= Required Time 59.903
- Arrival Time 0.243
= Slack Time 59.660
Clock Rise Edge 0.000
+ Clock Network Latency (Prop) 0.000
= Beginpoint Arrival Time 0.000
```

Load	Slew	Delay	Arrival Time	Cell	Arc	Pin
0.026	0.003	0.000			CK ^	R1/CK ->
0.011	0.063	0.187	0.187	DFF	CK ^ -> Q ^	R1/Q
0.003	0.042	0.056	0.243	BUF	A ^ -> Y ^	B1/Y
0.003	0.042	0.000	0.243	DFF	D ^	R2/D ->

```
tempus > report_timing -early -from R1 -rise_to R2/D
```

The following report is displayed:

```
Path 1: MET Hold Check with Pin R2/CK
Endpoint: R2/D (^) checked with leading edge of 'CLK_W_1'
Beginpoint: R1/Q (^) triggered by leading edge of 'CLK_W_1'
Path Groups: {CLK_W_1}
Other End Arrival Time 0.000
+ Hold -0.046
+ Phase Shift 0.000
- Cycle Adjustment 0.000 <- Adjusted 2 cycles
= Required Time -0.046
Arrival Time 0.243
Slack Time 0.289
Clock Rise Edge 0.000
```

```
+ Clock Network Latency (Prop) 0.000
= Beginpoint Arrival Time 0.000
```

Load	Slew	Delay	Arrival Time	Cell	Arc	Pin
0.026	0.003	0.000			CK ^	R1/CK ->
0.011	0.063	0.187	0.187	DFF	CK ^ -> Q ^	R1/Q
0.003	0.042	0.056	0.243	BUF	A ^ -> Y ^	B1/Y
0.003	0.042	0.000	0.243	DFF	D ^	R2/D ->

```
tempus > reset_path_exception -setup -rise_from [get_clocks CLK_W_1] -rise_through B1/A
tempus > report_timing -late -from R1 -rise_to R2/D
```

The following report is displayed:

```
Path 1: MET Setup Check with Pin R2/CK
Endpoint: R2/D (^) checked with leading edge of 'CLK_W_1'
Beginpoint: R1/Q (^) triggered by leading edge of 'CLK_W_1'
Path Groups: {CLK_W_1}
Other End Arrival Time 0.000
- Setup 0.097
+ Phase Shift 20.000 <-- Reset to single-cycle timing
= Required Time 19.903
- Arrival Time 0.243
= Slack Time 19.660
Clock Rise Edge 0.000
+ Clock Network Latency (Prop) 0.000
= Beginpoint Arrival Time 0.000
```

Load	Slew	Delay	Arrival Time	Cell	Arc	Pin
0.026	0.003	0.000			CK ^	R1/CK ->
0.011	0.063	0.187	0.187	DFF	CK ^ -> Q ^	R1/Q
0.003	0.042	0.056	0.243	BUF	A ^ -> Y ^	B1/Y
0.003	0.042	0.000	0.243	DFF	D ^	R2/D ->

```
tempus > report_timing -early -from R1 -rise_to R2/D
```

The following report is displayed:

```
Path 1: MET Hold Check with Pin R2/CK
Endpoint: R2/D (^) checked with leading edge of 'CLK_W_1'
```

```

Beginpoint: R1/Q (^) triggered by leading edge of 'CLK_W_1'
Path Groups: {CLK_W_1}
Other End Arrival Time 0.000
+ Hold -0.046
+ Phase Shift 0.000
- Cycle Adjustment 40.000 <-- Still 2 cycle adjustment
= Required Time -40.046 (from new setup edges)
Arrival Time 0.243
Slack Time 40.289
Clock Rise Edge 0.000
+ Clock Network Latency (Prop) 0.000
= Beginpoint Arrival Time 0.000

```

Load	Slew	Delay	Arrival Time	Cell	Arc	Pin
0.026	0.003	0.000			CK ^	R1/CK ->
0.011	0.063	0.187	0.187	DFF	CK ^ -> Q ^	R1/Q
0.003	0.042	0.056	0.243	BUF	A ^ -> Y ^	B1/Y
0.003	0.042	0.000	0.243	DFF	D ^	R2/D ->

- Consider the following `set_false_path` constraint:

```
tempus > set_false_path -from {u1/Y u2/Y} -to {u3/Y}
```

When you specify `reset_path_exception -from {u1/Y}` command without the `-exact` parameter, the command outputs the following:

```
tempus > set_false_path -from {u2/Y} -to {u3/Y}
```

The command `reset_path_exception -from {u1/Y} -exact` outputs the following:

```
tempus > set_false_path -from {u1/Y u2/Y} -to {u3/Y}
```

In this case (when `-exact` parameter is used), the path exception is not reset because differential matching is not allowed.

Related Topics

- [Path Exception Priorities](#)

Related Commands

- [set_false_path](#)

reset_path_group

```
reset_path_group  
[-help]  
[-name group_name]  
[-all]
```

Removes the specified path groups or completely removes all the groups.

Parameters

-help	Prints out the command usage.
-all	Removes all path groups.
-name <i>group_name</i>	Removes the named group completely.

Example

- The following command removes the group named GRPA:
`tempus > reset_path_group -name GRPA`

Related Commands

- `group_path`
- `get_path_groups`
- `report_path_groups`

reset_pll_timing

```
reset_pll_timing
[-help]
inst_name
-output_clock
clock name or ID
-output_pin output pin name
```

Resets the delay offset settings applied to a PLL (phase locked loop) output clock.

Parameters

-help	Prints out the command usage.
inst_name	Specifies the name of the instance on which reference pins, feedback pins, or output pins will be searched.
-output_clock <clock name or ID>	Specifies the name of the PLL output clock.
-output_pin <output pin name>	Specifies the name of the PLL output pin.

Related Commands

- [set_pll_timing](#)

reset_propagated_clock

```
reset_propagated_clock  
[-help]  
pin_clock_list
```

Removes the propagated clock assertion for the specified clock waveforms or pins.

The ideal or propagated status of clock endpoints in the fanout of the specified pin/clock list is determined by the remaining clock latency propagated clock assertions in the clock network.

When new analysis views are used, the new corners do not have the source latency updated as per the clock propagation (*update_io_latency*) settings, so a reset and re-calculation is required.

Note: You must specify *set_interactive_constraint_modes* constraint prior to running the *reset_propagated_clock* command.

Parameters

<i>-help</i>	Prints out the command usage.
<i>pin_clock_list</i>	Specifies a list of pins or clocks for which you want to reset the clock propagation mode to ideal mode. The <i>pin_clock_list</i> parameter can be a collection, however, no hierarchical pins are allowed in the list.

Examples

- The following command sets the CLK1 clock waveform to *propagated* mode:
tempus > set_propagated_clock CLK1
- The following command resets the CLK1 clock waveform back to *ideal* mode:
tempus > reset_propagated_clock CLK1
- The following commands set the CLK1 clock to *propagated* mode and then reports them:
tempus > set_propagated_clock [get_clocks CLK1]
tempus > report_clocks

The following report is displayed:

```
-----  
                                Clock Descriptions  
-----  
                                Attributes  
-----
```


Tempus Text Command Reference
Timing Constraint Commands

Clock Name	Source	Period	Lead	Trail	Generated	Propagated
CLK1	CLK1	20.000	0.000	10.000	n	y
CLK2	CLK2	25.000	0.000	12.500	n	y
CLK3	CLK3	10.000	0.000	6.000	n	y
GCLK3	u8/Q	20.000	0.000	10.000	y	y

Generated-Clock Descriptions

Name	Generated	Master	Master-clock	Invert	Freq.	Duty-Cycle	Edges
Edge-Shift	Source (pin)	Source (pin)			Multiplier		
GCLK3	u8/Q	CLK3	CLK3	n	1/2	-	- -

- The following commands reset the CLK1 clock to ideal mode and then reports them:

```
tempus > reset_propagated_clock [get_clocks CLK1]
```

```
tempus > report_clocks
```

The following report is displayed:

Clock Descriptions

Attributes						
Clock Name	Source	Period	Lead	Trail	Generated	Propagated
CLK1	CLK1	20.000	0.000	10.000	n	n
CLK2	CLK2	25.000	0.000	12.500	n	y
CLK3	CLK3	10.000	0.000	6.000	n	y
GCLK3	u8/Q	20.000	0.000	10.000	y	y

Generated-Clock Descriptions

Name	Generated	Master	Master-clock	Invert	Freq.	Duty-Cycle	Edges
Edge-Shift	Source (pin)	Source (pin)			Multiplier		
GCLK3	u8/Q	CLK3	CLK3	n	1/2	-	-

Related Commands

- `create_clock`
- `get_propagated_clock`
- `report_clocks`
- `set_clock_latency`
- `set_propagated_clock`
- `update_io_latency`
- `set_interactive_constraint_modes`

reset_property

```
reset_property
[-help]
[-object_type {object_type}]
{object_list}
property_name
```

Resets user-defined properties on objects set using the `set_property` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>object_list</code>	Specifies a list of objects for which the property is to be reset.
<code>-object_type {object_type}</code>	Specifies the type of object on which the property is to be reset.
<code>property_name</code>	Specifies the name of the property for which the user-defined property is to be reset.

Examples

- The following command resets user-defined property `slk` from pins `C1/A`:
`tempus > reset_property [get_pins C1/A] slk`

Related Commands

- `set_property`
- `get_property`
- `report_property`

reset_pulse_clock_max_transition

```
reset_pulse_clock_max_transition  
[-help]  
object_list  
[-transitive_fanout]  
[-rise]  
[-fall]
```

Resets the maximum pulse clock transition constraint specified (using the `set_pulse_clock_max_transition` command) on the pulse generator cells, pulse generator library cells, clocks, and/or current design.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-fall</code>	Resets transition limit value for falling transition.
<code>object_list</code>	Specifies a list of objects - pulse generator cells, pulse generator library cells, clocks, or current design - for which the transition value is to be reset.
<code>-rise</code>	Resets transition limit value for rising transition.
<code>-transitive_fanout</code>	Specifies the transition limit value for transitive fanout.

Related Commands

- `set_pulse_clock_max_transition`

reset_pulse_clock_max_width

```
reset_pulse_clock_max_width  
[-help]  
object_list  
[-transitive_fanout]
```

Resets the maximum pulse clock width constraint specified (using the `set_pulse_clock_max_width` command) on the pulse generator cells, pulse generator library cells, clocks, and/or current design.

Parameters

<code>-help</code>	Prints out the command usage.
<code>object_list</code>	Specifies a list of objects - pulse generator cells, pulse generator library cells, clocks, or current design - for which the maximum pulse clock width is to be reset.
<code>[-transitive_fanout]</code>	Specifies the pulse width value for transitive fanout.

Related Commands

- `set_pulse_clock_max_width`

reset_pulse_clock_min_transition

```
reset_pulse_clock_min_transition  
[-help]  
object_list  
[-transitive_fanout]  
[-rise]  
[-fall]
```

Resets the minimum pulse clock transition constraint specified (using the `set_pulse_clock_min_transition` command) on the pulse generator cells, pulse generator library cells, clocks, and/or current design.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-fall</code>	Resets transition limit value for falling transition.
<code>object_list</code>	Specifies a list of objects - pulse generator cells, pulse generator library cells, clocks, or current design - for which the minimum pulse clock transition value is to be reset.
<code>-rise</code>	Resets transition limit value for rising transition.
<code>-transitive_fanout</code>	Specifies the transition limit value for transitive fanout.

Related Commands

- `set_pulse_clock_min_transition`

reset_pulse_clock_min_width

```
reset_pulse_clock_min_width  
[-help]  
object_list  
[-transitive_fanout]
```

Resets the minimum pulse clock width constraint specified (using the `set_pulse_clock_min_width` command) on the pulse generator cells, pulse generator library cells, clocks, and/or current design.

Parameters

<code>-help</code>	Prints out the command usage.
<i>object_list</i>	Specifies a list of objects - pulse generator cells, pulse generator library cells, clocks, or current design - for which the minimum pulse clock width is to be reset.
<code>-transitive_fanout</code>	Specifies the pulse width value for transitive fanout.

Related Commands

- `set_pulse_clock_min_width`

reset_resistance

```
reset_resistance  
[help]  
net_list
```

Resets the existing `set_resistance` assertions specified on nets.

Parameters

<code>-help</code>	Prints out the command usage.
<code>net_list</code>	Specifies a list of nets for which the wire resistance is reset.

Related Commands

- [set_resistance](#)

reset_sense

```
reset_sense  
[-help]  
[-all | [[-clocks <clock_list>] <pin_or_port_list>]]
```

Resets clock sense constraint(s) specified using the `set_sense` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-all</code>	Resets all the specified <code>clock_sense</code> constraints.
<code>-clocks</code> <i>clock_list</i>	Specifies the list of clocks for which the clock sense constraints are to be removed. The clocks must have previously been defined using the <code>set_sense -clocks</code> parameter.
<i>pin_or_port_list</i>	Specifies the list or collection of cell output pins or ports for which the clock sense constraints will be removed.

Examples

- The following command resets all the previously defined `clock_sense` constraints from a design:

```
tempus > reset_sense -all
```

Related Commands

- [set_sense](#)

set_active_clocks

```
set_active_clocks  
[-help]  
active_clock_list | all_clocks
```

Defines a list of active clocks in the design.

When the `set_active_clocks` command is specified, only the specified clocks are made active while others remain inactive. Any clock created after using the `set_active_clocks` command is active by default. The following commands are impacted:

- `report_clocks` command output displays a new column showing active or inactive clocks in the design.
- `get_clocks` command will report all clocks including inactive clocks.
- `get_property` command allows you to query on active clocks using the `is_clock_active` property.

Parameters

<code>-help</code>	Prints out the command usage.
<code>active_clock_list all_clocks</code>	Specifies a list of active clocks.

Examples

- The following command makes the specified clocks active while others remain inactive (only `Iclk1` is active, while `Iclk2` is inactive):

```
tempus > set_active_clocks Iclk1
```

Related Commands

- `report_clocks`
- `get_clocks`
- `get_property`

set_annotated_check

```
set_annotated_check
[-help]
[-clock clock_check ]
[-cond sdf_condition ]
check_value
{-setup | -hold | -recovery | -removal | -nochange_high | -nochange_low}
[-rise]
[-fall]
[-min]
[-max]
[-incremental]
-from from_pins
-to to_pins
```

Annotates the setup, hold, recovery, or removal timing check value between two or more pins of a cell in the current design.

Parameters

-help	Prints out the command usage.
<i>check_value</i>	Specifies the timing check value.
-clock <i>clock_check</i>	Specifies clock rising or falling. Default: Sets checks for both clock rise and fall. Use the <i>clock_check</i> argument to specify rise or fall.
-cond S <i>sdf_condition</i>	Specifies the condition only if the library has a condition attached to the specified timing check arc. The <i>sdf_expression</i> must match the syntax of the condition specified in the library.
-fall	Specifies the data fall transition for the timing check. Default: Sets both -rise and -fall.
-from <i>from_pins</i>	Specifies start points of the timing arcs for which checks are annotated. Use the <i>from_pins</i> argument to specify a list of leaf cell clock pins.
-incremental	Sets incremental checks.

<code>-max</code>	Specifies the maximum timing check for both data rise and data fall transitions. Use this option only if the design uses minimum and maximum (<code>min-max</code>) operating conditions.
<code>-min</code>	Specifies the minimum timing check for both data rise and data fall transitions. Use this option only if the design uses minimum and maximum (<code>min-max</code>) operating conditions.
<code>-rise</code>	Specifies the data rise transition for the timing check. Default: Sets both rise and fall.
<code>-setup -hold -recovery -removal -nochange_high -nochange_low</code>	Specifies the timing check type. Make sure there is a corresponding timing arc between the <i>from_pins</i> and the <i>to_pins</i> arguments.
<code>-to to_pins</code>	Specifies endpoints of the timing arcs for which checks are annotated. Use the <i>to_pins</i> argument to specify a list of leaf cell clock pins.

Examples

- The following commands annotates a `nochange` check between data low and clock low. The `nochange` margin before the clock is 1.3 and the `nochange` margin after the clock is 2.4:

```
tempus > set_annotated_check -nochange_low 1.3 -clock fall -fall -from {I1/CP} -to {I1/EN}
tempus > set_annotated_check -nochange_low 2.4 -clock fall -rise -from {I1/CP} -to {I1/EN}
```
- The following command annotates a 2.0 setup time between the CP clock pin and the A data pin of the `<instance_name>` instance:

```
tempus > set_annotated_check -setup 2.0 -from <instance_name>/CP -to <instance_name>/A
```

Related Commands

- [reset_annotated_check](#)
- [set_annotated_delay](#)

set_annotated_delay

```
set_annotated_delay
[-help]
delay_value
[-cond expression]
[-delta_only]
[-fall]
[-increment]
[-max]
[-min]
[-rise]
{-net | -cell }
{[-from from_pins] [-to to_pins]}
```

Annotates delay to timing arcs. If you specify the `set_annotated_delay` command without any parameters, the delay value specified is assumed to include both the base delay and any delta (cross-talk) delay.

Parameters

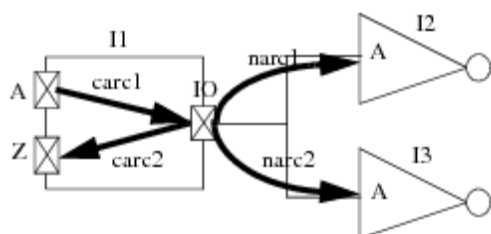
<code>-help</code>	Prints out the command usage.
<code>-cond expression</code>	Specifies the sdf condition of the selected timing arcs. <i>Default:</i> The delay applies to all the timing arcs selected using the <code>-from</code> and the <code>-to</code> options. The expression is ignored for net arcs. Note: You cannot specify the <code>-cond</code> parameter with <code>-net</code> . If specified, an error is displayed.
<code>delay_value</code>	Specifies the delay value to be annotated to the selected timing arcs.
<code>-delta_only</code>	Populates the incremental delay slot with the delay value, so that you can differentiate between the base net delay and a delay offset that is due to crosstalk. The software reports the incremental (or delta) delay value separately in the timing report using <code>report_timing -format incr_delay</code> , and maintains the separate value for CPPR calculations. If the base delay was previously calculated or annotated from the SDF, the <code>-delta_only</code> parameter adds the specified delay to the existing base delay. Note: This parameter only can be used with the <code>-net</code> parameter; you cannot specify it with the <code>-cell</code> parameter.

<code>-from</code> <code>from_pins</code>	Specifies timing arcs to which the delay applies. The selected timing arcs are those that have their source pins specified using this option and the sink pins specified using the <code>-to</code> option. If this option is not specified, only timing arcs whose sink pins are specified using the <code>-to</code> option are selected. Note: You cannot specify hierarchical pins. If specified, an error message is displayed.
<code>-increment</code>	Increments the specified delay value on top of the existing base delay value.
<code>-min</code> <code>-max</code>	Specifies the annotated delay for a particular operating corner. You can apply the <code>-min</code> parameter to the minimum operating corner and the <code>-max</code> parameter to the maximum operating corner.
<code>-net</code> <code>-cell</code>	Specifies whether the delay should be annotated to cell arcs or net arcs.
<code>-rise/-fall</code>	Specifies the transition to which the delay applies. For cell arc, this is the output transition. <i>Default:</i> The delay applies to both transitions.
<code>-to</code> <code>to_pins</code>	Specifies timing arcs to which the delay applies. The selected timing arcs are those that have their sink pins specified using this option and the source pins specified using the <code>-from</code> option. If this option is not specified, only timing arcs whose source pins are specified using the <code>-from</code> option are selected. Note: You cannot specify hierarchical pins. If specified, an error message is displayed.

Examples

- The following example shows the usage of `-cell` and `-net` options:

Figure: How to Use the -cell and -net Options



- Use the `-cell` option to select the cell arc `carc2` only:

```
tempus > set_annotated_delay 3.0 -cell -from {I1/IO}
```

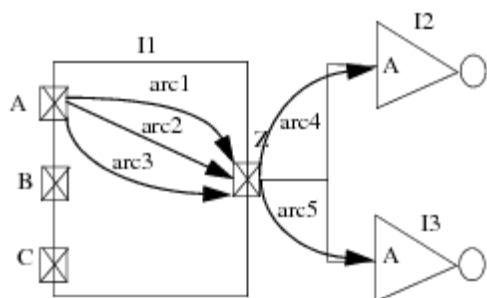
- Use the `-net arc` option to select net arcs `arc1` and `arc2` only:

```
tempus > set_annotated_delay 3.0 -net -from {I1/IO}
```

- This TLF description is used for the following examples:

```
CELL (NAND3 . . . // arc 1 Path( A => Z 10 01 DELAY(ioDelayRiseModel0)
SLEW(SlopeRiseModel0) ) Path( A => Z 01 10 DELAY(ioDelayFallModel0)
SLEW(SlopeFallModel0) ) // arc2 Path( A => Z 10 01 COND(B & C) SDF_COND(B & C)
DELAY(ioDelayRiseModel1) SLEW(SlopeRiseModel1) ) Path( A => Z 01 10 COND(B & C)
SDF_COND(B & C) DELAY(ioDelayFallModel1) SLEW(SlopeFallModel1) ) arc3 Path( A
=> Z 10 01 COND(~B | ~C) SDF_COND(!B | !C) DELAY(ioDelayRiseModel2) SLEW(SlopeRiseModel2) ) Path( A
=> Z 01 10 COND(~B | ~C) SDF_COND(!B | !C) DELAY(ioDelayFallModel2)
SLEW(SlopeFallModel2) ) . . . }
```

Figure: How to Use the -from and -to Options



- The following command annotates a delay of 3.0 to arcs `arc1`, `arc2`, and `arc3` for the rise and fall transition at the pin `I1/Z`, for the two corners `{min typ max}`:
tempus > set_annotated_delay 3.0 -from A -to Z -cell
- The following command annotates a delay of 2.5 to net arc `arc4` for both the rising and the falling transition and for the two corners `{min typ max}`:
tempus > set_annotated_delay 2.5 -from I1/Z -to I2/A -net
- The following command annotates a delay of 1.5 to `arc2` for the rising transition and the min corner:
tempus > set_annotated_delay 1.5 -from I1/A -to I1/Z -rise -min -cond "B & C" -cell
- The following commands annotate a delay of 3.0 to `arc1` and `arc2`, and a delay of 2.4 to the `arc3`:
tempus > set_annotated_delay 3.0 -from I1/A -to I1/Z -cell set_annotated_delay 2.4 -from I1/A -to I1/Z -cond {!B | !C} -cell
- The following commands first annotate a delay of 3.0 to `arc1`, `arc2` and `arc3`, then overwrite the

value set by the first `tempus > set_annotated_delay` command:

```
tempus > set_annotated_delay 2.4 -from I1/A -to I1/Z -cond {!B | !C} -cell
```

```
tempus > set_annotated_delay 3.0 -from I1/A -to I1/Z -cell
```

Related Commands

- `set_annotated_check`
- `reset_annotated_delay`

set_annotated_transition

```
set_annotated_transition
[help]
[-rise | -fall]
[-min | -max]
slew_value
port_or_pin_list
```

Sets the transition time to be annotated on specified ports or pins in the current design. The transition time that you specify using this command overrides the transition time calculated internally within the software. You can use this command to set the transition time of internal cell instance pins.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-min -max</code>	Specifies whether the transition value is set for minimum or maximum transition time. <i>Default:</i> Transition value is set for both transition times.
<code>port_or_pin_list</code>	Specifies a list of pins or ports to be annotated with the transition time that you specify using the <code>slew_value</code> parameter.
<code>-rise -fall</code>	Specifies whether the transition value is set for rising or falling transition time. <i>Default:</i> Transition value is set for both transitions.
<code>slew_value</code>	Specifies the transition time to be annotated to the selected pins or ports.

Examples

- The following command annotates a transition time value of 0.6 on port A:
tempus > set_annotated_transition 0.6 [get_ports A]
- The following command annotates a transition time value of 0.7 on pin I2/A:
tempus > set_annotated_transition 0.7 [get_pins I2/A]
- The following command annotates a transition value of 0.7 on pin I2/A for rising transition time:
tempus > set_annotated_transition -rise 0.7 [get_pins I2/A]
- The following command annotates a transition value of 0.7 on pin I2/A for rising transition time and minimum pvt corner:
tempus > set_annotated_transition -rise -min 0.7 [get_pins I2/A]

Related Commands

- `set_input_transition`
- `get_pins`
- `get_ports`
- `report_ports`
- `reset_annotated_transition`

set_aocv_interface_path_offset

```
set_aocv_interface_path_offset
[-help]
derateOffsetVal
[lib_cells]
[-cell]
[-delay_corner dc_corner_name]
[-net]
-power_domain power_domain_name_list
[-view view_name]
```

You can perform path-based inter power domain (IPD) timing analysis by specifying derate offset value for an entire power domain(s) and specific library cell(s) on paths crossing multiple power domains. This feature is supported for path-based analysis (PBA) mode only.

The `set_aocv_interface_path_offset` command allows you to specify the derate offset value for path segments crossing multiple power domains. The software adds the derate offset value to the AOCV derate value if the combined (AOCV + IPD offset) derate number causes pessimistic PBA timing slack value.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-cell</code>	Includes cells.
<code>-delay_corner</code> <code>dc_corner_name</code>	Specifies a list of delay corners.
<code>derateOffsetVal</code>	Specifies value of derate offset.
<code>lib_cells</code>	Specifies a list of library cells for a specific interface.
<code>-net</code>	Applies derate offset to net objects only. When this option is not specified, the derate offset value will be applied to both the cell and net type of objects.
<code>-power_domain</code> <code>power_domain_name_list</code>	Specifies a list of power domains.
<code>-view view_name</code>	Specifies a list of views.

Examples

- The following example specifies the offset derate value of -0.6 for the power domain {PD3 PD4}:

```
tempus > set_aocv_interface_path_offset -0.6 -power_domain {PD3 PD4} -delay_corner  
V1_PM1_dc
```

set_aocv_stage_weight

```
set_aocv_stage_weight
[-help]
stage_weight
port_list
[-early]
[-late]
[-input]
[-output]
```

While performing advanced on-chip variation (AOCV) analysis of a block in standalone mode, the command allows you to specify the stage weight seen at the port(s) when it is modeled as part of the top level design. The applied stage weights affect the internal weight count of the block for the IO paths. The stage weights can be applied only on ports.

You can view stage weights in the `report_timing` output under the `Stage Weighting` column.

The `set_aocv_stage_weight` command can be used in multi-mode multi-corner mode. The command can be used as part of SDC file.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-early</code>	Specifies weight for early paths.
<code>-input</code>	Specifies input weight for the ports.
<code>-late</code>	Specifies weight for late paths.
<code>-output</code>	Specifies output weight for port.
<code>port_list</code>	Specifies a list or collection of input, output, and bidi ports.
<code>stage_weight</code>	Specifies value of stage weight.

Examples

- The following example specifies the stage weight value as 25.5 for port `out0`:

```
tempus > set_aocv_stage_weight 25.5 out0
```

Related Commands

- `reset_aocv_stage_weight`
- `report_timing`

set_case_analysis

```
set_case_analysis
[-help]
{0 | 1 | zero | one | rising | falling | rise | fall | non_switching}
list_of_ports_or_pins
```

Sets constant or transitional values to a list of pins or ports for use by the timing engine. The specified constants or transitional values are valid only during timing analysis and do not alter the netlist. If there is a conflict between the specified `set_case_analysis` value and the logical value from the netlist, the value from the `set_case_analysis` is used in timing analysis.

Parameters

<code>-help</code>	Prints out the command usage.
<code>0 1 zero one</code>	Specifies a logic value of 0 or 1, which is propagated through combinational logic to disable or enable parts of logic.
<code>non_switching</code>	Marks the nets that are in a constant, non-transitioning state. This marking can be used when nets are considered static without the additional impact of logic state injection into the analysis, e.g., when modes are merged into a single constraint mode. The <code>report_cell_instance_timing</code> and <code>report_case_analysis</code> command reports show non-switching as constant values.
<code>rising falling rise fall</code>	Specifies a valid transition value to be considered by the timing engine. For example, if a rising value is specified at a certain pin or port, only the rising signal transition from the pin or port is considered for timing analysis and the opposite transition type (falling) is ignored.
<code>list_of_ports_or_pins</code>	Specifies a list of ports or pins on which to apply the constant or transitional value. Specify multiple pins by enclosing them with curly braces {} and separating the pin names with white space. The <code>list_of_ports_or_pins</code> argument can be a collection.

Examples

- The following command sets the value of pin i4/A to 1 for use by the timing engine:
tempus > set_case_analysis 1 i4/A
- The following example shows that the pins P1/P2/B and P1/P3/CI are considered only for a rising transition. The falling transition on these pins is ignored:

```
tempus > set_case_analysis rising {P1/P2/B P1/P3/CI}
```

Related Commands

- `check_timing -type constant_collision`
- `report_case_analysis`
- `report_timing`
- `reset_case_analysis`
- `set_disable_timing`
- `set_false_path`

set_clock_exclusivity

```
set_clock_exclusivity
[-help]
pin_name
[-exclude_opposite_polarity]
[-exclude_related_input_pins]
[-exclude_same_polarity]
[-group clock_list]
[-inputs <string>]
```

Controls filtering of mutually-exclusive clocks at strategic points in the design, e.g., the output of clock multiplexers.

Parameters

-help	Prints out the command usage.
-exclude_opposite_polarity	Isolates positive and negative phases of the clocks in the same group. By default, the positive and negative phases of the clocks in the same group that arrive at a strategic point will not be isolated from each other.
-exclude_related_input_pins	Allows comparison of clocks going through the same input pins of the specified gate.
-exclude_same_polarity	Allows comparison of clocks going through the same launch or capture paths.
-group clock_list	Defines a set of clocks that are exclusive to the clocks defined in all other groups.
-inputs <string>	Specifies a list or collection of input pins for which clock exclusivity is to be recognized.
pin_name	Specifies the name of the pin.

Examples

- The following command will cause downstream analysis from the mux to isolate interactions between the groups:

```
tempus > set_clock_exclusivity -group {CLK1 CLK2} -group {CLK3 CLK4} U1/Y
```

This assertion will cause downstream analysis from pin U1/Y to isolate interactions between the groups. The groups {CLK1 CLK2} and {CLK3 CLK4} are allowed, but all other combinations are considered logically exclusive.

Related Commands

- [reset_clock_gating_check](#)
- [create_clock](#)

set_clock_gating_check

```
set_clock_gating_check
[-help]
[object_list]
[-fall]
[-hold hold_value]
[-rise]
[-setup setup_value]
[-high | -low]
```

Specifies or overrides the default setup and hold values for clock gating checks.

Parameters

-help	Prints out the command usage.
-high -low	Specifies that the non-controlling value of the clock is high or low. Timing analysis performs the check on the non-controlling value of the clock. By default, the software determines the non-controlling value of the clock using information from the gate's logic. For example, for OR gates the non-controlling value is low. For complex gates, such as XOR and MUX, the clock is never controlling and it does not perform checks unless you specify either the <code>-high</code> option or the <code>-low</code> option. The specified value takes precedence over the default values. This option can only be used with pins or instances. Note: The clock pin value is controlling if it causes the gate output to be a constant (0 or 1).
-hold hold_value	Specifies the hold value. The default is 0.0.
object_list	Specifies a list of clocks, instances, or pins. If a clock is specified, the check applies to all the clock gating gates driven by this clock. If an instance is specified, the check applies to all input pins of the instance. If the <code>object_list</code> argument is not specified, the clock gating check is applied to the current design. Note: The SDC <code>set_clock_gating_check</code> asserts checks on ICG (Integrated Clock Gated) cell only if it is applied on either the cell itself or pins. It does not assert checks on ICG cells if <code>set_clock_gating_check</code> is applied on a design or clock object.
-rise -fall	Applies the specified setup (or hold) value to the rising (or falling) signals on the specified pins. If the clock is specified or the <code>object_list</code> variable is not specified, the value applies to rising or falling, setup or hold. If neither option is specified, the default is both rising and falling.

<code>-setup setup_value</code>	Specifies the setup value. The default is 0.0.
-------------------------------------	--

Note: The clock gating constraint is not propagated along the clock tree.

The setup check is performed with respect to the edge of the clock signal that changes the state of the clock pin from controlling to non-controlling. For example, for AND and NAND gates, the setup check is performed with respect to the rising edge of the clock input. For OR and NOR gates, the setup check is performed with respect to falling edge of the clock input.

The hold check is performed with respect to the edge of the clock signal that changes the state of the clock pin from non-controlling to controlling. For example, for AND and NAND gates, the hold check is performed with respect to the falling edge of the clock input. For OR and NOR gates, the hold check is performed with respect to rising edge of the clock input.

If the gate logic is unknown, setup and hold checks are performed with respect to both the rising edge and the falling edge of the clock signal and the timing analyzer uses the worst case values.

Priority Rules for Determining Setup or Hold Values

Timing analysis uses the following priority rules to determine the setup (or hold) value to use for a particular clock gating check:

1. Use the clock gating check assertion on the data pin if it exists
2. Otherwise, use the clock gating check assertion on the clock pin if it exists
3. Otherwise, use the clock gating check assertion on the clock waveform for the clock pin if it exists
4. Otherwise, use the global clock gating check assertion if it exists
5. If no clock gating check assertion is present, use the default setup (or hold) value of 0.0.

These rules are demonstrated in the example below:

Examples

Here are some examples.

- The following example shows the priority rules for multiple clock gating check constraints for the gated clock shown in figure below:

Gated Clock



Tcl Script for set_clock_gating_check

```
tempus > create_clock -name CLK1 -period 12 -waveform {0 6} clk
...
# clock-gating timing check assertions:
# On ideal clock waveform
set_clock_gating_check -setup 0.44 [get_clocks {CLK1}]
# On clock pin
set_clock_gating_check -hold 0.11 [get_pins {g1/A}]
# On data pin
set_clock_gating_check -setup 0.66 -rise [get_pins {g1/B}]
set_clock_gating_check -hold 0.22 -fall [get_pins {g1/B}]
```

Given the previous constraints, the following timing margins are used for slack calculation:

- SETUP g1/B rise: 0.66 -- from assertion on data pin g1/B ([rule 1](#))
- SETUP g1/B fall: 0.44 -- from assertion on ideal clock CLK1 ([rule 3](#))
- HOLD g1/B rise: 0.11 -- from assertion on clock pin g1/A ([rule 2](#))
- HOLD g1/B fall: 0.22 -- from assertion on data pin g1/B ([rule 1](#))

Related Commands

- [create_clock](#)
- [reset_clock_gating_check](#)

set_clock_groups

```
set_clock_groups
[-help]
[-comment <string>]
[-group <clock_list>]
[-name <name>]
[-logically_exclusive ] | [-physically_exclusive ] | [ [-asynchronous ] [-allow_paths ] ]
```

Defines clock groups with specified clock definitions. The software assigns all clocks that are defined after you use the `set_clock_groups` command to a default clock group. The software does not apply clock groupings on a master clock to the generated child clocks.

To report the defined clock groups, you can use the following command:

```
report_clocks -groups <name>
```

If there is an overlap in the clock group specification, the software uses the following precedence to resolve the behavior:

1. Physically exclusive
2. Asynchronous
3. Logically exclusive

The software drops any explicit false path assertions between clocks that you have set as either physically or logically exclusive or asynchronous using the `set_clock_groups` command.

An alternative to setting the clocks as physically or logically exclusive using the `set_clock_groups` command is to use the multi-mode mechanism. Generally in an implementation flow, sequential setting of active clocks cannot be used. For concurrent analysis of different modes, you can use an multi-mode based approach to drive IPO instead of using the `set_clock_groups` command. For more information on multi-mode multi-corner analysis, see *Performing Multi-Mode Multi-Corner Timing Analysis and Optimization* chapter in the *Tempus User Guide*.

Note: If the `set_clock_groups` constraint is added incrementally in SI mode, an explicit `update_timing -full` command is required (before `report_timing` or `report_constraint`) for SI full accuracy.

Parameters

-help	Prints out the command usage.
-------	-------------------------------

<code>-allow_paths</code>	Allows paths between asynchronous clock groups to be traced in timing analysis. These paths are blocked by default. When you use this option, the software still considers the paths to be asynchronous for signal integrity considerations. Signal integrity analysis does not support the <code>-allow_paths</code> parameter. Use the <code>-allow_paths</code> parameter with the <code>-asynchronous</code> parameter.
<code>-asynchronous</code>	Defines clock group with asynchronous clocks. Asynchronous clocks have no specified phase relationship. The software considers asynchronous nets to have infinite timing window overlap for signal integrity considerations and considers these as false paths for timing analysis. Use the <code>-allow_paths</code> parameter to trace these paths in timing analysis.
<code>-comment <i>string</i></code>	Allows you to insert comments. The comments are read but are ignored with a one-time warning message. They are not used during timing analysis reporting nor are they written out during saving of constraints.
<code>-group <i>clock_list</i></code>	Defines a set of clocks that are exclusive or asynchronous to the clocks defined in all other groups. If you define only one group, then all clocks not in the group implicitly belong to a default clock group. The software automatically adds any new clocks that you define after the <code>set_clock_groups</code> command to this default clock group.
<code>-logically_exclusive</code>	Defines clock group with clocks that do not have any functional paths between them, but coexist in the design at the same time. The software considers aggressor interactions between these clocks for signal integrity considerations, but considers these as false paths for timing analysis. Note: The <code>-logically_exclusive</code> parameter does not have any impact on SI analysis. For example, clocks selected via a clock mux select.
<code>-name <i>name</i></code>	Specifies the names of the clock groups to be added.

– physically_exclusive	Defines clock group with clocks that cannot exist in the design at the same time. The software does not consider aggressor interactions between clock groups that are physically exclusive for signal integrity considerations. From the perspective of SI analysis, all attackers that are physically exclusive with the victim net are ignored for both glitch and noise-on-delay analysis. Attacker nets that belong to physically exclusive groups cannot attack together. In this case, all sets of attackers belonging to all the physically exclusive groups are evaluated and the set of nets that has the highest impact on SI analysis is chosen. For timing analysis, the software considers these clock groups as false paths. Example: multiple clocks defined on the same source pin
-----------------------------------	--

Examples

Here are some examples.

- The following commands assert two different clocks on the same clock root port, and uses `set_clock_groups` to isolate them from each other:

```
tempus > create_clock -name SYSCLK -period 10 [get_ports sclk]
tempus > create_clock -name SCANCLK -period 100 -add [get_ports sclk]
tempus > set_clock_groups -name PHY_EX_SCLK -physically_exclusive -group {SYSCLK} -group {SCANCLK}
```
- The following command declares a single bus clock to be asynchronous to all of the other on board clocks in the design (which remain in the default clock group):

```
tempus > set_clock_groups -name ASYNCH_BUS -asynchronous -group {BUSCLK}
```

Assume a design with four clocks CLK1, CLK2, CLK3, and CLK4.
- The following command defines CLK1 physically exclusive to CLK2, CLK3 as well as CLK4.

```
tempus > set_clock_groups -name group1 -group CLK1 -physically_exclusive
```
- The following command defines CLK1 and CLK2 to be physically exclusive to CLK3 and CLK4.

```
tempus > set_clock_groups -name group2 -group {CLK1 CLK2} -physically_exclusive
```
- The following command defines CLK1 and CLK2 as physically exclusive to CLK3

```
tempus > set_clock_groups -name group3 -group {CLK1 CLK2} -group {CLK3} -physically_exclusive
```
- The following command defines the following:

```
tempus > set_clock_groups -name group4 -group {CLK1} -group {CLK2} -group {CLK3} -physically_exclusive
```


- CLK1 physically exclusive to CLK2 and CLK3.
- CLK2 physically exclusive to CLK1 and CLK3.
- CLK3 physically exclusive to CLK1 and CLK2.
- The following example shows that the clock group {clkA clkB} is asynchronous only to clock group {clkC clkD}:

```
tempus > set_clock_groups -asynchronous -group {clkA clkB} -group {clkC clkD}
```

If an extra clock clkE is defined in the design, then timing check will be allowed in between the first group to clkE and second group to clkE.

- In the following example since only one clock group is specified with the command `set_clock_groups` command, so the specified clock group is considered to be asynchronous to all the other clocks in the design:

```
tempus > set_clock_groups -asynchronous -group {clkA clkB}
```

```
tempus > set_clock_groups -asynchronous -group {clkC clkD}
```

Related Commands

- `create_clock`
- `reset_clock_groups`

set_clock_latency

```
set_clock_latency  
[-source [-early | -late]]  
[-rise]  
[-fall]  
[-jitter <jitter_val>]  
[-clock_gate]  
[-clock clock_list]  
[-min]  
[-max]  
latency  
pin_or_clock_list
```

Specifies ideal internal clock latency and external clock arrival delay.

Clock latency has two components:

- Source latency:
 - Specified using the `-source` parameter and is the delay from the external clock generator to the clock port of the design.
 - Applied when the clock is either ideal or propagated.
 - `set_clock_latency` set on a clock port has higher precedence than `set_clock_latency` set on a clock waveform.
 - Specified on clock waveforms with both the `set_input_delay` command and the `set_output_delay` command.
- Network latency (clock tree delay):
 - Default latency type and is the delay from the clock port to the register.
 - Applied only when the clock is ideal. When the clock is propagated, this delay is replaced by the actual clock tree delays.
 - Specified on a clock waveform or a clock port is ignored if the clock is propagated.
 - `set_clock_latency` set on a timing pin has higher precedence than `set_clock_latency` set on a waveform.
 - If there are multiple `set_clock_latency` and `set_propagated_clock` assertions between the clock waveform, clock root and clock endpoint, the closest assertion to the clock endpoint takes precedence.
 - Specified on internal circuit pins or hierarchical ports and used in the fanout cone of the network latency pin, when a clock is ideal.

Use the `set_clock_latency` command in two ways:

- To specify clock latency for a clock waveform:
 - Use the following syntax for source latency associated with a clock signal:
`set_clock_latency [-source] [-early | -late] [-min ][-max] latency pin_or_clock_list`
 - Use the following syntax for network latency associated with a clock signal:
`set_clock_latency [-min] [-max] latency pin_or_clock_list`

You cannot specify the `-late` and `-early` parameters for network latency.

Specifying latency on a port or pin overrides any previous latency specified on the port pin or waveform. The `pin_or_clock_list` argument is the list of the clock waveform names as specified by the `create_clock` command.

- To specify clock latency on clock pins:
 - Use the following syntax for source delay on a clock port:
`set_clock_latency -source [-early | -late] [-rise] [-fall]
insertion_delay pin_or_clock_list`
 - Use the following syntax for network delay on a clock pin:
`set_clock_latency [-min] [-max] [-rise] [-fall]
insertion_delay -pin pin_or_clock_list`

When a clock is ideal, the last specification along a path is used. This is useful when you want to intentionally skew the clock tree and model the physical hierarchical clock tree.

Parameters

<code>-clock clock_list</code>	Associates clock latency constraints, placed at the pin level, with a specific clock. This makes it possible to specify a different latency per clock. Note: You cannot use the <code>-clock</code> parameter with clock waveform objects.
<code>-clock_gate</code>	Allows setting the local clock latency value on pin or port objects. This setting overwrites the existing latency value of clock gating cells and is used when performing timing checks against the specified pins or ports.

<code>-early -late</code>	Specifies clock arrival time with respect to the early or the late time of the clock signal. In setup analysis, launch path is the late path and capture path is the early path. In hold analysis, launch path is the early path and the capture path is the late path. If neither parameter is specified, the default is both early and late. Use these parameters only with the <code>-source</code> parameter. You cannot use these parameters with network latency. Specify the clock latency at the given operating corner. Note: The <code>-min</code> and <code>-max</code> parameters refer to the operating corner and the <code>-early</code> and <code>-late</code> parameters refer to the clock arrival time at the source. Use the <code>-early</code> and <code>-late</code> parameters with the <code>-source</code> parameter to specify the four different clock latencies.
<code>-jitter</code> <code>jitter_val</code>	Specifies the portion of source clock latency value that can have cycle-to-cycle variation. When calculating CPPR adjustment this portion is removed from CPPR adjustment for different clock edge checks (e.g., setup), but it remains as part of cppr adjustment for same clock edge checks (e.g., hold).
<code>latency</code>	Specifies the value of the latency, in library units, as a floating point.
<code>pin_or_clock_list</code>	Specifies a list of clock waveform names or a list of pins to associate with the clock latency. The pin can be a port or an instance pin. Setting source latency on a pin is ignored.
<code>-max</code>	Specifies the clock latency for a particular operating corner. Apply the <code>-max</code> parameter to the maximum operating corner.
<code>-min</code>	Specifies the clock latency for a particular operating corner. Apply the <code>-min</code> parameter to the minimum operating corner.
<code>-rise/-fall</code>	Specifies the <code>-rise</code> or <code>-fall</code> parameter to the waveform at register clock pins for ideal latency and to the waveform at the source for source latency. Ideal network latency does not change polarity when crossing negative unate arcs.
<code>-source</code>	Specifies the source latency. If you do not use the <code>-source</code> parameter, the network latency will be applied. You can use the <code>timing_generated_clocks_inherit_ideal_latency</code> global variable to include the closest applied network latency for ideal inheritance. However, when the <code>-source</code> parameter is used, the software will completely override any inheritance.

Examples

- The following set of commands enable you to create a clock using the `create_clock` command and apply latency to the created clock using the `set_clock_latency` command:

```
tempus > create_clock CLK2 -period 20 [get_ports {clkA clkB clkD}]
tempus > set_clock_latency -source 2 {clkB clkD} # Define port settings
tempus > set_clock_latency -rise 5 {clkB clkD} # Define port settings
tempus > set_clock_latency 3 -early -late -source CLK1 # Define latency and waveform
settings
```

Related Commands

- `create_clock`
- `set_clock_transition`
- `set_clock_uncertainty`
- `set_input_delay`
- `set_output_delay`
- `set_propagated_clock`
- `reset_clock_latency`

set_clock_sense

```
set_clock_sense
[-help]
pin_or_port_list
[-clocks clock_list]
{-positive | -negative | -stop_propagation | -logical_stop_propagation | -stop type_list}
```

Defines the phases of a clock or clock source that can propagate beyond a port or instance pin. The clocks can be selectively or entirely blocked through the pin to all downstream logic. Additionally, data relating to clocks can be blocked through the pin.

Note: The `set_clock_sense` command is now obsolete. This command will continue to be supported in the current release, but will be removed in the next major release of the software.

Parameters

-help	Prints out the command usage.
-clocks <i>clock_list</i>	Specifies a list of clocks passing through a specified point to which the constraint applies. When this parameter is not specified, the filtering of clock phase will be applicable to all the clocks passing through that specified point. This parameter must be used with a specified pin name and one of the parameters : -positive, -negative, -stop_propagation, -logical_stop_propagation, or -stop.
-logical_stop_propagation	Stops the propagation of specified clocks passing through the specified pins.
-negative	Specifies that positive phase filtered clocks pass through pins in the <i>pin_list</i> .
<i>pin_or_port_list</i>	Specifies the list or collection of cell output pins. The unateness specified by the <code>set_clock_sense</code> command is applied to clocks propagating through the specified point.
-positive	Specifies that negative phase filtered clocks pass through pins in the <i>pin_list</i> .
-stop_propagation	Stops the propagation of specified clocks and clock source data paths passing through the specified pins.

`-stop type_list`

Applies clock sense to object types specified in the *type_list*. The type list options are - all, clock, data, clock_source_data, default_data.

These are explained below:

- **all:** Stops all phases from propagating. This is equivalent to specifying both the `clock` and `data` options.
- **clock:** Stops clock phases launched by clock(s) specified in the *clock_list* from propagating, and is equivalent to existing the - `logical_stop_propagation` parameter.
- **data:** Stops all data phases launched by clock(s) specified in *clock_list*, starting from the pins/ports specified in *pin_or_port_list*. The objects in the *pin_or_port_list* can be selected anywhere in the design - clock/data network.
- **clock_source_data:** Stops only clock source data phases from propagating. The combination of `clock_source_data` and `clock` is equivalent to the existing `-stop_propagation` option.
- **default_data:** Stops only default data phases (representing constrained paths) from propagation. This option cannot be specified along with the `-clock clock_list`.

The `-stop` parameter extends the capability of the `set_clocks_sense` command to selectively block data timing phases. This parameter provides a more generalized use model for selecting among all types of paths that should be filtered.

The `-stop` parameter is mutually exclusive with the `-positive`, `-negative`, `-stop_propagation`, and `-logical_stop_propagation` options.

Examples

- The following command stops the propagation of clock CLK1 through the pin CLKBUF1/Z:
`tempus > set_clock_sense -logical_stop_propagation -clocks CLK1 CLKBUF1/Z`
- The following command allows the positive phase clock CLK1 to pass through CLKBUF1/Z but filters out the negative phase clock CLK1:
`tempus > set_clock_sense -positive -clocks CLK1 CLKBUF1/Z`
- The following command stops clock phases starting from pin1:
`tempus > set_clock_sense pin1 -stop {clock}`
or
`tempus > set_clock_sense pin1 -logical_stop_propagation`
- The following command stops data phases based only on CK1 and CK2 starting from pin1:
`tempus > set_clock_sense pin1 -stop {data} -clocks {CK1 CK2}`
- The following command stops only the clock source data phases starting from pin1:
`tempus > set_clock_sense pin1 -stop {clock_source_data}`
- The following command stops the clock and clock source data from pin1:
`tempus > set_clock_sense pin1 -stop {clock clock_source_data}`
or
`tempus > set_clock_sense pin1 -stop_propagation`

Related Commands

- `reset_clock_sense`

set_clock_transition

```
set_clock_transition
[help]
[-rise]
[-fall]
[-min]
[-max]
slew_time
clock_list
```

Specifies the transition time of sequential endpoints of the ideal clock network.

This command only affects ideal clock networks, so typically it is used before the clock network is implemented. If no clock transition is explicitly specified for an ideal clock, the system's default transition time (often close to 0.0) is used. For propagated clocks, the calculated transition times are used, so you can set propagated clocks only after the final clock tree is constructed.

This command only affects sequential clock endpoints, typically the clock pin of registers in the combinational fanout of the clock. Any data endpoints that are also in the combinational fanout of the clock will not be affected. The software will always use the propagated transition times. You can consider turning off clock-source data paths until the clock tree is built or, you can use the [set_ideal_network](#) command for additional control.

Parameters

<code>-help</code>	Prints out the command usage.
<code>clock_list</code>	Specifies a list of clocks only with ideal mode in the design, which affects the slew times on all register clock pins in the transitive fanout of the specified clocks. Specified <code>set_clock_transition</code> values on register clock pins in the fanout of a clock with propagated latency are ignored.
<code>-rise/-fall</code>	Specifies the rising or falling edge of the register clock pins on which the slew time is asserted.
<code>-min/-max</code>	Specifies the clock slew time for minimum or maximum process corner slew time.
<code>slew_time</code>	Specifies the slew time (transition time) of the clock pins.

Examples

- The following commands specify a 2.0 rise slew time and a 1.0 fall slew time for register clock pins clocked by `CLKA`:

```
tempus > set_clock_transition -rise 2.0 [get_clocks CLKA]
```

```
tempus > set_clock_transition -fall 1.0 [get_clocks CLKA]
```

Related Commands

- `report_clocks`
- `reset_clock_transition`
- `set_clock_latency`
- `set_clock_uncertainty`
- `set_input_delay`
- `set_propagated_clock`

set_clock_uncertainty

```
set_clock_uncertainty
[-help]
uncertainty_value
[-setup]
[-hold]
[-half_cycle_jitter value | -full_cycle_jitter value]
[-rise | -fall]
{ {-from | -rise_from | -fall_from} clksig_from_list
  {-to | -rise_to | -fall_to} clksig_to_list
  | pin_or_clock_list
}
```

Specifies the clock uncertainty (skew) on the clock network.

If you set multiple constraints with the `set_clock_uncertainty` command, the constraints set with the `-from* clksig_from_list/-to* clksig_to_list` parameters take precedence over simple constraints set on clock objects.

Clock uncertainty set on pins takes highest precedence and overrides clock based uncertainty.

The `set_clock_uncertainty` command can also support clock uncertainty for period, pulse width, skew, and separation checks. To enable this feature, you need to set `timing_enable_uncertainty_for_clock_checks` global variable to true. By default, the global is set to false.

Parameters

<code>-help</code>	Prints out the command usage.
<code>uncertainty_value</code>	Specifies the uncertainty value.
<code>-from -rise_from -fall_from</code> <code>clksig_from_list</code>	Applies the uncertainty value to the specified list of launching clocks. Note: You must specify one of these parameters with one of the <code>-to</code> , <code>-rise_to</code> , or <code>-fall_to</code> parameters. You cannot use these parameters when <code>pin_or_clock_list</code> is specified.
	<code>-fall_from</code>: Applies the uncertainty value only to the falling edge of the launching clocks.
	<code>-from</code>: Applies the uncertainty value to both the rising and falling edges of the launching clocks.

	• -rise_from: Applies the uncertainty value only to the rising edge of the launching clocks.
<code>-half_cycle_jitter value</code> <code> -full_cycle_jitter value</code>	Specifies the clock source jitter for half-cycle and for full-cycle paths. These parameters can accept different values, that can be specified in the same constraint. Note: These parameters can be used on clocks only. You cannot use them when any of the <code>-from</code> and <code>-to</code> parameter pairs are specified.
<code>pin_or_clock_list</code>	Specifies a list of clocks or pins for target uncertainty. The uncertainty value is applied to the registers clocked by the target waveform, or register pins in the fanout of the specified pin. You can specify a list of clock names, or clock root pin or port names relative to the current module. You also can specify a collection of clocks, or clock root pins or ports. Note: You cannot use this parameter when any of the <code>-from</code> and <code>-to</code> parameter pairs are specified.
<code>-rise -fall</code>	Specifies the rising or falling edge at the capture clock pin on which the uncertainty is applied. <i>Default:</i> Applies uncertainty on both edges Note: You must use these parameters with the <code>-to</code> and <code>-from</code> parameters.
<code>-setup/-hold</code>	Specifies whether the clock uncertainty applies to setup or hold checks. <i>Default:</i> Applies to both setup and hold check
<code>-to -rise_to -fall_to</code> <code>clk_sig_to_list</code>	Applies the uncertainty value to the specified list of capture clocks. Note: You must specify one of these parameters with one of the <code>-from</code>, <code>-rise_from</code>, or <code>-fall_from</code> parameters. You cannot use these parameters if <code>pin_or_clock_list</code> is specified.
	• -fall_to: Applies the uncertainty value only to the falling edge of the capture clocks.
	• -rise_to: Applies the uncertainty value only to the rising edge of the capture clocks.

- | | |
|--|---|
| | <ul style="list-style-type: none">• <code>-to</code>: Applies the uncertainty value to both the rising and falling edges of the capture clocks. |
|--|---|

Examples

- The following command gives all paths starting from registers clocked by `clk1` to registers clocked by the rising edge of `clk2` a hold uncertainty of 2.0:

```
tempus > set_clock_uncertainty 2.0 -from clk1 -to clk2 -rise -hold
```

In case of multiple constraints, the actual uncertainty applied by the software depends on which constraint matches the specific path in question.

- The following command sets an uncertainty on the `clk_in` clock root pin that is applied to all the register pins in the fanout of `clk_in`:

```
tempus > set_clock_uncertainty 1.2 clk_in
```

- The following command specifies an inter-clock uncertainty of 0.2 between clock waveforms `clk1` and `clk2`:

```
tempus > set_clock_uncertainty 0.2 -from clk1 -to clk2
```

- The following command specifies a target uncertainty of 0.5 for all registers clocked by the clock waveform `ideal_clk1`:

```
tempus > set_clock_uncertainty 0.5 ideal_clk1
```

Related Commands

- `report_clocks`
- `set_clock_latency`
- `set_clock_transition`
- `reset_clock_uncertainty`

set_data_check

```
set_data_check
[-help]
<check_value>
[-clock <clock_object>]
[-hold]
[-setup]
{-from <pin_or_port_list> | -rise_from <pin_or_port_list> | -fall_from <pin_or_port_list>}
{-to <pin_or_port_list> | -rise_to <pin_or_port_list> | -fall_to <pin_or_port_list>}
```

Sets data-to-data checks using the specified values of setup and hold time. Data-to data checks are non-sequential checks that constrain a data signal with respect to another data signal. The reference (related) delay is defined by the `-from` pin/port and the target(s) by the `-to` pin/port. Note that these need not to be on the same component.

In a normal sequential check, a data signal is constrained with respect to a clock signal. In a sequential setup, the clock is incremented by a cycle between launch (`-from`) and capture (`-to`). In contrast, a data-to-data setup check is performed on the same edge as the launch. A clock can be specified to refine the triggering condition of the reference signal. You can use the `set_multicycle_path` constraint to adjust the cycle accounting rules.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-clock <i>clock_object</i></code>	Specifies the name of the clock that triggers the data event for the reference (<code>-from</code>) pin. In the case where there are multiple data events triggered by different clock phases arriving at the <code>-from</code> pin, the <code>-clock</code> parameter allows you to specify which data event should be used for the "related pin" side of the timing check.

<pre>{-from <pin_or_port_list> -rise_from <pin_or_port_list> -fall_from <pin_or_port_list>}</pre>	Specifies the pins or ports that are set as related pins for data-to-data checks. The <i>pin_or_port_list</i> argument can contain either object IDs or hierarchical names relative to the current module. Use only one <i>-from</i>, <i>-rise_from</i>, or <i>-fall_from</i> parameter per command. By default, the <i>-from</i> parameter checks both the rising and the falling edges. Using <i>-rise_from</i> parameter checks at the rising edge of the signal on the from pins. Using <i>-fall_from</i> parameter checks at the falling edge of the signal on the from pins. Note: You cannot use the <i>-from</i> parameter to specify multiple objects.
<pre>-setup -hold</pre>	Specifies whether the data check value is for setup or hold check. If you do not specify either <i>-setup</i> or <i>-hold</i> option, the value applies to both setup and hold.
<pre>{-to <pin_or_port_list> -rise_to <pin_or_port_list> -fall_to <pin_or_port_list>}</pre>	Specifies the pins or ports that are set as constrained pins for data-to-data checks. The <i>pin_or_port_list</i> argument can contain either object IDs or hierarchical names relative to the current module. Use only one <i>-to</i>, <i>-rise_to</i>, or <i>-fall_to</i> parameter per command. By default, the <i>-to</i> parameter constrains both the rising and the falling edges. Using <i>-rise_to</i> parameter constrains the rising edge of the signal on the to pins. Using <i>-fall_to</i> parameter constrains the falling edge of the signal on the to pins.
<pre>check_value</pre>	Specifies the value of the setup or hold time for the check.

Examples

- The following example demonstrates how the `set_data_check` command can be used to constrain signals:

Consider two control signals, say XE and SE, going to a macro block. Both of these are derived from the same state machine, and hence launched from the same clock domain. You want XE to reach the macro earlier than SE. The `set_data_check` constraint can meet this design intent, by applying data checks on non-sequential signals, without setting path specific limits using the following command:

```
tempus > set_data_check -from [get_pins {sub/SE}] -to [get_pins {sub/XE}] -setup 0.5
```

The `set_data_check` command establishes data-to-data checks using similar ideas of setup and hold time as in sequential checks, with the `-from` parameter denoting the reference signal (the clock in a traditional setup/hold timing check), and the `-to` parameter indicating the signal being constrained.

Related Commands

- `reset_data_check`

set_disable_clock_gating_check

```
set_disable_clock_gating_check  
[-help]  
object_list
```

Disables clock gating checks for the specified instances or pins.

The `set_disable_clock_gating_check` command has a character limit for the details that are logged into the SDC file (the `write_sdc` command output). This helps to prevent SDC file errors.

Parameters

<code>-help</code>	Prints out the command usage.
<code>object_list</code>	Specifies the list of pins or instances that are to be disabled.

Examples

- The following command disables clock gating checks for the specified instance:

```
tempus > set_disable_clock_gating_check [get_cells clockgate1]
```

Related Commands

- `reset_disable_clock_gating_check`

set_disable_timing

```
set_disable_timing
[-help]
[-from pin_name -to pin_name]
object_list
```

Disables timing propagation through the specified arcs, or a collection of instance-specific arcs that are created using the `get_arcs` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-from</code> <i>pin_name</i> <code>-to</code> <i>pin_name</i>	Specifies the arc that is to be disabled. Specify the <code>-from</code> and <code>-to</code> options together. You can specify the <code>-from</code> and <code>-to</code> parameters when the <i>object_list</i> contains instance or library cell objects. Do not specify the <code>-from</code> and <code>-to</code> parameters when the <i>object_list</i> includes instance pin or library cell pin objects. The <code>-from</code> and <code>-to</code> options are optional. If not specified, all cell arcs found in the instance or library cell objects are disabled.
<i>object_list</i>	Specifies the instance, instance pin, library cell, library cell pin objects, collection of library arcs, or collection of timing arcs. If the <i>object_list</i> includes instance pin or library cell pin objects, all cell arcs attached to the specified pin objects are disabled. If the <i>object_list</i> contains instance or library cell objects, and the <code>-from</code> and <code>-to</code> parameters are not specified, all arcs found in the specified instance or library cell objects are disabled.

Examples

Here are some examples.

- The following command disables the arcs that lead to the output pins of the `i1`, `i2`, and `i3` instances. Note that timing checks at the inputs of `i1`, `i2`, and `i3` are still active.

```
tempus > set_disable_timing [get_cells {i1 i2 i3}]
```

- The following command disables the arc from pin `B` to pin `Z` in the `i1` and `i2` instances, and the arc from pin `A` to pin `Z` in the `i3` instance:

```
tempus > set_disable_timing -from B -to Z [get_cells {i1 i2}]
```

```
tempus > set_disable_timing -from A -to Z [get_cells {i3}]
```

- The following example disables timing check arcs from pin `CLK` to pin `D` of the `ff1` instance:

```
tempus > set_disable_timing -from CLK -to D [get_cells ff1]
```
- The following example disables the outgoing cell arcs from instance input pin `i1/B`:

```
tempus > set_disable_timing [get_pins i1/B]
```
- The following example disables all arcs found in any instances whose reference cell is `AN2`:

```
tempus > set_disable_timing [get_lib_cells AN2]
```
- The following example disables a collection of timing arcs:

```
tempus > set arcs [get_lib_arcs -from typical/MX2X1/S0 -to typical/MX2X1/Y]  
tempus > set_disable_timing [index_collection $arcs 3]
```
- The following example disables arcs created using `filter_collection` on library timing arcs:

```
tempus > set arc [filter_collection $arcs {when == "(A)&(!(B))"}]  
tempus > set_disable_timing $arc
```

Related Commands

- `report_timing`
- `reset_disable_timing`

set_dont_touch

```
set_dont_touch
[-help]
object_list
[{true | false}]
```

Prevents the specified object from being modified during optimization.

When the `set_dont_touch` command is set to `true`, the specified nets cannot be buffered or become part of the logic cone restructuring. However, you can resize the instances connected to this net because the `set_dont_touch` attribute does not affect the connected instances.

Parameters

<code>-help</code>	Prints out the command usage.
<code>{true false}</code>	Sets or removes the <code>set_dont_touch</code> attribute for the specified list of objects. <i>Default:</i> <code>true</code>
<code>object_list</code>	Specifies the collection or list of instances, library cells, nets, or sub-modules.

Examples

- The following command prevents the specified instance from being modified:
tempus > `set_dont_touch [get_cells {TWA/FF1}]`
- The following commands prevent the specified net from being modified and check the setting with `get_property`:
tempus > `set_dont_touch clk1_net`
tempus > `get_property [get_nets clk1_net] dont_touch true`
- The following commands can be used to prevent any ECO inside `RESULTS_CONV_INST` hierarchical instance:
`set_dont_touch [get_cells RESULTS_CONV_INST/*] true`
`set_dont_touch [get_nets RESULTS_CONV_INST/*] true`

The `set_dont_touch RESULTS_CONV_INST/* true` command will return `[get_cells RESULTS_CONV_INST/*]` only. This is because the software looks into the first match (type of object). If a match is not found, then the successive objects are searched.

Related Commands

- `get_cells`
- `get_lib_cells`
- `get_nets`
- `get_property`

set_dont_touch_network

```
set_dont_touch_network  
[-help]  
object_list  
[-no_propagate]
```

Prevents the combinational path connected to the given clocks, pins, or ports from being modified.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-no_propagate</code>	Preserves all the nets driven by the specified pin up to the first leaf-cell input (combinational or sequential). If this parameter is omitted, then preserving stops at the first sequential cell in the network.
<i>object_list</i>	Specifies the objects - clocks, pins, or ports, to be preserved.

Examples

- The following command prevents the combinational path connected to the specified pin from being modified:

```
tempus > set_dont_touch_network [get_pins and1/z]
```

Related Commands

- `get_pins`

set_dont_use

```
set_dont_use  
[-help]  
object_list  
[true | false]
```

Prevents the specified library cells from being used during optimization and ECO operations.

Parameters

-help	Prints out the command usage.
object_list	Specifies the list of library cells.
true false	Sets or removes the <code>set_dont_use</code> attribute for the specified object list. <i>Default:</i> <code>true</code>

Examples

- The following command prevents the specified library cell from being used during optimization:
`tempus > set_dont_use [get_lib_cells {and2}]`

Related Commands

- `get_lib_cells`

set_drive

```
set_drive
[-help]
[-rise]
[-fall]
[-max]
[-min]
resistance_value
port_list
```

Sets the drive resistance on the specified input and/or bidirectional ports of the top cell.

The drive resistance models the cell driving the port. This resistance is used to compute the load dependent delay and the slew at the specified ports. The load-dependent delay is added to the arrival times.

The load-dependent delay and the slew are computed by multiplying the drive resistance by the wire capacitance of the net connected to the port.

Use the [report_ports](#) command to view drive resistances asserted on ports.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-fall</code>	Specifies that the drive resistance is to be used for the falling transition at the specified ports. By default, if neither the <code>-rise</code> nor the <code>-fall</code> option are specified, the drive resistance applies to both the rising transition and the falling transition.
<code>-min</code>	Specifies that the drive resistance is to be used to compute the early path delay. By default, if neither the <code>-min</code> option nor the <code>-max</code> option is specified, the drive resistance applies to both the early path delay and the late path delay.
<code>-max</code>	Specifies that the drive resistance is to be used to compute the late path delay. By default, if neither the <code>-min</code> option nor the <code>-max</code> option is specified, the drive resistance applies to both the early path delay and the late path delay.
<i>port_list</i>	Specifies a list of input or bidirectional ports on which the drive resistance is set.
<i>resistance_value</i>	Specifies the value of the drive resistance. This value must be greater than or equal to zero. The drive resistance is in units found in the target technology library.

<code>-rise</code>	Specifies that the drive resistance is to be used for the rising transitions at the specified ports. By default, if neither the <code>-rise</code> option nor the <code>-fall</code> option are specified, the drive resistance applies to both the rising transition and the falling transition.
--------------------	---

Example

- The following command sets a drive resistance of 2.0 for the early path, the late path, the rising and the falling transitions at the in1 and in2 ports and shows how drive resistances are reported using the `report_ports` command as shown in the example below:

```
tempus > set_drive 2.0 [get_ports {in1 in2}]
tempus > report_ports -type drive_resistance -pin [get_ports {in1 in2}]
```

The following report is displayed:

```
-----
Early Late
-----
Pin      Dir  Assertion      Clock  Rise  Fall  Rise  Fall
Name                                Name
-----
in1      IN   drive_resistance *(D) (P) 2.000  2.000  2.000  2.000
in2      IN   drive_resistance *(D) (P) 2.000  2.000  2.000  2.000
-----
```

Related Commands

- `all_inputs`
- `get_ports`
- `report_ports`
- `reset_drive`
- `set_driving_cell`
- `set_input_transition`
- `set_load`

set_driving_cell

```
set_driving_cell
[-help]
[-library library_name]
-lib_cell cell_name
[-pin pin_name]
[-from_pin from_pin_name]
[-rise]
[-fall]
[-min]
[-max]
[-multiply_by factor]
[-no_design_rule]
[-input_transition_rise rise_slew_value]
[-input_transition_fall fall_slew_value]
port_list
```

Models the drive capability of an external driver connected to the input port. Use this command only for timing analysis. The command does not affect the electrical properties of the design, which means that the capacitance of the output pin of the driving cell does not add to the total capacitance of the input port.

To model the drive capability of the external driver, the timing analyzer computes an offset to the arrival time of an input, and changes the slew time used to compute the delay for the next cell. The timing analyzer computes the offset by subtracting the zero load delay from the total delay of the cell. The `set_driving_cell` command uses the offset value to identify a cell from the library to drive the specified input port(s).

This command overrides the `set_drive` command per port or per net, if it is the last command applied. Also, if the `set_driving_cell` command is set for a port, the `set_input_transition` command is ignored for that port. If you do not define drive using the `set_driving_cell` command, the software uses the default input transition value, defined using the `delaycal_input_transition_delay` the variable. This is set in the initialization global variable file and has a default value of 0.1ps.

Parameters

-help	Prints out the command usage.
-fall	Annotates only the falling drive assertion. The default is to annotate both rise and fall delays.

<code>-from_pin</code> <code>from_pin_name</code>	Specifies the name of the input port of the driver cell that has an arc to the output port of the driver cell, which is connected to the input port of the design. This option is required only if there are multiple timing arcs in the driver cell from its input ports to its output ports, and the arcs from those pins have different drive characteristics. If you do not specify the <code>-from_pin</code> option, the default is to use the first timing arc found on the library cell.
<code>-input_transition_fall</code> <code>fall_slew_value</code>	Specifies the slew value for the falling signal at the input of the drive cell.
<code>-input_transition_rise</code> <code>rise_slew_value</code>	Specifies the slew value for the rising signal at the input of the drive cell.
<code>-lib_cell cell_name</code>	Specifies the name of the driver cell. This parameter supports wildcard patterns and collections.
<code>-library library_name</code>	Specifies the name of the library that contains the driving cell.
<code>-max</code>	Annotates only the late drive assertion. The default is to annotate both early and late.
<code>-min</code>	Annotates only the early drive assertion. The default is to annotate both early and late.
<code>-multiply_by factor</code>	Specifies a multiplier for the drive.
<code>-no_design_rule</code>	Does not allow transfer of design rules from the driving cell to the port. If you do not specify this option, the design rules (such as <code>max_capacitance</code>) of the library pin are applied to the port.
<code>-pin pin_name</code>	Specifies the name of the output port of the driver cell that drives the signal at the input port of the design. This option is required if there are multiple output ports on the driver. If there is only one output port on the driver cell, then using this option is not necessary.
<code>port_list</code>	Specifies the list of ports driven by the driver cell.
<code>-rise</code>	Specifies only the rising drive assertion is annotated. The default is to annotate both rise and fall delays.

Examples

- The following command sets 'AND' cell (arc B -> Y) as the driving cell for the input port IN1:

```
tempus > set_driving_cell -lib_cell AND -from_pin B -pin Y IN1
```

Related Commands

- `all_inputs`
- `report_ports`
- `reset_driving_cell`
- `set_drive`
- `set_input_transition`

set_false_path

```
set_false_path
[-help]
[-hold]
[-setup]
[-rise]
[-fall]
[{-from | -rise_from | -fall_from} from_list]
[{-through | -rise_through | -fall_through} through_list]
[{-to | -rise_to | -fall_to} to_list]
```

Identifies false paths in a design, and breaks or disables specific instance timing arcs in a design.

 To break combinational loops, use the `set_disable_timing` command instead of the `set_false_path` command.

The `set_false_path` command differs from the `set_disable_timing` command as follows:

- `set_false_path` command selectively disables the arrival time information depending on the precedence rules.
- `set_false_path` command does not affect constant values.
- `set_disable_timing` command physically snips a timing arc such that neither arrival times nor constant values propagate through it.
- `set_disable_timing` command is applied in all cases to valid timing arcs.

The `set_false_path -from fpin` command disables all timing paths whose source pin is an `fpin`. Similarly, the `set_false_path -to tpin` command disables all timing paths whose sink pin is a `tpin`. Using the `set_false_path -from fpin -to tpin` command disables all timing paths whose source is `fpin` and sink is `tpin`.

Specify a group of pins by enclosing the group with curly braces (`{}`) within one `set_false_path` command.

 To set a false path for any path beginning at `p1` or `p2` and terminating at `t1` or `t2`, group the pins as follows:
`set_false_path -from {p1 p2} -to {t1 t2}`

Note: This pin-grouping method reduces the number of exceptions. Also, having a large number of exceptions adversely impacts the runtime of the timing analysis commands.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-fall</code>	Applies the assertion at the falling edge of the signal on the path endpoints.
<pre>{-from - rise_from - fall_from} from_list</pre>	Specifies a list of timing paths start points. The <code>from_list</code> argument can contain either object IDs or hierarchical names relative to the current module. Intermediate hierarchical pins should not be specified as <code>-from</code> in path exceptions unless those are valid begin points. Even if the specified constraint is accepted, timing analysis will ignore such (<code>set_false_path/set_multicycle_path</code>) constraints. The <i>from_list</i> argument can be a collection. The valid start points are: • Input port • Bidirectional port • Clock pin of sequential cell ◦ Clock pin of sequential cell is one of the following: ◦ Structural clock pin that is part of a latch or flip-flop. ◦ Inferred clock pin in cells like macro cells. A pin is an inferred clock pin if it is associated with one of the following: ■ Reference end of setup check ■ Reference end of hold check ■ Reference end of removal check ■ Reference end of recovery check ■ Minimum period check ■ Minimum pulse width check • Pin associated with <code>set_input_delay</code> Specify one of the following:

	• -from: Applies the false path constraint to all paths originating from the <code>from_list</code> list. If a clock is specified as the <code>-from</code> object, all primary inputs and registers clocked by that clock are used as start points. By default, the <code>-from</code> option applies the constraint to both the rising and the falling edges.
	• -rise_from: Applies the false path constraint at the rising edge of the signal on the from pins.
	• -fall_from: Applies the false path constraint at the falling edge of the signal on the from pins.
<code>-hold/-setup</code>	Sets false paths for either setup or hold analysis. Note: If you do not specify <code>-setup</code> or <code>-hold</code>, the software sets false paths for both setup and hold analysis.
<code>-rise</code>	Applies the assertion at the rising edge of the signal on the path endpoints.
<pre>{-through - rise_through - fall_through} through_list</pre>	Specifies a list of timing paths through points. The <code>through_list</code> argument can contain either object IDs or hierarchical names relative to the current module. The pins can be on intermediate hierarchical boundaries. The <code>through_list</code> argument can be a collection. You can specify one of the following options: • -through: When you use this parameter without using the <code>-from</code> or <code>-to</code> options, the false path constraint is set for all paths that go through the specified pins. When you use it with both the <code>-from</code> and <code>-to</code> options, any path starting at any pin on the <code>-from</code> list and going through any point on the <code>-through</code> pin list and going to any point on the <code>-to</code> list is affected by the constraint. By default, the <code>-through</code> parameter applies constraints to both the rising and falling edges. • -rise_through: Applies the assertion at the rising edge of a signal on the through pins. • -fall_through: Applies the assertion at the falling edge of a signal on the through pins. Note: The <code>-through</code> parameter does not support clock paths.

<pre>{-to -rise_to -fall_to} to_list</pre>	Specifies a list of timing path endpoints. The to_list argument can contain either object IDs or hierarchical names relative to the current module. The pins can be on intermediate hierarchical boundaries. The to_list argument can be a collection. The valid endpoints are: • Output port • Bidirectional port • Data pin of the sequential cell Data pin of sequential cell is one of the following: ◦ Structural data pin that is part of latch or flip-flop. ◦ Inferred data pin in cells like macro cells. A pin is an inferred data pin if it is associated with one of the following: ■ Signal end of setup check ■ Signal end of hold check ■ Signal end of non-sequential setup check ■ Signal end of non-sequential hold check ■ Signal end of removal check ■ Signal end of recovery check • Pin associated with set_external_delay • Data pin of clock gating cell Select from one of the following:
	• -to: Applies the false path constraint to all paths ending in the -to pin list. If a clock is specified as the -to object, all primary outputs and registers clocked by that clock are used as endpoint. By default, the -to option applies the constraint to both the rising and the falling edges.
	• -rise_to: Applies the false path constraint at the rising edge of the signal on the to pins.
	• -fall_to: Applies the false path constraint at the falling edge of the signal on the to pins.

Examples

- The following command disables hold checking from `ff1` to `ff2`:

```
tempus > set_false_path -hold -from ff1 -to ff2
```
- The following command sets all timing paths from `ff1/CK` to `D` input of `ff2` that pass through `u1/Y` and one or more of `u2/Y` and `u3/Y` as false paths:

```
tempus > set_false_path -from ff1/CK -through {u1/Y} -through {u2/Y u3/Y} -to ff2/D
```
- The following command sets all timing paths from the rise of `ff1/CK` to `ff2/D` through one or more of `u1/Y` and `u4/Y` as false paths:

```
tempus > set_false_path -rise_from ff1/CK -through {u1/Y u4/Y} -to ff2/D
```

Related Topics

- [Path Exception Priorities](#)

Related Commands

- `set_disable_timing`
- `set_max_delay`
- `set_min_delay`
- `set_multicycle_path`
- `reset_path_exception`
- `report_path_exceptions`

set_fanout_load

```
set_fanout_load  
[-help]  
fanout_load  
port_list
```

Specifies the fanout load on a set of ports.

The library instance pins can have fanout loads defined in the technology library. The fanout does not have specific units so it is assumed that a fanout as used by the constraints is consistent with the fanout defined in the technology libraries. If a pin or port has no fanout attribute, either from the constraints or from the technology library, then the fanout load will be zero. The fanout load is used only for design rule checking and does not impact the timing results.

Parameters

<code>-help</code>	Prints out the command usage.
<code>fanout_load</code>	Specifies the fanout load value for the port or ports.
<code>port_list</code>	Specifies the list of ports for which the fanout load is applied.

Examples

- The following command sets the fanout load of port `out` to 2:

```
tempus > set_fanout_load 2 out
```
- The following command sets the fanout load of ports `in1` and `in2` to 1:

```
tempus > set_fanout_load 1 {in1 in2}
```
- The following command sets the fanout load of element 3 of port `out` to 12:

```
tempus > set_fanout_load 12 { out[3] }
```
- The following command sets the fanout load of all the ports to 2:

```
tempus > set_fanout_load 2 [get_ports *]
```

Related Commands

- `report_net`
- `report_ports`
- `set_max_fanout`
- `report_constraint`

set_ideal_latency

```
set_ideal_latency  
[-help]  
[-min]  
[-max]  
[-rise]  
[-fall]  
latency_value  
object_list
```

Specifies the ideal latency to use for input or output leaf-cell pins and top-level ports that are part of an ideal network. These set a point delay on the specified object.

Note: You should specify `set_ideal_latency` with the `set_ideal_network` command to view the ideal latency result.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-fall</code>	Specifies the ideal latency for the falling edge.
<code>latency_value</code>	Specifies the ideal latency value.
<code>-max</code>	Specifies the ideal latency for the maximum corner.
<code>-min</code>	Specifies the ideal latency for the minimum corner.
<code>object_list</code>	Specifies a list of objects - instance pins or ports.
<code>-rise</code>	Specifies the ideal latency for the rising edge.

Examples

- The following command adds an ideal latency of 1.5ns to the input pin A of cell FE_OFC413_scan_se:

```
tempus > set_ideal_latency 1.5 FE_OFC413_scan_se/A
```

Please note that the above command will only be valid if the pin is part of an ideal network.

Now, this ideal latency value can be observed in the output of the following [report_timing](#) command:

```
tempus > report_timing -through FE_OFC413_scan_se/A
```

```
Path 1: MET Setup Check with Pin RESULTS_CONV_INST/r697_reg[14]/CK
Endpoint: RESULTS_CONV_INST/r697_reg[14]/SE (v) checked with trailing edge of
'm_rcc_clk'
```

```
Beginpoint: scan_se (v) triggered by leading edge of 'm_ram_clk'
```

```
Path Groups: {m_rcc_clk}
```

```
Analysis View: scan_fast_RCMIN
```

```
Other End Arrival Time          20.679
```

```
- Setup                        1.123
```

```
+ Phase Shift                  0.000
```

```
+ Cycle Adjustment             40.000
```

```
- Uncertainty                  0.400
```

```
= Required Time                59.156
```

```
- Arrival Time                 8.000
```

```
= Slack Time                   51.156
```

```
Clock Rise Edge               0.000
```

```
+ Input Delay                  4.000
```

```
+ Ideal Latency                1.500
```

```
= Beginpoint Arrival Time      5.500
```

```
-----
Pin                Cell    Load    Slew    Delay Arc
                                   Annotation
-----
```

```
scan_se           0.000    0.000    Ideal
```

```
FE_OFC413_scan_se/Z BUF_X32 0.000    0.000    0.000 Ideal
```

```
FE_OFC247_scan_se/A BUF_X16 0.000    0.000    0.000 Ideal
```

```
FE_OFC247_scan_se/Z BUF_X16 0.000    0.000    0.000 Ideal
```

```
FE_OFC248_scan_se/A BUF_X16 0.000    0.000    0.000 Ideal
```

```
FE_OFC248_scan_se/Z BUF_X16 0.000    0.000    0.000 Ideal
```

```
FE_OFC249_scan_se/A BUF_X8  0.000    0.000    0.000 Ideal
-----
```

Related Commands

- `reset_ideal_latency`
- `report_timing`
- `set_ideal_network`

set_ideal_network

```
set_ideal_network  
[-help]  
object_list  
[-no_propagate]
```

Identifies driver pins or ports as sources of an ideal network. By default, all pins and nets in the transitive fanout of the specified port or driver pin are marked as ideal. Ideal network marking normally propagates through combinational cells and stops at sequential cells.

This command is not view/mode-based - it applies to all the active views.

The `set_ideal_network` command sets 0 slew for bi-directional pins. This ensures that unnecessary DRV violations do not occur.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-no_propagate</code>	Prevents propagation through transitive fanout.
<i>object_list</i>	Specifies a list of objects - instance pins, ports, or nets.

Examples

- The following command marks the fanout of the input port scan_se as an ideal network:

```
tempus > set_ideal_network scan_se
```

This can be observed in the output of below mentioned report_timing command where the whole fanout of 'scan_se' is considered as ideal :

```
tempus > report_timing -from scan_se
```

```
Path 1: MET Setup Check with Pin RESULTS_CONV_INST/r697_reg[12]/CK
Endpoint: RESULTS_CONV_INST/r697_reg[12]/SE (v) checked with trailing edge of
'm_rcc_clk'
Beginpoint: scan_se (v) triggered by leading edge of 'm_ram_clk'
Path Groups: {m_rcc_clk}
Analysis View: scan_fast_RCMIN
Other End Arrival Time 20.700
- Setup 1.122
+ Phase Shift 0.000
+ Cycle Adjustment 40.000
- Uncertainty 0.400
= Required Time 59.178
- Arrival Time 4.000
= Slack Time 55.178
Clock Rise Edge 0.000
+ Input Delay 4.000
= Beginpoint Arrival Time 4.000
```

Pin	Cell	Load	Slew	Delay	Arc Annotation
scan_se ->		0.000	0.000		Ideal
FE_OFC413_scan_se/Z	BUF_X32	0.000	0.000	0.000	Ideal
FE_OFC247_scan_se/Z	BUF_X16	0.000	0.000	0.000	Ideal
FE_OFC248_scan_se/Z	BUF_X16	0.000	0.000	0.000	Ideal
FE_OFC249_scan_se/Z	BUF_X8	0.000	0.000	0.000	Ideal
FE_OFC250_scan_se/Z	BUF_X16	0.000	0.000	0.000	Ideal
FE_OFC251_scan_se/Z	BUF_X4	0.000	0.000	0.000	Ideal
RE_IN/r67_reg[12]/SE	SDFF_X2	0.000	0.000	0.000	Ideal

Related Commands

- [reset_ideal_network](#)

set_ideal_transition

```
set_ideal_transition  
[-help]  
[-min]  
[-max]  
[-rise]  
[-fall]  
transition_time  
object_list
```

Specifies the ideal transition to use for input or output leaf-cell pins and top-level ports that are part of an ideal network. These set a point delay on the specified object.

Note: You should specify `set_ideal_transition` with `set_ideal_network` to view ideal transition result.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-fall</code>	Specifies the ideal transition for the falling edge.
<code>-max</code>	Specifies the ideal transition for the maximum corner.
<code>-min</code>	Specifies the ideal transition for the minimum corner.
<code>object_list</code>	Specifies a list of objects - instance pins or ports.
<code>-rise</code>	Specifies the ideal transition for the rising edge.
<code>transition_time</code>	Specifies the ideal transition time.

Examples

Here are some examples.

- The following command specifies the ideal transition of 0.5ns on pin `scan_se`:

```
tempus > set_ideal_transition 0.5 scan_se
```

Please note that this ideal transition would only be applicable if the pin is part of an ideal network, otherwise this value would be ignored.

This transition will be propagated through the ideal network as shown in the output of the following `report_timing` command:

```
tempus > report_timing -from scan_se
```

```
Path 1: MET Setup Check with Pin RESULTS_CONV_INST/r697_reg[14]/CK
Endpoint: RESULTS_CONV_INST/r697_reg[14]/SE (v) checked with trailing edge of
'm_rcc_clk'
Beginpoint: scan_se (v) triggered by leading edge of 'm_ram_clk'
Path Groups: {m_rcc_clk}
Analysis View: scan_fast_RCMIN
Other End Arrival Time 20.699
- Setup 1.277
+ Phase Shift 0.000
+ Cycle Adjustment 40.000
- Uncertainty 0.400
= Required Time 59.022
- Arrival Time 4.000
= Slack Time 55.022
Clock Rise Edge 0.000
+ Input Delay 4.000
= Beginpoint Arrival Time 4.000
```

Pin	Cell	Load	Slew	Delay	Arc Annotation
scan_se ->		0.000	0.500		Ideal
FE_OFC413_scan_se/Z	BUF_X32	0.000	0.500	0.000	Ideal
FE_OFC247_scan_se/Z	BUF_X16	0.000	0.500	0.000	Ideal
FE_OFC248_scan_se/Z	BUF_X16	0.000	0.500	0.000	Ideal
FE_OFC249_scan_se/Z	BUF_X8	0.000	0.500	0.000	Ideal
FE_OFC250_scan_se/Z	BUF_X16	0.000	0.500	0.000	Ideal
FE_OFC251_scan_se/Z	BUF_X4	0.000	0.500	0.000	Ideal
RE_IN/r67_reg[14]/SE	SDFF_X2	0.000	0.500	0.000	Ideal

Related Commands

- [reset_ideal_transition](#)

set_input_delay

```
set_input_delay
[-help]
[-clock clock_name]
[-clock_fall]
[-rise] [-fall]
[-max] [-min]
[-add_delay]
[-network_latency_included]
[-source_latency_included]
[-reference_pin pin_name]
[-level_sensitive]
delay_value
port_or_pin_list
```

Defines the arrival time relative to a clock edge on input ports or internal input pins. This input path delay models the delay from an external register to an input port of the module.

If the input delay is not specified on an input port, it is assumed to be zero.

Note: If the `set_input_delay` constraint is added incrementally in SI mode, an explicit `update_timing -full` command is required (before `report_timing` or `report_constraint`) for SI full accuracy.

Parameters

-help	Prints out the command usage.
-add_delay	Adds delay information to the existing input delay. The additional delay information can be for multiple paths relative to different clocks or clock edges leading to an input port or internal input pins. If you do not use the <code>-add_delay</code> option, the software uses the last <code>set_input_delay</code> related to a pin when there are multiple <code>set_input_delay</code> statements for the same pin.
-clock <i>clock_name</i>	Specifies that the delay is relative to the rising edge of the given clock. If you do not specify the <code>-clock</code> parameter, the delay is relative to time zero. To specify a delay relative to the falling edge of the clock, use the <code>-clock_fall</code> option with the <code>-clock</code> option. When you specify multiple clocks, an error occurs and the constraint is ignored.

<code>-clock_fall</code>	Specifies that the delay is relative to the falling edge of the clock. You must specify the <code>-clock</code> parameter with this parameter. <i>Default:</i> Delay is relative to the rising edge. You cannot specify the <code>-clock_fall</code> parameter without the <code>-clock</code> parameter.
<code>delay_value</code>	Specifies the value for the path delay.
<code>-fall</code>	Specifies that the input delay refers to the falling edge of the signal at specified ports or pins. If you do not specify the <code>-fall</code> option, the input delay applies to both the edges.
<code>-level_sensitive</code>	Models input delay that is launched by a positive level sensitive latch. Use the <code>-clock</code> parameter to model a positive level sensitive latch, or the <code>-clock_fall</code> parameter to model a negative level sensitive latch. You cannot specify the <code>-level_sensitive</code> parameter without the <code>-clock</code> parameter.
<code>-max</code>	Specifies that the delay refers to the setup analysis. If you do not specify the <code>-max</code> parameter, the input delay applies to both setup and hold analysis. Note: If for a specific clock the maximum delay is less than the minimum delay, then during analysis the software makes the maximum delay equal to the minimum delay.
<code>-min</code>	Specifies that the delay refers to the hold analysis. If you do not specify the <code>-min</code> parameter, the input delay applies to both setup and hold analysis.
<code>-network_latency_included</code>	Specifies that the clock network latency should not be added to the input delay value. If you do not specify this option, the clock network latency of the related clock is added to the input delay value. This parameter has no effect if the specified clock is in propagated mode.
<code>port_or_pin_list</code>	Specifies a list of input port or pin names. The <code>port_or_pin_list</code> argument can be a collection.

<code>-reference_pin pin_name</code>	Adds source and network latency of the specified reference pin to the input delay value in propagated mode. A reference pin is a leaf pin in the fanout cone of the clock source of the clock that you specify using the <code>-clock</code> parameter. You cannot use the <code>-reference_pin</code> parameter to specify multiple pins. If specified, the command issues a warning and accepts the first pin. You cannot use the <code>-reference_pin</code> parameter with the <code>-network_latency_included</code> or <code>-source_latency_included</code> parameters. You cannot use the <code>-reference_pin</code> parameter to specify hierarchical pins.
<code>-rise</code>	Specifies that input delay refers to the rising edge of the signal at the specified ports or pins. If you do not specify the <code>-rise</code> option, the input delay applies to the both edges.
<code>-source_latency_included</code>	Specifies that the clock source latency should not be added to the input delay value. With different delay values for <code>-rise/-fall</code> and <code>-min/-max</code>, only the last specified combination of <code>-network_latency_included</code> and <code>-source_latency_included</code> parameters is honored. The specified combination can be one of the following: a) Either <code>-network_latency_included</code> or <code>-source_latency_included</code> b) Both of these c) None of these

Examples

- The following command sets an input delay of 1.0 for all inputs in the current design relative to the rising edge of the `CLK1` clock:

```
tempus > set_input_delay -clock CLK1 1.0 [all_inputs]
```

- The following command sets an input delay of 2.0 for the input `IN1` port relative to the falling edge of the `CLK1` clock:

```
tempus > set_input_delay -clock CLK1 2.0 -clock_fall { IN1 }
```

- The following example shows an input `IN1` port that is constrained by both the rising edge and the falling edge of the same `CLK1` clock. The `-add_delay` option prevents the second `set_input_delay` statement from overwriting the first `set_input_delay`:

```
tempus > set_input_delay -clock CLK1 1.0 { IN1 }
```

```
tempus > set_input_delay -clock CLK1 -clock_fall -add_delay 2.0 { IN1 }
```

- When multiple `-min` and `-max` values are specified relative to the same clock using the `-add_delay` option at the same port or pin, the worst case value for each case is used. For the `-min` case the smallest value is retained, and for the `-max` case the largest value is retained. For example, the input delay value of 1.0 is stored for the shortest path and the value of 4.0 is stored for the longest path in the following example:

```
tempus > set_input_delay -min -clock CLK1 1.0 { IN1 }  
tempus > set_input_delay -min -clock CLK1 2.0 -add_delay { IN1 }  
tempus > set_input_delay -max -clock CLK1 3.0 -add_delay { IN1 }  
tempus > set_input_delay -max -clock CLK1 4.0 -add_delay { IN1 }
```

Related Commands

- [all_inputs](#)
- [create_clock](#)
- [report_ports](#)
- [reset_input_delay](#)
- [set_load](#)
- [set_output_delay](#)
- [report_path_exceptions](#)

set_input_transition

```
set_input_transition  
[-help]  
[-min]  
[-max]  
[-rise]  
[-fall]  
transition_time  
port_list
```

Specifies the slew time for a port on the top level module. This command is useful only in specifying a hard ramp signal as if the port were driven by a very powerful driver. The command does not change the arrival time of an input, but changes the slew time used to compute the delay of the cell on the sink of a net. If you do not define transition using the `set_input_transition` command, the software uses the default input transition value defined using the `delaycal_input_transition_delay` variable. This is set in the initialization global variable file and has a default value of 0.1ps.

The `set_input_transition` command specification for clock signals does not impact setup/hold timing calculations.

Note: The `set_input_transition` command does not support internal pins. For specifying transition on internal pins use the `set_annotated_transition` command.

Note: The most recent command settings (of `set_driving_cell`, `set_input_transition`, or `set_drive`) that are applied to a port will be considered - the software will override the previous commands.

The `set_input_threshold` command should be issued after the `set_lib_binding_by_voltage` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-fall/-rise</code>	Specifies that the slew time should be applied to the rising edge or the falling edge of the signal (data or clock). If neither time is specified, the default is to use both.
<code>-min/-max</code>	Indicates whether the slew times are with respect to the early (hold) or the late (setup) time checks of the data signal. If neither option is specified, the default is to use both.
<code>port_list</code>	Specifies the list of ports for which the slew time is specified. When you use both <code>set_input_transition</code> and <code>set_annotated_transition</code> commands for specifying transition on input ports, the worst value is used as the transition value. Note: You cannot use the <code>set_input_transition</code> command to specify transition on output ports.
<code>transition_time</code>	Specifies the transition time.

Examples

- The following command specifies the slew time of 0.5 on port `input3`.

```
tempus > set_input_transition 0.5 input3
```
- The following command specifies the input falling slew of 0.6ns on the port `input1` for minimum delay type:

```
tempus > set_input_transition -min -fall 0.6 input1
```

Related Commands

- `report_ports`
- `set_driving_cell`
- `set_drive`
- `set_clock_transition`

set_load

```
set_load  
[-help]  
[-max]  
[-min]  
[-pin_load]  
[-wire_load]  
[-subtract_pin_load]  
capacitance_value  
object_list
```

Sets the specified capacitance on the defined ports (or nets) of the top cell.

The specified capacitance overrides the capacitance from the wire load models and SPEF.

You can use the [report_ports](#) command to view capacitance of asserted ports. The pin load appears in the report as `port_cap` in the assertion column and the wire load appears as `port_wire_cap` in the assertion column.

In the timing report (generated using the [report_timing](#) command), the load value for ports is displayed as a sum of the `set_load` value on a port and the existing wire capacitance obtained from the SPEF file, and pin capacitance obtained from the library files.

Parameters

<code>-help</code>	Prints out the command usage.
<code>capacitance_value</code>	Specifies the value for the capacitance. The capacitance is in units from the target technology library.
<code>-min</code>	Specifies the minimum capacitance. The minimum capacitance is used to compute best-case delays in best-case worse-case (BcWc) analysis mode and early path delays in OCV analysis mode.
<code>-max</code>	Specifies the maximum capacitance. The maximum capacitance is used to compute worst-case delays in BcWc analysis mode and late path delays in OCV mode. It is also used to check the maximum capacitance design rule.
<code>object_list</code>	Specifies a list of ports or nets in the top cell.
<code>-pin_load</code>	Specifies the external pin capacitance for a port. You can use this parameter for ports only. For ports, the asserted capacitance is either the external pin capacitance when the <code>-pin_load</code> parameter is specified, or the external wire capacitance when the <code>-wire_load</code> parameter is specified, or the total external capacitance when neither option is specified.
<code>-subtract_pin_load</code>	This parameter can be used only if nets are specified in the object list. When this parameter is specified, the total net capacitance value will be the same as set by the <code>set_load capacitance_value object_list</code> command. When this parameter is not specified, the asserted load value will only change the wire capacitance value from the SPEF. The total net capacitance value will be the <code>set_load</code> capacitance value plus the pin capacitance from the library file.
<code>-wire_load</code>	Interprets the specified capacitance as the external wire capacitance of the port. Use this option only for ports. Use this option with the <code>port_net_list</code> argument if the list contains only ports.

Examples

Here are some examples.

- The following command sets an external minimum pin capacitance of 1.5 units on the `in1 in2` ports:

```
tempus > set_load 1.5 -min -pin_load [get_ports {in1 in2}]
```
- The following command sets an external minimum wire capacitance of 1.2 units on the `in1 in2`

ports:

```
tempus > set_load 1.2 -min -wire_load [get_ports {in1 in2}]
```

- The following command sets an external maximum pin capacitance of 2.6 units on the `in1 in2` ports:

```
tempus > set_load 2.6 -max -pin_load [get_ports {in1 in2}]
```

- The following command sets an external maximum wire capacitance of 3.0 units on the `in1 in2` ports:

```
tempus > set_load 3.0 -max -wire_load [get_ports {in1 in2}]
```

- The following command reports the capacitances set on the ports, as shown in the example below:

```
tempus > report_ports -pin [get_ports {in1}]
```

```
-----
Early Late
-----
```

Pin Name	Dir	Assertion	Clock Name	Rise	Fall	Rise	Fall
in1	IN	drive_resistance	*(D) (P)	2.000	2.000	2.000	2.000

```
-----
Pin    Dir    Assertion    Value
Name
-----
in1    IN      port_cap      ( 1.500 : : 2.600 )
in1    IN      port_wire_cap ( 1.200 : : 3.000 )
-----
```

Related Commands

- [all_outputs](#)
- [report_ports](#)
- [report_timing](#)
- [reset_load](#)
- [set_drive](#)
- [set_resistance](#)

set_logic_one

```
set_logic_one
-help
ports_pins
```

Sets ports or pins to logic 1 in the design.

Parameters

-help	Outputs a brief description that includes type and default information for each <code>set_logic_one</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man set_logic_one</code>
ports_pins	Specifies the list of the ports or pins that are to be set to logic 1.

Examples

- The following command sets the specified port to logic 1:
`tempus > set_logic_one [get_ports {din[1]}]`

Related Commands

- `get_ports`

set_logic_zero

set_logic_zero

-help

ports_pins

Sets ports or pins to logic 0 in the design.

Parameters

-help	Outputs a brief description that includes type and default information for each <code>set_logic_zero</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man set_logic_zero</code> .
ports_pins	Specifies the list of the ports or pins that are to be set to logic 0.

Examples

- The following command sets the specified port to logic 0:

```
tempus > set_logic_zero [get_ports {din[0]}]
```

Related Commands

- [get_ports](#)

set_max_capacitance

```
set_max_capacitance
[-help]
capacitance_limit
object_list
[-clock_path]
[-data_path]
[-fall]
[-override]
[-rise]
```

Sets the maximum capacitance limit on the specified instance pin, ports of the top cell, the specified designs, and/or clock waveforms. The pins of cell instances of timing libraries will have their own maximum constraint. The software will pick the more aggressive value of the two.

When set on a port, the total capacitance of the net connected to that port must be less than or equal to the specified limit. Maximum capacitance limits also come from library objects, which is the default maximum capacitance limit of libraries and the maximum capacitance limit on library cell pins. Pins in the design inherit the limits from libraries.

Note: The limit set on the top cell applies to its ports. Thus, if maximum capacitance limits are set on the top cell and its ports, the most constraining limit will apply to the ports.

When limits are set on a clock waveform, you can use `-clock_path`, `-data_path`, `-rise`, and `-fall` parameters to provide more specifications.

The `set_max_capacitance` constraint on clock objects will be honored only by the `report_constraint` command for DRV violations. The constraint will not be used in optimizations.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-clock_path</code>	Applies the maximum capacitance constraint only to pins in the clock network of the clocks defined in the object list.
<code>-data_path</code>	Applies the maximum capacitance constraint only to data path pins, where the path is launched by the clocks defined in the object list.
<code>-fall</code>	Applies the maximum capacitance constraint only to falling capacitance.
<code>-rise</code>	Applies the max capacitance constraint only to rising capacitance.
<code>capacitance_limit</code>	Specifies the value for the minimum capacitance limit. The limit must be greater than or equal to zero. The limit is in capacitance units from the target technology library.

<i>object_list</i>	Specifies a list of objects - ports, instance pins, library cell pins, designs, and/or clock waveforms.
<i>-override</i>	Overrides the capacitance limit from other sources, such as, library or constraints. This parameter can be used to force a specific constraint value for a specified list of instance pins/port/library cell pins - this overrides the maximum/minimum values set prior to this setting.

Examples

Here are some examples.

- The following example sets a maximum capacitance limit of 1.2 units on all the ports of the top cell and a maximum capacitance limit of 0.5 units on the port named `d_out`:

```
tempus > set_max_capacitance 1.2 [get_ports *]
tempus > set_max_capacitance 0.5 d_out
```

- The following example uses the `report_ports` command to show the maximum capacitance limit set on ports:

```
tempus > set_max_capacitance 0.9 [get_ports *]
tempus > report_ports -type port_cap_limit -pin [get_ports *]
```

The following report is displayed:

```
-----
Pin    Dir  Assertion      Value
Name
-----
CLK2   IN   port_cap_limit  ( : 0.900 )
CLK1   IN   port_cap_limit  ( : 0.900 )
in2    IN   port_cap_limit  ( : 0.900 )
in1    IN   port_cap_limit  ( : 0.900 )
out2   OUT   port_cap_limit  ( : 0.900 )
out1   OUT   port_cap_limit  ( : 0.900 )
-----
```

Related Commands

- `get_ports`
- `report_ports`
- `reset_max_capacitance`
- `set_max_transition`
- `set_max_fanout`
- `report_constraint`

set_max_delay

```
set_max_delay
[-help]
<value>
[-comment <string>]
[-fall]
[-fall_through <through_list>]
[-no_segment]
[-rise]
[-rise_through <through_list>]
[-through <through_list>]
[-combinational_from_to | -ignore_clock_latency ]
[-from <from_list> | -rise_from <from_list> | -fall_from <from_list>]
[-to <to_list>] | -rise_to <to_list> | -fall_to <to_list>]>
```

Specifies a maximum delay for a timing path. The `set_max_delay` command takes precedence over other path exceptions specified for the same path, except for the `set_false_path` command.

Use either the `-from`, `-to`, or `-through` option with the `set_max_delay` command to identify the paths. Specifying both the `-from` and `-to` options identifies a path with specific starting and ending points.

When you specify a sequential cell name for starting point of the timing path, the valid path starts from the clk pin. This means the latency and delay from clk to output Q is also included in the delay. For path segmentation, you can specify the path to start from Q output of the register by adding `/Q` after the cell name. In this case, path from clk to Q is not considered in the calculation.

Use only one `-from`, `-rise_from`, or `-fall_from`, `-to`, `-rise_to`, or `-fall_to` parameter per command. You can use multiple `-through` parameters in the same command.

Note: If the `set_max_delay` constraint is added incrementally in SI mode, an explicit `update_timing -full` command is required (before `report_timing` or `report_constraint`) for SI full accuracy.

Parameters

-help	Prints out the command usage.
-------	-------------------------------

- combinational_from_to	Defines a path defined by -from/to pins as combinational. When this parameter is used, the path between the -from/-to pins is treated as combinational, that is, both the source/capture clock latencies are excluded from slack computation. The presence of -from/-to pins is required for this option to take effect; the -from/-to pins can be sequential or combinational start/end pins. The -combinational_from_to parameter will be deprecated in a future release. The -ignore_clock_latency parameter will continue to work in the current release, but for compatibility with previous releases you should migrate your scripts to use the -ignore_clock_latency parameter. You can use the following global variables to match the semantics between the -ignore_clock_latency and -combinational_from_to parameters: • timing_constraint_path_delay_exclude_check_delay_from_ignore_clock_latency • timing_constraint_path_delay_exclude_io_delay_from_ignore_clock_latency
-comment <i>string</i>	Allows you to insert comments. The comments are read but are ignored with a time warning message. They are not used during timing analysis reporting nor are they written out during saving of constraints.
-fall	Apply the assertion at the falling edge of the signal on the path endpoints.

```
{-from |-rise_from |-  
fall_from} from_list
```

Specifies a list of timing paths start points. The *from_list* argument can contain object IDs or hierarchical names relative to the current module or clock name. Pins can be on intermediate hierarchical boundaries. The *from_list* argument can be a collection.

The valid start points are:

- Input port
- Bidirectional port
- Clock pin of sequential cell
 - Clock pin of sequential cell is one of the following:
 - Structural clock pin that is part of latch or flip-flop.
 - Inferred clock pin in cells like macro cells. A pin is an inferred clock pin associated with one of the following:
 - Reference end of setup check
 - Reference end of hold check
 - Reference end of removal check
 - Reference end of recovery check
 - Minimum period check
 - Minimum pulse width check
- Pin associated with `set_input_delay`

Specify one of the following:

- **-from:** Applies the maximum delay constraint to all paths originating from the *from_list* list. If a clock is specified as the `-from` object, all primary inputs and registers clocked by that clock are used as start points. By default, the `-from` option applies the constraint to both the rising and falling edges.
- **-rise_from:** Applies the maximum delay constraint at the rising edge of the signal on the from pins.
- **-fall_from:** Applies the maximum delay constraint at the falling edge of the signal on the from pins.

<code>-ignore_clock_latency</code>	When this parameter is specified, the software does not count the launch/cap latency delays into the timing path delay. The <code>-ignore_clock_latency</code> parameter is an industry-standard construct introduced in SDC 2.1 to control the handling of clock-related timing with respect to the <code>set_max_delay/set_min_delay</code> constraints.
<code>-rise</code>	Applies the assertion at the rising edge of the signal on the path endpoints.
<code>{-through -</code> <code>rise_through -</code> <code>fall_through}</code> <code>through_list</code>	Specifies a list of timing paths through points. The <code>through_list</code> argument can contain either object IDs or hierarchical names relative to the current module or clock names. The pins can be on intermediate hierarchical boundaries. The <code>through_list</code> argument can be a collection. Specify one of the following: • <code>-through</code>: When you use this parameter without using the <code>-from</code> or <code>-to</code> options, paths that go through the specified pins have the specified delay. When you use both the <code>-from</code> and <code>-to</code> options, any path starting at any pin on the <code>-from</code> list and going through any point on the <code>-through</code> pin list and going to any pin on the <code>-to</code> list is affected by the constraint. By default, the <code>-through</code> option applies the constraint to both the rising and falling edges. • <code>-rise_through</code>: Applies the assertion at the rising edge of the signal on the pins. • <code>-fall_through</code>: Applies the assertion at the falling edge of the signal on the pins.
<code>-no_segment</code>	Specifies the constraint as non-segmenting. When specified, the software allows path delay checks without path segmenting.

<pre>{-to -rise_to -fall_to} to_list</pre>	Specifies a list of timing paths end points. The <i>to_list</i> argument can contain either object IDs or hierarchical names relative to the current module or clock names. The pins can be on intermediate hierarchy boundaries. The <i>to_list</i> argument can be a collection. The valid end points are: • Output port • Bidirectional port • Data pin of sequential cell - Data pin of sequential cell is one of the following: ◦ Structural data pin that is part of latch or flip-flop. ◦ Inferred data pin in cells like macro cells. A pin is an inferred data pin if it is associated with one of the following: ■ Signal end of setup check ■ Signal end of hold check ■ Signal end of non-sequential setup check ■ Signal end of non-sequential hold check ■ Signal end of removal check ■ Signal end of recovery check • Pin associated with <i>set_external_delay</i> • Data pin of clock gating cell Select from one of the following: • <i>-to</i>: Applies the maximum delay constraint to all paths ending in the <i>-to</i> object. If a clock is specified as the <i>-to</i> object, all primary outputs and registers clocked by that clock are used as endpoint. By default, the <i>-to</i> option applies the constraint to both the rising and the falling edges. • <i>-rise_to</i>: Applies the maximum delay constraint at the rising edge of the signal to the pins. • <i>-fall_to</i>: Applies the maximum delay constraint at the falling edge of the signal to the pins.
<i>value</i>	Specifies the maximum delay value.

Examples

- The following command specifies that all paths from `in1` to `out1` must have delays less than 10 units:

```
tempus > set_max_delay 10 -from in1 -to out1
```

Related Topics

- [Path Exception Priorities](#)

Related Commands

- `create_clock`
- `set_false_path`
- `set_input_delay`
- `set_min_delay`
- `set_multicycle_path`
- `set_output_delay`
- `reset_path_exception`
- `report_path_exceptions`

set_max_fanout

```
set_max_fanout
[-help]
fanout_limit
object_list
[-override]
```

Sets the maximum fanout load limit constraint on the specified instance pins, library cell pins, ports of the top cell, clock waveforms, and/or specified modules. When pins of cell instances have their own maximum constraint defined in the library, the software will pick the more aggressive value of the two.

When set on a port, the fanout load of the net connected to the port must be less than or equal to the specified limit value to meet the constraint. The fanout load of a net is the sum of the fanout loads of the pins driven by the net. When set on a module, the limit applies to all the nets of that module. The libraries and library cell pins may also have the maximum fanout load limit. In this case, the highest constraints limit is applied to the nets.

Use the [report_ports -type fanout_load_limit](#) command to view the maximum fanout load limits set on ports of the top cell.

Fanout load limit can be asserted on input ports, output pins, bidirectional pins, or ports and library pins. You cannot apply the constraint on input pins and output ports.

Parameters

<code>-help</code>	Prints out the command usage.
<code>fanout_limit</code>	Specifies the value of the maximum fanout load limit constraint. This is a float number and it must be greater than or equal to zero.
<code>object_list</code>	Specifies a list of objects - input ports, instance output pins, designs, or library cell output pins.
<code>-override</code>	Overrides the fanout limit from other sources, such as, library or constraints. This parameter can be used to force a specific constraint value for a specified list of instance pins/port/library cell pins - this overrides the maximum/minimum values set prior to this setting.

Examples

- The following example sets a maximum fanout load limit of 5.0 units on all the ports of the top cell:

```
tempus > set_max_fanout 5.0 [get_ports *]
```

- The following example shows a report of the maximum fanout load limit set on ports using the `report_ports` command:

```
tempus > set_max_fanout 5.0 [get_ports *]  
tempus > report_ports -type fanout_load_limit -pin [get_ports *]
```

The following report is displayed:

Pin Name	Dir	Assertion	Value
CLK2	IN	fanout_load_limit	5.000
CLK1	IN	fanout_load_limit	5.000
in2	IN	fanout_load_limit	5.000
in1	IN	fanout_load_limit	5.000

Related Commands

- `get_ports`
- `report_ports`
- `reset_max_fanout`
- `set_fanout_load`
- `set_max_capacitance`
- `set_max_transition`
- `report_constraint`

set_max_time_borrow

```
set_max_time_borrow
[-help]
borrow_value
object_list
[-interface]
```

Specifies the maximum time that can be borrowed by one stage from the next logic stage following a latch to meet timing constraints. Time borrowing is sometimes called cycle stealing. Time borrowing is performed automatically when performing timing analysis.

Use the `set_max_time_borrow` assertion on pins or waveforms. If the assertion is placed on a clock waveform, all latches triggered by this clock signal get the specified maximum time borrow limit. If both the clock waveform arriving at the clock pin of a latch and the clock pin itself have time borrow limit constraints, the maximum borrow limit on the latch is decided by the assertion on the latch clock pin.

Parameters

<code>-help</code>	Prints out the command usage.
<code>borrow_value</code>	Specifies the maximum amount of time that can be borrowed. The minimum value allowed is 0, which disables time borrowing. The maximum allowed is equal to the pulse width minus the setup time: $\text{Max} = \text{pulse width} - \text{setup}$
<code>-interface</code>	Specifies the IO paths for which maximum time will be borrowed.
<code>object_list</code>	Specifies the list of instance pins, instances, or clocks on which the borrow value is applied. The <code>object_list</code> argument can be a collection.

Examples

Here are some examples.

- tempus > `set_max_time_borrow 2 [get_cells {ff1 ff2 ff3}]`
- tempus > `set_max_time_borrow 1.7 I4/g`
- tempus > `set_max_time_borrow 2.5 [get_clocks CLK1]`

Related Commands

- `get_clocks`
- `get_cells`
- `get_pins`
- `reset_max_time_borrow`

set_max_transition

```
set_max_transition
[-help]
transition_limit
object_list
[-clock_path]
[-data_path]
[-fall]
[-override]
[-rise]
```

Sets the maximum slew time limit (transition) on the specified instance pin, ports of the top cell, and/or clock waveforms. The pins of cell instances of timing libraries will have their own maximum constraint. The software will pick the more aggressive value of the two.

When set on a port, the slew time at the port must be less than or equal to the specified limit. Maximum transition limits on pins can come from library objects, which are the default maximum transition of the libraries or maximum transition on library cell pins. In this case, the most constraining limit, which is the smallest, will apply on pins and ports.

Use the `report_ports -type slew_limit` command to view the maximum transition limits set on ports of the top cell.

When limits are set on a clock waveform, you can use `-clock_path`, `-data_path`, `-rise`, and `-fall` parameters to provide more specifications.

Parameters

<code>-help</code>	Outputs a brief description that includes type and default information for each <code>set_max_transition</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man set_max_transition</code> .
<code>-clock_path</code>	Applies the max transition constraint only to pins in the clock network of the clocks defined in the object list.
<code>-data_path</code>	Applies the max transition constraint only to data path pins, where the path is launched by the clocks defined in the object list.
<code>-fall</code>	Applies the max transition constraint only to falling transitions.
<code>object_list</code>	Specifies the list of objects - ports, instance pins, library cell pins, clock waveforms, or current design.

<code>-override</code>	Overrides the transition limit from other sources, such as, library or constraints. This parameter can be used to force a specific constraint value for a specified list of instance pins/port/library cell pins - this overrides the maximum/minimum values set prior to this setting.
<code>-rise</code>	Applies the max transition constraint only to rising transitions.
<code>transition_limit</code>	Specifies the value of the maximum slew time limit (maximum transition), which must be greater than or equal to zero. The maximum slew time limit is in time units in the current session.

Examples

Here are some examples.

- The following command sets a maximum transition limit of 0.9 units on all the ports of the top cell:

```
tempus > set_max_transition 0.9 [get_ports *]
```

- The following example shows a report of the maximum transition limit set on ports using the `report_ports` command, as shown in the example below:

```
tempus > set_max_transition 0.9 [get_ports *]
```

```
tempus > report_ports -type slew_limit -pin [get_ports *]
```

The following report is displayed:

Pin Name	Dir	Assertion	Value
in1	IN	slew_limit	(: 0.9000)
in2	IN	slew_limit	(: 0.9000)
out1	OUT	slew_limit	(: 0.9000)
out2	OUT	slew_limit	(: 0.9000)
out3	OUT	slew_limit	(: 0.9000)
out4	OUT	slew_limit	(: 0.9000)

- The following command sets the maximum transition limit of 0.001 units on pins in the `all_clocks` collection:

```
tempus > set_max_transition 0.001 -clock_path [all_clocks]
```

- The following command sets the transitions to the specified value for your entire design:

```
tempus > set_max_transition value [current_design]
```

- The following command impacts `report_constraint` output as shown below:

```
set_max_transition 0.01 TDP_COE_INST/TDP_COE_GL_INST/p82
```

The following command shows output as shown below:

```
report_constraint -all_violators -drv_violation_type max_transition

#####
# format : frame 0 : split 0

Check type : max_transition

-----
Pin Name                                Required Actual  Slack    View
-----
TDP_COE_INST/TDP_COE_GL_INST/p82  0.010      1.138    -1.128   func_slow_RCMA
-----
```

Related Commands

- [get_ports](#)
- [report_ports](#)
- [reset_max_transition](#)
- [set_max_capacitance](#)
- [report_constraint](#)

set_min_capacitance

```
set_min_capacitance  
[-help]  
capacitance_limit  
object_list  
[-clock_path]  
[-data_path]  
[-fall]  
[-override]  
[-rise]
```

Sets the minimum capacitance limit on the specified ports of the top timing module and/or clock waveforms.

When set on a port, the total capacitance of the net connected to that port must be greater than or equal to the specified limit. When set on a module, the limit applies to all the nets of that module. The total capacitance of each net of the module must be greater than or equal the specified limit. The minimum capacitance limit also comes from library objects, which is the default minimum capacitance limit of libraries and minimum capacitance limit on library cell pins. Pins in the design inherit the limits from libraries. When limits are set on a module and the limit is inherited from libraries, the most constraining limit, which is the largest limit, is used.

Note: The limit set on the top timing module applies to its ports. Thus, if minimum capacitance limits are set on the top timing module and its ports, the most constraining limit will apply to the ports.

Use the `report_ports -type port_cap_limit` command to view the minimum capacitance limits set on ports.

When limits are set on a clock waveform, you can use `-clock_path`, `-data_path`, `-rise`, and `-fall` parameters to provide more specifications.

The `set_min_capacitance` constraint on clock objects will be honored only by the `report_constraint` command for DRV violations. The constraint will not be used in optimizations.

Note: This command syntax is applicable only when the flow-compatible mode is on.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-clock_path</code>	Specifies all pins that are in the network of the specific clocks in the object list.
<code>-data_path</code>	Specifies all pins that are in the data paths launched by the specific clocks in the object list.
<code>-fall</code>	Specifies falling capacitance.
<code>-rise</code>	Specifies rising capacitance for all selected pins.
<code>capacitance_limit</code>	Specifies the value for the minimum capacitance limit. The limit must be greater than or equal to zero. The limit is in capacitance units from the target technology library.
<code>object_list</code>	Specifies the list of objects - ports, instance pins, library cell pins, modules, and/or clock waveforms.
<code>-override</code>	Overrides the capacitance limit from other sources, such as, library or constraints. This parameter can be used to force a specific constraint value for a specified list of instance pins/port/library cell pins - this overrides the maximum/minimum values set prior to this setting.

Examples

- The following example sets a minimum capacitance limit of 0.02 units on all the ports of the top timing module and a minimum capacitance limit of 0.05 units on the top timing module:

```
tempus > set_min_capacitance 0.02 [get_ports *]
tempus > set_min_capacitance 0.05 [current_design]
```

The limit used for ports is 0.05, which is the most restrictive limit set on the top timing module (0.05) and the limit set on ports (0.02).

- The following example uses the `report_port` command to show the minimum capacitance limit set on ports:

```
tempus > set_min_capacitance 0.02 [get_ports *]
tempus > report_ports -type port_cap_limit -pin [get_ports *]
```

- The following command sets the minimum capacitance limit of 0.001 units on pins in the `all_clocks` collection (see example below):

```
tempus > set_min_capacitance 0.001 -data_path [all_clocks]
tempus > report_ports -type port_cap_limit -pin [get_ports *]
```

The following report is displayed:

Pin Name	Dir	Assertion	Value
CLK2	IN	port_cap_limit	(0.020 : 0.900)
CLK1	IN	port_cap_limit	(0.020 : 0.900)
in2	IN	port_cap_limit	(0.020 : 0.900)
in1	IN	port_cap_limit	(0.020 : 0.900)
out2	OUT	port_cap_limit	(0.020 : 0.900)
out1	OUT	port_cap_limit	(0.020 : 0.900)

Related Commands

- `report_ports`
- `reset_min_capacitance`
- `set_max_capacitance`
- `set_min_capacitance`
- `set_max_transition`
- `set_min_transition`
- `report_constraint`

set_min_delay

```
set_min_delay
[-help]
value
[-combinational_from_to | -ignore_clock_latency]
[{-from | -rise_from | -fall_from} from_list]
[-no_segment]
[{-through | -rise_through | -fall_through} through_list]
[{-to | -rise_to | -fall_to} to_list]
[-rise]
[-fall]
[-comment string]
```

Specifies a minimum delay for a timing path. The `set_min_delay` command takes precedence over other path exceptions specified for the same path, except for `set_false_path` command specifications.

Use either the `-from`, `-to`, or `-through` option with the `set_min_delay` command to identify the paths. Specifying both the `-from` and `-to` options identifies a path with specific starting and ending points. If a clock is specified as the `-to` object, all primary outputs and registers clocked by that clock are used as an endpoint.

When you specify a sequential cell name for starting point of the timing path, the valid path starts from the clk pin. This means the latency and delay from clk to output Q is also included in the delay. For path segmentation, you can specify the path to start from Q output of the register by adding `/Q` after the cell name. In this case, path from clk to Q is not considered in the calculation. For example, in the following set of commands, the valid path starts from different points:

```
set_min_delay 0.66 -from TR1/BR1 -to BLK/BR1/D
set_min_delay 0.66 -from TR1/BR1/Q -to BLK/BR1/D
```

For the first command, the valid path starts from clk, and for the second command the valid path starts from the Q output. In the second case, the path from clk to Q is not considered. For an example of the results generated for paths with and without path segmentation, see [Examples](#).

Use only one `-from`, `-rise_from`, or `-fall_from`, `-to`, `-rise_to`, or `-fall_to` parameter per command. You can use multiple `-through` parameters in the same command.

Parameters

<code>-help</code>	Prints out the command usage.
--------------------	-------------------------------

- combinational_from_to	Defines a path defined by -from/to pins as combinational. When this parameter is used, the path between the -from/-to pins is treated as combinational, that is, both the source/capture clock latencies are excluded from slack computation. The presence of -from/-to pins is required for this option to take effect; the -from/-to pins can be sequential or combinational start/end pins. The -combinational_from_to parameter will be deprecated in a future release. The -combinational_from_to parameter will continue to work in the current release, but for compatibility with future releases you should migrate your scripts to use the -ignore_clock_latency parameter. You can use the following global variables to match the semantics between the -ignore_clock_latency and -combinational_from_to parameters: • timing_constraint_path_delay_exclude_check_delay_from_ignore_clock_latency • timing_constraint_path_delay_exclude_io_delay_from_ignore_clock_latency
-comment <i>string</i>	Allows you to insert comments. The comments are read but are ignored with a warning message. They are not used during timing analysis reporting nor are they written out during saving of constraints.
-fall	Applies the assertion at the falling edge of the signal on the path endpoints.

<pre>-from -rise_from -fall_from from_list</pre>	Specifies a list of timing paths start points. The <i>from_list</i> argument can contain object IDs or hierarchical names relative to the current module or clock name. The <i>from_list</i> argument can be a collection. The valid start points are: • Input port • Bidirectional port • Clock pin of sequential cell - Clock pin of sequential cell is one of the following: ◦ Structural clock pin that is part of latch or flip-flop. ◦ Inferred clock pin in cells like macro cells. A pin is an inferred clock pin associated with one of the following: ■ Reference end of setup check ■ Reference end of hold check ■ Reference end of removal check ■ Reference end of recovery check ■ Minimum period check ■ Minimum pulse width check • Pin associated with set_input_delay Specify one of the following:
	• -from: Applies the minimum delay constraint to all paths originating from <i>from_list</i> list. If a clock is specified as the <i>-from</i> object, all primary inputs registers clocked by that clock are used as start points. By default, the <i>-from</i> option applies the constraints to both the rising and the falling edges.
	• -rise_from: Applies the minimum delay constraint at the rising edge of the clock on the from pins.
	• -fall_from: Applies the minimum delay constraint at the falling edge of the clock on the from pins.

<code>-ignore_clock_latency</code>	When this parameter is specified, the software does not count the launch/cap latency delays into the timing path delay. The <code>-ignore_clock_latency</code> parameter is an industry-standard construct introduced in SDC 2.1 to control the handling of clock-related timing with respect to the <code>set_max_delay/set_min_delay</code> constraints.
<code>-rise</code>	Applies the assertion at the rising edge of the signal on the path endpoints.
<code>-through -rise_through -fall_through</code> <code>through_list</code>	Specifies a list of timing paths through points. The <code>through_list</code> argument can contain either object IDs or hierarchical names relative to the current module or clock names. The pins can be on intermediate hierarchical boundaries. The <code>through_list</code> argument can be a collection. Specify one of the following:
	• <code>-through</code>: When you use this parameter without using the <code>-from</code> or <code>-to</code> options, paths that go through the specified pins have the specified delay. When you use both the <code>-from</code> and <code>-to</code> options, any path starting at any pin on the <code>-from</code> list and going through any point on the <code>-through</code> pin list and going to any pin on the <code>-to</code> list is affected by the constraint. By default, the <code>-through</code> option applies the constraint to both the rising and falling edges.
	• <code>-rise_through</code>: Applies the assertion at the rising edge of the signal on the pins.
	• <code>-fall_through</code>: Applies the assertion at the falling edge of the signal on the pins.
<code>-no_segment</code>	Specifies the constraint as non-segmenting. When specified, the software allows path delay checks without path segmenting.

<pre>-to -rise_to -fall_to to_list</pre>	Specifies a list of timing paths end points. The <i>to_list</i> argument can contain either object IDs or hierarchical names relative to the current module or clock names. The pins can be on intermediate hierarchy boundaries. The <i>pin_or_clock_list</i> argument can be a collection. The valid end points are: • Output port • Bidirectional port • Data pin of sequential cell - Data pin of sequential cell is one of the following: ◦ Structural data pin that is part of latch or flip-flop. ◦ Inferred data pin in cells like macro cells. A pin is an inferred data pin if it is associated with one of the following: ■ Signal end of setup check ■ Signal end of hold check ■ Signal end of non-sequential setup check ■ Signal end of non-sequential hold check ■ Signal end of removal check ■ Signal end of recovery check • Pin associated with <code>set_external_delay</code> • Data pin of clock gating cell Select from one of the following:
	• -to: Applies the minimum delay constraint to all paths ending in the <code>-to</code> pin. If a clock is specified as the <code>-to</code> object, all primary outputs and registers clock output pins are used as endpoint. By default, the <code>-to</code> option applies the constraint to both the rising and the falling edges of the clock.
	• -rise_to: Applies the minimum delay constraint at the rising edge of the clock to the <code>-rise_to</code> pins.
	• -fall_to: Applies the minimum delay constraint at the falling edge of the clock to the <code>-fall_to</code> pins.

<i>value</i>	Specifies the minimum delay value.
--------------	------------------------------------

Examples

- The following command specifies a minimum delay of 2.5 for paths from `in1` to `out1`:

```
tempus > set_min_delay 2.5 -from in1 -to out1
```
- The following commands show the difference in `report_timing` results when you specify a minimum delay with and without path segmentation:

```
tempus > set_min_delay 9 -from R1 -to R2/D
tempus > set_min_delay 9 -from R1/Q -to R2/D
```

The `report_timing` results for the first command with path segmentation are:

```
tempus > report_timing -early -from R1
```

```
Path 1: VIOLATED Hold Check with Pin R2/CK
Endpoint: R2/D (v) checked with leading edge of 'CLK_W_1'
Beginpoint: R1/Q (v) triggered by leading edge of 'CLK_W_1'
Path Groups: {CLK_W_1}
Other End Arrival Time 0.000
+ Hold -0.042
+ Path Delay 9.000
= Required Time 8.958
Arrival Time 0.212
Slack Time -8.745
Clock Rise Edge 0.000
+ Clock Network Latency (Ideal) 0.000
= Beginpoint Arrival Time 0.000
```

Pin	Cell	Delay	Arrival Time	Required Time	Instance
R1/CK ->	-	-	0.000	8.745	R1
R1/Q	DFF	0.144	0.144	8.889	R1
B1/Y	BUF	0.068	0.212	8.958	B1
R2/D	DFF	0.000	0.212	8.958	R2

- The `report_timing` results for the second command without path segmentation are:

```
tempus > report_timing -early -from R1/Q
```

```
Path 1: VIOLATED Hold Check with Pin R2/CK
Endpoint: R2/D (^) checked with leading edge of 'CLK_W_1'
Beginpoint: R1/Q (^) triggered by leading edge of 'CLK_W_1'
Path Groups: {CLK_W_1}
Other End Arrival Time 0.000
+ Hold -0.047
+ Path Delay 9.000
= Required Time 8.953
Arrival Time 0.056
Slack Time -8.897
Clock Rise Edge 0.000
= Beginpoint Arrival Time 0.000
```

Pin	Cell	Delay	Arrival Time	Required Time	Instance
R1/Q ->	-	-	0.000	8.897	R1
B1/Y	BUF	0.056	0.056	8.953	B1
R2/D	DFE	0.000	0.056	8.953	R2

Related Topics

- [Path Exception Priorities](#) in the *Tempus Text Command Reference*.

Related Commands

- `set_input_delay`
- `set_output_delay`
- `set_false_path`
- `set_max_delay`
- `set_multicycle_path`
- `reset_path_exception`
- `report_path_exceptions`

set_min_fanout

```
set_min_fanout
[-help]
fanout_limit
object_list
[-override]
```

Sets the minimum fanout load limit constraint on the specified input ports of the top cell.

When set on a port, the fanout load of the net connected to the port must be greater than or equal to the specified limit value to meet the constraint. The fanout load of a net is the sum of the fanout loads of the pins driven by the net. When set on a module, the limit applies to all the nets of that module. The libraries and library cell pins may also have the minimum fanout load limit. In this case the highest constraints limit is applied to the nets.

The constraint will not be used in optimizations.

Fanout load limit can be asserted on input ports, output pins, bidirectional pins, or ports and library pins. The constraint cannot be applied on input pins and output ports.

Parameters

<code>-help</code>	Prints out the command usage.
<code>fanout_limit</code>	Specifies the value of the minimum fanout load limit constraint. The specified value must be greater than or equal to zero.
<code>object_list</code>	Specifies a list of objects - ports, instance pins, modules, or library cell pins.
<code>-override</code>	Overrides the minimum fanout limit from other sources, such as, library or constraints. This parameter can be used to force a specific constraint value for a specified list of instance pins/port/library cell pins - this overrides the maximum/minimum values set prior to this setting.

Examples

- The following example sets a minimum fanout load limit of 0.001 units on all the ports of the top cell:

```
tempus > set_min_fanout 0.001 [get_ports *]
```
- The following example shows a report of the minimum fanout load limit set on ports using the `report_ports` command, as shown in the following example:

```
tempus > set_min_fanout 5.0 [get_ports *]
```



```
tempus > report_ports -type fanout_load_limit -pin [get_ports *]
```

The following report is displayed:

Pin Name	Dir	Assertion	Value
CLK2	IN	fanout_load_limit	5.000
CLK1	IN	fanout_load_limit	5.000
in2	IN	fanout_load_limit	5.000
in1	IN	fanout_load_limit	5.000

Related Commands

- `get_ports`
- `report_ports`
- `reset_min_fanout`
- `set_fanout_load`
- `set_max_capacitance`
- `set_max_transition`
- `report_constraint`

set_min_pulse_width

```
set_min_pulse_width
[-help]
[-low]
[-high]
[-waveform_aware_type {absolute | source_width_ratio}]
value
object_list
```

Selects the most restrictive value when multiple minimum pulse width checks are set.

Note: The `timing_disable_lib_pulsewidth_checks` global variable has no effect on pulse width checks created by the `set_min_pulse_width` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-high</code>	Applies the minimum pulse width check to high signal levels only. If neither the <code>-low</code> or <code>-high</code> option is specified, the value is applied to both low and high signal levels.
<code>list</code>	Specifies a list of pins or clock waveforms to which the minimum pulse width check is to be applied. If the list is not specified, the minimum pulse width check applies to all the pins in the clock tree of the current design. If a clock waveform is specified, all the pins driven by the clock are affected.
<code>-low</code>	Applies the minimum pulse width check to low signal levels only. If neither the <code>-low</code> or <code>-high</code> option is specified, the value is applied to both low and high signal levels.
<code>object_list</code>	Specifies list of objects: pins, ports, or clocks.
<code>value</code>	Specifies a non-negative floating minimum pulse width check value. No default value is assumed for minimum pulse width checks.
<code>- waveform_aware_type {absolute source_width_ratio}</code>	Specifies the type of waveform-aware pulse width check to apply as <code>absolute</code> or <code>source_width_ratio</code> .

Examples

- The following command sets a minimum pulse width, both low and high, check of 1.0 to the clock network of the current design:

```
tempus > set_min_pulse_width 1.0
```

- The following command sets a minimum pulse width low check of 2.0 to all the pins clocked by the CLK1 clock:

```
tempus > set_min_pulse_width -low 2.0 [get_clocks CLK1]
```

- The following command sets a minimum pulse width -high check of 1.0 to the CP clock pin of register ff1, as shown in the following example:

```
tempus > set_min_pulse_width 0.5 u55/CK -low
tempus > set_min_pulse_width 1.0 u55/CK -high
tempus > report_min_pulse_width -pin u55/CK
```

```
-----
SEQUENTIAL CLOCK PULSE WIDTH
-----
```

Pin	Required Pulse Width	Actual Pulse Width	Slack
u55/CK (low)	0.500	4.200	3.700
u55/CK (high)	1.000	5.800	4.800

```
tempus > report_timing -check_type pulse_width -to u55/CK
```

```
Path 1: MET User PulseWidth Check with Pin u55/CK
Ending Clock Edge: u55/CK (^) checked with leading edge of 'CLK4'
Beginning Clock Edge: u55/CK (v) triggered by trailing edge of 'CLK4'
Path Groups: {CLK4}
Other End Arrival Time 0.152
- PulseWidth 0.500
+ Phase Shift 10.000
= Required Time 9.651
- Arrival Time 5.951
= Slack Time 3.700
Clock Fall Edge 6.000
+ Clock Network Latency (Prop) -0.049
= Beginpoint Arrival Time 5.951
```

```
-----
Load    Slew    Delay  Arrival  Cell  Arc  Pin
      Time
-----
0.015  0.019    5.951  DFF      CK   v   u55/CK ->
-----
```

Related Commands

- `report_analysis_coverage`
- `report_min_pulse_width`
- `report_timing`
- `reset_min_pulse_width`

set_min_transition

```
set_min_transition  
[-help]  
transition_limit  
object_list  
[-clock_path]  
[-data_path]  
[-fall]  
[-override]  
[-rise]
```

Sets the minimum slew time limit (transition) on the specified ports of the top timing module and/or clock waveforms.

When set on a port, the slew time at the port must be greater than or equal to the specified limit. When set on a module, the limit applies to all the pins in the module and its ports if the module is the top timing module. The minimum transition limits on pins can come from library objects, which are the default minimum transition of the libraries or minimum transition on library cell pins. In this case, the constraints limit, which is the largest, will apply on pins and ports.

Use the `report_ports -type slew_limit` command to view the minimum transition limits set on ports of the top timing module.

When limits are set on a clock waveform, you can use `-clock_path`, `-data_path`, `-rise`, and `-fall` parameters to provide more specifications.

The `set_min_transition` constraint on clock objects will be honored only by the `report_constraint` command for DRV violations. The constraint will not be used in optimizations.

Note: This command syntax is applicable only when the flow-compatible mode is on.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-clock_path</code>	Specifies all pins that are in the network of the specific clocks in the object list.
<code>-data_path</code>	Specifies all pins that are in the data paths launched by the specific clocks in the object list.
<code>-fall</code>	Specifies falling capacitance.
<code>object_list</code>	Specifies the list of objects - ports, instance pins, library cell pins, modules, and/or clock waveforms.
<code>-override</code>	Overrides the transition limit from other sources, such as, library or constraints. This parameter can be used to force a specific constraint value for a specified list of instance pins/port/library cell pins - this overrides the maximum/minimum values set prior to this setting.
<code>-rise</code>	Specifies rising capacitance for all selected pins.
<code>transition_limit</code>	Specifies the value of the minimum slew time limit (minimum transition). The minimum slew time limit is in time units from the target technology library. Transition limit value (float, required) Min=-1000000 Max=1000000

Examples

- The following example sets a minimum transition limit of 0.9 units on all the ports of the top timing module, and a minimum transition limit of 0.7 units on the top timing module:

```
tempus > set_min_transition 0.9 [get_ports *]
tempus > set_min_transition 0.7 [current_design]
```

The minimum limit that applies to ports is 0.9, which is the largest constraints value.

- The following example shows a report of the minimum transition limit set on ports using the `report_ports` command, as shown in the example below:

```
tempus > set_min_transition 0.9 [get_ports *]
tempus > report_ports -type slew_limit -pin [get_ports *]
```

The following report is displayed:

```
-----
Pin      Dir  Assertion  Value
Name
-----
CLK2     IN   slew_limit  ( 0.900 : )
CLK1     IN   slew_limit  ( 0.900 : )
in2      IN   slew_limit  ( 0.900 : )
in1      IN   slew_limit  ( 0.900 : )
out2     OUT   slew_limit  ( 0.900 : )
out1     OUT   slew_limit  ( 0.900 : )
-----

tempus > report_constraint -drv_violation_type min_transition

Check type : min_transition
-----
Pin : CLK4

Transition Time: 0.003
- min_transition: 0.900
-----

slack : -0.897 ( VIOLATED )
```

Related Commands

- [set_max_capacitance](#)
- [reset_min_transition](#)
- [report_ports](#)
- [report_constraint](#)

set_mode

set_mode
[-help]
[-type cell]
list_of_modes
object_list

Specifies the Liberty timing library modes to be active for a specific instance.

A library mode is a subset of active timing arcs that is defined in the timing library. Library modes are similar to conditional timing arcs, but do not require the application of constants to set. Instead of changing a cell's mode by applying constants to its input pins, you can use the `set_mode` command to refer to sets of active arcs by name (such as `READ` or `WRITE`, for a memory cell). By default, all library modes in the timing library are active. If you specify one or more library modes using the `set_mode` command, only the timing arcs consistent with the specified mode(s) are active for the instance. All other library modes become inactive.

To reset the library modes for an instance, use the `reset_mode` command.

Parameters

-help	Prints out the command usage.
object_list	Specifies the list of instances.
list_of_modes	Specifies the name of the library mode value defined in the timing library. You can specify one or more mode values, but only one mode value per defined mode group in the Liberty library.
-type cell	Sets the library mode at the instance level. Note: Setting the library mode at the instance level is the only form of <code>set_mode</code> that is currently supported. This parameter exists to support constraints coming from external sources.

Examples

- The following command specifies that only the timing arcs consistent with the test_A library mode are active for instances of b1 (The test_A library mode is made inactive.):

```
tempus > set_mode test_mode [get_cells b1]
```

The following command reports the status of all library modes associated with b1 . It shows that test_A is active and test_B is inactive.

```
tempus > report_mode b1
```

```
-----  
Mode Group test_mode of b1  
-----
```

Mode Name	Status	Condition
--------------	--------	-----------

```
-----  
test_B   Inactive   !(Y)
```

```
test_A   ACTIVE    Y  
-----
```

If you specify the `report_inactive_arcs` command, the report shows the disabled arcs for the b1 instance are from the `test_mode` library mode:

```
-----  
Flags :  const propagated constant  
         snipped loop snipped arcs  
         disable set_disable_timing  
         disable_clock_gating set_disable_clock_gating  
         library_disable set_disable_cell_timing  
         Missing_Phase No Arc Phase Data
```

From	To	DisableType	ArcType	Reason
u8/CK ^	u8/D ^	const	hold_rising_clk_rise	u8/D = 1
u8/CK ^	u8/D v	const	hold_falling_clk_rise	u8/D = 1
u8/CK ^	u8/D ^	const	setup_rising_clk_rise	u8/D = 1
u8/CK ^	u8/D v	const	setup_falling_clk_rise	u8/D = 1

Related Commands

- [reset_mode](#)
- [report_mode](#)
- [report_inactive_arcs](#)
- [get_cells](#)

set_multicycle_path

```
set_multicycle_path
[-help]
<number_of_cycles>
[-comment <string>]
[-fall]
[-fall_through <through_list>]
[-hold]
[-rise]
[-rise_through <through_list>]
[-setup]
[-through <through_list>]
[-start | -end ]
[-from <from_list> | -rise_from <from_list> | -fall_from <from_list>]
[-to <to_list>] | -rise_to <to_list> | -fall_to <to_list>]>
```

Specifies multicycle paths between specific timing paths in a design or between clock domains. The multicycle paths are timing paths that require more than one clock period for execution. By default, all the paths are considered as one cycle paths for setup and hold check evaluation. The `set_multicycle_path` command allows you specify additional cycles for the paths.

Parameters

<code>-help</code>	Prints out the command usage.
<code>number_of_cycles</code>	Specifies the number of cycles allowed for the path.
<code>-comment string</code>	Allows you to insert comments. The comments are read but are ignored with a one-time warning message. They are not used during timing analysis reporting nor are they written out during saving of constraints.
<code>-fall</code>	Applies the assertion at the falling edge of the signal on the path endpoints.
<code>[-fall_through <through_list>]</code>	Applies the assertion at the falling edge of the signal on the through pins

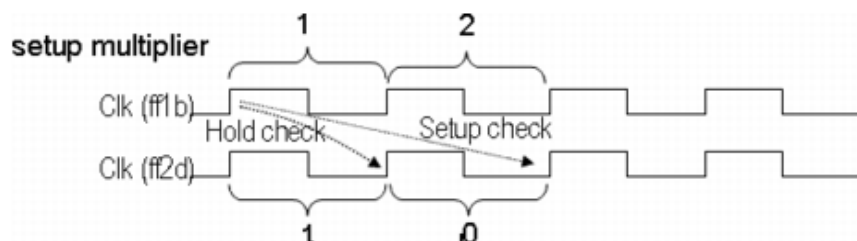
-from <from_list> - rise_from <from_list> - fall_from <from_list>	Specifies the clocks, ports, pins, or sequential instances that are at the start of multicycle paths. The port direction can be either input or inout. The pins must be valid path start points, such as sequential instance clock pins, latch data pins, or pins with specified input delays. Intermediate hierarchical pins should not be specified as -from in path exceptions unless those are valid begin points. Even if the specified constraint is accepted, timing analysis will ignore such (set_false_path/set_multicycle_path) constraints. The <i>from_list</i> argument can contain either collection IDs or object names. If the -from parameter is used without -to, all the paths originating from the <i>from_list</i> will be multicycle paths. Use either the -from, -to, or -through parameter to identify the paths. Specifying both the -from and -to parameters identifies a path with specific starting and ending points. You should use only one -from, -rise_from, or -fall_from parameter per command.
-hold -setup	Specifies whether the multicycle adjustment applies to the setup/hold check. The -setup parameter applies multicycles to the setup check. The hold check is affected, but the hold multiplier does not change. The -hold parameter applies multicycles to hold check only. <i>Default:</i> If you do not specify the -setup or -hold parameter, the software considers the specified multicycle value for the setup check and uses 0 for the hold check.
-rise	Applies the assertion at the rising edge of the signal on the path endpoints.
-rise_through <through_list>	Applies the assertion at the rising edge of the signal on the through pins.
-start -end	Specifies whether the multicycle adjustment is to be applied relative to the starting clock edge for the path or relative to the ending clock edge for the path. Use these options for multiple clocks designs. By default, the setup check is relative to the end clock and the hold check is relative to the start clock. Note: The -start and -end parameters are mutually exclusive; you cannot specify them together. You can set the <code>timing_enable_sdc_compatible_data_check_mcp</code> global variable to <code>true</code> to enable checking for SDC compatible data checks. When the -start parameter is specified, the parameter reverses the reference clock by one clock cycle.

<pre>-through <through_list></pre>	Applies the assertion to both the rising and the falling edges. This parameter specifies the ports, pins, instances, or nets that the path goes through. The pins can be on intermediate hierarchical boundaries. The <i>through_list</i> variable can contain either collection IDs or object names. When used without the <i>-from</i> or <i>-to</i> option, all paths that go through the <i>-through</i> pins are multicycle paths. When used with both the <i>-from</i> and <i>-to</i> options, any path starting at any pin on the <i>-from</i> list and going through any point on the <i>-through</i> pin list and going to any point on the <i>-to</i> list is affected by the assertion. You can use only one <i>-from</i> and <i>-to</i> option, but you can specify several <i>-through</i> options in the same command. By default, the <i>-through</i> option applies the assertion to both the rising and the falling edges. Using the <i>-rise_through</i> option applies the assertion at the rising edge of the signal on the through pins. Using the <i>-fall_through</i> option applies the assertion at the falling edge of the signal on the through pins.
<pre>-to <to_list> -rise_to <to_list> - fall_to <to_list></pre>	Specifies the clocks, ports, pins, or sequential instances that are at the end of the multicycle paths. The port direction can be either output or inout. Pins must be valid path end-points such as sequential instance data pins or pins on which output delays have been specified. Pins can be on intermediate hierarchical boundaries. The <i>to_list</i> variable can contain either collection IDs or object names. If the <i>-to</i> option is used without the <i>-from</i> option, all paths ending in the <i>-to</i> pin list are multicycle paths. Use only one <i>-to</i> option per command. Use only one <i>-to</i>, <i>-rise_to</i>, or <i>-fall_to</i> option per command.

Examples

- The following command sets all paths between ff1b + ff2d to two cycles for setup:

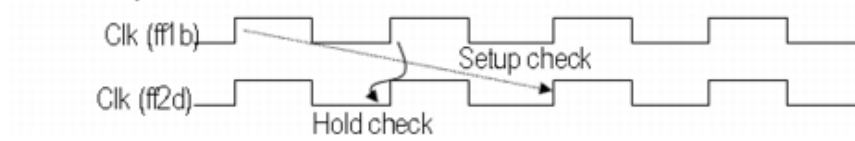
```
tempus > set_multicycle_path -setup 2 -from {ff1b} -to {ff2d}
```



- The following command moves hold check to start clock preceding edge:

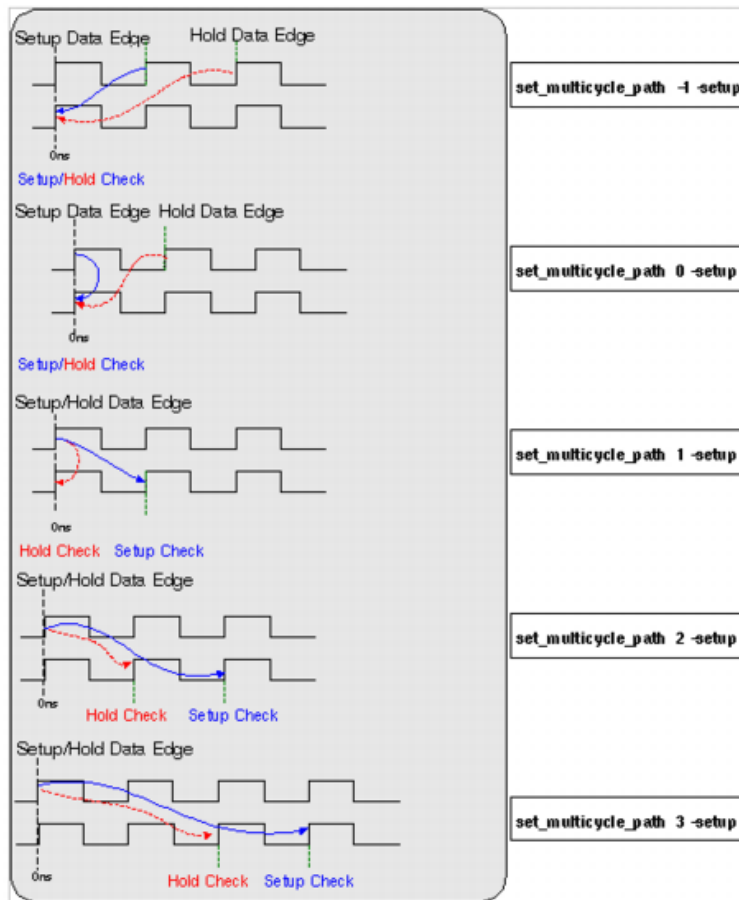
```
tempus > set_multicycle_path -hold 1 -from {ff1b} -to {ff2d}
```

hold multiplier



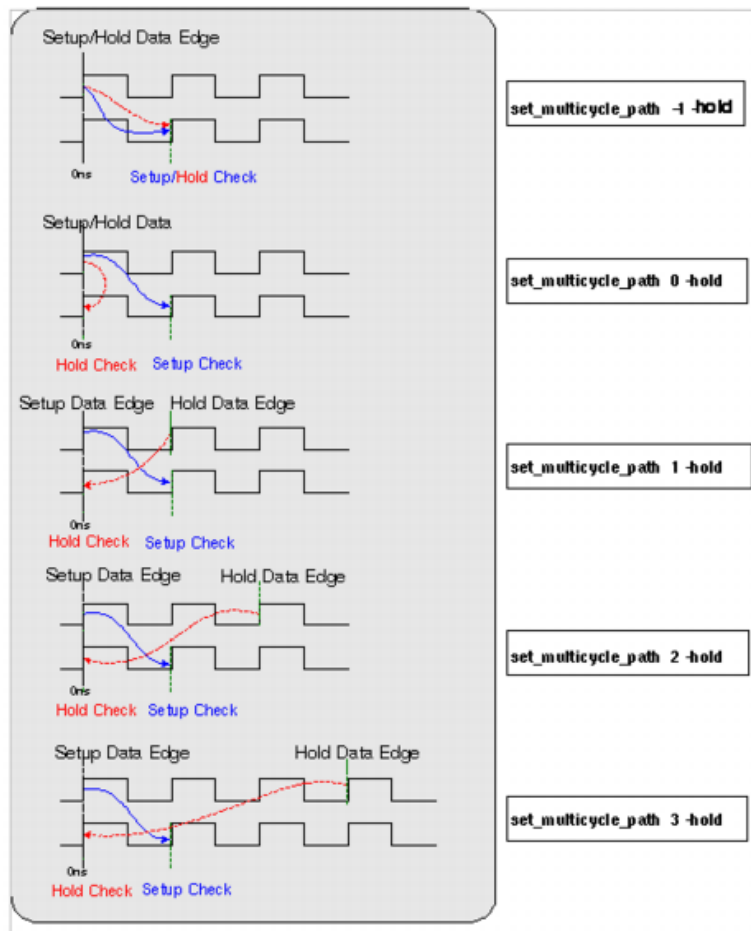
- The following commands show setup multiplier values from -1 to 3:

```
tempus > set_multicycle_path -1 -setup
tempus > set_multicycle_path 0 -setup
tempus > set_multicycle_path 1 -setup
tempus > set_multicycle_path 2 -setup
tempus > set_multicycle_path 3 -setup
```



- The following commands show hold multiplier values from -1 to 3:

```
tempus > set_multicycle_path -1 -hold
tempus > set_multicycle_path 0 -hold
tempus > set_multicycle_path 1 -hold
tempus > set_multicycle_path 2 -hold
tempus > set_multicycle_path 3 -hold
```



Related Topics

- [Path Exception Priorities](#)

Related Commands

- `set_false_path`
- `set_max_delay`
- `set_min_delay`
- `reset_path_exception`
- `report_path_exceptions`

set_normalized_driver_waveform_lib

```
set_normalized_driver_waveform_lib
[-help]
[-fall_waveform string]
[-rise_waveform string]
-ref_lib string
-libs string
```

Specifies normalized driver waveform for a library. The normalized driver waveform given in the reference library is applied to all the specified libraries.

Note: This command must be specified after the design is loaded, that is, after `set_top_module` command has been run. For this command to take effect, you should set the `timing_generate_normalized_driver_waveform` global variable to `false` before loading libraries.

Parameters

-help	Prints out the command usage.
-fall_waveform <i>string</i>	Specifies the name of the fall normalized driver waveform.
-libs <i>string</i>	Specifies a list of space separated libraries to which normalized driver waveform is to be applied.
-ref_lib <i>string</i>	Specifies the reference library from which normalized driver waveform is to be selected.
-rise_waveform <i>string</i>	Specifies the name of the rise normalized driver waveform.

Examples

- The following example shows that normalized driver waveform given in \$libDir/libs/ref.lib will be applied to all the cells in \$libDir/1.lib \$testDir/2.lib \$libDir/libs/3.lib:

```
tempus > set_normalized_driver_waveform_lib -libs "$libDir/1.lib $testDir/2.lib
$libDir/libs/3.lib" -ref_lib $libDir/libs/ref.lib
```


Related Commands

- `set_top_module`
- `timing_generate_normalized_driver_waveform`

set_output_delay

```
set_output_delay
[-help]
-clock clock_name
[-clock_fall]
[-rise]
[-fall]
[-max]
[-min]
[-add_delay]
[-network_latency_included]
[-source_latency_included]
[-reference_pin pin_name]
[-level_sensitive]
[-group_path group_name]
delay_value
port_or_pin_list
```

Specifies the data required time on output ports, hierarchical ports, or internal output pins.

Parameters

-help	Prints out the command usage.
-add_delay	Adds delay information to the existing output delay. The additional delay information can be for multiple paths relative to different clocks or clock edges leading to an input port or internal input pins. If you do not use the <code>-add_delay</code> option, the software uses the last <code>set_output_delay</code> related to a pin when there are multiple <code>set_output_delay</code> statements for the same pin.
-clock <i>clock_name</i>	Specifies that the delay is relative to the specified clock. You cannot use the <code>set_output_delay</code> command without the <code>-clock</code> parameter. Such constraints are accepted but are ignored during timing analysis. When you specify multiple clocks, an error occurs and the constraint is ignored.

<code>-clock_fall</code>	Specifies that the delays are with respect to the falling edge of the clock waveform. If you do not specify the <code>-clock_fall</code> option, the delay is with respect to the rising edge of the clock waveform. You cannot specify the <code>-clock_fall</code> parameter without the <code>-clock</code> parameter.
<code>delay_value</code>	Specifies the value for the path delay.
<code>-fall</code>	Specifies that the output delay refers to the falling edge of the signal at specified ports or pins. If you do not specify the <code>-fall</code> option, the output delay applies to both the edges.
<code>-group_path group_name</code>	Specifies the group name where paths ending at the specified ports or pins are to be added. The <code>-group_path</code> parameter is not honored in multi-mode multi-corner (MMMC) designs. If specified, a warning is issued and the constraints are applied as if the <code>-group_path</code> parameter is not specified.
<code>-level_sensitive</code>	Models output delay that is captured by a positive level sensitive latch. For setup checks, the timing system checks the arrival time at the output port with the opening edge of the latch. The timing system treats any arrival time exceeding the opening edge as negative slack. For hold checks, the timing system performs the checks with respect to the closing edge of the latch. Use the <code>-clock</code> parameter to model a positive level sensitive latch, or the <code>-clock_fall</code> parameter to model a negative level sensitive latch. You cannot use the <code>-level_sensitive</code> parameter without the <code>-clock</code> parameter.
<code>-max</code>	Specifies that the delay refers to the setup analysis. If you do not specify the <code>-max</code> parameter, the output delay applies to both setup and hold analysis.
<code>-min</code>	Specifies that the delay refers to the hold analysis. If you do not specify the <code>-min</code> parameter, the output delay applies to both setup and hold analysis.
<code>-network_latency_included</code>	Specifies that the clock network latency should not be added to the output delay value. If you do not specify this option, the clock network latency of the related clock is added to the output delay value. This parameter has no effect if the specified clock is in propagated mode.
<code>port_or_pin_list</code>	Specifies a list of output port or pin names. The <code>port_or_pin_list</code> argument can be a collection.

<code>-reference_pin pin_name</code>	Adds source and network latency of the specified reference pin to the output delay value in propagated mode. A reference pin is a leaf pin in the fanout cone of the clock source of the clock that you specify using the <code>-clock</code> parameter. You cannot specify multiple pins for <code>-reference_pin</code> parameter. The command issues a warning and accepts the first pin. You cannot use the <code>-reference_pin</code> parameter with the <code>-network_latency_included</code> or <code>-source_latency_included</code> parameters. You cannot use the <code>-reference_pin</code> parameter to specify hierarchical pins.
<code>-rise</code>	Specifies that output delay refers to the rising edge of the signal at the specified ports or pins. If you do not specify the <code>-rise</code> option, the output delay applies to the both edges.
<code>-source_latency_included</code>	Specifies that the clock source latency should not be added to the output delay value. With different delay values for <code>-rise/-fall</code> and <code>-min/-max</code>, only the last specified combination of <code>-network_latency_included</code> and <code>-source_latency_included</code> parameters is honored. The specified combination can be one of the following: - Either <code>-network_latency_included</code> or <code>-source_latency_included</code> - Both the parameters - None of these

Examples

- The following example sets the input and output delays for the bidirectional INOUT2 port. The input signal arrives at INOUT2 at 4.5 units after the falling edge of CLKB. The output signal is required at INOUT2 at 2.5 units before the rising edge of CLKD:

```
tempus > set_input_delay 4.5 -clock CLKB -clock_fall { INOUT2 }
tempus > set_output_delay 2.5 -clock CLKD { INOUT2 }
```

- The following example shows how to use the `-group_path` option to add ports into a named group:

```
tempus > set_output_delay 2.5 -max -clock CLK -group_path busC {busC[*]}
```

Related Commands

- `all_outputs`
- `create_clock`
- `report_ports`
- `reset_output_delay`
- `set_input_delay`
- `set_load`
- `report_path_exceptions`

set_pll_timing

```
set_pll_timing  
[-help]  
inst_name  
-feedback_pin feedback pin name  
-output_clock clock name  
-output_pin output pin name  
[-reference_edge {rise | fall | both}]  
-reference_pin reference pin name
```

Identifies phase locked loop (PLL) insertion delays and the associated generated clock(s) that require adjustment. To remove the effect of clock tree latency, PLLs can be used to adjust the delay on the PLL output clock waveform so that the effective edges of the clock at the clock tree endpoints are aligned with the reference edges of the input clock to the PLL.

When you use the `set_pll_timing` command, the software uses the settings to automatically calculate and apply a delay offset to a PLL output clock to compensate for internal clock tree latency.

The `set_pll_timing` command works in conjunction with a generated clock definition to provide the correct clock timing at the output of the PLL. To use this command, you must first assert a generated clock on the `set_pll_timing -output_pin`. The `-source` option of the `create_generated_clock` command must also match the `-reference_pin` specified by the `set_pll_timing` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>inst_name</code>	Specifies the instance name or the object ID of a PLL cell, on the basis of which the reference pins, feedback pins, or the output pins will be searched.
<code>-feedback_pin</code> <code>feedback pin name</code>	Specifies the name of the PLL feedback pin.
<code>-output_clock</code> <code>clock name</code>	Specifies the generated clock on the PLL output.
<code>-output_pin</code> <code>output pin name</code>	Specifies the PLL clock output pin name.
<code>-reference_edge</code> { <code>rise</code> <code>fall</code> <code>both</code> }	Defines adjustment edge(s). When this parameter is specified, the software selects the edges of the clocks at the reference and feedback pins that need to be aligned. If the <code>rise</code> or <code>fall</code> option is specified, then the clock latency from that reference edge to the related edge at the feedback pin will be used. In this case, both the rise and fall edges will use the same latency adjustment. If both of these options are specified, then the software will measure separate latency adjustments for both the rise and fall edges. <i>Default:</i> <code>rise</code>
<code>-reference_pin</code> <code>reference pin name</code>	Specifies the PLL reference pin name.

Examples

Here are some examples.

- The following example shows the defined clocks in a sample design:

```
tempus > create_clock [get_ports {clk_in}] -name CLK -period 6 -waveform {0 3}
tempus > create_generated_clock -name GCLK1 -source [get_pins {mypll/CLK}] -multiply_by 1 -master_clock [get_clocks {CLK}] -add [get_pins {mypll/GCLK}]
```

The following `report_timing` command shows the insertion delay values at various points in the timing report:

```
tempus > report_timing -from_rise mypll/GCLK -to_rise mypll/CEN -point_to_point
```

Path 1:Endpoint: mypll/CEN (^) (unconstrained output)

Beginpoint: mypll/GCLK (^) triggered by leading edge of 'GCLK1'

Arrival Time 0.496

Analysis View: default_analysis_view_setup

Pin	Cell	Delay	Arrival Time	Generated Clock Adjustment

mypll/CLK			0.112	
mypll/GCLK ->	PLL	0.133	0.246	GCLK1 Adj. = 0.000
clkTreeL0/Y	CLKBUF1	0.139	0.385	
clkTreeL1_1/Y	CLKBUF1	0.118	0.503	
mypll/CEN ->	PLL	-0.007	0.496	

From the above table, it can be seen that the rise insertion delay of clock CLK at pin mypll/CLK is 0.112ns.

Now, the following command will adjust the source latency of clock GCLK1, such that the edges at mypll/CLK and mypll/CEN are aligned, that is, the clock's rising edge should reach mypll/CEN at 0.112ns only:

```
tempus > set_pll_timing [get_cells mypll] -reference_pin CLK -feedback_pin CEN -
output_pin GCLK -output_clock [get_clocks GCLK1]
```

- The following `report_timing` command shows the source latency of GCLK clock as -0.138ns, and the clock reaches mypll/CEN pin at 0.112ns (same as on mypll/CLK):

```
tempus > report_timing -from_rise mypll/GCLK -to_rise mypll/CEN -point_to_point
```

Path 1:Endpoint: mypll/CEN (^) (unconstrained output)

Beginpoint: mypll/GCLK (^) triggered by leading edge of 'GCLK1'

Arrival Time 0.112

Analysis View: default_analysis_view_setup

+ Source Insertion Delay 0.112

- PLL Feedback Delay 0.250

= Beginpoint Arrival Time -0.138

Pin	Cell	Delay	Arrival Time	Generated Clock Adjustment

mypll/GCLK ->			-0.138	GCLK1 Pll Adj. = 0.000
clkTreeL0/Y	CLKBUF1	0.139	0.001	

Tempus Text Command Reference

Timing Constraint Commands

clkTreeL1_1/Y	CLKBUF1	0.118	0.120
mypll/CEN ->	PLL	-0.007	0.112

Related Commands

- [create_generated_clock](#)
- [create_clock](#)
- [reset_pll_timing](#)

set_propagated_clock

```
set_propagated_clock  
[-help]  
pin_clock_list
```

Puts the `propagated_clock` assertion on the specified pin, port, or clock object. This will cause all clock endpoints in the fanout of the specified object to receive propagated clock timing unless there is a `set_clock_latency` with higher precedence. If there are multiple `set_clock_latency` and `set_propagated_clock` assertions between the clock waveform, clock root and clock endpoint, the closest assertion to the clock endpoint takes precedence.

There are two methods of clock propagation: depending on whether the clock is ideal or propagated.

- When a clock is ideal, the delay from a clock port to a register clock pin (network insertion delay) comes from design constraints using the `set_clock_latency` command.
- When a clock is propagated, the network insertion delay is computed from the actual gates and interconnects in the clock network.

To set the `propagated_clock` property on all clock waveform objects, use the following command:

```
set_propagated_clock [all_clocks]
```

When `set_propagated_clocks [all_clocks]` command is asserted, it will apply to all the currently defined clock waveform objects. This setting will not apply to clocks created later in the script. Whether all the clock endpoints receive propagated or ideal timing depends on the precedence of `set_clock_latency` in the fanout.

The `set_propagated_clock [all_clocks]` command puts the `propagated_clock` assertion on all clocks except virtual clocks. When you put the assertion on a real clock that is used to define I/O timing (with `set_input_delay -clock` or `set_output_delay -clock`), the software does not take the specified network latency into account. Instead, the software calculates the network latency, which changes the I/O timing. You can change this behavior by using the following global variable:

```
set_global timing_io_use_clock_network_latency always
```

You can use this command after creating the clock with the `create_clock` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code><i>pin_clock_list</i></code>	Specifies the objects on which you want to put the <code>propagated_clock</code> assertion. The <code><i>pin_clock_list</i></code> argument can be a collection. When a pin is specified, it affects the propagation mode for all the registers in the transitive fanout (TFO) of the pin.

Examples

- The following command puts the propagated clock assertion on the clock waveform CLK1:

```
tempus > set_propagated_clock CLK1
```

- The following command puts the propagated clock assertion on all clock waveforms:

```
tempus > set_propagated_clock [all_clocks]
```

Related Commands

- `all_clocks`
- `get_propagated_clock`
- `report_clocks`
- `reset_propagated_clock`
- `set_clock_latency`
- `set_clock_uncertainty`

set_property

```
set_property  
[-help]  
[-object_type {object_type}]  
{object_list}  
[-quiet]  
property_name  
property_value
```

Sets user-defined properties on objects.

The `set_property` command needs a property to be defined using the [define_property](#) command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>object_list</code>	Specifies a list of objects on which the property is to be applied.
<code>-object_type {object_type}</code>	Specifies the type of object on which the property is to be defined.
<code>property_name</code>	Specifies the name of the property.
<code>property_value</code>	Specifies the value of the property.
<code>-quiet</code>	Specifies to exit quietly without issuing an error.

Examples

- The following command applies a user-defined property `slk` to pins `C1/A`:

```
tempus > set_property [get_pins C1/A] slk 2
```
- The following command applies `aocv_weight` property to library arcs:

```
tempus > set_property [get_lib_arcs -of_objects slow/INVX1] aocv_weight 8
```

Related Commands

- `define_property`
- `reset_property`
- `get_property`
- `report_property`

set_pulse_clock_max_transition

```
set_pulse_clock_max_transition
[-help]
transition_limit
object_list
[-transitive_fanout]
[-rise]
[-fall]
```

Sets the maximum pulse clock transition constraint on the specified pulse generator cells, pulse generator library cells, clocks, and/or current design.

Parameters

-help	Prints out the command usage.
-fall	Sets transition limit value for falling transition.
object_list	Specifies a list of objects - pulse generator cells, pulse generator library cells, clocks, or current design.
-rise	Sets transition limit value for rising transition.
transition_limit	Specifies the transition limit value.
-transitive_fanout	Specifies the transition limit value for transitive fanout.

Examples

The following example sets the max pulse clock transition constraint of 0.1ns on the input pins of all the instances of the lib cell PULSE_GEN_00:

```
tempus > set_pulse_clock_max_transition 0.1 PULSE_GEN_00
```

This constraint can be observed on the input pin 'A' of an instance 'pg_inst' of this lib cell using the following command:

```
tempus > report_constraint pg_inst/A -drv_violation_type pulse_clock_max_transition
```

Fragmented output:

```
Check type : pulse_clock_max_transition_rise
-----
-----
Pin Name Required Actual Slack
```

```
-----  
pg_inst/A 0.100 0.030 0.070  
Check type : pulse_clock_max_transition_fall  
-----  
-----  
Pin Name Required Actual Slack  
-----  
pg_inst/A 0.100 0.030 0.070
```

Related Commands

- `get_clocks`
- `get_cells`
- `reset_pulse_clock_max_transition`

set_pulse_clock_max_width

```
set_pulse_clock_max_width
[-help]
pulse_width
object_list
[-transitive_fanout]
```

Sets the maximum pulse clock width constraint on the specified pulse generator cells, pulse generator library cells, clocks, and/or current design.

To report the maximum pulse clock width constraint, you can use the [report_constraint](#) -
`check_type pulse_clock_max_width` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>object_list</code>	Specifies a list of objects - pulse generator cells, pulse generator library cells, clocks, or current design.
<code>pulse_width</code>	Specifies the pulse width value.
<code>- transitive_fanout</code>	Specifies the pulse width value for transitive fanout.

Examples

The following example sets the maximum pulse clock width constraint of 1 ns on the pulse generator instance 'pg_inst':

```
tempus > set_pulse_clock_max_width 1 pg_inst
```

This constraint can be observed using the following command:

```
tempus > report_constraint pg_inst/Y -check_type pulse_clock_max_width
```

Fragmented output:

```
-----
Pin Required Actual Pulse Slack Clock Type
Pulse Width Width
-----
pg_inst/Y (high) 1.000 0.366 0.634 PG
```

Related Commands

- `get_clocks`
- `get_cells`
- `reset_pulse_clock_max_transition`
- `reset_pulse_clock_max_width`

set_pulse_clock_min_transition

```
set_pulse_clock_min_transition
[-help]
transition_limit
object_list
[-transitive_fanout]
[-rise]
[-fall]
```

Sets the minimum pulse clock transition constraint on the specified pulse generator cells, pulse generator library cells, clocks, and/or current design.

Parameters

-help	Prints out the command usage.
-fall	Sets transition limit value for falling transition.
<i>object_list</i>	Specifies a list of objects - pulse generator cells, pulse generator library cells, clocks, or current design.
-rise	Sets transition limit value for rising transition.
<i>transition_limit</i>	Specifies the transition limit value.
- transitive_fanout	Specifies the transition limit value for transitive fanout.

Examples

The following example sets the min pulse clock transition constraint of 0.05ns on the input pins of all the instances of the lib cell PULSE_GEN_00:

```
tempus > set_pulse_clock_min_transition 0.05 PULSE_GEN_00
```

This constraint can be observed on the input pin 'A' of an instance 'pg_inst' of this lib cell using the following command:

```
tempus > report_constraint pg_inst/A -drv_violation_type pulse_clock_min_transition
```

Fragmented output:

```
Check type : pulse_clock_min_transition_rise
-----
Pin Name Required Actual Slack
-----
pg_inst/A 0.050 0.030 -0.020
```

Check type : pulse_clock_min_transition_fall

Pin Name	Required	Actual	Slack
----------	----------	--------	-------

pg_inst/A	0.050	0.030	-0.020
-----------	-------	-------	--------

Related Commands

- [reset_pulse_clock_min_transition](#)
- [get_clocks](#)
- [get_cells](#)

set_pulse_clock_min_width

```
set_pulse_clock_min_width  
[-help]  
pulse_width  
object_list  
[-transitive_fanout]
```

Sets the minimum pulse clock width constraint on the specified pulse generator cells, pulse generator library cells, clocks and/or current design.

When you set this constraint on a clock, the settings are applied to all the pulse generators that are driven by the clock. When this constraint is used for a design, the software applies the constraint to all the clock networks in the design.

The specified minimum pulse width is expected to be larger than the pulse width value. The width is defined when a clock object is created and its waveform is defined in the current design (using the `create_clock -waveform` parameter).

In case of conflicting minimum width constraints, the more restrictive constraint is used. When applied on a non-pulse generator object, or a non-clock target, this constraint has no effect.

Note: You must specify `set_interactive_constraint_modes` constraint prior to running the `set_pulse_clock_min_width` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>object_list</code>	Specifies a list of objects - pulse generator cells, pulse generator library cells, clocks, or current design.
<code>pulse_width</code>	Specifies the minimum pulse width value.
<code>-transitive_fanout</code>	Specifies the pulse width value for transitive fanout.

Examples

- The following example sets the min pulse clock width constraint of 0.3 ns on the pulse generator instance `pg_inst`:

```
tempus > set_pulse_clock_min_width 0.3 pg_inst
```

This constraint can be observed using the following command:

```
tempus > report_constraint pg_inst/Y -check_type pulse_clock_min_width
```

Fragmented output:

```
-----  
Pin Required Actual Pulse Slack Clock Type  
Pulse Width Width  
-----  
pg_inst/Y (high)      0.300 0.366 0.066 PG  
-----
```

Related Commands

- `report_constraint`
- `reset_pulse_clock_min_width`
- `set_pulse_clock_max_width`
- `timing_enable_uncertainty_for_pulsewidth_checks`
- `report_min_pulse_width`
- `set_interactive_constraint_modes`

set_resistance

```
set_resistance
-help
[-max]
[-min]
resistance_value
net_list
```

Sets the wire resistance value on the specified nets of the top cell, which is used to compute the net delay of the specified nets. The resistance that you specify using this command overrides the resistance from wire load models and SPEF.

Note: In the best-case worst-case (BC-WC) analysis mode, the minimum resistance is used to compute the early path delay and the late path delay for the minimum corner. The maximum resistance is used to compute the early path delay and the late path delay for the maximum corner.

Parameters

-help	Outputs a brief description that includes type and default information for each <code>set_resistance</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man set_resistance</code> .
-max	Specifies that the asserted resistance is the maximum resistance. <i>Default:</i> If the <code>-max</code> (or <code>-min</code>) parameter is not specified, the resistance value is used as both minimum and maximum resistance.
-min	Specifies that the asserted resistance is the minimum resistance. <i>Default:</i> If the <code>-min</code> (or <code>-max</code>) parameter is not specified, the resistance value is used as both minimum and maximum resistance.
<code>net_list</code>	Specifies a list of nets on which the wire resistance is set.
<code>resistance_value</code>	Specifies the value for the wire resistance. This value (in units) is found in the target technology library.

Examples

- The following example set the maximum wire resistance of 2.0 units on the net `n1`:
tempus > `set_resistance -max 2.0 [get_nets n1]`
- The following example set the minimum wire resistance of 1.2 units and a maximum wire resistance of 2.0 units on the net `n1`:

```
tempus > set_resistance 1.2 -min [get_nets n1]  
tempus > set_resistance 2.0 -max [get_nets n1]
```

Related Commands

- `report_net`
- `report_timing`
- `reset_resistance`
- `set_drive`
- `set_load`

set_sense

```
set_sense
[-help]
<pin_or_port_list>
[-clocks <clock_list>]
{[-positive | -negative | -stop_propagation | -clock_leaf]
[-type <type_list>]}
```

Selects which phase of the clock to filter at the specified point.

This command accepts timing arcs collection as a valid argument.

Parameters

-help	Prints out the command usage.
-clocks <i>clock_list</i>	Specifies a list of clocks passing through a specified point to which the constraint applies. When this parameter is not specified, the filtering of clock phase will be applicable to all the clocks passing through that specified point. This parameter must be used with a specified pin name and one of the parameters : -positive, -negative, -stop_propagation, or -clock_leaf.
-clock_leaf	When specified, the software propagates clock phases to the specified pin.
-negative	Specifies that positive phase filtered clocks pass through pins in the <i>pin_list</i> . Note: The -positive, -negative and -stop_propagation parameters are mutually exclusive.
<i>pin_or_port_list</i>	Specifies the list or collection of cell output pins. The unateness specified by the <i>set_sense</i> command is applied to clocks propagating through the specified point.
-positive	Specifies that negative phase filtered clocks pass through pins in the <i>pin_list</i> . Note: The -positive, -negative and -stop_propagation parameters are mutually exclusive.
- stop_propagation	Stops the propagation of the specified clocks passing through the specified pins, or any phases that are specified by the -type parameter. Note: The -positive, -negative and -stop_propagation parameters are mutually exclusive.

<code>-type type_list</code>	Applies clock sense to object types specified in the <i>type_list</i>. The type list options are - all, clock, data, and clock_source_data. These are explained below: • all: Stops all phases from propagating. This is equivalent to specifying both the <code>clock</code> and <code>data</code> options. • clock: Stops clock phases launched by clock(s) specified in the <i>clock_list</i> from propagating. • data: Stops all data phases launched by clock(s) specified in <i>clock_list</i>, starting from the pins/ports specified in <i>pin_or_port_list</i>. The objects in the <i>pin_or_port_list</i> can be selected anywhere in the design - clock/data network. • clock_source_data: Stops only clock source data phases from propagating. Notes: • The <code>-type clock</code> option should be used with the <code>-stop_propagation</code> parameter. When the <code>-type</code> parameter is not specified, then <code>set_sense - stop_propagation</code> will stop only the clock phases. • The <code>-type data</code> option should be used with the <code>-stop_propagation</code> and <code>-clock</code> parameters. • The <code>-type data</code> option is not supported with both <code>-positive</code> and <code>-negative</code> parameters. • The <code>write_sdc</code> command will automatically add <code>-type clock</code> when <code>set_sense</code> is specified with the <code>-stop_propagation</code> parameter (without <code>-type</code>).
------------------------------	--

Examples

- The following command allows the positive phase clock CLK1 to pass through CLKBUF1/Z and filters out the negative phase clock CLK1:

```
tempus > set_sense -positive -clocks CLK1 CLKBUF1/Z
```

- The following command stops clock phases starting from pin1:

```
tempus > set_sense pin1 -type {clock}
```

- The following command stops data phases based only on CK1 and CK2 starting from pin1:

```
tempus > set_sense pin1 -type {data} -clocks {CK1 CK2}
```

- The following command stops only the clock source data phases starting from pin1:

```
tempus > set_sense pin1 -type {clock_source_data}
```

- The following command stops the clock and clock source data phases:

```
tempus > set_sense pin1 -type {clock clock_source_data}
```

or

```
tempus > set_sense pin1 -stop_propagation
```

Related Commands

- [reset_sense](#)

write_annotated_transition

```
write_annotated_transition
[-help]
-file out_file
[-min]
[-max]
[-writeOutput]
[-direct]
[-noPort]
[-view view_name ]
```

Writes the value of transitions on each pin to an output file. You can either set the transitions using the `set_annotated_transition` command or the software computes them internally.

Parameters

-help	Prints out the command usage.
-direct	Refers the ports directly for which the annotated transition will be applied in the output SDC. This option removes the dependency of using the <code>get_ports</code> and <code>get_pins</code> commands.
-file out_file	Specifies the name of the output file.
-help	Outputs a brief description that includes type and default information for each <code>write_annotated_transition</code> parameter.
-min/-max	Specifies whether to write out minimum or maximum transition. <i>Default:</i> Both minimum and maximum transitions are written out.
-noPort	Skips writing the transition values for the primary ports (input and output).
-view view_name	Writes the transitions for the specified analysis view only. You can specify this parameter only when the software is in multi-mode multi-corner timing analysis mode.
-writeOutput	Writes the slews for input and output pins, and the output slew for bi-directional pins. If this option is not specified, the software writes the slew for input pins and the input slew of bi-directional pins.

Examples

Here are some examples.

- The following command generates the output file `tran.rpt`:

```
tempus > write_annotated_transition -file tran.rpt
```

The following file is generated:

```
set_annotated_transition -rise -min 0.0278 [get_ports {sig_out4000}]
set_annotated_transition -fall -min 0.0275 [get_ports {sig_out4000}]
set_annotated_transition -rise -max 0.0278 [get_ports {sig_out4000}]
set_annotated_transition -fall -max 0.0275 [get_ports {sig_out4000}]
set_annotated_transition -rise -min 0.001 [get_pins {buf_in2000/A}]
set_annotated_transition -fall -min 0.001 [get_pins {buf_in2000/A}]
set_annotated_transition -rise -max 0.001 [get_pins {buf_in2000/A}]
set_annotated_transition -fall -max 0.001 [get_pins {buf_in2000/A}]
set_annotated_transition -rise -min 0.001 [get_pins {buf_in2001/A}]
set_annotated_transition -fall -min 0.001 [get_pins {buf_in2001/A}]
set_annotated_transition -rise -max 0.001 [get_pins {buf_in2001/A}]
set_annotated_transition -fall -max 0.001 [get_pins {buf_in2001/A}]
set_annotated_transition -rise -min 0.0453 [get_pins {buf_out5000/A}]
set_annotated_transition -fall -min 0.0418 [get_pins {buf_out5000/A}]
set_annotated_transition -rise -max 0.0453 [get_pins {buf_out5000/A}]
set_annotated_transition -fall -max 0.0418 [get_pins {buf_out5000/A}]
```

- The following command generates the minimum transitions in the output file, `min_tran.rpt`:

```
tempus > write_annotated_transition -min -file min_tran.rpt
```

The following file is generated:

```
set_annotated_transition -rise -min 0.0278 [get_ports {sig_out4000}]
set_annotated_transition -fall -min 0.0275 [get_ports {sig_out4000}]
set_annotated_transition -rise -min 0.001 [get_pins {buf_in2000/A}]
set_annotated_transition -fall -min 0.001 [get_pins {buf_in2000/A}]
set_annotated_transition -rise -min 0.001 [get_pins {buf_in2001/A}]
set_annotated_transition -fall -min 0.001 [get_pins {buf_in2001/A}]
set_annotated_transition -rise -min 0.0453 [get_pins {buf_out5000/A}]
set_annotated_transition -fall -min 0.0418 [get_pins {buf_out5000/A}]
```

- The following command generates the maximum transitions in the output file, `max_tran.rpt`:

```
tempus > write_annotated_transition -max -file max_tran.rpt
```

The following file is generated:

```
set_annotated_transition -rise -max 0.0278 [get_ports {sig_out4000}]
set_annotated_transition -fall -max 0.0275 [get_ports {sig_out4000}]
set_annotated_transition -rise -max 0.001 [get_pins {buf_in2000/A}]
set_annotated_transition -fall -max 0.001 [get_pins {buf_in2000/A}]
set_annotated_transition -rise -max 0.001 [get_pins {buf_in2001/A}]
set_annotated_transition -fall -max 0.001 [get_pins {buf_in2001/A}]
set_annotated_transition -rise -max 0.0453 [get_pins {buf_out5000/A}]
set_annotated_transition -fall -max 0.0418 [get_pins {buf_out5000/A}]
```

Related Commands

- [set_annotated_transition](#)
- [get_ports](#)
- [get_pins](#)

write_load

```
write_load
[-help]
[-direct]
[-excludeZeroCap]
[-includePinCap]
[-wire_load]
-file out_file
```

Writes the capacitive loading on each net to an output file. By default, the capacitance value does not include the pin capacitance. You can either set the load using the [set_load](#) command or the software uses the load defined in the SPEF file.

Note: The command is currently not supported in multi-mode multi-corner mode.

Parameters

-help	Prints out the command usage.
-direct	Refers the ports directly for which the capacitive loading will be written. This option removes the dependency of using the get_nets command.
-excludeZeroCap	Excludes the nets not annotated in the design while writing out the capacitive loading on each net to the output file.
-file out_file	Specifies the name of the output file.
-help	Outputs a brief description that includes type and default information for each write_load parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man write_load</code> .
-includePinCap	Includes the pin capacitance while writing out the capacitive loading on each net to the output file.
-wire_load	Adds option '-wire_load' to "set_load" (that is, <code>set_load -wire_load</code>) only on the output (out) and bi-directional (inout) ports in the output file.

Examples

- The following command writes the capacitive load in the report file `load.rpt`:

```
tempus > write_load -file load.rpt
```

The following report is generated:

```
set_load 1e-05 [get_nets {sig_out4000}]
set_load 0.00011 [get_nets {sig_in1000}]
set_load 0.00011 [get_nets {sig_in1001}]
set_load 0.00015 [get_nets {node_out6000}]
```

Related Commands

- `set_load`
- `get_nets`

Distributed Static Timing Analysis Commands

- [distribute_create_scope_model](#)
- [distribute_launch_deferred_clients](#)
- [distribute_partition](#)
- [distribute_print_client_usage](#)
- [distribute_print_partition_stats](#)
- [distribute_start_clients](#)
- [report_noise](#) (DSTA)
- [set_glitch_threshold](#) (DSTA)

distribute_create_scope_model

```
distribute_create_scope_model  
[-help]  
[-cell string]  
[-cell_nohier string]  
-dir <string>  
[-exclude_cell string]  
[-exclude_cell_contents]  
[-exclude_instance string]  
[-ignore_scope_fanouts]  
[-instance string]  
[-instance_nohier string]  
[-nosettings]  
[-overwrite]
```

Creates the scope from a full-chip run in DSTA mode.

Parameters

-help	Prints out the command usage.
-cell <i>string</i>	Creates block-scope models for all instances of the specified cell
-cell_nohier <i>string</i>	Creates top scope models for all instances of a cell.
-dir <i>string</i>	Specifies the name of the directory for the writing scope.
-exclude_cell <i>string</i>	Excludes instances of the specified cells in the created scope.
-exclude_cell_contents	Does not include the scope cell SPEF and Verilog in the generated scope.
-exclude_instance <i>string</i>	Excludes the specified instances in the created scope.
- ignore_scope_fanouts	Ignores fanouts of the scope.
-instance <i>string</i>	Creates block scope models for the specified instances.
-instance_nohier <i>string</i>	Creates top scope models for the instances.
-overwrite	Allows overwrite of the existing directory.

Examples

- The following command creates a scope model for all instances of cell `hcell1` and writes the model files into the `./scope_dir` directory:

```
distributed_create_scope_model -dir ./scope_dir -overwrite -cell hcell1
```

Any existing directory contents are removed. If there are multiple instances of `hcell1`, a representative instance will be chosen for timing internal register-to-register paths.

- The following command creates a scope model for the single instance `hcell1_inst1` and writes the model files into the `./scope_dir` directory:

```
distributed_create_scope_model -dir ./scope_dir -instance hcell1_inst1
```

If the directory already exists, then an error is issued, and no scope model is created.

- The following command creates a scope model that includes the instances and nets in the top cell, along with the boundary paths of the hierarchical children:

```
distributed_create_scope_model -dir ./scope_dir -overwrite -cell_nohier top_cell
```

- The following command creates a scope model for `top_cell`, except that only the boundary paths are included for any child instances of the cell `hcell1`:

```
distributed_create_scope_model -dir ./scope_dir -overwrite -cell top_cell -exclude_cell hcell1
```

distribute_launch_deferred_clients

distribute_launch_deferred_clients
[-help]

Launches the clients and waits for them to become available.

When the `distribute_start_clients -defer_launch` parameter is specified, the clients are not launched until the launch command is run.

The `distribute_launch_deferred_clients` command can be run at any point after `distribute_start_clients -defer_launch` option is specified, but should be specified no later than immediately after the `distribute_partition` command. Most commands after this point will require clients to be running.

Parameters

-help	Prints out the command usage.
-------	-------------------------------

Related Commands

- `distribute_start_clients`
- `distribute_partition`

distribute_partition

```
distribute_partition  
[-help]  
[-block_list string]  
[-cell string]  
[-cell_nohier string]  
[-exclude_block_list string]  
[-instance string]  
[-instance_nohier string]
```

Controls the method by which design partitioning is performed, and determines the scope of DSTA. This command tells the master process to start using the clients for partitioning. It is recommended that you run the command after the parasitics are read in, so it can provide constraints, timing updates, and reports to be shared among all clients.

Parameters

-help	Outputs a brief description that includes type and default information for each <code>distribute_partition</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man distribute_partition</code> .
-block_list <i>string</i>	Specifies the list of blocks to be partitioned. It lists the blocks that are to be analyzed in the incremental flow. This enables block-scope analysis in DSTA; therefore, only the specified blocks are analyzed in the run, and all the other blocks are ignored.
-cell <i>string</i>	Specifies a list of cell instances for partitioning.
-cell_nohier <i>string</i>	Specifies a list of cell instances for partitioning, excluding hierarchical children.
- exclude_block_list <i>block_names</i>	Specifies the list of blocks to exclude from the DSTA analysis in the incremental flow. This enables top-scope analysis in DSTA. All blocks not on this list are analyzed, along with the top-level of the design.
-instance <i>string</i>	Specifies a list of instances for partitioning.
-instance_nohier <i>string</i>	Specifies a list of instances for partitioning, excluding hierarchical children.

Examples

The following command specifies to perform a full-flat (non-incremental) flow:

```
dsta> distribute_partition
```

distribute_print_client_usage

distribute_print_client_usage
[-help]

Reports CPU time, real time, current memory, and peak memory for all the clients. This command must be used instead of `report_resource`, which cannot run in DSTA mode.

Parameters

- help	Outputs a brief description that includes type and default information for each <code>distribute_print_client_usage</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man distribute_print_client_usage</code> .
-----------	--

distribute_print_partition_stats

```
distribute_print_partition_stats  
[-help]  
[-all]  
[-counts]  
[-resources]  
[-tcl_list]
```

Provides an estimate of the client memory requirements after partitioning, based on the instance count for each partition.

This command prints the partition statistics and estimated resource requirements after partitioning.

Parameters

-help	Prints out the command usage.
-all	Prints out instance and net counts by partition, overlap statistics, and resource requirements.
-counts	Prints out the instance and net counts by partition and overlap statistics.
-resources	Prints out the estimated memory and temp file size requirement by partition. This is the default behavior.
-tcl_list	Returns the requested statistics as a TCL list. Note: This parameter is supported with <code>-resources</code> .

Examples

- The following command shows the client resources:

```
distribute_print_partition_stats -resources
```

```
Active analysis views: 1
```

```
Partition 0 Estimated Memory: 50 GB. Estimated /tmp space: 60 GB.
```

```
Partition 1 Estimated Memory: 40 GB. Estimated /tmp space: 48 GB.
```

```
Partition 2 Estimated Memory: 50 GB. Estimated /tmp space: 60 GB.
```

```
Partition 3 Estimated Memory: 40 GB. Estimated /tmp space: 48 GB.
```

- The following command shows the instance count for each partition:

```
distribute_print_partition_stats -counts
```

Partition Stats:

Partition 0 instance-count: 13654609 extended instance-count: 4331968 (clustered: 4063874) outside clock instance-count: 320793

Partition 1 instance-count: 12983662 extended instance-count: 1153506 (clustered: 1045248) outside clock instance-count: 738683

Partition 2 instance-count: 14100713 extended instance-count: 4345271 (clustered: 4170905) outside clock instance-count: 308583

Partition 3 instance-count: 12959154 extended instance-count: 1166173 (clustered: 1064585) outside clock instance-count: 757964

Partition 0 net-count: 15498000 extended net-count: 1288107 (clustered: 1137204) outside clock net-count: 428712

Partition 1 net-count: 14730101 extended net-count: 345115 (clustered: 297528) outside clock net-count: 787647

Partition 2 net-count: 15970893 extended net-count: 1293759 (clustered: 1191587) outside clock net-count: 419663

Partition 3 net-count: 14751488 extended net-count: 350045 (clustered: 306878) outside clock net-count: 807687

Total nets: 59753864

Total partitioned nets: 59751874

Total instances: 52467900

Total partitioned instances: 52467832

Primary instance overlap: 2.34488%

Full instance overlap: 27.3563%(clustered: 26.113%)

When `-all` is specified, both the instance counts as well as client resources will be printed.

distribute_start_clients

```
distribute_start_clients  
[-help]  
[-client_nowait]  
[-count integer]  
[-defer_launch | -minHost integer]  
[-no_log]  
[-no_logv]  
[-outdir string]  
[-overwrite]  
[-startup_script <string>]
```

Starts Tempus on distributed clients. This command starts the client processes. It requests LSF or SGE for a machine and starts to create a copy of Tempus on it. The master machine will assign these client machines tasks.

Parameters

<code>-help</code>	Outputs a brief description that includes type and default information for each <code>distribute_start_clients</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man distribute_start_clients</code>.
<code>-client_nowait</code>	Specifies whether the master machine must wait for all the clients to be launched. When this parameter is specified, the master machine will continue processing and will not wait for all clients to be launched. However, the master will wait until partitioning occurs. Note: You cannot specify this parameter with the <code>-minHost</code> or <code>-defer_launch</code> parameter.
<code>-count <i>integer</i></code>	Specifies the number of clients. This parameter sets the number of clients when the <code>set_multi_cpu_usage</code> command does not specify a value.
<code>-defer_launch</code>	Defers the launch of any clients unless the <code>distribute_launch_deferred_clients</code> command is issued explicitly. Note: This parameter cannot be used with the <code>-minHost</code> parameter.
<code>-minHost <i>integer</i></code>	Specifies the minimum number of available remote hosts (servers) that can be accepted.
<code>-no_log</code>	Does not allow the software to create a client log file.
<code>-no_logv</code>	Does not allow the software to create a verbose client log file.
<code>-outdir <i>string</i></code>	Specifies the base name of the client output directory.
<code>-overwrite</code>	Overwrites an existing <code>.log</code> file (and associated <code>.cmd</code> and <code>.logv</code> files), and does not add an integer to make the files unique.
<code>- startup_script <string></code>	When specified, the software will source the flowtool file on clients for attribute and object initialization.

Examples

- The following command starts client processes on 4 distributed clients:

```
dsta> distribute_start_clients -count 4 -outdir run1_output_dir
```

Related Commands

- `distribute_launch_deferred_clients`

report_noise (DSTA)

report_noise

```
[-help]
[-bumpy_waveform]
[-cell_type {bbox | clock | data | latch}]
[-clock]
[-delay {min | max}]
[-expand_virtual_attacker_bus]
[-failure]
[-format string]
[-gui_victim_nets_file]
[-histogram]
[-level {vl | vh | vlu | vho}]
[-nets string]
[-output_format {single_line | multiple_line}]
[-overshoot_undershoot]
[-pins string]
[-quiet_nets]
[-report_virtual_attacker_constituents]
[-slack float]
[-sort_by {noise | receiver_peak | slack | cell_type}]
[-stdout]
[-style {default | extended}]
[-threshold float]
[-txtfile string]
[-view string]
[[>|>>]]
```

Generates text-based noise reports.

Note: You need to ensure that the `set_si_mode -enable_glitch_report` (to generate glitch report) or `set_si_mode -enable_delay_report` (to generate detailed delay report) commands are set to `true` before the `report_timing`, `update_timing`, or `update_glitch` commands are run.

For details about generating and interpreting report files, see "Signal Integrity Delay and Glitch Analysis" in the *Tempus User Guide*.

Parameters

-help	Prints out the command usage.
-------	-------------------------------

<code>-bumpy_waveform</code>	Flags nets with bumpy waveforms at the receiver. Note: This parameter will always show the maximum delay bumpy report (even if the <code>-delay min</code> option is specified).
<code>-cell_type {bbox clock data latch}</code>	Generates a glitch report for the user-specified cell type. You can specify one of the following options: • <code>bbox</code>: Generates a report for only the black-box receivers • <code>clock</code>: Generates a report for only the clock pin receivers • <code>data</code>: Generates a report for only the data pin receivers • <code>latch</code>: Generates a report for only the latch pin receivers
<code>-clock</code>	Prints the signal integrity report for the clock network.
<code>-delay {min max}</code>	Reports delay uncertainty instead of noise values. The only legal values are <code>min</code> or <code>max</code> for minimum or maximum path delay analysis. The <code>report_noise -clock -delay min max</code> command outputs the minimum/maximum delay report for clock nets.
<code>-expand_virtual_attacker_bus</code>	Reports virtual attacker components and expands bus bits in the noise reports. This parameter works only when the bus virtual attacker feature is enabled using the <code>set_si_mode -enable_virtual_attacker_constituent_report true</code> setting.

<code>-failure</code>	Reports receiver peak failures and DC tolerance failure nets. You can use this option only for glitch noise failure reports. The <code>report_noise</code>, without the <code>-failure</code> option, uses 10% of vdd as the default. When you specify the <code>-failure</code> option, you need to use <code>-threshold</code> with the lowest noise level saved; if not specified, it defaults to 5% of Vdd. Option: Directory Name: <code>-sort_by noise glitch</code> <code>-sort_by rcvr_peak glitch</code>
<code>-format <i>string</i></code>	Specifies the fields for the output glitch report format. The valid fields are: arc, attacker_edge, attacker_net, attacker_offset, attacker_peak, attacker_slew, attacker_source, attacker_status, attacker_vdd, attacker_xcap, cell_type, delay, glitch_slack, incr_delay, level, peak, receiver_cell, receiver_input_failure_threshold, receiver_input_pin, receiver_output_failure_threshold, receiver_output_peak, slack, slack_status, stage_info, thresh_source, victim_edge, victim_library, victim_net, victim_slew, victim_total_area, victim_total_cap, victim_total_cross_cap, victim_vdd, victim_width Each field corresponds to the following columns of the glitch report: • arc: Arc • attacker_edge: Edge • attacker_net: Net • attacker_offset: Offset (ps) • attacker_peak: Peak (mV) • attacker_slew: Slew (ps) • attacker_source: Constituents: Source • attacker_status: Statue • attacker_vdd Vdd (V) • attacker_xcap: Xcap (fF) • glitch_slack: Slack

- level: Level
- peak: Peak (mV)
- receiver_cell: Receiver cell
- receiver_input_failure_threshold: Receiver Input Threshold
- receiver_input_pin: ReceiverNet
- receiver_output_failure_threshold: Failure threshold
- receiver_output_peak: Receiver output peak Value
- slack_status: Status
- stage_info
- thresh_source: ReceiverThresholdType (Global | User | Library)
- victim_library: Library
- victim_net: VictimNet
- victim_total_area: TotalArea
- victim_total_cap: TotalCap (fF)
- victim_vdd: Vdd (V)
- victim_width: Width (ps)

The following options are not available with the single-line format:

- delay
- incr_delay
- slack
- thresh_source
- victim_edge
- victim_slew
- victim_total_cross_cap

To choose whether the report format should be on a single line or multiple lines, you can use the `output_format` parameter.

<code>-gui_victim_nets_file</code>	Generates the GUI format glitch reports (<code>aae</code> or <code>aae_<viewname></code> files under the reporting directory) and incremental delay reports (<code>aae_<max/min>_incDelay</code> or <code>aae_<viewname>_<max/min>_incDelay</code> files under the reporting directory) to be used in the Display Noise Net form.
<code>-histogram</code>	Generates the noise histogram.
<code>-level {vl vh vlu vho}</code>	Specifies the level for which a report should be generated when a single-level report is desired. You can specify one of these options - vl, vh, vlu, and vho.
<code>-nets string</code>	Reports the specified nets in the noise report.
<code>-pins string</code>	Reports on the specified list of pin names.
<code>-output_format {single_line multiple_line}}</code>	Specifies whether the output format should be single-line or multiple lines. You can specify one of the following options: • <code>single_line</code> • <code>multiple_line</code> If the <code>single_line</code> option is specified and the attacker info field is selected, all the attacker-related format options will be printed together. The <code>-output_format</code> parameter only works with the <code>-format</code> option and does not affect the default reporting output format. Default: <code>multiple_line</code>
<code>-overshoot_undershoot</code>	Reports the overshoot (VHO) and undershoot (VLU) glitches. The <code>set_si_mode -enable_overshoot_undershoot_glitch</code> must be set to <code>true</code> before the analysis is performed to generate the overshoot-undershoot glitch report. Default: If this option is not used during glitch reporting, then VL and VH types of glitches will be reported.
<code>-quiet_nets</code>	Generates a report of all the quiet attackers.
<code>-report_virtual_attacker_constituents</code>	Generates reports for virtual attacker constituents.

<code>-slack float</code>	Specifies the maximum slack value for a bumpy net.
<code>-sort_by {noise receiver_peak slack cell_type}</code>	Requests the sorting of entries. You can sort by: • noise • receiver_peak • slack • cell_type Tempus determines the default HTML output directories based on your chosen options. Depending on the sorting option you select, Tempus sorts the report fields in the following order: -sort_by Option Level 1 Level 2 Level 3 Level 4 rcvr_peak receiver peak noise netname level noise noise - netname level For example, if you use the <code>-sort_by noise</code> option, the first level of sorting will be based on noise, followed by receiver peak, netname, and level. Note: The report displays the <code>vl</code> noise level before the <code>vh</code> noise level. Default: receiver_peak
<code>-style {default extended}</code>	Generates reports in one of two styles - <code>extended</code> or <code>default</code>. • extended: Generates extended reports with victim and aggressor timing windows, dropped aggressor slews, and reasons for dropped aggressors. Additionally, the report also shows driver pins and driver library pins. • default: Generates the default report, which is similar to the text report generated using the <code>-text</code> parameter.

<code>-threshold float</code>	Specifies the minimum peak noise, absolute delay uncertainty, relative delay uncertainty, and receiver peak thresholds to report on. <i>Default:</i> • 10% of vdd for nets - when the glitch report is generated using the <code>report_noise -sort_by noise</code> option. • 10 (ps) for nets - when a detailed delay report is generated using the <code>report_noise -delay</code> parameter. Note: • The <code>-threshold</code> parameter accepts percentage values. • All <code>-sort_by</code> values that represent voltages will be in mV.
<code>-txtfile string</code>	Specifies the name of the text report file. If the specified file name ends in <code>.gz</code> extension, the file is gzipped. If you do not specify the <code>-txtfile</code> option: • Tempus automatically names the text report as 'glitch' for noise glitch reports and 'delay' for noise delay reports. • Tempus dumps the generated report to the console window for better usability. If the <code>-txtfile</code> parameter is not used, then Tempus will output the SI report to the screen. When the <code>-txtfile</code> parameter is specified, then Tempus uses the user-specified name. In case of MMMC, Tempus uses the <code><txtfile Name>_<view Name></code> naming convention.
<code>-view string</code>	Specifies the view name.
<code>[> >>]</code>	Stores the generated report in the specified file. You can add the <code>.gz</code> extension to the file name to generate a compressed report. Note: The file name must be the last parameter specified.

Examples

- The following command reports noise for the view named `default_emulate_view`:

```
tempus > report_noise -delay max -threshold 1600 -txtfile delay_max_format.rpt -format
{attacker_slew  attacker_xcap  attacker_status attacker_edge  attacker_net stage_info
slack  victim_slew  victim_total_cross_cap victim_net }
```

```
*****
Noise Delay Report (for view default_emulate_view)
*****
```

Report Options:

```
-----
-type delay
-threshold 1600 (ps)
-delay max
[-style default]
[-sortBy IncrDelay]
-----
```

```
-----
Slack(ps)   Slew(ps)   Total   Xcap(fF)   VictimNet
-2.973      0.635      277.574 seg3/u6/A   {seg3/n5}
```

Stage information:

```
      Cell      Pin
Driver:  INV      seg3/u5/Y
Receiver: INV      seg3/u6/A
```

Constituents:

```
Slew(ps)  Xcap(fF)  Status  Edge  Net
331.500   138.787   INF     F     seg1/n5
358.667   138.759   SY      F     seg2/n5
```

```
-----
Slack(ps)  Slew(ps)  Total Xcap(fF)  VictimNet
14.906     0.597    278.115      seg1/u6/A {seg1/n5}
```

Stage information:

```
      Cell      Pin
Driver :  INV      seg1/u5/Y
Receiver : INV      seg1/u6/A
```

Constituents:

```
Slew(ps)  Xcap(fF)  Status  Edge  Net
279.500   139.280   INF     F     w3
359.333   138.807   INF     F     seg3/n5
```

```
-----
Slack(ps) Slew(ps)  Total Xcap(fF)  VictimNet
15.737    0.587    270.604      seg1/u12/A {seg1/n11}
```

Stage information:

```
      Cell      Pin
```

```
Driver :   INV      seg1/u11/Y
Receiver : INV      seg1/u12/A
```

Constituents:

Slew(ps)	Xcap(fF)	Status	Edge	Net
276.500	135.654	INF	F	w5
337.167	134.517	INF	F	seg3/n11

- The following command reports noise for the specified reporting fields:

```
tempus > report_noise -delay max -format {delay incr_delay victim_slew victim_total_cap
victim_total_cross_cap victim_vdd victim_net attacker_offset attacker_slew attacker_xcap
attacker_vdd attacker_edge attacker_status attacker_net} -txtfile custom.rpt
```

```
-----
Delay(NomMax/Max)  IncrDelay(ps)  Slew(ps)  TotalCap(fF)  Total  Xcap(fF)  Vdd(V)  VictimNet

-43.400/2418.200  2461.600          12.372   639.088          599.796          1.62   i1/A {n1}
```

```
Type  IncrDelay(ps)  Delay(NomMax/Max)  VictimInst

R      3253.300          5457.100/8710.400  drv/Y {INV:A}
```

Constituents:

Offset(ps)	Slew(ps)	Xcap(fF)	Vdd(V)	Edge	Status	Net
-245.000	30.000	52.145	1.62	F	att1	
-199.000	120.000	26.751	0.00	F	INF	~n1

- The following command shows glitch values greater than 20% VDD as:

```
tempus > report_noise -threshold 0.20
Output report file name prefix is (null)
Info: Processing report for view default_emulate_view ...
#####
# Command: report_noise -threshold 0.20
# View: default_emulate_view
#####
```

Report Options:

```
-----
-type glitch
-threshold 20 % (of vdd)
```

- To generate a glitch report, you can use the following command:

```
tempus > report_noise -txtfile glitch.txt
```

- To generate a glitch report sorted by input noise, you can use the following command:

```
tempus > report_noise -txtfile sort_by_glitch_input.txt -sort_by noise
```

- To generate a glitch failure report, you can use the following command:

```
tempus > report_noise -txtfile glitch_failure.txt -failure
```

- To generate a glitch report for VL noises sorted by input noise, you can use the following command:

```
tempus > report_noise -sort_by noise -level vl
```

- To generate a glitch report for only latch receiver pin cell types, you can use the following command:

```
tempus > report_noise -txtfile latch_glitch.txt -cell_type latch
```

For more information on glitch reporting, refer to "[SI Glitch Reporting](#)" in the *Tempus User Guide*.

Note: DSTA supports only AAE specific reports so some of the options will be different from STA options. For more details on STA options, refer to [report_noise](#).

Related Commands

- [report_timing](#)
- [update_timing](#)
- [report_delay_calculation](#)

set_glitch_threshold (DSTA)

```
set_glitch_threshold  
[-help]  
[-cells string]  
[-failure_point {none | input | internal | last}]  
[-glitch_type {both | vl | vh | vlu | vho | vl_vh | vlu_vho }]  
[-nets string]  
[-pins string]  
[-pin_type {clock | data | latch | all}]  
[-threshold_type {failure | reporting}]  
-value float  
[-views string]
```

Allows for additional control of failure criteria for SI glitch analysis.

Parameters

-help	Prints out the command usage. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man set_glitch_threshold.</code>
-cells <i>string</i>	Specifies a list of library cell names.
-failure_point {none input internal last}	Specifies where the failure or reporting criteria apply. • <code>none</code>: Criteria does not apply. • <code>input</code>: Threshold applies when the failure point is at the input pin of the receiver cell. • <code>internal</code>: Threshold is applied when the failure point is internal to the cell. This occurs for multi-stage cells. The failure point will be the output of the first stage inside the cell. • <code>last</code>: Threshold is applied when the failure point is the output of the cell. This occurs for single-stage cells, such as inverters. Default: <code>none</code>
-glitch_type {both vl vh vlu vho vl_vh vlu_vho}	Specifies the glitch type to which the command applies. Here, the <code>both</code> option refers to <code>vl</code> and <code>vh</code> type of glitches. <code>vl_vh</code> : Includes both VL and VH noise. <code>vlu_vho</code> : Includes both VLU and VHO noise. Default: <code>both</code>

<code>-nets string</code>	Specifies the list of nets the command applies to. Default is not net-specific.
<code>-pins string</code>	Specifies the list of pins or primary ports. Default is not pin-specific.
<code>-pin_type</code> {clock data latch all}	Specifies the pin type to which the command applies. <i>Default:</i> all
<code>-threshold_type</code> {failure reporting}	Specifies whether the command options apply to failure or reporting criteria. In case of VL/VH glitch analysis, it is mandatory to specify the <code>-threshold_type</code> parameter. This is because when the computed glitch (ROP) at any given stage exceeds the failure threshold (with glitch propagation enabled), it is not propagated and is flagged as a failure in the noise report. With the <code>-threshold_type failure</code> setting, the propagation will stop as per the new threshold, and the failure flagging will change accordingly. However, with the <code>-threshold_type reporting</code> setting, the propagation threshold remains unchanged, but the failure reporting is updated. In case of VLU/VHO glitch analysis, it is optional to use the <code>-threshold_type</code> parameter because the above flow is not applicable. <i>Default:</i> failure
<code>-value float</code>	Specifies the threshold value as a ratio of the local VDD. <i>Default:</i> 0.1
<code>-views string</code>	Specifies a list of view names.

Examples

Here are some examples.

- The following command moves the failure point from ROP to RIP for `inst1/a1` and uses a RIP failure of $0.2 \times \text{ReceiverVDD}$:

```
set_glitch_threshold -failure_point input -pin inst1/a1 -threshold_type failure -value 0.2
```
- The following command sets clock failure criteria to $0.1 \times \text{VDD}$, latch failure criteria to $0.2 \times \text{VDD}$ and

all other nets to 0.3*VDD (set before update_glitch):

```
dsta> set_glitch_threshold -threshold_type failure -pin_type all -failure_point last -  
value 0.3  
dsta> set_glitch_threshold -threshold_type failure -pin_type clock -failure_point last -  
value 0.2  
dsta> set_glitch_threshold -threshold_type failure -pin_type latch -failure_point last -  
value 0.1
```

- The following command sets the failure threshold at the output of the 1st channel connect inside all the instances of macro1 to be 0.2 (set before update_glitch):

```
dsta> set_glitch_threshold -threshold_type failure -cell {macro1} -failure_point  
internal -value 0.2
```

- The following command changes the reporting criteria for latches to be 0.15*VDD (set after update_glitch):

```
dsta> set_glitch_threshold -threshold_type reporting -pin_type latch -value 0.15
```

If you need to set all the data failure criteria to be 0.3, then you need to make the following settings:

```
dsta> set_glitch_threshold -failure_point internal \  
-threshold_type failure -value 0.366 \  
-pin_type data
```

```
dsta> set_glitch_threshold -failure_point last \  
-threshold_type failure -value 0.366 \  
-pin_type data
```

Tcl Interface Commands

- [add_to_collection](#)
- [all_clocks](#)
- [all_connected](#)
- [all_fanin](#)
- [all_fanout](#)
- [all_inputs](#)
- [all_instances](#)
- [all_outputs](#)
- [all_registers](#)
- [append_to_collection](#)
- [compare_collections](#)
- [copy_collection](#)
- [define_proc_arguments](#)
- [define_property](#)
- [filter_collection](#)
- [foreach_in_collection](#)
- [get_activity](#)
- [get_arcs](#)
- [get_cells](#)
- [get_clocks](#)
- [get_designs](#)
- [get_generated_clocks](#)
- [get_lib_arcs](#)
- [get_lib_cells](#)
- [get_lib_pg_pins](#)

- [get_lib_pins](#)
- [get_libs](#)
- [get_nets](#)
- [get_object_name](#)
- [get_path_groups](#)
- [get_pg_nets](#)
- [get_pg_pins](#)
- [get_pins](#)
- [get_ports](#)
- [get_power](#)
- [get_property](#)
- [index_collection](#)
- [list_property](#)
- [parse_proc_arguments](#)
- [query_objects](#)
- [range_collection](#)
- [remove_from_collection](#)
- [report_property](#)
- [sizeof_collection](#)
- [sort_collection](#)

Overview

The following lists the Advanced Timing Tcl Scripting commands along with the description and examples.

all_* Commands

The `all_*` commands return a collection of all the items of a specific type. Several of the commands provide options to filter the collection based on certain properties:

- `all_clocks`
- `all_connected`

- `all_fanin`
- `all_fanout`
- `all_inputs`
- `all_instances`
- `all_outputs`
- `all_registers`

Examples

- To set all the clocks to propagated mode, use the following command:

```
set_propagated_clock [all_clocks]
```
- To report the worst timing path that begins at a register and ends at an output port, use the following command:

```
report_timing -from [all_registers] -to [all_outputs]
```
- To store a collection of pins that fanout from a specific port, use the following command:

```
set clk1Fanout [all_fanout -from clk1 -pin_levels 1]
```

get_* Commands

The `get_*` commands return a collection of items filtered by name pattern or object type. Additionally, you can use the `-filter` option for most of the `get_*` commands to filter the collection by property value. See the [get_property](#) command for a list of all the properties.

The `filter_expression` for the `-filter` option can be created by combining several object properties using the following relational and logical operators: `>`, `<`, `==`, `!=`, `<=`, `>=`, `&&`, `||`, `~=`, `!~`, `AND`, and `OR`. You also can use parentheses to create expressions. The `get_*` commands include the following:

- `get_arcs`
- `get_cells`
- `get_clocks`
- `get_designs`
- `get_generated_clocks`
- `get_lib_arcs`
- `get_lib_cells`
- `get_lib_pins`

- `get_libs`
- `get_nets`
- `get_object_name`
- `get_path_groups`
- `get_pins`
- `get_ports`

Examples

- To return a collection of all input ports (same as running `[all_inputs]`), use the following command:
`get_ports -filter "direction==in"`
- To return a collection of pins that match `*/D`, use the following command:
`get_pins -hier -filter "hierarchical_name =~ */D"`
- To return a collection of the libraries a cell belongs to, use the following command:
`get_lib_cells INVXL`

property Commands

The `list_property` command lists the properties for the different object types. This command, without arguments, will list all the properties for all object types. Use the `-type` option to list properties for a specific object type. Legal option types are: `cell`, `clock`, `design`, `lib`, `lib_cell`, `lib_pin`, `lib_timing_arc`, `net`, `path_group`, `pin`, `port`, `timing_arc`, `timing_path`, or `timing_point`. For example:

```
list_property -type lib_timing_arc
```

The above commad generates the following output:

```
object_type : lib_timing_arc
=====
property           | return_type
-----
from_lib_pin        | collection
has_ccs_noise       | boolean
has_socv            | boolean
is_disabled         | boolean
mode                | string
object_type         | string
output_signal_level_high | float
output_signal_level_low  | float
sdf_cond            | string
```

sense	string
timing_type	string
to_lib_pin	collection
when	string
when_end	string
when_start	string

Examples

- The [get_property](#) command returns the value of the specified property. To report the clock period of clk1, use the following command:

```
get_property [get_clocks clk1] period
```

- The [report_property](#) command returns the properties and their values for a specified object. To report the properties for clock clk1, use the following command:

```
report_property [get_clocks clk1]
```

- For timing related properties such as properties for timing arcs, paths, and points run `report_timing -collection`. This will return the results in a collection of timing paths.

- To report timing and store the top 10 paths as a collection in the variable `$paths`, use the following command:

```
set paths [report_timing -collection -max_paths 10]
```

- To filter `$paths` to include only those paths with slack between -0.806ns and 0ns, use the following command:

```
set newPaths [filter_collection $paths {(slack > -0.806 && slack < 0.0)}]
```

collection Commands

Once you have the collection defined you can manipulate, filter, cycle through, and query it using the collection related commands:

- `add_to_collection`
- `append_to_collection`
- `compare_collections`
- `copy_collection`
- `filter_collection`
- `foreach_in_collection`
- `index_collection`
- `query_objects`
- `remove_from_collection`

- `sizeof_collection`
- `sort_collection`

Examples

- To list the contents of a collection, use the following command:

```
query_objects $collectionName
```

- To cycle through the items of a collection, use the following command:

```
foreach_in_collection iCell [get_cells *] {  
  Puts "Cell = [get_property $iCell hierarchical_name]"  
}
```

add_to_collection

```
add_to_collection  
base_collection  
second_collection_or_list  
[-unique]
```

Creates a new collection by adding objects from one collection to another collection. The following rules apply:

- If all of the objects in the base collection are of the same type, only the objects in the second collection that are the same type as those in the base collection are added to the new collection. If you specify an object list instead, all of the objects in the list must be the same type as those in the base collection.
- If the base collection contains different types of objects, all objects in the second collection are added. In this situation, you cannot specify an object list.

If the software finds no objects to add, it returns an empty collection.

Parameters

<i>base_collection</i>	Specifies the base collection of objects. Objects from the second collection or object list are added to this collection to create the new collection.
<i>second_collection_or_list</i>	Specifies the second object collection, or a list of objects.
<i>-unique</i>	Removes any duplicate objects from the new collection.

Examples

- The following command returns a collection that contains both the output ports and the input ports:
tempus> add_to_collection [all_outputs] [all_inputs]

Output:

CLK4 CLK3 CLK2 CLK1 IN4 IN3 IN2 IN1 OUT4 OUT3 OUT2 OUT1 OUT5

Here, the input and output ports are as follows:

tempus> all_inputs

CLK4 CLK3 CLK2 CLK1 IN4 IN3 IN2 IN1

tempus> all_outputs

OUT4 OUT3 OUT2 OUT1 OUT5

all_clocks

```
all_clocks  
[-help]
```

Creates a collection of all the clocks in the current design. If no clocks were defined, an empty string "" is returned.

To filter clocks, use the `get_clocks` command to create a collection of clocks matching a specific pattern. These clocks can be assigned to a variable or passed using another command.

Parameters

-help	Prints out the command usage.
-------	-------------------------------

Example

- The following example applies the `set_propagated_clock` command to all the clocks in the design:

```
tempus > set_propagated_clock [all_clocks]
```

Related Commands

- `create_clock`
- `get_clocks`
- `set_propagated_clock`
- `all_inputs`

all_connected

```
all_connected
[-help]
[-leaf] single_object_collection_or_object
```

Returns a collection of all of the objects connected to the specified net, pin, or port.

Note: If you specify a pin or port object, the `all_connected` command will return the net connected to that pin or port. If you specify a net object, this command will return ports, pins, or hpins connected to that net.

Parameters

-help	Prints out the command usage.
-leaf	Returns all instance pins connected to the specified net. When specified, only the flattened connection points (leaf pins) for the specified net are reported. <i>Default:</i> Returns hierarchical ports connected to the specified net.
<i>single_object_collection_or_object</i>	Specifies the net, pin, or port for which a collection of connected objects is to be reported. You can specify an object name or a single-object collection.

Examples

- The following command returns a collection of all the objects connected to pin `b1_Sin1`:
`all_connected Sin1`

The software returns the following information:
`Sin1`
- The following command returns a collection of all the objects connected to a pin contained in the single collection object `i_block1/b1_Sin1`:
`all_connected [get_pins {i_block1/b1_Sin1}]`
The software returns the following information:
`Sin1`
- The following command returns a collection of all of the hierarchical ports connected to the net `net_1`:

```
all_connected [get_nets net_1]
```

The software returns the following information:

TL2/l2_in TL1/l1_out

- The following command returns a collection that contains only the flattened connection points for the net `net_1`:

```
all_connected -leaf [get_nets net_1]
```

The software returns the following information:

TL2/L1/RX/D TL1/RX/Q

all_fanin

```
all_fanin
[-help]
[-hpin]
[-only_cells]
-to {collection | object_list}
[-trace_through {case_disable | user_disable | all | clocks | loop_snipped | latches}]
[-view view_name]
[-levels value | -pin_levels value]
[-startpoints_only]
[ > | >> ]
```

Returns a collection of pins, ports, or cells that exist in the fanin cone of the specified objects.

Parameters

-help	Prints out the command usage.
> >>	Writes the returned collection to the specified file. You can add the .gz extension to the file name to generate a compressed report. Note: The file name must be the last parameter specified.
-hpin	Returns the hierarchical pins along with the flat pins.
-levels value	Specifies the number of gate levels to go back from a specific pin object to create the collection.
-only_cells	Returns a collection of cells in the fan-in zone.
-pin_levels value	Specifies the number of delay arcs to go back from a specific pin object to create the collection.
-startpoints_only	Returns a collection only of valid startpoints (sequential cells and input ports) in the fanin cone.

<pre>-trace_through {case_disable user_disable all clocks loop_snipped latches}</pre>	You can specify one or more of the following options: • case_disable: Allows traversal of arcs disabled using the <code>set_case_analysis</code> command. The arcs disabled using the <code>set_disable_timing</code> command are not traversed. • user_disable: Allows traversal of arcs disabled using <code>set_disable_timing</code> command, but arcs disabled using <code>set_case_analysis</code> command are not traversed. • all: Allows traversal through all the arcs. • clocks: Allows traversal through pins for which clock constraints (such as <code>create_clock</code>, <code>create_generated_clock</code>) are applied. The pins disabled with the <code>set_case_analysis</code> and <code>set_disable_timing</code> commands are not traversed. • loop_snipped: Allows trace through the loop snipped arcs. • latches: Allows traversal of arcs of the latch cells.
<pre>-to {collection object_list}</pre>	Specifies the objects for which to return the collection. You can specify a list of objects or a collection of objects.
<pre>-view view_name</pre>	Specifies the view name for which fan cone traversal is to be performed.

Examples

- The following command returns the collection of cells in the fan-in zone of the output port OUT1:

```
tempus > all_fanin -to OUT1 -only_cells
```
- The following command returns all the valid startpoints (sequential elements and/or input ports) in the fanin zone of OUT1, allowing the traversal through arcs disabled by `set_case_analysis` and `set_disable_timing`:

```
tempus > all_fanin -to OUT1 -startpoints_only -trace_through all
```

Related Commands

- [all_fanout](#)

all_fanout

```
all_fanout
[-help]
[-endpoints_only]
[-hpin]
[-only_cells]
[-trace_through {case_disable | user_disable | all | clocks | loop_snipped | latches}]
[-view view_name]
[-levels value | -pin_levels value]
{-from {collection | object_list} }
[ > | >> ]
```

Returns a collection of pins, ports, or cells that exist in the fanout cone of the specified objects.

Parameters

-help	Prints out the command usage.
-endpoints_only	Returns a collection of valid endpoints (sequential elements or output ports) in the fanout cone.
-hpin	Returns hierarchical pins along with the flat pins.
> >>	Writes a list of the returned collection elements to the specified file. You can add the .gz extension to the file name to generate a compressed report. Note: The file name must be the last parameter specified.
-from {collection object_list}	Specifies the objects for which to return the collection. You can specify a list of objects or a collection of objects.
-levels value	Specifies the number of gate levels to go forward from a specific pin object to create the collection.
-only_cells	Returns a collection only of the cells in the fanout zone.
-pin_levels value	Specifies the number of delay arcs to go forward from a specific pin object to create the collection.

<pre>-trace_through {case_disable user_disable all clocks loop_snipped latches}</pre>	Allows traversal through disabled arcs. You can specify one of the following options: • case_disable: Arcs that are disabled using the <code>set_case_analysis</code> command and through physical constants are traversed. The arcs disabled using <code>set_disable_timing</code> are not traversed. • user_disable: Arcs disabled using <code>set_disable_timing</code> command are traversed, but those disabled using <code>set_case_analysis</code> command are not traversed. • all: Enables trace through all the arcs. • clocks: Allows traversal through pins for which clock constraints (such as <code>create_clock</code>, <code>create_generated_clock</code>) are applied. The pins disabled with the <code>set_case_analysis</code> and <code>set_disable_timing</code> commands are not traversed. • loop_snipped: Allows trace through the loop snipped arcs. • latches: Allows traversal of arcs of the latch cells.
<pre>-view view_name</pre>	Specifies the view name for which fan cone traversal is to be performed.

Examples

- The following command returns the collection of cells in the fanout zone of the input port IN1:

```
tempus > all_fanout -from IN1 -only_cells
```
- The following command returns all the valid endpoints (sequential elements and/or output ports) in the fanout zone of IN1, allowing the traversal through arcs disabled by `set_case_analysis` and `set_disable_timing`:

```
tempus > all_fanout -from IN1 -endpoints_only -trace_through all
```

Related Commands

- `all_fanin`

all_inputs

```
all_inputs
[-help]
[-clock list_of_clocks]
[-no_clocks]
[-edge_triggered]
[-level_sensitive]
```

Returns a collection of input or output ports in the current design.

To filter inputs, use the `get_ports` command to create a collection of ports that match specific criteria. You can assign these ports to a variable or pass them using another command. You can limit the contents of the collection to ports with input delay relative to a specific clock.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-clock</code> <code>list_of_clocks</code>	Considers or limits the search to ports that have an input delay relative to a specified list of clocks.
<code>-edge_triggered</code>	Returns a collection of input ports (specified with the <code>set_input_delay -clock</code> constraint) that have edge-triggered input delay.
<code>-</code> <code>level_sensitive</code>	Returns a collection of input ports (specified with the <code>set_input_delay -level_sensitive</code> constraint) that have level-sensitive input delay.
<code>-no_clocks</code>	Returns primary input ports that do not have <code>create_clock</code> or <code>create_generated_clock</code> constraints.

Example

- The following command returns the names of all the input ports with input delay relative to an ideal_clk:

```
tempus > all_inputs -clock ideal_clk
```
- The following example lists all input ports in the current design:

```
tempus > all_inputs
```
- The following example sets the drive value of all the input ports on the current design to 10:

```
tempus > set_drive 10.0 [all_inputs]
```
- The following example marks with a multicycle value of 0 to all paths from inputs having level-sensitive input delay relative to PHI1 to level-sensitive registers clocked by CDNSCLK:

```
tempus > set_multicycle_path 0 \  
    -from [all_inputs -clock CDNSCLK -level_sensitive] \  
    -to [all_registers -data_pins -clock CDNSCLK]
```

Related Commands

- `get_ports`
- `report_ports`
- `set_input_delay`

all_instances

```
all_instances  
[-help]  
object  
[-hierarchical]
```

Returns a collection of all instances available in a design, or all instances of a specified library cell existing in the current design.

Note: You cannot use the `all_instances` command on the top level of the design (`all_instances [get_design top]`).

However, you can use the command for hierarchical modules inside the top level. For example, if `SUB` is a hierarchical module in the top level of the design, you can specify the following command to return all instances of that module:

```
all_instances [get_design SUB].
```

Parameters

<code>-help</code>	Prints out the command usage.
<code>- hierarchical</code>	Specifies that the patterns should be matched hierarchically. If the hierarchical name of an instance at a particular hierarchical level matches the patterns, it is included in the result.
<i>object</i>	Specifies the name of a design or a library cell in the current design. You can specify a pattern for the design or library cell name, or a single collection of a design or library cell.

Examples

- The following command returns a collection of all the instances of the library cell BUF:

```
tempus> all_instances BUF
```

- The following returns a collection of all instances of the sub-module `subblock`:

```
tempus > all_instances [get_design subblock]
```

```
subblock_inst1 subblock_inst2
```

all_outputs

```
all_outputs
[-help]
[-clock list_of_clocks]
[-edge_triggered]
[-level_sensitive]
```

Creates a collection of all output ports in the current design. You can limit the contents of a collection to ports with output delay relative to a specific clock. You can assign these ports to a variable or pass them using another command.

Parameters

-help	Prints the command usage.
-clock <i>list_of_clocks</i>	Considers only output ports with an output delay relative to a specific collection or a list of clocks.
-edge_triggered	Returns a collection of output ports with the <code>set_output_delay -clock</code> constraint.
-level_sensitive	Returns a collection of output ports with the <code>set_output_delay -level_sensitive</code> constraint.

Examples

- The following command returns a collection of output ports with an output delay relative to `ideal_clk`:

```
tempus > all_outputs -clock ideal_clk
```

Related Commands

- `get_ports`
- `report_ports`
- `set_output_delay`

all_registers

```
all_registers
[-help]
[-clock { clock_list }]
[-cells]
[-rise_clock { clock_list }]
[-fall_clock { clock_list }]
[-flops | -edge_triggered]
[-no_hierarchy]
[-latches | -level_sensitive]
[-macros]
[-master_slave]
[-data_pins]
[-clock_pins]
[-output_pins]
[-async_pins]
[-slave_clock_pins]
```

Returns a collection of instances and pins of flip-flops, latches, or sequential macros in the design. A cell is considered sequential if the library cell defines trigger or timing check arcs.

Note: The following options can be used only after building the timing graph:

- -clock
- -rise_clock
- -fall_clock

Parameters

<code>-help</code>	Prints out the command usage.
<code>-async_pins</code>	Returns a collection of async preset or clear pins.
<code>-cells</code>	Creates a collection of cells. This is the default option.
<code>-clock {clock_list}</code>	Specifies the name of a clock, a list of clocks, or a clock collection. The <code>all_registers</code> command returns a collection of registers clocked by the specified clock list.
<code>-clock_pins</code>	Returns a collection of register clock pins. You can use additional parameters to further filter this collection.
<code>-data_pins</code>	Returns a collection of register data input pins. You can use additional parameters to further filter this collection.
<code>-fall_clock {clock_list}</code>	Specifies the name of a clock, a list of clocks, or a clock collection. The <code>all_registers</code> command creates a collection of registers opened by the falling edge of the clocks specified in the clock list.
<code>-flops -edge_triggered</code>	Returns a collection of edge-triggered sequential instances. Both the <code>-flops</code> and <code>-edge_triggered</code> parameters return the same collection. The <code>-flops</code> parameter is provided for backward compatibility.
<code>-latches -level_sensitive</code>	Returns a collection of level-sensitive sequential instances. Both <code>-latches</code> and <code>-level_sensitive</code> parameters return the same collection. The <code>-latches</code> parameter is provided for backward compatibility.
<code>-macros</code>	Returns sequential instances that are neither flip-flops nor latches.
<code>-master_slave</code>	Returns a collection of master-slave register cells.
<code>-no_hierarchy</code>	Searches the current design hierarchy level; does not descend the hierarchy into sub-modules.
<code>-output_pins</code>	Returns a collection of register output pins.
<code>-rise_clock {clock_list}</code>	Specifies the name of a clock, a list of clocks, or a clock collection. The <code>all_registers</code> command creates a collection of registers opened by the rising edge of clocks specified in the clock list.
<code>-slave_clock_pins</code>	Returns a collection of slave clock pins of the master-slave registers. You can specify the slave clock pins as <code>clocked_on_also</code> in the library.

Example

- The following command reports the 10 worst paths going to registers in the design:
- The following script uses the `all_registers` command along with other TCL commands to get the number of flip-flops in the design:

```
tempus > report_timing -to [all_registers] -max_paths 10

tempus > set flopName [all_registers -flops]
tempus > set flopCount [sizeof_collection $flopName]
tempus > puts "Number of flops in the design is $flopCount"
```

Related Commands

- `get_cells`
- `get_pins`

append_to_collection

```
append_to_collection
[-help]
var_name
second_collection_or_list
[-unique]
```

Appends objects from one collection to another collection. The following rules apply:

- If all of the objects in the base collection are of the same type, only the objects in the second collection that are the same type as those in the base collection are appended to the base collection.
- If the base collection contains different types of objects, all objects in the second collection are appended to it. In this situation, you cannot specify an object list.

If the software finds no objects to add, it returns an empty collection.

Parameters

-help	Prints out the command usage.
var_name	Specifies a pointer to the base collection of objects. The objects in the second collection or object list are appended to this collection. If you specify a pointer that does not refer to an existing base collection, the software creates a new collection that uses the pointer as a reference.
second_collection_or_list	Specifies the second collection of objects, or an object list.
-unique	Retains any duplicate objects that already exist in the base collection, but does not append duplicate objects from the second collection to the base collection.

Examples

- The following command appends the nets in the specified collection to the collection `newVar`:
`append_to_collection newVar [get_nets {CLK1}]`
- The following command appends the specified nets to the collection `newVar`:
`append_to_collection newVar {Sin1 CLK1 Sout1}`
- You can use the following commands to create another collection:

```
tempus 6> set base_clock [get_clocks sbw_clk]
0xe50
tempus 7> set top_clock [get_clocks aclk]
0xe51
tempus 8> set base_top_clock ""
tempus 13> append_to_collection base_top_clock $base_clock
0xe54
tempus 14> append_to_collection base_top_clock $top_clock
0xe54
tempus 15> get_clocks $base_top_clock
sbw_clk aclk
0xe55
tempus 16> get_clocks $base_clock
sbw_clk
0xe56
tempus 17> get_clocks $top_clock
aclk
0xe57
```

Note: If a new collection variable is defined with an existing collection, then both point to the same object set. Any modifications made to either collection will be visible in both cases - because they reference each other.

compare_collections

```
compare_collections  
[-help]  
collection1  
collection2  
[-order_dependent]
```

Compares two collections and returns a value of 0 if all of the objects contained in both collections are the same. If the objects in both collections are different, the software returns 1.

Parameters

-help	Prints out the command usage.
collection1	Specifies the first collection.
collection2	Specifies the second collection.
-order_dependent	Specifies that the order of the objects in both collections must be the same for the collections to be considered the same.

Examples

- The following example shows 3 collections as defined below:

```
tempus> set coll1 [all_registers]  
tempus> set coll2 [get_cells -hierarchical * -filter "is_sequential==true"]  
tempus> set coll3 [get_cells -hierarchical * -filter "is_combinational==true"]
```

- The collections coll1 and coll2 are identical, while coll1 and coll3 are not, and the following commands can check this:

```
tempus> compare_collections $coll1 $coll2  
0  
tempus> compare_collections $coll1 $coll3  
1
```

copy_collection

```
copy_collection  
[-help]  
base_collection
```

Returns a collection that is an exact copy of the specified base collection.

Parameter

<code>-help</code>	Prints out the command usage.
<i>base_collection</i>	Specifies the base collection of objects.

define_proc_arguments

```
define_proc_arguments
[-help]
proc_name
[-info info_text]
[-define_args arg_defs]
[-hide]
```

Defines the arguments and attributes of a Tcl procedure. The arguments and attributes are displayed when you view the help associated with the procedure.

The `define_proc_arguments` command, when used with the `parse_proc_arguments` command, enables you to validate command usage. For invalid commands, appropriate error messages are displayed.

Parameters

<code>-help</code>	Prints out the command usage.
<code>proc_name</code>	Specifies the name of an existing Tcl procedure.
<code>-info</code> <code>info_text</code>	Provides help information for the procedure.
<code>-define_args</code> <code>arg_defs</code>	Defines all possible arguments of the Tcl procedure. It can also be used to define the help text and constraints for individual arguments of the procedure and generate the help for command options. The variable <code>arg_defs</code> is a list of lists, where each list element defines one argument. Each element in the list has the following format: <pre>"arg_name" "option_help" "value_help" "data_type" "attributes"</pre>
	<code>arg_name</code> : Specifies the name of the argument.
	<code>option_help</code> : Provides a short description of the argument.
	<code>value_help</code> : Specifies the help information for the associated option values.
	<code>data_type</code>: (Optional) Used for validation of options. Supported data types are <code>string</code>, <code>list</code>, <code>boolean</code>, <code>int</code>, <code>integer</code>, <code>float</code>, or <code>one_of_string</code>. Default: <code>string</code>

	<i>attributes:</i> (Optional) Used for validation of options. The attributes can be: • <code>required</code>: This argument must be specified. This is mutually exclusive with <code>optional</code> and <code>internal</code>. • <code>optional</code>: Specifying this argument is optional. This is mutually exclusive with <code>required</code>. • <code>internal</code>: Disables the display of help information for the option in the procedure help. This argument is mutually exclusive with <code>required</code>. • <code>value_help</code>: Indicates that the valid values for <code>one_of_string</code> argument should be listed whenever argument help is shown. • <code>values {allowable_values}</code>: This argument must be used with the <code>one_of_string</code> data type. • <code>merge_duplicates</code> : Ensures that when the option appears more than once in a command, its values are concatenated into a list of values. The default behavior is to use the rightmost value specified for the option. • <code>bind_option other_arg_name</code>: Binds the current option with the option specified as the <code>other_arg_name</code> variable. • <code>mutual_exclusive_group group_name</code>: Defines a group name and includes the current option in the <code>group_name</code> group. All options within a group are mutually exclusive. • <i>Default</i>: <code>optional</code>
<code>-hide</code>	Hides the procedure and its body.

Examples

The following Tcl procedure uses the `define_proc_arguments` command to define arguments and attributes, and to validate the command usage.

```
proc cmdProc {args} {
  parse_proc_arguments -args $args results
  foreach argname [array names results] {
    puts "$argname = $results($argname)"
  }
}
```

```
define_proc_arguments cmdProc -info "Command Description" \  
-define_args \  
{ {-string "String help" "string_val" string optional} \  
{ -integer "Int help" "int_value" int optional} \  
{ -list "List help" "list_value" list {optional bind_option -Bool}} \  
{ -boolean_a "BoolA help" "" boolean {optional mutual_exclusive_group \  
g0}} \  
{ -boolean_b "BoolB help" "" boolean {optional mutual_exclusive_group \  
g0}} \  
{ -float_num "Float help" "float_value" float} \  
{ -oosa "One of StringA help" "one_of_str_value" one_of_string \  
{optional value_help {values {v1 v2}}}} \  
{ -oosb "One of StringB help" "one_of_str_value" one_of_string \  
{optional {values {v4 v5}}}} \  
}
```

Now, if the `cmdProc` procedure is invoked with the `-help` option at the command prompt, the following help information is displayed.

```
tempus> cmdProc -help  
Usage: cmdProc # Command Description  
[-string string_val] (String help)  
[-integer int_value] (Int help)  
[-list list_value] (List help, Bind_Option=-Bool)  
[-boolean_a ] (BoolA help, Mutual_Exclusive_Group=g0)  
[-boolean_b ] (BoolB help, Mutual_Exclusive_Group=g0)  
[-float_num float_value] (Float help)  
[-oosa one_of_str_value] (One of StringA help, Values={v1 v2 })  
[-oosb one_of_str_value] (One of StringB help)
```

define_property

```
define_property
[-help]
-object_type string
-type data_type
[-lower_bound val]
[-upper_bound val]
[-string_values {values}]
property_name
```

Defines a new property.

Design objects can be associated with user-defined commands for analysis and debugging.

You can use the `define_property` command if a `define` statement is not used for the specified property value(s). To read property values from the library, the command must be issued before the timing library is read.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-lower_bound <i>val</i></code>	Specifies a lower range - integer, float, or double.
<code>-object_type <i>string</i></code>	Specifies the type of object on which the property is to be defined.
<code><i>property_name</i></code>	Specifies the name of the property.
<code>-string_values <i>values</i></code>	Specifies the values that can be defined for the property.
<code>-type <i>data_type</i></code>	Specifies the type of property.
<code>-upper_bound <i>val</i></code>	Specifies a higher range - integer, float, or double.

Examples

- The following example defines a property having an object type `pin` and a type of property `int`:
tempus > `define_property -object_type pin -type int slk`
- The following example defines a property on a library arc:
tempus > `define_property -object_type lib_arc -type int libArcProperty`
- The following example defines `aocv_weight` property on a library arc:
tempus > `define_property -type float -object_type lib_arc aocv_weight`

- The following example defines certain nets as critical, using the `critical_net` property and then setting it to `true` for those nets:

```
tempus > define_property critical_net -object_type net -type boolean
tempus > set_property [get_nets n2] critical_net true
```

You can view the report and query on critical nets, as shown below:

```
tempus > report_property [get_nets n2]
```

...

property	value

capacitance_max	0.002
capacitance_min	0.002
driver_pins	{...}
escaped_name	n2
hierarchical_name	n2
has_detailed_parasitics	false
is_dont_touch	false
is_hierarchical	false
load_pins	{...}
name	n2
object_type	net
pin_capacitance_max	0.002
pin_capacitance_max_rise	0.002
pin_capacitance_max_fall	0.002
pin_capacitance_min	0.002
pin_capacitance_min_rise	0.002
pin_capacitance_min_fall	0.002
total_capacitance_max_fall	0.002
total_capacitance_max_rise	0.002
total_capacitance_min_rise	0.002
total_capacitance_min_fall	0.002
wire_capacitance_max	0.000
wire_capacitance_min	0.000
wire_capacitance_max_rise	0.000
wire_capacitance_max_fall	0.000
wire_capacitance_min_rise	0.000
wire_capacitance_min_fall	0.000
critical_net	true

```
tempus > get_property [get_nets n2] critical_net
true
```

```
tempus > get_nets -filter "critical_net==true"
n2
0x1c
```

Related Commands

- `get_property`
- `report_property`
- `set_property`
- `reset_property`

filter_collection

```
filter_collection  
[-help]  
base_collection  
{filter_expression}  
[-nocase]  
[-quiet]  
[-regex]
```

Returns a new collection of objects that were filtered from the specified collection of objects based on user-specified criteria (*filter_expression*). The original collection remains unmodified.

The following rules apply:

- The *filter_expression* can be created by combining several object properties using the following relational and logical operators: >, <, ==, !=, <=, >=, &&, ||, =~, !~, AND, OR. You can also use parentheses to create expressions.
- You can use the `defined` and `undefined` existence operators to determine whether an attribute is defined for an object.
- All design object properties supported through the `get_property` command can be used for filtering the base collection.

Note: You can perform filtering operations by specifying the `-filter` parameter with the `get_*` commands.

Parameters

<code>-help</code>	Prints out the command usage.
<code>base_collection</code>	Specifies the collection of objects from which to filter objects for the new collection.
<code>filter_expression</code>	Specifies the criteria for filtering objects. The software adds any objects in the base collection that match <code>filter_expression</code> to the new collection. See " Examples " on how to create filter expressions.
<code>-nocase</code>	Specifies that pattern matching is not case sensitive. Note: You must specify <code>-regexp</code> to use this parameter.
<code>-quiet</code>	Suppresses all error and warning messages that are generated when the <code>filter_collection</code> command is run.
<code>-regexp</code>	Treats the specified patterns as regular expression patterns. <i>Default:</i> Treats the specified pattern as a wildcard.

Examples

- The following command returns a single-object collection containing the pin with the hierarchical name `lat_3/Q`:

```
filter_collection [get_pins] {hierarchical_name == lat_3/Q}
```
- The following command returns a collection of pins that have a maximum slew greater than 0.1:

```
filter_collection [get_pins] {slew_max_rise > 0.1}
```
- The following commands create a new collection containing all those paths from the path collection with a slack value greater than -0.806 and less than 0.0:

```
set paths [report_timing -collection -max_paths 10]
set newPaths [filter_collection $paths {(slack > -0.806) && (slack < 0.0)}]
```
- The following command returns a collection of object names that do not match `"*clk*"`:

```
filter_collection $abc {full_name !~ *clk*}
out in1 in2 in3 select TopBus[2] TopBus[1] TopBus[0] bidi
0x53
```
- The following command returns a collection of object names that are not `"*clk*"` only:

```
filter_collection $abc {full_name != *clk*}
```

```
out clk1 clk2 in1 in2 in3 select TopBus[2] TopBus[1] TopBus[0] bidi
0x59
```

- The following command returns a collection of object names that match “*clk*”:

```
filter_collection $abc {full_name =~ *clk*}
clk1 clk2
0x5f
```

- The following command creates a collection of pins with `user_constant_value`:

```
set constant_pins [filter_collection [get_pins] {defined(user_constant_value)}]
```

Related Commands

- `add_to_collection`
- `compare_collections`
- `copy_collection`
- `foreach_in_collection`

foreach_in_collection

```
foreach_in_collection  
[-help]  
var_name  
collection  
body
```

Iterates through all objects in the specified collection and executes the commands in the specified script body.

Parameters

<code>-help</code>	Prints out the command usage.
<code>body</code>	Specifies the script body of commands to start during each iteration.
<code>collection</code>	Specifies the collection of objects to iterate over.
<code>var_name</code>	Specifies the iterator variable. During each iteration, <code>var_name</code> is set to a collection of exactly one object. Any command that accepts collections can accept <code>var_name</code> as a value.

Examples

- The following command prints out the hierarchical names of all the cells in a design:

```
foreach_in_collection iCell [get_cells *] {  
  puts "Cell = [get_property $iCell hierarchical_name]"  
}  
Cell = ck_0  
Cell = ck_1  
Cell = ck_2  
Cell = ck_4
```

- The following example shows a more complex use of the `foreach_in_collection` command:

```
set cellCollection [get_cells templ_reg]  
set libCellCollection [get_lib_cells -of_objects $cellCollection]  
set libArc [get_lib_arcs -of_objects $libCellCollection]  
  
foreach_in_collection arc $libArc {  
  set fpin [get_property $arc from_lib_pin]  
  set tpin [get_property $arc to_lib_pin]
```

```
set sense [get_property $arc timing_type]

set from_pin_name [get_property $fpin name]
set to_pin_name [get_property $tpin name]

puts [format "%25s -> %25s %-20s" $from_pin_name $to_pin_name $sense]

    }
    CK -> D          setup_rising
    CK -> D          hold_rising
    CK -> CK         minimum_period
    CK -> CK         minimum_period
    CK -> CK         min_pulse_width

foreach_in_collection.txt

    CK -> CK         min_pulse_width
    RN -> RN         min_pulse_width
    CK -> RN         recovery_rising
    CK -> RN         removal_rising
    CK -> Q          falling_edge
    RN -> Q          clear
```

Related Commands

- [add_to_collection](#)
- [compare_collections](#)
- [copy_collection](#)
- [filter_collection](#)

get_activity

```
get_activity
[-help]
[-instance <string>]
[-list_of_nets_based_on_driver <string>]
[-list_of_nets_based_on_source_of_activity_info <string>]
[-net <string>]
[-outfile <string>]
[-pin <string>]
[-port <string>]
[-report_average_switching_activity]
[-report_cell_group_activity_summary_report]
[-summary]
[-tcl_list]
```

Determines the source of activity for the specified nets/pins/ports. The power analysis engine has various sources from which activities can be annotated, such as switching activity specification (input/sequential/global/clock gate), clock definitions/constants through SDC/TWF, activity files (VCD/TCF/SAF), user defined activity, and/or propagated values.

Parameters

-help	Outputs a brief description that includes the type and default information for each <code>get_activity</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man get_activity</code>
-instance <i>string</i>	Specifies the top/hierarchy instance or the leaf level instance name for which the average activity is to be reported.
-list_of_nets_based_on_driver <i>string</i>	Specifies to output the list of nets based on the specified net driver. You can specify one or more net drivers at a time, as shown below: <pre>get_activity -list_of_nets_based_on_driver primary_inputs combinational memory</pre> To specify multiple drivers, you can use / (means 'or') or & (means 'and') between two drivers, without a space.

- list_of_nets_based_on_source_of_activity_info string	Specifies to output the list of nets based on the specified activity source information.
-net string	Specifies the name of the net. Note: You can use the <code>get_nets</code> command to determine the activity source for a collection of nets.
-outfile string	Specifies the output file into which the activity sources are to be written.
-pin string	Specifies the name of the pin. Note: You can use the <code>get_pins</code> command to determine the activity source for a collection of pins.
-port string	Specifies the name of the port. Note: You can use the <code>get_ports</code> command to determine the activity source for a collection of ports.
-report_average_switching_activity	Specifies to report the average activity of the given instance/hierarchy/top-level.
-report_cell_group_activity_summary_report	Specifies to give a summary of cell group activity.
-summary	Generates a detailed activity-annotation report for the specified nets, pins, and ports. This report contains the following information: • Primary Inputs • Sequential Outputs • Memory/Macro Outputs • Tristate Outputs Note: Use the <code>get_activity -summary</code> command to report activity annotation summary without printing information for nets/pins/ports.

<code>-tcl_list</code>	Generates reports in TCL list format rather than tabular format. The <code>-tcl_list</code> and <code>-outfile</code> parameters are mutually exclusive; you cannot specify them together.
------------------------	--

The activity report for pins is displayed in the following format:

```
<Pin Name> <Source of the Activity Value> <Duty Cycle> <Toggles/Sec>
```

The report fields are described below:

- **Pin Name:** Name of the specified pins.
- **Source of the Activity Value:** Source of activity for the specified nets, pins, or ports.
- **Duty Cycle:** Probability of signal at 1.
- **Toggles/Sec:** Number of toggles divided by the total simulation time.

Examples

- The following example returns the activity sources of all nets that match the specified pattern and writes the output information to a file named `nets.rpt`:

```
get_activity -net [get_nets n*] -outfile nets.rpt
```

The following is the example output of the `nets.rpt` file:

```
n_14 user_defined_activity 0.5 6e+07
n_11 user_defined_activity 0.5 6e+07
n_25 user_defined_activity 0.5 6e+07
n_13 user_defined_activity 0.5 6e+07
n_26 user_defined_activity 0.5 6e+07
n_9 user_defined_activity 0.5 6e+07
n_27 user_defined_activity 0.5 6e+07
n_23 user_defined_activity 0.5 6e+07
n_20 user_defined_activity 0.5 6e+07
n_16 user_defined_activity 0.5 6e+07
```

- The following example returns the activity source of the pin `i_4/A1` and writes the output information to a power report file named `pin.rpt`:

```
get_activity -pin i_4/A1 -outfile pin.rpt
```

- The following command reports the average activity of all nets:

```
get_activity -report_average_switching_activity
```


The following is the example output:

Avg Switching activity of all nets 4.727e+17

- The following command gives a summary of cell group activity:

```
get_activity -report_cell_group_activity_summary_report
```

The following is the example output:

Group Activity Summary Report:

Source ->	activity file	set_switching_activity
propagated	default	Total
All nets	2	1
2	5	10
Break-up based on net driver:		
Primary Input	2 (40%)	0 (0%)
0 (0%)	3 (60%)	5
Sequential	0 (0%)	1 (33.33%)
0 (0%)	2 (66.67%)	3
Combinational	0 (0%)	0 (0%)
2 (100%)	0 (0%)	2
Memory	0 (0%)	0 (0%)
0 (0%)	0 (0%)	0
ICG	0 (0%)	0 (0%)
0 (0%)	0 (0%)	0
Black Box	0 (0%)	0 (0%)
0 (0%)	0 (0%)	0

- The following command gives the list of nets based on primary inputs:

```
get_activity -list_of_nets_based_on_driver primary_inputs
```

The following is the example output:

D2 D1 CP2 CP1 dummy0

- The following command gives the list of nets based on primary inputs or the sequential driver:

```
get_activity -list_of_nets_based_on_driver primary_inputs|sequential
```

The following is the example output:

Q2 Q1 D2 D1 CP2 CP1 n1 dummy0

- The following command gives the list of nets based on primary inputs or sequential driver for the instance `ff1b`:

```
get_activity -list_of_nets_based_on_driver primary_inputs|sequential -instance ff1b
```

- The following command gives the list of nets based on the propagated source for the instance `i2`:

```
get_activity -list_of_nets_based_on_source_of_activity_info propagated -instance i2
```

- The following command gives a summary of cell group activity for the instance `i2`:

```
get_activity -instance i2 -report_cell_group_activity_summary_report
```

get_arcs

```
get_arcs
[-help]
{ [-to to_list ] [-from from_list ]
  | -of_objects object_list }
[-filter expr ]
[-quiet]
```

Creates a collection of timing arcs. Assign these arcs to a variable, or pass them to another command. If no objects match the criteria, an empty collection is returned.

Note: The timing graph must be built before you can use this command. A warning message is issued indicating that the timing graph is not generated.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-filter <i>expr</i></code>	Filters a collection based on the specified expression. For any arcs that match the pattern, the software evaluates the expression based on the arc's attribute. For details on how to create filter expressions, refer to the <code>filter_collection</code> command.
<code>-from <i>from_list</i></code>	Adds all timing arcs that fan in directly to the specified pins or ports to the collection. You can specify a list of pins or a list of ports, but you cannot specify a list that contains both pins and ports. You can specify a pattern for the pin or port name, a collection of pins or ports, or a pattern and a collection. Note: You cannot specify this parameter with the <code>-of_objects</code> parameter.
<code>-of_objects <i>object_list</i></code>	Adds all timing arcs that are associated with the specified cells or nets to the collection. You can specify a list of cells or a list of nets, but you cannot specify a list that contains both cells and nets. You can specify a pattern for the cell or net name, a collection of cells or nets, or a pattern and a collection. Note: You cannot specify this parameter with the <code>-to</code> or <code>-from</code> parameters.
<code>-quiet</code>	Suppresses all error and warning messages generated when the <code>get_arcs</code> command is run.
<code>-to <i>to_list</i></code>	Adds all timing arcs that fan out directly from the specified list of pins or ports to the collection. You can specify a list of pins or a list of ports, but you cannot specify a list that contains both pins and ports. You can specify a pattern for the pin or port name, a collection of pins or ports, or a pattern and a collection. Note: You cannot specify this parameter with the <code>-of_objects</code> parameter.

Examples

- The following command returns a collection of the timing arcs fan in directly to the pin `dff1/CP`:

```
get_arcs -from dff1/CP
```
- The following command creates a collection of the timing arcs that are associated with the nets in the specified collection:

```
get_arcs -of_objects [get_nets {n1}]
```

get_cells

```
get_cells
[-help]
[-filter expr]
[-hierarchical]
[-hsc char]
[-leaf]
[-nocase]
[-quiet]
[-regexp]
[<patterns> | -of_objects object_list]
```

Creates a collection of instances in the current design whose name matches the supplied pattern list. Assign this collection to a variable or pass it as an argument in another command.

Parameters

-help	Prints out the command usage.
-filter <i>expr</i>	Filters the collection with the specified expression. For any cells that match the pattern, the software evaluates expression based on the cell's attribute. You can filter cells using any of the cell properties supported through the get_property command. For details on how to create filter expressions, refer to the filter_collection command.
-hierarchical	Specifies that the patterns should be matched hierarchically. If the flat name of an instance at a particular hierarchical level matches the patterns, it is included in the result. For example, if there is a top level hierarchical instance <code>Top</code> and it contains another sub-instance called <code>FF1</code>, then the following command can be used to get the instance <code>Top/FF1</code> (since the given pattern will be matched with the instance flat name <code>FF1</code>): <pre>get_cells -hierarchical FF*</pre> However, the following command will display an error: <pre>get_cells -hierarchical Top/FF*</pre>
-hsc <i>char</i>	Specifies the hierarchical delimiter for patterns. For example, if the hierarchical delimiter is <code>@</code>, the <code>sub/I1@I2</code> pattern matches instance <code>I2</code> within the hierarchy limited at <code>sub/I1</code>.

<code>-leaf</code>	Returns leaf cells only. When the <code>-leaf</code> parameter is not specified, the <code>get_cells</code> command returns the hierarchical instances present on a net along with the flat instances at the same hierarchical level as the specified object. Note: The <code>-leaf</code> parameter can only be specified with the <code>-of_objects</code> parameter.
<code>-nocase</code>	Specifies that pattern matching is not case sensitive. Note: You must specify <code>-regexp</code> to use this parameter.
<code>-of_objects</code> <i>object_list</i>	Creates a collection of all cells associated with the specified pins, nets, library cells, or design cells. You can specify a list of pins or a list of nets, but you cannot specify a list that contains both pins and nets. You can specify patterns for the pin or net names, a collection of pins or nets, or a combination of patterns and a collection. The <code>-of_objects</code> parameter supports <code>get_designs</code> and <code>get_lib_cells</code> as valid collection objects. The use model is as follows: <pre>get_cells -of_objects [get_designs object_list] get_cells -of_objects [get_lib_cells object_list]</pre>
<i>patterns</i>	Specifies patterns for instance names. Patterns include the <code>*</code> and <code>?</code> wildcard characters and collections of instances. The <code>get_cells</code> command returns a collection of instances that match the patterns. If no such instance is found, an empty collection is returned.
<code>-quiet</code>	Suppresses all error and warning messages generated when the <code>get_cells</code> command is run.

-regexp	Treats the specified patterns as regular expression patterns. When this parameter is specified, the given <code>start</code> and <code>end</code> patterns are assumed to match the start and end part of the complete object name. For example, the following can be used to retrieve the hierarchical instance <code>Top/block1/block2/FF1</code>: <pre>get_cells Top/*/*/FF1 get_cells -regexp Top/.*/FF1</pre> However, the following command will display an error : <pre>get_cells Top/*/FF1</pre> If this parameter is specified with the <code>-hierarchical</code> parameter, the given pattern will be matched with the complete hierarchical name of objects instead of only the flat name, and all the successful matches are included in the result. For example, if there is a top level hierarchical instance <code>Top</code> and it contains another sub-instance called <code>FF1</code>, then either of these can be used to retrieve the <code>Top/FF1</code> instance: <pre>get_cells -hierarchical FF1 get_cells -hierarchical -regexp Top/FF1</pre> However, the following commands will display an error : <pre>get_cells -hierarchical Top/FF1 get_cells -hierarchical -regexp FF1</pre>
----------------	--

Examples

- The following command searches for pins within the hierarchy limited to `B`:

```
tempus > get_cells -hsc @ B@*
```
- The following command searches for cells with a set pattern and filter:

```
tempus > get_cells "o*" -filter "@ref_lib_cell_name != FD2 && @is_black_box==false"
```

Related Commands

- `get_libs`
- `get_lib_cells`
- `get_lib_pins`
- `get_pins`

get_clocks

```
get_clocks
[-help]
[-filter expr]
[-regexp]
[-nocase]
[-quiet]
[patterns]
```

Creates a collection of clocks whose names match the supplied pattern list. This parameter assigns this collection to a variable or passes it as an argument to another command.

The `get_clocks` command reports all clocks, including inactive clocks.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-filter <i>expr</i></code>	Filters the collection with the specified expression. For any clocks that match the pattern, the software evaluates the expression based on the clock's attribute. You can filter clocks using any of the clock properties supported through the <code>get_property</code> command. For details on how to create filter expressions, refer to the <code>filter_collection</code> command.
<code>-nocase</code>	Specifies that pattern matching is not case sensitive. Note: You must specify <code>-regexp</code> to use this parameter.
<code><i>patterns</i></code>	Specifies a pattern (or a list of patterns) for clock names. A pattern can contain <code>*</code> and <code>?</code> wildcard characters, or a collection of instances and clocks. Note: If a pattern is not specified, <code>get_clocks</code> returns all clocks that meet other filters.
<code>-quiet</code>	Suppresses all error and warning messages generated when the <code>get_clocks</code> command is run.
<code>-regexp</code>	Treats the specified patterns as regular expression patterns. <i>Default:</i> Treats the specified pattern as a wildcard

Examples

- The following example returns all clocks beginning with the name `CK`:
`get_clocks CK*`
- The following example returns a collection of all clocks beginning with the names `CK` and `Clk`:
`get_clocks [list CK* Clk* ]`
- The following example returns a collection of all active clocks beginning with the name `CK`:
`get_clocks CK* -filter {is_active == true}`
- The following command returns all the clocks that are propagated:
`get_clocks -filter "is_propagated_clock == true"`

Related Commands

- `all_clocks`
- `create_clock`
- `get_pins`
- `get_ports`
- `report_clocks`

get_designs

get_designs
[-help]
[-quiet]
patterns

Creates a collection of modules and assigns this collection to a variable or passes it as an argument to another command. The software uses this information during timing analysis.

Parameters

-help	Prints out the command usage.
<i>patterns</i>	Specifies patterns for module names. Patterns include the * and ? wildcard characters and collections of modules. The <code>get_designs</code> command returns a collection of modules that match the patterns. If no such module is found, an empty collection is returned.
-quiet	Suppresses all error and warning messages generated when the <code>get_design</code> command is run.

Example

- The following command returns a complete list of modules:
`get_designs *`

get_generated_clocks

```
get_generated_clocks
[-help]
[patterns]
[-filter expr]
[-nocase]
[-regexp | -exact]
```

Creates a collection of generated clocks whose names match the supplied pattern list. This parameter assigns this collection to a variable or passes it as an argument to another command.

Parameters

-help	Prints out the command usage.
-exact	Performs exact pattern matching against the object name, instead of wildcard or regular expression matching. Use this parameter when the object name includes special characters such as *, ?, and +. Note: If you specify this parameter, you cannot specify <code>-regexp</code> .
-filter expr	Filters the collection with the specified expression. For any clocks that match the pattern, the software evaluates the expression based on the clock's attribute.
-nocase	Specifies that pattern matching is not case sensitive.
patterns	Specifies patterns for generated clock names. Patterns include the * and ? wildcard characters, and collections of generated clocks.
-regexp	Treats the specified pattern as a regular expression. <i>Default:</i> Treats the specified pattern as a wild card Note: If you specify this parameter, you cannot specify <code>-exact</code> .

Examples

- The following command returns the collection of all the generated clocks present in the design whose name starts with "CK":

```
tempus> get_generated_clocks CK*
```



```
CKGEN1 CKGEN2 CKGEN3
```

get_lib_arcs

```
get_lib_arcs
[-help]
[-filter expr]
[-quiet]
{[[-from <from_lib_pins>] [-to <to_lib_pins>]] | -of_objects {lib_cell_list | timing_arcs}}
```

Creates a collection of library timing arcs. Assign these library timing arcs to a variable, or pass them to another command. If no objects match the criteria, an empty collection is returned.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-filter <i>expr</i></code>	Filters the collection with the specified expression. For any library arcs that match the pattern, the software evaluates the expression based on the attribute of the library arc. For details on how to create filter expressions, refer to the <code>filter_collection</code> command.
<code>-from</code> <code><i>from_lib_pins</i></code>	Creates a collection of all library arcs that originate from the specified library pins. You can specify a pattern for the pin name, a collection of pins, or a pattern and a collection. If you specify this parameter with the <code>-to</code> parameter, the software creates a collection of all library arcs that exist between the "to" and "from" library pins. Note: You cannot specify this parameter with the <code>-of_objects</code> parameter.
<code>-of_objects</code> <code>{<i>lib_cell_list</i></code> <code> <i>timing_arcs</i>}</code>	Creates a collection of all library arcs that are associated with the specified library cells or timing arcs. You can specify a list of library cells or a list of timing arcs, but you cannot specify a list that contains both library cells and timing arcs. You can specify a pattern for the library cell or timing arc name, a collection of library cells or timing arcs, or a pattern and a collection. Note: You cannot specify this parameter with the <code>-to</code> or <code>-from</code> parameters.
<code>-quiet</code>	Suppresses all error and warning messages generated when the <code>get_lib_arcs</code> command is run.
<code>-to</code> <code><i>to_lib_pins</i></code>	Creates a collection of all library arcs that terminate at the specified library pins. You can specify a pattern for the pin name, a collection of pins, or a pattern and a collection. If you specify this parameter with the <code>-from</code> parameter, the software creates a collection of all library arcs that exist between the "to" and "from" library pins. Note: You cannot specify this parameter with the <code>-of_objects</code> parameter.

Examples

- The following command reports the timing type of all the timing arcs of the flop 'DFF':

```
tempus> get_property [get_lib_arcs -of_objects DFF] timing_type
```

Sample output:

```
minimum_period minimum_period min_pulse_width min_pulse_width setup_rising hold_rising  
rising_edge rising_edge
```

Related Commands

- `report_lib_arcs`
- `get_lib_cells`
- `get_lib_pg_pins`
- `get_lib_pins`
- `get_libs`

get_lib_cells

```
get_lib_cells  
[-help]  
[-filter expr]  
[-nocase]  
[-quiet]  
[-regexp]  
{pattern_list | -of_objects object_list}
```

Creates a collection of library cells from the loaded libraries whose name matches the supplied pattern list. Assign these library cells to a variable or pass them to another command. If no objects match the criteria, an empty collection is returned.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-filter <i>expr</i></code>	Filters the collection with the specified expression. For any library cells that match the pattern, the software evaluates the expression based on the library cell's attribute. You can filter library cells using any of the library cell properties supported through the <code>get_property</code> command. For details on how to create filter expressions, refer to the <code>filter_collection</code> command.
<code>-nocase</code>	Specifies that pattern matching is not case sensitive. Note: You must specify <code>-regexp</code> to use this parameter.
<code>-of_objects <i>object_list</i></code>	Creates a collection of all library cells associated with the specified library pins or cells, or libs. You can specify a list of library pins or a list of cells or a list of libs, but you cannot specify a list that contains a mix of library pins and cells, and libs. You can specify patterns for the library pin or cell or lib names, a collection of library pins or cells or libs, or a combination of patterns and a collection.
<code><i>pattern_list</i></code>	Returns only those libraries in the collection whose name matches one or more of the patterns specified in the <code>pattern_list</code> argument. The characters <code>?</code> and <code>*</code> are wildcards. The <code>?</code> character matches any one character, while the <code>*</code> character matches zero or more characters. If a library matches multiple patterns, it appears multiple times in the collection. A pattern can have only one hierarchical separator character. The left part of the pattern is the library name pattern, and the right part is the cell name pattern. If there is no separator character, the pattern is compared to cell names in all the libraries. A pattern can have only one hierarchical separator character, dividing the pattern into one or two parts. If the hierarchical separator is <code>/</code>, the alternatives are: • <code>cell_name_pattern</code> Selects all cells whose name matches the <code>cell_name_pattern</code> from all libraries. • <code>lib_name_pattern/cell_name_pattern</code> Selects all cells whose name matches the <code>cell_name_pattern</code> from all libraries whose name matches the <code>lib_name_pattern</code>.
<code>-quiet</code>	Suppresses all error and warning messages generated when the <code>get_lib_cells</code> command is run.
<code>-regexp</code>	Treats the specified patterns as regular expression patterns. <i>Default:</i> Treats the specified pattern as a wildcard

Examples

- The following command prints all cells for all libraries:

```
tempus > get_lib_cells *
```

- The following command prints a collection with all cells in the `lca500kv` library:

```
tempus > get_lib_cells lca500kv/*
```

- The following command prints a collection with all cells whose names start with `AN` in all libraries starting with `lca`:

```
tempus > get_lib_cells lca*/AN*
```

- The following command disables delay arcs on the `EF11L050032A` library cell. The `set_disable_cell_timing` command affects all instantiations of the specified cell, and lets you disable delay arcs from and to internal pins.

```
tempus > set_disable_timing -from WACLK -to RB01 [get_lib_cells {EF11L050032A}]
```

Related Commands

- `get_cells`
- `get_libs`
- `get_lib_pins`

get_lib_pg_pins

```
get_lib_pg_pins
[-help]
[-filter expr]
{-of_objects object_list}
```

Returns a collection of library-level PG pins from the loaded libraries. An empty collection is returned if no PG pin definitions are found in the loaded library.

Parameters

-help	Prints out the command usage.
-filter <i>expr</i>	Filters the collection with the specified expression. For any library pg pin that matches the specified expression, the software evaluates the expression based on the library pg pin's attribute. You can filter library pg pins along with the <code>-of_objects</code> parameter - using the library pg pin properties of the <code>lib_pg_pin</code> object type (supported through the get_property command). For details on how to create filter expressions, refer to the filter_collection command.
-of_objects <i>object_list</i>	Creates a collection of all library PG pins associated with the specified library cells. The <code>-of_objects</code> parameter supports <code>get_lib_cells</code> as valid objects. For example, <pre>get_lib_pg_pins -of_objects [get_lib_cells sc9_cmos32lp/DFFQ_X1M] VDD VSS get_lib_pg_pins *</pre> **ERROR: (TCLCMD-1420): Currently only -of_objects option is supported

Examples

- The following command gets the collection of all the library-level pg pins of the specified library cell:

```
tempus > get_lib_pg_pins -of_objects [get_lib_cells sc_cmoslp/A022X]
```
- The following command filters the collection of all library-level pg pins of the library cell to return only the library pg pins containing the string name defined with the `lib_pg_pin` name attribute:

```
tempus > get_lib_pg_pins -of_objects [get_lib_cells sc_cmoslp/A022X] -filter "name =~"
```

VDD "

get_lib_pins

```
get_lib_pins
[-help]
[-filter expr]
[-nocase]
[-quiet]
[-regexp]
{pattern_list | -of_objects object_list}
```

Creates a collection of library cell pins from the loaded libraries whose name matches the supplied pattern list. Assign these library cell pins to a variable or pass them to another command. If no objects match the criteria, an empty collection is returned.

Parameters

-help	Prints out the command usage.
-filter <i>expr</i>	Filters the collection with the specified expression. For any library cell pins that match the pattern, the software evaluates the expression based on the library cell pin's attribute. You can filter library pins using any of the library pin properties supported through the get_property command. For details on how to create filter expressions, refer to the filter_collection command.
-nocase	Specifies that pattern matching is not case sensitive. Note: You must specify <code>-regexp</code> to use this parameter.
-of_objects <i>object_list</i>	Creates a collection of all library pins associated with the specified library cells or pins. You can specify a list of cells or a list of pins, but you cannot specify a list that contains both cells and pins. You can specify patterns for the cell or pin names, a collection of cells or pins, or a combination of patterns and a collection.

<i>pattern_list</i>	Returns only those libraries in the collection whose name matches one or more of the patterns specified in the <i>pattern_list</i> argument. The characters ? and * are wild cards; The ? character matches any one character, while the * character matches zero or more characters. If a library cell matches multiple patterns, it appears multiple times in the collection. A pattern can have only two hierarchical separator characters, dividing the pattern into one, two, or three parts. If the hierarchical separator is /, the alternatives are: • <i>pin_name_pattern</i> Selects all pins whose name matches the <i>pin_name_pattern</i> from all cells in the libraries. • <i>cell_name_pattern/pin_name_pattern</i> Selects all pins whose name matches the <i>pin_name_pattern</i> from all cells whose name matches the <i>cell_name_pattern</i> in all libraries. • <i>lib_name_pattern/cell_name_pattern/pin_name_pattern</i> Select all pins whose name matches the <i>pin_name_pattern</i> from all cells whose name matches the <i>cell_name_pattern</i> from all libraries whose name matches the <i>lib_name_pattern</i>.
-quiet	Suppresses all error and warning messages generated when the <code>get_lib_pins</code> command is run.
-regexp	Treats the specified patterns as regular expression patterns. <i>Default:</i> Treats the specified pattern as a wild card

Examples

- The following command returns all pins in all cells for all libraries:
tempus > get_lib_pins *
- The following command returns a collection with all pins for cell AO1 in any library:
tempus > get_lib_pins AO1/*
- The following command returns a collection with all pins whose name starts with Z or Q for the AO1 cell in any library:
tempus > get_lib_pins {AO1/Z* AO1/Q*}
- The following command returns a collection with all pins for the AO1 cell in the lca500kv lib library:
tempus > get_lib_pins lca500kv/AO1/*
- The following command returns a collection with all pins starting with Z in all cells whose names

start with AN in all libraries starting with lca:

```
tempus > get_lib_pins lca*/AN*/Z*
```

Related Commands

- [get_libs](#)
- [get_lib_cells](#)

get_libs

```
get_libs
[-help]
[-filter expr]
[-nocase]
[-quiet]
[-regexp]
{-of_objects object_list | pattern_list}
```

Creates a collection of loaded libraries whose name matches those specified in the pattern list. Assign these libraries to a variable or pass them to another command. If no objects match the criteria, an empty collection is returned.

Parameters

-help	Prints out the command usage.
-filter <i>expr</i>	Filters the collection with the specified expression. For any libraries that match the pattern, the software evaluates the expression based on the library's attribute. You can filter libraries using any of the library properties supported through the get_property command. For details on how to create filter expressions, refer to the filter_collection command.
-nocase	Specifies that pattern matching is not case sensitive. Note: You must specify <code>-regexp</code> to use this parameter.
-of_objects <i>object_list</i>	Creates a collection of libraries that belong to the library cells in a given object list.
<i>pattern_list</i>	Returns only those libraries in the collection whose name matches one or more of the patterns specified in the <i>pattern_list</i> argument. The characters <code>?</code> and <code>*</code> are wild cards; The <code>?</code> character matches any one character, while the <code>*</code> character matches zero or more characters. If a library matches multiple patterns, it appears multiple times in the collection.
-quiet	Suppresses all error and warning messages generated when the <code>get_lib_cells</code> command is run.
-regexp	Treats the specified patterns as regular expression patterns. <i>Default:</i> Treats the specified pattern as a wild card

Examples

- The following command displays a list of loaded libraries starting with a or l:

```
tempus > get_libs {a* l*}
```

- The following command queries all libraries beginning with the letter l:

```
tempus > get_libs l*
```

- The following command creates a collection of libraries that are associated with the library cells in the specified object list:

```
tempus > get_libs -of_objects [get_lib_cells *]
```

Related Commands

- `get_lib_cells`
- `get_lib_pins`

get_nets

```
get_nets
[-help]
[-filter expr]
[-hierarchical]
[-hsc char]
[-nocase]
[-quiet]
[-regexp]
[patterns | -of_objects object_list]
```

Creates a collection of nets in the current design whose name matches the supplied pattern list. Assign this collection to a variable or pass it as an argument to another command. If no such net is found, an empty collection is returned.

When you use the `get_nets` command to specify hierarchical nets, the software matches the specified pattern and returns the top level net connected to it.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-filter <i>expr</i></code>	Filters the collection with the specified expression. For any nets that match the pattern, the software evaluates the expression based on the net's attribute. You can filter nets using any of the net properties supported through the <code>get_property</code> command. For details on how to create filter expressions, refer to the <code>filter_collection</code> command.
<code>-hierarchical</code>	Specifies that the patterns should match hierarchically. If the hierarchical name of a net at any hierarchical level matches the patterns, it is included in the result.
<code>-hsc <i>char</i></code>	Specifies the hierarchical delimiter for patterns.
<code>-nocase</code>	Specifies that pattern matching is not case sensitive. Note: You must specify <code>-regexp</code> to use this parameter.
<code>-of_objects <i>object_list</i></code>	Creates a collection of all nets associated with the specified cells, pins, or ports. You can specify a list of cells, a list of pins, or a list of ports, but you cannot specify a list that contains cells, pins, and ports together. You can specify patterns for the cell, pin, or port names, a collection of cells, pins, or ports, or a combination of patterns and a collection.
<code><i>patterns</i></code>	Specifies patterns for net names. Patterns include the <code>*</code> and <code>?</code> wildcard characters and collections of nets.
<code>-quiet</code>	Suppresses all error and warning messages generated when the <code>get_nets</code> command is run.
<code>-regexp</code>	Treats the specified patterns as regular expression patterns. <i>Default:</i> Treats the specified pattern as a wildcard

Examples

- The following command returns all nets within the hierarchy limited to A:

```
tempus > get_nets -hsc @ A@*
```
- The following command returns all patterns that match hierarchy and identifies nets to be excluded from optimization:

```
tempus > get_nets * -hierarchical -filter "@dont_touch== true"
```

Related Commands

- `get_pins`

get_object_name

```
get_object_name  
[-help]  
collection
```

Returns the name of the object(s) contained in the specified collections.

By default, the command hides escape characters in the output. To enable printing of escape characters in the output, you can set `timing_enable_get_object_escaped_name_backward_compatible` global variable to `true`.

Parameters

<code>-help</code>	Prints out the command usage.
<code>collection</code>	Specifies multiple collections that contain objects.

Examples

- The following command returns a TCL list of names, which can match the collection inputs:

```
tempus > get_object_name [get_cells and*]
```
- The following command defines a variable called `endpoint_list` to store endpoints with the ten worst slacks:

```
tempus > set endpoint_list [get_object_name [get_property \  
[report_timing -collection -max_points 10] capturing_point]]
```

Related Commands

- `query_objects`

get_path_groups

```
get_path_groups  
[-help]  
[-regexp]  
[-nocase]  
patterns
```

Creates a collection of path groups whose name matches those specified in the pattern list. This parameter assigns these path groups to a variable or passes them to another command. If no objects match the criteria, an empty collection is returned.

Parameters

-help	Prints out the command usage.
-nocase	Specifies that pattern matching is not case sensitive.
patterns	Creates a collection of path groups whose name matches one or more of the specified patterns. Patterns include the * and ? wildcard characters and collections of path groups. If a path group matches multiple patterns, it appears multiple times in the collection.
-regexp	Treats the specified patterns as regular expression patterns. <i>Default:</i> Treats the specified pattern as a wildcard

Examples

- The following command returns a collection of all the path groups present in the design:

```
tempus> get_path_groups *
```

Sample output:

```
reg2reg reg2gate
```

Related Commands

- [group_path](#)
- [reset_path_group](#)
- [report_path_groups](#)

get_pg_nets

```
get_pg_nets
[-help]
[-filter expr]
[-of_objects object_list]
```

Creates a collection of netlist-level instance PG nets of leaf cells/pins in the current design. An empty collection is returned if no PG net definitions/connections are found in the design netlist.

Parameters

-help	Prints out the command usage.
-filter <i>expr</i>	Specifies the filters for the collection with the specified expression. For any pg nets that match the pattern, the software evaluates the expression based on the net's attribute. You can filter pg nets along with the <code>-of_objects</code> parameter - using any of the pg net properties of object type <code>pg_net</code> (supported through the <code>get_property</code> command). For details on how to create filter expressions, refer to the <code>filter_collection</code> command.
-of_objects <i>object_list</i>	Creates a collection of all the PG nets, or PG nets associated with the specified cells or PG pins. You can specify a list of cells or a list of pg pins, but you cannot specify a list that contains cells and pg pins together. The <code>-of_objects</code> parameter supports <code>get_cells</code> and <code>get_pg_pins</code> as valid objects.

Examples

- The following command filters the collection of all pg nets of the cell instance object to return only the pg nets containing the string name defined with the `pg_net` hierarchical_name attribute:
tempus > get_pg_nets -of_objects [get_cells iA/out_reg] -filter "hierarchical_name =~ *VDD*"
- The following command gets the collection of all pg nets of the cell instance object:
tempus > get_pg_nets -of_objects [get_cells iA/out_reg]

- The following command gets the collection of all pg nets of the pg_pins of the cell instance object:
`tempus > get_pg_nets -of_objects [get_pg_pins -of_objects [get_cells iA/out_reg]]`

get_pg_pins

```
get_pg_pins
[-help]
[-filter expr]
[-of_objects object_list]
```

Returns a collection of netlist-level instance PG pins in the current design. An empty collection is returned if no PG pin definitions are found in the design netlist.

Parameters

-help	Prints out the command usage.
-filter <i>expr</i>	Specifies the filters for the collection with the given expression. For any pg pins that match the pattern, the software evaluates the expression based on the pin's attribute. You can filter pg pins along with the <code>-of_objects</code> parameter - using the pg pin properties of object type <code>pg_pin</code> (supported through the get_property command). For details on how to create filter expressions, refer to the filter_collection command.
-of_objects <i>object_list</i>	Specifies the collection of all the pins associated with the specified cells or nets. The <code>-of_objects</code> parameter supports <code>get_cells</code> as valid objects. For example: <pre>get_pg_pins -of_objects [get_cells iA/out_reg] iA/out_reg/VSS iA/out_reg/VDD get_pg_pins iA/out_reg/VDD **ERROR: (TCLCMD-1420): Currently only -of_objects option is supported.</pre>

Examples

- The following command filters the collection of all the `pg_pins` of the cell instance object to return only the pg pins containing the string name defined with the `pg_pin hierarchical_name` attribute:

```
tempus > get_pg_pins -of_objects [get_cells iA/out_reg] -filter "hierarchical_name =~ *VDD*"

```


- The following command gets the collection of all pg pins of the cell instance object:

```
tempus > get_pg_pins -of_objects [get_cells iA/out_reg]
```

get_pins

```
get_pins  
[-help]  
[-filter expr]  
[-hierarchical]  
[-hsc char]  
[-leaf]  
[-nocase]  
[-quiet]  
[-regexp]  
{patterns | -of_objects object_list}
```

Creates a collection of instance pins whose name matches the supplied pattern list. Assign this collection to a variable or pass it as an argument to another command. If no such pin is found, an empty collection is returned.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-filter <i>expr</i></code>	Filters the collection with the specified expression. For any pins that match the pattern, the software evaluates the expression based on the pin's attribute. If the expression is set to true, the pin is included in the result. You can filter pins using any of the pin properties supported through the <code>get_property</code> command. For details on how to create filter expressions, refer to the <code>filter_collection</code> command.
<code>-hierarchical</code>	Specifies that patterns should match hierarchically. If the hierarchical name of a pin at any hierarchical level matches the patterns, it is included in the result.
<code>-hsc <i>char</i></code>	Specifies the hierarchical delimiter for patterns.
<code>-leaf</code>	Returns leaf pins only. When the <code>-leaf</code> parameter is not used, the <code>get_pins</code> command returns the hierarchical pins present on a net along with the flat pins at the same hierarchical level as the specified object.
<code>-nocase</code>	Specifies that pattern matching is not case sensitive. Note: You must specify <code>-regexp</code> to use this parameter.
<code>-of_objects <i>object_list</i></code>	Creates a collection of all pins associated with the specified cells or nets. You can specify a list of cells or a list of nets, but you cannot specify a list that contains both cells and nets. You can specify patterns for the cell or net names, a collection of cells or nets, or a combination of patterns and a collection.
<code><i>patterns</i></code>	Specifies patterns for pin names. Patterns include the <code>*</code> and <code>?</code> wildcard characters and collections of pins.
<code>-quiet</code>	Suppresses all error and warning messages generated when the <code>get_pins</code> command is run.
<code>-regexp</code>	Treats the specified patterns as regular expression patterns. <i>Default:</i> Treats the specified pattern as a wildcard

Examples

- The following command returns all instance pins within the hierarchy, limited to A:

```
tempus > get_pins -hsc @ A@*
```

- The following command filters the collection for the `pin_direction` pin attribute:

```
tempus > get_pins "in*" -filter "@pin_direction == in"
```

Related Commands

- `create_clock`
- `get_cells`
- `get_ports`

get_ports

```
get_ports  
[-help]  
[-filter expr]  
[-nocase]  
[-quiet]  
[-regexp]  
[patterns | -of_objects object_list]
```

Creates a collection of ports whose name matches the supplied pattern list. This parameter assigns this collection to a variable or passes it as an argument to another command.

Note: The `get_ports` command always creates a collection of top level ports irrespective of the current instance set using the `current_instance` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-filter <i>expr</i></code>	Filters the collection with the specified expression. For any ports that match the pattern, the software evaluates the expression based on the port's attribute. If the expression is set to true, the port is included in the result. You can filter ports using any of the port properties supported through the <code>get_property</code> command. For details on how to create filter expressions, refer to the <code>filter_collection</code> command.
<code>-nocase</code>	Specifies that pattern matching is not case sensitive. Note: You must specify <code>-regexp</code> to use this parameter.
<code>-of_objects <i>object_list</i></code>	Creates a collection of all ports associated with the specified nets. You can specify patterns for the net names, a collection of nets, or a combination of patterns and a collection.
<code><i>patterns</i></code>	Specifies patterns for port names. Patterns include the <code>*</code> and <code>?</code> wildcard characters and port collections. The command returns a collection of ports that match the patterns. If no such port is found, an empty collection is returned.
<code>-quiet</code>	Suppresses all error and warning messages generated when the <code>get_ports</code> command is run.
<code>-regexp</code>	Treats the specified patterns as regular expression patterns. <i>Default:</i> Treats the specified pattern as a wildcard

Example

- The following example returns all ports beginning with the name `inb`:
`get_ports inb*`
- The following command filters the collection for the `port_direction` port attribute.
`get_ports "mode*" -filter {port_direction == in}`

Related Commands

- `all_inputs`
- `all_outputs`
- `get_clocks`
- `get_pins`

get_power

```
get_power
[-help]
[-outfile filename]
[-nets nets_list | -pins pins_list | -instances instances_list]
[-attribute attributes_list]
[-include_unit]
[-tcl_list ]
[-pg_net {pg_net_name_list | all}]
```

Queries various power-related properties of design objects like nets, pins, and instances. Using this command, you can retrieve power attributes, such as internal power, switching power, leakage power, total power, and toggle rate. This command is very useful for writing custom scripts.

Note: This command outputs toggle rates as N.A. if the clock domain information for a given net/pin/instance is not found.

Use this command after running the `report_power` or `restore_power_database` command.

Note: The `-nets`, `-pins`, and `-instances` parameters are mutually exclusive. You must specify only one of the parameters.

Parameters

<code>-attribute</code> <i>attributes_list</i>	Specifies a list of power attributes. If an attribute is not available for the specified design object, the software returns its value as NA. See List of Power Attributes for a mapping of the available attributes with the design objects. Note: This is a mandatory parameter.
<code>-help</code>	Outputs a brief description that includes the type and default information for each <code>get_power</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man get_power</code>
<code>-include_unit</code>	Specifies to include the units for the power attributes.
<code>-instances</code> <i>instances_list</i>	Specifies a list of instance names. The <code>-nets</code> , <code>-pins</code> , and <code>-instances</code> parameters are mutually exclusive. You must specify one, and only one, parameter. Note: You can use the <code>get_cells</code> command to determine the power attributes for a collection of instances.

<code>-nets <i>nets_list</i></code>	Specifies a list of net names. The <code>-nets</code>, <code>-pins</code>, and <code>-instances</code> parameters are mutually exclusive. You must specify one, and only one, parameter. Note: You can use the <code>get_nets</code> command to determine the power attributes for a collection of nets.
<code>-outfile <i>filename</i></code>	Specifies the output file into which the power-related properties are to be written. The <code>-tcl_list</code> and <code>-outfile</code> parameters are mutually exclusive; you cannot specify them together.
<code>-pg_net {<i>pg_net_name_list</i> all}</code>	Specifies to query various power-related properties of a specified power net for an instance. This parameter supports the argument <code>all</code> to query properties of all power nets connected to the instance. You must use the <code>-instances</code> parameter in conjunction with the <code>-pg_net</code> parameter. The <code>-pg_net</code> parameter supports the following attributes: <code>switching_power</code>, <code>internal_power</code>, <code>leakage_power</code>, <code>total_power</code>, and <code>dynamic_power</code>.
<code>-pins <i>pins_list</i></code>	Specifies a list of pin names. The <code>-nets</code>, <code>-pins</code>, and <code>-instances</code> parameters are mutually exclusive. You must specify one, and only one, parameter. Note: You can use the <code>get_pins</code> command to determine the power attributes for a collection of pins.
<code>-tcl_list</code>	Produces the report in Tcl list format rather than tabular format. The <code>-tcl_list</code> and <code>-outfile</code> parameters are mutually exclusive; you cannot specify them together.

The following power attributes can be queried for the design objects:

List of Power Attributes with Attribute Description

- `switching_power (net, inst)`: switching power of a net or instance
- `internal_power (inst)`: internal power of an instance
- `dynamic_power (inst)`: switching and internal power of an instance
- `leakage_power (inst)`: leakage power of an instance
- `total_power (inst)`: total power of an instance
- `toggle_rate (net, pin, inst)`: Toggle rate - Number of toggles in the specified VCD/TCF in the duration
- `duty_cycle (net, pin)`: Static probability of the signal to stay high
- `ref_clock (net, pin, inst)`: Reference clock name (Fastest clock for multi-clock domains)
- `ref_clock_period (net, pin, inst)`: Reference clock period
- `transition_density (net, pin)`: $\text{toggle_rate} / \text{ref_clock_period}$ or $\text{toggle_rate} * \text{ref_clock_frequency}$
- `loading_cap (net, pin, inst)`: Loading capacitance
- `voltage (net, pin, inst)`: Operating voltage for a net or pin. For MSMV instance, this attribute can be used to get a breakdown of domain-specific power.
- `rise_slew (net, pin)`: Rise slew
- `fall_slew (net, pin)`: Fall slew
- `glitch_rate (net, pin)`: Glitch rate - Number of glitches per duration. Applies only to the vector-driven flow.
- `glitch_power (net, pin, inst)`: Glitch power. Applies only to the vector-driven flow.
- `activity_source (net, pin)`: Source of activity annotation. The options are: propagated, user_annotated, and constant.
- `pin_direction (pin)`: [input | output | inout]
- `cell_type (inst)`: Lists cell type as memory, standard cell, IO, sequential, and so on
- `all_attributes (net, pin, inst)`: Lists all related power attributes for the specified design object.

Examples

- The following command returns the specified power attributes for the net clock1:

```
get_power -nets clock1 -attribute {switching_power internal_power toggle_rate}
```

The following is the sample output:

```
clock1 0.04 NA 2
```

- The following command returns the specified power attributes for all nets that match the specified pattern master*:

```
get_power -nets [get_nets master*] -attribute {switching_power toggle_rate  
activity_source duty_cycle} -include_unit
```

The following is the sample output:

```
masterin[1] 0.001mW 0.2 user_annotated 0.5  
masterin[2] 0.002mW 0.2 user_annotated 0.5  
masterin[3] 0.003mW 0.1 propagated 0.5  
masterin[4] 0.005mW 1 constant 1
```

- The following command returns power-related properties of all power nets connected to the instance clk_buf_1:

```
get_power -pg_net all -instances clk_buf_1 -attribute {switching_power internal_power  
leakage_power}
```

The following is the sample output:

```
clk_buf_1 VDD 0.002 0.001 0.000001  
clk_buf_1 VDDL 0.0001 0.0001 0.0000001
```

get_property

```
get_property  
[-help]  
var_name  
property  
[-clock clock_name]  
[-quiet]  
[-view view_name]
```

Retrieves the attribute value for the specified object(s) property.

You can query the following:

- [Cell Properties](#)
- [Clock Properties](#)
- [Design Properties](#)
- [Library Properties](#)
- [Library Cell Properties](#)
- [Library Pin Properties](#)
- [Library PG Pin Properties](#)
- [Library Timing Arc Properties](#)
- [Net Properties](#)
- [PG Net Properties](#)
- [Path Group Properties](#)
- [Pin Properties](#)
- [PG Pin Properties](#)
- [Port Properties](#)
- [Timing Arc Properties](#)
- [Timing Path Properties](#)
- [Timing Point Properties](#)
- [si_victim Properties](#)
- [si_attacker Properties](#)
- [User-Defined Properties](#)

The `get_property` command supports the [report_precision](#) global variable settings.

The `get_property` command supports DRV constraints (`set_min_fanout`, `set_max_fanout`, `set_min_transition`, `set_max_transition`, `set_min_capacitance`, and `set_max_capacitance`) applied on library pins.

The `get_property` command does not work with partial property names. You are now required to specify the full name of the property.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-clock</code> <i>clock_name</i>	Specifies the clock for which to query certain object properties.
<i>property</i>	Specifies the property for which to retrieve the attribute value.
<code>-quiet</code>	Suppresses all error and warning messages generated when the <code>get_property</code> command is run.
<i>var_name</i>	Specifies a pointer to a collection containing the object. Note: If the collection contains more than one object, the <code>get_property</code> command returns the property values for each object in the specified collection.
<code>-view</code> <i>view_name</i>	Specifies the analysis view in which to query the object attribute. You can query properties only for the following objects in an analysis view: library cells, library pins, ports, cells, pins, timing arcs, and nets. You can specify this parameter only when the software is in multi-mode multi-corner timing analysis mode. Note: The properties (described in the tables below), which are related to slack, delay, arrival, slew, transition, capacitance, uncertainty and latency, depend on the specified view. When the <code>-view</code> parameter is not specified, the maximum (or minimum) value for all the views is returned.

Cell Properties

area	Returns the area of the cell. If the cell is hierarchical, also includes the net area. <i>Return type:</i> float
clock_gating_integrated_cell	Returns the clock gating integrated cells present in a library. When specified, the software returns a value that is used to determine the integrated cell functionality to be used by clock gating tools. For example, <pre>get_property [get_cells <cell name>] clock_gating_integrated_cell</pre> <i>Return type:</i> string
early_cell_check_derate_factor	Returns the derating factor for early paths specified through the set_timing_derate command with the <code>-early</code> parameter. <i>Return type:</i> float
early_cell_check_mean_derate_factor	Returns the derating factor for early paths on the instance – when the <code>set_timing_derate -mean</code> parameter is specified. <i>Return type:</i> float
early_clock_check_incremental_derate_factor	Returns the early clock path instances' incremental derating factor specified through the <code>set_timing_derate -increment</code> parameter. <i>Return type:</i> float
early_data_check_incremental_derate_factor	Returns the early data path instance's incremental derating factor specified through the <code>set_timing_derate -increment</code> parameter. <i>Return type:</i> float
early_cell_check_sigma_derate_factor	Returns the derating factor for early paths on the instance - when the <code>set_timing_derate -sigma</code> parameter is specified. <i>Return type:</i> float

<code>early_clk_cell_derate_factor</code>	Returns the derating factor for early clock paths specified through the <code>set_timing_derate</code> command with the <code>-early</code> parameter. <i>Return type:</i> float
<code>early_clk_cell_mean_derate_factor</code>	Returns the derating factor for early clock paths – when the <code>set_timing_derate -mean</code> parameter is specified. <i>Return type:</i> float
<code>early_clk_cell_sigma_derate_factor</code>	Returns the derating factor for early clock paths – when the <code>set_timing_derate -sigma</code> parameter is specified. <i>Return type:</i> float
<code>early_data_cell_derate_factor</code>	Returns the derating factor for early paths specified through the <code>set_timing_derate</code> command with the <code>-early</code> parameter. <i>Return type:</i> float
<code>early_data_cell_mean_derate_factor</code>	Returns the derating factor for early paths – when the <code>set_timing_derate -mean</code> parameter is specified. <i>Return type:</i> float
<code>early_data_cell_sigma_derate_factor</code>	Returns the derating factor for early paths – when the <code>set_timing_derate -sigma</code> parameter is specified. <i>Return type:</i> float
<code>early_fall_cell_check_derate_factor</code>	Returns the early cell check derating factor specified through the <code>set_timing_derate</code> command with the <code>-fall</code> parameter. <i>Return type:</i> float
<code>early_fall_cell_check_mean_derate_factor</code>	Returns the early cell check derating factor – when the <code>set_timing_derate -cell_check</code> and <code>-mean</code> parameters are specified. <i>Return type:</i> float

<code>early_fall_cell_check_sigma_derate_factor</code>	Returns the early cell check derating factor – when the <code>set_timing_derate -cell_check</code> and <code>-sigma</code> parameters are specified. <i>Return type:</i> float
<code>early_fall_clk_cell_derate_factor</code>	Returns the early clock path derating factor specified through the <code>set_timing_derate</code> command with the <code>-fall</code> parameter. <i>Return type:</i> float
<code>early_fall_clk_cell_mean_derate_factor</code>	Returns the early clock path derating factor – when the <code>set_timing_derate -fall</code> and <code>-mean</code> parameters are specified. <i>Return type:</i> float
<code>early_fall_clk_cell_sigma_derate_factor</code>	Returns the early clock path derating factor – when the <code>set_timing_derate -fall</code> and <code>-sigma</code> parameters are specified. <i>Return type:</i> float
<code>early_fall_clock_incremental_derate_factor</code>	Returns the early clock path instances' incremental derating factor specified through the <code>set_timing_derate -increment</code> parameter - when the <code>-fall</code> parameter is specified. <i>Return type:</i> float
<code>early_fall_data_cell_derate_factor</code>	Returns the early data cell check derating factor – when the <code>set_timing_derate -fall</code> parameter is specified. <i>Return type:</i> float
<code>early_fall_data_cell_mean_derate_factor</code>	Returns the early data cell check derating factor – when the <code>set_timing_derate -fall</code> and <code>-mean</code> parameters are specified. <i>Return type:</i> float
<code>early_fall_data_cell_sigma_derate_factor</code>	Returns the early data cell check derating factor – when the <code>set_timing_derate -fall</code> and <code>-sigma</code> parameters are specified. <i>Return type:</i> float

early_fall_data_incremental_derate_factor	Returns the early data path instances' incremental derating factor specified through the <code>set_timing_derate -increment</code> parameter - when the <code>-fall</code> parameter is specified. <i>Return type:</i> float
early_rise_cell_check_derate_factor	Returns the early cell check derating factor specified through the <code>set_timing_derate</code> command with the <code>-rise</code> and <code>-cell_check</code> parameter. <i>Return type:</i> float
early_rise_cell_check_mean_derate_factor	Returns the early cell check derating factor – when the <code>set_timing_derate -rise</code> and <code>-mean</code> parameters are specified. <i>Return type:</i> float
early_rise_cell_check_sigma_derate_factor	Returns the early cell check derating factor – when the <code>set_timing_derate -rise</code> and <code>-sigma</code> parameters are specified. <i>Return type:</i> float
early_rise_clk_cell_derate_factor	Returns the early clock cell check derating factor specified through the <code>set_timing_derate</code> command with the <code>-rise</code> parameter. <i>Return type:</i> float
early_rise_clk_cell_mean_derate_factor	Returns the early clock cell check derating factor – when the <code>set_timing_derate -rise</code> and <code>-mean</code> parameters are specified. <i>Return type:</i> float
early_rise_clk_cell_sigma_derate_factor	Returns the early clock cell check derating factor – when the <code>set_timing_derate -rise</code> and <code>-sigma</code> parameters are specified. <i>Return type:</i> float

early_rise_clock_incremental_derate_factor	Returns the early clock path instances' incremental derating factor specified through the <code>set_timing_derate -increment</code> parameter - when the <code>-rise</code> parameter is specified. <i>Return type:</i> float
early_rise_data_cell_derate_factor	Returns the early data cell check derating factor specified through the <code>set_timing_derate</code> command with the <code>-rise</code> parameter. <i>Return type:</i> float
early_rise_data_cell_mean_derate_factor	Returns the early data cell check derating factor – when the <code>set_timing_derate -rise</code> and <code>-mean</code> parameters are specified. <i>Return type:</i> float
early_rise_data_cell_sigma_derate_factor	Returns the early data cell check derating factor – when the <code>set_timing_derate -rise</code> and <code>-sigma</code> parameters are specified. <i>Return type:</i> float
early_rise_data_incremental_derate_factor	Returns the early data path instances' incremental derating factor specified through the <code>set_timing_derate -increment</code> parameter - when the <code>-rise</code> parameter is specified. <i>Return type:</i> float
has_sdc_assertions	Returns a value of <code>true</code>, if the specified cell has SDC assertions applied on it. For example, <pre>get_property [get_cells <cell name>] has_sdc_assertions</pre> <i>Return Type:</i> Boolean

hierarchical_level	Returns the hierarchical level of the specified cell. For example, <pre>get_property [get_cell -hierarchical *] hierarchical_level</pre> <i>Return Type:</i> Integer
hierarchical_name	Returns the complete hierarchical name of the cell. <i>Return type:</i> string
instance_model	Reports the model type of an instance. The Instance model type can be ecsmn, ccsn, cdb, udn, or nldm. In case a noise model does not exist, the software returns 'NA'. For example, <pre>get_property [get_cells inst1/buf22] instance_model</pre> <i>Return type:</i> string
is_black_box	Returns a value of <code>true</code> if the cell is a black box cell. A cell is considered as a black box if the "function" definition of the cell is missing from the library. <i>Return type:</i> boolean
is_buffer	Returns a value of <code>true</code> if the associated library cell is recognized as a buffer cell. For example, <pre>get_property [get_cells mmu/IB1] is_buffer</pre> <i>Return type:</i> boolean
is_clock_gating_check	Returns a value of <code>true</code> if the cell is gated. <i>Return type:</i> boolean
is_clock_network_inst	Returns a value of <code>true</code> if a cell is part of the clock network. <i>Return type:</i> boolean

<code>is_combinational</code>	Returns a value of <code>true</code> if the cell is a combinational cell (not a sequential cell). <i>Return type:</i> boolean
<code>is_disable_timing</code>	Returns a value of <code>true</code> if the cell's timing has been disabled using the <code>set_disable_timing</code> command. <i>Return type:</i> boolean
<code>is_dont_touch</code>	Returns a value of <code>true</code> if the cell definition includes <code>dont_touch:true</code>. <i>Return type:</i> boolean
<code>is_fall_edge_triggered</code>	Returns a value of <code>true</code> if the cell is triggered by the falling edge of the clock. <i>Return type:</i> boolean
<code>is_hierarchical</code>	Returns a value of <code>false</code> if the cell is a leaf or library cell. <i>Return type:</i> boolean
<code>is_integrated_clock_gating_cell</code>	Returns a value of <code>true</code> if the cell is an integrated clock gating cell. <i>Return type:</i> boolean
<code>is_interface_timing</code>	Returns a value of <code>true</code> if a library cell has interface timing specified for that cell. <i>Return Type:</i> boolean
<code>is_macro_cell</code>	Returns a value of <code>true</code> if the cell is a macro cell. For example, <pre>get_property [get_cells macro1] is_macro_cell</pre> <i>Return type:</i> boolean
<code>is_memory_cell</code>	Returns a value of <code>true</code> if the cell is a memory cell. <i>Return type:</i> boolean

<code>is_msv</code>	Returns a value of <code>true</code> if the cell has multiple supply voltages. For example, <pre>get_property [get_cells msv_cell] is_msv</pre> <i>Return type:</i> boolean
<code>is_mux</code>	Returns a value of <code>true</code> if the cell is mux. For example, <pre>get_property [get_cells TDSP_CORE_INST/PORT_BUS_MACH_INST/p11358A] is_mux</pre> <i>Return type:</i> boolean
<code>is_negative_level_sensitive</code>	Returns a value of <code>true</code> if the cell is used as a negative level-sensitive sequential element, such as a negative level-sensitive latch. <i>Return type:</i> boolean
<code>is_pad_cell</code>	Returns a value of <code>true</code> if the Liberty library <code>is_pad_cell</code> attribute is set to <code>true</code> for this pad cell. <i>Return type:</i> boolean
<code>is_positive_level_sensitive</code>	Returns a value of <code>true</code> if the cell is used as a positive level-sensitive sequential element, such as a positive level-sensitive latch. <i>Return type:</i> boolean
<code>is_rise_edge_triggered</code>	Returns a value of <code>true</code> if the cell is triggered by the rising edge of the clock. <i>Return type:</i> boolean
<code>is_sequential</code>	Returns a value of <code>true</code> if the cell is a latch or flip-flop, or if the cell has sequential timing checks. <i>Return type:</i> boolean
<code>is_tristate</code>	Returns a value of <code>true</code> if the cell definition includes <code>three_state:true</code>. <i>Return type:</i> boolean

is_unmapped	Returns a value of true if the specified cell name does not have a defined library cell. For example, <pre>get_property [get_cells <cell name>] is_unmapped</pre> <i>Return Type:</i> boolean
late_cell_check_derate_factor	Returns the derating factor for late paths specified through the <code>set_timing_derate</code> command with the <code>-late</code> parameter. <i>Return type:</i> float
late_cell_check_mean_derate_factor	Returns the derating factor for late paths – when the <code>set_timing_derate -late</code> and <code>-mean</code> parameters are specified. <i>Return type:</i> float
late_cell_check_sigma_derate_factor	Returns the derating factor for late paths – when the <code>set_timing_derate -late</code> and <code>-sigma</code> parameters are specified. <i>Return type:</i> float
late_clk_cell_derate_factor	Returns the derating factor for late clock paths specified through the <code>set_timing_derate</code> command with the <code>-late</code> parameter. <i>Return type:</i> float
late_clk_cell_mean_derate_factor	Returns the derating factor for late clock paths – when the <code>set_timing_derate -late</code> and <code>-mean</code> parameters are specified. <i>Return type:</i> float
late_clk_cell_sigma_derate_factor	Returns the derating factor for late clock paths – when the <code>set_timing_derate -late</code> and <code>-sigma</code> parameters are specified. <i>Return type:</i> float

late_clock_check_incremental_derate_factor	Returns the late clock path instance's incremental derating factor specified through the <code>set_timing_derate -increment</code> parameter. <i>Return type:</i> float
late_data_cell_derate_factor	Returns the derating factor for late data paths specified through the <code>set_timing_derate</code> command with the <code>-late</code> parameter. <i>Return type:</i> float
late_data_cell_mean_derate_factor	Returns the derating factor for late data paths – when the <code>set_timing_derate -late</code> and <code>-mean</code> parameters are specified. For example, <pre>get_property [get_cells <cell name>] late_data_cell_mean_derate_factor</pre> <i>Return Type:</i> Float
late_data_cell_sigma_derate_factor	Returns the derating factor for late data paths – when the <code>set_timing_derate -late</code> and <code>-sigma</code> parameters are specified. <i>Return type:</i> float
late_data_check_incremental_derate_factor	Returns the late data path instances' incremental derating factor specified through the <code>set_timing_derate -increment</code> parameter. <i>Return type:</i> float
late_fall_cell_check_derate_factor	Returns the late cell check derating factor specified through the <code>set_timing_derate</code> command with the <code>-fall</code> parameter. <i>Return type:</i> float
late_fall_cell_check_mean_derate_factor	Returns the late cell check derating factor – when the <code>set_timing_derate -fall</code> and <code>-mean</code> parameters are specified. <i>Return type:</i> float

<code>late_fall_cell_check_sigma_derate_factor</code>	Returns the late cell check derating factor – when the <code>set_timing_derate -fall</code> and <code>-sigma</code> parameters are specified. <i>Return type:</i> float
<code>late_fall_clk_cell_derate_factor</code>	Returns the late clock cell check derating factor specified through the <code>set_timing_derate</code> command with the <code>-fall</code> parameter. <i>Return type:</i> float
<code>late_fall_clk_cell_mean_derate_factor</code>	Returns the late clock cell check derating factor – when the <code>set_timing_derate -fall</code> and <code>-mean</code> parameters are specified. <i>Return type:</i> float
<code>late_fall_clk_cell_sigma_derate_factor</code>	Returns the late clock cell check derating factor – when the <code>set_timing_derate -fall</code> and <code>-sigma</code> parameters are specified. <i>Return type:</i> float
<code>late_fall_clock_incremental_derate_factor</code>	Returns the late clock path instances' incremental derating factor specified through the <code>set_timing_derate -increment</code> parameter - when the <code>-fall</code> parameter is specified. <i>Return type:</i> float
<code>late_fall_data_cell_derate_factor</code>	Returns the late data cell check derating factor specified through the <code>set_timing_derate</code> command with the <code>-fall</code> parameter. <i>Return type:</i> float
<code>late_fall_data_cell_mean_derate_factor</code>	Returns the late data cell check derating factor – when the <code>set_timing_derate -fall</code> and <code>-mean</code> parameters are specified. <i>Return type:</i> float

<code>late_fall_data_cell_sigma_derate_factor</code>	Returns the late data cell check derating factor – when the <code>set_timing_derate -fall</code> and <code>-sigma</code> parameters are specified. <i>Return type:</i> float
<code>late_fall_data_incremental_derate_factor</code>	Returns the late data path instances' incremental derating factor specified through the <code>set_timing_derate -increment</code> parameter - when the <code>-fall</code> parameter is specified. <i>Return type:</i> float
<code>late_rise_cell_check_derate_factor</code>	Returns the late cell check derating factor specified through the <code>set_timing_derate</code> command with the <code>-rise</code> parameter. <i>Return type:</i> float
<code>late_rise_cell_check_mean_derate_factor</code>	Returns the late cell check derating factor – when the <code>set_timing_derate -rise</code> and <code>-mean</code> parameters are specified. <i>Return type:</i> float
<code>late_rise_cell_check_sigma_derate_factor</code>	Returns the late cell check derating factor – when the <code>set_timing_derate -rise</code> and <code>-sigma</code> parameters are specified. <i>Return type:</i> float
<code>late_rise_clk_cell_derate_factor</code>	Returns the late clock cell check derating factor specified through the <code>set_timing_derate</code> command with the <code>-rise</code> parameter. <i>Return type:</i> float
<code>late_rise_clk_cell_mean_derate_factor</code>	Returns the late clock cell check derating factor – when the <code>set_timing_derate -rise</code> and <code>-mean</code> parameters are specified. <i>Return type:</i> float
<code>late_rise_clk_cell_sigma_derate_factor</code>	Returns the late clock cell check derating factor – when the <code>set_timing_derate -rise</code> and <code>-sigma</code> parameters are specified. <i>Return type:</i> float

<code>late_rise_clock_incremental_derate_factor</code>	Returns the late clock path instances' incremental derating factor specified through the <code>set_timing_derate -increment</code> parameter - when the <code>-rise</code> parameter is specified. <i>Return type:</i> float
<code>late_rise_data_cell_derate_factor</code>	Returns the late data cell check derating factor specified through the <code>set_timing_derate</code> command with the <code>-rise</code> parameter. <i>Return type:</i> float
<code>late_rise_data_cell_mean_derate_factor</code>	Returns the late data cell check derating factor – when the <code>set_timing_derate -rise</code> and <code>-mean</code> parameters are specified. <i>Return type:</i> float
<code>late_rise_data_cell_sigma_derate_factor</code>	Returns the late data cell check derating factor – when the <code>set_timing_derate -rise</code> and <code>-sigma</code> parameters are specified. <i>Return type:</i> float
<code>late_rise_data_incremental_derate_factor</code>	Returns the late data path instances' incremental derating factor specified through the <code>set_timing_derate -increment</code> parameter - when the <code>-rise</code> parameter is specified. <i>Return type:</i> float
<code>lib_cell</code>	Returns a collection of all the library cell objects associated with the specified library pins/cells. For example, <pre>get_property [get_cells AAA] lib_cell -view VIEW1</pre> <i>Return type:</i> collection
<code>name</code>	Returns the leaf-level name of the cell. <i>Return type:</i> string
<code>object_type</code>	Returns the object type <code>cell</code>. <i>Return type:</i> string

pin_count	Returns the total number of signal pins. <i>Return type:</i> integer
power_domain	Returns the power domains of a cell. For example, <code>get_property [get_cells FE_OCPC5_psw_en2] power_domain</code> <i>Return type:</i> string
ref_lib_cell_name	Returns the name of reference module of the given cell. The reference module of the cell can be a Verilog module or a leaf library cell. <i>Return type:</i> string
retention_cell	Returns the value of the <code>retention_cell</code> attribute derived from Liberty. For example, <code>get_property [get_cells retention_cell] retention_cell</code> <i>Return type:</i> string
timing_model_type	Returns the model type for a given cell or instance. The supported values are <code>abstracted</code> , <code>extracted</code> , and <code>qtm</code> . <i>Return type:</i> string
x_coordinate_max	Returns the maximum x coordinate of the cell. <i>Return type:</i> float
x_coordinate_min	Returns the minimum x coordinate of the cell. <i>Return type:</i> float
y_coordinate_max	Returns the maximum y coordinate of the cell. <i>Return type:</i> float
y_coordinate_min	Returns the minimum y coordinate of the cell. <i>Return type:</i> float

is_always_on	Returns a value of <code>true</code> if a cell instance is an always-on cell. <i>Return type:</i> boolean For example, <code>get_property [get_cells mmu/U1] is_always_on</code>
is_inverter	Returns a value of <code>true</code> if a cell instance is an inverter cell. <i>Return type:</i> boolean For example, <code>get_property [get_cells mmu/INV1] is_inverter</code>
is_isolation_cell	Returns a value of <code>true</code> if a cell instance is an isolation cell. <i>Return type:</i> boolean For example, <code>get_property [get_cells mmu/U1] is_isolation_cell</code>
is_level_shifter	Returns a value of <code>true</code> if a cell instance is a level-shifter cell. <i>Return type:</i> boolean For example, <code>get_property [get_cells mmu/U1] is_level_shifter</code>
is_power_switch	Returns a value of <code>true</code> if a cell instance is a power switch cell. <i>Return type:</i> boolean For example, <code>get_property [get_cells mmu/U1] is_power_switch</code>
is_retention	Returns a value of <code>true</code> if a cell instance is a retention cell. <i>Return type:</i> boolean For example, <code>get_property [get_cells mmu/U1] is_retention</code>

level_shifter_type	Returns the type of level-shifter (e.g., LH, HL, or HL_LH) that a cell instance is. <i>Return type:</i> string For example, <pre>get_property [get_cells mmu/U1] level_shifter_type</pre>
pg_pins	Returns a collection of the PG pins that are present in a cell instance. <i>Return type:</i> collection For example, <pre>get_property [get_cells mmu/U1] pg_pins</pre>
power_switch_type	Returns the type of power switch (e.g., coarse-grain, fine-grain) that a cell instance is. <i>Return type:</i> string For example, <pre>get_property [get_cells mmu/U1] power_switch_type</pre>
std_cell_main_rail_name	Returns the name of the main power rail driving the power pins of a cell instance that is the <code>pg_pin</code>, which has the <code>std_cell_main_rail</code> library attribute set as <code>true</code> for the instance cell. <i>Return type:</i> string For example, <pre>get_property [get_cells mmu/U1] std_cell_main_rail_name</pre>

Clock Properties

child_clocks	Returns the first level of generated clocks. This property also works for nested generated clocks. For example, <pre>get_property [get_clocks CK] child_clocks</pre> <i>Return type:</i> collection
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clock_network_pins	Returns a collection of all pins on the clock network for the clock. <i>Return type:</i> collection
comment	Returns the comment set for the specified clock. For example, <code>get_property [get_clocks CK] comment</code> <i>Return Type:</i> String
delay_max_fall	Returns the maximum falling delay value for the clock. <i>Return type:</i> float
delay_max_rise	Returns the maximum rising delay value for the clock. <i>Return type:</i> float
delay_min_fall	Returns the minimum falling delay value for the clock. <i>Return type:</i> float
delay_min_rise	Returns the minimum rising delay value for the clock. <i>Return type:</i> float
divide_by	Returns dfactor set for the specified clock. For example, <code>get_property [get_clocks ck] divide_by</code> <i>Return Type:</i> Integer
driving_network_pins	Returns a collection of the network pins of the driving clock. For example, <code>get_property [get_clocks clk] driving_network_pins</code> <i>Return Type:</i> Collection

duty_cycle	Returns the duty cycle for the specified clock. For example, <code>get_property [get_clocks CLK] duty_cycle</code> <i>Return Type:</i> Float
edges	Return the edges that are set for a specified clock. For example, <code>get_property [get_clocks ck] edges</code> <i>Return Type:</i> String
edge_shift	Return the edge shift that is set for a specified clock. For example, <code>get_property [get_clocks ck] edge_shift</code> <i>Return Type:</i> String
generated_clocks_extended	Returns generated clocks derived from the regular clock. <i>Return type:</i> collection
hierarchical_name	Returns the complete hierarchical name of the clock. <i>Return type:</i> string
hold_uncertainty	Returns the most conservative uncertainty of all the possible uncertainty assertions associated with a pin, when the <code>set_clock_uncertainty -hold</code> parameter is specified. For example, <code>get_property [get_clocks <clock name>] hold_uncertainty</code> <i>Return Type:</i> Float

is_context_mapped	Returns a value of <code>true</code> if a clock is mapped to the top in a context. Additionally, this is also reported with the <code>report_clocks</code> command. For example, <pre>get_property [get_clocks <clock name>] is_context_mapped</pre> <i>Return Type:</i> Boolean
is_source_inverted	Returns a value of <code>true</code> if a specified clock source is inverted. For example, <pre>get_property [get_clocks clk] is_source_inverted</pre> <i>Return Type:</i> Boolean
ideal_transition_max_fall	Returns the ideal clock transition asserted on clock waveforms specified with the <code>set_clock_transition</code> command using the <code>-fall</code> and <code>-max</code> parameters. <i>Return type:</i> float
ideal_transition_max_rise	Returns the ideal clock transition asserted on clock waveforms specified with the <code>set_clock_transition</code> command using the <code>-rise</code> and <code>-max</code> parameters. <i>Return type:</i> float
ideal_transition_min_fall	Returns the ideal clock transition asserted on clock waveforms specified with <code>set_clock_transition</code> command using the <code>-fall</code> and <code>-min</code> parameters. <i>Return type:</i> float
ideal_transition_min_rise	Returns the ideal clock transition asserted on clock waveforms specified with <code>set_clock_transition</code> command using the <code>-rise</code> and <code>-min</code> parameters. <i>Return type:</i> float

is_active	Returns a value of <code>true</code> if the clock specified is active in the current view. <i>Return type:</i> boolean
is_combinational_source_path	Returns a value of <code>true</code> if <code>create_generated_clock - combinational</code> is specified for the creation of the generated clock. For example, <pre>get_property [get_clocks GCLK] is_combinational_source_path</pre> <i>Return type:</i> boolean
is_generated	Returns a value of <code>true</code> if the clock is a generated clock. <i>Return type:</i> boolean
is_inverted	Returns a value of <code>true</code>, if a specified clock is inverted. For example, <pre>get_property [get_clocks ck] is_inverted</pre> Return Type: Boolean
is_library_created	Returns a value of <code>true</code> if <code>generated_clock</code> is specified in a Liberty <code>generated_clock</code> group and not created by a <code>create_generated_clock</code> constraint. For example, <pre>get_property [get_clocks UMC/generated_clock_cell/clkint] is_library_created</pre> <i>Return type:</i> boolean
is_propagated_clock	Returns a value of <code>true</code> if the clock is in propagated mode. <i>Return type:</i> boolean

is_virtual	Returns a value of <code>true</code> if the specified clock is virtual. For example, <pre>get_property [get_clocks v_clk] is_virtual</pre> Return Type: Boolean
latency_fall_max	Returns the maximum fall latency for a clock, when the <code>set_clock_latency -max -fall</code> parameters are specified. <i>Return type: float</i>
latency_fall_min	Returns the minimum fall latency for a clock, when the <code>set_clock_latency -min -fall</code> parameters are specified. <i>Return type: float</i>
latency_rise_max	Returns the maximum rise latency for a clock, when the <code>set_clock_latency -max -rise</code> parameters are specified. For example, <pre>get_property [get_clocks <clock name>] latency_rise_max</pre> <i>Return Type: float</i>
latency_rise_min	Returns the minimum rise latency for a clock, when the <code>set_clock_latency -min -rise</code> parameters are specified. For example, <pre>get_property [get_clocks <clock name>] latency_rise_min</pre> <i>Return Type: float</i>
lead_edge_type	Returns an edge type (rise or fall) corresponding to the leading edge for a specified clock. For example, <pre>get_property [get_clocks <clock name>] lead_edge_type</pre> <i>Return Type: String</i>

master_clock	Returns a collection of the master clocks for a specified clock. For example, <pre>get_property [get_clocks clk] master_clock</pre> <i>Return Type:</i> Collection
master_source	Returns a collection of the master sources for a specified clock. For example, <pre>get_property [get_clocks ck] master_source</pre> <i>Return Type:</i> Collection
max_capacitance_clock_path_fall	Returns fall max_capacitance value on clock paths. <i>Return type:</i> float
max_capacitance_clock_path_rise	Returns rise max_capacitance value on clock paths. <i>Return type:</i> float
max_capacitance_data_path_fall	Returns fall max_capacitance value on data paths. <i>Return type:</i> float
max_capacitance_data_path_rise	Returns rise max_capacitance value on data paths. <i>Return type:</i> float
max_transition_clock_path_fall	Returns fall max_transition value on clock paths. <i>Return type:</i> float
max_transition_clock_path_rise	Returns rise max_transition value on clock paths. <i>Return type:</i> float
max_transition_data_path_fall	Returns fall max_transition value on data paths. <i>Return type:</i> float

max_transition_data_path_rise	Returns rise max_transition value on data paths. <i>Return type:</i> float
min_pulse_width_high	Returns the minimum pulse width for high signals for the specified clock, when the set_min_pulse_width -high parameter is specified. For example, <pre>get_property [get_clocks <clock name>] min_pulse_width_high</pre> <i>Return Type:</i> Float
min_pulse_width_low	Returns the minimum pulse width for low signals for the specified clock, when the set_min_pulse_width -low parameter is specified. For example, <pre>get_property [get_clocks <clock name>] min_pulse_width_low</pre> <i>Return Type:</i> Float
multiply_by	Returns mfactor set for the specified clock. For example, <pre>get_property [get_clocks clk] multiply_by</pre> <i>Return Type:</i> Integer
name	Returns the name of the clock. <i>Return type:</i> string
object_type	Returns the object type clock. <i>Return type:</i> string
period	Returns the clock period. <i>Return type:</i> float

setup_uncertainty	Returns the most conservative uncertainty of all the possible uncertainty assertions associated with pins, when the <code>set_clock_uncertainty -setup</code> parameter is specified for a clock. For example, <pre>get_property [get_clocks <clock name>] setup_uncertainty</pre> <i>Return Type:</i> Float
source_jitter_early_fall_max	Returns the amount of the maximum early fall source insertion delay that is due to cycle-to-cycle variation. <i>Return Type:</i> Float
source_jitter_early_fall_min	Returns the amount of the minimum early fall source insertion delay that is due to cycle-to-cycle variation. <i>Return Type:</i> Float
source_jitter_early_rise_max	Returns the amount of the maximum early rise source insertion delay that is due to cycle-to-cycle variation. <i>Return Type:</i> Float
source_jitter_early_rise_min	Returns the amount of the minimum early rise source insertion delay that is due to cycle-to-cycle variation. <i>Return Type:</i> Float
source_jitter_full_cycle_hold	Returns the early full cycle path jitter that is specified on the clock source. <i>Return Type:</i> Float
source_jitter_full_cycle_setup	Returns the late full cycle path jitter that is specified on the clock source. <i>Return Type:</i> Float
source_jitter_half_cycle_hold	Returns the early half cycle path jitter that is specified on the clock source. <i>Return Type:</i> Float

source_jitter_half_cycle_setup	Returns the late half cycle path jitter that is specified on the clock source. <i>Return Type:</i> Float
source_jitter_late_fall_max	Returns the amount of the maximum late fall source insertion delay that is due to cycle-to-cycle variation. <i>Return Type:</i> Float
source_jitter_late_fall_min	Returns the amount of the minimum late fall source insertion delay that is due to cycle-to-cycle variation. <i>Return Type:</i> Float
source_jitter_late_rise_max	Returns the amount of the maximum late rise source insertion delay that is due to cycle-to-cycle variation. <i>Return Type:</i> Float
source_jitter_late_rise_min	Returns the amount of the minimum late rise source insertion delay that is due to cycle-to-cycle variation. <i>Return Type:</i> Float
source_latency_early_fall_max	Returns the early source latency for the falling transition at the maximum corner, when the <code>set_clock_latency -source -early -fall -max</code> parameters are specified for the clock. For example, <pre>get_property [get_clocks <clock name>] source_latency_early_fall_max</pre> <i>Return Type:</i> float
source_latency_early_fall_min	Returns the early source latency for the falling transition at the min corner, when the <code>set_clock_latency -source -early -fall -min</code> parameters are specified for the clock. For example, <pre>get_property [get_clocks <clock name>] source_latency_early_fall_min</pre> <i>Return type:</i> float

source_latency_early_rise_max	Returns the early source latency for the rising transition at the max corner, when the <code>set_clock_latency -source -early -rise -max</code> parameters are specified for the clock. For example, <pre>get_property [get_clocks <clock name>] source_latency_early_rise_max</pre> <i>Return Type:</i> float
source_latency_early_rise_min	Returns the early source latency for the rising transition at the min corner, when the <code>set_clock_latency -source -early -rise -min</code> parameters are specified for the clock. For example, <pre>get_property [get_clocks <clock name>] source_latency_early_rise_min</pre> <i>Return type:</i> float
source_latency_late_fall_max	Returns the late source latency for the falling transition at the max corner, when the <code>set_clock_latency -source -late -fall -max</code> parameters are specified for the clock. For example, <pre>get_property [get_clocks <clock name>] source_latency_late_fall_max</pre> <i>Return type:</i> float
source_latency_late_fall_min	Returns the late source latency for the falling transition at the min corner, when the <code>set_clock_latency -source -late -fall -min</code> parameters are specified for the clock. For example, <pre>get_property [get_clocks <clock name>] source_latency_late_fall_min</pre> <i>Return type:</i> float

source_latency_late_rise_max	Returns the late source latency for the rising transition at the max corner, when the <code>set_clock_latency -source -late -rise -max</code> parameters are specified for the clock. For example, <pre>get_property [get_clocks <clock name>] source_latency_late_rise_max</pre> <i>Return type:</i> float
source_latency_late_rise_min	Returns the late source latency for the rising transition at the min corner, when the <code>set_clock_latency -source -late -rise -min</code> parameters are specified for the clock. For example, <pre>get_property [get_clocks <clock name>] source_latency_late_rise_min</pre> <i>Return type:</i> float
trail_edge_type	Returns the edge type (rise or fall) corresponding to the trailing edge of the specified clock. For example, <pre>get_property [get_clocks <clock name>] trail_edge_type</pre> <i>Return Type:</i> string
source_latency_pins	Returns a list of pin or port object(s), which are present between the target pins of a generated clock and master clock. For example, <pre>get_property [get_clocks GCLK] source_latency_pins</pre> <i>Return type:</i> collection
sources	Returns a collection of the clock's source pins or ports. <i>Return type:</i> collection

view_name	In multi-mode multi-corner mode, returns the analysis view name associated with a given clock. <i>Return type:</i> string
waveform	Returns the rising and falling edge times of the waveform. <i>Return type:</i> string

Design Properties

analysis_type	Returns the analysis mode: <code>single</code> , <code>bc_wc</code> , or <code>on_chip_variation</code> . <i>Return type:</i> string
boundary_model_analysis	Returns a value of <code>true</code> if the design is a boundary model. For example, <code>get_property [get_designs MPTC] boundary_model_analysis</code> <i>Return type:</i> boolean
capacitance_unit_in_farad	Returns the capacitance unit specified in the library. <i>Return type:</i> float
chip_size	Returns the chip size. The chip size can be specified with the <code>aocv_chip_size</code> global variable. If this variable is not set, then the software will calculate the chip size from the design node locations. The global variable setting takes precedence over the software setting. <i>Return type:</i> float
current_unit_in_amp	Returns the current unit specified in the library. <i>Return type:</i> float
hierarchical_name	Returns the complete hierarchical name of the design. <i>Return type:</i> string

is_dont_touch	Returns <code>true</code> if the design is specified through the <code>set_dont_touch</code> command. <i>Return type:</i> <code>boolean</code>
is_max_transition_asserted	Returns a value of <code>true</code> if a maximum transition value (set using the <code>set_max_transition</code> command) is defined in the design. <i>Return type:</i> <code>boolean</code>
is_min_transition_asserted	Returns a value of <code>true</code> if a minimum transition value (set using the <code>set_min_transition</code> command) is defined in the design. <i>Return type:</i> <code>boolean</code>
max_capacitance	Returns the maximum capacitance value for the design. <i>Return type:</i> <code>float</code>
max_fanout	Returns the maximum fanout value for the design. <i>Return type:</i> <code>float</code>
min_fanout	Returns the minimum fanout value for the design. <i>Return type:</i> <code>float</code>
max_transition	Returns the maximum transition design rule limit for the design. <i>Return type:</i> <code>float</code>
min_transition	Returns the minimum transition design rule limit for the design. <i>Return type:</i> <code>float</code>
min_capacitance	Returns the minimum capacitance value for the design. <i>Return type:</i> <code>float</code>
object_type	Returns the object type <code>design</code> . <i>Return type:</i> <code>string</code>
operating_condition_max	Returns the name of the maximum operating condition specified for the design. <i>Return type:</i> <code>string</code>

<code>operating_condition_min</code>	Returns the name of the minimum operating condition specified for the design. <i>Return type:</i> string
<code>process_max</code>	Returns the process value of the maximum operating condition specified for the design. <i>Return type:</i> float
<code>process_min</code>	Returns the process value of the minimum operating condition specified for the design. <i>Return type:</i> float
<code>resistance_unit_in_ohm</code>	Returns the resistance value from the library. <i>Return type:</i> float
<code>slew_threshold_percent_fall_high</code>	Returns the upper slew trip point for falling waveforms, specified as a percent. <i>Return type:</i> float
<code>slew_threshold_percent_fall_low</code>	Returns the lower slew trip point for falling waveforms, specified as a percent. <i>Return type:</i> float
<code>slew_threshold_percent_rise_high</code>	Returns the upper slew trip point for rising waveforms, specified as a percent. <i>Return type:</i> float
<code>slew_threshold_percent_rise_low</code>	Returns the lower slew trip point for rising waveforms, specified as a percent. <i>Return type:</i> float
<code>temperature_max</code>	Returns the temperature value of the maximum operating condition for the design. <i>Return type:</i> float
<code>temperature_min</code>	Returns the temperature value of the minimum operating condition for the design. <i>Return type:</i> float
<code>timing_context_analysis</code>	Returns a value of <code>true</code> if timing context analysis is enabled on a design. <i>Return type:</i> boolean

<code>time_unit_in_second</code>	Returns the time units of the design, in seconds. that is, time unit of the design is 1.000000ns, the property returns 1.000000e-09. <i>Return type:</i> float
<code>tree_type_max</code>	Returns the value of the <code>tree_type</code> attribute for the specified maximum operating condition. <i>Return type:</i> string
<code>tree_type_min</code>	Returns the value of the <code>tree_type</code> attribute for the specified minimum operating condition. <i>Return type:</i> string
<code>voltage_max</code>	Returns the voltage value of the maximum operating condition for the design. <i>Return type:</i> float
<code>voltage_min</code>	Returns the voltage value of the minimum operating condition for the design. <i>Return type:</i> float
<code>voltage_unit_in_volt</code>	Returns the voltage units specified in the library for the design. <i>Return type:</i> float

Library Properties

<code>default_max_transition</code>	Returns the default maximum transition time limit specified for a given library. <i>Return type:</i> string
<code>default_operating_conditions</code>	Returns the default operating conditions specified for a given library. For example, <pre>get_property [get_libs FreePDK45_lib_v1.0] default_operating_conditions</pre> <i>Return type:</i> string
<code>hierarchical_name</code>	Returns the hierarchical path name for the library. <i>Return type:</i> string

input_threshold_pct_fall	Returns the value of threshold point on an input pin signal falling from 1 to 0, which is used in modelling the delay of a signal transmitting from an input pin to an output pin. <i>Return type:</i> float
input_threshold_pct_rise	Returns the value of threshold point on an input pin signal rising from 0 to 1, which is used in modelling the delay of a signal transmitting from an input pin to an output pin. <i>Return type:</i> float
lib_file_path	Reports full on-disk path of a library. In case of a library database (LDB) file, the property will return the full on-disk path of the original library from which LDB was compiled. <i>Return type:</i> string
lib_version	Returns the software version in which the library file (LDB) was generated. <i>Return type:</i> string
nominal_process	Returns a value of the process either modeled as nominal process or as the process in default operating condition in the library. For example, <pre>get_property [get_libs sc_cmoslp] nominal_process</pre> <i>Return Type:</i> float
nominal_temperature	Returns a value of the temperature either modeled as nominal temperature or as the temperature in default operating condition in the library. For example, <pre>get_property [get_libs sc_cmoslp] nominal_temperature</pre> <i>Return Type:</i> float
nominal_voltage	Returns a value of the voltage either modeled as nominal voltage or as the voltage in default operating condition in the library. For example, <pre>get_property [get_libs sc_cmoslp] nominal_voltage</pre> <i>Return Type:</i> float
object_type	Returns the object type library. <i>Return type:</i> string

output_threshold_pct_fall	Returns the value of threshold point on an output signal falling from 1 to 0, which is used in modelling the delay of a signal transmitting from an input pin to an output pin. <i>Return type:</i> float
output_threshold_pct_rise	Returns the value of threshold point on an output signal rising from 0 to 1, which is used in the modelling of a signal transmitting from an input pin to an output pin. <i>Return type:</i> float
power_rails	Returns the name of the power rails along with their corresponding voltages as defined in the <code>voltage_map</code> attribute in the library. For example, <pre>get_property [get_libs sc_cmoslp] power_rails</pre> Output: Is a list of lists, each list showing the power rail name along with its associated voltage. <pre>{ {VDDG 1.045} {VDD 1.045} {VSS 0} }</pre> <i>Return Type:</i> string
slew_derate_from_library	Specifies how the transition times found in the library need to be derated to match the transition times between the characterization trip points. <i>Return type:</i> float
slew_lower_threshold_pct_fall	Returns the value of lower threshold point used in modeling the delay of a pin falling from 1 to 0. <i>Return type:</i> float
slew_lower_threshold_pct_rise	Returns the value of lower threshold point used to model the delay of a pin rising from 0 to 1. <i>Return type:</i> float
slew_upper_threshold_pct_fall	Returns the value of upper threshold point used to model the delay of a pin falling from 1 to 0. <i>Return type:</i> float
slew_upper_threshold_pct_rise	Returns the value of upper threshold point used to model the delay of a pin rising from 0 to 1. <i>Return type:</i> float

source_file_name	Returns the full path of all library files that contain the specified library name. The use model is given below: <pre>get_property [get_libs libname] source_file_name [-view viewName]</pre> If the <code>-view</code> parameter is specified, the property returns the path of the library files in the given analysis view. If this parameter is not specified, the command returns the path of all library files in all the analysis views. <i>Return Type:</i> string
threshold_voltage_group	Returns the threshold voltage group of a library. For example, <pre>get_property [get_lib FreePDK45_lib_v1.0] threshold_voltage_group</pre> <i>Return Type:</i> string

Library Cell Properties

area	Returns the area of the library cell. <i>Return Type:</i> float
cell_footprint	Returns the cell footprint against a library cell, which the same layout boundary. For example, <pre>get_property [get_lib_cells 1.08_CTS_HT.db/CTB_</pre> <i>Return Type:</i> String
clock_gating_integrated_cell	Returns the clock gating integrated cells present in a For example, <pre>get_property [get_lib_cells CTS_HT.db/ICG_F4_HV</pre> <i>Return Type:</i> String

<code>early_clock_check_incremental_derate_factor</code>	Returns the early clock path library cell check increment <code>-increment</code> parameter. <i>Return Type:</i> float
<code>early_data_check_incremental_derate_factor</code>	Returns the early data path library cell check increment <code>-increment</code> parameter. <i>Return Type:</i> float
<code>early_fall_clock_incremental_derate_factor</code>	Returns the early clock path library cell check increment <code>-increment</code> parameter - when the <code>-fall</code> parameter is present. <i>Return Type:</i> float
<code>early_fall_data_incremental_derate_factor</code>	Returns the early data path library cell check increment <code>-increment</code> parameter - when the <code>-fall</code> parameter is present. <i>Return Type:</i> float
<code>early_fall_mean_slew_derate_factor</code>	Returns the early fall mean transition derate factor of the library. For example, <code>get_property [get_lib_cells FreePDK45_lib_v1.0/early_fall_mean_slew_derate_factor]</code> <i>Return Type:</i> float
<code>early_fall_sigma_slew_derate_factor</code>	Returns the early fall sigma transition derate factor of the library. For example, <code>get_property [get_lib_cells FreePDK45_lib_v1.0/early_fall_sigma_slew_derate_factor]</code> <i>Return Type:</i> float
<code>early_rise_clock_incremental_derate_factor</code>	Returns the early clock path library cell check increment <code>-increment</code> parameter - when the <code>-rise</code> parameter is present. <i>Return Type:</i> float
<code>early_rise_data_incremental_derate_factor</code>	Returns the early data path library cell check increment <code>-increment</code> parameter - when the <code>-rise</code> parameter is present. <i>Return Type:</i> float
<code>early_rise_mean_slew_derate_factor</code>	Returns the early rise mean transition derate factor of the library. For example, <code>get_property [get_lib_cells FreePDK45_lib_v1.0/early_rise_mean_slew_derate_factor]</code> <i>Return Type:</i> float

early_rise_sigma_slew_derate_factor	Returns the early rise sigma transition derate factor (0.0 to 1.0). For example, <pre>get_property [get_lib_cells FreePDK45_lib_v1.0/] early_rise_sigma_slew_derate_factor</pre> <i>Return Type:</i> float
early_fall_slew_derate_factor	Returns the derate factor applied on a library cell when early fall is specified. <i>Return Type:</i> float
early_rise_slew_derate_factor	Returns the derate factor applied on a library cell when early rise is specified. For example, <pre>get_property [get_lib_cells CTS_HT.db/ICG_F4_HT] early_rise_slew_derate_factor</pre> <i>Return Type:</i> float
fall_input_switching_derate_factor	Returns the derate factor applied on a library cell when fall input switching is specified. For example, <pre>get_property [get_lib_cells PDK45_v1.0/IV_X32] fall_input_switching_derate_factor</pre> <i>Return Type:</i> float
has_ccs_noise	Returns a value of <code>true</code> if a library cell has a CCS noise model. For example, <pre>get_property [get_lib_cells tst/tst_nm6] has_ccs_noise</pre> <i>Return type:</i> boolean
has_socv	Returns a value of <code>true</code> if a library cell has a spatial variation model. For example, <pre>get_property [get_lib_cells PK4_lib_v1.0/A4_X2] has_socv</pre> <i>Return type:</i> boolean
has_spatial	Returns a value of <code>true</code> if a spatial derate exists for a library cell. <i>Return type:</i> boolean
hierarchical_name	Returns the complete hierarchical name of the library cell. For example, <pre>get_property [get_lib_cells -of [get_cell AB_IT]] hierarchical_name</pre> <i>Return Type:</i> string

<code>is_black_box</code>	Returns a value of <code>true</code> if the cell is a black box cell. A cell is considered as a black box if the "function" d <i>Return type:</i> boolean
<code>is_buffer</code>	Returns a value of <code>true</code> if the associated library cell i For example, <code>get_property [get_lib_cells -of [get_cell AB_IT</code> <i>Return type:</i> boolean
<code>is_combinational</code>	Returns a value of <code>true</code> if the library cell is a combin <i>Return type:</i> boolean
<code>is_disable_timing</code>	Returns a value of <code>true</code> if a cell's timing has been dis <i>Return Type:</i> boolean
<code>is_dont_touch</code>	Returns a value of <code>true</code> if the library cell definition in <i>Return Type:</i> boolean
<code>is_dont_use</code>	Returns a value of <code>true</code> if the library cell description i <i>Return Type:</i> boolean
<code>is_fall_edge_triggered</code>	Returns a value of <code>true</code> if the library cell is triggered <i>Return Type:</i> boolean
<code>is_instantiated</code>	Returns a value of <code>true</code> if library cells are instantiated For example, <code>get_property [get_lib_cells P4_lib_v1.0/N4_X2]</code> <i>Return type:</i> boolean
<code>is_interface_timing</code>	Returns a value of <code>true</code> if a library cell has interface t <i>Return type:</i> boolean
<code>is_level_shifter</code>	Returns a value of <code>true</code> if a cell instance is a level-sh <i>Return type:</i> boolean
<code>is_integrated_clock_gating_cell</code>	Returns a value of <code>true</code> if a cell is an integrated clock For example, <code>get_property [get_lib_cells CTS_HT.db/ICG_F7_HT</code> <i>Return type:</i> boolean

is_inverter	Returns a value of <code>true</code> if a cell instance is an inverter. For example, <pre>get_property [get_lib_cells -of [get_cells inv]]</pre> <i>Return type:</i> boolean
is_level_shifter	Returns a value of <code>true</code> if a cell instance is a level-shifter. For example, <pre>get_property [get_lib_cells -of [get_libs mk_1p]]</pre> <i>Return type:</i> boolean
is_macro_cell	Returns a value of <code>true</code> if a macro cell exists in a library. For example, <pre>get_property [get_lib_cells L_mro2/mro2y] is_macro_cell</pre> <i>Return type:</i> boolean
is_memory_cell	Returns a value of <code>true</code> if the cell is a memory cell. <i>Return type:</i> boolean Note: The library cells that have a memory group defined are as follows: <pre>cell () { : memory() { type : ram ; address_width : 10 ; word_width : 7 ; } }</pre>
is_msv	Returns a value of <code>true</code> if a cell is an MSV cell. For example, <pre>get_property [get_lib_cells mk_1pv_125c/L8MTR]</pre> <i>Return type:</i> boolean
is_negative_level_sensitive	Returns a value of <code>true</code> if the library cell is used as a negative level-sensitive cell. <i>Return type:</i> boolean

<code>is_pad_cell</code>	Returns a value of <code>true</code> if the Liberty library <code>is_pad_c</code> <i>Return type:</i> boolean
<code>is_pll_cell</code>	Returns a value of <code>true</code> if the Liberty library <code>is_pll_c</code> <i>Return type:</i> boolean
<code>is_positive_level_sensitive</code>	Returns a value of <code>true</code> if the library cell is used as a <i>Return type:</i> boolean
<code>is_rise_edge_triggered</code>	Returns a value of <code>true</code> if the library cell is triggered <i>Return Type:</i> boolean
<code>is_sequential</code>	Returns a value of <code>true</code> if the library cell is a latch or <i>Return type:</i> boolean
<code>is_tristate</code>	Returns a value of <code>true</code> if the library cell definition in <i>Return type:</i> boolean
<code>late_clock_check_incremental_derate_factor</code>	Returns the late clock path library cell check increme increment parameter. <i>Return Type:</i> float
<code>late_data_check_incremental_derate_factor</code>	Returns the late data path library cell check increme increment parameter. <i>Return Type:</i> float
<code>late_fall_clock_incremental_derate_factor</code>	Returns the late clock path library cell check increme increment parameter - when the <code>-fall</code> parameter is sp <i>Return Type:</i> float
<code>late_fall_data_incremental_derate_factor</code>	Returns the late data path library cell check increme increment parameter - when the <code>-fall</code> parameter is : <i>Return Type:</i> float
<code>late_fall_clock_period_derate_factor</code>	Returns the derate factor for the clock period when tl specified. <i>Return type:</i> float
<code>late_fall_clock_period_mean_derate_factor</code>	Returns the late/fall derate factor for the clock period <i>Return type:</i> float

<code>late_fall_clock_period_sigma_derate_factor</code>	Returns the late/fall derate factor for the clock period <i>Return type:</i> float
<code>late_fall_mean_slew_derate_factor</code>	Returns the late/fall mean transition derate factor on For example, <code>get_property [get_lib_cells FreePDK45_lib_v1.0/</code> <i>Return type:</i> float
<code>late_fall_pulse_width_derate_factor</code>	Returns the late/fall derate factor for the pulse width specified. <i>Return type:</i> float
<code>late_fall_pulse_width_mean_derate_factor</code>	Returns the late/fall derate factor for the pulse width <i>Return type:</i> float
<code>late_fall_pulse_width_sigma_derate_factor</code>	Returns the late/fall derate factor for the pulse width <i>Return type:</i> float
<code>late_fall_sigma_slew_derate_factor</code>	Returns the late/fall sigma transition derate factor on For example, <code>get_property [get_lib_cells FreePDK45_lib_v1.0/</code> <i>Return type:</i> float
<code>late_fall_slew_derate_factor</code>	Returns the derate factor applied on a library cell, w are specified. <i>Return type:</i> float
<code>late_rise_clock_incremental_derate_factor</code>	Returns the late clock path library cell check increme increment parameter - when the <code>-rise</code> parameter is : <i>Return Type:</i> float
<code>late_rise_data_incremental_derate_factor</code>	Returns the late data path library cell check increme increment parameter - when the <code>-rise</code> parameter is : <i>Return Type:</i> float
<code>late_rise_clock_period_derate_factor</code>	Returns the derate factor for the clock period when tl <i>Return type:</i> float
<code>late_rise_clock_period_mean_derate_factor</code>	Returns the late/rise derate factor for the clock perio <i>Return type:</i> float

<code>late_rise_clock_period_sigma_derate_factor</code>	Returns the late/rise derate factor for the clock period. <i>Return type:</i> float
<code>late_rise_mean_slew_derate_factor</code>	Returns the late/rise mean transition derate factor or For example, <code>get_property [get_lib_cells FreePDK45_lib_v1.0/</code> <i>Return type:</i> float
<code>late_rise_pulse_width_derate_factor</code>	Returns the derate factor for the pulse width check w specified. <i>Return type:</i> float
<code>late_rise_pulse_width_mean_derate_factor</code>	Returns the late/rise derate factor for the pulse width <i>Return type:</i> float
<code>late_rise_pulse_width_sigma_derate_factor</code>	Returns the late/rise derate factor for the pulse width <i>Return type:</i> float
<code>late_rise_sigma_slew_derate_factor</code>	Returns the late/rise sigma transition derate factor or For example, <code>get_property [get_lib_cells FreePDK45_lib_v1.0/</code> <i>Return type:</i> float
<code>late_rise_slew_derate_factor</code>	Returns the derate factor applied on a library cell, w are specified. For example, <code>get_property [get_lib_cells PK4_v1.0/IV_32] lat</code> <i>Return type:</i> float
<code>mis_alignments_factor_min_fall</code>	Returns the minimum fall tuning factor applied on a l <code>switching_alignment_factor</code> parameter is specified. <i>Return type:</i> float
<code>mis_alignments_factor_min_rise</code>	Returns the minimum rise tuning factor applied on a <code>switching_alignment_factor</code> parameter is specified. <i>Return type:</i> float

master_physical_variant_cell	Returns the master library cell of the specified physical variant. For example, <code>get_property [get_lib_cell cell_w/INV2] master_physical_variant_cell</code> <i>Return type:</i> string
name	Returns the leaf-level name of the library cell. <i>Return Type:</i> string
object_type	Returns the object type <code>lib_cell</code> . <i>Return Type:</i> string
physical_variant_cells	Returns the physical variant cells that are available in the library. For example, <code>get_property [get_lib_cells lca_md/SR1_md] physical_variant_cells</code> <i>Return Type:</i> string
pin_count	Returns the total number of signal pins. For example, <code>get_property [get_lib_cells PK45_v1.0/IN_X32] pin_count</code> <i>Return Type:</i> integer
receiver_capacitance_type	Returns the value of input capacitance group being used (n-piece model). <i>Return Type:</i> string
retention_cell	Returns the value of the retention_cell attribute specified for the library cell. <i>Return type:</i> string
rise_input_switching_derate_factor	Returns the derate factor applied on a library cell, with the specified input. For example, <code>get_property [get_lib_cells PK45_v1.0/DL_X2] rise_input_switching_derate_factor</code> <i>Return type:</i> float
signal_pins	Returns the signal pins of a cell, along with the library cell name. For example, <code>get_property [get_lib_cells PK45_v1.0/IN_X32] signal_pins</code> <i>Return Type:</i> collection

sync_clear	Returns the collection of synchronous clear pins. Note: For a synchronous clear pin: • The flip-flop's function definition should have the • The pin corresponding to the synchronous clear i For example, <pre>get_property [get_lib_cells tcbn05_bwph21016p51cnod_base_svtffgnp_0p825v_12 sync_clear</pre> <i>Return type:</i> collection
sync_enable	Returns the collection of synchronous enable pins. For example, <pre>get_property [get_lib_cells tcbn05_bwph21016p51cnod_base_ulvtllffgnp_0p825v sync_enable</pre> <i>Return type:</i> collection
sync_preset	Returns the collection of synchronous preset pins. For example, <pre>get_property [get_lib_cells tcbn05_bwph21016p51cnod_base_svtffgnp_0p825v_12 sync_preset</pre> <i>Return type:</i> collection
test_scan_enable	Returns the collection of the test_scan_enable pins. For example, <pre>get_property [get_lib_cell SR80_W_125_0.855_0.8</pre> <i>Return type:</i> collection
test_scan_enable_inverted	Returns the collection of the test_scan_enable_inve For example, <pre>get_property [get_lib_cells SR80_W_125_0.855_0. test_scan_enable_inverted</pre> <i>Return type:</i> collection

test_scan_in	Returns the collection of the <code>test_scan_in</code> pins. For example, <pre>get_property [get_lib_cells SR80_W_125_0.855_0.</pre> <i>Return type:</i> collection
test_scan_in_inverted	Returns the collection of the <code>test_scan_in_inverted</code> For example, <pre>get_property [get_lib_cell SR80_W_125_0.855_0.8</pre> <i>Return type:</i> collection
test_scan_out	Returns the collection of the <code>test_scan_out</code> pins. For example, <pre>get_property [get_lib_cells SR80_W_125_0.855_0.</pre> <i>Return type:</i> collection
test_scan_out_inverted	Returns the collection of the <code>test_scan_out_inverted</code> For example, <pre>get_property [get_lib_cell SR80_W_125_0.855_0.8 test_scan_out_inverted</pre> <i>Return type:</i> collection
threshold_voltage_group	Returns the voltage group defined at the library level (<code>set_threshold_voltage_group</code> command). <i>Return type:</i> string For example, <pre>get_property [get_lib_cells L40_HT.db/CG_F8_HT]</pre>
timing_model_type	Returns the model type for a given cell or instance. 7 <i>Return type:</i> string
ground_pins	Returns a collection of the ground pins of a library cell. For example, <pre>get_property [get_lib_cells sc_cmoslp/AO22X] gr</pre> <i>Return type:</i> collection

is_always_on	Returns a value of <code>true</code> if the <code>always_on</code> attribute of a always-on cell. For example, <code>get_property [get_lib_cells sc_cmoslp/AON22_X4M</code> <i>Return type:</i> boolean
is_isolation_cell	Returns a value of <code>true</code> if the library cell is an isolatic For example, <code>get_property [get_lib_cells sc_cmoslp/ISO22_X4M</code> <i>Return type:</i> boolean
is_power_switch	Returns a value of <code>true</code> when the <code>switch_cell_type</code> cell is a power-switch cell. For example, <code>get_property [get_lib_cells sc_cmoslp/AO22X] is</code> <i>Return type:</i> boolean
is_retention	Returns a value of <code>true</code> when the <code>retention_cell att</code> is a retention cell. For example, <code>get_property [get_lib_cells sc_cmoslp/RET22_X4M</code> <i>Return type:</i> boolean
level_shifter_type	Returns the type of level-shifter (e.g., LH, HL, or HL_ cell. For example, <code>get_property [get_lib_cells sc_cmoslp/A2LVLD_X4</code> <i>Return type:</i> string
lib_pg_pins	Returns a collection of all PG pins (both power and c For example, <code>get_property [get_lib_cells sc_cmoslp/AO22X] li</code> <i>Return type:</i> collection

power_pins	Returns a collection of the power pins of a library cell. For example, <pre>get_property [get_lib_cells sc_cmoslp/A022X] po</pre> <i>Return type:</i> collection
power_switch_type	Returns the type of power-switch (e.g., coarse-grain, library cells). For example, <pre>get_property [get_lib_cells sc_cmoslp/A022X] po</pre> <i>Return type:</i> string
std_cell_main_rail_name	Returns the name of a power rail that has the <code>std_cell</code> property. For example, <pre>get_property [get_lib_cells sc_cmoslp/A022X] st</pre> <i>Return type:</i> string

Library Pin Properties

all_q_pin_of_d_pin	Returns a collection of <code>all_q_pin_of_d_pin</code> pins. For example, <pre>get_property [get_lib_pin srpg/SDRFFSRPQ_X1/D] all_q_pin_of_d_pin</pre> <i>Return type:</i> collection
capacitance	Returns the capacitance for the library pin. <i>Return type:</i> float
capacitance_max_fall	Returns the maximum value of the falling capacitance range. <i>Return type:</i> float
capacitance_max_rise	Returns the maximum value of the rising capacitance range. <i>Return type:</i> float
capacitance_min_fall	Returns the minimum value of the falling capacitance range. <i>Return type:</i> float

capacitance_min_rise	Returns the minimum value of the rising capacitance range. <i>Return type:</i> float
cell_name	Returns the library cell name for the corresponding library pins. For example, <pre>get_property [get_lib_pins PK4_lib_v1.0/AOI211_X2/B] cell_name</pre> <i>Return type:</i> string
direction	Returns the direction of the library pin: in, out, or inout. <i>Return type:</i> string
drive_resistance_fall	Specifies the driving resistance of the cell for falling transitions. For example, <pre>get_property [get_lib_pins FreePDK45_lib_v1.0/BUF_X1/Z] drive_resistance_fall</pre> <i>Return type:</i> float
drive_resistance_rise	Specifies the driving resistance of the cell for rising transitions. For example, <pre>get_property [get_lib_pins FreePDK45_lib_v1.0/BUF_X1/Z] drive_resistance_rise</pre> <i>Return type:</i> float
fanout_load	Returns the fanout load value of the library pin. <i>Return type:</i> float
function	Returns the function for the corresponding library pins. For example, <pre>get_property [get_lib_pins PK4_lib_v1.0/AND2_X1/ZN] function</pre> <i>Return type:</i> string
hierarchical_name	Returns the hierarchical name of the library pin. <i>Return type:</i> string

input_signal_level	Returns the power rail name/signal that is defined for a library cell input pin. For example, <pre>get_property [get_lib_pins PK4_lib_v1.0/BUF_X32/A] input_signal_level</pre> <i>Return type:</i> string
input_signal_level_high	Returns the upper limit of partial voltage swing value that is defined for a library cell input pin. For example, <pre>get_property [get_lib_pins rvt_ss_0p9v/LVLL/A] input_signal_level_high</pre> <i>Return type:</i> float
input_signal_level_low	Returns the lower limit of partial voltage swing value that is defined for a library cell input pin. For example, <pre>get_property [get_lib_pins rvt_ss_0p9v/LVLL/A] input_signal_level_low</pre> <i>Return type:</i> float
is_analog	Returns a value of <code>true</code> if the library pin is an analog pin. For example, <pre>get_property [get_lib_pins PK4_lib_v1.0/AOI222_X2/ZN] is_analog</pre> <i>Return type:</i> boolean
is_async	Returns a value of <code>true</code> if the library pin is an asynchronous preset pin, or an asynchronous clear pin. <i>Return type:</i> boolean
is_clear	Returns a value of <code>true</code> if the pin is an asynchronous clear pin. <i>Return type:</i> boolean
is_clock	Returns a value of <code>true</code> if the library pin definition includes <code>clock:true</code>. <i>Return type:</i> boolean

is_clock_gating_clock	Returns a value of <code>true</code> if the library pin is defined as a clock pin of a clock gating cell. For example, <pre>get_property [get_lib_pins P40_CTS_HVT.db/ICG_HVT/CLKIN] is_clock_gating_clock</pre> <i>Return type:</i> boolean
is_clock_gating_enable	Returns a value of <code>true</code> if the library pin corresponds to the enable pin of a clock gating cell. For example, <pre>get_property [get_lib_pins P40_CTS_HVT.db/ICG_HVT/CLKIN] is_clock_gating_enable</pre> <i>Return type:</i> boolean
is_clock_gating_test	Returns a value of <code>true</code> if the pin corresponds to the test pin of a clock gating cell. For example, <pre>get_property [get_lib_pins */NAND2_X2/A2] is_clock_gating_test</pre> <i>Return type:</i> boolean
is_clock_pin	Returns a value of <code>true</code> if the library pin is a clock pin. For example, <pre>get_property [get_lib_pins srpg/SDRFFSRPQ_X1/CK] is_clock_pin</pre> <i>Return Type:</i> boolean
is_data	Returns a value of <code>true</code> if the library pin, with defined sequential arcs, is the data pin of a flip-flop. <i>Return type:</i> boolean
is_fall_edge_triggered_clock	Returns a value of <code>true</code> if the library pin is the clock pin of a falling edge triggered device. <i>Return type:</i> boolean
is_fall_edge_triggered_data	Returns a value of <code>true</code> if the library pin is the data pin of a falling edge triggered device. <i>Return type:</i> boolean

<code>is_iq_function</code>	Returns a value of <code>true</code> if the output is an IQ pin. For example, <pre>get_property [get_lib_pins */SDFFRS_X1/Q] is_iq_function</pre> <i>Return Type:</i> boolean
<code>is_iqn_function</code>	Returns a value of <code>true</code> if the output is an IQN pin. For example, <pre>get_property [get_lib_pins */SDFFRS_X1/QN] is_iqn_function</pre> <i>Return Type:</i> boolean
<code>is_isolation_cell_clock</code>	Returns a value of <code>true</code> if the clock pin is a pin of isolation cell clock. For example, <pre>get_property [get_lib_pins cell_w/INV/CLK] is_isolation_cell_clock</pre> <i>Return Type:</i> boolean
<code>is_negative_level_sensitive_clock</code>	Returns a value of <code>true</code> if the library pin is an enable pin of an active low level-sensitive device. <i>Return type:</i> boolean
<code>is_negative_level_sensitive_data</code>	Returns a value of <code>true</code> if the pin is a data pin of an active low level-sensitive device. <i>Return type:</i> boolean
<code>is_pad</code>	Returns a value of <code>true</code> if the library pin is a pad pin. For example, <pre>get_property [get_lib_pins *] is_pad</pre> <i>Return type:</i> boolean
<code>is_pll_feedback_pin</code>	Returns a value of <code>true</code> if the Liberty library <code>is_pll_feedback_pin</code> attribute is set to <code>true</code> for this library pin. <i>Return type:</i> boolean

<code>is_pll_output_pin</code>	Returns a value of <code>true</code> if the Liberty library <code>is_pll_output_pin</code> attribute is set to <code>true</code> for this library pin. <i>Return type:</i> boolean
<code>is_pll_reference_pin</code>	Returns a value of <code>true</code> if the Liberty library <code>is_pll_reference_pin</code> attribute is set to <code>true</code> for this library pin. <i>Return type:</i> boolean
<code>is_positive_level_sensitive_clock</code>	Returns a value of <code>true</code> if the library pin is an enable pin of an active high level-sensitive device. <i>Return type:</i> boolean
<code>is_positive_level_sensitive_data</code>	Returns a value of <code>true</code> if the pin is a data pin of an active high level-sensitive device. <i>Return type:</i> boolean
<code>is_preset</code>	Returns a value of <code>true</code> if the pin is a preset pin. <i>Return type:</i> boolean

<code>is_ref_clock_pin</code>	Returns a value of <code>true</code>: • If the library pin is a reference clock pin. • Valid instance pin object and on an active clock pin where a clock signal reaches, and sequential cell is not disabled by disabled timing arcs or case analysis. Any pin with clock check-arcs or trigger-arcs will also be reported, if an active clock phase is present on that pin. • Clock pin on an inferred clock gating cell. • Valid <code>lib_pin</code> object, if there is at least one instance of this <code>lib_pin</code> in the design, and if <code>is_clock_pin</code> property is set to <code>true</code>. Returns a value of <code>false</code>, if clock propagation is disabled with the <code>set_disable_timing</code> or <code>set_sense -stop_propagation</code> command. For example, <pre>get_property [get_lib_pins srpg/SDRFFSRPQ_X1/CK] is_ref_clock_pin</pre> <i>Return Type:</i> boolean
<code>is_rise_edge_triggered_clock</code>	Returns a value of <code>true</code> if the library pin is the clock pin of a rising edge triggered device. <i>Return type:</i> boolean
<code>is_rise_edge_triggered_data</code>	Returns a value of <code>true</code> if the library pin is the data pin of a rising edge triggered device. <i>Return type:</i> boolean
<code>is_tristate</code>	Returns a value of <code>true</code> if the library pin definition includes <code>three_state:true</code>. <i>Return type:</i> boolean
<code>max_capacitance</code>	Returns the maximum capacitance limit for the library pin. <i>Return type:</i> float
<code>max_fanout</code>	Returns the maximum fanout value for the library pin. <i>Return type:</i> float

max_transition	Returns the maximum transition time limit specified for a given library pin. If the limit is not specified for a library pin, then the limit specified at the corresponding library level will be used. <i>Return type:</i> float
min_transition	Returns the minimum transition time limit specified for a given library pin. If the limit is not specified for a library pin, then the limit specified at the corresponding library level will be used. <i>Return type:</i> float
min_capacitance	Returns the minimum capacitance limit for the library pin. <i>Return type:</i> float
min_fanout	Returns the minimum fanout value for the library pin. <i>Return type:</i> float
name	Returns the leaf-level name of the library cell pin. <i>Return type:</i> string
object_type	Returns the object type; in this case, lib_pin. <i>Return type:</i> string
related_ground_pin_name	Returns related ground pin names for the corresponding library pins. <i>Return type:</i> string For example, <pre>get_property [get_lib_pins sc_cmoslp/A022X/I1] related_ground_pin_name</pre>
related_power_pin_name	Returns related power pin names for the corresponding library pins. <i>Return type:</i> string
slew_threshold_percent_fall_high	Returns the upper slew trip point for falling waveforms, specified as a percent. <i>Return type:</i> float
slew_threshold_percent_fall_low	Returns the lower slew trip point for falling waveforms, specified as a percent. <i>Return type:</i> float

slew_threshold_percent_rise_high	Returns the upper slew trip point for rising waveforms, specified as a percent. <i>Return type:</i> float
slew_threshold_percent_rise_low	Returns the lower slew trip point for rising waveforms, specified as a percent. <i>Return type:</i> float
input_signal_level_voltage	Returns the voltage value corresponding to the power rail name/signal which is defined for the library-cell input pin. This voltage value is the full rail-to-rail voltage value that is associated with the relative power rail name through the <code>voltage_map</code> attribute defined in the library. For example, <pre>get_property [get_lib_pins sc_cmoslp/A022X/I1] input_signal_level_voltage</pre> <i>Return type:</i> float
is_always_on	Returns a value of <code>true</code> if the <code>always_on</code> attribute of a library signal pin is set to <code>true</code> indicating that the library cell signal pin is an always-on pin. For example, <pre>get_property [get_lib_pins sc_cmoslp/A022X/I1] is_always_on</pre> <i>Return type:</i> boolean
is_isolated	Returns a value of <code>true</code> if the <code>is_isolated</code> attribute of a library signal pin is set to <code>true</code> indicating that the library cell signal pin is an internally isolated pin. For example, <pre>get_property [get_lib_pins sc_cmoslp/A022X/I1] is_isolated</pre> <i>Return type:</i> boolean
is_isolation_cell_data	Returns a value of <code>true</code> if the <code>isolation_cell_data_pin</code> attribute of a library signal pin is set to <code>true</code> indicating that the library cell signal pin is a data pin of an isolation library cell. For example, <pre>get_property [get_lib_pins sc_cmoslp/A022X/I1] is_isolation_cell_data</pre> <i>Return type:</i> boolean

<code>is_isolation_cell_enable</code>	Returns a value of <code>true</code> if the <code>isolation_cell_enable_pin</code> attribute of a library signal pin is set to <code>true</code> indicating that the library cell signal pin is an enable or control pin of an isolation library cell. For example, <pre>get_property [get_lib_pins sc_cmoslp/A022X/I1] is_isolation_cell_enable</pre> <i>Return type:</i> boolean
<code>is_level_shifter_data</code>	Returns a value of <code>true</code> if the <code>level_shifter_data_pin</code> attribute of a library signal pin is set to <code>true</code>, indicating that the library cell signal pin is a data pin of a library level-shifter cell. For example, <pre>get_property [get_lib_pins sc_cmoslp/A022X/I1] is_level_shifter_data</pre> <i>Return type:</i> boolean
<code>is_level_shifter_enable</code>	Returns a value of <code>true</code> if the <code>level_shifter_enable_pin</code> attribute of a library signal pin is set to be <code>true</code> indicating that the library-cell signal pin is an enable pin of a library level shifter cell. For example, <pre>get_property [get_lib_pins sc_cmoslp/A022X/I1] is_isolation_cell_enable</pre> <i>Return type:</i> boolean
<code>is_power_switch_enable</code>	Returns a value of <code>true</code> if the <code>switch_cell_type</code> attribute of the corresponding library cell is defined and the <code>switch_pin</code> attribute of the given library-cell signal pin is defined to be <code>true</code>. For example, <pre>get_property [get_lib_pins sc_cmoslp/A022X/I1] is_power_switch_enable</pre> <i>Return type:</i> boolean

is_retention_cell_enable	Returns a value of <code>true</code> if the <code>retention_pin</code> attribute of a signal pin is defined. For example, <pre>get_property [get_lib_pins sc_cmoslp/A022X/I1] is_retention_cell_enable</pre> <i>Return type:</i> boolean
output_signal_level	Returns the power rail name/signal that is defined for a library cell output and inout pins. For example, <pre>get_property [get_lib_pins sc_cmoslp/A022X/I1] output_signal_level</pre> <i>Return type:</i> string
output_signal_level_high	Returns the upper/higher limit of partial voltage swing value that is defined for a library cell output and inout pins. For example, <pre>get_property [get_lib_pins sc_cmoslp/A022X/I1] output_signal_level_high</pre> <i>Return type:</i> float
output_signal_level_low	Returns the lower limit of partial voltage swing value that is defined for a library cell output and inout pins. For example, <pre>get_property [get_lib_pins sc_cmoslp/A022X/I1] output_signal_level_low</pre> <i>Return type:</i> float
pin_capacitance	Returns the pin capacitance. For example, <pre>get_property [get_lib_pins FreePDK45_lib_v1.0/BUF_X1/A] pin_capacitance</pre> <i>Return type:</i> float
pin_capacitance_max_fall	Returns the maximum value of the falling capacitance range of the corresponding library pin. For example, <pre>get_property [get_lib_pins FreePDK45_lib_v1.0/BUF_X1/A] pin_capacitance_max_fall</pre> <i>Return type:</i> float

pin_capacitance_max_rise	Returns the maximum value of the rising capacitance range of the corresponding library pin. For example, <pre>get_property [get_lib_pins FreePDK45_lib_v1.0/BUF_X1/A] pin_capacitance_max_rise</pre> <i>Return type:</i> float
pin_capacitance_min_fall	Returns the minimum value of the falling capacitance range of the corresponding library pin. For example, <pre>get_property [get_lib_pins FreePDK45_lib_v1.0/BUF_X1/A] pin_capacitance_min_fall</pre> <i>Return type:</i> float
pin_capacitance_min_rise	Returns the minimum value of the rising capacitance range of the corresponding library pin. For example, <pre>get_property [get_lib_pins FreePDK45_lib_v1.0/BUF_X1/A] pin_capacitance_min_rise</pre> <i>Return type:</i> float
related_ground_pin	Returns the related ground pins as a collection for the corresponding library cell pins. For example, <pre>get_property [get_lib_pins sc_cmoslp/A022X/I1] related_ground_pin</pre> <i>Return type:</i> collection
related_ground_pin_name	Returns the related ground pin names for the corresponding library cell pins. For example, <pre>get_property [get_lib_pins sc_cmoslp/A022X/I1] related_ground_pin_name</pre> <i>Return type:</i> string

related_ground_pin_rail_voltage	Returns the voltage value corresponding to the ground rail name defined for the <code>related_ground_pin</code> attribute of the library cell pin. This voltage value is the full rail-to-rail value that is associated with the relative ground rail name through the <code>voltage_map</code> attribute defined in the library. For example, <pre>get_property [get_lib_pins sc_cmoslp/A022X/I1] related_ground_pin_rail_voltage</pre> <i>Return type:</i> float
related_power_pin	Returns the related power pins as a collection for the corresponding library cell pins. For example, <pre>get_property [get_lib_pins sc_cmoslp/A022X/I1] related_power_pin</pre> <i>Return type:</i> collection
related_power_pin_name	Returns the related power pin names for the corresponding library cell pins. For example, <pre>get_property [get_lib_pins sc_cmoslp/A022X/I1] related_power_pin_name</pre> <i>Return type:</i> string
related_power_pin_rail_voltage	Returns the voltage value corresponding to the power rail name defined for the <code>related_power_pin</code> attribute of a library cell pin. This voltage value is the full rail-to-rail value that is associated with the relative power rail name through the <code>voltage_map</code> attribute defined in the library. For example, <pre>get_property [get_lib_pins sc_cmoslp/A022X/I1] related_power_pin_rail_voltage</pre> <i>Return type:</i> float

Library PG Pin Properties

is_std_cell_main_rail	Returns a value of <code>true</code> if the library power PG pin is the main rail of the library cell. It is determined by the <code>std_cell_main_rail</code> library attribute of the library PG pin. <i>Return type:</i> boolean For example, <pre>get_property [get_lib_pg_pins -of_objects [get_lib_cells sc_cmoslp/A022X]] is_std_cell_main_rail</pre>
name	Returns the name of the library PG pin as defined in the library and associated with the <code>pg_pin</code> library attribute. <i>Return type:</i> string For example, <pre>get_property [get_lib_pg_pins -of_objects [get_lib_cells sc_cmoslp/A022X]] name</pre>
object_type	Returns <code>lib_pg_pin</code> as the object type for a library PG pin. <i>Return type:</i> string For example, <pre>get_property [get_lib_pg_pins -of_objects [get_lib_cells sc_cmoslp/A022X]] object_type</pre>
pg_type	Returns the type of the library PG pin as associated with the <code>pg_type</code> attribute of the corresponding library PG pin. <i>Return type:</i> string For example, <pre>get_property [get_lib_pg_pins -of_objects [get_lib_cells sc_cmoslp/A022X]] pg_type get_property [get_lib_pg_pins -of_objects [get_lib_cells sc_cmoslp/A022X] -filter " name =~ *VDD*"] pg_type</pre>
rail_name	Returns the name of the rail to which the PG pin is connected and is associated with the <code>voltage_name</code> attribute of the library PG pin. <i>Return type:</i> string For example, <pre>get_property [get_lib_pg_pins -of_objects [get_lib_cells sc_cmoslp/A022X]] rail_name</pre>

rail_voltage	Returns the voltage from the voltage_map association of the rail to which the library PG pin is connected. <i>Return type:</i> float For example, <pre>get_property [get_lib_pg_pins -of_objects [get_lib_cells sc_cmoslp/A022X]] rail_voltage</pre>
tied_to	Returns the library PG pin which is connected to substrate-bias PG pin. <i>Return type:</i> string For example, <pre>get_property [get_lib_pg_pins -of_objects [get_lib_cells sc_cmoslp/A022X]] tied_to</pre>

Library Timing Arc Properties

from_lib_pin	Returns a collection of the source library pins of the library timing arc. <i>Return type:</i> collection
has_ccs_noise	Returns a value of <code>true</code> if a cell has CCS noise data. For example, <pre>get_property [get_lib_arc -from ccs_timing_si/AN2D1BWP20P90/A1 -to ccs_timing_si/AN2D1BWP20P90/Z] has_ccs_noise</pre> <i>Return type:</i> boolean
has_mis_fall	Returns a value of <code>true</code>, if MIS (multi-input switching) derate fall value is specified for library timing arcs. For example, <pre>get_property [get_lib_arc -from FreePDK45_lib_v1.0/AND2_X1/A1 -to FreePDK45_lib_v1.0/AND2_X1/ZN] has_mis_fall</pre> <i>Return type:</i> boolean
has_mis_rise	Returns a value of <code>true</code>, if MIS (multi-input switching) derate rise value is specified for library timing arcs. For example, <pre>get_property [get_lib_arcs -from FreePDK45_lib_v1.0/OR2_X1/A1 -to FreePDK45_lib_v1.0/OR2_X1/ZN] has_mis_rise</pre> <i>Return type:</i> boolean

has_socv	Returns a value of <code>true</code> if a timing library arc in the library has a spatial derate table (SOCV). <i>Return type:</i> boolean
is_disabled	Return a value of <code>true</code> if a <code>set_disable_timing</code> constraint exists for an arc. <i>Return type:</i> boolean
mode	Returns the name of the group if the timing arc is defined as part of a Liberty mode group. For example, <pre>get_property [get_lib_arc -from c065lesram76x2x1x96b_PttV1200T025_lib/c065lesram76x2x1x96b/CL -to c065lesram76x2x1x96b_PttV1200T025_lib/c065lesram76x2x1x96b/RX] mode</pre> <i>Return type:</i> string
object_type	Returns the object type <code>lib_timing_arc</code>. <i>Return type:</i> string
sdf_cond	Returns the <code>sdf_cond</code> expression specified for the timing library arc in the library. <i>Return type:</i> string
sense	Returns the <code>timing_sense</code> specified for the timing library arc in the library. <i>Return type:</i> string
to_lib_pin	Returns a collection of the sink library pins of the library timing arc. <i>Return type:</i> collection
timing_type	Returns the type specified for the timing library arc in the library. <i>Return type:</i> string
when	Returns the <code>when</code> condition specified for the library timing arc in the library. <i>Return type:</i> string
when_library_string	Returns the exact value of the Liberty 'when' attribute for the specified arc. For example, <pre>get_property [get_lib_arc -from FreePDK45_lib_v1.0/BUF_X1/A -to FreePDK45_lib_v1.0/BUF_X1/Z] when_library_string</pre> <i>Return type:</i> string

Net Properties

annotated_capacitance_max	Returns the maximum annotated capacitance of the net for late path analysis. For example, <pre>get_property [get_nets w2] annotated_capacitance_max</pre> <i>Return type:</i> float
annotated_capacitance_min	Returns the minimum annotated capacitance of the net for late path analysis. For example, <pre>get_property [get_nets w2] annotated_capacitance_min</pre> <i>Return type:</i> float
attackers	Reports the parasitic data on a net - capacitance, resistance, or cross-coupling. <i>Return type:</i> collection
capacitance_max	Returns the total capacitance of the net for maximum conditions. <i>Return type:</i> float
coupling_capacitance_max	Returns the maximum value of coupling capacitance of a net over all the views. <i>Return type:</i> float
capacitance_min	Returns the total capacitance of the net for minimum conditions. <i>Return type:</i> float
coupling_capacitance_min	Returns the minimum value of coupling capacitance of a net over all the views. <i>Return type:</i> float
delta_delay_max	Returns the delta delay for the maximum condition. <i>Return type:</i> float
delta_delay_max_fall	Returns the maximum falling delta delay value. <i>Return type:</i> float

<code>delta_delay_max_rise</code>	Returns the maximum rising delta delay value. <i>Return type:</i> float
<code>delta_delay_min</code>	Returns the delta delay for the minimum condition. <i>Return type:</i> float
<code>delta_delay_min_fall</code>	Returns the minimum falling delta delay value. <i>Return type:</i> float
<code>delta_delay_min_rise</code>	Returns the minimum rising delta delay value. <i>Return type:</i> float
<code>driver_pins</code>	Returns a collection of the driver pins of the net. <i>Return type:</i> collection
<code>early_fall_clk_net_derate_factor</code>	Returns the early fall clock net derating factor specified through <code>set_timing_derate</code> command. <i>Return type:</i> float
<code>early_fall_clk_net_delta_derate_factor</code>	Returns the early fall clock delta delay net derating factor specified through the <code>set_timing_derate</code> command. <i>Return type:</i> float
<code>early_fall_data_net_delta_derate_factor</code>	Returns the early fall data delta net derating factor specified through the <code>set_timing_derate</code> command. <i>Return type:</i> float
<code>early_fall_data_net_derate_factor</code>	Returns the early fall data net derating factor specified through <code>set_timing_derate</code> command. <i>Return type:</i> float
<code>early_rise_clk_net_delta_derate_factor</code>	Returns the early rise clock delta net derating factor specified through the <code>set_timing_derate</code> command. <i>Return type:</i> float
<code>early_rise_clk_net_derate_factor</code>	Returns the early rise clock net derating factor specified through the <code>set_timing_derate</code> command. <i>Return type:</i> float

<code>early_rise_data_net_delta_derate_factor</code>	Returns the early rise data delta net derating factor specified through the <code>set_timing_derate</code> command. <i>Return type:</i> float
<code>early_rise_data_net_derate_factor</code>	Returns the early rise data net derating factor specified through the <code>set_timing_derate</code> command. <i>Return type:</i> float
<code>has_detailed_parasitics</code>	Returns a value of <code>true</code> if the net, or one of its parts, has detailed parasitics associated with it from either RC extraction or SPEF annotation. <i>Return type:</i> boolean
<code>hierarchical_name</code>	Returns the complete hierarchical name of the net. <i>Return type:</i> string
<code>is_clock_source_latency</code>	Returns a value of <code>true</code> if the net is a part of clock source latency network. The clock source latency network consists of the clock network cells, its pins, and the nets connecting the master clock source pin and the generated clock source pin. For example, <code>get_property [get_nets A] is_clock_source_latency</code> <i>Return type:</i> boolean
<code>is_dont_touch</code>	Returns a value of <code>true</code> if a <code>set_dont_touch</code> constraint exists for the net. <i>Return type:</i> boolean
<code>is_hierarchical</code>	Returns a value of <code>true</code> if the net is a part of hierarchical module boundary. For example, <code>get_property [get_nets W1] is_hierarchical</code> <i>Return type:</i> boolean
<code>late_fall_clk_net_delta_derate_factor</code>	Returns the late fall clock delta net derating factor specified through the <code>set_timing_derate</code> command. <i>Return type:</i> float

<code>late_fall_clk_net_derate_factor</code>	Returns the late fall clock net derating factor specified through the <code>set_timing_derate</code> command. <i>Return type:</i> float
<code>late_fall_data_net_delta_derate_factor</code>	Returns the late fall data delta net derating factor specified through the <code>set_timing_derate</code> command. <i>Return type:</i> float
<code>late_fall_data_net_derate_factor</code>	Returns the late fall data net derating factor specified through the <code>set_timing_derate</code> command. <i>Return type:</i> float
<code>late_rise_clk_net_delta_derate_factor</code>	Returns the late rise clock delta net derating factor specified through the <code>set_timing_derate</code> command. <i>Return type:</i> float
<code>late_rise_clk_net_derate_factor</code>	Returns the late rise clock net derating factor specified through the <code>set_timing_derate</code> command. <i>Return type:</i> float
<code>late_rise_data_net_delta_derate_factor</code>	Returns the late rise data delta net derating factor specified through the <code>set_timing_derate</code> command. <i>Return type:</i> float
<code>late_rise_data_net_derate_factor</code>	Returns the late rise data net derating factor specified through the <code>set_timing_derate</code> command. <i>Return type:</i> float
<code>load_pins</code>	Returns a collection of the load pins of the net. <i>Return type:</i> collection
<code>lumped_capacitance_max</code>	Returns the maximum value of capacitance of a net based on the input library and parasitic model. <i>Return type:</i> float
<code>lumped_capacitance_min</code>	Returns the minimum value of capacitance of a net based on the input library and parasitic model. <i>Return type:</i> float

lumped_resistance_max	Returns the maximum resistance of the net for late path analysis. You can use <code>-index</code> to return the value for a specific view. For example, <pre>get_property [get_nets FE_SIG_N671_FE_OFCN486_ds_data_6_] lumped_resistance_max</pre> <i>Return type:</i> float
lumped_resistance_min	Returns the minimum resistance of the net for late path analysis. You can use <code>-index</code> to return the value for a specific view. For example, <pre>get_property [get_nets FE_SIG_N671_FE_OFCN486_ds_data_6_] lumped_resistance_min</pre> <i>Return type:</i> float
name	Returns the name of the net. <i>Return type:</i> string
num_driver_pins	Returns the number of driver pins on a net. <i>Return type:</i> integer
num_load_pins	Returns the number of load pins on the net. <i>Return type:</i> integer
object_type	Returns the object type <code>net</code>. <i>Return type:</i> string
pin_capacitance_max	Returns the sum of all pin capacitances of the net for maximum conditions. <i>Return type:</i> float
pin_capacitance_max_fall	Returns the sum of all pin capacitances of the net for the maximum value of the falling capacitance range. <i>Return type:</i> float

<code>pin_capacitance_max_rise</code>	Returns the sum of all pin capacitances of the net for the maximum value of the rising capacitance range. <i>Return type:</i> float
<code>pin_capacitance_min</code>	Returns the sum of all pin capacitances of the net for minimum conditions. <i>Return type:</i> float
Note: For all the <code>pin_capacitance_*</code> properties (described above), the pin capacitance matches with the capacitance displayed under the "pin_load" column of the <code>report_timing</code> output.	
<code>pin_capacitance_min_fall</code>	Returns the minimum fall capacitance present at the pin of a net. For example, <pre>get_property [get_nets UMC/w3] pin_capacitance_min_fall</pre> <i>Return type:</i> float
<code>pin_capacitance_min_rise</code>	Returns the minimum rise capacitance present at the pin of a net. For example, <pre>get_property [get_nets UMC/w3] pin_capacitance_min_rise</pre> <i>Return type:</i> float
<code>resistance_max</code>	Returns the maximum value of resistance of a net. <i>Return type:</i> float
<code>resistance_min</code>	Returns the minimum value of resistance of a net. <i>Return type:</i> float
<code>timing_constant</code>	Reports a constant value on the net. For example, <pre>get_property [get_nets UMC/constant_net] timing_constant</pre> <i>Return type:</i> string

timing_constant_source	Reports the source through which the timing constant is applied. For example, <pre>get_property [get_nets UMC/constant_net] timing_constant_source</pre> <i>Return type:</i> string
total_capacitance_max_fall	Returns the total falling capacitance of the net used for late path analysis. For example, <pre>get_property [get_nets FE_SIG_N643_ds_data_7_] total_capacitance_max_fall -view func_slow_RCMAX</pre> <i>Return type:</i> float
total_capacitance_max_rise	Returns the total rising capacitance of the net used for late path analysis. For example, <pre>get_property [get_nets FE_SIG_N643_ds_data_7_] total_capacitance_max_rise -view func_slow_RCMAX</pre> <i>Return type:</i> float
total_capacitance_min_fall	Returns the total falling capacitance of the net used for early path analysis. For example, <pre>get_property [get_nets FE_SIG_N643_ds_data_7_] total_capacitance_min_fall -view func_fast_RCMIN</pre> <i>Return type:</i> float
total_capacitance_min_rise	Returns the total rising capacitance of the net used for early path analysis. For example, <pre>get_property [get_nets FE_SIG_N643_ds_data_7_] total_capacitance_min_rise -view func_fast_RCMIN</pre> <i>Return type:</i> float
total_lumped_capacitance_max_fall	Returns the maximum value of falling capacitance of a net based on the input library and parasitic model. <i>Return type:</i> float

<code>total_lumped_capacitance_max_rise</code>	Returns the maximum value of rising capacitance of a net based on the input library and parasitic model. <i>Return type:</i> float
<code>total_lumped_capacitance_min_fall</code>	Returns the minimum value of falling capacitance of a net based on the input library and parasitic model. <i>Return type:</i> float
<code>total_lumped_capacitance_min_rise</code>	Returns the minimum value of rising capacitance of a net based on the input library and parasitic model. <i>Return type:</i> float
<code>vl_vh_analysis_status</code>	Reports the glitch analysis (VL/VH) status for a net. For example, <pre>get_property [get_nets -of [get_cells FE_SIG_C622_FE_OFCN517_digit_2_]] vl_vh_analysis_status</pre> <i>Return type:</i> string
<code>vlu_vho_analysis_status</code>	Reports the glitch analysis status for undershoot (VLU) and overshoot (VHO) glitch for a net. For example, <pre>get_property [get_nets FE_SIG_N642_ds_data_5_] vlu_vho_analysis_status</pre> <i>Return type:</i> string
<code>wire_capacitance_max</code>	Returns the wire capacitance of the net for maximum conditions. <i>Return type:</i> float
<code>wire_capacitance_max_fall</code>	Returns the wire capacitance of the net for the maximum value of the falling capacitance range. For example, <pre>get_property [get_nets FE_SIG_N643_ds_data_7_] wire_capacitance_max_fall -view func_slow_RCMAX</pre> <i>Return type:</i> float

wire_capacitance_max_rise	Returns the wire capacitance of the net for the maximum value of the rising capacitance range. For example, <pre>get_property [get_nets FE_SIG_N643_ds_data_7_] wire_capacitance_max_rise -view func_slow_RCMAX</pre> <i>Return type:</i> float
wire_capacitance_min	Returns the wire capacitance of the net for minimum conditions. <i>Return type:</i> float
wire_capacitance_min_fall	Returns the wire capacitance of the net for the minimum value of the falling capacitance range. For example, <pre>get_property [get_nets FE_SIG_N643_ds_data_7_] wire_capacitance_min_fall -view func_fast_RCMIN</pre> <i>Return type:</i> float
wire_capacitance_min_rise	Returns the wire capacitance of the net for the minimum value of the rising capacitance range. For example, <pre>get_property [get_nets FE_SIG_N643_ds_data_7_] wire_capacitance_min_rise -view func_fast_RCMIN</pre> <i>Return type:</i> float
wire_lumped_capacitance_max	Returns the portion of the net's total lumped capacitance which comes extracted or annotated wire lumped capacitance values. For example, <pre>get_property [get_nets FE_SIG_N643_ds_data_7_] wire_lumped_capacitance_max -view func_slow_RCMAX</pre> <i>Return type:</i> float
wire_lumped_capacitance_max_fall	Returns the lumped wire capacitance of the net for the maximum value of the falling capacitance range. For example, <pre>get_property [get_nets FE_SIG_N643_ds_data_7_] wire_lumped_capacitance_max_fall -view func_slow_RCMAX</pre> <i>Return type:</i> float

wire_lumped_capacitance_max_rise	Returns the lumped wire capacitance of the net for the maximum value of the rising capacitance range. For example, <pre>get_property [get_nets FE_SIG_N643_ds_data_7_] wire_lumped_capacitance_max_rise -view func_slow_RCMAX</pre> <i>Return type:</i> float
wire_lumped_capacitance_min	Returns the portion of the net's total lumped capacitance which comes extracted or annotated wire lumped capacitance values. For example, <pre>get_property [get_nets FE_SIG_N643_ds_data_7_] wire_lumped_capacitance_min -view func_fast_RCMIN</pre> <i>Return type:</i> float
wire_lumped_capacitance_min_fall	Returns the lumped wire capacitance of the net for the minimum value of the falling capacitance range. For example, <pre>get_property [get_nets FE_SIG_N643_ds_data_7_] wire_lumped_capacitance_min_fall -view func_fast_RCMIN</pre> <i>Return type:</i> float
wire_lumped_capacitance_min_rise	Returns the lumped wire capacitance of the net for the minimum value of the rising capacitance range. For example, <pre>get_property [get_nets FE_SIG_N643_ds_data_7_] wire_lumped_capacitance_min_rise -view func_fast_RCMIN</pre> <i>Return type:</i> float
x_coordinate_max	Returns the maximum value of the x coordinate of the net. For example, <pre>get_property [get_nets FE_SIG_N671_FE_OF486_ds_data_6_] x_coordinate_max</pre> <i>Return type:</i> float

<code>x_coordinate_min</code>	Returns the minimum value of the <code>x</code> coordinate of the net. For example, <pre>get_property [get_nets FE_SIG_N671_FE_OFCN486_ds_data_6_] x_coordinate_min</pre> <i>Return type:</i> float
<code>y_coordinate_max</code>	Returns the maximum value of the <code>y</code> coordinate of the net. For example, <pre>get_property [get_nets FE_SIG_N671_FE_OFCN486_ds_data_6_] y_coordinate_max</pre> <i>Return type:</i> float
<code>y_coordinate_min</code>	Returns the minimum value of the <code>y</code> coordinate of the net. For example, <pre>get_property [get_nets FE_SIG_N671_FE_OFCN486_ds_data_6_] y_coordinate_min</pre> <i>Return type:</i> float

PG Net Properties

<code>asserted_voltage_max</code>	Returns the maximum asserted voltage on the power ground net. <i>Return type:</i> string
<code>asserted_voltage_min</code>	Returns the minimum asserted voltage on the power ground net. <i>Return type:</i> string
<code>hierarchical_name</code>	Returns the name of the PG net connected to the corresponding PG pin. <i>Return type:</i> string For example, <pre>get_property [get_pg_nets -of_objects [get_cells iA/out_reg]] hierarchical_name get_property [get_pg_nets -of_objects [get_cells iA/out_reg] -filter "is_power == true"] hierarchical_name</pre>

is_ground	Returns a value of <code>true</code> if the PG net is a ground net. <i>Return type:</i> boolean For example, <pre>get_property [get_pg_nets -of_objects [get_cells iA/out_reg]] is_ground get_property [get_pg_nets -of_objects [get_pg_pins -of_objects [get_cells iA/out_reg]]] is_ground</pre>
is_power	Returns a value of <code>true</code> if the PG net is a power net. <i>Return type:</i> boolean For example, <pre>get_property [get_pg_nets -of_objects [get_cells iA/out_reg]] is_power get_property [get_pg_nets -of_objects [get_cells iA/out_reg] -filter "name =~ *VDD*"] is_power</pre>
is_primary	Returns a value of <code>true</code> if the PG net is primary. <pre>get_property [get_pg_nets VDD1] is_primary</pre> <i>Return type:</i> boolean
object_type	Returns <code>pg_net</code> as the object type for a library PG pin. <i>Return type:</i> string For example, <pre>get_property [get_pg_nets -of_objects [get_cells iA/out_reg]] object_type</pre>
pg_net_group	Returns the name of the PG net group for the <code>pg_net</code> object. <pre>get_property [get_pg_nets VDDR] pg_net_group</pre> <i>Return type:</i> string
parent_pg_net	Returns a collection of a parent power net that is the net powering the power switch that drives the PG net. In case of designs that do not have any power switches powering the instances, this property will always return the net directly powering those instances. It therefore be noted that the usage of this property is more relevant for designs in which the leaf cell instances are powered through power switch cells. <i>Return type:</i> Collection For example, <pre>get_property [get_pg_nets mmu/U1/vdd] parent_pg_net get_property [get_pg_nets -of_objects [get_cells iA/pwr_swch]] parent_pg_net</pre>

top_pg_net	Returns a collection of the top most un-switched power net of the current PG net. Even in the case of a cascading of power switches present in a design, the software traces through all power switches back to the un-switched top-level primary power net. The top-most un-switched power net will be returned - switched output power net will not be returned. <i>Return type:</i> Collection For example, <pre>get_property [get_pg_nets -of_objects [get_cells iA/pwr_swch]] top_pg_net get_property [get_pg_nets -of_objects [get_pg_pins -of_objects [get_cells iA/pwr_swch]]] top_pg_net</pre>
voltage_max	Returns the maximum voltage of a given PG net. This could be voltage associated with a driver term of a net, power intent or CPF, <code>related_pg_pins</code> attribute in Liberty, or the operating voltage of a power domain or system default voltage. <i>Return type:</i> float For example, <pre>get_property [get_pg_nets -of_objects [get_cells iA/out_reg]] voltage_max</pre>
voltage_min	Returns the minimum voltage of a given PG net. This could be voltage associated with a driver term of a net, power intent or CPF, <code>related_pg_pins</code> attribute in Liberty or the operating voltage of a power domain or system default voltage. <i>Return type:</i> float For example, <pre>get_property [get_pg_nets -of_objects [get_cells iA/out_reg]] voltage_min</pre>

Path Group Properties

name	Returns the name of the path group. <i>Return type:</i> string
hierarchical_name	Returns the hierarchical name of the path group. <i>Return type:</i> string
group_type	Returns the type of group, when <code>group_path</code> and <code>-name</code> , <code>-from</code> , <code>-to</code> parameters are specified for the path. For example, <code>get_property [get_path_groups] group_type</code> <i>Return Type:</i> string
object_type	Returns the object type <code>path_group</code> . <i>Return type:</i> string
view_name	Returns the view name for the specified internal clock group. For example, <code>get_property [get_path_groups m_clk -include_internal_groups] view_name</code> <i>Return type:</i> string

Pin Properties

actual_latency_early_fall_max	When specified with the <code>set_clock_latency -clock</code> parameter, worst actual early fall insertion delay in setup mode for the specified clock at the given pin. The latency reported is ideal or propagated, depending on the clock phase. When specified without the <code>-clock</code> parameter, returns the worst early fall insertion delay on the pin. If multiple clocks reach the pin, the reported latency is the worst among all the clocks. When specified with the <code>-view</code> parameter, returns the worst latency for the given view. <i>Return type:</i> float
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actual_latency_early_fall_min	When specified with the <code>set_clock_latency -clock</code> parameter the worst actual early fall insertion delay in hold mode for the specified clock at the given pin. The latency reported is ideal or propagated depending on the clock phase. When specified without the <code>-clock</code> parameter, returns the worst early fall insertion delay on the pin. If multiple clocks reach the reported latency is the worst among all the clocks. When specified with the <code>-view</code> parameter, returns the worst latency given view. <i>Return type:</i> float
actual_latency_early_rise_max	When specified with the <code>set_clock_latency -clock</code> parameter the worst actual early rise source insertion delay in setup mode for the specified clock at the given pin. The latency reported is ideal or propagated, depending on the clock phase. When specified without the <code>-clock</code> parameter, returns the worst early rise source insertion delay on the pin. If multiple clocks reach the reported latency is the worst among all the clocks. When specified with the <code>-view</code> parameter, returns the worst latency given view. <i>Return type:</i> float
actual_latency_early_rise_min	When specified with the <code>set_clock_latency -clock</code> parameter the worst actual early rise source insertion delay in hold mode for the specified clock at the given pin. The latency reported is ideal or propagated, depending on the clock phase. When specified without the <code>-clock</code> parameter, returns the worst early rise source insertion delay on the pin. If multiple clocks reach the reported latency is the worst among all the clocks. When specified with the <code>-view</code> parameter, returns the worst latency given view. <i>Return type:</i> float

actual_latency_late_fall_max	When specified with the <code>set_clock_latency -clock</code> parameter the worst actual late fall source insertion delay in setup mode for the specified clock at the given pin. The latency reported is ideal or propagated, depending on the clock phase. When specified without the <code>-clock</code> parameter, returns the worst late fall source insertion delay on the pin. If multiple clocks are specified the reported latency is the worst among all the clocks. When specified with the <code>-view</code> parameter, returns the worst late fall source insertion delay for the given view. <i>Return type:</i> float
actual_latency_late_fall_min	When specified with the <code>set_clock_latency -clock</code> parameter, returns the worst actual late fall source insertion delay in hold mode for the specified clock at the given pin. The latency reported is ideal or propagated, depending on the clock phase. When specified without the <code>-clock</code> parameter, returns the worst late fall source insertion delay on the pin. If multiple clocks are specified the reported latency is the worst among all the clocks. When specified with the <code>-view</code> parameter, returns the worst late fall source insertion delay for the given view. <i>Return type:</i> float
actual_latency_late_rise_max	When specified with the <code>set_clock_latency -clock</code> parameter the worst actual late rise source insertion delay in setup mode for the specified clock at the given pin. The latency reported is ideal or propagated, depending on the clock phase. When specified without the <code>-clock</code> parameter, returns the worst late rise source insertion delay on the pin. If multiple clocks are specified the reported latency is the worst among all the clocks. When specified with the <code>-view</code> parameter, returns the worst late rise source insertion delay for the given view. <i>Return type:</i> float

actual_latency_late_rise_min	When specified with the <code>set_clock_latency -clock</code> parameter the worst actual late rise source insertion delay in hold mode for the specified clock at the given pin. The latency reported is ideal or propagated, depending on the clock phase. When specified without the <code>-clock</code> parameter, returns the worst late rise source insertion delay on the pin. If multiple clocks are specified, the reported latency is the worst among all the clocks. When specified with the <code>-view</code> parameter, returns the worst late rise source insertion delay for the given view. <i>Return type:</i> float
arrival_clocks	Returns the data phases, clock source data phases, and latency phases. <i>Return type:</i> collection The following global variables support these phases: <code>timing_property_arrival_clocks_consider_clock_source_data_phase</code> Considers the clock source data phases. <code>timing_property_arrival_clocks_consider_data_phase</code> Considers the pure data phase. <code>timing_property_arrival_clocks_consider_latency_phase</code> Considers the latency phase. By default, these global variables are set to <code>true</code>.
arrival_max	Returns the value of the arrival max at specified pin. You can view the arrival max through <code>report_timing</code>. For example, <pre>get_property [get_pins <pin name>] arrival_max</pre> <i>Return Type:</i> float
arrival_max_fall	Returns the arrival time for the falling transition at the max corner. <i>Return type:</i> float
arrival_max_rise	Returns the arrival time for the rising transition at the max corner. <i>Return type:</i> float
arrival_mean_max_fall	Returns the mean component of the arrival time for the falling transition at the max corner. <i>Return type:</i> float

arrival_mean_max_rise	Returns the mean component of the arrival time for the rising to the max corner. <i>Return type:</i> float
arrival_mean_min_fall	Returns the mean component of the arrival time for the falling to the min corner. <i>Return type:</i> float
arrival_mean_min_rise	Returns the mean component of the arrival time for the rising to the min corner. <i>Return type:</i> float
arrival_min	Returns the value of the arrival min at specified pin. You can view through <code>report_timing</code>. For example, <pre>get_property [get_pins <pin name>] arrival_min</pre> <i>Return Type:</i> Float
arrival_sigma_max_rise	Returns the mean component of the arrival time for the rising to the max corner. <i>Return type:</i> float
arrival_sigma_min_fall	Returns the sigma component of the arrival time for the falling to the min corner. <i>Return type:</i> float
arrival_sigma_min_rise	Returns the sigma component of the arrival time for the rising to the min corner. <i>Return type:</i> float
arrival_skewness_max_fall	In SOCV analysis mode, this returns the skewness value of the falling arrival time to the specified pin across all the concurrent views. <pre>get_property [get_pins cell11/ZN] arrival_skewness_max_fa</pre> <i>Return type:</i> float

arrival_skewness_max_rise	In SOCV analysis mode, this returns the skewness value of the rising arrival time to the specified pin across all the concurrent views. <pre>get_property [get_pins cell1/ZN] arrival_skewness_max_rise</pre> <i>Return type:</i> float
arrival_skewness_min_fall	In SOCV analysis mode, this returns the skewness value of the falling arrival time to the specified pin across all the concurrent views. <pre>get_property [get_pins cell1/ZN] arrival_skewness_min_fa</pre> <i>Return type:</i> float
arrival_skewness_min_rise	In SOCV analysis mode, this returns the skewness value of the rising arrival time to the specified pin across all the concurrent views. <pre>get_property [get_pins cell1/ZN] arrival_skewness_min_ri</pre> <i>Return type:</i> float
arrival_window	Returns the arrival times for the earliest and latest rising and falling transitions in timing window format. <i>Return type:</i> string
attacker_fall_slew	Returns the minimum falling slew of an attacker driver pin. For example: <pre>get_property [get_pins inst1/y] attacker_fall_slew</pre> <i>Return type:</i> float
attacker_rise_slew	Returns the minimum rising slew of an attacker driver pin. For example: <pre>get_property [get_pins inst1/y] attacker_rise_slew</pre> <i>Return type:</i> float
capacitance_max	Returns the value of the maximum pin capacitance specified for the capacitance attribute in the library for the corresponding pin. For example, <pre>get_property [get_pins <pin_name>] capacitance_max</pre> <i>Return Type:</i> float

capacitance_max_fall	Returns the maximum value of the falling capacitance range of corresponding library pin. <i>Return type:</i> float
capacitance_max_rise	Returns the maximum value of the rising capacitance range of corresponding library pin. <i>Return type:</i> float
capacitance_min	Returns the value of the minimum pin capacitance specified for capacitance attribute in the library for the corresponding pin. For example, <pre>get_property [get_pins <pin_name>] capacitance_min</pre> <i>Return Type:</i> float
capacitance_min_fall	Returns the minimum value of the falling capacitance range of corresponding library pin. <i>Return type:</i> float
capacitance_min_rise	Returns the minimum value of the rising capacitance range of corresponding library pin. Note: For the <code>capacitance_max_*</code> and <code>capacitance_min_*</code> properties (described above), the capacitance value does not match with the <code>pin_load</code> (<code>report_timing -format</code>) in the timing report because pin load is a sum of all pin capacitances of a net. <i>Return type:</i> float
capture_clocks	Returns the capture clock of the pin. <pre>get_property [get_pins UC/ret_cell/Q] capture_clocks</pre> <i>Return type:</i> collection
cell_name	Returns the cell name for the corresponding pins. For example, <pre>get_property [get_pins addr4_4/A] cell_name</pre> <i>Return type:</i> string

clock_pulse_width_high	Returns the actual high pulse width for the specified pin. <pre>create_clock -name CK -period 30 -waveform {0 10} [get_p</pre> You can verify through <code>report_min_pulse_width -clock <clock</code> For example, <pre>get_property [get_pins <pin_name>] clock_pulse_width_high</pre> <i>Return Type:</i> float
clock_pulse_width_low	Returns the actual low pulse width for the specified pin. <pre>create_clock -name CK -period 30 -waveform {0 10} [get_p</pre> You can verify through <code>report_min_pulse_width -clock <clock</code> For example, <pre>get_property [get_pins <pin_name>] clock_pulse_width_low</pre> <i>Return Type:</i> float
clock_sources	Returns a collection of the source pins of all the clocks arriving <i>Return type:</i> collection
clocks	Returns a collection of all of the clocks arriving at the pin. This property will report clocks on hierarchical pins even before timing update is performed (with the <code>update_timing -full compile</code> <i>Return type:</i> collection
constrained_capture_clocks	Returns the constrained capture clocks at the specified pin. <pre>get_property [get_pins UC/ret_cell/Q] constrained_capture</pre> <i>Return type:</i> collection
constant_value	Returns a value of 0 or 1 if a logic state has been asserted or pulled to the pin. <pre>get_property [get_pins RM_128x16_INST/oe] constant_value</pre> <i>Return type:</i> string
delay_max_fall	Returns the maximum falling delay - specified using the <code>set_max_fall</code> command. <i>Return type:</i> float

delay_max_rise	Returns the maximum rising delay - specified using the <code>set_max_rise</code> command. <i>Return type:</i> float
delay_min_fall	Returns the minimum falling delay - specified using the <code>set_min_fall</code> command. <i>Return type:</i> float
delay_min_rise	Returns the minimum rising delay - specified using the <code>set_min_rise</code> command. <i>Return type:</i> float
direction	Returns the direction of the pin: in, out, or inout. <i>Return type:</i> string
driver_cell	Returns the driver cell name for the corresponding pins. For example, <pre>get_property [get_pins ON28_n_17/A] driver_cell</pre> <i>Return type:</i> string
driver_cell_name	Returns the driving cell instance for the specified pin. For example, <pre>get_property [get_pins <pin name>] driver_cell_name</pre> <i>Return Type:</i> string
driver_power_pin_voltage_max	Reports the maximum power pin voltage of a driver cell. <pre>get_property [get_pins UC/mcr_cell/a] driver_power_pin_v</pre> <i>Return type:</i> float
driver_power_pin_voltage_min	Reports the minimum power pin voltage of a driver cell. <pre>get_property [get_pins UC/mcr_cell/a] driver_power_pin_voltage_min</pre> <i>Return type:</i> float
driver_ref_lib_cell_name	Returns the library defined driving cell for the specified pin. For example, <pre>get_property [get_pins <pin name>] driver_ref_lib_cell_n</pre> <i>Return Type:</i> string

escaped_name	Returns the full hierarchical name, including escaped characters. <i>Return Type:</i> string
fanin	Returns the number of sources connected with net of the specified pin. <i>Return type:</i> integer
fanout	Returns the number of sinks connected with the net of the specified pin. <i>Return type:</i> integer
fanout_load	Returns the fanout load for the pin. <i>Return type:</i> float
fastest_clock	Returns the collection of the clock name defined for the specified pin. create_generated_clock -name ck -source CK -divide_by 2 For example, get_property [get_pins <pin_name>] fastest_clock <i>Return Type:</i> collection
fastest_clock_name	Returns the clock name defined for the specified pin. create_generated_clock -name clk -source CLK -multiply_by 3 name> For example, get_property [get_pins <pin name>] fastest_clock_name <i>Return Type:</i> string
fastest_clock_period	Returns the clock period of the clock defined for the specified pin. create_clock -name CLK -period 10 -waveform {0 5} [get_pins {CK2}] create_generated_clock -name clk -source CLK -multiply_by 3 <pin name> You can verify through report_clocks. For example, get_property [get_pins <pin_name>] fastest_clock_period <i>Return Type:</i> float

hierarchical_level	Returns the hierarchical level of the pin. For example, <pre>get_property [get_pins <pin name>] hierarchical_level</pre> <i>Return Type:</i> integer
hierarchical_name	Returns the complete hierarchical name of the pin. <i>Return type:</i> string
holding_resistance_high	Returns the highest holding resistance of a pin. For example, <pre>foreach pin \$pinList { set vicArray [get_property [get_pins \$pin] noise_victim] foreach_in_collection vic \$vicArray { get_property \$vic holding_resistance_high } }</pre> <i>Return type:</i> float
holding_resistance_low	Returns the lowest holding resistance of a pin. <i>Return type:</i> float
interp_info_max	Returns information for late analysis. When specified, the report output will show the data for interpolation, and/or matched information for late analysis, also library details. <i>Return type:</i> string
interp_info_min	Returns information for early analysis. When specified, the report output will show the data for interpolation, and/or matched information for early analysis, also the library details. <i>Return type:</i> string
is_always_on	Returns a value of <code>true</code> if an instance pin is an always-on pin. <i>Return type:</i> boolean For example, <pre>get_property [get_pins mmu/U1/AON] is_always_on</pre>
is_async	Returns a value of <code>true</code> if the pin is an asynchronous clear or pin. <i>Return type:</i> boolean

is_clear	Returns a value of <code>true</code> if the pin is an asynchronous clear pin. <i>Return type:</i> boolean
is_clock	Returns a value of <code>true</code> if the pin has the Liberty pin attribute: <code>CLOCK</code> . <i>Return type:</i> boolean
is_clock_root	Returns <code>true</code> for a pin or port if real clock or generated clock is the pin/port. <code>create_generated_clock -name clk -source CLK -multiply_by <name></code> For example, <code>get_property [get_pins <pin_name>] is_clock_root</code> <i>Return Type:</i> boolean
is_clock_gating	Returns a value of <code>true</code> if the pin is defined as a pin of a clock gating cell. <i>Return type:</i> boolean
is_clock_gating_clock	Returns a value of <code>true</code> if the pin is defined as a clock pin of a clock gating cell. <i>Return type:</i> boolean
is_clock_gating_enable	Returns a value of <code>true</code> if the pin corresponds to the enable pin of a clock gating cell. <i>Return type:</i> boolean
is_clock_root	Returns a value of <code>true</code> if a real or generated clock is defined at the pin and/or port. <i>Return type:</i> boolean
is_clock_source_latency	Returns a value of <code>true</code> if a pin, port or any of the leaf pins of a clock source latency network. <i>Return Type:</i> Boolean
is_clock_source_pin	Returns a value of <code>true</code> for a valid instance pin or port object, if declared with that clock source at pin/port. <i>Return Type:</i> Boolean

<code>is_clock_used_as_clock</code>	Returns a value of <code>true</code> if the pin lies in the clock network and if one of the clocks arriving at the pin is used as a clock in the down network of the pin. When the <code>-clock</code> parameter is specified, this property returns <code>true</code> if the above condition is satisfied for the specified clock. <i>Return type:</i> boolean
<code>is_clock_used_as_data</code>	Returns a value of <code>true</code> if the pin lies in the clock source path and if one of the clocks arriving on the pin is used as data in the down network of the pin. When the <code>-clock</code> parameter is specified, this property returns <code>true</code> if the above condition is satisfied for the specified clock. <i>Return type:</i> boolean
<code>is_constrained</code>	Returns a value of <code>true</code> if the specified pin is constrained. <code>get_property [get_pins UM/A] is_constrained</code> <i>Return type:</i> boolean
<code>is_data</code>	Returns a value of <code>true</code> if the pin is a data pin of a flip-flop (that is, not a clock pin). <i>Return type:</i> boolean
<code>is_disable_timing</code>	Returns a value of <code>true</code> if the pin's timing has been disabled. <i>Return type:</i> boolean Note: This property checks if the associated arc for the pin is a propagation and analysis. For accurate results, re-run the <code>update_timing</code> command.
<code>is_fall_edge_triggered_clock</code>	Returns a value of <code>true</code> if the pin is a clock pin of a flop, and is triggered by the falling edge of a clock. <i>Return type:</i> boolean
<code>is_fall_edge_triggered_data</code>	Returns a value of <code>true</code> if the pin corresponds to the data pin of a flop edge triggered device. <code>get_property [get_pins FFN/D] is_fall_edge_triggered_data</code> <i>Return value:</i> boolean
<code>is_hierarchical</code>	Returns a value of <code>true</code> if the pin is a pin of a hierarchical model. <i>Return type:</i> boolean

<code>is_isolation_cell_clock</code>	Returns <code>true</code> if the instance signal pin corresponding to the <code>isolation_cell_clock_pin</code> attribute of a library signal pin is set indicating that the library cell signal pin is a data pin of an isolation cell. Note : The <code>is_isolation_cell</code> attribute must be set to <code>true</code> in the library cell. <i>Return Type:</i> boolean
<code>is_isolation_cell_data</code>	Returns <code>true</code> if the instance signal pin corresponding to the <code>isolation_cell_data_pin</code> attribute of a library signal pin is set indicating that the library cell signal pin is a data pin of an isolation cell. Note : The <code>is_isolation_cell</code> attribute must be set to <code>true</code> in the library cell. For example, <pre>get_property [get_pins <pin_name>] is_isolation_cell_data</pre> <i>Return Type:</i> boolean
<code>is_isolation_cell_enable</code>	Returns <code>true</code> if the instance signal pin corresponding to the <code>isolation_cell_enable_pin</code> attribute of a library signal pin is set indicating that the library cell signal pin is an enable or control isolation library cell. Note : The <code>is_isolation_cell</code> attribute must be set to <code>true</code> in the library cell. For example, <pre>get_property [get_pins <pin_name>] is_isolation_cell_enable</pre> <i>Return Type:</i> boolean
<code>is_latency_network_pin</code>	Returns <code>true</code> if the specified pin is having generated or created by a clock. <code>create_generated_clock -name ck -source CK -divide_by 2</code> For example, <pre>get_property [get_pins <pin name>] is_latency_network_pin</pre> <i>Return Type:</i> boolean

is_level_shifter_data	Returns <code>true</code> if the instance signal pin corresponding to the <code>level_shifter_data_pin</code> attribute of a library signal pin is set to <code>true</code>, indicating that the library cell signal pin is a data pin of a library cell. Note: The <code>is_level_shifter</code> attribute must be set to <code>true</code> in the library cell. For example, <pre>get_property [get_pins <pin_name>] is_level_shifter_data</pre> <i>Return Type:</i> boolean
is_level_shifter_enable	Returns <code>true</code> if the instance signal pin corresponding to the <code>level_shifter_enable_pin</code> attribute of a library signal pin is set to <code>true</code>, indicating that the library cell signal pin is an enable pin of a library cell. <i>Return Type:</i> boolean
input_signal_level	Returns the power rail name associated with the input signal pin. If no power rail has been explicitly specified in the Liberty description of the cell, the <code>input_signal_level</code> attribute. Typically, this attribute can only be defined in overdrive situations where the expected input voltage does not match the <code>related_power_input</code>. In other cases, a value of NA will be returned. <i>Return type:</i> string For example, <pre>get_property [get_pins mmu/U1/I] input_signal_level</pre>
input_signal_level_voltage	If the <code>input_signal_level</code> has an explicit value, then this attribute returns the voltage of that power rail. This is the expected input voltage for the input pin. A value of NA will be returned if <code>input_signal_level</code> is undefined. <i>Return type:</i> float For example, <pre>get_property [get_pins mmu/U1/I] input_signal_level_voltage</pre>
is_multiple_clock_fanin_point	Returns a value of <code>true</code> if multiple clock phases converge at the input pin. <i>Return type:</i> boolean

<code>is_mux_select_pin</code>	Returns <code>true</code> if the specified pin is the mux select pin. For example, <pre>get_property [get_pins m1/S] is_mux_select_pin</pre> <i>Return Type:</i> boolean
<code>is_negative_level_sensitive_clock</code>	Returns a value of <code>true</code> if the library pin is an enable pin of an level-sensitive device. <i>Return type:</i> boolean
<code>is_negative_level_sensitive_data</code>	Returns a value of <code>true</code> if the pin is a data pin of an active low level-sensitive device. <i>Return type:</i> boolean
<code>is_pin</code>	Returns a value of <code>true</code>. The <code>is_pin</code> and <code>is_port</code> queries are provided for both pin and objects to provide a quick way of determining the object type or identifier. <code>is_port</code> always returns <code>false</code> for pins. <code>is_pins</code> always returns <code>false</code> for ports. <i>Return type:</i> boolean
<code>is_port</code>	Returns a value of <code>false</code>. The <code>is_pin</code> and <code>is_port</code> queries are provided for both pin and objects to provide a quick way of determining the object type or identifier. <code>is_port</code> always returns <code>false</code> for pins. <code>is_pins</code> always returns <code>false</code> for ports. <i>Return type:</i> boolean
<code>is_positive_level_sensitive_clock</code>	Returns a value of <code>true</code> if the library pin is an enable pin of an level-sensitive device. <i>Return type:</i> boolean
<code>is_positive_level_sensitive_data</code>	Returns a value of <code>true</code> if the pin is a data pin of an active high sensitive device. <i>Return type:</i> boolean

is_power_mode_disabled	Returns a value of <code>true</code> if power mode of the specified pin is disabled. <pre>get_property [get_pins MD_C/C_IS] is_power_mode_disabled</pre> <pre>view_func_8V_OFF</pre> <i>Return type:</i> boolean
is_preset	Returns a value of <code>true</code> if the pin is a preset pin. <i>Return type:</i> boolean
is_propagated_clock	Returns a value of <code>true</code> if there is an explicit <code>set_propagated_clock</code> assertion at the pin. <i>Return type:</i> boolean
is_ref_clock_pin	Returns a value of <code>true</code> for the following: - Valid instance pin object and on an active clock pin where a signal reaches, and sequential cell is not disabled by disabled arcs or case analysis. Any pin with clock check-arcs or triggered by an active clock phase is present on that pin. - Clock pin on an inferred clock gating cell. Returns a value of <code>false</code>, if clock propagation disabled with the <code>set_disable_timing</code> or <code>set_sense -stop_propagation</code> command. <i>Return Type:</i> Boolean
is_rise_edge_triggered_clock	Returns a value of <code>true</code> if the pin is a clock pin of a flop, and is triggered by the rising edge of a clock. <i>Return type:</i> boolean
is_rise_edge_triggered_data	Returns a value of <code>true</code> if the pin corresponds to the data pin of an edge-triggered device. <i>Return type:</i> boolean
is_tristate	Returns a value of <code>true</code> if the pin has the <code>three_state</code> attribute in the Liberty timing library. <i>Return type:</i> boolean
is_tristate_enable	Returns a value of <code>true</code> if the pin is the source pin of timing arc either the <code>three_state_enable</code> or <code>three_state_disable</code> attribute in the Liberty timing library. <i>Return type:</i> boolean

<code>is_tristate_output</code>	Returns a value of <code>true</code> if the pin corresponds to a three-state output. <i>Return type:</i> boolean
<code>latency_fall_max</code>	Returns the maximum fall insertion delay specified by an explicit <code>set_clock_latency</code> at the pin. <i>Return type:</i> float
<code>latency_fall_min</code>	Returns the minimum fall insertion delay specified by an explicit <code>set_clock_latency</code> at the pin. <i>Return type:</i> float
<code>latency_rise_max</code>	Returns the maximum rise insertion delay specified by an explicit <code>set_clock_latency</code> at the pin. <i>Return type:</i> float
<code>latency_rise_min</code>	Returns the minimum rise insertion delay specified by an explicit <code>set_clock_latency</code> at the pin. <i>Return type:</i> float
<code>lib_pins</code>	Returns the collection of the library pin for the specified pin. For example, <pre>get_property [get_pins <pin name>] lib_pins</pre> <i>Return Type:</i> collection
<code>max_capacitance</code>	Returns the maximum capacitance limit for the pin. <i>Return type:</i> float
<code>max_fanout</code>	Returns the maximum fanout load that the pin can drive. This value is determined by the <code>set_max_fanout</code> or the <code>default_max_fanout</code> library attribute. <i>Return type:</i> float
<code>max_time_borrow</code>	Returns the value of the maximum time borrowed for the specified pin when the <code>set_max_time_borrow</code> parameter is specified. For example, <pre>get_property [get_pins <pin name>] max_time_borrow</pre> <i>Return Type:</i> float

max_time_borrow_io	Returns the maximum time that a path arriving at a latch input from the next stage during latch-based analysis. <pre>get_property [get_pins mro_dc/LP/d] max_time_borrow_io</pre> <i>Return Type:</i> float
max_transition	Returns the maximum transition time limit specified for the pin. <i>Return type:</i> float
min_capacitance	Returns the value of the minimum capacitance for the specified pin. The <code>set_min_capacitance</code> parameter is specified. For example, <pre>get_property [get_pins <pin name>] min_capacitance</pre> <i>Return Type:</i> float
min_fanout	Returns the minimum fanout design rule limit of the corresponding pin. <i>Return type:</i> float
min_transition	Returns the value of the minimum transition for the specified pin. For example, <pre>get_property [get_pins <pin name>] min_transition</pre> <i>Return Type:</i> float
name	Returns the leaf-level name of the pin. For example, <pre>get_property [get_pins <pin name>] name</pre> <i>Return Type:</i> string
netlist_constant_value	Returns a constant logic value of 1'b0 or 1'b1 in the Verilog. <pre>get_property [get_pins UC/flp/pez] netlist_constant_value</pre>
net_name	Returns the name of the net connected to the specified pin. <i>Return type:</i> string
noise_victim	Returns a collection of <code>si_victim</code> objects that can be used to query attributes. <i>Return type:</i> collection

object_type	Returns the object type <code>pin</code> . <i>Return type:</i> string
output_signal_level	Returns the rail name of the <code>related_power_pin</code> attribute of the output/inout signal pin. <i>Return type:</i> string For example, <code>get_property [get_pins mmu/U1/Y] output_signal_level</code>
pin_capacitance_max	Reports the maximum capacitance of a pin. <code>get_property [get_pins UMC/buff/A] pin_capacitance_max</code> <i>Return type:</i> float
pin_capacitance_max_fall	Returns the maximum falling capacitance of a pin. <i>Return type:</i> float
pin_capacitance_max_rise	Returns the maximum rise capacitance of pin. <i>Return type:</i> float
pin_capacitance_min	Returns the minimum capacitance of pin. <i>Return type:</i> float
pin_capacitance_min_fall	Returns the minimum fall capacitance of Pin. <i>Return type:</i> float
pin_capacitance_min_rise	Returns the minimum rise capacitance of Pin. <i>Return type:</i> float
pin_direction	Reports the direction of a pin. <code>get_property [get_pins UC/buf/B] pin_direction</code> <i>Return type:</i> string
power_rail_voltage_bidir_input_max	Reports power rail voltage for input part of bidi pins/ports for m operating condition. <i>Return type:</i> float
power_rail_voltage_bidir_input_min	Reports power rail voltage for input part of bidi pins/ports for m operating condition. <i>Return type:</i> float

power_rail_voltage_max	Reports power rail voltage for maximum operating condition for bidirectional, input and output pins. For bidirectional pins, the voltage for output signal is reported. <i>Return type:</i> float
power_rail_voltage_min	Reports power rail voltage for minimum operating condition for bidirectional, input and output pins. For bidirectional pins, the voltage for output signal is reported. <i>Return type:</i> float
receiver_ccc_type	Returns the type of receiver cell model. The return values can be single_ccc, dual_ccc, multi_ccc, or black_box. This attribute applies to receiver pins only. For example, <pre>get_property [get_pins inst1/a] receiver_ccc_type</pre> <i>Return type:</i> string
ref_lib_pin_name	Returns the name of the library pin associated with the instance. <i>Return type:</i> string
related_ground_pin	Returns the related ground pins as a collection for the corresponding instance pins. <i>Return type:</i> collection For example, <pre>get_property [get_pins mmu/U1/I] related_ground_pin</pre>
related_power_pin	Returns the related power pins as a collection for the corresponding instance pins. <i>Return type:</i> collection For example, <pre>get_property [get_pins mmu/U1/I] related_power_pin</pre>
setup_uncertainty	Returns the most conservative uncertainty of all possible uncertainty assertions associated with the pin. <i>Return type:</i> float
si_slew_max	Returns the maximum slew from all the incoming signals after SI analysis. This value is a maximum of si_slew_max_rise and si_slew_max_fall. <i>Return type:</i> float

si_slew_max_fall	Reports the maximum falling slew considering the incremental signal integrity analysis. <pre>get_property [get_pins UC/buf/A] si_slew_max_fall</pre> <i>Return type:</i> float
si_slew_max_rise	Reports the maximum rising slew considering the incremental signal integrity analysis. <pre>get_property [get_pins UC/buf/A] si_slew_max_rise</pre> <i>Return type:</i> float
si_slew_min_fall	Reports the minimum falling slew considering the incremental signal integrity analysis. <pre>get_property [get_pins UC/buf/A] si_slew_min_fall</pre> <i>Return type:</i> float
si_slew_min_rise	Reports the minimum rising slew considering the incremental signal integrity analysis. <pre>get_property [get_pins UC/buf/A] si_slew_min_rise</pre> <i>Return type:</i> float
si_slew_mean_max_fall	Returns the mean component of maximum falling slew from the signals after SI analysis is performed. <i>Return type:</i> float
si_slew_mean_max_rise	Returns the mean component of maximum rising slew from the signals after SI analysis is performed. <i>Return type:</i> float
si_slew_mean_min_fall	Returns the mean component of minimum falling slew from the signals after SI analysis is performed. <i>Return type:</i> float
si_slew_mean_min_rise	Returns the mean component of minimum rising slew from the signals after SI analysis is performed. <i>Return type:</i> float
si_slew_min	Returns the minimum slew from all the incoming signals after SI analysis. This value is a minimum of <code>si_slew_min_rise</code> and <code>si_slew_min_fall</code>. <i>Return type:</i> float

<code>si_victim</code>	Returns a collection of the <code>si_victim</code> objects that can be used graph-based SI delay attributes. <i>Return type:</i> collection
<code>slack_max</code>	Returns the slack time for a timing path at the max corner. <i>Return type:</i> float
<code>slack_max_edge</code>	Returns the edge (rise or fall) of the worst slack-causing path at a specified pin in late mode. <i>Return type:</i> string
<code>slack_max_fall</code>	Returns the slack time for the falling transition at the max corner. <i>Return type:</i> float
<code>slack_max_rise</code>	Returns the slack time for the rising transition at the max corner. When the <code>-view</code> parameter is not specified, the worst slack for all views is returned. <i>Return type:</i> float
<code>slack_mean_max</code>	Returns the maximum mean component of the slack time for a path at the max corner. <i>Return type:</i> float
<code>slack_mean_min</code>	Returns the minimum mean component of the slack time for a path at the min corner. <i>Return type:</i> float
<code>slack_mean_max_fall</code>	Returns the mean component of the slack time for the falling transition at the max corner. <i>Return type:</i> float
<code>slack_mean_max_rise</code>	Returns the mean component of the slack time for the rising transition at the max corner. <i>Return type:</i> float
<code>slack_mean_min_fall</code>	Returns the mean component of the slack time for the falling transition at the min corner. <i>Return type:</i> float

slack_mean_min_rise	Returns the mean component of the slack time for the rising transition at the min corner. <i>Return type:</i> float
slack_min	Returns the slack time for a timing path at the min corner. <i>Return type:</i> float
slack_min_edge	Returns the edge (rise or fall) of the worst slack-causing path at the specified pin in early mode. <i>Return type:</i> string
slack_min_fall	Returns the slack time for the falling transition at the min corner. <i>Return type:</i> float
slack_min_rise	Returns the slack time for the rising transition at the min corner. <i>Return type:</i> float
slack_sigma_max	Returns the variation component of the worst slack across all timing paths for concurrent MMMC views for Setup-style late data path checks. <code>get_property [get_pins dff1/Q] slack_sigma_max</code> <i>Return type:</i> float
slack_sigma_min	Returns the variation component of the worst slack across all timing paths for concurrent MMMC views for Hold-style early data path checks. <code>get_property [get_pins dff1/Q] slack_sigma_min</code> <i>Return type:</i> float
slack_skewness_max_fall	Returns the skewness component of the worst falling slack across all timing paths for concurrent MMMC views for Setup-style late data path checks. <code>get_property [get_pins df1/Q] slack_skewness_max_fall</code> <i>Return type:</i> float
slack_skewness_max_rise	Returns the skewness component of the worst rising slack across all timing paths for concurrent MMMC views for Setup-style late data path checks. <code>get_property [get_pins df1/Q] slack_skewness_max_rise</code> <i>Return type:</i> float

slack_skewness_min_fall	Returns the skewness component of the worst falling slack across concurrent MMMC views for Hold-style early data path checks <pre>get_property [get_pins df1/Q] slack_skewness_min_fall</pre> <i>Return type:</i> float
slack_skewness_min_rise	Returns the skewness component of the worst rising slack across concurrent MMMC views for Hold-style early data path checks <pre>get_property [get_pins df1/Q] slack_skewness_min_rise</pre> <i>Return type:</i> float
slack_sigma_max_fall	Returns the sigma component of the slack time for the falling transition at the max corner. <i>Return type:</i> float
slack_sigma_max_rise	Returns the sigma component of the slack time for the rising transition at the max corner. <i>Return type:</i> float
slack_sigma_min_fall	Returns the sigma component of the slack time for the falling transition at the min corner. <i>Return type:</i> float
slack_sigma_min_rise	Returns the sigma component of the slack time for the rising transition at the min corner. <i>Return type:</i> float
slew_max	Returns the value of the maximum slew for the specified pin. Verify through <code>report_timing</code>. For example, <pre>get_property [get_pins <pin name>] slew_max</pre> <i>Return Type:</i> float
slew_max_fall	Returns the slew time for the falling transition at the max corner. <i>Return type:</i> float
slew_max_rise	Returns the slew time for the rising transition at the max corner. <i>Return type:</i> float

slew_mean_max_fall	Returns the mean component of the slew time for the falling transition at the max corner. <i>Return type:</i> float
slew_mean_max_rise	Returns the mean component of the slew time for the rising transition at the max corner. <i>Return type:</i> float
slew_mean_min_fall	Returns the mean component of the slew time for the falling transition at the min corner. <i>Return type:</i> float
slew_mean_min_rise	Returns the mean component of the slew time for the rising transition at the min corner. <i>Return type:</i> float
slew_min	Returns the value of the minimum slew for the specified pin. You can verify through <code>report_timing</code> . For example, <code>get_property [get_pins <pin name>] slew_min</code> <i>Return Type:</i> float
slew_min_fall	Returns the slew time for the falling transition at the min corner. <i>Return type:</i> float
slew_min_rise	Returns the slew time for the rising transition at the min corner. <i>Return type:</i> float
slew_sigma_max_fall	Returns the sigma component of the slew time for the falling transition at the max corner. <i>Return type:</i> float
slew_sigma_max_rise	Returns the sigma component of the slew time for the rising transition at the max corner. <i>Return type:</i> float
slew_sigma_min_fall	Returns the sigma component of the slew time for the falling transition at the min corner. <i>Return type:</i> float

slew_sigma_min_rise	Returns the sigma component of the slew time for the rising transition at the min corner. <i>Return type:</i> float
slew_skewness_max_fall	Returns the skewness component of the maximum slew time for transitions across all the concurrent MMMC views. <code>get_property [get_pins UMC/buff/A] slew_skewness_max_fall</code> <i>Return type:</i> float
slew_skewness_max_rise	Returns the skewness component of the maximum slew time for transitions across all the concurrent MMMC views. <code>get_property [get_pins UMC/buff/A] slew_skewness_max_rise</code> <i>Return type:</i> float
slew_skewness_min_fall	Returns the variation component of the minimum slew time for transitions across all the concurrent MMMC views. <code>get_property [get_pins UMC/buff/A] slew_skewness_min_fall</code> <i>Return type:</i> float
slew_skewness_min_rise	Returns the variation component of the minimum slew time for transitions across all the concurrent MMMC views. <code>get_property [get_pins UMC/buff/A] slew_skewness_min_rise</code> <i>Return type:</i> float
source_latency_early_fall_max	Returns the maximum early fall source insertion delay specified explicitly by <code>set_clock_latency</code> at the pin. <i>Return type:</i> float
source_latency_early_fall_min	Returns the minimum early fall source insertion delay specified explicitly by <code>set_clock_latency</code> at the pin. <i>Return type:</i> float
source_latency_early_rise_max	Returns the maximum early rise source insertion delay specified explicitly by <code>set_clock_latency</code> at the pin. <i>Return type:</i> float
source_latency_early_rise_min	Returns the minimum early rise source insertion delay specified explicitly by <code>set_clock_latency</code> at the pin. <i>Return type:</i> float

source_latency_late_fall_max	Returns the maximum late fall source insertion delay specified explicit <code>set_clock_latency</code> at the pin. <i>Return type:</i> float
source_latency_late_fall_min	Returns the minimum late fall source insertion delay specified explicit <code>set_clock_latency</code> at the pin. <i>Return type:</i> float
source_latency_late_rise_max	Returns the maximum late rise source insertion delay specified explicit <code>set_clock_latency</code> at the pin. <i>Return type:</i> float
source_latency_late_rise_min	Returns the minimum late rise source insertion delay specified explicit <code>set_clock_latency</code> at the pin. <i>Return type:</i> float
stolen_slack	Returns the slack stolen at the specified pin. This property can only for pins in late mode. Note: The specified pin must be the endpoint of a transparent arc. <i>Return type:</i> float
timing_constant	Reports the constant value specified on a pin. <code>get_property [get_pins UC/retent_cell/SZ] timing_constant</code> <i>Return type:</i> string
timing_constant_source	Reports the source from which the timing constant was applied. <code>get_property [get_pins UC/retent_cell/SZ] timing_constant_source</code> <i>Return type:</i> string
user_constant_value	Returns the constant values from constraints for the specified pins. <code>get_property [get_pins UC/retent_cell/SZ] user_constant_value</code> <i>Return type:</i> integer
waveform_max_fall	Returns the fall waveform for the specified input pins during late analysis. <i>Return type:</i> string
waveform_max_rise	Returns the rise waveform for the specified input pins during late analysis. <i>Return type:</i> string

waveform_min_fall	Returns the fall waveform for the specified input pins during ea analysis. <i>Return type:</i> string
waveform_min_rise	Returns the rise waveform for the specified input pins during e analysis. <i>Return type:</i> string
x_coordinate	Returns the x coordinates of the placed pin. <i>Return type:</i> float
y_coordinate	Returns the y coordinates of the placed pin. <i>Return type:</i> float

PG Pin Properties

hierarchical_name	Returns the complete hierarchical name of the PG pin. For example, <pre>get_property [get_pg_pins -of_objects [get_cells iA/out_reg]] hierarchical_name get_property [get_pg_pins -of_objects [get_cells iA/out_reg] -filter "pg_type == primary_power"] hierarchical_name</pre> <i>Return type:</i> string
net	Returns the immediate PG net(s) connected to the corresponding PG pin objects, as a collection. For example, <pre>get_property [get_pg_pins -of_objects [get_cells iA/out_reg]] net</pre> <i>Return type:</i> Collection
object_type	Returns <code>pg_pin</code> as the object type for a PG pin. <i>Return type:</i> string

pg_type	Returns the type of PG pin. For example, <pre>get_property [get_pg_pins -of_objects [get_cells iA/out_reg]] pg_type get_property [get_pg_pins -of_objects [get_cells iA/out_reg] -filter "hierarchical_name =~ *VDD*"] pg_type</pre> <i>Return type:</i> string
ref_lib_pin_name	Returns the name of the library PG pin associated with an instance PG pin. For example, <pre>get_property [get_pg_pins -of_objects [get_cells iA/out_reg]] ref_lib_pin_name</pre> <i>Return type:</i> string
voltage_max	Returns the maximum voltage of a given PG pin. This could be the voltage associated with a pin, power intent or CPF, <code>related_pg_pins</code> attribute in Liberty, or the operating voltage of a power domain or system default voltage. For example, <pre>get_property [get_pg_pins -of_objects [get_cells iA/out_reg]] voltage_max</pre> <i>Return type:</i> float
voltage_min	Returns the minimum voltage of a given PG pin. This voltage may be associated with a pin, power intent or CPF, <code>related_pg_pins</code> attribute in Liberty, or the operating voltage of a power domain or system default voltage. For example, <pre>get_property [get_pg_pins -of_objects [get_cells iA/out_reg]] voltage_min</pre> <i>Return type:</i> float

Port Properties

actual_latency_early_fall_max	Returns the maximum fall latency on the specified port caused due to source and network latency asserted using the <code>set_clock_latency</code> command for early analysis. <pre>get_property [get_ports CLK] actual_latency_early_fall_max</pre> <i>Return type:</i> float
actual_latency_early_fall_min	Returns the minimum fall latency on the specified port caused due to source and network latency asserted using the <code>set_clock_latency</code> command for early analysis. <pre>get_property [get_ports CLK] actual_latency_early_fall_min</pre> <i>Return type:</i> float
actual_latency_early_rise_max	Returns the maximum rise latency on the specified port caused due to source and network latency asserted using the <code>set_clock_latency</code> command for early analysis. <pre>get_property [get_ports CLK] actual_latency_early_rise_max</pre> <i>Return type:</i> float
actual_latency_early_rise_min	Returns the minimum rise latency on the specified port caused due to source and network latency asserted using the <code>set_clock_latency</code> command for early analysis. <pre>get_property [get_ports CLK] actual_latency_early_rise_min</pre> <i>Return type:</i> float
actual_latency_late_fall_max	Returns the maximum fall latency on the specified port caused due to source and network latency asserted using the <code>set_clock_latency</code> command for late analysis. <pre>get_property [get_ports CLK] actual_latency_late_fall_max</pre> <i>Return type:</i> float

actual_latency_late_fall_min	Returns the minimum fall latency on the specified port caused due to source and network latency asserted using the <code>set_clock_latency</code> command for late analysis. <pre>get_property [get_ports CLK] actual_latency_late_fall_min</pre> <i>Return type:</i> float
actual_latency_late_rise_max	Returns the maximum rise latency on the specified port caused due to source and network latency asserted using the <code>set_clock_latency</code> command for late analysis. <pre>get_property [get_ports CLK] actual_latency_late_rise_max</pre> <i>Return type:</i> float
actual_latency_late_rise_min	Returns the minimum rise latency on the specified port caused due to source and network latency asserted using the <code>set_clock_latency</code> command for late analysis. <pre>get_property [get_ports CLK] actual_latency_late_rise_min</pre> <i>Return type:</i> float
aocv_early_input_stage_weight	Returns the input stage weight of the specified port for AOCV mode early analysis. <pre>get_property [get_ports int] aocv_early_input_stage_weight</pre> <i>Return type:</i> float
aocv_early_output_stage_weight	Returns the output stage weight of the specified port for AOCV mode late analysis. <pre>get_property [get_ports vcop] aocv_early_output_stage_weight</pre> <i>Return type:</i> float
aocv_late_input_stage_weight	Returns the input stage weight of the specified for AOCV mode late analysis. <pre>get_property [get_ports int] aocv_late_input_stage_weight</pre> <i>Return type:</i> float

aocv_late_output_stage_weight	Returns the output stage weight of the specified port for AOCV mode late analysis. <pre>get_property [get_ports vcop] aocv_late_output_stage_weight</pre> <i>Return type:</i> float
arrival_clocks	Reports clocks that correspond to the arrival times at the port. <pre>get_property [get_ports out] arrival_clocks</pre> <i>Return type:</i> collection
arrival_max	Reports the maximum arrival time at a port. <pre>get_property [get_ports out] arrival_max</pre> <i>Return type:</i> float
arrival_max_fall	Returns the worst arrival on the port for a falling signal. <i>Return type:</i> float
arrival_max_rise	Returns the worst arrival on the port for a rising signal. <i>Return type:</i> float
arrival_mean_max_fall	In SOCV analysis mode, this returns the mean value of the latest falling arrival time to the specified port across all concurrent MMMC views. <pre>get_property [get_ports out] arrival_mean_max_fall</pre> <i>Return type:</i> float
arrival_mean_max_rise	In SOCV analysis mode, this returns the mean value of the latest rising arrival time to the specified port across all concurrent MMMC views. <pre>get_property [get_ports out] arrival_mean_max_rise</pre> <i>Return type:</i> float
arrival_mean_min_fall	In SOCV analysis mode, this returns the mean value of the earliest falling arrival time to the specified port across all concurrent MMMC views. <pre>get_property [get_ports out] arrival_mean_min_fall</pre> <i>Return type:</i> float

arrival_mean_min_rise	In SOCV analysis mode, this returns the mean value of the earliest rising arrival time to the specified port across all concurrent MMMC views. <pre>get_property [get_ports out] arrival_mean_min_rise</pre> <i>Return type:</i> float
arrival_min	Reports minimum arrival time at the port. <pre>get_property [get_ports out] arrival_min</pre> <i>Return type:</i> float
arrival_min_fall	Returns the best arrival on the port for a falling signal. <i>Return type:</i> float
arrival_min_rise	Returns the best arrival on the port for a rising signal. <i>Return type:</i> float
arrival_sigma_max_fall	In SOCV analysis mode, this returns the variation component of the latest falling arrival time to the specified port across all the concurrent MMMC views. <pre>get_property [get_ports out] arrival_sigma_max_fall</pre> <i>Return type:</i> float
arrival_sigma_max_rise	In SOCV analysis mode, this returns the variation component of the latest rising arrival time to the specified port across all the concurrent MMMC views. <pre>get_property [get_ports out] arrival_sigma_max_rise</pre> <i>Return type:</i> float
arrival_sigma_min_fall	In SOCV analysis mode, this returns the variation component of the earliest falling arrival time to the specified port across all the concurrent MMMC views. <pre>get_property [get_ports out] arrival_sigma_min_fall</pre> <i>Return type:</i> float

arrival_sigma_min_rise	In SOCV analysis mode, this returns the variation component of the earliest rising arrival time to the specified port across all the concurrent MMMC views. <pre>get_property [get_ports out] arrival_sigma_min_rise</pre> <i>Return type:</i> float
attacker_fall_slew	Reports fall slew of attackers seen for nets connected to the port. <pre>get_property [get_ports CLK] attacker_fall_slew</pre> <i>Return type:</i> float
attacker_rise_slew	Reports the rise slew of attackers seen for nets connected to the port. <pre>get_property [get_ports CLK] attacker_rise_slew</pre> <i>Return type:</i> float
arrival_window	Returns the arrival times for the earliest and latest rising and falling transitions in timing window format. <i>Return type:</i> string
capture_clocks	Reports capture clocks at the port. <i>Return type:</i> collection
clock_sources	Returns a collection of the source pins of all the clocks arriving at the port. <i>Return type:</i> collection
clocks	Returns a collection of all of the clocks arriving at the port. <i>Return type:</i> collection
constant_value	Returns a constant value of 0 or 1 , if a logic state has been asserted or propagated to the port. <pre>get_property [get_ports D] constant_value</pre> <i>Return type:</i> integer

constrained_capture_clocks	Returns the capture clocks, which are constrained, at the port. <pre>get_property [get_ports C] constrained_capture_clocks</pre> <i>Return type:</i> collection
capacitance_max	Returns the maximum pin capacitance for the port. <i>Return type:</i> float
capacitance_min	Returns the minimum pin capacitance for the port. <i>Return type:</i> float
delay_max_fall	Returns the maximum falling delay value for the port. <i>Return type:</i> float
delay_max_rise	Returns the maximum rising delay value for the port. <i>Return type:</i> float
delay_min_fall	Returns the minimum falling delay value for the port. <i>Return type:</i> float
delay_min_rise	Returns the minimum rising delay value for the port. <i>Return type:</i> float
direction	Returns the direction of the port: in, out, or inout. <i>Return type:</i> string
driving_cell_no_design_rule	Returns a value of <code>true</code> if there is no design rule transfer from the driving cell set on a port. For example, if a driving cell is defined with the following command, then this property will return <code>true</code>: <pre>set_driving_cell -lib_cell <lib_cell_name> - no_design_rule <port></pre> <i>Return type:</i> boolean
drive_resistance_fall_max	Returns the falling linear drive resistance at the port for the max condition. <i>Return type:</i> float

drive_resistance_fall_min	Returns the falling linear drive resistance at the port for the min condition. <i>Return type:</i> float
drive_resistance_rise_max	Returns the rising linear drive resistance at the port for the max condition. <i>Return type:</i> float
drive_resistance_rise_min	Returns the rising linear drive resistance at the port for the min condition. <i>Return type:</i> float
driving_cell_fall_max	Returns the name of the library cell used to compute max falling delays and transitions at the port. <i>Return type:</i> string
driving_cell_fall_min	Returns the name of the library cell used to compute min falling delays and transitions at the port. <i>Return type:</i> string
driving_cell_from_pin_fall_max	Returns the input pin of the driving cell used in the max fall delay calculation at the port. <i>Return type:</i> string
driving_cell_from_pin_fall_min	Returns the input pin of the driving cell used in the min fall delay calculation at the port. <i>Return type:</i> string
driving_cell_from_pin_rise_max	Returns the input pin of the driving cell used in max rise delay calculation at the port. <i>Return type:</i> string
driving_cell_from_pin_rise_min	Returns the input pin of the driving cell used in min rise delay calculation at the port. <i>Return type:</i> string
driving_cell_library_fall_max	Returns the name of the library used to compute max falling delays and transitions at the port. <i>Return type:</i> string

<code>driving_cell_library_fall_min</code>	Returns the name of the library used to compute min falling delays and transitions at the port. <i>Return type:</i> string
<code>driving_cell_library_rise_max</code>	Returns the name of the library used to compute max rising delays and transitions at the port. <i>Return type:</i> string
<code>driving_cell_library_rise_min</code>	Returns the name of the library used to compute min rising delays and transitions at the port. <i>Return type:</i> string
<code>driving_cell_max_fall_itrans_fall</code>	Returns the falling input transition of the <code>from_pin</code> of the driving cell, which is used for the max falling delay calculation at the port. <i>Return type:</i> float
<code>driving_cell_max_fall_itrans_rise</code>	Returns the falling input transition of the <code>from_pin</code> of the driving cell, which is used for the max rising delay calculation at the port. <i>Return type:</i> float
<code>driving_cell_max_rise_itrans_fall</code>	Returns the rising input transition of the <code>from_pin</code> of the driving cell, which is used for the max falling delay calculation at the port. <i>Return type:</i> float
<code>driving_cell_max_rise_itrans_rise</code>	Returns the rising input transition of the <code>from_pin</code> of the driving cell, which is used for the max rising delay calculation at the port. <i>Return type:</i> float
<code>driving_cell_min_fall_itrans_fall</code>	Returns the falling input transition of the <code>from_pin</code> of the driving cell, which is used for the min falling delay calculation at the port. <i>Return type:</i> float
<code>driving_cell_min_fall_itrans_rise</code>	Returns the falling input transition of the <code>from_pin</code> of the driving cell, which is used for the min rising delay calculation at the port. <i>Return type:</i> float

driving_cell_min_rise_itrans_fall	Returns the rising input transition of the <code>from_pin</code> of the driving cell, which is used for the min falling delay calculation at the port. <i>Return type:</i> float
driving_cell_min_rise_itrans_rise	Returns the rising input transition of the <code>from_pin</code> of the driving cell, which is used for the min rising delay calculation at the port. <i>Return type:</i> float
driving_cell_pin_fall_max	Returns the name of the output pin used for max fall delay calculation at the port. <i>Return type:</i> string
driving_cell_pin_fall_min	Returns the name of the output pin used for min fall delay calculation at the port. <i>Return type:</i> string
driving_cell_pin_rise_max	Returns the name of the output pin used for max rise delay calculation at the port. <i>Return type:</i> string
driving_cell_pin_rise_min	Returns the name of the output pin used for min rise delay calculation at the port. <i>Return type:</i> string
driving_cell_rise_max	Returns the name of the library cell used to compute max rising delays and transitions at the port. <i>Return type:</i> string
driving_cell_rise_min	Returns the name of the library cell used to compute min rising delays and transitions at the port. <i>Return type:</i> string
fanin	Returns the number of sources connected with net of the specified port. <i>Return type:</i> integer
fanout	Returns the number of sinks connected with the net of the specified port. <i>Return type:</i> integer

fanout_load	Returns the fanout load for ports. (For input ports, the value is the sum of all fanout loads of all connected pins. For output ports, the value is set using <code>set_fanout_load</code>.) <i>Return type:</i> float
fastest_clock	Reports the clock with maximum frequency associated with the port. <code>get_property [get_ports CLK] fastest_clock</code> <i>Return type:</i> collection
fastest_clock_name	Reports the name of the clock with maximum frequency associated with the port. <code>get_property [get_ports CLK] fastest_clock_name</code> <i>Return type:</i> string
fastest_clock_period	Reports the period of the clock with maximum frequency associated with the port. <code>get_property [get_ports CLK] fastest_clock_period</code> <i>Return type:</i> float
hierarchical_level	Returns the hierarchical level for an instance port. <code>get_property [get_ports C] hierarchical_level</code> <i>Return type:</i> integer
hierarchical_name	Returns the complete hierarchical name of the port. <i>Return type:</i> string
hold_uncertainty	Returns the most conservative uncertainty of all possible uncertainty assertions associated with the port. <i>Return type:</i> float
input_slew_max_fall	Returns the max falling slew for the input or inout ports. (This value is set with the <code>set_input_transition</code> command.) <i>Return type:</i> float
input_slew_max_rise	Returns the max rising slew for the input or inout ports. (This value is set with the <code>set_input_transition</code> command.) <i>Return type:</i> float

input_slew_min_fall	Returns the min falling slew for the input or inout ports. (This value is set with the <code>set_input_transition</code> command.) <i>Return type:</i> float
input_slew_min_rise	Returns the min rising slew for the input or inout ports. (This value is set with the <code>set_input_transition</code> command.) <i>Return type:</i> float
is_clock_root	Returns a value of <code>true</code> if a real or generated clock is defined on a port. <code>get_property [get_ports CLK] is_clock_root</code> <i>Return type:</i> boolean
is_clock_source_latency	Returns a value of <code>true</code> if the object is a part of the latency network. <code>get_property [get_ports C] is_clock_source_latency</code> <i>Return type:</i>
is_clock_source_pin	Returns a value of <code>true</code> if the object is a clock source pin. <code>get_property [get_ports C] is_clock_source_pin</code> <i>Return type:</i> boolean
is_clock_used_as_clock	Returns a value of <code>true</code> if the port lies in the clock network and at least one of the clocks arriving at the pin is used as a clock in the downstream network of the port. <code>get_property [get_ports C] is_clock_used_as_clock</code> <i>Return type:</i> boolean
is_clock_used_as_data	Returns a value of <code>true</code> if the port lies in the clock source path and at least one of the clocks arriving on the port is used as data in the downstream network of port. <code>get_property [get_ports C] is_clock_used_as_data</code> <i>Return type:</i> boolean

is_constrained	Returns a value of <code>true</code> if the specified port is constrained. <pre>get_property [get_ports Ip] is_constrained</pre> <i>Return type:</i> boolean
is_clock_network_port	Returns a value of <code>true</code> if a port is part of the clock network. <i>Return type:</i> boolean
is_disable_timing	Returns a value of <code>true</code> if the timing of the port has been disabled using the <code>set_disable_timing</code> command. <i>Return type:</i> boolean
is_hierarchical	Returns a value of <code>true</code> if the port is a port of a hierarchical module. <i>Return type:</i> boolean
is_latency_network_pin	Returns a value of <code>true</code> if the specified port has a generated clock. <pre>get_property [get_ports CLK] is_latency_network_pin</pre> <i>Return type:</i> boolean
is_pin	Returns a value of <code>true</code> for pin objects and <code>false</code> for port objects. <i>Return type:</i> boolean
is_port	Returns a value of <code>true</code> for port objects and <code>false</code> for pin objects. <i>Return type:</i> boolean
is_power_mode_disabled	Returns a value of <code>true</code> if power mode of the specified port is disabled. <pre>get_property [get_ports B_OFF] is_power_mode_disabled -view view_func_18V_OFF</pre> <i>Return type:</i> boolean
is_propagated_clock	Returns a value of <code>true</code> if there is an explicit <code>set_propagated_clock</code> assertion at the port. <i>Return type:</i> boolean

latency_fall_max	Returns the maximum fall insertion delay specified by an explicit <code>set_clock_latency</code> at the port. <i>Return type:</i> float
latency_fall_min	Returns the minimum fall insertion delay specified by an explicit <code>set_clock_latency</code> at the port. <i>Return type:</i> float
latency_rise_max	Returns the maximum rise insertion delay specified by an explicit <code>set_clock_latency</code> at the port. <i>Return type:</i> float
latency_rise_min	Returns the minimum rise insertion delay specified by an explicit <code>set_clock_latency</code> at the port. <i>Return type:</i> float
max_capacitance	Returns the maximum capacitance limit asserted on the specified port. <code>get_property [get_ports Zf] max_capacitance</code> <i>Return type:</i> float
max_fanout	Returns the maximum fanout load that the port can drive. (This value is set using <code>set_max_fanout</code> , or the <code>default_max_fanout</code> library attribute.) <i>Return type:</i> float
max_transition	Returns the maximum transition time limit specified for the port. <i>Return type:</i> float
netlist_constant_value	Returns a value if the port has a constant logic value of 1'b0 or 1'b1 in the Verilog. <code>get_property [get_ports constant] netlist_constant_value</code> <i>Return type:</i> string
net_name	Returns the name of the net connected to the specified port. <i>Return type:</i> string

noise_victim	Returns a collection of <code>si_victim</code> objects that can be used to query glitch attributes. <i>Return type:</i> collection
object_type	Returns the object type <code>port</code>. <i>Return type:</i> string
power_rail_voltage_max	Reports power rail voltage for maximum operating condition for bidirectional, input and output pins. <pre>get_property [get_ports in] power_rail_voltage_max -view Setup_m3C</pre> <i>Return type:</i> float
power_rail_voltage_min	Reports power rail voltage for minimum operating condition for bidirectional, input and output pins. <pre>get_property [get_ports in] power_rail_voltage_min -view Hold_m3C</pre> <i>Return type:</i> float
setup_uncertainty	Returns the most conservative uncertainty of all possible uncertainty assertions associated with the port. <i>Return type:</i> float
si_victim	Returns a collection of the <code>si_victim</code> objects that can be used to query graph-based SI delay attributes. <i>Return type:</i> collection
slack_max	Returns the worst slack across all the concurrent MMMC views for setup style late data path checks. For example, <pre>get_property [get_ports clk_en] slack_max</pre> <i>Return type:</i> float
slack_max_edge	Returns the edge (rise or fall) of the worst slack-causing path at the specified port in late mode. <i>Return type:</i> string
slack_max_fall	Returns the worst setup slack for a falling signal at the port endpoint. <i>Return type:</i> float

slack_max_rise	Returns the worst setup slack for a rising signal at the port endpoint. When the <code>-view</code> parameter is not specified, the worst slack for all the views is returned. <i>Return type:</i> float
slack_mean_max	In SOCV analysis, it will return the mean component of the worst slack across all concurrent MMMC views for setup-style late data path checks. For example, <pre>get_property [get_ports clk_en] slack_mean_max</pre> <i>Return type:</i> float
slack_mean_max_fall	Returns the mean component of the worst setup slack for a falling signal at the port endpoint. For example, <pre>get_property [get_ports scan_clk] slack_mean_max_fall</pre> <i>Return type:</i> float
slack_mean_max_rise	Returns the mean component of the worst setup slack for a rising signal at the port endpoint. For example, <pre>get_property [get_ports scan_clk] slack_mean_max_rise</pre> <i>Return type:</i> float
slack_mean_min	Returns the mean component of the worst hold slack at the port endpoint. For example, <pre>get_property [get_ports clk_en] slack_mean_min</pre> <i>Return type:</i> float
slack_mean_min_fall	Returns the mean component of the worst hold slack for a falling signal at the port endpoint. For example, <pre>get_property [get_ports clk_en] slack_mean_min_fall</pre> <i>Return type:</i> float

slack_mean_min_rise	Returns the mean component of the worst hold slack for a rising signal at the port endpoint. For example, <pre>get_property [get_ports clk_en] slack_mean_min_rise</pre> <i>Return type:</i> float
slack_min	Returns the worst hold slack at the port endpoint. For example, <pre>get_property [get_ports clk_en] slack_min</pre> <i>Return type:</i> float
slack_min_edge	Returns the edge (rise or fall) of the worst slack-causing path at the specified port in early mode. <i>Return type:</i> string
slack_min_fall	Returns the worst hold slack for a falling signal at the port endpoint. <i>Return type:</i> float
slack_min_rise	Returns the worst hold slack for a rising signal at the port endpoint. <i>Return type:</i> float
slew_max	Returns the maximum slew from all of the incoming signals on the port. For example, <pre>get_property [get_ports ahbsi_out[402]] slew_max</pre> <i>Return type:</i> float
slew_max_fall	Returns the maximum falling slew from all of the incoming signals on the port. <i>Return type:</i> float
slew_max_rise	Returns the maximum rising slew from all of the incoming signals on the port. <i>Return type:</i> float

slew_mean_max_fall	Returns the mean component of the maximum falling slew from all of the incoming signals on the port. For example, <pre>get_property [get_ports ahbsi_out[402]] slew_mean_max_fall</pre> <i>Return type:</i> float
slew_mean_max_rise	Returns the mean component of the maximum rising slew from all of the incoming signals on the port. For example, <pre>get_property [get_ports ahbsi_out[407]] slew_mean_max_rise</pre> <i>Return type:</i> float
slew_mean_min_fall	Returns the mean component of the minimum falling slew from all of the incoming signals on the port. For example, <pre>get_property [get_ports BG_scan_out_3] slew_mean_min_fall</pre> <i>Return type:</i> float
slew_mean_min_rise	Returns the mean component of the minimum rising slew from all of the incoming signals on the port. For example, <pre>get_property [get_ports BG_scan_out_3] slew_mean_min_rise</pre> <i>Return type:</i> float
slew_min	Returns the minimum slew from all of the incoming signals on the port. For example, <pre>get_property [get_ports BG_scan_out_3] slew_min</pre> <i>Return type:</i> float
slew_min_fall	Returns the minimum falling slew from all of the incoming signals on the port. <i>Return type:</i> float

slew_min_rise	Returns the minimum rising slew from all of the incoming signals on the port. <i>Return type:</i> float
source_latency_early_fall_max	Returns the maximum early fall source insertion delay specified by an explicit <code>set_clock_latency</code> at the port. <i>Return type:</i> float
source_latency_early_fall_min	Returns the minimum early fall source insertion delay specified by an explicit <code>set_clock_latency</code> at the port. <i>Return type:</i> float
source_latency_early_rise_max	Returns the maximum early rise source insertion delay specified by an explicit <code>set_clock_latency</code> at the port. <i>Return type:</i> float
source_latency_early_rise_min	Returns the minimum early rise source insertion delay specified by an explicit <code>set_clock_latency</code> at the port. <i>Return type:</i> float
source_latency_late_fall_max	Returns the maximum late fall source insertion delay specified by an explicit <code>set_clock_latency</code> at the port. <i>Return type:</i> float
source_latency_late_fall_min	Returns the minimum late fall source insertion delay specified by an explicit <code>set_clock_latency</code> at the port. <i>Return type:</i> float
source_latency_late_rise_max	Returns the maximum late rise source insertion delay specified by an explicit <code>set_clock_latency</code> at the port. <i>Return type:</i> float
source_latency_late_rise_min	Returns the minimum late rise source insertion delay specified by an explicit <code>set_clock_latency</code> at the port. <i>Return type:</i> float
timing_constant	Returns the constant value on a port. <code>get_property [get_ports D] timing_constant</code> <i>Return type:</i> string

timing_constant_source	Returns the source by which the constant value is applied on a port. <pre>get_property [get_ports constant] timing_constant_source</pre> <i>Return type:</i> string
total_capacitance_fall	Returns the total capacitance of ports for falling transition. <pre>get_property [get_ports out] total_capacitance_fall</pre> <i>Return type:</i> float
total_capacitance_rise	Returns the total capacitance of ports for rising transition. <pre>get_property [get_ports out] total_capacitance</pre> <i>Return type:</i> float
transition_max	Returns the maximum transition limit asserted on the specified port. <pre>get_property [get_ports Ip] transition_max</pre> <i>Return type:</i> float
user_constant_value	Returns the constant values from constraints for the specified ports. <pre>get_property [get_ports D] user_constant_value</pre> <i>Return type:</i> integer
wire_capacitance_max	Returns the wire capacitance of the port for the maximum value of the capacitance range. For example, <pre>get_property [get_ports port_pad_data_out[9]] wire_capacitance_max</pre> <i>Return type:</i> float

wire_capacitance_max_fall	Returns the wire capacitance of a port for the maximum value of the falling capacitance range For example, <pre>get_property [get_ports port_pad_data_out[9]] wire_capacitance_max_fall</pre> <i>Return type:</i> float
wire_capacitance_max_rise	Returns the wire capacitance of a port for the maximum value of the rising capacitance range. For example, <pre>get_property [get_ports port_pad_data_out[0]] wire_capacitance_max_rise</pre> <i>Return type:</i> float
wire_capacitance_min	Returns the wire capacitance of a port for the minimum value of the capacitance range. For example, <pre>get_property [get_ports tdigit[2]] wire_capacitance_min</pre> <i>Return type:</i> float
wire_capacitance_min_fall	Returns the wire capacitance of a port for the minimum value of the falling capacitance range. For example, <pre>get_property [get_ports tdigit[2]] wire_capacitance_min_fall</pre> <i>Return type:</i> float
wire_capacitance_min_rise	Returns the wire capacitance of a port for the minimum value of the rising capacitance range. For example, <pre>get_property [get_ports tdigit[2]] wire_capacitance_min_rise</pre> <i>Return type:</i> float

wire_lumped_capacitance_max	Returns the wire lumped capacitance of the net associated to port for maximum conditions. Returns the portion of a port's total lumped capacitance, which may be extracted or annotated wire lumped capacitance values. For example, <pre>get_property [get_ports tdigit[7]] wire_lumped_capacitance_max</pre> <i>Return type:</i> float
wire_lumped_capacitance_max_fall	Returns the lumped wire capacitance of a port for the maximum value of the falling capacitance range. For example, <pre>get_property [get_ports port_pad_data_out[9]] wire_lumped_capacitance_max_fall</pre> <i>Return type:</i> float
wire_lumped_capacitance_max_rise	Returns the lumped wire capacitance of a port for the maximum value of the rising capacitance range. For example, <pre>get_property [get_ports port_pad_data_out[0]] wire_lumped_capacitance_max_rise</pre> <i>Return type:</i> float
wire_lumped_capacitance_min	Returns the portion of a port's total lumped capacitance, which may be extracted or annotated wire lumped capacitance values. For example, <pre>get_property [get_ports tdigit[3]] wire_lumped_capacitance_min</pre> <i>Return type:</i> float
wire_lumped_capacitance_min_fall	Returns the lumped wire capacitance of a port for the minimum value of the falling capacitance range. For example, <pre>get_property [get_ports tdigit[3]] wire_lumped_capacitance_min_fall</pre> <i>Return type:</i> float

wire_lumped_capacitance_min_rise	Returns the lumped wire capacitance of a port for the minimum value of the rising capacitance range. For example, <pre>get_property [get_ports tdigit[3]] wire_lumped_capacitance_min_rise</pre> <i>Return type:</i> float
x_coordinate	Returns the x coordinates of the placed port. <i>Return type:</i> float
y_coordinate	Returns the y coordinates of the placed port. <i>Return type:</i> float

Timing Arc Properties

aocv_derate_capture_clock_early_fall	Returns graph based AOCV derate factors in AOCV mode for a given timing arc that is a part of early capture clock paths with fall sink pin transitions. <i>Return type:</i> float
aocv_derate_capture_clock_early_rise	Returns graph based AOCV derate factors in AOCV mode for a given timing arc that is a part of early capture clock paths with rise sink pin transitions. <i>Return type:</i> float
aocv_derate_capture_clock_late_fall	Returns graph based AOCV derate factors in AOCV mode for a given timing arc that is a part of late capture clock paths with fall sink pin transitions. <i>Return type:</i> float
aocv_derate_capture_clock_late_rise	Returns graph based AOCV derate factors in AOCV mode for a given timing arc that is a part of late capture clock paths with rise sink pin transitions. <i>Return type:</i> float
aocv_derate_data_early_fall	Returns AOCV derate values for a timing arc on an early data path with fall transitions. <i>Return type:</i> float

aocv_derate_data_early_rise	Returns AOCV derate values for a timing arc on an early data path with rise transitions. <i>Return type:</i> float
aocv_derate_data_late_fall	Returns AOCV derate values for a timing arc on a late data path with fall transitions. <i>Return type:</i> float
aocv_derate_data_late_rise	Returns AOCV derate values for a timing arc on a late data path with rise transitions. <i>Return type:</i> float
aocv_derate_launch_clock_early_fall	Returns graph based AOCV derate factors in AOCV mode for a given timing arc that is a part of early launch clock paths with fall sink pin transitions. <i>Return type:</i> float
aocv_derate_launch_clock_early_rise	Returns graph based AOCV derate factors in AOCV mode for a given timing arc that is a part of early launch clock paths with rise sink pin transitions. <i>Return type:</i> float
aocv_derate_launch_clock_late_fall	Returns graph based AOCV derate factors in AOCV mode for a given timing arc that is a part of late launch clock paths with fall sink pin transitions. <i>Return type:</i> float
aocv_derate_launch_clock_late_rise	Returns graph based AOCV derate factors in AOCV mode for a given timing arc that is a part of late launch clock paths with rise sink pin transitions. <i>Return type:</i> float
aocv_stage_count_capture_clock_early	Returns graph based AOCV stage count values in AOCV mode for a given timing arc that is a part of early capture clock paths. <i>Return type:</i> float
aocv_stage_count_capture_clock_late	Returns graph based AOCV stage count values in AOCV mode for a given timing arc that is a part of late capture clock paths. <i>Return type:</i> float

aocv_stage_count_data_early	Returns AOCV stage count values for a timing arc on an early data path. <i>Return type:</i> float
aocv_stage_count_data_late	Returns AOCV stage count values for a timing arc on a late data path. <i>Return type:</i> float
aocv_stage_count_launch_clock_early	Returns AOCV stage count values in AOCV mode for a given timing arc that is a part of early launch clock paths. <i>Return type:</i> float
aocv_stage_count_launch_clock_late	Returns AOCV stage count values in AOCV mode for a given timing arc that is a part of late launch clock paths. <i>Return type:</i> float
arc_type	Returns the Liberty <code>timing_type</code> attribute of the arc. <i>Return type:</i> string
delay_max	Returns the delay for the maximum corner. For example, <pre>get_property [get_arcs -of_objects [get_cells regbuf]] delay_max</pre> <i>Return type:</i> float
delay_max_fall	Returns the fall delay for the maximum corner. <i>Return type:</i> float
delay_max_rise	Returns the rise delay for the maximum corner. <i>Return type:</i> float
delay_mean_max_fall	Returns the mean component of the maximum falling delay value for the timing arc. <i>Return type:</i> float
delay_mean_max_rise	Returns the mean component of the maximum rising delay value for the timing arc. <i>Return type:</i> float

delay_mean_min_fall	Returns the mean component of the minimum falling delay value for the timing arc. <i>Return type:</i> float
delay_mean_min_rise	Returns the mean component of the minimum rising delay value for the timing arc. <i>Return type:</i> float
delay_min	Returns the delay for the minimum corner. For example, <pre>get_property [get_arcs -of_objects [get_cells clkbuf]] delay_min</pre> <i>Return type:</i> float
delay_min_fall	Returns the fall delay for the minimum corner. <i>Return type:</i> float
delay_min_rise	Returns the rise delay for the minimum corner. <i>Return type:</i> float
delay_sigma_max	Returns the 1-sigma variation component of the maximum delay for the specified timing arc. <i>Return type:</i> float
delay_sigma_max_fall	Returns the 1-sigma variation component of the maximum falling delay for the specified timing arc. <i>Return type:</i> float
delay_sigma_max_rise	Returns the 1-sigma variation component of the maximum rising delay for the specified timing arc. <i>Return type:</i> float
delay_sigma_min	Returns the 1-sigma variation component of the minimum delay for the specified timing arc. <i>Return type:</i> float
delay_sigma_min_fall	Returns the 1-sigma variation component of the minimum falling delay for the specified timing arc. <i>Return type:</i> float

delay_sigma_min_rise	Returns the 1-sigma variation component of the minimum rising delay for the specified timing arc. <i>Return type:</i> float
delta_delay_max	Returns the delta delay for the maximum corner. For example, <pre>get_property [get_arcs -of_objects [get_cells clkbuf]] delta_delay_max</pre> <i>Return type:</i> float
delta_delay_max_fall	Returns the delta fall delay for the max corner. <i>Return type:</i> float
delta_delay_max_rise	Returns the delta rise delay for the max corner. <i>Return type:</i> float
delta_delay_min	Returns the delta delay for the minimum corner. For example, <pre>get_property [get_arcs -of_objects [get_cells clkbuf]] delta_delay_min</pre> <i>Return type:</i> float
delta_delay_min_fall	Returns the delta fall delay for the minimum corner. <i>Return type:</i> float
delta_delay_min_rise	Returns the delta rise delay for the minimum corner. <i>Return type:</i> float
dynamic_delta_delay_max_fall	Returns the dynamic delay component of the maximum falling delay through the arc. For example, <pre>get_property -view func_worst_cmax_w_dynv [get_arcs -from i_0/A -to i_0/Y] dynamic_delta_delay_max_fall</pre> <i>Return type:</i> float

dynamic_delta_delay_max_rise	Returns the dynamic delay component of the maximum rising delay through the arc. For example, <pre>get_property -view func_worst_cmax_w_dynv [get_arcs -from i_0/A -to i_0/Y] dynamic_delta_delay_max_rise</pre>
dynamic_delta_delay_min_fall	Returns the dynamic delay component of the minimum falling delay through the arc. For example, <pre>get_property -view func_worst_cmax_w_dynv [get_arcs -from i_0/A -to i_0/Y] dynamic_delta_delay_min_fall</pre>
dynamic_delta_delay_min_rise	Returns the dynamic delay component of the minimum rising delay through the arc. For example, <pre>get_property -view func_worst_cmax_w_dynv [get_arcs -from i_0/A -to i_0/Y] dynamic_delta_delay_min_rise</pre>
has_mis_fall	Returns a value of <code>true</code>, if MIS (multi-input switching) derate falling value is specified for timing arcs in the library. <pre>get_property [get_arc -from UC/and1/A1 -to UC/and1/ZN] has_mis_fall</pre> <i>Return type:</i> boolean
has_mis_rise	Returns a value of <code>true</code>, if MIS (multi-input switching) derate rising value is specified for timing arcs in the library. <pre>get_property [get_arcs -from UC/or1/A1 -to UC/or1/ZN] has_mis_rise</pre> <i>Return type:</i> boolean
is_cellarc	Returns a value of <code>true</code> if the arc belongs to a cell. <i>Return type:</i> boolean
is_disabled	Returns a value of <code>true</code> if a <code>set_disable_timing</code> has been applied on the supplied library timing arc. <i>Return type:</i> boolean

mode	Returns the mode specified for the timing arc in the library. <i>Return type:</i> string
object_type	Returns the object type <code>timing_arc</code> . <i>Return type:</i> string
sdf_cond	Returns the <code>sdf_cond</code> expression specified for the timing arc in the library. <i>Return type:</i> string
sdf_cond_end	Returns the <code>sdf_cond_end</code> expression specified for the timing arc in the library. <i>Return type:</i> string
sdf_cond_start	Returns the <code>sdf_cond_start</code> expression specified for the timing arc in the library. <i>Return type:</i> string
sense	Returns the sense of the timing arc. <i>Return type:</i> string
sink_pin	Returns the sink pin of the delay arc, or the reference pin of the check arc. <i>Return type:</i> collection
source_pin	Returns the source pin of the delay arc, or the signal pin of the check arc. <i>Return type:</i> collection
ssi_derate_fall	Returns the Simultaneously Switching Input derate fall value specified on library cells. For example, <pre>get_property [get_arcs -of_objects [get_cells clkbuf]] ssi_derate_fall</pre> <i>Return type:</i> float

ssi_derate_rise	Returns the Simultaneously Switching Input derate rise value specified on library cells. For example, <pre>get_property [get_arcs -of_objects [get_cells clkbuf]] ssi_derate_rise</pre> <i>Return type:</i> float
when	Returns the <code>when</code> condition specified for the timing arc in the library. <i>Return type:</i> string

Timing Path Properties

applied_exception_type	Returns the type of applied exceptions for a timing path. This property can be used to query dominant timing exceptions affecting a timing path. You can use this property to query on constrained/unconstrained timing paths. The possible return values are - <code>multi_cycle</code>, <code>path_delay</code> and <code>false_path</code> - indicating multicycle path, minimum/maximum delay path, and timing false path, respectively. The software will return one of these values for pin-based timing exceptions. However, the software returns “NA” when false paths are defined for clock objects, or if the paths do not have any related timing exception. <i>Return Type:</i> String
arrival	Returns the arrival time at the timing path. <i>Return type:</i> float
arrival_mean	Returns the mean component of the arrival time for the timing path. <i>Return type:</i> float

arrival_sigma	Returns the sigma component of the arrival time for the timing path. <i>Return type:</i> float
arrival_skewness	Returns the skewness in the arrival time for a timing path. <i>Return type:</i> float
arrival_stddev	Returns the standard deviation of the arrival time for a timing path. <i>Return type:</i> float
arrival_wire_stddev	Returns the standard deviation of arrival time due to wire variation for the specified timing paths. get_property [report_timing -retime path_slew_propagation -split_delay - path_type full_clock -collection] arrival_wire_stddev <i>Return type:</i> float
capturing_arc	Returns the capturing arc for the specified timing path. get_property [report_timing -clock_to clk2 - collection] capturing_arc <i>Return type:</i> collection
capture_clock_path	Returns the capturing clock path of the timing path. This property can be used only for path_type full_clock. <i>Return type:</i> collection
capturing_clock	Returns the capturing clock of the timing path. <i>Return type:</i> collection
capturing_clock_close_edge_time	Returns the closing edge time of the capturing clock. <i>Return type:</i> float

capturing_clock_close_edge_type	Returns the capturing clock edge direction: <code>rise</code> or <code>fall</code>. <i>Return type:</i> string
capturing_clock_is_inverted	Returns a value of <code>true</code> if the phase of the clock at the end point of the capture clock path is opposite to the phase of the capturing clock at the source. <i>Return type:</i> boolean
capturing_clock_is_propagated	Returns a value of <code>true</code> if the capturing clock is in the propagated mode, at the clock end point. <i>Return type:</i> boolean
capturing_clock_latency	Returns the ideal or propagated latency of the capturing clock. <i>Return type:</i> float
capturing_clock_latency_mean	Returns the capturing clock mean latency. <pre>get_property [get_timing_paths] capturing_clock_latency_mean</pre> <i>Return type:</i> float
capturing_clock_latency_sigma	Returns the sigma variation for the capturing clock latency. <pre>get_property [report_timing -collection] capturing_clock_latency_sigma</pre> <i>Return type:</i> float
capturing_clock_latency_skewness	Returns the skewness variation for the capturing clock latency. <pre>get_property [report_timing -collection] capturing_clock_latency_skewness</pre> <i>Return type:</i> float
capturing_clock_latency_stddev	Returns the stddev variation for the capturing clock latency. <pre>get_property [report_timing -collection] capturing_clock_latency_stddev</pre> <i>Return type:</i> float

capturing_clock_open_edge_type	Returns the opening edge time of the capturing clock. <i>Return type:</i> string
capturing_clock_pin	Returns the name of the capturing clock pin. <i>Return type:</i> collection
capturing_point_name	Returns the name of the capturing point. <pre>get_property [report_timing -collection] capturing_point_name</pre> <i>Return type:</i> string
capturing_clock_source_arrival_time	Returns the capturing clock arrival time at the clock root pin/port of the capturing clock path. This value is a sum of the clock edge time and the source latency constraints specified at the clock root. Note: You must specify <code>report_timing -path_type full_clock</code> in order to return capturing clock arrival time information. <i>Return type:</i> float
capturing_clock_source_arrival_time_mean	Returns the mean component of the capturing clock arrival time at the clock root pin/port of the capturing clock path. <i>Return type:</i> float
capturing_clock_source_arrival_time_sigma	Returns the sigma component of the capturing clock arrival time at the clock root pin/port of the capturing clock path. <i>Return type:</i> float
capturing_point	Returns a pointer to the end point of the timing path. <i>Return type:</i> collection
capturing_point_is_level_sensitive	Returns a value of <code>true</code> if the end point of the data path is level sensitive. <i>Return type:</i> boolean

check_delay	Returns the check delay value for the timing path. <i>Return type:</i> float
check_delay_mean	Returns the mean component of the check delay value for a timing path. <i>Return type:</i> float
check_delay_sigma	Returns the sigma component of the check delay value for a timing path. <i>Return type:</i> float
check_delay_sigma_detail	Returns separate delay sigma variation due to correlated variation of current stage, the correlated variation of previous stage, and uncorrelated variation of the current stage on the timing paths, when the <code>set_delay_cal_mode - socv_accuracy_mode</code> parameter is set to medium, high, or ultra. <code>get_property [report_timing -collection] check_delay_sigma_detail</code> <i>Return type:</i> string
check_delay_skewness_detail	Returns separate delay skewness variation due to correlated variation of current stage, the correlated variation of previous stage, and uncorrelated variation of the current stage on the timing path when the <code>set_delay_cal_mode - socv_accuracy_mode</code> parameter is set to high or ultra. <code>get_property [report_timing -collection] check_delay_skewness_detail</code> <i>Return type:</i> string
check_delay_skewness	Returns the skewness component of the check delay value for a timing path. <i>Return type:</i> float
check_delay_stddev	Returns the check delay standard deviation component of the check delay value for a timing path. <i>Return type:</i> float

check_type	Returns the check type for the timing path. <i>Return type:</i> float
clock_path_end_point	Returns the last point in the capture clock path of the timing path. <i>Return type:</i> collection
clock_source_jitter	Returns the clock source jitter value applied on a specific clock. <i>Return type:</i> float
clock_uncertainty	Returns the clock uncertainty of the timing path. <i>Return type:</i> float
cppr_adjustment	Returns the CPPR adjustment of the timing path. <i>Return type:</i> float
cppr_adjustment_mean	Returns the mean component of CPPR adjustment of the timing path. <i>Return type:</i> float
cppr_adjustment_sigma	Returns the sigma component of CPPR adjustment of the timing path. <i>Return type:</i> float
cppr_adjustment_skewness	Returns the skewness component of CPPR adjustment of a timing path. <i>Return type:</i> float
cppr_adjustment_stddev	Returns the standard deviation component of CPPR adjustment of a timing path. <i>Return type:</i> float
cppr_branch_point	Returns the CPPR branch point of the given timing path. <i>Return Type:</i> collection

cumulative_manhattan_length	Returns the cumulative manhattan length for wires of timing paths. <pre>get_property [report_timing -collection] cumulative_manhattan_length</pre> <i>Return type:</i> float
cumulative_manhattan_length_x	Returns the cumulative horizontal length for wires of timing paths. <pre>get_property [report_timing -collection] cumulative_manhattan_length_x</pre> <i>Return type:</i> float
cumulative_manhattan_length_y	Returns the cumulative vertical length for wires of timing paths. <pre>get_property [report_timing -collection] cumulative_manhattan_length_y</pre> <i>Return type:</i> float
cycle_adjustment	Returns the cycle adjustment value for the timing paths. <pre>get_property [report_timing -early - collection] cycle_adjustment</pre> <i>Return type:</i> float
drive_adjustment	Returns the drive adjustment value for the timing path. Return type: float
drive_adjustment_mean	Returns the mean component of the drive adjustment value for the timing path. Return type: float
drive_adjustment_sigma	Returns the sigma component of the drive adjustment value for the timing path. Return type: float
external_delay	Returns the value of the set_output_delay constraint at the end point of the path. <i>Return type:</i> float

gba_slack	Returns the original slack time for the timing path – before PBA retiming is performed. <i>Return type:</i> float
gba_slack_mean	Returns the mean component of the slack time for the timing path – before PBA retiming is performed. <i>Return type:</i> float
gba_slack_sigma	Returns the sigma component of the slack time for the timing path – before PBA retiming is performed. <i>Return type:</i> float
gba_slack_skewness	Returns the skewness variation of slack for GBA analysis. get_property [report_timing -skip_io_paths -not_through [get_ports *] -collection -early] gba_slack_skewness <i>Return type:</i> float
gba_slack_stddev	Returns the standard deviation of slack for GBA analysis. get_property [report_timing -collection] gba_slack_stddev <i>Return type:</i> float
hold	Returns the hold time at the end point of the path. <i>Return type:</i> float
is_borrow_edge_adjust	Returns a value of <code>true</code> if borrow edge adjustment is set on the timing paths. get_property [report_timing -through micro_dec/LOCKUP/d -collection] is_borrow_edge_adjust <i>Return type:</i> boolean
is_clock_gating_hold	Returns a value of <code>true</code> if the path is a clock gating hold check. <i>Return type:</i> boolean

is_clock_gating_setup	Returns a value of <code>true</code> if the path is a clock gating setup check. <i>Return type:</i> boolean
is_coverage_violating_path	Returns a value of <code>true</code> if a timing path contains violations from the timing check coverage. <pre>get_property [report_timing -collection] is_coverage_violating_path</pre> <i>Return type:</i> boolean
is_path_borrowing	Returns a value of <code>true</code> if the specified path is a time borrowing path. <pre>get_property [report_timing -collection -to micro_dec/LOCKUP/d] is_path_borrowing</pre> <i>Return type:</i> boolean
is_retimed	Returns a value of <code>true</code> if the timing path is recalculated using path-based analysis. <pre>get_property [report_timing -retime path_slew_propagation -collection] is_retimed</pre> <i>Return type:</i> boolean
is_time_given	Returns a value of <code>true</code> if for the specified timing path time is applied to the starting point. <pre>get_property [report_timing -to lat/Q - collection] is_time_given</pre> <i>Return type:</i> boolean
is_transparent_latch	Returns a value of <code>true</code> if the path is a transparent latch path. <i>Return type:</i> boolean
launch_clock_path	Returns the launching clock path of the timing path. This property can be used only for <code>path_type</code> <code>full_clock</code>. <i>Return type:</i> collection

launching_clock	Returns the launching clock of the timing path. <i>Return type:</i> collection
launching_clock_is_inverted	Returns a value of true if the phase of the clock at the start point of the launch clock path is opposite to the phase of the launch clock at the source. <i>Return type:</i> boolean
launching_clock_is_propagated	Returns a value of true if the launching clock is in the propagated mode, at the clock end point. <i>Return type:</i> boolean
launching_clock_latency	Returns the ideal or propagated latency of the launching clock. <i>Return type:</i> float
launching_clock_latency_skewness	Returns the skewness variation for the launching clock latency. get_property [report_timing -clock_from clk2 -path_type full_clock -collection] launching_clock_latency_skewness <i>Return type:</i> float
launching_clock_latency_stddev	Returns the standard deviation for the launching clock latency. get_property [report_timing -clock_from clk2 -path_type full_clock -collection] launching_clock_latency_stddev <i>Return type:</i> float
launching_clock_root_open_edge_type	Returns the open edge type for the root of the launching clock for the specified timing path. get_property [report_timing -collection] launching_clock_open_edge_type <i>Return type:</i> string

launching_clock_source_arrival_time_skewness	Returns the skewness variation of the launching clock arrival time at the source. <pre>get_property [report_timing -clock_from glck -path_type full_clock -collection] launching_clock_source_arrival_time_skewness</pre> <i>Return type:</i> float
launching_clock_source_arrival_time_stddev	Returns the standard deviation of the launching clock arrival time at the source. <pre>get_property [report_timing -clock_from glck -path_type full_clock -collection] launching_clock_source_arrival_time_stddev</pre> <i>Return type:</i> float
launching_clock_open_edge_time	Returns the opening edge time of the launching clock. <i>Return type:</i> float
launching_clock_open_edge_type	Returns the launching clock's open edge type at the launching point of the timing path. In case of input delay path with respect to a clock it will return the clock edge type, which is triggering the path. <i>Return type:</i> string
launching_clock_source_arrival_time	Returns the launching clock arrival time at the clock route pin/port of the launching clock path. This value is the sum of the clock edge time plus any source latency constraint specified at the clock root. Note: You must specify <code>report_timing -path_type full_clock</code> in order to return launching clock arrival time information. <i>Return type:</i> float
launching_clock_source_arrival_time_mean	Returns the mean component of the launching clock arrival time at the clock route pin/port of the launching clock path. <i>Return type:</i> float

launching_clock_source_arrival_time_sigma	Returns the sigma component of the launching clock arrival time at the clock route pin/port of the launching clock path. <i>Return type:</i> float
launching_clock_latency_mean	Returns mean component of latency of the launching clock. <i>Return type:</i> float
launching_clock_latency_sigma	Returns sigma component of latency of the launching clock. <i>Return type:</i> float
launching_input_delay	Returns the input delay if the timing path starts from an input or output port. This value is set using <code>set_input_delay</code>. <i>Return type:</i> float
launching_point	Returns a pointer to the start point of the timing path. <i>Return type:</i> collection
launching_point_is_level_sensitive	Returns a value of <code>true</code> if the begin point of the data path is level sensitive. <i>Return type:</i> boolean
launching_point_name	Returns the name of the launching point. <pre>get_property [report_timing -collection] launching_point_name</pre> <i>Return type:</i> string
lending_path_d_pin	Returns the D pin for the specified lending timing path. <pre>get_property [report_timing -to lat/Q - collection -path_type full_clock] lending_path_d_pin</pre> <i>Return type:</i> string

nets	Returns the nets for the specified timing paths. For example, <pre>get_property [get_timing_paths] nets</pre> <i>Return type:</i> string
num_cell_arcs	Returns the number of cell arcs in the given timing path. For example, <pre>set path [report_timing -from <start_point> -to <end_point> -collection] get_property \$path num_cell_arcs</pre> <i>Return Type:</i> float
num_net_arcs	Returns the number of net arcs in the given timing path. For example, <pre>set path [report_timing -from <start_point> -to <end_point> -collection] get_property \$path num_net_arcs</pre> <i>Return Type:</i> float
object_type	Returns the object type <code>timing_path</code>. <i>Return type:</i> string
other_end_arrival	Returns the capture clock path's begin time, In case of SOCV analysis, mean+N*sigma quantile value is reported. <i>Return type:</i> float
other_end_arrival_mean	Returns the mean component of capture clock path's begin time. <i>Return type:</i> float
other_end_arrival_sigma	Returns the sigma component of capture clock path's begin time. <i>Return type:</i> float

other_end_arrival_sigma_detail	Returns separate other end arrival time variation due to correlated and uncorrelated effects on timing paths, when <code>set_delay_cal_mode -socv_accuracy_mode</code> is set to medium, high, or ultra. <pre>get_property [report_timing -collection] other_end_arrival_sigma_detail</pre> <i>Return type:</i> string
other_end_arrival_skewness	Returns the skewness of other end arrival time for the timing path. <pre>get_property [report_timing -collection] other_end_arrival_skewness</pre> <i>Return type:</i> float
other_end_arrival_skewness_detail	Returns separate other end arrival time skewness due to correlated and uncorrelated effects on timing path, when <code>set_delay_cal_mode -socv_accuracy_mode</code> is set to high or ultra. <pre>get_property [report_timing -collection] other_end_arrival_skewness_detail</pre> <i>Return type:</i> string
other_end_arrival_stddev	Returns the standard deviation of other end arrival time for a timing path. <pre>get_property [report_timing -collection] other_end_arrival_stddev</pre> <i>Return type:</i> float
path_adjust_value	Returns the path adjust value asserted on timing paths using the <code>set_path_adjust</code> command. <pre>get_property [report_timing -through UMC/and1 -collection] path_adjust_value</pre> <i>Return type:</i> float
path_cell_delay	Returns the cell delay for the timing path. <pre>get_property [report_timing -collection] path_cell_delay</pre> <i>Return type:</i> float

path_cell_delay_mean	Returns the mean cell delay for the timing path. <pre>get_property [report_timing -collection] path_cell_delay_mean</pre> <i>Return type:</i> float
path_cell_delay_sigma	Returns the sigma component of cell delay for the timing path. <pre>get_property [report_timing -collection] path_cell_delay_sigma</pre> <i>Return type:</i> float
path_delay_adjustment	Returns the delay adjustment asserted on paths using the set_max_delay/set_min_delay command. <pre>get_property [report_timing -through UMC/and1 -collection] path_delay_adjustment</pre> <i>Return type:</i> float
path_delay_wire_stddev	Returns the standard deviation of path delay due to wire variation for the specified timing path. <pre>get_property [report_timing -retime path_slew_propagation -split_delay - path_type full_clock -collection] path_delay_wire_stddev</pre> <i>Return type:</i> float
path_net_delay	Returns the net delay for the timing paths. <pre>get_property [report_timing -collection] path_net_delay</pre> <i>Return type:</i> float
path_net_delay_mean	Returns the mean net delay for the timing paths. <pre>get_property [report_timing -collection] path_net_delay_mean</pre> <i>Return type:</i> float

path_net_delay_sigma	Returns the sigma variation for path net delay for the specified timing path. <pre>get_property [report_timing -retime path_slew_propagation -split_delay - path_type full_clock -collection] path_net_delay_sigma</pre> <i>Return type:</i> float
path_delay	Returns the delay of data path component of a timing path. In case of SOCV analysis, mean+N*sigma quantile delay is reported. <i>Return type:</i> float
path_delay_mean	Returns the mean delay component of data path of a timing path. <i>Return type:</i> float
path_delay_sigma	Returns the sigma delay component of data path of a timing path. <i>Return type:</i> float
path_group	Returns a collection of path groups associated with the timing path. <i>Return type:</i> collection
path_group_name	Returns the name of the path group for the timing path. <i>Return type:</i> string
path_type	Returns a value of <code>max</code> if the path is for setup, or <code>min</code> if the path is for hold. <i>Return type:</i> string
period	Returns the time duration between the launching clock edge and capturing clock edge of the path. Returns the period of the clock for the timing path. <code>get_property [report_timing -collection] period</code> The launching clock edge indicates the launching clock time when the signal at the start point of the path is launched, and the capturing clock edge indicates the capturing clock time

when the signal at the path end point is captured. For both of these edges, their ideal clock cycle times are considered for calculating the path period.

For those paths, where either the launching or capturing clock is absent or asynchronous, the property prints NA. The property works for both setup (late) paths, and hold (early) paths.

Usually the path period is decided by a simple subtraction of effective launching edge from the effective capturing edge of the path. Under certain scenario, these edges are moved back or ahead in time, and accordingly the path period is also affected.

The following factors impact period:

- Multicycle paths: The launch or capture edge will move by the cycle factor depending on the type of multi cycle exception, and this cycle factor is taken into account while computing the period.
- Max delay paths: Any max_delay specification overrides the relationship between capturing and launching edges and sets it to the max_delay value. The period of the path is thus returned as equal to the specified max delay value for sequential max delay paths. Note that a combinational max delay path has at least one of the launch or capture clocks either absent or asynchronous, and hence the effective period is not calculated for them.
- Clock uncertainty: The uncertainties applicable to the interaction of launch and capture clocks for this particular path also affect the effective edges, but this factor is not taken into consideration while computing the period.
- Unconstrained paths: The unconstrained paths have no capture clock and hence period can not be calculated for them. The

	property will return NA. <i>Return value:</i> float For more information, see "Examples".
phase_shift	Returns the phase shift for the timing paths. <pre>get_property [report_timing -collection] phase_shift</pre> <i>Return type:</i> float
recovery	Returns the recovery time at the end point of the path. <i>Return type:</i> float
removal	Returns the removal time at the end point of the path. <i>Return type:</i> float
required_time	Returns the required time for the timing path. <i>Return type:</i> float
required_time_mean	Returns the mean component of the required time for the timing path. <i>Return type:</i> float
required_time_sigma	Returns the sigma component of the required time for the timing path. <i>Return type:</i> float
required_time_sigma_detail	Returns separate required time variation due to correlated and uncorrelated effects on timing paths, when <code>set_delay_cal_mode - socv_accuracy_mode</code> is set to medium, high, or ultra. <pre>get_property [report_timing -collection] required_time_sigma_detail</pre> <i>Return type:</i> string

required_time_skewness	Returns the skewness in required time of a timing path. <pre>get_property [report_timing -collection] required_time_skewness</pre> <i>Return type:</i> float
required_time_skewness_detail	Returns separate required time skewness due to correlated and uncorrelated effects on timing paths, when <code>set_delay_cal_mode - socv_accuracy_mode</code> is set to <code>high</code> or <code>ultra</code>. <pre>get_property [report_timing -collection] required_time_skewness_detail</pre> <i>Return type:</i> string
required_time_stddev	Returns the standard deviation of the required time for a timing path. <pre>get_property [report_timing -collection] required_time_stddev</pre> <i>Return type:</i> float
setup	Returns the setup time at the end point of the path. <i>Return type:</i> float
skew	Returns the latency difference between the launching clock latency and capturing clock latency of the path (after considering the effect of CPPR factor). <i>Return type:</i> float
slack	Returns the slack time for the timing path. <i>Return type:</i> float
slack_mean	Returns the mean component of the slack time for the timing path. <i>Return type:</i> float
slack_sigma	Returns the sigma component of the slack time for the timing path. <i>Return type:</i> float

slack_sigma_detail	Returns separate slack sigma variation due to correlated and uncorrelated effects on timing paths, when <code>set_delay_cal_mode - socv_accuracy_mode</code> is set to medium, high, or ultra. <pre>get_property [report_timing -collection] slack_sigma_detail</pre> <i>Return type:</i> string
slack_skewness	Returns slack skewness for the timing paths. <pre>get_property [report_timing -collection] slack_skewness</pre> <i>Return type:</i> float
slack_skewness_detail	Returns separate slack skewness variation due to correlated and uncorrelated effects on timing paths, when <code>set_delay_cal_mode - socv_accuracy_mode</code> is set to high or ultra. <pre>get_property [report_timing -collection] slack_skewness_detail</pre> <i>Return type:</i> string
slack_wire_stddev	Returns the standard deviation of slack due to wire variation for the specified timing paths. <pre>get_property [report_timing -retime path_slew_propagation -split_delay - path_type full_clock -collection] slack_wire_stddev</pre> <i>Return type:</i> float
slack_stddev	Returns standard deviation of the slack. <i>Return type:</i> float
spatial_distance	Reports the bounding box distance for spatial derating applied on a timing path. <i>Return type:</i> float
spatial_distance_cell	Reports the bounding box distance for cells. <i>Return type:</i> float

spatial_distance_net	Reports the bounding box distance for nets. <i>Return type:</i> float
time_borrowed	Returns the amount of time borrowed from the timing end point. <i>Return type:</i> float
time_lent	Returns the amount of time lent to the timing start point. <i>Return type:</i> float
timing_points	Returns a collection of the timing points that make up the path. <i>Return type:</i> collection
total_cell_delta_delay	Returns the total cell incremental delay present in the timing paths. <pre>get_property [report_timing -through clk_I1/ZN -unconstrained -collection] total_cell_delta_delay</pre> <i>Return type:</i> float
total_delta_delay	Returns the total incremental delay present in the timing paths. <pre>get_property [report_timing -through clk_L1_I1/ZN -unconstrained -collection] total_delta_delay</pre> <i>Return type:</i> float
total_net_delta_delay	Returns the total net incremental delay present in the timing paths. <pre>get_property [report_timing -through clk_L1_I1/ZN -unconstrained -collection] total_net_delta_delay</pre> <i>Return type:</i> float
underated_slack	Returns the underated slack time for the timing path. The underated slack value is available with timing path object only if it is generated using the <code>-derate</code> parameter. <i>Return type:</i> float

view_name	Returns the analysis view name associated with the timing path. <i>Return type:</i> string
wire_variation_capacitance_slack_delta	Returns a list of lists with slack shift due to capacitance of each layer. <i>Return type:</i> string
wire_variation_layer_info	Returns layer information for wire variation analysis. <i>Return type:</i> string
wire_variation_layer_slack_delta	Returns a list of lists with total slack shift per layer. <i>Return type:</i> string
wire_variation_resistance_slack_delta	Returns a list of lists with slack shift due to resistance of each layer. <i>Return type:</i> string
wire_variation_slack_shift	Returns a list of lists with slack shift. <i>Return type:</i> string
wire_variation_slack_shift_total	Returns a list of lists with total slack shift. <i>Return type:</i> string
worst_cell_delay	Returns the worst cell delay of the timing paths. get_property [report_timing -collection] worst_cell_delay <i>Return type:</i> float
worst_cell_delay_mean	Returns the worst mean cell delay of the timing paths. get_property [report_timing -collection] worst_cell_delay_mean <i>Return type:</i> float

worst_cell_delay_sigma	Returns the sigma component for worst cell delay in the timing paths. <pre>get_property [report_timing -collection] worst_cell_delay_sigma</pre> <i>Return type:</i> float
worst_cell_delay_skewness	Returns the skewness for worst cell delay in the timing paths. <pre>get_property [report_timing -collection] worst_cell_delay_skewness</pre> <i>Return type:</i> float
worst_cell_delay_stddev	Returns the standard deviation for worst cell delay in the timing paths. <pre>get_property [report_timing -collection] worst_cell_delay_stddev</pre> <i>Return type:</i> float
worst_cell_delta_delay	Returns the worst cell delta delay of the timing paths. <pre>get_property [report_timing -through clk_L16/ZN -unconstrained -collection] worst_cell_delta_delay</pre> <i>Return type:</i> float
worst_delay	Returns the worst delay for the timing paths. <pre>get_property [report_timing -collection] worst_delay</pre> <i>Return type:</i> float
worst_delay_mean	Returns the mean worst delay for the timing paths. <pre>get_property [report_timing -collection] worst_delay_mean</pre> <i>Return type:</i> float

worst_delay_sigma	Returns the sigma component for worst delay in the timing paths. <pre>get_property [report_timing -collection] worst_delay_sigma</pre> <i>Return type:</i> float
worst_delay_skewness	Returns the skewness for worst cell delay in the timing paths. <pre>get_property [report_timing -collection] worst_delay_skewness</pre> <i>Return type:</i> float
worst_delay_stddev	Returns the standard deviation for worst cell delay in the timing paths. <pre>get_property [report_timing -collection] worst_delay_stddev</pre> <i>Return type:</i> float
worst_delta_delay	Returns the worst delta delay for the timing paths. <pre>get_property [report_timing -through clk_L16/ZN -unconstrained -collection] worst_delta_delay</pre> <i>Return type:</i> float
worst_manhattan_length	Returns the worst manhattan length of the timing paths. <pre>get_property [report_timing -collection] worst_manhattan_length</pre> <i>Return type:</i> float
worst_manhattan_length_net_name	Returns the name of the net with worst manhattan length in the timing paths. <pre>get_property [report_timing -collection] worst_manhattan_length_net_name</pre> <i>Return type:</i> string
worst_net_delay	Returns the worst net delay for the timing paths. <pre>get_property [report_timing -collection] worst_net_delay</pre> <i>Return type:</i> float

worst_net_delay_mean	Returns the mean worst net delay for the timing paths. <pre>get_property [report_timing -collection] worst_net_delay _mean</pre> <i>Return type:</i> float
worst_net_delay_sigma	Returns the sigma variation for worst net delay for the specified timing paths. <pre>get_property [report_timing -retime path_slew_propagation -split_delay - path_type full_clock -collection] worst_net_delay_sigma</pre> <i>Return type:</i> float
worst_net_delay_skewness	Returns the skewness for the worst net delay for the specified timing paths. <pre>get_property [report_timing -retime path_slew_propagation -split_delay - path_type full_clock -collection] worst_net_delay_skewness</pre> <i>Return type:</i> float
worst_net_delay_stddev	Returns the standard deviation for the worst net delay for the specified timing paths. <pre>get_property [report_timing -retime path_slew_propagation -split_delay - path_type full_clock -collection] worst_net_delay_stddev</pre> <i>Return type:</i> float
worst_net_delta_delay	Returns the worst net incremental delay for the timing paths. <pre>get_property [report_timing -through clk_L16/ZN -unconstrained -collection] worst_net_delta_delay</pre> <i>Return type:</i> float

Timing Point Properties

adjustment_in_delay	Returns the adjustment in delay for the specified timing point of a timing path. <pre>get_property [get_property [get_property [report_timing -from dff0/Q -to dff2/D -path_type full_clock -collection] capture_clock_path] timing_points] adjustment_in_delay</pre> <i>Return type:</i> float
arrival	Returns the arrival time at the timing point. <i>Return type:</i> float
arrival_mean	Returns the mean component of the arrival time at the timing point. <i>Return type:</i> float
arrival_sigma	Returns the sigma component of arrival time at the timing point. <i>Return type:</i> float
arrival_skewness	Returns the skewness in arrival for a timing point. <pre>get_property [get_property [report_timing -collection] timing_points] arrival_skewness</pre> <i>Return type:</i> float
arrival_stddev	Returns the standard deviation of arrival time for a timing point. <pre>get_property [get_property [report_timing -collection] timing_points] arrival_stddev</pre> <i>Return type:</i> float
arrival_wire_stddev	Returns the standard deviation of arrival time due to wire variation for the specified timing point. <pre>get_property [get_property [report_timing -retime path_slew_propagation -split_delay -path_type full_clock - collection] timing_points] arrival_wire_stddev</pre> <i>Return type:</i> float
delay	Returns the delay value for the timing point. <i>Return type:</i> float
delay_mean	Returns the mean component of the delay value for the timing point. <i>Return type:</i> float

delay_sigma	Returns the sigma component of the delay value for the timing point. <i>Return type:</i> float
delay_skewness	Returns the standard deviation of slew due to wire variation for the specified timing points. <pre>get_property [get_property [report_timing -collection] timing_points] delay_skewness</pre> <i>Return type:</i> float
delay_stddev	Returns the standard deviation of delay for the timing points. <pre>get_property [get_property [report_timing -collection] timing_points] delay_stddev</pre> <i>Return type:</i> float
delay_wire_stddev	Returns the standard deviation of delay due to wire variation for the specified timing points. <pre>get_property [get_property [report_timing -retime path_slew_propagation -split_delay -path_type full_clock - collection] timing_points] delay_wire_stddev</pre> <i>Return type:</i> float
direction	Returns the direction of the timing points. <pre>get_property [get_property [report_timing -collection] timing_points] direction</pre> <i>Return type:</i> string
delta_delay	Returns the incremental delay on a timing point after SI analysis is performed. <i>Return type:</i> float
fanin_arc	Returns the delay arc for the corresponding timing point. <i>Return type:</i> collection
hierarchical_name	Returns the hierarchical name of the timing points. <pre>get_property [get_property [report_timing -collection] timing_points] hierarchical_name</pre> <i>Return type:</i> string

is_fanin_arc_cellarc	Returns a value of <code>true</code> if the fan in arc of the timing points is a cell arc. <pre>get_property [get_property [report_timing -through UMC/and1 -collection] timing_points] is_fanin_arc_cellarc</pre> <i>Return type:</i> boolean
is_hierarchical	Returns a value of <code>true</code> if the timing point is hierarchical. <pre>get_property [get_property [report_timing -through UMC/latch/D -unconstrained -collection] timing_points] is_hierarchical</pre> <i>Return type:</i> boolean
net	Returns the net for the timing points. <pre>get_property [get_property [report_timing -collection] timing_points] net</pre> <i>Return type:</i> collection
object_type	Returns the <code>timing_point</code> object type. <i>Return type:</i> string
pin	Returns a collection of pins or ports of the timing point. <i>Return type:</i> collection
retime_mis_delta	Returns the retime MIS (multi-input switching) delta delay for PBA paths. <i>Return type:</i> float
si_object	Returns the <code>si_victim</code> object type that can be used to query SI delay attributes. Note: To enable SI objects, you can set the <code>timing_report_enable_si_debug</code> global variable to <code>true</code>.
slew	Returns the path-specific slew of the timing point. <i>Return type:</i> float
slew_mean	Returns the mean component of the path-specific slew of the timing point. <i>Return type:</i> float
slew_sigma	Returns the sigma component of the path-specific slew of the timing point. <i>Return type:</i> float

slew_skewness	Returns the skewness of slew for the specified timing points. <pre>get_property [get_property [report_timing -collection] timing_points] slew_skewness</pre> <i>Return type:</i> float
slew_stddev	Returns the standard deviation of slew for the specified timing points. <pre>get_property [get_property [report_timing -collection] timing_points] slew_stddev</pre> <i>Return type:</i> float
slew_wire_stddev	Returns the standard deviation of slew due to wire variation for the specified timing points. <pre>get_property [get_property [report_timing -retime path_slew_propagation -split_delay -path_type full_clock -collection] timing_points] slew_wire_stddev</pre> <i>Return type:</i> float
spatial_derate	Returns the spatial derate on the timing points. <pre>get_property [get_property [report_timing -collection] timing_points] spatial_derate</pre> <i>Return type:</i> float
spatial_derate_sigma	Returns the sigma component of spatial derate on the timing points. <pre>get_property [get_property [report_timing -collection] timing_points] spatial_derate_sigma</pre> <i>Return type:</i> float
ssi_derate	Returns the path-specific SSI derate of the timing points. <pre>get_property [index_collection [get_property [report_timing - collection] timing_points] 4] ssi_derate</pre> <i>Return type:</i> float
total_derate	Returns the total derate on the timing points. <pre>get_property [get_property [report_timing -collection] timing_points] total_derate</pre> <i>Return type:</i> float

total_derate_sigma	Returns the sigma component of total derate on the timing points. <pre>get_property [get_property [report_timing -collection] timing_points] total_derate_sigma</pre> <i>Return type:</i> float
transition_type	Returns the transition type of the timing point - rise or fall. <i>Return type:</i> string
user_derate	Returns the derate set using the <code>set_timing_derate</code> command on the timing points. <pre>get_property [get_property [report_timing -collection] timing_points] user_derate</pre> <i>Return type:</i> float
user_derate_sigma	Returns the sigma variation derate set using the <code>set_timing_derate</code> command on the timing points. <pre>get_property [get_property [report_timing -collection] timing_points] user_derate_sigma</pre> <i>Return type:</i> float
voltage	Returns the voltage at a timing points. <pre>get_property [get_property [report_timing -collection] timing_points] voltage</pre> <i>Return type:</i> float
vtskew_mean_derate	Reports the mean derating factors of voltage threshold (VT) skew for a timing point. <i>Return type:</i> float
vtskew_sigma_derate	Reports the sigma derating factors of voltage threshold (VT) skew for a timing point. <i>Return type:</i> float

Signal Integrity Attributes

The signal integrity attributes can be queried using the following object types:

- `si_victim`
- `si_attacker`
- `noise_victim`: Returns a collection of `si_victim` objects that can be used to query glitch attributes.
Return type: collection

These are explained below.

si_victim Properties

The following information is saved under the `si_victim` object type:

Property Name	Description	Valid for Properties		
		si_victim	si_object	noise_victim
<code>annotated_glitch</code>	Reports the annotated glitch. <i>Return type: float</i>	N	N	Y
<code>attackers</code>	Returns a collection of <code>si_attacker</code> objects that can be used to query attacker attributes. <i>Return type: collection</i>	Y	Y	Y
<code>capacitance</code>	Returns the total capacitance of victim net. <i>Return type: float</i>	Y	Y	Y
<code>coupling_capacitance</code>	Returns the total coupling capacitance of a victim net. <i>Return type: float</i>	Y	Y	Y
<code>glitch_area</code>	Returns the glitch area. <i>Return type: float</i>	N	N	Y
<code>glitch_width</code>	Returns the glitch width. <i>Return type: float</i>	N	N	Y
<code>hierarchical_name</code>	Returns the hierarchical name of a victim net. <i>Return type: string</i>	Y	Y	Y

input_peak	Returns the Receiver Input Peak (RIP) value. <i>Return type: float</i>	N	N	Y
input_peak_slack	Returns the Receiver Input Peak (RIP) slack value. <i>Return type: float</i>	Y	N	N
input_failure_threshold	Returns the failure threshold at RIP. This will always be reported irrespective of whether failure happens at RIP or ROP. <i>Return type: float</i>	N	N	Y
level	Returns the noise level - VL or VH. <i>Return type: string</i>	N	N	Y
num_active_attackers	Returns the number of active attackers. <i>Return type: float</i>	Y	Y	Y
num_attackers	Returns the number of total attackers (distinct + accumulated small attacker - as 1). The attackers that are dropped from the accumulated small attackers will not be included with the total attackers. <i>Return type: float</i>	Y	Y	Y
object_type	Returns the object type <code>si_victim</code> . <i>Return type: string</i>	Y	Y	Y
prop_glitch	Reports the propagated glitch seen at the victim driver output. <i>Return type: float</i>	N	N	Y
receiver_peak	Returns the Receiver Output Peak (ROP) value. <i>Return type: float</i>	N	N	Y

receiver_peak_slack	Returns the Receiver Output Peak (ROP) slack value. <i>Return type:</i> string	Y	N	N
receiver_peak_threshold	Returns the failure threshold at ROP. This will always be reported irrespective of whether failure happens at RIP or ROP. <i>Return type:</i> float	N	N	Y
resistance	Returns the total resistance of a victim net. <i>Return type:</i> string	Y	Y	Y
transition	Returns the victim transition direction (rise or fall). <i>Return type:</i> string	Y	Y	N
view_type	Returns the type of analysis (early or late). <i>Return type:</i> string	Y	Y	N

si_attacker Properties

The following Information is saved under the `si_attacker` object type for each attacker of the victim net in the path:

Property Name (for each attacker retrieved from a victim)	Description
coupling_capacitance	Returns the coupling of a victim net. <i>Return type:</i> float
hierarchical_name	Returns the hierarchical name of an attacker. <i>Return type:</i> string
is_active	Returns whether an attacker is active or not. <i>Return type:</i> string

noise_peak	Returns the Xtalk delay bump induced by an attacker. <i>Return type:</i> float
object_type	Returns the <code>si_attacker</code> object type. <i>Return type:</i> string
slew	Returns the slew of attackers. <i>Return type:</i> float
state	Returns the status of attackers. The status descriptions are given below: • SY: Synchronous • INF: Attacker with infinite timing window • PE: Filtered due to physical exclusion • UF: Filtered by user • CN: Filtered because it is a constant net • SB: Filtered due to small bump • TA: Filtered because of timing window does not overlap • LC: Filtered due to logical correlation • NA: Attacker status is not known <i>Return type:</i> string

Examples

The following `get_property` example shows the usage of SI attributes to query noise information:

```
source pinlist.tcl
set chan [open attribute.log w]
foreach pin $pinList {
    set vicArray [get_property [get_pins $pin] noise_victim]
    foreach_in_collection vic $vicArray {
        set input_peak [ get_property $vic input_peak]
        #if { $input_peak > 250 } {}
    }
}
```

```

set vic_name [ get_property $vic hierarchical_name ]
set x [ get_property [get_net -of_objects $vic_name ] name ]
set level [ get_property $vic level]
set ccap [ get_property $vic capacitance]
set vdd [get_property [get_pins $vic_name] power_rail_voltage_max]
set libs [get_libs -of_objects [get_lib_cells -of_objects [get_cells -of_objects
[get_pins $vic_name]]]]
foreach_in_collection libName $libs {set library [get_property $libName
hierarchical_name]}
set prop_glitch [ get_property $vic prop_glitch]
set cap [format %.2f $ccap]
set annotated_glitch [ get_property $vic annotated_glitch]
set receiver_peak [ get_property $vic receiver_peak]
set input_peak_threshold [ get_property $vic input_failure_threshold ]
set receiver_peak_threshold [ get_property $vic receiver_peak_threshold]
puts $chan "-----"
-----"

puts $chan "Peak(mV) Level TotalCap(fF) Vdd(V) Library VictimNet"
puts $chan "$input_peak $level $cap $vdd $library $vic_name {$x}"

if { $receiver_peak > 0 } {
    puts $chan "Receiver output peak:"
    puts $chan "Value ReceiverNet"
    puts $chan "$receiver_peak ( $receiver_peak_threshold) $vic_name (CELL)"
}

if { $input_peak > 0 } {
    puts $chan "Receiver Input peak:"
    puts $chan "Value ReceiverNet"
    puts $chan "$input_peak ( $input_peak_threshold) $vic_name (CELL)"
}

puts $chan "\nConstituents:"
puts $chan "Source Peak(mV) Offset(ps) Slew(ps) Xcap(fF) Vdd(V) Edge Status Net"
set aggs [ get_property $vic attackers ]
foreach_in_collection current_agg $aggs {

    set hierarchical_name [ get_property $current_agg hierarchical_name ]
    set noise_peak [ get_property $current_agg noise_peak]
    set state [ get_property $current_agg state]
    set slew [ get_property $current_agg slew]
    set edge [ get_property $current_agg transition]
    if {[string match ~* $hierarchical_name]} {
        set vdd [get_property [get_pins $vic_name] power_rail_voltage_max]
    } else {
        set vdd [get_property [get_pins -of_objects [get_nets $hierarchical_name]]
power_rail_voltage_max]
    }
}

```

```

        set coupling_capacitance [ get_property $current_agg coupling_capacitance]

        puts $chan "Cpl: $noise_peak $slew $edge $vdd $coupling_capacitance $state
$hierarchical_name"
    }
    if { $prop_glitch > 0 } {
        puts $chan "Prp: $prop_glitch"
    }

}
close $chan

```

For more information on querying SI attributes, refer to [Signal Integrity Delay and Glitch Analysis](#) chapter in the *Tempus User Guide*.

User-Defined Properties

The user-defined properties allow you to define certain properties which can be interpreted by the software in a predefined way. When these properties are not specified, the software will revert to the default behavior, as applicable.

Property Name	Type	Description
capacitive_load_unit	lib	Provides the load unit mentioned in the library.
time_unit	lib	Provides the time unit mentioned in the library.
aocv_weight	lib_cell	Defines AOCV weight for a given cell.
k_input_switching_window_fall*	lib	Defines the switching window threshold for fall signal during the application of SSI (simultaneous switching input) derating feature. <i>Default: 1</i>
k_input_switching_window_rise*	lib	Defines the switching window threshold for rise signal during the application of SSI (simultaneous switching input) derating feature. <i>Default: 1</i>

* When the above properties are defined in SSI derating environment, the delay of the respective cell multiplied by the above factor will be compared with the difference in (rise/fall) signal arrivals at two inputs to determine if switching derate needs to be applied to cell delay.

Examples

- The following command returns the hierarchical name for the pin contained in the collection referenced by \$pin:

```
get_property $pin hierarchical_name
```
- The following example shows how you can query a user-defined property from stage_weighting library at the cell level:

```
set_property [get_lib_arcs -of fast/BUFX2] stage_weighting 3 // sets the value
get_property [get_lib_arcs -of fast/BUFX2] stage_weighting
```
- The following example shows power and ground pins for the corresponding library pins:

```
get_property [get_lib_pins CELLA/A] related_power_pin
get_property [get_lib_pins CELLA/A] related_ground_pin
```
- The following example shows active clocks:

```
get_property [get_clocks Iclk1] is_active true
```
- The following commands define the switching window threshold for rise signal:

```
define_property -type float -object_type lib k_input_switching_window_rise
set_property [get_libs slow] k_input_switching_window_rise 1.1
```
- The following example shows that the phase shift is 2.0 ns, as returned by the "period" property:

```
report_timing -clock_to CLK_W_3 -clock_from CLK_W_3

Path 1: MET Setup Check with Pin seg3/u14/CK
Endpoint: seg3/u14/D (^) checked with leading edge of 'CLK_W_3'
Beginpoint: seg3/u9/Q (^) triggered by leading edge of 'CLK_W_3'
Path Groups: {CLK_W_3}
Other End Arrival Time 0.621
- Setup 0.186
+ Phase Shift 2.000
= Required Time 2.434
- Arrival Time 2.062
= Slack Time 0.372
Clock Rise Edge 0.000
+ Clock Network Latency (Prop) 0.682
= Beginpoint Arrival Time 0.682
.....
tempus > set path [rt -clock_to CLK_W_3 -clock_from CLK_W_3 -collection]
0x14
tempus > get_property $path period
2.000
```
- In this following example, there is a clock reference for launch clock (Null phase- @), so Tempus does not show any clock period for this path:

```
report_timing -clock_to CLK_W_3 -early
```

```
Path 1: VIOLATED Hold Check with Pin seg3/u3/CK
Endpoint: seg3/u3/D (v) checked with leading edge of 'CLK_W_3'
Beginpoint: IN3 (v) triggered by leading edge of '@'
Path Groups: {CLK_W_3}
Other End Arrival Time 0.682
+ Hold -0.034
+ Phase Shift 0.000
= Required Time 0.648
Arrival Time 0.117
Slack Time -0.531
Clock Rise Edge 0.000
+ Input Delay 0.000
+ Drive Adjustment -0.014
= Beginpoint Arrival Time -0.014
```

```
-----
```

Load	Slew	Delay	Incr	Arc	Fanout	Cell	Arc	User	Arrival
Instance									
			Delay	Annotation				Derate	Time

0.013	0.029			WLM	1	IN3	v		-0.014
0.013	0.057	0.054	0.000	WLM	1	CLKBUF	A v -> Y v	1.000	0.040
seg3/u1									
0.003	0.050	0.077	0.000	WLM	1	CLKBUF	A v -> Y v	1.000	0.117
seg3/u2									
0.003	0.050	0.000	0.000	WLM		DFF	D v	1.000	0.117
seg3/u3									

- tempus > set path [report_timing -clock_to CLK_W_3 -collection -early]
0x16
tempus > get_property \$path period
NA

- The following example shows if a real or generated clock is defined on the pins or ports:

```
tempus > create_clock [get_ports {CLK_t}] -name clk -period 100 -waveform {0 50}
tempus > create_generated_clock -divide_by 2 -source CLK -name d_2 CLK_b
tempus > get_property [get_ports CLK_t] is_clock_root
true
```

```
tempus > get_property [get_pins CLK_b] is_clock_root  
true
```

Related Commands

- `report_property`
- `set_property`
- `reset_property`
- `list_property`

index_collection

```
index_collection  
[-help]  
base_collection  
index
```

Returns a collection containing the object that exists at the specified index of the specified object collection.

Parameters

<code>-help</code>	Prints out the command usage.
<code>base_collection</code>	Specifies the base collection.
<code>index</code>	Specifies the index number. You can specify a number from 0 to <code>collection_size-1</code> .

Examples

- The following command returns the object that exists at index number 2 in the object collection referenced by `$nets`:

```
index_collection $nets 2
```

list_property

```
list_property
[-help]
[-type {cell clock design lib lib_cell lib_pin lib_timing_arc net path_group pin port
timing_arc timing_path timing_point pg_net pg_pin lib_pg_pin si_victim si_attacker}]
```

Lists all of the properties associated with the specified object type.

If you do not specify any object type, this command lists all the supported object types and all the properties associated with those object types.

Parameters

-help	Prints out the command usage.
-type <i>object_type</i>	Lists the properties associated with the specified object type. The following object types are supported: cell, clock, design, lib, lib_cell, lib_pin, lib_timing_arc, net, path_group, pin, port, timing_arc, timing_path, timing_point, pg_net, pg_pin, lib_pg_pin, si_victim, si_attacker Additional SI properties can be included with the <code>timing_point</code> object. To enable SI properties, you can set the <code>timing_report_enable_si_debug</code> global variable to <code>true</code>.

Related Commands

- `define_property`
- `get_property`
- `report_property`
- `reset_property`
- `set_property`

parse_proc_arguments

```
parse_proc_arguments
-help
-args string
result_array
[forced_proc]
```

Supports argument validation within a TCL procedure and enables the use of the `-help` option to view the help information associated with the procedure.

Procedures using the `parse_proc_arguments` command will validate the arguments. If the required arguments are missing or have invalid values, the `parse_proc_arguments` command generates a TCL error, and the calling procedure terminates.

The `parse_proc_arguments` command must be the first command called in the procedure. This command cannot be used outside of a procedure.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-args <i>string</i></code>	Specifies the list of arguments passed to the TCL procedure.
<code>result_array</code>	Specifies the result array into which the parsed arguments are to be stored.
<code>forced_proc</code>	Specifies the name of the invoking procedure.

Examples

The following TCL procedure uses the `parse_proc_arguments` command for argument validation.

```
proc cmdProc {args} {
  parse_proc_arguments -args $args results
  foreach argname [array names results] {
    puts "$argname = $results($argname)"
  }
}

define_proc_arguments cmdProc -info "Command Description" \
-define_args \
{{-string "String help" "string_val" string optional}}
```

```
{-integer "Int help" "int_value" int optional}

{-list "List help" "list_value" list {optional bind_option -Bool}}

{-boolean_a "BoolA help" "" boolean {optional mutual_exclusive_group g0}}

{-boolean_b "BoolB help" "" boolean {optional mutual_exclusive_group g0}}

{-float_num "Float help" "float_value" float}

{-oosa "One of StringA help" "one_of_str_value" one_of_string {optional value_help {values {v1 v2}}}}

{-oosb "One of StringB help" "one_of_str_value" one_of_string {optional {values {v4 v5}}}}

}
```

Now, if the `cmdProc` procedure is invoked using the following command at the command prompt, argument validation takes place, and the following errors are reported.

```
tempus> cmdProc -boolean_a -boolean_b

**ERROR: (Tempus-39): Options '-boolean_a -boolean_b' are mutually exclusive for 'cmdProc'.

Usage: cmdProc [-string string_val] [-integer int_value] [-list list_value] [-boolean_a ]
[-boolean_b ] [-float_num float_value] [-oosa one_of_str_value {v1 v2 }] [-oosb
one_of_str_value]]

**ERROR: (Tempus-27): Errors in 'cmdProc'.
```

The following command to invoke the `cmdProc` procedure will also return the errors shown below.

```
tempus> cmdProc -oosa v5

**ERROR: (Tempus-42): The arguments specified (v5) in cmdProc -oosa are not valid. Refer to
the documentation for valid arguments.

Usage: cmdProc [-string string_val] [-integer int_value] [-list list_value] [-boolean_a ]
[-boolean_b ] [-float_num float_value] [-oosa one_of_str_value {v1 v2 }] [-oosb
one_of_str_value]]

**ERROR: (Tempus-27): Errors in 'cmdProc'.
```

query_objects

```
query_objects  
[-help]  
collection  
[-limit value]
```

Displays the names of the objects contained inside the specified collection.

Note: The `query_objects` command displays object names only. To return a TCL list of object names, you can use the `get_object_name` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>collection</code>	Specifies the object collection whose names to print.
<code>-limit value</code>	Specifies the maximum number of object names to print. Specify a value of 0 to query the names of all of the objects in the collection. <i>Default:</i> Prints the first 100 object names in the collection.

Examples

- The following commands create collections of pins and nets, add them together to create a new collection, and print the names of the objects in the collection:

```
tempus > set pins [get_pins {i_block1/b1_Sin1 i_block1/b1_CLK1 i_block1/b1_Sout1}]  
tempus > set nets [get_nets {Sin1 CLK1 Sout1}]  
tempus > add_to_collection newVar $pins  
tempus > add_to_collection newVar $cells  
tempus > query_objects $newVar
```

The software returns the following information:

```
i_block1/b1_Sin1 i_block1/b1_CLK1 i_block1/b1_Sout1 Sin1 CLK1 Sout1
```

Related Commands

- [get_object_name](#)

range_collection

```
range_collection
[-help]
base_collection
from
to
```

Returns a sub-collection of the base collection. The sub-collection range lies between the ‘from’ and ‘to’ (including ‘from’ and ‘to’) index numbers specification of this command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>base_collection</code>	Specifies the base collection.
<code>from</code>	Specifies the ‘from’ index number.
<code>to</code>	Specifies the ‘to’ index number.

The index number can be an integer or an expression, such as “end OR end-integer” (with no spaces within the expression).

If the specified to/from value is negative (less than 0), the software will set it to 0. If the value is greater than the “collection_size – 1”, then it will be changed to “collection_size – 1”.

Examples

- The following example sets `$points` as the base collection:

```
tempus > set points [get_cells *]
0x13
tempus > sizeof_collection $points
22
```
- The following example returns the sub-collection of the base collection `$points` between the index number range from 2 to 4:

```
tempus > range_collection $points 2 4
clk_buf2 clk_buf3 clk_buf4
0x16
tempus > range_collection $points 4 2
clk_buf2 clk_buf3 clk_buf4
0x17
```

- The following example returns a sub-collection starting from index number 0 till the end of the base collection:

```
tempus > range_collection $points 0 end
clk_buf1 pad1 clk_buf2 clk_buf3 clk_buf4 buf1 buf2 buf3 buf4 mem dff1 dff2 clkand2
pu_cell_1 pu_lat_1 u1 cg4 genLoop comboLoop pad2 dff3 latch1
0x18
```

- The following example returns a sub-collection starting from end-integers 4 to 2:

```
tempus > range_collection $points end-4 end-2
genLoop comboLoop pad2
0x19
```

- The following example shows a range starting from a negative value to a value greater than the collection size. The following warning messages are displayed:

```
tempus > range_collection $points -5 100
**WARN: (TCLCMD-1419): 'from' index with value -5 is less than start index, so it is
changed to 0.
**WARN: (TCLCMD-1419): 'to' index with value 100 is greater than end index, so it is
changed to 21.
clk_buf1 pad1 clk_buf2 clk_buf3 clk_buf4 buf1 buf2 buf3 buf4 mem dff1 dff2 clkand2
pu_cell_1 pu_lat_1 u1 cg4 genLoop comboLoop pad2 dff3 latch1
0x1a
```

- The following example shows the warning message displayed for a negative start index number:

```
tempus > range_collection $points end-25 end-20
**WARN: (TCLCMD-1419): 'from' index with value -4 is less than start index, so it is
changed to 0.
clk_buf1 pad1
0x1b
```

- The following command shows an error if the specified index number is not an integer:

```
tempus > range_collection $points s end
**ERROR: (TCLCMD-1418): Bad index 's': must be integer? or end?[-]integer?
```

Related Commands

- [index_collection](#)

remove_from_collection

```
remove_from_collection  
[-help]  
base_collection  
object_collection_or_list  
[-intersect]
```

Creates a new collection by removing the objects specified in one collection from another collection.

Parameters

<code>-help</code>	Prints out the command usage.
<i>base_collection</i>	Specifies the base collection of objects. Objects from the second collection or object list are removed from this collection to create the new collection.
<code>-intersect</code>	Provides the intersection of base collection elements with a second collection of objects or lists.
<i>object_collection_or_list</i>	Specifies the collection or list of objects that should be removed from the base collection to create a new collection. When specified, this parameter returns those elements that intersect and exist in both the <i>base_collection</i> and the second collection. Also, all the elements of <i>base_collection</i> that do not intersect with the second collection will be removed.

Examples

- The following commands create a collection of pins that does not include clock pins:

```
set allP [get_pins *]  
set all_pin_no_clock [remove_from_collection $allP [get_pins -filter "is_clock ==  
true"]]
```

- You can use the `sizeof_collection` command to see that the `all_pin_no_clock` collection contains fewer objects (that is, no clock pins) than the original `allP` collection:

```
sizeof_collection $allP  
50  
sizeof_collection $all_pin_no_clock  
47
```

report_property

```
report_property
[-help]
{collection | list_of_collections}
[-property_list list of properties]
[> | >>] filename
```

Reports the properties associated with each object in the specified collection.

For properties that return a populated collection, the command output displays {...}. For any empty collections,{} will be displayed.

Parameters

-help	Prints out the command usage.
collection list_of_collections	Specifies a collection, or a list of collections, to report. The collections can contain any of the following object types: cell, clock, design, lib, lib_cell, lib_pin, lib_pg_pin, pg_net, pg_pin, lib_timing_arc, net, path_group, pin, port, timing_arc, timing_path, or timing_point
-property_list list of properties	Specifies a list of valid property names to report. Note: If the -property_list parameter is not specified, then all the properties associated with a collection of objects will be reported.
> >> filename	When specified with >, the software writes the report to the specified file name. If the file already exists, the software overwrites it. When specified with >>, the software writes the report to the specified file name. If the file already exists, the software concatenates the report to the end of the file. You can add the .gz extension to the file name to generate a compressed report. Note: The file name must be the last argument in the list.

Examples

- The following command shows the values of the specified properties:
`report_property [get_pins TU11/A] -property_list {name direction clocks}`

property		value

name		CK
direction		in
clocks		{...}

Related Commands

- `get_property`
- `define_property`
- `list_property`
- `set_property`
- `reset_property`

sizeof_collection

```
sizeof_collection  
[-help]  
collection
```

Returns the total number of objects contained in the specified collection. If the collection contains objects of different types, `sizeof_collection` returns the sum of all the objects of different types in the collection.

Parameters

<code>-help</code>	Prints out the command usage.
<i>collection</i>	Specifies the collection.

sort_collection

```
sort_collection
[-help]
collection
[-dictionary]
{property_or_list_of_properties}
[-descending]
```

Returns a sorted collection of objects based on the property.

Parameters

-help	Prints out the command usage.
collection	Specifies the collection to sort. The collection must be homogeneous, consisting of any one of the following object types: cell, clock, design, lib, lib_cell, lib_pin, lib_timing_arc, net, path_group, pin, port, timing_arc, timing_path, or timing_point.
-descending	Sorts the objects in the collection in descending order. <i>Default:</i> Sorts the collection in ascending order.
-dictionary	Sorts the objects in a collection by dictionary order.
property_or_list_of_properties	Sorts the collection by the specified property. You can specify any property associated with the object type. You can specify one or more properties. If you specify more than one property, the software sorts the collection in the order in which you specified the properties.

Examples

- The following command returns a collection of all the input ports in the design, sorted based on their hierarchical names:

```
tempus > sort_collection [all_inputs] hierarchical_name
```



```
CLK1 CLK2 CLK3 CLK4 IN1 IN2 IN3 IN4
```

Signal Integrity Commands

- [analyze_jitter](#)
- [check_noise](#)
- [create_aging_stress_profile](#)
- [create_spice_deck](#)
- [define_bundled_bus](#)
- [define_glitch_holding_resistance](#)
- [define_parameter](#)
- [get_aging_activity](#)
- [get_aging_analysis_mode](#)
- [get_delay_cal_mode](#)
- [get_glitch_threshold](#)
- [get_multi_input_switching_mode](#)
- [get_si_mode](#)
- [get_via_variation_mode](#)
- [get_wire_variation_mode](#)
- [merge_noise_reports](#)
- [report_aging_activity](#)
- [report_aging_stress_profile](#)
- [report_bundled_bus](#)
- [report_double_clocking](#)
- [report_glitch_thresholds](#)
- [report_instance_cdb](#)
- [report_jitter](#)
- [report_mutex_condition](#)
- [report_noise](#)

- [report_noise_lib_pin](#)
- [report_noise_propagation](#)
- [report_parameter](#)
- [report_voltage_scaling](#)
- [report_voltage_swing](#)
- [reset_glitch_derate](#)
- [reset_noise_lib_pin](#)
- [set_advanced_variation_mode](#)
- [set_aging_activity](#)
- [set_aging_analysis_mode](#)
- [set_annotated_glitch](#)
- [set_cdb_binding](#)
- [set_delay_cal_mode](#)
- [set_glitch_derate](#)
- [set_glitch_threshold](#)
- [set_model_priority](#)
- [set_mutex_condition](#)
- [set_noise_lib_pin](#)
- [set_outbound_report](#)
- [set_quiet_attacker](#)
- [set_si_mode](#)
- [set_via_variation_mode](#)
- [set_wire_variation_mode](#)
- [update_glitch](#)
- [write_noise_eco](#)

analyze_jitter

```
analyze_jitter
[-help]
[-cycle_interval integer]
[-eiv_end_time float]
[-eiv_file string]
[-eiv_start_time float]
[-eiv_vdd_vss string]
[-enable_spectre_x {true | false}]
[-force_delete_output_dir]
[-from string]
[-from_fall string]
[-from_rise string]
[-levels_combine integer]
[-lsf_job_string string]
[-mode {period | cycle_to_cycle | time_interval_error | backward_compatible}]
[-model_file string]
[-no_hard_stop_at_to_pins {true | false}]
[-num_cycle integer]
[-outdir string]
[-run_simulation]
[-skip_stage_fanout string]
[-spectre string]
[-spectre_cmd_options string]
[-subckt_file string]
[-start_cycle integer]
[-through string]
[-to string]
[[ > | >> ]]
```

Allows the software to perform Spice jitter analysis on a clock network using a user-specified EIV database. The EIV database stores the effective instance voltages (EIV) for each cycle across different instances in the design.

The software reads the EIV files and identifies the worst two consecutive cycles for each instance where the voltage variation is maximum. The voltage levels from these identified cycles are used as the corresponding supplies for different instances in the generated Spice deck, which is simulated using the Spectre simulator to compute cycle-to-cycle IR-induced jitter.

Parameters

-help	Prints out the command usage.
-------	-------------------------------

<code>-cycle_interval</code>	Specifies the cycle interval to use when selecting the EIV values for each instance in jitter computation. The default value is 1, where all the EIV values are considered for jitter computation. When 'N' is specified, the <code>analyze_jitter</code> command considers only the EIV entries in step size of 'N', starting from the value specified by the <code>-start_cycle</code> parameter.
<code>-eiv_end_time float</code>	Specifies the end time for EIV reading. The voltages specified in the EIV database for timestamps beyond this value are ignored. <i>Default:</i> The last timestamp from the EIV file is considered as the end time.
<code>-eiv_file string</code>	Specifies the effective instance voltage files for accounting for IR Drop in jitter computation.
<code>-eiv_vdd_vss string</code>	Specifies the power and ground net names for each EIV file specified using the <code>-eiv_file</code> parameter. For example, "vdd1 vss1 vdd2 vss2", vdd1 and vss1 are for the first, and vdd2 and vss2 for the second EIV file defined using the <code>-eiv_file</code> parameter.
<code>-eiv_start_time float</code>	Specifies the start time for EIV reading. The voltages specified in the EIV database for time stamps before this value are ignored. <i>Default:</i> The first timestamp in the EIV file is considered the start time.
<code>-enable_spectre_x</code>	Enables SpectreX license checkout when launching Spectre.
<code>-force_delete_output_dir</code>	Completely removes the directory specified by the user using the <code>-outdir</code> parameter.
<code>-from string</code>	Creates a Spice deck of the clock network starting from the specified pin.
<code>-from_fall string</code>	Creates a Spice deck of the clock network starting from the falling transition of the specified pin.
<code>-from_rise string</code>	Creates a Spice deck of the clock network starting from the rising transition of the specified pin.

<code>-levels_combine integer</code>	Specifies the number of levels, going up from the leaf nodes, at which the sub-trees can be combined. By default, the whole clock network is simulated as a single Spice netlist for jitter analysis. But for large networks, running a single simulation can significantly increase run time. When you use this parameter, the software partitions the clock network into multiple subtrees and runs Spice simulations in parallel. For the following values, the behavior is as follows: • 0: Divides the network such that the path traversed from a clock source to each leaf node is part of a separate sub-tree. • 1: Each sub-tree contains the path traversed from a clock source to one level up of a leaf node. And all the leaf nodes in the fanout of the identified level are part of this sub-tree, and so on. Note: The <code>-max_spectre_lic</code> parameter must be set to a value greater than 1 to enable this parallelization.
<code>-lsf_job_string string</code>	Specifies the handle to the farm job used to launch the simulation.
<code>-max_spectre_lic integer</code>	Specifies the number of available Spectre licenses for simulation parallelization. <i>Default:</i> 1 license is assumed to be available, thereby disabling parallelization.
<code>-mode {period cycle_to_cycle time_interval_error backward_compatible}</code>	Runs the <code>analyze_jitter</code> command natively - with the specified mode. <i>Default:</i> <code>period</code>
<code>-model_file string</code>	Specifies the path of Spice model files that contain the Spice model information.
<code>-no_hard_stop_at_to_pins {true false}</code>	When set to <code>false</code>, the Spice deck is generated only for the traced paths starting from the <code>-from</code> pin and ending at the <code>-to</code> pin. It does not traverse through the fanout branches that do not lead to the <code>-to</code> specified pin. When set to <code>true</code>, traversal occurs through the whole fanout cone of the <code>-from</code> pin till it hits a clock pin of a flop, or any stop point specified with the <code>-to</code> parameter <i>Default:</i> <code>false</code>

<code>-num_cycle</code>	Specifies the number of cycles for which jitter computation needs to be performed. By default, all the available EIV cycles in the input EIV file are considered for jitter computation.
<code>-outdir <i>string</i></code>	Specifies the output directory for the created Spice decks and the simulation output files. <i>Default:</i> tempus_pathsim
<code>-run_simulation</code>	Enables jitter simulation and generates the Spice deck.
<code>-skip_stage_fanout <i>string</i></code>	Accepts two positive integers in the format "N1 N2", and scans up to N1 stages in the fan-in cone of each endpoint. The software skips the segments where a stage is found to have a fanout greater than N2. For example, a jitter path Clk_root1 -> ... -> B3 -> B2 -> B1 -> Reg2, B2 and B1 are the last 2 levels. 1) Assume B1's fanout is 100 and B2's fanout is 80, then • When N1 = 1, N2 = 80, B1 will be skipped. • When N1 = 2, N2 = 100, B1 will be skipped • When N1 = 2, N2 = 80, B2 & B1 will be skipped Note: B2 will always cause B1 to be skipped even when B1's fanout is less than 80. So the reduction continues to increase when the level changes from 1 to 2. 2) Assume B1's fanout is 80, and B2 fanout is 100, then, • When N1 = 2, N2 = 90, both B2 and B1 will be skipped.
<code>-spectre <i>string</i></code>	Specifies the location of the Spectre path.
<code>-spectre_cmd_options <i>string</i></code>	Specifies the command-line options for Spectre.
<code>-start_cycle <<i>integer</i>></code>	Specifies the EIV cycle to be used as the starting point for jitter computation. All EIV cycles before this point will be ignored. The default value is 1.
<code>-subckt_file <i>string</i></code>	Specifies the file(s) containing subckt definitions of all the instances for which the Spice simulation needs to be performed.
<code>-through <i>string</i></code>	Creates the Spice deck of a clock network that starts from the <code>-to</code> pin and traverses through the specified pin(s).

<code>-to string</code>	Creates the Spice deck of a clock network terminating at the specified pin(s).
<code>> filename</code>	Redirects the <code>-run_simulation</code> output to a file.
<code>>> filename</code>	Appends the <code>-run_simulation</code> output to a file.

Examples

- The following command generates the Spice deck for jitter analysis of the entire clock network starting from the `CLK` pin in the `JITTER_SPICE` directory, and runs the simulations:

```
tempus> analyze_jitter -from CLK -no_hard_stop_at_to_pins true -outdir JITTER_SPICE -
eiv_file ./eiv.txt -eiv_vdd_vss {vdd vss} -run_simulation
```

When the Spice simulations are completed, a summary of the jitter analysis will be saved in the output directory that contains the jitter values of all the endpoints, as shown below:

```
-----
-----
Start-point-> [through-point-> through-point] -> End-point Rise/Fall Cycle-to-Cycle
Jitter Value (ps)
-----
-----
CLK -> DFF_2/DFF/D (F) 10.782
CLK -> DFF_2/DFF/D (R) 11.072
CLK -> DFF_2/DFF/CP (R) 9.054
CLK -> DFF_2/DFF/CP (F) 9.438
```

- The following command runs jitter analysis for the clock network that originates from the `CLK` node, and traverses through an intermediate node `INV/ZN`:

```
tempus> analyze_jitter -from CLK -through INV/ZN -no_hard_stop_at_to_pins true -outdir
JITTER_SPICE -eiv_file ./eiv.txt -eiv_vdd_vss {vdd vss} -run_simulation -levels_combine
1 -max_spectre_lic 8
```

The specified clock network is not simulated as a single netlist, but is broken down into multiple sub-trees. All the leaf nodes driven by a single driver are part of one sub-tree. The simulations on these smaller netlists are scheduled in such a way that a maximum of 8 simulations are run in parallel at any given time.

Related Commands

- `report_jitter`

check_noise

```
check_noise
[-help]
[-all]
[-check_only {receiver_no_propagation_model | receiver_no_dc_tolerance |
driver_no_current_source_model | driver_no_model}]
[-instance_list string]
[-instance_only {true | false}]
[-pin_list string]
[-type {receiver_model | driver_model}]
[-unique {true | false}]
[-verbose]
[-view string]
[ [ > | >> ] filename]
```

Performs several consistency and completeness checks on noise models specified for a design. These checks help ensure that the correct set of models is available for the SI analysis engine to perform crosstalk noise analysis.

Note: You need to set `set_delay_cal_mode -siAware` to `true` before running the `check_noise` command.

The settings from both the `set_glitch_threshold` and `set_annotated_glitch` commands are taken into account when the `check_noise` command is issued.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-all</code>	Displays all the debug information.
<code>-check_only</code> {receiver_no_propagation_model receiver_no_dc_tolerance driver_no_current_source_model driver_no_model}	Checks the warning name. The following warnings are checked: • <code>receiver_no_propagation_model</code>: Shows how many design pins are missing the receiver model for propagation. • <code>receiver_no_dc_tolerance</code>: Shows how many pins have no DC VH/VL tolerance specified. • <code>driver_no_current_source_model</code>: Shows how many design pins are missing the current source model. • <code>driver_no_model</code>: Specifies how many design pins are missing the driver model.
<code>-instance_list string</code>	Specifies a list of instance names.
<code>-instance_only {true false}</code>	Prints each instance only once.
<code>-pin_list string</code>	Specifies the instance pin names.
<code>-type {receiver_model driver_model}</code>	Checks the presence of a valid model type - <code>receiver_model</code> or <code>driver_model</code> . If the specified model is missing, a warning will be issued. The valid receiver models are those with vivo tables, such as CDB, CCSN, and ECSMN. The valid driver models are CDB, CCS, ECSM, NLDM, and UDN.
<code>-unique {true false}</code>	Reports unique cells or pins for each power domain.
<code>-verbose</code>	Generates verbose details.
<code>-view string</code>	Specifies the view name.
<code>filename</code>	Redirects output to a file.
<code>>> filename</code>	Appends to an output file.

Examples

Here are some examples.

- The following command generates the output as shown below:

```
tempus> check_noise -type receiver_model

**WARN: (IMPESI-3125): AAE_WARN: The analysis view specified to check_noise is not found
or specified. Checking view default_emulate_view_setup.

-----
CHECK NOISE SUMMARY
-----
Warning Number of Warning
-----
receiver_no_propagation_model 17931
receiver_no_dc_tolerance 17931
-----
# CHECK NOISE WARNING DESCRIPTIONS:
# receiver_no_propagation_model: The receiver model for glitch
# propagation or ROP is not available, only RIP analysis will be performed.
# receiver_no_dc_tolerance: The input pin do not have vh/vl dc
# tolerance specified in the noise file. Default tolerences will be used.
-----
```

Related Commands

- `check_timing`
- `set_delay_cal_mode`
- `get_delay_cal_mode`

create_aging_stress_profile

```
create_aging_stress_profile
[-help]
[-views {views}]
[-early]
[-late]
[-bti_pmos_scale <scale_val>]
[-bti_nmos_scale <scale_val>]
[-hci_pmos_scale <scale_val>]
[-hci_nmos_scale <scale_val>]
{segment1 segment2 ...}
```

Defines the stress profile for aging-aware STA. The profile contains stress duration, stress temperature, and stress voltage per power rail.

You can apply different stress profiles to different views and to early/late timing.

Parameters

-help	Prints out the command usage.
-early	Specifies the early timing path.
-late	Specifies the late timing path.
-bti_pmos_scale <scale_val>	Specifies the BTI PMOS scale value.
-bti_nmos_scale <scale_val>	Specifies the BTI NMOS scale value.
-hci_pmos_scale <scale_val>	Specifies the HCI PMOS scale value.
-hci_nmos_scale <scale_val>	Specifies the BTI NMOS scale value.
{segment1 segment2 ...}	Specifies the stress duration, voltages, and temperature information. Here, the <i>segment1</i> , <i>segment2</i> ... specifies the stress duration, voltage, and temperature information, for example, {{duration <float>}}[{{voltage {<pg_net1> <float> <pg_net2> <float> ...}}} [{temperature <float>}}]}
-views {views}	Specifies the views for which the stress profile is to be created.

Examples

- The following command specifies a multi-segment stress profile consisting of stress duration, stress temperature, and stress voltage for view “view1”.

```
tempus > create_aging_stress_profile -view view1 \  
{{{ duration 3} {voltage {VDD 0.8 VSS 0.0}}{temperature 125}} } \  
{{duration 7} {voltage {VDD 0.45 VSS 0.0}}{temperature 25}}}
```

You can specify the initial 3 years of aging with one stress condition and the rest of the 7 years with different stress conditions - for a total of 10 years of aging:

- The following command specifies the stress profile along with BTI and HCI scales for PMOS and NMOS.

```
tempus > create_aging_stress_profile { { { duration 10 } {voltage {VDD 1.05 VPW 0.0 VNW \  
\ 1.05 VSS 0.0 }} {temperature 125 } } } -bti_pmos_scale 0.5 -bti_nmos_scale 0.5 \  
-hci_pmos_scale 1 -hci_nmos_scale 1
```

Related Commands

- `set_analysis_mode -aging`
- `set_aging_analysis_mode`
- `set_aging_activity`
- `get_aging_activity`
- `report_aging_activity`
- `create_spice_deck`

Related Information

"[Aging-Aware STA Analysis](#)" chapter in the *Tempus User Guide*.

create_spice_deck

```
create_spice_deck
[-help]
[-add_driver_for_constraint_check {true | false}]
[-add_side_load_measurement {true | false}]
[-add_voltage_for_missing_sensitization]
[-aging_correlation_type {full | timingonly}]
[-aging_stress_num_cycles <integer greater than 0>]
[-aging_stress_use_timing_slew {false | true}]
[-aginglevel_only <string>]
[-bbox_cells string]
[-constraint_check {true | false}]
[-depth integer]
[-enable_arrival_measurements {true | false}]
[-enable_fast_simulation {true | false}]
[-enable_mis {tied_input | independent_input | none}]
[-enable_spectre_x {true | false}]
[-exclude_side_load_cells string]
[-exclude_side_load_cells_for_missing_subckt {true | false}]
[-force_delete_output_dir]
[-file_prefix string]
[-format string]
[-from string]
[-from_fall string]
[-from_rise string]
[-generate_output_only {true | false}]
[-glitch_simulation_mode {input_output | output}]
[-ground string]
[-include_cell_variation {true | false}]
[-lsf_job_string string]
[-mc_simulation_init_args string]
[-mc_simulation_tran_args string]
[-measure_point {all | path | stage}]
[-merge_spice_deck {true | false}]
[-model_file string]
[-no_hard_stop_at_to_pins {true | false}]
[-omipath <path>]
[-outdir string]
[-power string]
[-receiver_pin string]
[-reliability_model_type {URI | OMI | TMI}]
[-reliability_commands string]
[-report_noise string]
[-report_timing string]
```

```

[-run_simulation]
[-scale_mode string]
[-segment {data_path | launch_path | launch_clock_path | capture_path all}]
[-segment_end string]
[-segment_start string]
[-side_path_level integer]
[-spectre string]
[-spectre_analysis_options <string>]
[-spectre_cmd_options string]
[-spectre_lang_escape_chars <string>]
[-spice_include string]
[-stimuli_file string]
[-subckt_file string]
[-supply_voltage_file string]
[-supply_voltage_file_stress <file>]
[-sweep_range float]
[-sweep_steps integer]
[-sweep_victim {true | false}]
[-temperature float]
[-through string]
[-to string]
[-uri_lib_file string]
[-uri_mode <string>]
[-use_control_param {true | false}]
[-use_distributed_load_for_side_receivers {true | false}]
[-user_defined_measure_statements {append | overwrite}]
[-user_defined_measure_statements_file string]
[-use_ndw_for_constraint_check {true | false}]
[-use_noise_model_when_cond {true | false}]
[-waveform_aware_pulse_width_checks {true | false}]
[-view string]
[-xtalk {delay | vl_glitch | vh_glitch | vlu_glitch | vho_glitch}]
[[ > | >> ]]

```

Generates the SPICE trace for a path.

While generating a Spice Deck, the `create_spice_deck` command looks for library cells required to run Spice and adds all cells as ".include" in the generated Spice netlist.

The output Spice deck includes:

- Active and passive devices such as field-effect transistors (FETs), capacitors, and resistors.
- Initial conditions and voltage sources
- Actual names of the gates, nets, and cell instances, which help the simulator interpret this information effectively. For example, consider the following excerpt, which shows an entry for a gate named `U0` in the Spice deck:

```
*Gate U0
```

```
*-----
*Cell ports in order : x a VSS VDD
*-----
XU0 U0/x U0/a 0 U0/VDD inv_ahvt_64
*-----
```

Note: In order to output a Spice deck that generates the same report output as the [report_timing](#) command, you must run the `create_spice_deck` command with the `-report_timing`, `-segment_start`, and `-segment_end` parameters.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-add_driver_for_constraint_check {true false}</code>	Enables constraint check deck generation with pre-dr
<code>-aging_correlation_type {full timingonly}</code>	Creates the aging-aware Spice deck. You can use on the following options: <code>timingonly</code>: Writes out the Spice deck for timing-or correlation mode. The Spice deck contains device degradation parameters computed by Tempus. <code>full</code>: Writes out the Spice decks for full correlation. This involves two Spice simulations. The first Spice simulation computes device degradation using str simulation. The second Spice simulation uses this data for transient simulation, where delays and sle are measured for timing correlation.
<code>-aging_stress_num_cycles <integer greater than 0></code>	Specifies the number of cycles in stress simulation in Spice correlation mode. When this parameter is specified, the periodic stress waveform propagates through the entire path on which correlation is required. It is recommended to use the value 6, which is based the path depth.
<code>-aging_stress_use_timing_slew {false true}</code>	Uses the slew from STA in the stress Spice deck for full Spice correlation, instead of using the default slew of

<code>-aginglevel_only <string></code>	Indicates the type of aging, BTI, HCI, or both. This parameter is required only for URI models. Note: The exact aging level value is foundry-specific. Specifies the list of cells to be considered as black bo in SPICE simulations.
<code>-add_side_load_measurement {true false}</code>	Adds the side-load measurement to the clock mesh S deck. By default, Tempus measures the latency, delays, and slews of arcs that are part of the clock mesh specified using the <code>-from/-to</code> parameters. When set to <code>true</code>, the software additionally measures latency and slews of the side receivers at each stage. <i>Default:</i> <code>false</code>
<code>-add_voltage_for_missing_sensitization</code>	Adds the voltage for missing sensitization.
<code>-bbox_cells string</code>	Specifies the list of cells to be considered as black bo in SPICE simulations. This parameter allows you to continue simulations ev a subckt is missing for one or more cells. All instantiat of these cells will be replaced by pin caps to replace t input pins' connectivity of black boxes, and Tempus-computed PWL waveforms will be applied at the outp pins of black boxes. The SPICE delays/slews reported for these black box will be the Tempus computed values.
<code>-constraint_check {true false}</code>	Enables constraint check deck generation.
<code>-depth integer</code>	Creates a clock-mesh Spice deck for the specified sta depth. Tempus only traces the clock mesh up to the user-specified depth, starting from the specified <code>-from</code> pin. If both the <code>-depth</code> and <code>-to</code> parameters are specified, th Tempus generates the deck till the depth, which is the minimum of the <code>-to</code> pin depth and the user-specified depth.
<code>-enable_arrival_measurements {true false}</code>	Allows adding arrival measurements.

<pre>-enable_fast_simulation {true false}</pre>	When set to <code>true</code>, the software enables the fast simulation mode. In this mode, additional clock cycles are not required to initialize sequential cells in the Spice deck which speeds up Spice simulations. All the sequential cells are initialized using the initial conditions, and on single clock edge is applied as input. By default, this parameter is set to <code>false</code>. In this mode two clock cycles are applied as inputs to the clock network, where the first triggering edge is used to initialize sequential cells, and the second edge is used for delay measurement. Note: For better performance with sequential cell simulations, you can set the <code>-enable_fast_simulation</code> parameter to <code>true</code> during constraint check generation
<pre>-enable_spectre_x {true false}</pre>	Enables the spectreX license checkout when launching Spectre.
<pre>-enable_mis {tied_input independent_input none}</pre>	Enables advanced multi-input switching (AMIS) during Spice deck creation. You can specify one of the following options: • <code>tied_input</code>: Multiple inputs of an instance will be tied together in a netlist. You can set the <code>set_analysis_mode -multi_input_switching_mode</code> parameter to <code>1</code> to enable this behavior. • <code>independent_input</code>: Allows the Spice deck to write input stimulus for side inputs for multi-input gates. switching times of each input depend on the timing windows. This option can be used when the <code>set_analysis_mode -multi_input_switching_mode</code> parameter is set to <code>2</code>. • <code>none</code>: Disables AMIS during the Spice deck creation <i>Default:</i> <code>none</code>
<pre>-exclude_side_load_cells <i>string</i></pre>	Specifies a list of cells for which the pin capacitance will be used for side loads instead of instantiating them.
<pre>- exclude_side_load_cells_for_missing_subckt {true false}</pre>	When set to <code>true</code>, the software will add side load pin capacitance for a missing Spice subckt cell instead of instantiating it.

<code>-file_prefix string</code>	Adds the specified prefix to the generated Spice deck files. If this parameter is not used, one of the following default names of the generated Spice decks will be used: • <code>path_<index>_setup.sp</code>: for setup paths • <code>path_<index>_hold.sp</code>: for hold paths • <code>path_<index>_vl_glitch.sp</code>: for VL glitch • <code>path_<index>_vh_glitch.sp</code>: for VH glitch The <code><index></code> in the above file names is controlled based on the number of decks that are generated using the <code>create_spice_deck</code> command. A user-specified string (specified using this parameter) used as a prefix to the above mentioned file names.
<code>-force_delete_output_dir</code>	Completely removes the directory specified by the user using the <code>-outdir</code> parameter.
<code>-format string</code>	Allows you to modify the internal reporting format for the run path simulation. The format columns should be compatible with the <code>report_timing_format</code> settings.
<code>-from string</code>	Creates the Spice deck with a mesh net starting from the specified pin.
<code>-from_fall string</code>	Creates the Spice deck with a mesh net starting from the falling transition of the specified from pin.
<code>-from_rise string</code>	Creates the Spice deck with a mesh net starting from the rising transition of the specified from pin.
<code>-generate_output_only {true false}</code>	Generates the report output only, without triggering any simulations. <i>Default:</i> false
<code>-glitch_simulation_mode {input_output output}</code>	Specifies the glitch simulation mode. When set to <code>input_output</code>, the software uses both the input and ROP for Spice calculations. When <code>output</code> is specified, the Spice ROP will be computed using the output only.

<code>-ground string</code>	Specifies a list of ground nodes.
<code>-include_cell_variation {true false}</code>	Writes out the Spice deck required to run simulations. When set to <code>true</code> , the software runs Monte Carlo simulations within Tempus. <i>Default:</i> <code>false</code>
<code>-lsf_job_string string</code>	Specifies a handle to the farm job used to launch the simulation.
<code>-mc_simulation_init_args string</code>	Passes the necessary Monte Carlo parameters that are required to run MC simulations. The Monte Carlo analysis refers to “statistics blocks”, where statistical distributions and correlations of netlist parameters are specified. This parameter initializes these parameters under the statistics block. Note: This parameter can be used when <code>-include_cell_variation</code> is set to <code>true</code> .
<code>-mc_simulation_tran_args string</code>	Initializes the full simulation transient time and the transient time step.
<code>-measure_point {all path stage}</code>	Specifies whether to add all, path, or stage measurements. <i>Default:</i> <code>all</code>
<code>-merge_spice_deck {true false}</code>	Allows you to enable the merged Spice deck.
<code>-model_file string</code>	Specifies the path to SPICE model files that contain SPICE model information. This information is necessary to get accurate Spice information. If sub-circuit and model files are not provided and <code>cdE</code> are available, Spice information is dumped using <code>cdE</code>
<code>-no_hard_stop_at_to_pins {true false}</code>	When set to <code>false</code> (default mode), the Spice deck is generated only for traced paths that start from the <code>-from</code> pin and end at the <code>-to</code> pin. It does not traverse fanout branches that do not lead to the <code>-to</code> specified pin. When set to <code>true</code> , traversal occurs through the whole fanout cone of the <code>-from</code> pin till it hits a clock pin of a flip-flop or any stop point specified with the <code>-to</code> parameter, or passes through a depth specified with the <code>-depth</code> parameter.

<code>-omipath <string></code>	Specifies the OMI (Open Model Interface) directory - required only for OMI.
<code>-outdir <i>string</i></code>	Specifies the name of the output directory that will contain the generated delay and/or glitch spice decks. If simulation is also performed using the <code>-run_simulation</code> parameter, the Spice simulation output will also be saved in this directory. <i>Default:</i> <code>tempus_pathsim</code>
<code>-power <i>string</i></code>	Specifies the list of global names for the power nets. If the <code>-power</code> and <code>-ground</code> parameters are not specified, the software obtains this information from the power/ground supplies set via other ways in the design, such as libraries/cdb, and so on. If the information is not available, the software uses the default list of power node names (VDD and VCC) and ground node names (GND and VSS).

<pre>-reliability_commands string</pre>	Allows you to specify the analysis type. Spice deck creation while using OMI models requires to provide the following strings using the <code>-reliability_commands</code> parameter, to direct the Spice simulator for the appropriate analysis. The following 3 Spice parameters need to be defined per the required analysis type: <table><tr><th>Analysis</th><th>bti_on</th><th>hci_on</th></tr><tr><td>btirec</td><td></td><td></td></tr><tr><td>BTI-Only</td><td>1</td><td>0</td></tr><tr><td>BTI-Only with Recovery</td><td>1</td><td>0</td></tr><tr><td>HCI-Only</td><td>0</td><td>1</td></tr><tr><td>BTI+HCI without Recovery</td><td>1</td><td>1</td></tr><tr><td>BTI+HCI with Recovery</td><td>1</td><td>1</td></tr></table> A few examples are given below: BTI-only without recovery: <pre>-reliability_commands ".param hci_on=0 bti_on=1 she_on=0 btirec_on=0"</pre> HCI-only: <pre>-reliability_commands ".param hci_on=1 bti_on=0 she_on=0 btirec_on=0"</pre> BTI+HCI without recovery: <pre>-reliability_commands ".param hci_on=1 bti_on=1 she_on=0 btirec_on=0"</pre> BTI-only with recovery: <pre>-reliability_commands ".param hci_on=0 bti_on=1 she_on=0 btirec_on=1"</pre> BTI+HCI with recovery: <pre>-reliability_commands ".param hci_on=1 bti_on=1 she_on=0 btirec_on=1"</pre> Note: HCI support for OMI is not available.	Analysis	bti_on	hci_on	btirec			BTI-Only	1	0	BTI-Only with Recovery	1	0	HCI-Only	0	1	BTI+HCI without Recovery	1	1	BTI+HCI with Recovery	1	1
Analysis	bti_on	hci_on																				
btirec																						
BTI-Only	1	0																				
BTI-Only with Recovery	1	0																				
HCI-Only	0	1																				
BTI+HCI without Recovery	1	1																				
BTI+HCI with Recovery	1	1																				
<pre>-reliability_model_type {URI OMI TMI}</pre>	Specifies the aging-aware deck type: URI, OMI, or TM																					

<code>-receiver_pin string</code>	Specifies the list of receiver pins for which glitch Spice decks need to be generated. When used with the <code>-run_simulation</code> parameter, the software writes out a comparison report comparing various glitch parameters between Tempus and Spice such as RIP, input glitch area, and ROP (only if the R node name is available in the noise models) for the g receiver pins.
<code>-report_noise string</code>	Specifies the <code>report_noise</code> command parameters as a list of arguments within brackets. When specified, the Spice deck will be generated for receiver pins reported by the <code>report_noise</code> command and the specified options. When specified with the <code>-run_simulation</code> parameter, software writes out a comparison report comparing various glitch parameters between Tempus and Spice such as RIP, input glitch area, and ROP (only if the R node name is available in the noise models).
<code>-report_timing string</code>	Specifies the <code>report_timing</code> command options as a list of arguments within curly brackets. You can use this option to define the paths for which to generate a Spice trace. To select the paths for generating the Spice trace, specify the <code>report_timing</code> command options, as shown: <pre>create_spice_deck -report_timing {-clock_from CL clock_to CPU_CLK -nworst 2}</pre> When the <code>-report_timing</code> option is not used, the software generates a Spice trace for the worst path identified by the <code>report_timing</code> command. If the <code>-path_type {full_clock}</code> is used, then the clock path will also be processed. See the <code>report_timing</code> command for a complete list of options that can be used as arguments with the <code>-report_timing</code> option. Note: The <code>end</code> and <code>summary</code> values of the <code>-path_type</code> option of <code>report_timing</code> are not supported.

<code>-run_simulation</code>	Enables path simulation in addition to generating the Spice Deck.
<code>-scale_mode string</code>	Allows you to specify the UMC_URI scale mode option. <i>Default:</i> effective
<code>-segment {data_path launch_path launch_clock_path capture_path all}</code>	Creates the Spice deck of a segment out of the path's deck. The following command/global variable settings have the following impact on the <code>-segment</code> parameter options: • <code>-path_type full_clock</code> is defined under the <code>create_spice_deck-report_timing</code> parameter, then <code>-segment</code> default value is <code>launch_path</code>. • <code>-path_type full_clock</code> is not specified, then the default value is <code>data_path</code>. • <code>timing_disable_retime_clock_path_slew_propagation</code> global variable is set to <code>false</code>, the default value is <code>launch_path</code> (irrespective of <code>-path_type</code> settings). • <code>timing_disable_retime_clock_path_slew_propagation</code> global variable is set to <code>true</code> (default), then <code>launch_clock_path</code> and <code>capture_path</code> options are supported only when the <code>-path_type full_clock</code> option of the <code>create_spice_deck -report_timing</code> parameter is used.
<code>-segment_end string</code>	Specifies the name of the segment end term.
<code>-segment_start string</code>	Specifies the name of the segment start term.
<code>-side_path_level integer</code>	Specifies the number of levels of side path stages to be included in the Spice trace. The maximum number of levels that can be specified is 4. <i>Default:</i> 1
<code>-spectre string</code>	Specifies the location of the Spectre version that should be used for simulation. If the path is not specified with an option, the software will check if you have the Spectre location defined in your path.

<code>-spectre_analysis_options <string></code>	Adds Spectre analysis options to the run. When specified, the software provides instructions to Spectre, for example, run a session with extra accuracy.
<code>-spectre_cmd_options <i>string</i></code>	Specifies the command-line options for Spectre.
<code>-spectre_lang_escape_chars <i>string</i></code>	Specifies a list of escape characters in Spectre.
<code>-spice_include <i>string</i></code>	Specifies the user-defined lines that should be written to the Spice deck. For example, <pre>create_spice_deck -spice_include {.include extra_cmds.sp}</pre> where <code>extra_cmds.sp</code> includes additional SPICE commands required for the run and not included in the SPICE model file loaded with the <code>-models</code> parameter.
<code>-stimuli_file <i>string</i></code>	Allows you to specify the file path containing stimuli to be used for the specified library cells. The cells' sensitization from the user-specified stimuli file takes precedence over Tempus-inferred sensitization. Syntax for delay Spice decks is given below: <pre>lib_cell <i>string</i> <from from_rise from_fall> <i>string</i> <to to_rise to_fall> <i>string</i> input_vector <i>string</i> num_clk_pulse <i>integer</i></pre> Syntax for glitch Spice decks is given below: <pre>lib_cell <i>string</i> glitch_type <VL VH> output_pin <i>string</i> input_vector <i>string</i> num_clk_pulse <i>integer</i></pre> For victim driver sensitization <pre>lib_cell <i>cell_name</i> glitch_type <VL VH> input_pin <i>string</i> input_vector <i>string</i> num_clk_pulse <i>integer</i></pre> For victim receiver sensitization The <code>num_clock_pulse <i>integer</i></code> is significant for sequential arcs and specifies where to align the data pin transition with respect to the clock transition. For example, a value of '1' (default setting) specifies that the data pin transition must be aligned with the last clock transition, where the delay measurement is intended. This value can be used for regular flip-flops. The value '2' specifies that the data pin transition needs to be aligned with respect to the second-last triggering clock edge. This value can be used for cascaded 2-bit

	flops. Additionally, for sequential arcs, the first pin in the use specified input vector must be the data pin, whose sig will be latched in the sequential cell. Here are some examples of stimuli file: • The following example shows the stimuli file containing sensitizations for CP(rise)->Q(fall) and CP(rise)->Q(rise) timing arcs of library cell FF1: <pre>lib_cell FF1 from_rise CP to_fall Q input_vec !D&CDN&!SE&!SI num_clk_pulse 2 lib_cell FF1 from_rise CP to_rise Q input_vec D&CDN&!SE&!SI num_clk_pulse 2</pre> • The following example shows the stimuli file containing sensitizations for the victim driver cell A for both VL and VH glitches: <pre>lib_cell AOI glitch_type VL output_pin Z input_vector !A&!B&!C lib_cell AOI glitch_type VH output_pin Z input_vector A&!B&C</pre>
<code>-subckt_file <i>string</i></code>	Specifies the path of the Spice sub-circuit file,, which contains the Spice information. This information is necessary to get accurate Spice information. If sub-circuit and model files are not provided and cdE are available, then Spice information is dumped using cdB.

<code>-supply_voltage_file <i>string</i></code>	Specifies the file containing user-specified power/gro voltages to be used for SPICE simulations. By default design operating voltages will be used. The format of the supply voltage file is as follows: GLOBAL POWER #Specifies the global supply for all pow pins GLOBAL GROUND #Specifies the global supply for all gro pins GLOBAL #Specifies the voltage to be used for the given pin for all instances POWER #Specifies the VDD to be used for the specified port PWLs GROUND #Specifies the VSS to be used for specified I/C port PWLs POWER / #Specifies the VDD used for the specified instanceAC/AEURAtms signal pin PWL or sensitizatic GROUND / #Specifies the VSS used for the specified instanceAC/AEURAtms signal pin PWL or sensitizatic / #Specifies the voltage used for the specified instanceAC/AEURAtms pg pin
<code>-supply_voltage_file_stress <file></code>	Specifies an optional extra file containing the stress voltages for supply or ground pins that are in the cell : circuit definition.
<code>-sweep_range <i>float</i></code>	Specifies the range to sweep the victim.
<code>-sweep_steps <i>integer</i></code>	Specifies the number of steps by which the victim is swept.
<code>-sweep_victim {true false}</code>	Enables victim sweep.
<code>-temperature <i>float</i></code>	Specifies a user-defined temperature to be used in SPICE simulations. <i>Unit:</i> Celsius
<code>-through <i>string</i></code>	Creates the Spice deck with mesh net going by <code>-thro</code> points.
<code>-to <i>string</i></code>	Creates the Spice deck with a mesh net ending at the specified to pins.

<code>-uri_lib_file <string></code>	Specifies the URI model file (required only for URI). <i>Default:</i> none
<code>-uri_mode <string></code>	Specifies the URI appendage option. <i>Default:</i> appendage
<code>-use_control_param {true false}</code>	Allows you to control the power/ground voltage levels the created Spice deck through user-specified parameters. When set to <code>true</code> , an additional file with the same name as the deck and the <code>.control</code> extension is created, containing the parameterized design voltage levels for the PG pins of all cells instantiated in the deck. You can modify the <code>*.control</code> file as needed without altering the original deck.
<code>-use_distributed_load_for_side_receivers {true false}</code>	Allows you to control whether to use the receiver output load as a distributed or lumped load. <i>Default:</i> false
<code>-user_defined_measure_statements {append overwrite}</code>	Specifies whether to overwrite the default measurement statements in the Spice deck by user-defined measurements specified using the <code>-user_defined_measure_statements_file</code> parameter, or append it with default measure statements. <i>Default:</i> append
<code>-user_defined_measure_statements_file string</code>	Allows you to add the delay/slew measurement statements for any given segment of the path for which the Spice deck is generated. The format of the file is given below: <MEASURE_TAG_NAME TRIGGER_PIN_NAME TARGET_PIN_NAME Multiple such lines with unique tags can be specified in the file. The measure statement is written in the Spice deck for delay measurement between <code>TRIGGER_PIN</code> and <code>TARGET_PIN</code> , and slew measurement at <code>TARGET_PIN</code> .
<code>-use_ndw_for_constraint_check {true false}</code>	Enables constraint check deck generation with NDW waveforms.

<code>-use_noise_model_when_cond {true false}</code>	Allows the use of the when condition from a pin noise model.
<code>-view <i>string</i></code>	Specifies the view name - if a glitch Spice deck is created for a view other than the default view.
<code>-waveform_aware_pulse_width_checks {true false}</code>	Generates the Spice deck for waveform-aware pulse width checks.
<code>-xtalk {delay vl_glitch vh_glitch vlu_glitch vho_glitch}</code>	Generates the SI Spice deck for the stage specified using the <code>report_timing</code> parameter. - A stage can start with the victim driver input to the victim receiver input. - A stage can start with the victim driver input to the victim receiver output. The stage will be specified in the <code>report_timing</code> command using the <code>-point_to_point</code> parameter. You can specify one of the following options: <code>delay</code>: Allows dumping the SI delay Spice deck. <code>vl_glitch vh_glitch</code>: Specifies the type of glitch (or VH) the Spice deck will be used for. This option must be used for glitch Spice deck generation. <code>vlu_glitch vho_glitch</code>: Supports simulation of vlu/vho glitches
<code>> <filename></code>	Redirects <code>-run_path_simulation/-run_simulation</code> output file.
<code>>> <filename></code>	Appends <code>-run_path_simulation/-run_simulation</code> to output file.

Examples

- The following set of commands generates a Spice deck for a path and outputs it in the `path1_spice` directory:

```
tempus > set rpt_tim_opt [ list -clock_from CLK -clock_to CPU_CLK -nworst 2 ]
tempus > create_spice_deck -outdir path1_spice -report_timing $rpt_tim_opt
-subckt_file /path/subckt.sp -model_file /path/models.sp
```
- The following command creates a Spice deck and simulates it using the Spectre simulator:

```
tempus > create_spice_deck -run_simulation -spectre /tools/spectre
-subckt_file /path/subckt.sp -model_file /path/models.sp
```

- The following command creates a Spice deck in `mydir` and keeps the instances up to three levels for each side path attached to the main path:

```
tempus > create_spice_deck -side_path_level 3 -outdir mydir
-subckt_file /path/subckt.sp -model_file /path/models.sp
```

- The following command creates a Spice deck for the best path:

```
tempus > create_spice_deck -report_timing { -early }
-subckt_file /path/subckt.sp -model_file /path/models.sp
```

- The following command creates a Crosstalk Spice Deck for the specified Stage:

```
tempus > create_spice_deck -xtalk delay -model_file /path/models.sp -subckt_file
/path/subckt.sp -report_timing {-point_to_point -from <victim_driver_input> -to
<victim_receiver_driver_output>} -output spice_output
```

- The following command dumps the VL spice deck `report_timing` command options so that the transition at the receiver pin is failing:

```
tempus > create_spice_deck -xtalk vl_glitch -report_timing < -from driver/input -to_fall
receiver/input -point_to_point > -outdir <output dir> -model_file <model file> -
subckt_file <subckt file>
```

- The following command dumps the VH Spice deck `report_timing` command options so that the transition at the receiver pin is rising:

```
tempus > create_spice_deck -xtalk vh_glitch -report_timing < -from driver/input -to_rise
receiver/input -point_to_point > -outdir <output dir> -model_file <model file> -
subckt_file <subckt file>
```

- The following command run simulations and then generates a report:

```
create_spice_deck -report_timing "-net -from clk -to ca53/as_12/ff/D -path_type
full_clock" -run_simulation -include_cell_variation true > spice.socv.rpt
```

- The following command generates a report (if simulations have already been run):

```
create_spice_deck -generate_output_only true -report_timing "-net -from clk -to
ca53/as_12/ff/D -path_type full_clock" -run_simulation -include_cell_variation_true -
format {arrival spice_arrival spice_arrival_detail arrival_error_detail hpin} >
spice.socv.rpt
```

- The following section in Spice deck specifies global supplies and the transient analysis window time.

```
*-----
*Command and Options
*-----
.GLOBAL vdd vbbnw
VOXvdd vdd 0 1.0
VOXvbbnw vbbnw 0 1.0
.GLOBAL vss vbbspw
```

```
V0Xvss vss 0 0
V0Xvbbpw vbbpw 0 0
.TEMP -40.0000
.TRAN 1p 10256p
*-----
```

- The following section shows an example of a net and its instance connections.

```
*-----
*Net Instances
*-----
*Net module1/inst_1/net
*-----
Xmodule1/inst_1/net module1/inst_1/q
+ module1/inst_2/a wc:module1/inst_1/q
*-----
```

- The following section shows an example of a gate instance and its port connections.

```
*-----
*Gate Instances
*-----
*Gate module1/inst_2
*-----
*Cell ports in order : a y vss vdd vbbpw vbbnw
*-----
Xmodule1/inst_2/a module1/inst_2/y
+ 0 module1/inst_2/vdd 0 module1/inst_2/vbbnw
+ inv1
*-----
```

- The following section shows the measurement statements for delay and slew.

```
*-----
*Measurement statements
*-----
.MEASURE TRAN Stage_0001_module1/inst_1/clk_DELAY_
module1/inst_1/q -----clk to q delay
+ TRIG V(module1/inst_1/clk) VAL=0.5 TD=167.9p RISE=1 of module1/inst_1
+ TARG V(module1/inst_1/q) VAL=0.5 TD=167.9p RISE=1
.MEASURE TRAN Stage_0001_module1/inst_1/q_SLEW
+ TRIG V(module1/inst_1/q) VAL=0.30 TD=167.9p RISE=1 -----Slew at
+ TARG V(module1/inst_1/q) VAL=0.70 TD=167.9p RISE=1 module1/inst_1/q pin
.MEASURE TRAN Stage_0002_module1/inst_1/q_DELAY_
module1/inst_2/a -----Interconnect Delay
+ TRIG V(module1/inst_1/q) VAL=0.5 TD=167.9p RISE=1
```

```
+ TARG V(module1/inst_2/a) VAL=0.5 TD=167.9p RISE=1
.MEASURE TRAN Stage_0002_module1/inst_2/a_SLEW
+ TRIG V(module1/inst_2/a) VAL=0.30 TD=167.9p RISE=1
+ TARG V(module1/inst_2/a) VAL=0.70 TD=167.9p RISE=1
```

and Input Slew
-----Interconnect Delay
and Input Slew

Note: The `DELAY` and `SLEW` keywords in the `MEASURE` statements indicate that these are the measurement statements for delay and slew respectively.

Also, Stage numbers (Stage_0001 and Stage_0002) in the `MEASURE` statement indicate the respective stages; first and second stage, in the timing path, which makes it easier for correlation with the `report_timing` report.

- The following shows an example for generating a VL glitch Spice deck:

```
tempus > create_spice_deck -xtalk vl_glitch -receiver_pin x1/x2/I -outdir SPICE/net1_vl
-model_file ./all_cells.model -subckt_file ./all_cells.sp
```

- The following shows an example for generating a VH glitch Spice deck:

```
tempus > create_spice_deck -xtalk vh_glitch -receiver_pin x1/x2/I -outdir SPICE/net1_vh
-model_file ./all_cells.model -subckt_file ./all_cells.sp
```

- The following command sets the simulation time stop criteria to be 13.40 ns and the step time to be 1ps:

```
tempus > create_spice_deck -mc_simulation_init_args "stop=1.34e-08 step=1p"
```

- The following command writes out a statistics block in a deck to run Monte Carlo simulations:

```
tempus > create_spice_deck -mc_simulation_init_args "numruns=3000 seed=1234
distribute=fork numprocesses=8 variations=mismatch scalarfile=MCDatasetLoad.txt dut=
[%s]"
simulator lang = spectre

mcl montecarlo numruns=3000 seed=1234 distribute=fork numprocesses=8 variations=mismatch
scalarfile=MCDatasetLoad.txt dut=[ XStage2.XCTS_ccl_a_inv_01730
XStage3.XCTS_ccl_inv_01726 XStage4.XCTS_ccl_inv_01718 XStage5.XCTS_ccl_a_inv_01712 \
XStage6.XCTS_cpc_sk_inv_03405 XStage7.XCTS_cpc_sk_inv_03406 XStage8.XCTS_cdb_inv_03303
XStage9.XCTS_cdb_inv_02837 \
XStage10.XCTS_ccl_inv_01669 XStage11.XCTS_ccl_a_inv_01540 XStage12.XCTS_cpc_sk_inv_03860
XStage13.XCTS_cpc_sk_inv_03859 \
XStage14.Xugenq_uaesflushu1_aes_flush_u1_reg XStage15.XPLACEOPTFE_RC_149_0
XStage16.XPLACEOPTFE_RC_150_0 \
XStage17.XU22704 XStage18.XU22707 XStage19.XU22857 XStage20.XPLACEOPTFE_OFC2135_n30081
XStage21.XU23049 \
XStage22.XU140015 XStage23.XU132065 XStage24.XU132066 XStage25.XU132710
XStage26.XU132716 XStage27.XU133277 \
XStage28.XU133278 XStage29.XU133279 XStage30.XU133285 ]
```

```
{
tran1 tran stop=2.88312e-08 step=1p
}
simulator lang = spice
```

- The following commands show constraint checks applied during setup and hold mode:

```
Setup: create_spice_deck -model_file $model_file -subckt_file $subckt_file -out_dir
output -spectre $spectre_path -spectre_cmd_options $spectre_opt -run_simulation -
enable_fast_simulation true -constraint_check true -report_timing {-late -retime
path_slew_propagation -to EP1}
```

```
Hold: create_spice_deck -model_file $model_file -subckt_file $subckt_file -out_dir
output -spectre $spectre_path -spectre_cmd_options $spectre_opt -run_simulation -
enable_fast_simulation true -constraint_check true -report_timing {-early -retime
path_slew_propagation -to EP1}
```

- A sample constraint check report is given below:

Path 1: SPICE based Setup Check

Clock: dff2/CK (^)

Data: dff2/D (^)

Deck: path_1_setup_iter11.sp -----> **Final Deck**

Slew: 1e-09(Data), 1e-09(Clock) -----> **Data and Clock Pin Transitions**

Setup: 1.59911e-10 (Spice), 1.57721e-10 (Tempus) -----> Setup Time

```
-----
Iter   Offset(D-CK) Pass/Fail  Deck
      (ps)
```

```
-----
0      -2500.000    Pass    path_1_setup_iter0.sp |
1       1249.999   Fail    path_1_setup_iter1.sp |
2       -625.000    Pass    path_1_setup_iter2.sp |
3        312.500   Fail    path_1_setup_iter3.sp |
4       -156.250   Fail    path_1_setup_iter4.sp |
5       -390.624    Pass    path_1_setup_iter5.sp |-----> PASS/FAIL Iterations
6       -273.438    Pass    path_1_setup_iter6.sp |
7       -214.843    Pass    path_1_setup_iter7.sp |
8       -185.548    Pass    path_1_setup_iter8.sp |
9       -170.900    Pass    path_1_setup_iter9.sp |
```

10	-163.574	Pass	path_1_setup_iter10.sp
11	-159.911	Pass	path_1_setup_iter11.sp -----> Final Deck

Related Commands

- `report_timing`
- `report_timing_format`
- `report_noise`
- `set_analysis_mode -aging`
- `set_aging_analysis_mode`
- `set_aging_activity`
- `report_aging_activity`

define_bundled_bus

```
define_bundled_bus  
[-help]  
[-add_members string]  
[-delete_members string]  
[-name string]
```

Allows you to control user-defined bus groups.

The Tempus software automatically identifies bus nets. However, in some cases, the software may not be able to mark bus bits properly (or treat marked bits as bus). For improved bus identification, the `define_bundled_bus` command allows you to control the nets that you want to add (or remove) from the list of recognized bus bits.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-add_members <i>string</i></code>	Specifies the name of the nets to be added to the group.
<code>-delete_members <i>string</i></code>	Specifies the name of the nets to be deleted from the group.
<code>-name <i>string</i></code>	Specifies the name of the group.

Examples

- The following example command will create a bus named bus1 with 3 net members x[1] and x[2]:

```
define_bundled_bus -name bus1 -add_members {x[1] x[2]}
report_bundled_bus -type user_defined
#####
```

Summary:

```
Total number of user defined bus groups reported = 1
Total number of tool inferred bus groups reported = 0
```

Group Name (user_defined) : bus1

Constituents:

```
- x[1]
- x[2]
```

- The following example command will add one more member x[3] and delete member x[2] to the bus 'bus1':

```
define_bundled_bus -name bus1 -add_members x[3] -delete_members x[2]
Added [1] net(s) to group 'bus1'.
Deleted [1] net(s) from user-defined group 'bus1'.
```

Related Commands

- `report_bundled_bus`

define_glitch_holding_resistance

```
define_glitch_holding_resistance  
[-help]  
[-glitch_type {both | vl | vh | vlu | vho | vl_vh | vlu_vho}]  
[-instance_pin string]  
[-lib_pin string]  
-value double  
[-views string]
```

Allows you to specify the holding resistance.

You can use this command to define a net-specific holding resistance instead of using the default holding resistance of 1000 ohms.

This command can be specified multiple times per net. If there is a duplicate specification, then the last setting is used.

Parameters

-help	Prints out the command usage.
-glitch_type {both vl vh vlu vho vl_vh vlu_vho}	Specifies the glitch type for the resistance Hold application. The option <code>both</code> applies the value to VL and VH.
-instance_pin <i>string</i>	Specifies a list of instance pin names.
-lib_pin <i>string</i>	Specifies the instance/pin name of the driver to which the holding resistance should be applied.
-value <i>double</i>	Specifies the value in ohms of the holding resistance to be used by glitch analysis.
-views <i>string</i>	Specifies a list of view names. If this parameter is not specified, all views are applied by default.

Examples

- The following command sets the holding resistance on the Y pin of inst1 to 233ohms for VL:

```
tempus > define_glitch_holding_resistance -glitch_type vl -lib_pin {inst1/Y} 233
```
- The following command sets the holding resistance on the Y pin of inst2 to 380ohms for both VL & VH:

```
tempus > define_glitch_holding_resistance -glitch_type both -lib_pin {inst1/Y} 233
```

define_parameter

```
define_parameter  
[-help]  
[<setup/hold>]  
[-correlation <string>]  
[-distribution <string>]  
[-moments <string>]  
[-name <string>]  
[-range <string>]  
[-related <string>]  
-scope <string>  
[-view <string>]
```

Provides variation information to Tempus. You can directly specify the range information, and Tempus will internally calculate moments and use them for wire variation calculation.

Parameters

<code>-help</code>	Prints out the command usage.
<code><setup/hold></code>	Defines a parameter that is valid for setup or hold.
<code>-correlation</code> <code><string></code>	Specifies the correlation between different parameters within the same scope; parameters must be predefined.
<code>-distribution</code> <code><string></code>	Specifies the parameter distribution. You can also use user-defined distribution.
<code>-moments</code> <code><string></code>	Specifies the parameter moments - mean or std-dev.
<code>-name <string></code>	Specifies the name of the parameter.
<code>-range <string></code>	Specifies the parameter minimum or maximum range - min or max.
<code>-related</code> <code><string></code>	Specifies the related information on the sub-scope of the parameter within the specified <code>-scope</code>. For back-end-of-line (BEOL), the allowed arguments are: • <code><layer name></code>: capacitance • <code><layer name></code>: resistance
<code>-scope <string></code>	Specifies the parameter scope - <code>vtskew</code> , <code>wire_variation</code> , or <code>via_variation</code> .
<code>-view <string></code>	Specifies the view name.

Examples

- The following command defines R/C range for normal distribution for M1 layer for setup for view view1:

```
define_parameter setup -view view1 -name param_M1_R -scope wire_variation -distribution gaussian -range {1 1.15}
define_parameter setup -view view1 -name -param_M1_C -scope wire_variation -distribution gaussian -range {0.95 1}
```

- The following command defines R/C moments for normal distribution for M1 layer for setup for all views:

```
define_parameter setup -name param_M1_R -scope wire_variation -distribution gaussian -moments {0.2 0.01}
define_parameter setup -name param_M1_C -scope wire_variation -distribution gaussian -moments {0.95 1}
```

- The following command defines correlation between metal layers - if M1, M2, and M3 are fully correlated:

```
define_parameter -scope wire_variation -correlation {{param_M1_C param_M2_C param_M3_C} {1.0}}
```

Related Commands

- [report_parameter](#)
- [set_wire_variation_mode](#)
- [get_wire_variation_mode](#)

Related Topics

- For more information, refer to "[Wire Variation Analysis](#)" section in the *Tempus User Guide*.

get_aging_activity

```
get_aging_activity  
[-help]  
-net <string>  
[-views <string>]
```

Returns the current settings of the `set_aging_activity` command.

Parameters

-help	Prints out the command usage.
-net <string>	Returns the net names.
-views <string>	Returns a list of the specified views.

Related Commands

- `set_analysis_mode -aging`
- `set_aging_analysis_mode`
- `set_aging_activity`
- `report_aging_activity`
- `create_spice_deck`

Related Information

"[Aging-Aware STA Analysis](#)" chapter in the *Tempus User Guide*.

get_aging_analysis_mode

```
get_aging_analysis_mode  
[-help]  
[-activity_propagation_mode]  
[-constraints]  
[-disable_aging]  
[-max_duty_cycle]  
[-max_toggle_rate]  
[-max_toggle_rate_limit <value>]  
[-min_duty_cycle]  
[-min_toggle_rate]  
[-param_priority_sets]  
[-recovery]  
[-sequential_activity_propagation]  
[-sequential_max_duty_cycle]  
[-sequential_max_toggle_rate]  
[-sequential_min_duty_cycle]  
[-sequential_min_toggle_rate]  
[-type]  
[-use_fixed_duty_cycle]  
[-views]
```

Returns the aging analysis-related global variable settings and defaults that have been made using the `set_aging_analysis_mode` command.

Parameters

Displays the following information about the specified `set_aging_analysis_mode` parameters in the Tempus log file and the Tempus console:

- Parameter name
- Current value
- Type (Boolean, string, and so on)
- Whether the user sets the current value

If you do not specify any parameters, the software displays information for all of the parameters.

Parameters

<code>-help</code>	Prints out the command usage.
<code>parameter_names</code>	Displays information for the specified parameters. You can specify one or more parameters. See set_aging_analysis_mode for descriptions of the parameters you can specify.

Related Commands

- [set_aging_analysis_mode](#)
- [create_aging_stress_profile](#)
- [set_aging_activity](#)
- [get_aging_activity](#)
- [report_aging_activity](#)

Related Information

- "[Aging-Aware STA Analysis](#)" chapter in the *Tempus User Guide*.

get_delay_cal_mode

```
get_delay_cal_mode  
[-help]  
[accuracy_level]  
[-advanced_pincap_mode]  
[-accurate_receiver_out_load]  
[-combine_mmmc]  
[-disable_randomcell_in_socv]  
[-early_irdrop_data_type]  
[-enable_accurate_lle_analysis]  
[-enable_corner_based_beol]  
[-enable_high_fanout]  
[-enable_input_slew_sensitivity_on_constraint]  
[-enable_quiet_receivers_for_hold]  
[-equivalent_waveform_model]  
[-equivalent_waveform_model_for_timing_check]  
[-honorSlewPropConstraint]  
[-library_interpolation_mode]  
[-nonDefault]  
[-reportOutBound]  
[-SIAware]  
[-skip_slew_merge_from_disabled_path]  
[-slewOutBoundLimitHigh]  
[-slewOutBoundLimitLow]  
[-socv_lvf_mode]  
[-socv_use_lvf_tables]  
[-quiet]
```

Returns the current settings of the `set_delay_cal_mode` command. The following information is displayed:

- Parameter name
- Current value of the parameter
- Type (Boolean, string, and so on)
- Information on whether the parameter value is user-defined or default.

If you do not specify any parameter, the software displays the current settings of all the `set_delay_cal_mode` command parameters.

Parameters

<code>-help</code>	Outputs a brief description that includes type and default information for each <code>set_delay_cal_mode</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man get_delay_cal_mode</code> .
<code>parameter_names</code>	Displays information for the specified parameters. See the <code>set_delay_cal_mode</code> command for descriptions of the parameters that you can specify.
<code>-quiet</code>	Displays the current settings of the specified parameters in Tcl list format. If you specify <code>-quiet</code> without any other parameters, the software displays the current settings of all the <code>set_delay_cal_mode</code> parameters in Tcl list format.

Examples

- The following command displays the current settings of the `set_delay_cal_mode -combine_mmmc` parameter:

```
tempus> get_delay_cal_mode -combine_mmmc
```

The following information is displayed:

```
-combine_mmmc early_late # enums={none early_late}, default=early_late
early_late
```

- The following command displays the current settings of all the `set_delay_cal_mode` parameters:

```
tempus> get_delay_cal_mode -quiet
```

Related Commands

- `set_delay_cal_mode`
- `report_delay_calculation`
- `report_timing`

Related Topics

- [Base Delay Analysis](#) chapter in the *Tempus User Guide*.

get_glitch_threshold

```
get_glitch_threshold
[-help]
[-cells string]
[-failure_point {none | input | internal | last}]
[-glitch_type {both vl vh vlu vho vl_vh vlu_vho}]
[-library_pins string]
[-nets <string>]
[-pin_type {all | clock | data | latch}]
[-pins string]
[-threshold_type {reporting | failure}]
[-views string]
```

Displays information for the specified [set_glitch_threshold](#) parameters for the current Tempus session. If you do not specify a parameter, the software displays information for all [set_glitch_threshold](#) parameters.

Parameters

-help	Prints out the command usage.
<i>parameter_names</i>	Displays information for the specified parameter. You can specify one or more parameters. See set_glitch_threshold for descriptions of the timing analysis mode parameters that you can specify.

Examples

- The following command shows the failure threshold for ROP for single-stage cells:
tempus 16 > get_glitch_threshold -failure_point last
Here's the command output:
value=0.4 -threshold_type reporting -glitch_type both -pin_type all -failure_point last -failure_dependency none
- The following command shows the failure threshold for the ROP for the internal net of multi-stage cells:
tempus 17 > get_glitch_threshold -failure_point internal
Here's the command output:
value=0.2 -threshold_type reporting -glitch_type both -pin_type all -failure_point internal -failure_dependency none

- The following command shows the failure threshold for the RIP for the input net of cells without CCSN/ECSMN/cdB

```
tempus 18 > get_glitch_threshold -failure_point input
```

Here's the command output:

```
value=0.5 -threshold_type reporting -glitch_type both -pin_type all -failure_point input  
-failure_dependency none
```

Related Commands

- `set_glitch_threshold`
- `report_glitch_thresholds`

get_multi_input_switching_mode

```
get_multi_input_switching_mode
[-help]
{[-disable_lib_cells ] [-switching_alignment_factor ] [-mis_alignment_mode]}
```

Reports the MIS (multi-input switching) analysis settings configured using the `set_multi_input_switching_mode` command.

Parameters

-help	Prints out the command usage.
-disable_lib_cells <string>	Reports all the user-specified library cells for which multi-input switching (MIS) mode support is disabled.
-mis_alignment_mode {0 1}	Returns the current value set (0 or 1) of the MIS alignment mode.
- switching_alignment_factor <float>	Returns the globally specified switching alignment factor.

Related Commands

- `set_multi_input_switching_mode`
- `report_delay_calculation`
- `report_timing`
- `create_spice_deck`

Related Information

- [Advanced Multiple-Input Switching Analysis](#) section in the *Tempus User Guide*.

get_si_mode

```
get_si_mode
[-help]
[-reset]
[-accumulated_small_attacker_factor]
[-accumulated_small_attacker_mode]
[-accumulated_small_attacker_threshold]
[-attacker_alignment]
[-clock_delta_delay_threshold]
[-clocks]
[-delay_only]
[-delta_delay_annotation_mode]
[-delta_delay_threshold]
[-double_switching]
[-enable_bus_attacker_correlation]
[-enable_delay_report]
[-enable_double_clocking_check]
[-enable_drv_with_delta_slew]
[-enable_glitch_propagation]
[-enable_glitch_propagation_spice_deck]
[-enable_glitch_report]
[-enable_lle_glitch]
[-enable_logical_correlation]
[-enable_overshoot_undershoot_glitch]
[-enable_two_stage_driver_weakening]
[-enable_virtual_attacker_constituent_report]
[-glitch_analysis_type]
[-glitch_accuracy_level]
[-hold_slack]
[-individual_attacker_simulation_filtering]
[-individual_attacker_clock_threshold]
[-individual_attacker_threshold]
[-input_glitch_full_propagation_threshold]
[-input_glitch_full_propagation_vh_threshold]
[-input_glitch_full_propagation_vl_threshold]
[-input_glitch_threshold]
[-input_glitch_vh_threshold]
[-input_glitch_vl_threshold]
[-net_signal_type]
[-nonlinear_attacker_slew]
[-num_si_iteration]
[-receiver_clk_peak_limit]
[-receiver_latch_peak_limit]
[-receiver_peak_limit]
```

```
[-report_max_virtual_attacker_constituents]  
[-secondary_attacker_decoupling_factor]  
[-separate_delta_delay_on_data]  
[-setup_slack]  
[-si_reselection]  
[-si_reselection_delay_threshold]  
[-skip_noise_model_check]  
[-skip_tw]  
[-switch_prob]  
[-unconstrained_net_use_inf_tw]  
[-use_infinite_TW]
```

Displays the following information about the specified `set_si_mode` parameters in the Tempus log file and the Tempus console:

- Parameter name
- Current value
- Type (Boolean, string, and so on)
- Whether the user sets the current value.

If you do not specify any parameters, the software displays information for all of the parameters.

Parameters

<code>-help</code>	Prints out the command usage.
<i>parameter_names</i>	Displays information for the specified parameters. You can specify one or more parameters. See set_si_mode for descriptions of the parameters you can specify.

Examples

- The following command gets the individual attacker threshold, which is 0.015.

```
tempus> get_si_mode -individual_attacker_threshold 0.015 # float, default=0.015
```
- The following command gets the number of timing window iterations for performing SI delay analysis, which is 2.

```
tempus> get_si_mode -num_si_iteration 2 # int, default=2
```
- The following command gets the attacker switching probability, which is 0.3.

```
tempus> get_si_mode -switch_prob 0.3 # float, default=0.3
```

Related Commands

- [set_si_mode](#)
- [report_delay_calculation](#)
- [report_timing](#)
- [report_noise](#)
- [update_glitch](#)

get_via_variation_mode

```
get_via_variation_mode
[-help]
[-corner_sigma_multiplier]
[-correlation_type]
[-reset]
[-separate_params_for_via]
[-quiet]
[-nonDefault]
```

Displays the following information about the specified `set_via_variation_mode` command parameters in the Tempus log file and the Tempus console:

- Parameter name
- Current value
- Type (Boolean, string, and so on)
- Whether the user sets the current value

If you do not specify any parameters, the software displays information for all of the parameters.

Parameters

<code>-help</code>	Prints out the command usage.
<i>parameter_names</i>	Displays information for the specified parameters. You can specify one or more parameters. See set_via_variation_mode for descriptions of the parameters you can specify.

Related Commands

- [set_via_variation_mode](#)

get_wire_variation_mode

```
get_wire_variation_mode  
[-help]  
[-corner_sigma_multiplier]  
[-correlation_type]  
[-reset]  
[-separate_params_for_via]  
[-quiet]  
[-nonDefault]
```

Displays the following information about the specified `set_wire_variation_mode` command parameters in the Tempus log file and the Tempus console:

- Parameter name
- Current value
- Type (Boolean, string, and so on)
- Whether the user set the current value

If you do not specify any parameters, the software displays information for all of the parameters.

Parameters

<code>-help</code>	Prints out the command usage.
<code>parameter_names</code>	Displays information for the specified parameters. You can specify one or more parameters. See set_wire_variation_mode for descriptions of the parameters you can specify.

Related Commands

- [set_wire_variation_mode](#)

Related Topics

- For more information, refer to "[Wire Variation Analysis](#)" section in the *Tempus User Guide*.

merge_noise_reports

```
merge_noise_reports
[-help]
-base_dir <string>
[-delay {max | min}]
[-double_switching]
[-failure]
[-reverse_voltage_change_threshold <float>]
[-sort_by {noise | receiver_peak}]
[-threshold <float>]
-txtfile <string>
[-views <string>]
```

Generates merged noise reports for the specified views.

Parameters

-help	Prints out the command usage.
-base_dir <string>	Specifies the name of the directory provided as input to the <code>report_noise -merge_dir</code> parameter.
-delay {max min}	Specifies the uncertainty type (maximum or minimum delay).
-double_switching	Enables double switching reporting.
-failure	Shows output glitch report for failure nets only.
-reverse_voltage_change_threshold <float>	Reports the minimum reverse swing threshold values.
-sort_by {noise receiver_peak}	Specifies the sorting of the report based on noise or receiver_peak.
-threshold <float>	Reports the glitch (in mV) or uncertainty (in ps) filter threshold values.
-txtfile <string>	Specifies the name of the output file.
-views <string>	Specifies the name of views to include in the merged output.

Related Commands

- `report_noise`

report_aging_activity

```
report_aging_activity  
[-help]  
[-duty_cycle]  
-net <string>  
[-out_file <string>]  
[-toggle_rate]  
[-views <string>]
```

Reports the duty cycles and toggle rates.

Parameters

-help	Prints out the command usage.
-duty_cycle	Specifies the dump duty cycles.
-net <string>	Specifies the net names.
-out_file <string>	Specifies the output file name.
-toggle_rate	Specifies the dump toggle rates.
-views <string>	Specifies the list of view names.

Examples

- The following command reports the toggle rate and duty cycle information on net, `net_dcu[36]`. The output file “`out.txt`” will contain the required information:

```
tempus > report_aging_activity -duty_cyclcy -toggle_rate -net net_dcu[36] -out_file  
out.txt
```

Related Commands

- `set_analysis_mode -aging`
- `set_aging_analysis_mode`
- `set_aging_activity`
- `create_aging_stress_profile`
- `create_spice_deck`

Related Information

"[Aging-Aware STA Analysis](#)" chapter in the *Tempus User Guide*.

report_aging_stress_profile

```
report_aging_stress_profile  
[-help]  
[-early]  
[-late]  
[-views <string>]
```

Reports the stress profiles for aging-aware STA, as defined by the `create_aging_stress_profile` command. The report contains stress duration, stress temperature, and stress voltage per power rail.

Parameters

-help	Prints out the command usage.
-early	Reports the stress profile associated with early paths.
-late	Reports the stress profile associated with late paths.
-views <string>	Specifies a list of view names associated with the aging stress profile.

Related Commands

- `create_aging_stress_profile`
- `set_analysis_mode -aging`
- `set_aging_analysis_mode`
- `set_aging_activity`
- `get_aging_activity`
- `report_aging_activity`
- `create_spice_deck`

Related Information

- "[Aging-Aware STA Analysis](#)" chapter in the *Tempus User Guide*.

report_bundled_bus

```
report_bundled_bus  
[-help]  
[-name string]  
[-nets string]  
[-type {user_defined | tool_inferred}]  
[[ > | >> ]]
```

Generates a report for bundled buses.

Parameters

-help	Prints out the command usage.
-name <i>string</i>	Specifies the name of the user-defined bus group(s) to be reported.
-nets <i>string</i>	Specifies the name of the net(s) to be reported.
-type {user_defined tool_inferred}	Reports the type of bundled bus(es) - user-defined or tool-inferred - to include in the report. By default, the <code>report_bundled_bus</code> command generates reports for both user-defined and software-inferred types of buses.
> <i>filename</i>	Redirects the output to the specified file.
>> <i>filename</i>	Appends to the specified output file.

Examples

- The following command shows the user-defined bus groups:

```
report_bundled_bus
```

```
Summary:
```

```
Total number of user defined bus groups reported = 3
```

```
Total number of tool inferred bus groups reported = 41
```

```
-----  
Group Name (user_defined) : grp1
```

```
Constituents:
```

```
- EJ_PartNumber[15]  
- EJ_PartNumber[14]  
- EJ_PartNumber[13]  
- EJ_PartNumber[12]  
- EJ_PartNumber[11]  
- EJ_PartNumber[10]  
- EJ_PartNumber[9]  
- EJ_PartNumber[8]  
- EJ_PartNumber[7]  
- EJ_PartNumber[6]  
- EJ_PartNumber[5]  
- EJ_PartNumber[4]  
- EJ_PartNumber[3]  
- EJ_PartNumber[2]  
- EJ_PartNumber[1]  
- EJ_PartNumber[0]  
-----
```

```
Group Name (user_defined) : grp2
```

```
Constituents:
```

```
- EJ_ManufID[10]  
- EJ_ManufID[9]  
- EJ_ManufID[8]
```

Related Commands

- `define_bundled_bus`

report_double_clocking

```
report_double_clocking  
[-help]  
[-file string]  
[-max_delta_delay float]  
[-nworst integer]  
[-reverse_slope_limit float]  
[-reverse_voltage_change_threshold float]  
[-view string]
```

Reports noise impact on clock nets by generating information on the double clocking, transition type, and reverse slope.

The report output displays violations in sorted order - based on the absolute value of the slope, a negative value indicates a reverse slope.

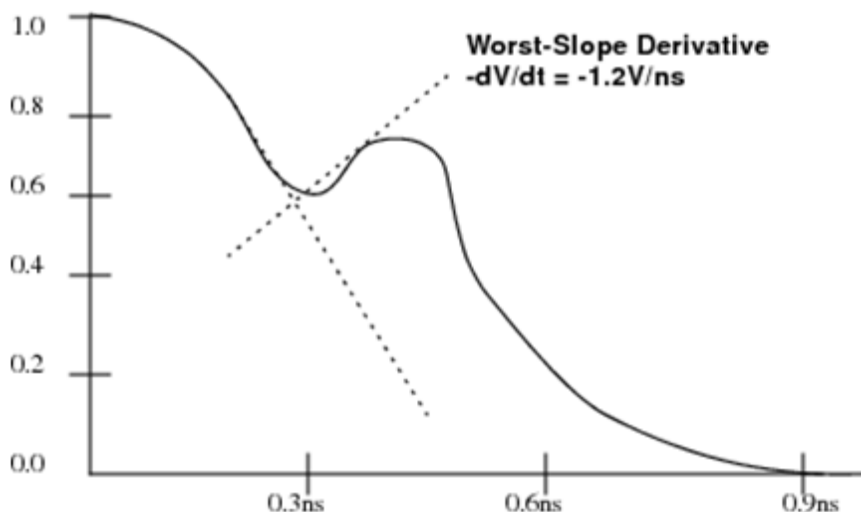
Note: The `report_double_clocking` command requires that clocks have been defined and applied prior to timing analysis with SI (`update_timing` or `report_timing`). The `report_double_clocking` command does not work with the glitch-only flow (`update_glitch`).

Double clocking occurs when the noise glitch attacks a transitioning signal. The multiple transition, or double clocking, check supplements the transition-time check performed in static timing analysis. Transition time is the time it takes a signal to cross from one threshold to another. However, transition time does not account for the waveform shape between the voltage thresholds.

For special signals such as clocks, it is important to analyze the waveform shape between the voltage threshold in case the signal transitions slowly or in the opposite direction (even for a short duration of time), causing multiple clock edges. This command enables you to report the worst-case slope (dV/dt for rising signal and $-dV/dt$ for falling signal) of a signal transitioning between the voltage thresholds. The slope derivative is measured in units of volts per nanosecond (V/ns). This value can be positive if the signal does not change direction, 0 if the signal is flat, and negative if the signal changes direction for even a short duration.

The following figure shows two falling signal transitions:

Figure: Negative Worst-Slope Derivative



In the above figure, the falling signal transitions have identical transition times. However, the worst-case slope derivatives for the two falling waveforms are different. If a clock signal exhibits a negative slope derivative, there is a possibility of a double clocking functional failure. A higher reverse slope results in a higher possibility of a double-clocking functional failure.

The double clocking report contains the following sections and their corresponding fields:

- **Double Clocking**
 - Rise Slope (V/ns): Specifies the worst reverse slope of the rising clock signal as a measure of volts per nanosecond.
 - Fall Slope (V/ns): Specifies the worst reverse slope for the falling clock signal as a measure of volts per nanosecond.
 - VictimNet: Specifies the name of the victim net.
- **Instance Pin:** Displays the name of the instance pin.
- **Transition Type:** Indicates the type of transition the net is undergoing – R for rising, F for falling.
- **Reverse Slope:** Indicates the slope that is opposite to the transition type, and the value reported is the worst reverse slope.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-file <i>string</i></code>	Specifies the output filename. <i>Default:</i> <code>{<i>\$parm_filename_prefix</i>}_clock.text</code>

<code>-max_delta_delay float</code>	Reports clock nets, for which the delay change caused by noise exceeds the specified number. <i>Default: 0</i>
<code>-max_slew float</code>	Reports clock nets for which the transition time, including SI, exceeds the specified number. <i>Default: 0</i>
<code>-nworst integer</code>	Specifies the number of clock nets to report in each category. <i>Default: 1</i>
<code>-reverse_slope_limit float</code>	Reports the clock nets for which the slope is below the specified number. <i>Default: 0</i>
<code>-reverse_voltage_change_threshold float</code>	Controls the reverse-slope failure criteria by allowing you to specify a minimum voltage change as a percentage of VDD. The reverse slope must be sustained in order for it to be considered as failing. <i>Default: 0.1</i>
<code>-view string</code>	Specifies the view name for which the report is to be generated.

Examples

- The following command reports clock nets that have a reverse slope less than -1:
tempus > report_double_clocking -reverse_slope_limit -1
- The following command reports 100 clock net violations of each category:
tempus > report_double_clocking -nworst 100
- The following command shows the double clocking report for default_emulate_view view:

```
tempus > report_double_clocking -view default_emulate_view

*****
Double Clocking Report For View: default_emulate_view
Generated: Thu Mar 12 08:43:01 2015
*****
Instance Pin                                Transition Type Reverse Slope
DTMF_INST/TDSP_CORE_INST/EXECUTE_INST/acc_reg_13/CK      R      -0.32465
DTMF_INST/TDSP_CORE_INST/EXECUTE_INST/acc_reg_13/CK      R      -0.32465
```

Related Commands

- `set_delay_cal_mode`
- `get_delay_cal_mode`

report_glitch_thresholds

```
report_glitch_thresholds
[-help]
[-failure_point {input | output}]
[-glitch_type <string>]
[-instance_pins <string>]
[-library_pins <string>]
[-list_unannotated_instance_pins]
[-list_unannotated_library_pins]
[-view <string>]
```

Reports user-defined glitch thresholds applied to library or instance pins.

Parameters

-help	Prints out the command usage.
-failure_point {input output}	Specifies if the threshold failure point is input or output. <i>Default:</i> input
-glitch_type <string>	You can specify one or more glitch types (space-separated) for reporting. The valid options are: vl, vh, vlu, vho, and all. When the -glitch_type parameter is specified, only the user-specified glitch types will be checked. <i>Default:</i> all
-instance_pins <string>	Specifies a list of instance pin names.
-library_pins <string>	Specifies a list of library pin names to be reported.
-list_unannotated_instance_pins	Reports list unannotated instance pins. Reports all the instance input pins that do not have a glitch threshold applied.
-list_unannotated_library_pins	Shows a list of all the library input pins that are not covered by the library pin-based/corresponding library cell-based glitch threshold.
-view <string>	Specifies the view name.

Examples

- The following report shows instance-pin-specific glitch threshold summary:

```

+-----+
|                                     |
|           Instance-Pin Glitch Thresholds Summary           |
|                                     |
+-----+-----+-----+-----+
| Glitch Type | Inst Pin Count | Annotated (%) | Not Annotated (%) |
+-----+-----+-----+-----+
| VL          | 153001         | 21075 13.77%  | 131926 86.23 %   |
+-----+-----+-----+-----+
| VH          | 153001         | 21075 13.77%  | 131926 86.23 %   |
+-----+-----+-----+-----+
| VLU         | 153001         | 0 0.00 %      | 153001 100.00%    |
+-----+-----+-----+-----+
| VHO         | 153001         | 0 0.00 %      | 153001 100.00%    |
+-----+-----+-----+-----+

```

The following section of the above report shows the library-cell/library-pin specific glitch threshold summary:

```

|                                     |
|           Library-Pin Glitch Thresholds Summary           |
|                                     |
+-----+-----+-----+-----+
| Glitch Type | Lib Pin Count | Annotated (%) | Not Annotated (%) |
+-----+-----+-----+-----+
| VL          | 742           | 0 0.00%       | 742 100.00%       |
+-----+-----+-----+-----+
| VH          | 742           | 0 0.00%       | 742 100.00%       |
+-----+-----+-----+-----+
| VLU         | 742           | 653 88.01%    | 89 11.99%         |
+-----+-----+-----+-----+
| VHO         | 742           | 653 88.01%    | 89 11.99%         |
+-----+-----+-----+-----+
...
...

```

- The following command generates a verbose report:

```

report_glitch_thresholds \
-list_unannotated_library_pins -list_unannotated_instance_pins

+-----+
|                                     |
|           Unannotated Instance Pins           |
|                                     |
+-----+-----+-----+
| Instance Pin | Unannotated Glitch Type |
+-----+-----+-----+

```



```
buf inst1/I          {VL, VH, VLU, VHO}
buf inst2/I          {VL, VHO}
...
...
+-----+
+-----+
Unannotated Library Pins
+-----+
Library Pin          Unannotated Glitch Type
+-----+
Lib_lV/BUF/I         {VL, VH, VLU, VHO}
Lib_OpV/BUF/I        {VL, VH}
...
...
+-----+
```

Related Commands

- `set_glitch_threshold`
- `get_glitch_threshold`

Related Information

- ["Signal Integrity Delay and Glitch Analysis"](#) chapter in the *Tempus User Guide*.

report_instance_cdb

```
report_instance_cdb  
[-help]  
[-analysis_type {early | late}]  
[{instance_list or collection}]  
[-output_file filename]  
[-quiet]  
[-view viewname]
```

Allows you to report the cdB for all instances or a subset of instances in a design.

Note: This command must be issued after `report_timing`, `update_timing`, or `update_glitch` commands.

Parameters

-help	Outputs a brief description that includes type and default information for each <code>report_instance_cdb</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man report_instance_cdb</code>
- analysis_type {early late}	Specify the analysis type for the query. <i>Default:</i> late
instance_list or collection	Specify the instance for which the cdB association is desired. By default, the cdB association for all the instances is reported.
-output_file filename	Specify the output filename for the instance report. By default, the report output is displayed on the screen.
-quiet	Suppresses warning messages for cells missing in CDB files.
-view {viewnames}	Specifies the view names. By default, all views are reported.

Examples

- The following command reports the cdB instance connection for all the instances in the design and writes it to the inst_cdb.rpt file:

```
tempus > report_instance_cdb -output_file inst_cdb.rpt
```

- The following command reports the cdB instance connection for the u5 instance to the screen:

```
tempus > report_instance_cdb u5
```

```
VIEW: default_emulate_view -- ANALYSIS TYPE: late
```

```
u5 : inv_t1 : test1.cdb.txt
```

- The following command reports the cdB instance connection for the early analysis type for the u5 and u6 instances

```
tempus > report_instance_cdb -analysis_type early {u5 u6}
```

```
VIEW: default_emulate_view -- ANALYSIS TYPE: early
```

```
u5 : inv_t1 : test1.cdb.txt
```

```
u6 : inv_t2 : test2.cdb.txt
```

Related Commands

- [read_lib](#)
- [read_view_definition](#)

report_jitter

```
report_jitter
[-help]
[-eiv_end_time float]
[-eiv_file string]
[-eiv_vdd_vss string]
[-eiv_start_time float]
[-from string]
[-from_fall string]
[-from_rise string]
[-mode {period | cycle_to_cycle | time_interval_error}]
[-outdir string]
[-through <string>]
[-to string]
[ [ > | >> ] ]
```

Provides a native jitter interface to report jitter through the native Tempus engine.

The existing `analyze_jitter` command will continue to be used for Spice-based jitter analysis.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-eiv_end_time float</code>	Specifies the end time in EIV.
<code>-eiv_file string</code>	Specifies the effective instance voltage files for accounting for IR Drop in Jitter computation.
<code>-eiv_vdd_vss string</code>	Specifies the power and ground net name for each EIV file. It is mandatory to provide the power and ground net name for each eiv file provided using the <code>-eiv_file</code> parameter. For example, "vdd1 vss1 vdd2 vss2", vdd1 and vss1 are for the first, and vdd2 and vss2 for the second EIV file - specified using the <code>-eiv_file</code> parameter.
<code>-eiv_start_time float</code>	Specifies the start time in EIV.
<code>-from string</code>	Creates the Spice deck with a mesh net starting from the pin.
<code>-from_fall string</code>	Creates the Spice deck with a mesh net starting from the falling transition of the specified from pin.
<code>-from_rise string</code>	Creates the Spice deck with a mesh net starting from the rising transition of the specified from pin.
<code>-mode {period cycle_to_cycle time_interval_error}</code>	Specifies the mode used by the <code>analyze_jitter</code> command. <i>Default:</i> period
<code>-outdir string</code>	Specifies the output directory for the created Spice.
<code>-through <string></code>	Creates the Spice deck with a mesh net going through the <code>-through</code> points.
<code>-to string</code>	Creates the Spice deck with a mesh net ending at two pins.
<code>> filename</code>	Redirects the <code>-run_path_simulation/-run_simulation</code> output to a file.
<code>>> filename</code>	Appends the <code>-run_path_simulation/-run_simulation</code> output to a file.

Examples

- The following command reports analysis jitter details (for `period` mode) in the specified output directory:

```
report_jitter -eiv_file eiv_file1 -outdir jitter_dir -mode period
```
- The following command reports analysis jitter details (for `cycle_to_cycle` mode) in the specified output directory:

```
report_jitter -eiv_file eiv_file1 -outdir jitter_dir -mode cycle_to_cycle
```

Related Commands

- `analyze_jitter`

report_mutex_condition

```
report_mutex_condition  
[-help]  
[-filename <string>]  
[-net <string>]
```

Reports mutex nets of a victim net.

When this command is used, all coupled nets participating in the mutex for a victim are reported. The coupled nets will be reported as a group of mutex nets (user-defined). By default, all the mutex conditions are reported.

Parameters

-help	Prints out the command usage.
-filename <string>	Specifies the name of the file that saves the command output results.
-net <string>	Reports the coupled nets mutex to the specified net name. You can specify multiple net names with multiple <code>report_mutex_condition</code> commands.

Examples

- The following example shows the mutex nets of a victim net. This report does not indicate the attackers not listed due to the mutex command:

```
Vic: Net1
Attackers: Att1, Att2, Att3
set_mutex_condition {Att1, Att2 }
set_mutex_condition {Att2, Att3 }
set_mutex_condition {Att1, Att3 }

report_mutex_condition -net Net1 will report:
```

```
- Victim Net: Net1
Mutex groups:
{Att1, Att2 },
{Att2, Att3 },
{Att1, Att3 }
```

- The following example shows that if a coupled net has an LC chain, then the entire chain will be reported along with other mutex group nets:

```
Vic: Net1
Attackers: Att1, Att2, Att3
Att1 chain: Agg1 Agg11 Agg111
Att2 chain: Agg2 Agg22 Agg222
Att3 chain: Agg3 Agg33 Agg333
set_mutex_condition {Att1, Att2 }
set_mutex_condition {Att2, Att3 }
set_mutex_condition {Att1, Att3 }

report_mutex_condition -net Net1 will report:
```

```
- Victim Net: Net1
Mutex groups:
{{ Agg1 Agg11 Agg111 }, { Agg2 Agg22 Agg222}}
{{ Agg2 Agg22 Agg222}, { Agg3 Agg33 Agg333}}
{{ Agg1 Agg11 Agg111 }, { Agg3 Agg33 Agg333}}
```

Related Commands

- [set_mutex_condition](#)

report_noise

```
report_noise
[-help]
[-bumpy_waveform]
[-cell_type {bbox | clock | data | latch}]
[-clock]
[-delay {min | max}]
[-double_clocking]
[-double_switching]
[-exclude_failures]
[-expand_virtual_attacker_bus]
[-failure]
[-format string]
[-gui_victim_nets_file]
[-histogram]
[-level {vl | vh | vlu | vho}]
[-max_threshold float]
[-merge]
[-merge_dir string]
[-nets string]
[-output_format string]
[-output_file string]
[-overshoot_undershoot]
[-pins string]
[-precision integer]
[-quiet_nets]
[-remove_status_row <string>]
[-report_virtual_attacker_constituents]
[-reverse_voltage_change_threshold <float>]
[-slack float]
[-skip_nets_or_pins_in_header]
[-sort_by string]
[-style default | extended]
[-threshold float]
[-type string]
[-txtfile filename[.gz]]
[-view string]
```

Generates text-based noise reports.

Note: You need to ensure that the `set_si_mode -enable_glitch_report` (to generate glitch report) or `set_si_mode -enable_delay_report` (to generate detailed delay report) commands are set to `true` before the `report_timing`, `update_timing`, or `update_glitch` commands are run.

For details about generating and interpreting report files, see "Signal Integrity Delay and Glitch Analysis" in the *Tempus User Guide*.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-bumpy_waveform</code>	Flags nets with bumpy waveforms at the receiver. Note: This parameter will always show the maximum delay bumpy report (even if the <code>-delay min</code> option is specified).
<code>-cell_type {bbox clock data latch}</code>	Generates a glitch report for the user-specified cell-type. You can specify one of the following options: <code>bbox</code>: Generates a report for only the black-box receivers <code>clock</code>: Generates a report for only the clock pin receivers <code>data</code>: Generates a report for only the data pin receivers <code>latch</code>: Generates a report for only the latch pin receivers
<code>-clock</code>	Prints the signal integrity report for the clock network.
<code>-delay {min max}</code>	Reports delay uncertainty instead of noise values. The only legal values are <code>min</code> or <code>max</code> for minimum or maximum path delay analysis. The <code>report_noise -clock -delay min max</code> command outputs the minimum/maximum delay report for clock nets. The attacker constituents are reported for nets which are critical and thereby reselected in the second SI iteration. For non-critical nets, the constituent section will be ignored. To report attacker constituents for such nets, the <code>report_delay_calculation -si</code> parameter can be used.
<code>-double_clocking</code>	Enables double clocking reporting.
<code>-double_switching</code>	Reports double-switching analysis data in noise reports.
<code>-exclude_failures</code>	Excludes all failing nets from the glitch report.

<code>-expand_virtual_attacker_bus</code>	Reports virtual attacker components and expands bus bits in the noise reports. This parameter works only when the bus virtual attacker feature is enabled using the <code>set_si_mode - enable_virtual_attacker_constituent_report true</code> setting.
<code>-failure</code>	Reports receiver peak failures and DC tolerance failure nets. Use this option only for glitch noise failure reports. The <code>report_noise</code> command without the <code>-failure</code> option uses 0.10 of vdd as the default. When you specify the <code>-failure</code> option, you need to use <code>-threshold</code> to the lowest noise level saved; if not specified, it defaults to 0.05 Vdd. Option: Directory Name: <code>-sort_by noise glitch</code> <code>-sort_by rcvr_peak glitch</code>
<code>-format <i>string</i></code>	Specifies the output report format fields. The valid fields are: arc, attacker_edge, attacker_logical_correlation_phase, attacker_net, attacker_offset, attacker_peak, attacker_slew, attacker_source, attacker_status, attacker_vdd, attacker_wire_length, attacker_xcap, cell_type, delay, glitch_slack, incr_delay, level, peak, receiver_cell, receiver_input_failure_threshold, receiver_input_pin, receiver_output_failure_threshold, receiver_output_peak, slack, slack_status, stage_info, thresh_source, verbose_logical_correlation_report, victim_edge, victim_library, victim_logical_correlation_phase, victim_net, victim_slew, victim_total_area, victim_total_cap, victim_total_cross_cap, victim_vdd, victim_width, victim_wire_length Note: The <code>victim_total_cross_cap</code> option is supported only for SI delay reporting. Each field corresponds to the following columns of the glitch report: • arc: Arc • attacker_edge: Edge

- attacker_net: Net
- attacker_offset: Offset (ps)
- attacker_peak: Peak (mV)
- attacker_slew: Slew (ps)
- attacker_source: Constituents: Source
- attacker_status: Statue
- attacker_vdd Vdd (V)
- attacker_wire_length
- attacker_xcap: Xcap (fF)
- glitch_slack: Slack
- level: Level
- peak: Peak (mV)
- receiver_cell: Receiver cell
- receiver_input_failure_threshold: Receiver Input Threshold
- receiver_input_pin: ReceiverNet
- receiver_output_failure_threshold: Failure threshold
- receiver_output_peak: Receiver output peak Value
- slack_status: Status
- stage_info
- thresh_source: ReceiverThresholdType (Global | User | Library)
- victim_library: Library
- victim_net: VictimNet
- victim_total_area: TotalArea
- victim_total_cap: TotalCap (fF)
- victim_vdd: Vdd (V)
- victim_width: Width (ps)
- victim_wire_length

The following options are not available with the single-line

	format: ◦ delay ◦ incr_delay ◦ slack ◦ thresh_source ◦ victim_edge ◦ victim_slew ◦ victim_total_cross_cap To choose whether the report format should be on a single line or multiple lines, you can use the <code>-output_format</code> parameter.
<code>-gui_victim_nets_file</code>	Generates the GUI format glitch reports (<code>aae</code> or <code>aae_<viewname></code> files under the reporting directory) and incremental delay reports (<code>aae_<max/min>_incDelay</code> or <code>aae_<viewname>_<max/min>_incDelay</code> files under the reporting directory) to be used in the Display Noise Net form.
<code>-histogram</code>	Generates the noise histogram.
<code>-level {vl vh vlu vho}</code>	Specifies the level for which a report needs to be generated when a single-level report is desired. You can specify one of these options: vl, vh, vlu, and vho.
<code>-max_threshold float</code>	Specifies the maximum peak noise or receiver peak thresholds to report on.
<code>-merge</code>	Generates a merged report that shows only the worst glitch Slack for each pin level combination across views.
<code>-merge_dir string</code>	Specifies the name of the merge database.
<code>-nets string</code>	Reports the specified nets in the noise report.
<code>-pins string</code>	Reports on the specified list of pin names.
<code>-precision integer</code>	Specifies the precision at which the glitch data is to be printed. The allowed values are 0, 1, 2, and 3. <i>Default: 2</i>

<code>-output_file string</code>	Specifies the name of the output file name prefix. The name of the output file is generated by combining the prefix and the specified view name. If you do not specify this parameter: • The software automatically names the text report as 'glitch' for noise glitch reports and 'delay' for noise delay reports. • The software dumps the generated report to the console window for better usability. Note: If the text report file has the .gz extension, the output file is gzipped to generate a compressed report.
<code>-output_format string</code>	Specifies whether the output format should be single-line or multi-line. You can specify one of the following options: • <code>single_line</code> • <code>multiple_line</code> If the <code>single_line</code> option is specified and the attacker info field is selected, all the attacker-related format options will be printed together. The <code>-output_format</code> parameter only works with the <code>-format</code> option and does not affect the default reporting output format. <i>Default:</i> <code>multiple_line</code>
<code>-overshoot_undershoot</code>	Reports the overshoot (VHO) and undershoot (VLU) glitches. The <code>set_si_mode -enable_overshoot_undershoot_glitch</code> must be set to <code>true</code> before the analysis is performed, in order to generate the overshoot-undershoot glitch report. <i>Default:</i> If this option is not used during glitch reporting, then VL and VH types of glitches will be reported.
<code>-quiet_nets</code>	Generates a report of all the quiet attackers.
<code>-remove_status_row <string></code>	Shows only the active attackers and filters out Sb, UF, and LC from the glitch report.
<code>-report_virtual_attacker_constituents</code>	Generates reports for virtual attacker constituents.

<code>-reverse_voltage_change_threshold</code> <code><float></code>	Reports the clock nets with reverse swing exceeding the user-specified threshold (ratio of VDD). By default, the threshold value is set to 0.1.
<code>-slack</code> <i>float</i>	Specifies the maximum slack value for a bumpy net. <i>Units:</i> Seconds
<code>-skip_nets_or_pins_in_header</code>	Generates the glitch report without a list of nets or pins in the report header.
<code>-sort_by</code> <i>string</i>	Requests the sorting of entries. You can sort by: • <code>noise</code> • <code>receiver_peak</code> • <code>slack</code> • <code>cell_type</code> Tempus determines the default HTML output directories based on your chosen options. Depending on the sorting option you select, Tempus sorts the report fields in the following order: -sort_by Option Level 1 Level 2 Level 3 Level 4 <code>rcvr_peak</code> receiver peak noise netname level noise noise - netname level
	For example, if you use the <code>-sort_by noise</code> option, the first level of sorting will be based on noise, followed by receiver peak, netname, and level. Note: The report displays the <code>v1</code> noise level before the <code>vh</code> noise level. <i>Default:</i> <code>receiver_peak</code>

<code>-style {default extended}</code>	Generates reports in one of two styles: <code>extended</code> or <code>default</code>. • <code>extended</code>: Generates extended reports with victim and aggressor timing windows, dropped aggressor slews, and reasons for dropped aggressors. Additionally, the report also shows driver pins and driver library pins. • <code>default</code>: Generates the default report, which is similar to the text report generated using the <code>-text</code> parameter.
<code>-threshold float</code>	Specifies the minimum peak noise, absolute delay uncertainty, relative delay uncertainty, and receiver peak thresholds to report on. <i>Default:</i> • 0.10 of vdd for nets - when the glitch report is generated using the <code>report_noise -sort_by noise</code> option. • 10 (ps) for nets - when a detailed delay report is generated using the <code>report_noise -delay</code> parameter. Note: • The <code>-threshold</code> parameter accepts percentage values. • All <code>-sort_by</code> values that represent voltages will be in mV.
<code>-type string</code>	Specifies the type of noise to be reported - glitch, delay, or slew.

<code>-txtfile filename[.gz]</code>	Specifies the name of the text report file. If the specified file name ends in <code>.gz</code>, the file is gzipped. If you do not specify the <code>-txtfile</code> option: - Tempus automatically names the text report as 'glitch' for noise glitch reports and 'delay' for noise delay reports. - Tempus dumps the generated report to the console window for better usability. If the <code>-txtfile</code> parameter is not used, then Tempus will output the SI report to the screen. When the <code>-txtfile</code> parameter is specified, then Tempus uses the user-specified name. In case of MMMC, Tempus uses the <code><txtfile Name>_<view Name></code> naming convention.
<code>-view string</code>	Specifies the view name.

Examples

- The following command reports noise for the view named `default_emulate_view`:

```
tempus > report_noise -delay max -threshold 1600 -txtfile delay_max_format.rpt -format
{attacker_slew attacker_xcap attacker_status attacker_edge attacker_net stage_info
slack victim_slew victim_total_cross_cap victim_net }

*****
Noise Delay Report (for view default_emulate_view)
*****

Report Options:
-----
-type delay
-threshold 1600 (ps)
-delay max
[-style default]
[-sortBy IncrDelay]
-----

Slack(ps)    Slew(ps)    Total    Xcap(fF)    VictimNet
-2.973      0.635      277.574  seg3/u6/A   {seg3/n5}

Stage information:
      Cell      Pin
Driver:  INV      seg3/u5/Y
Receiver: INV      seg3/u6/A
```

Constituents:

Slew(ps)	Xcap(fF)	Status	Edge	Net
331.500	138.787	INF	F	seg1/n5
358.667	138.759	SY	F	seg2/n5

Slack(ps)	Slew(ps)	Total Xcap(fF)	VictimNet
14.906	0.597	278.115	seg1/u6/A {seg1/n5}

Stage information:

	Cell	Pin
Driver :	INV	seg1/u5/Y
Receiver :	INV	seg1/u6/A

Constituents:

Slew(ps)	Xcap(fF)	Status	Edge	Net
279.500	139.280	INF	F	w3
359.333	138.807	INF	F	seg3/n5

Slack(ps)	Slew(ps)	Total Xcap(fF)	VictimNet
15.737	0.587	270.604	seg1/u12/A {seg1/n11}

Stage information:

	Cell	Pin
Driver :	INV	seg1/u11/Y
Receiver :	INV	seg1/u12/A

Constituents:

Slew(ps)	Xcap(fF)	Status	Edge	Net
276.500	135.654	INF	F	w5
337.167	134.517	INF	F	seg3/n11

- The following command reports noise for the specified reporting fields:

```
tempus > report_noise -delay max -format {delay incr_delay victim_slew victim_total_cap
victim_total_cross_cap victim_vdd victim_net attacker_offset attacker_slew attacker_xcap
attacker_vdd attacker_edge attacker_status attacker_net} -txtfile custom.rpt
```

Delay (NomMax/Max)	IncrDelay(ps)	Slew(ps)	TotalCap(fF)	Total Xcap(fF)	Vdd(V)	VictimNet
-43.400/2418.200	2461.600	12.372	639.088	599.796	1.62	i1/A {n1}

Type	IncrDelay(ps)	Delay (NomMax/Max)	VictimInst
R	3253.300	5457.100/8710.400	drv/Y {INV:A}

Constituents:

Offset(ps)	Slew(ps)	Xcap(fF)	Vdd(V)	Edge	Status	Net
------------	----------	----------	--------	------	--------	-----

```
-245.000    30.000    52.145    1.62    F    att1
-199.000    120.000    26.751    0.00    F    INF    ~n1
```

- The following command shows glitch values greater than 0.20 VDD as:

```
tempus > report_noise -threshold 0.20
Output report file name prefix is (null)
Info: Processing report for view default_emulate_view ...
#####
# Command: report_noise -threshold 0.20
# View: default_emulate_view
#####
```

Report Options:

```
-type glitch
-threshold 20% (of vdd)
```

- To generate a glitch report, you can use the following command:
tempus > report_noise -txtfile glitch.txt
- To generate a glitch report sorted by input noise, you can use the following command:
tempus > report_noise -txtfile sort_by_glitch_input.txt -sort_by noise
- To generate a glitch failure report, you can use the following command:
tempus > report_noise -txtfile glitch_failure.txt -failure
- To generate a glitch report for VL noises sorted by input noise, you can use the following command:
tempus > report_noise -sort_by noise -level vl
- To generate a glitch report for only latch receiver pin cell types, you can use the following command:
tempus > report_noise -txtfile latch_glitch.txt -cell_type latch

For more information on glitch reporting, refer to "[SI Glitch Reporting](#)" in the *Tempus User Guide*.

Related Commands

- [set_si_mode](#)
- [report_timing](#)
- [update_timing](#)
- [report_delay_calculation](#)

report_noise_lib_pin

```
report_noise_lib_pin  
[-help]  
[> filename]
```

Generates a report that provides information on all user-specified noise mappings in the design.

Parameters

-help	Prints out the command usage.
> filename	Writes the report to the specified file name. If the file already exists, the software will overwrite it.

Examples

- The following command generates the output report ‘noise_mapping.rpt’, which contains all the noise mappings present in the design:

```
tempus> report_noise_lib_pin > noise_mapping.rpt  
  
noise_mapping.rpt  
  
-----  
from_pin                to_pin  
-----  
library1/inverter/I      library2/macro/A  
library1/buffer/I        library3/memory/ADD[0]  
-----
```

Related Commands

- [set_noise_lib_pin](#)
- [reset_noise_lib_pin](#)

report_noise_propagation

```
report_noise_propagation  
[-help]  
[-level {VL | VH | worst | all}]  
[-max_paths integer]  
[-max_slack float]  
[-nets string]  
[-nworst integer]  
[-pins string]  
[-source_depth integer]  
[-style {default | extended}]  
[-views string]  
[ [ > | >> ] ]
```

Prints text-based noise propagation reports that show glitch-source tracing for all the user-specified or all-violating receiver pins. The report allows you to view full paths showing the glitch propagating from the source to the violating receiver pin. This helps to easily identify and fix glitch violations.

Note: You need to set the `set_si_mode -enable_glitch_report` parameter to `true` to enable glitch reporting.

Parameters

<code>-help</code>	Prints out the command usage. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man report_noise_propagation</code>
<code>-level {VL VH worst all}</code>	Reports paths for the specified glitch level. You can specify one of these options: VL, VH, worst, or all.
<code>-max_paths integer</code>	Reports the specified number of paths.
<code>-max_slack float</code>	Reports paths with a glitch Slack equal to or less than the specified value.
<code>-nets string</code>	Reports paths with the endpoint being all the input pins connected to the specified net.
<code>-nworst integer</code>	Specifies the maximum number of sources per endpoint for which paths need to be reported.
<code>-pins string</code>	Reports the paths for the specified endpoint pins.
<code>-source_depth integer</code>	Specifies the maximum number of sources to be traced in any given branch in the fanin cone of violation.
<code>-style {default extended}</code>	Provides an enhanced report format that shows the UDN and User-annotated fields. You can specify the following options: <code>default</code>: Prints the ROP column and VDD (in mV) in the output report. <code>extended</code>: Reports the drivers with annotated glitches as propagation sources.
<code>-views string</code>	Reports the paths for the specified view.
<code>> filename</code>	Redirects output to the specified file.
<code>>> filename</code>	Appends output to the specified output file.

Related Commands

- `set_si_mode`
- `report_noise`

report_parameter

```
report_parameter
[-help]
[-correlation]
[-name <string>]
```

Generates a report showing the settings for the `define_parameter` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-correlation</code>	Reports the parameter correlation.
<code>-name <string></code>	Reports the parameter name.
<code>-scope {vtskew wire_variation beol lle layout_effect}</code>	Reports the parameter scope. Allowed options are: • vtskew • wire_variation • beol • lle layout_effect
<code>-view <string></code>	Shows the name of the specified view name.

Examples

- The following command reports the moments information of the M2 layer resistance:

```
tempus > define_parameter -name param_M2_R -scope wire_variation -distribution gaussian
-range {1, 1.15}
tempus > report_parameter -scope wire_variation -view func_spg_m4c_cmax
```

```
-----
PARAMETERS ( SCOPE: wire_variation )
-----
Name           Mean-shift   Std-dev      Distribution   Analysis Type
-----
param_M2_R     0.075        0.025        Normal        Setup
param_M2_R     0.075        0.025        Normal        Hold
-----
```

Related Commands

- `define_parameter`
- `set_wire_variation_mode`
- `get_wire_variation_mode`

Related Topics

- For more information, refer to "[Wire Variation Analysis](#)" section in the *Tempus User Guide*.

report_voltage_scaling

```
report_voltage_scaling
[-help]
[objectListOrCollection]
[-analysis_type {early | late}]
[-view string]
[-output_file string]
```

Reports instances whose timing data is interpolated due to voltage mismatch. When a cell is operating at a voltage different from those at which it is characterized in the timing libraries, delay calculation (DC) interpolates/derives the timing data for the operating voltage.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-analysis_type {early late}</code>	Specifies whether early or late views have to be checked for voltage scaling. If unspecified, only 'late' views are checked.
<code>-output_file string</code>	Specifies the name of the output file to which the report will be written. When the specified file name includes a directory path, the directory hierarchy is automatically created if it does not already exist. If this option is not specified, the report is written to standard output (screen).
<code>objectListOrCollection</code>	Specifies the name(s) or a collection of leaf-level instances for which voltage scaling needs to be checked and reported. If not specified, all instances of the design are checked.
<code>-view string</code>	Specify the name(s) of view(s) to be checked for voltage scaling. If unspecified, all active analysis views are checked.

Examples

The following example shows a sample `report_voltage_scaling` report:

```
tempus 2 > report_voltage_scaling i_6 -analysis_type late -view func_worst_cmax

VIEW: func_worst_cmax -- ANALYSIS TYPE: late
<pin-name> : <cell-name> : <actual-voltage> : <low-lib-name> : <high-lib-name>
i_6/Y      : BUFX5MTH   : 0.820           : elc_08v          : elc_09v
```

The report is generated with the `func_worst_cmax` view. You can use the `-view` parameter to report for other views. You can run the `report_analysis_views` command to list views.

report_voltage_swing

```
report_voltage_swing
[-help]
-high_threshold float
-low_threshold float
[-output_file string]
[-required_width {all | clock}]
[-view string]
```

Reports the voltage swings that are shorter than the user-specified thresholds for both clock and data signals.

Parameters

-help	Outputs a brief description of the <code>report_voltage_swing</code> command. For a detailed description of the command, use the <code>man</code> command: <code>man report_voltage_swing</code>
-high_threshold <i>float</i>	Specifies the high threshold percentage of the pin supply voltage.
-low_threshold <i>float</i>	Specifies the low threshold percentage of the pin supply voltage.
-output_file <i>string</i>	Specifies the name of the output file.
-required_width {all clock}	Reports the required high and low pulse widths corresponding to the user-specified high and low thresholds. You can specify one of the following options: <code>clock</code>: Reports the required widths for all clock pins in the design. <code>all</code>: Reports the required pulse widths for all input pins in the design. Default: <code>all</code>
-view <i>string</i>	Specifies the name of the view being defined. Note: If this option is not specified, it will dump a report for all views.

Examples

An output file in the text format containing violating pin information is presented below:

```
View Name: default_setup_view1
```

Tempus Text Command Reference
Signal Integrity Commands

Hpin	Type_of_Voilation	Vdd	Vss	Slews (Rise/fall)	Margin (threshold)
X/buf/A	(upper/lower)	0.9v	0.0V	210ps/240ps	745mv (810mv)

reset_glitch_derate

```
reset_glitch_derate
[-help]
[-derate_height]
[-derate_width]
[-glitch_type string]
[-instance_pin string]
[-offset]
[-pin string]
[-view string]
```

Removes the derating factors for glitch waveforms that were previously set using the `set_glitch_derate` command.

Parameters

<code>-help</code>	Prints out the command usage.
<code>-derate_height</code>	Resets the derate height.
<code>-derate_width</code>	Resets the derate width.
<code>-glitch_type <i>string</i></code>	Specifies the glitch type: <code>vl</code> , <code>vh</code> , or <code>both</code> .
<code>-instance_pin <i>string</i></code>	Specifies the instance pin names on which derate factors are to be reset.
<code>-offset</code>	Resets the derate offset.
<code>-pin <i>string</i></code>	Specifies the library pin names on which derate factors are to be reset.
<code>-view <i>string</i></code>	Specifies the view name.

Related Commands

- `set_glitch_derate`

reset_noise_lib_pin

```
reset_noise_lib_pin  
[-help]  
[-all]  
[-to string]
```

Allows you to reset any noise property mapping specified previously using the `set_noise_lib_pin` command.

Parameters

-help	Prints out the command usage.
-all	Resets all the noise mappings in the design.
-to <i>string</i>	Specifies the library pin name for which the user-specified noise mapping (if any) needs to be reset. Note: The <code>-to</code> parameter accepts library cell pin names (that can be obtained using the <code>get_lib_pins</code> command). This parameter does not honor collections or instance pin names.

Examples

- The following command resets any noise mapping previously applied to the library pin `library1/macro_cell/A` :

```
tempus> reset_noise_lib_pin -to library1/macro_cell/A
```

- The following command resets all the existing noise mappings in the design:

```
tempus> reset_noise_lib_pin -all
```

Related Commands

- `set_noise_lib_pin`
- `report_noise_lib_pin`

set_advanced_variation_mode

```
set_advanced_variation_mode
[-help]
[-disable_arcs <string>]
-scope {vtskew | aging | wire_variation | beol | lle | layout_effect | lde}
```

Sets the advanced variation scope to configure generic effect-specific settings.

Parameters

-help	Prints out the command usage.
-disable_arcs <string>	Disables the generic effect for any arc types, such as constraint: all, constraint: setup, constraint: hold.
-scope {vtskew aging wire_variation beol lle layout_effect lde}	Provides variation-specific settings.

Related Commands

- `set_wire_variation_mode`
- `set_vt_skew_derate`

set_aging_activity

```
set_aging_activity
[-help]
[-max_duty_cycle <float>]
[-max_toggle_rate <(number of transitions per second)>]
[-min_duty_cycle <float>]
[-min_toggle_rate <float>]
-net <string>
[-views <string>]
```

Defines the duty cycles on nets.

By default, all signals in STA are assumed to have a 50% duty cycle at primary pins. The duty cycle for clock waveforms can be specified in the SDC. The same duty cycle is assumed at every clock pin associated with that waveform. You can override the duty cycle defaults for primary ports and force duty cycles on specific nets in the design.

Starting from the initial default values (and user overrides), Tempus propagates the duty cycle throughout the logic. Tempus gives higher priority to user-defined duty cycles than to those obtained from the SDC, using either constants or clock waveforms.

Parameters

-help	Prints out the command usage.
-max_duty_cycle <float>	Specifies the maximum duty cycle. Min: 0, Max: 1
-max_toggle_rate <(number of transitions per second)>	Specifies the maximum toggle rate. Min: 0, Max: 3.40282346638529e+38
-min_duty_cycle <float>	Specifies the minimum duty cycle. Min: 0, Max: 1
-min_toggle_rate <float>	Specifies the minimum toggle rate (number of transitions per second). Min: 0, Max: 3.40282346638529e+38
-net <string>	Specifies the net names.
-views <string>	Specifies the list of view names.

Examples

- The following command specifies the maximum duty cycle of 0.5 and minimum duty cycle of 0.3 on net, net_ops[0] for view view1:

```
tempus > set_aging_activity -max_duty_cycle 0.5 -min_duty_cycle 0.3 -net net_ops[0] -  
view view1
```

- The following command specifies the maximum and minimum toggle rate on net, net_pfs[45]:
- ```
tempus > set_aging_activity -min_toggle_rate 1e6 -max_toggle_rate 3e9 -net net_pfs[45]
```

## Related Commands

- `get_aging_activity`
- `set_analysis_mode -aging`
- `set_aging_analysis_mode`
- `report_aging_activity`
- `create_aging_stress_profile`
- `create_spice_deck`

## Related Information

- "[Aging-Aware STA Analysis](#)" chapter in the *Tempus User Guide*.



# set\_aging\_analysis\_mode

```
set_aging_analysis_mode
[-help]
[-activity_propagation_mode {probabilistic | worst_case}]
[-constraints {true | false}]
[-disable_aging]
[-max_duty_cycle <value>]
[-max_toggle_rate <value>]
[-max_toggle_rate_limit <value>]
[-min_duty_cycle <value>]
[-min_toggle_rate <value>]
[-param_priority_sets <string>]
[-recovery {true | false}]
[-sequential_activity_propagation {true | false}]
[-sequential_max_duty_cycle <value>]
[-sequential_max_toggle_rate <value>]
[-sequential_min_duty_cycle <value>]
[-sequential_min_toggle_rate <value>]
[-type {bti | hci | all}]
[-use_fixed_duty_cycle {true | false}]
[-views <string>]
```

Sets the aging analysis-related global variable settings and defaults.

By default, all signals in STA are assumed to have a 50% duty cycle at primary pins. The duty cycle for clock waveforms can be specified in the SDC. The same duty cycle is assumed at every clock pin associated with that waveform. You can override the duty cycle defaults for primary ports and force duty cycles on specific nets in the design.

Starting from the initial default values (and user overrides), Tempus propagates the duty cycle throughout the logic.

## Parameters

|                                                            |                                                                                                                                          |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| -help                                                      | Prints out the command usage.                                                                                                            |
| -activity_propagation_mode<br>{probabilistic   worst_case} | Specifies the activity (duty cycle and toggle rate) propagation mode - probabilistic or worst_case.<br><br><i>Default:</i> probabilistic |
| -constraints {true   false}                                | Enables the aging constraints.<br><br><i>Default:</i> false                                                                              |

|                                                              |                                                                                                                                                                                                       |
|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-disable_aging</code>                                  | Disables aging analysis.                                                                                                                                                                              |
| <code>-max_duty_cycle &lt;value&gt;</code>                   | Specifies the default maximum duty cycle.<br><i>Default:</i> 1<br><i>Min:</i> 0, <i>Max:</i> 1                                                                                                        |
| <code>-max_toggle_rate &lt;value&gt;</code>                  | Specifies the default maximum toggle rate (number of transitions per second).<br><i>Default:</i> 100000000<br><i>Min=</i> 0, <i>Max=</i> 3.40282346638529e+38                                         |
| <code>-max_toggle_rate_limit &lt;value&gt;</code>            | Specifies the maximum toggle rate limit.<br><i>Default=</i> 3.40282346638529e+38<br><i>Min=</i> 0, <i>Max=</i> 3.402                                                                                  |
| <code>-min_duty_cycle &lt;value&gt;</code>                   | Specifies the default minimum duty cycle.<br><i>Default:</i> 0<br><i>Min:</i> 0, <i>Max:</i> 1)                                                                                                       |
| <code>-min_toggle_rate &lt;value&gt;</code>                  | Specifies the default minimum toggle rate (number of transitions per second).<br><i>Default:</i> 0<br><i>Min:</i> 0, <i>Max:</i> 3.40282346638529e+38)                                                |
| <code>-param_priority_sets &lt;string&gt;</code>             | Specifies an ordered list of lists of additional aging parameters to be used across all views, e.g., { {p1 p2 p3} {q1 q2} }; parameters from the aging library will always have the highest priority. |
| <code>-recovery {true   false}</code>                        | Enables the aging recovery analysis.<br><i>Default:</i> true                                                                                                                                          |
| <code>-sequential_activity_propagation {true   false}</code> | Enables the activity propagation through sequentials.<br><i>Default:</i> false                                                                                                                        |
| <code>-sequential_max_duty_cycle &lt;value&gt;</code>        | Specifies the default sequential maximum duty cycle.<br><i>Default:</i> 1<br><i>Min:</i> 0, <i>Max:</i> 1                                                                                             |

|                                                            |                                                                                                                                                                                 |
|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-sequential_max_toggle_rate<br/>&lt;value&gt;</code> | <p>Specifies the default sequential maximum toggle rate (number of transitions per second).</p> <p><i>Default=100000000</i></p> <p><i>Min: 0, Max: 3.40282346638529e+38</i></p> |
| <code>-sequential_min_duty_cycle<br/>&lt;value&gt;</code>  | <p>Specifies the default sequential minimum duty cycle.</p> <p><i>Default: 0</i></p> <p><i>Min: 0, Max: 1</i></p>                                                               |
| <code>-sequential_min_toggle_rate<br/>&lt;value&gt;</code> | <p>Specifies the default sequential minimum toggle rate (number of transitions per second).</p> <p><i>Default: 0</i></p> <p><i>Min: 0, Max=3.40282346638529e+38)</i></p>        |
| <code>-type {bti   hci  <br/>all}</code>                   | <p>Specifies the type of aging analysis - bti, hci, or all.</p> <p><i>Default: all</i></p>                                                                                      |
| <code>-use_fixed_duty_cycle {true  <br/>false}</code>      | <p>Enables the fixed duty cycle instead of the propagated duty cycle for aging</p> <p><i>Default: false</i></p>                                                                 |
| <code>-views &lt;string&gt;</code>                         | <p>Specifies a list of view names for the defined settings.</p>                                                                                                                 |

## Examples

- The following command sets the minimum and maximum toggle rate on ports and special nets. This should be specified when HCI analysis is enabled in the flow:  

```
tempus > set_aging_analysis_mode -min_toggle_rate 1e+09 -max_toggle_rate 1e+09
```
- The following command sets the toggle rate on sequential cells' output pins. This should be specified when HCI analysis is enabled in the flow:  

```
tempus > set_aging_analysis_mode -sequential_min_toggle_rate 1e+09 -
sequential_max_toggle_rate 1e+09
```
- The following command models recovery for OMI and URI Spice models that are recovery-aware:  

```
tempus > set_aging_analysis_mode -recovery true
```
- The following command specifies the minimum and maximum toggle rate and duty cycle for probabilistic propagation mode:  

```
tempus > set_aging_analysis_mode -activity_propagation_mode probabilistic \
-min_toggle_rate 1e9 -max_toggle_rate 1e9 -min_duty_cycle 0 -max_duty_cycle 0.2
```

## Related Commands

- `set_analysis_mode -aging`
- `set_aging_activity`
- `create_aging_stress_profile`
- `get_aging_analysis_mode`
- `create_spice_deck`

## Related Information

- "[Aging-Aware STA Analysis](#)" chapter in the *Tempus User Guide*.

## set\_annotated\_glitch

```
set_annotated_glitch
[-help]
[-port string]
[-vh_glitch string]
[-vho_glitch string]
[-view string]
[-vl_glitch string]
[-vlu_glitch string]
```

Allows annotating glitches at any specified pin/port. This command setting will override the VH/VL glitch with the specified value on the given point and use that glitch value to propagate downstream.

If the defined annotated glitch waveforms do not start and end at 0, then the software will not consider those waveforms, and a warning will be issued. In the following example, both VL and VH waves should start and end with 0.0 values. Since the VH wave is not ending at 0, this waveform will be ignored.

```
set_annotated_glitch -vl_glitch {0 0 100e-12 0.01 200e-12 0.04 300e-12 0.03 600e-12 0} -port
dff1/CK -vh_glitch {0 0 100e-12 0.1 200e-12 0.4 300e-12 0.3 600e-12 0.1} .
```

The software does not save (and report) waveforms below 5% of VDD. Consider, for example, the annotated glitch defined on a pin as:

```
set_annotated_glitch -port dff1/CK -vh_glitch {0 0 100e-12 0.01 200e-12 0.04 300e-12 0.03
600e-12 0}
```

In this case, the glitch peak is 0.04 (40 mV), which is below 5% of VDD (1080 mV) - equal to 54 mV. Since this annotated glitch waveform peak is very low, the [report\\_noise](#) command will not report it. However, if the failure criteria is lower than 5%, then the software will save and report small glitch violations.

## Parameters

|                                           |                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                        | Prints out the command usage.                                                                                                                                                                                                                                                                                                                             |
| <code>-port</code><br><i>string</i>       | Specifies port or pin names on which you need to apply glitches.<br><br>This parameter works only when <code>set_si_mode -enable_glitch_propagation true</code> is set in the <code>report_timing/update_timing</code> flow.<br><br>In the <code>update_glitch</code> flow, the <code>-enable_glitch_propagation</code> parameter is already set to true. |
| <code>-vh_glitch</code><br><i>string</i>  | Specifies a list of VH glitch wave points. You must specify at least 3 time value pairs.                                                                                                                                                                                                                                                                  |
| <code>-vho_glitch</code><br><i>string</i> | Specifies a list of VHO glitch wave points.                                                                                                                                                                                                                                                                                                               |
| <code>-view</code><br><i>string</i>       | Specifies a list of view names.                                                                                                                                                                                                                                                                                                                           |
| <code>-vl_glitch</code><br><i>string</i>  | Specifies a list of VL glitch wave points. You must specify at least 3 time value pairs.                                                                                                                                                                                                                                                                  |
| <code>-vlu_glitch</code><br><i>string</i> | Specifies a list of VLU glitch wave points.                                                                                                                                                                                                                                                                                                               |

## Examples

- The following example annotates the specified VL glitch wave points:

```
tempus > set_annotated_glitch -vl_glitch {0 0.0 0.2e-9 0.540 0.405e-9 0} -vh_glitch
{0.1e-9 0 0.205e-9 0.50 0.405e-9 0.200 0.6e-9 0.0} -port {A}
```

- The following command overrides the glitch value at RIP with the annotated glitch, because the glitch is annotated at the receiver input:

```
tempus > set_annotated_glitch -vl_glitch {0 0 100e-12 0.1 200e-12 0.4 300e-12 0.3 600e-
12 0} -port seg2/u9/D
tempus > update_glitch
tempus > report_noise -nets seg2/n8
```

```

Peak(mV) Level TotalCap(fF) TotalArea Width(ps) Vdd(mV) Library VictimNet
399.924 VL 144.11 110.00 550.104 1080 cell_w seg2/u9/D {seg2/n8}
```

Receiver Input Threshold:

```
Value (Threshold) ReceiverNet Receiver failure_type
399.924 (431.244) seg2/u9/D (DFF) [bbox]
```

Constituents:

| Source | Peak (mV) | Offset (ps) | Slew (ps) | Xcap (fF) | Vdd (mV) | Edge | Status | Net       |
|--------|-----------|-------------|-----------|-----------|----------|------|--------|-----------|
| Prp:   |           | 399.924     | --        | --        | --       | --   | --     | seg2/u8/Y |

-----

- In the following example, the impact of aggressors coupled with the victim stage (along with the annotated glitch) is also considered because annotation is done at the driver output:

```
tempus > set_annotated_glitch -vl_glitch {0 0 100e-12 0.1 200e-12 0.4 300e-12 0.3 600e-
12 0} -port seg2/u8/Y
tempus > update_glitch
tempus > report_noise -nets seg2/n8
```

-----

| Peak (mV) | Level | TotalCap (fF) | TotalArea | Width (ps) | Vdd (mV) | Library | VictimNet           |
|-----------|-------|---------------|-----------|------------|----------|---------|---------------------|
| 548.208   | VL    | 144.11        | 159.30    | 581.172    | 1080     | cell_w  | seg2/u9/D {seg2/n8} |

Receiver Input Threshold:

| Value (Threshold) | ReceiverNet | Receiver | failure_type |
|-------------------|-------------|----------|--------------|
| 548.208 (432.648) | seg2/u9/D   | (DFF)    | [bbox]       |

Constituents:

| Source | Peak (mV) | Offset (ps) | Slew (ps) | Xcap (fF) | Vdd (mV) | Edge | Status | Net       |
|--------|-----------|-------------|-----------|-----------|----------|------|--------|-----------|
| Cpl:   |           | 80.148      | 371.000   | 340.000   | 68.138   | 1080 | R ACT  | seg3/n8   |
| Cpl:   |           | 75.159      | 381.000   | 348.500   | 62.007   | 1080 | R ACT  | seg1/n8   |
| Prp:   |           | 392.904     | --        | --        | --       | --   | --     | seg2/u8/Y |

-----

## Related Commands

- [report\\_noise](#)
- [update\\_glitch](#)
- [set\\_si\\_mode](#)
- [report\\_timing](#)

# set\_cdb\_binding

```
set_cdb_binding
[-help]
-from <string>
-to <string>
```

Provides an overriding mechanism that allows a UDN-to-Liberty mapping to be overridden at the Tempus command-file level.

This command helps correct any mismapped cases by enabling the software to select a user-specified cdB-to-Liberty mapping.

## Parameters

|                |                                            |
|----------------|--------------------------------------------|
| -help          | Prints out the command usage.              |
| -from <string> | Specifies the CDB file name and cell name. |
| -to <string>   | Specifies the library file name.           |

## Examples

When loading multiple cdB's with the same PVT, but different secondary voltages, you can use the `set_cdb_binding` command to ensure that they get bound correctly, as shown below:

```
set_cdb_binding -from pvt1_1.cdb -to pvt1_1.lib
set_cdb_binding -from pvt1_2.cdb -to pvt1_2.lib
```



## set\_delay\_cal\_mode

```
set_delay_cal_mode
[-help]
[-reset]
[-accuracy_level level]
[-advanced_pincap_mode {0 | 1 | 2}]
[-accurate_receiver_out_load {true | false}]
[-combine_mmmc {none | early_late}]
[-disable_randomcell_in_socv {true | false}]
[-enable_accurate_lle_analysis {true | false}]
[-enable_corner_based_beol {true | false}]
[-early_irdrop_data_type {best_average | worst | best | average | worst_average}]
[-enable_high_fanout {true | false}]
[-enable_input_slew_sensitivity_on_constraint {true | false}]
[-enable_quiet_receivers_for_hold {true | false}]
[-equivalent_waveform_model {none | no_propagation | propagation}]
[-equivalent_waveform_model_for_timing_check {true|false}]
[-ewm_type {moments | simulation}]
[-honorSlewPropConstraint {true | false}]
[-irdrop_data_type {best_average | worst | best | average | worst_average}]
[-irDrop_window_based {none | late | early | both}]
[-late_irdrop_data_type {best_average | worst | best | average | worst_average}]
[-library_interpolation_mode {linear | non_linear}]
[-ptr_effort <value>]
[-reportOutBound {true | false}]
[-SIAware {true | false}]
[-skip_slew_merge_from_disabled_path {false | true}]
[-slewOutBoundLimitHigh value]
[-slewOutBoundLimitLow value]
[-socv_accuracy_mode {low | medium | high | ultra}]
[-socv_lvf_mode {moments | early_late}]
[-socv_machine_learning_level {0 | 1 | 3}]
[-socv_use_lvf_tables {all | delay | slew | constraint}]
[-wire_variation_correlation_mode {statistical | binomial}]
[-wire_variation_rc_correlation <value>]
```

Sets the global parameters for delay calculation.

By default, the software uses linear interpolation for timing check delay calculation.

## Parameters

|                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                                      | <p>Outputs a brief description that includes type and default information for each <code>set_delay_cal_mode</code> parameter.</p> <p>For a detailed description of the command and all of its parameters, use the <code>man set_delay_cal_mode</code> command.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <code>-accuracy_level level</code>                      | <p>Sets the accuracy level for delay calculations.</p> <p>You can use the <code>-accuracy_level</code> parameter to enable new enhancements for different process nodes. This setting replaces earlier use models where the <code>set_design_mode -process</code> parameter was used to enable delay calculation accuracy settings.</p> <p>The following commands show the mapping of old and new options to use for specific nodes:</p> <pre>set_design_mode -process 16 == set_delay_cal_mode -accuracy_level 0 set_design_mode -process 10 == set_delay_cal_mode -accuracy_level 1 set_design_mode -process 7 == set_delay_cal_mode -accuracy_level 2 set_design_mode -process 5 == set_delay_cal_mode -accuracy_level 3</pre> <p><i>Default: 3</i></p> |
| <code>-accurate_receiver_out_load {true   false}</code> | <p>When set to <code>true</code>, Tempus uses a distributed interconnect model as the receiver output load to compute an accurate miller effect.</p> <p>In this mode, Tempus uses a reduced interconnect model for receivers instead of lumped capacitance as the receiver load. This improves the computation of the back-miller effect and the accuracy of delay estimation.</p> <p>When set to <code>false</code>, Tempus uses lumped capacitance as the receiver's load.</p> <p><i>Default: false</i></p>                                                                                                                                                                                                                                              |

|                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>-advanced_pincap_mode {0   1   2}</pre>            | <p>Simulates the current stage and receiver stages together, dynamically modeling the back-miller effect. You can use the following options:</p> <ul style="list-style-type: none"> <li>• 0: Receiver current sources are computed using ECSM/CCS capacitance tables with the estimated input slew.</li> <li>• 1: Receivers are no longer modeled as simple non-linear current source models. This significantly increases the accuracy of the delay calculation as the modeling of the back-miller effect improves.</li> <li>• 2: Uses an advanced simulation-based approach to model the receiver back-miller effect.</li> </ul> <p><i>Default:</i> 2</p>                                           |
| <pre>-combine_mmmc {none   early_late}</pre>            | <p>Specifies whether delay calculation runs are to be combined for delay calculation simulations. This parameter provides users with control over improving AAE runtime across all analysis types by combining early and late simulations into a single run. The views with common delay corners may also be combined into a single delay-calculation simulation for improved runtime.</p> <p>When <code>none</code> is specified, the software does not combine any delay calculation runs. When <code>early_late</code> is specified, the software combines early and late simulations of a single corner into one delay calculation simulation.</p> <p><i>Default:</i> <code>early_late</code></p> |
| <pre>-disable_randomcell_in_socv {true   false}</pre>   | <p>Disables random cell sensitivity in SOCV mode.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <pre>-enable_accurate_lle_analysis {true   false}</pre> | <p>Enables accurate LLE analysis.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-enable_corner_based_beol {true   false}</code>                                        | <p>Enables anchor-corner-based BEOL (back-end-of-line) analysis.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>-early_irdrop_data_type {best_average   worst   best   average   worst_average}</code> | <p>Allows you to set the early IR drop EIV values for different EIV voltages.</p> <p><i>Default:</i> <code>worst</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>-enable_high_fanout {true   false}</code>                                              | <p>When this parameter is set to <code>true</code>, the default net delay will be annotated on high fanout nets. A net will be treated as a high fanout net when the fanout is more than the specified limit. The default value of this command is <code>1000</code> fanouts. The default net delay is also specified using this command. The default net delay is <code>1ns</code>.</p> <p>The cell delay will use the lumped capacitance specified by the <code>delaycal_default_net_load</code> global variable for the table lookup. The default net load is <code>0.5pF</code>. The driver slew will use the default value of <code>0ps</code>, and there will be no slew degradation along the interconnect.</p> <p>The default value of this parameter is <code>false</code>.</p> |
| <code>-enable_input_slew_sensitivity_on_constraint {true   false}</code>                     | <p>Enables the input slew sensitivity on constraints.</p> <p>When set to <code>true</code>, the input slew sensitivities at constrained and related pins will be considered by the timing check sigma calculation.</p> <p>When set to <code>false</code>, the timing check arc is derived based on the nominal or mean slews only.</p> <p><i>Default:</i> <code>true</code></p>                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>-enable_quiet_receivers_for_hold {true   false}</code>                                 | <p>When set to <code>true</code>, Tempus forces some receivers to be static (“quiet”). To compute the worst-case delays for early analysis, the software assumes that the receivers are not switching. This yields more conservative early delays.</p> <p>When set to <code>false</code>, the software assumes receivers may switch.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>-equivalent_waveform_model {none  </code>                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

```
equivalent_waveform_model {none |
no_propagation | propagation}
```

Controls which equivalent waveform model to use. You can specify one of the following options:

- **none:** When this option is specified, the waveform shape beyond slew measurement points does not impact delays. For example,  

```
set_delay_cal_mode -
equivalent_waveform_model none
```

This mode is equivalent to setting the following options in the previous releases:

```
set_delay_cal_mode -
equivalent_waveform_model_type none
set_delay_cal_mode -
equivalent_waveform_model_propagation
false
```

- **no\_propagation:** In this mode, the impact of receiver waveform shape on the next stage delay is estimated and adjusted the net delay of the current stage. For example,  

```
set_delay_cal_mode -
equivalent_waveform_model no_propagation
set_delay_cal_mode -ewm_type moments
```

This mode is equivalent to setting the following options in the previous releases:

```
set_delay_cal_mode -
equivalent_waveform_model_type ecdsm
set_delay_cal_mode -
equivalent_waveform_model_propagation
false
```

- **propagation:** When specified, the impact of input waveform shape is considered during delay calculation, and the waveforms are propagated across each stage of the entire path. For example,  

```
set_delay_cal_mode -
equivalent_waveform_model propagation
set_delay_cal_mode -ewm_type moments
```

|                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                       | <p>This mode is equivalent to setting the following options in the previous releases:</p> <pre>set_delay_cal_mode - equivalent_waveform_model_type ecsm set_delay_cal_mode - equivalent_waveform_model_propagation true</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <pre>-equivalent_waveform_model_for_timing_check {true   false}</pre> | <p>Models the waveform distortion effect for timing check value lookups at both the data and clock pins.</p> <p>The model uses simulations to convert the distorted waveforms into equivalent waveforms (pre-driver waveforms in the libraries) by matching the pin CCB output waveforms. This feature requires the CCSN library.</p> <p><i>Default:</i> false</p>                                                                                                                                                                                                                                                                                                                       |
| <pre>-ewm_type {moments   simulation}</pre>                           | <p>Specifies the equivalent waveform model (EWM) to be used:</p> <p>The options are described below:</p> <ul style="list-style-type: none"> <li>• <b>moments:</b> Enables moment-based equivalent waveform model.</li> <li>• <b>simulation:</b> Enables simulation-based equivalent waveform model. The simulation-based EWM is required for predicting delay in noisy waveforms.</li> </ul> <p>The <b>simulation</b> option requires noise models (ECSMN/CCSN). If noise models are unavailable, the software defaults to a moments-based EWM.</p> <p>It is recommended to use simulation-based EWM as it is more accurate than moment-based EWM.</p> <p><i>Default:</i> simulation</p> |

|                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>-honorSlewPropConstraint {true   false}</pre>                                        | <p>Determines whether to propagate the slew from the disabled timing arcs to the output pin. When this parameter is set to <code>false</code>, the slew from the disabled timing arc affects the delay of the next stage.</p> <p>When set to <code>true</code>, the timing constraints defined using commands, such as <code>set_disable_timing</code> and <code>set_case_analysis</code> impact the slew propagation, and the slew from the disabled timing arcs is not propagated.</p> <p><i>Default:</i> <code>true</code></p> |
| <pre>-irdrop_data_type {best_average   worst   best   average   worst_average}</pre>      | <p>Allows you to set the early/late IR drop EIV values for different EIV voltages.</p> <p><i>Default:</i> <code>worst</code></p>                                                                                                                                                                                                                                                                                                                                                                                                  |
| <pre>-irDrop_window_based {none   late   early   both}</pre>                              | <p>Searches window-based EIV values from the EIV database.</p> <p><i>Default:</i> <code>both</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <pre>-late_irdrop_data_type {best_average   worst   best   average   worst_average}</pre> | <p>Allows you to set the late IR drop EIV values for different EIV voltages.</p> <p><i>Default:</i> <code>worst</code></p>                                                                                                                                                                                                                                                                                                                                                                                                        |
| <pre>-library_interpolation_mode {linear   non_linear}</pre>                              | <p>Controls library scaling mode - <code>linear</code> or <code>non_linear</code> - for the interpolation flow.</p> <p><i>Default:</i> <code>linear</code></p>                                                                                                                                                                                                                                                                                                                                                                    |
| <pre>-ptr_effort &lt;value&gt;</pre>                                                      | <p>Specifies the PTR analysis effort. This value should be between 0 and 1, where 0 is the lowest accuracy and runtime, and 1 is the highest accuracy and runtime.</p> <p><i>Default:</i> <code>0.5</code></p>                                                                                                                                                                                                                                                                                                                    |

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>-reportOutBound {true   false}</pre> | <p>Generates a report that contains a list of index values (input transition) in the delay tables that are beyond the index range. The index values detected by this parameter can lead to inaccurate delay values being calculated. The report generated by this parameter can be used for debugging.</p> <p>When this parameter is set to <code>true</code>, extrapolation information will be dumped into the following two files:</p> <ul style="list-style-type: none"> <li>• <code>&lt;design-name&gt;.dc.outbound.slew.early.&lt;view-name&gt;.txt</code></li> <li>• <code>&lt;design-name&gt;.dc.outbound.slew.late.&lt;view-name&gt;.txt</code></li> </ul> <p>The software writes a warning message in the log files, noting that there are input transition values that are out of the index range of the timing table. The warning message also provides the file name for viewing the detailed report.</p> <p>When set to <code>true</code>, the <code>-reportOutBound</code> parameter will generate reports that can be controlled using <code>set_outbound_report</code> command.</p> <p><i>Default:</i> <code>false</code></p> |
| <pre>-reset</pre>                         | <p>Resets the parameters to their default values. The <code>-reset</code> parameter must precede the parameters whose values you want to reset. If you specify the <code>-reset</code> parameter by itself, the software resets all <code>set_delay_cal_mode</code> parameters to their default values.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <pre>-SIAware {true   false}</pre>        | <p>Enables SIAware delay calculation that includes cross-talk-induced delays.</p> <p>When set to <code>true</code>, this parameter enables signal integrity (SI) analysis.</p> <p>When set to <code>false</code>, this parameter disables SI analysis.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |



|                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>-skip_slew_merge_from_disabled_path {false<br/>  true}</pre> | <p>Avoids merging slews from unconstrained, segmented paths and black-box output pin.</p> <p>When set to <code>true</code>, the software prevents merging transitions from disabled paths.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <pre>-slewOutBoundLimitHigh value</pre>                           | <p>Limits the maximum slew used in the analysis to the specified value.</p> <p><i>Min:</i> 1</p> <p><i>Max:</i> 3.40282346638529e</p> <p><i>Data type:</i> float</p> <p><i>Default:</i> 3.40282346638529e+38</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <pre>-slewOutBoundLimitLow value</pre>                            | <p>Limits the minimum slew used in the analysis to the specified value.</p> <p><i>Min:</i> 0.001</p> <p><i>Max:</i> 1</p> <p><i>Data type:</i> float</p> <p><i>Default:</i> 0.5</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <pre>-socv_accuracy_mode {low   medium   high  <br/>ultra}</pre>  | <p>Allows control over levels of accuracy.</p> <p>This parameter specifies the level-based accuracy effort for delays and slews. You can use one of the following options:</p> <ul style="list-style-type: none"> <li>• <code>low</code>: Ignores correlation between slews and delays. This is the default setting.</li> <li>• <code>medium</code>: Models the correlation between immediate slew and delay.</li> <li>• <code>high</code>: Considers the skewness of a non-Gaussian distribution into account for computing quantiles. The skewness is local to a single stage.</li> <li>• <code>ultra</code>: Considers the skewness of a non-Gaussian distribution, and also propagates skewness to the next stage in PBA analysis.</li> </ul> <p><i>Default:</i> <code>low</code></p> |

|                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>-socv_lvf_mode {moments   early_late}</pre>                     | <p>Controls the interpretation of LVF data.</p> <p><i>Default:</i> <code>moments</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <pre>-socv_machine_learning_level {0   1   3}</pre>                  | <p>Enables the SOCV machine learning mode.</p> <p>For accurate SOCV analysis, the Tempus software creates several statistical models that are not available in the standard LVF library, such as the correlation between cell delay and output slew, the conversion of moments from late/early sigma tables, and the effects of waveform distortion on sigma and mean shift.</p> <p>You can specify one of the following options:</p> <ul style="list-style-type: none"> <li>• 1: Enables several machine learning algorithms to generate such models internally.</li> <li>• 3: Enabling this mode uses improved modeling of slew-delay correlation using advanced machine learning techniques. This option enables accurate quantile modeling by computing quantiles from mean, stddev, and skewness values.</li> </ul> <p><b>Note:</b> The default setting of 0 is recommended for nodes 110, 40, 28, and 22nm. For other technology nodes, option 3 can be used. For this setting, you must enable the ultra mode using <code>-socv_accuracy_mode ultra</code> option.</p> <p><i>Default:</i> 0</p> |
| <pre>-socv_use_lvf_tables {all   delay   slew   constraint}</pre>    | <p>Controls the types of variations considered during analysis.</p> <p><i>Default:</i> <code>all</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <pre>-wire_variation_correlation_mode {statistical   binomial}</pre> | <p>Specifies wire variation inter-layer correlation. When set to <code>statistical</code>, all the layers are independent of each other</p> <p><i>Default:</i> <code>statistical</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                                                           |                                                                                                                                                                                |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-wire_variation_rc_correlation &lt;value&gt;</code> | <p>Specifies the same R and C layer correlations for wire variation. When set to 1, the R and C sensitivities are considered as fully correlated.</p> <p><i>Default:</i> 1</p> |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Examples

- The following command enables waveform propagation along the path for the equivalent waveform model:  

```
set_delay_cal_mode -equivalent_waveform_model propagation
```
- The following command specifies that the software should not combine any delay calculation runs:  

```
tempus> set_delay_cal_mode -combine_mmmc none
```
- If you have set the value of the `-SIAware` parameter to `true`, use the following command to reset the value of the parameter to the default value, which is `false`, to disable SI analysis during delay calculation:  

```
tempus> set_delay_cal_mode -reset -SIAware
```

Now when you retrieve the value of the `parameter` using the `get_delay_cal_mode` command, the software returns the following:

```
tempus> get_delay_cal_mode -SIAware

-SIAware false # enums={true false}, default=false
false
```

## Related Commands

- [get\\_delay\\_cal\\_mode](#)
- [report\\_delay\\_calculation](#)
- [report\\_timing](#)

## Related Topics

- [Base Delay Analysis](#)" chapter in the *Tempus User Guide*.

# set\_glitch\_derate

```
set_glitch_derate
[-help]
[-derate_height float]
[-derate_width float]
[-glitch_type string]
[-instance_pin string]
[-offset float]
[-pin string]
[-view string]
```

Sets the derating factors for glitch waveforms and determines the effect of on-chip variation on noise.

## Parameters

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                              | Prints out the command usage.                                                                                                                                                                                                                                                                                                                                                                                                                               |
| -<br>derate_height<br><i>float</i> | Specifies the derate height value.<br><br>This parameter scales all the glitch waveforms on the specified pins.<br><br>A value of 1.0 indicates that no glitch waveform derating will take place. A value greater than 1.0, such as 1.1, indicates a 10 percent increase in the magnitude of the complete glitch waveform, which increases propagated noise and noise failure. A value less than 1.0 reduces the magnitude of the complete glitch waveform. |
| -derate_width<br><i>float</i>      | Specifies the derate width value.                                                                                                                                                                                                                                                                                                                                                                                                                           |
| -glitch_type<br><i>string</i>      | Specifies the glitch type - <i>v1</i> , <i>vh</i> , or <i>all</i> .                                                                                                                                                                                                                                                                                                                                                                                         |
| -instance_pin<br><i>string</i>     | Specifies the instance pin names on which derate is to be applied.                                                                                                                                                                                                                                                                                                                                                                                          |
| -offset <i>float</i>               | Specifies the derate offset value.                                                                                                                                                                                                                                                                                                                                                                                                                          |
| -pin <i>string</i>                 | Specifies the library pin names on which the derate value is to be applied.                                                                                                                                                                                                                                                                                                                                                                                 |
| -view <i>string</i>                | Specifies the name of the view.                                                                                                                                                                                                                                                                                                                                                                                                                             |

## Related Commands

- `reset_glitch_derate`
- `update_glitch`

# set\_glitch\_threshold

```
set_glitch_threshold
[-help]
[-cells string]
[-failure_point {none | input | internal | last}]
[-glitch_type {both | vl | vh | vlu | vho | vl_vh | vlu_vho}]
[-library_pins <string>]
[-nets string]
[-pins string]
[-pin_type {clock | data | latch | all}]
[-threshold_type {failure | reporting | propagation}]
-value float
[-views string]
```

Allows for additional control of failure criteria for SI glitch analysis.

## Parameters

|                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                                                                   | Prints out the command usage. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man set_glitch_threshold</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| -cells <i>string</i>                                                    | Specifies a list of library cell names.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| -failure_point<br>{none   input  <br>internal  <br>last}                | <p>Specifies where the failure or reporting criteria apply.</p> <ul style="list-style-type: none"> <li>• <b>none:</b> Criteria does not apply.</li> <li>• <b>input:</b> Threshold applies when the failure point is at the input pin of the receiver cell.</li> <li>• <b>internal:</b> Threshold is applied when the failure point is internal to the cell. This occurs for multi-stage cells. The failure point will be the output of the first stage inside the cell.</li> <li>• <b>last:</b> Threshold is applied when the failure point is the output of the cell. This occurs for single-stage cells, such as inverters.</li> </ul> <p><i>Default:</i> <code>internal</code> and <code>last</code></p> |
| -glitch_type<br>{both   vl   vh<br>  vlu   vho  <br>vl_vh  <br>vlu_vho} | <p>Specifies the glitch type to which the command applies.</p> <ul style="list-style-type: none"> <li>• <b>both:</b> Refers to the <code>vl</code> and <code>vh</code> types of glitches.</li> <li>• <b>vl_vh:</b> Includes both VL and VH noise.</li> <li>• <b>vlu_vho:</b> Includes both VLU and VHO noise.</li> </ul> <p><i>Default:</i> <code>both</code></p>                                                                                                                                                                                                                                                                                                                                           |

|                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-library_pins</code><br><i>string</i>                               | Specifies a list of library pin names.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>-nets</code> <i>string</i>                                          | Specifies the list of nets the command applies to.<br>Default is not net-specific.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>-pins</code> <i>string</i>                                          | Specifies the list of pins or primary ports.<br>Default is not pin-specific.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>-pin_type</code><br>{clock  data  <br>latch  all}                   | Specifies the pin type to which the command applies.<br><i>Default:</i> all                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>-threshold_type</code><br>{failure  <br>reporting  <br>propagation} | <p>Specifies whether the command options apply to failure, reporting, or propagation criteria.</p> <p>In case of VL/VH glitch analysis, it is mandatory to specify the <code>-threshold_type</code> parameter. This is because when the computed glitch (ROP) at any given stage exceeds the failure threshold (with glitch propagation enabled), it is not propagated and is flagged as a failure in the noise report.</p> <p>With the <code>-threshold_type failure</code> setting, the propagation will stop as per the new threshold, and the failure flagging will change accordingly. However, with the <code>-threshold_type reporting</code> setting, the propagation threshold remains unchanged, but the failure reporting is updated.</p> <p>When <code>-threshold_type propagation</code> is used, the software allows for failing nets with ROP values less than the propagation threshold to propagate through single-stage cells.</p> <p>In case of VLU/VHO glitch analysis, it is optional to use the <code>-threshold_type</code> parameter because the above flow is not applicable.</p> <p><i>Default:</i> failure</p> |
| <code>-value</code> <i>float</i>                                          | Specifies the threshold value as a ratio of the local VDD.<br><i>Default:</i> 0.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <code>-views</code> <i>string</i>                                         | Specifies a list of view names.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

## Examples

Here are some examples.

- The following command moves the failure point from ROP to RIP for `inst1/a1` and uses a RIP failure of  $0.2 \times \text{ReceiverVDD}$ :

```
set_glitch_threshold -failure_point input -pin inst1/a1 -threshold_type failure -value 0.2
```

- The following command sets clock failure criteria to  $0.1 \times \text{VDD}$ , latch failure criteria to  $0.2 \times \text{VDD}$  and all other nets to  $0.3 \times \text{VDD}$  (set before `update_glitch`):

```
tempus> set_glitch_threshold -threshold_type failure -pin_type all -failure_point last -value 0.3
```

```
tempus> set_glitch_threshold -threshold_type failure -pin_type clock -failure_point last -value 0.2
```

```
tempus> set_glitch_threshold -threshold_type failure -pin_type latch -failure_point last -value 0.1
```

- The following command sets the failure threshold at the output of the 1st channel connect inside all the instances of `macro1` to be 0.2 (set before `update_glitch`):

```
tempus> set_glitch_threshold -threshold_type failure -cell {macro1} -failure_point internal -value 0.2
```

- The following command changes the reporting criteria for latches to be  $0.15 \times \text{VDD}$  (set after `update_glitch`):

```
tempus> set_glitch_threshold -threshold_type reporting -pin_type latch -value 0.15
```

If you need to set all the data failure criteria to be 0.3, then you need to make the following settings:

```
tempus> set_glitch_threshold -failure_point internal \
-threshold_type failure -value 0.3 \
-pin_type data
```

```
tempus> set_glitch_threshold -failure_point last \
-threshold_type failure -value 0.3 \
-pin_type data
```

## Related Commands

- `get_glitch_threshold`
- `report_glitch_thresholds`
- `update_glitch`



# set\_model\_priority

```
set_model_priority
[-help]
[-objects string]
[-priority_list string]
```

Allows you to define the priority order of models.

## Parameters

|                              |                                                                                    |
|------------------------------|------------------------------------------------------------------------------------|
| -help                        | Prints out the command usage                                                       |
| -objects <i>string</i>       | Specifies a list of library cells or library pins for which priority is specified. |
| -priority_list <i>string</i> | Specifies a list of pin model priorities.<br>Default: {ecsm ccs nldm cdb udn}      |

## Examples

- To force CCSN to be used, you can set the `timing_library_read_ccs_noise_data` global variable to `true` before reading the timing libraries and then use the following command before using the `update_glitch` command:  

```
set_model_priority -priority_list ccs
```

## Related Commands

- `update_glitch`

# set\_mutex\_condition

```
set_mutex_condition
[-help]
[netNames]
```

Allows the software to use one of the specified attackers - each attacker is evaluated and the worst one (for each victim net) is selected.

This command applies a mutex group to a set of attackers (for a victim) to reduce pessimism. A mutex group is a group of nets of which only one can transition in a given cycle.

The `set_mutex_condition` command will allow nets to be specified in multiple mutex commands when the `set_si_mode -enable_multi_mutex` parameter is set to `true`.

For example:

```
set_mutex_condition {Net1 Net2}
set_mutex_condition {Net2 Net3}
```

## Parameters

|                       |                                             |
|-----------------------|---------------------------------------------|
| <code>-help</code>    | Prints out the command usage.               |
| <code>netNames</code> | Specifies the net names of the mutex group. |

## Examples

- The following example shows that only one of the specified nets, between `abc[0-3]`, can switch at any given point in time:

```
tempus> set_mutex_condition {abc[0] abc[1] abc[2] abc[3]}
```

Each specified attacker will be evaluated, and the worst will be used.

## Related Commands

- `report_mutex_condition`

# set\_noise\_lib\_pin

```
set_noise_lib_pin
[-help]
-from string
-to string
```

Allows you to map the noise properties of an instance pin to those of a different library cell pin.

## Parameters

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                  | Prints out the command usage.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| -from<br><i>string</i> | Specifies the library pin name whose definition will be mapped from the given pin name.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| -to<br><i>string</i>   | <p>Specifies the library pin name whose definition will be mapped to the given pin name.</p> <p>When multiple <code>set_noise_lib_pin</code> commands are run on the same <code>-to</code> pin, the software will honor the first command only.</p> <p><b>Note:</b> The <code>-from</code> and <code>-to</code> parameters also accept library cell pin names (that can be retrieved using the <code>get_lib_pins</code> command). These parameters do not honor collections or instance pin names.</p> |

## Examples

- The input pin A of the `macro_cell` cell inherits the noise characteristics of the input pin I of the library cell, `ccs_noise_lib/inv`. This information is used to calculate the pin's noise immunity, `library1/macro_cell/A`.  
tempus> set\_noise\_lib\_pin -to library1/macro\_cell/A -from ccs\_noise\_lib/inv/I
- The following syntax specifies the library pin name whose definition will be mapped from the given pin name to the given pin name:  
tempus> set\_noise\_lib\_pin -from [from\_lib/from\_cell/from\_pin] -to [to\_lib/to\_cell/to\_pin]
- The following syntax specifies that a bidi pin is mapped to two pins; that is, -1 input pin and 1 output pin.  
tempus> set\_noise\_lib\_pin -to library1/macro\_cell/PAD -from "ccs\_noise\_lib/inv/I ccs\_noise\_lib/inv/Z"

## Related Commands

- `report_noise`

# set\_outbound\_report

```
set_outbound_report
[-help]
-delay {early | late | both}
[-filename_prefix string]
[-report_constant_nets {true | false}]
[-type {lower | upper | both}]
```

Controls what extrapolation information is to be saved in the log file(s). The slew extrapolation information can be stored in separate files.

By default, the software reports slew extrapolation for both minimum and maximum analysis when outbound reporting is enabled by setting `set_delay_cal_mode -reportOutBound` to `true`.

## Parameters

|                                                   |                                                                                                                     |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                                | Prints out the command usage.                                                                                       |
| <code>-delay {early   late   both}</code>         | Control whether extrapolation information is to be shown for min or max (or both) analyses.<br><i>Default:</i> both |
| <code>-filename_prefix <i>string</i></code>       | Specifies the prefix to be used in the name(s) of the report file(s).<br><i>Default:</i> design-name                |
| <code>-report_constant_nets {true   false}</code> | Controls whether extrapolation information is shown for constant nets.                                              |
| <code>-type {lower   upper   both}</code>         | Controls whether extrapolation information is to be shown for lower or upper (or both) outbound slews.              |

## Examples

- The following command will dump slew extrapolation information for maximum analysis into the myreport.dc.outbound.slew.late.<view-name>.txt file:

```
tempus> set_outbound_report -delay late -filename_prefix myreport
```

## Related Commands

- `set_delay_cal_mode`

# set\_quiet\_attacker

```
set_quiet_attacker
[-help]
[-analysis_type {early | late}]
[-attacker string]
[-transition {rise | fall}]
[-victim string]
[-view string]
```

Specifies one or more coupled nets as non-attackers (silent) for the delay and glitch analysis of a given victim net.

You can run the `update_glitch` or `update_timing` command and view the results in the `report_noise` or `report_timing` command output.

## Parameters

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                         | Prints out the command usage.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>-analysis_type {early   late}</code> | Specifies the analysis type for which the attackers will be assumed quiet. If no value is specified, attackers are skipped for both <code>early</code> / <code>late</code> analysis.                                                                                                                                                                                                                                                                                                               |
| <code>-attacker <i>string</i></code>       | Specifies the attackers that should be made quiet for the given victim net(s). If no attacker is specified, all the aggressors of the given victim(s) will be ignored.<br><b>Note:</b> If no victim is specified, the specified attackers are made quiet for all victim nets in the design.                                                                                                                                                                                                        |
| <code>-transition {rise   fall}</code>     | Specifies the victim transition for which attackers will be assumed quiet. If no value is specified, attackers are skipped for both <code>rise</code> and <code>fall</code> transitions of the victim net.                                                                                                                                                                                                                                                                                         |
| <code>-victim <i>string</i></code>         | Specifies a list of victim nets for which attackers should be made quiet. Specific attackers can be skipped using the <code>-attacker</code> option of this command. If the <code>-attacker</code> option is not specified, all the attackers of the given nets are made silent, and SI analysis will not be performed on these nets.<br><b>Note:</b> If this option is not specified, the attackers specified in the <code>-attacker</code> option are made quiet for all the nets in the design. |
| <code>-view <i>string</i></code>           | Specifies the view for which attackers should be made quiet. If no view is specified, attackers are assumed quiet in all views.                                                                                                                                                                                                                                                                                                                                                                    |

## Examples

Consider a design having an attacker `agg` to victim nets `vic1` and `vic2`. The following command sets the attacker `agg` as quiet for both the victim nets `vic1` and `vic2`.

```
tempus > set_quiet_attacker -attacker {agg}
```

- The following command sets the attackers `A1` and `A2` as quiet when `V1` is rising during late analysis.

```
tempus > set_quiet_attacker -victim V1 -attacker A1 A2 -transition rise -analysis_type
late
```

- The following command sets the attacker `A1` quiet for victim `V1` in the analysis view `VIEW1`. In this case, `A1` may be a valid attacker during SI analysis for victim `V1` in views other than `VIEW1`.

```
tempus > set_quiet_attacker -victim V1 -attacker A1 -view {VIEW1}
```

Consider a design having victim `vic` with attackers `agg1` and `agg2`:

- The following command sets both the attackers, `agg1` and `agg2`, as quiet for the victim net `vic`.

```
tempus > set_quiet_attacker -victim {vic}
```

- The following command sets the attacker `agg1` as quiet for the given victim net `vic`

```
tempus > set_quiet_attacker -attacker {agg1} -victim {vic}
```

## Related Commands

- [report\\_noise](#)
- [report\\_timing](#)

## set\_si\_mode

```
set_si_mode
[-help]
[-reset]
[-accumulated_small_attacker_factor double]
[-accumulated_small_attacker_mode {cap | current | zero_mean}]
[-accumulated_small_attacker_threshold double]
[-attacker_alignment {path | path_overlap | timing_aware_edge | input_arrival}]
[-clock_delta_delay_threshold double]
[-clocks {asynchronous | synchronous}]
[-constrained_input_threshold_failure_point {input | both}]
[-delay_only {true | false}]
[-delta_delay_annotation_mode {lumpedOnNet | arc}]
[-delta_delay_threshold double]
[-double_switching {none | clock | data_sequential | all}]
[-enable_advanced_flow {true | false}]
[-enable_bus_attacker_correlation {true | false}]
[-enable_delay_report {true | false}]
[-enable_double_clocking_check {true | false}]
[-enable_drv_with_delta_slew {true | false}]
[-enable_dynamic_receiver_peak_limits {true | false}]
[-enable_glitch_propagation {true | false}]
[-enable_glitch_propagation_spice_deck {true|false}]
[-enable_glitch_report {true | false}]
[-enable_lle_glitch {true | false}]
[-enable_logical_correlation {true | false}]
[-enable_overshoot_undershoot_glitch {true | false}]
[-enable_two_stage_driver_weakening {true | false}]
[-enable_user_defined_bus_merging {true | false}]
[-enable_virtual_attacker_constituent_report {true | false}]
[-glitch_accuracy_level <int>]
[-glitch_analysis_type {source | full_propagation}]
[-hold_slack double]
[-individual_attacker_simulation_filtering {true | false}]
[-individual_attacker_clock_threshold float]
[-individual_attacker_threshold double]
[-input_glitch_full_propagation_threshold double]
[-input_glitch_full_propagation_vh_threshold double]
[-input_glitch_full_propagation_vl_threshold double]
[-input_glitch_threshold double]
[-input_glitch_vh_threshold double]
[-input_glitch_vl_threshold double]
[-nonlinear_attacker_slew {true | false}]
[-num_si_iteration int]
```



```
[-receiver_clk_peak_limit double]
[-receiver_latch_peak_limit double]
[-receiver_peak_limit double]
[-report_max_virtual_attacker_constituents int]
[-secondary_attacker_decoupling_factor double]
[-separate_delta_delay_on_data {true | false}]
[-setup_slack double]
[-si_reselection {delta_delay | slack}]
[-si_reselection_delay_threshold double]
[-skip_noise_model_check pins]
[-skip_tw <nets>]
[-switch_prob double]
[-unconstrained_net_use_inf_tw {true | false}]
[-use_infinite_TW {true | false}]
[-virtual_inter_die_attacker_slew <value>]
```

Sets signal integrity configuration parameters used for crosstalk analysis and repair.

## Parameters

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                                     | Outputs a brief description that includes type and default information for each <code>set_si_mode</code> parameter. For a detailed description of the command and one of its parameters, use the man command: <code>man set_si_mode</code>                                                                                                                                                                                                                                                                                                                                                                                 |
| -reset                                    | Resets all or specified options to their default values                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| -accumulated_small_attacker_factor double | Specifies the multiplication factor that controls the effect of the cap mode on the accumulated small attacker. An accumulated small attacker is an imaginary attacker that represents the sum of weak attackers on a specified net. To reduce pessimism when using the accumulated small attacker, adjust this value to anything between 0 and 1. A value of 0 means that the cap mode accumulated by a small attacker will not be used.<br><br><i>Default:</i> 1<br><br><b>Note:</b> This option does not work with the current mode accumulated small attackers, which is the default for process sizes 45nm and below. |

|                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>-accumulated_small_attacker_mode {cap   current   zero_mean}</pre> | <p>Specifies whether to use the current matching based method for accumulated small attacker can specify the following options:</p> <ul style="list-style-type: none"><li>• <code>cap</code>: Uses the accumulated small attacker method, which matches the total coupling capacitance of an attacker with the small attackers that are being virtualized. This is default mode for all process nodes above .</li><li>• <code>current</code>: Uses the current matching-based method for accumulated small attackers. V this method, the transitions of virtual attack are modeled using Piece Wise Linear (PW waveforms. The PWL waveforms are gene so that the current induced by the transition matches the total current induced by all accumulated small attacker components.</li><li>• <code>zero_mean</code>: Uses the accumulation method where a series of small attackers (that is, attackers with less than the individual attac threshold value), are assumed to transition opposing directions and cancel one another aggregate.</li></ul> <p><i>Default:</i> <code>current</code> - for 45nm and below proces nodes.</p> |
| <pre>-accumulated_small_attacker_threshold &lt;double&gt;</pre>         | <p>Changes the accumulated small attacker glitch threshold. If the virtual attacker produces a gli whose peak is below this threshold, the comp coupling capacitors of the accumulated small attacker are split into two ground capacitors a there is no accumulated small attacker for the victim. The value is specified as a ratio of VDI</p> <p>To disable the accumulated small attacker, se value greater than or equal to 1.</p> <p><i>Default:</i> 10.01</p> <p>By default, this option is disabled.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

```
-attacker_alignment {path | path_overlap |
timing_aware_edge | input_arrival}
```

Specifies the attacker alignment. You can use following options:

- **path**: The victim worst arrival edge is used determining the attackers. Crosstalk impact computed according to the latest/earliest a edge of victim for max/min analysis.
- **path\_overlap**: The full timing window of victim considered while determining the attacker. Crosstalk impact is calculated according to the victim edge, lying anywhere in timing window which may experience maximum impact.
- **timing\_aware\_edge**: Realistic analysis similar to path-based analysis is performed. Each attacker's impact is annotated on victim transition where attacker timing window edge aligns with the victim. However, to ensure it bounds PBA analysis, the timing-aware edge mode handles some nets based on their timing properties in GBA analysis.
- **input\_arrival**: The attackers use timing windows formed by 2-sigma arrivals regardless of the analysis sigma multiplier. The victim input arrivals will be used for SI computation. Also, the mean victim timing window and edge will be used for SI computation in GBA and modes, respectively.

This behavior allows pessimism reduction in GBA SI analysis with narrower timing window. And provides a more accurate PBA SI analysis with no dependency of driver cell GBA delay. For more information, refer to the [Attacker Selection Based on Alignment](#) section in the *Tempus User Guide*.

**Default:** `input_arrival`

|                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <pre>-clock_delta_delay_threshold double</pre>                           | <p>Specifies the value above which the clock delay should be calculated.</p> <p><i>Default:</i> -1.</p> <p>The <code>-clock_delta_delay_threshold</code> parameter provides a way to specify different delta delay thresholds for clock nets only.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <pre>-clocks {asynchronous   synchronous}</pre>                          | <p>Treats the clocks either synchronous or asynchronous to each other.</p> <p><i>Default:</i> synchronous</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <pre>-constrained_input_threshold_failure_point {input<br/>  both}</pre> | <p>Allows you to control the glitch failure criteria where a user specifies an object-specific RIP threshold.</p> <p>When set to <code>input</code>, the software skips ROP analysis and performs RIP-based failure check on the receiver pins for cases where object-specific RIP threshold is set using the <code>set_glitch_threshold_failure_point input -net (or -pin)</code> command.</p> <p>When set to <code>both</code>, the software performs both ROP and RIP failure checks, wherever possible. If specified RIP failure criteria is violated, then it will be flagged as a RIP violation. If RIP criteria pass and receiver noise models exist, then the ROP is computed and checked against the ROP failure threshold for failure determination.</p> <p>For the following cases where the object-specific RIP thresholds are not specified, the command behavior remains unchanged:</p> <ul style="list-style-type: none"> <li>• For receiver pins with noise models present, ROP-based failure check is performed. The RIP-based failure criteria is not checked at all.</li> <li>• For receiver pins with noise models missing, RIP-based failure criteria is checked for violations.</li> </ul> <p><i>Default:</i> input</p> |

|                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <pre>-delta_delay_annotation_mode {lumpedOnNet   arc}]</pre> | <p>Calculates the delay for each arc (cell arc and arc) in a path. By default, AAE calculates the c including SI for each arc and if the user chooses to view the SI delay separately, it would show annotated to each arc.</p> <p>Alternatively, you can choose to annotate the delay solely to the net. In this case, the worst-arc is used to calculate the SI delay and it is annotated solely to the net.</p> <p><b>lumpedOnNet:</b> Lumps the delta delay on a the r is broken out.</p> <p><b>arc:</b> Separates the delta delay on each cell ar net arc, as calculated.</p> <p><b>Default:</b> arc</p>                                                                                                                                                                                                                                                                                                                                                                                               |
| <pre>-delta_delay_threshold &lt;double&gt;</pre>             | <p>Specifies a threshold value for a design below which the crosstalk impact will be removed or given net. Using this parameter, Tempus removes the SI impact on a stage if the impact is less than the specified threshold. The threshold value is specified in seconds.</p> <p><b>Default:</b> -1</p> <p>By default, the software uses the <code>-delta_delay_threshold</code> settings. If you set the <code>clock_delta_delay_threshold</code> parameter, then it will be used for the clock nets.</p> <p>When a threshold value using <code>-delta_delay_threshold</code> parameter is specified (and the default value of <code>-delta_delay_threshold</code> parameter is used), then Tempus will use the <code>-delta_delay_threshold</code> value for both data and clock.</p> <p>If you define values for both <code>-delta_delay_threshold</code> and <code>-clock_delta_delay_threshold</code> parameters, then Tempus will use the user-defined settings for data and clock, respectively.</p> |

|                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| <pre>-double_switching {none   clock   data_sequential<br/>  all}</pre> | <p>Specifies the type of data (clock, data sequential) for which double switching analysis will be performed.</p> <p><i>Default:</i> none</p>                                                                                                                                                                                                                                                                                                                                                                         |
| <pre>-delay_only {true   false}</pre>                                   | <p>Performs noise-on-delay analysis exclusively means that when you use this parameter, the software does not perform glitch analysis and performs noise-on-delay analysis. In addition, parameter performs SI analysis.</p> <p>To enable glitch noise, you can use the <code>-enable_glitch_report</code> parameter or run the <code>update_glitch</code> command.</p> <p>Note that <code>-delay_only</code> cannot be used with <code>-enable_glitch_report</code>.</p> <p><i>Default:</i> false</p>                |
| <pre>-enable_advanced_flow {true   false}</pre>                         | <p>Enables advanced SI analysis.</p> <p>When set to <code>true</code>, the advanced SI analysis flow is enabled, in which attacker slews are computed before establishing a signoff-accurate base delay calculation before SI analysis begins. Apart from accuracy benefits, it also provides runtime benefit through enhanced reuse from signoff-accurate base iteration.</p> <p>When set to <code>false</code>, a fast pre-SI analysis is performed for attacker slew computation.</p> <p><i>Default:</i> false</p> |

|                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| <pre>-enable_bus_attacker_correlation {true   false}</pre> | <p>Enables the software to treat bus bits as correlated attackers.</p> <p>When set to <code>true</code>, the bits of bus type nets are grouped together based on their cumulative voltage swing.</p> <p>The bus bits that act as attackers to a specific victim (bits of bus) are treated as a group. All the attackers are grouped into different bundles where normal (non-bus) attackers form one group and bits of the bus form multiple other groups. In case of a violation that has a number of small couplings to bits of bus, the combined effect of a full bus can be used to decide whether an attacker is significant.</p> <p><i>Default:</i> <code>false</code></p> |
| <pre>-enable_double_clocking_check {true   false}</pre>    | <p>Enables double clocking check in AAE.</p> <p>This option enables double clocking analysis. The <code>report_double_clocking</code> command can be run to print a report.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <pre>-enable_delay_report {true   false}</pre>             | <p>Enables SI delay reports.</p> <p>For generating the SI delay report, use the following:</p> <pre>set_si_mode -enable_delay_report true; report_noise -delay max -txtfile max_delay.txt; report_noise -delay min -txtfile min_delay.txt;</pre> <p>To run <code>report_noise -delay</code>, you need to set <code>update_timing</code> parameter prior to <code>update_timing</code>.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                                             |
| <pre>-enable_drv_with_delta_slew {true   false}</pre>      | <p>When set to <code>true</code>, the <code>max_transition</code> violation is reported based on the SI slews. Also, this parameter is mandatory to allow SI slew reporting/fixing by the <code>eco_opt_design</code> command and should be applied prior running the <code>write_eco_opt_db</code> command.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                                                                                                                       |

|                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| <pre>-enable_dynamic_receiver_peak_limits {true   false}</pre>  | <p>Enables automatic computation of ROP threshold value of two-stage cells, using the CCS-noise on timing arcs.</p> <p>When set to <code>true</code>, the software computes the threshold for the last stage from the table of D currents.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                          |
| <pre>-enable_lle_glitch {true   false}</pre>                    | <p>Enables SI glitch analysis for LLE.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                                                                                                                                                                                                              |
| <pre>-enable_logical_correlation {true   false}</pre>           | <p>Enables the AAE-SI logical correlation. Logical correlation refers to the correlation of logic between (aggressor) nets that are connected to each other. A simple case would be input and output nets of an inverter never switch in the same direction, so they cannot be attackers to the same net simultaneously.</p> <p><i>Default:</i> <code>false</code></p> <p><b>Note:</b> You can also use this parameter for Glitch Analysis.</p>                                                   |
| <pre>-enable_glitch_propagation {true   false}</pre>            | <p>Enables glitch propagation through single-stage cells and driver weakening through single-stage and dual-stage cells. When this attribute is set to <code>true</code>, the software does not delete the glitch waveform saved on instance input pins. The waveform is later used by Spice deck to propagate a glitch at the next stage receiver pin.</p> <p><i>Default:</i> <code>false</code></p> <p><b>Note:</b> The software does not support glitch propagation for multi-stage cells.</p> |
| <pre>-enable_glitch_propagation_spice_deck {true   false}</pre> | <p>Enables the driver input glitch waveform to be printed in the SPICE deck, such that SPICE can also perform single-stage glitch propagation. When this option is set to <code>true</code>, the software does not delete the glitch waveform saved on instance input pins. The waveform is later used by Spice Deck to propagate a glitch at the next stage receiver pin.</p> <p><i>Default:</i> <code>false</code></p>                                                                          |



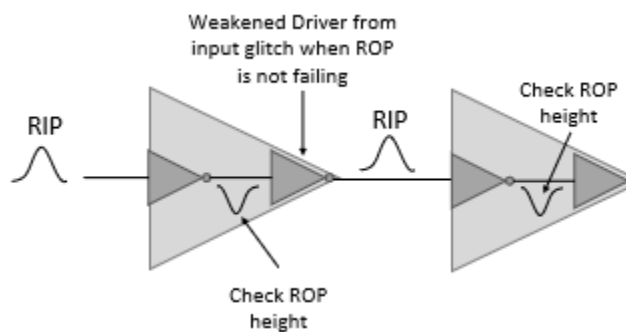
|                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <pre>-enable_glitch_report {true   false}</pre>               | <p>Enables AAE SI glitch analysis.</p> <p>For generating a glitch report, use the following:</p> <pre>set_si_mode -enable_glitch_report true;<br/>report_noise -txtfile glitch.txt</pre> <p>Note that you cannot use <code>-enable_glitch_report</code> with <code>-delay_only</code> because the <code>-delay_only</code> option does not perform glitch analysis, and as a result, no glitch is reported.</p> <p><i>Default:</i> <code>false</code></p> <p><b>Note:</b> For SI glitch fixing, you must set the <code>-enable_glitch_report</code> parameter to <code>true</code>. When the <code>false</code> setting is detected, the software will display a warning during glitch fixing.</p> |
| <pre>-enable_overshoot_undershoot_glitch {true   false}</pre> | <p>Enables overshoot/undershoot analysis and reporting.</p> <p>The lifespan of transistors is characterized by overshoot and undershoot values. If a transistor is consistently exposed to a certain percentage overshoot/undershoot, it may have a reduced lifespan.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                                                                                                                                                                                |

```
-enable_two_stage_driver_weakening {true | false}
```

Enables the two-stage driver weakening mod

A non-failing significant glitch at the input of a stage driver can propagate through the first st and weaken the second stage. This might cau the crosstalk-induced glitch to amplify at the n stage receiver (though it may not be significant enough to propagate). When two-stage driver weakening is enabled, the software considers weakened holding resistance for the two-stag driver for glitch analysis.

The following diagram illustrates two-stage dr weakening:



The dual-stage driver weakening honors the propagation threshold criteria.

**Note:** This feature is supported only with CCS models.

*Default:* true

```
-enable_user_defined_bus_merging {true | false}
```

Enables bus merging feature - if a net is in two groups, then the groups are merged.

*Default:* false

```
-enable_virtual_attacker_constituent_report {true | false}
```

Enables virtual attacker constituent reporting.

The virtual attacker constituent reporting feature is useful for identifying significant buses and the - this information can be used for glitch fixing.

This feature is enabled when the `set_si_mode` `enable_bus_attacker_correlation` parameter is set to true.

|                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <code>-glitch_accuracy_level &lt;int&gt;</code>                | <p>Specifies the accuracy levels for glitch analysis</p> <p><i>Default:</i> 0</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>-glitch_analysis_type {source   full_propagation}</code> | <p>Specifies the glitch analysis type - <code>source</code> or <code>full_propagation</code>.</p> <ul style="list-style-type: none"> <li><code>full_propagation</code>: Propagates glitch noise through single and 2-stage cells with CCS (example, <code>inv</code> or <code>buf</code>), and stops glitch propagation at any sequential cell, or 3 or stage cell input.</li> <li><code>source</code>: Propagates glitch through 1-stage only, and performs driver weakening through stage cells with CCSN.</li> </ul> <p><i>Default:</i> <code>source</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>-hold_slack &lt;double&gt;</code>                        | <p>Specifies the slack reselection threshold for hold</p> <p><i>Default:</i> <code>-1e-12</code> seconds</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>-individual_attacker_clock_threshold float</code>        | <p>Sets the glitch tolerance for individual attackers on victim clock nets. Tempus estimates the noise contributed by an individual attacker to the clock victim. If the peak of this noise exceeds the value of <code>individual_attacker_clock_threshold</code>, the attacker is significant and is individually reported in the constituents section of the report. Otherwise, the aggressor becomes eligible for the virtual attacker. If the virtual attacker peak does not exceed the value of <code>accumulated_small_attacker_threshold</code>, the aggressor nets in the virtual attacker are grounded, and there is no virtual attacker. If <code>individual_attacker_clock_threshold</code> is not specified, then clock nets will use the <code>individual_attacker_threshold</code> value instead.</p> <p>Note: Minimum allowed individual attacker for clock nets to be kept is 0.1% of vdd, which is equivalent to an <code>individual_attacker_clock_threshold</code> of 0.001.</p> <p><i>Default:</i> 1.5% or 0.015</p> |

|                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| <pre>-individual_attacker_threshold &lt;double&gt;</pre>                | <p>Sets the glitch tolerance for individual attacker on the victim net. Tempus estimates the noise contributed by an individual attacker to the victim net. If the peak of this noise exceeds the value of <code>individual_attacker_threshold</code>, the attacker is significant and is individually reported in the constituents section of the report. Otherwise, the attacker becomes eligible for the virtual attacker. If the virtual attacker peak does not exceed the value of <code>accumulated_small_attacker_threshold</code>, the aggressor nets in the virtual attacker are grounded, and there is no virtual attacker.</p> <p><b>Note:</b> The minimum allowed individual attacker count is 0.1% of vdd, which is equivalent to an <code>individual_attacker_threshold</code> of 0.001.</p> <p><i>Default:</i> 1.5% or 0.015</p> |
| <pre>-individual_attacker_simulation_filtering<br/>{true   false}</pre> | <p>Uses electrical filtering to reduce pessimism by preserving small attackers when their simulation value is below the individual attacker threshold.</p> <p><i>Default:</i> true</p> <p><b>Note:</b> This option can only be used when the accumulated small attacker is turned off (<code>accumulated_small_attacker_threshold</code> is 1.0 or larger).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <pre>-input_glitch_full_propagation_threshold double</pre>              | <p>Specifies the receiver input glitch threshold beyond which Tempus will not propagate the glitch in propagation mode, and an RIP violation will be flagged.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <pre>-<br/>input_glitch_full_propagation_vh_threshold double</pre>      | <p>Specifies the receiver VH input glitch threshold beyond which Tempus will not propagate the glitch in full-propagation mode, and an RIP violation will be flagged.</p> <p>This parameter takes precedence over <code>-input_glitch_full_propagation_threshold</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

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| <pre>- input_glitch_full_propagation_vl_threshold double</pre> | <p>Specifies the receiver VL input glitch threshold beyond which Tempus will not propagate the in full-propagation mode, and an RIP violation be flagged.</p> <p>This option takes precedence over <code>-input_glitch_full_propagation_threshold</code>.</p>                                                                                                                                                                                                               |
|                                                                | <p><b>Note:</b> These parameters can be used only when propagation mode is enabled using the <code>-glitch_analysis_type full_propagation</code> parameter.</p>                                                                                                                                                                                                                                                                                                             |
| <pre>-input_glitch_threshold &lt;double&gt;</pre>              | <p>Specifies the input glitch failure threshold value during AAE analysis.</p> <p><i>Default:</i> 0.4</p>                                                                                                                                                                                                                                                                                                                                                                   |
| <pre>-input_glitch_vh_threshold &lt;double&gt;</pre>           | <p>Specifies the input glitch <code>vh</code> failure threshold value during AAE analysis.</p> <p><i>Default:</i> 0.4</p>                                                                                                                                                                                                                                                                                                                                                   |
| <pre>-input_glitch_vl_threshold &lt;double&gt;</pre>           | <p>Specifies the input glitch <code>vl</code> failure threshold value during AAE analysis.</p> <p><i>Default:</i> 0.4</p>                                                                                                                                                                                                                                                                                                                                                   |
| <pre>-nonlinear_attacker_slew {true   false}</pre>             | <p>Uses a non-linear attacker. If set to <code>false</code>, it uses linear attacker.</p> <p><i>Default:</i> <code>true</code></p>                                                                                                                                                                                                                                                                                                                                          |
| <pre>-num_si_iteration value</pre>                             | <p>Specifies the maximum number of timing window iterations that should be performed during SI analysis. Performing delay analysis results in changes to the timing windows, potentially altering the victim/attacker net combinations. Timing window iterations enable the software to reach a point where the delay changes are negligible.</p> <p><i>Default:</i> 2</p> <p><b>Note:</b> The minimum allowed number of timing window iterations for SI analysis is 2.</p> |

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| <code>-receiver_clk_peak_limit double</code>                      | <p>Defines the glitch check limit for clock inputs in receiver peak mode. This parameter is specified as a percentage of the power net voltage.</p> <p><i>Default:</i> 0.05 (5 percent of VDD)</p>                                                                                                                                                                                         |
| <code>-receiver_latch_peak_limit double</code>                    | <p>Defines the glitch check limit for latch inputs in receiver peak mode. This parameter is specified as a ratio of vdd.</p> <p><i>Default:</i> 0.1 (10 percent of VDD)</p>                                                                                                                                                                                                                |
| <code>-receiver_peak_limit double</code>                          | <p>Defines the receiver peak/glitch peak limit at all nodes. The value specified for this parameter is a ratio of vdd.</p> <p><i>Default:</i> 0.15</p>                                                                                                                                                                                                                                     |
| <code>-report_max_virtual_attacker_constituents int</code>        | <p>Specifies the maximum number of virtual attacker constituents to report.</p> <p>This feature works only when bus virtual attacker constituent reporting is enabled using the <code>set_si_mode -enable_virtual_attacker_constituent_reporting</code> setting.</p> <p><i>Default:</i> 5</p>                                                                                              |
| <code>-secondary_attacker_decoupling_factor &lt;double&gt;</code> | <p>Specifies decoupling factor for secondary aggressors.</p> <p><i>Default:</i> 1</p>                                                                                                                                                                                                                                                                                                      |
| <code>-separate_delta_delay_on_data {true   false}</code>         | <p>Separates the delta delay on data (the delta delay on clock is always separated for the clock).</p> <p>Works with <code>-delta_delay_annotation_mode</code> to determine how the separated delay shows in the timing report. Default is the delta delay, which is not separated out.</p> <p>Using this parameter, you can increase the runtime/memory.</p> <p><i>Default:</i> false</p> |
| <code>-setup_slack &lt;double&gt;</code>                          | <p>Specifies the slack reselection threshold for setup.</p> <p><i>Default:</i> -1e-12 seconds</p>                                                                                                                                                                                                                                                                                          |

|                                                     |                                                                                                                                                                                                                                                                                                                                                                       |
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| <code>-si_reselection {delta_delay   slack}</code>  | <p>Specifies the SI reselection criteria during timing window iterations.</p> <p><b>delta_delay:</b> Reselects a net for calculation if delta delay on the net is greater than the delta value.</p> <p><b>slack:</b> Reselects nets with failing slack values during timing-window iterations.</p> <p><b>Default:</b> <code>slack</code></p>                          |
| <code>-si_reselection_delay_threshold double</code> | <p>Allows you to change the delay reselection criteria. You can specify <code>-si_reselection_delay_threshold</code> with the <code>-delta_delay</code> parameter only.</p> <p><b>Default:</b> <code>1e-11</code></p> <p>Units in seconds</p>                                                                                                                         |
| <code>-skip_noise_model_check pins</code>           | <p>Specifies a list of pins for which the noise model check will not take place.</p> <p><b>Default:</b> <code>" "</code></p>                                                                                                                                                                                                                                          |
| <code>-skip_tw &lt;nets&gt;</code>                  | <p>Specifies the list of nets for which the timing window is to be skipped. The timing window becomes infinite for the net when it is both a victim and aggressor.</p> <p>When specified, only the last setting is retained. The <code>set_si_mode -skip_tw</code> command settings to the previous setting, are ignored.</p> <p><b>Default:</b> <code>" "</code></p> |
| <code>-switch_prob &lt;double&gt;</code>            | <p>Sets the attacker switching probability. If you specify a value less than 0.0 or more than 1.0, the switching probability is automatically set to the default value of 0.3.</p> <p><b>Default:</b> <code>0.3</code></p>                                                                                                                                            |
| <code>-use_infinite_TW {true   false}</code>        | <p>Uses infinite timing windows in a single SI run.</p> <p><b>Default:</b> <code>false</code></p>                                                                                                                                                                                                                                                                     |

|                                                             |                                                                                                                                                                                                      |
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| <code>-unconstrained_net_use_inf_tw {true   false}</code>   | Specifies that unconstrained nets will have an infinite timing window. To turn this off and use constrained timing window, set this to <code>false</code> .<br><br><i>Default:</i> <code>true</code> |
| <code>-virtual_inter_die_attacker_slew &lt;value&gt;</code> | Specifies the virtual inter-die attacker slew value.<br><br><i>Default:</i> <code>2.5e-11</code> (float)                                                                                             |

## Examples

- The following command runs AAE, keeping all individual attackers larger than  $0.01 \times VDD$  and not using any accumulated small attackers:

```
tempus> set_si_mode -individual_attacker_threshold 0.01 -
accumulated_small_attacker_threshold 10.01
```

- The following command sets the glitch peak limit at  $0.5 \times vdd$  at data nodes,  $0.1 \times vdd$  at clock nodes, and  $0.2 \times vdd$  at latch nodes.

```
tempus> set_si_mode -receiver_peak_limit 0.5 -receiver_latch_peak_limit 0.2 -
receiver_clk_peak_limit 0.1
```

- To keep all attackers less than `set_si_mode -individual_attacker_threshold`, whose accumulated glitch bump exceeds 1% of VDD, you can use the following command

```
tempus> set_si_mode -accumulated_small_attacker_threshold 0.01
```

## Related Commands

- `get_si_mode`
- `report_delay_calculation`
- `report_timing`
- `report_noise`
- `update_glitch`



## set\_via\_variation\_mode

```
set_via_variation_mode
[-help]
[-reset]
[-corner_sigma_multiplier <std::string>]
[-correlation_type {corner}]
[-reset {true | false}]
[-separate_params_for_via {true | false}]
```

Allows you to scale via resistance from SPEF based on a user-specified scaling factor.

This command allows you to specify a per-via sigma multiplier and handles advanced via variation scaling.

### Parameters

|                                                |                                                                                            |
|------------------------------------------------|--------------------------------------------------------------------------------------------|
| -help                                          | Prints out the command usage                                                               |
| -<br>corner_sigma_multiplier<br><std::string>  | Provides the scaling multiplier for corner scaling.<br><br><i>Default:</i> " "             |
| -correlation_type<br>{corner}                  | Specifies inter-layer correlation values.<br><br><i>Default:</i> corner                    |
| -reset                                         | Resets all or specified options to their default value.                                    |
| -reset {true   false}                          | Resets via variation modes to their default value.<br><br><i>Default:</i> false            |
| -<br>separate_params_for_via<br>{true   false} | Allows the creation of separate parameters for via variation.<br><br><i>Default:</i> false |

## Examples

- The following command uses correlation corner for scaling purposes:

```
set_via_variation_mode -view view1 -correlation_type corner -corner_sigma_multiplier
{{{param_V1 -3} {param_V2 -3}..}}
```

where V1 and V2 are via names and “-3” is used as a scaling factor.

- The following command will scale the via resistance as shown below:

```
set_via_variation_mode -rc_variation false -corner_sigma_multiplier {{{param_V1 2}
{param_V2 2}..}}
```

Via Calculation below:

```
Via1_Rnew = Via1_Rspef *(1 + 0.021 + 2*0.110)
Via2_Rnew = Via2_Rspef *(1 + 0.010 + 2*0.120)
```

```
 Via1:
table_meanshift: 0.021
table_stdev : 0.110
 Via2:
table_meanshift: 0.010
table_stdev : 0.120
```

## Related Commands

- [get\\_via\\_variation\\_mode](#)

## set\_wire\_variation\_mode

```
set_wire_variation_mode
[-help]
[-reset]
[-corner_sigma_multiplier <std::string>]
[-correlation_type {statistical | correlated | binomial | corner}]
[-disable_positive_slack_shift {true | false}]
[-enable_seperate_wire_var_sigma_multiplier {true | false}]
[-layer_avg_rpt {true | false}]
[-rc_correlation_factor <int>]
[-reset {true | false}]
[-use_nsigma_coupling_capacitance {true | false}]
[-use_nsigma_slew_for_timing_check {true | false}]
[-wire_cell_variation_correlation {true | false}]
[-wire_var_reporting_sigma_multiplier <double>]
```

Defines wire variation analysis settings.

### Parameters

|                                                                  |                                                                                                                                                                                                                                          |
|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                                                            | Prints out the command usage.                                                                                                                                                                                                            |
| -reset                                                           | Resets all or the specified options to their default value.                                                                                                                                                                              |
| -corner_sigma_multiplier <std::string>                           | Provides a scaling multiplier for corner scaling.<br><i>Default:</i> ""                                                                                                                                                                  |
| -correlation_type {statistical   correlated   binomial   corner} | Provides control to handle inter-layer correlation. The accepted values are: <ul style="list-style-type: none"> <li>• statistical:</li> <li>• binomial:</li> <li>• corner:</li> <li>• correlated:</li> </ul> <i>Default:</i> statistical |
| -disable_positive_slack_shift {true   false}                     | Disables positive slack shift caused by back-end-of-line (BEOL) variation.<br><i>Default:</i> false                                                                                                                                      |

|                                                                        |                                                                                                                                                                                                          |
|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-enable_seperate_wire_var_sigma_multiplier {true   false}</code> | <p>Allows a separate reporting sigma multiplier for wire variations.</p> <p>This parameter should be used along with <code>-wire_var_reporting_sigma_multiplier</code>.</p> <p><i>Default:</i> false</p> |
| <code>-layer_avg_rpt {true   false}</code>                             | <p>Enables layer averaging reporting.</p> <p><i>Default:</i> false</p>                                                                                                                                   |
| <code>-rc_correlation_factor &lt;int&gt;</code>                        | <p>Allows the same R and C layer correlations. The accepted values are 1 and -1.</p> <p><i>Default:</i> 1</p>                                                                                            |
| <code>-reset {true   false}</code>                                     | <p>Resets wire variation modes to the default value.</p> <p><i>Default:</i> false</p>                                                                                                                    |
| <code>-use_nsigma_coupling_capacitance {true   false}</code>           | <p>Allows to use of mean and nSigma coupling capacitance.</p> <p><i>Default:</i> false</p>                                                                                                               |
| <code>-use_nsigma_slew_for_timing_check {true   false}</code>          | <p>Enables the use of wire-variation-aware slews for timing-check awareness.</p> <p><i>Default:</i> false</p>                                                                                            |
| <code>-wire_cell_variation_correlation {true   false}</code>           | <p>Controls whether to use RSS or add random-cell and wire variations for arrivals and slack.</p> <p><i>Default:</i> false</p>                                                                           |
| <code>-wire_var_reporting_sigma_multiplier &lt;double&gt;</code>       | <p>Specifies deterministic sigma multiplier for reporting BEOL variations.</p> <p><i>Default:</i> 0</p>                                                                                                  |

## Examples

- The following command enables binomial mode:  

```
tempus> set_wire_variation_mode -correlation_type binomial
```
- The following command can be used to consider the wire variation and random-cell variation to be correlated, which will linearly add the wire and cell sigma instead of adding statistically:  

```
tempus> set_wire_variation_mode -wire_cell_variation_correlation 1
```
- The following command can be used to perform bounding analysis to use n\*wireSigma clock and

data pins slews at the endpoint to look up the timing checks:

```
tempus> set_wire_variation_mode -use_nsigma_slew_for_timing_check true
```

- The following command configures wire variation analysis for statistical mode:

```
tempus> set_wire_variation_mode -correlation_type statistical -rc_correlation_factor -1
```

## Related Commands

- [read\\_spef](#)
- [report\\_timing](#)
- [get\\_wire\\_variation\\_mode](#)

## Related Topics

- For more information, refer to the "[Wire Variation Analysis](#)" section in the *Tempus User Guide*.

# update\_glitch

```
update_glitch
[-help]
[-effort {medium | high}]
```

Allows you to run glitch analysis without running SI delay analysis. This command provides the ability to run glitch analysis on all the nets in a design or on a subset of nets.

Using the `update_glitch` command without `report_timing` or `update_timing` will only run glitch analysis.

**Note:** It is not recommended to run the `update_glitch` command after the `report_timing` or `update_timing` command.

## Parameters

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                   | Outputs a brief description that includes type and default information for each <code>update_glitch</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man update_glitch</code> .                                                                                                                                    |
| <code>-effort {medium   high}</code> | <p>Specifies the effort level for the update glitch.</p> <p>When set to <code>high</code>, glitch analysis uses more accurate modeling of the glitch for small attackers, thereby reducing pessimism.</p> <p><b>Note:</b> The run time in <code>high</code> mode may be slightly higher and is largely dependent on the distribution of coupling.</p> <p><i>Default:</i> <code>medium</code></p> |

## Examples

- The following command will run glitch analysis without SI delay:

```
tempus> update_glitch
```

## Related Commands

- `report_noise`
- `report_timing`
- `update_timing`
- `set_si_mode`
- `get_si_mode`

## write\_noise\_eco

```
write_noise_eco
[-help]
[-filename string]
[-problem_nets string]
```

Tempus performs ECO analysis for noise failures. The default ECO file is named `glitch`.

**Note:** The `write_noise_eco` command should always be called after glitch analysis has been performed.

### Parameter

|                                                              |                                                                                                                                   |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                                           | Prints out the command usage.                                                                                                     |
| <code>-filename</code><br><i>string</i>                      | Specifies the name for the ECO file.<br>If this parameter is not specified, the report is written to standard output (screen).    |
| <code>-</code><br><code>problem_nets</code><br><i>string</i> | Reports the specified list of net names. The software provides the capability to generate ECO for the user-specified set of nets. |

### Examples

The following command generates an ECO file for all the glitch failures.

```
tempus> write_noise_eco -filename glitch_eco.txt
```

### Related Commands

- `report_noise`
- `set_delay_cal_mode`
- `get_delay_cal_mode`



---

# Timing Model Commands

---

- [Extracted Timing Model - ETM](#)
- [Interface Logic Models - ILM](#)
- [Prototype Timing Model - PTM](#)

## Extracted Timing Model - ETM

- [compare\\_model\\_timing](#)
- [do\\_extract\\_model](#)
- [merge\\_model\\_timing](#)
- [write\\_model\\_timing](#)

## compare\_model\_timing

```
compare_model_timing
[-help]
[-absolute_cap_tolerance <float>]
[-absolute_tolerance <float>]
[-absolute_tolerance_optimistic <float>]
[-absolute_tolerance_pessimistic <float>]
[-absolute_tran_tolerance <float>]
-compare <string>
[-ignore_view]
-outFile <string>
[-percent_cap_tolerance <float>]
[-percent_tolerance <float>]
[-percent_tolerance_optimistic <float>]
[-percent_tolerance_pessimistic <float>]
[-percent_tran_tolerance <float>]
-ref <string>
```

Compares two reports that contain interface timing characteristics generated by the [write\\_model\\_timing](#) command.

This command compares the interface timing report generated by using the [write\\_model\\_timing](#) command on the original netlist with the interface timing report generated by using the [write\\_model\\_timing](#) command on the model extracted by the [do\\_extract\\_model](#) command for the same design.

### Parameters

|       |                                                                                                                                                                                                                                                                   |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help | Outputs a brief description that includes type and default information for each <a href="#">compare_model_timing</a> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man compare_model_timing</code> . |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-absolute_cap_tolerance value</code>                 | <p>Specifies the tolerance value for capacitance (as an absolute value) while comparing two report files. This parameter specifies the permissible difference, as an absolute value, between the capacitance values of two report files. If the difference in capacitance exceeds the specified absolute tolerance value, the comparison status is set to <code>FAIL</code>.</p> <p><b>Note:</b> When this parameter is not specified, the software uses the absolute value specified with the <code>-absolute_tolerance</code> parameter. If the <code>-absolute_tolerance</code> parameter is not specified, the value defaults to 0.</p> <p><b>Note:</b> When the <code>-percent_cap_tolerance</code> and <code>-absolute_cap_tolerance</code> parameters are used together, the comparison status is set to <code>FAIL</code> only when the difference in the value of capacitance exceeds the specified absolute as well as percentage tolerance values.</p> |
| <code>-absolute_tolerance tolerance_value</code>           | <p>Specifies the absolute tolerance value. The tolerance value is used when there is a difference between the timing characteristics of the reference file (<code>-ref</code>) and the file that is being compared (<code>-compare</code>) to determine the <code>PASS</code> or <code>FAIL</code> status of the comparison. If the difference in the value of a timing characteristic exceeds the absolute tolerance value, the comparison status is set to <code>FAIL</code>.</p> <p><b>Note:</b> When used with the <code>-percent_tolerance</code> option, the difference in the value of a timing characteristic should exceed both absolute and percentage tolerance values for the comparison result status of an arc to be <code>FAIL</code>.</p> <p><i>Default:</i> 0</p>                                                                                                                                                                                |
| <code>-absolute_tolerance_optimistic &lt;float&gt;</code>  | <p>Specifies the absolute tolerance limit for flagging an optimistic failure. An optimistic path violating both absolute and percentage tolerance limits will be flagged as <code>FAIL</code> in the report output.</p> <p><i>Default:</i> 15ps</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>-absolute_tolerance_pessimistic &lt;float&gt;</code> | <p>Specifies the absolute tolerance limit for flagging a pessimistic failure. A pessimistic path violating both absolute and percentage tolerance limit will be flagged as <code>FAIL</code> in the report output.</p> <p><i>Default:</i> 15ps</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-absolute_tran_tolerance value</code> | <p>Specifies the tolerance value for transition time at ports (as an absolute value) while comparing two report files. This parameter specifies the permissible absolute difference in transition time between the ports of two report files. If the difference in transition time exceeds the specified absolute tolerance value, the comparison status is set to <code>FAIL</code>.</p> <p>When the <code>-percent_tran_tolerance</code> and <code>-absolute_tran_tolerance</code> parameters are used together, the comparison status is set to <code>FAIL</code> only when the difference in the value of transition exceeds the specified absolute as well as percentage tolerance values.</p> <p><b>Note:</b> When this parameter is not specified, the software uses the absolute value specified with the <code>-absolute_tolerance</code> parameter. If the <code>-absolute_tolerance</code> parameter is not specified, the value defaults to 0.</p> |
| <code>-compare filename</code>              | <p>Specifies the name of the report file that contains the interface timing characteristics extracted from the model created using the <code>do_extract_model</code> command. The timing characteristics of this file are compared with the timing characteristics of the reference file that you specify with the <code>-ref</code> option.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>-ignore_view</code>                   | <p>Compares the non-MMMC design <code>write_model_timing</code> output report with the MMMC design output report (of the <code>write_model_timing</code> command).</p> <p>When you specify this parameter, the software ignores the view information while comparing the reports. So you do not need to create dummy views for the timing model validation flow.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>-outFile output_filename</code>       | <p>Specifies the name of the output file in which the comparison results will be generated. This file contains the <code>PASS</code> or <code>FAIL</code> status for the timing data comparison.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-percent_cap_tolerance value</code>                 | <p>Specifies the tolerance value for capacitance (as a percentage) while comparing two report files. This parameter specifies the permissible percentage difference between the capacitance values of two report files. If the percentage difference in capacitance exceeds the specified tolerance, the comparison status is set to <code>FAIL</code>.</p> <p>When the <code>-percent_cap_tolerance</code> and <code>-absolute_cap_tolerance</code> parameters are used together, the comparison status is set to <code>FAIL</code> only when the difference in the value of capacitance exceeds the specified absolute as well as percentage tolerance values.</p> <p><b>Note:</b> When this parameter is not specified, the software uses the percentage value specified with the <code>-percent_tolerance</code> parameter. If the <code>-percent_tolerance</code> parameter is not specified, the value defaults to 0.</p> |
| <code>-percent_tolerance tolerance_value</code>           | <p>Specifies the percentage tolerance value for comparing the timing data of two report files. The percentage tolerance value is used when there is a difference between the timing characteristics of the reference file (<code>-ref</code>) and the file that is being compared (<code>-compare</code>) to determine the <code>PASS</code> or <code>FAIL</code> status of the comparison. If the percentage difference in the value of a timing characteristic exceeds the percentage tolerance value, the comparison status is set to <code>FAIL</code>.</p> <p><b>Note:</b> When used with the <code>-absolute_tolerance</code> option, the difference in the value of a timing characteristic should exceed both percentage and absolute tolerance values for the comparison result status of an arc to be <code>FAIL</code>.</p> <p><i>Default:</i> 0</p>                                                                 |
| <code>-percent_tolerance_optimistic &lt;float&gt;</code>  | <p>Specifies the percentage tolerance limit for flagging an optimistic failure. An optimistic path violating percentage, as well as an absolute tolerance limit, will be flagged as <code>FAIL</code> in the report output.</p> <p>Default: 3%</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>-percent_tolerance_pessimistic &lt;float&gt;</code> | <p>Specifies the percentage tolerance limit for flagging a pessimistic failure. A pessimistic path-violating percentage as well as an absolute tolerance limit will be flagged as <code>FAIL</code> in the report output.</p> <p>Default: 3%</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><code>-percent_tran_tolerance value</code></p> | <p>Specifies the tolerance value for transition time at ports (as a percentage) while comparing two report files. This parameter specifies the permissible difference, in percentage, between the transition time at the ports of two report files. If the percentage difference in the transition time exceeds the specified percentage tolerance value, the comparison status is set to <code>FAIL</code>.</p> <p>When the <code>-percent_tran_tolerance</code> and <code>-absolute_tran_tolerance</code> parameters are used together, the comparison status is set to <code>FAIL</code> only when the difference in the value of transition exceeds the specified absolute as well as percentage tolerance values.</p> <p><b>Note:</b> When this parameter is not specified, the software uses the percentage value specified with the <code>-percent_tolerance</code> parameter. If the <code>-percent_tolerance</code> parameter is not specified, the value defaults to 0.</p> |
| <p><code>-ref reference_filename</code></p>       | <p>Specifies the name of the reference file that is used for comparison. This is the report file generated by using the <code>write_timing_model</code> command on the original design.</p> <p>The file specified with the <code>-compare</code> option is compared with this file.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

## Example

- The following command compares the timing characteristics of the `sample_compare.rpt` file with the `sample_reference.rpt` file and generates the comparison results in the `sample_output.rpt` file:

```
tempus > compare_model_timing -ref sample_reference.rpt -compare sample_compare.rpt
-outFile sample_output.rpt
```

```
#####
```

```
compare_model_timing report
```

```
Design : netlist = topcell =
```

```
Version : 07.10-dev.indfe_cm
```

```
Analysis Mode : setup
```

```
Date : Wed Mar 14 15:56:59 IST 2007
```

```
#####
```

```
#Section 1: Worst-case Arc-Delay
```

```
#Tolerance Percent: 0.0%
```

```
#Tolerance Absolute: 0.0
```

```
#NOTE: The various columns list values in the form 'X[Y]', where:
```

```
X = value that is compared.
```

```
Y = reference value.

#NOTE2: If a comparison fails (ie. status = FAIL), then the clauses
(ie. 'X[Y]' pairs) that caused the failure are identified with
a '*' in front of them.
#From To Arc-Type Transition Arc-Delay abdDiff %Diff Status
#####
in CLK3 setup fall/rise 0.134[0.134] 0.000 0.000% PASS
in CLK3 setup rise/rise -0.005[-0.005] 0.000 -0.000% PASS
#####
#Section 2: Pin-capacitances
#Tolerance: 0.0%
#NOTE: The various columns list values in the form 'X[Y]', where:
X = value that is compared.
Y = reference value.

#NOTE2: If a comparison fails (ie. status = FAIL), then the clauses
(ie. 'X[Y]' pairs) that caused the failure are identified with
a '*' in front of them.
#
#Pin C(total)[rise] C(total)[fall] Status
#####
clk 0.002[0.002] 0.002[0.002] PASS
in 0.002[0.002] 0.002[0.002] PASS
out 0.000[0.000] 0.000[0.000] PASS
#####
#Section 3: Transition-times
#Tolerance: 0.0%
#NOTE: The various columns list values in the form 'X[Y]', where:
X = value that is compared.
Y = reference value.
#
#NOTE2: If a comparison fails (ie. status = FAIL), then the clauses
(ie. 'X[Y]' pairs) that caused the failure are identified with
a '*' in front of them.
#
#Pin Rise Fall Status
#####
```



```
clk 0.000[0.000] 0.000[0.000] PASS
in 0.000[0.000] 0.000[0.000] PASS
out 0.056[0.056] 0.051[0.051] PASS
```

```
#####
#Section 4: Design Rules
#Tolerance: 0.0%
#NOTE: The various columns list values in the form 'X[Y]', where:
X = value that is compared.
Y = reference value.

#NOTE2: If a comparison fails (ie. status = FAIL), then the clauses
(ie. 'X[Y]' pairs) that caused the failure are identified with
a '*' in front of them.
#
#Pin MaxCap MinCap MaxTrans MaxFanout FanoutLoad Status
#####
clk NA[NA] NA[NA] NA[NA] NA[NA] NA[NA] PASS
in NA[NA] NA[NA] NA[NA] NA[NA] NA[NA] PASS
out NA[NA] NA[NA] NA[NA] NA[NA] NA[NA] PASS

#####
SUMMARY
#
Check-Type Passed Failed N/A Total
#####
Worst-Slacks 2 0 0 2
Capacitances 3 0 0 3
Trans-Times 3 0 0 3
Design-Rules 3 0 0 3
```

## Related Commands

- [do\\_extract\\_model](#)
- [write\\_model\\_timing](#)

## do\_extract\_model

```
do_extract_model
[-help]
[-abs_tol_val float]
[-abs_tol_val_optimistic float]
[-abs_tol_val_pessimistic float]
[-lib_name lib_name]
[-cell_name cell_name]
[-check]
[-clock_slews {clk_slew1 clk_slew2 ...}]
[-format string]
[-include_aocv_weights]
[-include_power]
[-input_slews {input_slew1 input_slew2 ...}]
[-output_loads {output_load1 output_load2 ...}]
[-precision value]
[-gain integer]
[-resolution value]
[-tolerance value]
[-assertions constraint filename]
[-validate string]
[-verilog_shell_file filename]
[-verilog_shell_module top_module_name]
[-max_num_slews value]
[-max_num_loads value]
[-per_tol_val float]
[-per_tol_val_optimistic float]
[-per_tol_val_pessimistic float]
[-pg]
[-view viewname]
model_filename
```

Builds a Liberty (.lib) format model for the top cell, which is the timing model equivalent of the original design. The model is extracted for the given analysis modes, PVT conditions, and parasitics. If any one of these parameters changes, then you must extract the model again. The model is extracted for a range of input slews and load capacitances. If you do not specify these ranges, the software automatically computes the default values. The extracted model defines all of the top-level ports of the design as pins. Additionally, all other design pins that are targets of `create_clock` or `create_generated_clock` are retained as internal pins.

✔ For model extraction, your design must be properly constrained. For example, no clock or data conflicts can be present. Use the `check_timing` command and correct all errors prior to model extraction.

## Parameters

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                           | Prints out the command usage.                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>-abs_tol_val float</code>              | <p>Specifies the absolute tolerance value to be used during the validation of the extracted timing model. For more details, refer to the <a href="#">compare_model_timing -absolute_tolerance</a> command option.</p> <p>This parameter can be used only when the <code>-validate</code> option is specified.</p> <p>The integer value specified with this option is read as ns/ps depending upon the time unit.</p> <p><i>Default:</i> 3ps</p> |
| <code>-abs_tol_val_optimistic float</code>   | <p>Specifies the absolute tolerance limit for flagging an optimistic failure. An optimistic path violating the absolute as well as percentage tolerance limit will be flagged as FAIL in the <a href="#">compare_model_timing</a> command output report.</p> <p>You can use this parameter when the <code>-validate</code> is used with model extraction.</p> <p><i>Default:</i> 15ps</p>                                                       |
| <code>-abs_tol_val_pessimistic float</code>  | <p>Specifies the absolute tolerance limit for flagging a pessimistic failure. A pessimistic path violating the absolute as well as percentage tolerance limit will be flagged as FAIL in the <a href="#">compare_model_timing</a> command output report.</p> <p>You can use this parameter when the <code>-validate</code> is used with model extraction.</p> <p><i>Default:</i> 15ps</p>                                                       |
| <code>-assertions constraint filename</code> | <p>Dumps the constraints for a model, which is used in the validation flow.</p> <p><i>Default:</i> <code>model.asrt</code></p>                                                                                                                                                                                                                                                                                                                  |
| <code>-cell_name cell_name</code>            | <p>Specifies the cell name used in the timing model.</p> <p><i>Default:</i> The name of the top cell to be extracted.</p>                                                                                                                                                                                                                                                                                                                       |
| <code>-check</code>                          | Checks the extracted timing model for any possible liberty-related issue. When specified, the command displays detailed error and warning messages, and their summary.                                                                                                                                                                                                                                                                          |

|                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-clock_slews {clk_slew1<br/>clk_slew2...}</code>      | <p>Specifies a range of slew values used at all clock pins. The list of values must be enclosed by curly braces. If this parameter is not specified, the software automatically computes the appropriate values.</p> <p>This parameter accepts a maximum of 32 float values.</p>                                                                                                                                                                                                                                                                                                                   |
| <code>-format string</code>                                 | <p>Specifies the format for the extracted model in tlf, dotlib, or syntech.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>-gain integer</code>                                  | <p>Represents an increase in the number of delay arcs if a pin is removed. This option reduces the extracted model size in scenarios where removing a pin can result in increased model size. The gain at an internal pin can be defined as:</p> $(\text{number of incoming delay arcs} * \text{number of outgoing delay arcs}) - (\text{number of incoming delay arcs} + \text{number of outgoing delay arcs})$ <p>If an internal pin has a gain greater than the specified number, then that pin needs to be preserved in order to reduce the extracted model size.</p> <p><i>Default:</i> 0</p> |
| <code>-include_aocv_weights</code>                          | <p>Enables the software to write stage weights on individual arcs. Before using this option, it is mandatory that the AOCV libraries are read in and AOCV analysis is enabled (by using <code>set_analysis_mode -aocv true</code>).</p>                                                                                                                                                                                                                                                                                                                                                            |
| <code>-include_power</code>                                 | <p>Writes information about the power consumed by a block.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>-input_slews {input_slew1<br/>input_slew2 ...}</code> | <p>Specifies a range of slew values used at all input pins. Enclose the list of values with curly braces. If this parameter is not specified, the software automatically computes the appropriate values.</p> <p>This parameter accepts a maximum of 32 float values.</p>                                                                                                                                                                                                                                                                                                                          |
| <code>-lib_name lib_name</code>                             | <p>Specifies the library name used in the timing model. Write extracted timing models in the Liberty format by specifying the <code>.lib</code> file suffix.</p> <p><i>Default:</i> The name of the top cell to be extracted.</p>                                                                                                                                                                                                                                                                                                                                                                  |
| <code>-max_num_loads value</code>                           | <p>Specifies the maximum number of output load values used for delay arc characterization. A low value of 1 or 2 improves runtime and memory usage but may significantly degrade model accuracy. Use this parameter only if the model size or model extraction time is critical.</p> <p>This parameter accepts values between 1 and 32.</p> <p><i>Default:</i> 8</p>                                                                                                                                                                                                                               |

|                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-max_num_slews value</code>                                 | <p>Specifies the maximum number of input slews or clock slews used for delay and check arc characterization. Specifying a low value of 1 or 2 improves runtime and memory but may significantly degrade model accuracy. Using the <code>-max_num_slews</code> parameter has a larger impact on runtime and memory than the <code>-max_num_loads</code> parameter. Use this parameter only if the model size or the model extraction time is critical.</p> <p>The <code>max_num_slews</code> and <code>max_num_loads</code> parameters specify only the maximum limits on how many slew and load points are chosen. The software scans the gates connected to the input and output ports to calculate the number of points.</p> <p>This parameter accepts values between 1 and 32.</p> <p><i>Default:</i> 8</p> |
| <code>model_filename</code>                                       | <p>Specifies the name of the output file.</p> <p>Avoid problems with rebind by using the <code>-lib_name</code> and <code>-cell_name</code> parameters instead of using the default names.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>-output_loads<br/>{output_load1output_load2<br/>...}</code> | <p>Specifies a range of load values used at all output pins. Enclose the list of values in curly braces. If this parameter is not specified, the software automatically computes the appropriate values.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>-per_tol_val &lt;float&gt;</code>                           | <p>Specifies the percentage tolerance value to be used during the validation of the extracted ETM. This option can be used only when the <code>-validate</code> option is also specified.</p> <p>For more details, refer to <a href="#">compare_model_timing</a> <code>-percent_tolerance</code>.</p> <p><i>Default:</i> 3%</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <code>-per_tol_val_optimistic<br/>float</code>                    | <p>Specifies the percentage tolerance limit for flagging an optimistic failure. An optimistic path violating percentage, as well as an absolute tolerance limit, will be flagged as FAIL in the <a href="#">compare_model_timing</a> command output report.</p> <p>You can use this parameter when the <code>-validate</code> is used with model extraction.</p> <p><i>Default:</i> 3%</p>                                                                                                                                                                                                                                                                                                                                                                                                                     |

| <code>-per_tol_val_pessimistic</code><br><i>float</i> | <p>Specifies the percentage tolerance limit for flagging a pessimistic failure. A pessimistic path-violating percentage, as well as an absolute tolerance limit, will be flagged as FAIL in the <code>compare_model_timing</code> command output report.</p> <p>You can use this parameter when the <code>-validate</code> is used with model extraction.</p> <p><i>Default:</i> 3%</p> |           |                    |   |        |   |       |   |      |   |     |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------|---|--------|---|-------|---|------|---|-----|
| <code>-pg</code>                                      | <p>Allows power or ground-related pin information to be extracted inside the extracted model.</p>                                                                                                                                                                                                                                                                                       |           |                    |   |        |   |       |   |      |   |     |
| <code>-precision value</code>                         | <p>Specifies the floating-point precision used for the timing model.</p> <p>This parameter accepts values between 1 and 10.</p> <p>For example, the floating point precision for 3.0089 can be changed as follows:</p>                                                                                                                                                                  |           |                    |   |        |   |       |   |      |   |     |
|                                                       | <table> <thead> <tr> <th>Precision</th><th>Data Value in .lib</th></tr> </thead> <tbody> <tr> <td>4</td><td>3.0089</td></tr> <tr> <td>3</td><td>3.009</td></tr> <tr> <td>2</td><td>3.01</td></tr> <tr> <td>1</td><td>3.0</td></tr> </tbody> </table> <p><i>Default:</i> 4</p>                                                                                                           | Precision | Data Value in .lib | 4 | 3.0089 | 3 | 3.009 | 2 | 3.01 | 1 | 3.0 |
| Precision                                             | Data Value in .lib                                                                                                                                                                                                                                                                                                                                                                      |           |                    |   |        |   |       |   |      |   |     |
| 4                                                     | 3.0089                                                                                                                                                                                                                                                                                                                                                                                  |           |                    |   |        |   |       |   |      |   |     |
| 3                                                     | 3.009                                                                                                                                                                                                                                                                                                                                                                                   |           |                    |   |        |   |       |   |      |   |     |
| 2                                                     | 3.01                                                                                                                                                                                                                                                                                                                                                                                    |           |                    |   |        |   |       |   |      |   |     |
| 1                                                     | 3.0                                                                                                                                                                                                                                                                                                                                                                                     |           |                    |   |        |   |       |   |      |   |     |
| <code>-resolution value</code>                        | <p>Specifies the magnitude of delay difference that can be ignored by the extractor.</p> <p><i>Default:</i> 0.001</p> <p><i>Type:</i> Float</p>                                                                                                                                                                                                                                         |           |                    |   |        |   |       |   |      |   |     |
| <code>-tolerance value</code>                         | <p>Specifies the accuracy level of the extractor in percent. For example, 5.0 represents a plus or a minus 5.0 percent allowable error in the extracted model. The higher the tolerance, the faster the extraction time.</p> <p><i>Default:</i> 1.0</p> <p><i>Type:</i> Float</p>                                                                                                       |           |                    |   |        |   |       |   |      |   |     |

|                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-validate <i>string</i></code>                          | <p>Enables the validation of the extracted timing model generated for the corresponding view and displays the number of failures corresponding to setup and hold checks.</p> <p>You are required to specify the name of the validation directory, for example, <code>do_extract_model -validate <i>valDir</i></code>, which contains data for <code>write_model_timing</code>, <code>compare_model_timing</code>, and other summary reports.</p> |
| <code>-view <i>viewname</i></code>                            | Specifies the current active view.                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>-verilog_shell_file<br/><i>filename</i></code>          | <p>Generates a Verilog shell file that instantiates the timing model. This parameter is useful for validating the model. Use the <code>-verilog_shell_module</code> parameter to specify the name of the top module in this Verilog file.</p>                                                                                                                                                                                                    |
| <code>-verilog_shell_module<br/><i>top_module_name</i></code> | <p>Specifies the top module name of the Verilog shell file.</p> <p><i>Default:</i> <code>test_shell</code></p>                                                                                                                                                                                                                                                                                                                                   |

### Examples:

- The following command builds a `.lib` model (`block.lib`) for the block:  

```
tempus > do_extract_model block.lib
```
- The following command creates the timing model for the `test_model` cell belonging to the `test_lib` library, and writes the model into the `test.lib` liberty file:  

```
tempus > do_extract_model -cell_name test_model -lib_name test_lib test.lib
```

### Related Commands

- `compare_model_timing`
- `set_analysis_mode`

## merge\_model\_timing

```
merge_model_timing
-help
-library_file fileName
-modes mode_name
-mode_group group_name
[-tolerance float]
[-keep_separate_arc]
[-pin_cap_tolerance float]
[-outfile fileName]
```

Merges extracted timing models (ETM) for various analysis views into a single output.

The `merge_model_timing` command should be the last in the run script. Also, this command should be run separately in a Tempus session when a design is not loaded.

When the DRV values are different in the libraries that are merged, then Tempus saves the worst value of the DRV with a warning message.

If the DRV values are not equal, the greater value is stored for the following:

- Minimum transition
- Minimum capacitance
- Minimum fanout

The smaller value is stored for the following, in case of unequal DRVs:

- Maximum transition
- Maximum capacitance
- Maximum fanout
- Fanout load



## Parameters

|                                       |                                                                                                                                     |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                    | Prints out the command usage.                                                                                                       |
| <code>-keep_separate_arc</code>       | Enables creating separate timing arcs with respective modes.                                                                        |
| <code>-library_file fileName</code>   | Specifies the input library files to be merged.                                                                                     |
| <code>-modes mode_name</code>         | Specifies the required mode names for merging libraries. The number of mode names should be equal to the number of input libraries. |
| <code>-mode_group group_name</code>   | Specifies the name of the mode group.                                                                                               |
| <code>-outfile fileName</code>        | Specifies the name of the merged library output file. By default, the output file name is <code>merge.lib</code> .                  |
| <code>-pin_cap_tolerance float</code> | Specifies the pin capacitance tolerance for comparing tolerances during library merging.                                            |
| <code>-tolerance float</code>         | Specifies the tolerance value used for tolerance comparisons during library merging.                                                |

## Examples

- The following command merges library files `model1.view1.lib` and `model1.view1.lib.ref` into a single output file:

```
tempus > merge_model_timing -library_file "model1.view1.lib model1.view1.lib.ref"
-modes "v1 v2" -mode_group v1_v2
```

## Related Commands

- [do\\_extract\\_model](#)

## Related Information

- See "[Extracted Timing Models](#)" in the *Tempus User Guide*.

## write\_model\_timing

```
write_model_timing
[-help]
[model_timing_report_file]
[-inputs string]
[-nworst integer]
[-outputs string]
[-slew_propagation {worst | path_based}]
[-type {arc | slack}]
[-verbose]
[-view string]
-setup string | -hold string
```

Writes the interface timing characteristics of the design in the specified timing model report file.

The report file contains the following sections:

- **Worst-Case Arc/Slack:** Contains details about the extracted arcs and the slack or delay across them, depending on the argument specified with the `-type` option.
- **Transition Time:** Contains information about transition time at ports.
- **Capacitance:** Contains the capacitance values for the ports.
- **Design Rules:** Contains the design rules applied to the ports.

The command reports unconstrained arcs' delays by default.

## Parameters

|                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                                   | Outputs a brief description that includes type and default information for each <code>write_model_timing</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man write_model_timing</code> .                                                                                                                                                                     |
| <code>-hold string</code>                            | Writes the paths corresponding to early analysis.                                                                                                                                                                                                                                                                                                                                                                              |
| <code>-inputs {port_list}</code>                     | Restricts the <code>write_model_timing</code> report to the input ports specified with this parameter. As a result, the software considers only the timing paths associated with the specified input ports when generating the report.<br><br>If this parameter is not specified, all input ports in the design are included when generating the report.                                                                       |
| <code>model_timing_report_file</code>                | Specifies the name of the interface timing model report file that will be generated with the interface timing characteristics of the loaded design.                                                                                                                                                                                                                                                                            |
| <code>-nworst value</code>                           | Specifies the number of paths to be enumerated for each endpoint.<br><br><b>Note:</b> The extracted model will contain a greater number of setup/hold paths per endpoint because it will have fewer endpoints compared to the original netlist. Therefore, it is recommended that you use a greater <code>-nworst</code> value such as 40, while writing a timing report for the extracted model to derive better correlation. |
| <code>-outputs {port_list}</code>                    | Restricts the <code>write_model_timing</code> report to the output ports specified with this parameter. As a result, the software considers only the timing paths associated with the specified output ports when generating the report.<br><br>If this parameter is not specified, all output ports in the design are included when generating the report.                                                                    |
| <code>-setup string</code>                           | Writes the paths corresponding to late analysis.                                                                                                                                                                                                                                                                                                                                                                               |
| <code>-slew_propagation {worst   path_based }</code> | Sets the slew-propagation mode to use when dumping the design's timing characteristics.                                                                                                                                                                                                                                                                                                                                        |
| <code>-type { arc   slack }</code>                   | Determines whether to report arc delay values ( <code>arc</code> ) or the worst-slack values ( <code>slack</code> ).                                                                                                                                                                                                                                                                                                           |
| <code>-view view_name</code>                         | Specifies a view name. The report can be generated for a specific view.                                                                                                                                                                                                                                                                                                                                                        |
| <code>-verbose</code>                                | Prints the command progress details.                                                                                                                                                                                                                                                                                                                                                                                           |

## Example

- The following command extracts the interface timing characteristics of the currently loaded design and writes out the report file named `sample.rpt` with the arc delay information:

```
tempus > write_model_timing -type arc sample.rpt
```

**Output:**

```
#####
write_model_timing report
Design : netlist = ../case1.v topcell = set_propagated_clock_ttc2
Version : 07.10-dev.indfe_cm
Analysis Mode : setup
Date : Wed Mar 14 15:29:57 IST 2007
Command : write_model_timing -type arc sample.rpt

#####
Section 1 : Worst Case Arc
Analysis Mode : setup
Description : worst case arcs
#
From To Arc Type Transition Arc-Delays
#####
in CLK3 setup fall/rise 0.134
in CLK3 setup rise/rise -0.005
#####
Section 2 : Transition time
Analysis Mode : setup
Description : Actual transition times on the ports
#
Port RiseTran FallTran
#####
in 0.000 0.000
clk 0.000 0.000
out 0.056 0.051
#####
Section 3 : Capacitance
Analysis Mode : setup
Description : Total Cap on the ports
#
Port CTotal(Rise) CTotal(Fall)
#####
in 0.002 0.002
clk 0.002 0.002
out 0.000 0.000
#####
Section 4 : Design Rules
Analysis Mode : setup
```

```
Description : Design Rules on the ports
#
Port MaxCap MinCap MaxTrans MaxFanout Fanout Limit
#####
in NA NA NA NA NA
clk NA NA NA NA NA
out NA NA NA NA NA
```

- In MMMC mode, the `write_model_timing` command displays the following report output:

```
#####
write_model_timing report
Design : netlist =
./../../../../tc/tempus/testcases/tempus_ilm/DATA/mmmc/dtmf_chip/dtmf_chip.Route.opt.IL
M.enc.dat/dtmf_chip.v topcell = dtmf_chip
Analysis Mode : setup
Date : Mon Sep 21 22:07:40 PDT 2009
Command : write_model_timing -type arc sample.rpt
#####
#####
Section 1 : Worst Case Arcs
Analysis Mode : setup
Description : worst case arcs
#
From To Arc Type Transition Arc-Delays View
#####
vclk2 tdigit[0] max_sequential rise/fall 2.676 mission+worst-rcMax
vclk2 tdigit[0] max_sequential rise/rise 2.831 mission+worst-rcMax
vclk2 tdigit[1] max_sequential rise/fall 2.413 mission+worst-rcMax
vclk2 tdigit_flag max_sequential rise/rise 2.58 mission+worst-rcMax
vclk2 tdigit[0] max_sequential rise/fall 2.676 test+worst-rcMax
vclk2 tdigit[0] max_sequential rise/rise 2.831 test+worst-rcMax
vclk2 tdigit[1] max_sequential rise/fall 2.413 test+worst-rcMax
#####
Section 2 : Transition time
Analysis Mode : setup
Description : Actual transition times on the ports
View : default
#
Port RiseTran FallTran
#####
reset 0.120 0.120
int 0.120 0.120
port_pad_data_in[15] 0.120 0.120
port_pad_data_in[0] 0.120 0.120
scan_en 0.120 0.120
scan_clk 0.120 0.120
```

## Tempus Text Command Reference

### Timing Model Commands

---

```
scan_in_1 0.120 0.120
scan_in_2 0.120 0.120
test_mode 0.120 0.120
spi_data 0.120 0.120
spi_fs 0.120 0.120
refclk 0.120 0.120
vcom 1.025 1.138
#####
Section 3 : Capacitance
Analysis Mode : setup
Description : Total Cap on the ports
View : default
#
Port CTotal(Rise) CTotal(Fall)
#####
reset 3.078 3.078
port_pad_data_in[0] 3.078 3.078
scan_en 3.078 3.078
scan_clk 3.078 3.078
scan_in_1 3.078 3.078
vcop 0.007 0.007
#####
Section 4 : Design Rules
Analysis Mode : setup
Description : Design Rules on the ports
View : default
#
Port MaxCap MinCap MaxTrans MaxFanout Fanout Limit
#####
reset NA NA 2.700 15.000 NA
int NA NA 2.700 15.000 NA
port_pad_data_in[15] NA NA 2.700 15.000 NA
port_pad_data_in[14] NA NA 2.700 15.000 NA
port_pad_data_in[13] NA NA 2.700 15.000 NA
port_pad_data_in[12] NA NA 2.700 15.000 NA
```

## Related Commands

- [compare\\_model\\_timing](#)
- [do\\_extract\\_model](#)

## Interface Logic Models - ILM

- [read\\_ilm](#)
- [set\\_ilm\\_mode](#)
- [specify\\_ilm](#)
- [unspecify\\_ilm](#)
- [write\\_ilm](#)
- [import\\_ilm\\_data](#)

## read\_ilm

```
read_ilm
[-help]
[-noSpef]
[-noSI]
```

Specifies the Interface Logic Module (ILM) or the eXtended Interface Logic Module (XILM) directory for the specified block.

You can run the `read_ilm` command only once during a session. If there are multiple ILM blocks, you must specify them in a list. The `-cell` and `-dir` parameters accept a list instead of a single value only. The order of the cells and directories must match, for example:

```
ilmview> read_ilm -cell {cell1 cell2} -dir {dir1 dir2}
```

If ILM data is generated using previous releases in which both the module definition in the netlist and the `.lib` for the ILM module are removed, then you must set the following variable before `read_design` and `set_top_module` to load the top-level design in the new design flow.

```
set_var load_netlist_ignore_undefined_cell 1
```

If an ILM model is created using the current release, you need not set the `set_var load_netlist_ignore_undefined_cell` variable.

**Note:** The ILM is not supported for Signoff in Tempus.

### Parameters

|         |                                                                                                                                                                                                                |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help   | Prints out the command usage.                                                                                                                                                                                  |
| -noSpef | Loads the ILM or XILM netlist, core <code>.sdc</code> , and/or <code>.xtwf</code> file without loading the ILM or XILM SPEF file, even if the top module SPEF is loaded before running <code>read_ilm</code> . |
| -noSI   | Does not read the SI ILM data.                                                                                                                                                                                 |

### Related Commands

- `specify_ilm`
- `unspecify_ilm`
- `write_ilm`



## set\_ilm\_mode

```
set_ilm_mode
[-help]
[-reset]
[-keepAsync {true | false}]
[-keepFlatten {true | false}]
[-keepHighFanoutPorts {true | false}]
[-keepLoopBack {true | false}]
```

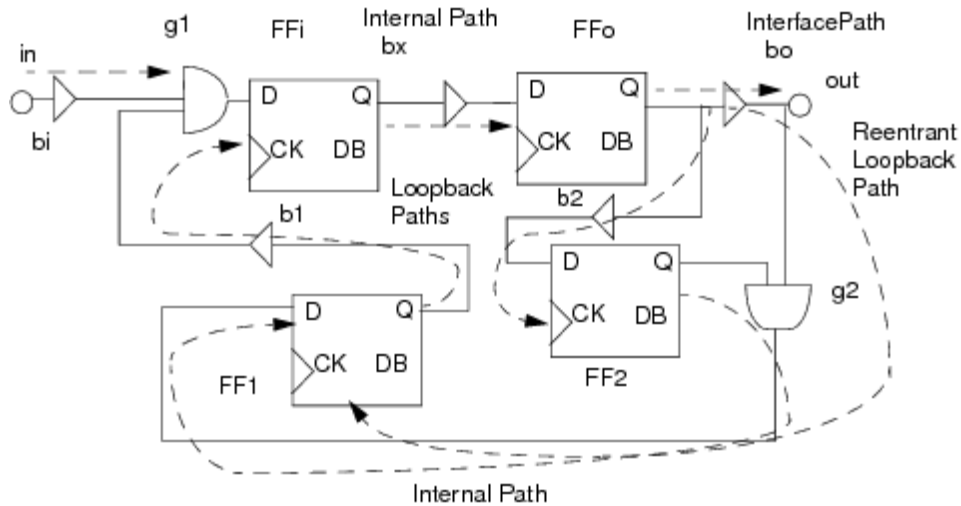
Controls certain behaviors of ILM commands.

**Note:** The ILM is not supported for Signoff in Tempus.

### Parameters

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                               | <p>Outputs a brief description that includes type and default information for each <code>set_ilm_mode</code> parameter.</p> <p>For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man set_ilm_mode</code>.</p>                                                                                                                                                                                                                                                                                                          |
| -keepAsync {true   false}           | <p>Keeps paths from asynchronous input ports when ILMs are created by <code>write_ilm</code>. For example, paths from an input port to the reset pin of interface sequential instances, but not to the internal sequential instances.</p> <p><i>Default:</i> false</p>                                                                                                                                                                                                                                                                                                            |
| -keepFlatten {true   false}         | <p>Instructs the software to keep in a flattened state.</p> <p><i>Default:</i> true</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| -keepHighFanoutPorts {true   false} | <p>Keeps high-fanout input port paths when ILMs are created by <code>write_ilm</code>.</p> <p><i>Default:</i> true</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| -keepLoopBack {true   false}        | <p>Keeps loop back paths when ILMs are created by <code>write_ilm</code>.</p> <p>When set to <code>true</code>, any path that touches a net used for the ILM model is also included. This is done to improve model accuracy, so that the actual capacitance at an output port can affect the delay on a net feeding an internal path that is normally not included in the ILM model. This also degrades the reduction ratio (<code>#ILM_insts/#all_inst</code>) for the block.</p> <p>For more details, see the "Loop-Back Paths" section below.</p> <p><i>Default:</i> false</p> |

## Loop-Back Paths



## Examples

- The following command does not keep high fanout ports when ILMs are created by `write_ilm`:  

```
ilmview> set_ilm_mode -keepHighFanoutPorts false
```
- The following command resets the `-highFanoutPort` parameter to its original value:  

```
ilmview> set_ilm_mode -reset -keepHighFanoutPorts
```
- The following command resets all `set_ilm_mode` parameters to their default values:  

```
ilmview> set_ilm_mode -reset
```

## specify\_ilm

```
specify_ilm
[-help]
-cell cellName
-dir dirName
```

Specifies the ILM data directory for the specified block. You can run `specify_ilm` multiple times in the same session. Each time you specify `specify_ilm`, the software overwrites the previous setting for the block. If master/clones exist in the design, the cell name will match the master partition's name.

**Note:** The ILM is not supported for Signoff in Tempus.

### Parameters

|                       |                                                         |
|-----------------------|---------------------------------------------------------|
| -help                 | Prints out the command usage.                           |
| -cell <i>cellName</i> | Specifies the block name.                               |
| -dir <i>dirName</i>   | Specifies the directory for the files you want to read. |

### Examples

- The following commands specifies directory `dir_dsp_core` for block `tdsp_core`, and directory `dir_onemore` for block `onemore`:

```
ilmview> specify_ilm -cell tdsp_core -dir dir_dsp_core
ilmview> specify_ilm -cell onemore -dir dir_onemore
```

### Related Commands

- `unspecify_ilm`

## unspecify\_ilm

```
unspecify_ilm
[-help]
-cell cellName
```

Unspecifies the ILM cell.

**Note:** The ILM is not supported for Signoff in Tempus.

### Parameters

|                       |                                               |
|-----------------------|-----------------------------------------------|
| -help                 | Prints out the command usage.                 |
| -cell <i>cellName</i> | Specifies the name of the block to unspecify. |

### Related Commands

- [specify\\_ilm](#)

## write\_ilm

```
write_ilm
[-help]
-dir dirname
[-modelType {timing|si|all}]
[-noCoreSdc]
[-noSpef]
[-noXtwf]
[-ignore_ports ListOfPorts]
[-include_object ListOfInstancesorNets]
[-validation_dir]
[-writeSDF]
```

Creates the specified directory containing the Interface Logic Model (ILM) data for timing analysis and the Extended Interface Logic Model (XILM) data for SI analysis.

The information is saved in the following directories:

For Timing:

- \$dir/ilm\_data
  - \*.v
  - \*.sdc
  - \*.spef

For SI:

- \$dir/si\_ilm\_data
  - \*.v
  - \*.sdc
  - \*.spef
  - \*.xtwf

**Note:** The ILM is not supported for Signoff in Tempus.

## Parameters

|                                                                                         |                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                                                                      | Prints out the command usage.                                                                                                                                                                                                                                                                                                     |
| <code>-dir <i>dirname</i></code>                                                        | Specifies the location of the ILM and XILM files, which are generated in the <code>ilm_data</code> and the <code>si_ilm_data</code> sub-directories, respectively.                                                                                                                                                                |
| <code>-include_object</code><br><i>ListOfInstancesorNets</i>                            | Accepts a collection of instances and/or nets. All instances and nets specified are preserved in the generated ILM. When instances are preserved, their driving and load nets are also preserved. When nets are preserved, only the driving instances are preserved.                                                              |
| <code>-ignore_ports</code><br><i>ListOfPorts</i>                                        | The given ports and the associated interface path(s) that start or end at these ports will not be included in the ILM model.                                                                                                                                                                                                      |
| <code>-modelType</code><br>{ <code>timing</code>   <code>si</code>   <code>all</code> } | Generates ILM and XILM models. <ul style="list-style-type: none"> <li>• The <code>-modelType timing</code> parameter creates only ILM models.</li> <li>• The <code>-modelType si</code> parameter creates only XILM models.</li> <li>• The <code>-modelType all</code> parameter creates both the ILM and XILM models.</li> </ul> |
| <code>-noCoreSdc</code>                                                                 | Does not create the core SDC data for ILM/XILM models.                                                                                                                                                                                                                                                                            |
| <code>-noSpef</code>                                                                    | Does not generate the SPEF data for ILM and XILM models.                                                                                                                                                                                                                                                                          |
| <code>-noXtwf</code>                                                                    | Does not create the xtwf data for ILM/XILM models.<br><br><b>Note:</b> When the <code>-noCoreSdc</code> and <code>-noXtwf</code> parameters are specified, <code>write_ilm</code> does not call any delay or timing calculation, and skips creating both the SDC and XTWF files for ILM/XILM models.                              |
| <code>-validation_dir</code>                                                            | Creates a script and the necessary data used to validate ILM/XILM.                                                                                                                                                                                                                                                                |
| <code>-writeSDF</code>                                                                  | Creates an SDF file that is based only on the logic kept in the ILM model after the netlist is trimmed. In the MMMC mode, one SDF file is generated per view.                                                                                                                                                                     |

## Example

- The following command generates the ILM, ignoring certain ports and including certain objects in the `BlockB` directory. It also writes out the ILM SDF file under the `BlockB` directory.  

```
ilmview> write_ilm -writeSDF -ignore_ports $ignorePorts include_object $includeObjects -
dir BlockB
```
- In SMSC designs, the following command generates ILM files in the three subdirectories

(ilm\_setup, ilm\_hold, and si\_ilm\_setupHold) under the directory BlockB:

```
ilmview> write_ilm -dir BlockB
```

- In MMMC designs, the following command generates ILM files in one subdirectory (MMMC) under the directory BlockB:

```
ilmview> write_ilm -dir BlockB
```

## Related Commands

- `read_ilm`

## import\_ilm\_data

```
import_ilm_data
[-help]
-cell string
[-incr]
-model_type {timing | cts | si}
[-overwrite]
[-xtwf string]
{-dir string | -cell_view string}
[-verilog string [-cell_prefix string -uniquify_netlist]]
[-sdc string -timing_view string]
[-spef string -rc_corner string]
[-optStage string]
```

Creates an ILM directory from existing data. It can only create non-merged models. This command is useful in situations where an ILM is to be created through external inputs or using data from third party models into a format that could be recognized by Tempus.

If the data files already exist in the specified directory, the tool stops and generates an error message. You can overwrite existing files by specifying the `-overwrite` option. You can add files to the directory by specifying `-incr`.

The command also creates a TCL script (`cell_name_ILM.tcl`) that you can source in your command file. The script contains the following command:

```
specify_ilm -cell cell_name -dir dir_name
```

**Note:** The ILM is not supported for Signoff in Tempus.

### Parameters

|                              |                                                                                                                                                                                                                                                                                |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                        | Outputs a brief description that includes the type and default information for each <code>import_ilm_data</code> parameter.<br><br>For a detailed description of the command and all of its parameters, use the <code>man</code> command:<br><code>man import_ilm_data.</code> |
| -cell <i>string</i>          | Specifies the name of the block.                                                                                                                                                                                                                                               |
| -cell_prefix <i>string</i>   | Specifies the cell prefix.                                                                                                                                                                                                                                                     |
| -<br>cell_view <i>string</i> | Specifies the library name, cell name, and view name in the OpenAccess database format.                                                                                                                                                                                        |



|                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-dir string</code>                                                                 | Stores the ILM data in the specified directory: the absolute directory path or the path relative to the current directory.                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>-model_type</code><br>{ <code>timing</code>   <code>cts</code>   <code>si</code> } | Specifies the type of ILM model, i.e., timing, cts, or si.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>-incr</code>                                                                       | Let's you add files in addition to the existing files to the directory.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>-optStage string</code>                                                            | Specifies the optimization stage, preCTS or postCTS.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <code>-overwrite</code>                                                                  | Overwrites existing files in the directory.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <code>-rc_corner</code><br><i>string</i>                                                 | Specifies the RC corner name for MMMC analysis. It identifies the specified corner to be used with <code>-spef</code> .                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>-sdc string</code>                                                                 | Specifies an existing .sdc file that you want to include in the directory. It identifies the SDC file to be used with the specified view.                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>-spef string</code>                                                                | Specifies an existing .spef file that you want to include in the directory. It identifies the SPEF file to be used with the specified corner.                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>-timing_view</code><br><i>string</i>                                               | Specifies the view name for the ILM module. It identifies the specified view to be used with <code>-sdc</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>-uniquify_netlist</code>                                                           | <p>Uniquifies the input Verilog netlist. This ensures that there are no errors when ILMs are being loaded.</p> <p>Using this parameter, each sub-module for an ILM has a unique name, which is composed of the time stamp and its original name.</p> <p>For example, in <code>top_sm WedJun31138212009\$data_path</code>, <code>top_sm</code> specifies the top-level design, <code>WedJun31138212009</code> specifies the time stamp, and <code>\$data_path</code> specifies the original module name.</p> <p><i>Default:</i> <code>Off</code></p> |
| <code>-verilog string</code>                                                             | Specifies the Verilog file you want to include in the ILM data directory.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>-xtwf string</code>                                                                | Specifies the .xtwf file to include in the directory.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

## Examples

- The following commands create two module directories:

```
import_ilm_data -cell blk1 -dir A -model_type timing -verilog V1.v -spef 1.spef -
rc_corner corner1 -sdc sdcfile1 -timing_view view1
```

```
import_ilm_data -cell blk2 -dir A -model_type si -verilog V2.v -spef 2.spef -rc_corner
corner2 -sdc sdcfile2 -timing_view view2
```

- The command creates two TCL files in the current directory. The `blk1_ILM.tcl` file contains the following command:

```
specify_ilm -dir A -cell blk1
```

The `blk2_ILM.tcl` file contains the following command:

```
specify_ilm -dir A -cell blk2
```

## Prototype Timing Model - PTM

- `create_ptm_constraint_arc`
- `create_ptm_delay_arc`
- `create_ptm_drive_type`
- `create_ptm_generated_clock`
- `create_ptm_load_type`
- `create_ptm_model`
- `create_ptm_path_type`
- `create_ptm_port`
- `report_ptm_model`
- `save_ptm_model`
- `set_ptm_design_rule`
- `set_ptm_global_parameter`
- `set_ptm_port_drive`
- `set_ptm_port_load`

## create\_ptm\_constraint\_arc

```
create_ptm_constraint_arc
-setup
-hold
-from clock_port_name]
[-to port_name]
-edge { rise | fall }
-path_type pathTypeName
- path_factor value
[-value constraint_value]
```

Sets the timing check from the clock input port to the data input port.

To create a new timing check arc that does not already exist in the timing model, the clock port that you specify as `-from` should be created as a clock type by the `create_ptm_port` command.

**Note:** The `create_ptm_constraint_arc` command can be specified only after you create a PTM model using the `create_ptm_model` command.

## Parameters

|                                          |                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-edge { rise   fall }</code>       | Specifies which transition on the clock input is relevant to the timing check arc definition.                                                                                                                                                                                                                                             |
| <code>-from<br/>clock_port_name</code>   | Specifies the name of the clock port, which can be an input, bidirectional, or internal port.<br><br><b>Note:</b> This parameter allows you to specify a pattern for port names using wildcard characters.                                                                                                                                |
| <code>-hold</code>                       | Specifies that the check arc to be created is of <code>hold</code> type.                                                                                                                                                                                                                                                                  |
| <code>-path_type<br/>pathTypeName</code> | Specifies the reference time arc created using the <code>create_ptm_path_type</code> command. This parameter is mutually exclusive to the <code>-value</code> parameter currently available. Either the <code>-value</code> parameter or the <code>-path_type</code> parameter must be specified.                                         |
| <code>-path_factor<br/>value</code>      | Used as an additional scaling value for the delay/slew tables.<br><br><i>Default:</i> 1.0                                                                                                                                                                                                                                                 |
| <code>-setup</code>                      | Specifies that the check arc to be created is of the <code>setup</code> type.                                                                                                                                                                                                                                                             |
| <code>-to port_name</code>               | Specifies the name of the input or bidirectional port that will serve as the signal pin for the check arc. If you do not specify a port name, the software sets the timing check for all input and bidirectional data ports.<br><br><b>Note:</b> This parameter allows you to specify a pattern for port names using wildcard characters. |
| <code>-value<br/>constraint_value</code> | Specifies a value for the timing check.                                                                                                                                                                                                                                                                                                   |

## Example

- The following command sets the setup timing check value of 0.3 with respect to the rising edge at the clock port `clock_in` to the data input port `data_in`:

```
tempus> create_ptm_constraint_arc -setup -value 0.3 -from clock_in -to data_in -edge
rise
```

- The following command specifies the reference check arc by specifying the path type as `SETUP_TYPE1` and path factor as 1.03:

```
tempus> create_ptm_constraint_arc -from clock_gen_1 -to data_in -path_type SETUP_TYPE1 -
path_factor 1.03
```

## Related Commands

- `create_ptm_port`

## create\_ptm\_delay\_arc

```
create_ptm_delay_arc
[-from port_name]
[-path_type pathTypeName]
[-path_factor value]
[-to port_name]
[-edge {rise | fall}]
[-value delay_value]
```

Sets the delay for combinational or sequential paths from input ports to the output ports. If you do not specify the drive type on the output port, a constant timing table is created. In a constant timing table, the delay is constant. It does not vary with respect to input slew and output capacitance.

To create a new sequential timing arc that does not already exist in the timing model, the port you specify as `-from` should be created as a clock type by the `create_ptm_port` command.

**Note:** The `create_ptm_delay_arc` command can be specified only after you create a PTM model using the `create_ptm_model` command.

### Parameters

|                                          |                                                                                                                                                                                                                                                                               |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-edge {rise<br/>  fall}</code>     | Specifies the edge on the clock input relevant to the sequential delay arc definition. By default, the software uses a rising clock transition.                                                                                                                               |
| <code>-from<br/>port_name</code>         | Specifies the name of the input port or the clock port.<br><b>Note:</b> This parameter allows you to specify a pattern for port names using wildcard characters.                                                                                                              |
| <code>-path_type<br/>pathTypeName</code> | Specifies the reference time arc created using the <code>create_ptm_path_type</code> command. This parameter is mutually exclusive to the <code>-value</code> parameter. Either the <code>-value</code> parameter or the <code>-path_type</code> parameter must be specified. |
| <code>-path_factor<br/>value</code>      | Used as an additional scaling value for the delay/slew tables.<br><i>Default:</i> 1.0                                                                                                                                                                                         |
| <code>-to<br/>port_name</code>           | Specifies the name of the output port or the bidirectional port.<br><b>Note:</b> This parameter allows you to specify a pattern for port names using wildcard characters.                                                                                                     |
| <code>-value<br/>delay_value</code>      | Specifies the value for the delay.                                                                                                                                                                                                                                            |

## Example

- The following command defines a sequential delay arc with respect to the rising edge at the clock port `clock_in` to the output port `data_out` and specifies the delay value as 1.0:

```
tempus> create_ptm_delay_arc -from clock_in -to data_out -edge rise -value 1.0
```

- The following command specifies the reference check arc by specifying the path type as `SETUP_TYPE1` and path factor as 1.03:

```
tempus> create_ptm_path_type SEQ_TYPE -type sequential -lib_cell DFFHQX1 -relative_pin
CK -target_pin Q
```

```
tempus> create_ptm_delay_arc -from ptm_clk -to ptm_reg_out -path_type SEQ_TYPE
```

## Related Commands

- `create_ptm_port`



## create\_ptm\_drive\_type

```
create_ptm_drive_type
-lib_cell cell_name
[-input_slew_fall value]
[-input_slew_rise value]
[-input_pin pin_name]
[-output_pin pin_name]
drive_type_name
```

Creates drive types to specify the drive strength equivalent to existing library cells. Optionally, you can also specify the rise or fall transition value(s) at the driver port using this command.

This command can be specified only after you create a PTM model using the `create_ptm_model` command.

**Note:** Use the `set_ptm_port_drive` command to associate the drive type with existing ports.

## Parameters

|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>drive_type_name</i>                        | Specifies the drive type name to use for the timing model.                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>-lib_cell</code><br><i>cell_name</i>    | Specifies the reference cell used to determine the drive value.                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>-input_pin</code><br><i>pin_name</i>    | <p>Specifies the input pin of the timing arc for the reference cell specified with the <code>-lib_cell</code> parameter. The combinational arcs starting from this pin will be used to create the drive type.</p> <p>When only one of the <code>-input_pin/-output_pin</code> parameters is specified, the tool chooses the first random combinational arc at the specified pin.</p> <p>When none of these options are specified, the tool chooses the first random combinational arc.</p> |
| <code>-input_slew_fall</code><br><i>value</i> | Specifies the fall transition value.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>-input_slew_rise</code><br><i>value</i> | Specifies the rise transition value.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>-output_pin</code><br><i>pin_name</i>   | <p>Specifies the output pin of the timing arc for the reference cell specified with the <code>-lib_cell</code> option. The combinational arcs ending at this pin will determine the drive type.</p> <p>When only one of the <code>-input_pin/-output_pin</code> parameters is specified, the tool chooses the first random combinational arc at the specified pin.</p> <p>When none of these options are specified, the tool chooses the first random combinational arc.</p>               |

## Example

- The following command defines the drive type name as `stdDrive` with drive resistance of library cell `INVX1`:  

```
tempus> create_ptm_drive_type -lib_cell INVX1 stdDrive
```
- The following command defines the drive type name as `stdDrive` with drive resistance of the library cell `BUF1`, and input and output pins as `B` and `Y`, respectively:  

```
tempus> create_ptm_drive_type -lib_cell BUF1 -input_pin B -output_pin Y stdDrive
```

## Related Commands

- `create_ptm_model`
- `set_ptm_port_drive`

## create\_ptm\_generated\_clock

```
create_ptm_generated_clock
-source clock_pin
[-divide_by value | -multiply_by value]
clock_name
```

Creates an internal pin on a timing model or uses an existing internal pin (defined by the `create_ptm_port` command), and creates a generated clock on this internal pin.

The following points apply to the usage of this command:

- The internal pin created with this command will be treated as a clock port. This internal pin can, therefore, be used to create timing check arcs using the `create_ptm_constraint_arc` command and sequential delay arcs using the `create_ptm_delay_arc` command.
- The latency of the generated clock is set to 0.
- Any timing arc related to the internal pin must include the clock latencies defined on both the internal pin and the master pin.
- The generated clock appears in the timing model generated by the `save_ptm_model` command.
- The internal pin and the generated clock are persistent, that is, they cannot be deleted using PTM commands.

## Parameters

|                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>clock_name</code>                            | Specifies the name of the generated clock defined on the internal pin.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>-divide_by value   -multiply_by value</code> | <p>Specifies the frequency division or multiplication factor.</p> <p>When you use the <code>-divide_by</code> option, the generated clock period is equal to the master clock period multiplied by the value specified with the <code>-divide_by</code> option. For example, if the value is 2, the generated clock period is two times the master clock period.</p> <p>When you use the <code>-multiply_by</code> option, the generated clock period is equal to the master clock period divided by the value specified with the <code>-multiply_by</code> option. For example, if the value is 3, the generated clock period is one-third the master clock period.</p> |
| <code>-source clock_pin</code>                     | <p>Specifies the source from which the derived clock is generated.</p> <p>The source can be an existing clock port or the one created by using the <code>create_ptm_port</code> command, and a clock should be available on this port.</p> <p>This clock port is set at the master pin for the generated clock.</p>                                                                                                                                                                                                                                                                                                                                                      |

## Example

- The following command creates a generated clock with the name `clock_gen` on internal pin `clock_gen` (also creates the pin if it does not already exist), using the clock port `clock_in` as the source pin for the generated clock, and the frequency division factor of 2:

```
tempus> create_ptm_generated_clock -source clock_in -divide_by 2 clock_gen
```

## Related Commands

- `create_ptm_constraint_arc`
- `create_ptm_delay_arc`

## create\_ptm\_load\_type

```
create_ptm_load_type
-lib_cell lib_cell_name
[-input_pin lib_pin_name]
load_type_name
```

Creates a load type, which is identified by a unique *load\_type\_name*. Typically, the load type represents a capacitance value from a library input or a bidirectional pin. You can create multiple load types for a PTM model, provided you use different `create_ptm_load_type` commands for each load type.

**Note:** The load type can be associated with PTM ports by using the `set_ptm_port_load` command.

### Parameters

|                                                |                                                                                                                                                                                                                                                 |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-lib_cell</code><br><i>lib_cell_name</i> | Specifies the reference cell that will be used to derive the load characteristics.                                                                                                                                                              |
| <code>-input_pin</code><br><i>lib_pin_name</i> | Specifies the name of the input or bidirectional pin that will be used to derive the load characteristics.<br><br><b>Note:</b> When this option is not used, the software randomly selects an input or bidirectional pin from the library cell. |
| <i>load_type_name</i>                          | Specifies a unique name for the load type.                                                                                                                                                                                                      |

### Example

- The following command defines a load type called `stdLoad` for the `INVX1` cell and the input pin `A`:

```
tempus> create_ptm_load_type -lib_cell INVX1 -input_pin A stdLoad
```

### Related Commands

- `set_ptm_port_load`

## create\_ptm\_model

```
create_ptm_model model_name
```

Initiates the creation of a Prototype Timing Model (PTM) model. A PTM model contains preliminary information about the block, such as ports, delay arcs, and timing check arcs.

**Note:** You must save the existing model details using the `save_ptm_model` command before creating another model. This means that you cannot generate multiple models at the same time.

### Parameters

|                   |                                     |
|-------------------|-------------------------------------|
| <i>model_name</i> | Specifies a name for the PTM model. |
|-------------------|-------------------------------------|

### Example

- The following command creates a PTM model named `sample_model`:

```
tempus> create_ptm_model sample_model
```

### Related Commands

- `report_ptm_model`
- `save_ptm_model`

## create\_ptm\_path\_type

```
create_ptm_path_type
[-help]
-type arc_type
-lib lib_name
-lib_cell lib_cell_name
-relative_pin pin_name
-target_pin pin_name
path_type_name
```

Uniquely identifies the worst time arc of the given type between a pin pair for a given library cell when `-target_pin` and `-relative_pin` have been specified. If only either of `-target_pin` or `-relative_pin` is specified, the tool finds out the worst library arc seen with the type as provided with the `-type` parameter. If no pin is specified, then the first worst arc with the type specified with the `-type` parameter is selected. The created path type name is used later with the `create_ptm_constraint_arc` and `create_ptm_delay_arc` commands to create the reference time arc inside the PTM model. The newly created arc will inherit the rise/fall delay/ slew LUTs after applicable scale specified by `-path_factor` in `create_ptm_constraint_arc/create_ptm_delay_arc` and the unateness property from the reference arc.

### Parameters

|                                      |                                                                                                                                                                                                                                                                                       |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                   | Prints out the command usage.                                                                                                                                                                                                                                                         |
| <code>path_type_name</code>          | Specifies the name to uniquely identify a reference timing arc specification (path type), which can later be used for multiple PTM creations.                                                                                                                                         |
| <code>-type arc_type</code>          | Provides the arc type from which the reference library is to be picked up. This reference library will be used as the reference arc for this path type. Possible type values are: <code>setup</code> , <code>hold</code> , <code>combinational</code> , and <code>sequential</code> . |
| <code>-lib lib_name</code>           | Specifies the reference library name.                                                                                                                                                                                                                                                 |
| <code>-lib_cell lib_cell_name</code> | Specifies the reference cell name.                                                                                                                                                                                                                                                    |
| <code>-relative_pin pin_name</code>  | Specifies the relative pin (that is, the reference pin for check arcs and source pin for delay arcs) for picking the reference arc.                                                                                                                                                   |
| <code>-target_pin pin_name</code>    | Specifies the target pin (that is, the signal pin for check arcs and the sink pin for delay arcs) for picking the reference arc.                                                                                                                                                      |



## Example

- The following command specifies the reference check arc from relative pin `CLK1` to the target pin `D1`:

```
tempus> create_ptm_path_type SETUP_TYPE1 -lib_cell DFFX1 \
-relative_pin CLK -target_pin D1 -type setup
```

## Related Commands

- `create_ptm_constraint_arc`
- `create_ptm_delay_arc`

## create\_ptm\_port

```
create_ptm_port
[-help]
[port_list]
[-clock]
[-type {output | inout | input | internal | clock}]
```

Creates a PTM port for the current model.

### Parameters

|                                                   |                                                                                                                                                                                                                                        |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                                             | Prints out the command usage.                                                                                                                                                                                                          |
| -clock                                            | Specifies that the provided port is a clock port.<br><br>The software allows clock creation from an internal pin of PTM, with another internal pin as the source.                                                                      |
| port_list                                         | Specifies a port name of the type specified with the -type option.                                                                                                                                                                     |
| -type {output   inout   input   internal   clock} | Specifies the type of port to create. You can use one of the following values to specify the type of port: <ul style="list-style-type: none"><li>• output</li><li>• inout</li><li>• input</li><li>• internal</li><li>• clock</li></ul> |

## Examples

- The following command creates a bidirectional port named `bidia`:  

```
tempus > create_ptm_port -type inout bidia
```
- The following command will create a port with the name `clk` as a clock with direction `internal`:  

```
tempus > create_ptm_port -type internal clk -clock
```

A sample script is given below:

```
pin(clk) {
 clock : true;
 direction : internal;
 capacitance : 0.011040;
 max_transition : 0.800000;
}
```

## Related Commands

- `create_ptm_load_type`

## report\_ptm\_model

```
report_ptm_model
[-path_types]
[-ports]
[-delay_param]
[-arcs]
[-arc_type]
```

Generates a report on ports, arcs, and global delay parameter details for the current PTM model. This command must be used after the `create_ptm_model` command and before the `save_ptm_model` command.

### Parameters

|              |                                                                                                                                                 |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| -arcs        | Displays the timing arc details for the current model, such as the input port, output port, and the delay value associated with the timing arc. |
| -arc_type    | Reports the type of arc. When specified, the command output displays the <code>Arc Type</code> column.                                          |
| -delay_param | Displays the global delay parameter details set using the <code>set_ptm_global_parameter</code> command.                                        |
| -path_types  | Reports only the path type details.<br><i>Default:</i> Includes the details of path types created.                                              |
| -ports       | Displays the port details from the current PTM model that were set using PTM commands.                                                          |

### Example

- The following command displays a report on the arcs of the `dma` PTM model:

```
tempus> report_ptm_model -arcs
```

```
ModelName : dma
```

| Arc                        | From        | To          | Delay | PathType | PathFactor |
|----------------------------|-------------|-------------|-------|----------|------------|
| clock_in_to_data_in        | clock_in    | data_in     | 0.3   | -        | -          |
| clock_in_to_data_in        | clock_in    | data_in     | 0.4   | -        | -          |
| clock_in_to_data_in        | clock_in    | data_in     | 0.1   | -        | -          |
| clock_in_to_data_in        | clock_in    | data_in     | 0.2   | -        | -          |
| clock_in_to_data_out       | clock_in    | data_out    | 0.05  | -        | -          |
| data_test_1_to_data_test_2 | data_test_1 | data_test_2 | 1.0   | -        | -          |

```
data_test_2_to_data_out data_test_2 data_out 1.5 - -
```

- The following command displays a report on the port details of the `dma` PTM model:

```
tempus> report_ptm_model -ports
```

```
ModelName : dma
```

```
DriveType Cell
```

```

```

```
stdDrive3 AND2XL
```

```
stdDrive4 AND2XL
```

```
stdDrive ADDFHx1
```

```
stdDrive2 AND2XL
```

| Port        | Direction | BusRange | LoadValue | LoadType | DriveValue | DriveType |
|-------------|-----------|----------|-----------|----------|------------|-----------|
| ----        | -----     | -----    | -----     | -----    | -----      | -----     |
| bcstop      | inout     | -        | -         | stdLoad  | -          | -         |
| bcstop_2    | in        | -        | -         | stdLoad  | -          | -         |
| bcstop_3    | inout     | -        | -         | stdLoad2 | -          | -         |
| bdis1       | in        | -        | 4.0       | -        | -          | -         |
| clock_in    | clock     | -        | -         | -        | -          | -         |
| data_in     | in        | -        | -         | -        | -          | -         |
| data_out    | out       | -        | -         | -        | -          | stdDrive  |
| data_out_2  | out       | -        | -         | -        | -          | stdDrive2 |
| data_out_3  | out       | -        | -         | -        | -          | stdDrive3 |
| data_out_4  | out       | -        | -         | -        | -          | stdDrive4 |
| data_test_1 | clock     | -        | -         | -        | -          | -         |
| data_test_2 | inout     | -        | -         | -        | -          | -         |

- The following command displays a report on the global delay parameters of the `dma` PTM model:

```
tempus> report_ptm_model -delay_param
```

```
ModelName : dma
```

```
DelayParameter Delay
```

```

```

```
setup 1.2
```

```
hold 1.5
```

```
clk_to_output 1.6
```

- The following command reports the path type details for the `SETUP_TYPE1` path type.

```
tempus> create_ptm_model <name>
:
tempus> create_qtm_port -type clock CLOCK
tempus> create_qtm_port -type input IN
:
tempus> create_ptm_path_type SETUP_TYPE1 -lib_cell typ_1.2/DFFX1 \
-relative_pin CLK -target_pin D1
tempus> create_ptm_path_type HOLD_TYPE1 -lib_cell typ_1.2/DFF2X1 \
-relative_pin CLK -target_pin D2
tempus> create_ptm_constraint_arc -from clock_gen_1 -to data_in -setup \
-edge fall -path_type SETUP_TYPE1
```

## Related Commands

- [create\\_ptm\\_model](#)
- [save\\_ptm\\_model](#)

## save\_ptm\_model

```
save_ptm_model
[-help]
[-output filename_prefix]
[-library_cell]
```

Saves the PTM model, including the timing arc specifications in the `.lib` library file format. The file name contains the PTM cell, the pin and bus definitions, and various constraints and delay arcs created by the `create_ptm_*` command set.

### Parameters

|                                          |                                                                                                                                                         |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                       | Prints out the command usage.                                                                                                                           |
| <code>-library_cell</code>               | Specifies the reference cell which will be used to derive the model characteristics.                                                                    |
| <code>-output<br/>filename_prefix</code> | Specifies the output file name.<br><br>If a file name is not specified, the model name defined using the <code>create_ptm_model</code> command is used. |

### Examples

- The following command specifies a PTM model file name:

```
tempus > save_ptm_model -output ptm.lib
```

### Related Commands

- `create_ptm_model`
- `report_ptm_model`

## set\_ptm\_design\_rule

```
set_ptm_design_rule
{-max_transition <max_tran> | -max_capacitance <max_cap> | -max_fanout <max_fanout>}
[-port port_list]
```

Specifies a value for the `max_transition`, `max_capacitance`, and `max_fanout` design constraints. These values can be applied to all the ports or a selected list of ports.

### Parameters

|                                                 |                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-max_capacitance</code><br><i>max_cap</i> | Specifies the <code>max_capacitance</code> value for the ports in the PTM model.                                                                                                                                                                                                                                             |
| <code>-max_fanout</code><br><i>max_fanout</i>   | Specifies the <code>max_fanout</code> value for the ports in the PTM model.                                                                                                                                                                                                                                                  |
| <code>-max_transition</code><br><i>max_tran</i> | Specifies the <code>max_transition</code> value for the ports in the PTM model.                                                                                                                                                                                                                                              |
| <code>-port port_list</code>                    | <p>Specifies a list of PTM port names on which the value specified with the <code>-max_capacitance</code>, <code>-max_fanout</code>, or <code>-max_transition</code> parameters should be applied.</p> <p><b>Note:</b> This parameter allows you to specify a list of patterns for port names using wildcard characters.</p> |

### Example

- The following commands specify a global `max_transition` value of 1.0 for all ports except `trigger`, which will use the `max_transition` value of 1.1:

```
tempus> set_ptm_design_rule -max_transition 1.0
tempus> set_ptm_design_rule -max_transition 1.1. -port trigger
```



## set\_ptm\_global\_parameter

```
set_ptm_global_parameter
-param {setup | hold | clk_to_output}
-value real_number
```

Defines global values for setup check arcs, hold check arcs, or sequential delay arcs across the PTM model. This global value is added to the arc-specific delay value. As a result, the final arc's delay is calculated as a sum of the global parameter value and the arc delay specified using the `create_ptm_constraint_arc` or the `create_ptm_delay_arc` command.

**Note:** The global parameters must be specified before defining the first arc in the current PTM model.

### Parameters

|                                                    |                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-param {setup   hold   clk_to_output}</code> | Specifies the name of a global parameter, which can be one of the following: <ul style="list-style-type: none"> <li>• <code>setup</code> for setup check arcs.</li> <li>• <code>hold</code> for hold check arcs.</li> <li>• <code>clk_to_output</code> for sequential delay arcs.</li> </ul> |
| <code>-value <i>real_number</i></code>             | Defines a value for the specified global parameter name.                                                                                                                                                                                                                                     |

### Example

- The following command specifies `setup` as the global parameter and assigns the value `1.2` to it:  
tempus> set\_ptm\_global\_parameter -param setup -value 1.2

### Related Commands

- `create_ptm_constraint_arc`
- `create_ptm_delay_arc`

## set\_ptm\_port\_drive

```
set_ptm_port_drive
-type drive_type
[-input_slew_rise rise_slew_value]
[-input_slew_fall fall_slew_value]
port_list
```

Sets the drive for the output and bidirectional ports. The *drive\_type* specifies the type of cell within the model as drivers on the interface nets. You cannot specify two different drivers for two timing arcs ending at the same output port.

### Parameters

|                                                   |                                                                                                                                                                           |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>port_list</i>                                  | Specifies the name of the output or bidirectional port.<br><b>Note:</b> This parameter allows you to specify a list of patterns for port names using wildcard characters. |
| <i>-input_slew_fall</i><br><i>fall_slew_value</i> | Overrides the fall transition value specified with the <i>-input_slew_fall</i> option of the <code>create_ptm_drive_type</code> command.                                  |
| <i>-input_slew_rise</i><br><i>rise_slew_value</i> | Overrides the rise transition value specified with the <i>-input_slew_rise</i> option of the <code>create_ptm_drive_type</code> command.                                  |
| <i>-type drive_type</i>                           | Specifies the drive type to use for the timing model.                                                                                                                     |

### Examples

- The following command sets the `stdDrive` drive to be used on the output data port `data_out`:  
tempus> set\_ptm\_port\_drive -type stdDrive data\_out

### Related Commands

- `create_ptm_drive_type`

## set\_ptm\_port\_load

```
set_ptm_port_load
-type load_type_name | -value real_number
[-multiply_by multiply_factor]
port_list
```

Defines a load for PTM model ports by either specifying a capacitance value or by using the load type defined by the `create_ptm_load_type` command.

### Parameters

|                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-type</code><br><code>load_type_name</code>         | Specifies the load type name created by the <code>create_ptm_load_type</code> command.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <code>-multiply_by</code><br><code>multiply_factor</code> | Specifies the multiplication factor that will be used to calculate the load on the specified ports, as shown:<br><br>Load on port(s) = {(load_value) * (factor)}<br><br>Where: <ul style="list-style-type: none"> <li><code>load_value</code> is the capacitance value derived from the cell name specified with the <code>-lib_cell</code> option of the <code>create_ptm_load_type</code> command when you use the <code>-type</code> option.</li> <li><code>factor</code> is the multiplication factor.</li> </ul> |
| <code>-value</code><br><code>real_number</code>           | Specifies the value to use as load/capacitance for the specified list of ports. The value specified with this option overrides the load value that is derived by the <code>-type</code> option.<br><br><b>Note:</b> This option cannot be used with the <code>-multiply_by</code> option.                                                                                                                                                                                                                             |
| <code>port_list</code>                                    | Specifies the list of input or bidirectional ports that exist in the PTM model defined by the <code>create_ptm_port</code> command.<br><br><b>Note:</b> This parameter allows you to specify a list of patterns for port names using wildcard characters.                                                                                                                                                                                                                                                             |

### Example

- The following command defines a load for the PTM model port `data_in_2` by specifying a capacitance value of 0.15:  
tempus> set\_ptm\_port\_load -value 0.15 data\_in\_2
- The following command specifies a load type called `stdLoad` and a multiplication factor of 2 for the

data\_in port:

```
tempus> set_ptm_port_load -type stdLoad -multiply_by 2 data_in
```

## Related Commands

- `create_ptm_load_type`
- `create_ptm_port`

---

# Timing Debug Commands

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- [analyze\\_paths\\_by\\_basic\\_path\\_group](#)
- [analyze\\_paths\\_by\\_bottleneck](#)
- [analyze\\_paths\\_by\\_clock\\_domain](#)
- [analyze\\_paths\\_by\\_critical\\_false\\_path](#)
- [analyze\\_paths\\_by\\_drv](#)
- [analyze\\_paths\\_by\\_hier\\_port](#)
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- [load\\_path\\_categories](#)
- [load\\_timing\\_debug\\_report](#)
- [save\\_path\\_categories](#)
- [write\\_category\\_summary](#)
- [write\\_text\\_timing\\_report](#)
- [load\\_failure\\_report](#)

# analyze\_paths\_by\_basic\_path\_group

```
analyze_paths_by_basic_path_group
[-help]
[-master master_category_name]
[-predefined]
```

Categorizes the timing paths based on basic path groups. The basic path groups are fundamental to timing analysis and help the user easily observe the timing information of different basic path groups. For instance, if user is only concerned about the paths from registers to registers, the user can analyze the paths in the category REG -> REG. The basic path groups categories are:

- Register to register
- Input to register
- Register to output
- Input to output

The following categories are created:

- REG->REGe
- IN->REG
- REG->OUT
- IN->OUT

Timing debug can also convert user defined path groups into path categories. The command can be applied wherever the user loads the machine readable timing report.

**Note:** If your design does not have paths belonging to a type of path group, the software will not create that category.

## Parameters

|                                 |                                                                                                                                                                                                                                                                                                                 |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                           | <p>Outputs a brief description that includes type and default information for each <code>analyze_paths_by_basic_path_group</code> parameter.</p> <p>For a detailed description of the command and all of its parameters, use the <code>man</code> command:</p> <pre>man analyze_paths_by_basic_path_group</pre> |
| -master<br>master_category_name | <p>Specifies that the analysis should be performed only for the specified master category.</p>                                                                                                                                                                                                                  |
| -predefined                     | <p>Creates path categories for user defined path groups.</p>                                                                                                                                                                                                                                                    |

## Examples

- The following command create path categories based on basic path groups:

```
tempus> analyze_paths_by_basic_path_group
```

The following categories are created: REG->REG, IN->REG, REG->OUT, IN->OUT.

- The following command create path categories based on predefined path group "critical" :

```
group_path -name critical -from {ff*/CLK } -to {ff*/D} #create predefined path group
```

```
tempus> analyze_paths_by_basic_path_group -predefined
```

The category "critical" will be created based on the predefined path group name.

## Related Commands

- [load\\_timing\\_debug\\_report](#)
- [create\\_path\\_category](#)

## analyze\_paths\_by\_bottleneck

```
analyze_paths_by_bottleneck
[-help]
{-category n | -incr_delay {worst|total}}
[-path_num m]
[-max_slack float]
[-min_slack float]
[-master master_category_name]
```

Creates categories according to the bottleneck analysis of the critical paths and fixes the violation according to the bottleneck information. Bottleneck Analysis detects the instances that are often involved in the critical paths in the design. Any change in timing for these instances impacts multiple critical paths in the design. Therefore you can use this information to change your design (floorplan, placement, constraints etc.) around these instances. The bottleneck analysis computes the number of occurrences that an instance appears in the paths that you display in the Timing Debug window, and provides the following output:

- Colors most used instance based on number of occurrences.
- Creates *n* number of path categories for the instances with the highest number of occurrences. *n* is the number you specify using the `-category` parameter.

Users can also perform bottleneck analysis of the critical paths based on incremental delay. Categories will be created based on the worst *m* paths with:

- The biggest worst `incr_delay` contribution
- The biggest total `incr_delay` contribution

The command can be applied wherever the user loads the machine readable timing report.



## Parameters

|                                           |                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                        | Outputs a brief description that includes type and default information for each <code>analyze_paths_by_bottleneck</code> parameter.<br><br>For a detailed description of the command and all of its parameters, use the <code>man</code> command:<br><code>man analyze_paths_by_bottleneck</code> |
| <code>-category n</code>                  | Specifies the number of categories to be created for most occurring instances.                                                                                                                                                                                                                    |
| <code>-incr_delay {worst total}</code>    | Specifies that the categories are created based on worst <code>incr_delay</code> contribution or total <code>incr_delay</code> contribution.                                                                                                                                                      |
| <code>-master master_category_name</code> | Specifies that the analysis should be performed only for the specified <code>master</code> category.                                                                                                                                                                                              |
| <code>-max_slack float</code>             | Specifies the max slack value of the critical paths for bottleneck analysis based on <code>incr_delay</code> .                                                                                                                                                                                    |
| <code>-min_slack float</code>             | Specifies the minimum slack value of the critical paths for bottleneck analysis based on <code>incr_delay</code> .                                                                                                                                                                                |
| <code>-path_num m</code>                  | Specifies the number of critical paths to be analyzed when performing bottleneck analysis based on <code>incr_delay</code> .                                                                                                                                                                      |

## Examples

- The following command creates path categories based on bottleneck analysis of the critical paths on the instances:

```
tempus> analyze_paths_by_bottleneck -category 2
```

The following categories are created: `BOT_instance1` and `BOT_instance2`. You can change the number of categories.

- The following command creates path categories based on bottleneck analysis of the critical 20 paths on the worst `incr_delay`:

```
tempus> analyze_paths_by_bottleneck -path_num 20 -incr_delay worst
```

The following category is created: `BOT_SI_WORST`.

## Related Commands

- `load_timing_debug_report`
- `create_path_category`

# analyze\_paths\_by\_clock\_domain

```
analyze_paths_by_clock_domain
[-help]
[-include_edges]
[-include_views]
[include_mode]
[-master master_category_name]
```

To find the possible timing issues in clock combinations, the command creates categories according to launch clocks - capture clock combinations.

Following categories are created:

- Paths with clock fully contained in a single domain, clk1.
- Paths with clocks starting one clock domain and ending at another, clk1-> clk2.

The command can be applied wherever the user loads the machine readable timing report.

## Parameters

|                                 |                                                                                                                                                                                                                                                                                                       |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                           | Outputs a brief description that includes type and default information for each <code>analyze_paths_by_clock_domain</code> parameter.<br><br>For a detailed description of the command and all of its parameters, use the <code>man</code> command:<br><code>man analyze_paths_by_clock_domain</code> |
| -include_edges                  | Considers leading or trailing edge of the clock while defining categories.                                                                                                                                                                                                                            |
| -include_mode                   | Considers constraint mode while defining categories.                                                                                                                                                                                                                                                  |
| -include_views                  | Considers analysis views while defining categories.                                                                                                                                                                                                                                                   |
| -master<br>master_category_name | Specifies that the analysis should be performed only for the specified master category.                                                                                                                                                                                                               |

## Examples:

- The following command creates path categories based on clock domain analysis of the critical paths , and considers the edges of the clock:  
`tempus> analyze_paths_by_clock_domain -include_edges`
- The following command creates path categories based on clock domain analysis of the critical paths , and considers the analysis views:

```
tempus> analyze_paths_by_clock_domain -include_views
```

## Related Commands

- `load_timing_debug_report`
- `create_path_category`

# analyze\_paths\_by\_critical\_false\_path

```
analyze_paths_by_critical_false_path
[-help]
[-critical_false_paths {false_path_only | exclude_false_path}]
[-master string]
```

Identifies violating false paths, which are paths that appear as violations but cannot be exercised due to the structure of the netlist, and creates a category that contains them. With this command the user can, in advance, categorize the critical false paths together to analyze them.

The command can be applied wherever the user loads the machine readable timing report.

## Parameters

|                                    |                                                                                                                                                                                                                                                                                                                           |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                 | Outputs a brief description that includes type and default information for each <code>analyze_paths_by_critical_false_path</code> parameter.<br><br>For a detailed description of the command and all of its parameters, use the <code>man</code> command:<br><code>man analyze_paths_by_critical_false_path</code>       |
| <code>-critical_false_paths</code> | Specifies which paths to analyze. You can select one of the following options:<br><br><code>exclude_false_paths</code> : Excludes false paths when creating the category.<br><br><code>false_path_only</code> : Includes only false paths when creating the category.<br><br><i>Default:</i> <code>false_path_only</code> |
| <code>-master <i>string</i></code> | Specifies that the analysis should be performed only for the specified master category.                                                                                                                                                                                                                                   |

## Examples:

- The following command creates path categories based on the critical false paths:  
`tempus> analyze_paths_by_critical_false_path -critical_false_path\ false_path_only`
- The following category is created: `default_report_false_path_only`

## Related Commands

- `load_timing_debug_report`

# analyze\_paths\_by\_drv

```
analyze_paths_by_drv
[-help]
[[-load_cap_file capacitance_file] | [-generate_cap_file fileName]]
[[-load_tran_file transition_file] | [-generate_tran_file fileName]]
[-load_fanout_file fanout_file | [-generate_fanout_file fileName]]
[-master master_category_name]
```

Reads or generates a report containing max transition, max capacitance, and max fanout violations. Paths that are affected by the selected DRV type are grouped into a category. This command can be used if the user wants to debug the timing issues caused by DRV.

The command can be applied wherever the user loads the machine readable timing report.

## Parameters

|                                           |                                                                                                                                                                                                                                                                                             |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                                     | <p>Outputs a brief description that includes type and default information for each <code>analyze_paths_by_drv</code> parameter.</p> <p>For a detailed description of the command and all of its parameters, use the <code>man</code> command:<br/><code>man analyze_paths_by_drv</code></p> |
| -load_cap_file<br><i>capacitance_file</i> | Reads a file containing a list of paths containing max capacitance violations.                                                                                                                                                                                                              |
| -load_fanout_file<br><i>fanout_file</i>   | Reads a file containing a list of paths containing max fanout violations.                                                                                                                                                                                                                   |
| -load_tran_file<br><i>transition_file</i> | Reads a file containing a list of paths containing max transition violations.                                                                                                                                                                                                               |
| -generate_cap_file<br><i>fileName</i>     | Generates a file containing a list of paths containing max capacitance violations.                                                                                                                                                                                                          |
| -generate_fanout_file<br><i>fileName</i>  | Generates a file containing a list of paths containing max fanout violations.                                                                                                                                                                                                               |
| -generate_tran_file<br><i>fileName</i>    | Generates a file containing a list of paths containing max transition violations.                                                                                                                                                                                                           |
| -master<br><i>master_category_name</i>    | <p>Specifies that the analysis should be performed only for the specified master category.</p> <p><b>Note:</b> This option cannot be used standalone.</p>                                                                                                                                   |

## Examples:

The following command generates DRV report files, then creates path categories based on the critical paths which are affected by DRV:

```
tempus> analyze_paths_by_drv -generate_cap_file capViolation.rpt
-generate_tran_file tranViolation.rpt
-generate_fanout_file fanoutViolation.rpt
```

## Related Commands

- [load\\_timing\\_debug\\_report](#)
- [create\\_path\\_category](#)

# analyze\_paths\_by\_hier\_port

```
analyze_paths_by_hier_port
[-help]
[-gen_hier_file string]
[-load_hier_file string]
[-master string]
[-out_file string]
```

Helps you to automatically group paths based on each hierarchical port of a given instance. You can use this command to perform the timing results analysis of a hierarchical design. For example, while looking at the setup/hold time at the same time of each port.

The command can be applied wherever the user loads the machine readable timing report.

## Parameters

|                                            |                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                         | Outputs a brief description that includes type and default information for each <code>analyze_paths_by_hier_port</code> parameter.<br><br>For a detailed description of the command and all of its parameters, use the <code>man</code> command:<br><code>man analyze_paths_by_hier_port</code> |
| <code>-gen_hier_file <i>string</i></code>  | Generates hier ports fanin/fanout file when the database is available based on the given InstanceName_list. Then uses that file to create the categories.                                                                                                                                       |
| <code>-load_hier_file <i>string</i></code> | Uses a previously saved hier ports fanin/fanout file when the database is not available.                                                                                                                                                                                                        |
| <code>-master <i>string</i></code>         | Specifies that the timing paths should be categorized only for the specified master category.                                                                                                                                                                                                   |
| <code>-out_file <i>string</i></code>       | Specifies the generated hier ports fanin/fanout file name.                                                                                                                                                                                                                                      |

## Examples

The following command categorizes paths based on hierarchical port of the instance

DTMF\_INST/TDSP\_CORE\_INST:

```
tempus> analyze_paths_by_hier_port -gen_hier_file DTMF_INST/TDSP_CORE_INST
```



## Related Commands

- `load_timing_debug_report`

# analyze\_paths\_by\_hierarchy

```
analyze_paths_by_hierarchy
[-help]
[-master string]
{-fp_file string [-include_macros]} | {-use_current_floorplan [-all] [-types string] [-
port2port] [-port2macro] [-port2bbox] [-port2hinst] [-port2stdcell] [-port2instgrp] [-
macro2port] [-macro2macro] [-macro2bbox] [-macro2hinst] [-macro2stdcell] [-macro2instgrp
] [-bbox2port] [-bbox2macro] [-bbox2bbox] [-bbox2hinst] [-bbox2stdcell] [-bbox2instgrp]
[-hinst2port] [-hinst2macro] [-hinst2bbox] [-hinst2hinst] [-hinst2stdcell] [-
hinst2instgrp] [-stdcell2port] [-stdcell2macro] [-stdcell2bbox] [-stdcell2hinst] [-
stdcell2stdcell] [-stdcell2instgrp] [-instgrp2port] [-instgrp2macro] [-instgrp2bbox] [-
instgrp2hinst] [-instgrp2stdcell] [-instgrp2instgrp]} | {-partitions string | -
macros string}
```

Creates categories according to the hierarchical characteristics of the paths. User can use either a user specified floorplan file or current floorplan file of the design in memory and can create categories according to the hierarchy of the design. This way, the paths can be narrowed down to small group.

## Parameters

|                                 |                                                                                                                                                                                                                                                                                                   |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                           | <p>Outputs a brief description that includes type and default information for each <code>analyze_paths_by_hierarchy</code> parameter.</p> <p>For a detailed description of the command and all of its parameters, use the <code>man</code> command:</p> <pre>man analyze_paths_by_hierarchy</pre> |
| -fp_file<br>floorplan_file      | <p>Specifies the name of the floorplan file.</p> <p><i>Default:</i> Uses the current floorplan file.</p>                                                                                                                                                                                          |
| -include_macros                 | <p>Includes macros in addition to guides and fences for creating categories.</p> <p><i>Default:</i> Macros are not considered.</p>                                                                                                                                                                |
| -macros macro_list              | <p>Specifies the names of the macros to be used. Use this parameter with the <code>-partitions</code> parameter.</p>                                                                                                                                                                              |
| -master<br>master_category_name | <p>Specifies that the timing paths should be categorized only for the specified master category.</p>                                                                                                                                                                                              |
| -partitions<br>partition_list   | <p>Specifies the names of the partitions to consider for extracting hierarchical information.</p>                                                                                                                                                                                                 |
| -<br>use_current_floorplan      | <p>Uses the current floorplan.</p>                                                                                                                                                                                                                                                                |

|                  |                                                                                                         |
|------------------|---------------------------------------------------------------------------------------------------------|
| -all             | Creates categories of paths within all types<br><i>Default:</i> All types are categorized.              |
| -types type_list | Creates categories of paths within the type_list.<br>Valid types are: port, macro, bbox, hinst, stdcell |
| -port2port       | Creates categories of paths from port to port.                                                          |
| -port2macro      | Creates categories of paths from port to macro.                                                         |
| -port2bbox       | Creates categories of paths from port to black box.                                                     |
| -port2hinst      | Creates categories of paths from port to hierarchical instance.                                         |
| -port2stdcell    | Creates categories of paths from port to std cell.                                                      |
| -macro2port      | Creates categories of paths from macro to port.                                                         |
| -macro2macro     | Creates categories of paths from macro to macro.                                                        |
| -macro2bbox      | Creates categories of paths from macro to black box.                                                    |
| -macro2hinst     | Creates categories of paths from macro to hierarchical instance.                                        |
| -macro2stdcell   | Creates categories of paths from macro to standard cell.                                                |
| -bbox2port       | Creates categories of paths from black box to port.                                                     |
| -bbox2macro      | Creates categories of paths from black box to macro.                                                    |
| -bbox2bbox       | Creates categories of paths from black box to black box.                                                |
| -bbox2hinst      | Creates categories of paths from black box to hierarchical instance.                                    |
| -bbox2stdcell    | Creates categories of paths from black box to standard cell.                                            |
| -hinst2port      | Creates categories of paths from hierarchical instance to port.                                         |
| -hinst2macro     | Creates categories of paths from hierarchical instance to macro.                                        |
| -hinst2bbox      | Creates categories of paths from hierarchical instance to black box.                                    |
| -hinst2hinst     | Creates categories of paths from hierarchical instance to hierarchical.                                 |
| -hinst2stdcell   | Creates categories of paths from hierarchical instance to standard cell.                                |
| -stdcell2port    | Creates categories of paths from standard cell to port.                                                 |
| -stdcell2macro   | Creates categories of paths from standard cell to macro.                                                |

|                  |                                                                           |
|------------------|---------------------------------------------------------------------------|
| -stdcell2bbox    | Creates categories of paths from standard cell to black box.              |
| -stdcell2hinst   | Creates categories of paths from standard cell to hierarchical instance.  |
| -stdcell2stdcell | Creates categories of paths from standard cell to standard cell.          |
| -bbox2instgrp    | Creates categories of paths from black box to instance group.             |
| -hinst2instgrp   | Creates categories of paths from hierarchical instance to instance group. |
| -instgrp2bbox    | Creates categories of paths from instance group to black box.             |
| -instgrp2hinst   | Creates categories of paths from instance group to hierarchical instance. |
| -instgrp2instgrp | Creates categories of paths from instance group to instance group.        |
| -instgrp2macro   | Creates categories of paths from instance group to macro.                 |
| -instgrp2port    | Creates categories of paths from instance group to port.                  |
| -instgrp2stdcell | Creates categories of paths from instance group to standard cell.         |
| -macro2instgrp   | Creates categories of paths from macro to instance group.                 |
| -port2instgrp    | Creates categories of paths from port to instance group.                  |
| -stdcell2instgrp | Creates categories of paths from standard cell to instance group.         |

## Examples:

- By default, it categorizes all paths inter/intra the 5 hierarchical entities by following command :  
tempus> analyze\_paths\_by\_hierarchy -use\_current\_floorplan -all
- The following command categorizes all paths from "port" or to "port":  
tempus> analyze\_paths\_by\_hierarchy -use\_current\_floorplan -type port

## Related Commands

- [load\\_timing\\_debug\\_report](#)
- [create\\_path\\_category](#)

# analyze\_paths\_by\_view

```
analyze_paths_by_view
[-help]
[-prefix string]
[-master master_category_name]
```

Categorizes the timing paths based on views that you create during multi-mode multi-corner analysis. By arranging the timing information of different views quickly, the command helps in analyzing timing paths.

The command can be applied wherever the user loads the machine readable timing report.

## Parameters

|                                 |                                                                                                                                                                                                                                                                                       |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                           | Outputs a brief description that includes type and default information for each <code>analyze_paths_by_view</code> parameter.<br><br>For a detailed description of the command and all of its parameters, use the <code>man</code> command:<br><code>man analyze_paths_by_view</code> |
| -master<br>master_category_name | Specifies that the timing paths should be categorized only for the specified master category.                                                                                                                                                                                         |
| -prefix string                  | Adds a prefix to the category name.                                                                                                                                                                                                                                                   |

## Examples:

The following command categorizes paths based on views that you created during mmmc analysis :

```
tempus> analyze_paths_by_view
```

## Related Commands

- `load_timing_debug_report`
- `create_path_category`

## create\_path\_category

```
create_path_category
[-help]
[-capture_latency string]
[-check_type string]
[-clock string]
[-comment string]
[-critical_false_path string]
[-delay_of_any_inst string]
[-delay_of_any_inst_not_of_celltype string]
[-delay_of_any_inst_not_of_name string]
[-delay_of_any_inst_of_celltype string]
[-delay_of_any_inst_of_name string]
[-delay_of_any_net string]
[-delay_of_any_net_not_of_name string]
[-delay_of_any_net_of_name string]
[-delay_of_every_inst string]
[-delay_of_every_inst_not_of_celltype string]
[-delay_of_every_inst_not_of_name string]
[-delay_of_every_inst_of_celltype string]
[-delay_of_every_inst_of_name string]
[-delay_of_every_net string]
[-delay_of_every_net_not_of_name string]
[-delay_of_every_net_of_name string]
[-file string]
[-from_cell string]
[-from_clock string]
[-from_clock_edge string]
[-from_inst string]
[-from_inst_group string]
[-from_inst_pointer string]
[-from_pd string]
[-from_pin string]
[-from_port string]
[-from_port_pointer string]
[-incr_delay_of_any_net string]
[-incr_delay_of_every_net string]
[-launch_latency string]
[-master string]
[-name string]
[-not_from_cell string]
[-not_from_clock string]
[-not_from_clock_edge string]
[-not_from_inst string]
```

`[-not_from_pd string]`  
`[-not_from_pin string]`  
`[-not_from_port string]`  
`[-not_through_cell string]`  
`[-not_through_inst string]`  
`[-not_through_net string]`  
`[-not_through_pd string]`  
`[-not_through_pin string]`  
`[-not_through_port string]`  
`[-not_to_cell string]`  
`[-not_to_clock string]`  
`[-not_to_clock_edge string]`  
`[-not_to_inst string]`  
`[-not_to_pd string]`  
`[-not_to_pin string]`  
`[-not_to_port string]`  
`[-number_of_insts string]`  
`[-number_of_insts_not_of_celltype string]`  
`[-number_of_insts_not_of_name string]`  
`[-number_of_insts_of_celltype string]`  
`[-number_of_insts_of_name string]`  
`[-overwrite]`  
`[-sdc string]`  
`[-skew string]`  
`[-slack string]`  
`[-through_cell string]`  
`[-through_inst string]`  
`[-through_net string]`  
`[-through_pd string]`  
`[-through_pin string]`  
`[-through_port string]`  
`[-to_cell string]`  
`[-to_clock string]`  
`[-to_clock_edge string]`  
`[-to_inst string]`  
`[-to_inst_group string]`  
`[-to_inst_pointer string]`  
`[-to_pd string]`  
`[-to_pin string]`  
`[-to_port string]`  
`[-to_port_pointer string]`  
`[-total_delay_of_insts string]`  
`[-total_delay_of_insts_not_of_celltype string]`  
`[-total_delay_of_insts_not_of_name string]`  
`[-total_delay_of_insts_of_celltype string]`  
`[-total_delay_of_insts_of_name string]`

```
[-total_delay_of_level_of_chain_of_insttype string]
[-total_delay_of_nets string]
[-total_delay_of_nets_not_of_name string]
[-total_delay_of_nets_of_name string]
[-total_delay_of_path string]
[-total_incr_delay_of_path string]
[-uncertainty string] [-view string]
```

Creates path categories using conditions for grouping paths according to analyzed timing results. The command can be applied wherever the user loads the machine readable timing report.

For more information on timing debug, see the '[Appendix - Timing Debug GUI](#)' chapter in the *Tempus User Guide*.

## Parameters

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                                 | <p>Outputs a brief description that includes type and default information for each <code>create_path_category</code> parameter.</p> <p>For a detailed description of the command and all of its parameters, use the <code>man</code> command:</p> <pre>man create_path_category</pre>                                                                                                                                                                                                             |
| -clock {both   launch   capture}      | <p>Helps you search the Clock Tree instead of the data path. If this option is not specified, the user searches the data path by default. You can use one of the following values:</p> <ul style="list-style-type: none"> <li>• <b>Both</b>: searches on launch or capture clock paths.</li> <li>• <b>Launch</b>: searches only on launch clock paths</li> <li>• <b>Capture</b>: searches only on capture clock paths</li> </ul> <p><b>Note:</b> Supports simultaneous criteria on data path.</p> |
| -comment { <i>commentText</i> }       | <p>Specifies a comment that is associated with the category.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| -master <i>existing_category_name</i> | <p>Specifies that the new category should be created as a nested category of the specified master category. The master category should be an existing category.</p> <p><i>Default:</i> If you do not specify this parameter, the category is created as a new category at the top level.</p>                                                                                                                                                                                                      |



|                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-name <i>group_name</i></code>                                                                                                                                                                                                                                                                                                         | Specifies the name of the category to be created.                                                                                                                                                                                                    |
| <code>-overwrite</code>                                                                                                                                                                                                                                                                                                                      | Overwrites any predefined category with the same name in your category list.                                                                                                                                                                         |
| <code>-sdc {<i>file_name line_number</i>}</code>                                                                                                                                                                                                                                                                                             | Creates a category based on the SDC constraint as per the specified SDC file name and line number. Creating a path category based on an SDC constraint helps you to analyze the affect of an SDC constraint by grouping the affected paths together. |
| <code>-check_type {Setup Check   Hold Check<br/>  Clock Gating Setup Check<br/>  Library Clock Gating Setup Check<br/>  Clock Gating Hold Check<br/>  Library Clock Gating Hold Check<br/>  External Delay Assertion<br/>  Latch Borrowed Time Check<br/>  PulseWidth Check   ClockPeriod Check<br/>  Recovery Check   Removal Check}</code> | Indicates that the check type matches the specified option.                                                                                                                                                                                          |
| <code>-critical_false_path</code>                                                                                                                                                                                                                                                                                                            | Specifies the critical false path.                                                                                                                                                                                                                   |
| <code>-file <i>string</i></code>                                                                                                                                                                                                                                                                                                             | Specifies the file name.                                                                                                                                                                                                                             |
| <code>-from_inst   -from_inst_pointer   -<br/>from_pin   -from_cell   -from_clock   -<br/>from_port   -from_port_pointer   -from_pd<br/><i>string</i></code>                                                                                                                                                                                 | Indicates that the path must be from the specified instance, pin, cell, clock, power domain, or port pattern.                                                                                                                                        |
| <code>-from_inst_group <i>string</i></code>                                                                                                                                                                                                                                                                                                  | Specifies the source instance group from which timing paths needs to begin. When this parameter is specified, the software considers only those timing paths whose start points belong to the specified instance group.                              |
| <code>-from_clock_edge</code>                                                                                                                                                                                                                                                                                                                | Indicates that the path must be from the specified clock edge.                                                                                                                                                                                       |
| <code>-incr_delay_of_any_net <i>float</i></code>                                                                                                                                                                                                                                                                                             | Indicates that the incremental delay of at least a net that matches the specified <i>float</i> value.                                                                                                                                                |
| <code>-incr_delay_of_every_net <i>float</i></code>                                                                                                                                                                                                                                                                                           | Indicates that the incremental delay of every net that matches the specified <i>float</i> value.<br><br><i>Type:</i> String, optional                                                                                                                |

|                                                                                                                                                   |                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-not_from_inst   -not_from_pin<br/>  -not_from_cell   -not_from_clock<br/>  -not_from_port   -not_from_pd string</code>                     | Indicates that the path must not be from the specified instance, pin, cell, clock, or port pattern.                                             |
| <code>-through_inst   -through_pin<br/>  -through_cell   -through_net<br/>  -through_port   -through_pd string</code>                             | Indicates that the path must be through the specified instance, pin, cell, net, power domain, or port pattern.                                  |
| <code>-not_from_clock_edge</code>                                                                                                                 | Indicates that the path must not be from the specified clock edge.                                                                              |
| <code>-not_through_inst   -not_through_pin<br/>  -not_through_cell   -not_through_net<br/>  -not_through_port   -not_through_pd<br/>string</code> | Indicates that the path must not be through the specified instance, pin, power domain, cell, net, or port pattern.                              |
| <code>-not_to_clock_edge</code>                                                                                                                   | Indicates that the path must not be to the specified clock edge.                                                                                |
| <code>-to_inst   -to_inst_pointer   -to_pin   -<br/>to_cell   -to_clock   -to_port   -<br/>to_port_pointer   -to_pd string</code>                 | Indicates that the path must be the specified instance, pin, cell, clock, power domain, or port pattern.                                        |
| <code>-not_to_inst   -not_to_pin<br/>  -not_to_cell   -not_to_clock<br/>  -not_to_port   -not_to_pd string</code>                                 | Indicates that the path must not be to the specified instance, pin, cell, clock, power domain, or port pattern.                                 |
| <code>-slack float</code>                                                                                                                         | Indicates that the slack of the path must match the specified <i>float</i> value.                                                               |
| <code>-skew float</code>                                                                                                                          | Indicates that the skew of the path must match the specified <i>float</i> value.                                                                |
| <code>-launch_latency float</code>                                                                                                                | Indicates that the launch latency of the path must match the specified <i>float</i> value.                                                      |
| <code>-capture_latency float</code>                                                                                                               | Indicates that the capture latency of the path must match the specified <i>float</i> value.                                                     |
| <code>-uncertainty float</code>                                                                                                                   | Indicates that the uncertainty of the path must match the specified <i>float</i> value.                                                         |
| <code>-number_of_insts integer</code>                                                                                                             | Indicates that the number of instances in the data segment (with launch and capture clock paths) must match the specified <i>integer</i> value. |

|                                                                       |                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-number_of_insts_of_name<br/>string_plus_integer</code>         | Indicates that the number of instances matching the specified instance name match the specified <i>integer</i> value.                                                                                             |
| <code>-number_of_insts_not_of_name<br/>string_plus_integer</code>     | Indicates that the number of instances not matching the specified instance name match the specified <i>integer</i> value.                                                                                         |
| <code>-number_of_insts_of_celltype<br/>string_plus_integer</code>     | Indicates that the number of instances matching the specified celltype match the specified <i>integer</i> value.                                                                                                  |
| <code>-number_of_insts_not_of_celltype<br/>string_plus_integer</code> | Indicates that the number of instances not matching the specified celltype match the specified <i>integer</i> value.                                                                                              |
| <code>-to_clock_edge</code>                                           | Indicates that the path must be directed to the specified clock edge.                                                                                                                                             |
| <code>-to_inst_group string</code>                                    | Specifies the destination instance group where timing paths need to end. When this parameter is specified, the software includes only those timing paths whose end points belong to the specified instance group. |
| <code>-total_delay_of_insts float</code>                              | Indicates that the total delay of instances matches the specified <i>float</i> value.                                                                                                                             |
| <code>-<br/>total_delay_of_level_of_chain_of_insttype</code>          | <i>Type:</i> String, optional                                                                                                                                                                                     |
| <code>-total_delay_of_path float</code>                               | Indicates that the total delay of path matches the specified <i>float</i> value.                                                                                                                                  |
| <code>-total_delay_of_nets float</code>                               | Indicates that the total delay of nets matches the specified <i>float</i> value.                                                                                                                                  |
| <code>-total_delay_of_insts_of_name<br/>string_plus_float</code>      | Indicates that the total delay of instances matching the specified name matches the specified <i>float</i> value.                                                                                                 |
| <code>-total_delay_of_insts_not_of_name<br/>string_plus_float</code>  | Indicates that the total delay of instances not matching the specified name matches the specified <i>float</i> value.                                                                                             |
| <code>-total_delay_of_insts_of_celltype<br/>string_plus_float</code>  | Indicates that the total delay of instances matching the specified celltype matches the specified <i>float</i> value.                                                                                             |

|                                                                          |                                                                                                                                      |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <code>-total_delay_of_insts_not_of_celltype<br/>string_plus_float</code> | Indicates that the total delay of instances not matching the specified celltype matches the specified <i>float</i> value.            |
| <code>-total_delay_of_nets_of_name<br/>string_plus_float</code>          | Indicates that the total delay of nets matching the specified name matches the specified <i>float</i> value.                         |
| <code>-total_delay_of_nets_not_of_name<br/>string_plus_float</code>      | Indicates that the total delay of nets not matching the specified name matches the specified <i>float</i> value.                     |
| <code>-total_incr_delay_of_path float</code>                             | Indicates that the total incremental delay of path matches the specified <i>float</i> value.                                         |
| <code>-delay_of_any_inst float</code>                                    | Indicates that the delay of any instance in the design matches the specified <i>float</i> value.                                     |
| <code>-delay_of_any_net float</code>                                     | Indicates that the delay of any net in the design matches the specified <i>float</i> value.                                          |
| <code>-delay_of_every_inst float</code>                                  | Indicates that the delay of every instance in the design matches the specified <i>float</i> value.                                   |
| <code>-delay_of_every_net float</code>                                   | Indicates that the delay of every instance in the design matches the specified <i>float</i> value.                                   |
| <code>-delay_of_any_inst_of_name<br/>string_plus_float</code>            | Indicates that the delay of any instance in the design matching the specified name matches the specified <i>float</i> value.         |
| <code>-delay_of_any_inst_not_of_name<br/>string_plus_float</code>        | Indicates that the delay of any instance in the design not matching the specified name matches the specified <i>float</i> value.     |
| <code>-delay_of_any_inst_of_celltype<br/>string_plus_float</code>        | Indicates that the delay of any instance in the design matching the specified celltype matches the specified <i>float</i> value.     |
| <code>-delay_of_any_inst_not_of_celltype<br/>string_plus_float</code>    | Indicates that the delay of any instance in the design not matching the specified celltype matches the specified <i>float</i> value. |
| <code>-delay_of_any_net_of_name<br/>string_plus_float</code>             | Indicates that the delay of any net in the design matching the specified name matches the specified <i>float</i> value.              |

|                                                                         |                                                                                                                                        |
|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <code>-delay_of_any_net_not_of_name<br/>string_plus_float</code>        | Indicates that the delay of any net in the design not matching the specified name matches the specified <i>float</i> value.            |
| <code>-delay_of_every_inst_of_name<br/>string_plus_float</code>         | Indicates that the delay of every instance in the design matching the specified name matches the specified <i>float</i> value.         |
| <code>-delay_of_every_inst_not_of_name<br/>string_plus_float</code>     | Indicates that the delay of every instance in the design not matching the specified name matches the specified <i>float</i> value.     |
| <code>-delay_of_every_inst_of_celltype<br/>string_plus_float</code>     | Indicates that the delay of every instance in the design matching the specified celltype matches the specified <i>float</i> value.     |
| <code>-delay_of_every_inst_not_of_celltype<br/>string_plus_float</code> | Indicates that the delay of every instance in the design not matching the specified celltype matches the specified <i>float</i> value. |
| <code>-delay_of_every_net_of_name<br/>string_plus_float</code>          | Indicates that the delay of every net in the design matching the specified name matches the specified <i>float</i> value.              |
| <code>-delay_of_every_net_not_of_name<br/>string_plus_float</code>      | Indicates that the delay of every net in the design not matching the specified name matches the specified <i>float</i> value.          |
| <code>-view</code>                                                      | Specifies the view name.                                                                                                               |

## Examples

- The following command categorizes paths whose source clock is `vclk1`:  

```
create_path_category -name vclk1 -from_clock vclk1
```

The category `vclk1` is created.
- The following command categorizes paths whose skew is larger than 0.2:  

```
create_path_category -name worst_skew -skew {> 0.2}
```

The category `worst_skew` is created.

# delete\_path\_category

```
delete_path_category
[-help]
category_name
```

Deletes the selected category name.

The command can be applied when categories are created in Global Timing Debug.

For more information on timing debug categories, see the '[Appendix - Timing Debug GUI](#)' chapter in the *Tempus User Guide*.

## Parameter

|               |                                                                                                                                                                                                                                                                        |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help         | Outputs a brief description that includes type and default information for each delete_path_category parameter.<br><br>For a detailed description of the command and all of its parameters, use the <code>man</code> command:<br><code>man delete_path_category</code> |
| category_name | Specifies the name of the category to be deleted.                                                                                                                                                                                                                      |

## Examples:

The following command deletes the category `worst_paths`:

```
tempus> delete_path_category worst_paths
```

## Related Commands

- [load\\_timing\\_debug\\_report](#)

# dump\_histogram\_view

```
dump_histogram_view
[-help]
-name string
[-type string]
```

Saves (dumps) the image of the histogram shown in *Global Timing Debug* automatically during the process.

## Parameters

|                        |                                                                                                                                                                                                                                                                       |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                  | Outputs a brief description that includes type and default information for each <code>dump_histogram_view</code> parameter.<br><br>For a detailed description of the command and all of its parameters, use the man command:<br><code>man dump_histogram_view.</code> |
| -name<br><i>string</i> | Specifies the name of the image file to save the histogram.                                                                                                                                                                                                           |
| -type<br><i>string</i> | Specifies the type of file (image format) to be saved. Options are <code>bmp/jpg/gif/tif/csv</code> .<br><br><i>Default:</i> <code>bmp</code>                                                                                                                         |

## Examples

The following commands automatically generates the `gtd2.bmp` image file of the histogram:

```
tempus 1> start_gui
Loading (super_filter)
tempus 2> Parsing file top.mtarpt...
tempus 2> INFO: processing SDC file "../design/base.sdc" for SDC<->STA report cross-
probing....
INFO(../design/base.sdc, line 10,10) : Command "set_ideal_network" will not be registered for
SDC cross-probing, because it is unsupported, or contained an unsupported command in one of
its options.
INFO: processed 6 commands in 11 lines from file "../design/base.sdc", with 1 errors.
INFO: registered 5 SDC constraints for cross-probing.
INFO: The following commands were not registered for SDC cross-probing, for reasons
detailed previously in the logfile, or because they are not relevant to SDC:
cross-probing. set_ideal_network : 1
```

```
tempus 3> dump_histogram_view -name gtd2 1
```

## Related Commands

[load\\_timing\\_debug\\_report](#)



# highlight\_timing\_report

```
highlight_timing_report
[-help]
[-append]
[-color_index integer]
[-file filename]
[-noarrow]
[-path path_number | -all]
```

Highlight the selected paths in the user interface.

For more information on timing debug categories, see the '[Appendix - Timing Debug GUI](#)' chapter in the *Tempus User Guide*.

## Parameters

|                      |                                                                                                                                                                                                                                                                                           |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                | Outputs a brief description that includes type and default information for each <code>highlight_timing_report</code> parameter.<br><br>For a detailed description of the command and all of its parameters, use the <code>man</code> command:<br><code>man highlight_timing_report</code> |
| -all                 | Highlights all the path in the Global Timing Debug through the user interface.                                                                                                                                                                                                            |
| -append              | Appends a path to the current highlight.                                                                                                                                                                                                                                                  |
| -color_index integer | Specifies the path's color index on a scale of 1 to 8.                                                                                                                                                                                                                                    |
| -file filename       | Name of the file. The option is of type string and optional.                                                                                                                                                                                                                              |
| -noarrow             | Removes the highlighted arrow. The option is of type boolean and optional.                                                                                                                                                                                                                |
| -path path_number    | Specify the number of the path. The option is of type string and required.                                                                                                                                                                                                                |

## Examples

The following commands generate the worst 1000 paths to a machine readable report, and then highlight the top five paths in Global Timing Debug:

```
report_timing -machine_readable -max_points 1000 > foo.mtarpt
load_timing_debug_report foo.mtarpt
highlight_timing_report -file foo.mtarpt -path "1 2 3 4 5" -noarrow
```

## Related Commands

- `load_timing_debug_report`

# load\_path\_categories

```
load_path_categories
[-help]
-filename filename
```

Loads the path category file in the Timing Debug form. The path category file contains category definitions that you saved in an earlier timing debug analysis. To save path categories, use the [save\\_path\\_categories](#) command.

For more information on timing debug categories, see the '[Appendix - Timing Debug GUI](#)' chapter in the *Tempus User Guide*.

## Parameter

|                       |                                                                                                                                                                                                                                                                                     |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                 | Outputs a brief description that includes type and default information for each <code>load_path_categories</code> parameter.<br><br>For a detailed description of the command and all of its parameters, use the <code>man</code> command:<br><code>man load_path_categories</code> |
| -filename<br>filename | Specifies the name of the category file.                                                                                                                                                                                                                                            |

## Examples

The following command loads the path category file `category.cat`:

```
tempus> load_path_categories -filename category.cat
```

## Related Commands

- [load\\_timing\\_debug\\_report](#)

# load\_timing\_debug\_report

```
load_timing_debug_report
[-help]
{{<file> [-name <string>]}}{{ -proto [-additional_slack_past_wns <float>] [-num_path
<integer>]}}
```

Loads the violation report in Tempus for debugging timing results.

 The violation report that you specify using the `load_timing_debug_report` command has a special output binary format, which you can generate through the `binary` value of the `report_timing -output_format` command parameter.

**Note:** The violation report that you specify in this command has a special machine readable format. Use the `report_timing -machine_readable` command to generate the machine readable violation report.

For more information on timing debug categories, see the '[Appendix - Timing Debug GUI](#)' chapter in the *Tempus User Guide*.

## Parameter

|                                         |                                                                                                                                                                                                                                                                                             |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                      | Outputs a brief description that includes type and default information for each <code>load_timing_debug_report</code> parameter.<br><br>For a detailed description of the command and all of its parameters, use the <code>man</code> command:<br><code>man load_timing_debug_report</code> |
| <code>-additional_slack_past_wns</code> | Specifies the the additional specified slack.                                                                                                                                                                                                                                               |
| <code>filename</code>                   | Specifies the name of the timing debug file.                                                                                                                                                                                                                                                |
| <code>-name</code>                      | Specifies the report name to load the machine readable timing report files incrementally.                                                                                                                                                                                                   |
| <code>-num_path</code>                  | Specifies the number of paths to be reported in path violations.                                                                                                                                                                                                                            |
| <code>-proto</code>                     | Specifies the FlexModel prototyping.                                                                                                                                                                                                                                                        |

## Examples:

- The following command loads the machine readable timing report `top.btarpt`:  

```
tempus> load_timing_debug_report top.btarpt
```
- The following commands load two machine readable timing reports `top1.btarpt` and `top2.btarpt` together:  

```
tempus> load_timing_debug_report -name default_report top1.btarpt
tempus> load_timing_debug_report -name default_report top2.btarpt
```

GTD will first load the report `top1.btarpt`, then append `top2.btarpt` incrementally to the current report.

**Note:** The `name` can be specified by user with any string, it is not the file name.

## Related Commands

- `report_timing`

## save\_path\_categories

```
save_path_categories
[-help]
-filename filename
```

Saves the selected categories in a path category file. You can use the category file to load saved category definitions in a new timing debug session. To load the saved category definitions, use the [load\\_path\\_categories](#) command.

For more information on timing debug categories, see the '[Appendix - Timing Debug GUI](#)' chapter in the *Tempus User Guide*.

### Parameter

|                                                 |                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                              | <p>Outputs a brief description that includes type and default information for each <code>save_path_categories</code> parameter.</p> <p>For a detailed description of the command and all of its parameters, use the <code>man</code> command:</p> <pre>man save_path_categories</pre> |
| <code>-filename</code><br><code>filename</code> | <p>Specifies the name of the category file.</p>                                                                                                                                                                                                                                       |

### Examples:

The following command saves the path category file `category.cat`:

```
tempus> save_path_categories -filename category.cat
```

### Related Commands

- [load\\_timing\\_debug\\_report](#)

## write\_category\_summary

```
write_category_summary [-help] -report <string> [[-ascend | -descend] [-byTNS | -byWNS]]
[-csv | {-no_frame_fix_width [-no_split]}]
-csv
```

Writes a text file containing the following information:

- Category name
- Total number of paths
- Number of passing paths
- Number of failing paths
- Worst negative slack
- Total negative slack

## Parameters

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>               | <p>Outputs a brief description that includes type and default information for each <code>write_category_summary</code> parameter.</p> <p>For a detailed description of the command and all of its parameters, use the <code>man</code> command:<br/> <code>man write_category_summary</code></p>                                                                                                                                                                                                            |
| <code>-ascend</code>             | Specifies that the report in ascending order, should specify option <code>-byTNS</code> or <code>-byWNS</code> .                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>-byTNS</code>              | Specifies that the TNS value specified as the sorting key, should specify option <code>-ascend</code> or <code>-descend</code> .                                                                                                                                                                                                                                                                                                                                                                            |
| <code>-byWNS</code>              | Specifies that the WNS value specified as the sorting key, should specify option <code>-ascend</code> or <code>-descend</code> .                                                                                                                                                                                                                                                                                                                                                                            |
| <code>-csv</code>                | Reports entries in csv format.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>-descend</code>            | Specifies that the report in descending order should specify option <code>-byTNS</code> or <code>-byWNS</code> .                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>-no_frame_fix_width</code> | <p>Specifies that the table columns have a fixed minimum width, but no maximum width (the data does not wrap). Column data that fits within the column's minimum width will left-justify and pad its location to fill the specified column space.</p> <p>Column data that is longer than the column's minimum width will take as much space in the table as needed. If a column is overwritten by the data from the preceding column, its data will begin on a new line at the correct column location.</p> |
| <code>-no_split</code>           | <p>Specifies that each column follows the preceding column without splitting to a new line, if the data in one column overruns the next column. The wrapping behavior for data is controlled by the size of the display window.</p> <p>This format is useful for machine parsing the report.</p> <p><b>Note:</b> You must specify <code>-no_frame_fix_width</code> in order to use this parameter.</p>                                                                                                      |
| <code>-report fileName</code>    | <p>Specifies the name of the category report file.</p> <p><i>Default:</i> <code>category.rpt</code></p>                                                                                                                                                                                                                                                                                                                                                                                                     |

## Examples

The following command writes a category summary file `category.rpt`:



```
tempus> write_category_summary -report category.rpt
```

## Related Commands

- `load_timing_debug_report`

## write\_text\_timing\_report

```
write_text_timing_report
[-help]
[-mtarpt file.mtarpt |
-category category_name]
[-format {{column1 width1} {column2 width2} ...}]
[-path_specification path_specification]
[-summary] |
-report file.tarpt
```

Writes a text file containing information about paths that have been loaded from a `.mtarpt` file from Timing Debug, either with the `load_timing_debug_report` command or in the *Display/Generate Timing Report* form in the GUI.

Paths are selected as follows:

- If only one `.mtarpt` file is loaded and no other parameters are specified, then the tool writes all the paths contained in the loaded `.mtarpt` file.
- If there are several `.mtarpt` files loaded, you must specify the `.mtarpt` file from which the tool can select the paths, except if a category is specified.

You can use this report for performing timing analysis.

If only one machine-readable report file is loaded, then the tool writes all paths from that file into the output report. If more than one file is loaded, then the tool issues an error message.

**Note:** The generated report cannot be read back into the timing debug environment.

## Parameters

|                                                                                                                        |                                                                                                                                                                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                                                                                                     | <p>Outputs a brief description that includes type and default information for each <code>write_text_timing_report</code> parameter.</p> <p>For a detailed description of the command and all of its parameters, use the <code>man</code> command:</p> <pre>man write_text_timing_report</pre> |
| <code>-category</code><br><code>category_name</code>                                                                   | Specifies the category of paths to write to the text file. You cannot specify <code>-mtarpt</code> if you specify this parameter.                                                                                                                                                             |
| <code>-format</code> <code>{{column1</code><br><code>width1}</code> <code>{column2</code><br><code>width2}</code> ...} | Formats the report according to the specified column name and its width. The column-width pair specifies the column to display in the report and the width of that column. Multiple column names can be specified.                                                                            |
| <code>-mtarpt</code><br><code>file.mtarpt</code>                                                                       | Specifies the name of the <code>.mtarpt</code> file to read and convert to a text file. You cannot specify this parameter if you specify <code>-category</code> .                                                                                                                             |
| <code>-</code><br><code>path_specification</code><br><code>path_specification</code>                                   | Writes a single number or path range separated by blank space. A path range is two numbers separated by a minus "-" sign. You can use this parameter only if a single file is specified or loaded.                                                                                            |
| <code>-report</code> <code>file.tarpt</code>                                                                           | Specifies the name of the text output file.                                                                                                                                                                                                                                                   |
| <code>-summary</code>                                                                                                  | Writes the file header only. Use this parameter if you do not need a detailed table.                                                                                                                                                                                                          |

## Examples

- The following command loads machine-readable report `top.mtarpt`:  

```
tempus> load_timing_debug_report top.mtarpt
```
- The following command reads the `top.mtarpt` report and reports paths 1 through 50 in the `top_tarpt.txt` file.  

```
tempus> write_text_timing_report -mtarpt top.mtarpt -path_specification {1-50} -report top_tarpt.txt
```

## Related Commands

- `load_timing_debug_report`

# load\_failure\_report

```
load_failure_report
[-help]
-file filename
```

Loads the failure report generated during `eco_opt_design`. This failure report contains various reasons that limited timing closure during automated signoff ECO. This command reads the Failure Catalog File (FCF) in the format used by the Failure Catalog Browser (FCB).

## Parameters

|                    |                                                                                                                                                                                                                                                                                    |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help              | Outputs a brief description that includes the type and default information for each <code>load_failure_report</code> parameter.<br><br>For a detailed description of the command and all its parameters, use the <code>man</code> command:<br><code>man load_failure_report</code> |
| -<br>file filename | Specifies the name of the file describing the failure category. This is a mandatory parameter.                                                                                                                                                                                     |

## Related Commands

```
eco_opt_design
```

---

# Manual ECO Commands

---

- [add\\_inst](#)
- [add\\_net](#)
- [add\\_repeater](#)
- [attach\\_module\\_port](#)
- [attach\\_net](#)
- [attach\\_term](#)
- [change\\_cell](#)
- [change\\_inst\\_name](#)
- [delete\\_inst](#)
- [delete\\_net](#)
- [delete\\_repeater](#)
- [detach\\_module\\_port](#)
- [detach\\_net](#)
- [detach\\_term](#)
- [edit\\_delete](#)
- [load\\_eco](#)
- [set\\_eco\\_mode](#)
- [write\\_eco](#)

# add\_inst

```
add_inst
-help
<list_of_instances>
<lib_cell_name>
```

Adds an instance and places it in the design. The syntax is order-dependent.

After this command, the software will perform a full timing update when you run the next command that reports timing, for example, [report\\_timing](#).

**Note:** The software supports this command for flattened designs only.

## Parameters

|                                |                                                                                                                                                                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>             | Outputs a brief description that includes type and default information for each <code>add_inst</code> parameter.<br><br>For a detailed description of the command and all of its parameters, use the man command: <code>man add_inst</code> |
| <code>lib_cell_name</code>     | Specifies the name of a cell or the names of a collection of cells used as a reference for the new instances added. You will need to specify the library when you specify the cell.                                                         |
| <code>list_of_instances</code> | Specifies a list of instances to create.                                                                                                                                                                                                    |

## Examples

The following example creates two instances, `inst1` and `inst2`, of library cell `LIB/BUF2`:

```
tempus> add_inst "inst1 inst2" [get_lib_cells LIB/BUF2]
```

## Related Commands

- `add_net`
- `attach_net`
- `attach_term`
- `change_inst_name`
- `delete_inst`
- `delete_net`
- `detach_net`
- `detach_term`

# add\_net

```
add_net
-help
<list_of_nets>
-module_based <string>
```

Adds a net to the design.

After this command, the software will perform a full timing update when you run the next command that reports timing, for example, [report\\_timing](#).

**Note:** The tool supports this command for flattened designs only.

## Parameters

|                           |                                                                                                                                                                                                                                           |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                     | Outputs a brief description that includes type and default information for each <code>add_net</code> parameter.<br><br>For a detailed description of the command and all of its parameters, use the man command: <code>man add_net</code> |
| <list_of_nets>            | Specifies the list of nets to add.                                                                                                                                                                                                        |
| -module_based<br><string> | Adds the new net to the specified Verilog module.<br><br>If the module name contains special characters, you need to start the module name with a backslash '\' and end it with a space.                                                  |

## Examples

- The following command adds nets `n1`, `n2`, and `n3`:  

```
tempus> add_net "n1 n2 n3"
```
- The following command adds net with base name `ant_net[1]` to module `H[1]`:  

```
tempus> add_net {\ant_net[1] } -module_based {\H[1] }
```

## Related Commands

- [add\\_inst](#)
- [attach\\_net](#)
- [attach\\_term](#)
- [change\\_inst\\_name](#)



- `delete_inst`
- `delete_net`
- `detach_net`
- `detach_term`

## add\_repeater

```
add_repeater
[-help]
[-name <string>]
[-new_net_name <string>]
[-relativeDistToSink <float>]
[-spread_count <integer>]
[-spread_dist <float>]
[-spread_prefix <string>]
{-net <string> | -term <string>}
[-loc {{x1 y1} {x2 y2}...} [-buforient <string>]]
[-sort_by {slack trans area} [-max_cells <integer>]]
[-cell <string> [-evaluate_only]] [-evaluate_only | -evaluate_all] [[[-evaluate_all] [-
evaluate_only]] [[[-view <string>] [-early] [-late] [-fall] [-rise] [-output_as_tcl] [-
sort_by {slack trans area}]]]] [-early | -late]
[-first_spread_dist <float>]
```

Adds either a single buffer or two inverters to a net. In the Multi-mode Multi-corner (MMMC) mode, this command adds a repeater and updates timing in all views. Optionally, you can specify the location of the buffer or inverters relative to the sink pin.

**Note:** You do not need to provide the physical information (LEF/DEF) to perform ECO using this command.

### Parameters

|                   |                                                                                                                                                                                                                                                                                                                                                          |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help             | <p>Outputs a brief description that includes type and default information for each <code>add_repeater</code> parameter.</p> <p>For a detailed description of the command and all of its parameters, use the man command: <code>man add_repeater</code></p>                                                                                               |
| -buforient string | <p>Specifies the legalized orientation for the repeater instance.</p> <p><b>Note:</b> You can specify this parameter with the <code>-loc</code> parameter only.</p>                                                                                                                                                                                      |
| -cell string      | <p>Specifies the library cell to be used as a repeater. If the cell is an inverter, two inverters in series will be added.</p>                                                                                                                                                                                                                           |
| -early            | <p>Specifies that the timing will be evaluated for early analysis. In such a case, only the active hold analysis views will be considered. If this parameter is not specified, timing will be evaluated for late analysis.</p> <p><b>Note:</b> This parameter must be specified with <code>-evaluate_only</code> or <code>-evaluate_all</code> only.</p> |

|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-evaluate_all</code>                      | Specifies to only evaluate timing for all buffer/inverter/delay cells with matching footprint (or all footprints) without any changes in the netlist.                                                                                                                                                                                                                                                 |
| <code>-evaluate_only</code>                     | Specifies to only evaluate timing for the given buffer/inverter/delay cells without any changes in the netlist.                                                                                                                                                                                                                                                                                       |
| <code>-fall</code>                              | <p>Specifies that the timing will be evaluated for the fall transition. If this parameter is not specified, timing will be evaluated for both rise and fall transitions, and the worst result is reported.</p> <p><b>Note:</b> This parameter must be specified with <code>-evaluate_only</code> or <code>-evaluate_all</code> only.</p>                                                              |
| <code>-first_spread_dist</code><br><i>float</i> | <p>Specifies the distance from the driver after which the first repeater is to be added.</p> <p><b>Note:</b> This parameter must be specified with <code>-spread_dist</code>.</p>                                                                                                                                                                                                                     |
| <code>-late</code>                              | <p>Specifies that timing will be evaluated for late analysis. In such a case, only the active setup analysis views will be considered. If this parameter is not specified, timing will still be evaluated for late analysis.</p> <p><b>Note:</b> This parameter must be specified with <code>-evaluate_only</code> or <code>-evaluate_all</code> only.</p>                                            |
| <code>-loc {x y   x1 y1<br/>x2 y2}</code>       | <p>Specifies the x and y coordinates for the location of the buffer or inverter pair. If you do not specify this parameter, the tool determines the location.</p> <p>If the cell is a buffer and a location is specified, the newly created buffer is placed at that location. If the cell is an inverter and only one location is specified, both inverters will be placed at the same location.</p> |
| <code>-max_cells</code> <i>integer</i>          | Specifies the maximum number of improving and degrading cells (each) to be reported for evaluation.                                                                                                                                                                                                                                                                                                   |
| <code>-name</code> <i>inst_name</i>             | Specifies the base names of the new instances to be added.                                                                                                                                                                                                                                                                                                                                            |
| <code>-net</code> <i>net_name</i>               | Specifies the name of the net on which the buffer will be added. All sinks of the net are buffered when this parameter is specified.                                                                                                                                                                                                                                                                  |
| <code>-new_net_name</code><br><i>netname</i>    | Specifies the base name(s) for the new net(s) to be added.                                                                                                                                                                                                                                                                                                                                            |
| <code>-output_as_tcl</code>                     | Displays the output as a TCL list rather than as text.                                                                                                                                                                                                                                                                                                                                                |

|                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-relativeDistToSink value</code>   | Specifies the location of the buffer instance based on its (relative) distance from the sink pin. The value is a number between 0 and 1 (both inclusive). A low value (such as 0.1) places the buffer near the sink. A high value (such as 0.9) places the buffer near the driver. You can use the value 0 to do sink term buffering and the value 1 to do driver term buffering. The fraction is based on the length of the wire. Fractional values can only be specified for two-pin nets. |
| <code>-rise</code>                       | Specifies that the timing will be evaluated for the rise transition. If this parameter is not specified, timing will be evaluated for both rise and fall transitions, and the worst result is reported.<br><br><b>Note:</b> This parameter must be specified with <code>-evaluate_only</code> or <code>-evaluate_all</code> only.                                                                                                                                                            |
| <code>-sort_by {slack trans area}</code> | Specifies the sorting order for evaluation.                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>-spread_count integer</code>       | Specifies the number of repeaters to be added for buffering.<br><br><i>Minimum:</i> 1<br><i>Maximum:</i> INT_MAX                                                                                                                                                                                                                                                                                                                                                                             |
| <code>-spread_dist float</code>          | Specifies the distance (starting from the driver) after which repeaters are to be added.<br><br><i>Minimum:</i> 2.2250738585072e-308<br><i>Maximum:</i> 1073741.823                                                                                                                                                                                                                                                                                                                          |
| <code>-spread_prefix string</code>       | Specifies the prefix used for new instances and nets added during spread buffering.                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>-term pin_or_ports</code>          | Specifies the list of pins (with one common net) that the new buffer or inverters would drive.                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>-view string</code>                | Specifies the active analysis views for which timing must be evaluated. If this parameter is not specified, timing will be evaluated for all active analysis views and results reported for the worst view.<br><br><b>Note:</b> This parameter must be specified with <code>-evaluate_only</code> or <code>-evaluate_all</code> only.                                                                                                                                                        |

## Examples

- The following command inserts a repeater with cell `BUF4` on the net connected to terminal `DTMF_INST/RESULT_CONV_INST/RE_PHC369_r1336_0/Y`:  

```
tempus> add_repeater -term DTMF_INST/RESULT_CONV_INST/RE_PHC369_r1336_0/Y -cell BUF4
```

- The following command inserts a repeater with cell `BUFX4` and attaches it to (driver)

**terminal** DTMF\_INST/RESULT\_CONV\_INST/RE\_PHC369\_r1336\_0\_/Y:

```
tempus> add_repeater -term DTMF_INST/RESULT_CONV_INST/RE_PHC369_r1336_0_/Y -cell BUFX4 -
relativeDistToSink 1
```

- The following command inserts a repeater with cell `BUFX4` and attaches it to (sink)

**terminal** DTMF\_INST/RESULT\_CONV\_INST/RE\_PHC369\_r1336\_0\_/A:

```
tempus> add_repeater -term DTMF_INST/RESULT_CONV_INST/RE_PHC369_r1336_0_/A -cell BUFX4 -
relativeDistToSink 0
```

- The following command inserts a repeater with cell `BUFX4` on net `n123`, places it at the given location, and names the new repeater instance `ECO_inst`:

```
tempus> add_repeater -net n123 -cell BUFX4 -loc {23.45 98.12} -name {ECO_inst}
```

- The following command inserts a repeater with cell `BUFX4` at a relative distance of 0.3 from the sink terminal of the two-pin net `n123`:

```
tempus> add_repeater -net n123 -cell BUFX4 -relativeDistToSink 0.3
```

- The following command reports the evaluation results of buffering pin, `EXECUTE_INST/ar0_reg_9/SE`, by adding each of all the buffer/inverter/delay cells in the library:

```
tempus> add_repeater -term EXECUTE_INST/ar0_reg_9/SE -evaluate_all
```

```

```

| Library-cell                                           | Area    | Timing-slack | Worst-view   |
|--------------------------------------------------------|---------|--------------|--------------|
| Transition    Trans-slack    Worst-Trans-Sink          |         |              |              |
| -----                                                  |         |              |              |
| Reference                                              | 0.0000  | -0.1181      | io+typ-rcMax |
| rise            2.9024       EXECUTE_INST/ar1_reg_9/SE |         |              |              |
| INVXL(inv_pair)                                        | 6.6528  | -0.2273      | io+typ-rcMax |
| rise            2.8825       EXECUTE_INST/ar0_reg_9/SE |         |              |              |
| INVX8(inv_pair)                                        | 19.9584 | -0.2505      | io+typ-rcMax |
| rise            2.7712       EXECUTE_INST/ar1_reg_9/SE |         |              |              |
| INVX4(inv_pair)                                        | 13.3056 | -0.2000      | io+typ-rcMax |
| rise            2.8560       EXECUTE_INST/ar1_reg_9/SE |         |              |              |
| INVX3(inv_pair)                                        | 13.3056 | -0.2093      | io+typ-rcMax |
| rise            2.8692       EXECUTE_INST/ar1_reg_9/SE |         |              |              |
| INVX2(inv_pair)                                        | 9.9792  | -0.2056      | io+typ-rcMax |
| rise            2.8842       EXECUTE_INST/ar1_reg_9/SE |         |              |              |
| INVX20(inv_pair)                                       | 63.2016 | -0.3922      | io+typ-rcMax |

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|                     |      |        |                           |              |
|---------------------|------|--------|---------------------------|--------------|
|                     | rise | 2.8415 | EXECUTE_INST/ar1_reg_9/SE |              |
| INVX1(inv_pair)     |      |        | 6.6528      -0.2119       | io+typ-rcMax |
|                     | rise | 2.9038 | EXECUTE_INST/ar0_reg_9/SE |              |
| INVX16(inv_pair)    |      |        | 56.5488      -0.3908      | io+typ-rcMax |
|                     | rise | 2.8518 | EXECUTE_INST/ar1_reg_9/SE |              |
| INVX12(inv_pair)    |      |        | 43.2432      -0.3977      | io+typ-rcMax |
|                     | rise | 2.8632 | EXECUTE_INST/ar1_reg_9/SE |              |
| DLY4X1(buf)         |      |        | 23.2848      -0.8877      | io+typ-rcMax |
|                     | rise | 2.8231 | EXECUTE_INST/ar0_reg_9/SE |              |
| DLY3X1(buf)         |      |        | 23.2848      -0.6581      | io+typ-rcMax |
|                     | rise | 2.8397 | EXECUTE_INST/ar0_reg_9/SE |              |
| DLY2X1(buf)         |      |        | 19.9584      -0.4660      | io+typ-rcMax |
|                     | rise | 2.8523 | EXECUTE_INST/ar0_reg_9/SE |              |
| DLY1X1(buf)         |      |        | 19.9584      -0.3181      | io+typ-rcMax |
|                     | rise | 2.8632 | EXECUTE_INST/ar0_reg_9/SE |              |
| CLKINVL(inv_pair)   |      |        | 9.9792      -0.2642       | io+typ-rcMax |
|                     | rise | 2.8451 | EXECUTE_INST/ar0_reg_9/SE |              |
| CLKINVL8(inv_pair)  |      |        | 19.9584      -0.2647      | io+typ-rcMax |
|                     | rise | 2.7866 | EXECUTE_INST/ar1_reg_9/SE |              |
| CLKINVL4(inv_pair)  |      |        | 13.3056      -0.2062      | io+typ-rcMax |
|                     | rise | 2.8653 | EXECUTE_INST/ar1_reg_9/SE |              |
| CLKINVL3(inv_pair)  |      |        | 9.9792      -0.2062       | io+typ-rcMax |
|                     | rise | 2.8772 | EXECUTE_INST/ar1_reg_9/SE |              |
| CLKINVL2(inv_pair)  |      |        | 9.9792      -0.2179       | io+typ-rcMax |
|                     | rise | 2.8893 | EXECUTE_INST/ar1_reg_9/SE |              |
| CLKINVL20(inv_pair) |      |        | 96.4656      -0.4696      | io+typ-rcMax |
|                     | rise | 2.8205 | EXECUTE_INST/ar1_reg_9/SE |              |
| CLKINVL1(inv_pair)  |      |        | 9.9792      -0.2460       | io+typ-rcMax |
|                     | rise | 2.8718 | EXECUTE_INST/ar0_reg_9/SE |              |
| CLKINVL16(inv_pair) |      |        | 83.1600      -0.4635      | io+typ-rcMax |
|                     | rise | 2.8403 | EXECUTE_INST/ar1_reg_9/SE |              |
| CLKINVL12(inv_pair) |      |        | 63.2016      -0.4383      | io+typ-rcMax |
|                     | rise | 2.8601 | EXECUTE_INST/ar1_reg_9/SE |              |
| CLKBUFXL(buf)       |      |        | 13.3056      -0.2006      | io+typ-rcMax |
|                     | rise | 2.9064 | EXECUTE_INST/ar1_reg_9/SE |              |
| CLKBUFX8(buf)       |      |        | 23.2848      -0.2301      | io+typ-rcMax |
|                     | rise | 2.8591 | EXECUTE_INST/ar1_reg_9/SE |              |

**Tempus Text Command Reference**  
Manual ECO Commands

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|                |        |                           |         |         |              |
|----------------|--------|---------------------------|---------|---------|--------------|
| CLKBUF4 (buf)  |        |                           | 16.6320 | -0.2111 | io+typ-rcMax |
| rise           | 2.8965 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| CLKBUF3 (buf)  |        |                           | 13.3056 | -0.2111 | io+typ-rcMax |
| rise           | 2.9005 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| CLKBUF2 (buf)  |        |                           | 13.3056 | -0.2058 | io+typ-rcMax |
| rise           | 2.9019 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| CLKBUF20 (buf) |        |                           | 79.8336 | -0.2979 | io+typ-rcMax |
| rise           | 2.7343 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| CLKBUF1 (buf)  |        |                           | 13.3056 | -0.2129 | io+typ-rcMax |
| rise           | 2.9002 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| CLKBUF16 (buf) |        |                           | 63.2016 | -0.2758 | io+typ-rcMax |
| rise           | 2.7814 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| CLKBUF12 (buf) |        |                           | 53.2224 | -0.2592 | io+typ-rcMax |
| rise           | 2.8028 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| BUF4 (buf)     |        |                           | 13.3056 | -0.2163 | io+typ-rcMax |
| rise           | 2.8830 | EXECUTE_INST/ar0_reg_9/SE |         |         |              |
| BUF8 (buf)     |        |                           | 29.9376 | -0.2183 | io+typ-rcMax |
| rise           | 2.8450 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| BUF4 (buf)     |        |                           | 16.6320 | -0.1963 | io+typ-rcMax |
| rise           | 2.8891 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| BUF3 (buf)     |        |                           | 13.3056 | -0.1988 | io+typ-rcMax |
| rise           | 2.8953 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| BUF2 (buf)     |        |                           | 13.3056 | -0.2027 | io+typ-rcMax |
| rise           | 2.9008 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| BUF20 (buf)    |        |                           | 53.2224 | -0.2577 | io+typ-rcMax |
| rise           | 2.7673 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| BUF1 (buf)     |        |                           | 13.3056 | -0.2096 | io+typ-rcMax |
| rise           | 2.9043 | EXECUTE_INST/ar0_reg_9/SE |         |         |              |
| BUF16 (buf)    |        |                           | 43.2432 | -0.2449 | io+typ-rcMax |
| rise           | 2.7935 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| BUF12 (buf)    |        |                           | 33.2640 | -0.2318 | io+typ-rcMax |
| rise           | 2.8191 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |

- The following command reports evaluation results (sorted in increasing order of area) of buffering pin, EXECUTE\_INST/ar0\_reg\_9/SE by adding each of all the buffer/inverter/delay cells in the library:

```
tempus> add_repeater -term EXECUTE_INST/ar0_reg_9/SE -evaluate_all -sort_by area
```

**Tempus Text Command Reference**  
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| Library-cell                                  | Area    | Timing-slack              | Worst-view   |
|-----------------------------------------------|---------|---------------------------|--------------|
| Transition    Trans-slack    Worst-Trans-Sink |         |                           |              |
| Reference                                     | 0.0000  | -0.1181                   | io+typ-rcMax |
| rise                                          | 2.9024  | EXECUTE_INST/ar1_reg_9/SE |              |
| INVXL(inv_pair)                               | 6.6528  | -0.2273                   | io+typ-rcMax |
| rise                                          | 2.8825  | EXECUTE_INST/ar0_reg_9/SE |              |
| INVX1(inv_pair)                               | 6.6528  | -0.2119                   | io+typ-rcMax |
| rise                                          | 2.9038  | EXECUTE_INST/ar0_reg_9/SE |              |
| INVX2(inv_pair)                               | 9.9792  | -0.2056                   | io+typ-rcMax |
| rise                                          | 2.8842  | EXECUTE_INST/ar1_reg_9/SE |              |
| CLKINVXL(inv_pair)                            | 9.9792  | -0.2642                   | io+typ-rcMax |
| rise                                          | 2.8451  | EXECUTE_INST/ar0_reg_9/SE |              |
| CLKINVX3(inv_pair)                            | 9.9792  | -0.2062                   | io+typ-rcMax |
| rise                                          | 2.8772  | EXECUTE_INST/ar1_reg_9/SE |              |
| CLKINVX2(inv_pair)                            | 9.9792  | -0.2179                   | io+typ-rcMax |
| rise                                          | 2.8893  | EXECUTE_INST/ar1_reg_9/SE |              |
| CLKINVX1(inv_pair)                            | 9.9792  | -0.2460                   | io+typ-rcMax |
| rise                                          | 2.8718  | EXECUTE_INST/ar0_reg_9/SE |              |
| INVX4(inv_pair)                               | 13.3056 | -0.2000                   | io+typ-rcMax |
| rise                                          | 2.8560  | EXECUTE_INST/ar1_reg_9/SE |              |
| INVX3(inv_pair)                               | 13.3056 | -0.2093                   | io+typ-rcMax |
| rise                                          | 2.8692  | EXECUTE_INST/ar1_reg_9/SE |              |
| CLKINVX4(inv_pair)                            | 13.3056 | -0.2062                   | io+typ-rcMax |
| rise                                          | 2.8653  | EXECUTE_INST/ar1_reg_9/SE |              |
| CLKBUFXL(buf)                                 | 13.3056 | -0.2006                   | io+typ-rcMax |
| rise                                          | 2.9064  | EXECUTE_INST/ar1_reg_9/SE |              |
| CLKBUF3(buf)                                  | 13.3056 | -0.2111                   | io+typ-rcMax |
| rise                                          | 2.9005  | EXECUTE_INST/ar1_reg_9/SE |              |
| CLKBUF2(buf)                                  | 13.3056 | -0.2058                   | io+typ-rcMax |
| rise                                          | 2.9019  | EXECUTE_INST/ar1_reg_9/SE |              |
| CLKBUF1(buf)                                  | 13.3056 | -0.2129                   | io+typ-rcMax |
| rise                                          | 2.9002  | EXECUTE_INST/ar1_reg_9/SE |              |
| BUFXL(buf)                                    | 13.3056 | -0.2163                   | io+typ-rcMax |



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|                     |      |        |                           |              |
|---------------------|------|--------|---------------------------|--------------|
|                     | rise | 2.8830 | EXECUTE_INST/ar0_reg_9/SE |              |
| BUFX3 (buf)         |      |        | 13.3056      -0.1988      | io+typ-rcMax |
|                     | rise | 2.8953 | EXECUTE_INST/ar1_reg_9/SE |              |
| BUFX2 (buf)         |      |        | 13.3056      -0.2027      | io+typ-rcMax |
|                     | rise | 2.9008 | EXECUTE_INST/ar1_reg_9/SE |              |
| BUFX1 (buf)         |      |        | 13.3056      -0.2096      | io+typ-rcMax |
|                     | rise | 2.9043 | EXECUTE_INST/ar0_reg_9/SE |              |
| CLKBUFX4 (buf)      |      |        | 16.6320      -0.2111      | io+typ-rcMax |
|                     | rise | 2.8965 | EXECUTE_INST/ar1_reg_9/SE |              |
| BUFX4 (buf)         |      |        | 16.6320      -0.1963      | io+typ-rcMax |
|                     | rise | 2.8891 | EXECUTE_INST/ar1_reg_9/SE |              |
| INVX8 (inv_pair)    |      |        | 19.9584      -0.2505      | io+typ-rcMax |
|                     | rise | 2.7712 | EXECUTE_INST/ar1_reg_9/SE |              |
| DLY2X1 (buf)        |      |        | 19.9584      -0.4660      | io+typ-rcMax |
|                     | rise | 2.8523 | EXECUTE_INST/ar0_reg_9/SE |              |
| DLY1X1 (buf)        |      |        | 19.9584      -0.3181      | io+typ-rcMax |
|                     | rise | 2.8632 | EXECUTE_INST/ar0_reg_9/SE |              |
| CLKINVX8 (inv_pair) |      |        | 19.9584      -0.2647      | io+typ-rcMax |
|                     | rise | 2.7866 | EXECUTE_INST/ar1_reg_9/SE |              |
| DLY4X1 (buf)        |      |        | 23.2848      -0.8877      | io+typ-rcMax |
|                     | rise | 2.8231 | EXECUTE_INST/ar0_reg_9/SE |              |
| DLY3X1 (buf)        |      |        | 23.2848      -0.6581      | io+typ-rcMax |
|                     | rise | 2.8397 | EXECUTE_INST/ar0_reg_9/SE |              |
| CLKBUFX8 (buf)      |      |        | 23.2848      -0.2301      | io+typ-rcMax |
|                     | rise | 2.8591 | EXECUTE_INST/ar1_reg_9/SE |              |
| BUFX8 (buf)         |      |        | 29.9376      -0.2183      | io+typ-rcMax |
|                     | rise | 2.8450 | EXECUTE_INST/ar1_reg_9/SE |              |
| BUFX12 (buf)        |      |        | 33.2640      -0.2318      | io+typ-rcMax |
|                     | rise | 2.8191 | EXECUTE_INST/ar1_reg_9/SE |              |
| INVX12 (inv_pair)   |      |        | 43.2432      -0.3977      | io+typ-rcMax |
|                     | rise | 2.8632 | EXECUTE_INST/ar1_reg_9/SE |              |
| BUFX16 (buf)        |      |        | 43.2432      -0.2449      | io+typ-rcMax |
|                     | rise | 2.7935 | EXECUTE_INST/ar1_reg_9/SE |              |
| CLKBUFX12 (buf)     |      |        | 53.2224      -0.2592      | io+typ-rcMax |
|                     | rise | 2.8028 | EXECUTE_INST/ar1_reg_9/SE |              |
| BUFX20 (buf)        |      |        | 53.2224      -0.2577      | io+typ-rcMax |
|                     | rise | 2.7673 | EXECUTE_INST/ar1_reg_9/SE |              |

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|                     |        |                           |         |         |              |
|---------------------|--------|---------------------------|---------|---------|--------------|
| INVS16(inv_pair)    |        |                           | 56.5488 | -0.3908 | io+typ-rcMax |
| rise                | 2.8518 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| INVS20(inv_pair)    |        |                           | 63.2016 | -0.3922 | io+typ-rcMax |
| rise                | 2.8415 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| CLKINVS12(inv_pair) |        |                           | 63.2016 | -0.4383 | io+typ-rcMax |
| rise                | 2.8601 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| CLKBUFS16(buf)      |        |                           | 63.2016 | -0.2758 | io+typ-rcMax |
| rise                | 2.7814 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| CLKBUFS20(buf)      |        |                           | 79.8336 | -0.2979 | io+typ-rcMax |
| rise                | 2.7343 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| CLKINVS16(inv_pair) |        |                           | 83.1600 | -0.4635 | io+typ-rcMax |
| rise                | 2.8403 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| CLKINVS20(inv_pair) |        |                           | 96.4656 | -0.4696 | io+typ-rcMax |
| rise                | 2.8205 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |

- This command reports the evaluation results (sorted in increasing order of slack) of buffering pin, EXECUTE\_INST/ar0\_reg\_9/SE, by adding each of all the buffer/inverter/delay cells in the library:

```
tempus> add_repeater -term EXECUTE_INST/ar0_reg_9/SE -evaluate_all -sort_by slack
```

| Library-cell                            | Area    | Timing-slack | Worst-view   |
|-----------------------------------------|---------|--------------|--------------|
| Transition Trans-slack Worst-Trans-Sink |         |              |              |
| DLY4X1(buf)                             | 23.2848 | -0.8877      | io+typ-rcMax |
| rise 2.8231 EXECUTE_INST/ar0_reg_9/SE   |         |              |              |
| DLY3X1(buf)                             | 23.2848 | -0.6581      | io+typ-rcMax |
| rise 2.8397 EXECUTE_INST/ar0_reg_9/SE   |         |              |              |
| CLKINVS20(inv_pair)                     | 96.4656 | -0.4696      | io+typ-rcMax |
| rise 2.8205 EXECUTE_INST/ar1_reg_9/SE   |         |              |              |
| DLY2X1(buf)                             | 19.9584 | -0.4660      | io+typ-rcMax |
| rise 2.8523 EXECUTE_INST/ar0_reg_9/SE   |         |              |              |
| CLKINVS16(inv_pair)                     | 83.1600 | -0.4635      | io+typ-rcMax |
| rise 2.8403 EXECUTE_INST/ar1_reg_9/SE   |         |              |              |
| CLKINVS12(inv_pair)                     | 63.2016 | -0.4383      | io+typ-rcMax |
| rise 2.8601 EXECUTE_INST/ar1_reg_9/SE   |         |              |              |

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|                    |        |                           |         |         |              |
|--------------------|--------|---------------------------|---------|---------|--------------|
| INVS12(inv_pair)   |        |                           | 43.2432 | -0.3977 | io+typ-rcMax |
| rise               | 2.8632 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| INVS20(inv_pair)   |        |                           | 63.2016 | -0.3922 | io+typ-rcMax |
| rise               | 2.8415 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| INVS16(inv_pair)   |        |                           | 56.5488 | -0.3908 | io+typ-rcMax |
| rise               | 2.8518 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| DLY1X1(buf)        |        |                           | 19.9584 | -0.3181 | io+typ-rcMax |
| rise               | 2.8632 | EXECUTE_INST/ar0_reg_9/SE |         |         |              |
| CLKBUF20(buf)      |        |                           | 79.8336 | -0.2979 | io+typ-rcMax |
| rise               | 2.7343 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| CLKBUF16(buf)      |        |                           | 63.2016 | -0.2758 | io+typ-rcMax |
| rise               | 2.7814 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| CLKINVS8(inv_pair) |        |                           | 19.9584 | -0.2647 | io+typ-rcMax |
| rise               | 2.7866 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| CLKINVS1(inv_pair) |        |                           | 9.9792  | -0.2642 | io+typ-rcMax |
| rise               | 2.8451 | EXECUTE_INST/ar0_reg_9/SE |         |         |              |
| CLKBUF12(buf)      |        |                           | 53.2224 | -0.2592 | io+typ-rcMax |
| rise               | 2.8028 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| BUF20(buf)         |        |                           | 53.2224 | -0.2577 | io+typ-rcMax |
| rise               | 2.7673 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| INVS8(inv_pair)    |        |                           | 19.9584 | -0.2505 | io+typ-rcMax |
| rise               | 2.7712 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| CLKINVS1(inv_pair) |        |                           | 9.9792  | -0.2460 | io+typ-rcMax |
| rise               | 2.8718 | EXECUTE_INST/ar0_reg_9/SE |         |         |              |
| BUF16(buf)         |        |                           | 43.2432 | -0.2449 | io+typ-rcMax |
| rise               | 2.7935 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| BUF12(buf)         |        |                           | 33.2640 | -0.2318 | io+typ-rcMax |
| rise               | 2.8191 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| CLKBUF8(buf)       |        |                           | 23.2848 | -0.2301 | io+typ-rcMax |
| rise               | 2.8591 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| INVS1(inv_pair)    |        |                           | 6.6528  | -0.2273 | io+typ-rcMax |
| rise               | 2.8825 | EXECUTE_INST/ar0_reg_9/SE |         |         |              |
| BUF8(buf)          |        |                           | 29.9376 | -0.2183 | io+typ-rcMax |
| rise               | 2.8450 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| CLKINVS2(inv_pair) |        |                           | 9.9792  | -0.2179 | io+typ-rcMax |
| rise               | 2.8893 | EXECUTE_INST/ar1_reg_9/SE |         |         |              |
| BUFS1(buf)         |        |                           | 13.3056 | -0.2163 | io+typ-rcMax |

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|                    |      |        |                           |              |
|--------------------|------|--------|---------------------------|--------------|
|                    | rise | 2.8830 | EXECUTE_INST/ar0_reg_9/SE |              |
| CLKBUF1 (buf)      |      |        | 13.3056      -0.2129      | io+typ-rcMax |
|                    | rise | 2.9002 | EXECUTE_INST/ar1_reg_9/SE |              |
| INV1 (inv_pair)    |      |        | 6.6528      -0.2119       | io+typ-rcMax |
|                    | rise | 2.9038 | EXECUTE_INST/ar0_reg_9/SE |              |
| CLKBUF4 (buf)      |      |        | 16.6320      -0.2111      | io+typ-rcMax |
|                    | rise | 2.8965 | EXECUTE_INST/ar1_reg_9/SE |              |
| CLKBUF3 (buf)      |      |        | 13.3056      -0.2111      | io+typ-rcMax |
|                    | rise | 2.9005 | EXECUTE_INST/ar1_reg_9/SE |              |
| BUF1 (buf)         |      |        | 13.3056      -0.2096      | io+typ-rcMax |
|                    | rise | 2.9043 | EXECUTE_INST/ar0_reg_9/SE |              |
| INV3 (inv_pair)    |      |        | 13.3056      -0.2093      | io+typ-rcMax |
|                    | rise | 2.8692 | EXECUTE_INST/ar1_reg_9/SE |              |
| CLKINV4 (inv_pair) |      |        | 13.3056      -0.2062      | io+typ-rcMax |
|                    | rise | 2.8653 | EXECUTE_INST/ar1_reg_9/SE |              |
| CLKINV3 (inv_pair) |      |        | 9.9792      -0.2062       | io+typ-rcMax |
|                    | rise | 2.8772 | EXECUTE_INST/ar1_reg_9/SE |              |
| CLKBUF2 (buf)      |      |        | 13.3056      -0.2058      | io+typ-rcMax |
|                    | rise | 2.9019 | EXECUTE_INST/ar1_reg_9/SE |              |
| INV2 (inv_pair)    |      |        | 9.9792      -0.2056       | io+typ-rcMax |
|                    | rise | 2.8842 | EXECUTE_INST/ar1_reg_9/SE |              |
| BUF2 (buf)         |      |        | 13.3056      -0.2027      | io+typ-rcMax |
|                    | rise | 2.9008 | EXECUTE_INST/ar1_reg_9/SE |              |
| CLKBUFXL (buf)     |      |        | 13.3056      -0.2006      | io+typ-rcMax |
|                    | rise | 2.9064 | EXECUTE_INST/ar1_reg_9/SE |              |
| INV4 (inv_pair)    |      |        | 13.3056      -0.2000      | io+typ-rcMax |
|                    | rise | 2.8560 | EXECUTE_INST/ar1_reg_9/SE |              |
| BUF3 (buf)         |      |        | 13.3056      -0.1988      | io+typ-rcMax |
|                    | rise | 2.8953 | EXECUTE_INST/ar1_reg_9/SE |              |
| BUF4 (buf)         |      |        | 16.6320      -0.1963      | io+typ-rcMax |
|                    | rise | 2.8891 | EXECUTE_INST/ar1_reg_9/SE |              |
| Reference          |      |        | 0.0000      -0.1181       | io+typ-rcMax |
|                    | rise | 2.9024 | EXECUTE_INST/ar1_reg_9/SE |              |

-----  
-----

- This command evaluates buffering of pin, EXECUTE\_INST/ar0\_reg\_9/SE, by adding each of all the buffer/inverter/delay cells in the library, and report (sorted in increasing order of slack) the top 5

best and bottom 5 worst cells, along with the 'reference':

```
tempus> add_repeater -term EXECUTE_INST/ar0_reg_9/SE -evaluate_all -sort_by slack -
max_cells 5
```

```


Library-cell Area Timing-slack Worst-view
 Transition Trans-slack Worst-Trans-Sink

DLY4X1 (buf) 23.2848 -0.8877 io+typ-rcMax
 rise 2.8231 EXECUTE_INST/ar0_reg_9/SE
DLY3X1 (buf) 23.2848 -0.6581 io+typ-rcMax
 rise 2.8397 EXECUTE_INST/ar0_reg_9/SE
CLKINX20 (inv_pair) 96.4656 -0.4696 io+typ-rcMax
 rise 2.8205 EXECUTE_INST/ar1_reg_9/SE
DLY2X1 (buf) 19.9584 -0.4660 io+typ-rcMax
 rise 2.8523 EXECUTE_INST/ar0_reg_9/SE
CLKINX16 (inv_pair) 83.1600 -0.4635 io+typ-rcMax
 rise 2.8403 EXECUTE_INST/ar1_reg_9/SE
Reference 0.0000 -0.1181 io+typ-rcMax
 rise 2.9024 EXECUTE_INST/ar1_reg_9/SE


```

- The following command reports the evaluation results (worst setup timing across all views) of buffering pin, MPY\_32\_INST/M16X16\_INST/i\_910/B0, by adding each buffer/inverter/delay cell in the library that matches the footprint of cell, BUF4:

```
tempus> add_repeater -term MPY_32_INST/M16X16_INST/i_910/B0 -cell BUF4 -evaluate_all
```

```


Library-cell Area Timing-slack Worst-view Transition Trans-
slack Worst-Trans-Sink

Reference 0.0000 -0.1242 core+typ-rcMax rise 2.8485
 MPY_32_INST/M16X16_INST/i_910/B0
CLKBUF4 (buf) 13.3056 -0.1785 core+typ-rcMax rise 2.9125
 MPY_32_INST/M16X16_INST/i_910/B0


```

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|                                  |         |         |                |      |        |
|----------------------------------|---------|---------|----------------|------|--------|
| CLKBUF8 (buf)                    | 23.2848 | -0.2197 | core+typ-rcMax | rise | 2.9580 |
| MPY_32_INST/M16X16_INST/i_910/B0 |         |         |                |      |        |
| CLKBUF4 (buf)                    | 16.6320 | -0.1928 | core+typ-rcMax | rise | 2.9497 |
| MPY_32_INST/M16X16_INST/i_910/B0 |         |         |                |      |        |
| CLKBUF3 (buf)                    | 13.3056 | -0.1892 | core+typ-rcMax | rise | 2.9453 |
| MPY_32_INST/M16X16_INST/i_910/B0 |         |         |                |      |        |
| CLKBUF2 (buf)                    | 13.3056 | -0.1854 | core+typ-rcMax | rise | 2.9311 |
| MPY_32_INST/M16X16_INST/i_910/B0 |         |         |                |      |        |
| CLKBUF20 (buf)                   | 79.8336 | -0.4212 | core+typ-rcMax | rise | 2.9477 |
| MPY_32_INST/M16X16_INST/i_910/B0 |         |         |                |      |        |
| CLKBUF1 (buf)                    | 13.3056 | -0.2002 | core+typ-rcMax | rise | 2.9081 |
| MPY_32_INST/M16X16_INST/i_910/B0 |         |         |                |      |        |
| CLKBUF16 (buf)                   | 63.2016 | -0.3583 | core+typ-rcMax | rise | 2.9509 |
| MPY_32_INST/M16X16_INST/i_910/B0 |         |         |                |      |        |
| CLKBUF12 (buf)                   | 53.2224 | -0.3147 | core+typ-rcMax | rise | 2.9542 |
| MPY_32_INST/M16X16_INST/i_910/B0 |         |         |                |      |        |
| BUF8 (buf)                       | 13.3056 | -0.2028 | core+typ-rcMax | rise | 2.8799 |
| MPY_32_INST/M16X16_INST/i_910/B0 |         |         |                |      |        |
| BUF4 (buf)                       | 29.9376 | -0.2222 | core+typ-rcMax | rise | 2.9595 |
| MPY_32_INST/M16X16_INST/i_910/B0 |         |         |                |      |        |
| BUF4 (buf)                       | 16.6320 | -0.1836 | core+typ-rcMax | rise | 2.9545 |
| MPY_32_INST/M16X16_INST/i_910/B0 |         |         |                |      |        |
| BUF3 (buf)                       | 13.3056 | -0.1805 | core+typ-rcMax | rise | 2.9502 |
| MPY_32_INST/M16X16_INST/i_910/B0 |         |         |                |      |        |
| BUF2 (buf)                       | 13.3056 | -0.1818 | core+typ-rcMax | rise | 2.9358 |
| MPY_32_INST/M16X16_INST/i_910/B0 |         |         |                |      |        |

- The following command reports the evaluation results (worst setup timing across all views) of buffering pin, MPY\_32\_INST/M16X16\_INST/i\_910/B0, by adding cell, BUF4:

```
tempus> add_repeater -term MPY_32_INST/M16X16_INST/i_669/A -cell BUF4 -evaluate_only
```

```

Library-cell Area Timing-slack Worst-view Transition Trans-
slack Worst-Trans-Sink

```

|                                  |         |         |                |      |        |
|----------------------------------|---------|---------|----------------|------|--------|
| Reference                        | 0.0000  | -0.1242 | core+typ-rcMax | rise | 2.6766 |
| MPY_32_INST/M16X16_INST/i_639/A1 |         |         |                |      |        |
| BUFX4 (buf)                      | 16.6320 | -0.1816 | core+typ-rcMax | rise | 2.6555 |
| MPY_32_INST/M16X16_INST/i_639/A1 |         |         |                |      |        |

- The following command reports the evaluation results (worst setup timing across all views) of buffering net, MPY\_32\_INST/M16X16\_INST/n\_647, by adding cell, CLKBUF20:

```
tempus> add_repeater -net MPY_32_INST/M16X16_INST/n_647 -cell CLKBUF20 -evaluate_only
```

|              |                  |              |            |            |
|--------------|------------------|--------------|------------|------------|
| Library-cell | Area             | Timing-slack | Worst-view | Transition |
| Trans-slack  | Worst-Trans-Sink |              |            |            |

|                |                                 |         |                |      |  |
|----------------|---------------------------------|---------|----------------|------|--|
| Reference      | 0.0000                          | -0.0745 | core+typ-rcMax | rise |  |
| 2.8849         | MPY_32_INST/M16X16_INST/i_388/A |         |                |      |  |
| CLKBUF20 (buf) | 79.8336                         | -0.2270 | core+typ-rcMax | rise |  |
| 2.9586         | MPY_32_INST/M16X16_INST/i_388/A |         |                |      |  |

- The following command reports the evaluation results (worst setup timing across all views) of buffering pin, MPY\_32\_INST/M16X16\_INST/i\_669/A, by adding cell, BUFX4:

```
tempus> add_repeater -term MPY_32_INST/M16X16_INST/i_669/A -cell BUFX4 -evaluate_only -
output_as_tcl
```

```
{MPY_32_INST/M16X16_INST/n_917 {Reference ref 0.0000 -0.1242 core+typ-rcMax rise 2.6766
MPY_32_INST/M16X16_INST/i_639/A1} {BUFX4 buf 16.6320 -0.1816 core+typ-rcMax rise 2.6555
MPY_32_INST/M16X16_INST/i_639/A1}}
```

- The following command reports the evaluation results (hold timing analysis for only core+typ-rcMin view) of buffering pin, MPY\_32\_INST/M16X16\_INST/i\_699/A1, by adding cell, BUFX16:

```
tempus> add_repeater -term MPY_32_INST/M16X16_INST/i_699/A1 -cell BUFX16 -evaluate_only
-view core+typ-rcMin -early
```

|              |                  |              |            |            |        |
|--------------|------------------|--------------|------------|------------|--------|
| Library-cell | Area             | Timing-slack | Worst-view | Transition | Trans- |
| slack        | Worst-Trans-Sink |              |            |            |        |

|             |                                  |        |                |      |
|-------------|----------------------------------|--------|----------------|------|
| Reference   | 0.0000                           | 2.1814 | core+typ-rcMin | rise |
| 2.6919      | MPY_32_INST/M16X16_INST/i_699/A1 |        |                |      |
| BUFX16(buf) | 43.2432                          | 2.3008 | core+typ-rcMin | fall |
| 2.4436      | MPY_32_INST/M16X16_INST/i_731/B  |        |                |      |

- The following commands add one instance of inverter `INVX1` after every 50 um distance, starting from the net's driver (polarity of some or all sinks may get reversed when odd number of instances are added):

```
set_eco_mode -leq_check false
add_repeater -net n123 -cell INVX1 -spread_dist 50
```

- The following commands add one instance of inverter `INVX1` after every 50 um distance, starting from the net's driver, and if required, one instance of this inverter at a sink to preserve its polarity:

```
set_eco_mode -leq_check true
add_repeater -net n123 -cell INVX1 -spread_dist 50
```

- The following command adds 10 instances of buffer `BUFX1`, equi-distant from each other, such that only the specified sinks are buffered:

```
add_repeater -term {sink1 sink3} -cell BUFX1 -spread_count 10
```

- The following command adds one instance each of buffers `BUFX1`, `BUFX2`, `BUFX3`, and `BUFX4`, equi-distant from each other, such that all the sinks of this net are buffered [total 4 instances added]:

```
add_repeater -net n123 -cell {BUFX1 BUFX2 BUFX3 BUFX4} -spread_count 4
```

- The following commands add two instances each of inverters `INVX1` and `INVX3`, and one instance each of buffers `BUFX2` and `BUFX4`, each buffer or inverter equi-distant from each other, such that only the specified sinks are buffered [total 6 instances added]:

```
set_eco_mode -leq_check true
add_repeater -term {sink1 sink3} -cell {INVX1 BUFX2 INVX3 BUFX4} -spread_count 4
```

- The following command adds one instance of buffer `BUFX1` after every 50 um distance starting from the net's driver:

```
add_repeater -net n123 -cell BUFX1 -spread_dist 50 -spread_prefix {ab}
```

The added instances are named as `ab_inst_0`, `ab_inst_1`, ... and the added nets are named as `ab_net_0`, `ab_net_1`, ...

- The following command adds four instances of buffer `BUFX1`, equidistant from each other so that all sinks of the net are buffered:

```
add_repeater -net n123 -cell BUFX1 -spread_count 4 -spread_prefix {mn}
```

The added instances are named as `mn_inst_0`, `mn_inst_1`, ... and the added nets are named as `mn_net_0`, `mn_net_1`, ...

**Note:** The suffix number automatically starts after the last buffer or inverter added earlier.

- The following command adds the first buffer (`BUFFD4BWP`) after 20 micron distance from the driver term, and subsequent buffers after 50 micron distance from the previous buffer.

```
add_repeater -net iso -cell BUFFD4BWP -spread_dist 50 -first_spread_dist 20
```



## Related Commands

- `change_cell`
- `delete_repeater`
- `set_eco_mode`

# attach\_module\_port

```
attach_module_port
[-help]
module_name
port_name
net_name
[-no_new_port]
```

Attaches a port in the specified instance (or top level) to a net.

## Parameters

|                  |                                                                                                                                                                                                                                                                                                                                                                    |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help            | Outputs a brief description that includes the type and default information for each <code>attach_module_port</code> .<br><br>For a detailed description of the command and all of its parameters, use the man command:<br><code>man attach_module_port</code>                                                                                                      |
| module_name      | Specifies the module to which you want to attach the port. To specify the top module, enter <code>`-'</code> .                                                                                                                                                                                                                                                     |
| net_name         | Specifies the net on the module.                                                                                                                                                                                                                                                                                                                                   |
| port_name        | Specifies the port on the module.                                                                                                                                                                                                                                                                                                                                  |
| -<br>no_new_port | Prevents creation of hierarchical ports if a module port is attached. If the module port does not connect to the net in a different hierarchy through the existing ports, the software displays an error message, and the command stops running.<br><br><i>Default:</i> If you do not specify this parameter, the software creates hierarchical ports as required. |

## Examples

- The following command attaches port `in` on the top module to `net123`:

```
tempus> attach_module_port - in net123
```

## Related Commands

- `add_inst`
- `add_net`
- `attach_term`
- `change_inst_name`
- `delete_inst`
- `detach_module_port`
- `delete_net`
- `detach_net`
- `detach_term`

# attach\_net

```
attach_net
-help
net_name
list_of_pins
```

Attaches nets to pins. The syntax is order-dependent.

After this command, the software will perform a full timing update when you run the next command that reports timing, for example, [report\\_timing](#).

**Note:** The tool supports this command for flattened designs only.

## Parameters

|              |                                                                                                                                                                                                                                                    |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help        | Outputs a brief description that includes type and default information for each <code>attach_net</code> parameter.<br><br>For a detailed description of the command and all of its parameters, use the man command:<br><code>man attach_net</code> |
| list_of_pins | Specifies a list of pins/ports to attach to the specified net or nets.                                                                                                                                                                             |
| net_name     | Specifies the net to attach to the pins.                                                                                                                                                                                                           |

## Examples

The following example attaches `net26` to pins `i1/A`, `i2/B`, and `i3/Z`:

```
tempus> attach_net net26 "i1/A i2/B i3/Z"
```

The following command is equivalent to the above example

```
tempus> attach_net [get_nets net26] [get_cells "i1/A i2/B i3/Z"]
```

## Related Commands

- `add_inst`
- `add_net`
- `attach_term`
- `change_inst_name`
- `delete_inst`
- `delete_net`
- `detach_net`
- `detach_term`

## attach\_term

```
attach_term
-help
inst_name
term_name
net_name
[-noNewPort]
[-port port_name | -pin ref_inst_name ref_pin_name]
```

Attaches a terminal to a net. The syntax is order-dependent.

After this command, the software will perform a full timing update when you run the next command that reports timing, for example, [report\\_timing](#).

**Note:** The tool supports this command for flattened designs only.

## Parameters

|                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                                           | <p>Outputs a brief description that includes type and default information for each <code>attach_term</code> parameter.</p> <p>For a detailed description of the command and all of its parameters, use the man command: <code>man attach_term</code></p>                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>inst_name</code>                                       | Specifies the instance or list of instances containing the specified terminal or terminals.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>net_name</code>                                        | Specifies the net or list of nets to attach to a specified terminal of a specified instance.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>-noNewPort</code>                                      | Prohibits the tool from creating hierarchical ports when it attaches a terminal. If the terminal cannot connect to the net via existing ports, the tool displays an error message, and the command stops.                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <code>-pin</code><br><code>ref_inst_name ref_pin_name</code> | Specifies the pin <code>refPinName</code> on instance <code>refInstName</code> connected to the net that the tool connects to the terminal. You cannot specify the <code>-port</code> parameter when using this parameter.                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>-port port_name</code>                                 | <p>Specifies the hierarchical port used to connect the terminal with the net. If you specify this parameter, the hierarchical port must exist in the module that contains the instance. The hierarchical port must connect to the net. This parameter lets you use a specific port to maintain the same netlist topology, which simplifies equivalence checking later in the design flow. You cannot specify the <code>-pin</code> parameter if you use this parameter.</p> <p><i>Default:</i> If you do not specify this parameter, the tool uses existing ports or creates new hierarchical ports as necessary to connect the terminal to the net.</p> |
| <code>term_name</code>                                       | Specifies the name of the terminal that the tool connects to the specified net.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## Examples

The following example attaches `net1` to terminal `A` on `inst1`:

```
tempus> attach_term inst1 A net1
```

## Related Commands

- `add_inst`
- `add_net`
- `attach_net`
- `change_inst_name`
- `delete_inst`
- `delete_net`
- `detach_net`
- `detach_term`



## change\_cell

```
change_cell
[-help]
[-pinmap <string>]
{-cell <string> | -upsize | -downsize | -evaluate_all }
[-evaluate_only | -evaluate_all]
[-early | -late]
[-loc {x y} [-orient <string>]]
[-sort_by {slack delay trans area} [-max_cells <integer>]]
{-inst <string> | -from <string> | -from_rise <string> | -from_fall <string>}
{-inst <string> | -to <string> | -to_rise <string> | -to_fall <string>} [[{-from <string>] [-from_rise <string>] [-from_fall <string>] [-to <string>] [-to_rise <string>] [-to_fall <string>]] [[{-timing_type {combinational combinational_fall combinational_rise falling_edge hold_falling hold_rising recovery_falling recovery_rising removal_falling removal_rising rising_edge setup_falling setup_rising}} [-view <string>] [-early] [-late] [-output_as_tcl] [-sort_by {slack delay trans area}]]] [[{-evaluate_all } [-evaluate_only }} [[{-from <string>] [-from_rise <string>] [-from_fall <string>] [-to <string>] [-to_rise <string>] [-to_fall <string>] [-timing_type {combinational combinational_fall combinational_rise falling_edge hold_falling hold_rising recovery_falling recovery_rising removal_falling removal_rising rising_edge setup_falling setup_rising}} [-view <string>] [-early] [-late] [-output_as_tcl }} [-sort_by {slack delay trans area}]]]
```

Changes the cell of an instance. In multi-mode multi-corner (MMMC) mode, this command changes the cell and updates timing in all views. This command upsizes or downsizes the specified instance based on the available footprint. Cell mapping is also possible as long as the output pins are functionally equivalent. This command also checks for and observes power domains.

This command has the following features:

- Changes the cell
- Performs incremental timing updates of the paths passing through the cell
- Supports spief, LEF, and DEF-based flows

### Note:

You do not need to provide physical information (LEF/DEF) to perform ECO with this command.

## Parameters

|                                |                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>             | <p>Outputs a brief description that includes type and default information for each <code>change_cell</code> parameter.</p> <p>For a detailed description of the command and all of its parameters, use the <code>man</code> command:</p> <p><code>man change_cell</code>.</p>                                                        |
| <code>-cell string</code>      | <p>Specifies the name of the new cell of the given instance. The function of the original cell must match the function of the new cell for all the outputs used by the old cell. The number of input pins must be the same for both the original and new cells.</p>                                                                  |
| <code>-downsize</code>         | <p>Downsizes the specified instance to the next available cell in the same footprint unless the specified cell is already of the lowest driving strength. Downsizing an instance on the noncritical path can reduce the loading of its driver on the critical path.</p>                                                              |
| <code>-early</code>            | <p>Specifies that timing will be evaluated for early analysis. In such a case, only the active hold analysis views will be considered. If this parameter is not specified, timing will be evaluated for late analysis.</p> <p><b>Note:</b> This parameter must be specified with <code>-evaluate_only/-evaluate_all</code> only.</p> |
| <code>-evaluate_all</code>     | <p>Allows evaluating new timing(s) within the instance using all functionally equivalent library cell(s) for the instance.</p>                                                                                                                                                                                                       |
| <code>-evaluate_only</code>    | <p>Allows evaluating new timing within the instance using the specified library cell for the instance.</p>                                                                                                                                                                                                                           |
| <code>-from tring</code>       | <p>Specifies the input pin for the instance to be evaluated for resize. Both the rise and fall transitions are considered in the timing report.</p> <p><b>Note:</b> This parameter must be specified with <code>-evaluate_only/-evaluate_all</code> only.</p>                                                                        |
| <code>-from_fall string</code> | <p>Specifies the input pin for the instance to be evaluated for resize. Only the fall transition is considered in the timing report.</p> <p><b>Note:</b> This parameter must be specified with <code>-evaluate_only/-evaluate_all</code> only.</p>                                                                                   |
| <code>-from_rise string</code> | <p>Specifies the input pin for the instance to be evaluated for resize. Only the rise transition is considered in the timing report.</p> <p><b>Note:</b> This parameter must be specified with <code>-evaluate_only/-evaluate_all</code> only.</p>                                                                                   |
| <code>-inst string</code>      | <p>Specifies the name of the instance to be resized.</p>                                                                                                                                                                                                                                                                             |

|                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-late</code>                                                             | <p>Specifies that timing will be evaluated for late analysis. In such a case, only the active setup analysis views will be considered. If this parameter is not specified, timing will still be evaluated for late analysis.</p> <p><b>Note:</b> This parameter must be specified with <code>-evaluate_only/-evaluate_all</code> only.</p>                                                                                                                                                                                                                                                                                                                   |
| <code>-loc {x y}</code>                                                        | <p>Specifies the legalized location where the instance will be placed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>-max_cells</code><br><code>&lt;integer&gt;</code>                        | <p>Specifies the maximum number of improving and degrading cells (each) to be reported for evaluation.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>-orient string</code>                                                    | <p>Specifies the legalized orientation for the instance.</p> <p>You can specify this parameter with the <code>-loc</code> parameter only.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <code>-output_as_tcl</code>                                                    | <p>Returns the evaluation result as a TCL list. By default, the command returns a text-based format.</p> <p><b>Note:</b> This parameter must be specified with <code>-evaluate_only/-evaluate_all</code> only.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>-pinmap string</code>                                                    | <p>Specifies the pin mapping for the new cell based on the old cell. This option is required if the new master cell has different pin names than the original cell. The net connected to <i>oldpin1</i> of the original cell should be connected to <i>newpin1</i> of the new cell, and so forth.</p>                                                                                                                                                                                                                                                                                                                                                        |
| <code>-sort_by {slack</code><br><code>delay trans</code><br><code>area}</code> | <p>Specifies the sorting order for evaluation output.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>-timing_type</code><br><code>&lt;timing_type&gt;</code>                  | <p>Specifies the type of the timing arc to be used for timing evaluation between the specified <code>-from/-from_rise/-from_fall</code> and <code>-to/-to_rise/-to_fall</code> pins. The software will check for valid timing_types:</p> <ul style="list-style-type: none"> <li>• If only one arc is available, it will proceed.</li> <li>• If multiple arcs are available, you need to specify which timing_type to be considered via this parameter.</li> </ul> <p><b>Note:</b> This parameter must be specified with <code>-evaluate_only/-evaluate_all</code> only. For valid timing types, refer to the section, <b>List of Valid Timing-types</b>.</p> |
| <code>-to string</code>                                                        | <p>Specify one of the following:</p> <ul style="list-style-type: none"> <li>• The output pin of the instance to be evaluated for resize, or</li> <li>• The input pin for the instance driven by the instance to be evaluated for resize.</li> </ul> <p><b>Note:</b> This parameter must be specified with <code>-evaluate_only/-evaluate_all</code> only.</p>                                                                                                                                                                                                                                                                                                |

|                              |                                                                                                                                                                                                                                                                                                                                              |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-to_fall string</code> | Specifies the name of the target pin of the cell to be resized, or the input pin of the next-stage cell. Only the fall transition is considered in the timing report.                                                                                                                                                                        |
| <code>-to_rise string</code> | Specifies the name of the target pin of the cell to be resized, or the input pin of the next-stage cell. Only the rise transition is considered in the timing report.                                                                                                                                                                        |
| <code>-upsize</code>         | Upsizes the specified instance with the next available cell in the same footprint unless the specified cell is already of the highest driving strength. Upsizing an instance that drives a large load can improve the driver delay and the transition time at the receivers.                                                                 |
| <code>-view string</code>    | <p>Specifies the active analysis views for which timing must be evaluated. If this parameter is not specified, timing will be evaluated for all active analysis views and results reported for the worst view.</p> <p><b>Note:</b> This parameter must be specified with <code>-evaluate_only</code> or <code>-evaluate_all</code> only.</p> |

## Examples

- The following command substitutes the cell for instance `DTMF_INST/RESULTS_CONV_INST/FE_PHC268_r1477_1_` with cell `DLY3X2`. Pin A maps to pin I, and pin Y maps to pin O of the new cell.  

```
tempus> change_cell -inst DTMF_INST/RESULTS_CONV_INST/FE_PHC268_r1477_1_ -cell DLY3X2 -pinmap {A I Y O}
```
- The following command substitutes the cell for instance `DTMF_INST/RESULTS_CONV_INST/FE_PHC268_r1477_1_` with cell `BUF2`.  

```
tempus> change_cell -inst DTMF_INST/RESULTS_CONV_INST/FE_PHC268_r1477_1_ -cell BUF2
```
- The following command upsizes the cell, for instance `DTMF_INST/RESULTS_CONV_INST/FE_PHC268_r1477_1_` and places the instance at the given location.  

```
tempus> change_cell -inst DTMF_INST/RESULTS_CONV_INST/FE_PHC268_r1477_1_ -upsized -loc {23.45 98.12}
```
- The following command evaluates timing for instance, `buf2`, using library cell `BUF2` (prints delay for A to Y arc, slack at A pin, and slew at Y pin, corresponding to the worst timing path) :

```
tempus> change_cell -cell BUF2 -from buf2/A -to buf2/Y -evaluate_only
From Pin: buf2/A
To Pin: buf2/Y
```

```


Library-cell From To Arc-type Delay Timing-slack Trans-slack Worst-view

*BUF2 rise rise combinational 0.1036 -1.9976 4.4498 view1
BUF2 fall fall combinational 0.0963 -1.8642 4.8364 view1


```

(Note: A '\*' preceding an entry indicates data for the present library cell of the instance.)

- The following command evaluates timing for instance, `TDSP_CORE_INST/TDSP_CORE_GLUE_INST/p1929A341`, using library cell `INR2D2` for combinational timing arc and prints the delay, estimated slack and slew (for the worst timing path across all views)  

```
tempus> change_cell -cell INR2D2 -from TDSP_CORE_INST/TDSP_CORE_GLUE_INST/p1929A341/B1 -to TDSP_CORE_INST/TDSP_CORE_GLUE_INST/p1929A341/ZN -evaluate_only -timing_type combinational
```

```
From Pin: TDSP_CORE_INST/TDSP_CORE_GLUE_INST/p1929A341/B1
To Pin: TDSP_CORE_INST/TDSP_CORE_GLUE_INST/p1929A341/ZN
```

```

Library-cell From To Arc-type Delay Timing-slack Trans-slack Worst-
view

*INR2D1 fall rise combinational 0.1793 1.4719 0.7492 view1
INR2D2 fall rise combinational 0.1206 1.5431 0.8401 view1

```

- The following command evaluates timing for instance, TDSP\_CORE\_INST/TDSP\_CORE\_GLUE\_INST/p1929A341, using library cell INR2D2 for view1 and prints the delay, estimated slack and slew (for the worst slack timing path corresponding to the specified fall/rise transitions) -from\_fall TDSP\_CORE\_INST/TDSP\_CORE\_GLUE\_INST/p1929A341/B1 -to\_rise TDSP\_CORE\_INST/TDSP\_CORE\_GLUE\_INST/p1929A341/ZN

```
tempus> change_cell -cell INR2D2 -from TDSP_CORE_INST/TDSP_CORE_GLUE_INST/p1929A341/B1 -
to TDSP_CORE_INST/TDSP_CORE_GLUE_INST/p1929A341/ZN -evaluate_only -view view1
```

```
From Pin: TDSP_CORE_INST/TDSP_CORE_GLUE_INST/p1929A341/B1
To Pin: TDSP_CORE_INST/TDSP_CORE_GLUE_INST/p1929A341/ZN
```

```

Library-cell From To Arc-type Delay Timing-slack Trans-slack
Worst-view

*INR2D1 fall rise combinational 0.1793 1.4719 0.7492 view1
INR2D2 fall rise combinational 0.1206 1.5431 0.8401 view1

```

- The following command evaluates timing for instance, TDSP\_CORE\_INST/TDSP\_CORE\_GLUE\_INST/p1929A341, using library cell, INR2D2:

```
tempus> change_cell -cell INR2D2 -from
```

```

TDSP_CORE_INST/TDSP_CORE_GLUE_INST/p1929A341/B1 -to
TDSP_CORE_INST/TDSP_CORE_GLUE_INST/p1929A341/ZN -evaluate_only -output_as_tcl
#####
TDSP_CORE_INST/TDSP_CORE_GLUE_INST/p1929A341/B1
TDSP_CORE_INST/TDSP_CORE_GLUE_INST/p1929A341/ZN {{*INR2D1 fall rise combinational 0.1793
1.4719 0.7492 view1} {INR2D2 fall rise combinational 0.1206 1.5431 0.8401 view1}}
#####

```

- The following command evaluates timing(s) using all functionally equivalent library cell(s) for instance, TDSP\_CORE\_INST/TDSP\_CORE\_GLUE\_INST/p1929A341, and prints the delay, estimated slack and slew (for worst timing path for each individual cell):

```

tempus> change_cell -from TDSP_CORE_INST/TDSP_CORE_GLUE_INST/p1929A341/B1 -to
TDSP_CORE_INST/TDSP_CORE_GLUE_INST/p1929A341/ZN -evaluate_all
From Pin: TDSP_CORE_INST/TDSP_CORE_GLUE_INST/p1929A341/B1
To Pin: TDSP_CORE_INST/TDSP_CORE_GLUE_INST/p1929A341/ZN

```

```

Library-cell From To Arc-type Delay Timing-slack Trans-slack
Worst-view

*INR2D1 fall rise combinational 0.1793 1.4719 0.7492
view1
INR2D4 rise fall combinational 0.0972 1.5457 0.8742
view1
INR2D2 fall rise combinational 0.1206 1.5431 0.8401
view1
INR2D4HVT fall rise combinational 0.1472 1.4934 0.8635
view1
INR2D2HVT rise fall combinational 0.1749 1.4763 0.7981
view1
INR2D1HVT fall rise combinational 0.2645 1.3490 0.6334
view1


```

- The following command reports evaluation results (for A to Y arc) of resizing instance, TDSP\_CORE\_GLUE\_INST/FE\_OFC29\_mdr\_3\_, with each of all the equivalent cells in the library:

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---

```
change_cell -from TDSP_CORE_GLUE_INST/FE_OFC29_mdr_3_/A -to
TDSP_CORE_GLUE_INST/FE_OFC29_mdr_3_/Y -evaluate_all
```

From Pin: TDSP\_CORE\_GLUE\_INST/FE\_OFC29\_mdr\_3\_/A

To Pin: TDSP\_CORE\_GLUE\_INST/FE\_OFC29\_mdr\_3\_/Y

| Library-cell |             |         | From           | To   | Arc-type      | Delay  |
|--------------|-------------|---------|----------------|------|---------------|--------|
| Timing-slack | Trans-slack | Area    | Worst-view     |      |               |        |
| <hr/>        |             |         |                |      |               |        |
| <hr/>        |             |         |                |      |               |        |
| *BUF1        |             |         | rise           | rise | combinational | 0.5189 |
| -0.2121      | 1.9871      | 13.3056 | core+typ-rcMax |      |               |        |
| CLKBUF1      |             |         | fall           | fall | combinational | 0.5434 |
| -0.2203      | 2.1958      | 13.3056 | core+typ-rcMax |      |               |        |
| CLKBUF8      |             |         | rise           | rise | combinational | 0.1443 |
| 0.1037       | 2.8471      | 23.2848 | core+typ-rcMax |      |               |        |
| CLKBUF4      |             |         | rise           | rise | combinational | 0.2021 |
| 0.0668       | 2.7276      | 16.6320 | core+typ-rcMax |      |               |        |
| CLKBUF3      |             |         | rise           | rise | combinational | 0.2449 |
| 0.0287       | 2.6377      | 13.3056 | core+typ-rcMax |      |               |        |
| CLKBUF2      |             |         | rise           | rise | combinational | 0.3028 |
| -0.0242      | 2.4827      | 13.3056 | core+typ-rcMax |      |               |        |
| CLKBUF20     |             |         | rise           | rise | combinational | 0.1262 |
| -0.0181      | 2.9200      | 79.8336 | core+typ-rcMax |      |               |        |
| CLKBUF1      |             |         | rise           | rise | combinational | 0.4874 |
| -0.1931      | 2.0677      | 13.3056 | core+typ-rcMax |      |               |        |
| CLKBUF16     |             |         | rise           | rise | combinational | 0.1297 |
| 0.0317       | 2.9166      | 63.2016 | core+typ-rcMax |      |               |        |
| CLKBUF12     |             |         | rise           | rise | combinational | 0.1259 |
| 0.0571       | 2.9097      | 53.2224 | core+typ-rcMax |      |               |        |
| BUF1         |             |         | rise           | rise | combinational | 0.6868 |
| -0.3586      | 1.6156      | 13.3056 | core+typ-rcMax |      |               |        |
| BUF8         |             |         | rise           | rise | combinational | 0.1216 |
| 0.1078       | 2.8491      | 29.9376 | core+typ-rcMax |      |               |        |
| BUF4         |             |         | rise           | rise | combinational | 0.1798 |
| 0.0781       | 2.7308      | 16.6320 | core+typ-rcMax |      |               |        |
| BUF3         |             |         | rise           | rise | combinational | 0.2199 |



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|         |        |         |                         |        |
|---------|--------|---------|-------------------------|--------|
| 0.0459  | 2.6492 | 13.3056 | core+typ-rcMax          |        |
| BUFX2   |        |         | rise rise combinational | 0.2963 |
| -0.0193 | 2.4806 | 13.3056 | core+typ-rcMax          |        |
| BUFX20  |        |         | rise rise combinational | 0.0954 |
| 0.0408  | 2.9108 | 53.2224 | core+typ-rcMax          |        |
| BUFX16  |        |         | rise rise combinational | 0.1016 |
| 0.0669  | 2.9011 | 43.2432 | core+typ-rcMax          |        |
| BUFX12  |        |         | rise rise combinational | 0.1096 |
| 0.0896  | 2.8839 | 33.2640 | core+typ-rcMax          |        |

- The following command reports (sorted in increasing order of area) evaluation results (for A to Y arc) of resizing instance , TDSP\_CORE\_GLUE\_INST/FE\_OFC29\_mdr\_3\_, with each of all the equivalent cells in the library:

```
change_cell -from TDSP_CORE_GLUE_INST/FE_OFC29_mdr_3_/A -to
TDSP_CORE_GLUE_INST/FE_OFC29_mdr_3_/Y -evaluate_all -sort_by area
```

From Pin: TDSP\_CORE\_GLUE\_INST/FE\_OFC29\_mdr\_3\_/A

To Pin: TDSP\_CORE\_GLUE\_INST/FE\_OFC29\_mdr\_3\_/Y

| Library-cell                  | From       | To      | Arc-type                | Delay  |
|-------------------------------|------------|---------|-------------------------|--------|
| Timing-slack Trans-slack Area | Worst-view |         |                         |        |
| *BUFX1                        |            |         | rise rise combinational | 0.5189 |
| -0.2121                       | 1.9871     | 13.3056 | core+typ-rcMax          |        |
| CLKBUFXL                      |            |         | fall fall combinational | 0.5434 |
| -0.2203                       | 2.1958     | 13.3056 | core+typ-rcMax          |        |
| CLKBUFX3                      |            |         | rise rise combinational | 0.2449 |
| 0.0287                        | 2.6377     | 13.3056 | core+typ-rcMax          |        |
| CLKBUFX2                      |            |         | rise rise combinational | 0.3028 |
| -0.0242                       | 2.4827     | 13.3056 | core+typ-rcMax          |        |
| CLKBUFX1                      |            |         | rise rise combinational | 0.4874 |
| -0.1931                       | 2.0677     | 13.3056 | core+typ-rcMax          |        |
| BUFXL                         |            |         | rise rise combinational | 0.6868 |
| -0.3586                       | 1.6156     | 13.3056 | core+typ-rcMax          |        |

**Tempus Text Command Reference**  
Manual ECO Commands

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|           |        |         |                         |        |
|-----------|--------|---------|-------------------------|--------|
| BUFX3     |        |         | rise rise combinational | 0.2199 |
| 0.0459    | 2.6492 | 13.3056 | core+typ-rcMax          |        |
| BUFX2     |        |         | rise rise combinational | 0.2963 |
| -0.0193   | 2.4806 | 13.3056 | core+typ-rcMax          |        |
| CLKBUFX4  |        |         | rise rise combinational | 0.2021 |
| 0.0668    | 2.7276 | 16.6320 | core+typ-rcMax          |        |
| BUFX4     |        |         | rise rise combinational | 0.1798 |
| 0.0781    | 2.7308 | 16.6320 | core+typ-rcMax          |        |
| CLKBUFX8  |        |         | rise rise combinational | 0.1443 |
| 0.1037    | 2.8471 | 23.2848 | core+typ-rcMax          |        |
| BUFX8     |        |         | rise rise combinational | 0.1216 |
| 0.1078    | 2.8491 | 29.9376 | core+typ-rcMax          |        |
| BUFX12    |        |         | rise rise combinational | 0.1096 |
| 0.0896    | 2.8839 | 33.2640 | core+typ-rcMax          |        |
| BUFX16    |        |         | rise rise combinational | 0.1016 |
| 0.0669    | 2.9011 | 43.2432 | core+typ-rcMax          |        |
| CLKBUFX12 |        |         | rise rise combinational | 0.1259 |
| 0.0571    | 2.9097 | 53.2224 | core+typ-rcMax          |        |
| BUFX20    |        |         | rise rise combinational | 0.0954 |
| 0.0408    | 2.9108 | 53.2224 | core+typ-rcMax          |        |
| CLKBUFX16 |        |         | rise rise combinational | 0.1297 |
| 0.0317    | 2.9166 | 63.2016 | core+typ-rcMax          |        |
| CLKBUFX20 |        |         | rise rise combinational | 0.1262 |
| -0.0181   | 2.9200 | 79.8336 | core+typ-rcMax          |        |

- The following command reports (sorted in increasing order of slack) evaluation results (for A to Y arc) of resizing instance , TDSP\_CORE\_GLUE\_INST/FE\_OFC29\_mdr\_3\_, with each of all the equivalent cells in the library.

```
change_cell -from TDSP_CORE_GLUE_INST/FE_OFC29_mdr_3_/A -to
TDSP_CORE_GLUE_INST/FE_OFC29_mdr_3_/Y -evaluate_all -sort_by slack
```

From Pin: TDSP\_CORE\_GLUE\_INST/FE\_OFC29\_mdr\_3\_/A

To Pin: TDSP\_CORE\_GLUE\_INST/FE\_OFC29\_mdr\_3\_/Y

---

| Library-cell | From        | To   | Arc-type   | Delay |
|--------------|-------------|------|------------|-------|
| Timing-slack | Trans-slack | Area | Worst-view |       |

---

**Tempus Text Command Reference**  
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|           |        |         |                         |        |
|-----------|--------|---------|-------------------------|--------|
| BUFXL     |        |         | rise rise combinational | 0.6868 |
| -0.3586   | 1.6156 | 13.3056 | core+typ-rcMax          |        |
| CLKBUFXL  |        |         | fall fall combinational | 0.5434 |
| -0.2203   | 2.1958 | 13.3056 | core+typ-rcMax          |        |
| *BUFX1    |        |         | rise rise combinational | 0.5189 |
| -0.2121   | 1.9871 | 13.3056 | core+typ-rcMax          |        |
| CLKBUFX1  |        |         | rise rise combinational | 0.4874 |
| -0.1931   | 2.0677 | 13.3056 | core+typ-rcMax          |        |
| CLKBUFX2  |        |         | rise rise combinational | 0.3028 |
| -0.0242   | 2.4827 | 13.3056 | core+typ-rcMax          |        |
| BUFX2     |        |         | rise rise combinational | 0.2963 |
| -0.0193   | 2.4806 | 13.3056 | core+typ-rcMax          |        |
| CLKBUFX20 |        |         | rise rise combinational | 0.1262 |
| -0.0181   | 2.9200 | 79.8336 | core+typ-rcMax          |        |
| CLKBUFX3  |        |         | rise rise combinational | 0.2449 |
| 0.0287    | 2.6377 | 13.3056 | core+typ-rcMax          |        |
| CLKBUFX16 |        |         | rise rise combinational | 0.1297 |
| 0.0317    | 2.9166 | 63.2016 | core+typ-rcMax          |        |
| BUFX20    |        |         | rise rise combinational | 0.0954 |
| 0.0408    | 2.9108 | 53.2224 | core+typ-rcMax          |        |
| BUFX3     |        |         | rise rise combinational | 0.2199 |
| 0.0459    | 2.6492 | 13.3056 | core+typ-rcMax          |        |
| CLKBUFX12 |        |         | rise rise combinational | 0.1259 |
| 0.0571    | 2.9097 | 53.2224 | core+typ-rcMax          |        |
| CLKBUFX4  |        |         | rise rise combinational | 0.2021 |
| 0.0668    | 2.7276 | 16.6320 | core+typ-rcMax          |        |
| BUFX16    |        |         | rise rise combinational | 0.1016 |
| 0.0669    | 2.9011 | 43.2432 | core+typ-rcMax          |        |
| BUFX4     |        |         | rise rise combinational | 0.1798 |
| 0.0781    | 2.7308 | 16.6320 | core+typ-rcMax          |        |
| BUFX12    |        |         | rise rise combinational | 0.1096 |
| 0.0896    | 2.8839 | 33.2640 | core+typ-rcMax          |        |
| CLKBUFX8  |        |         | rise rise combinational | 0.1443 |
| 0.1037    | 2.8471 | 23.2848 | core+typ-rcMax          |        |
| BUFX8     |        |         | rise rise combinational | 0.1216 |

0.1078          2.8491          29.9376          core+typ-rcMax

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-----

- The following command evaluates the resize of instance, TDSP\_CORE\_GLUE\_INST/FE\_OFC29\_mdr\_3\_, with each of all the equivalent cells in the library, and report (sorted in increasing order of slack) the best and worst cell, along with the reference:

```
change_cell -from TDSP_CORE_GLUE_INST/FE_OFC29_mdr_3_/A -to
TDSP_CORE_GLUE_INST/FE_OFC29_mdr_3_/Y -evaluate_all -sort_by slack -max_cells 1
```

From Pin: TDSP\_CORE\_GLUE\_INST/FE\_OFC29\_mdr\_3\_/A

To Pin:     TDSP\_CORE\_GLUE\_INST/FE\_OFC29\_mdr\_3\_/Y

-----  
-----

| Library-cell |             |      | From | To         | Arc-type | Delay |
|--------------|-------------|------|------|------------|----------|-------|
| Timing-slack | Trans-slack | Area |      | Worst-view |          |       |

-----  
-----

|         |        |         |      |                |               |        |
|---------|--------|---------|------|----------------|---------------|--------|
| BUFXL   |        |         | rise | rise           | combinational | 0.6868 |
| -0.3586 | 1.6156 | 13.3056 |      | core+typ-rcMax |               |        |

|         |        |         |      |                |               |        |
|---------|--------|---------|------|----------------|---------------|--------|
| *BUFX1  |        |         | rise | rise           | combinational | 0.5189 |
| -0.2121 | 1.9871 | 13.3056 |      | core+typ-rcMax |               |        |

|        |        |         |      |                |               |        |
|--------|--------|---------|------|----------------|---------------|--------|
| BUFX8  |        |         | rise | rise           | combinational | 0.1216 |
| 0.1078 | 2.8491 | 29.9376 |      | core+typ-rcMax |               |        |

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## List of Valid Timing-types

- combinational
- combinational\_fall
- combinational\_rise
- falling\_edge
- hold\_falling
- hold\_rising
- recovery\_falling
- recovery\_rising
- removal\_falling
- removal\_rising
- rising\_edge
- setup\_falling
- setup\_rising

## Related Commands

- [add\\_repeater](#)
- [delete\\_repeater](#)
- [set\\_eco\\_mode](#)

# change\_inst\_name

```
change_inst_name
-help
-inst inst_name
-newBaseName base_name
```

Changes the base name of the specified instance to the given base name.

**Note:** The `change_inst_name` command does not change the path of hierarchy.

## Parameters

|                           |                                                                                                                                                                                                                                                                |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                     | Outputs a brief description that includes type and default information for each <code>change_inst_name</code> parameter.<br><br>For a detailed description of the command and all of its parameters, use the man command:<br><code>man change_inst_name</code> |
| -inst inst_name           | Specifies the name of the instance that needs to be changed.                                                                                                                                                                                                   |
| -newBaseName<br>base_name | Specifies the new base name for the specified instance.                                                                                                                                                                                                        |

## Example

The following example changes the instance name from `h3/inv3` to `h3/h2/inv1`:

```
tempus> change_inst_name -inst h3/inv3 -newBaseName h2/inv1
```

## Related Commands

- `add_inst`
- `add_net`
- `attach_net`
- `attach_term`
- `delete_inst`
- `delete_net`
- `detach_net`
- `detach_term`



# delete\_inst

```
delete_inst
[-help]
<list_of_instances>
[-module_based <string>]
```

Deletes specified instances. The tool checks whether the instance is physical or logical, and deletes the instance accordingly.

After this command, the software will perform a full timing update when you run the next command that reports timing, for example, [report\\_timing](#).

**Note:** The tool supports this command for flattened designs only.

## Parameters

|                                                   |                                                                                                                                                                                                                                                          |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                                | Outputs a brief description that includes type and default information for each <code>delete_inst</code> parameter.<br><br>For a detailed description of the command and all of its parameters, use the man command:<br><br><code>man delete_inst</code> |
| <code>list_of_instances</code>                    | Specifies the instances to be removed.                                                                                                                                                                                                                   |
| <code>-module_based</code><br><code>string</code> | Specifies the name of the block in which you want to delete the instances.                                                                                                                                                                               |

## Examples

- The following command deletes instance `i1/i2`:  

```
tempus> delete_inst i1/i2
```
- The following command deletes all instances whose names begin with `i`:  

```
tempus> delete_inst [get_cells i*]
```

## Related Commands

- [add\\_inst](#)
- [add\\_net](#)
- [attach\\_net](#)
- [attach\\_term](#)



- `change_inst_name`
- `delete_net`
- `detach_net`
- `detach_term`

# delete\_net

```
delete_net
-help
{<net_name_list> | -all}
-module_based <string>
```

Logically deletes nets from a design. If a net is routed, the wire segments associated with the net are also deleted.

You can use wildcards (\*?) to specify the nets you want Tempus to delete.

## Parameters

|                           |                                                                                                                                                                                                                                                                     |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                     | Outputs a brief description that includes type and default information for each <code>delete_net</code> parameter.<br><br>For a detailed description of the command and all of its parameters, use the <code>man</code> command:<br><br><code>man delete_net</code> |
| -all                      | Deletes all nets from a design.                                                                                                                                                                                                                                     |
| -module_based<br><string> | Deletes the net from the specified Verilog module.                                                                                                                                                                                                                  |
| net_name_list             | Specifies the name of the net to delete.                                                                                                                                                                                                                            |

## Example

- The following command deletes `net1`:  
`tempus> delete_net net1`
- The following command deletes all nets whose name begins with `net1`:  
`tempus> delete_net net1*`
- The following command deletes all nets from the design:  
`tempus> delete_net -all`
- The following command deletes the net with base name `ant_net[1]` from module `H[1]`:  
`tempus> delete_net {\ant_net[1] } -module_based {\H[1] }`

## Related Commands

- `add_inst`

- `add_net`
- `attach_net`
- `attach_term`
- `change_inst_name`
- `delete_inst`
- `detach_net`
- `detach_term`

# delete\_repeater

delete\_repeater

```
[-help] { -inst <string> | -inv_pair <string> } [-early | -late] [[-evaluate_only] [[[-view <string>] [-early] [-late] [-fall] [-rise] [-output_as_tcl]]]]]
```

Deletes a specified buffer or pair of inverters connected back-to-back. In multi-mode multi-corner (MMMC) mode, this command deletes a repeater and updates timing in all views. The command searches backward, then forward, for another inverter pair to delete. This command merges parasitics by default. You might see an unconnected wire after committing the change; however, it has no impact on timing results.

## Note:

You do not need to provide physical information (LEF/DEF) to perform ECO with this command.

## Parameters

|                        |                                                                                                                                                                                                                                                                                                                                                      |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                  | <p>Outputs a brief description that includes type and default information for each <code>delete_repeater</code> parameter.</p> <p>For a detailed description of the command and all of its parameters, use the <code>man</code> command:</p> <pre>man delete_repeater</pre>                                                                          |
| -early                 | <p>Specifies that timing will be evaluated for early analysis. In such a case, only the active hold analysis views will be considered. If this parameter is not specified, timing will be evaluated for late analysis.</p> <p><b>Note:</b> This parameter must be specified with <code>-evaluate_only</code> or <code>-evaluate_all</code> only.</p> |
| -evaluate_only         | <p>Specifies that only timing be evaluated, without any changes to the netlist.</p>                                                                                                                                                                                                                                                                  |
| -fall                  | <p>Specifies that the timing will be evaluated for the fall transition. If this parameter is not specified, timing will be evaluated for both the rise and fall transitions, and the worst result is reported.</p> <p><b>Note:</b> This parameter must be specified with <code>-evaluate_only</code> only.</p>                                       |
| -inv_pair<br><string>  | <p>Specifies pairs of inverters to be deleted/evaluated for deletion.</p>                                                                                                                                                                                                                                                                            |
| -inst inst1<br>[inst2] | <p>Specifies a buffer or inverter (pair) for deletion. Two inverters are deleted only if they are tied together, and both can be deleted.</p> <p>To delete an inverter pair, specify only one of the two inverters.</p>                                                                                                                              |

|                                |                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-late</code>             | <p>Specifies that timing will be evaluated for late analysis. In such a case, only the active setup analysis views will be considered. If this parameter is not specified, timing will still be evaluated for late analysis.</p> <p><b>Note:</b> This parameter must be specified with <code>-evaluate_only</code> or <code>-evaluate_all</code> only.</p> |
| <code>-output_as_tcl</code>    | <p>Displays the output as a TCL list instead of as text.</p>                                                                                                                                                                                                                                                                                               |
| <code>-rise</code>             | <p>Specifies that the timing will be evaluated for the rise transition. If this parameter is not specified, timing will be evaluated for both rise and fall transitions, and the worst result is reported.</p> <p><b>Note:</b> This parameter must be specified with <code>-evaluate_only</code> only.</p>                                                 |
| <code>-view</code><br><string> | <p>Specifies the active analysis views for which timing must be evaluated. If this parameter is not specified, timing will be evaluated for all active analysis views and results reported for the worst view.</p> <p><b>Note:</b> This parameter must be specified with <code>-evaluate_only</code> or <code>-evaluate_all</code> only.</p>               |

## Examples

- Consider three inverters connected in series: `instA -> inv1 -> inv2 -> inv3 -> instB` (where instances `instA` and `instB` are not inverters).
  - The following command deletes inverters `inv1` and `inv2`: -  

```
tempus> delete_repeater -inst inv1
```
  - The following command deletes inverters `inv2` and `inv3`: -  

```
tempus> delete_repeater -inst inv3
```
  - If `inv1` can be deleted, the following command deletes inverters `inv1` and `inv2` (even if `inv3` can also be deleted): -  

```
tempus> delete_repeater -inst inv2
```
  - If `inv1` cannot be deleted but `inv3` can be deleted, the following command deletes inverters `inv2` and `inv3`: -  

```
tempus> delete_repeater -inst inv2
```
- The following command deletes instances `DTMF_INST/RESULTS_CONV_INST/FE_PHC845_r1644_2_` and `DTMF_INST/RESULTS_CONV_INST/FE_PHC845_r1643_3_`:  

```
tempus> delete_repeater -inst {DTMF_INST/RESULTS_CONV_INST/FE_PHC845_r1644_2_
DTMF_INST/RESULTS_CONV_INST/FE_PHC845_r1643_3_}
```
- The following command reports the evaluation results (worst setup timing across all views) of deleting buffer, `EXECUTE_INST/FE_OFC17_sel_op_a_2_`:

```
tempus> delete_repeater -inst EXECUTE_INST/FE_OFC17_sel_op_a_2_ -evaluate_only
```

Evaluating deletion of the given buffer 'EXECUTE\_INST/FE\_OFC17\_sel\_op\_a\_2\_'.

```

```

| Type<br>Sink                 | Timing-slack | Worst-view     | Transition | Trans-slack | Worst-Trans- |
|------------------------------|--------------|----------------|------------|-------------|--------------|
| -----                        |              |                |            |             |              |
| Reference                    | -0.1242      | core+typ-rcMax | rise       | 2.8061      |              |
| TDSP_CORE_GLUE_INST/i_6126/A |              |                |            |             |              |
| Target                       | -0.0331      | core+typ-rcMax | rise       | 2.8575      |              |
| TDSP_CORE_GLUE_INST/i_6126/A |              |                |            |             |              |

- The following command reports the evaluation results (worst setup timing across all views) of deleting buffer, TDSP\_CORE\_GLUE\_INST/FE\_OFC24\_n\_410:

```
tempus> delete_repeater -inst {TDSP_CORE_GLUE_INST/FE_OFC24_n_410 } -evaluate_only -output_as_tcl
```

Evaluating deletion of the given buffer 'TDSP\_CORE\_GLUE\_INST/FE\_OFC24\_n\_410'.

```
{TDSP_CORE_GLUE_INST/FE_OFC24_n_410 buf {Reference -0.1242 core+typ-rcMax rise 2.0936
TDSP_CORE_GLUE_INST/i_685/A1} {Target -0.5868 core+typ-rcMax rise 1.0583
TDSP_CORE_GLUE_INST/i_5408983/A1N}}
```

- The following command reports the evaluation results (worst setup timing across all views) of deleting these two buffers separately (i.e. mutually exclusive):

```
tempus> delete_repeater -inst {TDSP_CORE_GLUE_INST/FE_OFC24_n_410
EXECUTE_INST/FE_OFC17_sel_op_a_2_} -evaluate_only -output_as_tcl
```

Evaluating deletion of the given buffer 'TDSP\_CORE\_GLUE\_INST/FE\_OFC24\_n\_410'.

Evaluating deletion of the given buffer 'EXECUTE\_INST/FE\_OFC17\_sel\_op\_a\_2\_'.

```
{TDSP_CORE_GLUE_INST/FE_OFC24_n_410 buf {Reference -0.1242 core+typ-rcMax rise 2.0936
TDSP_CORE_GLUE_INST/i_685/A1} {Target -0.5868 core+typ-rcMax rise 1.0583
TDSP_CORE_GLUE_INST/i_5408983/A1N}} {EXECUTE_INST/FE_OFC17_sel_op_a_2_ buf {Reference -
0.1242 core+typ-rcMax rise 2.8061 TDSP_CORE_GLUE_INST/i_6126/A} {Target -0.0331
core+typ-rcMax rise 2.8575 TDSP_CORE_GLUE_INST/i_6126/A}}}
```

- The following command reports the evaluation results (hold timing analysis for only 'core+typ-rcMin' view) of deleting buffer, TDSP\_CORE\_GLUE\_INST/FE\_OFC24\_n\_410:

```
tempus> delete_repeater -inst {TDSP_CORE_GLUE_INST/FE_OFC24_n_410 } -evaluate_only -view
core+typ-rcMin -early
```

Evaluating deletion of the given buffer 'TDSP\_CORE\_GLUE\_INST/FE\_OFC24\_n\_410'.

| Type                              | Timing-slack | Worst-view     | Transition | Trans-slack | Worst-Trans-Sink |
|-----------------------------------|--------------|----------------|------------|-------------|------------------|
| Reference                         | 1.0837       | core+typ-rcMin | rise       | 2.0925      |                  |
| TDSP_CORE_GLUE_INST/i_5408983/A1N |              |                |            |             |                  |
| Target                            | 1.1376       | core+typ-rcMin | rise       | 1.7090      |                  |
| TDSP_CORE_GLUE_INST/i_5408983/A1N |              |                |            |             |                  |

- The following command reports the evaluation results (worst setup timing across all views) of deleting buffer, TDSP\_CORE\_GLUE\_INST/FE\_OFC24\_n\_410:

```
tempus> delete_repeater -inst {TDSP_CORE_GLUE_INST/FE_OFC24_n_410 } -evaluate_only -late
```

Evaluating deletion of the given buffer 'TDSP\_CORE\_GLUE\_INST/FE\_OFC24\_n\_410'.

| Type                              | Timing-slack | Worst-view     | Transition | Trans-slack | Worst-Trans-Sink |
|-----------------------------------|--------------|----------------|------------|-------------|------------------|
| Reference                         | -0.1242      | core+typ-rcMax | rise       | 2.0936      |                  |
| TDSP_CORE_GLUE_INST/i_685/A1      |              |                |            |             |                  |
| Target                            | -0.5868      | core+typ-rcMax | rise       | 1.0583      |                  |
| TDSP_CORE_GLUE_INST/i_5408983/A1N |              |                |            |             |                  |

## Related Commands

- [add\\_repeater](#)
- [change\\_cell](#)
- [set\\_eco\\_mode](#)

# detach\_module\_port

```
detach_module_port
[-help]
hinst_name
port_name
```

Detaches the net connected to the specified port on the specified instance.

## Parameters

|            |                                                                                                                                                                                                                                    |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help      | Outputs a brief description that includes the type and default information for each detach_module_port.<br><br>For a detailed description of the command and all of its parameters, use the man command:<br>man detach_module_port |
| hinst_name | Specifies the name of the hierarchical instance.                                                                                                                                                                                   |
| port_name  | Specifies the port on the module.                                                                                                                                                                                                  |

## Example

- The following command detaches port p1 from moduleA:  
tempus> detach\_module\_port moduleA p1



## Related Commands

- `add_inst`
- `add_net`
- `attach_module_port`
- `attach_term`
- `change_inst_name`
- `delete_inst`
- `delete_net`
- `detach_net`
- `detach_term`

# detach\_net

```
detach_net
[-help]
net_name
{list_of_pins | -all }
```

Disconnects a net and connects it to new pins or ports. The syntax is order-dependent.

After this command, the software will perform a full timing update when you run the next command that reports timing, for example, [report\\_timing](#).

**Note:** The software supports this command only for flattened designs.

## Parameters

|              |                                                                                                                                                                                                                                                        |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help        | Outputs a brief description that includes type and default information for each <code>detach_net</code> parameter.<br><br>For a detailed description of the command and all of its parameters, use the man command:<br><br><code>man detach_net</code> |
| -all         | Detaches all terminals connected to the net. You cannot specify a pin/port list if you specify this parameter.                                                                                                                                         |
| net_name     | Specifies the net to connect.                                                                                                                                                                                                                          |
| list_of_pins | Specifies the pins from which the specified net is to be detached.                                                                                                                                                                                     |

## Examples

- The following command detaches `net26` from pins `i1/A`, `i2/B`, and `i2/Z`:

```
tempus> detach_net net26 "i1/A i2/B i3/Z"
```

- The following command is equivalent to the above example.

```
tempus> detach_net [get_nets net26] [get_pins "i1/A i2/B i3/Z"]
```

## Related Commands

- [add\\_inst](#)
- [add\\_net](#)
- [attach\\_net](#)

- `attach_term`
- `change_inst_name`
- `delete_inst`
- `delete_net`
- `detach_term`

# detach\_term

detach\_term  
-help  
inst\_name  
term\_name  
net\_name

Disconnects a terminal from a net. The syntax is order-dependent.

After this command, the software will perform a full timing update when you run the next command that reports timing, for example, [report\\_timing](#).

**Note:** The tool supports this command for flattened designs only.

## Parameters

|           |                                                                                                                                                                                                                                |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help     | Outputs a brief description that includes type and default information for each detach_term parameter.<br><br>For a detailed description of the command and all of its parameters, use the man command:<br><br>man detach_term |
| inst_name | Specifies the instance that contains the terminal you want to detach.                                                                                                                                                          |
| net_name  | Specifies the net to disconnect from the terminal.                                                                                                                                                                             |
| term_name | Specifies the terminal to disconnect from the net. If the terminal does not exist on the specified instance, the software displays an error message, and the command stops.                                                    |

## Examples

The following command detaches net1 from terminal A on inst1:

```
tempus> detach_term inst1 A net1
```

## Related Commands

- [add\\_inst](#)
- [add\\_net](#)
- [attach\\_net](#)
- [attach\\_term](#)

- `change_inst_name`
- `delete_inst`
- `delete_net`
- `detach_net`

# edit\_delete

```
edit_delete
[-help]
[-net <string>]
```

Deletes physical wires, any vias connected to the wires, and violation markers associated with the wires.

**Note:** The `edit_delete` command deletes only those vias that are connected to wires.

## Parameters

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>               | <p>Outputs a brief description that includes type and default information for each <code>edit_delete</code> parameter.</p> <p>For a detailed description of the command and all of its parameters, use the <code>man</code> command:</p> <pre>man edit_delete</pre>                                                                                                                                                             |
| <code>-net &lt;string&gt;</code> | <p>Specifies the net or list of nets from which to delete wires. You can use wildcard values in this field. For example, if you specify <code>V*</code>, it deletes wires from nets <code>VDD</code> and <code>VSS</code>. If you are specifying multiple nets, enclose them in braces.</p> <p><b>Note:</b> By default, the wires and vias shielding the nets specified with <code>edit_delete -net</code> are not deleted.</p> |

## Examples

- The following command deletes the wires from the specified net:  

```
tempus> edit_delete -net net1
```
- The following command deletes the wires from all nets with a name beginning with 'net':  

```
tempus> edit_delete -net net*
```

## Related Commands

- [delete\\_net](#)

# load\_eco

```
load_eco <fileName>
-help
fileName
```

Loads a file that was generated in db format using the `write_eco -format db` command.

## Parameters

|                 |                                                                             |
|-----------------|-----------------------------------------------------------------------------|
| <i>fileName</i> | Specifies the name of the file generated by <code>write_eco</code> to load. |
|-----------------|-----------------------------------------------------------------------------|

## Examples

The following example reads `myfile` generated by the `write_eco` command.

```
tempus> load_eco myfile
```

## Related Commands

`write_eco`

# set\_eco\_mode

set\_eco\_mode

```
[-help]
[-reset]
[-batch_mode {true | false}]
[-honor_dont_touch {true | false}]
[-honor_dont_use {true | false}]
[-honor_fixed_net_wire {true|false}]
[-honor_fixed_status {true | false}]
[-honor_power_intent {true|false}]
[-leq_check {true | false}]
[-prefix_name prefix]
[-refine_place {true|false}]
[-si_effort {low | medium}]
[-update_timing {true | false}]
```

Controls timing updates during ECO changes, checks for dont touch and dont use cells, and allows swapping non-equivalent cells.

## Parameters

|                                  |                                                                                                                                                                                                                                                                     |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                            | Outputs a brief description that includes type and default information for each <code>set_eco_mode</code> parameter.<br><br>For a detailed description of the command and all of its parameters, use the <code>man</code> command:<br><code>man set_eco_mode</code> |
| -batch_mode {true   false}       | Enables or disables the batch mode.                                                                                                                                                                                                                                 |
| -honor_dont_touch {true   false} | Checks for dont touch on insts, cells, and nets.<br><i>Default:</i> true                                                                                                                                                                                            |
| -honor_dont_use {true   false}   | Checks for dont use on cells.<br><i>Default:</i> true                                                                                                                                                                                                               |



|                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>- honor_fixed_net_wire {true false}</pre> | <p>Checks for fixed net wires.</p> <p>It prevents the addition or deletion of fixed nets, so the wires' fixed status remains unchanged.</p> <p><i>Default:</i> <code>true</code></p> <p>When <code>-honor_fixed_net_wire</code> is set to <code>false</code>, <code>add_repeater</code> can insert a cell on the wires with the FIXED/COVER status and change their status from FIXED/COVER to ROUTED.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <pre>-honor_fixed_status {true   false}</pre>  | <p>Does not allow preplaced, fixed instances to be resized.</p> <p><i>Default:</i> <code>true</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <pre>-honor_power_intent {true   false}</pre>  | <p>Performs Multi Supply Voltage Design (MSV) checks during ECO. When this parameter is set to <code>true</code>:</p> <ul style="list-style-type: none"> <li>• <code>change_cell</code> does not allow: <ul style="list-style-type: none"> <li>◦ Resizing a regular cell with an always-on cell (and vice-versa).</li> <li>◦ Resizing with cells belonging to different power domains.</li> </ul> </li> <li>• <code>add_repeater</code> does not allow: <ul style="list-style-type: none"> <li>◦ Adding a buffer to a power domain that it doesn't belong to.</li> <li>◦ Buffering a cross-power domain net can cause MSV violations.</li> <li>◦ Adding a regular buffer to a net that requires an always-on buffer.</li> <li>◦ Adding an always-on buffer to a net that does not require an always-on buffer.</li> </ul> </li> <li>• <code>delete_repeater</code> does not allow deletion of a buffer or an inverter-pair that will cause: <ul style="list-style-type: none"> <li>◦ A redundant isolation cell.</li> <li>◦ A CPF-constrained net to be deleted.</li> </ul> </li> </ul> <p><i>Default:</i> <code>true</code></p> |
| <pre>-leq_check {true   false}</pre>           | <p>Allows the swapping of non-equivalent cells.</p> <p><i>Default:</i> <code>true</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <pre>-prefix_name prefix</pre>                 | <p>Adds a prefix name to be used by ECO commands.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-refine_place {true false}</code>    | <p>Legalizes placement whenever <code>add_repeater</code> or <code>change_cell</code> are used. When set to <code>false</code>, you must run <code>-refine_place</code> after ECO to legalize placement.</p> <p><i>Default:</i> <code>true</code></p> <p><b>Note:</b> Specifying <code>-refine_place true</code> can affect performance when large numbers of cells are added or modified.</p>                                                  |
| <code>-reset</code>                        | Resets all or specified options to the default value.                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>-si_effort {low   medium}</code>     | <p>Specifies the effort level to be used for SI analysis during ECO. This parameter allows you to choose between SI analysis accuracy and run time levels.</p> <ul style="list-style-type: none"> <li><code>low</code>: An approximate SI analysis will be done in this mode.</li> <li><code>medium</code>: More accurate SI analysis using the two SI iterations will be done in this mode.</li> </ul> <p><i>Default:</i> <code>low</code></p> |
| <code>-update_timing {true   false}</code> | <p>Enables or disables the timing update. If this option is set to <code>false</code>, ECO commands do not perform the timing update.</p> <p><i>Default:</i> <code>true</code></p>                                                                                                                                                                                                                                                              |

The batch mode command also performs timing updates (base delay and noise analysis) on the `add_repeater`, `change_cell`, and `delete_repeater` commands only when the batch mode is closed.

The software keeps track of instances where the `add_repeater`, `change_cell`, and `delete_repeater` commands were applied in batch mode. When batch mode is closed, the base delay for these instances is updated, and the incremental noise analysis is triggered for the impacted nets.

The use model of the command is as follows:

```
#start a batch
set_eco_mode -batch_mode true

#perform add_repeater, change_cell, delete_repeater operations
add_repeater -net ...
change_cell -upsized ...
delete_repeater -inst ...

#close the batch
set_eco_mode -batch_mode false
```

At this point, the software automatically triggers the timing updates and incremental signal integrity analysis.

### Note:

- It is recommended to use this mode for improved ECO performance.
- Only ECO commit operations are allowed inside the ECO batch mode. Evaluations are not supported in batch mode.

### Examples

- The following command resets all `set_eco_mode` parameters to their default values:  

```
tempus> set_eco_mode -reset
```
- The following command resets the `-honor_dont_use` parameter to its default value;  

```
tempus> set_eco_mode -reset -honor_dont_use
```
- The following command allows the `change_cell` command to swap non-equivalent cells:  

```
tempus> set_eco_mode -leq_check false
```
- The following command disables the timing update during ECO operations:  

```
tempus> set_eco_mode -update_timing false
```

## write\_eco

```
write_eco
-help
-format {innovus tempus}
-hierarchical
-output <string>
-prefix <string>
[-skip_non_eco_commands]
```

Logs ECO commands into an output file per file format. You can later load the files in the following ways:

- Source the innovus format file in the Innovus tool
- Source the `tempus` format file into the Tempus tool

The following commands can be appended to the report file:

- `add_net`
- `delete_net`
- `add_inst`
- `delete_inst`
- `attach_net`
- `detach_net`
- `attach_term`
- `detach_term`
- `change_cell`
- `add_repeater`
- `delete_repeater`

## Parameters

|                                           |                                                                                                                                                                                                                                                               |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                        | Outputs a brief description that includes type and default information for each <code>write_eco</code> parameter.<br><br>For a detailed description of the command and all of its parameters, use the <code>man</code> command:<br><code>man write_eco</code> |
| <code>-format {innovus<br/>tempus}</code> | Generates an ECO file in the specified format.                                                                                                                                                                                                                |
| <code>-hierarchical</code>                | Generate an ECO file for optimized partitions.                                                                                                                                                                                                                |
| <code>-output &lt;string&gt;</code>       | Specifies the name of the ECO file generated using this command.                                                                                                                                                                                              |
| <code>-prefix &lt;string&gt;</code>       | Specifies the prefix to be used in naming the output ECO file.                                                                                                                                                                                                |
| <code>-<br/>skip_non_eco_commands</code>  | Removes all non-ECO commands, which are not specified by the user, from the automatically generated ECO files.                                                                                                                                                |

## Examples

- The following command logs ECO commands into file `Tempus.eco`, in the `tempus` format:  
`tempus> write_eco -format tempus -output Tempus.eco`
- The following command logs the ECO command into the file `innovus.eco`, in the `innovus` format:  
`tempus> write_eco -format innovus -output innovus.eco`
- The following command removes all non-ECO commands from the automatically generated ECO file, `opt_eco_tempus.tcl`:  
`write_eco -format tempus -output opt_eco_tempus.tcl -skip_non_eco_commands`

## Related Commands

- `load_eco`

---

## Signoff ECO Commands

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- [eco\\_opt\\_design](#)
- [get\\_eco\\_opt\\_mode](#)
- [merge\\_hierarchical\\_def](#)
- [read\\_partition](#)
- [set\\_eco\\_opt\\_mode](#)
- [set\\_filler\\_mode](#)
- [set\\_place\\_mode](#)
- [specifyCellEdgeSpacing](#)
- [specifyCellEdgeType](#)
- [write\\_eco\\_opt\\_db](#)

# eco\_opt\_design

```
eco_opt_design
[-help]
[-area]
[-drv]
[-dynamic]
[-hold]
[-leakage]
[-power]
[-setup]
[-update_timing]
```

Triggers ECO. Depending on the `set_eco_opt_mode` settings, it will either auto-generate the ECO Timing DB or load an existing one, then proceed to the timing fixing phase and finally output an ECO file and detailed diagnostics in a log file.

**Note:** This command requires a 64-bit executable.

## Parameters

|       |                                                                                                                                                                                                                                                                        |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help | Outputs a brief description that includes type and default information for each <code>eco_opt_design</code> parameter.<br><br>For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man eco_opt_design</code> . |
| -area | Reclaims area on the netlist without causing any DRV/Setup/Hold degradation.                                                                                                                                                                                           |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -drv     | <p>Optimizes DRV violations. DRV optimization fixes violations in the following order:</p> <ol style="list-style-type: none"> <li>1. glitch</li> <li>2. drv (max_transition/max_capacitance violations)</li> <li>3. xtalk delta delay</li> </ol> <p><b>Note:</b></p> <ul style="list-style-type: none"> <li>• DRV optimization supports buffer addition and resizing.</li> <li>• CellEM violations are fixed only when <code>set_eco_opt_mode -fix_cell_em</code> is set to <code>true</code>.</li> </ul> |
| -dynamic | <p>Prints the initial and final power numbers. It optimizes the dynamic power to reduce the total power consumption.</p>                                                                                                                                                                                                                                                                                                                                                                                  |
| -hold    | <p>Optimizes hold violations.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| -leakage | <p>Optimizes leakage power. Leakage optimization supports only VT swapping.</p> <p><b>Note:</b> To optimize leakage using resizing in addition to VT swapping, use the following:</p> <pre>set_eco_opt_mode -power_opt_focus eco_opt_design -power</pre>                                                                                                                                                                                                                                                  |
| -power   | <p>Optimizes the total power in the design. During total power optimization, the software performs swapping and sizing on both combinational and sequential cells, and removes buffers when enabled. It reduces leakage, internal, and switching power in the design.</p>                                                                                                                                                                                                                                 |
| -setup   | <p>Optimizes the setup timing violations. The setup optimization supports buffer addition, VT swapping, and resizing.</p>                                                                                                                                                                                                                                                                                                                                                                                 |



|                    |                                                                                                                                                                                                                                            |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -<br>update_timing | <p>Generates updated timing data for use during optimization.</p> <p><b>Note:</b> This parameter applies only to the Tempus Paradime flow or Certus. It is only valid if <code>allow_multiple_incremental</code> is <code>true</code>.</p> |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Examples

- The following commands are used to annotate the parasitics of three active RC corners in the design using SPEF files before running timing analysis or `eco_opt_design`:

```
tempus> read_spef -rc_corner rc_max max.spef.gz
tempus> read_spef -rc_corner rc_min1 min1.spef.gz
tempus> read_spef -rc_corner rc_min2 min2.spef.gz
```

- The following command performs hold fixing while not degrading setup timing:

```
tempus> eco_opt_design -hold
```

- The following command recovers area by performing cell downsizing and buffer deletion, while ensuring no new Setup/Hold/DRV violations are created:

```
tempus> eco_opt_design -area
```

- The following command runs multiple types of optimization in incremental mode and generates a new ECO DB (assuming `-load_eco_opt_db` is not used). It also generates the new timing data to be used during power and setup optimizations:

```
set_eco_opt_mode -allow_multiple_incremental true
eco_opt_design -drv
eco_opt_design -setup
eco_opt_design -hold
eco_opt_design -power -update_timing
eco_opt_design -setup -update_timing
```

## Related Commands

- [set\\_eco\\_opt\\_mode](#)
- [write\\_eco\\_opt\\_db](#)
- [set\\_filler\\_mode](#)
- [set\\_place\\_mode](#)

## get\_eco\_opt\_mode

```
get_eco_opt_mode
[-help]
[-reset]
[-add_inst {true|false}]
[-add_load {true|false}]
[-allow_multiple_incremental {true|false}]
[-allow_skewing {true|false}]
[-along_route_buffering {true|false}]
[-buffer_cell_list cell_list]
[-check_type {early | late | both}]
[-clock_cell_list {}]
[-clock_max_level <INT>]
[-delete_inst {true | false}]
[-drv_margin <FLOAT>]
[-disable_geometry_checks {true|false}]
[-eco_file_prefix name]
[-fix_clock_drv {true|false}]
[-fix_data_drv {true|false}]
[-fix_glitch {true|false}]
[-fix_hold_allow_setup_optimization {true|false}]
[-fix_hold_allow_setup_tns_degrade {true|false}]
[-fix_hold_with_margin <FLOAT>]
[-fix_ir_drop {true|false}]
[-fix_max_cap {true|false}]
[-fix_max_tran {true|false}]
[-fix_si_slew {true|false}]
[-fix_slack_in_range_lower_limit]
[-fix_slack_in_range_upper_limit]
[-fix_xtalk {true|false}]
[-hold_target_slack number]
[-hold_xtalk_delta_threshold <DOUBLE>]
[-hold_xtalk_slack_threshold <DOUBLE>]
[-ignore_drv_checks {true|false}]
[-keep_temp_files {true|false}]
[-legal_only {true|false}]
[-load_cell_list <cell_list>]
[-load_eco_opt_db name]
[-load_irdrop_db name]
[-max_cap_margin <float>]
[-max_slack <FLOAT>]
[-max_paths <paths>]
[-max_run_time <minutes>]
[-max_tran_margin <float>]
```

```
[-nworst <INT>]
[-optimize_core_only {true|false}]
[-optimize_sequential_cells {true|false}]
[-optimize_replicated_modules {true|false}]
[-partition_list_file <string>]
[-pba_effort {medium|high}]
[-power_opt_focus {total|leakage|dynamic|leakageOnly|dynamicOnly}]
[-post_sta_tcl file]
[-pre_sta_tcl file]
[-prefix_name string]
[-pruned_block_name string]
[-resize_inst {true|false}]
[-retime {none| aocv | path_slew_propagation | aocv_path_slew_propagation}]
[-routing_congestion_aware {true|false}]
[-save_eco_opt_db <dir_name>]
[-select_hold_endpoints string]
[-select_drv_net_file <FILE>]
[-select_setup_endpoints string]
[-setup_recovery {true|false}]
[-setup_target_slack number]
[-setup_xtalk_delta_threshold <DOUBLE>]
[-setup_xtalk_slack_threshold <DOUBLE>]
[-skip_drv_net_file <FILE>]
[-specify_hold_endpoints_margin string]
[-specify_setup_endpoints_margin string]
[-swap_inst {true | false}]
[-verbose {true | false}]
[-quiet]
```

Displays the following information about a specified parameter in the Tempus log file and in the Tempus console:

- Parameter name
- Current value
- Type (Boolean, string, and so on)
- Whether the user set the current value

If you do not specify a parameter, the software displays information for all of the parameters.

## Parameters

|                              |                                                                                                                                                                                                                                                                               |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>           | <p>Outputs a brief description that includes type and default information for each <code>get_eco_opt_mode</code> parameter.</p> <p>For a detailed description of the command and all of its parameters, use the <code>man</code> command:</p> <pre>man get_eco_opt_mode</pre> |
| <code>parameter_names</code> | <p>Displays information for the specified parameters. You can specify one or more parameters.</p> <p>See <code>set_eco_opt_mode</code> for descriptions.</p>                                                                                                                  |
| <code>-quiet</code>          | <p>Displays the current settings for the specified parameters in Tcl list format only.</p> <p>If you specify <code>-quiet</code> without any parameters, the software displays the current settings of all <code>set_eco_opt_mode</code> parameters in Tcl list format.</p>   |

## Examples

The following command displays the parameter name as "`-add_inst`", the default value as "`true`", and the current value set in the tool as "`true`".

```
tempus> get_eco_opt_mode -add_inst
```

The output is displayed as shown below:

```
-add_inst true # bool, default=true
true
```

## Related Commands

`set_eco_opt_mode`

## merge\_hierarchical\_def

```
merge_hierarchical_def
[-help]
defFiles <...>
[-add_blockage]
[-delete_blockages_over_partition_macro]
[-ignore_drcfill]
[-incremental_ilm_def]
[-preserve_shape]
[-rdl_def string]
[-rdl_orientation orientation]
[-rdl_placement {x y}]
[-secondary {{suffix def_file_list}...}]
[-skipNets]
[-skip_filler]
[-skip_pg]
[-top_scope]
[-topcell_orientation orientation]
[-topcell_placement {x y}]
[-transform_file string]
[-skip_routing {[pg|signal] [regular|special] [high_layer_routing]}... [-
to_layer routing_layer_name]]
[-use_top_def_die_area]
[-modules <vCell>+ | -hInsts <hInst>+]
```

Merges DEF data from various blocks to a single chip database.

The `merge_hierarchical_def` command provides the following features:

- Takes the TOP DEF file and a set of block DEF files and merges them into a single chip database.
- The Verilog files for each of the blocks and TOP need to be loaded first through `read_verilog`.
- The block and TOP Verilog files (or their DEF files) can be specified in any order. However, if there are empty Verilog module stubs for blocks in some parent Verilog files, ensure that the parent Verilog files are specified before the child Verilog files.
- There is no need to specify the LEF files for the blocks.
- During the course of this command, TOP rows will be cut around the block boundary, and all block rows will be brought back as is.
- Partition definitions are created for each of the assembled blocks as per the DIE AREA statement in the block DEF.
- The wire at the block boundaries will be stitched with the electrically connected net's wires at the TOP.

- All orientations of blocks are supported for assembly. Repeated blocks are also supported. Non-R0 master is also supported.

**Note:** The `save_design` command does not write out physical data added by the `merge_hierarchical_def` command unless you include the Innovus license. You can add one of the possible Innovus licenses by using the `tempus -lic_startup_extra` option when Tempus is started.

**Note:** To prevent ECO timing optimization on hierarchical modules for which the DEF file was not loaded, you can use the following command:

```
set_dont_touch [get_cells <hier_module_name>] true
```

## Parameters

|                                                     |                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                                  | Outputs a brief description that includes the type and default information for each <code>merge_hierarchical_def</code> parameter.<br><br>For a detailed description of the command and its parameters, use the <code>man</code> command:<br><code>man merge_hierarchical_def</code> |
| <code>-add_blockage</code>                          | Specifies that the missing instances from the netlist should be converted into placement blockages.                                                                                                                                                                                  |
| <code>defFiles &lt;...&gt;</code>                   | Specifies DEF files for TOP and blocks to be provided one by one in any order.                                                                                                                                                                                                       |
| <code>-delete_blockages_over_partition_macro</code> | Deletes the top design placement and routing (both rectangular and rectilinear) blockages overlapping partition modules.                                                                                                                                                             |
| <code>-hInsts &lt;hInst&gt;+</code>                 | Specifies the list of hierarchical instances for which physical data needs to be loaded. The tool then reads data from def file(s) associated with the specified hInstance(s).                                                                                                       |
| <code>-ignore_drcfill</code>                        | Ignores the special routing with the DRCFILL shape attribute.                                                                                                                                                                                                                        |
| <code>-incremental_ilm_def</code>                   | Incrementally flatten the ILM def.                                                                                                                                                                                                                                                   |
| <code>-modules &lt;vCell&gt;+</code>                | Specifies the list of modules for which physical data needs to be loaded. The tool then reads data only from def file(s) associated with the specified module(s).                                                                                                                    |
| <code>-preserve_shape</code>                        | Preserves shapes and geometries when DEF data from various blocks is merged into a single chip database.                                                                                                                                                                             |

|                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-rdl_def string</code>                                                       | Specifies RDL def file.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>-rdl_orientation orientation</code>                                          | Specifies orientation for RDL def.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>-rdl_placement {x y}</code>                                                  | Specifies the offset of the RDL def.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <code>-secondary {{suffix def_file_list}...}</code>                                | Specifies the secondary DEF file(s) to load.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>-skip_filler</code>                                                          | Skip merging filler and decap cells.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <code>-skip_pg</code>                                                              | Skips merging routing on pg nets.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>-skipNets</code>                                                             | Skips merging signal nets section.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>-skip_routing {[pg signal] [regular special] [high_layer_routing]}...</code> | <p>Skips merging routing based on the settings specified. This setting can be any combination of the three optional keywords from the following set:</p> <ul style="list-style-type: none"> <li>• <code>[pg signal]</code>: Specifies the type of the net on which the routing is to be skipped. <code>pg</code> - pg net, <code>signal</code> - signal net, <code>both</code> if not specified.</li> <li>• <code>[regular special]</code>: Specifies the type of routing to be skipped. <code>regular</code> - regular routing, <code>special</code> - special routing, <code>both</code> if not specified.</li> <li>• <code>[high_layer_routing]</code>: Specifies that only the routing on high layers is to be skipped, where the layers are calculated automatically. If not specified, the routing on all layers is skipped. The layers can be specified with the <code>-to_layer</code> parameter. In that case, the routing on the layers from the top to the specified layer is skipped.<br/><b>Note:</b> You need to specify the <code>-to_layer</code> parameter explicitly if <code>high_layer_routing</code> is specified in <code>-skip_routing</code>; otherwise, <code>high_layer_routing</code> is not applicable.</li> </ul> <p>For example, you can specify multiple sets of arguments to this parameter, as shown below:</p> <pre>-skip_routing {{pg regular} {signal special high_layer_routing}}</pre> |

|                                               |                                                                                                                                                                                       |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-to_layer routing_layer_name</code>     | Skips the routing of layers from the top to the specified layer (inclusive) when the <code>high_layer_routing</code> option is specified in the <code>-skip_routing</code> parameter. |
| <code>-top_scope</code>                       | Merges routing on top scope nets only.                                                                                                                                                |
| <code>-topcell_orientation orientation</code> | Specifies orientation for topcell.                                                                                                                                                    |
| <code>-topcell_placement {x y}</code>         | Specifies placement for topcell.                                                                                                                                                      |
| <code>-transform_file string</code>           | Reports the final x and y offsets and orientation in the top-level of each LEF block that got replaced by DEF.                                                                        |
| <code>-use_top_def_die_area</code>            | Enables the use of TOP DEF DIE AREA.                                                                                                                                                  |

## Examples

The following command sequence merges DEF data of a top (DTMF\_CHIP) and a block (tdsp\_core) into a single chip database:

### # load timing libraries first

```
tempus> read_lib {./LIBS/tlf/pllclk_slow_4.3.tlf ./LIBS/tlf/ram_128x16A_slow_4.3.tlf
./LIBS/tlf/ram_256x16A_slow_4.3.tlf ./LIBS/tlf/rom_512x16A_slow_4.3.tlf
./LIBS/tlf/slow_4.3.tlf ./LIBS/tlf/tpz973gwc-lite_slow_4.3.tlf ./LIBS/tlf/fast_4.3.tlf
./LIBS/tlf/pllclk_fast_4.3.tlf ./LIBS/tlf/ram_128x16A_fast_4.3.tlf
./LIBS/tlf/ram_256x16A_fast_4.3.tlf ./LIBS/tlf/rom_512x16A_fast_4.3.tlf ./LIBS/tlf/tpz973gbc-
lite_fast_4.3.tlf }
```

### # load LEF files of std cells. No need to load the LEF file of the block to be assembled (tdsp\_core)

```
tempus> read_lib -lef {./LIBS/lef/all.lef.save ./DESIGN/row/DTMF_CHIP/arb.partition.lef}
```

### # load verilog files of top (DTMF\_CHIP) and block (tdsp\_core), in any order.

```
tempus> read_verilog PTN/DTMF_CHIP/DTMF_CHIP.v.gz
tempus> read_verilog PTN/tdsp_core/tdsp_core.v.gz
```

### # specify the top verilog module now

```
tempus> set_top_module DTMF_CHIP
```

### # Assemble top and block DEF files into a single Tempus database

```
tempus> merge_hierarchical_def {PTN/DTMF_CHIP/DTMF_CHIP.def.gz
PTN/tdsp_core/tdsp_core.def.gz}
```



# read\_partition

```
read_partition
[-help]
-verilog fileName
-def fileName
```

The `read_partition` command allows you to load the Verilog/DEF files for a module that is currently treated as a black box (represented by LEF/LIB). This command is needed for hierarchical designs to assemble the top-level database with all the block-level databases. The tool will automatically create a fence for every partition loaded.

**Note:** It is mandatory to provide LEF files for every partition when loading the top-level database.

## Parameters

|                      |                                                                                                                                                                                                                                                            |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                | Outputs a brief description that includes type and default information for each <code>read_partition</code> parameter.<br><br>For a detailed description of the command and all of its parameters, use the man command:<br><code>man read_partition</code> |
| -verilog<br>fileName | Specifies the Verilog file name.                                                                                                                                                                                                                           |
| -def fileName        | Specifies the DEF file name.                                                                                                                                                                                                                               |

## Examples

- The following command assembles the ptn1 block-level database to the top level database already present in the memory.

```
tempus> read_partition -verilog ptn1.v -def ptn1.def
```

## set\_eco\_opt\_mode

```
set_eco_opt_mode
[-help]
[-reset]
[-add_inst {true|false}]
[-add_load {true|false}]
[-allow_multiple_incremental {true|false}]
[-allow_skewing {true|false}]
[-along_route_buffering {true|false}]
[-buffer_cell_list cell_list]
[-check_type {early | late | both}]
[-clock_cell_list {}]
[-clock_max_level <INT>]
[-create_smart_mmmc_db {true | false}]
[-delete_inst {true | false}]
[-dont_touch_partition_list <string>]
[-drv_margin <FLOAT>]
[-disable_geometry_checks {true|false}]
[-eco_file_prefix name]
[-em_type {avg | peak | rms | all}]
[-enable_high_capacity_eco {true | false}]
[-fix_cell_em {true|false}]
[-fix_clock_drv {true|false}]
[-fix_data_drv {true|false}]
[-fix_glitch {true|false}]
[-fix_hold_allow_setup_degrade {none|low|medium}]
[-fix_hold_allow_setup_optimization {true|false}]
[-fix_hold_allow_setup_tns_degrade {true|false}]
[-fix_hold_variation_aware {true | false}]
[-fix_hold_with_margin <FLOAT>]
[-fix_ir_drop {true|false}]
[-fix_max_cap {true|false}]
[-fix_max_tran {true|false}]
[-fix_osus_glitch {true|false}]
[-fix_si_slew {true|false}]
[-fix_slack_in_range_lower_limit <FLOAT>]
[-fix_slack_in_range_upper_limit <FLOAT>]
[-fix_xtalk {true|false}]
[-fix_xtalk_preserve_hold {true|false}]
[-fix_xtalk_preserve_setup {true|false}]
[-high_effort_cell_list <list of cells>]
[-hold_target_slack number]
[-hold_target_slack_per_view {viewName1 slack1 viewName2 slack2 ... }]
[-hold_xtalk_delta_threshold <DOUBLE>]
```

```
[-hold_xtalk_slack_threshold <DOUBLE>]
[-ignore_drv_checks {true|false}]
[-include_hold_path_groups <string>]
[-include_setup_path_groups <string>]
[-inter_partition <string>]
[-inter_partition_exclusive {{blockA blockB} {blockC blockD}}]
[-intra_partition <string>]
[-keep_temp_files {true|false}]
[-legal_only {true|false}]
[-load_cell_list <cell_list>]
[-load_eco_opt_db name]
[-load_irdrop_db name]
[-load_smart_mmmc_db <dirName>]
[-max_cap_margin <float>]
[-max_local_density <float>]
[-max_opt_run_time <minutes>]
[-max_slack <FLOAT>]
[-max_paths <paths>]
[-max_run_time <minutes>]
[-max_tran_margin <float>]
[-nworst <INT>]
[-optimize_core_only {true|false}]
[-optimize_sequential_cells {true|false}]
[-optimize_replicated_modules {true|false}]
[-partition_list_file <string>]
[-path_group_file_name <FILE>]
[-pba_effort {medium|high}]
[-post_mask {true|false}]
[-post_sta_tcl file]
[-power_aware {true|false}]
[-power_opt_focus {total|leakage|dynamic|leakageOnly|dynamicOnly}]
[-pre_sta_tcl file]
[-prefix_name <string>]
[-resize_inst {true|false}]
[-retime {none| aocv | path_slew_propagation | aocv_path_slew_propagation}]
[-retime_depth {default|infinite}]
[-retime_depth_early {default|infinite}]
[-retime_mode {none|path|exhaustive}]
[-route_congestion_threshold threshold_value]
[-routing_congestion_aware {true|false}]
[-save_eco_opt_db <dir_name>]
[-select_hold_endpoints <string>]
[-select_drv_net_file <FILE>]
[-select_drv_pin_file <FILE>]
[-select_setup_endpoints <string>]
[-setup_recovery {true|false}]
```

```
[-setup_target_slack number]
[-setup_target_slack_per_view {viewName 1 slack1 viewName2 slack2 ... }]
[-setup_xtalk_delta_threshold <DOUBLE>]
[-setup_xtalk_slack_threshold <DOUBLE>]
[-skew_core_only {true|false}]
[-skip_drv_net_file <FILE>]
[-smart_mmmc_effort {low|medium|high}]
[-smart_mmmc_force_keep_hold_view_list <string>]
[-smart_mmmc_force_keep_setup_view_list <string>]
[-smart_mmmc_max_view_count {{setup <maxViewCount>} {hold <maxViewCount>} {drv
<maxViewCount>}}]
[-smart_mmmc_reduce_clients {true | false}]
[-specify_hold_endpoints_margin <string>]
[-specify_setup_endpoints_margin <string>]
[-swap_inst {true | false}]
[-touch_partition_list <string>]
[-upsample_in_power_opt {true|false}]
[-use_ga_filler_list {<list_of_Gate_Array_filler_cells_name>}]
[-use_pfc_decap_list <cell_list>]
[-use_pfc_filler_list <cell_list>]
[-use_pfc_regular_list <cell_list>]
[-verbose {true | false}]
```

Sets the global parameters for ECO. Parameters that you specify with `set_eco_opt_mode` are then used automatically when you run ECO optimization by running the `eco_opt_design` command.

## Parameters

|                                     |                                                                                                                                                                                                                                                                                                   |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                  | <p>Outputs a brief description that includes type and default information for each <code>set_eco_opt_mode</code> parameter.</p> <p>For a detailed description of the command and its parameters, use the <code>man</code> command: <code>man set_eco_opt_mode</code>.</p>                         |
| <code>-reset</code>                 | <p>Resets all or the specified parameters to their default value.</p>                                                                                                                                                                                                                             |
| <code>-add_inst {true false}</code> | <p>Specifies whether the timing optimizer is allowed to add buffer/delay cell instances. This parameter applies to setup, hold, and DRV optimization.</p> <p><b>Default:</b> <code>true</code></p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p> |

|                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <pre>-add_load {true false}</pre>                   | <p>Specifies whether the Hold optimizer can add dummy cell instances.</p> <p><i>Default:</i> false</p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <pre>-allow_multiple_incremental {true false}</pre> | <p>Allows running multiple types of optimization in incremental mode. Note that there should not be any <code>report_*</code> commands between successive <code>eco_opt_design</code> commands. The recommended order to fix violations is DRV, Hold and Setup. Hold and Setup can be interchanged but DRV optimization must be performed first. You must specify a unique ECO file prefix before each optimization step. You can then source all the ECO files in the same order as they were generated to run the What-If analysis later.</p> <p><i>Default:</i> false</p> <p><b>Note:</b></p> <p>If set, this parameter must be applied before <code>eco_opt_design</code>.</p> |
| <pre>-allow_skewing {true false}</pre>              | <p>Allows clock skew to be applied during Setup timing optimization to improve Setup slack further.</p> <p><i>Default:</i> false</p> <p><b>Note:</b></p> <ul style="list-style-type: none"> <li>• If set, this parameter must be applied before <code>eco_opt_design</code>.</li> <li>• When HC-ECO (High Capacity ECO) is enabled, the <code>-set_eco_opt_mode -allow_skewing true</code> setting must be set before the design is loaded (that is, before the <code>distribute_read_design</code> command).</li> </ul>                                                                                                                                                           |

|                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <code>-along_route_buffering {true false}</code> | <p>When set to <code>true</code>, it uses the nodes' location provided through the parasitic data to insert buffers along the routing topology, not only at the start or end of the net. This helps to better balance the slew or the load for long nets, which is typically needed for Setup timing optimization or DRV fixing.</p> <p><i>Default:</i> <code>true</code>.</p> <p><b>Note:</b></p> <ul style="list-style-type: none"><li>• The prerequisite of this parameter is that parasitic with the node location is provided to <code>eco_opt_design</code>. Further, the node location parasitic should be read using the <code>-starN</code> parameter (For example: <code>read_spef -starN</code>).</li><li>• If set, this parameter must be applied before <code>eco_opt_design</code>.</li></ul> |
| <code>-buffer_cell_list cell_list</code>         | <p>Specifies a list of buffer/delay cells that can be used for the timing optimizer, regardless of whether those cells have a <code>dont_use</code> attribute. In the Setup fixing mode, this parameter is ignored to give the tool greater flexibility to insert any usable buffer cells. When the list is empty, the tool will automatically identify the list of usable buffer/delay cells.</p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                                                                                                                                           |

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| <pre>-check_type {early   late   both}</pre>  | <p>Specifies the type of ECO Timing DB to be generated in PBA mode. <code>early</code> specifies hold ECO Timing DB, and <code>late</code> specifies setup ECO Timing DB. When set to “<code>both</code>”, the ECO Timing DB generation will be done in PBA for both Setup and Hold timing.</p> <p><i>Default:</i> both</p> <p><b>Note:</b></p> <ul style="list-style-type: none"> <li>• This parameter is only active when the <code>-retiming</code> option is being set.</li> <li>• If set, this parameter must be applied before the <code>write_eco_opt_db</code>.</li> </ul> |
| <pre>-clock_cell_list {}</pre>                | <p>Restricts the list of buffers/inverters/gating or any other cells that are allowed to be used by ECO on the clock tree.</p> <p><i>Default:</i> empty</p> <p><b>Note:</b> If set, this parameter must be applied before the <code>eco_opt_design</code>.</p>                                                                                                                                                                                                                                                                                                                     |
| <pre>-clock_max_level &lt;INT&gt;</pre>       | <p>Specifies the maximum levels in clock skewing, thus limiting the number of clock tree levels and avoiding the creation of an unbalanced clock tree. This parameter sets an upper boundary when adding a buffer or pair of inverters to the clock nets. The software counts the number of levels in the clock tree branch being modified to ensure that the maximum level will be equal to or below the value set by this parameter.</p> <p><i>Default:</i> 0</p> <p><b>Note:</b> If set, this parameter must be applied before the <code>eco_opt_design</code>.</p>             |
| <pre>-create_smart_mmmc_db {true false}</pre> | <p>Enables saving the additional compact timing information in the ECO DB directory during the STA session. When set to <code>true</code>, the parameter must be applied before the <code>write_eco_opt_db</code> command and does not affect the <code>eco_opt_design</code> command.</p>                                                                                                                                                                                                                                                                                         |

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| <pre>-delete_inst {true   false}</pre>               | <p>Enables buffer deletion when doing timing, power, or area reduction. This option is not honored by the <code>-leakage</code> parameter of <code>eco_opt_design</code>, since it only performs swapping.</p> <p><i>Default:</i> <code>true</code></p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                                               |
| <pre>-dont_touch_partition_list &lt;string&gt;</pre> | <p>Selects a list of partition module names not to be touched during optimization(s). For paths that traverse across the specified hierarchies, optimization would only be applied to a portion of the paths that lie on the non-dont_touch side.</p> <p><b>Notes:</b></p> <ul style="list-style-type: none"> <li>• This parameter is exclusive to <code>touch_partition_list</code>.</li> <li>• This parameter must be applied before <code>eco_opt_design</code> and can be changed between each call of <code>eco_opt_design</code>.</li> </ul>                   |
| <pre>-disable_geometry_checks {true false}</pre>     | <p>Performs Vth swapping without checking the pin and geometry of a design. You can relax/stop/avoid the geometry checks during Vth swapping so that Tempus only checks the cells while performing Vth swapping. This will lead to more Vth swaps. The counterpart to this is that <code>ecoRouting</code> will be needed once the ECOs are applied, since pin geometries might always match between old and new cells.</p> <p><i>Default:</i> <code>false</code></p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p> |



|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <pre>-drv_margin &lt;FLOAT&gt;</pre>         | <p>Specifies the ratio by which the DRV check would be relaxed or tightened. It scales the maximum capacitance and maximum transition constraints based on the specified margin decimal value.</p> <p>For example, specifying a value of 0.2 for this parameter tightens the margin (makes it more pessimistic) by 20 percent during optimization, and specifying a value of -0.2 relaxes the margin (makes it more optimistic) by 20 percent.</p> <p>This margin is honored for DRV optimization.</p> <p><i>Default:</i> 0</p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p> |
| <pre>-eco_file_prefix name</pre>             | <p>Specifies a prefix name for the ECO files generated.</p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <pre>-em_type {avg   peak   rms   all}</pre> | <p>Fixes the electromigration (EM) type during DRV optimization in standalone Tempus ECO. You can specify any one of the following options:</p> <ul style="list-style-type: none"> <li>• <code>avg</code>: Specifies the average current limit violations.</li> <li>• <code>peak</code>: Specifies the peak current limit violations.</li> <li>• <code>rms</code>: Specifies the Root-Mean-Square (RMS) current limit violations.</li> <li>• <code>all</code>: Specifies the worst of the <code>avg/peak/rms</code> violations.</li> </ul> <p><i>Default:</i> <code>all</code></p>                                             |

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| <pre>-enable_high_capacity_eco {true   false}</pre> | <p>Enables database size reduction on the manage session for high-capacity ECO timing/power closure, based on timing violations, a list of partition names, or user-defined path_group. You must apply this parameter before the <code>write_eco_opt_db</code> command, whereas it is not required during the ECO fixing session. Note that in the Tempus Paradime flow or Certus, this parameter is replaced by <code>set_distribute_mmmc_mode -enable_high_capacity_eco</code>, and is set before <code>distribute_read_design</code>.</p> <p><i>Default:</i> false</p>            |
| <pre>-fix_cell_em {true false}</pre>                | <p>Fixes cell-level signal electromigration violation during DRV by resizing and applying additional transformational techniques.</p> <p><i>Default:</i> false</p> <p><b>Note:</b></p> <ul style="list-style-type: none"> <li>• If set, this parameter must be applied before <code>eco_opt_design -drv</code>.</li> <li>• If set during DRV fixing, the tool will fix only the cell EM violations. This implies that fixing of max_transition/max_capacitance/SI glitch/Xtalk/IR drop will be disabled.</li> </ul>                                                                  |
| <pre>-fix_clock_drv {true   false}</pre>            | <p>Fixes max_cap/max_tran violations on the clock networks. This is performed only when the <code>eco_opt_design -drv</code> command is run.</p> <p><i>Default:</i> false</p> <p><b>Note:</b></p> <ul style="list-style-type: none"> <li>• For clock networks DRV fixing, only the cells listed in the clock cell list (specified using the <code>set_eco_opt_mode -clock_cell_list</code> parameter) will be used. The DRV fixing is not performed if a cell list is not provided.</li> <li>• If set, this parameter must be applied before <code>eco_opt_design</code>.</li> </ul> |

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| <pre>-fix_data_drv {true   false}</pre>                    | <p>Fixes max_cap/max_tran violations on the data networks. This is performed only when the <code>eco_opt_design -drv</code> command is run.</p> <p><i>Default:</i> <code>true</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <pre>-fix_glitch {true false}</pre>                        | <p>Performs SI glitch fixing during DRV fixing using resizing and buffering techniques.</p> <p><i>Default:</i> <code>false</code></p> <p><b>Note:</b></p> <ul style="list-style-type: none"> <li>• If set, this parameter must be applied before <code>write_eco_opt_db</code>.</li> <li>• The glitch fixing is done before max_tran/max_cap fixing.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <pre>-fix_hold_allow_setup_degrade {none low medium}</pre> | <p>Guides the Hold fixing engine in determining the effort required to resolve timing violations, and the flexibility it has to affect the Setup timing.</p> <p><i>Default:</i> <code>none</code></p> <ul style="list-style-type: none"> <li>• When set to <code>low</code>, the Hold optimization engine will buffer sink pins that have large Setup positive slack, even though the net Setup slack is barely meeting timing.</li> <li>• When set to <code>medium</code>, the Hold optimization engine will behave the same as in the <code>low</code> mode but will also allow Setup TNS to degrade, if required.</li> </ul> <p><b>Note:</b></p> <ul style="list-style-type: none"> <li>• This parameter is honored by the Hold optimization engine.</li> <li>• If set, this parameter must be applied before <code>eco_opt_design</code>.</li> <li>• When <code>-fix_hold_allow_setup_degrade</code> is set <code>low</code> or <code>medium</code>, the <code>-fix_hold_allow_setup_tns_degrade</code> parameter is ignored.</li> </ul> |

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| <pre>-fix_hold_allow_setup_optimization {true false}</pre> | <p>When set to <code>true</code>, it performs incremental Setup fixing during Hold fixing to create more Setup margins for Hold violated nodes that could not be fixed earlier due to Setup timing conflicts.</p> <p><i>Default:</i> <code>false</code></p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                            |
| <pre>-fix_hold_allow_setup_tns_degrade {true false}</pre>  | <p>When active, the setup's total negative slack can be degraded during hold time violation fixing while still maintaining the setup's worst negative slack in every setup active view.</p> <p><i>Default:</i> <code>false</code></p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                                                  |
| <pre>-fix_hold_variation_aware {true false}</pre>          | <p>When set to <code>true</code>, more sizing and Vth swapping are performed during Hold fixing to reduce the cost of buffer insertion. This is particularly effective when the Hold timing paths have incremental delay due to SI or large SOCV-impacted instances.</p> <p><i>Default:</i> <code>false</code></p> <p><b>Note:</b></p> <ul style="list-style-type: none"> <li>• This parameter is only honored by <code>eco_opt_design -hold</code>.</li> <li>• If set, this parameter must be applied before <code>eco_opt_design</code>.</li> </ul> |
| <pre>-fix_hold_with_margin &lt;FLOAT&gt;</pre>             | <p>Specifies a slack margin value in nanoseconds for hold-time violation fixing. The purpose is to overcome any hold violations by the amount of the margin value. This margin is not applied to nets that are already meeting the hold target slack.</p> <p><i>Default:</i> <code>0</code></p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                        |

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| <code>-fix_ir_drop {true false}</code>                     | <p>Fixes the IR drop voltages for all instances in the design.</p> <p><i>Default:</i> false</p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                              |
| <code>-fix_max_cap {true false}</code>                     | <p>Fixes max_cap violations only when fixing DRV</p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                         |
| <code>-fix_max_tran {true false}</code>                    | <p>Fixes max_tran violations only when fixing DRV</p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                        |
| <code>-fix_osus_glitch {true false}</code>                 | <p>Fix nets due to overshoot and undershoot glitch during DRV optimization.</p> <p><i>Default:</i> false</p>                                                                                                                                                                                                                                |
| <code>-fix_si_slew {true false}</code>                     | <p>Reports SI slews violations and fixes during DR optimization. This is run as the first step of optimization before proceeding with any other incremental optimization.</p> <p><i>Default:</i> false</p> <p><b>Note:</b> If set, this parameter must be applied before <code>write_eco_opt_db</code> and <code>eco_opt_design</code>.</p> |
| <code>-fix_slack_in_range_lower_limit &lt;FLOAT&gt;</code> | <p>Sets the lower slack limit for a violated path to be considered during timing closure.</p> <p><i>Default:</i> -1e+30</p>                                                                                                                                                                                                                 |
| <code>-fix_slack_in_range_upper_limit &lt;FLOAT&gt;</code> | <p>Sets the upper slack limit for a violated path to be considered during timing closure.</p> <p><i>Default:</i> 1e+30</p>                                                                                                                                                                                                                  |
| <code>-fix_xtalk {true false}</code>                       | <p>Reduces crosstalk on the net that violates the thresholds set by you.</p> <p><i>Default:</i> false</p> <p><b>Note:</b> If set, this parameter must be applied before <code>write_eco_opt_db</code> and <code>eco_opt_design</code>.</p>                                                                                                  |

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| <pre>-fix_xtalk_preserve_hold {true false}</pre>  | <p>When this parameter is set to <code>true</code>, the Xtalk fixin engine is not allowed to degrade Hold timing.</p> <p><i>Default:</i> <code>false</code></p> <p><b>Note:</b></p> <ul style="list-style-type: none"><li>• This parameter is only honored by <code>eco_opt_design -drv</code> when Xtalk fixing is enabled.</li><li>• If set, this parameter must be applied before <code>eco_opt_design</code>.</li></ul>  |
| <pre>-fix_xtalk_preserve_setup {true false}</pre> | <p>When this parameter is set to <code>true</code>, the Xtalk fixin engine is not allowed to degrade Setup timing.</p> <p><i>Default:</i> <code>false</code></p> <p><b>Note:</b></p> <ul style="list-style-type: none"><li>• This parameter is only honored by <code>eco_opt_design -drv</code> when Xtalk fixing is enabled.</li><li>• If set, this parameter must be applied before <code>eco_opt_design</code>.</li></ul> |

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| <code>-high_effort_cell_list &lt;list of cells&gt;</code> | <p>Enables the controlled utilization of high-leakage or high-power cells during signoff optimization. These cells are typically classified as <code>dont_use</code> <code>dont_touch</code> due to their power characteristics; however, they may be beneficial in specific scenarios. When this parameter is specified, ensure that the <code>don't_use</code> constraint is set to <code>false</code> for cells included in <code>-high_effort_cell_list</code>.</p> <ul style="list-style-type: none"><li>◦ In setup timing optimization, the cells are considered only if they are essential to meet timing requirements without causing a significant increase in power consumption.</li><li>◦ In power optimization, the cells may be inserted if their usage leads to an overall decrease in power usage, such as enhancements in fanin/fanout cone.</li></ul> <p>This parameter permits considering the specified cells as high-effort resources, using them only when necessary to enhance timing or power efficiency, while maintaining the broader design objective of reducing power impact.</p> <p><i>Default:</i> <code>empty</code></p> |
| <code>-hold_target_slack number</code>                    | <p>Specifies a target slack value in nanoseconds for hold timing.</p> <p><i>Default:</i> <code>0</code></p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

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| <pre>-hold_target_slack_per_view {viewName 1 slack1 viewName2 slack2 ... }</pre> | <p>When enabled, this parameter allows for the for of a specific hold target slack per view. If a given Hold active view is not listed in this parameter, the software will use the global Hold target slack set by the <code>-hold_target_slack</code> parameter for that view.</p> <p><i>Default:</i> "" (empty string)</p> <p><b>Note:</b></p> <ul style="list-style-type: none"> <li>• This parameter is honored by Setup, Hold, Area, Power, and Leakage optimization engines.</li> <li>• If set, this parameter must be applied before <code>eco_opt_design</code>.</li> </ul> |
| <pre>-hold_xtalk_delta_threshold &lt;DOUBLE&gt;</pre>                            | <p>Fixes the nets with the crosstalk delta delay value equal to or greater than a specified threshold value for hold views.</p> <p><i>Default:</i> 0.3 ns</p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                                                                                                                                                         |
| <pre>-hold_xtalk_slack_threshold &lt;DOUBLE&gt;</pre>                            | <p>Fixes the nets with the slack threshold value equal to or less than a specified threshold value for hold views.</p> <p><i>Default:</i> 1000 ns</p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                                                                                                                                                                 |
| <pre>-ignore_drv_checks {true false}</pre>                                       | <p>Ignores DRV checks for the <code>max_tran</code> and <code>max_cap</code> degradation during the start of an optimization session and for any subsequent optimizations when running in incremental mode.</p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                                                                                                       |
| <pre>-include_hold_path_groups &lt;string&gt;</pre>                              | <p>Allows you to select which path group to consider for Hold timing during <code>eco_opt_design</code>. It is set before <code>write_eco_opt_db</code>, and also specified before <code>eco_opt_design</code> in case of a different session.</p>                                                                                                                                                                                                                                                                                                                                   |



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| <pre>-include_setup_path_groups &lt;string&gt;</pre> | <p>Allows you to select which path group to have the interface for Setup timing during <code>eco_opt_design</code> is set before <code>write_eco_opt_db</code>, and also specifies before <code>eco_opt_design</code> in case of a different session.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <pre>-inter_partition &lt;string&gt;</pre>           | <p>Selects a list of partition module names to be optimized for the interface logic of the specified partition. During the High Capacity ECO flow, it will reduce the database to have the interface logic of the listed partitions only. Each listed partition is treated separately, thus all from/to paths for each partition are included. This parameter must be applied before <code>write_eco_opt_db</code> and <code>eco_opt_design</code>. In case of the Tempus Paradime flow or Certus, it must be applied before <code>distribute_read_design</code>.</p> <p><b>Note:</b></p> <p>It is mandatory to list the partition name inside the file specified with the <code>-partition_list_file</code> parameter.</p> <ul style="list-style-type: none"> <li>• For High Capacity ECO flow, the <code>-partition_list_file</code> parameter should be specified before <code>distribute_read_design</code>.</li> <li>• For non-High Capacity ECO flow, the <code>-partition_list_file</code> parameter should be specified before the first optimization.</li> </ul> <p>However, any partition name, which is a part of <code>inter_partition</code> and missing in <code>-partition_list_file</code> will result in an error, as shown below:</p> <pre>**ERROR: (IMPESO-616): Partition 'tdsp_core' is missing in -partition_list_file file. Partition specified for inter/intra partition fixing should be present in -partition_list_file list as well.</pre> <p>In case of Certus, since <code>-partition_list_file</code> is not required, the software will use <code>-distributed_opt_block_restore_file</code> to identify the partition names.</p> |

```
-inter_partition_exclusive {{blockA blockB}
{blockC blockD}}
```

Selects a list of partition module pair names to be optimized for the interface logic between the specified partitions. During the High Capacity ECO flow, the tool will reduce the database to have the interface logic of the listed partition pair only. Each listed partition pair is treated separately, and all from/to paths between partition pairs are included.

#### Notes:

- This parameter must be applied before `write_eco_opt_db` and `eco_opt_design`. However, in the case of the Tempus Paradigm flow or Certus, it must be applied before `distribute_read_design`.
- It is mandatory to list the partition name(s) inside the file specified using the `-partition_list_file` parameter:
  - For High Capacity ECO flow, the `-partition_list_file` parameter must be specified before `distribute_read_design`.
  - For non-High Capacity ECO flow, the `-partition_list_file` parameter must be specified before the first optimization.

However, an error is issued when any partition name, which is a part of `-inter_partition_exclusive` is missing in `-partition_list_file`, as shown below:

```
**ERROR: (IMPESO-616): Partition 'tdsp_core'
is missing in -partition_list_file file.
Partition specified for
inter/intra/inter_partition_exclusive
partition fixing should be present in -
partition_list_file list as well.
```

In case of Certus, since `-partition_list_file` is not required, the software will use `-distributed_opt_block_restore_file` to identify the partition names.

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| <pre>-intra_partition &lt;string&gt;</pre> | <p>Selects a list of partition module names to be optimized for the core logic of the specified partition. During the High Capacity ECO flow, it will reduce the database having the core logic of the listed partitions only. This parameter must be applied before <code>write_eco_opt_db</code> and <code>eco_opt_design</code>. In case of the Tempus Paradim flow or Certus, it must be applied before <code>distribute_read_design</code>.</p> <p><b>Note:</b></p> <p>It is mandatory to list the partition name inside the file specified with the <code>-partition_list_file</code> parameter.</p> <ul style="list-style-type: none"> <li>• For High Capacity ECO flow, the <code>-partition_list_file</code> parameter should be specified before <code>distribute_read_design</code>.</li> <li>• For non- High Capacity ECO flow, the <code>-partition_list_file</code> parameter should be specified before first optimization.</li> </ul> <p>However, any partition name, which is a part of <code>intra_partition</code> and missing in <code>-partition_list_file</code> will result in an error :</p> <pre>**ERROR: (IMPESO-616): Partition 'tdsp_core' is missing in -partition_list_file file. Partition specified for inter/intra partition fixing should be present in -partition_list_file list as well.</pre> <p>In case of Certus, since <code>-partition_list_file</code> is not required, the software will use <code>-distributed_opt_block_restore_file</code> to identify the partition names.</p> |
| <pre>-keep_temp_files {true false}</pre>   | <p>Keeps all the temporary files generated by DMMMC and Tempus clients for debugging purposes.</p> <p><i>Default:</i> false</p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

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| <pre>-legal_only {true false}]</pre>         | <p>Enables the tool to use the physical data to search for legal location while doing any buffer insertion or instance movement. This ensures that there is no instance overlap at the end of the ECO fixing stage, and after sourcing the ECO file in the implementation tool, you can directly run ECO routing.</p> <p>When this parameter is set to <code>false</code>, <code>eco_opt_design</code> will not check for legal location, although physical data is provided. In <code>eco_opt_design</code> mode, <code>eco_opt_design</code> behaves in a logical mode flow, which means that physical data will be ignored, and this might create an instance overlap, and it will not check local density while inserting buffers. Also, the routing congestion checks are skipped when this parameter is set to <code>false</code>. So it is recommended that you first run ECO fixing in the legal-only mode (which is the default mode) and then, depending on the requirement, you may perform an incremental routing by setting <code>-legal_only</code> to <code>false</code>.</p> <p><i>Default:</i> <code>true</code></p> <p><b>Note:</b></p> <ul style="list-style-type: none"> <li>• The <code>-legal_only</code> parameter is also honored by the Setup/Hold/DRV fixing engines.</li> <li>• If set, this parameter must be applied before <code>eco_opt_design</code>.</li> </ul> |
| <pre>-load_cell_list &lt;cell_list&gt;</pre> | <p>Specifies the list of load cells to be inserted during Setup/Hold fixing, regardless of whether those cells have a <code>dontUse</code> attribute.</p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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| <code>-load_eco_opt_db name</code>               | <p>Specifies the path of the directory where previously generated ECO Timing DB files are kept. If not specified, the ECO Timing DB files will be generated.</p> <p><i>Default=""</i></p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                                                                                                                     |
| <code>-load_irdrop_db name</code>                | <p>Specifies the path of the directory where the command, <code>debug_irdrop</code>, generated IR drop DB files are kept.</p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                                                                                                                                                                                 |
| <code>-load_smart_mmmc_db &lt;dirName&gt;</code> | <p>Points Tempus ECO to the ECO DB directory containing the additional compact timing information during the ECO fixing session. This parameter must be set before the design is loaded and must point to an ECO DB directory, which was previously generated through <code>write_eco_opt_db</code> using the setting, <code>set_eco_opt_mode -create_smart_mmmc true</code>.</p> <p><b>Note:</b> Before <code>eco_opt_design</code>, you still need to point to the ECO DB directory using the setting, <code>set_eco_opt_mode -load_eco_opt_db &lt;dirName&gt;</code>.</p> |
| <code>-max_cap_margin &lt;float&gt;</code>       | <p>Specifies the ratio by which the max_capacitance constraints would be relaxed or tightened. The parameter takes values between the range of -1 and 0.99</p> <p><i>Default:0</i></p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                                                                                                                        |

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| <code>-max_local_density &lt;float&gt;</code>  | <p>Applies a local placement density threshold that Tempus will honor during timing optimization. When the local density for a given area has overpassed the threshold value, the software stops adding or moving more instances in that local area.</p> <p>The default value of this parameter is <code>-1</code>, which means that the software will not perform any local placement density checks because by default it is running with <code>-legal_only true</code>.</p> <p>This parameter accepts values greater than 0. For example, to allow a maximum local placement density of 98%, set the value to 0.98.</p> <p><i>Default:</i> <code>-1</code></p> <p><b>Note:</b></p> <ul style="list-style-type: none"><li>• This parameter is honored by DRV, Setup, and Hold optimization engines.</li><li>• If set, this parameter must be applied before <code>eco_opt_design</code>.</li><li>• The <code>max_local_density</code> (greater than 1) and <code>legal_only false</code> settings can cause some placement in later stages while doing <code>eco_route</code>.</li></ul> |
| <code>-max_opt_run_time &lt;minutes&gt;</code> | <p>Specifies the timeout period for the timing or power optimization phase of the <code>set_eco_opt_mode</code> command. When the value of this parameter is set to a positive value, the software restricts the optimization runtime by the value, specified in minutes.</p> <p><i>Default:</i> <code>0</code></p> <p>This means that there is no limit for the timeout period. When the timeout period is reached, the software will exit the optimization phase, and displays the Tempus prompt with a stable DB.</p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

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| <p><code>-max_slack &lt;FLOAT&gt;</code></p> | <p>Specifies the maximum slack value to be used during ECO Timing DB generation when using PBA. Setting a positive <code>-max_slack</code> value forces the tool to retime positive GBA slack paths so that ECO optimization engine can benefit from an extra PBA Setup slack credit when doing Hold fixing or Power recovery. A positive <code>-max_slack</code> value implies larger runtime during ECO Timing DB generation, but tool will retime only as many paths as what is specified through the <code>-max_paths</code> parameter.</p> <p><i>Default:</i> 0</p> <p><b>Note:</b></p> <ul style="list-style-type: none"> <li>• This parameter is only active when the <code>-retime</code> parameter is being set.</li> <li>• If set, this parameter must be applied before <code>write_eco_opt_db</code>.</li> </ul>                                                                                         |
| <p><code>-max_paths &lt;paths&gt;</code></p> | <p>Specifies the worst path found for each endpoint up to the number specified by the <code>-max_paths</code> parameter during ECO Timing DB generation. By default, the tool will automatically set a value for <code>-max_paths</code> that would lead to good PBA coverage of the design within reasonable runtime. You can overwrite this by setting a specific <code>max_paths</code> value, but it is recommended to keep larger than 1 million. In any case, you must ensure that <code>-max_paths</code> value is superior or equal than the <code>-nworst</code> value multiplied by the number of GBA violated endpoints.</p> <p><i>Default:</i> 5000000</p> <p><b>Note:</b></p> <ul style="list-style-type: none"> <li>• This parameter is only active when the <code>-retime</code> parameter is set.</li> <li>• If set, this parameter must be applied before <code>write_eco_opt_db</code>.</li> </ul> |

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| <code>-max_run_time &lt;minutes&gt;</code>  | <p>Specifies the time out period (in minutes) after which <code>eco_opt_design</code> will exit the session.</p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>-max_tran_margin &lt;float&gt;</code> | <p>Specifies the ratio by which the <code>max_transition</code> constraints would be relaxed or tightened. The parameter takes values between the range of -1 and 0.99.</p> <p><i>Default:</i> 0.</p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                                                                                                                                                                |
| <code>-nworst &lt;INT&gt;</code>            | <p>Specifies the number of paths to be enumerated for each endpoint. By default, the tool will automatically set a value for <code>-nworst</code> that would lead to a good PBA coverage of the design with reasonable run time. You can overwrite this by setting a specific <code>-nworst</code> value, but it is recommended to keep it larger than 50.</p> <p><i>Default:</i> 50</p> <p><b>Note:</b></p> <ul style="list-style-type: none"><li>• This parameter is only active when the <code>-retime</code> parameter is set.</li><li>• If set, this parameter must be applied before <code>write_eco_opt_db</code>.</li></ul> |



|                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <pre>-optimize_core_only {true false}</pre>          | <p>When set to <code>true</code>, enables the tool to automatically skip the timing optimization on the IO paths in order to focus only on the internal register to register paths. However, when set to <code>true</code> for leakage and area optimization, the tool doesn't touch any instances, which would be part of an path in order to avoid creating any new timing violations on the IO paths.</p> <p><i>Default:</i> <code>false</code></p> <p><b>Note:</b></p> <ul style="list-style-type: none"> <li>• This parameter is not applicable for DRV optimization.</li> <li>• If set, this parameter must be applied before <code>eco_opt_design</code>.</li> </ul>                                                                                                                                                                                                                               |
| <pre>-optimize_replicated_modules {true false}</pre> | <p>By enabling the <code>-optimize_replicated_modules</code> parameter, the tool will be able to optimize timin in the replicated modules. Any netlist change wi be checked against each clone's timing and physical context to ensure that no violations are created.</p> <p>In this mode, only one ECO file is generated per group of clones. It is mandatory to list the cell name of the clones inside the file specified with the <code>-partition_list_file</code> parameter.</p> <p><i>Default:</i> <code>false</code></p> <p><b>Note:</b></p> <ul style="list-style-type: none"> <li>• The design must be assembled in Tempus through the <code>merge_hierarchical_def</code> command. You can also load an assembled DB from Innovus. This parameter works only the physical mode optimization.</li> <li>• If set, this parameter must be applied before <code>eco_opt_design</code>.</li> </ul> |

|                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <pre>-optimize_sequential_cells {true false}</pre> | <p>Enables the timing optimizer to optimize the sequential cells. In addition, it performs VT swapping and resizing of sequential cells.</p> <p><i>Default</i> : true</p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <pre>-partition_list_file &lt;string&gt;</pre>     | <p>Enables the software to select the hierarchical instances that should be treated as partitions. The file specified with this parameter must contain a list of module names of those hierarchical instances. The <code>eco_opt_design</code> command will generate an ECO file for the top module and the partitions (with relative placement locations for cells). The default ECO files are generated as mentioned below:</p> <pre>top_eco_tempus.tcl and top_eco_innovus.tcl;<br/>blockA_eco_tempus.tcl and<br/>blockA_eco_innovus.tcl and so on.</pre> <p>where the top ECO file contains all ECOs, which is a union of top and partition ECOs.</p> <p><i>Default:</i> empty</p> <p><b>Notes:</b></p> <ul style="list-style-type: none"> <li>• It is mandatory that the modules that are treated as partitions are of type FENCE.</li> <li>• To avoid performing ECO in a given partition you need to apply <code>dont_touch</code> attribute on the module.</li> <li>• To perform top-level buffering only, you need to apply <code>dont_touch</code> on all the partitions.</li> <li>• If set, this parameter must be applied before <code>eco_opt_design</code>.</li> </ul> <p><b>Example:</b></p> <p>Consider a design having the following module:</p> <ul style="list-style-type: none"> <li>• A module, <code>blockA</code>, being replicated twice with the hierarchical instances name, <code>top/blockA</code> and <code>top/blockA_2</code>.</li> </ul> |

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|                                           | <ul style="list-style-type: none"> <li>• A second module, <code>blockB</code>, being replicated twice with hierarchical instances name, <code>top/blockB_1</code> and <code>top/blockB_2</code>.</li> <li>• A third module, <code>blockC</code>, which has only one instance, <code>top/blockC_1</code>.</li> </ul> <p>In this case, the partition list file must contain the given below modules.</p> <pre>blockA blockB blockC</pre>                                                                                                                                                                                                                                                                                                                                                             |
| <pre>-path_group_file_name filename</pre> | <p>Allows you to pass a TCL file, which will be sourced on each Tempus client, after they load their respective saved session, which contains the path group definitions to be used by the tool when running in High Capacity ECO mode. Those path groups can then be referred to <code>set_eco_opt_mode</code></p> <pre>-include_setup_path_groups</pre> <p>and <code>set_eco_opt_mode</code></p> <pre>-include_hold_path_groups</pre> <p>applied on the manager session.</p> <p><b>Note:</b> This parameter is only applicable for Tempus Paradime flow or Certus and must be applied before <code>distribute_read_design</code>.</p>                                                                                                                                                            |
| <pre>-pba_effort {medium high}</pre>      | <p>Creates a high or medium effort PBA mode honored by all optimization engines. When set to <code>high</code>, all optimization engines perform more aggressive timing closure or area/power recovery with respect to PBA timing and the expectation is to get better quality of results.</p> <p>In some cases, when large amount of ECOs are done in this mode, the final timing signoff might show some setup timing degradation, which can be recovered by running a second ECO loop doing only setup timing recovery with the <code>set_eco_opt_mode -setup_recovery</code> parameter set to <code>true</code>.</p> <p><b>Default:</b> <code>high</code></p> <p><b>Note:</b> If set, this parameter must be applied before <code>write_eco_opt_db</code> and <code>eco_opt_design</code>.</p> |

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| <pre>-post_mask {true false}</pre> | <p>Enables the timing closure engine to perform the following netlist changes without touching the base layer mask:</p> <ul style="list-style-type: none"> <li>• Inserts the Gate Array buffers or an inverter p in place of existing Gate Array filler cells.</li> <li>• Resizes regular cells to Gate Array cells. It al resize Gate Array cells to Gate Array cells.</li> <li>• Deletes a regular cell.</li> </ul> <p>The software automatically identifies the Gate Array cells by checking which library instances have the same SITE name as the Gate Array filler cells specified by the user. Here, empty spaces will be filled up by the Gate Array filler cells to ensure 100% placement density.</p> <p><b>Note:</b></p> <ul style="list-style-type: none"> <li>• This is supported by DRV, Setup, and Hold fixing.</li> <li>• When <code>setTieHiLoMode -cell {{&lt;tieLo_name&gt; &lt;tieHi_name&gt;}}</code> setting is used, the Metal EC flow will insert the TieLo cells to connect dangling input pins, in case the existing TieL cells in the floorplan are extremely far or too overloaded.</li> </ul> <p><i>Default:</i> false</p> |
| <pre>-post_sta_tcl file</pre>      | <p>Specifies the file containing the TCL code that w be executed immediately after signoff timing analysis per view.</p> <p><i>Default:</i> " " (empty string)</p> <p><b>Note:</b> If set, this parameter must be applied befo <code>write_eco_opt_db</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

`-power_aware {true|false}`

Specifies power awareness in all optimizers and transforms.

Tempus ECO provides the ability to take power part of the overall cost function when doing timer closure.

When this parameter is enabled before the start an optimization session, the current optimization and any subsequent optimizations (if running in incremental mode) checks for power changes for all optimizers and transforms during any change in the design.

In any of the above cases, the first priority remain getting best final timing but enabling this option will get there with possibly slighter ECO although resulting in lesser final power.

For example, while fixing a Setup timing path violation, by default, the tool chooses to size one instance to a high leakage cell, while with the `-power_aware` mode, the tool tries to get to same timing by doing multiple sizing to non-high leakage cells.

**Note :**

- This feature requires power activity to be annotated on the netlist with the `report_power` command, either through external activity file (recommended) or using propagated activity based on some default toggling rate.
- This feature honors the `-power_opt_focus` parameter, which means that you can decide force the tool in reducing more the total or leakage, or dynamic power increase.

*Default:* `false`

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| <pre>-power_opt_focus {total leakage dynamic leakageOnly dynamicOnly}</pre> | <p>Specifies whether a given netlist should be optimized only for leakage or dynamic power reduction, or more for leakage or dynamic power while still reducing the overall total power during total power optimization.</p> <p>The options are described below:</p> <ul style="list-style-type: none"> <li>• <b>total</b>: The power optimization engine focuses on getting the optimal total power in the design.</li> <li>• <b>leakage</b>: The power optimization engine optimizes total power but focuses more on leakage power reduction than on dynamic power reduction.</li> <li>• <b>dynamic</b>: The power optimization engine optimizes total power but focuses more on dynamic power reduction than on leakage power reduction.</li> <li>• <b>leakageOnly</b>: The power optimization engine focuses only on leakage power reduction.</li> <li>• <b>dynamicOnly</b>: The power optimization engine focuses only on dynamic power reduction.</li> </ul> <p><i>Default:</i> total</p> <p><b>Note:</b></p> <ul style="list-style-type: none"> <li>• This parameter is honored by <code>eco_opt_design -power</code> and also by <code>eco_opt_design -drv/-setup/-hold</code> when the <code>-power_aware</code> parameter is set to <code>true</code>.</li> <li>• If set, this parameter must be applied before <code>eco_opt_design</code>.</li> </ul> |
| <pre>-pre_sta_tcl file</pre>                                                | <p>Specifies a file containing the TCL code that will be executed immediately prior to signoff timing analysis per view.</p> <p><i>Default:</i> " " (empty string)</p> <p><b>Note:</b> If set, this parameter must be applied before <code>write_eco_opt_db</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

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| <code>-prefix_name &lt;string&gt;</code>                                                      | <p>Specifies the prefix to be assigned to the new instances inserted.</p> <p><i>Default:</i> ESO</p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                                             |
| <code>-resize_inst {true   false}</code>                                                      | <p>Enables or disables the resizing of cells. This parameter is applicable to setup, hold, and DRV optimization. In setup optimization, resizing also includes delay-to-buffer conversion, whereas in hold optimization, resizing is limited to buffer-to-delay conversion only.</p> <p><i>Default:</i> true</p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p> |
| <code>-retime &lt;none   aocv   path_slew_propagation   aocv_path_slew_propagation&gt;</code> | <p>Supports PBA-based optimization for all optimizers.</p> <p>For detailed information of the parameter option see <a href="#">report_timing</a>.</p> <p><i>Default:</i> path_slew_propagation</p> <p><b>Note:</b> If set, this parameter must be applied before <code>write_eco_opt_db</code> and <code>eco_opt_design</code>.</p>                                                                             |
| <code>-retime_depth {default infinite}</code>                                                 | <p>Retimes depth for exhaustive path-based analysis.</p> <p><i>Default:</i> default</p>                                                                                                                                                                                                                                                                                                                         |
| <code>-retime_depth_early {default infinite}</code>                                           | <p>Retimes depth for exhaustive path-based early analysis.</p> <p><i>Default:</i> default</p>                                                                                                                                                                                                                                                                                                                   |
| <code>-retime_mode {none path exhaustive}</code>                                              | <p>Retimes mode for path-based analysis.</p> <p><i>Default:</i> path.</p>                                                                                                                                                                                                                                                                                                                                       |

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| <pre>-route_congestion_threshold threshold_value</pre> | <p>Allows changing the threshold value at which the software considers a local area for routing congestion buffering, to further allow any ECO change. This parameter accepts a value between 0 (minimum) and 1 (maximum).</p> <p>To tighten the routing congestion checks, apply value between 0 and 0.7. To relax the routing congestion checks, apply a value between 0.7 and 1.</p> <ul style="list-style-type: none"><li>• <i>Default:</i> 0.7</li><li>• <i>Min value:</i> 0</li><li>• <i>Max value:</i> 1</li></ul> <p><b>Note:</b></p> <ul style="list-style-type: none"><li>• This parameter is honored by DRV, Setup, Hold, Area, and Power optimization engines</li><li>• If set, this parameter must be applied before <code>eco_opt_design</code>.</li></ul> |
| <pre>-routing_congestion_aware {true false}</pre>      | <p>Prevents doing ECO change in the design area where routing congestion is very high. This will avoid causing any new routing DRC violations due to ECO buffering/resizing and routing detour for new ECO nets.</p> <p><i>Default:</i> false</p> <p><b>Note:</b></p> <ul style="list-style-type: none"><li>• This parameter may lead to worse timing closure when the design has routing congestion, but it will limit the potential Setup timing degradation and routing DRC increases after full place and route loop.</li><li>• If set, this parameter must be applied before <code>eco_opt_design</code>.</li></ul>                                                                                                                                                 |



|                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>-save_eco_opt_db &lt;dir_name&gt;</pre>     | <p>Path of the directory where the generated database files are kept.</p> <p><b>Default:</b> ecoTimingDB</p> <p><b>Note:</b> If set, this parameter must be applied before <code>write_eco_opt_db</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <pre>-select_hold_endpoints &lt;string&gt;</pre> | <p>Specifies a file containing the list of selected endpoints for hold fixing.</p> <p><b>Note:</b></p> <ul style="list-style-type: none"> <li>When including endpoints, the <code>optimize_core_only</code> parameter must be set to <code>false</code>.</li> <li>This parameter is set during ECO optimization.</li> </ul> <p>For more details, see section, <a href="#">Optimization using Endpoint Control</a> in the <i>Tempus User Guide</i>.</p>                                                                                                                                                                                                                                                                                                                                                                                              |
| <pre>-select_drv_net_file &lt;FILE&gt;</pre>     | <p>Specifies a file containing the list of nets to be optimized during DRV fixing. The fixing of <code>max_transition</code>, <code>max_capacitance</code>, SI glitch, SI Xtalk, and SI Slew violations occurs only on the specified nets and the other violating nets are ignored. The file specified includes one net per line. When this parameter is used, the software prints the DRV summary only for the selected nets/terms under the headings, <i>Selected Signal Nets</i> and <i>Selected Clock nets</i>, respectively. It also specifies the reasons for the selected nets only in the failure summary.</p> <p><b>Note:</b></p> <ul style="list-style-type: none"> <li>Wild cards and regular expressions are not supported in this file.</li> <li>If set, this parameter must be applied before <code>eco_opt_design</code>.</li> </ul> |
| <pre>-select_drv_pin_file &lt;FILE&gt;</pre>     | <p>Allows you to provide a list of instance pins that are optimized during <code>eco_opt_design -drv</code>. It is specified before <code>write_eco_opt_db</code> for the High Capacity ECO flow and also specified before <code>eco_opt_design</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-select_setup_endpoints &lt;string&gt;</code> | <p>Specifies a file containing the list of selected endpoints for setup fixing.</p> <p><b>Note:</b></p> <ul style="list-style-type: none"><li>• When including endpoints, the <code>-optimize_core_only</code> parameter must be set to <code>false</code>.</li><li>• If set, this parameter must be applied before <code>eco_opt_design</code>.</li></ul> <p>For more details, see the section, <a href="#">Optimization using Endpoint Control</a> in the <i>Tempus User Guide</i>.</p>                                    |
| <code>-setup_recovery {true false}</code>           | <p>Recovers the Setup timing, which might have been degraded after a large ECO change during Power optimization. When this parameter is enabled, the Setup optimization will be done with Vth swapping only, which means that the same size and the same pin geometry cell sizing. Because of this, there is no need to re-route the design, and final timing can be generated.</p> <p><i>Default:</i> <code>false</code></p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p> |
| <code>-setup_target_slack number</code>             | <p>Specifies a target slack value in nanoseconds for setup timing.</p> <p><i>Default:</i> <code>0</code></p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                                                                                                                                                  |

|                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>-setup_target_slack_per_view {viewName1 slack1 viewName2 slack2 ... }</pre> | <p>When enabled, this parameter allows for forcing specific setup target slack per view. If a given Setup active view is not listed in this parameter, the software will use the global Setup target slack set by the <code>-setup_target_slack</code> parameter for the view.</p> <p><b>Note:</b></p> <ul style="list-style-type: none"> <li>• This parameter is honored by Setup, Hold, Area, Power, and Leakage optimization engines.</li> <li>• If set, this parameter must be applied before <code>eco_opt_design</code>.</li> </ul> <p><i>Default:</i> "" (empty string)</p> |
| <pre>-setup_xtalk_delta_threshold &lt;DOUBLE&gt;</pre>                           | <p>Fixes the nets with the crosstalk delta delay value equal to or greater than a specified threshold value for setup views.</p> <p><i>Default:</i> 0.3 ns</p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                                                                                                                                                      |
| <pre>-setup_xtalk_slack_threshold &lt;DOUBLE&gt;</pre>                           | <p>Fixes the nets with the slack threshold value equal to or less than a specified threshold value for setup views.</p> <p><i>Default:</i> 1000 ns</p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                                                                                                                                                              |
| <pre>-skew_core_only {true false}</pre>                                          | <p>When set to <code>true</code>, it does not allow the skewing engine to skew sequential elements that are part of an IO boundary timing path.</p> <p><i>Default:</i> false</p> <p><b>Note:</b></p> <ul style="list-style-type: none"> <li>• This parameter is only honored by <code>eco_opt_design -setup</code>.</li> <li>• If set, this parameter must be applied before <code>eco_opt_design</code>.</li> </ul>                                                                                                                                                               |

`-skip_drv_net_file <FILE>`

Specifies a file containing the list of nets to be excluded during DRV fixing. The max\_transition and max\_capacitance violations fixing occurs only on the specified nets, and the other violating nets are ignored. The file specified includes one net per line. When this parameter is used, the software prints the DRV summary only for the selected nets/terms under the headings, *Selected Signal Nets* and *Selected Clock nets*, respectively. It also specifies the reasons for the selected nets only in the failure summary.

*Default=* " " (empty string)

**Note:**

- Wild cards and regular expressions are not supported in this file.
- If set, this parameter must be applied before `eco_opt_design`.

|                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>-smart_mmmc_effort {low medium high}</pre>                  | <p>Controls the trade-off between memory or runtime and timing QoR for timing or power ECO closure flow. This means that the higher the effort, the less memory or runtime, but the lesser timing QoR while running the <code>eco_opt_design</code> command.</p> <ul style="list-style-type: none"> <li>• <code>low</code>: Is the recommended (default) setting and ensures there is no timing QoR loss.</li> <li>• <code>medium</code>: Reduces memory and runtime significantly with minimal timing QoR loss.</li> <li>• <code>high</code>: Reduces memory and runtime further when compared to the medium mode, but with additional timing QoR loss.</li> </ul> <p>In the <code>low</code> effort mode, which is the default and recommended setting, there is no compromise on timing QoR.</p> <p>The <code>medium</code> and <code>high</code> effort modes can be used for initial rounds of ECO timing/power ECO closure flow.</p> <p><b>Note:</b> This parameter must be set before the design is loaded.</p> <p><i>Default:</i> <code>low</code></p> |
| <pre>-smart_mmmc_force_keep_hold_view_list &lt;string&gt;</pre>  | <p>List of hold views, which have to be retained after SmartMMMC view compaction. These views represent the minimum hold views retained by the system, on top of which additional views can be retained.</p> <p><i>Default:</i> <code>" "</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <pre>-smart_mmmc_force_keep_setup_view_list &lt;string&gt;</pre> | <p>List of setup views, which have to be retained after SmartMMMC view compaction. These views represent the minimum setup views retained by the system, on top of which additional views can be retained.</p> <p><i>Default:</i> <code>" "</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

```
-smart_mmmc_max_view_count {{setup
<maxViewCount>} {hold <maxViewCount>} {drv
<maxViewCount>}}
```

Allows you to force the software to collect only a given number of views to cover Setup timing, Hold timing, and DRV. It acts as a limit to prevent smartMMMC from selecting more views than what is specified by the user. This parameter is especially useful in combination with `-smart_mmmc_reduce_clients`, as it enables a user-defined limit on hardware resources for Tempus clients.

**Note:**

- This parameter is only honored by Tempus Paradime or Certus, and should be set before design loading.
- When `-smart_mmmc_max_view_count` is specified, `-smart_mmmc_effort` is ignored.
- When `-smart_mmmc_max_view_count` is specified, you need to provide `maxViewCount` each Setup, Hold, and DRV field.
- `maxViewCount` should be equal to or greater than 1 for setup and hold, but can be 0 for drv only if you do not specify `set_analysis_view` earlier.

*Default:* ""

|                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>-smart_mmmc_reduce_clients {true   false}</pre>      | <p>When set to <code>true</code>, the SmartMMMC engine determines which views should be retained and made available to Tempus clients. This helps to reduce the number of Tempus clients started by Tempus Paradime or Certus.</p> <p><i>Default:</i> <code>false</code></p> <p><b>Note:</b></p> <ul style="list-style-type: none"> <li>• This parameter is only honored by Tempus Paradime or Certus, and must be set before design loading.</li> <li>• The Tempus STA sessions must have been saved with <code>set_eco_opt_mode - create_smart_mmmc_db true</code>.</li> <li>• SmartMMMC must be enabled through <code>set_distribute_mmmc_mode - enable_smart_mmmc</code>.</li> </ul> |
| <pre>-specify_hold_endpoints_margin &lt;string&gt;</pre>  | <p>Specifies the end point-specific margin for hold fixing.</p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <pre>-specify_setup_endpoints_margin &lt;string&gt;</pre> | <p>Specifies endpoint-specific margin for setup fixing.</p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p> <p>For more information on the endpoint parameter above, refer to the <i>Optimization using Endpoint Control</i> section in the ECO chapter of the <i>Tempus User Guide</i>.</p>                                                                                                                                                                                                                                                                                                                                              |
| <pre>-swap_inst {true false}</pre>                        | <p>Enables or disables the VT swap resizing. This parameter applies to all optimization engines, except for Leakage optimization, which cannot be disabled.</p> <p><i>Default:</i> <code>true</code></p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                  |

|                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <pre>-upsize_in_power_opt {true false}</pre>                                      | <p>Allows the power optimization engine to perform cell up sizing to recover more power.</p> <p><i>Default:</i> false</p> <p><b>Note :</b></p> <ul style="list-style-type: none"> <li>• This parameter is only honored by <code>eco_opt_design -power</code>.</li> <li>• If set, this parameter must be applied before <code>eco_opt_design</code>.</li> </ul>                                                                                                                                                |
| <pre>-use_ga_filler_list<br/>{&lt;list_of_Gate_Array_filler_cells_name&gt;}</pre> | <p>Enables the software to list all Gate Array filler cells that can be deleted or reinserted.</p> <p><b>Note:</b> This parameter is applicable only in the Metal ECO mode.</p>                                                                                                                                                                                                                                                                                                                               |
| <pre>-use_pfc_decap_list cell_list</pre>                                          | <p>Lists all the programmable decap cells that are allowed to be removed/reinserted in post-mask mode.</p> <p><i>Default:</i> ""</p>                                                                                                                                                                                                                                                                                                                                                                          |
| <pre>-use_pfc_filler_list cell_list</pre>                                         | <p>Lists all programmable filler cells that are allowed to be removed in post-mask mode.</p> <p><i>Default:</i> ""</p>                                                                                                                                                                                                                                                                                                                                                                                        |
| <pre>-use_pfc_regular_list cell_list</pre>                                        | <p>Lists all the programmable regular logic/ECO cells that are allowed to be inserted in place of filler/decap cells during optimization in post-mask mode.</p> <p><i>Default:</i> ""</p>                                                                                                                                                                                                                                                                                                                     |
| <pre>-touch_partition_list &lt;string&gt;</pre>                                   | <p>Selects a list of partition module names to be considered as touch-only during optimizations. Any partition not specified with this parameter will be considered as don't_touch during optimization.</p> <p><b>Notes:</b></p> <ul style="list-style-type: none"> <li>• This parameter is exclusive to <code>dont_touch_partition_list</code>.</li> <li>• This parameter must be applied before <code>eco_opt_design</code> and can be changed between each call of <code>eco_opt_design</code>.</li> </ul> |



|                                  |                                                                                                                                                                                                           |
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| <pre>-verbose {true false}</pre> | <p>Specifies whether the command should output a log file with verbosity.</p> <p><i>Default:</i> false</p> <p><b>Note:</b> If set, this parameter must be applied before <code>eco_opt_design</code>.</p> |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Examples

- The following command resets all `set_eco_opt_mode` parameters to their default values:

```
tempus> set_eco_opt_mode -reset
```

- The following example shows how to run `eco_opt_design` to fix hold-timing violations up to +150ps while ensuring no setup timing degradation below +100ps :

```
tempus> set_eco_opt_mode -setup_target_slack 0.1 -hold_target_slack 0.15
tempus> eco_opt_design -hold
```

- The following example shows how to run `eco_opt_design` to fix maximum capacitance and maximum transition violations that are 10 percent above their respective maximum constraints:

```
tempus> set_eco_opt_mode -drv_margin -0.1
tempus> eco_opt_design -drv
```

- The following example shows how to run `eco_opt_design` to optimize leakage power without allowing any setup timing degradation:

```
tempus> set_eco_opt_mode -setup_target_slack 0
tempus> eco_opt_design -leakage
```

- The following example shows how to perform hold fixing and treat every cell listed in the `partition.txt` file as a partition.

```
tempus> set_eco_opt_mode -partition_list_file partition.txt
tempus> eco_opt_design -hold
```

- The following commands will perform Hold fixing, and the replicated modules of the cell listed in the partition list file will be treated as clones:

```
tempus> set_eco_opt_mode -optimize_replicated_modules true
tempus> set_eco_opt_mode -partition_list_file partitions.txt
tempus> eco_opt_design -hold
```

- The following example shows an incremental optimization flow:

```
tempus> set_eco_opt_mode -load_eco_opt_db <>
tempus> set_eco_opt_mode -allow_multiple_incremental true
tempus> set_eco_opt_mode -eco_file_prefix FULL_DRV
tempus> eco_opt_design -drv
tempus> set_eco_opt_mode -eco_file_prefix FULL_HOLD
tempus> eco_opt_design -hold
tempus> set_eco_opt_mode -eco_file_prefix FULL_SETUP
tempus> eco_opt_design -setup
```

- The following command is used to perform Hold fixing in routing congestion-aware mode.

```
tempus> set_eco_opt_mode -routing_congestion_aware true
tempus> eco_opt_design -hold
```

- The following command performs Hold fixing via regular buffering/resizing/swapping along with inserting a dummy load cell, namely, dummyX1 cell.

```
tempus> set_eco_opt_mode -add_load true -load_cell_list {dummyX1}
tempus> eco_opt_design -hold
```

- The following command fixes IR Drop violations in a C-MMMC session:

```
tempus> analyze_rail -results_directory ./dynamic_rail -type domain all
tempus> debug_irdrop -domain all -output_directory my_ir_db -state_directory
dynamic_rail/all_25C_dynamic_1 -eco_report -nworst_instances 1000000
tempus> set_eco_opt_mode -save_eco_opt_db ECODB
tempus> write_eco_opt_db
tempus> set_eco_opt_mode -load_eco_opt_db ECODB
tempus> set_eco_opt_mode -fix_ir_drop true
tempus> set_eco_opt_mode -load_irdrop_db my_ir_db
```

```
tempus> set_eco_opt_mode -fix_max_tran false -fix_max_cap false
tempus> eco_opt_design -drv
```

- The following command enables Hold fixing in the metal ECO mode by allowing the listed Gate Array filler cells to be deleted or inserted during the ECO process.

```
tempus> set_eco_opt_mode -post_mask true
tempus> set_eco_opt_mode -use_ga_filler_list {GFILL1BWP GFILL2BWP GFILL4BWP GFILL10BWP}
tempus> eco_opt_design -hold
```

- The following command saves the regular timing and additional compact timing information in the ECO DB directory named `EcoTimingDB` in the STA SMSC session.

```
tempus> set_eco_opt_mode -save_eco_opt_db EcoTimingDB
tempus> set_eco_opt_mode -create_smart_mmmc_db true
tempus> write_eco_opt_db
```

- The following command loads the design with a minimized set of data to reduce the memory or runtime during an ECO fixing stage when many timing views are enabled, and performs Setup timing closure in the ECO MMMC session.

```
tempus > set_eco_opt_mode -load_smart_mmmc_db EcoTimingDB
tempus> <load_design>
tempus> set_eco_opt_mode -load_eco_opt_db EcoTimingDB
tempus> eco_opt_design -setup
```

- The following commands allow Hold timing closure in the post-mask mode using programmable filler cells.

```
tempus> set_eco_opt_mode -post_mask true
tempus> set_eco_opt_mode -use_pfc_filler_list {ln10_ecofiller40x ... ln10_ecofiller4x}
tempus> set_eco_opt_mode -use_pfc_decap_list {ln10_ecofillsgcap40x ...
ln10_ecofillsgcap4x}
tempus> set_eco_opt_mode -use_pfc_regular_list {ln10_ecodlybt01 ln10_ecobuft04 ...
ln10_ecoand02 ... ln10_ecoinv04 ...}
tempus> eco_opt_design -hold
```

- The following commands perform clock skewing during setup fixing. Only `CKBUFX1` and `CKBUFX4` can be added to the clock tree.

```
tempus> set_eco_opt_mode -clock_cell_list {CKBUFX1 CKBUFX4}
tempus> set_eco_opt_mode -allow_skewing true
```

```
tempus> eco_opt_design -setup
```

- The following commands enable cell EM fixing for all types of EM violations.

```
tempus> set_eco_opt_mode -fix_cell_em true
tempus> set_eco_opt_mode -em_type all
tempus> eco_opt_design -drv
```

- The following command specifies the number of views for setup timing, hold timing, and drv:

```
tempus> set_eco_opt_mode -smart_mmmc_max_view_count {{setup 3} {hold 2} {drv 1}}
```

- The following command considers the specified cells (BUF8\_L3\_P48\_ELVT and BUF6\_L3\_P48\_ELVT) as high-effort resources to enhance timing or power efficiency during signoff optimization:

```
tempus> set_eco_opt_mode -high_effort_cell_list {BUF8_L3_P48_ELVT BUF6_L3_P48_ELVT}
tempus> eco_opt_design -setup
```

## Related Commands

- [eco\\_opt\\_design](#)
- [get\\_eco\\_opt\\_mode](#)
- [write\\_eco\\_opt\\_db](#)

## set\_filler\_mode

```
set_filler_mode
[-help]
[-check_trim_rule {true|false}]
[-double_height_filler_insertion {true|false}]
[-enable_restore_filler_flow {true|false}]
[-reset]
[-core {{list-of-cells1} {list-of-cells2} ...}]
[-corePrefix prefix]
[-vt_swap_out_dir {vtSwapReportFileDirectory}]
[-vt_swap_out_file {file name}]
[-vt_swap_to_fix_filler_violations {true|false}]
```

Controls how the software deletes filler cells during [eco\\_opt\\_design](#). The mode setting is persistent and saved along with the database when the [save\\_design](#) command is run. It is important to ensure that fillers can be removed by [eco\\_opt\\_design](#); otherwise, there would be no legal location to insert buffers or resize instances.

## Parameters

|                                                             |                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                                          | <p>Outputs a brief description that includes type and default information for each <code>set_filler_mode</code> parameter.</p> <p>For a detailed description of the command and all of its parameters, use the man command: <code>man set_filler_mode</code>.</p> |
| <code>-check_trim_rule {true false}</code>                  | <p>Controls trim rule checking during the filler insertion. When set to true, the trim rule is checked, and the runtime increases.</p> <p><i>Default:</i> false</p>                                                                                               |
| <code>-double_height_filler_insertion {true false}</code>   | <p>Enables double-height filler during filler insertion.</p>                                                                                                                                                                                                      |
| <code>-enable_restore_filler_flow {true false}</code>       | <p>When this option is set to false, fillers are not saved in the memory (which is the default, i.e, the save and restore filler insertion flow is default ON).</p> <p><i>Default:</i> true</p>                                                                   |
| <code>-reset</code>                                         | <p>Resets all or specified options to their default value.</p>                                                                                                                                                                                                    |
| <code>-core {{list-of-cells1} {list-of-cells2} ...}</code>  | <p>Specifies the filler cells. Enclose the filler cell names in quotation marks (") or curly braces ({ }). You can specify multiple cell lists at one time, using curly braces ({ }) to separate each one.</p> <p><i>Default:</i> "" (empty string)</p>           |
| <code>-corePrefix prefix</code>                             | <p>Specifies the prefix for the filler cells.</p> <p><i>Default:</i> "" (empty string)</p>                                                                                                                                                                        |
| <code>-vt_swap_out_dir vtSwapReportFileDirectory</code>     | <p>Specifies the directory of the vt swap report file,</p> <p><i>Default:</i> Current directory ./</p>                                                                                                                                                            |
| <code>-vt_swap_out_file fileName</code>                     | <p>Specifies the file name of the vt swap report.</p> <p><i>Default:</i> ""</p>                                                                                                                                                                                   |
| <code>-vt_swap_to_fix_filler_violations {true false}</code> | <p>Changes the VT of the functional cell to fix filler gaps.</p> <p><i>Default:</i> false</p>                                                                                                                                                                     |

## Examples

- The following command specifies that filler instances of cells FILL64 and FILL8 have a prefix of ABC in the design.

```
tempus> set_filler_mode -corePrefix ABC -core "FILL64 FILL8"
```

- The following command lets `eco_opt_design` delete all filler cells that have a prefix FILL in their instance name.

```
tempus> set_filler_mode -corePreFix FILL -core <List of Filler cells>
tempus> eco_opt_design -setup
```

## Related Commands

- `get_eco_opt_mode`
- `set_place_mode`

# set\_place\_mode

```
set_place_mode
[-help]
[-reset]
[-place_check_report_fixed_inst_on_blockage {true|false}]
[-place_3d_use_existing_bump_tsv {true|false}]
[-place_design_enable_3d {true|false}]
[-place_design_integrity_ir_fix_effort {low|medium|high}]
[-place_design_refine_macro {true|false}]
[-place_detail_allow_border_pin_abut {true|false}]
[-place_detail_allow_single_height_row_symmetry_x siteName]
[-place_detail_check_route {true|false}]
[-place_detail_color_aware_legal {true|false}]
[-place_detail_context_aware_legal {none|all|optional|required|user}]
[-place_detail_honor_inst_pad {true|false}]
[-place_detail_pad_fixed_insts {true|false}]
[-place_detail_PGFTV_insertion_cell_list listOfCells]
[-place_detail_swap_eeq_cells {true|false}]
[-place_detail_use_diffusion_transition_fill {true|false}]
[-place_global_adaptive_subdl {true|false}]
```

Controls certain aspects of how the software places cells. This command is enabled only in physical mode and primarily used to control placement settings during the `eco_opt_design` command.

## Parameters

|                                                                      |                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                                                   | <p>Outputs a brief description that includes type and default information for each <code>set_place_mode</code> parameter.</p> <p>For a detailed description of the command and all of its parameters, use the <code>man</code> command:</p> <p><code>man set_place_mode.</code></p> |
| <code>-reset</code>                                                  | <p>Resets all specified options back to default values.</p>                                                                                                                                                                                                                         |
| <code>-place_check_report_fixed_inst_on_blockage {true false}</code> | <p>When <code>true</code>, <code>checkPlace</code> reports placement blockage violations for fixed instances under placement blockages.</p> <p><i>Default:</i> <code>false</code></p>                                                                                               |



|                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-place_3d_use_existing_bump_tsv {true false}</code>            | Enables 3D placement to use preplaced but unassigned bump/TSV in the design for signal net assignment. The preplaced bump/TSV must be in the specified cells for 3D concurrent bump/TSV insertion and legalization.                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>-place_design_enable_3d {true false}</code>                    | Enable 3D placement in the Integrity 3D-IC platform. You need to set this to true before invoking concurrent macro placement, standard cell placement, and bump/TSV insertion ( <code>place_bump_tsv_detail</code> ) in a 3D design.<br><i>Default:</i> false                                                                                                                                                                                                                                                                                                                                                                            |
| <code>-place_design_integrity_ir_fix_effort {low medium high}</code> | Sets the power integrity placement IR fixing effort.<br><i>Default:</i> low                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>-place_design_refine_macro {true false}</code>                 | Specifies whether to legalize the macros based on constraints after the concurrent timing-driven placement of macros and standard cells.<br><i>Default:</i> false                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>-place_detail_allow_border_pin_abut {true false}</code>        | Some libraries have cells with pin geometry that is not set back by half the routing pitch from the normal PR-boundary of that cell. When the pin geometries touch the cell PR_boundary, placing an instance of this cell can cause a DRV violation with the normal pin geometry of other cells. To avoid these DRVs, placer will add one Site padding to the cell-side during pre-processing, thus satisfying the min-spacing rules. The <code>-place_detail_allow_border_pin_abut</code> option disables this pre-emptive DRV avoidance behavior, and no auto-padding is done when the option is set to true.<br><i>Default:</i> false |

|                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>-place_detail_allow_single_height_row_symmetry_x &lt;siteName&gt;</pre>   | <p>Specifies a single-height row tech-site allowing symmetry X during detail placement.</p> <p><i>Default:</i> ""</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <pre>-place_detail_check_route {true false}</pre>                              | <p>Considers pre-routed <code>FIXED</code> signal wires and avoids creating DRC violations between <code>FIXED</code> signal wires and instance pins. When this parameter is not specified, <code>refinePlace</code> ignores <code>FIXED</code> signal wires.</p> <p><i>Default:</i> <code>false</code></p> <p><b>Note:</b> Using this parameter might result in longer run time.</p> <p>If you specify <code>refinePlace - place_detail_io_pin_blockage</code>, you do not need to specify <code>- place_detail_check_route</code> for signal pins because <code>refinePlace</code> will treat the signal pins similarly to supply pins for blockages. However, signal wires still require this parameter.</p> |
| <pre>-place_detail_color_aware_legal {true false}</pre>                        | <p>Enables the placer to honor and fix double pattern constraints between adjacent cells.</p> <p><i>Default:</i> <code>false</code></p> <p><b>Note:</b> This option does not affect cell to pre-route color DRC checks.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <pre>-place_detail_context_aware_legal {none all optional required user}</pre> | <p>Instructs placer to support cell edge to cell edge spacing constraints as defined in the LEF files.</p> <p>The <code>optional</code>, <code>required</code>, <code>user</code> argument values can be used separately or together, but they are exclusive to <code>all</code> and <code>none</code>.</p> <p><i>Default:</i> <code>all</code></p>                                                                                                                                                                                                                                                                                                                                                             |

|                                                                       |                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-place_detail_honor_inst_pad {true false}</code>                | <p>Honors padding from <code>specifyInstPad</code> in the detailed placement. Currently, only left and right padding is supported.</p> <p><i>Default:</i> <code>false</code></p> <p>When this option is set to <code>true</code>, instance padding is also considered in <code>refinePlace</code> and <code>checkPlace</code>.</p> |
| <code>-place_detail_pad_fixed_insts {true false}</code>               | <p>Specifies whether the <code>refinePlace</code> command honors cell padding on <code>FIXED</code> instances.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                       |
| <code>-place_detail_PGFTV_insertion_cell_list listOfCells</code>      | <p>Specifies the list of cells as PG FTV cells.</p> <p><i>Default:</i> <code>" "</code></p>                                                                                                                                                                                                                                        |
| <code>-place_detail_swap_eeq_cells {true false}</code>                | <p>Determines whether master cells can be replaced by EEQ cells during detailed placement to improve routability. After detailed placement, the software reports the number of replacements. EEQ cells are defined in the LEF file.</p> <p><i>Default:</i> <code>false</code></p>                                                  |
| <code>-place_detail_use_diffusion_transition_fill {true false}</code> | <p>Specifies that the design has fillers with jogged diffusion obs. For designs where fillers with jogged diffusion obs are to be inserted between cells with varying diffusion widths, please specify this option.</p> <p><i>Default:</i> <code>false</code></p>                                                                  |
| <code>-place_global_adaptive_subd1 {true false}</code>                | <p>When set to <code>true</code>, placement will call Opt API to get a list of cells not usable for subD1 and disable the usage of cells in the placement engine.</p> <p>The <code>set_place_mode</code> command can be used independently to get the list of subD1 cells.</p> <p><i>Default:</i> <code>false</code></p>           |

## Examples

The following command resets all `set_place_mode` parameters to their default values:

```
tempus> set_place_mode -reset
```

The following is a Tempus script where Tempus ECO legal placement code honors the double pattern constraints between adjacent cells:

```
<Load LEF/Verilog/DEF/MMMC>
```

```
tempus> set_place_mode -place_detail_color_aware_legal false # bool, default=false
```

```
tempus> eco_opt_design -setup
```

## Related Commands

- [get\\_eco\\_opt\\_mode](#)

## specifyCellEdgeSpacing

```
specifyCellEdgeSpacing
[-help]
{<type1> <type2> <spacing>
[-exceptExactSpacing distance][-exceptAbutted | -forbiddenSpacing][-
exceptNonFillerInBetween]} | [-reset]
```

Creates the cell edge spacing rule as in `CELLEDGESPACINGTABLE` in LEF.

 The tool, by default, supports the latest TCL settings. If a user tries to change the edge spacing, the tool will display a warning message that resetting edge spacing would overwrite the old value.

## Parameters

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                        | <p>Outputs a brief description that includes type and default information for each <code>specifyCellEdgeSpacing</code> parameter.</p> <p>For a detailed description of the command and all of its parameters, use the <code>man</code> command:</p> <pre>man specifyCellEdgeSpacing</pre>                                                                                                                                 |
| <code>-exceptAbutted</code>               | Specifies that two abutted macros of the given edge types are also allowed.                                                                                                                                                                                                                                                                                                                                               |
| <code>-exceptExactSpacing distance</code> | Specifies an exception distance to the minimum spacing.                                                                                                                                                                                                                                                                                                                                                                   |
| <code>-exceptNonFillerInBetween</code>    | <p>Specifies that the edge spacing is only for adjacent instances.</p> <p>If two MACROs (cells) have an edge-spacing rule (for example, 0.8 microns), the rule normally applies regardless of the presence of another macro (cell) in the gap between. When the <code>-exceptNonFillerInBetween</code> option is used, the rule is exempted when a non-filler cell is present between the two constrained cell edges.</p> |
| <code>&lt;type1&gt;</code>                | Specifies edge <code>edgeType1</code> for edge spacing.                                                                                                                                                                                                                                                                                                                                                                   |
| <code>&lt;type2&gt;</code>                | Specifies edge <code>edgeType2</code> for edge spacing.                                                                                                                                                                                                                                                                                                                                                                   |
| <code>-forbiddenSpacing</code>            | Forbids <code>edgeType1</code> and <code>edgeType2</code> with the space.                                                                                                                                                                                                                                                                                                                                                 |
| <code>-help</code>                        | <p>Outputs a brief description that includes type and default information for each <code>specifyCellEdgeSpacing</code> parameter.</p> <p>For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man specifyCellEdgeSpacing</code></p>                                                                                                                               |
| <code>-reset</code>                       | Resets the cell edge spacing to LEF.                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>spacing</code>                      | <p>Specifies edge spacing between <code>edgeType1</code> and <code>edgeType2</code>.</p> <p>If there is no spacing defined between two edge types, it indicates that there is no spacing constraint (or 0.0 <math>\mu\text{m}</math>) between them.</p>                                                                                                                                                                   |

## Example

The following command creates a 0.18  $\mu\text{m}$  spacing between `grp1` and `grp2`:

```
specifyCellEdgeSpacing grp1 grp2 0.18
```

## Related Command

- `specifyCellEdgeType`

# specifyCellEdgeType

```
specifyCellEdgeType
[-help]
{-left edgeType | -right edgeType | -top edgeType | -bottom edgeType | -reset }
[{-left edgeType | -right edgeType} [-cellRow rowNumber]]
[{-left edgeType | -right edgeType} [-half {top | bottom}]]
[{-top edgeType | -bottom edgeType} [-range {x y}]]
[{-cell cellName} [-reset]]
```

Specifies edge type for cell edges as `EDGETYPE` in macro LEF. It allows specifying one edge at a time for a particular cell.

To ensure the correct edge type specification, edge types asserted by the `specifyCellEdgeType` command should be predefined by the `specifyCellEdgeSpacing` command.

For example:

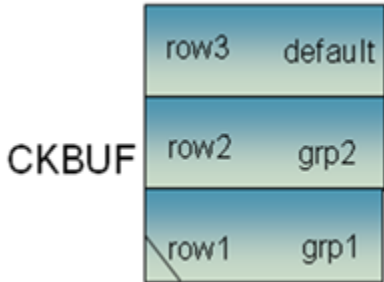
```
specifyCellEdgeSpacing grp1 grp2 0.18
specifyCellEdgeType -cell BUF1X -left grp1
```

If the edge type is specified without a corresponding spacing rule, the tool issues a warning and ignores the edge type specification.

## Parameters

|                  |                                                                                                                                                                                                                                                                        |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help            | <p>Outputs a brief description that includes type and default information for each <code>specifyCellEdgeType</code> parameter.</p> <p>For a detailed description of the command and all of its parameters, use the man command:</p> <pre>man specifyCellEdgeType</pre> |
| -bottom edgeType | Specifies an edge type on the bottom side.                                                                                                                                                                                                                             |
| -cell cellName   | Specifies the cell to which to add an edge constraint.                                                                                                                                                                                                                 |



|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>-<br/>cellRow <i>rowNumber</i></p> | <p>Specifies the row number for the horizontal edge type.</p> <p>You can define different edge types for each row of multiple height cells using -cellRow.</p> <p>For example, there is a triple-height cell CKBUF.</p>  <p>The following commands define different edge types for different rows of the CKBUF cell.</p> <pre>specifyCellEdgeType -cell CKBUF -right grp1 -cellRow 1 specifyCellEdgeType -cell CKBUF -right grp2 -cellRow 2</pre> <p>In the above example, <i>grp1</i> is applied to the first row (bottom row) of the right edge, <i>grp2</i> is applied to the second row of the right edge, and no specific edge type is defined for the third row of the right edge and is of the DEFAULT type.</p> <p>The legal value of -cellRow is from 1 to n, where n is the number of cells in the row.</p> |
| <p>-half {top<br/>bottom}</p>         | <p>Defines the edge types corresponding to the top or bottom half of the cell.</p> <p>The following example associates the edge type 'grp1' with the bottom row, top half of the multi-height cell SDF8:</p> <pre>specifyCellEdgeType -cell SDF8 -right grp1 -cellRow 1 -half top</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <p>-help</p>                          | <p>Outputs a brief description that includes type and default information for each <code>specifyCellEdgeType</code> parameter.</p> <p>For a detailed description of the command and all of its parameters, use the man command: <code>man specifyCellEdgeType</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <p>-left <i>edgeType</i></p>          | <p>Specifies an edge type on the left side.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

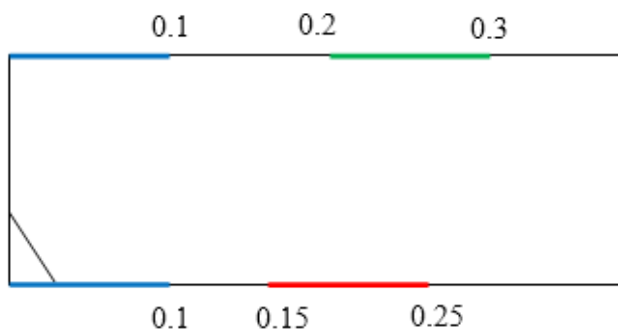
`-range {x y}`

Specifies the range for the vertical edge type.

The range for the vertical edge type can also be negative.

The following example defines a partial range of an edge that the type should be labeled. This could only be defined on the TOP or BOTTOM edges.

```
specifyCellEdgeType -cell A -top GROUP1 -range 0 0.1
specifyCellEdgeType -cell A -bottom GROUP1 -range 0 0.1
specifyCellEdgeType -cell A -top GROUP2 -range 0.2 0.3
specifyCellEdgeType -cell A -top GROUP3 -range 0.15 0.25
```



Blue line indicates GROUP1 edge  
Green line indicates GROUP2 edge  
Red line indicates GROUP3 edge

Example of  
MACRO A

```
...
 PROPERTY LEF58_EDGETYPE “
 EDGETYPE TOP GROUP1 RANGE 0 0.1 ;
 EDGETYPE BOTTOM GROUP1 RANGE 0 0.1 ;
 EDGETYPE TOP GROUP2 RANGE 0.2 0.3 ;
 EDGETYPE BOTTOM GROUP3 RANGE 0.15 0.25 ; “ ;
...
END A
```

|                                     |                                                         |
|-------------------------------------|---------------------------------------------------------|
|                                     |                                                         |
| <code>-reset</code>                 | Resets the specified cell constraint to the LEF status. |
| <code>-right <i>edgeType</i></code> | Specifies an edge type on the right side.               |
| <code>-top <i>edgeType</i></code>   | Specifies an edge type on the top side.                 |

## Example

The following commands specify one edge at a time for the `BUF1X` cell

- `specifyCellEdgeType -cell BUF1X -left grp1`
- `specifyCellEdgeType -cell BUF1X -right grp1 -cellRow 1`
- `specifyCellEdgeType -cell BUF1X -right grp1 -cellRow 2`

## write\_eco\_opt\_db

write\_eco\_opt\_db

[-help]

Saves an ECO Timing DB after performing timing analysis, in a non-distributed mode or in a distributed-MMMC session. When several views are active at the same time, the ECO Timing DB for each of them is saved. The data is saved in the directory pointed to by the `set_eco_opt_mode -save_eco_opt_db` parameter.

**Note:** This command requires a 64-bit executable.

### Examples

The following command saves an ECO Timing DB in the `mydb` directory.

```
tempus> set_eco_opt_mode -save_eco_opt_db mydb
```

```
tempus> write_eco_opt_db
```

### Related Commands

- `set_eco_opt_mode`
- `eco_opt_design`

# Low Power Commands

- [infer\\_pgv\\_power\\_intent](#)
- [report\\_voltage](#)
- [set\\_lib\\_binding\\_by\\_voltage](#)
- [set\\_msv\\_mode](#)

## infer\_pgv\_power\_intent

```
infer_pgv_power_intent
[-help]
[-default_net string]
```

Initiates power intent inferencing after loading the design with the template view definition file. This command creates an individual power domain for each power net, similar to the SMSC PGV flow. It also updates the view definition file and `delay_corners` with the power domain information. This command creates library sets for each power domain based on PVT. Power domain PVT is inferred based on the voltage from the associated power net of the domain, and the process and temperature of design-level PVT for the corner.

### Parameters

|                                              |                                                                                                                                                                                                                                                                                      |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                           | Outputs a brief description that includes type and default information for each <code>infer_pgv_power_intent</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command:<br><br><code>man infer_pgv_power_intent</code> |
| <code>-<br/>default_net <i>string</i></code> | Specifies the default power net.                                                                                                                                                                                                                                                     |

### Related Commands

[report\\_voltage](#)

# report\_voltage

```
report_voltage
[-help]
[-early]
[-inst string]
[-late]
[-net string]
[-term string]
[-quiet]
[-view string]
```

This command reports the voltage for the net/pin (term)/inst in the specific view under a specific corner.

## Parameters

|                     |                                                                                                                                                                                                                                                                          |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help               | Outputs a brief description that includes the type and default information for each <code>report_voltage</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command:<br><br><code>man report_voltage</code> |
| -early              | Specifies that the voltage is reported in the early mode.                                                                                                                                                                                                                |
| -inst <i>string</i> | Specifies the instance power domain voltage (PVT).                                                                                                                                                                                                                       |
| -late               | Specifies that the voltage is reported in the late mode.                                                                                                                                                                                                                 |
| -net <i>string</i>  | Specifies the net voltage (driver's pin voltage).                                                                                                                                                                                                                        |
| -term <i>string</i> | Specifies the term voltage.                                                                                                                                                                                                                                              |
| -quiet              | Displays only the pin voltage-related information without other additional unnecessary information to avoid huge log size.                                                                                                                                               |
| -view <i>string</i> | Specifies the analysis view.                                                                                                                                                                                                                                             |

## Related Commands

```
infer_pgv_power_intent
```

# set\_lib\_binding\_by\_voltage

```
set_lib_binding_by_voltage
[-help]
[-default_power_net string]
[-early_dynamic_power_net_voltages {pwr_net1@volt1 pwr_net2@volt2 ...}]
[-late_dynamic_power_net_voltages string]
[-process float]
[-temperature float] -power_net_voltages {pwr_net1@volt1 pwr_net2@volt2 ...} | {-
early_power_net_voltages {pwr_net1@volt1 pwr_net2@volt2 ...} -late_power_net_voltages
{pwr_net1@volt1 pwr_net2@volt2 ...}} | {-data_early_power_net_voltages {pwr_net1@volt1
pwr_net2@volt2 ...} -data_late_power_net_voltages {pwr_net1@volt1 pwr_net2@volt2 ...} -
clock_early_power_net_voltages {pwr_net1@volt1 pwr_net2@volt2 ...} -
clock_late_power_net_voltages {pwr_net1@volt1 pwr_net2@volt2 ...}}
```

Initiates auto-inferencing of power intent, library binding, and view definition. It also provides voltages for each power net in the design. Use this command to assert voltages for power rails. This command creates the view definition file for single corner analysis and assigns the operating voltage for each power domain accordingly. It also creates library sets for each power domain based on the assigned PVT and associates the corresponding library set for the power domains.

## Parameters

|                                                                                  |                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                                                                            | Outputs a brief description that includes type and default information for each <code>set_lib_binding_by_voltage</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command:<br><br><code>man set_lib_binding_by_voltage</code> |
| -<br>clock_early_power_net_voltages { <i>pwr_net1@volt1 pwr_net2@volt2 ...</i> } | Assigns power net voltage for early corner of the clock path.                                                                                                                                                                                                                                |
| -clock_late_power_net_voltages { <i>pwr_net1@volt1 pwr_net2@volt2 ...</i> }      | Assigns power net voltage for late corner of the clock path.                                                                                                                                                                                                                                 |
| -data_early_power_net_voltages { <i>pwr_net1@volt1 pwr_net2@volt2 ...</i> }      | Assigns power net voltage for early corner of the data.                                                                                                                                                                                                                                      |
| -data_late_power_net_voltages { <i>pwr_net1@volt1 pwr_net2@volt2 ...</i> }       | Assigns power net voltage for late corner of the data.                                                                                                                                                                                                                                       |
| -default_power_net <i>string</i>                                                 | Specifies the default power net.                                                                                                                                                                                                                                                             |

|                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>-early_dynamic_power_net_voltages {pwr_net1@volt1 pwr_net2@volt2 ...}</pre> | <p>Specifies whether the software uses power-net voltage values during early-stage dynamic power analysis. When this parameter is specified, the software considers power-net voltages while estimating dynamic power in early stages of the flow.</p> <div style="border: 1px solid #fde725; padding: 10px; margin-top: 10px;"> <p> This parameter is intended for early power analysis scenarios where approximate power-net voltage information is available.</p> </div> |
| <pre>-early_power_net_voltages {pwr_net1@volt1 pwr_net2@volt2 ...}</pre>         | <p>Assigns power net voltage for early corner.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <pre>-late_dynamic_power_net_voltages {pwr_net1@volt1 pwr_net2@volt2 ...}</pre>  | <p>Specifies whether the software uses power-net voltage values during late-stage dynamic power analysis. When this parameter is specified, the software considers power-net voltages while estimating dynamic power in later stages of the flow.</p> <div style="border: 1px solid #fde725; padding: 10px; margin-top: 10px;"> <p> This parameter is intended for late power analysis scenarios where approximate power-net voltage information is available.</p> </div> |
| <pre>-late_power_net_voltages {pwr_net1@volt1 pwr_net2@volt2 ...}</pre>          | <p>Assigns power net voltage for late corner.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <pre>-power_net_voltages {pwr_net1@volt1 pwr_net2@volt2 ...}</pre>               | <p>Assigns power net voltage.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <pre>-process float</pre>                                                        | <p>Defines filtering condition for the timing library characterized process.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <pre>-temperature float</pre>                                                    | <p>Defines filtering condition for timing library characterized temperature.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |



## Related Commands

`set_input_transition`

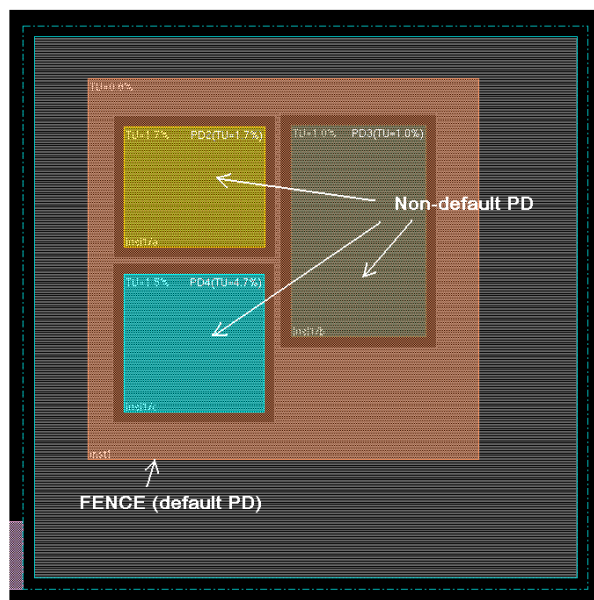
# set\_msv\_mode

```
set_msv_mode
[-help]
[-allowNestedDefaultDomain {true | false}]
[-allowPowerDomainMinGapZero {1 | 0}]
[-checkAllNetsForDomainCrossing {true | false}]
[-handle_old_lib_with_signal_level {true | false}]
[-handleBackToBackIsolation {true | false}]
[-handlePD {true | false}]
[-handlePDComplex {true | false}]
[-honorCpfGlobalConnectionSpecForAoBuffer {true | false}]
[-honorCpfGlobalConnectionSpecForShifter {true | false}]
[-honorDotLibRelatedPGPin {true | false}]
[-honorEffectiveDomainForIsoLsInsertion {true | false}]
[-markIsoEnablePinAsAlwaysOn {true | false}]
[-notUseTopFTermDomainForVoltage {true | false}]
[-reset]
[-upf_allow_shifter_voltage_mismatch {true | false}]
[-upf_insert_on_floating_pins {true | false}]
```

Sets the special handling for inter-domain nets and to use effective power domains for ISO/LS insertion.

## Parameters

|                                          |                                                                                                                                                                                                                                                                                                     |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                                    | <p>Outputs a brief description that includes type and default information for each <code>setMvsMode</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command:</p> <pre>man setMvsMode</pre>                                          |
| -allowNestedDefaultDomain {true   false} | <p>Allows a non-default power domain member to be logically nested in a default power domain. This default power domain member has fence constraint.</p> <p>By default, due to grouping, the fence constraint is lost for the non-default power domain member. Set <code>true</code> to enable.</p> |



In the netlist: modules `HInst11` and `HInst12` are under `HInst1`; and `HInst1` is under the top.

You have the following:

```
create_power_domain -instances HInst1
create_power_domain -default -instances
{HInst11 HInst12}...
```

That means `HInst11` and `HInst12` are cut from `HInst1` to default power domain. If there is a fence/region constraint for `HInst11` and `HInst12` in default power domain, then please turn this global on. Otherwise, the fence/region for `HInst11` and `HInst12` will be lost.

`-allowPowerDomainMinGapZero {1 | 0}`

Allows a non-default power domain member to be logically nested in a default power domain.

*Default:* 0

`-checkAllNetsForDomainCrossing {true | false}`

Checks all the nets for domain crossings. Set `true` to enable.

|                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-handle_old_lib_with_signal_level {true   false}</code>        | <p>Allows using the old-style <code>.lib</code> with <code>signal_level</code> and <code>power_rail</code> in the MSV cell identification and term voltage.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                                                                                                      |
| <code>-handleBackToBackIsolation {true   false}</code>               | <p>Supports back-to-back isolation insertion along one cross-domain net based on the CPF ISO rule. Set <code>true</code> to enable.</p>                                                                                                                                                                                                                                                                                                                                                                        |
| <code>-handlePD {true   false}</code>                                | <p>Routes intra-power domain and global nets using the specifications of the power domains. <a href="#">On using this option</a>, <code>earlyGlobalRoute</code> routes intra-power domain nets only within their power domains. Global nets that do not connect with a power domain do not get routed through that power domain.</p> <p><i>Default:</i> <code>true</code></p>                                                                                                                                  |
| <code>-handlePDComplex {true   false}</code>                         | <p>Routes nets belonging to more than one power domain so that the routing does not violate power domain boundaries, in addition to routing intra-power domain and global nets using the specifications of the power domains.</p> <p>For example, when you specify <code>-handlePDComplex true</code>, if a net belongs to power domains PD1 and PD3, but not PD2, <code>earlyGlobalRoute</code> routes the net without going over the PD2 power domain boundary.</p> <p><i>Default:</i> <code>true</code></p> |
| <code>-honorCpfGlobalConnectionSpecForAoBuffer {true   false}</code> | <p>Honors connection specs for AO buffer based on the CPF rules. Set <code>true</code> to enable.</p>                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>-honorCpfGlobalConnectionSpecForShifter {true   false}</code>  | <p>Honors connection specs for shifters based on the CPF rules. Set <code>true</code> to enable.</p>                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>-honorDotLibRelatedPGPin {true   false}</code>                 | <p>Honors library related PG pins. Set <code>true</code> to enable.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>-honorEffectiveDomainForIsoLsInsertion {true   false}</code>   | <p>Honors effective domains for ISO/LS insertions. Set <code>true</code> to enable.</p>                                                                                                                                                                                                                                                                                                                                                                                                                        |

|                                                                 |                                                                                                                             |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <code>-markIsoEnablePinAsAlwaysOn {true   false}</code>         | Insert ISO/LS for all the inter-domain nets using effective domain based on the CPF rules. Set <code>true</code> to enable. |
| <code>-notUseTopFTermDomainForVoltage {true   false}</code>     | Controls voltage using top fterm domain for IO pin voltage.                                                                 |
| <code>-reset</code>                                             | Resets the MSV mode to the default value.                                                                                   |
| <code>-upf_allow_shifter_voltage_mismatch {true   false}</code> | When set to false, it prevents insertion of level shifters by allowing voltage tolerance.                                   |
| <code>-upf_insert_on_floating_pins {true   false}</code>        | Allows controlling of low power cell insertion on the floating pins.<br><i>Default:</i> <code>true</code>                   |

## Examples

- In the following example, allows the non-default power domain members nested in the default power domain member are allowed logically and physically and this default power domain has the fence constraint:  
`set_msv_mode -allowNestedDefaultDomain true`
- In the following example, all the nets for the iso/ls insertion are checked:  
`set_msv_mode -checkAllNetsForDomainCrossing true`
- In the following example, back-to-back isolation along with one net are inserted:  
`set_msv_mode -handleBackToBackIsolation true`
- In the following example, the CPF specification (`create_global_connection -domain`) are used for the ISO/LS PG pin connection:  
`set_msv_mode -honorCpfGlobalConnectionSpecForShifter true`
- In the following example, the pin power domain is assigned based on the related power pin specified in `*.lib`:  
`set_msv_mode -honorDotLibRelatedPGPin true`
- In the following example, effective domain for the iso/ls insertion is used:  
`set_msv_mode -honorEffectiveDomainForIsoLsInsertion true`
- In the following example, the iso enable based on the tag (power modes) is marked:  
`set_msv_mode -markIsoEnablePinAsAlwaysOn false`



---

# Clock Tree Debugger Commands

---

- `close_ctd_win`
- `ctd_save_histogram`
- `ctd_save_view`
- `ctd_show_skew_group`
- `ctd_trace`
- `ctd_win`
- `get_ctd_win_id`
- `get_ctd_win_title`
- `set_ctd_win_title`
- `gui_zoom_ctd`

# close\_ctd\_win

```
close_ctd_win
[-help]
[-all | -id WindowIDName]
```

Closes open CTD windows. When no parameter is specified, the command closes the current active CTD window.

## Parameters

|                            |                                                                                                                                                                                                                                                  |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                      | Outputs a brief description that includes type and default information for each <code>close_ctd_win</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man close_ctd_win</code> . |
| -all                       | Closes all open CTD windows.                                                                                                                                                                                                                     |
| -id<br><i>WindowIDName</i> | Closes the CTD window with the specified window ID.                                                                                                                                                                                              |

## Examples

- The following command closes all CTD windows in the current session:  
`tempus> close_ctd_win -all`
- The following command closes the CTD window with the ID, `user_id2`:  
`tempus> close_ctd_win -id user_id2`

## Related Commands

- `ctd_win`
- `get_ctd_win_title`
- `get_ctd_win_id`
- `set_ctd_win_title`



## ctd\_save\_histogram

```
ctd_save_histogram
[-help]
[-csv]
-file fileName
[-id windowID]
```

Saves the CTD histogram to a file. By default, the file type is PNG. You can also specify that the file be saved in the CSV format.

Before using this command to save the histogram view, first open and configure the histogram in the GUI, including skew group and delay corners.

### Parameters

|                             |                                                                                                                                                                                                                                                              |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                       | Outputs a brief description that includes type and default information for each <code>ctd_save_histogram</code> parameter. For a detailed description of the command and all of its parameters, use the man command:<br><code>man ctd_save_histogram.</code> |
| -csv                        | Saves the histogram in CSV format. This parameter is optional. By default, the histogram is saved as a PNG file.                                                                                                                                             |
| -file <i>fileName</i>       | Specifies the name of the file where you want to save the histogram. This parameter is required.                                                                                                                                                             |
| -<br>id <i>windowIDName</i> | Specifies the ID of the active CTD window.<br>The default value is the ID of the current active window.                                                                                                                                                      |

### Example

The following command saves the histogram in the `histogram1.csv` file.

```
gui_ctd_save_histogram -file histogram1 -csv
```

## Related Information

- `close_ctd_win`
- `ctd_save_view`
- `ctd_win`
- `get_ctd_win_id`
- `get_ctd_win_title`
- `gui_zoom_ctd`
- `set_ctd_win_title`
- The Clock Menu chapter of the *Tempus Menu Reference*.

## ctd\_save\_view

```
ctd_save_view
[-help]
-file fileName
[-id windowID]
[-legend_on]
[-type {PNG | GIF}]
```

Saves the current snapshot of the clock tree viewer into a file.

### Parameters

|                             |                                                                                                                                                                                                                                                  |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                       | Outputs a brief description that includes type and default information for each <code>ctd_save_view</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man ctd_save_view</code> . |
| -file <i>fileName</i>       | Specifies the name of the file where you want to dump the display image. This parameter is required.                                                                                                                                             |
| -<br>id <i>windowIDName</i> | Specifies the window ID whose display is to be saved.<br>The default value is the returned value of command <code>get_ctd_win_id</code> .                                                                                                        |
| -legend_on                  | Saves the whole window including the panels that are open.                                                                                                                                                                                       |
| -type {PNG  <br>GIF}        | Specifies whether the format of the image should be Gif or PNG                                                                                                                                                                                   |

### Example

The following command saves the view snapshot in the `snapshot1.png` file.

```
ctd_save_view -file snapshot1 -type PNG
```

## Related Information

- `close_ctd_win`
- `ctd_win`
- `get_ctd_win_id`
- `get_ctd_win_title`
- `gui_zoom_ctd`
- `set_ctd_win_title`
- The Clock Menu chapter of the *Tempus Menu Reference*.

# ctd\_show\_skew\_group

```
ctd_show_skew_group
[-help]
[-id window_id_name]
-skew_group skew_group_name
```

Shows the specified skew group in the CTD window.

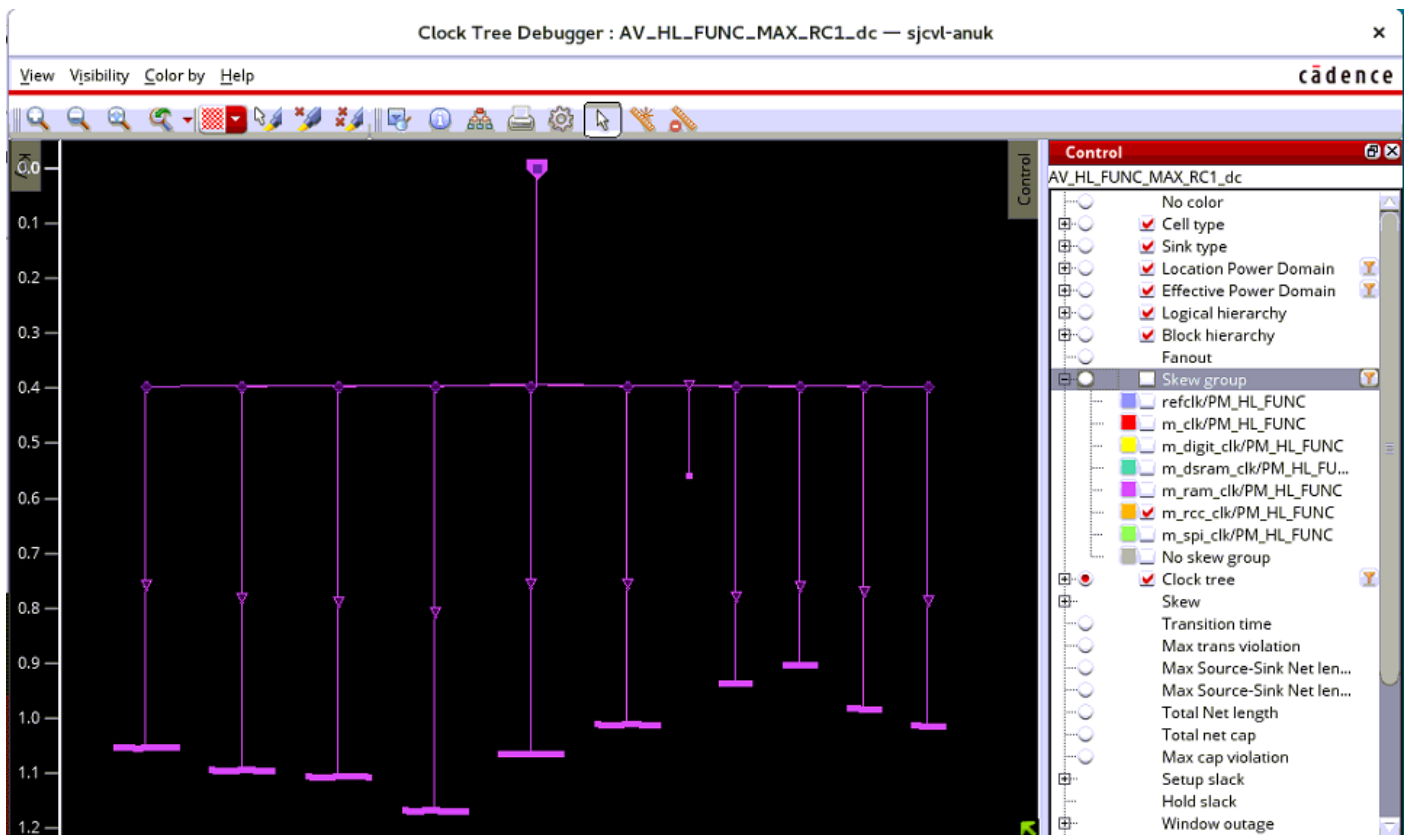
## Parameters

|                                |                                                                                                                                                                                                                                                                              |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                          | Outputs a brief description that includes type and default information for each <code>ctd_show_skew_group</code> parameter.<br>For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man ctd_show_skew_group</code> . |
| -id<br>window_id_name          | Specifies the CTD window ID. The default value is the ID of the current active CTD window.                                                                                                                                                                                   |
| -skew_group<br>skew_group_name | Specifies the name of the skew group. This is a required parameter.                                                                                                                                                                                                          |

## Example

Use the following command to show the specified skew group in the CTD window:

```
ctd_show_skew_group -skew_group m_rcc_clk/PM_HL_FUNC
```



## Related Commands

- `ctd_win`
- `close_ctd_win`
- `get_ctd_win_id`
- `get_ctd_win_title`
- `gui_zoom_ctd`
- `set_ctd_win_title`

## ctd\_trace

```
ctd_trace
[-help]
[-color colorname]
[-file fileName]
[-from rootclock]
[-index colorindex]
[-to sink | -through instancename]
```

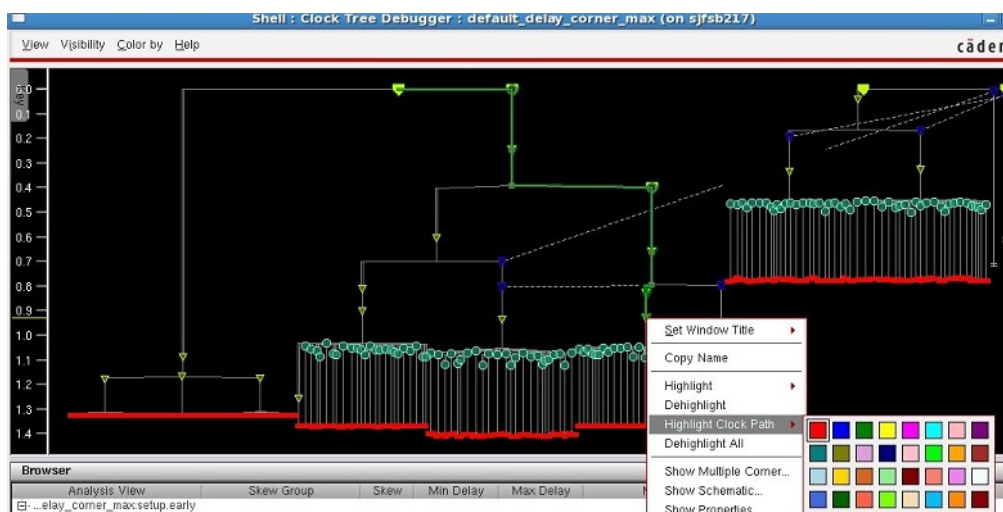
Highlights the clock tree path from the clock root to the pin or from one pin to the other if the two pins are on the same path using the specified color or color index. The highlighted clock tree path can be viewed in the Clock Tree Viewer of the CCOpt Clock Tree Debugger (CTD). You can also use the Highlight Clock Path option in the context menu of the CTD to highlight the complete clock tree path of a selected pin to the clock root.

## Parameters

| <code>-help</code>                           | <p>Outputs a brief description that includes type and default information for each <code>ctd_trace</code> parameter.</p> <p>For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man ctd_trace</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |            |             |          |             |          |     |       |         |  |       |  |             |  |      |       |       |        |           |  |       |  |  |  |    |       |       |        |            |  |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------|-------------|----------|-------------|----------|-----|-------|---------|--|-------|--|-------------|--|------|-------|-------|--------|-----------|--|-------|--|--|--|----|-------|-------|--------|------------|--|
| <code>-color</code><br><i>colorname</i>      | <p>Specifies the color to be used for highlighting the clock tree path that is being traced.</p> <p><i>Default:</i> red</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |            |             |          |             |          |     |       |         |  |       |  |             |  |      |       |       |        |           |  |       |  |  |  |    |       |       |        |            |  |
| <code>-file</code><br><i>fileName</i>        | <p>Specifies the name of the file in which the physical information for each instance and net in the clock path being traced will be dumped.</p> <p>The following information is saved in the file:</p> <table><thead><tr><th>Name</th><th>Cell</th><th>Status</th><th>Location</th><th>Wire Length</th></tr></thead><tbody><tr><td>CLK_ROOT</td><td>PAD</td><td>Fixed</td><td>(20,40)</td><td></td></tr><tr><td>net_1</td><td></td><td>don't_touch</td><td></td><td>12.5</td></tr><tr><td>Buf_1</td><td>BUFX1</td><td>placed</td><td>(100,180)</td><td></td></tr><tr><td>net_2</td><td></td><td></td><td></td><td>35</td></tr><tr><td>Buf_2</td><td>BUFX2</td><td>placed</td><td>(200, 120)</td><td></td></tr></tbody></table> <p><i>Name</i> is the instance name or the net name</p> <p><i>Cell</i> is the instance cell type. This information is only for instances</p> <p><i>Status</i> is the place status for an instance and constrain status for a net. For dont_touch nets, the status is 'dont_touch', otherwise this column is left blank</p> <p><i>Location</i> is the physical location of the instance or the coordinate of that instance</p> <p><i>Wire Length</i> is only for nets</p> | Name        | Cell       | Status      | Location | Wire Length | CLK_ROOT | PAD | Fixed | (20,40) |  | net_1 |  | don't_touch |  | 12.5 | Buf_1 | BUFX1 | placed | (100,180) |  | net_2 |  |  |  | 35 | Buf_2 | BUFX2 | placed | (200, 120) |  |
| Name                                         | Cell                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Status      | Location   | Wire Length |          |             |          |     |       |         |  |       |  |             |  |      |       |       |        |           |  |       |  |  |  |    |       |       |        |            |  |
| CLK_ROOT                                     | PAD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Fixed       | (20,40)    |             |          |             |          |     |       |         |  |       |  |             |  |      |       |       |        |           |  |       |  |  |  |    |       |       |        |            |  |
| net_1                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | don't_touch |            | 12.5        |          |             |          |     |       |         |  |       |  |             |  |      |       |       |        |           |  |       |  |  |  |    |       |       |        |            |  |
| Buf_1                                        | BUFX1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | placed      | (100,180)  |             |          |             |          |     |       |         |  |       |  |             |  |      |       |       |        |           |  |       |  |  |  |    |       |       |        |            |  |
| net_2                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |            | 35          |          |             |          |     |       |         |  |       |  |             |  |      |       |       |        |           |  |       |  |  |  |    |       |       |        |            |  |
| Buf_2                                        | BUFX2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | placed      | (200, 120) |             |          |             |          |     |       |         |  |       |  |             |  |      |       |       |        |           |  |       |  |  |  |    |       |       |        |            |  |
| <code>-from</code><br><i>clockroot</i>       | <p>Specifies the starting point of the highlight. If this parameter is not specified, the starting point of the highlight is the clock root of the specified sink.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |             |            |             |          |             |          |     |       |         |  |       |  |             |  |      |       |       |        |           |  |       |  |  |  |    |       |       |        |            |  |
| <code>-index</code><br><i>colorindex</i>     | <p>Specifies the range of colors that can be used to highlight the clock tree path. Minimum is 1 and maximum is 32.</p> <p>The range corresponds to the colors in the color picker provided in the <i>Highlight Clock Path</i> option in the context menu of the Clock Tree Viewer.</p> <p>The number 1 means red, 2 means blue, and so on. An example is shown in the image below.</p> <p><i>Default:</i> 1 or red.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |            |             |          |             |          |     |       |         |  |       |  |             |  |      |       |       |        |           |  |       |  |  |  |    |       |       |        |            |  |
| <code>-through</code><br><i>instancename</i> | <p>Specifies the instance or pin through which the path should be traced.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |            |             |          |             |          |     |       |         |  |       |  |             |  |      |       |       |        |           |  |       |  |  |  |    |       |       |        |            |  |
| <code>-to</code> <i>sink</i>                 | <p>Specifies the end point of the highlighted path.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             |            |             |          |             |          |     |       |         |  |       |  |             |  |      |       |       |        |           |  |       |  |  |  |    |       |       |        |            |  |



The image below shows the clock tree path highlighted in green, which corresponds to the color index 3. It also shows the color picker with the range of colors included in the color index.

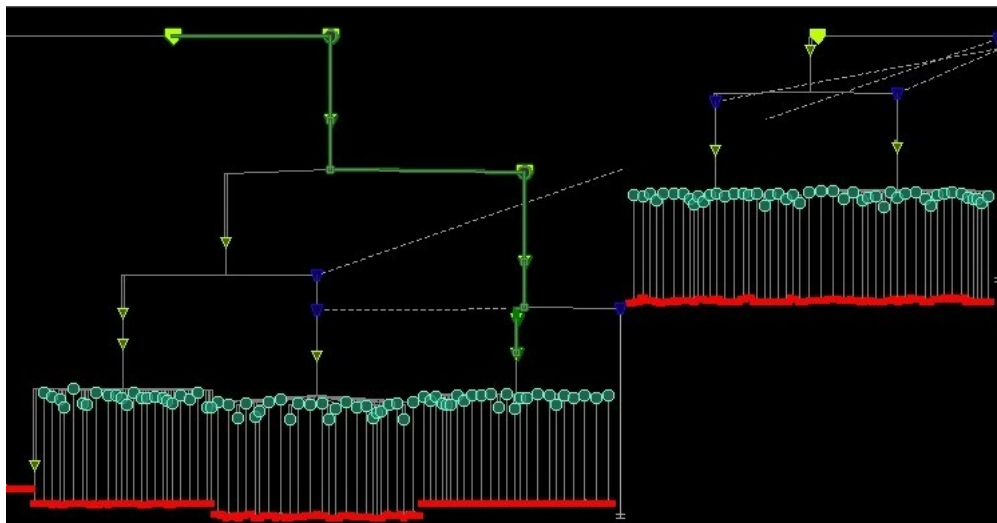


## Examples

The following command highlights the clock tree path for the specified pin, `ff1`, from the clock root in green color:

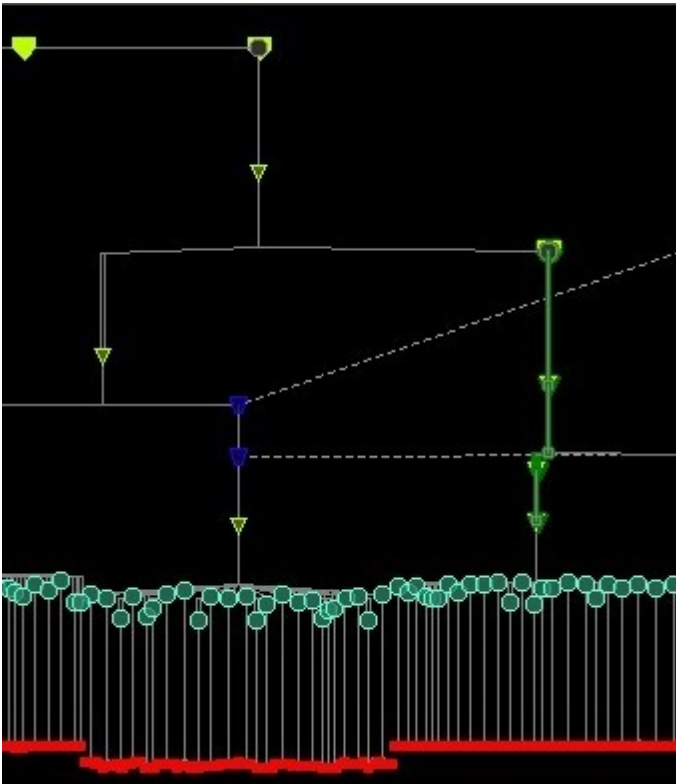
```
tempus> ctd_trace -to ff1 -index 3
```

The CTD shows the following:



The following command highlights the path between two sinks that are on the same path, in green color:

```
tempus> ctd_trace -to ff1 -from u1 -index 3
```



## Related Topics

- Clock Menu chapter in the *Tempus Menu Reference Guide*
  - Context Menu of the Clock Tree Debugger

## Related Command

- `ctd_win`

# ctd\_win

```
ctd_win
[-help]
[-id windowIDName]
[-include_reporting_only_skew_groups]
[-title CTDWindowID]
[-unit_delay]
```

Opens a Clock Tree Debugger (CTD) window with the user-defined window ID and title.

When run without specifying the ID and title, the command opens a CTD window with a system default window title. By default, the window title is the name of the current analysis corner, which shows in the CTD window. For example, "Clock Tree Debugger - func\_rcbest".

By default, all CTD windows have their unique window ID. The mechanism of the default window ID is ctdMainWin\_1, ctdMainWin\_2, ctdMainWin\_3....and so on. So, by default, the window ID of a new opened Clock Tree Debugger window is ctdMainWin\_1.

The information in the title comprises the following:

```
Session title: Window type: Window title
```

where,

Window type is always “Clock Tree Debugger”.

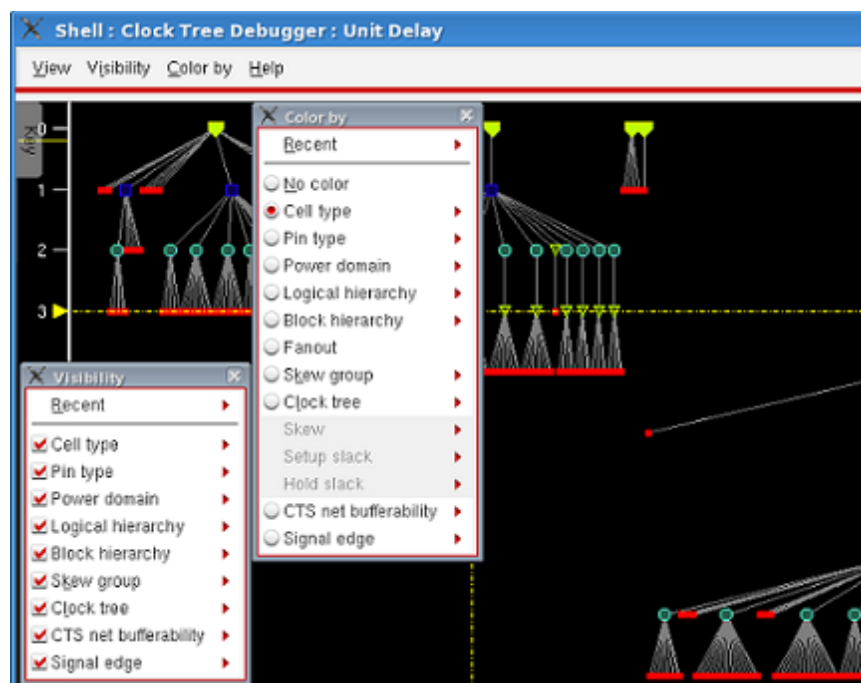
If the user-specified session title is “my\_test.tcl”, and the default primary analysis corner is “primary\_corner”, then with ctd\_win, the total set of the window title line will be:

```
my_test.tcl : Clock Tree Debugger : primary_corner
```

## Parameters

|                                     |                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                               | Outputs a brief description that includes type and default inforr parameter. For a detailed description of the command and all c the man command: man ctd_win.                                                                                                                                                                                                               |
| -id windowIDName                    | Opens a Clock Tree Debugger window with a user-specified w the window IDs will be ctdMainWin_0, or ctdMainWin_1, and so                                                                                                                                                                                                                                                      |
| -include_reporting_only_skew_groups | <p>When this parameter is specified, the GUI will include the repo By default, this setting if off, which means the reporting-only sk omitted from the CTD.</p> <p><b>Note:</b> Once the CTD is started, it will not be possible to add or only skew groups from those that are displayed. You will need window specifying the new skew group filtering that is requirec</p> |

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><code>-title CTDWindowID</code></p> | <p>Specifies the user-defined title for the CTD window. Open a CTD window with a user specified title. For example, if you specify <code>my_title</code>, when session title is <code>my_test.tcl</code> and the primary analysis is <code>primary_corner</code>, then the total set of the window title line will be</p> <pre>my_test.tcl : Clock Tree Debugger : my_title</pre> <p><b>Note:</b> The window's ID if not specified using the <code>-id</code> parameter is the window ID. To view the default window ID, use the <code>get_ctd_win_id</code> command.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <p><code>-unit_delay</code></p>        | <p>Opens the CTD in the unit delay mode. The unit delay lets you view clock tree objects easily. This is useful when you have large designs or complex clock trees. When debugging clock trees, you can use this view to check if clock trees are balanced by looking at the depth of the clock trees. A unit delay mode can also be enabled using the <i>Unit delay</i> submenu in the CTD but this is only valid after running clock tree synthesis, and then invoke timing calculation and extraction, which are time consuming operations. The <code>-unit_delay</code> parameter to open the CTD in unit delay mode even if the design has not performed CTS.</p> <p>When the CTD is opened in this mode, the following options in the CTD are disabled:</p> <p><i>Visibility</i> menu:</p> <ul style="list-style-type: none"> <li>• Timing windows</li> <li>• Delays</li> <li>• Unit delay</li> </ul> <p><i>Color by</i> menu and the <i>Control</i> panel:</p> <ul style="list-style-type: none"> <li>• Transition time</li> <li>• Total net cap</li> <li>• Max cap violation</li> <li>• Skew</li> <li>• Setup slack</li> <li>• Hold slack</li> <li>• Timing windows</li> </ul> <p>This is shown in below image.</p> |



## Examples

- The following command creates a CTD window ID with the user-specified name, `my_test`:

```
tempus> ctd_win -id my_test
```

When you use the `get_ctd_win_id` command to list all window IDs, you get the above window ID in the list:

```
tempus> get_ctd_win_id -all
```

```
ctdMain_0 my_test
```

- The following command creates a CTD window ID, `my_test` and CTD window title of `myTest.tcl`:

```
tempus> ctd_win -title myTest.tcl
```

When you use the `-detail` parameter of the `get_ctd_win_id` command, the following information is returned:

```
tempus> get_ctd_win_id -detail
```

```
ctdMain_1 "Clock Tree Debugger - func_rcbest"
```

where, `ctdMain_01` is the default window ID, window type is Clock Tree Debugger, and `func_rcbest` is the name of the current analysis corner.

## get\_ctd\_win\_id

```
get_ctd_win_id
[-help]
[-all]
[-detail]
```

Retrieves the IDs of the CTD windows. If no parameters are specified, the command retrieves the window ID of the current active CTD window. For example, if you create a window with the ID, `user_id`, and then run this command without specifying any parameters, the software displays the following information:

```
tempus> ctd_win -title my_title -id user_id
tempus> get_ctd_win_id
user_id
```

**Note:** Before using this command, ensure that a CTD window is open. You can open a CTD window using the `ctd_win` command.

### Parameters

|         |                                                                                                                                                                                                                                                                    |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| - help  | Outputs a brief description that includes type and default information for each <code>get_ctd_win_id</code> parameter.<br>For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man get_ctd_win_id</code> . |
| -all    | Retrieves the IDs of all CTD windows.                                                                                                                                                                                                                              |
| -detail | Retrieves the IDs of CTD windows, with a mapping to their titles.                                                                                                                                                                                                  |

### Examples

- The following command retrieves the ID of all CTD windows in the current session:

```
tempus> get_ctd_win_id -all
user_id1 user_id2 user_id3
```

- The following command retrieves the IDs of CTD windows, with a mapping of their titles:

```
tempus> get_ctd_win_id -detail
user_id2 : Clock Tree Debugger : title2
```

where, `user_id2` is the window ID, `Clock Tree Debugger` is the window type, and `title2` is the mapping title for this window.

## Related Commands

- `ctd_win`

# get\_ctd\_win\_title

```
get_ctd_win_title
[-help]
[-all | -id WindowIDName]
```

Retrieves the titles of the CTD windows. If no parameters are specified, the command retrieves the title of the current active CTD window.

**Note:** Before using this command, ensure that a CTD window is open. You can open a CTD window using the `ctd_win` command.

## Parameters

|                            |                                                                                                                                                                                                                                                          |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                      | Outputs a brief description that includes type and default information for each <code>get_ctd_win_title</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man get_ctd_win_title</code> . |
| -all                       | Retrieves the IDs of all CTD windows.                                                                                                                                                                                                                    |
| -id<br><i>WindowIDName</i> | Retrieves the title of the CTD window with the specified ID.                                                                                                                                                                                             |

## Examples

- The following command retrieves the titles of all CTD windows in the current session:

```
tempus> get_ctd_win_title -all

{user_id1 : Clock Tree Debugger : title1} {user_id2 : Clock Tree Debugger : title2} {user_id3 :
Clock Tree Debugger : title3}
```

- The following command retrieves the title of the CTD window with the ID, `user_id2`:

```
tempus> get_ctd_win_title -id user_id2

user_id2 : Clock Tree Debugger : title2
```

where, `user_id2` is the window ID, Clock Tree Debugger is the window type, and `title2` is the mapping title for this window.

## Related Commands

- `ctd_win`
- `set_ctd_win_title`



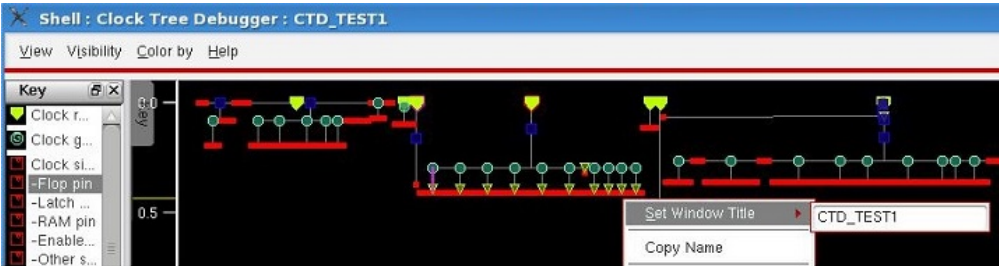


# set\_ctd\_win\_title

```
set_ctd_win_title
[-help]
title
[-id WindowIDName]
```

Specifies the title of the CTD window. By default, this command is used to set the title of the current active CTD window either if a unique ID has not been specified for this window or if you want to change the existing window ID.

You can also set the window title from the context menu of the Clock Tree Viewer in the CTD. Click Set Window Title, type the title for the CTD session, and press enter. The new title appears in the top bar of the CTD. This is shown in below image.



## Parameters

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| - help           | Outputs a brief description that includes type and default information for each <code>set_ctd_win_title</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man set_ctd_win_title</code> .                                                                                                                                                                                                                                                                                                        |
| -id WindowIDName |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                  | Specifies the ID of the window whose title is to be set. If there are several CTD windows open in the current session, with window IDs, <code>user_id1</code> , <code>user_id3</code> , <code>user_id3</code> , and so on, with corresponding window titles, <code>title1</code> , <code>title2</code> , <code>title3</code> , and so on, then the following command will only change the <code>title1</code> of window with ID, <code>user_id1</code> , to <code>my_new_title</code> .<br><br>tempus> <code>set_ctd_win_title my_new_title -id user_id1</code> |
| title            | Specifies the title to be given to the specified window. This parameter is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

## Examples

- The following command sets the window title of the current CTD window, `my_title`, to `my_new_title`:

```
tempus> set_ctd_win_title my_new_title
```

The active CTD window title will be changed from:

```
my_test.tcl : Clock Tree Debugger : my_title
```

to:

```
my_test.tcl : Clock Tree Debugger : my_new_title
```

- The following command retrieves the window IDs of CTD windows, with a mapping of their window titles:

```
tempus> get_ctd_win_id -detail
```

```
user_id2 : Clock Tree Debugger : title2
```

where, `user_id2` is the window ID, Clock Tree Debugger is the window type, and `title2` is the mapping title for this window.

## Related Commands

- `ctd_win`
- `get_ctd_win_title`

# gui\_zoom\_ctd

```
gui_zoom_ctd
[-help]
[-id string]
[-in | -out | -selected]
[-vertical_range {ymin ymax}]
```

Zooms into or out of the CTD window.

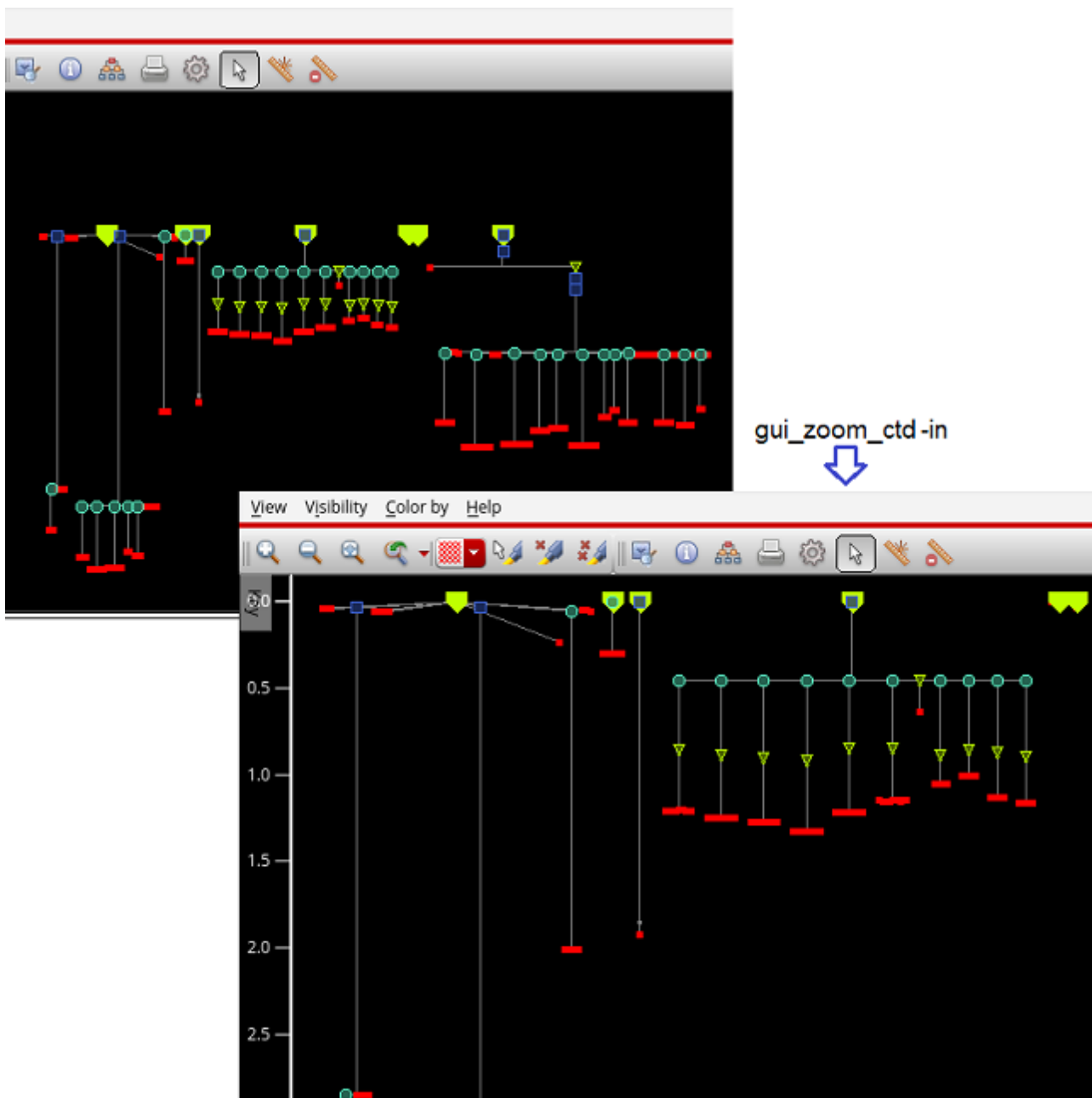
## Parameters

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                                | Prints a brief description of the <code>gui_zoom_ctd</code> command.<br><br>For a detailed description of the command, use the <code>man</code> command: <code>man gui_zoom_ctd</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| -id <i>string</i>                    | Specifies the window ID of the CTD window in which to zoom in or zoom out.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| -in                                  | Zoom into the CTD window.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| -out                                 | Zoom out of the CTD window.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| -selected                            | Zoom to enclose all selected objects.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| -vertical_range { <i>ymin ymax</i> } | <p>Zoom to a specific vertical latency range. If CTD is in timing-based mode, the window will zoom to the clock tree parts where latency is between <i>ymin</i> and <i>ymax</i>, and the unit is the same as global timing unit. If CTD is in unit delay mode, the window will zoom to the clock tree parts where stage depth is between <i>ymin</i> and <i>ymax</i>.</p> <p>For example, if CTD is in timing-based mode, then the command, <code>gui_zoom_ctd -vertical_range {0 1.5}</code> will zoom to the clock tree parts where latency is between 0 and 1.5, the unit is the same as <code>get_time_unit</code>. If CTD is in unit delay mode, then the command, <code>gui_zoom_ctd -vertical_range {0 1.5}</code> will zoom to the clock tree parts where stage depth is between 0 and 1.5.</p> <p><b>Note:</b> The window only zooms in vertical.</p> |

## Examples

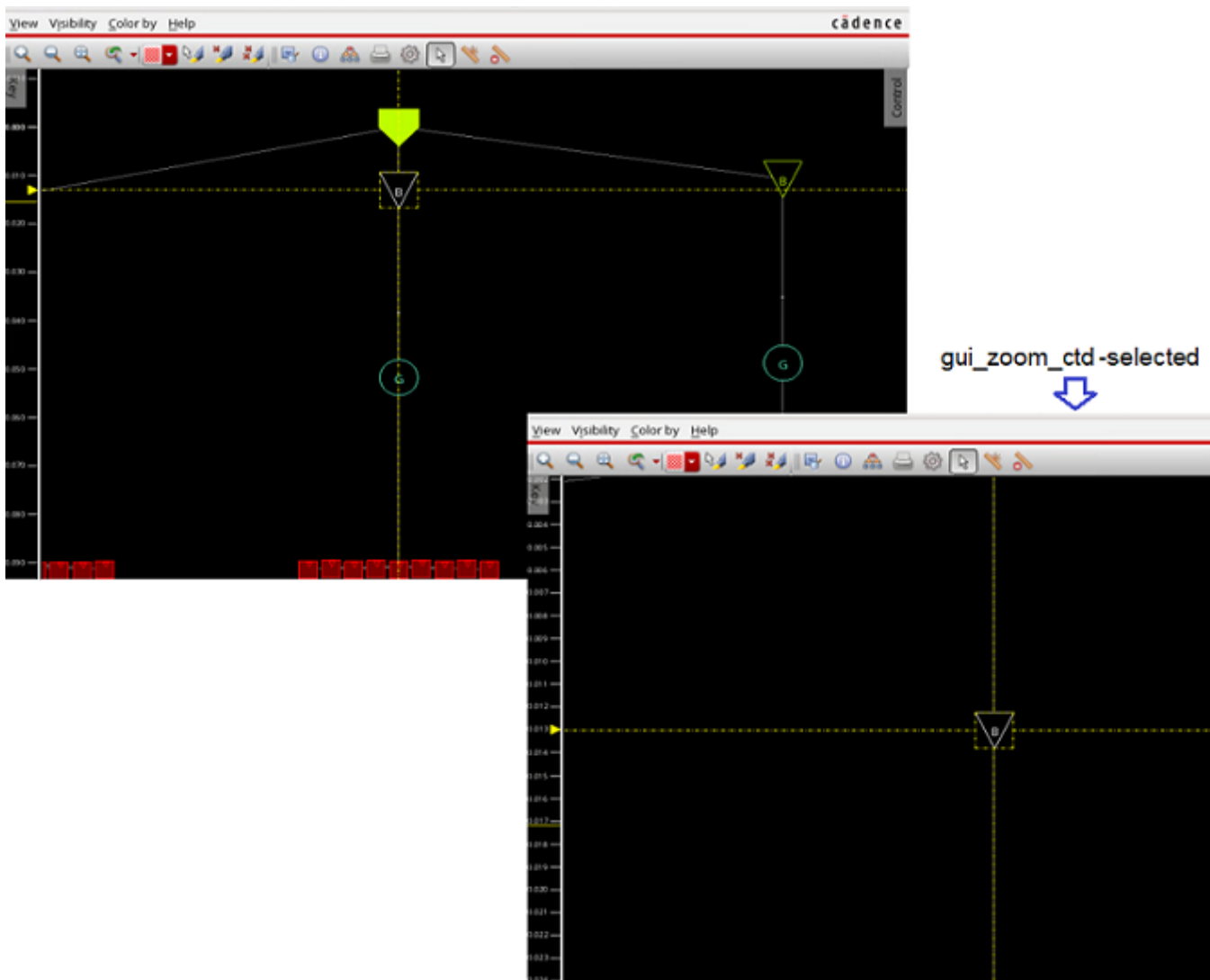
Use the following command to zoom in to the CTD:

```
gui_zoom_ctd -in
```



Select an instance and use the following command to zoom into the selected instance:

```
gui_zoom_ctd -selected
```



## Related Commands

- `close_ctd_win`
- `ctd_win`
- `gui_ctd_open`
- `gui_ctd_close`
- `gui_ctd_highlight`



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# Auxiliary Commands

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- [RC Extraction Commands](#)
- [Power Calculation Commands](#)

## RC Extraction Commands

- [extract\\_rc](#)
- [get\\_extract\\_rc\\_mode](#)
- [get\\_qrc\\_tech\\_file](#)
- [rc\\_out](#)
- [read\\_parasitics](#)
- [report\\_net\\_parasitics](#)
- [report\\_rcdb](#)
- [report\\_unit\\_parasitics](#)
- [set\\_extract\\_rc\\_mode](#)
- [set\\_qrc\\_tech\\_file](#)



## extract\_rc

extract\_rc  
[-help]

Extracts resistance and capacitance for the interconnects and stores the results in an RC database. Optionally, an ASCII report file can be generated. Use this command after performing trial routing or final routing, and after using the [set\\_extract\\_rc\\_mode](#) command.

### Parameter

|           |                                                                                                                                                                   |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -<br>help | Outputs a brief description of the <code>extract_rc</code> command. For a detailed description of the command, use the man command: <code>man extract_rc</code> . |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Example

The following command extracts interconnect resistance and capacitance:

```
tempus> extract_rc
```

RC database is created.

### Related Command

- [set\\_extract\\_rc\\_mode](#)

## get\_extract\_rc\_mode

```
get_extract_rc_mode
[-help]
[-capFilterMode { relOnly | relAndCoup | relOrCoup }]
[-cerebrus_license_only {true | false}]
[-compressOptMemRCDB {true | false}]
[-coupled {true | false}]
[-coupling_c_th value_in_fF]
[-lefTechFileMap file_name]
[-qrcCmdFile file_name]
[-qrcCmdType {auto | partial | custom}]
[-qrcRunMode {concurrent | sequential}]
[-relative_c_th value]
[-scOpTemp value]
[-total_c_th value_in_fF]
[-tsvSubcktFile file_name]
[-turboReduce {true | false | auto}]
[-quiet]
```

Displays the following information about a `set_extract_rc_mode` parameter in the Tempus log file and in the Tempus console:

- Parameter name
- Current value
- Type (Boolean, string, and so on)
- Whether the current value was set by user

If you do not specify a parameter, the software displays information for all of the `set_extract_rc_mode` parameters.

## Parameters

|                              |                                                                                                                                                                                                                                                              |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>           | Outputs a brief description that includes type and default information for each <code>get_extract_rc_mode</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man get_extract_rc_mode</code> . |
| <code>parameter_names</code> | <p>Displays information for the specified parameters. You can specify one or more parameters.</p> <p>See <code>set_extract_rc_mode</code> for descriptions of the extraction setup parameters you can specify.</p>                                           |

|                     |                                                                                                                                                                                                                                                                     |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-quiet</code> | Displays the current settings for the specified parameters in Tcl list format only.<br>If you specify <code>-quiet</code> without any parameters, the software displays the current settings of all <code>set_extract_rc_mode</code> parameters in Tcl list format. |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Examples

- The following command displays the current setting for the `-coupled` parameter:

```
tempus> get_extract_rc_mode -coupled
```

The software displays the following information:

```
-coupled true # bool, default=true
true
```

- The following command displays the current settings for the `-capFilterMode` parameter:

```
tempus> get_extract_rc_mode -capFilterMode
```

The software displays the following information:

```
-capFilterMode relAndCoup # enuns = (relOnly, relAndCoup, relOrCoup),
default=relAndCoup
```

- The following command displays the current setting for the `-coupled` parameter in Tcl list format only:

```
tempus> get_extract_rc_mode -coupled -quiet
```

The software displays the following information:

```
true
```

- The following command displays the current settings for all `set_extract_rc_mode` parameters in Tcl list format only:

```
tempus> get_extract_rc_mode -quiet
```

## Related Command

`set_extract_rc_mode`

## get\_qrc\_tech\_file

```
get_qrc_tech_file
[-help]
[-techfile]
[-quiet]
```

Returns the name of the Quantus QRC technology file used for standalone Quantus QRC extraction in a single corner design. You can set these values using the [set\\_qrc\\_tech\\_file](#) command.

### Parameters

|           |                                                                                                                                                                                                                                                          |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help     | Outputs a brief description that includes type and default information for each <code>get_qrc_tech_file</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man get_qrc_tech_file</code> . |
| -quiet    | Returns an abbreviated output that only shows the names or current settings of the parameters set with the <a href="#">set_qrc_tech_file</a> command.                                                                                                    |
| -techfile | Returns the name of the Quantus QRC technology file.                                                                                                                                                                                                     |

### Example

- The following command specifies the name of the Quantus QRC technology file:

```
tempus> get_qrc_tech_file -techfile

#techfile 'simple.tech' (string, user setting)

simple.tech
```

### Related Command

[set\\_qrc\\_tech\\_file](#)

## rc\_out

```
rc_out
[-help]
[-excludeIlm]
[-noRes]
{{{ -spef file_name } [-excNetFile file_name | -net file_name | -netName list_of_nets]}}
[-useEEQCellWithLibertyInfo]
[-view view_name | -rc_corner rc_corner_name]
[-cUnit {pF | fF}]
```

Reads the parasitic database and outputs the parasitics in the Standard Parasitic Exchange Format (SPEF).

## Parameters

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                      | Outputs a brief description that includes type and default information for each <code>rc_out</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man rc_out</code> .                                                                                                                                                                                                                                   |
| <code>-cUnit unit</code>                | Specifies the unit of capacitance as <code>fF</code> or <code>pF</code> in the output SPEF file.<br><i>Default:</i> <code>pF</code>                                                                                                                                                                                                                                                                                                                                  |
| <code>-excludeIlm</code>                | Writes out only the top-level SPEF while excluding the ILM parasitics. For example,<br><br><code>rc_out -excludeIlm -spef abc.spef -rc_corner corner1</code>                                                                                                                                                                                                                                                                                                         |
| <code>-excNetFile file_name</code>      | Specifies the name of the file that contains nets to be excluded from the output file.                                                                                                                                                                                                                                                                                                                                                                               |
| <code>-net file_name</code>             | Specifies the name of the input file that contains the net names to be extracted.<br><i>Default:</i> Extracts all net names                                                                                                                                                                                                                                                                                                                                          |
| <code>-netName list_of_nets</code>      | Specifies the names of the nets to be written in the SPEF file.<br><i>Default:</i> Writes all nets                                                                                                                                                                                                                                                                                                                                                                   |
| <code>-noRes</code>                     | Generates a SPEF/SPF file without the resistance information. The SPEF/SPF file generated by this parameter can only be used to improve the performance of third-party simulator tools in early stages of the design.<br><br><b>Caution:</b> Most tools expect the resistance information to be present in the SPEF/SPF file for performing extraction. Therefore, SPEF/SPF files with no resistance information should not be used with ostrich for RC correlation. |
| <code>-rc_corner rc_corner_name</code>  | Specifies the RC corner that is used to generate the parasitic output file in the multi-corner flow.                                                                                                                                                                                                                                                                                                                                                                 |
| <code>-spef file_name</code>            | Specifies the name of the SPEF output file.                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>-useEEQCellWithLibertyInfo</code> | Writes EEQ master cell in case the EEQ cells are present.                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>-view view_name</code>            | Specifies that the RC corner associated with the specified view is used to generate the output SPEF file in the multi-corner flow.                                                                                                                                                                                                                                                                                                                                   |

## Example

- Reads the parasitic database created during RC extraction and writes the interconnect capacitance in the SPEF format :

```
tempus> rc_out -spef design.spef
```

## read\_parasitics

```
read_parasitics
[-help]
[-force]
{[[[[-rc_corner string]]]]}
[-starN]
[-via_variation_file viaVariationFile]
```

Reads SPEF and RCDB-based RC parasitics information and already created RCDBs in Tempus. For hierarchical designs, use this command to read top-level as well as block-level parasitic information.

**Note:** The following are important aspects to be considered for using this command:

- The RCDB parasitic file must have the `.rcdb.d` extension.
- The RCDB-based parasitic data that is provided as input to this command must be the one that was saved in the same major release of the software. The command will not read the RCDB data saved in a previous release. For example, if you provide RCDB saved in 10.1.x release as input to the `read_parasitics` command in 11 release, then it cannot be read.
- Only the RCDB parasitic data on systems that are same as those used for saving the data can be read. For example, if the data is saved on Linux in 32bit mode, you can only restore it on Linux 32 bit mode.

## Parameters

|                                                                                                               |                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                                                                                                         | Outputs a brief description that includes type and default information for each <code>read_parasitics</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man read_parasitics</code> .                            |
| -all_rc_corner <i>string</i>                                                                                  | Specifies the multi-corner RCDB database to be read. For hierarchical designs, this parameter lists multi-corner RCDBs for both top and blocks levels.                                                                                                                          |
| -force                                                                                                        | Allows reading of parasitic data even when block level RCDB/ SPEF is missing for a corner.<br><br><b>Note:</b> This parameter is only applicable for hierarchical designs. It is required that the top cell should have parasitic data specified for all the active RC corners. |
| <i>-rc_corner list of RCcorner specific RCDBs/SPEFs for hierarchical design   RCcorner specific RCDB/SPEF</i> |                                                                                                                                                                                                                                                                                 |



|                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                            | <p>Specifies the RC corner name along with a corner-specific RCDB or SPEF for both top and block levels. For hierarchical designs, this parameter lists the corner-specific SPEFs/RCDBs for top and block levels.</p> <p>To read multi-value SPEFs, specify the <code>spef_list</code>. The <code>spef_list</code> provides the list of SPEFs and <code>spef_field</code> provides the corresponding SPEF fields to read for the specified RC corner.</p> <p>In the example below, for RC corner <code>RC1</code>, <code>spef_field 2</code> from <code>&lt;spef1&gt;</code> and <code>spef_field 3</code> from <code>&lt;spef2&gt;</code> will be read.</p> <pre>read_parasitics \ -rc_corner RC1 {spef_list {&lt;spef1&gt; &lt;spef2&gt;} spef_field {2 3}}</pre> <p>Similarly, you can specify this for multiple RC corners.</p> <pre>-rc_corner RC2 {spef_list {&lt;spef3&gt; &lt;spef4&gt;} spef_field {1 3}} \ -rc_corner RC3 .... \</pre> <p><b>Note:</b> A corner-specific RCDB parasitic file can only be provided as input to the RC corner for which it was generated. It cannot be provided as input to other RC corners.</p> |
| <p><code>-starN</code></p>                                                 | <p>Specifies whether or not to read the <code>*N</code> statements in the SPEF file. The <code>*N</code> statements define the location of the RC nodes.</p> <p>When this parameter is specified, the <code>*N</code> statements are read, otherwise the <code>*N</code> statements are ignored.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p><code>-</code><br/><code>via_variation_file viaVariationFile</code></p> | <p>Specifies the name of the file for modeling via variation. The contents of the file include all the via layer names with the corresponding <code>MEAN_SHIFT</code> and <code>STDDEV</code> so that all the VIA resistance in SPEF can be mapped to their own variation multiplier. This option is used to specify the <code>via_variation</code> file only when the design has a single corner.</p> <p>For example,</p> <pre>read_parasitics -rc_corner rcCornerName \ -via_variation_file viaVariationFileName</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

## Examples

- The example below is for hierarchical design with Top (T), Block 1(B1) and Block 2(B2), and two active RC corners, C1 and C2.
  - The following command specifies the RC corner names along with corner-specific RCDB and SPEF for both top and block levels:
- The example below is for hierarchical design with Top (T), Block 1(B1) and Block 2(B2), and three active RC corners, C1, C2, and C3.

```
tempus> read_parasitics \
-rc_corner C1 Top_C1.spef Block1_C1.spef Block2_C1.spef.gz \
-rc_corner C2 Top_C2.rcdb.d Block1_C2.spef Block2_C2.rcdb.d
```

- The following command specifies the multi-corner RCDB database to be read for both top and block levels and a single RCDB will be generated for the complete design:

```
tempus> read_parasitics \
-rc_corner C1 Top_C1.spef.gz \
-rc_corner C2 Top_C2.compressed.rcdb.d \
-rc_corner C3 Top_C3.rcdb.d
```

## report\_net\_parasitics

```
report_net_parasitics
net_name
[-help]
-rc_corner rc_corner_name
[-file_name output_file_name]
```

Reports the RC parasitics of the specified net.

The `report_net_parasitics` command reports a list of attackers in the order of worst coupling capacitance by default. The software determines and reports the probable top N worst attackers for a victim. The command output will show worst coupling capacitance, in addition to RC parasitics.

### Parameters

|                                                          |                                                                                                                                                                                                                                                                  |
|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                                       | Outputs a brief description that includes type and default information for each <code>report_net_parasitics</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man report_net_parasitics</code> . |
| <code>-file_name</code><br><code>output_file_name</code> | Specifies the name of the output file.                                                                                                                                                                                                                           |
| <code>-</code><br><code>rc_corner rc_corner_name</code>  | Specifies the RC corner name for which the RC parasitics are to be reported.                                                                                                                                                                                     |
| <code>net_name</code>                                    | Specifies the name of the net for which the RC parasitics are to be reported.                                                                                                                                                                                    |

## Example

- The following command reports the RC parasitics for net, NET1 for RC corner, `rc_best`:

```
report_net_parasitics -rc_corner rc_best NET1
```

```
Net: NET1
```

```
RC Corner: rc_best
```

```
Total Cap = 0.473082 ff
```

```
Total Res = 2.104041 ohms
```

```
Total XCap = 0.334035 ff
```

```
Cap Distribution:
```

```
Total Ground Cap = 0.139046 ff
```

```
XCap With Nets:
```

```
NET3 = 0.059964 ff
```

```
NET4 = 0.040277 ff
```

```
NET5 = 0.233794 ff
```

# report\_rcdb

```
report_rcdb
[-help]
rcdb_dir_name
```

Displays the contents of the RCDB being read.

**Note:**

- 1. This command is supported only for reading flat single RCDB. Multiple RCDB's cannot be read simultaneously using this command.
- 2. The RCDB being read must be generated on the same platform and architecture on which it is being read. For example, if you have generated the RCDB on Linux platform, then the `report_rcdb` command cannot report its content on the IBM AIX platform. Similarly, an RCDB generated on a 64-bit system cannot be reported on a 32-bit system.

**Parameters**

|               |                                                                                                                                                                                                                                              |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help         | Outputs a brief description that includes type and default information for each <code>report_rcdb</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man report_rcdb</code> . |
| rcdb_dir_name | Specifies the location of the RCDB directory.                                                                                                                                                                                                |

**Example**

- The following command reports the contents of the RCDB being read in the `report.rcdb.d` directory:

```
write_rcdb report.rcdb.d
report_rcdb report.rcdb.d/

RCDB VERSION : 11.15
OS-bit : 64
OS-endianness : little-endian
RCDB Compression : off
RCDB Source : Innovus
DESIGN : dtmf_recvr_core
RC-Corners: : corner_worst_RCMAX, corner_worst_RCMIN
Coupled : yes
Node-loc : no
Statistical : no
```

## report\_unit\_parasitics

```
report_unit_parasitics
[-help]
-layer metal_layer_name
-rc_corner rc_corner_name
[-space wire_space_in_um]
[-via_res via_resistance]
[-width wire_width_in_um]
```

Reports unit parasitics for individual metal layers and vias. The report prints the per unit resistance and capacitance for the specified layer. The resistance values are reported in ohms per micron for min width and min spacing of the layer. The capacitance values are reported in femtofarad (fF) per micron for min width and min spacing of the layer.

Via parasitics are printed considering the layer mentioned in the command as the lower layer of the via. The default via specified in the LEF file will be considered in this case.

### Parameters

|                                     |                                                                                                                                                                                                                                                                    |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                               | Outputs a brief description that includes type and default information for each <code>report_unit_parasitics</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man report_unit_parasitics</code> . |
| -layer<br><i>metal_layer_name</i>   | Specifies the name of the metal layer for which the unit parasitics are to be reported. This parameter is required.                                                                                                                                                |
| -rc_corner<br><i>rc_corner_name</i> | Specifies the name of the RC corner for which the unit parasitics are to be reported. This parameter is required.                                                                                                                                                  |
| -space<br><i>wire_space_in_um</i>   | Specifies the wiring space in microns.<br><i>Default:</i> minimum spacing                                                                                                                                                                                          |
| -<br><i>via_res via_resistance</i>  | When this parameter is specified, the resistances of all vias, whose upper layer is specified using the <code>-layer</code> parameter, are reported.                                                                                                               |
| -width<br><i>wire_width_in_um</i>   | Specifies the wiring width in microns.<br><i>Default:</i> minimum width                                                                                                                                                                                            |

## Examples

- The following command reports the unit parasitics for metal layer, `Metal5`, for the specified RC corner, `rc_worst`:

```
report_unit_parasitics -layer Metal5 -rc_corner rc_worst
```

The software gives the following information:

```
Unit parasitics for layer Metal5 (in fF,ohm)
 RC-Corner = rc_worst
 Cap = 0.213204
 Res = 0.900873
Parasitics for default Via "VIA5_0_VH"
 Via Cap = 0.017901
 Via Res = 8.859576
```

- The following commands reports the unit parasitics for `Metal5` with spacing two times the minimum spacing:

```
report_unit_parasitics -layer Metal5 -rc_corner rc_worst -space [expr 2 * [dbGet [dbGet
head.layers.extName Metal5 -p].minSpacing]]
```

The software gives the following information:

```
Unit parasitics for layer Metal5 (in fF,ohm)
 RC-Corner = rc_worst
 Cap = 0.188657
 Res = 0.900873
Parasitics for default Via "VIA5_0_VH"
 Via Cap = 0.017901
 Via Res = 8.859576
```

- The following commands reports the unit parasitics for `Metal5` with width two times the minimum width:

```
report_unit_parasitics -layer Metal5 -rc_corner rc_worst -width [expr 2 * [dbGet [dbGet
head.layers.extName Metal5 -p].minWidth]]
```

The software gives the following information:

```
Unit parasitics for layer Metal5 (in fF,ohm)
 RC-Corner = rc_worst
 Cap = 0.198657
 Res = 0.555333
Parasitics for default Via "VIA5_0_VH"
 Via Cap = 0.017901
 Via Res = 8.859576
```

- The following commands reports the unit parasitics for vias with top layer of Metal5:

```
report_unit_parasitics -layer Metal5 -rc_corner rc_worst -via_res
```

The software gives the following information:

```
Unit parasitics for layer Metal5 (in fF,ohm)
```

```
RC-Corner = rc_worst
Cap = 0.213204
Res = 0.900873
```

```
Parasitics for default Via "VIA5_0_VH"
```

```
Via Cap = 0.017901
Via Res = 8.859576
Other Via res
VIA45_1C 8.859576
VIA45_1C_H 8.859576
VIA45_1C_V 8.859576
VIA45_1ST_N 8.859576
VIA45_1ST_S 8.859576
VIA45_2C_E 4.429788
VIA45_2C_W 4.429788
VIA45_2C_N 4.429788
VIA45_2C_S 4.429788
VIA45_2ST_N 4.429788
VIA45_2ST_S 4.429788
VIA45_4C 2.214894
```



# set\_extract\_rc\_mode

```
set_extract_rc_mode
[-help]
[-reset]
[-capFilterMode { relOnly | relAndCoup | relOrCoup }]
[-cerebrus_license_only {true | false}]
[-compressOptMemRCDB {true | false}]
[-coupled {true | false}]
[-coupling_c_th value_in_fF]
[-lefTechFileMap file_name]
[-qrcCmdFile file_name]
[-qrcCmdType {auto | partial | custom}]
[-qrcRunMode {concurrent | sequential}]
[-relative_c_th value]
[-scOpTemp value]
[-total_c_th value_in_fF]
[-tsvSubcktFile file_name]
[-turboReduce {true | false | auto}]
```

Sets the RC extraction mode. Use this command before using the `extract_rc` command.

## Parameters

|       |                                                                                                                                                                                                                                                                     |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help | Outputs a brief description that includes type and default information for each <code>set_extract_rc_mode</code> parameter.<br><br>For a detailed description of the command and all of its parameters, use the man command: <code>man set_extract_rc_mode</code> . |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>-capFilterMode { relOnly   relAndCoup   relOrCoup }</pre> | <p>Specifies the capacitance filtering mode. The filtering modes are conditions which define how to use the <code>-coupling_c_th</code> and <code>-relative_c_th</code> values, or their combination. Based on the filtering mode specified, the software determines the nets in the design, which will have their coupling capacitance lumped to the ground.</p> <p><b>Note:</b> The behavior of the <code>-total_c_th</code> filter is independent of the <code>-capFilterMode</code> parameter setting. Nets that have a total capacitance value lower than the value specified with the <code>-total_c_th</code> parameter will have their coupling capacitances grounded regardless of the <code>-capFilterMode</code> setting.</p> <p>You can use one of the following modes:</p> <ul style="list-style-type: none"> <li>• <code>relOnly</code>: Grounds the coupling capacitance of nets based on the threshold value set by the <code>-relative_c_th</code> parameter.</li> <li>• <code>relAndCoup</code>: Grounds the coupling capacitance of nets only when it is lower than the threshold value specified with the <code>-relative_c_th</code> parameter as well as the <code>-coupling_c_th</code> parameter. In this mode, you enforce maximum restriction on grounding the coupling capacitances.</li> <li>• <code>relOrCoup</code>: Grounds the coupling capacitance of nets if it is lower than the threshold value specified with one of the <code>-relative_c_th</code> or <code>-coupling_c_th</code> parameters.</li> </ul> <p><i>Default:</i> For standalone QRC extraction, if the process node is 130nm or below, the default value is <code>relAndCoup</code>. If the process node is above 130nm, the default value is <code>relOnly</code>.</p> |
| <pre>-cerebrus_license_only {true   false}</pre>               | <p>This parameter is applicable to the Cerebrus-Quantus flow.</p> <p>When set to <code>true</code>, only the Cerebrus license will be considered and Tempus will not fall back to use Quantus license when Cerebrus license(s) is/are not available. By default, the option is set to <code>false</code>, which means the regular Quantus license will be considered when Cerebrus license(s) is/are not available.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <pre>-compressOptMemRCDB {true   false }</pre>                 | <p>Generates compressed RCDB data.</p> <p><i>Default:</i> <code>false</code></p> <p><b>Note:</b> This functionality reduces the disk usage but leads to marginal increase in run time.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-coupled {true   false}</code>    | <p>Separately outputs grounded and coupling capacitance. The <code>-coupled false</code> value is typically used for static timing analysis. Signal integrity analysis requires the coupled mode.</p> <p><i>Default:</i> <code>true</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <code>-coupling_c_th value_in_fF</code> | <p>Specifies the threshold value that determines when the extractor lumps a net's coupling capacitance to ground. This parameter is applicable only when the <code>-coupled</code> parameter is set to <code>true</code>.</p> <p>The software decouples the coupling capacitance of nets when the total coupling capacitance between the pair of nets is lower than the threshold specified with this parameter. The actual decoupling also depends on the setting of the <code>-capFilterMode</code> parameter.</p> <p><i>Default:</i> <code>0.1fF</code>. If the <code>-process process_node</code> parameter of the <code>set_design_mode</code> command is used, the software assigns the recommended default value for this parameter based on the specified process node.</p> <div style="border: 1px solid #c6e0b4; padding: 10px; margin-top: 10px;"> <p>✔ See the <code>set_design_mode</code> command for the list of default values that are applied based on the process node setting.</p> </div> |
| <code>-lefTechFileMap fileName</code>   | <p>Specifies the location of the layer map file for the standalone QRC engine.</p> <p><b>Note:</b> The layer map file must be in the Common Command Language (CCL) format.</p> <p>The following is an example of the layer map file format:</p> <pre>extraction_setup \ -technology_layer_map \ # &lt;LEF/DEF Name&gt; #&lt;Tech File Name&gt; M1 METAL_1 \ VIA1 VIA_1 \ M2 METAL_2 \</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>-qrcCmdFile fileName</code>       | <p>Specifies the user created Quantus (QRC) command file to be used for running standalone Quantus. This command file should be in the new CCL syntax.</p> <p><b>Note:</b> This parameter is only valid if <code>-qrcCmdType</code> <code>partial</code> or <code>custom</code> is selected.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>-qrcCmdType {auto  </code>        | <p>Specifies whether to derive the settings for running QRC from the Tempus</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

`partial | custom}`

settings, or from user created command file using the `-qrcCmdFile` parameter.

When set to `auto`, derives the Quantus settings automatically from the Tempus parameter settings.

When set to `partial`, appends user commands from the `qrcCmdFile` to the settings derived from the Tempus parameter settings. You must provide the partial mode command file in CCL syntax.

Use the partial command file only when you add extraction directives for those options that cannot be set in Tempus.

Following are some important restrictions:

- Do not specify the following options in the partial mode cmd file:
  - Threshold setting
  - Coupled/decoupled mode
  - QRC techfile (s)
  - Operating temperature
- Do not specify `"-directory_name"` and `"-temporary_directory_name"` using `output_setup` command in partial cmd file
- Do not specify output file using `output_setup -file_name` command in partial cmd file

Neither the syntax nor the content of this file can be checked. This file will be appended as is to the automatically generated Quantus command file.

When set to `custom`, gets the complete Quantus settings from the user created command file. Except for input and output, no settings from Tempus parameters are used. You must provide the complete Quantus command file in CCL syntax.

Following are some important restrictions:

- Do not specify an input DEF/OA DB in the custom mode cmd file
- Do not specify `"-directory_name"` and `"-temporary_directory_name"` in `output_setup` section of the command file.
- Specify output SPEF with the following convention for the name:  
`<design>.spef`.

In MMC, the output SPEF files will be named:

`<design>_Temperature_corner.spef`.

Here "design" refers to the top cell name.

|                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                  | <p><b>Note:</b> Do not prefix or postfix the output file name.</p> <ul style="list-style-type: none"> <li>Specify the Quantus techfile(s) to be used. In MMMC, the corner names should match the names used in the <code>create_rc_corner</code> command</li> <li>Provide Quantus techfile layermapping information in custom command file</li> <li>If GDS is provided for cell contents, provide a Quantus techfile to GDS layer number matching</li> </ul> <p><b>Note:</b> There is no checking for either the syntax or the content of the command file prior to running extraction. The command file will be used as is. If the output file is not specified correctly, the resulting parasitic file will not be read into the Tempus RC database.</p>                                           |
| <pre>-qrcRunMode {concurrent   sequential}</pre> | <p>Specifies the Quantus run mode in MMMC designs.</p> <p>When set to <code>concurrent</code>, runs Quantus for all active corners concurrently.</p> <p>When set to <code>sequential</code>, runs Quantus for all active corners sequentially.</p> <p><i>Default: concurrent</i></p> <div> <p> For MMMC designs, in Quantus partial command type, do not specify the output SPEF file name and RC corner name in the command file to run Quantus in concurrent or sequential mode.</p> <p>For MMMC designs, in Quantus custom command type, do not specify the output SPEF file name and RC corner name together in the command file. Quantus sequential mode is not supported in custom command type.</p> </div> |

|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><code>-relative_c_th value</code></p>        | <p>Sets a ratio threshold value that determines when the extractor lumps a net's coupling capacitance to ground. This parameter is applicable only when the <code>-coupled</code> parameter is set to <code>true</code>.</p> <p>If the total coupling capacitance between a pair of nets is less than the percentage (specified with this parameter) of the total capacitance of the net with the smaller total capacitance in the pair, the coupling capacitance between these two nets will be considered for grounding. The actual decoupling also depends on the setting of the <code>-capFilterMode</code> parameter.</p> <p><i>Default:</i> 1.0.</p> <p>The range of this value is 0 to 1. If the <code>set_design_mode -process process_node</code> command is used, the software assigns the recommended default value for this parameter based on the process node specified.</p> <div style="border: 1px solid #c6e0b4; padding: 10px; margin-top: 10px;"> <p>✔ See the <code>set_design_mode</code> command for the list of default values that are applied based on the process node settings.</p> </div> |
| <p><code>-reset</code></p>                      | <p>Resets parameters to their default values. The <code>-reset</code> parameter must be the first parameter specified. If you specify <code>-reset</code> by itself, the software resets all <code>set_extract_rc_mode</code> parameters to their default values. If you specify parameters after <code>-reset</code>, the software resets only those parameters to their default values.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <p><code>-scOpTemp value</code></p>             | <p>Specifies the temperature derating value to be used in single-corner analysis mode. The derating value is specified in centigrade.</p> <p><i>Default:</i> 25C</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <p><code>-total_c_th<br/>value_in_fF</code></p> | <p>Specifies the threshold value (femtoFarads) that determines when the extractor lumps a net's coupling capacitance to ground. This parameter is applicable only when the <code>-coupled</code> parameter is set to <code>true</code>.</p> <p>The software grounds the coupling capacitances for the nets which have a total capacitance value less than the value specified with this parameter.</p> <p><i>Default:</i> 0 fF. If the <code>set_design_mode -process process_node</code> command is used, the software assigns the recommended default value for this parameter based on the process node specified.</p> <div style="border: 1px solid #c6e0b4; padding: 10px; margin-top: 10px;"> <p>✔ See the command for the list of default values that are applied based on the process node settings.</p> </div>                                                                                                                                                                                                                                                                                               |

|                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><code>-tsvSubcktFile</code><br/><i>file_name</i></p> | <p>Specifies the name of the Through Silicon Via (TSV) subckt file to be used. The file contains the .subckt model information for TSV. The file has a Spice format. A sample file is shown below.</p> <pre>.subckt TSV top bottom R1 top n2 0.03075 R2 n2 bottom 0.03075 C1 n2 0 28f .ends</pre> <p><b>Where,</b></p> <p><code>top</code> and <code>bottom</code> refer to the terminal nodes on the top and bottom layers of the TSV</p> <p><code>n2</code> is the internal node between the top and bottom layers</p> <p><code>R1 top n2 0.03075</code> means that a resistor with a resistance of 0.03075 Ohms is added between the <code>top</code> and <code>n2</code> nodes</p> <p><code>R2 n2 bottom 0.03075</code> means that a resistor with a resistance of 0.03075 Ohms is added between the <code>bottom</code> and <code>n2</code> nodes</p> <p><code>C1 n2 0 28f</code> means that a ground capacitance of 28 fF is added on the <code>n2</code> node</p> <p><b>Note:</b> The IQuantus engine might not keep the subckt connectivity as it is. The optimization of RES network will remove node <code>n2</code> (if there is no other connections) and keep a single resistor between the top and bottom nodes with an accumulated resistance. For example, in the above example, the accumulated resistance will be 0.0615 Ohms. Capacitance will also be moved to a “higher layered node”. In the above example, this layer is <code>top</code>.</p> <p><i>Default:</i> "" (empty string)</p> |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

`-turboReduce`  
`{true|false|auto}`

Controls the turbo reduction feature for standalone Quantus extraction.

*Default:* `auto`

When set to `auto`, the software automatically turns `on` turbo reduction and proceeds when the required Quantus QRC license is available. However, if the required Quantus QRC license is not installed in the license server, the software automatically turns off Turbo reduction and proceeds with a warning message. If the license is installed in the license server but temporarily not available, then two scenarios are possible. First, if the license wait mode is not turned on, then Quantus will look for the license right away and error out because the license is not available. Second, if the license wait mode is turned on, then Quantus will wait and check for the license periodically until timeout. If the license is available before the timeout, turbo reduction will be performed but if the license is not available before timeout, QRC will error out.

When set to `true`, the turbo reduction engine is `on`. However, if the required Quantus QRC license is not installed in the license server, then standalone Quantus will error out with a message and the run will stop. It is because setting the option to `true` means the user really wants to perform turbo reduction.

When set to `false`, the turbo reduction engine is switched `off`.

For details, see the table titled, Behavior of Turbo Reduction, below.



## Behavior of Turbo Reduction

| Option Selected –<br>extract_rc_turbo_reduction {1   0   auto} | Availability of Quantus QRC AA License                                                                                                                                    |                                                                                                                                                                                                                                                                                         |                                                                    |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
|                                                                | <u>Quantus QRC AA Installed and Available</u><br>(IQRC gets the license right away even in the wait mode because sufficient licenses are available in the license server) | <u>Quantus QRC AA Installed but not Available</u><br>(If license wait mode is not turned on, then Quantus looks for the license. If the wait mode is turned on and if the license is temporarily not available, then IQRC waits and checks for the license periodically until timeout.) | <u>Quantus QRC AA Not Installed on License Server</u>              |
| <b>auto</b>                                                    | perform turbo reduction                                                                                                                                                   | 1. error out if wait mode is not on or if the specified wait time is exceeded<br>2. perform turbo reduction if wait mode is on and the license is available within the specified time.                                                                                                  | print warning and do not perform turbo reduction                   |
| <b>1</b>                                                       | perform turbo reduction                                                                                                                                                   | error out                                                                                                                                                                                                                                                                               | error out because the user explicitly wants to use turbo reduction |
| <b>0</b>                                                       | do not perform turbo reduction                                                                                                                                            | do not perform turbo reduction                                                                                                                                                                                                                                                          | do not perform turbo reduction                                     |

## Examples

- The following command specifies the capacitance filtering mode to relative only. In this case, the software grounds the coupling capacitance of nets based on the threshold value set by the `relative_c_th` parameter:

```
tempus> set_extract_rc_mode -capFilterMode relOnly
```

- The following command sets the coupled mode to `true` for SI analysis:

```
tempus> set_extract_rc_mode -coupled true
```

- The following command resets the `relative_c_th` parameter to its default value:

```
tempus> set_extract_rc_mode -reset relative_c_th
```

## Related Commands

- `create_rc_corner`
- `extract_rc`
- `get_extract_rc_mode`
- `set_design_mode`



## set\_qrc\_tech\_file

```
set_qrc_tech_file
[-help]
techFileName
```

Specifies the Quantus (QRC) technology file.

**Note:** Use this command for single corner extraction only. For multi-corner extraction, use the following command:

```
tempus> create_rc_corner -name rcCornerName -qx_tech_file fileName
```

### Parameters

|                           |                                                                                                                                                                                                                                                          |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>        | Outputs a brief description that includes type and default information for each <code>set_qrc_tech_file</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man set_qrc_tech_file</code> . |
| <code>techFileName</code> | Specifies the name of the Quantus technology file. The technology file contains the interconnect models that standalone Quantus extraction uses.                                                                                                         |

### Example

- The following command performs standalone Quantus extraction using the `qrcTechfile` technology file:

```
tempus> set_qrc_tech_file qrcTechfile
```

### Related Commands

- `create_rc_corner`
- `get_qrc_tech_file`

## Power Calculation Commands

- [analyze\\_ir\\_paths](#)
- [dump\\_unannotated\\_nets](#)
- [encrypt\\_thermal\\_file](#)
- [get\\_default\\_switching\\_activity](#)
- [get\\_dynamic\\_power\\_simulation](#)
- [get\\_power\\_analysis\\_mode](#)
- [map\\_activity\\_file](#)
- [propagate\\_activity](#)
- [read\\_activity\\_file](#)
- [report\\_instance\\_power](#)
- [report\\_power](#)
- [report\\_power\\_domain](#)
- [reset\\_power\\_activity](#)
- [restore\\_power\\_database](#)
- [set\\_default\\_switching\\_activity](#)
- [set\\_dynamic\\_power\\_simulation](#)
- [set\\_inst\\_temperature\\_file](#)
- [set\\_power](#)
- [set\\_power\\_analysis\\_mode](#)
- [set\\_power\\_calc\\_temperature](#)
- [set\\_power\\_include\\_file](#)
- [set\\_power\\_output\\_dir](#)
- [set\\_switching\\_activity](#)
- [set\\_twf\\_attribute](#)
- [set\\_virtual\\_clock\\_network\\_parameters](#)
- [write\\_power\\_constraints](#)
- [write\\_tcf](#)

## analyze\_ir\_paths

```
analyze_ir_paths
[-help]
-clock_period <float>
[-hierarchical_activity_file <string>]
[-max_slack <float>]
[-pre_simulation_period <float>]
-seq_activity <float>
-simulation_period <float>
[-user_defined_critical_paths <string>]
```

Generates a physical-aware multi-cycle switching activity file containing toggle information of voltage sensitive paths as well as timing critical paths in the design for Tempus Power Integrity analysis. This command generates 'swsrc.txt' file in the run directory, which then is fed back to the software using the [read\\_activity\\_file](#) command.

### Parameters

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                                   | Prints out the command usage.                                                                                                                                                                                                                                                                                                                                                                                                   |
| -clock_period <float>                   | Specifies the period of dominant clock of the design in library time units. Tempus PI refers to this specified period for multi-cycle activity generation.                                                                                                                                                                                                                                                                      |
| -hierarchical_activity_file<br><string> | Provides a mechanism for different sequential activity for several partitions in the design. The following is the required format:<br><pre>&lt;hierarchical instance 1&gt; &lt;activity&gt; &lt;hierarchical instance 2&gt; &lt;activity&gt; ...</pre>                                                                                                                                                                          |
| -max_slack <float>                      | Specifies the maximum slack limit in library time units, above which the timing paths are not considered for activity generation.                                                                                                                                                                                                                                                                                               |
| -pre_simulation_period<br><float>       | Specifies the pre-simulation duration that will be used by Tempus PI for activity generation.<br><br>This parameter is useful when package model information is read-in using <a href="#">set_package</a> command, as it allows the package oscillations to settle down.<br><br><b>Note:</b> The value specified must match to value specified under <a href="#">set_rail_analysis_mode -record_results_start_time</a> setting. |
| -seq_activity <float>                   | Specifies the activity of sequential endpoints in the design. This value will be used to control the number of IR sensitive paths, along with their fanout cone, to toggle in each cycle.                                                                                                                                                                                                                                       |

|                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>-simulation_period &lt;float&gt;</pre>                     | <p>Specifies the simulation period in library time units that Tempus PI dynamic power analysis will use. The larger the simulation period, the more will be the number of paths covered in the analysis.</p> <p>The following formula can be used to set the simulation period:<br/> <math display="block">sim\_cycles = (2/seq\_act)*N</math></p> <p>Where,</p> <ul style="list-style-type: none"> <li>• <i>seq_act</i> is the input sequential activity (e.g., 0.05, 0.1 etc.)</li> <li>• <i>N</i> is the number of nworst paths to be covered per endpoint</li> <li>• <i>sim_period</i> is equal to the <i>sim_cycles</i>*&lt;time period of dominant clock&gt;</li> </ul> <p>It is recommended to have <math>N \geq 1</math>, that is, at least one path per endpoint should be scheduled in the analysis.</p> |
| <pre>-<br/>user_defined_critical_paths<br/>&lt;string&gt;</pre> | <p>Specifies a collection of timing paths that are to be prioritized for toggling while generating Tempus PI activity.</p> <p>This feature is useful when designers have prior information on certain timing paths that may switch together and can potentially cause IR-induced STA violations.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

## Examples

- The following command generates the activity file for design with dominant clock of 2GHz, with input sequential activity being 0.1 and simulation duration being 20ns:  

```
tempus > analyze_ir_paths -seq_activity 0.1 -simulation_period 20 -clock_period 0.5
```
- The following example specifies different sequential activity for different hierarchical partitions, with 0.2 sequential activity for unspecified partitions and top:  

```
tempus > analyze_ir_paths -seq_activity 0.2 -simulation_period 30 -clock_period 0.5 -
hierarchical_activity_file hact.txt
```

Contents of 'hact.txt':

```
hierA/inst1 0.25
hierB/inst2 0.15
```
- The following command specifies the pre-simulation duration, particularly useful for package oscillations to settle when package models have been read-in:  

```
tempus > analyze_ir_paths -seq_activity 0.1 -simulation_period 30 -clock_period 0.5 -
pre_simulation_period 10
```

## dump\_unannotated\_nets

```
dump_unannotated_nets
[-help]
[-file filename]
-type annotationType
```

Shows the percentage of nets annotated by a specific type and also prints the names of all nets that are not annotated by that type.

Annotation information is stored on a per-file basis for activity, VCD, TCF, and user commands to provide detailed annotation reports when requested.

`dump_unannotated_nets` can be invoked multiple times with different report types (`-type`). You must specify this command before `report_power`.

## Parameters

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-file filename</code>       | Specifies the text file name that includes the annotation report information. If no file is specified, the report will be written to the default output file <code>annotation_type.log</code> , where <i>type</i> is taken from the <code>-type</code> argument specified.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>-help</code>                | Outputs the command usage and a brief description about the command parameters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>-type annotationType</code> | <p>The type can be any of the following:</p> <ul style="list-style-type: none"> <li>• <code>all</code><br/>Shows annotation reports for all types.</li> <li>• <code>activity</code><br/>Shows annotation data for all types of activity specifications including TCF, VCD, SDC, TWF, and user commands that set activity.</li> <li>• <code>cap</code><br/>Shows annotations from the SPEF files.</li> <li>• <code>fsdb</code><br/>Shows annotations from the FSDB files.</li> <li>• <code>saif</code><br/>Shows annotations from the SAIF files.</li> <li>• <code>tcf</code><br/>Shows annotations from the TCF files.</li> <li>• <code>timing</code><br/>Shows annotations from the TWF files. This type is applicable only to the dynamic power analysis flow.</li> <li>• <code>vcd</code><br/>Shows annotation from the VCD files.</li> </ul> |

## Example

- The following command dumps the nets in the netlist that are missing in the vcd file:  

```
tempus> dump_unannotated_nets -type vcd -file vcd.log
```

The net names are written in the file `vcd.log`.
- The following command dumps the nets in the netlist which do not have an activity annotated from any source (VCD/TCF/SDC/TWF/user defined) and writes the net names in the file `unannotatednets.log`:



```
tempus> dump_unannotated_nets -type all -file unannotatednets.log
```

## Related Commands

- `set_power_analysis_mode`
- `read_activity_file`

# encrypt\_thermal\_file

```
encrypt_thermal_file
-input ASCII.txt
-output $encrypted_conductivity.txt
```

Specifies to encrypt the file containing the thermal conductivity value for layers, and to protect foundry's thermal conductivity parameters used for power map generation. This encrypted file is then specified with the `report_power -thermal_conductivity_inputs` parameter for generating a tile-based power map file.

## Parameters

|                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-input ASCII.txt</code>                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                   | <p>Specifies the name of the ASCII thermal file that needs to be encrypted. This file contains the thermal conductivity parameters used for power map generation.</p> <p>The <code>-input</code> parameter supports both layer type and layer name based thermal conductivity file.</p> <p>The following is a snippet of the input thermal conductivity file content (layer type based file):</p> <pre>Metal 430 Silicon 250 Dielectric 0.6</pre> |
| <code>-output \$encrypted_conductivity.txt</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                   | <p>Specifies the name of the encrypted output thermal file.</p> <p>The following is a snippet of the encrypted thermal conductivity file content:</p> <pre>Metal \$encryption_value Silicon \$encryption_value Dielectric \$encryption_value</pre>                                                                                                                                                                                                |

## Examples

- The following command encrypts the specified input file (`format_1.txt`) to generate the encrypted output file (`format_1_encr.txt`):

```
encrypt_thermal_file -input encrypt/format_1.txt -output encrypt/format_1_encr.txt
```

The encrypted file (`format_1_encr.txt`) is then specified for generating a tile-based power map file, as shown below:

```
report_power \
-outfile static.rpt \
-thermal_power_map_tile {5 5}\
-thermal_power_map_format stack \
-thermal_power_map_file powermap_format1_encr_in.txt \
-thermal_conductivity_inputs ./encrypt/format_1_encr.txt
```

## get\_default\_switching\_activity

```
get_default_switching_activity
[-help]
[-black_box_density]
[-black_box_duty]
[-block]
[-duty]
[-global_activity]
[-hier]
[-input_activity]
[-period]
[-seq_activity]
[-reset]
[-reset_type]
[-clock_gates_enable]
[-icg_ratio]
[-comb_clockgate_ratio]
[-clock_gates_output]
[-clock_gates_output_ratio]
[-clip_activity_to_domain_freq]
[-name]
[-macro_activity]
```

Displays the following information about a `set_default_switching_activity` parameter in the system log file and in the system console:

- Parameter name
- Current value

If you do not specify a parameter, the software displays information for all of the `set_default_switching_activity` parameters. The `get_default_switching_activity` command prints only the user-specified value for the parameters. If a parameter has not been set using the `set_default_switching_activity` command, the `get_default_switching_activity` command does not output any information in the console.

## Parameters

|                              |                                                                                                                                                                                                                                                                                                 |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>           | Outputs a brief description that includes type and default information for each <code>get_default_switching_activity</code> parameter. For a detailed description of the command and all of its parameters, use the <code>man</code> command: <code>man set_default_switching_activity</code> . |
| <code>parameter_names</code> | <p>Displays information for the specified parameters. You can specify one or more parameters.</p> <p>See <code>set_default_switching_activity</code> for description of the parameters you can specify.</p>                                                                                     |

## Examples

The following command displays the current setting of the `-period` parameter:

```
get_default_switching_activity -period
```

The software displays the following information:

```
-period {20}
```

## get\_dynamic\_power\_simulation

```
get_dynamic_power_simulation
[-help]
[-activity_pattern]
[-period]
[-resolution]
```

Displays the following information about a `set_dynamic_power_simulation` parameter in the system log file and in the system console:

- Parameter name
- Current value

If you do not specify a parameter, the software displays information for all of the `set_dynamic_power_simulation` parameters. The `get_dynamic_power_simulation` command prints only the user-specified value for the parameters. If a parameter has not been set using the `set_dynamic_power_simulation` command, the `get_dynamic_power_simulation` command does not output any information in the console.

### Parameters

|                              |                                                                                                                                                                                                                                                                                |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>           | Outputs a brief description that includes type and default information for each <code>get_dynamic_power_simulation</code> parameter. For a detailed description of the command and all of its parameters, use the man command: <code>man set_dynamic_power_simulation</code> . |
| <code>parameter_names</code> | Displays information for the specified parameters. You can specify one or more parameters.<br><br>See <code>set_dynamic_power_simulation</code> for description of the parameters you can specify.                                                                             |

### Examples

The following command displays the current setting of the `-period` parameter:

```
get_dynamic_power_simulation -period
```

The software displays the following information:

```
-period {20}
```

## get\_power\_analysis\_mode

get\_power\_analysis\_mode  
[-quiet]

Returns the following information about a [set\\_power\\_analysis\\_mode](#) parameter in the log file and in the software console:

- Parameter name
- Current value
- Type (Boolean, string, and so on)
- Whether the current value was set by user

If you do not specify a parameter, the software returns information for all of the [set\\_power\\_analysis\\_mode](#) parameters.

### Parameters

|                 |                                                                                                                                                                                 |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parameter_names | Displays information for the specified parameters. You can specify one or more parameters.<br><br>See <a href="#">set_power_analysis_mode</a> for descriptions you can specify. |
| -quiet          | Returns the current settings of all <a href="#">set_power_analysis_mode</a> parameters in Tcl list format.                                                                      |

### Example

- The following command returns the current settings for all [set\\_power\\_analysis\\_mode](#) parameters:

```
tempus> get_power_analysis_mode \
-honor_negative_energy true # bool, default=true \
-ignore_control_signals true # bool, default=true \
-leakage_scale_factor_for_temp 1 # float, default=1 \
-off_pg_nets {} # string, default="" \
-x_transition_factor 0.5 # float, default=0.5 \
-z_transition_factor 0.25 # float, default=0.25
```

- The following command returns the current settings for all [set\\_power\\_analysis\\_mode](#) parameters in Tcl list format only:

```
tempus> get_power_analysis_mode -quiet
```

The software returns the following information:

```
{honor_negative_energy true} {ignore_control_signals true}
{leakage_scale_factor_for_temp 1} {off_pg_nets {}} {x_transition_factor 0.5}
{z_transition_factor 0.25}
```

## Related Commands

- `set_power_analysis_mode`
- `report_power`



## map\_activity\_file

```
map_activity_file
[-golden {rtl | gate}]
[-gate_block block_name]
[-reset]
-rtl2gate mapping_file
[-rtl_block block_name]
[-two_column_mapping_file mapping_file]
```

Specifies instance name mapping between RTL netlist and GATE level netlist. Power analysis uses this instance name mapping to annotate the output nets of this instance from the pin based RTL VCD or TCF.

You must specify this command before `report_power`.

### Parameters

|                                     |                                                                                                                                                                                                                                                                                      |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-gate_block block_name</code> |                                                                                                                                                                                                                                                                                      |
|                                     | Specifies the GATE activity block name to map the block-level RTL mapping files to the top-level RTL activity files. The GATE block name is appended before the GATE level entries in the mapping file to match the entries in the top-level RTL activity file.                      |
| <code>-golden {rtl   gate}</code>   |                                                                                                                                                                                                                                                                                      |
|                                     | Specifies whether RTL or Gate is Golden. The one not specified as golden will be considered as Revised. Currently the Conformal MatchPoint file does not specify this.<br><i>Default:</i> rtl                                                                                        |
| <code>-reset</code>                 | Specifies to reset and ignore all the previous mapping files specified using the <code>map_activity_file</code> command.<br><i>Default:</i> All the previous mapping files specified using the <code>map_activity_file</code> command are loaded and used.                           |
| <code>-rtl2gate mapping_file</code> |                                                                                                                                                                                                                                                                                      |
|                                     | Specifies instance name mapping between RTL netlist and GATE level netlist. RTL level VCD or TCF are obtained from simulation on RTL Verilog netlist. Power analysis uses this instance name mapping to annotate the output nets of this instance from the pin based RTL VCD or TCF. |

|                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-rtl_block <i>block_name</i></code>                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                           | <p>Specifies the RTL activity block name to map the block-level RTL mapping files to the top-level RTL activity files. The RTL block name is appended before the RTL level entries in the mapping file to match the entries in the top-level RTL activity file.</p>                                                                                                                                                                                                                                                                                    |
| <code>-two_column_mapping_file <i>mapping_file</i></code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                           | <p>Specifies a file that contains the instance name mapping between the RTL netlist and GATE level netlist.</p> <p>This parameter can be used to specify a mapping file instead of the LEC generated mapping file (<code>-rtl2gate <i>mapping_file</i></code>) to map points from RTL to Gate netlist. The format of the two column mapping file is:</p> <pre>#Gate name =&gt; # RTL name gate_inst_name1 =&gt; rtl_inst_name1 ...</pre> <p>Here, the first column corresponds to the gate name and the second column corresponds to the RTL name.</p> |

## Example

- The following command specifies a mapping file `rtl_map`:  
`map_activity_file -rtl2gate rtl_map`
- The following command performs instance name mapping between RTL netlist and GATE level netlist, wherein the GATE level netlist is Golden and the RTL netlist is Revised; it specifies the mapping file `MapFile.alt`  
`map_activity_file -rtl2gate MapFile.alt -golden gate`
- The following commands appends the RTL block name and GATE block name before the RTL and GATE level entries in the mapping file to match the entries in the top-level RTL activity file:

```
map_activity_file -rtl2gate LEC.top \
 -rtl_block "core/block_top" \
 -gate_block "core/block_top"

map_activity_file -rtl2gate LEC.tile.top \
 -rtl_block "core/tile_instance\[0\]/tile_inst" \
 -gate_block "core/tile_instance_0_tile_inst"
```

An example of the mapping file output from Conformal is described as follows:

```
Mapped points: SYSTEM class
1-th mapped points:
```

```
(G) + 1 PI /x_ick
(R) + 595 PI /x_ick
2-th mapped points:
(G) + 2 PI /x_jreset_cp_p
(R) + 594 PI /x_jreset_cp_p
3-th mapped points:
(G) + 3 PI /x_mreset_cp_p
(R) + 593 PI /x_mreset_cp_p
.....
60-th mapped points:
(G) + 4582 DFF /cpexec0/cpddecls0/gi_opls/q_reg_reg[35]
(R) + 3856 DFF /cpexec0/cpddecls0/gi_opls/q_reg_reg_35_
61-th mapped points:
(G) + 4583 DFF /cpexec0/cpddecls0/gi_opls/q_reg_reg[34]
(R) + 3855 DFF /cpexec0/cpddecls0/gi_opls/q_reg_reg_34_
62-th mapped points:
(G) + 4633 DFF /cpexec0/cpddecex0/gi_opex/q_reg_reg[21]
(R) + 3805 DFF /cpexec0/cpddecex0/gi_opex/q_reg_reg_21_
63-th mapped points:
(G) + 4634 DFF /cpexec0/cpddecex0/gi_opex/q_reg_reg[20]
(R) + 3804 DFF /cpexec0/cpddecex0/gi_opex/q_reg_reg_20_
64-th mapped points:
(G) + 4635 DFF /cpexec0/cpddecex0/gi_opex/q_reg_reg[19]
(R) + 3803 DFF /cpexec0/cpddecex0/gi_opex/q_reg_reg_19_
65-th mapped points:
(G) + 4636 DFF /cpexec0/cpddecex0/gi_opex/q_reg_reg[18]
(R) + 3802 DFF /cpexec0/cpddecex0/gi_opex/q_reg_reg_18_
```

where, **G** refers to the Golden netlist name and **R** refers to the Revised netlist name.

By default, the tool assumes that the Golden netlist name refers to the RTL net name and Revised netlist refers to the GATE level net name.

## Related Topics

- ["RTL Activity File Flow in Static Power Calculation"](#) in the *Voltus User Guide*

## propagate\_activity

```
propagate_activity
[-set_net_freq {true | false}]
```

Propagates the activity file in the database after it is read in using the [read\\_activity\\_file](#) command. This command propagates the activity for all primary inputs, nets, and other devices in the design whose activity has not been previously defined through user attributes.

`propagate_activity` can be used to propagate activities in the design and check for activity annotation before running power calculation. This is not a mandatory step as execution of `report_power` will also do the same and then do power calculation based on the propagated activities.

### Parameters

```
-set_net_freq {true | false}
```

Specifies that activity from the generated TCF file needs to be used to capture net frequency for signal EM analysis. When this parameter is set, you do not have to read the TCF file to set net frequency.

### Example

- The following commands read and propagate the activity file in the database:

```
tempus> read_activity_file
tempus> propagate_activity
```

### Related Topics

- "Signal ElectroMigration" in the *Voltus User Guide*

### Related Commands

- [read\\_activity\\_file](#)
- [report\\_power](#)

# read\_activity\_file

```
read_activity_file name
[-block block_name]
[-cell cell_name]
[-end { time1 time2 ...timen }]
[-format { VCD | TCF | SAIF | FSDB | PHY | SHM}]
[-hier_separator separator]
[-name_mapping_rule file]
[-reset]
[-rtl {true|false}]
[-scale_duration scalefactor]
[-scope scope_name]
[-set_net_freq {true | false}]
[-start { time1 time2 ...timen }]
[-start_time_shift value]
[-weight value]
[-zero_delay {true|false}]
```

Specifies the name and type of activity file to be used as input. This command must be used when specifying activity files for static or dynamic power calculation.

## Parameters

|                          |                                                                                                                                                                                                                                                                                                                             |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -block <i>block_name</i> |                                                                                                                                                                                                                                                                                                                             |
|                          | <p>Specifies the name of the FSDB/SAIF/TCF/VCD block to map a sub-block activity file with a top level Verilog file. Provides the ability to support multiple activity files based upon block instance names.</p> <p>The block argument is added to the FSDB/SAIF/TCF/VCD name before comparing it to the Verilog file.</p> |
| -cell <i>cell_name</i>   |                                                                                                                                                                                                                                                                                                                             |

Specifies the master cell name for the activity file. This parameter allows you to support master-cell-based activity specification. That is, you need to specify only the master cell name for the activity file, instead of specifying all the instances that were instantiated from a master cell.

### Example

Let us consider a design that has a cell named "super\_filter" with four block instances "super\_filter1", "super\_filter2", "super\_filter3", and "super\_filter4":

#### Instance-Based Use Model:

```
read_activity_file -format VCD -block super_filter1
read_activity_file -format VCD -block super_filter2
read_activity_file -format VCD -block super_filter3
read_activity_file -format VCD -block super_filter4
```

#### Master-Cell-Based Use Model:

```
read_activity_file -format VCD -cell super_filter
```

```
-end {time1 time2 ...timen }
```

When you use the vector-driven method of performing dynamic instance-based power-consumption calculation, you can specify the part of the activity file from which you want power analysis to use for the power calculation. This option specifies the end time of this range and is used in conjunction with the `-start` option.

Specifies the end time as time-value pairs to report the average power across non-overlapping multiple windows specified in the activity file for a block, level of hierarchy, or other part of the design that you want to analyze. Units are in seconds (s), milliseconds (ms), microseconds (us), nanoseconds (ns), or picoseconds (ps).

*Default*: The time unit is defined in the first .lib file read during design import.

You can specify the `$worst_power_window_end` variable with the `-end` parameter to take the worst case end time from the last vector profiler run. Not required for TCF and SAIF.

**Note**: You can read multiple windows in a single VCD file by specifying multiple pairs of the start time and end time for the non-overlapping multiple windows specified in the activity file. The use model for reading multiple windows is:

```
read_activity_file -format VCD design.vcd -start {start1 start2 start3} -end {end1
end2 end3}
```

The `start1/2/3` and `end1/2/3` values should be in the ascending order.

Multiple current waveforms representing multiple VCD windows are merged into a single continuous current waveform. To capture the waveform for signals at the edge of user windows, a buffer time is added between the windows being stitched using the `set_power_analysis_mode -settling_buffer` parameter. This buffer time ensures that the stitched waveform is contiguous. The generated waveform hence has the duration of user windows and the intervening settling buffers. The default settling buffer period is 400ps.

`-format { VCD | TCF | SAIF | FSDB | PHY | SHM }`

Specifies the net activity of the design using an activity file in the VCD, TCF, FSDB, SAIF, PHY, or SHM format. The arguments specified with the `-format` parameter are case-insensitive.

- The VCD file format uses the toggle count information as the basis for the power consumption calculation. These files can be in compressed (zipped) format using the .gz extension.
- The TCF file format contains data on the switching activity of the nets in the design. The switching activity includes the toggle count information and the probability of the net or pin being in the logic 1 state.  
These files can be in compressed (zipped) format using the .gz extension.

Supports extended TCF which contains transition density for X and Z as shown in the following extended TCF example:

"out" : "0.0 0" "0.005000 2" "0.0 0"

The fields are defined as follows:

Field 1: net or pin name

Field 2: duty and transition density for transition of 0-1

Field 3: duty and transition density for transition of X(0-X-1)

Field 4: duty and transition density for transition of Z(0-Z-1)

- The SAIF file format contains aggregate switching activity information on the nets and/or ports in the design. It does not contain time-based information. This file format is used for quick average power analysis of the design.
- The FSDB file format specifies the net activity of the design using an activity file in FSDB format. Voltus supports the FSDB version up to 5.8.

#### Use Model

```
read_activity_file -format FSDB fsdbfile
```

```
report_power
```

(or)

```
report_vector_profile
```

#### Example

```
read_activity_file -format FSDB test.fsdb -scope top
```

- The PHY (Palladium database format generated in the Palladium environment) format contains the simulation activity data generated during emulator-based simulation. The PHY format can be used in all vector-based power analysis flows (vector profiler, static power analysis and dynamic vector-based power analysis).

#### Use Model

```
read_activity_file-format PHY <path to PHY database directory>
```

#### Example

```
read_activity_file -format PHY ./phy2/trace.phy -scope test
```

#### Notes:

- The xeDebug software must be present in the `PATH` environment. Add xeDebug in the path using the following command:  

```
set path = (<install_dir>/bin $path)
```

The xeDebug software version should be the same as the one using which the PHY database is generated.
- The PHY database generated from the following UXE/VXE version is supported:  
UXE18.6.0 onwards  
VXE18.6.0 onwards
- When reading a PHY format activity file, you require a DPA



(DPA\_ToggleCountFormat) license.

- For information on error messages displayed while reading the PHY database, refer to the `xe.msg` file in the PHY database directory.
- Make sure that the PHY database is not locked before invoking the Voltus software.
- Make sure that the xeDebug software is not already running on the machine.
- Ensure that the following directories are present in the top-level PHY database directory: `*.phy`, `PDB`, `cellList`, `QTDB`, `dbFiles`, and `.design`.
- It is recommended that the write permission is assigned on the entire PHY database. In case the write permission cannot be granted, you must set the `SWFV_FREEZE_PHY` environment variable to 1.
- If you want to run parallel Voltus executables on the same top-level PHY directory with multiple `*.phy` databases, it is suggested to create local copies of the multiple top-level directories, with links to the original PHY database. This will allow local copies of the PHY database to acquire separate locks.
- The Simulation History Manager (SHM) database format is a record of the data signal changes that occur during design simulation. The SHM database consists of a directory, typically with the `".shm"` suffix. The directory contains two disk files, namely `*.trn` and `*.dsn`. For more information about the SHM format, refer to the *SimVision Analysis Environment* manual available on the Cadence Online Support (COS) website.

#### Example

```
read_activity_file -format SHM ../shm/ADDER.trn -scope adder_e/u1 -start 0ns -
end 60ns
```

`-hier_separator separator`

Specifies the separator character in the hierarchical net names, bus names, and pin names for the block or other part of the design that you want to analyze. It must be the same as the separator character in the design netlist.

**Default:** slash (/)

`-name_mapping_rule file`

|                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | <p>Specifies to enable rule-based mapping to map RTL vectors to gate-level netlist. This parameter helps in resolving name mismatch between RTL and gate level netlists.</p> <p>The name mapping rule file format is:</p> <pre>Name_Mapping_Rule_file # RTL          # Gate &lt;rtlstring&gt; &lt;gatestring&gt;</pre> <p>where, each line is a rule. The string in the first column(RTL) is what we want to replace with the string in the second column(Gate). The rules in every line are added onto the previous rules.</p> <p>For example, to replace all the square [ ] brackets with under score _ for RTL-to-Gate matching, the rule file will be:</p> <pre>#RTL      #GATE [          _ ]          _</pre> |
| -<br>reset                               | <p>If <code>-reset</code> is specified, all previous activity files specified using the <code>read_activity_file</code> command will be reset and ignored.</p> <p><b>Default:</b> All the previous activity files specified using the <code>read_activity_file</code> command are loaded and used (They are additive).</p>                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>-rtl {true false}</code>           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                          | <p>Specifies to process the vector as an RTL vector and perform state propagation. The propagation behavior is global and the full design is propagated. RTL vectors are zero-delay vectors so delay annotation is always done for signals from this vector file.</p> <p><b>Default:</b> false</p>                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>-scale_duration scalefactor</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                          | <p>Scales the FSDB/SAIF/TCF/VCD duration value by the specified scale factor. For example, an FSDB scale of 2 will double the FSDB duration value.</p> <p><b>Note:</b> The <code>-start</code> and <code>-end</code> parameters of the <code>read_activity_file</code> command are post scaled.</p>                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>-scope scope_name</code>           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

Specifies the FSDB/SAIF/TCF/VCD scope, that is, the name of the module within the activity file associated with the block or other part of the design that you want to analyze.

The scope argument is removed from the FSDB/SAIF/TCF/VCD name before comparing it to the Verilog file.

 If the scope value has backslash '\' or square brackets '[]', you must specify the scope within curly brackets {{.....}} to ensure that '\' or '[]' are correctly processed by the Tcl Interpreter. In the following example, the scope value is specified within curly brackets as it contains '\' and '[]':

```
read_activity_file -scope {{reg_based_buffer.reg_inst\sv_xpol_trans_q_reg[0]
[14][7][sv_re][4]}}
```

If '\' and '[]' are not there, curly brackets are not required.

```
-set_net_freq {true | false}
```

This option should be set when an activity from a TCF or VCD file needs to be used to capture net frequency for signal EM analysis (`verify_AC_limit`). In addition, you must use the `verify_AC_limit -use_db_freq` option to read the toggle rates from the activity file.

**Default:** false

#### Example

The following example reads a TCF file which is used to capture net frequency for signal EM analysis:

```
read_activity_file -format TCF -set_net_freq true design.tcf
verify_AC_limit -use_db_freq -report signal_em.rpt
```

```
-start { time1 time2 ... timen }
```

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | <p>When you use the vector-driven method of performing dynamic instance-based power-consumption calculation, you can specify the part of the activity file from which you want power analysis to use for the power calculation. This option specifies the start time of this range and is used in conjunction with the <code>-end</code> option.</p> <p>Specifies the start time as time-value pairs to report the average power across non-overlapping multiple windows specified in the VCD file for a block, level of hierarchy, or other part of the design that you want to analyze. Units are in seconds (s), milliseconds (ms), microseconds (us), nanoseconds (ns), or picoseconds (ps).</p> <p><i>Default:</i> The time unit is defined in the first .lib file read during design import.</p> <p>You can specify the <code>\$worst_power_window_start</code> variable with the <code>-start</code> parameter to take the worst case start time from the last vector profiler run. Not required for TCF and SAIF.</p> <p><b>Note:</b> You can read multiple windows in a single VCD file by specifying multiple pairs of the start time and end time for the non-overlapping multiple windows specified in the activity file. The use model for reading multiple windows is:</p> <pre>read_activity_file -format VCD design.vcd -start {start1 start2 start3} -end {end1 end2 end3}</pre> <p>The <code>start1/2/3</code> and <code>end1/2/3</code> values should be in the ascending order.</p> <p>Multiple current waveforms representing multiple VCD windows are merged into a single continuous current waveform. To capture the waveform for signals at the edge of user windows, a buffer time is added between the windows being stitched using the <code>set_power_analysis_mode -settling_buffer</code> parameter. This buffer time ensures that the stitched waveform is contiguous. The generated waveform hence has the duration of user windows and the intervening settling buffers. The default settling buffer period is 400ps.</p> |
|             | <pre>-start_time_shift value</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|             | <p>Specifies to shift the start time of the specified activity files.</p> <p>When you specify the <code>-scale_duration</code> parameter with the <code>-start_time_shift</code> parameter, it scales the start time shift along with the start/end time.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <i>name</i> | <p>The name of the activity file.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|             | <pre>-weight value</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | <p>Specifies the weight number of the TCF file. 1.0 is the default value. The value should be a floating number that is smaller than 1.0. This parameter allows you to merge multiple TCF files at different function modes for static power estimation of a design.</p> <p>In the following example, TCF <i>file1</i>, <i>file2</i>, and <i>file3</i> has the weight numbers 0.3, 0.2, and 0.4, respectively.</p> <pre>read_activity_file -format TCF file1 -weight 0.3 read_activity_file -format TCF file2 -weight 0.2 read_activity_file -format TCF file3 -weight 0.4</pre> <p>Here, <i>file1</i>, <i>file2</i>, and <i>file3</i> will be merged together to represent the toggle and duty for the nets/pins in the design. The relative weight of 0.3/0.9, 0.2/0.9, and 0.4/0.9 would be used for static power calculation. In this example, 0.9 is the sum of weights.</p> |
| -zero_delay {true false} |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                          | <p>Specifies to process the vector as a zero-delay vector and perform delay annotation.</p> <p><b>Default:</b> false</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## Activity Precedence

The following commands can define activity:

```
read_activity_file file
 [-format { VCD | TCF | SAIF | FSDB }]
 [-hier_separator separator]
 [-start time]
 [-end time]
 [-reset]
 [-scope scope_name]
 [-block block_name]
```

```
set_default_switching_activity
 [-input_activity factor]
 [-seq_activity factor]
 [-period value]
 [-duty value]
 [-global_activity factor]
 [-hier hierarchy _name]
 [-clock_gates_output activity_factor]
```

```
set_switching_activity
 [-reset]
 [-clock clock_name]
 [-period value]
 [-unclocked]
```

```
[-activity factor | -density transition_density]
[-net net_name | -port port_name | -pin pin_name]
[-duty value]
[-inst]
```

The precedence of activity is as follows:

1. User-Defined: Activities applied to specific nets/pins/ports
2. Activity File: Activity specification through a VCD/FSDB/SAIF/TCF format
3. Clock Gates Output Activity: Activity specification to the output of all clock gates
4. SDC/TWF constants: Set constants to a list of pins or ports for use by the timing engine (timing analysis)
5. Hierarchical Global Activity: Activity specification for a specific hierarchy
6. Global Activity: Activity specification for all instances that are part of the data network.
7. Sequential Element Activity: Activity specification applied to the output of all sequential elements.
8. Primary Input Activity: Activity specification for all primary inputs

If you specify multiple activity data for the same items, the power engine will use the one with the higher precedence.

## Examples

- The following command reads the compressed `ap_wait_test_pll1_mod.vcd.gz` VCD file that contains the toggle count information for the power consumption calculations, specifies the `crm_ap` module within the VCD file as the module associated with the part of the design being analyzed, specifies the slash character (/) as the hierarchical separator for hierarchical net names, bus names, and pin names for the part of the design being analyzed, and specifies the start (180002ns) and end (189802ns) times from the VCD file for the part of the design being analyzed:

```
read_activity_file -format VCD \
-scope testbench/top/ap/mcu_platform/crm_ap \
-hier_separator \
-start 180002ns \
-end 189802ns \
ap_wait_test_pll1_mod.vcd.gz
```

- The following command reads the compressed `A.vcd.gz` VCD file that contains the toggle count information for the power consumption calculations, specifies `A1` as the block to map a sub-block VCD file with a block instance at the top-level, and specifies the start (180002ns) and end (189802ns) times from the VCD file for the part of the design being analyzed:

```
read_activity_file -format VCD \
-block ap/A1 \

```

```
-start 180002ns \
-end 189802ns \
A.vcd.gz
```

- The next example specifies a VCD activity file called `dmac_mac.vcd` for the scope `top/dma_dut`.

```
read_activity_file -format VCD -start 10ns -end 20ns \
-scope top/dma_dut dmac_mac.vcd
```

- The following command specifies multiple pairs of the start time and end time for the multiple windows in the VCD file:

```
read_activity_file \
-format VCD \
-scope adder/u1 \
-start {0ps 2000ps} \
-end {1100ps 3100ps} \
-block ../vcd/dut_5buf_full_chip.vcd
```

- The following command specifies to read the RTL-level VCD activity file:

```
read_activity_file -format VCD -scope scope_top -start 300ns -end 700ns
/DesignData/VCD/rtl.vcd -block inst_C1 -rtl true
set_power_analysis_mode -method vector_profile -vector_profile_mode transient -
worst_window_type full -power_grid_library /DesignData/PGV/library_pv.cl
```

## Related Topics

- "[Dynamic Power and IRDrop Analysis](#)" in the *Voltus User Guide*

## Related Commands

- `set_power_analysis_mode`
- `report_power`

## report\_instance\_power

```
report_instance_power
[-help]
instance [-file filename]
[-inst_file <string>]
```

Generates a text based report for the specified instance. This command enables you to determine how the different components of static power are calculated. You can use the report to debug static power numbers after performing power analysis (`report_power`).

This command is available through both the static and dynamic power engines. This command is supported in the dynamic power engine by including this command in the power include file through the command `set_power_include_file`.

**Note:** For dynamic power engine, the `report_instance_power` command does not generate text reports incrementally. If the dynamic run has been performed and you need the debug information for a static power number, you should re-run power analysis.

### Parameters

|                                            |                                                                                                                                                                                                                 |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-file<br/>filename</code>            | Writes out the static power analysis report into the file that you specify.                                                                                                                                     |
| <code>-help</code>                         | Outputs the command usage and a brief description about the command parameters.                                                                                                                                 |
| <code>instance</code>                      | Specifies the instances to include in the instance power report.                                                                                                                                                |
| <code>-inst_file<br/>&lt;string&gt;</code> | <p>Specifies a file that includes a list of instances. The parameter allows you to report and debug power numbers for multiple instances.</p> <p>A sample file is given below:</p> <pre>INST1 INST2 INST3</pre> |

### Example

- The following command specifies an instance power report file `debug.rep` for the instance `top/i_1`:  
tempus> report\_instance\_power -file debug.rep top/i\_1

### Related Topics

- "[Static Power, IRdrop and EM Analysis](#)" in the *Voltus User Guide*
  - "Debugging Instance Power"



## Related Commands

- `report_power`
- `set_power_include_file`

## report\_power

```
report_power
[-block block_name]
[-cap]
[-cap_unit unit]
[-cell_fit_file filename]
[-cell_list_file filename]
[-cell { cell_list }]
[-cell_type {all | {macro io combinational sequential clock_combinational clock_sequential}
}]
[-clock_domain names]
[-clock_gating_efficiency]
[-clock_network {all | { clock_list } }}]
[-cluster_gating_efficiency]
[-comb_seq_power]
[-compress compression_ratio]
[-count_seq_elements_in_clock_network]
[-die_instance_name dieinstname]
[-distribute]
[-exclude_cells {cell_list}]
[-exclude_cells_file filename]
[-exclude_instances {instance_list}]
[-exclude_instances_file filename]
[-float_precision value]
[-format { simple | detailed }]
[-include_group_instance_count]
[-inst_list_file filename]
[-instances { instance_list }]
[-group_type user_defined_group_name]
[-hierarchy {all | hierarchy_level}]
[-hierarchical_instances {inst1 inst2 inst3 ...}]
[-leakage]
[-net [-nworst number_of_nets]]
[-no_wrap]
[-o directory]
[-outfile filename]
[-output directory]
[-output_cell_fit_file filename]
[-pg_net {all | pg_net_name_list }}]
[-pg_pin]
[-power_domain {all | { power_domain_list } }]
[-power_db_directory directory]
[-power_unit unit]
[-power_density_tiles value]
[-power_density_tiles_row_col {value1 value2}]
```

```
[-power_density_tiles_size {value1 value2}]
[-print_memory_power]
[-register_gating_efficiency]
[-report_prefix prefix]
[-sort {internal | switching | leakage | total}]
[-stat]
[-thermal_leakage_temp {temp1 temp2 temp3 ...}]
[-thermal_power_map_file file_name]
[-thermal_power_map_tile {Xint Yint}]
[-thermal_power_map_top_layer layer_name]
[-thermal_power_map_format {simple | stack }]
[-thermal_conductivity_inputs file_name]
[-thermal_power_map_header_include_file filename]
[-thermal_leakage_temperature_scale_table_file filename]
[-thermal_leakage_temperature_scale_table {$temp1 $scalefactor1 $temp2 $scalefactor2 $temp3
$scalefactor3}]
[-thermal_material_file filename]
[-thermal_power_map_bottom_layer layer_name]
[-thermal_power_map_SI_unit {true | false}]
[-thermal_power_map_powertable_tile {Xint Yint}]
[-thermal_power_map_densitytable_tile {Xint Yint}]
[-thermal_power_map_format_version {new|default}]
[-threshold value]
[-threshold_voltage_group { all | group_name }]
[-time_unit unit]
[-time_based_report]
[-toggle_rate]
[-use_geometry_cell_size {true | false}]
[-view view_name]
```

Reports global as well as specific instance power, clock network power, clock domain power, and net switching power. It also reports the power for power domains and specific power nets. Units are reported in milliwatts.

This command starts the execution of power analysis (both static and dynamic).

 The parameters of the `report_power` command, except `-cap`, `-clock_gating_efficiency`, `-register_gating_efficiency`, and `-cluster_gating_efficiency`, are not supported in the dynamic only mode (`set_power_analysis_mode -disable_static true`). These parameters are only supported in the static mode. However, the `-power_domain` parameter is not supported even if the static power mode is enabled. You can use the `set_power_include_file` command to pass the `report_power` command options in the dynamic analysis mode.

The `report_power` command allows generation of multiple power reports with a single `report_power` command. This allows you to generate all the static reports during dynamic analysis, and removes the

need to run incremental `report_power` in static power analysis using a Verilog netlist. By default, a detailed power summary report, and a short summary will be printed at the beginning of all reports. To enable this flow, you need to specify the following parameters:

- `-output/o`
- `-report_prefix`

The following `report_power` command parameters are supported by the multi-report flow, both in static and dynamic power analysis:

- `-cell`
- `-clock_gating_efficiency`
- `-register_gating_efficiency`
- `-cluster_gating_efficiency`
- `-net`
- `-instances`
- `-pg_net`
- `-clock_network`
- `-hierarchy`

The following parameters are not supported in the multi-report flow:

- `-cell_type`
- `-power_domain`
- `-clock_domain`
- `-clock_gating_efficiency` and `-clock_network` in the same run



In the static power analysis flow, you can run the `report_power` command after the `restore_power_database` command without loading the design. The following command is an example of the use model:

```
restore_power_database -file power.db
report_power -output same -report_prefix report -clock_network {all} -instances {*}
```

The following parameters of the `report_power` command are supported without design loading:

- `-cell`
- `-cell_type`
- `-instances`
- `-clock_network`
- `-clock_domain`
- `-pg_net`
- `-pg_net -format detailed`
- `-hierarchy`
- `-hierarchical_instances`

Parameters

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-block <i>block_name</i></code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                       | <p>Specifies to generate the different power analysis reports for a block. This parameter is used for debugging power consumption issues for a specific block after the power run is complete</p> <p><b>Example:</b></p> <p>The following command will create the summary and <code>pg_net</code> reports for the design block</p> <pre>report_power -block {A/B} -outfile blockB.rpt -pg_net -pg_pin {all} -instances {</pre> <p>Following are the sample block-specific reports:</p> <pre>blockB.rpt.avppower blockB.rpt.pgnetpwr blockB.rpt.pgpinpwr blockB.rpt.instpwr</pre> |
| <code>-cap</code>                     | Dumps out the list of net, pin, and total capacitances in three different columns.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>-cap_unit <i>unit</i></code>    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                       | Sets the capacitance unit to be used for power report generation. The default unit is pF (p                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-cell { cell_list }</code>                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                                                                          | Specifies the cells to include in the power report. Accepts wildcards (*)                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>-cell_fit_file filename</code>                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                                                                          | <p>Specifies a file that accepts a cell-based FIT input table with two columns: cell name and parameter is used to generate an instance-level FIT report based on the cell-level data.</p> <p>Instance-based FIT report is supported only in the XP mode.</p> <p>The format of the file is:</p> <p><b><u>File Format Example for the FIT Input File:</u></b></p> <pre>##Cell Name          FIT ----- CellA                 .25 CellB                 .5</pre> |
| <code>-cell_list_file filename</code>                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                                                                          | Specifies a file containing a list of cells that are to be included in the power reports.                                                                                                                                                                                                                                                                                                                                                                     |
| <code>-cell_type {all   {macro io combinational sequential clock_combinational clock_sequential}}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                                                                          | <p>Specifies the cell type to include in the power report.</p> <p><i>Default:</i> all</p> <p>When you specify the cell types, <code>-cell_type combinational</code> or <code>sequential</code>, the power report includes the clock combinational or clock sequential cells.</p>                                                                                                                                                                              |
| <code>-clock_domain names</code>                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                                                                          | <p>Specify clock domains in power reports.</p> <p><b>Usage</b></p> <pre>report_power -clock_domain name -outfile filename</pre>                                                                                                                                                                                                                                                                                                                               |
| <code>-clock_gating_efficiency</code>                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                     | <p>Generates a report that includes the Clock Gating Efficiency (CGE) for all clock gating instances in different hierarchies in the design. Average CGE for different hierarchies is also reported.</p> <p>Clock Gating Efficiency is calculated as:</p> $\text{CGE} = (\text{toggles at clock gate output} / \text{toggles at clock gate input})$ $\text{Average CGE} = (\text{avg of all CGEs above})$ <p>When you generate a CGE report using the command <code>report_power -clock_gating_efficiency cge.rpt</code>, the following information will be displayed:</p> <ul style="list-style-type: none"> <li>• <b>Clock-gating Efficiency Report</b> - for each clock domain, it includes the toggle rate, number of clock gates, average clock toggle at registers, average toggle savings at registers, and toggle savings histogram.</li> <li>• <b>Hierarchical View of Average Toggle Savings</b> - number of clock gates and average toggle savings per hierarchical module in the design</li> </ul> |
| <code>-clock_network {all   { clock_list } }</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                     | <p>Reports the power consumption of the clock network, including the power for generated clock signals.</p> <p>Use <code>clock_list</code> to report the power consumed by specific clocks.</p> <p><i>Default:</i> all</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>-cluster_gating_efficiency</code>             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                     | <p>Specifies to generate the cluster efficiency report in the RGE report. This report gives the toggle rates at the registers at the fanout of each clock gate instance in the design.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>-comb_seq_power</code>                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                     | <p>Specifies to generate a report that gives the ratio of combinational versus sequential power for each hierarchy. You must specify this parameter with the <code>-instances</code> parameter.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>-compress compression_ratio</code>            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                     | <p>Specifies to compress the power analysis report files in the GNU Zip format. The compressed files have the suffix <code>.gz</code>.</p> <p><code>compression_ratio</code> is from 0-9. 0 indicates that the report files are not to be compressed. 1 provides the fastest compression speed but at a lower ratio, so the file size is larger. 9 provides the highest compression ratio but at a lower speed.</p> <p><i>Default:</i> 0</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>-count_seq_elements_in_clock_network</code>   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 | Includes the leaf flip-flop power in the clock network power report.<br><i>Default:</i> The leaf flip-flop power will not be included in the clock network power report.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| - distribute                                    | Enables standalone distributed power analysis. This parameter allows you to perform dynamic power analysis in the multi-CPU mode without running rail analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>-die_instance_name dieinstname</code>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                 | Specifies the die instance name for multi-die analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>-exclude_cells {cell_list}</code>         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                 | Specifies a list of cells that should be excluded from power analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>-exclude_cells_file filename</code>       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                 | Specifies a file containing a list of cells (one per line) which should be excluded from power analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>-exclude_instances {instance_list}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                 | Specifies a list of instances that should be excluded from power analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>-exclude_instances_file filename</code>   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                 | Specifies a file containing a list of instances (one per line) which should be excluded from power analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>-float_precision value</code>             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                 | Sets the precision value to be used for power report generation. The default value is 4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>-format { simple   detailed }</code>      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                 | <p>Specifies the type of report summary required for static power reporting. The possible arguments are:</p> <ul style="list-style-type: none"> <li><code>simple</code> - reports the amount of power consumed by all power nets in a condensed form based on the switching/internal/leakage components.</li> <li><code>detailed</code> - reports the amount of power consumed by all power nets in a detailed form includes additional information, such as output pin capacitance (<code>Cpin Sum</code>), output net Total Load Capacitance (<code>Cpin Sum + Cnet Sum = Total Load_Cap</code>), maximum frequency output pins (<code>Max_Clk_Freq</code>), maximum activity factor at the input and output pins (<code>Max_Activity</code>), the instance (<code>Cell Name</code>), clock or data cell (C/D), and library used in power analysis for each cell. The libraries are numbered in the header section.</li> </ul> <p>The name of the generated report file is <code>report.*.detailed.rpt</code>. Currently, the <code>-format detailed</code> is available for the <code>-pg_net</code> report.</p> |



| <code>-hierarchical_instances {inst1 inst2 inst3 ...}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                |                 |                |                 |                |             |                |           |       |  |  |  |  |  |  |  |          |           |      |          |         |       |       |             |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------------|----------------|-----------------|----------------|-------------|----------------|-----------|-------|--|--|--|--|--|--|--|----------|-----------|------|----------|---------|-------|-------|-------------|
|                                                              | Specifies to report all the power content for the specified hierarchical instances/ blocks. It (a breakup of power components) and <code>Group</code> (a breakup of power by cell type) of the specified hierarchy. The name of the report can be specified with the <code>-outfile</code> parameter ( <code>&lt;outfile&gt;</code> ). A prefix can be specified with the <code>-report_prefix</code> parameter. The report is written in the specified directory.                                        |                |                 |                |                 |                |             |                |           |       |  |  |  |  |  |  |  |          |           |      |          |         |       |       |             |
| <code>-hierarchy {all   hierarchy_level }</code>             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                |                 |                |                 |                |             |                |           |       |  |  |  |  |  |  |  |          |           |      |          |         |       |       |             |
|                                                              | Specifies the hierarchy level that is included in the power report. You can specify the <code>hierarchy_level</code> number that corresponds to a specific hierarchy level in the design. A <code>hierarchy_level 0</code> level.<br><br><i>Default:</i> all                                                                                                                                                                                                                                              |                |                 |                |                 |                |             |                |           |       |  |  |  |  |  |  |  |          |           |      |          |         |       |       |             |
| <code>-include_group_instance_count</code>                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                |                 |                |                 |                |             |                |           |       |  |  |  |  |  |  |  |          |           |      |          |         |       |       |             |
|                                                              | Specifies to report the instance count for each cell type. When specified, you can view the report for the sequential, macro, IO, combinational, clock sequential, and clock combinational cell types in the report (power.rpt). The parameter is supported in both the static and dynamic power analysis. The parameter allows you to verify if there are any missing instances.                                                                                                                         |                |                 |                |                 |                |             |                |           |       |  |  |  |  |  |  |  |          |           |      |          |         |       |       |             |
| <code>-instances {instance_list}</code>                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                |                 |                |                 |                |             |                |           |       |  |  |  |  |  |  |  |          |           |      |          |         |       |       |             |
|                                                              | Specifies the instances to include in the power report. The power report provides the amount of power consumed by a specified list of hierarchical or leaf-level instances. <code>instance_list</code> accepts hierarchical level instances only. An instance with 0 power will not be listed in the power report.                                                                                                                                                                                        |                |                 |                |                 |                |             |                |           |       |  |  |  |  |  |  |  |          |           |      |          |         |       |       |             |
| <code>-inst_list_file filename</code>                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                |                 |                |                 |                |             |                |           |       |  |  |  |  |  |  |  |          |           |      |          |         |       |       |             |
|                                                              | Specifies a file containing a list of instances that are to be included in the power reports.                                                                                                                                                                                                                                                                                                                                                                                                             |                |                 |                |                 |                |             |                |           |       |  |  |  |  |  |  |  |          |           |      |          |         |       |       |             |
| <code>-leakage</code>                                        | Reports only leakage power of the design.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                |                 |                |                 |                |             |                |           |       |  |  |  |  |  |  |  |          |           |      |          |         |       |       |             |
| <code>-net</code>                                            | Reports the net switching power, as well as the load capacitance, toggle rate, and switching delay in the design.                                                                                                                                                                                                                                                                                                                                                                                         |                |                 |                |                 |                |             |                |           |       |  |  |  |  |  |  |  |          |           |      |          |         |       |       |             |
| <code>-no_wrap</code>                                        | Displays the <code>report_power -cell_type all</code> output on a single line in the report output file.<br><br><b>Example</b><br><table><tr><th>Instance</th><th>Toggle</th><th>Internal Power</th><th>Switching Power</th><th>Leakage Power</th><th>Total Power</th><th>Percentage (%)</th><th>Cell Name</th></tr><tr><td colspan="8">-----</td></tr><tr><td>dma/ram0</td><td>7.102e+08</td><td>5.58</td><td>0.002307</td><td>0.02424</td><td>5.607</td><td>18.19</td><td>/data/cellA</td></tr></table> | Instance       | Toggle          | Internal Power | Switching Power | Leakage Power  | Total Power | Percentage (%) | Cell Name | ----- |  |  |  |  |  |  |  | dma/ram0 | 7.102e+08 | 5.58 | 0.002307 | 0.02424 | 5.607 | 18.19 | /data/cellA |
| Instance                                                     | Toggle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Internal Power | Switching Power | Leakage Power  | Total Power     | Percentage (%) | Cell Name   |                |           |       |  |  |  |  |  |  |  |          |           |      |          |         |       |       |             |
| -----                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                |                 |                |                 |                |             |                |           |       |  |  |  |  |  |  |  |          |           |      |          |         |       |       |             |
| dma/ram0                                                     | 7.102e+08                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 5.58           | 0.002307        | 0.02424        | 5.607           | 18.19          | /data/cellA |                |           |       |  |  |  |  |  |  |  |          |           |      |          |         |       |       |             |
| <code>-nworst number_of_nets</code>                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                |                 |                |                 |                |             |                |           |       |  |  |  |  |  |  |  |          |           |      |          |         |       |       |             |

|                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                | <p>Reports the number of nets with the highest net switching power.</p> <p><b>Note:</b> You must use the <code>-nworst</code> parameter in conjunction with the <code>-net</code> parameter.</p>                                                                                                                                                                                                                                                                                                                                                               |
| <code>-outfile filename</code>                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                | <p>Specifies the name of the report file in which the top-level summary of the power consumption is written. You can include the directory name.</p> <p>You can convert a power report file to the gzip file format by specifying <code>-outfile filename.gz</code>.</p> <p><b>Note:</b> If you do not specify this parameter, the software directs the power report to the console log file, and no power report is written to a separate file.</p>                                                                                                           |
| <code>-output/-o directory</code>              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                | <p>Specifies the name of the power analysis output directory that saves the power reports. This parameter replaces the need to specify <code>set_power_output_dir</code>; if both <code>output</code> and <code>set_power_output_dir</code> are specified, <code>output</code> has the higher priority.</p>                                                                                                                                                                                                                                                    |
| <code>-output_cell_fit_file filename</code>    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                | <p>Specifies the name of the output file for the generated FIT report, which includes three columns: instance name, cell name, and computed FIT value. This parameter is specified with the <code>-output_cell_fit_file</code> parameter. The output file is generated in the power output directory.</p> <p><b>File Format Example for the FIT Output File:</b></p> <pre>#Instance Name           Cell Name           FIT ##### inst_A/al_inst/g5        INV_X1A_A9TL        0.25 inst_A/al_inst/g8        INV_X1B_A9TL        0.25 .... Total FIT: 115</pre> |
| <code>-pg_net {all   pg_net_name_list }</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                | <p>When set to <code>all</code>, reports the amount of power consumed by all power nets.</p> <p>If you specify a list, reports the power consumed by each instance that is connected to the nets.</p> <p><b>Note:</b> When using this parameter, the static power calculation engine always performs post-layout power calculation.</p> <p><b>Default:</b> <code>all</code></p>                                                                                                                                                                                |
| <code>-pg_pin</code>                           | <p>Reports power consumed by all PG pins in a design.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>-power_db_directory directory</code>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                  | Specifies the location of the power database that the Power Calculation engine uses for power reports.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>-power_domain {all   { <i>power_domain_list</i> } }</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                                  | Specifies all power domains or lists power domains to be included in report. It prints a summary down for each domain before listing power for each instance in the domain.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>-power_density_tiles <i>value</i></code>                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                                  | Specifies the number of tiles used for power density calculation.<br><i>Default:</i> 100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>-power_density_tiles_row_col {<i>value1 value2</i>}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                                  | <p>Defines the number of tiles by rows * columns. <i>value1</i> is the number of tile rows and <i>value2</i> is the number of tile columns.</p> <p>The following example specifies to perform power density aware vector profiling using 2 tile rows:</p> <pre>set_power_analysis_mode -method vector_profile -vector_profile_mode power_density_aware report_power -power_density_tiles_row_col {2 2} -time_based_report -outfile tile_report.txt</pre> <p>The following is the vector profile report for this command:</p> <pre>Tile 0_0_0 5.06 5.33 14.95 14.555 Tile 1_0_1 14.95 5.33 24.84 14.555 Tile 2_1_0 5.06 14.555 14.95 23.78 Tile 3_1_1 14.95 14.555 24.84 23.78 ... ...</pre> |
| <code>-power_density_tiles_size {<i>value1 value2</i>}</code>    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                                  | Defines the absolute tile size. <i>value1</i> is the horizontal width of a tile (in micron) and <i>value2</i> is the vertical height of a tile (in micron).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>-power_unit <i>unit</i></code>                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                                                                  | Sets the power unit to be used for power report generation. The default unit is mW (milliwatt).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>-print_memory_power</code>                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

Specifies to report power values for the memory and macro cells under separate groups (The following is a snapshot of the power report showing separate groups for memory and

| Group                 | Internal<br>Power | Switching<br>Power | Leakage<br>Power |      |
|-----------------------|-------------------|--------------------|------------------|------|
| Sequential            | 0.4031            | 0.1283             | 0.006583         |      |
| Memory                | 0.1624            | 0.004345           | 0.011            |      |
| IO                    | 0                 | 0                  | 0                |      |
| Physical-Only         | 0                 | 0                  | 2.91e-06         | 2.91 |
| Combinational         | 0.8711            | 1.05               | 0.01579          |      |
| Clock (Combinational) | 0.002302          | 0.01018            | 7.654e-05        | 0.0  |
| Clock (Sequential)    | 0.01177           | 0.008387           | 0.000347         | 0.   |
| Macro (non-mem)       | 0.2317            | 0.007259           | 0.001234         | 0.   |
| Total                 | 1.682             | 1.209              | 0.03503          |      |

-group\_type user\_defined\_group\_name

Specifies to report power (internal/switching/leakage/total) for a user-defined collection of cells under a power group in the power summary report. You can use the following Tempus command to define power groups:

- `define_property` - define a property.
- `set_property` - assign the defined property (group type) and its value (power group) to a collection of cells.

The use model of the `-group_type` parameter is:

```
report_power -group_type user_defined_group_type
```

Here, `user_defined_group_type` is the user-defined power group type applied to a list of cells. When the `-group_type` parameter is specified, the command reports power for the power groups specified group type in the power summary report.

### Example:

The following command defines the `poly_bias` property (group type) on library cells:

```
define_property poly_bias -object_type lib_cell -type string
```

The following commands apply the property `poly_bias` with the group name `P0` and `P10` to a collection of cells:

```
set_property [get_lib_cells -of_objects [get_cells -hier * -filter {ref_name=~*P0}] poly_bias
set_property [get_lib_cells -of_objects [get_cells -hier * -filter {ref_name=~*P10}] poly_bias
```

The following command reports power for the user-defined power groups that are associated with the property `poly_bias`:

```
report_power -outfile basic_group_type.rpt -group_type poly_bias
```

The following is a snippet of the power report that shows the user-defined power groups associated with the property `poly_bias`, in addition to the existing groups (sequential, macro, IO, combinational, clock (combinational), clock (sequential), and total).

| User Group | Internal Power | Switching Power | Leakage Power | Total Power | Percentage (%) |
|------------|----------------|-----------------|---------------|-------------|----------------|
| P0         | 0.01498        | 0.0002017       | 0.01765       | 0.03283     | 15.63%         |
| P10        | 0.1268         | 0.04026         | 0.01011       | 0.1772      | 84.37%         |
| Total      | 0.1418         | 0.04047         | 0.02776       | 0.21        | 100%           |

| Group                 | Internal Power | Switching Power | Leakage Power | Total Power | Percentage (%) |
|-----------------------|----------------|-----------------|---------------|-------------|----------------|
| Sequential            | 0.02958        | 0.0003768       | 0.01458       | 0.04453     | 16.62          |
| Macro                 | 0              | 0               | 0             | 0           | 0              |
| IO                    | 0              | 0               | 0             | 0           | 0              |
| Combinational         | 0.008217       | 0.002633        | 0.01025       | 0.0211      | 7.877          |
| Clock (Combinational) | 0.1239         | 0.04421         | 0.007419      | 0.1755      | 65.51          |
| Clock (Sequential)    | 0.01773        | 0.008783        | 0.0002459     | 0.02676     | 9.988          |
| Total                 | 0.1794         | 0.056           | 0.03249       | 0.2679      | 100            |

`-register_gating_efficiency`

|                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                             | <p>Generates a report that includes the Register Gating Efficiency (RGE) and Data Aware Gating Efficiency (DAGE) for all sequential cell instances as well as different hierarchies in the design. Average RGE for each hierarchy is reported. It also displays a section on ICG cluster information, that shows the number of registers at the fanout of each ICG instance along with their RGE metrics.</p> <p>Register Gating Efficiency is calculated as:</p> $RGE = 1 - (\text{toggles at register clock pin} / \text{root clock toggles})$ <p><b>Note:</b> Root clock means the clock defined in SDC containing register timing constraints, not the global generated clock. It can be back traced from the register clock pin to the clock definition point.</p> <p>When you generate an RGE report using the command <code>report_power -register_gating_efficiency</code>, the following information will be displayed:</p> <ul style="list-style-type: none"> <li>• <b>Register-gating Efficiency Report</b> - for each clock domain, it includes the toggle rate, number of clock gates, average clock toggle at registers, average toggle savings at registers, and a toggle savings histogram.</li> <li>• <b>Register Gating Opportunities Report</b> - this report is sorted based on the Q/CLK toggle rate. If a register's Q/CLK ratio is greater than .25 (25%), Q toggles every other clock cycle. However, if the ratio is less than 25%, there is a gating opportunity to reduce power.</li> </ul> <p>Data Aware Gating Efficiency is calculated as:</p> $DAGE = 1 - ((\text{toggles at clock pin} - 2 * \text{toggles at Q pin}) / (\text{root clock toggles}))$ |
| <code>-report_prefix prefix</code>                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                             | <p>Specifies a prefix for the report in which the top-level summary of the power consumption is generated. All the new report names will be <code>prefix.*.rpt</code>, and the detailed summary will be <code>prefix.summary.rpt</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>-sort {internal   switching   leakage   total}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                             | <p>Sorts the instance-based power consumption data by <code>internal</code>, <code>switching</code>, <code>leakage</code>, or <code>total</code>.</p> <p><b>Default:</b> <code>total</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>-stat</code>                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

Reports a set of statistics on the net status after propagation, for example, duty, transition parameter is useful in viewing and debugging user-defined activity factors after the power

**Example:**

```
report_power -time_based_report -distribute -outfile power.rpt -stat
```

**-- Net status after propagation --**

| Net   | Transition | Density | Duty | Activity | Ref. Freq |
|-------|------------|---------|------|----------|-----------|
| VDDA1 | 0          | 1       | 0    | 1.25e+08 |           |
| VSSA1 | 0          | 0       | 0    | 1.25e+08 |           |
| ...   |            |         |      |          |           |

```
-thermal_conductivity_inputs file_name
```

Specifies a file containing the thermal conductivity value for layers. You can use this parameter to specify the thermal conductivity value of the following layers in the power map file:

- Silicon
- Metal
- Dielectric
- Micro Bump (optional for 3DIC design)
- Fill material in micro bump layer (optional for 3DIC design)

The thermal conductivity file format is:

**Section 1 (layer type based thermal conductivity value)**

```
#default
$layer_type1 $thermal_conductivity_parameter1
$layer_type2 $thermal_conductivity_parameter2
...
```

**Section 2 (layer name based thermal conductivity value)**

```
#perlayer
$layer1_name $thermal_conductivity_parameter1 $dielectric_thermal_conductivity_1
$layer2_name $thermal_conductivity_parameter2 $dielectric_thermal_conductivity_2
...
```

Here,

- You can either specify both Section 1 (layer type based) and Section 2 (layer name based) or only one of the two sections. If both sections are specified, then layer name based section takes precedence over the layer type based section.
- The layer name based section has thermal conductivity value and its dielectric layer's thermal conductivity value for each metal/via layer.

- If section 2 has no parameters defined, the software will use the default parameters de

**Example:**

###default

Metal 430  
Silicon 250  
Dielectric 0.6

###Perlayer

Metal1 430 0.4  
Via1 410 0.5  
Metal2 390 0.5  
Via2 410 0.5

...

For information on the thermal conductivity input file and power map file, refer to the "Stat  
EM Analysis" chapter in *Voltus User Guide*.

-thermal\_leakage\_temp {temp1 temp2 temp3 ...}

Defines the temperature to calculate leakage for the power map file.

-thermal\_leakage\_temperature\_scale\_table {\$temp1 \$scalefactor1 \$temp2 \$scalefactor2 \$temp3 \$:

Specifies the global scale factor for a temperature value to enable scaling of the leakage from the input dotlib file. This parameter allows you to scale the power values only at cert temperatures.

This parameter allows you to generate a multi-temperature leakage power map table for file in the power map generation flow. -thermal\_leakage\_temperature\_scale\_table and -thermal\_leakage\_temperature\_scale\_table\_file are mutually exclusive.

-thermal\_leakage\_temperature\_scale\_table\_file filename



|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>Specifies a file containing the user-defined temperature values and the corresponding scale factor for each library, cell, or instance. The format of the file is:</p> <pre>#Type Name temp1 temp2 temp3 temp4 Lib    library_name    scalefactor@temp1    scalefactor@temp2    scalefactor@temp3 Cell   cell_name       scalefactor@temp1    scalefactor@temp2    scalefactor@temp3 inst   instance_name   scalefactor@temp1    scalefactor@temp2    scalefactor@temp3</pre> <p>This parameter allows you to generate a multi-temperature leakage power map table from a file in the power map generation flow. <code>-thermal_leakage_temperature_scale_table</code> and <code>thermal_leakage_temperature_scale_table_file</code> are mutually exclusive.</p> <p><b>File Format Example for the Type Lib:</b></p> <pre>#Type Name          25 50 125 150 Lib    fast          1 2 10 100 Lib    bufao          1 2 10 100 Lib    header         1 2 10 100 Lib    pso_ring_wc    1 2 10 100</pre> |
|  | <code>-thermal_material_file filename</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|  | <p>Specifies the name of a file containing the user-specified substrate layer thickness. The layer thickness in the power map generation flow can either be a user-specified value or derived from the thermal analysis. If the values are available, the user-specified value takes precedence.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|  | <code>-thermal_power_map_bottom_layer layer_name</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|  | <p>Writes zero metal density tables for layers lower than the user-specified layer in the power map generation flow. This parameter prevents duplication of metal density tables when merging the top and block thermal maps.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|  | <code>-thermal_power_map_densitytable_tile {Xint Yint}</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|  | <p>Specifies the number of tiles in the X and Y directions for the metal density table.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|  | <code>-thermal_power_map_file file_name</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|  | <p>Generates the power map file with the specified name in the output directory.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|  | <code>-thermal_power_map_top_layer layer_name</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|  | <p>Writes zero metal density tables for layers higher than the user-specified layer in the power map generation flow. This parameter prevents duplication of metal density tables when merging the top and block thermal maps.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|  | <code>-thermal_power_map_format {simple   stack }</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  | <p>Generates a tile-based power map file in the simple or stack format to perform accurate thermal analysis. You must use the stack format to support multiple dies in a package for thermal analysis.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|  | <code>-thermal_power_map_format_version {new default}</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>Determines whether to generate the default or the new power map file format. When set to the new format, the power map file format with non-uniform tile numbers for the power (<code>-thermal_power_map_powertable_tile</code>) and density (<code>-thermal_power_map_densitytable_tile</code>) can be generated.</p> <p>The following are the features of the new report format:</p> <ul style="list-style-type: none"> <li>• version number added to the header "Version : 2020.0"</li> <li>• nx/ny under the "DIE_AREA" section has been removed</li> <li>• XCOR/YCOR for each layer has been removed</li> <li>• nx/ny has been added for the POWER and METAL_DENSITY sections</li> </ul>                                                                                         |
|  | <code>-thermal_power_map_header_include_file filename</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|  | <p>Specifies a file that contains comments or additional information to be included in the header of the power map file.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|  | <code>-thermal_power_map_powertable_tile {Xint Yint}</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|  | <p>Specifies the number of tiles in the X and Y directions for the power table.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|  | <code>-thermal_power_map_SI_unit {true   false}</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|  | <p>Specifies to change chip thermal power map units to SI in the thermal power map file. This option was introduced to support the thermal power map file format used by foundry. When this parameter is set to true, the values for all the power map parameters (length, power, and thermal conductivity) will be in SI units.</p> <ul style="list-style-type: none"> <li>• <b>Length Unit:</b> default is <math>\mu\text{m}</math> and SI is <math>\text{m}</math></li> <li>• <b>Power Unit:</b> default is <math>\text{mW}</math> and SI is <math>\text{W}</math></li> <li>• <b>Metal/Silicon Thermal Conductivity Unit:</b> default is <math>\text{mW}/(\mu\text{m} * \text{C})</math> and SI is <math>\text{W}/(\text{m} * \text{C})</math></li> </ul> <p><i>Default:</i> true</p> |
|  | <code>-thermal_power_map_tile {Xint Yint}</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|  | <p>Defines the tile number in the X and Y direction. By default 10*10 will be used.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|  | <code>-threshold value</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|  | <p>Filters out the instances whose total power dissipation is less than the specified value. Useful for filtering out small power consuming components.</p> <p><i>Default:</i> 0</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|  | <code>-threshold_voltage_group { all   group_name }</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                     | <p>Reports leakage by voltage threshold group in the power report. This parameter supports (default value) or any other string, such as hvt or lvt.</p> <p><i>Default:</i> all</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>-time_based_report</code>                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                     | <p>Writes out a detailed VCD profiling report into the file that you specify.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>-time_unit unit</code>                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                     | <p>Sets the time unit to be used for power report generation. The default unit is ns (nanosecond).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>-toggle_rate</code>                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                     | <p>Specifies to dump out instance-based activity to the outfile. The toggle rate for each instance is the clock frequency to which it belongs. If multiple clocks are reaching the instance, then the first one is used as a reference.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>-use_geometry_cell_size {true   false}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                     | <p>Specifies whether to use the actual geometry or placement bounding box for macro PGV generation of tile-based power map file. This parameter is used when a macro's power is distributed across multiple tiles. When this parameter is set to <code>true</code>, the geometry-based bounding box will be used for power generation and power will be distributed evenly across tiles.</p> <p><i>Default:</i> false</p>                                                                                                                                                                                                                                                                                                     |
| <code>-view view_name</code>                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                     | <p>Specifies the active timing view that was created using the multi-mode multi-corner (MMM) <code>-setup -hold</code> command. Power analysis is performed for the specified view. The command name in the output file even if you do not specify the <code>-view</code> parameter.</p> <p>If <code>report_power -view view_name</code> is set, it will override the view set in the <code>set_power_analysis</code> command. However, if you specify the <code>set_analysis_view -dynamic</code> and <code>-leakage</code> parameters using these <code>set_analysis_view</code> command parameters has higher priority over the <code>report_power</code> command.</p> <p><i>Default:</i> Uses current view in Voltus.</p> |

## Examples

- The following command reports the amount of power consumed by the clock nets `clk` and

`test_clk` (including leaf flip-flop power) and writes the output information to a power report file named `clk_pwr.rep`:

```
report_power -clock_network{clk test_clk} \
-count_seq_elements_in_clock_network -outfile clk_pwr.rep
```

- The following command reports the amount of power consumed down to hierarchy level 3 from the top and writes the output information to a power report file named `hier.rep`:

```
report_power -hierarchy 3 -outfile hier.rep
```

- The following command compresses the power report file `pg_net.rep` with gzip:

```
report_power -outfile pg_net.rep.gz
```

- The following command reports the amount of power consumed by leaf-level instances that match `ECO*` and writes the output information to a power report file named `inst.rep`:

```
report_power -instances ECO* -outfile inst.rep
```

- The following command reports the amount of power consumed by all cells that match `buf*6` and writes the output information to a power report file named `cell.rep`:

```
report_power -cell buf*6 -outfile cell.rep
```

- The following command reports the amount of power consumed by all I/O and sequential cells and writes the output information to a power report file named `cell_type.rep`:

```
report_power -cell_type {io sequential} -outfile cell_type.rep
```

- The following command sorts the instance-based power consumption data by `leakage` power and writes the output information to a power report file named `sort.rep`:

```
report_power -sort leakage -outfile sort.rep
```

- The following command reports the total power of all instances whose total power dissipation is more than 1 milliwatt and writes the output information to a power report file named `threshold.rep`:

```
report_power -threshold 1 -outfile threshold.rep
```

- The following command reports the net switching power, load capacitance, toggle rate, and switching values for each net in the design, reports the top 100 nets with the highest net switching

power, and writes the output information to a power report file named `net.rep`:

```
report_power -net -nworst 100 -outfile net.rep
```

- The following command reports the amount of power consumed by each power domain and writes the output information to a power report file named `pd_all.rep`:

```
report_power -power_domain all -outfile pd_all.rep
```

- The following command reports the amount of power consumed by each power domain `PD2` and writes the output information to a power report file named `pd.rep`:

```
report_power -power_domain PD2 -outfile pd.rep
```

- The following command reports the amount of power consumed by view `view_max-hp-max-lp1` and writes the output information to a power report file named `view.rep`:

```
report_power -view view_max-hp-max-lp1 -outfile view.rep
```

- The following command reports the amount of power consumed by each instance connected to power nets `vdd` and `vdd1` and writes the output information to a power report file named `pg_net.rep`:

```
report_power -pg_net {vdd vdd1} -outfile pg_net.rep
```

- The following command reports the leakage for the threshold voltage group `HVT`:

```
report_power -leakage -threshold_voltage_group { HVT }
```

- The following command writes out a detailed vector profiling report for the combinational and macro cell types in the `vectorprofile.rpt` file:

```
report_power -time_based_report -outfile vectorprofile.rpt -cell_type {combinational macro}
```

- The following command is an example of generating multiple reports at the same time:

```
report_power -output static_power -report_prefix design -net {name1} -instance {inst_name}
```

The following reports will be generated:

```
static_power/design.rpt
static_power/design.netpwr.rpt
static_power/design.instpwr.rpt
```

- The following command specifies to report power for combinational cells on the clock network.

```
report_power -cell_type clock_combinational -output static_power -report_prefix
```

power\_clk

- The following command specifies to report the power content for the `inst1` hierarchical instance in the `report.hierinstpwr.rpt` report:

```
report_power -output static -report_prefix report -clock_network {all} -instances * -
no_wrap -hierarchy {all} -hierarchical_instances inst1 -cell_type {all}
```

- The following command generates a power map file in the simple format for the specified tile:

```
report_power -outfile powermeter.txt -thermal_power_map_file powermap.txt -
thermal_power_map_tile { 4 6 } -thermal_power_map_format simple
```

### Example of the output of the power report.

```
*-----
* Version 64-bit
* Date & Time: 2008-Jan-27 15:49:51 (2008-Jan-27 23:49:51 GMT)
*-----
* Design: dtmf_chip
*
* Liberty Libraries used:
* /sev//DTMF_CHIP/TIMING/max/scmetropmk_cmos10lp_rvt_ss_1p08v_125c.lib
* /sev/DTMF_CHIP/TIMING/max/scmetropmk_cmos10lp_rvt_ss_0p8v_1p08v_125c.lib
* /sev/DTMF_CHIP/TIMING/max/scmetropmk_cmos10lp_rvt_ss_0p8v_125c.lib
*
*
*
* Power Domain used:
*
* Rail: VDD_AO Voltage: 1.32
* Rail: VDD_TDSPCore Voltage: 1.32
* Rail: VDD_TDSPCore_R Voltage: 1.32
*
* Parasitic Files used:
* /sev/DTMF_CHIP/SPEF/dtmf_chip.decoup.spef
*
* DEF Files used:
* /sev/DTMF_CHIP/DEF/dtmf_chip.def
*
* Power Units = 1mW
* Time Units = 1e-09 secs
* report_power
*-----
*
* Total Power
*-----
*
* Total Internal Power: 0.01943(62.6774%)
```

**Tempus Text Command Reference**  
Auxiliary Commands

Total Switching Power: 0.01057 (34.0968%)  
Total Leakage Power: 0.001 (3.2258%)  
Total Power: 0.031

| Group         | Internal<br>Power | Switching<br>Power | Leakage<br>Power | Total<br>Power | Percentage<br>(%) |
|---------------|-------------------|--------------------|------------------|----------------|-------------------|
| Sequential    | 0.0006738         | 0.0002215          | 4.045e-05        | 0.0009358      | 3.019             |
| Macro         | 0.0004916         | 1.515e-05          | 0.0002609        | 0.0007677      | 2.477             |
| IO            | 0.01743           | 0.009085           | 0.0003172        | 0.02683        | 86.56             |
| Combinational | 0.0008324         | 0.001247           | 0.0003815        | 0.002461       | 7.941             |
| Total         | 0.01943           | 0.01057            | 0.001            | 0.031          | 100               |

| Rail           | Voltage | Internal<br>Power | Switching<br>Power | Leakage<br>Power | Total<br>Power | Percentage<br>(%) |
|----------------|---------|-------------------|--------------------|------------------|----------------|-------------------|
| VDD_AO         | 1.32    | 0.0008238         | 0.0003404          | 0.0002694        | 0.001434       | 4.625             |
| VDD_TDSPCore   | 1.32    | 0.001126          | 0.001022           | 0.0003931        | 0.002541       | 8.197             |
| VDD_TDSPCore_R | 1.32    | 5.833e-06         | 3.253e-07          | 8.371e-07        | 6.996e-06      | 0.02257           |

| Clock       | Internal<br>Power | Switching<br>Power | Leakage<br>Power | Total<br>Power | Percentage<br>(%) |
|-------------|-------------------|--------------------|------------------|----------------|-------------------|
| m_digit_clk | 0                 | 0                  | 0                | 0              | 0                 |
| m_ram_clk   | 0                 | 0                  | 0                | 0              | 0                 |
| m_dsram_clk | 0                 | 0                  | 0                | 0              | 0                 |
| m_spi_clk   | 1.018e-06         | 1.09e-06           | 2.572e-08        | 2.134e-06      | 0.006884          |
| m_rcc_clk   | 1.282e-05         | 1.513e-05          | 2.672e-07        | 2.821e-05      | 0.09103           |
| m_clk       | 5.039e-05         | 7.278e-05          | 7.658e-07        | 0.0001239      | 0.3998            |
| refclk      | 1.11e-05          | 3.017e-05          | 2.488e-07        | 4.152e-05      | 0.134             |
| Total       | 7.533e-05         | 0.0001192          | 1.307e-06        | 0.0001958      | 0.6317            |

\* Power Distribution Summary:

\* Highest Average Power: IOPADS\_INST/Prefclkip (PBCSCT2B):0.003727

\* Highest Leakage Power: DTMF\_INST/ROM\_512x16\_0\_INST (rom\_512x16A): 8.899e-05

\* Total Cap: 2.46221e-10 F

\* Total instances in design: 8525

```
* Total instances in design with no power: 24
* Total instances in design with no activty: 24
* Total Fillers and Decap: 8

```

- The following command creates a leakage report (see example report below):

```
report_power -leakage
*-----
*
* Date & Time: 2008-Aug-24 21:34:17 (2008-Aug-25 04:34:17 GMT)
*
*-----
*
* Design: dma_mac
*
* Liberty Libraries used:
* DATA/timing_libs//tcbn65lp_LVL_c060217wc0d90d9.lib
* DATA/timing_libs//tcbn65lpcgwc.lib
* ...
*
* Power Domain used:
* Rail: VDDlu Voltage: 0.84
* Rail: VDDm Voltage: 1.08
* Rail: VDD Voltage: 1.08
*
* Power Units = 1mW
*
* Time Units = 1e-09 secs
*
* report_power -outfile Leakage.rep -leakage
*
*-----
Total Power

Total Leakage Power: 0.1879

Group Leakage Percentage
 Power (%)
```



**Tempus Text Command Reference**  
Auxiliary Commands

|               |         |       |
|---------------|---------|-------|
| Sequential    | 0.0127  | 6.756 |
| Macro         | 0.1556  | 82.8  |
| IO            | 0       | 0     |
| Combinational | 0.01962 | 10.44 |
| -----         |         |       |
| Total         | 0.1879  | 100   |

|       |         |                  |                   |
|-------|---------|------------------|-------------------|
| Rail  | Voltage | Leakage<br>Power | Percentage<br>(%) |
| ----- |         |                  |                   |
| VDDau | 0       | 0                | 0                 |
| VDDlu | 0.84    | 0.06477          | 34.47             |
| VDDm  | 1.08    | 0.003587         | 1.909             |
| VDD   | 1.08    | 0.1195           | 63.62             |

|             |                  |                   |
|-------------|------------------|-------------------|
| Clock       | Leakage<br>Power | Percentage<br>(%) |
| -----       |                  |                   |
| phy_rxclk_2 | 0                | 0                 |
| phy_txclk_2 | 0.0004786        | 0.2547            |
| phy_rxclk_1 | 0                | 0                 |
| phy_txclk_1 | 3.39e-05         | 0.01804           |
| clk         | 0.001723         | 0.9171            |
| -----       |                  |                   |
| Total       | 0.002236         | 1.19              |
| -----       |                  |                   |

```

* Power Distribution Summary:
* Highest Average Power: ethernet_mac_2/tx_fifo_module_from_dma/fifo1/dual_port_
 ram_4_tx_fifo/ram2P1024x32/ram (TSDN65LPA1024X32M8F): 0.02424
* Highest Leakage Power: ethernet_mac_2/tx_fifo_module_from_dma/fifo1/dual_port_
 ram_4_tx_fifo/ram2P1024x32/ram (TSDN65LPA1024X32M8F): 0.02424
* Total Cap: 1.01625e-10 F
* Total instances in design: 29235

```

```
* Total instances in design with no power: 3596
* Total instances in design with no activity: 115
* Total Fillers and Decap: 0
```

-----

Total leakage power = 0.187903 mW

Cell usage statistics:

```
Library tsdn65lpa128x16m8f_1v08, 1 cells (0.003900%) , 0.01016 mW (5.407063%)
Library tsdn65lpa1024x32m8f_1v08, 6 cells (0.023402%), 0.145427 mW (77.394568%)
Library tcbn65lp_LVL_c060217wc0d90d9 , 120 cells (0.468037%), 0.000175605 mW (
0.093455%)
```

```
Library unknown, 3881 cells (15.137096%), 0.00630072 mW (3.353178%)
Library tcbn65lpcgwc, 118 cells (0.460236%), 0.000625092 mW (0.332667%)
Library tcbn65lplvtwc, 3762 cells (14.672959%), 0.0110529 mW (5.882233%)
Library tcbn65lpwc, 17751 cells (69.234370%), 0.0141619 mW (7.536835%)
```

## Related Commands

- `set_power`
- `set_power_analysis_mode`
- `set_power_include_file`

## report\_power\_domain

```
report_power_domain
[-help]
[-bindLib]
[-file <string>]
[-inst <inst1 inst2 ...>]
[-isoInst]
[-module <module1 module2 ...>]
[-net {net1 net2 ...}]
[-pgNet]
[-pin {pin1 pin2 ...}]
[-powerDomain <string>]
[-shifter]
[-verbose]
[-voltage]
```

Reports information about power domains. The report includes the following attributes:

- Power domain name
- Power domain bounding box
- Timing library
- Mingap
- RS extension
- Row type
- Row spacing

The command also reports whether the power domain is always on.

### Parameters

|                              |                                                                                                                                                                                                                                     |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-bindLib</code>        | Reports the timing libraries available for instance binding including voltage scaling. However, the actual libraries that are used by the AAE delay calculation might be a subset of the timing libraries reported by this command. |
| <code>-file outFile</code>   | Specifies the name of the output file.<br><i>Default:</i> <code>designName-powerdomain.rpt</code>                                                                                                                                   |
| <code>-inst inst_name</code> | Reports the power domain-related information for the specified instance: the input, output, and power/ground connections.                                                                                                           |

|                                               |                                                                                                                                                                                  |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-isoInst</code>                         | Reports all isolation cells in the power domain.                                                                                                                                 |
| <code>-module {list<br/>of modules}</code>    | Checks whether the specified instances or hierarchical instances are in the power domain.<br><br><i>Default:</i> The software does not perform this check.                       |
| <code>-net</code>                             | Reports the input and output connection names along with the related power domain names for the specified net.                                                                   |
| <code>-powerDomain<br/>powerDomainName</code> | Specifies the power domain whose attributes you want to report.<br><br><i>Default:</i> If you do not specify the power domain, this command report data about all power domains. |
| <code>-pgNet</code>                           | Reports all the power/ground nets in the power domain.                                                                                                                           |
| <code>-pin {pin1 pin2<br/>...}</code>         | Specifies the list of pins.                                                                                                                                                      |
| <code>-shifter</code>                         | Reports all shifters in the power domain.                                                                                                                                        |
| <code>-verbose</code>                         | Prints detailed information related to the pins.                                                                                                                                 |
| <code>-voltage</code>                         | Reports domain voltages.                                                                                                                                                         |

## Examples

- The following command reports all shifter cells, all isolation cells, and all power/ground nets in power domain PD1. The command also checks whether HInst `instA` is contained in PD1, and reports all results to `myDesign-powerdomain.rpt`.

```
tempus> report_power_domain -powerDomain PD1 -file myDesign-powerdomain.rpt \
 -module {instA} -shifter -isoInst -pgNet
```

- The following example shows how output file `outfile` contains the binding between the timing libraries and instances in power domain VCORE:

```
tempus> report_power_domain -bindLib -powerDomain VCORE -file outfile
```

```
Timing library for instance(s):
```

```
=====
```

```
INST = uLSCVSocDownVCore/uACLKEND/uLVHL (1v1_slow_1v08_1v08)
INST = uLSCVSocDownVCore/uACLKENI/uLVLHL (1v1_slow_1v08_1v08)
INST = uLSCVSocDownvCore/uaCLKENP/uLVLHL (1v1_slow_1v08_1v08)
INST = uLSCVSocDownvCore/uaCLKENRW/uLVLHL (1v1_slow_1v08_1v08)
INST = uLSCVSocDownvCore/uaBIGENDINIT/uLVH1L (1v1_slow_1v08_1v08)
```

```
INST = uLSCVSocDownvCore/uaCLKIN/uLVHL (lvl_slow_1v08_1v08)
INST = uLSCVSocDownvCore/uCP15SDISABLE/uLVHL (lvl_slow_1v08_1v08)
```

- The following command reports the power domain-related information about instance

PM\_INST/state\_inst/g45:

```
tempus> report_power_domain -inst PM_INST/state_inst/g45
```

### Result:

```
Analyzing PM_INST/state_inst/g45 (Cell: AN2D1) power domain:A0
-----from side -----
Pin A2 => Net PM_INST/state_inst/n_28
PM_INST/state_inst/g71/ZN in power domain A0
Pin A1 => Net PM_INST/state_inst/n_25
PM_INST/state_inst/g47/Z in power domain A0
-----to side -----
Pin Z => Net PM_INST/state_inst/n_36
PM_INST/state_inst/fsm_reg[1]/qi_reg/D in power domain A0
=====PG pins =====
VDD connected to Net: VDD
VSS connected to Net:VSS
```

## Related Commands

- `read_power_domain`

## reset\_power\_activity

reset\_power\_activity

Resets all activity in the database. You can specify this command to reset all activity and power information in a given session.

 You can reset activity from an activity file using `read_activity_file -reset`, reset activity for nets, pins, and ports using `set_switching_activity -reset`, and reset the default switching activities using `set_default_switching_activity -reset`.

### Parameters

None

### Related Commands

- `set_power_analysis_mode`
- `read_activity_file`

## restore\_power\_database

```
restore_power_database
[-help]
-file {file1 file2 file3 ...}
[-hierarchy {prefix1 prefix2 prefix3 ...}]
[-hierarchy_separator "/"]
```

Restores any `power.db` file in the Voltus/Innovus session. This command can be used after restoration of a design to view the static power results or incremental report generation without having to do power analysis again.

**Note:** You will be able to restore the power database that you created previously, only if you created the power database using the `set_power_analysis_mode -create_binary_db true` command. Also, the command loads power databases created in the 9.1 release only, and not in the older releases.

- The command can map a hierarchical instance to a flat netlist.
- It can restore multiple power dbs for blocks and top level, that are generated separately.
- When the information for a net or an instance is present in multiple power dbs, the last one will overwrite.

### Parameters

|                                                       |                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                                    | <p>Outputs a brief description that includes the type and default information for each <code>restore_power_database</code> parameter.</p> <p>For a detailed description of the command and all of its parameters, use the <code>man</code> command <code>man restore_power_database</code>.</p>                                |
| <code>-file {file1 file2 file3 ...}</code>            | <p>Specifies the name of the power database file.</p>                                                                                                                                                                                                                                                                          |
| <code>-hierarchy {prefix1 prefix2 prefix3 ...}</code> | <p>Specifies to map each instance in the file with a <code>prefixN/instance</code> (for example, <code>prefixA/inst</code>) in the flat netlist, where <code>/</code> is the hierarchy separator.</p> <p>This is an optional parameter.</p> <p><i>Default:</i> <code>all</code></p>                                            |
| <code>-hierarchy_separator "/"</code>                 | <p>Specifies the separator character in the hierarchical net names, bus names, and pin names for the block or other part of the design that you want to analyze. It must be the same as the separator character in the design netlist.</p> <p>This is an optional parameter.</p> <p><i>Default:</i> slash (<code>/</code>)</p> |

## Use Model

```
tempus> read_design
tempus> restore_power_database -file power.db
tempus> report_power # for text based power reports
```

## Example

For each instance in file 'file2' a prefix "A" is added to instance name such as instC, A/instC. No prefix is added to 'file1' since prefix1 is none for file1.

```
tempus> restore_power_database -file {file1 file2} -hierarchy {none A}
```

## Related Commands

- `report_power`
- `read_design`



# set\_default\_switching\_activity

```
set_default_switching_activity
[-black_box_density value]
[-black_box_duty value]
[-block master_cell_name]
[-clip_activity_to_domain_freq {true | false}]
[-clock_gates_enable {activity_factor}]
[-clock_gates_output {activity_factor}]
[-clock_gates_output_ratio num]
[-comb_clockgate_ratio num]
[-duty value]
[-global_activity factor]
[-hier hierarchy_name]
[-icg_ratio num]
[-input_activity factor]
[-macro_activity factor]
[-name activity_name]
[-period value]
[-pg_net netname]
[-reset]
[-reset_type { global_activity | seq_activity ... }]
[-seq_activity factor]
```

Specifies the switching activity for all primary inputs, nets, and other devices in the design whose activity has not been previously defined through user attributes, the toggle count format (TCF) file, the simulation vectors file (VCD, SHM, and so on), or the tracing of the clock network.

You can use this command to specify user-defined activities at the design level.

## Parameters

|                          |                                                                                                                                                                                                                                                                                                                                   |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -black_box_density value |                                                                                                                                                                                                                                                                                                                                   |
|                          | <p>Specifies to set the default transition density on all the output pins of all bl the design.</p> <p>The -black_box_density and -black_box_duty parameters must be used t one or both of these parameters are not specified, then the software will u average transition density from all the input pins of the blackboxes.</p> |
| -black_box_duty value    |                                                                                                                                                                                                                                                                                                                                   |

|                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                           | <p>Specifies to set the default duty on all the output pins of all blackboxes in the design.</p> <p>The <code>-black_box_density</code> and <code>-black_box_duty</code> parameters must be used together. If one or both of these parameters are not specified, then the software will use the average transition density from all the input pins of the blackboxes.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>-block master_cell_name</code>                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                           | <p>Specifies the name of the master cell for activity setting. The parameter allows you to define the user-defined switching activity for a block partition. This parameter is used for assigning block-based activity.</p> <p>The <code>-block</code> parameter is applicable only to the block-based distributed power analysis.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>-clip_activity_to_domain_freq {true   false}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                           | <p>Specifies to perform domain-based activity clipping. If a data path net activity is above 1x the clock frequency of that path due to activity propagation, the net will be clipped and set to 1x frequency. For clock nets, clipping will be based on the clock frequency.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>-clock_gates_enable {activity_factor}</code>        | Specifies the activity at the enable pin(s) of the clock gating cells.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>-clock_gates_output {activity_factor}</code>        | Specifies the activity at the output pin(s) of the clock gating cells. This works for ICGs and combinational clock gating cells.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>-clock_gates_output_ratio num</code>                | <p>Specifies to control transition density at the output pin(s) of the integrated combinational clock gating cells. This parameter is equivalent to specifying the parameters <code>-icg_ratio</code> and <code>-comb_clockgate_ratio</code> together with the same values.</p> <p>For example, <code>set_default_switching_activity -clock_gates_output_ratio 0.5</code> is equivalent to <code>set_default_switching_activity -icg_ratio 0.6 -comb_clockgate_ratio 0.6</code>.</p> <p><b>Note:</b> This parameter has a cascading effect when there are multiple levels of clock gates on the clock path.</p> <p><b>For example,</b> when <code>-clock_gates_output_ratio</code> is set to 0.5, the first level ICG cell will have 0.5 ratio of the root clock's toggle density, the second level ICG cell will have <math>0.5 \times 0.5 = 0.25</math> ratio of the root clock's toggle density, and the third level ICG cell will have <math>0.5 \times 0.5 \times 0.5 = 0.125</math> ratio of the root clock's toggle density, and so on.</p> |

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-comb_clockgate_ratio<br/>num</code> | <p>Specifies to set the propagation ratio for combinational clock gate cell out specified, this parameter sets the output activity of any instance that is ide combinational clock gating cell to <i>num</i> times the fastest transition density a the instance's clock inputs.</p> <p><b>Note:</b> This parameter does not apply to combinational clockgates with a c input, that is, a select pin tied low.</p>    |
| <code>-duty value</code>                   | <p>Specifies the duty cycle, which is the probability that the signal is a logical</p> <p><b>Note:</b> You must specify the <code>-period</code> parameter when assigning a duty.</p> <p><i>Default:</i> 0.5 (equivalent to a 50 percent duty cycle)</p>                                                                                                                                                              |
| <code>-global_activity factor</code>       | <p>Specifies the average number of times that all unset nodes switch in a clo not specified the fastest domain frequency will be used unless one turns c domains.</p>                                                                                                                                                                                                                                                 |
| <code>-hier hierarchy_name</code>          | <p>A hierarchical name that the global activity is being assigned to. If not pro whole design is assumed.</p>                                                                                                                                                                                                                                                                                                         |
| <code>-icg_ratio num</code>                | <p>Specifies to set the propagation ratio for integrated clock gate (ICG) cell o specified, this parameter sets the output pin of all identified ICG instances the fastest transition density arriving on clock inputs of the instance.</p>                                                                                                                                                                           |
| <code>-input_activity factor</code>        | <p>Specifies the switching activity of primary inputs in a clock cycle. It can be number.</p> <p><i>Default:</i> 0.2 (Switches once every five clock cycles.)</p>                                                                                                                                                                                                                                                     |
| <code>-macro_activity factor</code>        |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                            | <p>Specifies the activity factor for macros in a design. The allowed values for parameter are between 0 to 1.</p> <p>If you set the macro activity to .30, 30% of macros in the design would be switching in a given cycle during dynamic simulation. When <code>-macro_activ</code> specified for the static power analysis, all output pins of the macros are as specified activity.</p> <p><i>Default:</i> 0.2</p> |
| <code>-name activity_name</code>           |                                                                                                                                                                                                                                                                                                                                                                                                                       |

|                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                         | <p>Allows you to set the activity level for the specified jitter activity name. You can set a low/high activity level for jitter analysis.</p> <p><b>Examples:</b></p> <pre>set_default_switching_activity -name low -clock_gates_output 0 - global_activity 0.01 set_default_switching_activity -name high -clock_gates_output 2 - global_activity 0.2</pre>                                                                                                                                                                                                                                                                                                                                                   |
| <code>-period value</code>                                              | <p>Specifies the default operating period (1/frequency).</p> <p>Specifies the period that the activity is referenced to. If not specified, the software uses the <i>dominant</i> clock in the design.</p> <p>If no clock is specified, the software generates an error and prompts you to specify a clock. Units in seconds (s), milliseconds (ms), microseconds (us), nanoseconds (ns), and picoseconds (ps).</p> <p><b>Note:</b> If you enter the period as a negative number, the software reverts to the default frequency.</p> <p><i>Default:</i> The time unit is defined in the first .lib file read during design import.</p>                                                                           |
| <code>-pg_net netname</code>                                            | <p>Specifies the name of the power net for domain-based switching activity analysis. The parameter allows you to define different switching activities for different domains within the same DEF block during stable flop scheduling.</p> <p>When this parameter is specified, the <code>voltus_power.stateprop.swichsrc</code> and <code>voltus_power.stateprop.swichdst</code> are affected because the flop toggle values will change.</p> <p>The parameter is applicable only to the dynamic vectorless flow. The <code>-pg_net</code> parameter must be specified with <code>set_power_analysis_mode -honor_default_switching_activity_in_scheduling true</code> for domain-based activity assignment.</p> |
| <code>-seq_activity factor</code>                                       | <p>Specifies the activity of outputs of the sequential logic. This value will be used for the output ports of flops for propagating activity.</p> <p>If the sequential activity is not specified, propagation will be done for the first flop starting from the input ports.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <code>-reset</code>                                                     | <p>Resets all specified options back to default values.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>-reset_type {<br/>global_activity  <br/>seq_activity ... }</code> | <p>Specifies to selectively reset the activity specifications.</p> <p><b>Note:</b> You must use the <code>-reset_type</code> parameter in conjunction with the <code>-pg_net</code> parameter.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

## Activity Precedence

See "[Activity Precedence](#)".

## Examples

- The following command specifies that the primary inputs in the design will switch once every ten clock cycles, with a clock period of 7520 ns and a duty cycle of 50 percent:

```
tempus> set_default_switching_activity -input_activity 0.10 -period 7520ns -duty .5
```

- The following command specifies that all unswitched nodes in the design will switch once every ten clock cycles, with a clock period of 7520 ns and a duty cycle of 50 percent:

```
tempus> set_default_switching_activity -global_activity 0.10 -period 7520ns -duty .5
```

- The next example defines the switching activity, sequential activity, and the period:

```
tempus> set_default_switching_activity -input_activity 0.2 -period 10 \
 -seq_activity 0.1
```

- The next example sets the default global activity to 0.5 for the hierarchy dma\_dut:

```
tempus> set_default_switching_activity -global_activity 0.5 -hier dma_dut
```

- The following command resets the sequential and global activity specification:

```
tempus> set_default_switching_activity -reset -reset_type
{seq_activity,global_activity}
```

## Related Commands

- [read\\_activity\\_file](#)
- [get\\_activity](#)
- [propagate\\_activity](#)
- [reset\\_power\\_activity](#)
- [set\\_switching\\_activity](#)

## set\_dynamic\_power\_simulation

```
set_dynamic_power_simulation
[-resolution value]
[-reset]
[-activity_pattern {time1 activity_name1 time2 activity_name2 }]
[-period value]
```

Specifies the period and resolution for a dynamic power simulation. This command is used to specify the simulation period in the dynamic vectorbased/vectorless and power-up flows.

### Parameters

|                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -activity_pattern {time1 activity_name1 time2 activity_name2 .... } |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                                     | <p>Divides the total simulation time based on the windows given in the <code>activity_pattern</code> parameter. For a particular window, it uses the defined activity specified with the <code>set_default_switching_activity</code> command.</p> <p>When the user-defined low cycle and high cycle time is specified for the input/sequential activity, all the clocks will be in the low or high cycle until completely past the cycle transition time.</p> <p>Following is the use model of this parameter:</p> <ul style="list-style-type: none"> <li>• <code>set_default_switching_activity -name low -input_activity 0.05 -seq_activity 0.1 -clock_gates_output 0.2</code></li> <li>• <code>set_default_switching_activity -name high -input_activity 0.1 -seq_activity 0.2 -clock_gates_output 2.0</code></li> <li>• <code>set_dynamic_power_simulation -resolution 50ps -activity_pattern {0 high 5ns low 10ns high}</code></li> </ul> <p>Here, high activity is used for 0ns to 5ns, low activity is used for 5ns to 10ns, and high activity is used for 10ns to 15ns.</p> |
| - period value                                                      | Specifies the simulation period that dynamic power analysis will use. The supported time units are <code>s</code> , <code>ms</code> , <code>us</code> , <code>ns</code> , and <code>ps</code> . If no explicit time unit is specified, the default <code>ns</code> (nanosecond) will be applied.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| -reset                                                              | Resets all specified options back to default values.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| -resolution value                                                   | Specifies the transient time step size or resolution. The supported time units are <code>s</code> , <code>ms</code> , <code>us</code> , <code>ns</code> , and <code>ps</code> . If no explicit time unit is specified, the default <code>ps</code> (picosecond) will be applied.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

## Examples

- The following command sets up the resolution for a dynamic simulation:

```
tempus> set_dynamic_power_simulation -resolution 100ps
```

## Related Commands

- `set_power_analysis_mode`

## set\_inst\_temperature\_file

```
set_inst_temperature_file
file
[-reset]
```

Specifies the instance temperature file to support thermal-aware leakage. You can use this command to do thermal-aware power calculation. Based on the temperature specified in the file, the software will calculate temperature and leakage power.

In the presence of a trilib set, the internal and leakage power will be interpolated as the per the given temperature specification.

### Parameters

|             |                                                      |
|-------------|------------------------------------------------------|
| <i>file</i> | Specifies the name of the instance temperature file. |
| -reset      | Resets the temperature file.                         |

### Related Commands

- `report_power`
- `set_power_analysis_mode`



## set\_power

```
set_power
[value]
[-ascii_power_file filename]
[-cap value]
[-cell cell_name value]
[-clock {all|clock_name}]
[-custom_macro_pwl filename]
[-dynamic_async_mode { mode_name }]
[-default_dynamic_pgv_mode { mode_name }]
[-dynamic_switch_event { switching_events }]
[-dynamic_switch_pattern pattern]
[-exclude_clock]
[-exclude_ddv_modes {pattern_list}]
[-fanout_limit value]
[-force]
[-include_ddv_modes {pattern_list}]
[-instance inst_list value]
[-internal]
[-leakage value]
[-no_propagation]
[-pin pin_name pin_power]
[-pg_pin {pg_pin}]
[-pg_net railname]
[-pwl]
[-repeat]
[-reset]
[-static_mode { custom_label_name | mode_name }]
[-scale_factor value]
[-signal_net netname]
[-slew value]
[-sticky]
[-switching value]
[-trigger_time_adjustment time]
```

Specifies the amount of power for all or part of the design. This command is used to define user-specified power on cells and instances. You must specify this command before `report_power`.

The `set_power` command can accept a cell/instance list using the `get_cells/get_lib_cells` commands. You can use the `get*` commands to create a collection of cells/instances whose name matches the supplied pattern list. Therefore, providing you with the flexibility in providing lists through use of regular filter patterns. This feature is supported in both static and dynamic analysis engines.

## Parameters

|                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-ascii_power_file filename</code>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                          | <p>Specifies the name of an instance list file. This file contains the list of instances in the following format:</p> <pre>instance_name power_value &lt;pin_name&gt;</pre> <p>Here,</p> <ul style="list-style-type: none"> <li>• <code>instance_name</code> is the instance name to which power is to be distributed.</li> <li>• <code>power_value</code> is the power value for the instance in watts.</li> <li>• <code>pin_name</code> is an optional argument to specify the pin name. When specified, allows you to assign power to the specific pin of the instance.</li> <li>• The three fields should be in one line, and each instance should be in a new line.</li> </ul> |
| <code>-cap value</code>                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                          | Specifies the capacitance value to all instances with nets more than or equal to the fanout limit ( <code>-fanout_limit</code> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>-cell cell_name value</code>       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                          | <p>Specifies the cell name. If the cell or instance name is not specified, the power will be distributed to all rails.</p> <p>If you want to specify power for a group of cells/instances, use the <code>get_cells/get_lib_cells</code> commands. The <code>-cell</code> parameter can accept these commands to create a collection of cells/instances whose name matches the supplied pattern list. For example, <code>-cell[get_lib_cells NAND*] 2mw</code></p>                                                                                                                                                                                                                   |
| <code>-clock<br/>{all clock_name}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                          | <p>Specifies the power value for clock networks. You can specify the power value for a specific clock network or the clock network power value for the whole design.</p> <p><b>Note:</b> This parameter cannot be used with the <code>-cell</code>, <code>-instance</code> and <code>-pg_net</code> parameters.</p>                                                                                                                                                                                                                                                                                                                                                                 |
| <code>-custom_macro_pwl filename</code>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                         | <p>Specifies to use a custom trigger file to apply instance or cell PWL waveforms. Given one master cell PWL waveform, the software can automatically generate scaled PWL waveforms of the other companion cells defined in the trigger file proportional to the cell internal power.</p> <p><b>Note:</b> This parameter must be used with <code>set_power -dynamic_switch_pattern</code> in the vectorless flow.</p> <p>For more information on the trigger file format, see <a href="#">Trigger File Format</a>.</p>                                                                                                                                                                                                                                                                                                                                    |
| <code>-default_dynamic_pgv_mode { mode_name }</code>    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                         | <p>Specifies the current characterization mode to be used during dynamic power and jitter analyses. This parameter can be specified when using a PGV with multiple functional modes.</p> <p>When this parameter is specified, the Power Calculation engine uses this preferred mode in scheduling the switching events for the macro blocks, and saves these events in the trigger file.</p> <p>The use model of this parameter is:</p> <pre>set_power -cell {}   -instance {} -default_dynamic_pgv_mode { mode_name }</pre> <p>This parameter is mutually exclusive to the <code>-dynamic_switch_pattern</code> parameter. If a cell or instance has both the <code>-dynamic_switch_pattern</code> and <code>-default_dynamic_pgv_mode</code> parameters defined in the same Tcl script, <code>-dynamic_switch_pattern</code> takes higher priority.</p> |
| <code>-dynamic_async_mode { mode_name }</code>          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                         | <p>Specifies to trigger asynchronous macros based on the length of the PWL waveform to create a single continuous waveform. This parameter allows you to specify the asynchronous mode for an instance or a cell. The start and end of the PWL saved in the PGV must have similar current value in order to avoid sudden jump in the current.</p> <p><b>Use model:</b></p> <pre>set_power -dynamic_async_mode {mode} [-instance {name}   -cell {name}]</pre> <p><b>Example:</b></p> <pre>set_power -cell pll -dynamic_asynchronous_mode {READ_C}</pre> <p>Here, power analysis will trigger the <code>READ_C</code> mode and automatically splice the waveform based on its length.</p>                                                                                                                                                                   |
| <code>-dynamic_switch_event { switching_events }</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                  | <p>Specifies the switching events for cells, instances, and pins along with their rise and fall time. <i>switching_events</i> is a string of toggle time followed by “r” or “f”, where “r” is for rise toggle and “f” is for fall toggle. The <code>-dynamic_switch_event</code> parameter should not be specified with the <code>-repeat</code> parameter.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>-dynamic_switch_pattern<br/>pattern</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                  | <p>Specifies stimulus pattern on the pins of an instance and cell. Specify this parameter only in the dynamic mode for the type <i>instance</i> and <i>cell</i>.</p> <p>The <i>pattern</i> is applied to all pins (input and output) on the instance/cell. It is a string.</p> <p><i>pattern</i> specifies the switching pattern. If there is more than one clock domain with timing windows on a pin, only the windows on the fastest clock will be used, and the pattern will be applied to that clock period. It is a string of 0, 1, r, and f values. 0 for cycles not to switch and 1 for cycles to switch. r is for rise only and f for fall only.</p> <p>The pattern string should include enough digits to cover all of the clock cycles in the simulation period. If there are not enough digits, the pattern will be padded with 0 values at the end. If there are too many, the pattern will be truncated to the number of applicable cycles. Transitions will be scheduled in the center of each applicable timing window.</p> <p><b>Notes</b></p> <ul style="list-style-type: none"> <li>• 0 and 1 is supported only in the probability-based vectorless flow. It is not supported in the state-propagation-based vectorless flow and the vector-based flow.</li> <li>• In case of macros, you can specify the mode names, that is, which mode switches in which clock cycle <code>-dynamic_switch_pattern { model - mode2 - }</code>.</li> </ul> <p>You can specify “-” to indicate that instances will not switch in a particular clock cycle. If “-” is the first value in the string of values specified within {}, you must insert a space before “-”, as shown below:</p> <ul style="list-style-type: none"> <li>• no switching: <code>-dynamic_switch_pattern { - - }</code></li> <li>• no switching in the 1st clock period: <code>-dynamic_switch_pattern { - r }</code></li> </ul> <p><u>Examples:</u></p> |

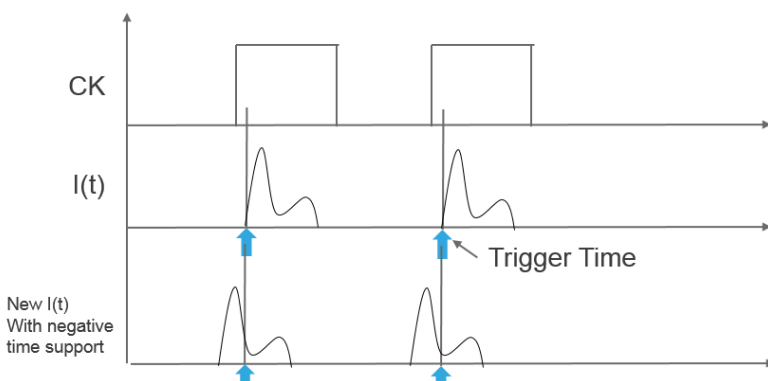
|                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                | <p>In the following example, instance will not switch in the 4th clock cycle:<br/> <pre>set_power -dynamic_switch_pattern { f r f - r }</pre></p> <p>In the following example, the mode <code>write</code> will switch in the 1st clock cycle, the mode <code>read</code> will switch in the 2nd and 4th clock cycle, and none of the modes will switch in the 3rd clock cycle:<br/> <pre>set_power -dynamic_switch_pattern { write read - read }</pre></p> <p><b>Note:</b> You can use the wildcard ( <code>*</code> ) character for the <code>-instance</code> and <code>-cell</code> parameters to specify a pattern of instance/cell names to which the specified switching pattern is to be applied.</p> |
| <code>-exclude_clock</code>                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                                | <p>Excludes clock networks when specifying a scale factor or power specification for the whole design.</p> <p>This parameter cannot be used with the <code>-cell</code>, <code>-instance</code> and <code>-pg_net</code> parameters.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>-exclude_ddv_modes</code><br><code>{pattern_list}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                                | <p>Excludes the current characterization (or Dynamic Detailed View) modes that match with the specified patterns. This parameter allows you to control the modes to be switched, instead of switching all the current characterization modes defined in the power-grid view.</p> <p><b>Use Model</b></p> <pre>set_power -instance -cell name -exclude_ddv_modes {pattern_list}</pre> <p>Here:</p> <ul style="list-style-type: none"> <li>• <code>pattern_list</code> is a list of mode names, separated by space or comma</li> <li>• the instance, cell, and pattern names can contain wildcard characters</li> </ul>                                                                                         |
| <code>-fanout_limit value</code>                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                                | <p>Specifies the fanout limit for nets. If an instance has nets more than or equal to the fanout limit, its switching/internal/leakage power is taken to be zero. When this parameter is specified, the user-defined power value is applied to all the driver instances that have nets more than or equal to the fanout limit.</p>                                                                                                                                                                                                                                                                                                                                                                            |
| <code>-force</code>                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                                | <p>Specifies to apply the <code>set_power</code> command even if there are Verilog mismatches. For dynamic or DEF-based static analysis, this parameter allows you to apply <code>set_power</code> to the DEF instances that do not exist in Verilog.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                               | Outputs the command usage.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| -include_ddv_modes<br>{ <i>pattern_list</i> } |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                               | <p>Includes only the current characterization (or DDV) modes that match with the specified patterns. This parameter allows you to control the modes to be switched, instead of switching all the current characterization modes defined in the power-grid view.</p> <p><b>Use Model</b></p> <pre>set_power -instance -cell <i>name</i> -include_ddv_modes {<i>pattern_list</i>}</pre> <p>Here:</p> <ul style="list-style-type: none"> <li>• <i>pattern_list</i> is a list of mode names, separated by space or comma</li> <li>• the instance, cell, and pattern names can contain wildcard characters</li> </ul> |
| -internal                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                               | <p>Specifies the target internal power of the design, cell, or instance. To specify internal power for cells and instances, you can use the <code>-internal</code> parameter in conjunction with the <code>-cell</code> or <code>-instance</code> parameter.</p>                                                                                                                                                                                                                                                                                                                                                 |
| -instance<br><i>inst_list</i> value           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                               | Specifies the instance name. If the cell or instance name is not specified, the power will be distributed to all rails.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| -leakage                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                               | <p>Specifies the target leakage power of the design, cell, instance, or power net.</p> <p>To specify leakage power for cells and instances you can use <code>-leakage</code> option in conjunction with <code>-cell</code> or <code>-instance</code> option.</p> <p>To specify leakage power of a certain power net in the design you can use <code>-pg_net</code> option along with <code>-leakage</code> option.</p>                                                                                                                                                                                           |
| -no_propagation                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | <p>Specifies that the user-defined switch pattern is applied only after state-propagation is complete. The use model of the command parameter is:<br/> <code>set_power -dynamic_switch_pattern {pattern} -no_propagation</code></p> <p>The <code>-no_propagation</code> parameter must be specified with the <code>-dynamic_switch_pattern</code> parameter. When the <code>-no_propagation</code> parameter is specified, the dynamic power engine performs state-propagation first, and then overrides the single instance switching event by the user-defined dynamic switch pattern without any further propagation.</p> |
| <code>-pg_net railname</code>        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                      | <p>Optional. The specified power or pwl for the cell, instance or design is applied to the specified rail. When not specified, the power or pwl is applied to all the rails associated with the instance. This option is generally used for MSMV cells.</p> <p>When this option is not specified the behavior is different for pwl and power options. When not specified and power option is used, the power is applied for the whole design, cell, or instance. When not specified and pwl option is used, the pwl is applied to every rail of the cell or instance.</p>                                                    |
| <code>-pg_pin {pg_pin}</code>        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                      | <p>Specifies the switch pattern for a specific PG pin of an instance. This parameter must be specified with the <code>-dynamic_switch_pattern</code> parameter.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>-pin pin_name pin_power</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                      | <p>Specifies pin-wise power numbers for a specific instance. This parameter allows you to control the power assigned to power ground pins of an instance. Following is the use model of this parameter:</p> <pre>set_power -instance &lt;inst_name&gt; -pin &lt;VDDA&gt; &lt;pin_power&gt; set_power -instance &lt;inst_name&gt; -pin &lt;VDDb&gt; &lt;pin_power&gt;</pre>                                                                                                                                                                                                                                                   |
| <code>-pwl</code>                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                      | <p>Specifies that power argument (<code>power</code>   <code>pwl</code>) is a waveform rather than a power number.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>-repeat</code>                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                      | <p>Repeats dynamic switch pattern of all the clock cycles in the simulation period.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>-reset</code>                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                             | <p>This parameter causes all previous <code>set_power</code> commands to be ignored.</p> <p>Note: You can selectively reset the instance/cell specifications. In addition, you can also selectively reset the cell and instance specifications of a list of objects through the <code>get_cells</code> and <code>get_lib_cells</code> commands.</p>                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>-scale_factor value</code>                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                             | <p>Scales the power value by the specified scale factor. This parameter allows you to set a user-defined scale factor in the trigger file to scale the dynamic current of the PGV.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>-signal_net netname</code>                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                             | <p>Specifies the signal net name for power assignment. When this parameter is specified, the driver instances connected to the signal net will be assigned the user-defined power value.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>-slew value</code>                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                             | <p>Specifies the slew value to all instances with nets more than or equal to the fanout limit (<code>-fanout_limit</code>).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>-static_mode { custom_label_name   mode_name }</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                             | <p>Specifies the mode name or the custom label name associated with a cell or instance for computing instance current from PGV with static multi-voltage multi-frequency (MvMf) modes. This current is used to compute the instance's internal power, and its switching and leakage power components would be set to zero.</p> <p>The use model of this parameter is given below:<br/> <code>set_power -instance -static_mode READ</code></p> <p>If you specify custom label of a static MvMf mode, currents would be taken from it directly. Alternatively, if only mode name is specified for a cell/instance, the software snaps to the nearest custom label defined within this mode, based on the instance's voltage and frequency to get the currents.</p> |
| <code>-sticky</code>                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                             | <p>Applies the simulation based PWL waveform as is, to the specified cell/instance for dynamic power analysis.</p> <p>Use this option only if you have specified <code>-pwl</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>-switching value</code>                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |



|                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                   | <p>Specifies the target switching power of the design, cell, or instance. To specify switching power for cells and instances, you can use the <code>-switching</code> parameter in conjunction with the <code>-cell</code> or <code>-instance</code> parameter.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>-trigger_time_adjustment</code> <i>time</i> | <p>Specifies to adjust the trigger time of the current waveforms for the various functional modes (read/write/idle) of the custom macros. This parameter allows you to start the waveform before the trigger time (negative time support) to account for the setup time for the operation, as shown below:</p>  <p>The specified time will be applied to the computed trigger time for each mode in the trigger files generated by power analysis. For example, if the trigger time (from timing window/vector) is computed as 1 ns, 2 ns, and 3 ns, and if you specify <code>set_power -trigger_time_adjustment -100ps</code>, the software will adjust the trigger time to .9 ns, 1.9 ns, and 2.9 ns.</p> |
| <i>value</i>                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

Specifies power in milliwatts during static power calculation or piecewise linear (`-pwl`) current waveform, during dynamic power calculation for cell or instance. The command accepts a unit specification (e.g. 10w).

The format of the PWL waveform is:

```
{time1 current1 time2 current2 ...time_n current_n}
```

Time is in ns and current in mA.

**Example:**

```
set_power -pwl -pg_net {} -cell BUFX2MTL {0 0 10 0 10.1 5 10.5 0}
```

Here, at 0ns its value is 0mA, at 10ns its value is 0mA, at 10.1ns its value is 5mA, and so on.

## Examples

- The following command sets power for vdd rail to 10 milliwatts:

```
tempus> set_power -pg_net vdd 10mw
```

- The following command sets a power value of .005 milliwatts on power net vdd for instance `clkin_neg_L14_I1`:

```
tempus> set_power -pg_net vdd -instance Top/A/clkin_neg_L14_I1 .005
```

- The following command sets a power value of .005 milliwatts for any cell named `INVX2` in the design:

```
tempus> set_power -cell INVX2 .005mw
```

- The following command defines a piecewise linear waveform of {0 0 10 0 10.1 5 10.5 0} for the cell of type `BUFX2MTL` for all power rails.

```
tempus> set_power -pwl -pg_net {} -cell BUFX2MTL {0 0 10 0 10.1 5 10.5 0}
```

- The following command specifies the leakage power for the cell `DFFRX1`.

```
tempus> set_power -leakage -cell DFFRX1 10mw
```

- The following commands specify the target leakage power for power nets `VDD1` and `VDD2`.

```
tempus> set_power -leakage -pg_net VDD1 100mw
```

```
tempus> set_power -leakage -pg_net VDD2 200mw
```

- The following command specifies the PWL description of current for the instance `SEHD130_1024X32X1CM8`. The `-sticky` option tells the software to use the specified PWL simulation based waveforms as is.

```
tempus> set_power -pwl -instance SEHD130_1024X32X1CM8 -sticky {0 0 1 1 2 2 7 0}
```

- The following command scales the total power of the instance `i_1` by a scale factor of 2:

```
tempus> set_power -scale_factor 2 -instance i_1
```

- The following command sets the total power of the design to 20mw while keeping the clock network power the same:  
tempus> set\_power 20mw -exclude\_clock {all}
- The following command assigns a switching power of 3mw to all instances of the cell AND1:  
tempus> set\_power -cell AND1 -switching 3mw
- The following command assigns 2mw power to all cells that match the specified pattern:  
tempus> set\_power -cell[get\_lib\_cells NAND\*] 2mw
- The following command specifies that mode1 switches in the first cycle, mode2 switches in the third cycle and no switching in others:  
tempus> set\_power -dynamic\_switch\_pattern { mode1 0 mode2 0 }
- The following command resets power of all cells that match the specified pattern:  
tempus> set\_power -reset -cell[get\_cells A/\*]
- The following command is an example of cell-based switching:  
tempus> set\_power -cell HS65\_LS\_\* -dynamic\_switch\_pattern {1101}
- The following command specifies that the switching pattern value 1010 will be applied to all the instance names starting with m\_inst:  
tempus> set\_power -instance m\_inst\* -dynamic\_switch\_pattern {1010}

## Trigger File Format

For specifying a custom trigger file, you must use the `-custom_macro_pwl` parameter. The format of the trigger file is:

| Trigger File Format                                                                   | Description                                                                                                 |
|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>#Cell names</b>                                                                    |                                                                                                             |
| MASTER CELL <cellname1>                                                               | Name of the reference cell.                                                                                 |
| COMPANION_CELLS <cellname1<br>cellname2 ... >                                         | Name of the cells that will use scaled PWL from the master cell.                                            |
| <b>#Mode and conditional details for trigger</b>                                      |                                                                                                             |
| MODE_NAME <mode1><br>CONDITIONAL_INPUT <conditional<br>input statement for cellname1> | Mode name and conditional input statement for the cell.<br>Conditional input of current for the given mode. |

|                                                |                                                                                                                                                                                  |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CONDITIONAL_PIN <pin_name><br>(rise fall both) | Conditional pin statement for the cell. Signal pin for which the tool should rise or fall.                                                                                       |
| CONDITIONAL_STIMULUS_FILE<br><usim1.txt>       | Simulation vector detail in the given mode.                                                                                                                                      |
| USER_PWL_FILE <pwl1.txt>                       | User-specified PWL file that contains information about the PWL waveforms (power/ground currents).<br><br>For an example of a PWL file, see <a href="#">PWL Format Example</a> . |
| END_MODE                                       | End mode section                                                                                                                                                                 |
| <b>#End cell section</b>                       |                                                                                                                                                                                  |
| END                                            | End cell section                                                                                                                                                                 |

### Example of a Custom Macro Power Trigger File

```
MASTER_CELL SRAM_A1
COMPANION_CELLS { SRAM_A2, SRAM_A3, SRAM_A4, SRAM_A5 }
 MODE_NAME WRITE
 CONDITIONAL_INPUT = (!SD & !SLP & !CEB & !WEB)
 CONDITIONAL_PIN = CLK { RISE }
 USER_PWL_FILE write_rise.txt
 END_MODE
 MODE_NAME READ
 CONDITIONAL_INPUT = (!SD & !SLP & !CEB & WEB)
 CONDITIONAL_PIN = CLK { RISE }
 USER_PWL_FILE read_rise.txt
 END_MODE
 ...
END
```

### Example of “USER\_PWL\_FILE” (Multiple regions)

```
UNIT CURRENT mA
UNIT TIME ns
 NET VSS
 REGION { 0 0 0.1 -1 0.2 -2 0.3 -3 0.4 -10 0.5 -50 0.6 -40 0.7 -30 0.8 -20 1 0
2 0 3 0 3.9 0 4 -0.1 4.5 -0.2 5 -4 5.5 -3 6 -2 7 0 }
 NET VDD
 REGION { 0 0 0.1 1 0.2 2 0.3 3 0.4 10 0.5 50 0.6 40 0.7 30 0.8 20 1 0 2 0 3 0
3.9 0 4 0.1 4.5 0.2 5 4 5.5 3 6 2 7 0 }
 NET VDDM
 REGION { 0 0 0.1 1 0.2 2 0.3 3 0.4 10 0.5 50 0.6 40 0.7 30 0.8 20 1 0 2 0 3 0
3.9 0 4 0.1 4.5 0.2 5 4 5.5 3 6 2 7 0 }
```

## Related Commands

- `get_power`
- `get_cells`
- `get_lib_cells`

## set\_power\_analysis\_mode

```
set_power_analysis_mode
[-reset]
[-add_simulation {true | false}]
[-adjust_input_activity_in_iterations {true | false}]
[-adjust_macro_activity_in_iterations {true | false}]
[-annotation_detail_report {true | false}]
[-average_rise_fall_cap {true | false}]
[-binary_db_name filename]
[-block_independent_peakpower {true | false}]
[-boundary_gate_leakage_file filename]
[-boundary_gate_leakage_report {true | false}]
[-boundary_leakage_cell_property_file filename]
[-boundary_leakage_edge_annotation_file filename]
[-boundary_leakage_multi_pgpin_support {true | false}]
[-boundary_leakage_pessimism_removal {true | false}]
[-boundary_leakage_pxe_support {true | false}]
[-bulk_pins {bulk_pin_list}]
[-capacity {low|medium|high}]
[-case_insensitive_mapping {true | false}]
[-clock_source_as_clock {true | false}]
[-comprehensive_automapping {true | false}]
[-constant_override {true | false}]
[-corner {min|max}]
[-create_binary_db {true | false}]
[-create_driver_db {true | false}]
[-create_gui_db {true | false}]
[-current_generation_method { avg | peak }]
[-decap_cell_list cell_list]
[-default_frequency value]
[-default_slew value]
[-default_supply_voltage value]
[-disable_clock_gate_clipping {true | false}]
[-disable_leakage_scaling {true | false}]
[-disable_static {true | false}]
[-distribute_switching_power {true | false}]
[-distributed_combine_report_format {none | detailed | reduced}]
[-distributed_setup file]
[-domain_based_clipping {true | false}]
[-dynamic_glitch_filter value]
[-dynamic_vectorless_ranking_methods {load | clock | vector_activity}]
[-enable_auto_mapping {true | false}]
[-enable_duty_prop_with_global {true | false}]
[-enable_dynamic_current_slew_load_interpolation {true | false}]
[-enable_dynamic_scaling {true | false}]
```

```
[-enable_flop_state_propagation {true | false}]
[-enable_generated_clock {true | false}]
[-enable_input_net_power {true | false}]
[-enable_mt_in_vectorbasedflow {true | false}]
[-enable_pba_for_tempus_pi {true | false}]
[-enable_power_target_flow {true | false}]
[-enable_rtl_vectorbased_dynamic_analysis {true | false}]
[-enable_scan_report {true | false}]
[-enable_slew_based_ccs_pin_cap {true | false}]
[-enable_state_propagation {true | false}]
[-enable_tempus_pi {true | false}]
[-enable_xp {true | false}]
[-enhanced_blackbox_avg {true | false}]
[-enhanced_blackbox_max {true | false}]
[-equivalent_annotation {true | false}]
[-event_based_leakage_power {true | false}]
[-external_load_config_file filename]
[-extraction_tech_file filename]
[-extractor_include filename]
[-fanout_limit value]
[-flatten_xpgv_block_instances filename]
[-force_library_merging {true | false}]
[-from_x_transition_factor value]
[-from_z_transition_factor value]
[-generate_activity_mapping_report {true | false}]
[-generate_current_for_rail railnames]
[-generate_flop_ranking_data directoryname]
[-generate_leakage_power_map_based_on_calculated_leakage {true | false}]
[-generate_static_report_from_state_propagation {true | false}]
[-handle_glitch {true|false}]
[-handle_tri_state {false|true}]
[-hier_delimiter character]
[-honor_combinational_logic_on_clock_net {true | false}]
[-honor_negative_energy {true | false}]
[-honor_net_activity {true | false}]
[-hybrid_analysis {true | false}]
[-ignore_control_signals {true | false}]
[-ignore_data_phase_for_clk {true | false}]
[-ignore_end_toggles_in_profile {true | false}]
[-ignore_glitches_at_same_time_stamp {true | false}]
[-ignore_inout_pin_cap {true | false}]
[-include_seq_clockpin_power {true | false}]
[-include_timing_in_current_file {true | false}]
[-ir_derated_timing_view view_name]
[-keep_clock_gate_ratio_in_iterations {true | false}]
[-leakage_scale_factor_for_temp scale]
```

```
[-library_preference {voltage | ecdm_ccsp}]
[-mbff_toggle_behavior { simultaneous | independent | sbff }]
[-merge_switched_net_currents {true | false}]
[-method { static | dynamic_vectorless | dynamic_vectorbased | dynamic_mixed_mode |
event_based | vector_profile}]
[-min_leaf_count value]
[-multi_scenario_simulation {true | false}]
[-off_pg_nets net _list]
[-output_current_data_prefix prefix]
[-partition_twf {true | false}]
[-pin_based_twf {true | false}]
[-power_grid_library { library _list }]
[-power_include_initial_x_transitions {true | false}]
[-power_match_state_for_logic_x value]
[-precision value]
[-pre_simulation_empty_period period]
[-pre_simulation_period period]
[-pre_simulation_power_exclude_period period]
[-quit_on_activity_coverage_threshold threshold_value]
[-read_rcdb {true | false}]
[-relax_arc_match {true | false}]
[-report_black_boxes {true | false}]
[-report_idle_instances {true | false}]
[-report_instance_switching_info {all | output_logic | none}]
[-report_instance_switching_list filename]
[-report_library_usage {true | false}]
[-report_missing_bulk_connectivity {true | false}]
[-report_missing_input {true | false}]
[-report_missing_nets { true|false }]
[-report_scan_chain_stats {true | false}]
[-report_stat {true | false}]
[-report_time_display_fraction_digits value]
[-report_twf_attributes {detailed | summary}]
[-reuse_flop_ranking_data directoryname]
[-reuse_flop_ranking_data_hier blockname]
[-scale_to_sdc_clock_frequency {true | false}]
[-scan_chain_activity {design_name chain_name activity_number} | {vectorless
activity_number}]
[-scan_chain_name {design_name chain_name pattern}]
[-scan_control_file filename]
[-scan_mbff_chain_type type]
[-settling_buffer value]
[-smart_window {true | false}]
[-split_bus_power {true | false}]
[-start_time_alignment {true |false}]
[-state_dependent_leakage {true | false}]
```



```
[-stateprop_ignore_unannot_pins {true | false}]
[-static_multi_mode_scenario_file filename]
[-static_netlist {verilog | def}]
[-switching_power_on_rise_only {true | false}]
[-thermal_input_file file]
[-thermal_leakage_temperature_scale_table_file filename]
[-to_x_transition_factor value]
[-to_z_transition_factor value]
[-transition_factor_based_duty {true | false}]
[-transition_time_method {min | avg | max}]
[-twf_delay_annotation {min | avg | max}]
[-twf_load_cap {min | avg | max}]
[-use_cell_leakage_power_density {true | false}]
[-use_fastest_clock_for_dynamic_scheduling {true | false}]
[-use_lef_for_missing_cells {true | false}]
[-use_zero_delay_vector_file {true | false}]
[-vector_profile_mode {activity | event_based | power_density | transient}]
[-worst_case_vector_activity {true | false}]
[-worst_step_size value]
[-worst_window_count value]
[-worst_window_coverage { top | tile }]
[-worst_window_reports {full | worst | both}]
[-worst_window_size value]
[-worst_window_type { power | activity | delta_power | delta_activity | ir |
critical_inst_power }]
[-write_boundary_leakage_edge_annotation {true | false}]
[-write_default_uti {true | false}]
[-write_dynamic_currents {true | false}]
[-write_profiling_db {true | false}]
[-write_simulation_db {true | false}]
[-write_static_currents {true | false}]
[-x_count_transition_using_3_states {true | false}]
[-x_transition_factor value]
[-z_transition_factor value]
[-zero_delay_vector_toggle_shift value]
```

Sets the parameters for doing the power analysis.

## Parameters

```
-analysis_view mmmc_view
```

|                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                    | <p>Specifies the power analysis library view (for MMMC setup). You can specify only one analysis view at a time.</p> <p><b>Note:</b> You can also use <code>-view</code> instead of <code>-analysis_view</code> to specify the power analysis library view.</p> <p>If <code>report_power -view view</code> is specified, this view will override <code>-analysis_view</code> setting.</p> <p>There is no default.</p>                                                               |
| <code>-average_rise_fall_cap {true   false}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                    | <p>When set to true, the software uses the average of rise and fall capacitance from the liberty file.</p> <p><i>Default:</i> false</p>                                                                                                                                                                                                                                                                                                                                             |
| <code>-binary_db_name filename</code>              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                    | <p>Specifies the name of the binary power database file (<code>filename.db</code>).</p> <p>You can specify this parameter in scenarios where more than one db is generated and you need to track which one was generated in which mode when restoring the db, or if your design has several blocks and analysis is done on one block all the time and all the databases are named <code>power.db</code>.</p>                                                                        |
| <code>-bulk_pins {bulk_pin_list}</code>            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                    | <p>Defines power and ground bulk LEF pins for the design.</p> <p>Use this parameter when library cells have bulk pins defined in LEF but the liberty file does not contain power associated with these pins. If a liberty file does not have body bias definition, power analysis does not distribute power to the body bias domain.</p> <p><b>Note:</b> The <code>-bulk_pins</code> parameter cannot be reset using the <code>set_power_analysis_mode -reset</code> parameter.</p> |
| <code>-capacity {low medium high}</code>           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                       | <p>Enables segmented power processing. When this parameter is specified, the software performs concurrent processing of vector simulation annotation, power calculation, and report construction.</p> <p>This parameter enables performance optimization with some trade-off on memory consumption and runtime. The arguments of the <code>-capacity</code> parameter are:</p> <ul style="list-style-type: none"> <li>• <code>low</code> - reduces memory for a flat run.</li> <li>• <code>medium</code> - further reduces memory with moderate increase in runtime (1.2X).</li> <li>• <code>high</code> - maximum memory reduction at about 2X runtime.</li> </ul> <p>The <code>-capacity</code> parameter is used for reducing memory when processing long simulation vectors. In case of the FSDB and SHM format activity file, the runtime can increase up to 2X.</p> |
| <code>-case_insensitive_mapping {true   false}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                       | <p>Specifies whether instance name mapping between the RTL netlist and GATE level netlist in the mapping file is case sensitive or not. When set to <code>true</code>, the name mappings are not case sensitive.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>-clock_source_as_clock {true   false}</code>    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                       | <p>Specifies to use the correct clock frequency for activity calculation. If you stop the arrival of the clock signal by setting <code>set_case_analysis</code> to 0, the transition density at the output net of the flop is calculated using the correct clock frequency instead of the default frequency.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>-constant_override {true   false}</code>        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                       | <p>Specifies to give higher precedence to propagated constants defined using the <code>set_case_analysis</code> command. It allows propagated constants to override global activity.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>-corner {min max}</code>                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                        | <p>Defines the library corner (for non-MMMC setup).</p> <ul style="list-style-type: none"> <li>• <code>min</code> corner means power calculator will use min timing libraries for the power calculation.</li> <li>• <code>max</code> corner means power calculator will use max timing libraries for the power calculation.</li> </ul> <p><i>Default:</i> <code>max</code></p>                                                                                                                                                                                                                                                                                      |
| <code>-create_binary_db {true   false}</code>          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                        | <p>A value of <code>true</code> creates a binary database of power results.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <code>-create_driver_db {true   false}</code>          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                        | <p>Specifies to determine all the drivers/receivers on the ground net.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>-current_generation_method { avg   peak }</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                        | <p>Specifies the current waveform generation method during dynamic power analysis. The two possible arguments of this parameter are:</p> <ul style="list-style-type: none"> <li>• <code>avg</code> - Specifies to use the average current value of each time step. The output current filename is <code>filename.ptiavg</code>. The default value is <code>avg</code>.</li> <li>• <code>peak</code> - Specifies to use the peak current value of each time step. The output current filename is <code>filename.ptipeak</code>. It is recommended to use the <code>peak</code> method when using the Composite Current Source Power (CCSP) Liberty files.</li> </ul> |
| <code>-default_frequency value</code>                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                        | <p>Specifies to set the default frequency in the MHz unit for net not annotated by TWF. You can specify a numerical value without unit.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <code>-default_slew value</code>                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                        | <p>Specifies to set the default slew in the ns unit for net not annotated by TW. You can specify a numerical value without unit.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <code>-default_supply_voltage value</code>             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                                                        | <p>Specifies to set a default voltage for power nets in a scenario where the power engine cannot determine the voltage of these nets. The default value of this parameter is <code>1.0</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

|                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <code>-disable_clock_gate_clipping {true   false}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                          | <p>Specifies to enable clock gate output clipping so that the output transition density of an ICG cell does not exceed the input clock pin transition density.</p> <p><i>Default:</i> <code>true</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>-disable_leakage_scaling {true   false}</code>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                          | <p>Specifies to exclude leakage power during scaling factor computation when target power is specified using the <code>set_power</code> command. When <code>set_power_analysis_mode -disable_leakage_scaling</code> is set to <code>true</code>, the leakage power does not get scaled, and only the internal and switching components are scaled to meet the user-specified total power.</p> <p>This parameter is relevant only for the <code>set_power</code> commands for which total power is specified, and has no effect for the <code>set_power</code> command specified with the <code>-internal</code>, <code>-switching</code>, or <code>-leakage</code> parameters.</p>                                                                                                                                                                                                                                        |
| <code>-disable_static {true   false}</code>              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                          | <p>When set to <code>true</code> in the dynamic vector-based or dynamic vectorless flows the command performs only dynamic analysis and turns-off static power calculation.</p> <p><i>Default:</i> <code>true</code></p> <p><b>Note:</b> When you perform static power calculation in the dynamic power analysis flow, you can use either the <code>-disable_static</code> or <code>-write_static_currents</code> parameter to enable static power calculation and write static current file. If only <code>-disable_static</code> is set to <code>false</code> and <code>-write_static_currents</code> is not set, the tool automatically sets <code>-write_static_currents</code> to <code>true</code>. Similarly, if <code>-write_static_currents</code> is set to <code>true</code>, and <code>-disable_static</code> is not set, the tool automatically sets <code>-disable_static</code> to <code>false</code>.</p> |
| <code>-distributed_setup file</code>                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                    | <p>Specifies a file containing the customized setup details for running power analysis in the distributed mode.</p> <p>The specified file has 4 columns:</p> <pre>INST/CELL &lt;name&gt; hier/block &lt;power.inc&gt;</pre> <p>where,</p> <ul style="list-style-type: none"> <li>• <b>INST/CELL:</b> Keyword indicating cell or instance to be run in the distributed mode. Cell can be useful for repeated DEF blocks.</li> <li>• <b>name:</b> Name of the hierarchical block or instance.</li> <li>• <b>hier/block:</b> Keyword indicating hierarchy or block. <b>hier</b> implies that the full hierarchy will be preserved by including the DEFs from top to the block, and <b>block</b> implies that only the lower level DEF will be included</li> <li>• <b>power.inc:</b> name of the power include file. This column is optional.</li> </ul> <p>The unspecified instance/cell will be run with the top level as a single group. Each distributed block/instance should have corresponding TWF and SPEF to improve runtime.</p> |
| <code>-decap_cell_list cell_list</code>            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                    | <p>Specifies the physical only cells, such as decap cells, to include in the power report.</p> <p><b>Note:</b> This is only supported for Static Power Analysis.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>-domain_based_clipping {true   false}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                    | <p>Determines the frequency to be used to clip propagated activity that is too high. When set to <code>false</code>, the frequency of the fastest clock in the design is used. If you set this parameter to <code>true</code>, the fastest clock domain on the net used to clip propagated activity.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>-dynamic_glitch_filter value</code>          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                    | <p>Sets the absolute threshold time (in ns) for filtering glitches in net toggles. When you specify a time value for this parameter, glitches in net toggles that are equal to or smaller in width than the specified time value are removed.</p> <p><i>Default:</i> <code>-1</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>-dynamic_power_view view_name</code>         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

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|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                 | <p>Specifies a separate dynamic analysis view for power calculation.</p> <p>You can set the <code>-dynamic_power_view</code> as the analysis view for dynamic, ne and total power calculation. If you do not specify this parameter, the 1st setup view will be assumed as dynamic view and will be used to calculate power.</p> <p>Ensure that the specified views are well defined from both a Power and Timing point of view.</p> <p><b>Note:</b> You can specify the same or different views as the leakage and dynamic views for optimization and signoff.</p> <ul style="list-style-type: none"> <li>• If the leakage and dynamic view is to be the same, then run the following command:<br/> <pre>set_power_analysis_mode -leakage_power_view power_view_name -dynamic_power_view power_view_name</pre> </li> <li>• If the leakage and dynamic view is to be different, then run the following command:<br/> <pre>set_power_analysis_mode -leakage_power_view leakage_view_name -dynamic_power_view dynamic_view_name</pre> </li> </ul> <p>For more information, refer to the “Specifying the Correct Power Views for Optimization” section in the <b>Optimizing Timing</b> chapter of <i>Innovus User Guide</i>.</p> |
| <pre>-dynamic_vectorless_ranking_methods {load   clock   vector_activity}</pre> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                                                 | <p>Specifies to change the ranking method from the default to the following:</p> <ul style="list-style-type: none"> <li>• <code>load</code> - Prioritizes scheduling cells with higher fanout</li> <li>• <code>clock</code> - Prioritizes scheduling cells associated with faster clocks</li> <li>• <code>vector_activity</code> - Prioritizes scheduling cells with higher activity</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <pre>-enable_auto_mapping {true   false}</pre>                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|                                                                                 | <p>Specifies to automatically perform instance name mapping between the RTL netlist and GATE level netlist. This parameter allows you to map the RTL vector to the Gate level netlist without a mapping file.</p> <p>You can use this parameter with the <code>read_activity_file -name_mapping_rule &lt;filename&gt;</code> command parameter to specify a file containing customized or design-specific rules.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <pre>-enable_duty_prop_with_global {true   false}</pre>                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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|                                                           | Specifies to propagate duty cycle with the global switching activity setting for all data nets in the dynamic vectorless analysis flow.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>-enable_dynamic_scaling {true   false}</code>       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                           | <p>Enables <code>set_power</code> scaling for dynamic power analysis. The power and scaling specifications are used to scale dynamic current on both the power and ground currents of the specified instance/cell.</p> <p>This parameter honors the user-defined average (static) power for leaf-level instances and hierarchies (instances/cells/modules) along with power net specification in the dynamic current scaling flows such that the average of the dynamic current is equal to the assigned average power. When you specify this parameter, the average of the dynamic current of these leaf-level instances will match the scaled static power numbers.</p> <div style="border: 1px solid #f0e68c; padding: 10px; margin: 10px 0;"> <p> This parameter is not supported in the state-propagation-based vectorless flow (<code>set_power_analysis_mode -enable_state_propagation true</code>). It is applicable only to the probability-based vectorless flow.</p> </div> <p>Default: <code>false</code></p> |
| <code>-enable_generated_clock {true   false}</code>       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                           | Specifies to get generated clock frequency during activity propagation for static power analysis. The default value of this parameter is <code>true</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>-enable_input_net_power {true   false}</code>       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                           | <p>Specifies to calculate the switching power of input nets.</p> <p>Default is <code>false</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>-enable_mt_in_vectorbasedflow {true   false}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                           | <p>Specifies to enable multi-threading for the VCD/FSDB based dynamic vector-based flows.</p> <p>Default is <code>false</code>. To enable multi-threading in the vector-based flows, set this parameter to <code>true</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>-enable_pba_for_tempus_pi {true   false}</code>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |



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|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                      | <p>Specifies to enable the Path-Based Analysis (PBA) for slack calculation in the Tempus Power Integrity (Tempus PI) flow. By default, the Graph-Based Analysis (GBA) is used. Default is <code>false</code>.</p> <p>This parameter is only applicable to the state-propagation-based vectorless flow (<code>set_power_analysis_mode -method dynamic_vectorless -enable_state_propagation true</code>).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>-enable_rtl_vectorbased_dynamic_analysis {true   false}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                                      | <p>Specifies to take an RTL or partial VCD/FSDB file as input and use that for dynamic vector-based flow. This parameter allows you to estimate current even for instances which are missing from the activity file.</p> <p>You must use this parameter in conjunction with the <code>-method dynamic_vectorbased</code> parameter. Default is <code>false</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <code>-enable_scan_report {true   false}</code>                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                                      | <p>Specifies to generate all the scan mode analysis reports. The following reports will be generated:</p> <ul style="list-style-type: none"> <li>• <code>voltus_power.scanff.duplicated.rpt</code> - flip-flops that are specified in multiple chains</li> <li>• <code>voltus_power.scanff.missing.rpt</code> - flip-flops that are missing in the scan control file but exist in the design</li> <li>• <code>voltus_power.scanff.nonassigned.rpt</code> - flip-flops that do not exist in the design and are dropped from the scan control file</li> <li>• <code>voltus_power.scanff.nonscan.rpt</code> - assigned flip-flops that have no scan-out pin</li> <li>• <code>voltus_power.scanff.nonswitched.rpt</code> - non-switching flip-flops in the scan control file</li> <li>• <code>voltus_power.scanff.rpt</code> - detailed scan chain report containing global statistics for all scan chains and statistics for each scan chain in the design</li> <li>• <code>voltus_power.scanff.stateconflict.rpt</code> - flip-flops that have conflicting state assignment</li> </ul> <p><i>Default:</i> <code>false</code></p> |
| <code>-enable_state_propagation {true   false}</code>                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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|                                                    | <p>Specifies to enable the vectorless state-propagation-based dynamic analysis flow. Default is <code>false</code>.</p> <p>For more information, refer to the <b>State-Propagation-Based Vectorless Methodology</b> section in the "<i>Dynamic Power and IRDrop Analysis</i>" chapter of the <i>Voltus User Guide</i>.</p>                                                                                                                                                                                                                       |
| <code>-enable_tempus_pi {true   false}</code>      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                    | <p>Specifies to enable the Tempus PI analysis flow. When this parameter is set to <code>true</code>, data path scheduling is timing-aware, that is, scheduling is based on timing slacks. Default is <code>false</code>.</p> <p>This parameter is only applicable to the state-propagation-based vectorless flow (<code>set_power_analysis_mode -method dynamic_vectorless -enable_state_propagation true</code>).</p> <p>For more information, refer to the <i>Tempus Power Integrity Analysis</i> chapter in the <i>Tempus User Guide</i>.</p> |
| <code>-enable_xp {true   false}</code>             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                    | <p>Specifies to run standalone report generation in the XP mode.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>-enhanced_blackbox_avg {true   false}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                    | <p>Specifies to use the average toggle rate of related inputs for enhanced blackbox propagation.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>-enhanced_blackbox_max {true   false}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                    | <p>Specifies to use the maximum toggle rate of related inputs for enhanced blackbox propagation.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>-extraction_tech_file filename</code>        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                    | <p>Specifies the extraction technology file to be used for top-level power-grid extraction.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>-fanout_limit value</code>                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                    | <p>Specifies the fanout limit for nets. If an instance has nets more than or equal to the fanout limit, its switching/internal/leakage power is taken to be zero.</p>                                                                                                                                                                                                                                                                                                                                                                            |

|                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <code>-flatten_xpgv_block_instances filename</code>                        | <p>Specifies a file that includes a list of block instances that use DEF instead of xPGV for power analysis, that is, specify the instances that are not to be used as xPGV. This parameter allows you to specify a few instances of the same block that will use DEF, and the others that will use xPGV at the top-level. If you do not specify this parameter, and if both DEF and xPGV are given for a block, xPGV takes precedence over DEF.</p> <p><b><u>Instance List File Format:</u></b></p> <pre>INST CELL  &lt;instance_name/cell_name&gt; &lt;block_instance_name All None&gt;</pre> <p>In this file, you can specify if a block is using a DEF by specifying the block instance name, or with the <code>All</code> or <code>None</code> keywords. <code>All</code> indicates that all instances are using DEF, and <code>None</code> indicates that none of the instances are using DEF.</p> <p><b><u>Example:</u></b></p> <pre>INST instA  /top/x1/x12/instA</pre> <p>For a design with three blocks, if you want to use the DEF and xPGV as following:</p> <p>Block_1 -&gt; A1 (xPGV ) &amp; A2 (DEF)<br/> Block_2 -&gt; B1 &amp; B2 (xPGV for both instances)<br/> Block_3 -&gt; C1 &amp; C2 (DEF for both instances)</p> <p>The specified file will contain the following information:</p> <pre>INST Block_1 A2 INST Block_2 None INST Block_3 All</pre> |
| <code>-from_x_transition_factor value</code>                               | <p>Specifies how transitions from X to 0/1 are counted. The default value is 0.5.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>-from_z_transition_factor value</code>                               | <p>Specifies how transitions from Z to 0/1 are counted. The default value is 0.25.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <code>-generate_current_for_rail railnames</code>                          | <p>Generates current files for the specified rail.</p> <p>By default, the command generates current files for all power rails.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>-generate_static_report_from_state_propagation {true   false}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

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|                                                                     | <p>Specifies to directly generate the static power analysis result based on the RTL-VCD input and state-propagation. When this parameter is specified, the RTL vector power analysis flow does not require the generation of a TCF/FSDB file. This parameter is applicable only to the RTL vector-based dynamic power analysis flow.</p> <p><i>Default:</i> false</p>                                                                                                                                                                                                                                                                                                                              |
| <code>-handle_glitch {true   false}</code>                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                                     | <p>A transition is defined as a glitch when the time difference between two sequential toggles are less than half of the rise transition time + fall transition time in the vector-based static power calculation.</p> <p>If set to <code>true</code>, glitch power will be identified and calculated according to this definition and it will be reported in a separate column. Switching or internal power calculation does not include power due to glitches.</p> <p>If set to <code>false</code>, glitch power will not be identified, and the respective power will be calculated as normal signal switches. Hence, the total power will be a little higher.</p> <p><i>Default:</i> false</p> |
| <code>-handle_tri_state {false true}</code>                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                                     | <p>When set to <code>true</code>, all tristate device enable pin values will be taken into account when determining the propagation of the activity through tristate gates.</p> <p><i>Default:</i> false</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>-hier_delimiter character</code>                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                                     | <p>Specifies the hierarchical delimiter in the DEF file. The default hierarchical delimiter is a forward slash (/) but can be changed by setting the <code>-hier_delimiter</code> parameter. You can use this parameter to override the existing hierarchical delimiter.</p>                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>-honor_combinational_logic_on_clock_net {true   false}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                                     | <p>Specifies the behavior of combination logic on clock net.</p> <ul style="list-style-type: none"> <li><code>true</code> (default) - specifies to honor activity propagation through combinational logic.</li> <li><code>false</code> - specifies that the combination logic will be treated like a combinational clockgate.</li> </ul>                                                                                                                                                                                                                                                                                                                                                           |

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| <code>-honor_negative_energy {true   false}</code>              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                                 | <p>A value of <code>true</code> specifies that Voltus will keep negative internal energy numbers from the <code>.lib</code> internal power table. When set to <code>false</code>, negative power is treated as 0.</p> <p><i>Default:</i> <code>true</code></p>                                                                                                                                                                                                                                                                                                       |
| <code>-honor_net_activity {true   false}</code>                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                                 | <p>Specifies to use activity from the <code>Net</code> section of the Switching Activity Interchange Format (SAIF) file for power calculation and optimization, and to ignore the pin-based state-dependent path delay (SDPD) activity. The default value of this parameter is <code>true</code>.</p> <p>When this parameter is set to <code>false</code>, the power calculation engine can use both the net-based and pin-based activity. In this case, precedence will be given to activity from net or pin that appears later in the SAIF file.</p>               |
| <code>-ignore_control_signals {true   false}</code>             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                                 | <p>A value of <code>true</code> specifies that control signals will be ignored when propagating activity.</p> <p><i>Default:</i> <code>true</code></p>                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>-ignore_data_phase_for_clk {true   false}</code>          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                                 | <p>Specifies to ignore data frequency in the clock path for the clock instances in the design. When this parameter is set to <code>true</code>, the software propagates only the fastest clock pin frequency instead of the data pin frequency for activity propagation through clock network. The default value is <code>false</code>. In the default mode, the software picks the fastest frequency between the data phase and clock phase on that pin.</p>                                                                                                        |
| <code>-ignore_glitches_at_same_time_stamp {true   false}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                                 | <p>Specifies to ignore glitches at the same transient time in the vector-based static and dynamic power analysis flow. When –</p> <p><code>ignore_glitch_at_same_time_stamp</code> is set to <code>true</code>, only the last signal value change is honored, and all the preceding value changes at the same time stamp are ignored. The last signal value will overwrite the previous value for the same signal at the same time stamp, so the last value will represent the state of the signal at that time stamp.</p> <p><i>Default:</i> <code>false</code></p> |
| <code>-ignore_inout_pin_cap {true   false}</code>               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

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|                                                            | <p>When set to <code>true</code>, ignores the bidirectional pin capacitances (<code>direction : inout</code>) defined for I/O cells in the .lib file, when calculating switching power and internal power; this also includes the default capacitance value for <code>inout</code> pins.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>-ignore_end_toggles_in_profile {true   false}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                            | <p>By default, the software profiles the toggles in the user-selected vector time window by steps including the toggles at the start/end boundary of the window. In each step, the toggles on the start boundary edge of the step is counted in the current step; the toggles on the end boundary edge is counted in the next step.</p> <p>When this parameter is set to <code>false</code>, the toggles on the end boundary edge is also counted in the last step, as there is no next step to it. This may lead to double count of the power profile for the last step, especially for zero-delay vector profiling. The default of this parameter is <code>false</code>.</p> <p>When this parameter is set to <code>true</code>, the software ignores the toggle on the end boundary of the last step.</p> |
| <code>-include_seq_clockpin_power {true   false}</code>    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                            | <p>Reports the clock pin power of flip-flops as part of the clock network power.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <code>-ir_derated_timing_view view_name</code>             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                            | <p>Specifies the name of the timing view used to calculate timing slack and to rank the flops in the Tempus PI flow. This parameter must be specified if there are multiple views.</p> <p>The <code>-ir_derated_timing_view</code> parameter defines the timing view used to detect the IR-sensitive paths using a lower voltage than the actual analysis. This lower voltage simulates the IR-drop for each instance in the design. The software uses this view to prioritize the end-points that need to be prioritized for switching.</p> <p>This parameter is only applicable to the state-propagation-based vectorless flow (<code>set_power_analysis_mode -method dynamic_vectorless -enable_state_propagation true</code>).</p>                                                                       |
| <code>-leakage_power_view view_name</code>                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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|                                                         | <p>Specifies a separate leakage analysis view for power calculation.</p> <p>You can set the <code>-leakage_power_view</code> as the analysis view for leakage power calculation. If you do not specify this parameter, the 1st setup view will be assumed as leakage view and will be used to calculate power.</p> <p>The leakage view can be inactive only if <code>-state_dependent_leakage</code> is set <code>false</code>. Ensure that the specified views are well defined from both a Power and Timing point of view.</p> <p><b>Note:</b> You can specify the same or different views as the leakage and dynamic views for optimization and signoff.</p> <ul style="list-style-type: none"> <li>• If the leakage and dynamic view is to be the same, then run the following command:<br/> <pre>set_power_analysis_mode -leakage_power_view power_view_name -dynamic_power_view power_view_name</pre> </li> <li>• If the leakage and dynamic view is to be different, then run the following command:<br/> <pre>set_power_analysis_mode -leakage_power_view leakage_view_name -dynamic_power_view dynamic_view_name</pre> </li> </ul> <p>For more information, refer to the “Specifying the Correct Power Views for Optimization” section in the <b>Optimizing Timing</b> chapter of <i>Innovus User Guide</i>.</p> |
| <code>-leakage_scale_factor_for_temp scale</code>       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                                                         | <p>Performs a linear scaling of the leakage power for temperature in all libraries. This replaces the k-factor for temperature in the .lib file. For process and voltage, the scaling will continue to be based on the corresponding k-factors in the .lib file. A value of 0.8 scales the leakage power in the library to 80 percent. A value of 1.2 scales the leakage power in the library to 120 percent.</p> <p><i>Default:</i> 1</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <code>-library_preference {voltage   ecs_m_ccsp}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

|                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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|                                                                                               | <p>Instances of the same cell can be attached to different power domains in a design. If you have specified different PVT libraries, then the software will by default bind each instance to the closest library definition.</p> <p>You can use this parameter to select either the closest Effective Current Source Model (ECSM)/Composite Current Source-power (CCSP), or the closest voltage library for the instance:</p> <ul style="list-style-type: none"> <li>• <code>voltage</code>: when specified, enables library binding to the closest voltage library.</li> <li>• <code>ecsm_ccsp</code>: when specified, enables library binding to the closest ECSM/CCSP library.</li> </ul> <p>The default is <code>voltage</code>.</p> |
| <pre>-mbff_toggle_behavior { simultaneous   independent   sbff }</pre>                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                                                               | <p>Specifies the toggle behavior of multi-bit flip-flops (MBFF). The possible arguments are:</p> <ul style="list-style-type: none"> <li>• <code>simultaneous</code>: The multiple bits of the MBFF will toggle or not toggle at the same time. The MBFF will be considered as one flip-flop.</li> <li>• <code>independent</code>: The multiple bits of the MBFF will toggle independently. The MBFF will be considered as one flip-flop.</li> <li>• <code>sbff</code>: The MBFF will be considered as multiple Single-Bit Flip-Flops (SBFFs). Each bit will be considered as one flip-flop, and will toggle independently.</li> </ul> <p>The default value is <code>independent</code>.</p>                                              |
| <pre>-merge_switched_net_currents { true   false }</pre>                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|                                                                                               | <p>Specifies to merge switched net currents into always-on currents. For designs with a large number of switched nets, this parameter helps improve performance by merging switched net instance currents with always-on instance currents. However, when this parameter is set to <code>true</code> you will not be able to use the generated current files to run power-up analysis.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                                                                                                     |
| <pre>-method { static   dynamic_vectorless   dynamic_vectorbased   dynamic_mixed_mode }</pre> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |



Specifies the type of analysis to be performed. You can perform two types of power analysis, namely static and dynamic.

- `static` - specifies to perform static power analysis to calculate average power.

Dynamic power analysis can be a vector-based or vectorless approach.

- `dynamic_vectorbased` - uses the full VCD output of a logic simulator to identify the instances that are switching and when they switch.
- `dynamic_vectorless` - uses the timing arrival window information from a static timing analysis tool to determine when instances switch. A timing arrival window describes when a signal can change within a clock cycle. An additional algorithm then determines the instances that switch.
- `dynamic_mixed_mode` - specifies to combine the vector-based and vectorless (mixed) dynamic analysis. This flow can be used in the following cases:

- **case 1:** vectors are not available for every block of a full-chip design. In this case, the software uses the vectorless methodology for blocks with incomplete vectors.

- **case 2:** blocks with vectors having unannotated combinational/sequential logic. In this case, the software performs state propagation for the missing combinational/sequential logic. It uses the vectorless methodology for a logic cone that does not have the primary inputs or flops annotated from the vector, and then performs state propagation.

To enable the dynamic mixed mode analysis flow, you must specify the following commands:

```
set_default_switching_activity -clock_gates_output 2.0 -
global_activity <activity>
set_power_analysis_mode -method dynamic_mixed_mode
read_activity_file -scope <> -block <> -> (for available blocks)
```

**Default:** `static`

`-min_leaf_count value`

|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 | <p>Specifies the minimum number of leaf instances required for writing the hierarchy data in the profiling database. It allows you to restrict the number of hierarchies for which waveforms are reported in the profiling database when using both the <code>-write_profiling_db</code> and <code>report_power -hierarchy</code> parameters.</p> <p>The default value of this parameter is 0. Any hierarchy whose leaf instance count in its complete sub-tree is less than the <code>-min_leaf_count</code> value will be skipped from getting reported in the profiling database. This parameter can be used to skip very small hierarchies and as a result save on the database writing time and disk space.</p> |
| <code>-off_pg_nets net_list</code>              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                 | <p>The name of the power rails that will be turned off. The net name can contain instance hierarchy if appropriate. This data will be taken into account when figuring out the list of power gates which are switched off.</p> <p><b>Note:</b> A net is turned off if all connected power gates are turned off by control signals. When a net is off, all of the instances connected to the net will only contribute leakage current into the connecting nets.</p>                                                                                                                                                                                                                                                   |
| <code>-output_current_data_prefix prefix</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                 | <p>Specifies a prefix to the static or dynamic current files generated by the software. When power analysis is done in the multi-CPU mode, the output current data directory name and the underlying current files will also use the same prefix by default.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <code>-partition_twf {true   false}</code>      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                 | <p>Specifies to read the input TWF data (flat or hierarchical TWF files), and generate block-level TWFs.</p> <p>The default is <code>false</code>. This parameter must be specified when the block/partition level TWF files are not available.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>-pin_based_twf {true   false}</code>      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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|                                                                  | <p>Specifies whether to generate pin-based TWF or net-based TWF. The default value is <code>false</code>.</p> <ul style="list-style-type: none"> <li>When set to <code>false</code>, a net-based TWF file is generated. Net-based TWF file is smaller in size and therefore results in faster processing.</li> <li>When set to <code>true</code>, a pin-based TWF file is generated. Pin-based TWF file is bigger in size and requires longer processing time, however, is more accurate because the slews numbers are pin-based.</li> </ul> <p><b>Note:</b> This parameter must be used with the <code>read_sdc</code> command.</p>                                             |
| <code>-power_grid_library { library_list }</code>                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                                  | <p>Specifies the name of the Cadence power cell libraries (<code>.cl</code>). You can also specify a list of defined power-grid libraries having decap cells tagged in them.</p> <p><b>Note:</b> When specifying a list of PGVs to power analysis using the <code>set_power_analysis_mode</code> command, you must specify the technology library as the first PGV library in the list, followed by the standard and macro PGV libraries. The selection between the standard and macro PGV library is order based. Therefore, if multiple power-grid libraries are specified for a cell, the power engine maps the cell with the first library after the technology library.</p> |
| <code>-power_include_initial_x_transitions {true   false}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                                  | <p>Controls how the initial X state toggle should be counted. The initial X state means the state at time 0.</p> <p>The default value is <code>true</code>. This means that the software counts the power caused by the <code>X-&gt;0</code> or <code>X-&gt;1</code> toggle. When set to <code>false</code>, the software does not count the power caused by the initial X state toggle.</p>                                                                                                                                                                                                                                                                                     |
| <code>-power_match_state_for_logic_x value</code>                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                                  | <p>Controls how the logic X will be evaluated in the boolean function of the 'when' state of a power table. The default value is 0.</p> <p>The parameter uses the following arguments:</p> <ul style="list-style-type: none"> <li>0 - regards X as 0</li> <li>1 - regards X as 1</li> <li>x/X - regards X as neither 0 nor 1, that is, whenever there is any X logic in it, the boolean function will be evaluated false.</li> </ul>                                                                                                                                                                                                                                             |

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| <code>-precision value</code>                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                                   | Specifies precision of decimal range [1-8] to ensure consistent decimal places are displayed for each power component. The default value is 8. This parameter is used for precision handling in <code>report_instance_power</code> and <code>report_power</code> .                                                                                                                                                                                                                               |
| <code>-quit_on_activity_coverage_threshold threshold_value</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                                   | Specifies the activity (VCD/FSDB/TCF) coverage threshold value. This parameter allows you to control the acceptable activity coverage that you are expecting and enforce the software to quit when it does not meet the criteria. When you specify the threshold value, the software exits if the real activity coverage is below the threshold.                                                                                                                                                 |
| <code>-read_rcdb {true   false}</code>                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                                   | Specifies to write a SPEF file containing only the total C for signal nets and pass it to the Dynamic Power engine. Dynamic power analysis requires only the total C for signal nets. Therefore, instead of writing a large SPEF file (with distributed RC) from the RCDB, this parameter specifies to create and pass a SPEF file with only total C for all the signal nets. This significantly reduces the parasitic annotation time in the Dynamic Power engine.<br><br><i>Default:</i> false |
| <code>-report_black_boxes {true   false}</code>                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                                                                   | Specify this parameter to report cells that are used as black boxes.<br><br>A black box is a cell which is either missing from the liberty file or has no functionality defined in the liberty file.<br><br><i>Default:</i> false<br><br>This is an optional parameter.                                                                                                                                                                                                                          |
| <code>-report_idle_instances {true   false}</code>                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

Specifies to generate a report (`idleinstance.rpt`) that lists all the idle or non-switching instances.

The default value is `false`.

The following is a snippet of the `idleinstance.rpt` report:

```
DF22/DF2 F
DF42/DF2 F
...
```

Here, the first column is the instance name and the second column is the type of instance (`F` indicates flop and `C` indicates combinational).

```
-report_instance_switching_info {all | output_logic | none}
```

- `all` - Specifies to report instance toggle information for both input and output pins. When this parameter is set to `all`, the switching of both the input and output pins of all the instances are reported in the `voltus_power.stateprop.switchinst` file and the switching of flops are reported in the `voltus_power.stateprop.switchsrc` file.
- `output_logic` - Specifies to report instance toggle information for the output pins only. When this parameter is set to `output_logic`, the switching of the output pins of all the instances are reported in the `voltus_power.stateprop.switchinst` file and the switching of flops are reported in the `voltus_power.stateprop.switchsrc` file.
- `none` - Disables generation of instance toggle report. This is the default option. When this parameter is set to `none`, only switching of flops are reported in the `voltus_power.stateprop.switchsrc` file. The file does not report switching of instances.

The following is a snippet of the `voltus_power.stateprop.switchinst` report:

Here,

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`-report_instance_switching_list filename`

Specifies to generate a report (`voltus_power.stateprop.switchlist`) containing all the switching times for the specified instances when the `-enable_state_propagation` parameter is set to `true`.

The format of the specified file is:

```
instname1
instname2
...
```

The following is the format of the `voltus_power.stateprop.switchlist` report:

```
*-----
*-----
*Instance Name Pin Name Pin Direction Frequency Switch Tim
*-----
*
*
```

Here,

- `Instance Name`: instance name that can be either combinational or sequential
- `Pin Name`: name of the input/output pin of the instance
- `Pin Direction`: indicates whether the pin is an INPUT (I) or OUTPUT (O) pin
- `Frequency`: clock frequency
- `Switch Time`: gives the switching pattern for the instance, that is, `<time1> <logic> <time2> <logic>`. The logic of the pin can be "r" or "f". r indicates rise and f indicates fall.

The following is a snippet of the `voltus_power.stateprop.switchlist` report:

```
*-----
*-----
*Instance Name Pin Name Pin Direction Frequency Switch Tim
*-----
*
*
* DFF11/DFF QN O 100 0.106 f
10.107 r
* DFF11/DFF Q O 100 0.097 r
10.099 f
```

|                                                             | <div>DFF11/DFFCPI1000.059 r5.056 f 10.059 r 15.056 ...</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |             |            |             |            |     |           |     |     |      |                 |   |  |  |  |     |      |      |      |            |         |   |  |  |  |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|------------|-------------|------------|-----|-----------|-----|-----|------|-----------------|---|--|--|--|-----|------|------|------|------------|---------|---|--|--|--|
| <div>-report_library_usage {true   false}</div>             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |             |            |             |            |     |           |     |     |      |                 |   |  |  |  |     |      |      |      |            |         |   |  |  |  |
|                                                             | <div><p>Specifies to report library usage statistics in a text report. The report contains information such as, cell name, library format used (CCSP/ECSM/NLPM) for each cell, library name, and so on. This parameter allows you to perform library integrity checking for debugging purpose.</p><p><i>Default:</i> false</p><p>The format of the text report with sample content is given below:</p><p><u>Example:</u></p><table><thead><tr><th>#cell_name</th><th>net_name</th><th>op_voltage</th><th>lib_voltage</th><th>lib_format</th></tr></thead><tbody><tr><td>INV</td><td>VDD, VDDA</td><td>0.9</td><td>1.0</td><td>NLDM</td></tr><tr><td>leakage, timing</td><td>A</td><td></td><td></td><td></td></tr><tr><td>INV</td><td>VDDC</td><td>0.75</td><td>0.75</td><td>CCSP, NLDM</td></tr><tr><td>leakage</td><td>B</td><td></td><td></td><td></td></tr></tbody></table></div> | #cell_name | net_name    | op_voltage | lib_voltage | lib_format | INV | VDD, VDDA | 0.9 | 1.0 | NLDM | leakage, timing | A |  |  |  | INV | VDDC | 0.75 | 0.75 | CCSP, NLDM | leakage | B |  |  |  |
| #cell_name                                                  | net_name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | op_voltage | lib_voltage | lib_format |             |            |     |           |     |     |      |                 |   |  |  |  |     |      |      |      |            |         |   |  |  |  |
| INV                                                         | VDD, VDDA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0.9        | 1.0         | NLDM       |             |            |     |           |     |     |      |                 |   |  |  |  |     |      |      |      |            |         |   |  |  |  |
| leakage, timing                                             | A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |             |            |             |            |     |           |     |     |      |                 |   |  |  |  |     |      |      |      |            |         |   |  |  |  |
| INV                                                         | VDDC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.75       | 0.75        | CCSP, NLDM |             |            |     |           |     |     |      |                 |   |  |  |  |     |      |      |      |            |         |   |  |  |  |
| leakage                                                     | B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |             |            |             |            |     |           |     |     |      |                 |   |  |  |  |     |      |      |      |            |         |   |  |  |  |
| <div>-report_missing_bulk_connectivity {true   false}</div> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |             |            |             |            |     |           |     |     |      |                 |   |  |  |  |     |      |      |      |            |         |   |  |  |  |
|                                                             | <div><p>Specifies to report missing bulk pins in the missing input report (-report_missing_input).</p></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |             |            |             |            |     |           |     |     |      |                 |   |  |  |  |     |      |      |      |            |         |   |  |  |  |
| <div>-report_missing_input {true   false}</div>             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |             |            |             |            |     |           |     |     |      |                 |   |  |  |  |     |      |      |      |            |         |   |  |  |  |
|                                                             | <div><p>Specifies to report missing netlist information in input files, such as library LEF/DEF, PGV, SPEF, TWF, and missing logical connectivity for cell instance (PGNET).</p><p>The name of the report file is: &lt;prefix&gt;_missingdata.&lt;spef lefdef lib pgv twf pgnet&gt;</p><p>Following is the use model:</p><pre>set_power_analysis_mode -report_missing_input true report_power -output &lt;dirname&gt; -report_prefix &lt;prefix&gt;</pre></div>                                                                                                                                                                                                                                                                                                                                                                                                                       |            |             |            |             |            |     |           |     |     |      |                 |   |  |  |  |     |      |      |      |            |         |   |  |  |  |
| <div>-report_missing_nets { true false }</div>              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |            |             |            |             |            |     |           |     |     |      |                 |   |  |  |  |     |      |      |      |            |         |   |  |  |  |



|                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                         | <p>Controls reporting of missing nets in both static and dynamic power analysis.</p> <p>This parameter controls reporting for TWF and activity files in static analysis, and controls reporting for TWF/SDC/SPEF and activity files in dynamic analysis.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>-report_scan_chain_stats {true   false}</code>    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                         | <p>Specifies to generate a report file called <code>voltus_power.schain.stat</code> during scan mode analysis that contains scan chain statistics.</p> <p>The default value is <code>false</code>.</p> <p>The report has 2 parts:</p> <ul style="list-style-type: none"> <li>• <b>Global Statistics:</b> this section has the cumulative statistics of all scan chains in the design</li> <li>• <b>Chain Statistics:</b> this section has the statistics for each scan chain in the design</li> </ul> <p>The Global and Chain Statistics sections have the following information:</p> <ul style="list-style-type: none"> <li>• <b>Total:</b> total instances specified in the control file or in a scan chain</li> <li>• <b>AssignedFF:</b> flip-flops defined in the scan control file which exist in the design</li> <li>• <b>SwitchedFF:</b> flip-flops that have switching event at the output</li> <li>• <b>Non-ScanFF:</b> flip-flops that have no scan-out pin</li> <li>• <b>DuplicatedFF:</b> flip-flops that are specified in multiple chains</li> <li>• <b>StateConflictFF:</b> flip-flops that have conflicting state assignment</li> </ul> |
| <code>-report_stat {true   false}</code>                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                         | <p>Reports a set of statistics on the instance power, instance power density, clock power, or transition density. This parameter is useful in viewing and debugging user defined activity factors.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>-report_time_display_fraction_digits value</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

Sets the number of fractional digits or decimal places displayed for the instance switching time values in the `voltus_power.stateprop.switchinst` file. This parameter displays fractional digits that show the exact time difference if there are two events (rise/fall) of an instance switching at the same time.

*Default:* -1

`-report_twf_attributes {detailed | summary}`

Controls the generation of the TWF attribute report based on the TWF attributes that are specified using the `set_twf_attribute` command.

The possible arguments are:

- **summary** - generates a summarized static timing analysis report “`voltus_power.twf_attribute.summary`” that displays the number of user-defined TWF attributes (delay, slew, and slack values for both the rise and fall transitions) and the number of annotated TWF attributes for dynamic power analysis. The following is a snippet of the report:

```
STATISTICS:
shift_arrival_time User Defined Annotated Annotated/Total (
rise_slew 5 4 80
rise_slew 2 2 100
fall_slew 2 2 100
rise_delay 5 4 80
fall_delay 5 4 80
rise_slack 2 2 100
fall_slack 2 2 100
```

- **detailed** - generates a detailed static timing analysis report “`voltus_power.twf_attribute.detailed`” that includes the summary report, and a detailed report of the rise and fall transitions for the TWF attributes of each annotated instance/pin. It also includes the shift arrival time report.

This argument also allows you to generate a report `voltus_power.twf.rpt` containing the TWF statistics to highlight clock-frequency and delay used for the static and dynamic power calculation.

The TWF statistics report is generated only if:

- the `read_twf` command is specified before `set_power_analysis_mode`
- the `set_power_analysis_mode -method static -static_netlist de` or `set_power_analysis_mode -method dynamic_vectorless | dynamic_vectorbased` command is specified.

|                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                           | <p>The format of the <code>voltus_power.twf.rpt</code> report is:</p> <pre># Voltus TWF Statistics # Units: Freq (Hz), Delay (second), Slew (second) # TWF File List: #     TWF_file1 #     TWF_fileN</pre> <p>USER DEFINED TWF ATTRIBUTE STATISTICS<br/>&lt;Clock Freq&gt; &lt;Min Rise_Delay&gt; &lt;Rise Delay&gt; &lt;Max Rise Delay&gt; &lt;Min Fall Delay&gt; &lt;Fall Delay&gt; &lt;Max Fall Delay&gt; &lt;Net&gt;</p> <p>The USER DEFINED TWF ATTRIBUTE STATISTICS section is included in the report only if the TWF attributes are defined using the <code>set_twf_attribute</code> command.</p> <p><b>Example:</b></p> <pre>#check all instances of inst_C1 get shifted by 3ns set_twf_attribute -pin inst_C1 -shift_arrival_time 3</pre> <pre>#check all instances of inst_D1 get shifted by 3ns set_twf_attribute -net inst_D1 -shift_arrival_time 3</pre> <pre>set_power_analysis_mode -method dynamic_vectorless - create_binary_db false -disable_static false - enable_state_propagation true -current_generation_method peak - power_grid_library ../../library_pv.cl -report_twf_attributes summar</pre> |
| <code>-reset</code>                                       | <p>Resets all or specified parameters to their default values.</p> <p>When you specify only the <code>-reset</code> parameter, the software resets all parameters for the <code>set_power_analysis_mode</code> command.</p> <p>When resetting specific parameters, the <code>-reset</code> parameter must be the first followed by the specific parameter. For example, you can specify the following command to reset only the <code>-method</code> parameter while keeping all other parameters' values unchanged:</p> <pre>set_power_analysis_mode -reset -method</pre> <p><b>Note:</b> When resetting a specific parameter, you must not specify any arguments for that parameter.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>-scale_to_sdc_clock_frequency {true   false}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

Specifies to enable the VCD clock frequency to be scaled up to the SDC clock frequency. This allows you to match the VCD clock to the SDC clock to run analysis. By default, the clock frequency is scaled based on the activity file (Default: `false`). When you set the –

`scale_to_sdc_clock_frequency` parameter to `true`, the power report and current waveforms apply the after-scaled clock frequency.

The default behavior for clock frequency scaling in the RTL and Gate level VCD flows is different, as described below:

- **RTL VCD** - By default, the clock frequency is scaled to the SDC clock frequency (`-scale_to_sdc_clock_frequency true`) when the RTL VCD flow is enabled (`set_power_analysis_mode -enable_rtl_vectorbased_dynamic_analysis true`).

You can disable clock frequency scaling by setting the

`set_power_analysis_mode -scale_to_sdc_clock_frequency` parameter `false`.

- **Gate VCD** - By default, scaling of the clock frequency is disabled (`-scale_to_sdc_clock_frequency false`).

You can enable the VCD clock frequency to be scaled up to the SDC clock frequency by setting the `set_power_analysis_mode -scale_to_sdc_clock_frequency` parameter to `true`. This works only when `set_power_analysis_mode -use_zero_delay_vector_file` is set to `true`, which means that the gate VCD is a zero-delay gate VCD.

`-scan_chain_name {design_name chain_name pattern}`

|                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                          | <p>Specifies to automatically generate a scan-chain control file from a scan-chain based DEF file for the shift mode scan-chain analysis. This parameter allows you to create a control file for a large-sized DEF containing long chains or for multiple-DEF files. The generated scan-chain control file is saved in the power output directory.</p> <p>The use model of the parameter is:</p> <ul style="list-style-type: none"> <li>• <code>-scan_chain_name {all pattern}</code> - generate scan control file for all scan chains.<br/><b>Example:</b> <code>-scan_chain_name {all 11010}</code></li> <li>• <code>-scan_chain_name {design_name chain_name pattern}</code> - generate scan control file for a specific chain.<br/><b>Example:</b> <code>-scan_chain_name {counter AutoChain_1_seg1_clk_rising 10010}</code></li> </ul> <p>Using this parameter, you can generate the control file for both flat and hierarchical DEFs.</p> |
| <code>-scan_control_file filename</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                          | <p>Specifies the scan control file name required to run the Scan Mode Analysis flow.</p> <p>The scan control file contains details about the scan analysis mode used for testing. For more information, refer to <a href="#">Scan Control File Format</a>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>-scan_mbff_chain_type type</code>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                          | <p>Specifies the type of multi-bit scan flip-flop.</p> <ul style="list-style-type: none"> <li>• <code>liberty</code> - specifies to use the syntax in Liberty to decide the multi-bit flip-flop type.</li> <li>• <code>serial</code> - specifies to pick the first flop as the starting flop, and treat the scan chain as serial, regardless of .lib.</li> <li>• <code>parallel</code> - specifies to treat the scan chain as parallel, regardless of .lib.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>-settling_buffer value</code>      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                              | <p>Specifies the settling buffer time (in ps unit) between multiple windows of VCD file. This parameter is used when you specify multiple pairs of the start time and end time for the non-overlapping multiple windows specified in an activity file (using the <code>read_activity_file</code> command).</p> <p>Multiple current waveforms representing multiple VCD windows are merged into a single continuous current waveform. To capture the waveform for signals at the edge of user windows, a buffer time is added between the windows being stitched. This buffer time ensures that the stitched waveform is contiguous. The generated waveform hence has the duration of user windows and the intervening settling buffers. The default settling buffer period is 400ps. You must specify the buffer value without unit.</p> <p>The following is an example of this feature:</p> <pre>set_power_analysis_mode -verbosity true \     -settling_buffer 200 -report_missing_nets true  read_activity_file \     -format VCD -scope adder/u1 \     -start {0ps 2000ps} -end {1100ps 3100ps} \     -block ../vcd/dut_5buf_full_chip.vcd</pre> |
| <p><code>-split_bus_power {true   false}</code></p>          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                              | <p>A value of <code>false</code> applies the internal power number to each individual bit.</p> <p>A value of <code>true</code> divides the internal power number by bus width, before applying it to the bus bit.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p><code>-state_dependent_leakage {true   false}</code></p>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                                              | <p>When set to <code>false</code>, power analysis performs state independent leakage power calculation.</p> <p><i>Default:</i> <code>true</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p><code>-static_multimode_scenario_file filename</code></p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                 | <p>Specifies a file with information on instance scaling scenario for the static multi-mode IR analysis flow. When this parameter is specified, Power Analysis creates the multi-mode static current file that is used during rail analysis. The format of the input file is:</p> <pre>INST_NAME      PIN_NAME      SCALE_FACTOR_1      .... SCALE_FACTOR_N</pre> <p>Here, the first column indicates the macro instance name, the second column indicates the pin name, followed by multiple current scaling factor. Each column of the scaling factor is referred to as a scenario. You must specify positive values only for scale factors.</p> <p>The following is a snippet from the specified input file:</p> <pre>top/macroA      VDD      1      0.5      0.5      1 top/macroA      VDDM     0.5      0.5      0.5      1 top/macroB      VDD      0.5      1      0.5      0.5</pre> <p>If scale factors are not specified for certain scenarios, the tool will automatically use the instance's last stored scale factor value to complete current file generation.</p> <p>In the following example, scale factor value 0.45 will be used in the 3rd and 4th scenario for instanceC, and scale factor value 0.35 will be used in the 4th scenario for instanceD:</p> <pre>instanceA VDD 0.45 0.55 0.65 0.31 instanceB VDD 0.31 0.5 0.67 0.21 instanceC VDD 0.25 0.45 instanceD VDD 0.31 0.25 0.35</pre> |
| <pre>-static_netlist {verilog   def}</pre>      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                                                 | <p>Specifies to perform static power analysis using a Verilog or DEF only netlist.</p> <p>By default, the static power analysis is performed using the Verilog netlist as the primary netlist. Using this parameter, you can specify DEF when a DEF netlist with logical connectivity is available, and use it as primary and not consider the Verilog netlist.</p> <p>This parameter is also enabled when <code>set_rail_analysis_mode -report_power_in_parallel</code> is <code>true</code> for static power computation.</p> <p><b>Default:</b> verilog</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <pre>-start_time_alignment {true   false}</pre> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | <p>Specifies to align all the signals of the activity file at time zero. When set to <code>true</code>, the simulation period is <code>max(endtime - starttime)</code>. When it is <code>false</code> the simulation period is <code>max(endtime) - min(starttime)</code>.</p> <p><b>Default:</b> <code>true</code></p> <p><b>Note:</b> This parameter is not supported in the vector profiling flow. In the vector profiling flow, the start time will always be the <code>min(starttime)</code> and the end time will always be <code>max(endtime)</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <p><code>-thermal_input_file file</code></p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                              | <p>Specifies to read a power map file to perform thermal analysis. Thermal input file contains die area divided into tiles, and temperature value for each tile. This temperature of each tile is used to calculate power of all the instances that fall in that tile's bounding box.</p> <p>The format of the thermal input file is:</p> <pre> DIE_TEMPERATURE_MAP DIE_AREA    xmin(um) ymin(um) xmax(um) ymax(um) nx(int) ny(int)  NUMBER_OF_LAYERS    nLayers(int) LAYER    layer_name(string) TEMPERATURE XCOORD    x0(0)    x0(1)    x0(2) ... ..x0(nx-1) YCOORD    y0(0)    y0(1)    y0(2) ... ..y0(ny-1)  T0(0,0)    T0(1,0)    ...    T0(nx-1,0) T0(0,1)    T0(1,1)    ...    T0(nx-1,1)  : T0(0,ny-1)    T0(1,ny-1)    T0(nx-1,ny-1) END_TEMPERATURE END_LAYER END_DIE_TEMPERATURE_MAP </pre> <p>where,</p> <ul style="list-style-type: none"> <li>• <code>DIE_AREA</code> specifies the coordinates of the four points of die outline as well as the tile number in X and Y direction.</li> <li>• <code>NUMBER_OF_LAYERS</code> is the number of layers in each die instance</li> <li>• <code>LAYER</code> specifies the LEF layer name</li> <li>• <code>XCOORD</code> and <code>YCOORD</code> are the coordinates of each tile</li> <li>• <code>T0(0,0) ....</code> is the temperature value assigned to a tile at a specific coordinate</li> </ul> |



|                                                                       |                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-to_x_transition_factor value</code>                            |                                                                                                                                                                                                                                                                                                          |
|                                                                       | Specifies how 0/1 to X transitions are counted. The default value is 0.5.                                                                                                                                                                                                                                |
| <code>-to_z_transition_factor value</code>                            |                                                                                                                                                                                                                                                                                                          |
|                                                                       | Specifies how 0/1 to Z transitions are counted. The default value is 0.25.                                                                                                                                                                                                                               |
| <code>-transition_time_method {min   avg   max}</code>                |                                                                                                                                                                                                                                                                                                          |
|                                                                       | Specifies the minimum, maximum, or average transition time method that will be used with the integrated timer or the external TWF.<br><i>Default:</i> max                                                                                                                                                |
| <code>-twf_delay_annotation {min   avg   max}</code>                  |                                                                                                                                                                                                                                                                                                          |
|                                                                       | Allows you to choose between min, max, or avg arrival times from the timing window file (TWF). The arrival time is annotated to the vectors processed from the RTL VCD/FSDb Zero-Delay flows. The default value is avg.                                                                                  |
| <code>-twf_load_cap {min   avg   max}</code>                          |                                                                                                                                                                                                                                                                                                          |
|                                                                       | Allows you to select minimum, maximum, or average value of TWF external load or capacitance for power calculation.<br><i>Default:</i> max                                                                                                                                                                |
| <code>-use_cell_leakage_power_density {true   false}</code>           |                                                                                                                                                                                                                                                                                                          |
|                                                                       | Specifies to use library leakage density times area (default_leakage_power_density), if defined, instead of the library's default cell leakage power (default_cell_leakage_power). This is used when leakage power for cells are missing or not defined.<br>The default value of this parameter is true. |
| <code>-use_fastest_clock_for_dynamic_scheduling {true   false}</code> |                                                                                                                                                                                                                                                                                                          |
|                                                                       | Specifies to use only the respective fastest clock associated with each net or pin to schedule events for a given simulation period. The default value of this parameter is false.                                                                                                                       |
| <code>-use_lef_for_missing_cells {true   false}</code>                |                                                                                                                                                                                                                                                                                                          |

|                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                        | <p>Allows a combination of LEF and PGV in the dynamic analysis flow. The default value of this parameter is <code>false</code>.</p> <p>If there are cells that are common to LEF and PGV, the PGV cell will get the precedence.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>-use_zero_delay_vector_file {true   false}</code>                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                                                        | <p>Enables zero delay mode in vector-based dynamic analysis to avoid pessimism in current estimation. When set to <code>true</code>, the software reads the cycle accurate (zero delay) VCD/FSDB file and adds the timing/delay information from the Timing Analyzer or external TWF file to the VCD/FSDB file.</p> <p>Default: <code>false</code></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>-vector_profile_mode {activity   event_based   power_density   transient}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                                                        | <p>Specifies the method used to perform vector profiling. The possible arguments are:</p> <ul style="list-style-type: none"> <li>• <code>activity</code> - performs activity profiling. This parameter generates only the worst activity report, and does not generate the average power report. The name of the file is as specified using the <code>report_power -outfile</code> parameter.</li> <li>• <code>event_based</code> - generates the average power report. This is the average power of all events in the user-specified window. This report includes a summary based on the power components, cell type, power rail, and clock network. The worst window report is captured in a file specified with the <code>report_power -outfile</code> parameter, and the report is suffixed by <code>`.avgpower'</code>. When <code>-outfile</code> is not specified, the <code>vectorprofile.report.avgpower</code> report file is created.</li> <li>• <code>power_density</code> - performs area or power density profiling. For this method, power is reported for each tile of the design, where design can be considered to be physically divided into equal area tiles. By default a design is divided in 10x10 tiles.</li> <li>• <code>transient</code> - generates the event-based average power report and current files for the worst power window.</li> </ul> <p>This parameter must be specified with the <code>-method vector_profile</code> parameter.</p> <p><i>Default:</i> <code>event_based</code></p> |

|                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-worst_case_vector_activity {true   false}</code>                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                                | Specifies to use the worst activity value when multiple vectors are specified in the static power calculation flow. The default value is <code>false</code> .                                                                                                                                                                                                                                                                                                              |
| <code>-worst_step_size value</code>                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                                | <p>Specifies the step size or resolution for vector profiling. Activity/ Power profiling is carried out for every step size that you specify.</p> <p>When this parameter is used with <code>-worst_window_size</code>, the specified step size is the resolution at which window sliding is performed. The supported time units are s, ms, us, ns, and ps. If no explicit time unit is specified, the default ns will be applied.</p>                                      |
| <code>-worst_window_count value</code>                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                                | <p>Specifies the number of worst windows from the last vector profile run to be used for dynamic power analysis. It provides n top windows with maximum activity/power/IR. This parameter is used when you need to combine dynamic power analysis and vector profiling for multiple worst-case windows in a single run.</p> <p><i>Default:</i> 1 (implements single worst window)</p>                                                                                      |
| <code>-worst_window_size value</code>                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                                                                                | <p>Specifies the duration of dynamic analysis. This parameter is used for sliding window vector profiling that involves sliding the profiling windows by using a very small resolution value to get the window with the worst current profile. <code>-worst_window_size</code> must be specified with <code>-worst_step_size</code>.</p> <p>The supported time units are s, ms, us, ns, and ps. If no explicit time unit is specified, the default ns will be applied.</p> |
| <code>-worst_window_type { power   activity   delta_power   delta_activity   ir   critical_inst_power }</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                    | <p>Specifies the window type to be used for multi-window analysis. The possible arguments are:</p> <ul style="list-style-type: none"> <li>• <code>power</code> - worst instance power</li> <li>• <code>activity</code> - worst instance activity</li> <li>• <code>delta_power</code> - worst dp/dt (rate of change of power)</li> <li>• <code>delta_activity</code> - worst da/dt (rate of change of activity)</li> <li>• <code>ir</code> - worst IR</li> <li>• <code>critical_inst_power</code> - worst power for critical/IR-sensitive instance (worst timing and IR)</li> </ul> <p><i>Default:</i> <code>power</code></p> |
| <code>-write_default_util {true   false}</code>    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                    | <p>For instances that are not hooked to any power/ground rail, the current for these instances are reflected in the default static current files:</p> <ul style="list-style-type: none"> <li>• <code>static_default_ground_rail.ptiavg</code></li> <li>• <code>static_default_power_rail.ptiavg</code></li> </ul> <p><i>Default:</i> <code>true</code></p>                                                                                                                                                                                                                                                                   |
| <code>-write_profiling_db {true   false}</code>    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                    | <p>Writes out profiling databases which can later be viewed as histograms using the SimVision interface. You can specify this parameter during both the activity and power vector profiling.</p> <p>The profiling databases support instance level and power/ground net histograms. It creates an output file name with the <code>*.trn</code> extension.</p> <p><i>Default:</i> <code>false</code></p>                                                                                                                                                                                                                      |
| <code>-write_static_currents {true   false}</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                    | <p>A value of <code>true</code> tells Voltus to generate the current data files per net.</p> <p><i>Default:</i> <code>false</code></p> <p><b>Note:</b> When you perform static power calculation in the dynamic power analysis flow, you can use either the <code>-disable_static</code> or <code>-write_static_currents</code> parameter to enable static power calculation and write static current file. If only <code>-disable_static</code> is set to <code>false</code> and <code>-write_static_currents</code> is not set, the tool automatically sets <code>-write_static_currents</code> to <code>true</code>. Similarly, if <code>-write_static_currents</code> is set to <code>true</code>, and <code>-disable_static</code> is not set, the tool automatically sets <code>-disable_static</code> to <code>false</code>.</p> |
| <code>-x_transition_factor value</code>            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                    | <p>When using VCD/FSDB, one can specify how transitions to and from X are counted. These transitions are defined as shown in Table 18-1. Transition to and from the X state will be treated as full transitions multiplied by the specified factor.</p> <p><b>Note:</b> 0 -&gt; X -&gt; 1 or 1 -&gt; X -&gt; 0 count as full transitions and they are not controlled by the parameter.</p> <p><i>Default:</i> <code>0.5</code></p>                                                                                                                                                                                                                                                                                                                                                                                                      |
| <code>-z_transition_factor value</code>            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                                    | <p>When using VCD/FSDB, one can specify how transitions to and from Z are counted. These transitions are defined as shown in Table 18-1. Transition to and from the Z state will be treated as full transitions multiplied by the specified factor.</p> <p><b>Note:</b> 0 -&gt; XZ -&gt; 1 or 1 -&gt; Z -&gt; 0 count as full transitions and they are not controlled by the parameter.</p> <p><i>Default:</i> <code>0.25</code></p>                                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>-zero_delay_vector_toggle_shift value</code> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

Specifies to shift the current waveform in the zero-delay VCD flow. This parameter is useful in removing the idle period for the current waveform result with a long idle period (small flat current) before the first high current peak because of delay annotation. The parameter allows to offset the idle period from the delay values.

The use model is given below:

```
set_power_analysis_mode -zero_delay_vector_toggle_shift -3.0ns
```

The default value is 0ns. The supported time units are s, ms, us, ns, and p. The default unit is ns. You can specify a negative or a positive value. A positive value means right shift in the timeline or add more delay to the original arrival time, and a negative value means left shift in the timeline to trim the unwanted idle period.

The `-zero_delay_vector_toggle_shift` parameter enables you to make a more realistic current waveform that can be easily aligned with the other current waveforms.

**Note:** This parameter is similar to the `set_twf_attribute -shift_arrival_time` parameter, with the only difference that `-zero_delay_vector_toggle_shift` supports both positive and negative values while `-shift_arrival_time` supports only a positive value.

Table 18-1 Transitions from/to X and Z values and

| Transition | Definition           | Default |
|------------|----------------------|---------|
| to/from X  | [ 0 or 1 or Z] <-> X | 0.5     |
| to/from Z  | [ 0 or 1 ] <-> Z     | 0.25    |

## Scan Control File Format

The use model for the Scan Mode Analysis flow is:

```
set_power_analysis_mode -method dynamic_vectorbased -scan_control_file filename
```

The format of the scan control file is:

```
MODE = [SHIFT | VECTORLESS]
VECTORLESS_GLOBAL_ACTIVITY = <factor> [Between 0 - 1]
<chain_name> <flop_inst> <initial_state> <master_cell_name> <scale> <mode> <scan_in_pin>
<scan_out_pin> <scan_en_pin>
```

Note that `<chain_name>` is optional, `<initial_state>` is optional in vector-less mode, and

<master\_cell\_name>, <scale>, <mode>, <scan\_in\_pin>, <scan\_out\_pin>, <scan\_en\_pin> are optional and only needed for the first internal scan flop inside one macro.

Where,

- Mode specifies the mode used for scan design analysis. The two modes are: `SHIFT` and `VECTORLESS`.
- `VECTORLESS_GLOBAL_ACTIVITY` specifies the activity specification used in the `VECTORLESS` mode.
- `chain_name` is the name of the scan chain.
- `initial_state` is the initial state of all scan flops.
- `master_cell_name` is the name of the macro.
- `scale` is the scaling factor applied to the current.
- `mode` is mode name of the scan mode in the PGV view.
- `scan_in_pin` is the macro's `scan_in` pin name.
- `scan_out_pin` is the macro's `scan_out` pin name.
- `scan_en_pin` is the macro's `scan_enable` pin name.

## Examples

- The following command uses the negative internal power from the .lib files during analysis, specifies that the transitions that are to and from the Z state will be multiplied by .2, and specifies that the transitions that are to and from the X state will be multiplied by .45:

```
tempus> set_power_analysis_mode -honor_negative_energy true \
-z_transition_factor .2 \
-x_transition_factor .45
```

- The following command resets the `-honor_negative_energy` parameter to its default value:  

```
tempus> set_power_analysis_mode -reset -honor_negative_energy
```
- The following command resets all `set_power_analysis_mode` parameters to their default values:  

```
tempus> set_power_analysis_mode -reset
```
- The following command ignores leakage power, and scales only the switching and internal power by the specified scale factor:  

```
tempus> set_power -instance top.block2 2mW
tempus> set_power_analysis_mode -disable_leakage_scaling true
```
- The following command sets up dynamic power analysis.  

```
tempus> set_power_analysis_mode \
-method dynamic_vectorless \
-corner max \
```

```
-create_binary_db true \
-write_static_currents true \
-honor_negative_energy true \
-ignore_control_signals false \
-power_grid_library { fast_allcells.cl }
```

- The following command adds a prefix SM\_ to the current data directory and the underlying current files:

```
tempus> set_power_analysis_mode -output_current_data_prefix SM_
```

The following current files are generated:

```
SM_dynamic_VDD.ptiavg
./SM_PTIData/SM_dynamic_1_VDD.ptiavg
./SM_PTIData/SM_dynamic_2_VDD.ptiavg
```

- The following command is another example of dynamic power analysis.

```
tempus> set_power_analysis_mode \
-method dynamic_vectorless \
-analysis_view AVmac2off \
-off_pg_nets VDDlu \
-create_binary_db true \
-write_static_currents true \
-honor_negative_energy true \
-ignore_control_signals false \
-power_grid_library {accurate_stdcells.cl TSDN65LPA1024X32M8F.cl}
```

- The following command is an example of the state-propagation-based dynamic vectorless flow, and also specifies to generate a report (voltus\_power.stateprop.switchlist) containing all the switching times for the instances specified in the inst\_list file:

```
tempus> set_power_analysis_mode -method dynamic_vectorless -enable_state_propagation
true -report_instance_switching_list inst_list
```

- The following is an example of performing standalone power report generation in the XP mode:

```
tempus> set_power_analysis_mode -method dynamic_vectorless \
-enable_state_propagation true \
-power_grid_library { /pgv_dir/tech_pgv/techonly.cl /pgv_dir/stdcell_pgv/stdcells.cl
/pgv_dir/macro_pgv/macros_pll.cl }\ -extraction_tech_file /qrcTechFile -enable_xp true
...
tempus> report_power
```



## Related Commands

- `get_power_analysis_mode`
- `report_power`
- `read_activity_file`

# set\_power\_calc\_temperature

set\_power\_calc\_temperature  
temp

Sets the temperature for static power calculation.

In the presence of a trilib set, the internal and leakage power will be interpolated as the per the given temperature specification.

## Parameters

|      |                                                                                                                                                                                                                                                                                                                                                                                                          |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| temp | <p>Specifies the temperature for static power calculation.</p> <p><b>Note:</b> The static power calculation software gets the temperature value of each instance from the following sources (in order of precedence):</p> <ul style="list-style-type: none"><li>• Instance temperature file</li><li>• set_power_calc_temperature command</li><li>• The current operating condition temperature</li></ul> |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Related Commands

- report\_power
- set\_inst\_temperature\_file

## set\_power\_include\_file

```
set_power_include_file
file
-block <string>
[-reset]
```

Specifies a power analysis include file. You can use the `set_power_include_file` command to specify certain commands of the dynamic engine which do not have a Voltus equivalent. For more information on these commands, refer to the "Power Analysis Options" chapter in *Voltus Text Command Reference*.

**Note:** The `set_power_include_file` command is used to pass these legacy power analysis commands and `report_power` command options only in the dynamic mode (vectorbased and vectorless). If `set_power_analysis_mode` is set to `static`, the `set_power_include_file` command is ignored.

### Parameters

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| file               | Name of the include file. This file will include all commands that can not be translated to Voltus commands. In addition, you can also include the <code>report_power</code> command options to generate different power reports.                                                                                                                                                                                                 |
| -block<br><string> | Specifies the master cell name to which the powermeter options are to be applied. The parameter allows you to apply certain commands of the dynamic engine only to a specific block partition. This parameter allows you to apply the include options to a specific block instead of applying them globally to all blocks.<br><br>The <code>-block</code> parameter is applicable only to the block-based distributed power flow. |
| -reset             | Resets all options back to default values.                                                                                                                                                                                                                                                                                                                                                                                        |

### Examples

- The following command specifies a power include file of `power.inc`:  

```
tempus> set_power_include_file power.inc
```
- The following example shows the contents of the `power.inc` file.  

```
power.inc file must use the pre-defined design and power objects as "design" # and "ChipPwr" respectively. The power output object is not supported.
design calCellsToIgnore BUFX1
ChipPwr calRailVoltage VDD 1.2
```

## Related Commands

- `report_power`
- `set_power_analysis_mode`

# set\_power\_output\_dir

```
set_power_output_dir dir
[-reset]
```

Specifies the name of the directory that power information will be output to.

## Parameters

|        |                                                                                            |
|--------|--------------------------------------------------------------------------------------------|
| dir    | Specifies the name of the directory in which power analysis output information is written. |
| -reset | Resets all options back to default values.                                                 |

## Examples

- The following command specifies the output directory called `static_power_AvallOn` in which the power information is written.  
tempus> set\_power\_output\_dir static\_power\_AVallOn

## Related Commands

- `report_power`
- `set_power_analysis_mode`

## set\_switching\_activity

```
set_switching_activity
[-reset]
[-activity factor | -density transition_density]
[-clock clock_name]
[-duty value]
[-inst instance_name]
[-net net_name | -port port_name | -pin pin_name | -input_port port_name | -output_port
port_name | -bidir_port port_name]
[-period value]
[-unclocked]
[-icg_ratio num]
[-comb_clockgate_ratio num]
[-hier hierarchy_name]
[-scale_factor value]
[-cell cell_name]
[-force]
[-quiet]
```

Specifies activity for nets, pins, and ports. This command is used to specify user-defined activities at specific pins/nets/ports. You must specify this command before `report_power`.

The `set_switching_activity` command can accept a net/pin/port list using the `get_pins/get_ports/get_nets` commands. You can use the `get*` commands to create a collection of nets/pins/ports whose name matches the supplied pattern list. Therefore, providing you with the flexibility in providing lists through use of regular filter patterns. This feature is supported in both static and dynamic analysis engines.

### Parameters

|                                           |                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-activity <i>factor</i></code>      | Specifies the activity for a net, pin, or top-level port. The <i>activity_factor</i> specifies the number of times the net, pin, or port switches in a clock cycle. Enter any positive number. If the activity is set to 0.5, the activity happens once for every two clock cycles. If the activity is set to 2.0, the activity happens two times per clock cycle. There is no default value. |
| <code>-bidir_port <i>port_name</i></code> | Specifies the name of the bidirectional port. The port names are in the following format: <code>bidirectional-<i>port</i></code> . This parameter supports wildcards (*).                                                                                                                                                                                                                     |
| <code>-clock <i>clock_name</i></code>     | Assigns global activity factor to all nets in domain of <i>clock_name</i> if and only if <i>clock_name</i> is the fastest clock in the domain set.<br><br>To use this option, <code>-clock</code> and <code>-activity</code> must be used without any other options.                                                                                                                          |

|                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-cell cell_name</code>             | Specifies the activity on all output pins for all instances of the specified cell.                                                                                                                                                                                                                                                                                                                                                                                                    |
| <code>-comb_clockgate_ratio num</code>   | <p>Specifies to set the propagation ratio for combinational clock gate cell outputs. When specified, this parameter sets the output activity of any instance that is identified as a combinational clock gating cell to <i>num</i> times the fastest transition density arriving on the instance's clock inputs.</p> <p>You can specify this ratio for a net, pin, port, or clock.</p>                                                                                                |
| <code>-density transition_density</code> | <p>Specifies the transition density for a specific net, pin, or top-level port. Transition density specifies the number of transitions per second. It can be any positive number. There is no default value.</p> <p><b>Note:</b> If you specify the <code>-density</code> parameter in conjunction with the <code>-period</code> parameter, the software issues a warning and the <code>-period</code> parameter is ignored.</p>                                                      |
| <code>-duty value</code>                 | <p>Specifies the duty cycle, which is the probability that the signal is a logical 1. The <code>-period</code> parameter must be specified to add a duty. If not specified, the <code>set_switching_activity</code> command picks this value from the external TWF or SDC through the Common Timing Engine (CTE).</p> <p><i>Default:</i> Uses the duty specified by the <code>set_default_switching_activity</code> command if the <code>-duty</code> parameter is not specified.</p> |
| <code>-force</code>                      | <p>Specifies to apply user-defined activities for specific nets, pins, and ports, without having to load a verilog netlist using the <code>set_top_module</code> command.</p> <p>When the <code>set_top_module</code> command is not specified, the <code>set_switching_activity</code> command does not support the <code>-cell</code> parameter, and the net/pin/port list using the <code>get_pins/get_ports/get_nets</code> commands.</p>                                         |
| <code>-hier hierarchy_name</code>        | A hierarchical name that an activity is being assigned to.                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>-icg_ratio num</code>              | <p>Specifies to set the propagation ratio for integrated clock gate (ICG) cell outputs. When specified, this parameter sets the output pin of all identified ICG instances to <i>num</i> times the fastest transition density arriving on clock inputs of the instance.</p> <p>You can specify this ratio for a net, pin, port, or clock.</p>                                                                                                                                         |
| <code>-input_port port_name</code>       | Specifies the name of the input port. The port names are in the following format: <i>input-port</i> . This parameter supports wildcards (*)                                                                                                                                                                                                                                                                                                                                           |
| <code>-inst instance_name</code>         | Specifies the activity on a specific instance.                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>-net net_name</code>               | Specifies the name of the net. This parameter supports wildcards (*).                                                                                                                                                                                                                                                                                                                                                                                                                 |

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-output_port<br/>port_name</code> | Specifies the name of the output port. The port names are in the following format: <i>output-port</i> . This parameter supports wildcards (*)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>-period value</code>              | <p>Specifies the period that the activity is referenced to. If not specified, a default frequency is used. If a negative number is specified, the static power calculation engine reverts back to the default frequency. Units in seconds (s), milliseconds (ms), microseconds (us), nanoseconds (ns), or picoseconds (ps).</p> <p><b>Note:</b> If you specify the <code>-period</code> parameter in conjunction with the <code>-density</code> parameter, the software issues a warning and the <code>-period</code> parameter is ignored.</p> <p><i>Default:</i> The time unit is defined in the first <code>.lib</code> file read during design import. If the time unit is not defined, the period specified in the <code>set_default_switching_activity</code> command is used.</p> |
| <code>-pin pin_name</code>              | Specifies the name of a pin. The pin names are in the following format: <i>instance-name + default-hierarchy-separator + pin</i> . This parameter supports wildcards (*)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <code>-port port_name</code>            | Specifies the name of a top-level port. The port names are in the following format: <i>top-level-port</i> . This parameter supports wildcards (*)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <code>-quiet</code>                     | Suppresses all error and warning messages generated when the <code>set_switching_activity</code> command is run.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <code>-reset</code>                     | <p>Resets all specified options back to default values. The reset option should be the first option specified, otherwise all options set prior to this will be reset.</p> <p><b>Note:</b> You can selectively reset the pin/port/net specifications. In addition, you can also selectively reset the pin/port/net specifications of a list of objects through the <code>get_pins/get_ports/get_nets</code> commands.</p>                                                                                                                                                                                                                                                                                                                                                                 |
| <code>-scale_factor value</code>        | <p>Specifies to scale frequencies of the specified clock domains. This parameter allows you to scale transition densities of all instances associated with the clock domain by the specified scale factor to compute internal and switching power.</p> <p>This parameter must be specified with the <code>-clock</code> parameter. The <code>-scale_factor</code> parameter is only applicable to static power and dynamic vectorless power analysis.</p>                                                                                                                                                                                                                                                                                                                                |



|            |                                                                                                                                                                                                                        |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -unclocked | <p>Applies <i>activity</i> (with reference <i>period</i>) to all nets that have no clock domain (-unclocked).</p> <p>To use this option, -unclocked must be used with -activity and -period with no other options.</p> |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Activity Precedence

See "[Activity Precedence](#)".

## Examples

- The following command specifies that net n189 will switch once every ten clock cycles, with a clock period of 7520ns and a duty cycle of 50 percent:  
tempus> set\_switching\_activity -net n189 -activity 0.10 -period 7520ns -duty .5
- The following command sets the transition density of port out\_cpu\_top to 20000 transitions per second:  
tempus> set\_switching\_activity -port out\_cpu\_top -density 20000
- The following command applies a transition density of 1e+06 to all input ports that match the specified pattern:  
tempus> set\_switching\_activity -port[get\_ports "mode\*" -filter {port\_direction==in}] -density 1e+06
- The following command resets the previously defined activity on all nets that match the specified pattern:  
tempus> set\_switching\_activity -reset -net[get\_nets n23\*]
- The following command specifies the clock gate ratio for the clock domain sys\_clk:  
tempus> set\_switching\_activity -clock sys\_clk -icg\_ratio 0.8 -comb\_clockgate\_ratio 0.5
- The following command specifies to set the default switching activity for all other clock domains of the design:  
tempus> set\_default\_switching\_activity -icg\_ratio 1.0 -comb\_clockgate\_ratio 1.0

## Related Commands

- `report_power`
- `get_pins`
- `get_ports`
- `get_nets`

## set\_twf\_attribute

```
set_twf_attribute
[-help]
[-clock clock_name]
[-clock_edge {rise | fall}]
[-clock_freq value]
[-fall_delay {min:max or single_value}]
[-fall_slack {min:max or single_value}]
[-fall_slew {min:max or single_value}]
[-net net_name]
[-pin pin_name]
[-reset]
[-rise_delay {min:max or single_value}]
[-rise_slack {min:max or single_value}]
[-rise_slew {min:max or single_value}]
[-type {data | clock}]
[-shift_arrival_time time]
[-clear_twf_attr]
```

Specifies static timing analysis information. The command allows you to override timing parameters, such as slew and clock, on desired pins/nets from an existing timing windows file (TWF).

### Parameters

|                           |                                                                                                                                                                            |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -clear_twf_attr           |                                                                                                                                                                            |
|                           | Specifies to reset the TWF file values for a specific pin or net. The use model of this parameter is:<br>set_twf_attribute -clear_twf_attr -pin/-net <name>                |
| -clock <i>clock_name</i>  |                                                                                                                                                                            |
|                           | Specifies the clock name. This indicates that the nets and pins that are timing constrained have arrival times (timing windows) caused by this clock.                      |
| -clock_edge {rise   fall} |                                                                                                                                                                            |
|                           | Specifies the capturing clock edge direction: rise or fall.                                                                                                                |
| -clock_freq <i>value</i>  |                                                                                                                                                                            |
|                           | Specifies the clock frequency. You must specify the -clock parameter along with the -clock_freq parameter to assign the clock for which the frequency needs to be updated. |

|                                                    |                                                                                                          |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <code>-fall_delay {min:max OR single_value}</code> |                                                                                                          |
|                                                    | Specifies the fall delay for the pin. This parameter is not supported in the static power analysis flow. |
| <code>-fall_slack {min:max or single_value}</code> |                                                                                                          |
|                                                    | Specifies the fall slack for the pin.                                                                    |
| <code>-fall_slew {min:max or single_value}</code>  |                                                                                                          |
|                                                    | Specifies the fall slew for the pin.                                                                     |
| <code>-net net_name</code>                         |                                                                                                          |
|                                                    | Specifies the net name. The <code>-net</code> and <code>-pin</code> parameters are mutually exclusive.   |
| <code>-pin pin_name</code>                         |                                                                                                          |
|                                                    | Specifies the pin name. The <code>-net</code> and <code>-pin</code> parameters are mutually exclusive.   |
| <code>-reset</code>                                | Resets all specified options back to default values.                                                     |
| <code>-rise_delay {min:max OR single_value}</code> |                                                                                                          |
|                                                    | Specifies the rise delay for the pin. This parameter is not supported in the static power analysis flow. |
| <code>-rise_slack {min:max OR single_value}</code> |                                                                                                          |
|                                                    | Specifies the rise slack for the pin.                                                                    |
| <code>-rise_slew {min:max OR single_value}</code>  |                                                                                                          |
|                                                    | Specifies the rise slew for the pin.                                                                     |
| <code>-shift_arrival_time time</code>              |                                                                                                          |

Specifies to filter the idle period of the current waveform in the dynamic power analysis flow. This parameter is useful in removing the idle period of the current waveform result with long idle period (small current) before the 1st high current peak because of TWF delay annotation. The parameter allows to offset idle period from the TWF delay values.

The use model is given below:

```
set_twf_attribute -shift_arrival_time 3.0ns
```

The default value is 0ns. The supported time units are s, ms, us, ns, and ps. The default unit is ns. You must specify a positive value only. A positive value means shift right in the time line or add more delay to the original arrival time.

You can specify the `-clock` parameter to limit the changes to a clock domain. In addition, you can further add the `-pin/-net` parameter to limit the changes to a block. The following are examples of shifting different delays to different clock domains because different clocks may have different clock insertion delays at the top level:

```
set_twf_attribute -shift_arrival_time 1.2ns -clock clk_a
```

```
set_twf_attribute -shift_arrival_time 2.3ns -clock clk_b
```

If a clock domain is not specified, the delay will be applied to all the specified nets/pins.

**Note:** The final rise/fall delays for the annotated nets/pins after the shift has been applied are reported under the “[Shift Arrival Time Report](#)” section of the detailed TWF Attributes report. This report is generated using the `set_power_analysis_mode -report_twf_attributes detailed` parameter.

The following example illustrates a sample command and report snippet:

#### Command:

```
set_twf_attribute -shift_arrival_time 3.0ns
set_power_analysis_mode -method dynamic_vectorless ... -
report_twf_attributes detailed
```

#### Report:

Clock domain annotated

- CK1\_1
- CK1\_2
- CK2\_1
- CK2\_2

| Rise Delay | Fall Delay | Net Name |
|------------|------------|----------|
| 1e-09      | 1e-09      | CK1_1    |
| 1e-09      | 1e-09      | CK1_2    |
| 1e-09      | 1e-09      | CK2_1    |
| ...        |            |          |

|                                                                        |
|------------------------------------------------------------------------|
| -type {data   clock}                                                   |
| Specifies whether the net or pin is part of the clock or data network. |

## Examples

- The following command uses the following TWF attributes for the net `netB1`:  

```
set_twf_attribute -net netB1 -rise_delay 2.1978:2.1978 -rise_slew 0.05:0.05 -rise_slack 0.2741:3.6961 -fall_delay 2.1819:2.1819 -fall_slew 0.05:0.05 -fall_slack 0.237:3.6948 -clock CK1_1 -clock_freq 100 -type data
```
- The following commands reset the TWF values for the "DFF11/DFF/Q" pin to default, and then sets the rise/fall delay:  

```
set_twf_attribute -clear_twf_attr -pin "DFF11/DFF/Q"
set_twf_attribute -pin "DFF11/DFF/Q" -rise_delay 0.35 -fall_delay 0.45 -clock CLK1
```

## set\_virtual\_clock\_network\_parameters

```
set_virtual_clock_network_parameters
[-help]
[-cell cell_name]
[-clock list_of_clocks]
[-library library_name]
[-max_fanout value]
[-wire_load_model wireload_model]
[-reset]
```

Specifies to estimate power for a virtual clock tree. When the command is specified, the software reports the power of the clock network based on the buffers inserted in the clock tree. It also reports the number of buffers used for implementing the clock tree and the depth of the clock tree.

You can use this command for fast power estimation of clock network power of a pre-CTS netlist.

This command should be invoked before running the `report_power` command.

### Parameters

|                                              |                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-help</code>                           | Outputs a brief description that includes the type and default information for each <code>set_virtual_clock_network_parameters</code> parameter.<br><br>For a detailed description of the command and all of its parameters, use the <code>man</code> command <code>man set_virtual_clock_network_parameters</code> . |
| <code>-cell cell_name</code>                 | Specifies the name of the buffer or inverter cell that has to be used to build the clock tree.                                                                                                                                                                                                                        |
| <code>-clock list_of_clocks</code>           | Specifies the list of clocks for which the clock tree needs to be implemented. The default is for all the clocks.                                                                                                                                                                                                     |
| <code>-library library_name</code>           | Specifies the name of the library which contains the cell.                                                                                                                                                                                                                                                            |
| <code>-max_fanout value</code>               | Specifies the maximum fanout that the buffer cell can drive.                                                                                                                                                                                                                                                          |
| <code>-wire_load_model wireload_model</code> | Specifies the wire load model used to compute the wire load model for the nets in the clock network.                                                                                                                                                                                                                  |
| <code>-reset</code>                          | Resets all specified parameters back to default values.                                                                                                                                                                                                                                                               |

## Example

The following command specifies to use the cell `BUFX12`, with a maximum fanout value of 2, to build a virtual clock tree:

```
tempus> set_virtual_clock_network_parameters -cell { BUFX12 } -library typical -max_fanout 2
-wire_load_model BOX0
```

## Related Topics

- "[Static Power, IRdrop and EM Analysis](#)" in the *Voltus User Guide*

## Related Commands

- `report_power`
- `set_power_analysis_mode`



# write\_power\_constraints

```
write_power_constraints
[-help]
[-outfile filename]
```

Writes out all the power constraints such as the power analysis mode, activity file specifications, simulation period specifications, and so on, into a file. You can use this command at any time during the session to dump out the power commands/options specified till that point.

You can source this file into other Voltus/Tempus/Innovus sessions in order to use the same settings.

## Parameters

|                      |                                                                                                                                                                                                                                                                                |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -help                | Outputs a brief description that includes the type and default information for each <code>write_power_constraints</code> parameter.<br><br>For a detailed description of the command and all of its parameters, use the man command <code>man write_power_constraints</code> . |
| -outfile<br>filename | Specifies the output file into which the power constraints are to be written.                                                                                                                                                                                                  |

## Example

```
tempus> write_power_constraints -outfile test1.tcl
```

## Related Commands

- `set_power_analysis_mode`
- `read_activity_file`

## write\_tcf

```
write_tcf file
[-block block_name]
[-exclude_hier_insts list_of_instances]
[-primary_input]
[-seq_output]
[-pin]
[-macro_output]
```

Writes out the propagated switching activity information to a toggle count format (TCF) file. The TCF file contains switching activity information, toggle count information, and the probability of the net or pin being in the logic1 state.

You must specify this command before `report_power`.

### Parameters

|                                                           |                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-block <i>block_name</i></code>                     | <p>Specifies to generate the block-level TCF from the top-level design. When specified, it writes out the block-level duty and activity in the generated TCF file.</p> <p>The use model of this parameter is:</p> <pre>write_tcf -pin -block <i>blockname</i> <i>TCF_filename</i></pre>                      |
| <code>-exclude_hier_insts <i>list_of_instances</i></code> | <p>Excludes the specified hierarchical modules or instances when writing out a pin-based TCF. This parameter allows you to ignore the activity generated for the excluded instances.</p> <p>The use model of this parameter is:</p> <pre>write_tcf -pin -exclude_hier_insts {<i>list_of_instances</i>}</pre> |
| <code>-macro_output</code>                                | <p>Specifies to generate a TCF file that includes all the activities for the macro outputs in the design.</p>                                                                                                                                                                                                |
| <code>-pin</code>                                         | <p>Specifies to write out pin-based TCF.</p>                                                                                                                                                                                                                                                                 |
| <code>-primary_input</code>                               | <p>Generates a TCF file that includes only the primary inputs in the design.</p>                                                                                                                                                                                                                             |
| <code>-seq_output</code>                                  | <p>Generates a TCF file that includes all the activities for the sequential outputs in the design.</p>                                                                                                                                                                                                       |
| <code><i>file</i></code>                                  | <p>Specifies the name of the output TCF file.</p>                                                                                                                                                                                                                                                            |

## Example

- The following command writes the propagated switching activity information to TCF file mmm.tcf:

```
tempus> write_tcf mmm.tcf
```

- The following command writes a pin-based TCF for the whole design:

```
tempus> write_tcf -pin pin_design.tcf
```

- The following command writes out a TCF which only includes activity information for the primary inputs and sequential outputs in the design:

```
tempus> write_tcf -primary_input -seq_output Power.tcf
```

## Related Commands

- [report\\_power](#)

# Appendix - Preference Variables

## Preference Variables

The following table lists the preference variables you can set using the `set_preference` command and their GUI equivalent in the Preferences form:

| Variable Name           | GUI Equivalent in Preferences Form |                                                                                                                                                                              | Type    | Range/Default                                                                                |
|-------------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------------------------------------------------------------------------------------------|
|                         | Tab Name                           | Option Name                                                                                                                                                                  |         |                                                                                              |
| DesignName              | <b>Design</b>                      | <i>Design Name</i>                                                                                                                                                           | String  | /spc/spc4/FE_                                                                                |
| DesignHierChar          |                                    | <i>Design Hierarchical Character</i>                                                                                                                                         | String  | /                                                                                            |
| DEFHierChar             |                                    | <i>DEF Hierarchical Character</i>                                                                                                                                            | String  | /                                                                                            |
| PDEFBusDelim            |                                    | <i>PDEF Bus Delimiter</i>                                                                                                                                                    |         | []                                                                                           |
| LecDofile               |                                    | <i>Write Conformal/LEC dofile When Design is Saved</i>                                                                                                                       | Boolean | 0 - <b>False</b><br>1 - True                                                                 |
| CmdLogMode              |                                    | <i>Command Log Mode</i> <ul style="list-style-type: none"> <li>◦ <i>Command Only</i></li> <li>◦ <i>Command and Log</i></li> <li>◦ <i>Command, Log, and Screen</i></li> </ul> | Enum    | 0 - <i>Command Only</i><br>1 - <b>Command and Log</b><br>2 - <i>Command, Log, and Screen</i> |
| LogTypeInCmd            |                                    | <i>Log Type-In Command</i>                                                                                                                                                   | Boolean | 0 - <b>False</b><br>1 - True                                                                 |
| logviewer               |                                    | <i>Invoke viewLog at Start Up</i>                                                                                                                                            | Boolean | 0 - <b>False</b><br>1 - True                                                                 |
| EnableRectilinearDesign |                                    | <i>Enable Rectilinear Design</i>                                                                                                                                             | Boolean | 0 - <b>False</b><br>1 - True                                                                 |

**Tempus Text Command Reference**  
Appendix - Preference Variables

|                      |                |                                                |         |                                    |
|----------------------|----------------|------------------------------------------------|---------|------------------------------------|
| ShowFPObjInPlace     | <b>Display</b> | <i>Show Floorplan Objects in Physical View</i> | Boolean | <b>0 - False</b><br>1 - True       |
| ShowAllFence         |                | <i>Show All Constraint</i>                     | Boolean | <b>0 - False</b><br>1 - True       |
| DisplayPtnPin        |                | <i>Display Partition Pin</i>                   | Boolean | 0 - False<br><b>1 - True</b>       |
| ShowAttrPopup        |                | <i>Show Attributes Popup</i>                   | Boolean | 0 - False<br><b>1 - True</b>       |
| ShowUnplacedInst     |                | <i>Display Unplaced Instances</i>              | Boolean | <b>0 - False</b><br>1 - True       |
| ShowRectilinearPad   |                | <i>Show Rectilinear Pad Cell</i>               | Boolean | <b>0 - False</b><br>1 - True       |
| DisplayCellPin       |                | <i>Display Instance Pin</i>                    | Boolean | <b>0 - False</b><br>1 - True       |
| ShowLefLayerName     |                | <i>Show Lef Layer Name</i>                     | Boolean | 0 - False<br><b>1 - True</b>       |
| AutoRedraw           |                | <i>Auto-redraw after Layer Change</i>          | Boolean | 0 - False<br><b>1 - True</b>       |
| EnlargeLogicalPin    |                | <i>Enlarge Logical Pin In Fit View</i>         | Boolean | <b>0 - False</b><br>1 - True       |
| InstDisplayThreshold |                | <i>Instance Display Threshold</i>              | Real    | <b>0.0</b>                         |
| WireDisplayThreshold |                | <i>Wire Display Threshold</i>                  | Integer | <b>0</b>                           |
| WireDisplayCheck2D   |                | <i>Apply Threshold in 2D</i>                   | Boolean | 0 - False<br><b>1 - True</b>       |
| percentageOfPan      |                | <i>Percentage of Window to Pan</i>             | Enum    | 10, 20, 30, 40, 50, 60, 70, 80, 90 |
| scaleOfZoom          |                | <i>Scale of Zoom In/Out</i>                    | Real    | <b>2.0</b>                         |
| zoomPrevCount        |                | <i>Number of Previous Zoom Saved</i>           |         | <b>1</b> to 6                      |

**Tempus Text Command Reference**  
Appendix - Preference Variables

|                     |             |                                                                                                                                                                                                            |         |                                                      |
|---------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------------------------------------------------------|
| MinFPModuleSize     |             | <i>Min. Floorplan Module Size</i>                                                                                                                                                                          | Integer | <b>100</b>                                           |
| ShowParentModule    |             | <i>Show Parent Module With Level</i>                                                                                                                                                                       | Integer | <b>0</b>                                             |
| ShowChildPartition  |             | <i>Show Child Partition With Level</i>                                                                                                                                                                     | Integer | <b>0</b>                                             |
| AutoDetailDisplay   |             | <i>Automatic Detail Display for Large Design</i> <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li>Special Wire</li> <li>Regular Wire</li> </ul> </li> </ul>                  | Boolean | <b>0 - False</b><br>1 - True                         |
| DetailDisplayFactor |             | <i>Factor</i>                                                                                                                                                                                              | Integer | <b>16</b>                                            |
| DisplayRelFPlan     | <b>Edit</b> | <i>Honor Relative Floorplan Constraints</i>                                                                                                                                                                | Boolean | 0 - False<br><b>1 - True</b>                         |
| SingleLayerMode     |             | <i>Single Layer Mode For Routing Blockage, Ptn Pin Blockage and Ptn Feed</i>                                                                                                                               | Boolean | <b>0 - False</b><br>1 - True                         |
| ShowDelBox          |             | <i>Show Delete Confirmation Box</i>                                                                                                                                                                        | Boolean | <b>0 - False</b><br>1 - True                         |
| StretchRestriction  |             | <i>Box Stretch Restrictions</i> <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li>No Restriction</li> <li>Maintain Area</li> <li>Maintain Aspect Ratio</li> </ul> </li> </ul> | Enum    | <b>0 - No Restr:</b><br>1 - Maintain<br>2 - Maintain |
| StretchRectilinear  |             | <i>Rectilinear Stretch Restrictions</i> <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li>No Restriction</li> <li>Maintain Area</li> </ul> </li> </ul>                        |         | <b>0 - No Restr:</b><br>1 - Maintain                 |
| MoveRestriction     |             | <i>Move Restrictions</i> <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li>No Restrictions</li> <li>Orthogonal</li> </ul> </li> </ul>                                         |         | <b>0 - No Restr:</b><br>1 - Orthogonal               |

|                       |                  |                                                                                                                                                                                                                                                                                     |         |                                                                           |
|-----------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------------------------------------------------------------------------|
| GuideSnapRule         | <b>Floorplan</b> | <b>Snap Guides/Regions/Fences to</b> <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Std Row and Metal 2 Pitch</i></li> <li><i>Manufacture Grid</i></li> <li><i>User-defined Grid</i></li> <li><i>Block Placement Grid</i></li> </ul> </li> </ul> | Enum    | <b>stdRow - Std</b><br><i>user-define</i>                                 |
| BlockSnapRule         |                  | <b>Snap Macros/Blackboxes to</b> <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Manufacture Grid</i></li> <li><i>Std Row and Metal 2 Pitch</i></li> <li><i>User-defined Grid</i></li> <li><i>Block Placement Grid</i></li> </ul> </li> </ul>     | Enum    | <b>0 - Manufacture</b><br>1 - Std Row &<br>2 - User-defi<br>3 - Placement |
| SnapAllCorners        |                  | <b>Snap All Corners to Grid</b>                                                                                                                                                                                                                                                     | Boolean | <b>0 - False</b><br>1 - True                                              |
| ConstraintUserXGrid   |                  | <b>User-defined Grid:</b> <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>X</i></li> </ul> </li> </ul>                                                                                                                                            | Real    | <b>0.0</b>                                                                |
| ConstraintUserXOffset |                  | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Offset</i></li> </ul> </li> </ul>                                                                                                                                                                 | Real    | <b>0.0</b>                                                                |
| ConstraintUserYGrid   |                  | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Y</i></li> </ul> </li> </ul>                                                                                                                                                                      | Real    | <b>0.0</b>                                                                |
| ConstraintUserYOffset |                  | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Offset</i></li> </ul> </li> </ul>                                                                                                                                                                 | Real    | <b>0.0</b>                                                                |

**Tempus Text Command Reference**  
Appendix - Preference Variables

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|                          |                                                                                                                                                                                                |         |                                |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------------------------------|
| UserGridUnit             | <i>Unit</i> <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Micron</i></li> <li><i>Track</i></li> </ul> </li> </ul>                                          | Boolean | <b>0 - Micron</b><br>1 - Track |
| MoveMacrosWithGuide      | <i>Descendant Macros Move with Their Ancestor Modules for Constraints</i> <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Guide</i></li> </ul> </li> </ul>   | Boolean | <b>0 - False</b><br>1 - True   |
| MoveMacrosWithRegion     | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Region</i></li> </ul> </li> </ul>                                                                            | Boolean | <b>0 - False</b><br>1 - True   |
| MoveMacrosWithFence      | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Fence</i></li> </ul> </li> </ul>                                                                             | Boolean | 0 - False<br><b>1 - True</b>   |
| MoveStdCellWithGuide     | <i>Descendant Std Cell Move with Their Ancestor Modules for Constraints</i> <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Guide</i></li> </ul> </li> </ul> | Boolean | <b>0 - False</b><br>1 - True   |
| MoveStdCellWithRegion    | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Region</i></li> </ul> </li> </ul>                                                                            | Boolean | <b>0 - False</b><br>1 - True   |
| MoveStdCellWithFence     | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Fence</i></li> </ul> </li> </ul>                                                                             | Boolean | <b>0 - False</b><br>1 - True   |
| MovePreplacedStdCellOnly | <i>Move Preplaced Cell Only</i>                                                                                                                                                                | Boolean | <b>0 - False</b><br>1 - True   |
|                          |                                                                                                                                                                                                |         |                                |



**Tempus Text Command Reference**  
Appendix - Preference Variables

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|                          |                  |                                                                                                                                                                                     |         |                                      |
|--------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------------------------------------|
| SelectByArea             | <b>Selection</b> | <i>Select by Area</i> <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Enclosed by Box</i></li> <li><i>Intersecting Box</i></li> </ul> </li> </ul> | Boolean | <b>0 - Enclosed</b><br>1 - Intersect |
| SelectByLine             |                  | <i>Select by Line</i>                                                                                                                                                               | Boolean | <b>0 - False</b><br>1 - True         |
| SelectStickyMode         |                  | <i>Sticky Mode</i>                                                                                                                                                                  | Boolean | <b>0 - False</b><br>1 - True         |
| SelectNetWhenSelectPin   |                  | <i>Hilite Net When Selecting a Pin</i>                                                                                                                                              | Boolean | <b>0 - False</b><br>1 - True         |
| HiliteNetWhenQueryObj    |                  | <i>Hilite Nets When Querying an Instance</i>                                                                                                                                        | Boolean | 0 - False<br><b>1 - True</b>         |
| HiliteShapeWhenSelectNet |                  | <i>Hilite Pin Shapes When Selecting a Net</i>                                                                                                                                       | Boolean | <b>0 - False</b><br>1 - True         |
| QueryWireNet             |                  | <i>Query Nets When Wires are Selected</i>                                                                                                                                           | Boolean | <b>0 - False</b><br>1 - True         |
| QuerySkipInst            |                  | <i>Skip Objects by Auto Query</i> <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Cell</i></li> </ul> </li> </ul>                                 | Boolean | <b>0 - False</b><br>1 - True         |
| QuerySkipInstObs         |                  | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Obs in Cell</i></li> </ul> </li> </ul>                                                            | Boolean | <b>0 - False</b><br>1 - True         |
| QuerySkipInstPin         |                  | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Pin in Cell</i></li> </ul> </li> </ul>                                                            | Boolean | <b>0 - False</b><br>1 - True         |
| QuerySkipRegular         |                  | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Regular Wire</i></li> </ul> </li> </ul>                                                           | Boolean | <b>0 - False</b><br>1 - True         |
|                          |                  |                                                                                                                                                                                     |         |                                      |

|                      |                                                                                                                        |         |                              |
|----------------------|------------------------------------------------------------------------------------------------------------------------|---------|------------------------------|
| QuerySkipSpecial     | <ul style="list-style-type: none"><li>•<ul style="list-style-type: none"><li>◦ <i>Special Wire</i></li></ul></li></ul> | Boolean | <b>0 - False</b><br>1 - True |
| WinSelectMargin      | <i>Window Select Margin</i>                                                                                            | Integer | <b>8</b>                     |
| HighlightColorNumber | <i>Number of Highlight Color</i>                                                                                       | Enum    | <b>8, 12, 16</b>             |

**Tempus Text Command Reference**  
Appendix - Preference Variables

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|                           |                          |                                                |         |                              |
|---------------------------|--------------------------|------------------------------------------------|---------|------------------------------|
| MinFlightLineWidth        | <b><i>Flightline</i></b> | <i>Minimum Flightline Width</i>                | Integer | <b>1</b>                     |
| MaxFlightLineNetTerms     |                          | <i>Maximum Flightline Net Terms</i>            | Integer | <b>500</b>                   |
| DisplayPlaceFlightLine    |                          | <i>Show Flightline In Physical View</i>        | Boolean | 0 - False<br><b>1 - True</b> |
| ShowNumberBlockConnection |                          | <i>Show Number of Connections For Block</i>    | Boolean | <b>0 - False</b><br>1 - True |
| FlightLineInMove          |                          | <i>Draw Flightline When Moving Instance(s)</i> | Boolean | 0 - False<br><b>1 - True</b> |
| ShowConnectionInOutNumber |                          | <i>Show Input/Output Connection Number</i>     | Boolean | <b>0 - False</b><br>1 - True |
| ShowFlightLineTermMark    |                          | <i>Show Pin's Input/Output Mark</i>            | Boolean | <b>0 - False</b><br>1 - True |
| ShowBothInputConnection   |                          | <i>Show Both Input Pins Connection</i>         | Boolean | <b>0 - False</b><br>1 - True |
| ShowNetWeightConnection   |                          | <i>Show Connection With Netweight 0</i>        | Boolean | 0 - False<br><b>1 - True</b> |
| ShowFlightLineThruPtnPin  |                          | <i>Show Flightlines Through Partition Pins</i> | Boolean | <b>0 - False</b><br>1 - True |
| noInsideMacro             |                          | <i>Don't Show Flightline to Inside Macro</i>   | Boolean | <b>0 - False</b><br>1 - True |
| SkipBufferFlightline      |                          | <i>Skip Buffer/Invert Cell</i>                 | Boolean | <b>0 - False</b><br>1 - True |
| NoClockFlightline         |                          | <i>Don't Show Clock Net</i>                    | Boolean | <b>0 - False</b><br>1 - True |
| OnlyClockFlightline       |                          | <i>Show Clock Net Only</i>                     | Boolean | <b>0 - False</b><br>1 - True |
|                           |                          |                                                |         |                              |

|                                                      |                                                                                                                                                                                                  |                                                                                                                                                                                                    |                                                        |                                                        |               |        |             |               |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------|---------------|--------|-------------|---------------|
| DrawFlightlineLast                                   | <div></div>                                                                                                                                                                                      | <i>Draw Flightline on Top of Selection</i>                                                                                                                                                         | Boolean                                                | <b>0 - False</b><br>1 - True                           |               |        |             |               |
| OnlyBundleNetFlight                                  |                                                                                                                                                                                                  | <i>Only Show Net Bundle Flight Line</i>                                                                                                                                                            | Boolean                                                | <b>0 - False</b><br>1 - True                           |               |        |             |               |
| ExclusiveFlight                                      |                                                                                                                                                                                                  | <i>Show Connection Only Between Selected Objects</i>                                                                                                                                               | Boolean                                                | 0 - False<br><b>1 - True</b>                           |               |        |             |               |
| ConnectionColorType                                  |                                                                                                                                                                                                  | <i>Single Color</i>                                                                                                                                                                                | Enum                                                   | <b>0 - Single Color</b>                                |               |        |             |               |
|                                                      |                                                                                                                                                                                                  | <i>Different Color by Type of Connections</i>                                                                                                                                                      |                                                        |                                                        |               |        |             |               |
|                                                      |                                                                                                                                                                                                  | <i>Different Color by Number of Connections</i>                                                                                                                                                    |                                                        |                                                        |               |        |             |               |
| InputConnectionColor                                 |                                                                                                                                                                                                  | <table><tr><td><i><b>Different Color by Type of Connections</b></i></td><td><i><b>Different Color by Number of Connections</b></i></td></tr><tr><td><i>Input</i></td><td>Small</td></tr></table>   | <i><b>Different Color by Type of Connections</b></i>   | <i><b>Different Color by Number of Connections</b></i> | <i>Input</i>  | Small  |             | <b>green</b>  |
| <i><b>Different Color by Type of Connections</b></i> |                                                                                                                                                                                                  | <i><b>Different Color by Number of Connections</b></i>                                                                                                                                             |                                                        |                                                        |               |        |             |               |
| <i>Input</i>                                         |                                                                                                                                                                                                  | Small                                                                                                                                                                                              |                                                        |                                                        |               |        |             |               |
| OutputConnectionColor                                |                                                                                                                                                                                                  | <table><tr><td><i><b>Different Color by Type of Connections</b></i></td><td><i><b>Different Color by Number of Connections</b></i></td></tr><tr><td><i>Output</i></td><td>Medium</td></tr></table> | <i><b>Different Color by Type of Connections</b></i>   | <i><b>Different Color by Number of Connections</b></i> | <i>Output</i> | Medium |             | <b>yellow</b> |
| <i><b>Different Color by Type of Connections</b></i> | <i><b>Different Color by Number of Connections</b></i>                                                                                                                                           |                                                                                                                                                                                                    |                                                        |                                                        |               |        |             |               |
| <i>Output</i>                                        | Medium                                                                                                                                                                                           |                                                                                                                                                                                                    |                                                        |                                                        |               |        |             |               |
| InoutConnectionColor                                 | <table><tr><td><i><b>Different Color by Type of Connections</b></i></td><td><i><b>Different Color by Number of Connections</b></i></td></tr><tr><td><i>Inout</i></td><td>Large</td></tr></table> | <i><b>Different Color by Type of Connections</b></i>                                                                                                                                               | <i><b>Different Color by Number of Connections</b></i> | <i>Inout</i>                                           | Large         |        | <b>pink</b> |               |
| <i><b>Different Color by Type of Connections</b></i> | <i><b>Different Color by Number of Connections</b></i>                                                                                                                                           |                                                                                                                                                                                                    |                                                        |                                                        |               |        |             |               |
| <i>Inout</i>                                         | Large                                                                                                                                                                                            |                                                                                                                                                                                                    |                                                        |                                                        |               |        |             |               |
|                                                      |                                                                                                                                                                                                  |                                                                                                                                                                                                    |                                                        |                                                        |               |        |             |               |

| MixtureConnectionColor  | <table><tr><th>Single Color</th><th>Different Color by Type of Connections</th></tr><tr><td>&lt;color&gt;</td><td>Inout</td></tr></table>                                 | Single Color | Different Color by Type of Connections | <color> | Inout |  | blue |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------------------------|---------|-------|--|------|
| Single Color            | Different Color by Type of Connections                                                                                                                                    |              |                                        |         |       |  |      |
| <color>                 | Inout                                                                                                                                                                     |              |                                        |         |       |  |      |
| ShowConnectionWithWidth | Different Width by Number of Connections                                                                                                                                  | Boolean      | 0 - False<br>1 - True                  |         |       |  |      |
| FLWidthThresholdLow     | Number of Connections <ul style="list-style-type: none"><li>Small - Fewer than specified low threshold</li></ul>                                                          | Integer      | 20                                     |         |       |  |      |
| FLWidthThresholdHigh    | <ul style="list-style-type: none"><li>Medium - Equal to or between than specified low and high thresholds</li><li>Large - Greater than specified high threshold</li></ul> | Integer      | 80                                     |         |       |  |      |
| NoInstFlightLine        | Don't Show Connection to Unselected Object <ul style="list-style-type: none"><li>Instance</li></ul>                                                                       | Boolean      | 0 - False<br>1 - True                  |         |       |  |      |
| NoBlockFlightLine       | <ul style="list-style-type: none"><li>Block</li></ul>                                                                                                                     | Boolean      | 0 - False<br>1 - True                  |         |       |  |      |
| NoModuleFlightLine      | <ul style="list-style-type: none"><li>Module</li></ul>                                                                                                                    | Boolean      | 0 - False<br>1 - True                  |         |       |  |      |
| NoBlackBlobFlightLine   | <ul style="list-style-type: none"><li>Black Blob</li></ul>                                                                                                                | Boolean      | 0 - False<br>1 - True                  |         |       |  |      |

**Tempus Text Command Reference**  
Appendix - Preference Variables

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|                           |             |                                                                                                                                                                             |         |                              |
|---------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------------------------------|
| NoBlackBoxFlightLine      |             | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Black Box</i></li> </ul> </li> </ul>                                                      | Boolean | <b>0 - False</b><br>1 - True |
| InstanceText              | <b>Text</b> | <i>Instance Text</i> <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Instance Name</i></li> <li><i>Master Name</i></li> </ul> </li> </ul> |         | <b>Instance Master</b>       |
| ShowNetFullName           |             | <i>Show Full Net Name</i>                                                                                                                                                   | Boolean | <b>0 - False</b><br>1 - True |
| ShowNetNameWithLayerColor |             | <i>Show Net Name With Layer Color</i>                                                                                                                                       | Boolean | <b>0 - False</b><br>1 - True |
| ShowModuleText            |             | <i>Object Text Display</i> <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Module</i></li> </ul> </li> </ul>                              | Boolean | 0 - False<br><b>1 - True</b> |
| ShowAmoebaModuleText      |             | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Module in Amoeba</i></li> </ul> </li> </ul>                                               | Boolean | 0 - False<br><b>1 - True</b> |
| ShowRowSiteText           |             | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Row Site</i></li> </ul> </li> </ul>                                                       | Boolean | 0 - False<br><b>1 - True</b> |
| ShowIoPadText             |             | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>IO Pad</i></li> </ul> </li> </ul>                                                         | Boolean | 0 - False<br><b>1 - True</b> |
| ShowInstanceText          |             | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Instance</i></li> </ul> </li> </ul>                                                       | Boolean | 0 - False<br><b>1 - True</b> |
| ShowInstancePinText       |             | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Instance Pin</i></li> </ul> </li> </ul>                                                   | Boolean | 0 - False<br><b>1 - True</b> |
| ShowBlackBlobText         |             | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Black Blob</i></li> </ul> </li> </ul>                                                     | Boolean | 0 - False<br><b>1 - True</b> |

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|                      |                                                                                                                                                                                          |         |                              |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------------------------------|
| ShowIoPinText        | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>IO Pin</i></li> </ul> </li> </ul>                                                                      | Boolean | 0 - False<br><b>1 - True</b> |
| ShowGroupText        | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Group</i></li> </ul> </li> </ul>                                                                       | Boolean | 0 - False<br><b>1 - True</b> |
| ShowBumpText         | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Bump</i></li> </ul> </li> </ul>                                                                        | Boolean | 0 - False<br><b>1 - True</b> |
| ShowClockTreeText    | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Clock Tree</i></li> </ul> </li> </ul>                                                                  | Boolean | 0 - False<br><b>1 - True</b> |
| ShowChannelText      | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Channel</i></li> </ul> </li> </ul>                                                                     | Boolean | 0 - False<br><b>1 - True</b> |
| ShowLefPortNumText   | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Lef Port Num</i></li> </ul> </li> </ul>                                                                | Boolean | 0 - False<br><b>1 - True</b> |
| ShowMacroSitePtnText | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Macro Site Ptn</i></li> </ul> </li> </ul>                                                              | Boolean | 0 - False<br><b>1 - True</b> |
| ShowSIPFingerText    | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>SIP Finger</i></li> </ul> </li> </ul>                                                                  | Boolean | 0 - False<br><b>1 - True</b> |
| ShowNetText          | <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li><i>Net</i></li> </ul> </li> </ul>                                                                         | Boolean | 0 - False<br><b>1 - True</b> |
| TextDisplaySize      | <p><i>Text Display Size</i></p> <ul style="list-style-type: none"> <li> <ul style="list-style-type: none"> <li>Small</li> <li>Medium</li> <li>Large</li> <li>Auto</li> </ul> </li> </ul> | Enum    | <b>s, m, l, a</b>            |

|              |                               |      |                |
|--------------|-------------------------------|------|----------------|
| TermCrossPix | <i>Term Cross Symbol Size</i> | Enum | <b>4, 6, 8</b> |
|--------------|-------------------------------|------|----------------|



# Innovus to Tempus Command Mapping

The following table lists the Innovus commands that are supported in Tempus and their equivalent Tempus commands.

| Innovus Command                       | Equivalent Tempus Command                      |
|---------------------------------------|------------------------------------------------|
| <b>Import and Export Commands</b>     |                                                |
| checkDesign                           | <a href="#"><u>check design</u></a>            |
| checkTimingLibrary                    | <a href="#"><u>check library</u></a>           |
| commitConfig                          | <a href="#"><u>read design</u></a>             |
| commitCPF                             | <a href="#"><u>read power domain</u></a>       |
| defIn                                 | <a href="#"><u>read def</u></a>                |
| freeDesign                            | <a href="#"><u>free design</u></a>             |
| loadConfig                            | <a href="#"><u>read design</u></a>             |
| loadCPF                               | <a href="#"><u>read power domain</u></a>       |
| loadLefFile                           | <a href="#"><u>read_lib -lef file_name</u></a> |
| loadTimingCon                         | <a href="#"><u>read sdc</u></a>                |
| unloadTimingCon                       | <a href="#"><u>reset sdc</u></a>               |
| restoreDesign                         | <a href="#"><u>read design</u></a>             |
| restoreOaDesign                       | <a href="#"><u>restore oa design</u></a>       |
| saveDesign                            | <a href="#"><u>save design</u></a>             |
| setIrDropInstVoltage                  | <a href="#"><u>read instance voltage</u></a>   |
| <b>Design Rules Checking Commands</b> |                                                |
| reportCapViolation                    | <a href="#"><u>report design rule</u></a>      |
| reportFanoutViolation                 | <a href="#"><u>report design rule</u></a>      |

|                                      |                                                 |
|--------------------------------------|-------------------------------------------------|
| reportTranViolation                  | <a href="#"><u>report_design_rule</u></a>       |
| setMaxCapPerFreq                     | <a href="#"><u>set_max_cap_per_freq</u></a>     |
| setMaxTranPerFreq                    | <a href="#"><u>set_max_tran_per_freq</u></a>    |
| <b>Timing Commands</b>               |                                                 |
| createUserDisableForCombLoop         | <a href="#"><u>write_loop_break_sdc</u></a>     |
| getAnalysisMode                      | <a href="#"><u>get_analysis_mode</u></a>        |
| getDelayCalMode                      | <a href="#"><u>get_delay_cal_mode</u></a>       |
| reportDelayCalculation               | <a href="#"><u>report_delay_calculation</u></a> |
| reportTimingDerate                   | <a href="#"><u>report_timing_derate</u></a>     |
| setAnalysisMode                      | <a href="#"><u>set_analysis_mode</u></a>        |
| setDelayCalMode                      | <a href="#"><u>set_delay_cal_mode</u></a>       |
| setOpCond                            | <a href="#"><u>set_op_cond</u></a>              |
| setTimingLibrary                     | <a href="#"><u>set_timing_library</u></a>       |
| spefIn                               | <a href="#"><u>read_spef</u></a>                |
| timeDesign                           | <a href="#"><u>report_timing</u></a>            |
| writeTimingCon                       | <a href="#"><u>write_sdc</u></a>                |
| <b>General Commands</b>              |                                                 |
| setLicenseCheck                      | <a href="#"><u>set_license_check</u></a>        |
| setMessageLimit                      | <a href="#"><u>set_message_limit</u></a>        |
| suppressMessage                      | <a href="#"><u>suppress_message</u></a>         |
| unsetMessageLimit                    | <a href="#"><u>unset_message_limit</u></a>      |
| unsuppressMessage                    | <a href="#"><u>unsuppress_message</u></a>       |
| <b>Multi-CPU Processing Commands</b> |                                                 |
| get_distribute_host                  | <a href="#"><u>get_distribute_host</u></a>      |
| getMultiCpuUsage                     | <a href="#"><u>get_multi_cpu_usage</u></a>      |

|                            |                            |
|----------------------------|----------------------------|
| setDistributeHost          | <u>set_distribute_host</u> |
| setMultiCpuUsage           | set_multi_cpu_usage        |
| <b>Manual ECO Commands</b> |                            |
| deleteNet                  | delete_net                 |
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**Note:** For details of the Database Access Tcl Commands see the chapter *Basic Database Access Tcl Commands* in the *Innovus Text Command Reference Guide*.

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timing\_library\_scale\_aocv\_to\_socv\_to\_n\_sigma  
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unspecify ilm

unsuppress\_message  
update\_constraint\_mode  
update\_glitch  
update\_library\_set  
update\_rc\_corner

update\_analysis\_view  
update\_delay\_corner  
update\_io\_latency  
update\_model\_context  
update\_timing

## W

write\_annotated\_transition  
write\_def  
write\_eco\_opt\_mode  
write\_global\_slack\_report  
write\_ilm  
write\_ldb  
write\_loop\_break\_sdc  
write\_model\_timing  
write\_path\_descriptions  
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write\_timing\_windows

write\_category\_summary  
write\_eco  
write\_flow\_template  
write\_global\_slack\_worst\_trigger\_path\_on\_clocks  
write\_ipd\_scope  
write\_load  
write\_model\_context  
write\_noise\_eco  
write\_power\_constraints  
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